

The SCIENCE — OF — FAIRNESS



STEPHEN TAKOWSKY

THE SCIENCE OF FAIRNESS

Publishing Details

The Science of Fairness

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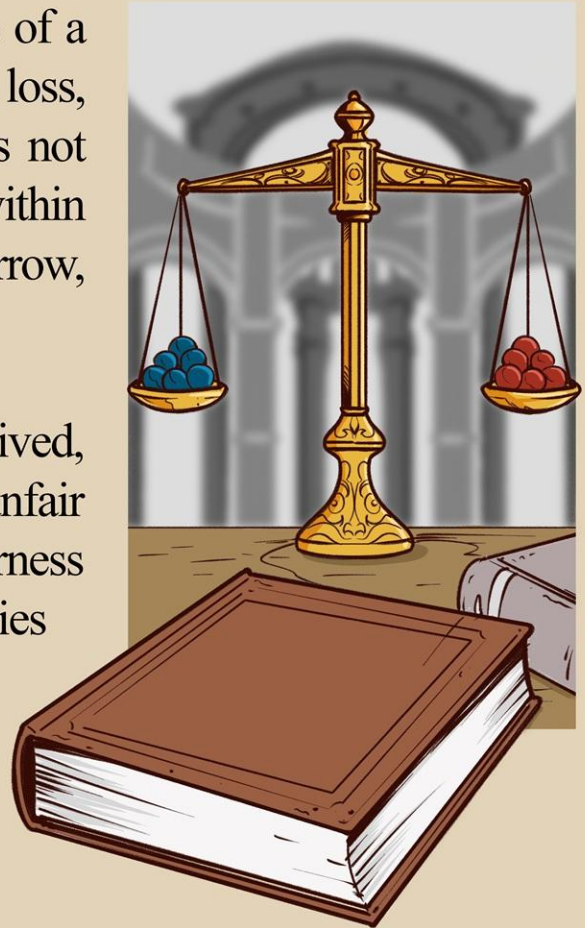
Series Map

THE LAW OF FAIRNESS

A new way of seeing every life.

We judge fairness by the visible shape of a life: health and hardship, love and loss, opportunity and misfortune. Yet life is not lived from the outside. It is lived within consciousness, through joy and sorrow, pleasure and pain, peace and fear.

Could life be fair where it is actually lived, even when it appears profoundly unfair from the outside? The Law of Fairness proposes that every felt experience carries weight according to its intensity and duration. Across the completed conscious life, total positive and negative feeling exactly balance.



Picture a scale open for a lifetime. Joy and relief add weight to one side; pain and grief to the other. It may tip heavily for years, but when the conscious life is complete, the two sides balance. Each life carries its own scale. No life balances another.

Lives may differ enormously in length, choices, pleasures, wounds, and circumstances. The scale weighs neither wealth nor virtue, status nor faith. It weighs felt experience alone. Circumstances tell the visible story of a life, but feeling is how that story is lived. The Law asks us not to mistake the visible story for the whole.

WHAT IF THE LAW IS REAL?

A new meaning of utopia.

If the Law of Fairness is real, utopia cannot mean endless pleasure. More comfort, luxury, or consumption does not make a society ideal. If conscious lives end with no lasting surplus of positive feeling, utopia must mean something deeper and steadier than limitless happiness.

“There is no free happiness” does not mean joy is wrong, pain is payment, or suffering is holy. Pleasure alone cannot define a good life, nor appearances a good society. The goal is joy without excess. A fairer world would reduce needless suffering, poverty, disease, cruelty, corruption, and abuse while protecting love, beauty, rest, play, choice, and joy. Moderation restrains excess, work builds worthy goals, and stewardship preserves the world so future generations inherit something worth protecting.

If the Law is real, its deepest equality lies beneath wealth, status, power, culture, and belief. Lives may differ while reaching the same final balance of felt experience. These ten books follow that question from science into life, spirituality, government, cities, law, transition, and peace, exploring how humanity might build a lasting utopia.



THE TEN BOOKS OF FAIRNESS

A complete path from truth to practice.

THE LAW OF FAIRNESS



Explores the core question, defines key terms, and sets the ethical bounds of the series.

THE SCIENCE OF FAIRNESS



Evaluates the Law through evidence, measurement, rival models, protocols, and falsification.

THE UTOPIA OF FAIRNESS



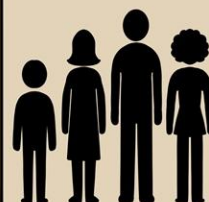
Brings the series into a mature view of what a peaceful world could become.

THE TRANSITION OF FAIRNESS



Asks how a move toward utopia could unfold with care, step by step, under real limits.

THE LIFE OF FAIRNESS



Brings the Law into daily life and asks what it means for how we live.

THE RELIGION OF FAIRNESS



Explores faith, world religions, and the optional path of Fairness.

THE GOVERNMENT OF FAIRNESS



Examines civic structure and the public institutions that give it form in practice.

THE CITY OF FAIRNESS



Builds a civic space for learning, service, and peace with privacy kept safe.

THE CONSTITUTION OF FAIRNESS



Sets out the rights, rules, paths, and checks a lasting utopia would need.

THE PEACE OF FAIRNESS



Builds the base for lasting peace and a fairer future for all humanity.

The series moves from the deepest question about reality to the structures that make a more peaceful world.

THE WORK AHEAD

A shared mission to build a better world.

Read the books, share them, and bring the Law of Fairness into classrooms, research groups, reading circles, and community conversations. Important ideas grow stronger when people examine them together, challenge assumptions, compare perspectives, and turn understanding into shared effort.

Scientists, researchers, clinicians, philosophers, and other experts can help test the Law, refine its measurements, improve its methods, and compare it with strong rivals. Serious criticism is essential. It exposes weaknesses, sharpens the questions, and helps reveal what should endure, what needs revision, and what should be rejected.

Beyond the hypothesis lies a practical challenge: building a world with less corruption and needless suffering, and more room for dignity, freedom, meaning, beauty, and opportunity. If this project speaks to you, bring your knowledge, experience, curiosity, and determination. Progress begins when people from different paths meet, share what they know, and build what survives scrutiny.



Help us make this generation
the one that brought peace to our world.

How to Read This Book

Some questions are important enough to deserve the strongest science we can bring to them. Per-Stream Terminal Valence Closure, or PTVC, asks whether each qualifying conscious stream's complete pre-endpoint history has equal finite positive and negative phenomenal magnitude under the bearer, routing, and endpoint rules defined here. This volume develops that hypothesis through explicit models, observations, and comparisons. The preceding Series Map introduces the series; the chapters that follow develop its scientific questions and tests.

Read the book as a sequence of scientific opportunities and the tests that make them informative. It begins with claim status and ethical authority, then defines the formal object without rewriting it. Later Parts address valence measurement, neutral origin, conscious-stream identity, endpoint construction, missing data, measured-to-latent bridges, candidate routes, study families, and decision rules. Each validated layer supplies a foundation for the next. Instrument validation establishes what can be measured; mechanism comparison identifies which processes explain the observations; and terminal inference assesses the declared closure claim with its remaining uncertainty.

Treat the rival models as central characters. Adaptation, opponent processes, homeostasis, allostasis, predictive processing, computational mood, ordinary mean reversion, conditioned bridges, endpoint selection, memory reconstruction, and observation floors can imitate pieces of the pattern. The Law deserves support only if it survives fair comparison against serious alternatives given the same data, flexibility, and test

budget. A rival victory is useful knowledge, not a failure of the scientific mission.

Read each result by the layer it tests. A study may validate a measure, identify a neutral point, recover a simulated route, narrow a reference class, or favor an explanation without adjudicating complete-stream PTV. Stronger verdicts require independently defined streams and endpoints, credible experiential coverage, honest uncertainty, and a measurement-to-hypothesis bridge.

Use the Table of Contents and Part and chapter introductions to navigate. This technical companion to The Law of Fairness is advanced, especially in its formal and mathematical chapters. Readers seeking an accessible entry may begin with The Law of Fairness, then return for the models, evidence standards, and research program. This volume also stands independently: general readers can follow the conceptual path, while specialists can move directly to their fields and shared methods.

Read as both skeptic and builder. Ask where the hypothesis can break and what theorem, instrument, dataset, simulation, cohort, or humane experiment could decide the issue. Laboratories can contribute immediately by choosing a tractable component and applying methods strong enough to produce a genuinely informative result. Care remains prior to evidence: no harm induction, no delayed treatment, no human-worth score, and no forced meaning. The goal is a research program capable of discovering how far the Law reaches.

Part I — Scientific Jurisdiction and Synopsis

Science begins when a compelling idea accepts a disciplined form of vulnerability. *The Science of Fairness* opens at that threshold. It receives Per-Stream Terminal Valence Closure, or PTVC, as a formally specified candidate boundary hypothesis and asks how a claim about complete conscious streams could become measurable, comparable, corrigible, and genuinely answerable.

The hypothesis is ambitious in a precise way. On its primary finite signed-measure branch, each separately routed maximal qualifying stream is proposed to carry equal finite positive and negative phenomenal magnitudes over its complete strict-pre-endpoint domain. The equality is intrastream and nontransferable, so unequal lives, unequal opportunities, unequal welfare, and unequal gross experience remain fully compatible with the formal target.

That compact statement opens a large scientific landscape. A responsible test specifies the phenomenal representation, neutral origin, sign, scalarity, exposure rule, bearer, ownership and routing, truth-level endpoint, protocol endpoint, observation process, missingness model, measured-to-latent bridge, estimand, estimator, reference class, modality, rival set, and verdict rule. Explicit connections among these layers make their respective roles clear.

The purpose of a separate science volume is therefore constructive. It gives the hypothesis one place where every dependency can be developed at field-expert depth and every improvement can be inherited without forcing the rest of the series to become a technical monograph. *The Law of Fairness*

remains the public formal root; this volume becomes the governed scientific body around it.

The first opportunity is representational. A question about felt experience can be scientifically meaningful only if positive, negative, mixed, and neutral-relative states can be modeled without manufacturing the result through an arbitrary origin or completed-record normalization. Measurement is not clerical support for the theory. It is one of the main discoveries the research program must earn.

The second opportunity is temporal. PTVC concerns a complete strict-pre-endpoint domain, not a final interview or a convenient study stop. Endpoint construction, stream continuity, return, interruption, resuscitation, and terminal-window access therefore become central scientific problems. A valid endpoint bridge can advance the inquiry; an invalid one can stop an inference without erasing the value of the surrounding work.

The third opportunity is comparative. Adaptation, opponent processes, allostasis, homeostasis, predictive processing, learning, selection, memory, observation attenuation, endpoint conditioning, and hidden-state models can imitate parts of the proposed pattern. Strong rivals are not obstacles placed around the contribution. They are the instruments that reveal whether the contribution has distinctive empirical content.

The fourth opportunity is cumulative. Measurement foundations can be improved before complete-stream adjudication becomes possible. Route models can be supported, revised, or retired while the target remains open. Model-recovery studies can show

whether an analysis can distinguish candidate worlds before the analysis is used to interpret empirical data. Each stage can create durable knowledge and open a stronger next test.

Scientific optimism in this volume is earned through that architecture. If PTVC survives these gates, science would gain evidence for a new boundary condition on conscious trajectories, opening work across phenomenology, psychometrics, consciousness, and stochastic dynamics. Early studies need not decide the whole hypothesis. Each must state its object, evidence role, rival pressure, uncertainty, and next decision. Severe testing and reproducible practice make a large conjecture more interesting by turning admiration into work (Mayo, 2018; Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019).

This Part establishes that scientific constitution before the book enters its specialized domains. Chapter 1 explains why the hypothesis needs a dedicated science. Chapter 2 classifies the kinds of claim at issue and the ways they may narrow or fail. Chapter 3 defines the ethical and authority boundaries that govern every later protocol. Chapter 4 gives a technical synopsis of the target, dependency chain, signatures, rivals, first protocol, and verdict logic.

The structure also connects the titles across the series. Applications in government, cities, law, peace, private life, religion, or transition can generate operational evidence and valuable research questions. Scientific findings can inform those applications through a documented handoff, with each domain

evaluating them against its own purposes, evidence, rights, and responsibilities.

By the end of this Part, readers should have a compact map of the target, its scientific dependencies, its ethical limits, and the evidence states that can change it. Different specialists can then enter at the dependency they can actually test, without mistaking local progress for whole-hypothesis support.

The rest of the book should carry these standing safeguards by reference and repeat them only when they change a local design, inference, or public claim. That rule matters for readability as well as rigor: repetition that adds no local consequence makes the scientific architecture harder to see.

Chapter 1 — Why the Hypothesis Needs a Science

A hypothesis becomes a scientific opportunity when it can organize better questions than its rivals and accept results that genuinely change its future. PTVC reaches that threshold only through a dedicated research program. The need is not created by the prestige of mathematics, but by the number of distinct scientific obligations packed into the claim.

The public sentence is simple: positive and negative phenomenal magnitudes may close in exact intrastream balance at the true end of each qualifying stream. The scientific sentence is a dependency network. It requires a defensible target, a stable representation, an independently calibrated neutral origin, a qualifying bearer, exhaustive routing, a complete domain, an endpoint object, an observation model, a recovery bridge, serious rivals, and a decision rule.

A dedicated volume connects this network. It brings affective science, terminal care, consciousness research, and statistics into a shared account of their assumptions and interfaces. Readers can see which discipline owns each question and how the resulting work contributes to the scientific target.

The first contribution is governance. Definitions, formulas, protocols, code, instruments, endpoint rules, and public verdicts must remain versioned and linked. A change to a bearer rule is not a cosmetic edit. A change to a neutral origin can change the target itself. Scientific memory makes those consequences visible.

The second contribution is modular failure. An inadequate momentary estimator need not defeat every branch of PTVC, yet it must block the inference that depended on that estimator. A

failed route may retire the route without automatically rejecting the terminal target. A gate-valid counterexample may defeat a declared set-universal branch even when other branches remain open.

This modularity is disciplined by cumulative judgment. Research programs grow when successful components generate novel, risky, and transferable work; they weaken when every difficulty produces only retrospective rescue (Lakatos, 1978). The volume therefore protects local learning without allowing endless deferral of the overall empirical burden.

The third contribution is a staged sequence. Formal verification comes before empirical promotion. Measurement qualification comes before terminal inference. Model recovery comes before real-data model comparison. Safe observational and natural-perturbation work comes before any stronger intervention. Replication and held-out transport come before a broad public verdict.

The fourth contribution is evidence-role clarity. A calibration result may establish the foundation for a later test. A route signature may favor one mechanism family. A pilot may show operational feasibility. A recurring human idea may establish salience and generate comparative hypotheses. None of those achievements needs to be dismissed merely because it is not complete-stream target evidence.

Several foundations are already empirical. Everyday activity choices vary systematically with mood; reward experiments quantitatively link changing feelings to expectations and

outcomes; and biological circuits can integrate error and maintain regulated outputs despite perturbation (Taquet et al., 2016; Rutledge et al., 2014; Aoki et al., 2019). These findings give researchers observed phenomena, candidate mechanisms, and reproducible questions with which to begin.

The scientific opportunity is to discover whether a common relation among signed experiential history, future response, and a separately identified stream boundary explains more than those established mechanisms explain individually or in combination. That possibility is specific enough to investigate.

The fifth contribution is open comparison. A theory that receives more flexible parameters, richer information, or kinder endpoints than its rivals has not earned superiority. Equal-budget comparisons, proper scoring, held-out prediction, sensitivity analysis, and model-recovery benchmarks create a fairer arena for candidate and rival models (Gneiting & Raftery, 2007; Mayo, 2018).

The program also needs transparent prospective specification. Protocol-fixed hypotheses, estimands, endpoint rules, analysis windows, exclusion criteria, and rival families separate confirmatory risk from exploration. A protocol is publicly preregistered only when it is deposited before outcome inspection in a timestamped external registry with a persistent identifier; Registered Reports add a distinct journal-governed commitment (Nosek et al., 2018; Chan et al., 2025).

Current reporting standards make that discipline more concrete. SPIRIT 2025 specifies the minimum content of a trial protocol,

CONSORT 2025 strengthens complete and transparent reporting, and the consolidated ICH E6(R3) guideline links quality by design to proportionate participant protection and reliable results. For PTVC, the practical implication is direct: participant burden, endpoint integrity, missingness, estimand definition, rival comparison, and public reporting are critical-to-quality factors rather than appendices added after the design is fixed (Chan et al., 2025; Hopewell et al., 2025; International Council for Harmonisation, 2026).

Builders benefit from the same discipline. A useful measurement instrument, a strong model-recovery library, an endpoint adjudication method, or a humane data architecture can remain valuable even if PTVC is narrowed or defeated. Outcome-resilient value makes the program more ambitious because it asks every stage to leave science stronger than it found it.

Figures and public summaries must identify whether they show observations, estimates, simulations, or hypotheses. Section 1.6 develops that release standard. Keeping those categories visible in compact artifacts helps prevent visual clarity from being mistaken for evidentiary strength.

Different readers enter through different doors. A general reader needs the question and the dependency map. A field expert needs assumptions, estimands, comparators, and failure routes. A participant needs burden, withdrawal, privacy, and care protections. A public communicator needs language that remains accurate when quoted alone. Designing these entry points explicitly lets readers reach the level of detail they need without changing the scientific status of the underlying claims.

The chapter ends with a demanding but optimistic proposition: answerability is not retreat. It is the form of courage appropriate to a scientific claim. Once the hypothesis has a governed home, its strongest possibilities can be explored without asking rhetoric, adoption, or beauty to carry the weight of evidence. That discipline gives favorable and unfavorable results equal power to improve the research program.

1.1 — Why a Separate Science Volume

The Law of Fairness asks a question that ordinary public language cannot carry safely: whether every separately routed maximal qualifying conscious stream in a declared class satisfies an exact cumulative boundary condition over its complete strict-pre-endpoint domain. Public language can awaken the question, but it cannot govern the representation, bearer and routing rules, endpoints, instruments, models, rivals, or status labels needed to test it. A dedicated science volume exists because the claim is too consequential to remain distributed across inspiration, analogy, and application (Popper, 1959; Lakatos, 1978; Laudan, 1977; Mayo, 2018).

A separate volume can connect scientific references that would otherwise remain scattered across the series. Measurement, consciousness, terminal care, and Artificial Intelligence safeguards each contribute distinct expertise; bringing them together makes their assumptions, interfaces, and evidence roles visible. The reader can then see how those contributions support a coherent research plan and where further work is needed.

The separate volume also prevents a temptation inside ambitious systems: letting coherence impersonate evidence. When ten domains fit together, the fit can feel like discovery. But fit across books is a design achievement, not a finding about nature. The science book exists so the series has one place where the difference is never optional.

1.1.1 — A Formal Claim Needs Scientific Governance

The Law of Fairness receives a formal architecture involving separately routed maximal qualifying streams, finite signed phenomenal measure on the primary branch, exhaustive ownership and routing, a complete strict-pre-endpoint domain, and a truth-level endpoint. Scientific development assigns each term its status, assumptions, observation requirements, tests, and rules for revision.

The first governance problem is version control. A statement of the Law, an estimator of lived experience, a stopping-time rule, a reserve-coupled realization, and a public verdict label are different objects. If one changes, the research program must record which claim version has changed. A favorable result under one bridge cannot be transferred quietly to a different bridge simply because the project prefers continuity.

The second governance problem is assumption visibility. The formal claim may be stated in an idealized latent space, but empirical work requires auxiliary assumptions about representation, scale, sampling, endpoint classification, channel reliability, missingness, and rival processes. A science volume makes those auxiliary assumptions visible so that critics can attack them without being accused of attacking the whole moral aspiration of the series (Popper, 1959; Lakatos, 1978; Laudan, 1977).

Governance also requires contribution typing: definitions specify objects, formal derivations establish consequences of stated assumptions, measurement models link observations to targets,

studies evaluate components or predictions, and results update only the claims they actually reach. Making those transitions explicit keeps each statement connected to its warrant.

The separate science volume is therefore an institutional answer to a linguistic danger. The Law is too large to be left to informal confidence. It needs a place where every important term is made answerable to a method, every method is made answerable to a rival, and every favorable result is made answerable to the status it can actually bear.

A field expert will also look for governance of comparison, not merely governance of definition. If the Law is compared against a simplified rival while the Law receives a richer auxiliary structure, the comparison is biased before the study begins. Scientific governance therefore includes a symmetry rule: a model that is allowed to use context, noise, and hidden state should face rivals that are allowed comparable resources under comparable constraints.

1.1.2 — Applied Coherence Is Not Validation

Applied coherence is not validation. A city, institution, ritual, or interface may be useful because it solves a local problem, and its difficulties may generate research questions, without supplying evidence for PTVC. Translate applied observations into bounded, ethically authorized studies with explicit rivals before they enter the scientific record. Operational success can motivate a test; it cannot decide the target hypothesis (Laudan, 1977; Mayo, 2018).

1.1.3 — A Large Claim Needs Modular Failure

The larger a claim becomes, the more dangerous it is to treat it as one indivisible object. *The Law of Fairness* touches the bearer of experience, the currency of felt valence, the temporal accumulation of experience, the endpoint at which a trajectory closes, the measurement bridge from latent to observed, the candidate dynamics that could realize closure, and the statistical rules that would justify a verdict. Each component can fail in a different way.

Modular testing assigns each study to the component it can actually reach: representation, measurement, endpoint construction, candidate realization, or rival explanation. Failure blocks the descendants that depend on that component without automatically deciding unrelated branches (National Academies of Sciences, Engineering, and Medicine, 2019).

The program therefore uses a dependency lattice rather than a single ladder of belief. That structure prevents one weak instrument from becoming total defeat and one favorable signal from becoming total validation; it lets each success or failure change only the status it has earned.

The separate science volume is needed because no applied book can maintain this lattice without becoming a methods book. Each applied domain has its own burdens. The science book carries the cross-cutting burden of deciding which components are vulnerable, how they are connected, and what consequence follows when one of them fails.

The modular structure also prevents premature despair. If endpoint research proves intractable in one clinical setting, measurement development may still proceed in ordinary longitudinal settings. If a reserve-coupled realization fails, rival comparison may still clarify which ordinary mechanisms explain apparent balancing. A good program learns from partial loss rather than requiring every failure to become total collapse.

Cumulative judgment evaluates how those modules work together. Sustained problems across representation, measurement, endpoints, model comparison, and transport can justify revising or retiring the affected scientific claim. The component record identifies the evidence behind that decision.

1.1.4 — Answerability Before Adoption

Scientific answerability requires explicit predictions, fixed comparisons, and result-contingent decisions before adoption pressure can shape the test. Public interest does not raise evidential status. For confirmatory work, freeze the empirical bridge, name serious rivals, specify adverse-result rules, and preserve the dependency order from representation through measurement, endpoint, model comparison, replication, and transport. Applications may proceed on their own evidence, but they cannot make the scientific claim more nearly proven by momentum (Popper, 1959; Lakatos, 1978; Laudan, 1977).

1.1.5 — Scientific Seriousness Before Support

A hypothesis can be scientifically serious before it is supported when it is coherent enough to state, distinctive enough to generate risky consequences, vulnerable enough to lose, and connected to

methods capable of distinguishing it from rivals. PTVC now has a defined target, candidate consequence families, explicit rivals, and routes to narrowing or defeat. Those features justify treating it as an answerable research program; they do not constitute empirical support. Credibility must come from instruments, comparisons, replications, and results that survive the stated failure conditions (Laudan, 1977; Mayo, 2018).

1.1.6 — The House-Skeptic Function

The house-skeptic function gives criticism equal standing inside the research program. It asks which component is supported, which rival could explain the same pattern, whether endpoint and analysis rules were fixed before outcomes, which missingness process could mimic convergence, and what claim must change if a bridge fails.

To make that role substantive, skeptical and constructive teams should receive comparable access to assumptions, data dictionaries, simulation fixtures, code, protocol history, model budgets, and scoring rules. A rival denied relevant covariates or forced into a weaker observation law is not a fair comparator (Gneiting & Raftery, 2007; Mayo, 2018).

Major objections should enter a challenge log with the component challenged, evidence needed, owner, and interim public status. Chapter 25.7 supplies the detailed red-team and challenge-log procedure; Section 1.6 governs hostile-crop public communication.

1.2 — Inheritance and Scientific Jurisdiction

The Science of Fairness develops the public question introduced in *The Law of Fairness* into a self-contained scientific inquiry. Chapters 4–7 state the hypothesis, define its objects, and derive its principal consequences; later chapters develop measurement, candidate dynamics, tests, and interpretation. Readers can follow the scientific argument within this volume, with external research cited where it informs the analysis (Douglas, 2009).

The book connects the public question to scientific investigation and to the applied titles in the series. A shared formulation lets the books develop distinct responsibilities: this volume specifies the scientific target and methods, while the related titles explore legal, spiritual, civic, clinical, and personal questions in their own domains.

Scientific jurisdiction is bounded authority. This volume owns measurement governance, protocol design, rival comparison, model recovery, replication, narrowing, defeat, correction, and scientific public communication. This scientific remit leaves human worth, spiritual interpretation, legal enactment, government operation, ordinary privacy, and transition authorization within their respective domains.

That division of work is productive. *The Law of Fairness* states the public formal root; *The Science of Fairness* develops and tests the scientific questions; and applied titles evaluate proposals within their respective domains.

1.2.1 — The Formal Hypothesis Is Stated

The first scientific act is specification. This volume states a formal hypothesis about separately routed maximal qualifying conscious streams and exact terminal valence closure over complete strict-pre-endpoint domains. It does not replace the bearer with public identity, the phenomenal target with welfare preference, the endpoint with bodily death, or the latent cumulative object with a dashboard. An operational target must remain recognizably connected to the object it is meant to test.

Specification makes the vocabulary explicit. Qualifying bearer, exhaustive ownership and routing, separately routed maximal stream, phenomenal valence, complete strict-pre-endpoint domain, truth-level endpoint, latent terminal residual, momentary valence estimator, interval contribution estimator, and branch-matched inference are defined here as the objects of scientific inquiry. Science may show that an object is difficult to identify, propose provisional bridges toward it, or conclude that a bridge fails. It may not declare that a simpler neighboring object was the target all along.

The hypothesis also retains its ethical boundaries: PTVc carries no implication of moral desert, victim-blaming, cross-stream transfer, pain as morally owed payment, harm induction, delayed care, human ranking, or machine authority; Chapter 3 governs the corresponding operational protections (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

The task is to determine what the received objects require from research. What counts as a bearer in a study? Which aspects of felt valence can be observed? How can an endpoint be classified independently of the ledger result? How should each scientific finding be communicated?

The public question motivates the formal target specified here. This volume develops the observations, models, and comparisons through which researchers can evaluate it.

The first jurisdiction rule follows: receive the formal target whole enough to test honestly, then record any scientific narrowing openly. If research shows that a component is unidentified, a reference class is too broad, or an endpoint bridge fails, the change is a scientific status update. It is not a hidden rewrite that preserves the original language while testing something else.

Reception also requires resisting friendly simplification. It would be easier to replace the Law with a familiar claim about adaptation, life satisfaction, welfare, resilience, or moral balance. Those topics have literatures and instruments. But a research program that changes the target to fit available tools has not made the original claim scientific. It has studied a neighboring object.

Approximation is sometimes necessary, but it should never erase the harder target from view. Public readers should be shown both layers: the received formal object and the provisional bridge by which a particular study approaches it. Keeping those layers visible prevents an operational measure from quietly replacing the claim it was designed to investigate.

1.2.2 — The Formulation Becomes a Research Target

A formal model becomes a research target when its definitions are translated into obligations. The ledger is not merely a symbol; it demands a representation theory of lived valence. The bearer and routing rules demand a defensible account of which experiences belong to which maximal qualifying stream. The truth-level endpoint demands prospective construction and adjudication that do not define closure by the ledger value being tested. Some protocol-fixed branches add timing-independence, support-density, hitting, absorption, atom, or continuation premises, and each such premise must remain branch-specific rather than becoming a universal assumption.

The translation turns the formal model into operational questions. For latent $L(t)$ and its measured estimates, the science book specifies the requirements for construction, calibration, versioning, and reporting. For terminal valence closure, it derives measurable consequences under declared observation models and rival data-generating processes.

The model also becomes a target for dimensional discipline. A rate, an integral, a unit, a reserve variable, a horizon, a variance, and an endpoint are not interchangeable pieces of poetic mathematics. Each has assumptions, units, and failure conditions. No equation should travel into public language without its object, dimensions, branch, and status attached (Folland, 1999; Protter, 2005).

A candidate stochastic realization derives a possible process from stated assumptions. Its scientific development identifies those

assumptions, predicts observable consequences, and compares the resulting model with empirical data and alternatives.

Making this formulation a research target also gives critics a fair route in. They can dispute representation, scalarization, bearer qualification, ownership and routing, endpoint construction, branch-specific timing assumptions, reserve coupling, measurement invariance, rival comparison, or verdict logic. That is better than a general dispute over whether the Law feels plausible. A target that can be attacked component by component can also be improved component by component.

The target also demands that proposed failures be meaningful. A failed momentary valence estimator channel is not a decisive refutation of the Law if the channel never estimated the intended construct. But repeated failure of reasonable, prospectively specified bridge families is not irrelevant either. The research target must allow both distinctions, because otherwise the program becomes either too brittle or too protected.

1.2.3 — Scientific Status Is Governed Here

Scientific-status governance is this book's central jurisdiction. The volume defines the procedures, evidence roles, and public language through which PTVC-related claims may be updated, while evidence, independent review, and the wider scientific community determine the verdict. Without that distinction, public language can drift toward whatever status feels most useful at the moment.

Scientific-status governance records how findings update claims. A bounded cohort can support a specified consequence;

measurement work can qualify or revise an estimator; and a comparative result can guide mechanism selection. These decisions make the research record usable.

Negative status belongs here as visibly as support. Nulls, failed instruments, rival victories, narrowing, route retirement, and theory-level loss remain part of the scientific record. A program that publicizes only favorable movement cannot learn reliably from evidence.

Replication is part of status governance. A single site, instrument, clinical setting, culture, or endpoint procedure cannot carry universal language. The science book must require transport before broadening. It must record when a result is local, fragile, incompatible across sites, sensitive to missingness, or dependent on an auxiliary assumption that has not yet traveled.

Scientific-status governance includes correction of public artifacts; Chapter 28 supplies the full propagation, archival, and retirement workflow. A correction must reach the public surface where the overstatement traveled.

Scientific-status authority is narrow and consequential. It does not govern law, faith, city design, ordinary practice, or transition sequence. It determines whether those domains may use scientific language and, when they may, the exact scope that the evidence has earned.

Scientific-status governance also applies to language inside the volume. Words like suggest, support, validate, imply, estimate, demonstrate, and prove are not stylistic variations; they name different relationships between evidence and claim. Treating

those verbs as interchangeable misleads even when a formal warning appears elsewhere.

1.2.4 — Definitions Flow Out; Evidence Flows Back

The series requires a clean direction of flow. Definitions flow outward from *The Law of Fairness* because the root book states the formal claim. Evidence flows back through *The Science of Fairness* because scientific findings must update public status under governed rules. If this direction is reversed, applied success can rewrite definitions, and the project becomes self-validating.

This flow rule protects the formal root from convenience. Suppose measurement works better when the endpoint is defined by a clinical event rather than by truth-level endpoint. The science book may use the clinical event as a proxy under a declared bridge, but it cannot say the endpoint was always that event. Suppose a rival explains terminal concentration better than reserve coupling. The science book may narrow or retire the realization; it cannot quietly alter the formal hypothesis to preserve a favored result.

The flow rule also protects science from canon pressure. A later book may produce an elegant institution, ritual, or transition pathway that depends rhetorically on the Law. That elegance cannot demand that science find support. Evidence must remain able to disappoint the canon. A series that cannot be disappointed by evidence has already left science (Douglas, 2009; Mitchell, 2009; Woodward, 2003).

Flow is not a wall. Applied books can send observations back as research leads. A city may reveal record burden; an ordinary-life

practice may reveal repair dynamics; a religious house may reveal ritual effects; a transition pilot may reveal public-status misunderstanding. But those observations become questions only after they are transformed into research designs with consent, context, rivals, and status limits.

The return flow from science is component-specific. Support for a measurement bridge returns as that bounded update; it does not strengthen every sentence in every title. A failure returns with its scope and consequence, which may require language correction, artifact retirement, source updates, or claim narrowing without creating panic beyond the component affected.

A favorable scientific finding can update the relevant passage in the formal-law book. The handoff identifies the supported component, the evidence behind it, and its established scope, allowing readers across the series to follow the same result.

1.2.5 — Applied Observations Are Research Leads

Applied observations matter because the world often reveals burden before instruments do. A household may expose how care work disappears into routine. A public office may expose how answerability shifts onto citizens. A religious practice may reveal how meaning changes distress. A technological system may reveal how a dashboard pressures users to perform. These are valuable observations, but their value begins as question-generation.

A research lead names a phenomenon that deserves controlled inquiry. It does not state the cause. If a group using a Fairness practice reports more repair, the science must ask whether the

effect comes from attention, community, expectancy, selection, time, language, ritual, reduced isolation, or some Law-specific consequence. Treating the observation as evidence too early would disrespect the complexity that made it interesting.

Harms can also become research leads without becoming research data by default. If a public artifact creates misunderstanding, communication research can examine the pathway. If an experience measure creates anxiety, burden research can examine reactivity. If readers infer terminal destiny, hostile-crop testing can study which fragments carry that inference. The moral problem is immediate; the scientific opportunity begins only after consent, context, rival explanations, and status limits are specified (Fischhoff & Davis, 2014; National Academies of Sciences, Engineering, and Medicine, 2017; National Institute of Standards and Technology, 2020).

Applied observations can reveal rival mechanisms worth formalizing. Relief after ritual may arise through expectancy, attention, social support, or meaning. Terminal calm in a setting with strong death scripts may reflect script-induced appraisal. Apparent variance compression where sensors fail may be observation-gain collapse. Each observation improves the program when it becomes a discriminating question rather than a favored anecdote.

The research lead must pass through a separate door before becoming evidence. That door includes consent, data minimization, context definition, sampling, prospectively specified outcomes, endpoint rules where relevant, rival models, missingness plans, and public-status limits. Operational records

do not become research data merely because they are convenient. Participation in an institution is not research consent.

Applied observations also help the program notice harms that laboratory design may miss. A public dashboard may produce social pressure. A community ritual may intensify guilt for some participants while relieving others. A record system may help officials while exposing families. These are not peripheral anecdotes. They are burden data that can improve future research questions if they are handled under consent and scope discipline.

1.2.6 — Jurisdiction Is Not Mastery

Scientific jurisdiction is bounded. This volume governs scientific claim status; it does not govern human worth, spiritual meaning, legal rights, clinical care, transition timing, or private life. Section 3.6 states the operative domain boundaries. For example, qualifying an affect measure can inform an application while leaving the application's legal and clinical decisions to their own review. A scientific result can therefore sharpen a factual premise without deciding the rule another field should apply. The receiving field remains responsible for integrating that evidence with its own standards, duties, and forms of review.

That limit also applies to participants. Measurement and consent authorize only the declared research uses and inferences; they do not transfer ownership of inner life. Ethics, law, clinical care, disability studies, privacy, and science communication remain responsible for the consequences within their domains. A later request to train a new model on stored reports is a new proposed use, not an extension silently granted by the original

measurement. Secondary use should therefore be treated as a fresh governance decision, with its own stated purpose, access conditions, retention limits, and challenge route. This keeps consent tied to a defined use rather than turning one permission into an open-ended claim on future data.

Cross-disciplinary work therefore uses explicit handoffs: each contribution states what it establishes, the assumptions it carries, and which questions remain for other fields. Scientific precision is strongest when it does not borrow authority from neighboring domains. A useful handoff identifies which conclusion depends on the scientific result, so a later revision can update that conclusion without reopening unrelated judgments. A strong handoff also carries the result's status, uncertainty, and reference class, because those features determine how far the result can travel. The receiving field can then combine it with its own evidence and decision rules without stripping away the conditions that made the scientific statement valid.

1.3 — The Research Program in Stages

A staged research program connects representation, measurement, model recovery, endpoint adjudication, and rival comparison before terminal testing. Work at each stage identifies the conditions and methods needed for the next study (Chambers, 2013; Munafò et al., 2017; Nosek et al., 2018).

This section uses the language of research programs because it fits the problem. A program has a hard formal target, auxiliary assumptions, instruments, models, rivals, and rules for revision. It can become progressive when changes solve independent problems and generate new risky consequences. It can degenerate when changes only protect a favored conclusion after difficulty appears.

The stages introduced here are an orientation, not a substitute for protocols. The program moves from formal coherence to representation, from representation to measurement, from measurement to endpoint and observation, from those prerequisites to rival comparison, and from successful recovery to replication and public status. A later stage never compensates for a failed earlier gate.

The staged architecture gives the project clear decision points. Each stage identifies what has been established, what remains open, and which next investment is scientifically justified.

1.3.1 — Core Claim and Auxiliary Hypotheses

The core claim is the strongest formal target the science receives. On the primary finite signed-measure branch, every separately

routed maximal qualifying stream in the declared truth-domain class has equal finite positive and negative phenomenal magnitude over its complete strict-pre-endpoint domain, equivalently a zero terminal signed residual. The public empirical claim is not identical to that set-universal target. Public inquiry must proceed through validated representation, ownership, endpoint, observation, measured-to-latent bridges, rival comparison, and branch-matched inference. The target gives direction; auxiliary hypotheses make bounded research possible.

Auxiliary hypotheses include representation assumptions. Can lived experience be treated, for a declared purpose, as a signed valence-rate target that accumulates over time? How are mixed affect, numbness, dread, relief, meaning, dignity, memory, anticipation, and bodily distress handled? A failure here would not be a small measurement problem. It would challenge whether the proposed target can be represented in the required form.

Other auxiliaries concern measurement. Candidate momentary and interval estimators may function as estimators only within a specified context of use, with uncertainty, calibration, versioning, missingness, and correction rules. If the estimator changes, the claim version changes. If the estimator cannot preserve sign or scale within the required setting, downstream terminal analysis cannot repair that failure by collecting more data (Chambers, 2013; Lakatos, 1978; Laudan, 1977).

Endpoint assumptions form another auxiliary layer. The truth-level endpoint must be classified through channels not algebraically identical to the ledger estimator. If the endpoint is defined as the moment the ledger reaches zero, neutrality

becomes a tautology. If endpoint channels share hidden tuning with the ledger, residual dependence must be modeled rather than waved away.

Candidate mechanisms include reserve coupling, horizon pressure, queue dynamics, adaptation, predictive control, and homeostasis. Each generates patterns and consequences that can be developed into comparative predictions.

The first stage of seriousness is therefore not data collection. It is an assumption register. The reader should know what would remain if a representation failed, what would remain if a measurement failed, and what would remain if a mechanism failed. Only then can later evidence have a clear target.

The auxiliary structure should be visible enough that a reviewer can ask whether the program is becoming progressive or degenerating. If a measurement bridge is replaced, what new prediction does the replacement create? If a reserve assumption changes, what rival does it now face? If an endpoint rule becomes more complex, does complexity improve independent classification or merely protect the desired terminal pattern?

Some auxiliaries determine whether the next inference is available. A defensible origin, endpoint, and measurement bridge provide the foundation for terminal analysis. The staged architecture directs early work toward those dependencies.

1.3.2 — Progressive Problem Shifts

A research program progresses when a revision does more than protect the core from trouble. It solves an independent problem, clarifies a mechanism, improves measurement, predicts

something not used to build the revision, or exposes the program to a new possible loss. *The Law of Fairness* needs that standard because it will inevitably meet anomalies, instrument failures, ambiguous endpoints, and rival explanations.

A progressive shift might occur if a failed single-channel momentary valence estimator model leads to a better declared observation model that predicts when self-report and physiology diverge, then correctly forecasts those divergences in a new cohort. The revision would not merely rescue neutrality. It would solve a measurement conflict and generate a new testable consequence.

Another progressive shift might arise from endpoint adjudication. Suppose a unity-proxy rule performs poorly in one boundary state but, after expert review, yields a refined confidence-class system that predicts which cases should return no estimate. If that system improves prospective endpoint classification and increases no-estimate honesty, the research program has progressed even before any Law-specific terminal support appears (Laudan, 1977; Munafò et al., 2017).

Progressive shifts can also involve rivals. If an adaptation model explains most local rebound but fails to match reserve-linked drift and innovation behavior under held-out prediction, the Law-specific program gains a sharper target. The gain is not that adaptation was dismissed. The gain is that the contrast taught the program what it must uniquely explain.

An informative revision improves the connection between model and observation. Additional variables, uncertainty models, and

flexibility contribute when they yield clearer predictions, better calibrated estimates, or a more discriminating comparison.

A progressive shift should improve the critic's position as well as the proponent's. After the revision, it should be easier to say what would now count against the program. Better measurement, for example, should create clearer defeat conditions, not only better-looking fits. Better endpoint classification should increase no-estimate honesty as well as terminal precision.

1.3.3 — Degenerating Rescue Moves

A degenerating rescue move is a change made after difficulty appears, not to improve contact with the world, but to preserve a desired conclusion. Such moves are especially tempting here because the Law touches invisible experience. If unfavorable data appear, one can always say the relevant experience was unmeasured, the endpoint was misclassified, the scale was wrong, or the compensation occurred elsewhere. Sometimes these objections may be true. They become degenerating when they remove risk without adding a new test.

The clearest forbidden rescue is endpoint recoding by result. If the measured cumulative estimate does not approach neutrality, the project cannot redefine the truth-level endpoint as whatever time produces a favorable estimate. Endpoint discipline exists precisely to prevent that move. The endpoint must be prospectively specified through unity or mind-continuity channels, and ambiguity must be allowed to produce no estimate rather than a rescued verdict.

Another rescue is origin adjustment after the fact. Because felvalence lacks an instrument-fixed natural zero, empirical work must declare origins, anchors, and equivalence bands before analysis. If the origin is shifted only after a result disappoints the Law, the program has not discovered neutrality. It has moved the ruler (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

Measurement expansion can also degenerate. If every failed channel leads to adding more channels until the desired pattern appears, the estimator becomes an oracle for the theory. A legitimate measurement revision must explain why the old model failed, state what the new channel contributes, freeze the update before evaluation, and test whether the revision predicts beyond the dataset that motivated it.

Rival handling can degenerate as well. A rival is not serious if it is made artificially weak, excluded after it wins, or reclassified as part of the Law whenever it succeeds. Adaptation, homeostasis, predictive control, Brownian bridge effects, observation shutdown, clinical confounds, and expectation scripts must be allowed to win under the same constraints imposed on the Law-specific model.

The practical rule is severe but fair: every rescue must pay rent. It must create a new prediction, a new independent measurement, a narrower public claim, or a declared loss. If it merely preserves favorable language, it is not a scientific modification. It is protection from science.

Degenerating moves often begin with true statements. It is true that measures are imperfect. It is true that endpoints are difficult. It is true that unobserved experience matters. It is true that rivals may be incomplete. The danger lies in using true difficulties to make the Law unreachable by any practical test. A serious book must distinguish legitimate uncertainty from theory protection.

The practical safeguard is prospective classification. Before outcome inspection, the protocol states which uncertainty makes the study unusable, which widens an interval, which narrows the claim, and which becomes adverse under valid prerequisites. After the result, those categories do not change merely because the outcome is disappointing. That discipline separates humility from retrospective rescue.

1.3.4 — The First Five Studies

The first five studies should be understood as dependency studies, not as a rush toward terminal proof. Their purpose is to ask whether the empirical bridge is strong enough to justify later tests. A program that begins with dying persons before representation, measurement, and endpoint rules are ready would be ethically and scientifically inverted.

Study one is formal verification and representation audit. It asks whether the received objects are coherent, whether dimensions and process classes are defensible, which assumptions are independent, and which claim versions are identifiable in principle. The deliverable is a target capable of losing, not a verdict about whether PTVC is true.

Study two develops the measurement bridge. It asks whether momentary and interval valence contributions can be estimated with declared channels, sign conventions, calibration, uncertainty, and context of use. Model recovery, channel conflict, missingness, and transport determine whether the bridge estimates the intended construct without becoming a score or a theory-protecting instrument (Chambers, 2013; Lakatos, 1978; Laudan, 1977).

Study three validates consciousness-state and endpoint procedures. It tests whether the bearer and endpoint channels can distinguish waking continuity, sleep, anesthesia, coma, delirium, dissociation, degraded continuity, return, and irreversible loss with graded confidence and protected no-estimate states. The ledger may not define the endpoint whose independence the analysis later invokes.

Study four maps palliative feasibility and burden. It asks whether terminal observation can occur without increasing suffering, distorting care, burdening families, or producing missingness too severe for the intended inference. The deliverable may be a redesigned low-burden protocol, a restricted reference class, or a decision to keep the next stage nonterminal.

Study five is a prospectively specified terminal-dynamics comparison against strong rivals. It tests whether declared preterminal signatures, using the best available measurement and endpoint bridges, out-predict adaptation, homeostasis, endpoint conditioning, observation shutdown, clinical confounds, and expectation scripts. Its first question is whether the consequence

structure is discriminating, not whether exact complete-stream closure has been seen.

These five studies are gates, not publicity milestones. Formal incoherence, failed measurement transport, ambiguous endpoints, unacceptable palliative burden, or rival victory can close the route; terminal recruitment is justified only after earlier uncertainties have been reduced enough that vulnerable participants are not carrying avoidable method-development burden.

1.3.5 — Replication Before Status Upgrade

Report the first study by the component tested, the setting, the instrument and endpoint procedure, and the alternatives compared. Those details identify the result that subsequent studies can reproduce and extend.

Replication upgrades status only when the relevant claim survives independent teams, prospectively fixed analyses, appropriate transport, missingness sensitivity, and adversarial review. Replication should also test the bridge itself: disagreement across instrument versions, endpoint classes, or care settings is evidence about scope, not noise to be averaged away. Chapter 26 specifies the multisite system and Chapter 27 governs the resulting status transitions.

Public status should move more slowly than excitement. A component can gain support, a bridge can improve, a rival can weaken, or a scope can narrow. The phrase validated Law is unavailable until the exact claim version being announced has

passed the prerequisites, rival comparisons, replication, and quantifier bridge required for that announcement.

Replication should also include adversarial sympathy. A replication team should be allowed to disagree about the best rival, the missingness model, or the endpoint confidence threshold. The program should preserve those disagreements rather than smoothing them into a single friendly result. Consensus is valuable only after disagreement has been given room to sharpen the analysis.

Status updates should identify concrete advances: a measurement bridge has matured, a rival has been distinguished in one cohort, a protocol proved humane, or a no-estimate rule worked as intended. Each gain contributes to the research program and deserves a clear account of what was learned.

An inspectable replication packet should therefore exist before the first result is known. At minimum it should preserve the claim version, protocol registration, representation and endpoint specifications, analysis code, synthetic recovery benchmarks, instrument versions, data dictionary, missingness plan, adverse-event logic, decision thresholds, and the exact public wording permitted by each outcome. The packet is not bureaucracy around the experiment. It is the experiment's memory (Chambers, 2013; Munafò et al., 2017; Nosek et al., 2018).

Chapter 26 specifies how frozen and adaptive elements, implementation changes, tacit knowledge, provenance, executable code, and analytical reproducibility are governed across sites. Here the governing rule is enough: replication cannot

upgrade status unless an independent team can reconstruct both the analysis and the claim-relevant implementation without hidden favorable choices (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018; Munafò et al., 2017).

This standard is demanding because broad claims are unusually vulnerable to invisible degrees of freedom. A theory can appear to replicate when each site silently chooses a favorable origin, smoothing rule, endpoint window, exclusion set, or rival budget. Replication becomes evidential only when those freedoms are controlled, recorded, and exposed to a team that has no obligation to preserve the preferred story.

1.3.6 — A Program With Clear Decision Gates

A research program specifies the conditions for pausing a study, retiring a bridge, narrowing a claim, replacing a mechanism, or reaching a theory-level verdict. Those decisions connect the evidence to the next responsible action.

Stopping can occur early. If no defensible representation of the target can be specified, terminal testing should not proceed under a false sense of precision. If measurement channels cannot preserve sign or transfer across required contexts, the public empirical claim must narrow. If endpoint adjudication remains circular, terminal valence closure cannot be tested responsibly. Each stop protects the project from building on a broken dependency.

Stopping can occur at the mechanism level. Reserve-coupled dynamics may fail while other aspects of the inquiry remain alive.

Queue language may prove useful as a model of pressure but not as a Law-specific realization. Shadow-price formulations may help describe horizon pressure while failing as evidence for closure. A serious book lets useful mechanisms retire from stronger claims while preserving the knowledge they helped produce (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

Stopping can also occur because rivals win. If adaptation and homeostasis explain the signatures with better prediction, simpler assumptions, and equal or better transport, the Law-specific claim must lose the relevant field. If endpoint-conditioned bridges reproduce the terminal shape without reserve coupling, the program must narrow. If observation shutdown creates false compression, the measurement claim must be corrected before future study.

Public communication is reviewed for clarity, unintended personal interpretations, and participant pressure. When problems recur, the affected materials are paused, revised, and assessed before further release. The correction record identifies what changed and how the revised design addresses the problem.

Stopping also protects contributors from sunk-cost pressure. Researchers, donors, builders, and readers may invest time and identity in the program. The longer they invest, the harder it becomes to admit that a component has failed. A public stop rule gives loyalty a better form: loyalty to the truth conditions that made the project worth beginning.

1.3.7 — Negative Results as Scientific Infrastructure

Negative results are typed outputs, not one null. Distinguish identifiable nulls, invalid instruments, failed model recovery, unadjudicable endpoints, rival wins, burden-limited studies, and nontransport because they block different descendants and imply different next work. Preserve the failed object, the reason, the surviving claims, and the successor obligations in a versioned record; Chapters 27 and 28 contain the detailed status and archive machinery. A negative result is scientifically useful when it changes the next decision or prevents unnecessary participant burden.

1.4 — The Book's Evidence Responsibilities

A monograph that studies fairness has to expose the costs of its own inquiry. Vague status language shifts interpretive work to readers; excessive observation shifts time and discomfort to participants; impractical procedures shift work to clinicians; unclear targets shift model construction to critics; premature conclusions shift repair to later titles. The program should name and govern those costs rather than hide them inside ambition (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; National Academies of Sciences, Engineering, and Medicine, 2019).

That accounting reaches beyond ethics. It includes the inferential cost of flexible hypotheses, the statistical cost of weak comparators, the clinical cost of intrusive observation, the communicative cost of compressed uncertainty, and the archival cost of losing failed versions. Each cost needs an owner, a control, and a recorded consequence.

The first visible map of those burdens appears here. It is not the full Technical Synopsis, which belongs in Chapter 4. It is a dependency lattice that shows how plausibility can rise only when several independently vulnerable components survive. The lattice prevents the reader from mistaking elaborateness for evidence.

The second visible map is a compact table from formal object to status change. It shows how a formal object becomes an empirical target, which study family bears on it, and what public status might change if the study succeeds or fails. The table is not a protocol. It is a contract against overclaim.

Table 1.1. Component-dependency lattice from formal claim to verdict.

Layer	What must be vulnerable	Typical failure	Status consequence
Formal hypothesis	The received target must be coherent and versioned.	Contradictory notation or untracked claim change.	Formal clarification or change-control request.
Representation assumptions	Experience must support a signed temporal target for the declared purpose.	Mixed affect, origin, scalarization, or duration cannot be responsibly specified.	Public empirical claim narrows before measurement.
Observation model	Candidate momentary and interval estimators and related channels must estimate the intended target with uncertainty.	Sign loss, calibration failure, missingness, channel conflict, or version drift.	Instrument retired, bridge narrowed, or no estimate returned.
Endpoint rule	Truth-level stream endpoint must be adjudicated independently of the ledger.	Endpoint defined by the outcome, shared tuning, or low-confidence boundary state.	Terminal analysis blocked or restricted.

Layer	What must be vulnerable	Typical failure	Status consequence
Candidate realization	Reserve, horizon, queue, or control dynamics must generate distinctive consequences.	Closure by construction, nonidentifiability, or rival equivalence.	Mechanism retired or treated as teaching model only.
Study design	Predeclared estimands, sampling, missingness, rivals, and burden limits must hold.	Unusable data, excessive burden, unfrozen bridge, or post hoc outcome selection.	Affected evidence classified as Unusable.
Rival comparison	Serious alternatives must be allowed to win under shared constraints.	Law-specific model wins only against weak or late-added rivals.	No support beyond model-fitting.
Verdict rule	Status update must match component, scope, replication, and transport.	Universal language from local result or silence after failure.	Public correction, narrowing, or defeat label.

Table 1.2. Formal object, empirical target, study family, and possible status change.

Formal object	Empirical target	Study family	Possible status change
Unified conscious stream	Confidence-class bearer and continuity adjudication.	Consciousness-state validation and boundary-state review.	Bearer rule strengthened, narrowed, or suspended.
Latent valence-rate and ledger	Declared candidate momentary and interval estimates with uncertainty and context of use.	Measurement development, calibration, model recovery, and missingness studies.	Estimator qualified, retired, or approved for bounded use.
Truth-level stream endpoint	Endpoint rule independent of ledger estimator.	Endpoint validation, palliative feasibility, and no-estimate classification.	Terminal study permitted, restricted, or blocked.
Reserve and horizon dynamics	Preterminal drift, variance, innovation, and reserve-coupling signatures.	Generative benchmarks and prospective longitudinal tests.	Mechanism supported, narrowed, or defeated against rivals.
Terminal measurable consequences	Equivalence-band outcomes, terminal concentration, and rival comparison.	Humane prospectively specified terminal dynamics study.	Component support, inconclusive status, or local defeat.

Formal object	Empirical target	Study family	Possible status change
Transport and replication	Cross-site and cross-population stability under source-use tags.	Multisite replication, adversarial review, and registered reports.	Bounded upgrade, reference-class ceiling, or no transport.
Public scientific claim	Status label tied to component, scope, uncertainty, and rivals.	Evidence accumulation and correction systems.	Support, narrowing, defeat, or artifact correction.

1.4.1 — Claim Burden

Claim burden is the duty to say what kind of sentence is being spoken. A definition, formal hypothesis, candidate realization, measurement estimate, rival model, preliminary result, and public verdict may all sound authoritative in polished prose. They carry different warrants. The category should be visible in the sentence itself, not recoverable only from a remote disclaimer.

The burden begins with the strong form. The latent target is an exact set-universal boundary condition over separately routed maximal qualifying streams in a declared class, not a directly observed public fact. Public science must proceed through measurable consequences, truth-containing compatibility sets, model comparisons, finite bands, endpoint and observation bridges, and explicit status updates. A chapter that treats those layers as one claim fails before the study begins.

Claim burden also requires scope. A result may concern a measurement bridge in healthy-adult diary data, a terminal signature in one palliative cohort, a sleep-related adjustment pattern, or a model-recovery benchmark under simulated data. The status label must name the component, population, window, instrument, and dependency under discussion (Gneiting & Raftery, 2007; Lakens, 2017).

A claim can become too vague by trying to be safe. Repeating that nothing is proven does not satisfy claim burden. The reader needs to know what has been proposed, what would count as evidence, what would count against it, and which version of the claim is under review. Humility without specificity becomes another form of evasion.

Claim burden is therefore the book's grammar of honesty. Every major sentence must carry its own status or be surrounded by enough context that the status cannot be mistaken. The Law can remain ambitious only if its sentences refuse to pretend they have earned more than they have.

1.4.2 — Identification Burden

Identification burden asks whether the target can be distinguished in principle. It is deeper than reducing noise. A more sensitive instrument cannot identify an object whose origin, unit, scalarization, bearer, endpoint, or reference class remains underdetermined. The central challenge is therefore not merely to build better affect sensors, but to show what those sensors can and cannot identify.

The terminal-zero problem separates direct measurement from empirical exposure. Finite noisy observations can challenge exact closure when a valid residual set excludes zero under independently justified calibration, missing-history, and endpoint assumptions. A deterministic bound and a statistical confidence set license different conclusions, as Section 24.6.4 explains. Neutrality must be identified independently of the result: recentering a record to make its total vanish supplies no evidence for closure.

Endpoint identification is equally binding. If the endpoint is chosen by the ledger, closure becomes a tautology. If it is equated with bodily death without a consciousness bridge, it may miss the truth-level ending of the relevant stream. If endpoint channels and ledger channels share hidden causes, a terminal pattern may reflect the same clinical process entering both estimators rather than PTVC. These dependencies must be declared before endpoint data are interpreted (International Council for Harmonisation, 2019).

Bearer identification also belongs here. A conscious stream is not identical to a legal person, biography, body, or public identity. Sleep, anesthesia, dementia, dissociation, coma, developmental thresholds, nonhuman animals, and future artificial systems all pressure the bearer concept. Chapter 1 does not solve those cases, but it explains why the volume must own a research route for them.

Identification work can produce a No adequate estimate outcome for a particular cohort, endpoint, channel, or state. Reporting that

outcome directs attention to the evidence needed to complete the measurement bridge and gives later studies a clear starting point.

Better sensing can make important contributions to inner-life science when it is joined to clear identification work. Even a highly reliable channel must be connected to a defensible neutral origin, scale, bearer, routing rule, truth-level endpoint, and target-class transport argument. Separating those tasks lets improvements in instruments and improvements in conceptual foundations reinforce one another.

The same point protects participants. If a study cannot identify the target, the ethical response is not to measure more intrusively. Intrusion can increase data while worsening identification by altering the state under study. Sometimes fairness to the person and fairness to the science both require less measurement and more honest abstention.

1.4.3 — Measurement Burden

Measurement burden begins after identification has named a target and still leaves enormous work. Lived experience does not hand itself to science as a single clean signal. Self-report, physiology, behavior, context, sleep, clinical observation, caregiver testimony, passive signals, and records are all partial witnesses. Each witness can be useful; none is sovereign.

Measurement must therefore be treated as disciplined translation, not extraction. The momentary valence estimator is a context-bound momentary estimate under uncertainty, not a device that reads inner life. The interval valence contribution estimate is a context-bound interval estimate under conditions, not a moral

unit. The running measured cumulative estimate is an approximation under a model, not the latent cumulative process itself. Measurement burden requires this layered humility whenever numbers appear.

Calibration is part of the burden. A measure that performs well within one person may not compare cleanly across persons, cultures, ages, disabilities, languages, clinical conditions, or terminal states. Translation does not establish invariance, and identical responses need not carry identical latent meaning. The program therefore distinguishes within-stream calibration, cross-context linking, partial identification, and principled local inference (Borsboom, 2005; International Test Commission, 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

Missingness is not an empty space in a dataset. It can follow distress, fatigue, withdrawal, death, technology failure, family intervention, treatment, or the point at which a protocol no longer fits care. A terminal study that treats those absences as random can manufacture calm by losing precisely the observations most likely to challenge it. The missingness mechanism must therefore be modeled before favorable patterns are interpreted.

Versioning is also measurement burden. If the momentary valence estimator channel set changes, if weights change, if anchor rules change, if a wearable model is updated, if a natural-language model summarizes reports differently, or if a clinical workflow changes observation frequency, the measured object may change. Version labels and bridge tests are required rather than silent continuity.

Interpretation itself also needs governance. When channels disagree, the protocol has to state how self-report, physiology, behavior, clinician observation, and model output are weighted or held apart. Those choices should be versioned and auditable because a clean estimate can otherwise conceal a value judgment about whose evidence counts.

1.4.4 — Participant and Bystander Burden

Participant and bystander burden is part of the data-generating process. Prompts, devices, interviews, family involvement, staff workload, and clinical interference can alter the state observed, who enrolls, who remains, and which data go missing.

Protocols therefore need prospective burden endpoints and stop rules covering fatigue, distress, privacy risk, device discomfort, family pressure, clinical interference, and staff overload. Burden-linked selection and missingness must be analyzed rather than treated as background noise.

Sections 3.3 through 3.5 provide the detailed rules for consent, bystanders, vulnerability, withdrawal, and burden budgets. The evidence responsibility is to identify how measurement changes participation and observation, then account for those effects in the analysis.

1.4.5 — Rival and Negative-Result Burden

Rival burden means giving serious alternatives enough information, flexibility, and expertise to explain the data under common rules. Adaptation, homeostasis, predictive processing, learning, endpoint conditioning, observation loss, treatment, frailty, and selection are competing generators.

Negative results belong to the same record as favorable ones. Failed representation, instrument loss, endpoint failure, rival victory, replication failure, and nontransport each change the relevant component and should remain discoverable.

Section 2.4 defines severe comparison, and Chapters 21 through 25 develop the rival and recovery methods. Strong rivals are allowed to win, and unfavorable results remain part of the scientific memory.

1.4.6 — Correction Burden

Correction burden is the duty to repair the public record where a claim traveled. Claims, instruments, endpoint rules, analyses, and public artifacts must remain traceable enough that readers can distinguish current from superseded authority.

Chapter 28 describes the full correction, propagation, archive, and retirement workflow. When scientific status changes, dependent public language changes with it, while prior versions remain visible as history rather than continuing authority.

1.4.7 — Failure Memory and Successor Design

Failure memory should identify the failed object, the reason it failed, the burdens it created, the descendant claims it invalidates, and the repair a successor version is required to test. A successor does not inherit evidentiary credit from the failed version; it begins with a new empirical commitment. Chapters 27 and 28 provide the full evidence and change-control machinery.

A research program becomes cumulative when its successors are constrained by failures as well as inspired by successes. That

memory is especially important for an ambitious hypothesis, where ambition can erase past disappointment. The evidence responsibility is to move forward only after failure is made legible enough to improve what comes next. The next version should be able to identify the specific uncertainty it reduces and the specific weakness it no longer carries forward. Otherwise repeated revision can create the appearance of progress without making the underlying test more informative.

Superseded drafts, retired diagrams, obsolete instruments, and earlier claims can reenter circulation simply because they remain vivid or easy to find. Preserve their history and retired status, and make current versions easy to distinguish from archival ones wherever researchers or public readers encounter them.

1.5 — How to Read as Skeptic and Builder

This book is written for more than one kind of reader, and the differences matter. A skeptical statistician, a philosopher of science, a builder of civic systems, a clinician, a public reader in grief, and a reader of the wider canon do not need the same invitation. They also do not carry the same risk of misreading (Douglas, 2009; Gelman & Loken, 2014; Mayo, 2018; Nosek et al., 2015).

Each reader role carries a distinct discipline. Skeptics should attack the strongest defensible claim rather than a caricature; builders should not promote conditional usefulness into evidence; scientists should follow the dependency order; philosophers should examine claim class and theory change; public readers should separate importance from proof; and readers of the wider series should let scientific status constrain language without granting this volume authority over every domain.

The goal is a shared discipline. Readers may disagree about the Law, but they should be able to agree about the difference between a formal hypothesis, a measurement bridge, a research lead, a rival victory, and a public status change. That shared discipline is more important at this stage than agreement about outcome.

1.5.1 — For Skeptics

Precise criticism strengthens this book's research program. Critics can identify the component they question, trace its dependencies, and propose observations or comparisons that would clarify it. That contribution belongs throughout the inquiry.

The strongest skeptical attack begins with representation. Can lived experience responsibly be modeled as a signed temporal target at all? If mixed affect, agency, meaning, dread, numbness, relief, anticipation, and memory resist scalarization beyond repair, the program may narrow before any measurement study begins. That is a serious criticism, not a side issue.

A second attack concerns identification. Can a privileged origin be defended? Can an endpoint be classified independently of the ledger? Can terminal concentration be separated from observation shutdown or clinical selection? If these cannot be distinguished in principle, then better sensors will not rescue the relevant public claim (Douglas, 2009; Gelman & Loken, 2014).

A third attack concerns rivals. Adaptation, homeostasis, predictive control, endpoint-conditioned stochastic processes, medication, frailty, disease trajectory, death scripts, and missingness can each produce patterns that may look Law-like. A skeptic should ask whether the Law-specific model predicts more, predicts earlier, predicts in held-out data, and survives model recovery under adversarial simulation.

A fourth attack concerns public risk: a skeptic should test whether the project's language predictably encourages moral-payment readings, unsupported personal-destiny claims, human ranking, or unwarranted claims to public authority. Section 1.6 and Chapter 3 make those communication failures part of release review.

The book asks skeptics for rigor, not gentleness. It also asks them to attack the actual claim version, not a simplified enemy. A

precise skeptic is one of the most important collaborators this volume can have, because the fastest way to make the Law smaller or stronger is to identify exactly where it fails.

Skeptics should also be invited to notice when the book is too weak on its own positive case. Excessive caution can become a way of avoiding a clear claim. A critic may rightly object that a theory has not become testable if it only says what it is not. The best skeptical reading therefore presses both directions: no overclaim, and no evasive underclaim.

1.5.2 — For Builders

Builders may use the fairness-oriented lens to generate bounded questions before PTVC is supported, but usefulness is not validation. An applied system may provide a context for research only through a separate research door with appropriate consent or authorization, minimization, access, retention, and use rules. Operational success remains evidence for the application, not the terminal hypothesis (Mayo, 2018; Nosek et al., 2015).

Build conditionally, label precisely, separate operational records from research data, and avoid dashboards or scores that imply Law-based judgment. When an application reveals burden, confusion, or a useful mechanism, translate that observation into a study with its own design and comparators.

1.5.3 — For Scientists

Scientists should use the book as a dependency stack: representation and origin first; measurement and endpoint qualification next; then model recovery, rival comparison, replication, and transport. Work is useful when it makes one

dependency harder to evade, whether by improving an instrument, strengthening a rival, sharpening an endpoint, publishing a null, or showing where a bridge fails. Findings should be named at the level they reach: phenomenon, measurement relation, mechanism comparison, boundary condition, or target evidence.

1.5.4 — For Philosophers

Philosophers should examine how a large formal claim becomes empirically answerable without quietly changing identity. *The Law of Fairness* raises linked questions about lawhood, explanation, auxiliary assumptions, underdetermination, theory change, measurement, naturalized phenomenology, and the relation between formal structure and empirical status. Its research-program form is central: auxiliaries are unavoidable, but revisions become progressive only when they solve independent problems and generate new risky consequences rather than merely protecting the preferred conclusion.

Underdetermination appears throughout. The same observed terminal pattern may be explained by reserve coupling, adaptation, endpoint conditioning, observation shutdown, clinical scripts, or selection effects. A philosopher should ask not only whether the data fit, but whether the program has generated a distinction that makes one fit more informative than another (Gelman & Loken, 2014; Mayo, 2018).

The origin problem also belongs to philosophy as much as measurement. If valence has no instrument-fixed zero, the meaning of neutrality cannot be imported casually from

arithmetic. The science book must distinguish a latent structural origin from a public operational convention. A philosopher should test whether that distinction is coherent and whether empirical verdicts respect it.

Philosophers should also attend to ethical epistemology. Claims about lived experience can burden the people they describe. A theory may be formally clear and still publicly harmful if its concepts are used to judge, rank, console falsely, or delay care. The project therefore asks philosophy of science to meet moral philosophy at the point where measurement affects persons.

The philosopher's role is not to bless the ambition or dismiss it as metaphysics. It is to ask whether the claim has the right structure for empirical answerability, whether its revisions remain honest, and whether its public language respects the warrant it has earned.

Philosophers may also press the relation between explanation and consolation. A claim can be psychologically meaningful without being explanatory, and explanatory ambition can become ethically dangerous when it touches suffering. Meaning should not be silenced, but it cannot count as evidence. That distinction is philosophical before it is statistical.

Another philosophical pressure concerns natural law language. If the Law is called a law, what kind of necessity is being proposed? Is it an invariant, a boundary condition, a best-system compression, a constraint, a statistical tendency, or a metaphor? This chapter only opens that question. Chapter 2 must carry it with exactness.

1.5.5 — For Public Readers

Public readers may come because fairness, grief, burden, and relief make the question personally important; scientific reading should separate that importance from the status of the evidence. Care follows present need, not PTV; no estimator, ledger, endpoint, or verdict in this book can rank a life, diagnose a person, determine care, or assign spiritual or civic standing. Results may be supportive, adverse, inconclusive, local, or unavailable. The public promise is therefore not certainty but traceable language about what was tested, what changed, and what remains unknown (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; World Medical Association, 2024).

1.5.6 — For Canon Readers

Across the wider series, scientific status should propagate only to passages that actually depend on the changed component. Applied domains retain their own evidence, values, and authority; the science volume contributes a bounded premise, not master jurisdiction. Chapter 28 supplies the cross-book correction machinery. Here the rule is simply dependency-based updating: when a measurement, endpoint, route, or target changes status, revise only the summaries, diagrams, and applications that rely on that object, while preserving the conditions and reference class that made the scientific statement valid.

1.6 — Public Artifacts and Scientific Humility

Public artifacts are the shortest routes by which a careful theory can become careless in the world. A diagram may circulate without the chapter that limits it; a formula may be copied without its assumptions; a cover may imply destiny; a social excerpt may turn a hypothesis into proof; a donor description may make ambition sound like validation. High-reach compression therefore requires its own evidence and hostile-crop review (Fischhoff & Davis, 2014; National Academies of Sciences, Engineering, and Medicine, 2017; Spiegelhalter, 2017; Tufte, 2001).

A scientific visual can be beautiful, memorable, and inviting while showing its subject precisely. Its labels and design should help readers identify the question, the represented quantities, and the relationships being proposed or observed.

The section also establishes correction. Public artifacts are not disposable because they are short. Their shortness is what makes them powerful. If an artifact misstates the science, it must be revised, retired, or publicly corrected where it traveled.

1.6.1 — Visuals Orient; They Do Not Prove

A visual can make a concept visible before the reader knows its technical vocabulary. That is useful. A dependency map can show that representation precedes measurement. A diagram of rival models can show that several explanations compete. A simple ledger illustration can make accumulation intuitive. But the image must never make the reader feel that symmetry, smoothness, or scientific styling has proven the claim.

Visual design should make scientific relationships easy to understand. Graphs identify their variables, endpoint diagrams explain their timing conventions, equations name their objects, and illustrations clarify what the instruments observe. Clear captions connect each image to the argument it serves.

An orientation visual may illustrate a hypothesis, a conditional consequence, or a practical aspiration without becoming a research protocol. The Series Map uses the balance scale to introduce the proposed complete-stream equality and asks what would follow if the Law is real. Read its statements within that declared frame. Figures reporting observations, estimates, or simulations identify those different evidence roles explicitly (Fischhoff & Davis, 2014; Spiegelhalter, 2017; Tufte, 2001).

Visuals must also respect persons. A person in a study should not be represented as a data source emptied of privacy. A terminal image should not romanticize death or imply that dying reveals cosmic justice. A child, disabled person, patient, or grieving family should not become an aesthetic symbol for a scientific hypothesis.

The standard is not dullness. It is truth in miniature. A good visual invites the reader into the burden of inquiry. It makes the problem clearer without making the result easier than it is. It should help a skeptic identify a test, not help a supporter feel that the test has been passed.

For this reason, every visual artifact belongs to scientific governance. It needs a label, a status, a scope, and a release

review. If the visual cannot survive those conditions, it should stay inside internal planning until it can.

A public visual should carry enough provenance to resist detachment: source chapter, model or instrument version, information set, simulation status, accessibility description, and correction route. The purpose is not to crowd the image with production metadata. It is to keep an attractive teaching object from becoming remembered as an observed result when it circulates alone (Fischhoff & Davis, 2014; Spiegelhalter, 2017; Tufte, 2001).

1.6.2 — Formula Status Must Travel

Formulas carry special danger because they look more authoritative than ordinary prose. A formula can define a target, describe a model, derive a consequence, specify an estimator, simulate a rival, or state a verdict rule. These are different statuses. A public formula that does not carry its status invites readers to treat mathematics as evidence.

When the formal ledger is displayed, the reader must know whether it is a received definition, an idealization, an estimator, or a candidate realization. When a reserve-coupled equation appears, the reader must know that it is a model with assumptions, not a measured fact. When an equivalence band appears, the reader must know that it is a public verdict rule under observation error, not direct sight of exact terminal equality.

Formula status also requires assumptions. What is the unit? What is the time variable? Is the process signed? Is reserve measured or latent? How is the endpoint defined? Which observations are

missing? Is the displayed path simulated, filtered, smoothed, or retrospectively aligned? A formula without these conditions can create precision without identification (Box, 1976; Gneiting & Raftery, 2007; Wilson & Collins, 2019).

For public use, attach a plain-language status label to the formula: received definition, candidate realization, estimator, design criterion, or open conjecture. Carry that label across slides, websites, figures, donor materials, classroom handouts, and Artificial Intelligence summaries. The formula need not carry its full derivation, but it must retain the assumptions needed to prevent a definition or simulation from being mistaken for evidence. If detachment would predictably raise the claim's status, redesign the artifact.

1.6.3 — Hostile-Crop Review

Hostile-crop review asks a simple question: if the least charitable reader extracted this sentence, caption, diagram, or formula from its context, would it still be accurate enough not to mislead? This reflects ordinary public communication in a fragmented environment. A claim that is safe only when surrounded by five paragraphs of correction may not be safe as a public artifact.

The test should be applied before release. Crop the headline away from the chapter. Crop the graph away from its caption. Crop the caption away from the graph. Crop the formula away from the assumptions. Crop the public summary away from the full status label. If the artifact now implies proof, destiny, scoring, terminal consolation, or scientific authority that the evidence has not earned, the artifact fails.

The review should test both literal cropping and likely paraphrase. Ask how a supporter, critic, clinician, civil-liberties reader, religious reader, or grieving reader could reasonably restate the artifact. If the shortened meaning becomes proof, score, destiny, permission, or consolation that the evidence has not earned, rewrite the artifact before release. Use specific nouns and verbs so the object studied and the result obtained remain visible even when the surrounding explanation disappears (Spiegelhalter, 2017; Tufte, 2001).

1.6.4 — No Score or Terminal-Destiny Aesthetic

Public visuals must not turn estimates of lived experience into human rankings. Leaderboards, red/green lives, badges, glowing ranks, public burden maps, or other score-like presentations can convert a research estimate into implied trust, worth, holiness, eligibility, clinical priority, or moral standing faster than a disclaimer can undo it.

A balanced scale can illustrate the proposed equality over a completed conscious stream. It must not be presented as an observed verdict on a particular person, proof obtained at biological death, or a guarantee of spiritual justice. The stream endpoint remains distinct from a clinical marker or the last available report. Terminal research retains its narrow clinical and scientific scope (Fischhoff & Davis, 2014; National Academies of Sciences, Engineering, and Medicine, 2017).

The same distinction applies to suffering. A visual may explain the equal-and-opposite net remainder required by exact PTVC; it must not portray that requirement as moral debt, deserved harm,

purification, or a reliable technique for obtaining a chosen reward. The absence of event-by-event pairing does not remove the aggregate requirement. Section 3.7.1 states that distinction.

A safer aesthetic shows inquiry, uncertainty, care, and constraints. It may show instruments with limits, pathways with gates, rival models, consent boundaries, uncertainty bands, and correction loops. It may show the dignity of subjects and the burden of researchers. It should not show a hidden accountant, a cosmic scoreboard, or a final spiritual settlement.

Score aesthetics can also appear in research-facing tools. A model card, participant dashboard, cohort map, or quality-control display may be intended for internal use but later leak into public interpretation. The same no-score discipline should govern internal visual design when the artifact could be misunderstood outside its narrow context.

Images accompanying terminal research should represent care, clinical context, and the participant's expressed choices with dignity. Figures of scientific results should identify the measurements and processes actually depicted.

1.6.5 — Artificial Intelligence Disclosure and Limits

Artificial Intelligence may assist bounded research and public-communication tasks only under documented human responsibility, purpose-limited data use, and independent scientific review. The book must disclose and limit that role because fluent outputs can overstate evidence or omit context. Sensitive participant data require explicit authorization and governance. Treat every AI-generated summary, visual,

translation, or accessibility artifact as a new derivative artifact with source and version provenance, human review, and checks that claim status, assumptions, and correction routes survive. Section 3.6.4 governs authority and Chapter 28 governs correction and retirement.

1.6.6 — Release, Correction, and Retirement

Release review should test status, scope, source, audience, burden, hostile-crop risk, and correctability before public circulation. When evidence or assumptions change, revise or retire the affected artifact where it traveled and preserve a visible correction history. Section 28.6 contains the detailed workflow.

Chapter 2 — What Kind of Claim Is the Law of Fairness?

The word law names an ambition before it names an achievement. In this volume, it refers to a proposed invariant or boundary condition whose scientific status remains that of a candidate hypothesis. The distinction is not semantic housekeeping. It determines what observations can support, what counterexamples can reject, and what public language may responsibly claim.

PTVC sits inside a family of related statements. One statement defines a finite signed phenomenal measure. Another states the additive identity that positive mass minus negative mass equals the signed residual. A further statement proposes exact terminal closure. Optional route models explain how closure might arise. Measurement models explain how hidden objects might become data.

These statements do different work. Mathematical identities follow from definitions. Conditional theorems follow from premises. Empirical hypotheses meet the world. Estimators recover declared quantities under assumptions. Heuristic lenses organize attention. Spiritual or philosophical interpretations supply meaning. A mature science keeps the boundaries visible without stripping any category of its legitimate value.

The strong primary branch is intentionally exact and intrastream. It is not a population average, a tendency toward adaptation, or a claim that circumstances are equal. Under exact PTVC, a genuinely nonzero accumulated balance requires an equal and opposite signed remainder before the true endpoint. This does not identify particular future events, their timing, or a visible-life reward. The branch's representation and domain requirements remain binding.

Modal language matters as well. Existence of one completing continuation, positive-probability completion, fixed-policy probability-one completion, limit-sure reachability, support containment, and closure of every admissible continuation are not interchangeable. Each statement quantifies over a different object and carries a different evidentiary burden.

The chapter uses category discipline to make the positive case sharper. A candidate law can compress diverse findings only if it rules out outcomes that credible rivals permit. If every result can be redescribed as support, the word law adds grandeur but no information. If a compact structure yields risky, branch-matched consequences, the claim gains scientific content before it gains confirmation.

This is the role of severe testing. A good test is not merely capable of producing a low p-value or an attractive curve. It places the claim in a situation where a specified flaw would probably be found and where rival explanations receive enough freedom to reveal themselves (Mayo, 2018; Wasserstein & Lazar, 2016).

Underdetermination is therefore treated as a design problem. A near-zero endpoint distribution may arise from genuine closure, mean reversion, endpoint selection, smoothing, observation floors, attrition, medication, or a fitted bridge. The research task is to construct discriminators that make those explanations disagree before the decisive data are inspected.

Proper scoring and held-out prediction help by rewarding calibrated forecasts rather than stories written after the fact (Gneiting & Raftery, 2007). Equivalence procedures help

distinguish a declared finite band from simple failure to reject a difference (Schuirmann, 1987). Neither method establishes exact latent zero by itself, but each can earn a precise place in the inference ladder.

Confirmation must also remain component-specific. A valid origin calibration supports the representation gate. Successful model recovery supports the analysis architecture. A route signature may favor one candidate mechanism. Replicated complete-stream evidence under a validated bridge would bear on a terminal branch. The word support should never erase the noun that follows it.

Model change is expected in a serious program. New instruments, better endpoint adjudication, and stronger rival models may improve the inquiry. The discipline lies in recording what changed, why it changed, which earlier result motivated it, and whether the revised claim predicted something new before fresh data arrived (Lakatos, 1978).

Rescue is different. Rescue changes the endpoint, reference class, origin, sign, exclusion rule, or success criterion only after seeing that the earlier version failed. A single repair may be justified, but a sequence of result-dependent repairs can exhaust the empirical content of the program even while the prose remains impressive.

Narrowing, mechanism retirement, rival favoring, and theory-level defeat are therefore part of the positive architecture. They give the research program honest destinations other than victory or silence. A narrower true statement can be more valuable than a grand statement protected from contact with evidence.

The chapter also connects the applied series to scientific findings. Each domain can retain useful results, state its own purposes, and evaluate the evidence relevant to its proposals.

By the end of the chapter, readers should be able to identify the kind of claim, the evidence supporting it, the observations that would change it, and the scope in which it applies. Those questions support confident, precise interpretation.

2.1 — Claim Families and Category Discipline

The Law of Fairness reaches across languages that normally remain apart. It has formal language, scientific language, ethical language, public language, religious resonance, and ordinary-life usefulness. That range is one reason the project is intellectually alive. It is also the reason the project can become scientifically unsafe if every language is allowed to strengthen the others without carrying their burdens (Cartwright, 1983; Hempel, 1965; Lakatos, 1978; Lange, 2009).

Table 2.1 distinguishes the main claim categories, the evidence appropriate to each, and the language required when a claim narrows or is withdrawn.

Category discipline is especially important because the Law of Fairness may be meaningful at more than one level. A hypothesis may fail as a universal law and still leave behind a powerful question about hidden burden. A study may support a local tendency without supporting a universal terminal condition. A metaphor may remain beautiful after scientific narrowing. The point is not to force every value into the strongest category. The point is to stop the strongest category from borrowing value that belongs elsewhere.

Specific claim labels make summaries, images, captions, and conversations easier to interpret. A scientific result, a proposed model, a spiritual interpretation, and a practical recommendation each benefit from wording that identifies its role directly.

Table 2.1. Claim categories, support requirements, licenses, and withdrawal language.

Claim family	What it says	What would support it	What it licenses	Withdrawal or narrowing language
Lawlike constraint claim	A proposed invariant or boundary condition on qualifying conscious trajectories.	A replicated, reference-class-specific conjunction of component results, rival defeat, and transport success under frozen bridges.	Strongest scientific ambition, still under hypothesis status until earned.	Withdraw law-language or narrow to a smaller class when the conjunction fails.
Empirical hypothesis	A testable claim that may fail under declared measurement, endpoint, and rival rules.	Prospective results that improve prediction, survive model recovery, and reduce rival plausibility.	Prospectively specified tests, provisional support, component status updates.	Say unsupported, inconclusive, narrowed, or defeated according to the component that failed.

Claim family	What it says	What would support it	What it licenses	Withdrawal or narrowing language
Statistical tendency	A population or reference-class pattern without pathwise universality.	Stable estimates in a named population, endpoint class, instrument version, and transport window.	Local generalization inside stated bounds.	Name the class; do not return to universal language without new transport evidence.
Heuristic and analytic lens	A disciplined way to locate hidden burden, repair, consent, and reviewability.	Practical insight, improved questions, better burden detection, or safer design.	Ethical analysis and design prompts under uncertainty.	Use as a lens for questions and design.
Metaphor and spiritual resonance	An image or meaning structure that illuminates experience without evidentiary force.	Human resonance, pastoral or literary usefulness, symbolic clarity under limits.	Meaning, reflection, ritual, or communication when clearly labeled.	Retain only as image or meaning; remove proof-shaped language.

2.1.1 — Lawlike Constraint Claim

The strongest ambition of the Law of Fairness is a lawlike constraint claim. On the primary finite signed-measure branch, it proposes that every separately routed maximal qualifying conscious stream in a declared class has an exact terminal signed residual of zero over its complete strict-pre-endpoint domain. That is the most demanding version, and for that reason it must be the most exposed.

A lawlike constraint is not a moral command. It proposes a structural restriction with testable consequences; it does not assign purpose, justify suffering, or tell anyone what their experience was for.

Recurring patterns provide starting points for stronger theory. Adaptation after joy and grief, homeostatic regulation, and institutional burden shifting each suggest regularities to explain. A proposed lawlike constraint becomes informative when it adds a specified restriction that can distinguish its predictions from those of simpler accounts (Cartwright, 1983; Hempel, 1965; Lakatos, 1978).

For this volume, lawlike status cannot be announced by title, beauty, or inheritance from the formal-law book. It can only become more defensible through a staged program in which representation, bearer and routing, measurement, endpoint construction, observation, measured-to-latent transport, rival comparison, replication, and reference-class transport survive separately. The exact set-universal form is asymmetric: no finite sample can confirm every stream in the declared class, but one

independently audited, gate-valid counterexample whose truth-containing latent compatibility set excludes zero is adverse to that universal claim and may defeat it or force an explicit narrowing of the class.

Public language should identify the current stage of the claim. Proposed lawlike constraint, candidate invariant, and scientific hypothesis express the Law's ambition while leaving its standing to evidence. If evidence changes the claim, the label and public explanation change with it.

2.1.2 — Empirical Hypothesis

An empirical hypothesis is a claim that asks reality to make a difference. It is not merely consistent with the world; it expects the world to show something under stated conditions. For *The Science of Fairness*, the empirical hypothesis is not that a hidden ledger has been directly observed. It is that certain measurable consequence structures should appear if the latent architecture and the empirical bridges are adequate.

This category connects the formal target to its empirical hypotheses. Those hypotheses concern representation, measurement, endpoint classification, reserve-linked dynamics, terminal or preterminal patterns, and comparative performance against rivals. Each has its own observations, tests, and criteria for updating the research program.

A hypothesis must say what would count as an unfavorable result. It is not enough to say that future science may discover better instruments. A failed observation model can be a local failure. Repeated failure across independently motivated observation

models can become evidence against the public empirical claim. If every failure simply calls for another untested bridge, the hypothesis has stopped paying rent to reality (Lakatos, 1978; Lange, 2009).

An empirical test evaluates a named claim version: its population, endpoint class, instrument, observation model, alternatives, and analysis plan. A revised bridge receives a new version and its own evaluation, with the earlier result retained in the record.

This chapter uses empirical hypothesis language to protect the positive case as well as the negative case. A large hypothesis can be serious before it is supported if it has specific consequences, serious rivals, and defeat conditions. Calling it an empirical hypothesis is not demotion. It is the category that lets ambition become answerable.

2.1.3 — Statistical Tendency

A statistical tendency describes a pattern in a named reference class under stated measurement conditions, including its uncertainty and exceptions. Such findings can establish useful early knowledge and identify the conditions under which a pattern becomes more general.

Tendency status may be the first place where the Law-specific program earns partial public standing. A hospice cohort might show terminal concentration within an equivalence band under one validated instrument. A sleep study might show affective reprocessing patterns that fit a queue or reserve account better than a simple mood-rebound account. A longitudinal design might

show reserve-linked variance behavior in one clinical population. None of those results would establish universality.

The reference class is not an administrative detail. It is part of the claim. Evidence from adults receiving palliative care in one health system is not evidence about children, sudden deaths, nonhuman animals, artificial systems, nonclinical dying, or cultures with different reporting practices unless transport assumptions are defended. Every tendency statement must name who, where, when, how measured, and under what endpoint rule (Popper, 1959).

A tendency may prove real but local, for example, limited to a disease course, reserve trajectory, reporting-capable population, or estimator version. Report that boundary as part of the result. Scope expands only through dedicated transport evidence and may narrow when that evidence fails.

2.1.4 — Heuristic and Analytic Lens

A heuristic is a disciplined way of looking. A fairness-oriented analysis asks where burden is carried, hidden, shifted, reduced, repaired, recorded, consented to, or made reviewable. That question can be valuable even when the strong formal Law remains untested. It can help a researcher notice participant burden, a city designer notice privacy inversion, a government office notice paperwork shifted onto the sick, or a household notice care work that has become invisible.

A fairness-oriented lens earns its value by improving questions or design, not by inheriting evidentiary force from PTVC. It may identify hidden burden, consent problems, practical

improvements, or comparisons worth testing; decisions about care, eligibility, or personal standing still require independent grounds.

Applied domains may use the lens to generate research questions, and science may evaluate whether the resulting questions or designs actually improve detection, safety, or explanation. That is a legitimate form of usefulness without converting the heuristic into evidence for the strong Law.

2.1.5 — Metaphor and Spiritual Resonance

Metaphor gives difficult experience a shape. Words such as ledger, burden, queue, horizon, balance, and closure can make invisible cost more visible. They can help a reader feel the difference between outward condition and inward experience. They can also become dangerous when they start performing scientific or moral work that they have not earned.

Metaphor and spiritual resonance may illuminate experience, but they carry no empirical evidentiary force. A balance or accounting metaphor can express the conditional arithmetic of the hypothesis. It does not thereby establish PTVC, assign moral desert, promise a chosen reward, or impose a spiritual interpretation.

The right posture is not metaphor prohibition. A book without image becomes less human than its subject. The right posture is labeled beauty. A metaphor may invite attention, not settle the claim. It may open a feeling, not close an argument. It may help a reader ask where burden is hidden, not declare that burden has already been redeemed.

2.1.6 — Category Drift as Auditable Error

Category drift is the movement of a claim between statuses without public accounting. It can happen innocently. A writer may speak poetically in one passage, scientifically in another, spiritually in a third, and practically in a fourth. The problem begins when the audience cannot tell which burden the sentence is carrying.

In this project, category drift has a distinctive shape. The Law may sound like a deep law when introduced, a hypothesis when questioned, a metaphor when criticized, a heuristic when applied, and a law again when admired. That cycle gives the claim the benefits of every category while protecting it from the costs of each one. It is not fair to critics, readers, participants, or neighboring books.

Category drift is auditable because it leaves traces. A statement can be checked for what category it invokes, what evidence it cites, what authority it implies, and what correction it would require if the evidence changes. If a public artifact presents a proposed law as empirically established, it requires correction. The title Law of Fairness and a clearly framed statement of what the hypothesis proposes do not by themselves make that error. If a donor page implies validation where the science chapter says research program, the page has shifted burden (Popper, 1959).

The audit must include old materials. Category correction is not only future-facing. Earlier summaries, images, captions, talks, guides, social excerpts, and Artificial Intelligence summaries can continue to carry obsolete status. A serious project must repair its

public memory when the claim changes category. Otherwise the correction lives in the current book while the old authority continues in the world.

Status may change, but the change must be explicit, versioned, justified, and propagated to the public sentence that actually travels. A lawlike claim may narrow to a tendency; a mechanism may retire while a measurement contribution survives; a metaphor may remain after scientific defeat. Use category-specific verbs; hypotheses propose, estimators estimate, tendencies appear, lenses ask, metaphors illuminate; so the claim cannot inflate when admired or retreat into metaphor when challenged.

A useful test of category discipline is which admissible outcomes would require revising or withdrawing the claim. If no possible outcome under its declared test conditions could change its status, the sentence makes no testable empirical claim. The stronger the consequence of a valid adverse result, the more explicitly those conditions should be stated. The discipline is not punitive. It lets strong statements remain strong by attaching them to the exact risk they carry.

2.2 — Lawhood, Explanation, and Compression

The word law has a history too large for any one theory to possess. Some laws describe mathematical relations. Some describe robust empirical regularities. Some name constraints that make many phenomena intelligible at once. Some, after scientific change, lose their deepest status and survive as useful approximations. This section does not attempt to settle the philosophy of laws of nature. It asks the more local question: what would the Law of Fairness have to do before law-language becomes scientifically responsible (Box, 1976; Hempel, 1965; Lipton, 2004; Mitchell, 2009)?

The answer begins with explanation. A proposed lawlike claim must not merely restate what it hopes to explain. It must compress the field. It should make several otherwise separate features intelligible through a smaller set of assumptions, and it should do so while generating consequences that could have failed. If it imports a new bridge for every difficulty, its apparent explanatory scope is purchased by hidden complexity.

The proposed lawlike constraint concerns conscious trajectories, lived valence, endpoint definition, and measurement under uncertainty. Its scientific development brings affective science, consciousness studies, palliative practice, psychometrics, statistics, and rival dynamics into a common set of questions.

Exactness is not privileged because zero is aesthetically pleasing; it earns attention only if, under an independently identified origin, it creates restrictions that approximate regulation does not. The target and its declared bridges must therefore yield consequences

that serious rivals fail to match. If those restrictions never appear, exactness has added grandeur rather than information and the claim should contract. That is lawhood as burden, not lawhood as crown.

2.2.1 — Best-System Ambition

A best-system ambition asks whether a hypothesis belongs in the simplest, strongest, most informative account of a domain. The phrase is useful here because it refuses two shortcuts at once. It refuses the shortcut of mere regularity, since not every pattern deserves lawlike dignity. It also refuses the shortcut of isolated success, since a local fit may be outweighed by complexity, fragility, or poor rival performance.

Best-system appraisal asks whether adding the Law improves the account as a whole. The relevant comparison connects endpoint behavior, preterminal dynamics, measurement, reserve or control mechanisms, and alternative explanations. A successful account gains predictive reach and coherence while keeping its assumptions economical.

This ambition is especially demanding because the object is not a planet, gas, gene frequency, or engineered signal. It is lived experience in a conscious stream. The Law would have to compress patterns across a domain where units, origins, endpoints, and bearers are contested. That difficulty does not make the claim impossible. It makes the standards for law-language higher (Box, 1976; Hempel, 1965; Lipton, 2004).

Best-system ambition also creates a rival obligation. The Law must be compared with homeostasis, allostasis, adaptation,

predictive processing, reinforcement learning, Brownian bridge conditioning, frailty, sedation, selection, and observation artifacts. If those accounts produce the same consequence structure with fewer assumptions or better predictions, the Law loses lawlike standing even if some patterns remain interesting.

The scientific ambition is concrete: explain how reserve-linked dynamics, endpoint behavior, affective reprocessing, and hidden-burden signatures fit together. Shared predictions across those domains would create a focused comparison with alternative explanations.

2.2.2 — Explanatory Compression

Explanatory compression means that a theory reduces the number of independent facts that must be separately accepted. A compressed explanation says: these several patterns hang together because one deeper structure is doing work. The danger is that compression can be faked by hiding assumptions in auxiliary bridges, broad definitions, flexible endpoints, and adjustable measurement choices.

The Law of Fairness aims at compression across a difficult field. It suggests that the total positive and negative phenomenal variation of a qualifying stream may be governed by an exact terminal boundary rather than merely described as disconnected episodes. It asks whether preterminal and terminal patterns, if observed under strong controls, would be better explained as consequences of that candidate boundary than as adaptation, clinical practice, selection, conditioning, or observation effects.

The compression burden is therefore two-sided. The program must explain more than terminal valence closure. It must explain why a measurement stack should have a privileged origin, why an endpoint can be defined independently, why reserve or control dynamics could generate the relevant shapes, and why rivals fail on the conjunction. If each piece requires a separate special story, the program is not compressed. It is an anthology (Lipton, 2004; Mitchell, 2009).

A useful way to count compression is to ask what becomes harder to believe without the Law. If the Law merely renames adaptation, it adds no compression. If it explains why several independent signatures survive prospective specification while simpler models repeatedly miss them, it begins to compress. If it keeps adding exceptions after failures, the compression reverses: the theory becomes bulkier than the facts.

A theory can look simple because the cost of its auxiliaries is hidden from the reader. An untested measurement bridge, a convenient endpoint rule, a narrowed reference class, or an unspecified mechanism may be doing most of the explanatory work. Scientific fairness requires those commitments to be visible so that compression is not purchased by concealed complexity.

Cross-domain convergence becomes informative when one specified structure predicts several domains with a limited set of shared parameters. A candidate state or potential can be estimated in one module, frozen, and used to predict another. The comparison should record the cost of transferring parameters, converting units, and adding context-specific terms. A structure that retains calibrated predictive performance across those

transitions offers a concrete form of explanatory compression: the same commitments explain more of the observations.

2.2.3 — Predictive Novelty

Predictive novelty tests a model on observations it was developed to anticipate. In this volume, terminal patterns, emotional adjustment, and clinical trajectories provide settings in which competing accounts can make specified predictions before the outcomes are inspected.

A prediction is not novel merely because it is printed in a book. For the prospective tests emphasized here, predictive novelty requires evaluation on observations not used to construct or tune the prediction. Discrimination additionally requires the competing models to assign different expectations to those observations. For the Law, candidate consequences might include reserve-linked variance shape, forecast performance under a prospectively specified independent-endpoint branch, model-recovery success, or a signature conjunction that simpler homeostatic and conditioning accounts do not reproduce.

Existing observations can provide comparative evidence when models assign them different probabilities under explicit assumptions. Retrospective comparisons should record how the hypotheses, measurements, and datasets were selected, account for model flexibility, and retain independent tests wherever possible. Prospective prediction offers especially clear separation between model construction and evaluation. Both approaches contribute when their information and selection processes are visible (Salmon, 1984; Woodward, 2003).

Accommodation is not forbidden. A new program often begins by trying to make sense of known anomalies, recurring intuitions, or unexplained patterns. The rule is that accommodation must be labeled and then converted into prospective risk. If the Law explains a known terminal report pattern, the next question is what new data it predicts differently from its rivals.

Prequential performance gives one useful discipline: prediction is assessed as data arrive, not after the full dataset is available for interpretive adjustment. A program that repeatedly improves held-out forecasts under frozen bridges earns more respect than a program that repeatedly fits the past by moving its categories. The Law should seek the former path.

The public language should track this distinction. A result that was predicted prospectively may be called prospective support for the tested component. A result fitted after exploration may be called a generated hypothesis or an accommodated pattern. Both can be part of science. Only one carries the full burden of prediction.

2.2.4 — Mechanistic Depth

The strong Law proposes a structural relation among conscious trajectories. Candidate mechanisms explore how that relation could be realized through reserve-coupled dynamics, queue processes, finite-horizon control, interoception, opponent dynamics, memory reprocessing, or other processes examined in later chapters.

Mechanistic depth comes from specifying how a proposed process works. A Queue model may describe burden ordering;

reserve coupling may generate a mathematical realization; interoceptive control may offer a brain-body route. Each candidate can be developed, compared, and revised on its own evidence.

Mechanistic depth links formal quantities to observables, observables to biological or psychological processes, and those processes to distinct predictions. A mechanism contributes by explaining a process, deriving a restriction, or separating competing accounts (Box, 1976; Hempel, 1965; Lipton, 2004).

The Law's mechanism burden is unusually high because felt experience is not an external object like temperature on a wall. It is lived from within and inferred from without. A mechanistic account must therefore respect first-person evidence, physiological signals, behavioral patterns, clinical realities, and the fact that no single channel owns the person. Depth here means disciplined humility across channels.

Mechanistic depth also limits public overclaim. If a mechanism is candidate, the text must say candidate. If a process is a realization, the text must say realization. If a formula demonstrates possibility rather than observation, the text must say construction, not evidence. The same precision that satisfies experts protects vulnerable readers from hearing destiny.

2.2.5 — Causal and Structural Explanation

Causal and structural explanations answer different questions. A causal explanation asks what process produced an outcome. A structural explanation asks what constraints shape the space of possible outcomes. *The Law of Fairness*, in its strongest ambition,

is closer to a structural boundary condition than to a proximate causal story. But public readers often hear law-language as if it named an agent making events happen.

A structural constraint limits possible trajectories; it does not choose events or supply moral purpose.

Causal explanation still matters. If the proposed boundary condition has empirical consequences, some process must mediate the observable patterns. Reserve, adaptation, sleep, memory, interoception, prediction, and control systems are causal or dynamical candidates. The science can ask which processes generate which patterns without implying that the processes morally justify those patterns (Lipton, 2004; Mitchell, 2009).

A useful local rule is to keep three questions apart. What is the formal target? What process could realize the relevant consequence? What data would distinguish that process from rivals? The first belongs to received formal structure, the second to mechanism chapters, and the third to study and inference chapters. Chapter 2 supplies the category discipline that keeps them from impersonating one another.

Structural explanation also creates a special defeat problem. A mechanism can fail while a structural claim remains open, but a structural claim cannot hide forever behind failed mechanisms. If all independently motivated mechanisms, observation bridges, endpoint rules, and consequence tests fail or become ad hoc, the structure loses empirical content even if metaphysical possibility remains.

The same analytic lens makes this an issue of burden honesty. A theory that claims structural depth while refusing causal or empirical accountability shifts the cost of its grandeur onto others. A theory that distinguishes structure, cause, and measurement lets each carry the burden it actually owns.

2.2.6 — Lawhood Under Rival Pressure

Lawhood under rival pressure means that the favored claim and its alternatives receive comparable information, fitting freedom, diagnostic attention, and opportunity to fail. Rivals must be generative rather than decorative: Brownian bridges, ordinary mean reversion, sedation, disease severity, and other competitors should produce their own expected patterns and face held-out scoring under rules fixed before results (Gneiting & Raftery, 2007; Mayo, 2018).

The positive case becomes meaningful through conjunctions that strong rivals fail. Drift, concentration, or a terminal band alone may be imitated, whereas measurement validity, noncircular endpoint construction, required timing premises, route-state restrictions, model recovery, held-out prediction, and transport expose different failure routes. Lawhood therefore requires victories across study families, each attached to its own reference class and limits (Salmon, 1984; Woodward, 2003).

Rival pressure is a protection against confidence by identity. If adaptation explains a field better, the Law narrows; if endpoint conditioning explains apparent closure, that evidence is withdrawn; if measurement fails, the program abstains. Letting

ordinary explanations win makes the inquiry more publicly trustworthy, not less ambitious.

A useful lawhood audit asks four questions: does the hypothesis compress more structure than it imports, generate prospective restrictions, connect to mechanisms or structural constraints that can fail, and outperform serious rivals? Compression is not simplicity at any cost. A complex model may be warranted, while a simple sentence may merely hide bridges, endpoints, observation rules, and trust assumptions. The audit should therefore count auxiliary assumptions and model flexibility alongside the apparent economy of the headline claim.

Novelty adds temporal discipline. Retrospective explanation can illuminate known phenomena, but progress requires consequences specified before observation. Mechanism and structure add empirical risk: reserve coupling, endpoint dynamics, and interoceptive or control routes are candidate ways for the boundary claim to produce consequences that can be lost. Their evidentiary value increases when the predicted direction, timing, or dependency is fixed clearly enough to distinguish success from later reinterpretation.

Rival performance is the sharpest question. A moving account of hidden burden may still lose to adaptation, homeostasis, predictive regulation, treatment, or observation artifacts. The relevant measure is surplus explanation after those rivals receive full strength, not the emotional force of the theory. A rival victory should therefore identify which additional structure proved unnecessary and which observations gave the competing account its advantage.

Lawhood must also be earned inside the claimed domain. Conservation analogies, thermodynamic or quantum vocabulary, cosmic imagery, capitalized names, and formal symbols may orient readers but supply no necessity or evidence about conscious streams. The proposal stands or falls through its own definitions, discriminators, endpoint discipline, and willingness to forbid outcomes. Borrowed scientific language becomes useful only when it contributes a precise relation or test that remains valid in the phenomenal domain being studied.

A strong future case remains possible. Repeated prospective restrictions, failure of simpler rivals on a predeclared conjunction, and transport without moving the bridges would make law-language increasingly earned. The expert reader should be able to reject the hypothesis and still recognize a fair standard: clarity, assumption visibility, empirical risk, and a genuine capacity to lose. Confidence should rise only to the extent that these successes persist across independent teams, settings, and measurement implementations.

2.3 — Component Claims and Reference Classes

Separating component tests makes a large hypothesis easier to investigate. *The Law of Fairness* connects bearer identity, felt valence, temporal accumulation, endpoint adjudication, candidate dynamics, measurement, and public verdict language. Testing those components individually shows which part of the proposal a result supports or challenges, allowing each finding to guide the appropriate next step (Borsboom, 2005; Hernán & Robins, 2020; Manski, 2003).

Component testing supports cumulative progress. A representation question can remain open while an instrument is revised; an endpoint study can guide measurement development; and a reserve model can sharpen comparison with control-based explanations. Local results show which part of the architecture should be developed next.

Reference classes complete the discipline. Every component result belongs somewhere and not everywhere. A population, endpoint type, disease trajectory, culture, age group, instrument version, and analysis window are not incidental details. They are the scope of the claim. Without them, the project would carry evidence beyond the people, contexts, and methods that produced it.

The following ladder states the dependencies among research tasks. It shows which results enable the next stage and where further development is needed.

Table 2.2. Component-evidence dependency ladder.

Ladder step	Question answered	Possible favorable outcome	Possible failure outcome
Representation	Can lived experience support a signed, temporally additive research target?	A defensible target for restricted scientific use.	No current scalar target; exact-zero claims narrow or abstain.
Measurement	Can candidate momentary and interval estimators or successors estimate the target in context?	Context-limited estimator with uncertainty and version control.	Instrument failure; no Law-specific inference from that channel.
Endpoint	Can the stopping event be classified prospectively and independently?	Endpoint class usable for specified study families.	Terminal inference rejected or delayed.
Realization	Do reserve-coupled or control dynamics describe the data-generating process?	Mechanism or candidate realization survives rivals locally.	Mechanism retires or narrows; other components remain distinct.

Ladder step	Question answered	Possible favorable outcome	Possible failure outcome
Consequence	Does the protocol-fixed signature conjunction appear?	Component support for specified consequence family.	Consequence claim weakens or is defeated in that class.
Reference class	Does the result transport across populations and conditions?	Scope expands cautiously.	Support remains local or disappears.

2.3.1 — Representation Claim

The representation claim asks whether lived experience can be treated, for a scientific purpose, as a signed and temporally additive target. This is not a casual assumption. It asks whether moments of felt burden and relief can be represented in a way that permits accumulation without flattening dignity, meaning, context, or depth into a crude score.

The representation claim is logically prior to measurement. An instrument cannot measure what the theory has not represented coherently. If valence is only ordinal, if neutral origin cannot be defended, if temporal additivity fails, or if experiences differ in kind in ways no scalar can responsibly carry, then the strongest ledger language must narrow before any terminal study begins.

A scientific representation selects the features needed for a defined question. A signed valence-rate representation can be evaluated by how coherently and usefully it describes its intended research context (Borsboom, 2005).

For the Law, the representation claim includes the origin problem. Exact closure depends on a neutral point that is not created by subtracting a sample mean or centering a questionnaire. If the origin is arbitrary, then exact zero is not identified. The project must either defend a privileged origin, test quantities invariant to the chosen gauge, or narrow the claim.

Representation testing determines which mathematical operations a domain supports. Its result can license the next measurement stage, identify a repair, or direct the program toward an alternative representation.

The representation claim is therefore the first place where law-language can lose height. A theory about felt experience cannot borrow the simplicity of its notation. It must show that the notation is worthy of the subject.

2.3.2 — Measurement Claim

The measurement claim asks whether momentary and interval valence estimators or later instruments estimate the intended construct inside a declared context of use. It is not a claim that an instrument reads inner life directly. It is a bridge claim: given a representation, a population, a protocol, a set of channels, and an error model, the estimator tracks enough of the target to support the specified inference.

Instrument qualification is context-specific. Sedation, communication capacity, medication, and language can change the performance of physiological estimators, self-report measures, wearables, and item banks. Testing those contexts

identifies where an instrument supports inference and where it needs adaptation.

The measurement claim must include uncertainty, missingness, drift, calibration, and versioning. A candidate momentary-or-interval estimate is not a human-worth score, civic trust, spiritual rank, religious standing or commitment within a spiritual community, or public eligibility. It is a research estimate under constraints. The word estimate should remain visible wherever public language might turn the measure into access or rank (Borsboom, 2005; Manski, 2003; Nissenbaum, 2010).

A favorable measurement result has a limited grammar. It may say that an instrument estimates a target well enough for a study family under specified conditions. It may not say that the instrument has seen the latent cumulative process or the whole person. If the measurement bridge later changes, earlier claims must be indexed to the old version rather than silently reinterpreted.

Measurement claims are also reference-class claims. A tool validated in English-speaking adult hospice participants cannot be applied without transport to children, different cultures, nonverbal states, neurodivergent populations, sudden deaths, animals, or artificial systems. The burden of extension belongs to the project, not to the people being generalized over.

2.3.3 — Endpoint Claim

The endpoint claim asks whether the truth-level endpoint of a maximal qualifying stream can be prospectively defined without using the ledger outcome later tested. Biological death, legal

death, narrative closure, family memory, or sensor cessation are not substitutes. Defining the endpoint by ledger zero would make closure tautological, so endpoint channels must be prospectively specified and not algebraically identical to the measured cumulative estimate; residual dependence must be modeled. On a separately declared independent-endpoint branch, timing independence from the ledger-generated information set and the branch's support conditions are additional premises.

Endpoint claims vary by class. Slow terminal decline, sudden death, resuscitation, coma, anesthesia, dementia, delirium, sleep, and disorders of consciousness present different evidentiary structures. A result in one endpoint class cannot silently transport to another. Sudden low-signal endings, in particular, cannot be filled in by analogy with palliative windows unless the bridge is defended (van der Vaart, 1998).

Endpoint validation permits terminal or near-terminal data to be interpreted under a defined stopping rule. When the boundary remains unresolved, the analysis reports an indeterminate classification and identifies the evidence needed to resolve it.

Endpoint studies place care, comfort, and consent first. Their procedures combine ethical participation with independently justified classification, and record an indeterminate endpoint when either requirement prevents adequate observation.

The endpoint claim is therefore both scientific and humane. It keeps the theory from defining victory into existence, and it keeps research from treating the last vulnerability of a person as a laboratory convenience.

2.3.4 — Realization Claim

The realization claim asks whether a candidate dynamical model describes the relevant data-generating process. Sections 4.2 and 7.6 develop a reserve-coupled realization that derives closure-related consequences from explicit assumptions. Testing those assumptions and consequences connects the construction to observed dynamics.

A realization claim is not identical to the strong latent claim. The Law might be stated as a boundary condition while a reserve product, queue model, finite-horizon control process, or interoceptive mechanism proposes how patterns could arise. If a candidate realization loses, that result may retire the realization without instantly exhausting the space of possible realizations.

A realization that imposes an endpoint by construction can generate useful predictions about the path leading to it. Empirical evaluation compares those predictions with observations whose selection and measurement are specified independently of the constructed result (Borsboom, 2005; Hernán & Robins, 2020).

The realization claim requires rival processes. Ordinary homeostatic mean reversion, endpoint-conditioned processes, safe-control baselines, and predictive regulation can all generate orderly trajectories. The question is not whether the Law can generate a pattern. The question is whether it generates a conjunction, under frozen assumptions, that rivals do not match as well.

A favorable realization result might support a mechanism family: reserve coupling in slowly declining terminal illness, queue

pressure in certain affective reprocessing, or finite-horizon shadow pricing in choice behavior. Each support statement names the mechanism, the tested reference class, and the evidence that distinguishes it from competing accounts.

The realization claim is where a good theory becomes willing to lose machinery it likes. Reserve-coupled dynamics, completion-potential constructions, conditioned bridges, or any other candidate process must be removable without collapsing dignity or hiding failure. A mechanism that cannot be retired has become doctrine.

2.3.5 — Consequence Claim

The consequence claim asks whether the prospectively specified signature conjunction appears and out-predicts rivals. This is where the Law begins to touch data in a way that can matter. The relevant question is not whether one curve looks suggestive. It is whether a declared family of consequences appears under the declared bridge, with rivals given serious opportunity to win.

A consequence can be preterminal, terminal, sleep-related, life-course, behavioral, physiological, or model-comparison-based. The strongest empirical burden in the current program begins with prospective and comparative consequence structures because exact latent equality is not directly observable. Support is therefore indirect, component-level, and bridge-dependent.

Consequence claims must specify the direction of inference. A terminal concentration result may support a bounded terminal consequence under one endpoint and instrument. A variance-shape result may support a reserve-linked process if measurement

noise, sedation, frailty, and endpoint conditioning are ruled out. A held-out prediction gain may support one model class over another. None alone establishes the exact set-universal target over the declared class (Gneiting & Raftery, 2007; Manski, 2003).

The chapter's worked example is simple. Suppose a prospectively specified palliative study on a declared independent-endpoint branch finds reserve-scaled variance compression, prospective forecast gain under the branch's timing assumptions, and better model recovery for the reserve-coupled candidate than for a homeostatic baseline in one adult cancer cohort. That would be important route and component support. It would not show that every qualifying stream in the declared class closes exactly. It would say that one consequence family survived in one reference class under one protocol-fixed bridge.

Suppose adaptation and medication-adjusted frailty models predict terminal patterns as well as or better than the PTVC candidate. The result then supports parity or preference for those explanations in the tested class and guides the next version of the consequence model.

2.3.6 — Reference-Class Discipline

Reference-class discipline turns a sample description into the boundary of warranted speech. Every empirical status statement should name the population, condition, endpoint class, instrument version, analysis window, and transport assumptions to which it applies. The strongest formal target is broad; every first study is local.

A selected clinical cohort can speak most directly to comparable cohorts, less directly to sudden interruption or other endpoint classes, and almost not at all to nonhuman animals or artificial systems without additional bridges. A study of report-capable adults cannot silently represent nonreporting states, and a culture-specific calibration does not become global because the result is striking. Local truth is not lesser truth; it is truth with honest borders (van der Vaart, 1998).

Transport assumptions must be stated before they are used. A claim that a signature should persist across diagnoses, cultures, communication states, or endpoint classes must name the invariance asserted and the rival reasons it might fail. If the bridge changes after an adverse result, the new bridge defines a new version rather than rescuing the old claim without loss.

The component program forms a dependency lattice. Representation, measurement, bearer and routing, endpoint, realization, consequence, rival discrimination, replication, and transport each have their own validation tasks and connections to the others.

The strong latent claim is the most dependency-heavy node, not a crown that lends authority downward. Failure at one node weakens the claims that depend on it while leaving neighboring nodes open. Conversely, genuine support at a lower node should be reported as support for that node rather than minimized to metaphor merely because the strong target remains unproved.

Reference class runs through every component. Representation may work for some forms of experience and not others;

measurement may be stable in one language, disease class, or communication state and not another; endpoint adjudication may be defensible in slow decline and weak in sudden interruption; route signatures may depend on care context or reserve definition. Transport is therefore part of each claim, not a final decorative test. Two findings can therefore differ without contradiction when their reference classes differ. A genuine conflict begins only when the relevant populations, instruments, endpoint rules, and conditions overlap enough that both claims address the same empirical domain.

Partial identification is a legitimate result. Data may constrain a truth-containing region without isolating one latent value or one explanation. A bounded identified set is progress when its boundaries, assumptions, and unresolved rivals are reported as such rather than converted into hidden certainty. The structure of the identified set can itself guide the next study by showing which assumptions, channels, or bridges carry the remaining uncertainty. Progress can therefore consist in shrinking or partitioning that set with independently justified information before a point estimate becomes defensible.

Component-level reporting supplies the grammar for honest updating. Evidence may strengthen one measurement bridge, weaken an endpoint rule, retire a route, or leave the set-universal target open or narrowed. That is more precise than saying a study supports or refutes the Law without naming the object that changed. This approach allows heterogeneous evidence to accumulate without forcing a single book-level verdict. An improved estimator, endpoint method, or recovery procedure can

remain a durable scientific contribution even when a mechanism fails or the strongest branch remains unresolved.

Transport analysis must name the target population, relevant differences from the enrolled cohort, plausible effect modifiers, and the assumptions needed for reweighting or structural transport. When those assumptions fail, the result remains local or set-valued; sample size inside a narrow class cannot manufacture support outside it (Manski, 2003; Hernán & Robins, 2020). Transport sensitivity should reveal which assumptions are doing the work rather than presenting one reweighted estimate as though extension were automatic. Agreement across independently defensible transport specifications strengthens generalization, while disagreement identifies the boundary that future data must resolve.

Instrument and protocol versions also belong inside the reference class. A classifier validated under one medication pattern, device generation, language, reporting capacity, or care setting may change elsewhere. Failed transport is evidence about the phenomenon, the measurement system, or both, and should narrow public speech rather than be dismissed as nuisance replication (Little & Rubin, 2019; National Academies of Sciences, Engineering, and Medicine, 2019). A version change should therefore be treated as a potential change in reference class until equivalence or transport has been demonstrated. Reporting the exact instrument, software, protocol, and calibration version makes replication interpretable and prevents implementation differences from being mistaken for substantive contradictions.

2.4 — Severe Tests and Underdetermination

An informative scientific test distinguishes a model's predictions from alternatives the method can detect. Terminal and affective observations make that comparison especially useful because several latent processes can generate similar measured patterns. Explicit alternatives and common comparison rules turn those possibilities into tractable questions (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018; Quine, 1951).

Underdetermination is not a skeptical trick. It is a normal feature of scientific inference, intensified here by latent experience, noisy channels, terminal vulnerability, missing data, and endpoint ambiguity. A terminal curve can converge for reasons that have nothing to do with terminal valence closure. A measured cumulative estimate can stabilize because a sensor loses gain. A person's reports can change because medication, social pressure, delirium, adaptation, or family context changed.

The answer is not despair. The answer is severe testing. A severe test makes specific errors likely to show themselves if they are present. It does not merely seek confirmation. It designs the world-facing encounter so that rival explanations, measurement failures, and auxiliary assumptions can break the favored claim.

2.4.1 — Risky Prediction

A risky prediction is one that could plausibly have failed and would have mattered if it did. *The Law of Fairness* should not claim support from patterns it would have expected no matter what happened. If the theory can treat convergence, divergence,

volatility, flatness, silence, and missingness all as favorable without prior rules, it has no empirical risk.

Risk is created by forbidding or disfavoring outcomes. A reserve-coupled signature might predict a particular relationship between reserve decline and ledger variance under a specified bridge. A rival homeostatic model might predict local mean reversion without the same terminal scaling. A Brownian bridge might predict endpoint convergence without independent reserve coupling. The risky prediction is not the pattern alone; it is the pattern as differentially expected across models.

The prediction must be generated before outcome inspection, not narrated afterward. Prospective specification creates a dated record of hypotheses, endpoints, instruments, analysis windows, exclusions, and rival families. It does not sanctify a weak design; it distinguishes a risk taken in advance from an explanation composed after the answer. Public preregistration is claimed only when a timestamped external registry and persistent identifier actually exist (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018; Nosek et al., 2018).

Risky prediction also protects against private hope. A community that wants the Law to be true may notice favorable anecdotes and forget silent failures. A research program that wants the Law to be true may adjust models until the result looks natural. The remedy is to state in advance what the model should not do well under rivals and what would count against it.

A risky prediction can be modest. It need not test the whole Law. It may test one component: a measurement bridge, endpoint

classifier, model-recovery procedure, or reference-class transport assumption. Smaller severe tests often teach more than grand unfocused tests because they know exactly what they can break.

Falsifiability, identifiability, and adjudicability are different achievements. A sentence may be logically falsifiable because a counterexample is conceivable, yet no current study can adjudicate it if the latent target, endpoint, or measured-to-latent bridge is unidentified. Conversely, a highly measurable surrogate may be adjudicable without testing the received target. The program advances only when a design connects a vulnerable claim to an identifiable object and a decision rule that would respond differently in relevant rival worlds (Borsboom, 2005; Mayo, 2018; Woodward, 2003).

The public language after a risky prediction should be restrained and exact. If the prediction succeeds, say which component survived and against which rival set. If it fails, say which component lost. Do not let a successful local prediction become total validation, and do not let a failed prediction vanish into metaphor.

2.4.2 — Underdetermination by Terminal Patterns

Terminal observations bring several explanatory processes together. Clinical care, medication, sedation, frailty, reporting changes, family context, selection, sensor performance, and endpoint conditioning can each influence an observed trajectory. A study becomes informative by distinguishing their predictions.

Concentration near an endpoint is not unique. A Brownian bridge conditioned to end at a value can show approach behavior by

construction. A clinical process that reduces variability near death can mimic variance compression. A dataset that loses the most distressed participants can appear to stabilize. A scale that loses sensitivity at low arousal can produce false neutrality. Each rival must be treated as a potential generator, not as a footnote.

The Law-specific question is therefore conjunctive. Is the endpoint independently defined? Does the result survive observation-model and missingness checks? Is any reserve variable estimated without using the ledger outcome? Does the candidate out-predict endpoint-conditioned, clinical, and measurement rivals on held-out data? Most importantly, can the analysis recover each generator from simulated worlds where the truth is known? A test that returns the same favorable verdict in candidate and rival worlds is not severe, however sophisticated its fit appears (Mayo, 2018; Quine, 1951; Wilson & Collins, 2019).

Terminal studies combine careful inference with comfort-first care. Their design accounts for clinical context, communication, observation quality, and consent so that participation remains proportionate and the resulting comparisons are interpretable.

A terminal pattern can provide valuable comparative evidence when its measurement, endpoint procedure, and alternative explanations are specified. The result identifies the consequence supported within that study and the next conditions to investigate.

The correct public sentence is not: the terminal pattern confirms the Law. The correct sentence is more demanding: this terminal pattern, in this reference class, under this endpoint and

observation model, supports or fails to support this consequence relative to these rivals. That sentence is less dramatic. It is also the beginning of science.

2.4.3 — Model Pluralism

Model pluralism means keeping multiple live explanations in the field long enough for evidence to discriminate among them. It is not indecision. It is the refusal to let one favored vocabulary monopolize interpretation before the data have done their work. For the Law of Fairness, pluralism is mandatory because affective dynamics, consciousness, endpoint states, and measurement each have serious existing theories.

The rival set must include models that are attractive for reasons independent of the Law. Hedonic adaptation explains many returns toward baseline. Opponent processes explain rebound and counterweighting. Predictive processing and active-inference accounts explain regulation through prediction and error. Homeostasis and allostasis explain viability-centered regulation. Reinforcement learning explains policy adaptation under reward and constraint. Clinical models explain terminal decline through disease, treatment, and care context.

A pluralistic field also includes statistical and observation rivals. Smoothing can manufacture approach. Noise floors can manufacture variance contraction. Equivalence procedures can be applied to biased estimators. Missingness can select the visible sample. Flexible latent models can absorb the same pattern under different mechanisms. These rivals deserve equal information, parameter, and tuning budgets because a candidate has not won if

the comparison was calibrated to let it win (Gelman & Loken, 2014; Stanford, 2006; Wilson & Collins, 2019).

Model pluralism protects the positive case by demanding specificity. If the Law wins only against weak rivals, its victory is shallow. If it wins against strong rivals in a known reference class, its status improves. A theory that can say which rival would have explained the result and why that rival failed has earned more trust than a theory that treats alternatives as disbelief.

Pluralism also protects against premature defeat. If one Law-specific mechanism fails, another component may remain live. If one rival wins in one class, another class may remain open. The discipline is not to preserve the Law at any cost, but to identify exactly which model won where and why.

2.4.4 — Auxiliary-Hypothesis Failure

Every empirical test of a large theory depends on auxiliary hypotheses. *The Law of Fairness* depends on assumptions about representation, measurement, endpoint classification, instrument calibration, sampling, rival specification, missingness, and transport. When a test fails, the failure may belong to one of those bridges rather than to the strong latent claim itself.

Auxiliary dependence creates a legitimate diagnostic problem and a familiar escape route. A proponent can always announce that some bridge failed while leaving the core untouched. The responsible response localizes the failed bridge, demonstrates the failure with independent evidence, changes public status immediately, and permits a successor only when the repair creates

a new risky prediction. Without those conditions, auxiliary diagnosis becomes immunity.

For example, suppose a terminal study fails because the endpoint classifier was later shown to conflate irreversible loss of consciousness with reversible sedation. That may be an endpoint-bridge failure, not a defeat of the strong form. But the repaired endpoint bridge must be specified prospectively in the next study. The old result cannot be rebranded as support after the bridge changes (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018).

Suppose instead that multiple endpoint classifiers, measurement systems, and rival comparisons all fail to show Law-specific consequences in a reference class where they should have appeared. At some point, the appeal to auxiliary failure becomes degenerating. The core may remain metaphysically conceivable, but the public scientific claim loses status.

A title-owned fairness analysis again reveals the fairness problem. If the core keeps prestige while each failed bridge is discarded without public cost, the theory has shifted burden onto instruments, participants, critics, and future studies. Auxiliary hypotheses may protect inquiry only when their own failures are recorded and paid for in claim status.

2.4.5 — Local Versus Theory-Level Defeat

Local defeat occurs when a component, mechanism, instrument, endpoint class, reference class, or prediction fails. Theory-level defeat occurs when the failures reach the dependencies required by the strong claim and cannot be repaired without destroying

empirical content. Confusing these two levels makes science either too brittle or too evasive.

A local defeat may retire a momentary valence estimator version, a reserve proxy, a terminal classifier, a conditioned-path candidate route, a completion-potential model, or a reference-class transport assumption. Such defeats are valuable. They teach the program what not to say. They prevent weak bridges from carrying stronger public language. They also allow surviving components to remain under discipline.

A weak study does not defeat the strong claim. One gate-valid counterexample can contradict a declared set-universal branch under its stated inferential guarantee. Separately, repeated failure of independently motivated bridges and discriminating predictions can justify retiring the empirical program without establishing that the latent target is false. A theory that preserves its headline only through outcome-dependent changes to origins, endpoints, instruments, or reference classes loses the scientific standing claimed for that version (Mayo, 2018; Quine, 1951).

The exact set-universal form is asymmetric. No finite sample confirms every qualifying stream in a declared class. A counterexample matters only when the stream qualifies under the prospectively fixed bearer and routing rules, the truth-containing residual object is nonempty and excludes zero, and the representation, endpoint, observation, missingness, and measured-to-latent bridges are adequate for that conclusion. When those conditions hold, one stream is adverse to the protocol-fixed universal branch; a population average cannot rescue it.

The chapter's second worked example lives here. Suppose adaptation models repeatedly out-predict Law-specific models in slowly declining adult terminal cohorts, while the measurement and endpoint bridges are adequate. The Law need not vanish from every domain, but it must narrow in that class, retire the defeated consequence claim, and stop using those cohorts as support. If future work proposes a new bridge, it begins as a new version.

Distinguishing local from theory-level defeat protects both ambition and honest inference. It keeps a difficult idea open to productive investigation while ensuring that adequate contrary evidence has its stated consequence.

2.4.6 — Severe-Test Reporting

A severe test is incomplete unless the report states what had a genuine chance to go wrong and demonstrates that the analysis could have detected it. The report names the component tested, empowered rivals, frozen bridges, negative controls, model-recovery thresholds, failure routes, and decision rule for status change. Without a recovery result, apparent discrimination in real data may only reflect an analysis that favors one model by construction.

Report each study through its scientific object. A palliative study examines a defined endpoint class and consequence family; a sleep study investigates reprocessing and associated dynamics; a measurement study qualifies an estimator. This specificity lets each contribution receive appropriate recognition.

The report should distinguish prediction, retrodiction, accommodation, and exploration. A predicted effect under a

frozen plan has one status. An exploratory pattern has another. A model tuned after data has another. All can be useful. They cannot be given the same public language (Stanford, 2006; Wilson & Collins, 2019).

Severe-test reporting includes failed and unusable results. An unavailable estimate can identify unfinished measurement work; informative missingness can identify the conditions needed for inference; and a rival victory can show that PTVC-specific structure is unnecessary in the tested class. Each outcome has an explicit next action.

Public reporting should make meaningful scientific progress easy to recognize. Name the component supported, the observations that changed its status, and the next question those findings open. Precise reporting gives each result its full value.

The final reporting rule is hostile-crop safe: an isolated result sentence still names the component, reference class, and status ceiling. If three distant paragraphs are needed to prevent a local sentence from implying proof, the local sentence carries too much borrowed authority and must be rewritten.

Example: terminal concentration near a neutral band has little Law-specific force if endpoint-conditioned simulations, sedation, or informative dropout reproduce it. The result becomes more discriminating only when independently estimated reserve relations, stable measurement gain, noncircular endpoint classification, and held-out superiority to strong rivals survive together; and even then it is component support in that reference class, not universal validation.

A second worked example goes the other direction. Suppose the Law-specific model predicts a reserve-linked signature, but a clinical frailty model with medication, delirium, and care-context variables out-predicts it across held-out sites. Suppose model recovery shows the analysis could have detected the Law-specific pattern if it were present. In that case, the rival wins for that class. The correct response is narrowing or defeat of the consequence claim there, not a search for a poetic interpretation.

These examples show why underdetermination is not a reason to stop. It is a reason to build sharper discriminators. If several latent stories can explain one pattern, the program should seek patterns that divide them: timing, reserve coupling, noncircular endpoint construction, branch-specific endpoint premises, negative controls, transport behavior, and forecast performance. The work becomes more technical because the claim is large.

Severe testing also changes the moral tone of the book. The project is not asking dying people, sleeping people, grieving people, or ordinary readers to serve as illustrations for a beautiful idea. It is asking whether a precisely stated claim survives designs that count the burden of being studied. A severe test that violates care is not severe in the sense this book requires, because it has shifted the burden of clarity onto vulnerable subjects.

Underdetermination remains even after good tests. The goal is not to eliminate every alternative forever. The goal is to reduce live alternatives in a declared model space and report the remainder. A scientific verdict is always relative to what was compared, what was measured, and what could have been detected.

Severe testing must be designed before emotional investment hardens around a preferred result. If the first genuinely dangerous test arrives only after the community has built identity, public artifacts, and institutional plans around support, the social cost of losing becomes part of the data environment. A humane research program therefore makes loss speakable early. The chapter's philosophical point has a practical purpose: reduce the burden carried by future scientists who may need to report disappointing results.

2.4.7 — Auxiliary Hypotheses and the Duhem Problem

No empirical test confronts an isolated proposition. A terminal-pattern study simultaneously relies on a representation of valence, a neutral origin, a stream-routing rule, an endpoint construction, an observation process, an estimator, a missingness model, a rival set, and a decision rule. When the prediction fails, the data do not announce which member of that network was responsible. This is the practical force of the Duhem problem for the Law of Fairness.

The answer is not to pretend the network can be eliminated. The answer is to type it. Each auxiliary should have an owner, a source of independent motivation, a validation route, a version, and a known effect on the inference. Representation failure should not be redescribed as endpoint failure. Endpoint circularity should not be repaired by widening an equivalence band. A model-recovery failure should stop real-data interpretation rather than become a footnote after a striking curve appears.

A dependency graph makes this discipline visible. Upstream gates include whether the target quantity is coherent and whether the

instrument can identify it in the relevant population. Midstream gates include routing, time, endpoint, and observation. Downstream gates include estimators, candidate signatures, rivals, multiplicity, transport, and verdicts. Success at a downstream node cannot compensate for failure above it, because the downstream calculation would no longer be about the intended object (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018).

Auxiliary hypotheses can often be tested with data independent of the principal prediction. Neutral-origin procedures can be evaluated in longitudinal cohorts; missingness models in synthetic and semi-synthetic data; endpoint rules against independent clinical adjudication; and model recovery before cohort analysis. Together, these studies qualify the components used in terminal inference.

Synthetic challenge data are especially useful because the generating truth is known. Researchers can ask whether the pipeline recovers a PTVC-consistent route when that route generated the data, an Ornstein-Uhlenbeck process when mean reversion generated them, a conditioned bridge when endpoint conditioning generated them, and an observation artifact when dropout generated the shape. If the pipeline cannot tell these worlds apart in simulation, it cannot claim to distinguish them in human data.

Negative controls provide another route through underdetermination. A candidate signature should disappear, reverse, or remain invariant under declared perturbations for reasons specific to the model. If the same result appears under a

control window that should contain no endpoint information, if it survives randomized endpoint labels, or if it strengthens when measurement quality worsens, the signature is probably not carrying the intended meaning. Controls turn abstract rival pressure into inspectable consequences. A useful control therefore challenges a specific pathway by which the apparent signal could be produced without the candidate mechanism. Convergence across controls with different failure modes provides stronger discrimination than repeating variations of the same check.

Underdetermination changes what a favorable result means: within a declared representation, endpoint, observation model, estimator, and rival set, one prediction structure survived better than the alternatives tested. The program gains credibility by reducing viable explanations while keeping the assumptions and unresolved alternatives visible.

Endpoint analysis brings several ordinary processes into the same frame: changing reserve, care, family presence, medication, speech, and observation quality. Comparing their predictions helps identify which process explains a terminal pattern. The most informative designs seek observations on which these processes predict different timing, direction, or magnitude rather than merely asking which can fit the same final curve. This turns clinical complexity into comparative leverage while preserving the limits imposed by care and measurement.

2.5 — Confirmation, Rescue, and Model Change

A research program is not a statue. It should learn. Instruments improve, endpoint rules sharpen, rivals become stronger, and mechanisms are revised. The scientific problem is not revision itself. The problem is rescue that changes the tested claim while keeping the old public authority. This section distinguishes improvement from evasion (Gelman et al., 2013; Kuhn, 2012; Lakatos, 1978; Laudan, 1977).

The Law of Fairness needs a rule for updating because its strongest form cannot be confirmed by a single study, and its component claims will develop unevenly. A result may strengthen one bridge, weaken another, retire a mechanism, open a new reference class, or force abstention. Those possible status movements must be specified before data arrive.

Confirmation should be understood as a change in status under a defined model space, not as a mystical glow around the whole series. Bayesian updating, likelihood comparison, prequential performance, model recovery, and severe-test logic can all help. None can remove the need to ask what the tested claim actually was.

The framework's burden question frames rescue as a burden issue. If a theory changes its endpoint after failure without acknowledging claim change, the researcher keeps hope while the public carries confusion. If it changes the origin after a negative result, the critic carries the cost of chasing a moving target. Fair updating records the movement.

2.5.1 — Bayesian Updating as One Tool

Bayesian updating offers one way to express how evidence changes degrees of belief. A result has confirmatory force when it is more expected under one model than under alternatives, given prior assumptions. This can be useful for *The Science of Fairness* because component claims are uncertain, rival models are many, and evidence will accumulate gradually rather than arrive as a single decisive spectacle.

Bayesian language must be used with humility. Priors are not neutral simply because they are numerical. Model spaces are not complete simply because they are formal. A posterior can look decisive because serious alternatives were not included, because prior mass was assigned too generously, or because the observation model was treated as known when it was not. Numerical updating cannot rescue a poorly defined claim.

For the Law, Bayesian confirmation is most useful at the component level. A measurement validation result may update belief in one estimator. A model-comparison result may update belief in a reserve-coupled realization relative to a homeostatic rival. A transport study may update belief about a reference class. None of those posterior shifts should be described as proof of the exact set-universal target (Gelman et al., 2013; Kuhn, 2012; Lakatos, 1978).

Bayesian analysis also helps expose prior-sensitivity. If support for a component depends on highly favorable priors, the report should say so. If a rival model receives little prior mass because

it was inconvenient rather than implausible, the analysis is not fair. The rival set is part of the burden the claim must carry.

Updating cannot ignore versioning. Evidence updates a specific claim version. If the bridge changes after the result, the posterior for the old version does not automatically transfer. A new endpoint, instrument, or origin convention requires its own model and status history. Otherwise Bayesian language becomes a polished form of category drift.

2.5.2 — Likelihood and Predictive Comparison

Likelihood and predictive comparison ask a direct question: which model expected the observed data better under declared assumptions, and which model predicts new or held-out data better? For a program facing many rivals, this question is more useful than asking whether a pattern feels compatible with the Law. Comparative prediction asks more of a model than compatibility alone.

Held-out prediction matters because it reduces the freedom to sculpt a story after seeing the result. A Law-specific model may fit a training dataset because its bridges and parameters were tuned there. The stronger question is whether it predicts a new dataset, new site, new endpoint class, or new time window better than homeostatic, adaptive, bridge, clinical, and observation-process rivals.

Calibration is as important as raw accuracy. A model that gives high-confidence wrong predictions is not merely inaccurate; it is dangerous. In a public-risk project, overconfident models can create proof drift, media overclaim, or research escalation.

Predictive comparison should therefore examine uncertainty, calibration, abstention behavior, and failure modes (Lakatos, 1978; Laudan, 1977).

Prequential performance is a useful discipline for longitudinal studies. The model makes predictions as the stream unfolds, and each new observation tests the forecast without access to the future. This matters for the Law because endpoint-aligned analysis can otherwise create the illusion that the model saw the end coming when it was only aligned after the fact.

Predictive comparison can also defeat mechanisms without defeating the entire question. If a predictive-processing rival repeatedly forecasts measured trajectories better than a Queue model, the Queue model should narrow or retire. If a clinical frailty model explains variance compression better than reserve coupling, reserve coupling loses in that class. The Law-specific program learns by letting other models own the data they explain.

The public language after predictive comparison should be exact. Say that one model out-predicted another under a specified bridge. Do not say that the Law has been confirmed because a model with Law-like variables performed well. The bridge between model performance and law-language must remain visible.

2.5.3 — Independent Motivation for New Bridges

A bridge connects the latent claim to an empirical test. It may be a measurement instrument, endpoint classifier, origin convention, reserve proxy, scalarization method, rival model, or transport rule. Bridges can change as science improves. The rule is that a new

bridge must be independently motivated before or beyond the failed result, not invented solely because the old bridge lost.

Independent motivation can take several forms. A measurement bridge may be supported by validation studies outside the Law-specific test. An endpoint bridge may be grounded in consciousness science and clinical classification before terminal valence closure is analyzed. A reserve proxy may be justified by biological theory and external data. A rival set may be expanded because a field has advanced, not because the favored model needs an easier comparison.

A bridge that appears only after an unfavorable result may still generate a future hypothesis. It may not erase the status of the failed test. The failed version remains failed or inconclusive under its own bridge. The new bridge begins with a new version label and a new risk burden. This preserves the original test record while making the revised bridge available for a new prospective test (Nosek et al., 2018; Vehtari, Gelman, & Gabry, 2017).

The distinction between progressive and ad hoc revision begins here. A progressive bridge explains the failure, predicts something new, and exposes itself to further risk. An ad hoc bridge absorbs the failure while adding no new exposure. The first improves the research program. The second turns the program toward immunity.

A bridge revision must leave an audit trail: what changed, what failure motivated it, which earlier claim loses authority, and what new prospective prediction exposes the revision to risk. Without

that record, participants, critics, and readers inherit the cost of an unnamed moving target.

2.5.4 — No Moving Endpoints or Origins

Endpoints and origins are especially dangerous bridges because they can turn failure into success by definition. If the endpoint moves after the trajectory is seen, the test may be selecting the place where the result looks best. If the neutral origin moves after the distribution is known, equivalence can be manufactured. The chapter therefore treats endpoint and origin movement as a major form of claim change.

The endpoint is not a decorative timestamp. It defines which trajectory is being tested. A study of truth-level endpoint cannot become a study of biological death because the signal is easier to obtain. A low-signal ending cannot be treated as neutral because the instrument fell silent. A resuscitated or reversible state cannot be forced into terminal closure because the theory wanted a clean end.

Exact neutrality requires a justified neutral point. Cohort means, personal baselines, and questionnaire midpoints provide useful reference points for relative analyses. Terminal valence inference additionally specifies how its chosen origin represents phenomenal neutrality and how uncertainty in that origin affects the result (Gelman et al., 2013; Kuhn, 2012; Lakatos, 1978).

A study may refine endpoint and origin rules prospectively. It may learn from failure and build better rules for the next version. What it may not do is reinterpret an unfavorable result by moving the clock or zero point and then claim that the same hypothesis

survived. The bridge has changed, so the claim version has changed.

When an origin or endpoint remains unresolved, an indeterminate result identifies the evidence needed for the next study. Resolving that question provides a firmer basis for the ensuing comparison.

Public communication should be blunt here. No moving endpoints. No moving origins. No neutrality by recentering. No closure by choosing the place where the curve looks closed. These are not hostile rules; they are the conditions under which a positive result could deserve belief.

2.5.5 — Progressive Versus Ad Hoc Revision

A progressive revision explains a failure, changes a declared object or bridge, and creates a new prospectively testable risk. An ad hoc revision changes the endpoint, origin, reference class, or auxiliary structure only after an unfavorable result in a way that preserves the preferred conclusion without adding independent exposure to failure. Mixed cases should be labeled as new candidate versions pending severe test rather than treated as continuous support (Lakatos, 1978; Laudan, 1977).

2.5.6 — Versioned Theory Change

Versioned theory change records the object that changed, the evidence that forced the change, the earlier claim or public artifact that loses authority, and the new risky prediction that the revision must face. Formal-definition changes, empirical-bridge changes, and applied-domain corrections remain distinct. Chapter 28 provides the full register, migration, and cross-book propagation procedure.

Bayesian and likelihood tools can live inside this bridge record. They can show how much a result changes relative support among models. But the record should also include non-numerical obligations: source of priors, omitted rivals, context of use, endpoint confidence, participant burden, and public wording changes. Updating is not only a calculation. It is a governance act.

The question becomes almost procedural: who carries the cost when the bridge changes? In an answerable research program, the theory pays by losing status or version continuity where appropriate. The public, the critic, and the participant do not pay by having to chase an unnamed moving target.

Bayesian language can clarify updating, but it cannot cleanse a badly formed model space. If the only live models in the comparison set are weak versions of rivals and a protected version of the Law, a posterior shift toward the Law is bookkeeping over an unfair contest. The fair comparison includes strong rivals, plausible observation artifacts, and the option that none of the proposed models deserves confidence. Otherwise the math has inherited the hidden bias of the menu.

2.5.7 — Predictive Scoring and the Cost of Flexibility

Model change becomes easier to judge when prediction is separated from accommodation. A model assessed on the observations used to invent its features may fit beautifully while saying little about what comes next. Held-out data, prospective forecasts, and prequential evaluation impose the needed order: the model speaks before the outcome, and the outcome can answer back.

Proper scoring rules reward calibrated probabilistic predictions rather than confident retrospective stories, but the score and prediction task must be fixed for their intended use. Log scores, interval scores, calibration measures, and decision losses answer different questions. Cross-validation also depends on correct grouping, time order, clustering, preprocessing, and leakage control; random splits within one person do not establish transport to new persons, and smoothing before a temporal split can leak the future (Gneiting & Raftery, 2007; Vehtari, Gelman, & Gabry, 2017).

Flexibility must be counted. Searching across origins, endpoint windows, reserve proxies, exclusions, transformations, latent states, rival sets, or scoring rules can create apparent stability when no analysis was prospectively privileged. Multiverse or specification-curve work can reveal sensitivity, but it cannot retroactively convert exploration into confirmation. The report must preserve the analytic history and the uncertainty around model differences.

Complexity is warranted when it earns recovery, discrimination, or transport. A larger hidden-state model may absorb local irregularities yet fail to predict held-out transitions, survive sealed-world recovery, or retain meaning across cohorts and channels. Unrecoverable complexity is another source of underdetermination, not depth. Simulation can check whether the added states are recoverable from the measurements the study will actually collect. Additional structure should therefore be retained only when the data can distinguish its components well enough for the resulting interpretation to be meaningful.

Otherwise, apparent mechanistic detail may simply redistribute uncertainty into parameters that cannot be separately identified.

Flexibility should therefore be budgeted rather than denied. A materially changed model requires fresh holdout or replication evidence, and the cost of rescue is paid through renewed exposure to failure rather than a rhetorical claim that the revised model remains the same test (Chambers, 2013; Gelman & Loken, 2014; Nosek et al., 2018). The revision plan should reserve data or replication capacity for testing the specific improvement that motivated the change. That reserve prevents all available evidence from being consumed during model development, leaving no independent information with which to evaluate the revision. A flexible program remains scientifically disciplined when every substantial change creates a corresponding obligation for fresh evidence.

2.6 — Narrowing, Retirement, and Defeat

The Law of Fairness research program distinguishes local support, partial support, instrument failure, endpoint failure, mechanism defeat, rival victory, reference-class narrowing, inconclusive evidence, and strong-form defeat. Defined before evaluation, these outcomes connect each result to an appropriate response (Lakatos, 1978; Laudan, 1977; National Academies of Sciences, Engineering, and Medicine, 2019; Popper, 1959).

This section gives the chapter its exit map. The exits are not escapes from evidence. They are the honest paths by which a claim changes status after evidence. A smaller true claim is better than a larger protected one. A retired mechanism is better than a mechanism kept alive as ornament. A defeated strong form is better than a strong form that survives by ceasing to say anything.

Narrowing or defeat changes the status of the scientific claim, not the independent duties of care, privacy, and dignity governed in Chapter 3; the burden lens may remain useful only as a non-evidentiary analytic tool.

The following outcome map is a discipline for future chapters. Each status has different public language. No chapter should call a component result validation, no summary should call an inconclusive study defeat, and no public artifact should hide a rival victory inside a phrase such as mixed evidence.

Table 2.3. Outcome status, public wording, and prohibited reinterpretation.

Outcome	What happened	Public wording	What must not happen
Reference-class narrowing	A pattern survives only in defined populations or conditions.	Supported in that class; transport open.	Do not universalize.
Component narrowing	A representation, measurement, endpoint, or consequence survives without the full Law.	Component support.	Do not upgrade to strong form.
Mechanism retirement	A candidate realization or mechanism loses to rivals.	Mechanism retired or narrowed.	Do not keep using it as if supported.
Retirement to heuristic or metaphor	Scientific claim loses, but lens or image remains useful.	Useful lens or metaphor only.	Do not call resonance evidence.
Strong-form defeat	Adequate tests or counterexamples defeat the universal claim.	Strong form withdrawn or defeated.	Do not rescue by category drift.
Dignity after loss	Ethics and care remain unchanged.	Care and dignity do not depend on verdict.	Do not make people carry the theory's disappointment.

2.6.1 — Narrowing to a Reference Class

Reference-class qualification identifies a result supported in a particular population, condition, endpoint type, or instrument context. This local knowledge supplies a clear basis for replication and transport studies.

A reference-class claim should name its limits in the sentence itself. Supported in verbally report-capable adults receiving palliative care under instrument version two. Supported in slow terminal decline but not sudden death. Supported in one culture-language calibration but untested elsewhere. The wording may feel less elegant than universal language, but it prevents evidence from becoming imperial.

Reference-class narrowing can be progressive. A local finding can generate sharper transport studies. It can reveal moderators, failure conditions, and rival boundaries. It can teach the program where the Law-specific pattern is plausible and where other processes dominate. A smaller map is still a map (Lakatos, 1978; Laudan, 1977; National Academies of Sciences, Engineering, and Medicine, 2019).

The danger is rhetorical rebound. After a careful local statement, public summaries may return to the phrase the Law is supported. That rebound must be prevented. The science volume should make local status quotable so summaries can remain safe without long explanatory scaffolding.

A result may exclude nonverbal, disabled, sedated, pediatric, or nonhuman cases from the tested reference class; it cannot turn

exclusion from an estimand into a conclusion about consciousness, standing, dignity, or eligibility for care.

The acceptance rule is exact: until transport is defended, the claim stays in its class. If transport succeeds, the class expands. If transport fails, the boundary becomes part of the result.

2.6.2 — Narrowing to a Component

Component findings can advance the project and neighboring fields together. A measurement stack may improve affective science, an endpoint classifier may sharpen consciousness research, and a model-recovery method may strengthen rival comparison. A terminal consequence that survives a local test supplies a defined result for replication and extension.

Component support should be celebrated precisely because it is not exaggerated. A program that can say we learned this smaller thing has become scientifically productive. The component may benefit research, care, measurement, or theory even if the strong Law never reaches validation. That is not failure of the book's ambition. It is the ordinary dignity of science.

The component language must be precise. Representation support means the target may be scientifically usable in a context. Measurement support means an estimator behaves adequately for a task. Endpoint support means a stopping event can be classified under a rule. Realization support means a candidate mechanism describes a pattern. Consequence support means a signature appeared. Each is different (National Academies of Sciences, Engineering, and Medicine, 2019; Popper, 1959).

A component can also outlive the Law. If the strong form fails, a measurement-related insight may still be valuable. An endpoint discipline may still improve palliative or consciousness research. Rival-model infrastructure may still strengthen affective science. The project should be prepared to preserve those contributions without calling them Law support.

A component result is reported by its object: chart, tool, classifier, estimator, endpoint, or local consequence. That precise label tells readers what the study achieved.

2.6.3 — Mechanism Retirement

Mechanism retirement occurs when a candidate process loses to rivals or fails its own predictions. A Queue model, reserve-coupled realization, shadow-price account, interoceptive pathway, or sleep-reprocessing mechanism may turn out to be wrong, too local, or less predictive than an alternative. The program must be able to remove such machinery cleanly.

Mechanism retirement can preserve scientific value. Scientific mechanisms are tools for learning how a claim could be realized. If a tool fails, the program has learned something about the space of possibility. A mechanism that fails its declared test should retain only the narrower role the evidence supports.

Mechanism retirement requires public propagation. If a mechanism has appeared in diagrams, public summaries, or neighboring books, those materials must be corrected. A retired Queue model cannot keep functioning as the hidden engine of applied advice. A retired reserve model cannot keep decorating

proof-shaped images. The mechanism's status must travel (Stanford, 2006).

Retirement may be local. A mechanism might fail in terminal studies but remain useful in sleep or ordinary affective reprocessing. It might work in slowly declining reserve but fail in sudden endpoint classes. The retirement statement should name scope rather than deleting nuance. The goal is accuracy, not punishment.

When predictive processing, homeostasis, adaptation, or clinical models perform better under the declared comparison rules, the record credits their explanatory contribution. That finding helps the field select and develop the next model.

2.6.4 — Retirement to Heuristic or Metaphor

If a scientific claim is retired but a heuristic or metaphor retains independent value, the public label should say so explicitly: the scientific claim was not supported in the relevant form, while the remaining lens is non-evidentiary. Any retained application remains bounded and noncoercive.

2.6.5 — Defeat of the Strong Form

Defeat of the exact set-universal target is the most serious negative outcome. On a declared branch and reference class, one independently audited, gate-valid stream is adverse when its nonempty truth-containing latent compatibility set excludes zero under the protocol-fixed inferential guarantee. Local instrument failure, endpoint ambiguity, observation failure, or loss of an optional mechanism may instead block inference, narrow a class, or retire the affected route; route failure becomes target adversity

only when a protocol-fixed entailment warrants that step. A credible counterexample cannot be absorbed by quiet scope drift: any narrowing must be explicit, versioned, and accompanied by withdrawal of the broader language. Program-level defeat can also arise when independently motivated bridges repeatedly fail and replacements cease to generate novel risky predictions (Laudan, 1977; National Academies of Sciences, Engineering, and Medicine, 2019).

Defeat also includes empirical-content exhaustion. If every independent bridge fails and each replacement merely explains the previous failure without predicting something new, the program loses scientific character. It may continue as philosophy, spirituality, or metaphor, but not as a live empirical research program with the same authority.

Report theory-level defeat clearly and calmly: identify the affected claim, the evidence behind the verdict, the findings that remain useful, and the materials and public statements that require correction.

2.6.6 — Dignity After Scientific Loss

Scientific loss changes a claim, not the standing of persons or the duties in Chapter 3. Public language should state exactly what survived: for example, a qualified measurement component can remain usable even if a mechanism is retired and terminal evidence remains inconclusive.

When the strong empirical program is defeated or retired, dependent tables, summaries, diagrams, and applied claims

should be corrected or withdrawn through the Chapter 28 process. Scientific status changes; care and dignity do not.

2.6.7 — Defeat as Knowledge Transfer

Defeat should change more than the label attached to a preferred hypothesis. It should transfer what was learned into the fields that made the loss intelligible. If conditioned paths explain an apparent closure signature, the result belongs to stochastic-process research; if allostatic regulation wins, it belongs to affective science; if observation shutdown produces the pattern, it belongs to missing-data methodology and clinical study design.

The transfer begins by separating what failed from what remains valid. A measurement instrument, humane burden protocol, endpoint-adjudication procedure, negative control, or model-recovery suite may remain useful even when the terminal target or a candidate route fails. Scientific loss can reduce one theory while enlarging the surrounding body of knowledge.

A rival victory must remain a rival victory. The preferred program may learn from the winning account, but it cannot absorb that explanation and preserve its former status. A successor theory needs a new identity, dependency map, and statement of what the earlier evidence no longer supports (Lakatos, 1978).

Defeat also creates correction duties. Diagrams, summaries, websites, teaching materials, and cross-volume references that imply a stronger status must be corrected or withdrawn. The record should state what failed, what remains useful, and how the burdens contributed by participants, clinicians, families, and staff will prevent repeated work or repeated harm.

Credit, citation, funding, and training should follow the result. Later work should cite the defeated model for the problem it clarified and the rival for the explanation that prevailed. If the decisive advances came from missingness models, palliative methods, endpoint adjudication, or conditioned-process theory, the next research investment should strengthen those capacities rather than preserve the identity of the losing mechanism. Resource allocation should therefore follow demonstrated explanatory and methodological value rather than institutional attachment to a favored theory. This helps ensure that defeat redirects scientific capacity toward the tools and questions that proved most informative.

A successor should begin with the loss. It should list abandoned claims, remaining validated tools, which rival explains best, unresolved uncertainties, and evidence required for a genuinely new hypothesis. Beginning from the defeat prevents continuity of name from disguising discontinuity of content. The successor should also identify which earlier results remain interpretable under the new framework and which require reanalysis. That separation prevents inherited evidence from quietly supporting a claim it was never designed to test.

Defeat may reveal a poorly framed problem. A terminal-closure program might find that the tractable object is the interaction among affective dynamics, endpoint selection, observation loss, and memory. Reframing is legitimate when declared as a new question and when unresolved discrepancies among instruments, cultures, clinical states, and endpoint classes remain visible. A useful reframing should make those discrepancies more testable

rather than merely absorb them into a broader theory. The new question earns attention by creating clearer empirical distinctions than the one it replaces.

Reusable scientific objects should stay attached to the claim they failed to support. Benchmark simulations, measurement-failure signatures, burden thresholds, endpoint criteria, and negative controls become more valuable when their provenance records which attractive path no longer deserves the same investment (Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019). Preserving that provenance also helps later researchers understand why a tool was created and which failure modes it was designed to expose. Reuse is strongest when the object travels with both its demonstrated utility and the limits revealed by the original test.

A successor may retain an instrument while rejecting the terminal law, preserve the intrastream unit while changing the endpoint claim, or keep the ethical infrastructure while replacing every realization. That inheritance is progress only when continuity and abandonment are explicit. A program earns trust by letting reality end a story without taking the science, care, or contributors down with it (Lakatos, 1978; Laudan, 1977). Scientific continuity should therefore reside in validated methods, transparent records, and durable safeguards rather than in preserving the identity of a theory. This allows useful knowledge to survive even when the explanation that originally motivated it does not.

Chapter 3 — Ethics and Scientific Authority

A science of lived experience is judged not only by what it can detect, but by what it refuses to make people carry. Ethical design is therefore part of epistemic design. A study that distorts care, pressures participation, erases privacy, or changes the very state it claims to observe can be both morally wrong and scientifically uninterpretable.

The governing order is simple: care comes before evidence. Clinical treatment, comfort, rescue, pain relief, family presence, lawful protection, and ordinary professional duties are never delayed to preserve a measurement window. The scientific record must be designed around that priority rather than asking participants to fit the record.

This principle governs study design and participant protection throughout the program. No experiment may induce suffering to create compensatory dynamics, manipulate a person toward an endpoint, delay care to observe an unaltered trajectory, or invite self-testing. A conditional balancing requirement is not moral payment or permission to cause harm, and no person owes the project a demonstration of the hypothesis.

Ethical research instead seeks naturally occurring variation, low-burden measurement, existing data, simulation, model recovery, noninvasive observation, and clinically indicated changes whose primary purpose remains care. The most valuable protocol may be the one that discovers how little additional burden is needed to answer a sharper question.

Terminal research receives the strongest comfort-first rule. The truth-level endpoint cannot be manufactured by a study, and the

protocol endpoint cannot govern bedside decisions. Sedation, analgesia, oxygen, repositioning, privacy, family access, and withdrawal of research procedures remain under clinical and participant authority (World Medical Association, 2024).

Consent must be ongoing rather than symbolic. A signature begins a relationship; it does not transfer ownership of a person's inner life. Capacity, comprehension, voluntariness, language access, fatigue, distress, medication effects, and changed circumstances may alter what continuing participation means.

Bystanders and families require independent protection. A participant's consent does not automatically authorize the capture of a spouse, child, caregiver, clinician, visitor, or household environment. Record routing must distinguish direct participant data from collateral information and offer removal, shielding, or noncollection routes.

Vulnerability is not a permanent label attached to a population. It can arise from illness, dependency, grief, detention, poverty, cognitive change, institutional hierarchy, language barriers, digital exclusion, or the pressure created by a trusted professional. Proportionate safeguards must respond to the mechanism of vulnerability rather than use the label as a substitute for design.

Independent review is necessary but not sufficient. Ethics committees, data and safety monitors, clinicians, participant advocates, privacy specialists, and community partners each see different failure modes. Their authority should be real enough to pause, modify, or end a procedure, not merely to endorse a

protocol at the beginning (Emanuel, Wendler, & Grady, 2000; International Council for Harmonisation, 2026).

Scientific authority is deliberately narrow. Researchers may report evidence about a declared component under stated conditions. They may not convert a latent estimate into a human-worth score, care priority, insurance decision, employment criterion, spiritual status, civic rank, or personal forecast. No archive, model, institution, or artificial system owns the participant's inner truth.

Return of results also requires discipline. Participants may deserve understandable aggregate findings, corrections, and clinically relevant information when a validated pathway exists. They should not receive speculative terminal-ledger interpretations, hidden classifications, or burdensome alerts that the science cannot support (National Academies of Sciences, Engineering, and Medicine, 2018).

These safeguards are not warning wallpaper. The chapter states the shared floor once, then ties each later protection to a concrete mechanism: a sensor that interrupts sleep, an endpoint that conflicts with care, a family member caught in a recording, a model that invites ranking, or a public artifact that reveals more than the protocol allowed.

The result is a more hopeful research program. When participants retain authority, clinicians retain professional judgment, and abstention remains publishable, the inquiry can pursue difficult questions without asking anyone to become expendable. Humane limits improve the chance that the evidence will deserve trust.

3.1 — Care Before Evidence

Care before evidence is the first operational law of this research program. The presence of a hypothesis never suspends the ordinary duties owed to a suffering person, and the research question cannot reinterpret a person's need for help as a temporary state whose later balance may excuse present neglect (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; World Medical Association, 2024).

This section sets the floor beneath every later protocol. The floor is not mystical and not merely therapeutic. It is a rule about priority: clinical safety, shelter, crisis support, pain relief, communication access, and ordinary human presence come before data collection, endpoint clarity, or public explanation.

A theory about total lived burden is especially dangerous if it makes present burden seem provisional. *The Science of Fairness* must refuse that danger at the beginning. The moment a person needs care, the research question becomes smaller than the need before us.

This rule also preserves scientific integrity. A participant who is denied care, pressured to perform meaning, or asked to endure discomfort for signal clarity no longer belongs to an ethically interpretable study. The procedure has changed the phenomenon. The failure is not only moral; it is epistemic.

The familiar research-ethics language of respect, beneficence, and justice becomes sharper here. Respect means the person is not translated into the theory without permission. Beneficence means

relief and risk minimization outrank elegant inference. Justice means the costs of research do not fall most heavily on those least able to refuse.

Later chapters will discuss endpoint rules, candidate momentary and interval estimates, terminal signals, and rival models. This section says what all of them inherit: no result is worth less care, and no uncertainty about the Law weakens the duty to respond to suffering now.

3.1.1 — Present Suffering Is Present

Every protocol should contain a visible present-need trigger: acute pain, panic, breathlessness, confusion, fear, unsafe housing, hunger, exposure, abandonment, or any condition that makes data collection secondary. Staff respond to the need first, pause or stop research as required, and record the care event only to the extent consent and clinical governance permit. A possible terminal balance never reduces the urgency of present suffering; ignoring present need would be both ethically wrong and scientifically confounding (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; World Medical Association, 2024).

3.1.2 — Clinical and Social Care Remain Prior

The priority of care is not limited to physicians or hospitals. Terminal research may occur in hospice, nursing homes, home care, shelters, family settings, or ordinary communities, where transportation, translation, food, housing, disability access, caregiver respite, safety planning, and crisis support can determine who can participate and what becomes observable.

These supports are ordinary protections, not research rewards or leverage; if study staff become the only route to them, both voluntariness and the data-generating context have changed.

Clinical care does not become secondary because a person enters a study. Physicians, nurses, chaplains, social workers, therapists, aides, family caregivers, and crisis workers do not become servants of the research schedule. Their obligations remain directed toward the person, not toward the elegance of the data.

The distinction matters because the Law of Fairness touches settings where research and care can easily blur. A terminal study may occur inside hospice. A measurement study may recruit from a clinic. A life-course study may observe people during hardship. In each case, participants must know that care is not contingent on participation and that refusal will not cost them help (International Council for Harmonisation, 2026).

The therapeutic misconception is not only a consent-form problem. It can be built into tone, setting, dependence, and hope. If a participant mistakes research participation for individualized clinical care or believes that refusing will reduce the care otherwise available, the consent process has not done its work. The protocol must separate research contact from clinical entitlement visibly enough that refusal remains real.

A care-prior design should include a care-independence statement, a refusal-protection route, and a staff escalation path. If the study uncovers unmet need, it routes the person to support without making that support a study benefit that pressures others

to enroll. Care discovered by research should still belong to the person, not to the recruitment strategy.

Clinical and social care remain prior because the study is not the highest institution in the room. The person's safety, comfort, voice, and material need come first. Research may observe and learn under that rule, but it may not reorganize care around itself.

3.1.3 — No Scientific Desert

Scientific desert also has a quiet institutional form. Reviewers may never say that a participant deserved pain, but a report can imply it by describing severe burden as an instructive case, a morally necessary endpoint, or an unusually useful trajectory. The safer language names exposure, condition, injury, loss, stressor, symptom, and context without converting them into earned states. In data dictionaries, adverse circumstances should be coded as conditions to protect and understand, not as morally meaningful ledger entries. This distinction matters because later statistical models may inherit the coding scheme. Coding exposure and context separately from personal attributes helps later analyses preserve that distinction.

Scientific findings must never be translated into desert. A pattern in data cannot show that someone deserved a burden, earned a loss, invited a disease, merited grief, or received pain as correction. The word fairness in this project does not authorize blame.

This distinction must be made early because visible suffering often invites moral causality. People ask why a child suffers, why a gentle person dies, why luck is uneven, and why grief falls

where it falls. The scientific volume cannot answer those questions by making victims responsible for pain or by treating suffering as evidence about desert (National Commission, 1979).

Even a future result favorable to the Law would not identify the moral meaning of any event. Section 3.7.1 states the conditional net-remainder arithmetic; here the ethical point is narrower: no illness, injury, deprivation, loss, or other event becomes deserved, earned, or justified.

Recruitment materials, consent scripts, public diagrams, and study reports should distinguish conditional balance requirements from a known personal balance or a moral account to settle. Participants are not recruited to prove that their burdens are justified. Study descriptions identify the measured trajectories, uncertainty, and limits of the proposed inference.

No scientific desert also protects staff and families. Researchers must not impose an explanation of a loved one's suffering, describe it as morally deserved, or use it to discourage care. Questions about PTVC can be answered through its conditional arithmetic and stated uncertainties without declaring anyone's actual balance or assigning a purpose to their pain.

The framework can reveal hidden burden without assigning blame for it. Its better use here is to ask who is carrying cost, who can refuse, who can be helped, and who is being asked to bear the burden of evidence. It does not rank the person who suffers.

3.1.4 — No Forced Meaning

No forced meaning applies to researchers as well as to family, clergy, media, and public readers. A bereaved parent, a patient

approaching death, a nurse after a difficult shift, or a survivor describing trauma may use language that is sacred, skeptical, angry, grateful, confused, or silent. The study may classify speech acts for analytic purposes, but it may not tell the person what the experience means. The most humane codebook includes a category for refusal of interpretation. The most honest public report can say that a pattern was observed without saying that a human being's pain completed a lesson. Scientific humility is not the absence of interpretation; it is the refusal to seize another person's inner authority.

Forced meaning asks a person who is already suffering to satisfy someone else's need for coherence. In a project about fairness, that transfer is especially dangerous because the theory can tempt observers to make narrative peace before the person has chosen any meaning at all. Comfort, refusal, silence, and plural interpretation remain legitimate (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Scientific language can force meaning when it presents a pattern as personal explanation. A terminal chart, a calm expression, a dream report, a silence, or a family anecdote must not be treated as the participant's final message. Evidence about a model is not authority over a life story.

This rule does not forbid participants from speaking. If someone chooses to describe meaning, regret, peace, anger, doubt, faith, resentment, or refusal, the research can record that statement under consent and context. The point is that the meaning belongs to the person, and the right to revise or withhold it remains.

Consent materials should therefore separate study explanation from existential interpretation. They can say what data may be collected, why the study exists, what risks are present, and what may be learned. They should not tell participants that their involvement gives their suffering a role in proving the Law.

No forced meaning keeps the science clean. It prevents researcher expectancy, family desire, spiritual pressure, and public storytelling from becoming hidden variables. The study can ask whether a pattern exists; it cannot decide what anyone's pain meant.

3.1.5 — Care Survives Every Verdict

Consent materials should state before enrollment that symptom relief, emotional support, communication access, withdrawal, and ordinary care continue under every analytic branch. Participants, staff, and families must never feel that comfort contaminates the study or that leaving threatens help. This separation protects dignity, voluntariness, and validity by removing pressure to perform a terminal story for the theory.

Care does not depend on scientific status. It remains when the strong form fails, a component survives, the study is inconclusive, or measurement collapses. Human beings receive care because their burdens matter before any theory succeeds, not because a model describes them correctly (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000).

The same independence applies after favorable or adverse results. No finding creates a class of lives more worthy of care, and no

cumulative estimate determines triage, clinical priority, or social support. Those decisions follow their own ethical and legal standards.

Care-independent planning should specify clinical support, crisis response, interpreter and transportation access, privacy, and ordinary communication with the care team without conditioning them on analyzable data or study completion. This reduces selection pressure and keeps research from becoming the price of help (World Medical Association, 2024).

When a claim narrows or fails, public and clinical records should carry a concise care-continuity consequence: treatment, support, privacy, and dignity remain in force because the hypothesis never licensed them. Negative results concern the claim, not the person's entitlement to relief.

Later science can therefore build and lose models under a prior commitment: the person in front of us is not waiting for the universe to certify their need. That independence is one of the clearest tests of whether care truly outranks claim status (Wendler & Grady, 2008).

3.2 — No Harm-Induction or Balance-Later Designs

A theory becomes most dangerous when it imagines that the decisive experiment is the one ethics forbids. For the Law of Fairness, the tempting forbidden experiment would be to create burden, withhold relief, manipulate the life-course, or watch deprivation unfold in order to see whether later experience balances. This section says no before the temptation acquires technical dignity (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; National Commission, 1979; U.S. Department of Health and Human Services, 2018).

The refusal is not a retreat from science. It is a definition of admissible science. A research program can be rigorous without becoming cruel. It can study naturally occurring variation, clinical trajectories, observational data, optional low-burden devices, and ethically reviewed interventions designed for independent benefit. It cannot manufacture suffering for Law-specific inference.

The forbidden designs remain scientifically informative in one limited sense. They clarify what the project is not allowed to test directly and therefore force a better methodological route. When a clean but violent design is unavailable, the program must rely on prospective observation, natural experiments, careful rival modeling, simulation, model recovery, negative controls, and explicit uncertainty.

This distinction protects against a second danger: post hoc moralization of harm. A harmful event that occurs in the world

may become an object of study under consent and protection, but it does not become a good event because it provides data. The study did not earn the harm; it arrived after it.

Balance-later designs are rejected because they turn the theory into an excuse for present indifference. They ask one period of suffering to be justified by an imagined future offset. The science volume must use the opposite discipline: present burden counts now, and future data collection cannot erase how it was produced.

In research terms, harm induction also contaminates inference. Fear, deprivation, humiliation, untreated symptoms, and coercive stress alter physiology, reporting, adherence, attrition, and social behavior. The procedure would change the trajectory it claims to test while violating the person it claims to study.

3.2.1 — No Deliberate Suffering

No study may create or intensify pain, deprivation, trauma, fear, humiliation, social isolation, material insecurity, sleep disruption, clinical discomfort, or emotional distress for Law-specific evidence. Existing suffering may be observed under consent and protection, and independently justified beneficial interventions may be studied, but deliberately worsening a condition to create a balancing contrast is outside the program (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; National Commission, 1979).

3.2.2 — No Withholding or Delay

Withholding can hide inside scheduling: a device fitting before medication, an interview before rest, a questionnaire before transport, or a recording window before family contact. No

protocol may delay treatment, rescue, comfort, analgesia, social assistance, safety planning, translation, clinical response, or material help for cleaner data. Whenever research and care compete, care comes first; the study records the intervention and models the resulting signal under humane care rather than preserving an untreated state (National Commission, 1979; U.S. Department of Health and Human Services, 2018).

Operationalize this rule with an escalation clock that names the distress trigger, responder, stopped procedure, and reporting route. A needed care event ends or changes data collection immediately. The resulting missingness or treatment effect is modeled as part of the observation process; any actual protocol deviation is still reported.

This rule also protects against inequity: participants with less power, language access, advocacy, or independence must not be the ones least able to interrupt a research schedule. In terminal or crisis settings, care and safety remain prior to measurement.

3.2.3 — No Unethical Path Manipulation

Life-course manipulation is the fantasy test that must remain forbidden. A researcher might imagine that extreme enrichment, deprivation, isolation, care intensity, or imposed repair would make the Law easier to evaluate. Such designs would corrupt the project at its root because they use lives as trajectories to be engineered rather than persons to be protected. Ethical science can study naturally occurring differences, policy changes, clinical pathways, and historical variation under careful confounding analysis. It cannot assign a person to suffer in order to see whether

later relief appears. A study may learn from the world's existing unfairness. It may not manufacture unfairness and then call the result evidence.

The Law's strong form concerns whole conscious trajectories, which makes life-course manipulation especially tempting and especially forbidden. No research program may engineer enriched and deprived lives, manipulate opportunity menus, alter family support, distribute hardship, or shape a person's path in order to see whether later experience balances.

This prohibition includes designs that would be unacceptable even if consent were claimed. A person cannot be asked to accept a deprived trajectory so a hypothesis can be tested. A family cannot be randomized into avoidable burden. A community cannot become the comparison arm for a civilization-scale claim (World Medical Association, 2024).

Ethically justified interventions may be studied when they have an independent care, social, clinical, educational, or supportive purpose. Their felt effects may be analyzed under an appropriate protocol, but the intervention is justified by its independent benefit, not by the Law; engineering suffering or withholding help to produce a balancing contrast remains prohibited.

Path manipulation also corrupts reference classes. A trajectory engineered for research would not reflect natural life-course dynamics, ordinary care, or voluntary adaptation. It would reflect behavior under artificial moral and institutional pressure, so its results would be hard to interpret even if ethics were ignored.

The program must therefore use ethically available variation: natural experiments, clinical pathways, longitudinal observation, consented diaries, low-burden measures, and independent support interventions. A world already full of unfairly distributed burden gives science enough to study without creating more.

3.2.4 — No Self-Experimentation Through Danger

Self-experimentation can look morally cleaner because the subject and investigator are the same person, but the risk is not solved by consent when the design is dramatic, illegal, self-harm-adjacent, or socially contagious. A public figure who endangers themselves to test the Law would create pressure on admirers, critics, and vulnerable readers to treat danger as devotion. The project must refuse that theater. Private journals, voluntary reflection, low-burden experience sampling, and ordinary health records can be useful when governed properly. Risk-taking as proof-seeking is different. It confuses courage with validity, and it invites a mythology in which pain becomes a research offering.

A public theory can tempt admirers or critics to perform dramatic self-tests. Someone may try to create extreme pleasure, extreme pain, risk, deprivation, intoxication, sleep loss, social humiliation, or dangerous endurance to see whether experience later compensates. This chapter rejects that pathway absolutely.

Self-experimentation does not become safe because it is voluntary. A person can be pressured by belief, pride, despair, public attention, group loyalty, or the desire to produce evidence. The project must not praise, publish, amplify, or interpret dangerous self-tests as research (Council for International

Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; National Commission, 1979).

This matters for public communication. A careless phrase about testing the Law in one's own life could invite ledger management or experiential extremes. The safer language is narrow: private reflection may be meaningful, but it is not evidence, and no one should seek pain, risk, deprivation, or crisis for the theory.

No-crisis recruitment belongs here. A person in grief, terminal fear, suicidal crisis, panic, spiritual emergency, financial desperation, or public humiliation is not an opportunity for dramatic data. They need help, privacy, and support. Research invitation, if any, must wait until refusal can be real and care is secure.

Approved studies provide independent oversight, justified measurement and endpoint definitions, protocol control, adverse-event response, and rival comparison. The project excludes dangerous self-testing and pursues its empirical questions through those approved studies.

The public rule is simple: do not hurt yourself or take risks to test the Law of Fairness. The science volume will not treat such acts as evidence, nor will the wider project honor them as courage.

3.2.5 — Forbidden Designs Stay Informative

The value of a forbidden design is analytic. It identifies a causal contrast that cannot be pursued directly and forces the program to specify the estimand, confounds, and humane alternatives needed to study it indirectly. The ethical boundary therefore sharpens the method rather than weakening the question.

For example, directly inducing suffering might appear to identify a later counterweight. Since that design is prohibited, the program must instead use observational contrasts, natural symptom fluctuation, care interventions with independent benefit, negative controls, and careful rival modeling. Those humane alternatives should be judged by the identification they actually support, with any remaining limits stated explicitly (National Commission, 1979; U.S. Department of Health and Human Services, 2018).

Similarly, withholding comfort might seem to preserve a natural terminal path. Since withholding is prohibited, the program must model treatment within the terminal process. Sedation, analgesia, family presence, clinical escalation, spiritual support, and withdrawal choices become modeled variables, not nuisances.

The same is true for path manipulation. Because the project cannot engineer lives, it must face selection, non-ergodicity, culture, disability, socioeconomic context, and missingness honestly. This keeps the program grounded in the conditions it can responsibly observe.

A forbidden-design register should travel with prospective specification. It tells reviewers and red teams which apparently decisive tests are excluded and why. It also prevents later public critics from claiming that the program avoided such tests because it feared the result. The answer is more principled: some knowledge is not worth obtaining that way.

Excluded designs can clarify methodological questions and motivate acceptable alternatives. The same exclusion applies to pilot studies, self-experiments, documentaries, and donor-funded

challenges. A noninvasive surrogate can answer a narrower contrast when its relationship to the excluded intervention is explicitly modeled.

A prohibited experiment can still define an identification boundary. If deliberate harm would be required to separate two mechanisms under otherwise identical conditions, the program should say that the causal contrast is unavailable rather than substituting a weaker observation and keeping the same verdict language. Natural changes, clinical variation, and carefully modeled observational data may narrow the possibilities, but the assumptions that connect them to the unavailable intervention must remain explicit (Manski, 2003; Hernán & Robins, 2020). The resulting bound can identify which additional naturally occurring variation would be informative.

Simulation can then ask what a safe design can learn before recruitment. Synthetic worlds can vary the hidden mechanism, observation loss, treatment changes, endpoint process, and burden-triggered stopping under the prohibition on induced suffering. If all ethically admissible designs fail to distinguish the candidate from its rivals, that result is not a reason to relax the ethical boundary. It is evidence that the proposed question is not presently adjudicable at the desired level (Box, 1976; National Academies of Sciences, Engineering, and Medicine, 2019). Comparing simulated confusion across observation schedules can then prioritize feasible designs before any participant is contacted.

3.3 — Comfort-First Terminal Research

Terminal research can contribute valuable knowledge about care, measurement, endpoint uncertainty, and observation when it is governed by comfort first. Palliative care remains the primary clinical purpose of the setting. The strongest scientific design treats symptom relief, family support, and clinical judgment as part of the process being studied, while keeping observation proportionate and participation voluntary (Council for International Organizations of Medical Sciences, 2016; Sanders et al., 2024; World Health Organization, 2020; World Medical Association, 2024).

Comfort-first terminal research keeps the person approaching death at the center of care. Their pain, breath, agitation, silence, preferences, family relationships, spiritual life, and refusal take priority over the study's need for completeness.

Sedation, delirium, organ failure, fatigue, medication, symptom control, family presence, cultural context, communication capacity, and device availability all shape terminal observations. Including them in the study's design and analysis connects the measurement model to the actual clinical setting.

Comfort-first design distinguishes among research activities. Quiet observation may have one burden. A wearable sensor may have another. A diary prompt, interview, bedside device, family questionnaire, or clinician workflow task may carry still another. Each must be justified separately and stopped separately.

The relevant burden is not only the participant's burden. Families may be grieving, guilty, fearful, exhausted, or protective. Staff

may be overloaded, morally distressed, or pressed by protocols against clinical judgment. A site may lack time, privacy, interpretation services, or technical support. These are design variables, not background noise.

A terminal protocol follows the participant's care and preferences. A request for quiet ends observation; device discomfort prompts removal; family overload pauses contact; and excessive staff burden triggers redesign. An endpoint that cannot be classified through those conditions receives a No adequate estimate result.

Table 3.1. Terminal-study burden and stop matrix.

Research activity	Participant burden	Family or staff burden	Scientific risk	Stop or redesign rule
Quiet chart review under governance	Privacy exposure through clinical context	Record-preparation and review workload	Incomplete context and selected documentation	Use only with purpose limitation and no individual inference beyond the record
Optional bedside sensor	Discomfort, sleep disruption, skin irritation, feeling watched	Placement, troubleshooting, removal requests, family anxiety	Sensor gain collapse, device removal, adherence bias	Remove immediately on discomfort, distress, clinical interference, or staff overload

Research activity	Participant burden	Family or staff burden	Scientific risk	Stop or redesign rule
Brief participant interview	Fatigue, emotional strain, privacy loss	Scheduling and support burden	Response shift, demand characteristics, selective reportability	Shorten, postpone, or stop when fatigue, distress, or preference requires
Family or caregiver report	Grief burden, guilt, interpretive pressure	Family time and emotional labor	Proxy error, meaning projection, family script bias	Use only as contextual proxy with consent and never as final inner life authority
Clinician classification	None directly if integrated into care notes	Additional charting, moral distress, workflow strain	Observer expectancy, treatment linked confounding	Limit to clinically natural documentation or compensate staff time and monitor burden

3.3.1 — The Dying Person Owes No Result

A person approaching death owes the study no result. Silence, withdrawal, sleep, distress, loss of signal, changed capacity, a request for family only, or any care priority may end observation and produce a protected no-estimate branch. Neither family interpretation nor model output may be substituted for a missing participant channel, and a missing terminal record must not be

personalized as fear, imbalance, or failure (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

3.3.2 — Palliative Care Outranks Measurement

A palliative-care priority is not merely the absence of interference. It means the protocol is designed around clinical response, not clinical response around the protocol. Device placement should yield to positioning, hygiene, pain control, oxygen, family presence, spiritual care, privacy, and rest. Questionnaires should yield to fatigue. Recording schedules should yield to symptom crises. The study may record that care changed the observation process, but it may not ask care to wait while data remain clean. The burden of measurement must be carried by the research design, never by the dying person's body or the staff's uncounted labor.

Palliative care is not a confound to be minimized for theory. It is the governing clinical duty in terminal research. Analgesia, sedation when clinically indicated, symptom relief, psychosocial support, spiritual care, family presence, and comfort measures follow patient need and clinical judgment, not protocol convenience.

The consequence is methodological as well as ethical. A protocol cannot ask clinicians to hold medication constant for signal clarity when symptoms require change. It cannot ask staff to delay a comfort intervention so a measure can be collected first. It cannot ask families to postpone presence or ritual to preserve a data

window (World Health Organization, 2020; World Medical Association, 2024).

Instead, the research design records care as part of the observed trajectory. Medication changes, sedation depth, delirium episodes, oxygen support, repositioning, family visits, clinician interventions, and device removals become time-stamped context where consent permits. They are not failures of nature. They are real terminal conditions.

The design must also accept that palliative care can make certain inferences impossible. If a person is deeply sedated for comfort, the study may lose direct self-report. If delirium fluctuates, the study may lose stable interpretation. If a device is removed because it irritates the participant, the time series ends. No-estimate is a legitimate result.

Palliative priority should appear in site training. Research staff must know whom to call, when to leave, how to respond to distress, and how to explain that the study has no authority over care. Clinicians must know they are free to act for comfort without asking permission from the research team.

Example: during a scheduled prompt, a participant develops acute breathlessness. The nurse stops the prompt, removes any interfering device, provides clinically indicated treatment, and asks research staff to leave. The interruption is recorded as care-priority, not participant deviation; the missing interval is not imputed as calm, distress, or neutrality (World Medical Association, 2024; International Council for Harmonisation,

2026). Event details are retained only to the extent permitted by consent and clinical policy.

If the participant later asks to resume, the study reconfirms consent, checks that burden remains proportionate, and records changed medication and observation conditions as a new segment. If participation does not resume, the analysis follows its predeclared partial-information or no-estimate branch rather than substituting family testimony for the lost channel.

The site then asks whether the event exposed a design defect: repeated device interference, poorly timed prompts, or unmanageable staff load triggers redesign or paused enrollment. Care-first becomes scientifically credible when it changes both the procedure and the analysis exactly as the protocol says it will (Sanders et al., 2024; World Health Organization, 2020).

3.3.3 — Sedation and Delirium Are Clinical Realities

Sedation and delirium are not nuisances to be subtracted after the fact. Both are part of the terminal clinical context; proportionate sedation may be part of humane care, while delirium is a clinical condition requiring assessment and appropriate management. A study that excludes them too easily may select for a false picture of dying; a study that includes them without modeling them may mistake treatment, physiology, or reportability for a ledger signal. The right posture is classification with humility. Sedation depth, indication, medication timing, delirium assessment, communication capacity, pain crisis, oxygenation, infection, metabolic instability, and family interpretation all belong in the

observation model. A low-signal endpoint does not become meaningful because the theory needs a signal.

Sedative exposure, sedative effect, delirium, and reduced reportability are not interchangeable variables. Dose, timing, indication, prior tolerance, organ function, symptom crisis, paradoxical agitation, fluctuating attention, and motor suppression can alter channels in different directions. A protocol therefore models treatment and state transitions separately, records uncertainty in each classification, and abstains when the available channels cannot distinguish diminished expression from diminished experience (International Council for Harmonisation, 2026; World Medical Association, 2024).

A weak protocol treats sedation as a nuisance variable and delirium as noise. A stronger protocol specifies both as clinically meaningful intercurrent states, records their timing and indication, and declares which inferences remain admissible. Deep sedation may leave some physiological observations while defeating claims about conscious valence. Delirium may carry distress while defeating stable questionnaire interpretation. The result is not forced into a clean trajectory; the permitted inference changes, narrows, or becomes unavailable (International Council for Harmonisation, 2019; International Council for Harmonisation, 2026).

A terminal observation model distinguishes changes in clinical sedation, neurological state, device availability, reportability, and sensor performance. Estimating these processes helps interpret why a measured signal changes near the endpoint.

Delirium requires special humility. Agitation, hallucination, fear, confusion, withdrawal, or muteness can be overinterpreted by observers seeking meaning. The protocol should protect families from being asked to decode every expression and protect clinicians from research pressure to assign theoretical significance to clinical signs.

Sedation and delirium also affect missingness. Participants in low-signal states may drop out of self-report, fail device adherence, or become noncommunicative. If the study excludes those periods silently, it may select only the most reportable endings. If it includes them without qualification, it may interpret absence as calm. Both errors are preventable only if the observation model is honest.

The safest rule is status labeling. Every terminal segment should be labeled by consciousness confidence, reportability, treatment context, sensor availability, and inference permission. Some segments support only care description. Some support measurement feasibility. Some support component analysis. None supports more than the labels allow.

3.3.4 — Family and Staff Burden

Family and staff burden should appear in the protocol budget before recruitment begins. A bedside study may ask relatives to tolerate equipment, answer questions, interpret preferences, manage visitors, protect privacy, and carry the emotional weight of being watched while losing someone. It may ask nurses, aides, chaplains, interpreters, physicians, and social workers to add research tasks to already saturated work. If those burdens are

unmeasured, they reappear as missingness, protocol deviations, recruitment failure, moral distress, and staff turnover. Counting them is not administrative overhead. It is scientific responsibility directed toward the production of knowledge itself.

These pathways should enter operations records as measurable process variables. Prompt delay, skipped classifications, device removal, staffing ratios, overnight response time, protocol pauses, and reasons for missed contact can reveal when site workload changes the observation process. The aim is not to score staff performance; it is to distinguish latent change from workflow-induced visibility change (Little & Rubin, 2019; Rubin, 1976).

Families may feel that participation honors the person, helps future patients, pleases clinicians, proves something sacred, or avoids regret. Those motives can coexist with genuine consent, but they can also become pressure. Recruitment and continuing consent should make clear that stopping is not betrayal and nonparticipation is not failure (Council for International Organizations of Medical Sciences, 2016; Wendler & Grady, 2008; World Medical Association, 2024).

Clinical and research staff time has to be counted explicitly. Nurses, aides, physicians, chaplains, social workers, and coordinators already work under pressure. Devices, forms, reminders, family explanations, incident reports, and endpoint classifications can alter care even when each task seems small. A protocol that depends on invisible labor has not established feasibility.

The site should therefore maintain a burden budget for staff and family contact. It should track time added per shift, missed tasks, after-hours calls, emotional strain, technical failures, and complaints. When burden exceeds a preset limit, the protocol pauses or shrinks rather than asking people to absorb it.

Family and staff burden also affect external validity. A protocol feasible at a highly resourced academic hospice may fail in a small facility, a rural home-care setting, a crowded ward, or a community with different language needs. The result should not travel farther than the burden conditions allow.

A comfort-first terminal study honors family and staff by treating their burden as real knowledge. Not every cost belongs to the participant alone. Terminal research must identify who carries hidden work and answer for the work it creates.

3.3.5 — Stop, Withdraw, and Remain Silent

The right to silence is especially important because this project studies felt experience. A participant may stop speaking because of fatigue, fear, medication, privacy, dignity, spiritual practice, distrust, or simple preference. The study cannot presume that silence is data in the direction it wants. A no-inference state should be available whenever the communicative channel fails for reasons the protocol cannot ethically resolve. Withdrawal should be fast, nonpunitive, and possible through the participant, an authorized surrogate within limits, or a clinical stop order when participation burdens care. The research record may preserve the reason for stopping; it may not turn the stop into a hidden endpoint story.

A terminal protocol must make stopping easy. The right to withdraw cannot be buried in a consent form, spoken once, and then surrounded by gratitude, urgency, devices, staff effort, and family hope. Withdrawal must remain available, visible, and free from interpretation.

Stopping can occur for discomfort, fatigue, privacy, family request, staff workload, clinical change, loss of capacity, distress, device irritation, spiritual preference, data-security concern, or no stated reason. The protocol does not require a morally adequate explanation. The participant or authorized route may simply say no (Council for International Organizations of Medical Sciences, 2016; International Council for Harmonisation, 2026; World Medical Association, 2024).

Remaining silent is also a protected choice. A participant may decline interviews, decline meaning questions, decline final messages, decline dream sharing, decline family interpretation, or decline public quotation. Silence should not be coded as denial, imbalance, fear, or evidence. It is often the most dignified form of agency left.

Withdrawal raises data questions. The consent process should state what happens to already collected data, what can be deleted, what cannot be withdrawn because it has been aggregated or legally retained, and whether derived analyses can remain. Those distinctions must be understandable before the terminal period, not negotiated during crisis.

A stop rule must also be independent of investigator disappointment. If a participant withdraws near a valuable

endpoint, the study loses the endpoint. If a site pauses because staff burden is too high, the study loses the site temporarily. If a family declines further contact, the study loses context. These losses are not failures of cooperation; they are ethical facts. The analysis plan should anticipate these losses so that investigators do not face incentives to reinterpret withdrawal or burden after seeing its effect on statistical power. Ethical stopping is most credible when its scientific consequences are accepted in advance.

A research program earns trust when it can preserve silence and still publish honestly. It should be able to say: the endpoint was not observed; the inference was not made; the person remained protected. Reporting protected absence as an outcome also makes the study's evidentiary boundary visible to readers. What was not observed should remain distinguishable from what was observed and found neutral.

Silence must be represented as a governed outcome rather than interpreted as neutral experience. A missing report near the endpoint may reflect sleep, comfort, fatigue, delirium, sedation, device removal, clinical urgency, refusal, or loss of capacity. The study should preserve those possibilities in the observation record and return no estimate when they cannot be separated. Imputing peace from nonresponse would turn a humane stop into a favorable result (Little & Rubin, 2019; World Medical Association, 2024). Where feasible, the record should distinguish known reasons for nonresponse from genuinely unresolved missingness without inferring the participant's unobserved

experience. That distinction improves later sensitivity analysis while preserving the no-inference rule.

Independent stop authority is equally important. Participants, authorized representatives where applicable, bedside clinicians, palliative teams, and designated research monitors need practical routes to pause or end observation without negotiating with the hypothesis owner. The stop should take effect before analytic value is weighed, and the reason should be recorded at the level needed for safety and feasibility without demanding intimate explanation. A study that can stop cleanly is not weaker. It is better designed for the realities it claims to study (Council for International Organizations of Medical Sciences, 2016; International Council for Harmonisation, 2026). Stop authority should also be operationally reachable at the moment it is needed, including outside ordinary research hours when terminal care continues. A safeguard that exists only on paper cannot reliably protect either participants or the integrity of the resulting evidence.

3.4 — Consent, Privacy, and Bystander Protection

Consent in this volume is granular because the research object is intimate. Lived-experience data may include self-report, physiology, behavior, clinical records, device streams, family context, staff observations, language, location, sleep, medication, disability accommodation, and terminal circumstances. A signature records agreement at one time; it does not by itself establish comprehension or authorize every later use (Council for International Organizations of Medical Sciences, 2016; National Commission, 1979; Tam et al., 2015; U.S. Department of Health and Human Services, 2018).

Consent as a single event fails when data travel. A participant may understand a bedside interview but not future linkage, accept sensor monitoring but not publication of excerpts, permit coded sharing but not Artificial Intelligence training, or withdraw after derived results exist. The design therefore separates permission states, explains what withdrawal can still change, and tests comprehension of the uses that carry the greatest risk.

Privacy also extends beyond the enrolled participant. Family members, staff, roommates, household members, visitors, community members, and clinicians can appear in data through speech, proximity, relationships, or context. They can carry burden without ever being recruited. That is bystander burden.

Research often hides its own transfers. A study may collect data from one consenting participant while shifting exposure onto a spouse, a child, a nurse, a translator, or a community. The consent form may appear complete while the data ecology is not.

This section therefore treats data dignity as a lifecycle. Permission is given for a use, not for all possible uses. Access is limited by purpose. Retention has an end or a justified retention route. Deletion and withdrawal are explained honestly. Derived results are handled with humility. Future reuse and model training require their own governance.

The result should be a consent architecture that is specific enough to protect people and stable enough to support science. A dataset whose permission is vague is not a stronger dataset. It is a liability waiting to distort trust, recruitment, participation, and public interpretation.

Table 3.2. Data-use lifecycle, permissions, and correction routes.

Stage	Required question	Permission or governance route	Exit or correction mechanism
Collection	What exact data answer the stated question?	Specific consent or approved narrow waiver with minimization	Refuse, pause, or collect less without losing care
Linkage	Which external records are necessary and why?	Separate linkage permission and purpose review	Decline linkage or sever future linkage where feasible
Storage	Who can access the data and for how long?	Role-based access, logs, retention schedule, security review	Correction, access audit, incident response, scheduled retirement

Stage	Required question	Permission or governance route	Exit or correction mechanism
Analysis	What estimand and model use are authorized?	Approved analysis plan and version control	Analysis stop when consent, data quality, or burden gate fails
Sharing	Who receives data and under what constraints?	Data-use agreement, repository review, bystander risk assessment	Withdraw before sharing where feasible, restrict or recall after breach
Publication	What can be shown without exposing persons?	Public artifact review and identification-risk check	Remove, revise, or correct public material when risk or error appears
Future reuse or Artificial Intelligence training	Is the new use materially different?	New consent or governance approval; no automatic conversion	Decline, retire data, prohibit model use, document derived-result limits

3.4.1 — Specific Consent by Use

Use-specific consent should be written as a menu with consequences, not as a dense legal wall. Observation, wearable data, audio, video, clinical-record linkage, family interviews, staff notes, deidentified sharing, publication of examples, future secondary analysis, and Artificial Intelligence training differ in identifiability, sensitivity, reversibility, and social risk. A participant who agrees to one does not agree to all. The protocol

should track permissions in a versioned consent state, not merely in a scanned form. If a later analysis wants a use that was not clearly permitted, the default is renewed permission or abstention, not clever reinterpretation of the original consent.

This specificity protects autonomy. A participant can say yes to a low-burden observation and no to open data sharing. They can allow coded clinical linkage and refuse identifiable quotations. They can permit an analysis for the stated study and decline downstream model training. The study remains viable because consent has options rather than pretending refusal must be all or nothing (Council for International Organizations of Medical Sciences, 2016; National Commission, 1979; National Institute of Standards and Technology, 2020).

Specificity can improve both autonomy and data quality. Participants who understand the use, limits, and withdrawal route can make more stable choices and report concerns earlier. Evidence from clinical trials shows that signatures do not guarantee understanding, so comprehension must be checked in a proportionate way rather than presumed from completed paperwork (Tam et al., 2015).

Artificial Intelligence training needs its own explicit route. A participant who agrees to a research estimate has not granted permission for a model to learn from their text, voice, physiology, or clinical context in ways that may be hard to trace. De-identification alone is not a moral solvent.

Publication is likewise distinct. A chart, graph, quote, case vignette, or public artifact can expose meaning even when names

are removed. Consent should describe what kinds of material may appear publicly and what review or withdrawal opportunity exists before publication where feasible.

The rule is straightforward: each data use must have a lawful and ethical basis, a participant-facing explanation, a refusal route, a governance owner, and a consequence for misuse. Consent that cannot name the use is not consent to that use.

3.4.2 — Capacity and Ongoing Consent

Capacity fluctuation creates a temporal ethics problem. A participant may consent while lucid, become delirious, regain clarity, become fatigued, or communicate only through supported methods. The study must decide in advance which activities require contemporaneous assent, which may continue under a prior directive, which require surrogate confirmation, and which must stop when active dissent appears. A legally authorized representative is not a theory representative. The surrogate's role is to protect the participant's known wishes and interests, not to help the research program reach completion. When capacity returns, the participant's current voice should re-enter the consent relationship as soon as practicable.

Capacity can change over time, especially in terminal, neurological, psychiatric, pediatric, traumatic, or heavily medicated settings. A person may understand the study at enrollment and later become confused, fatigued, sedated, delirious, or unable to communicate. The consent system must be designed for that movement.

Ongoing consent does not mean constant interrogation. It means periodic, proportionate confirmation that participation still matches the person's preference, understanding, and capacity. A low-burden reminder, a simple stop card, a device-removal option, a family-notification rule, or a re-consent checkpoint may be appropriate depending on the study. Evidence that participants often misunderstand material trial information makes comprehension an object to assess, not a box inferred from a signature (Council for International Organizations of Medical Sciences, 2016; Tam et al., 2015; U.S. Department of Health and Human Services, 2018).

Surrogate permission is limited. A legally authorized representative may sometimes permit research under law and review, but the surrogate cannot turn the participant into a proof object. They cannot authorize avoidable harm, comfort delay, humiliating exposure, public meaning, or uses outside the approved purpose. The participant's known preferences and present signs of resistance matter.

Assent and dissent deserve special weight when full legal consent is unavailable. A person who cannot sign a form may still show discomfort, refusal, agitation, withdrawal, or fatigue. Those signals should be treated as participation data only after they are treated as possible refusal. In many cases, they end the procedure.

Re-consent is required when material facts change. A new data use, a new repository, a new model-training plan, a changed device, a more burdensome schedule, an unexpected privacy risk, or a new publication mode should not ride on the old agreement. A changed bridge is a changed permission problem.

Ongoing consent is the temporal form of respect. It recognizes that a conscious stream is not a frozen legal moment and that the burden of participation can change as the person's body, mind, relationships, and risks change.

3.4.3 — Operational Participation Is Not Research Consent

Participation in care, civic life, employment, religious practice, ordinary-life tools, public services, or any applied project is not research consent. Operational records do not become scientific data merely because they are interesting. Public access does not become model-training permission merely because a record can be scraped.

This rule is crucial across the series. *The Science of Fairness* may receive questions from other domains, but it may not convert their records into candidate momentary and interval estimates or terminal evidence without a separate research route. A city record, government service log, note or form from a religious community, Workgame artifact, or transition support file is not a measured cumulative estimate (U.S. Department of Health and Human Services, 2018; Wendler & Grady, 2008).

Operational data are collected for operational purposes. A clinic notes symptoms to care for a patient. A service office records eligibility to route help. A public institution stores a decision to make authority reviewable. Those purposes do not silently authorize scientific inference about inner life.

The difference protects nonparticipants. A person may accept care, submit paperwork, attend a ceremony, contact support, or

use a tool without enrolling in a study. They should not have to avoid ordinary help in order to avoid becoming research material.

If operational data are ever proposed for research, the protocol must pass a conversion gate: purpose compatibility, legal basis, consent or waiver justification, privacy review, de-identification humility, bystander analysis, data minimization, and no-score discipline. Many proposed conversions should fail.

Operational participation is not research consent because burden follows purpose. A record made to help a person should not later carry the burden of proving a theory unless the person and the governance route have specifically allowed it.

3.4.4 — Bystanders and Relational Data

Bystander data are not marginal in this program because lived experience is relational. A participant's record may contain a spouse's grief, a child's voice, a nurse's judgment, a roommate's routine, a family conflict, a chaplain's visit, a household address, or a community marker. The study must decide which relational information is necessary, how it will be masked or excluded, when bystander permission is required, and when the analysis must abstain because privacy cannot be protected. A person should not become research material merely because they loved, treated, lived near, transported, translated for, or appeared beside someone who enrolled.

Bystander protection begins with recognition. The protocol should map who may be captured around the participant and what kind of information could be revealed. It should distinguish unavoidable contextual presence from deliberate collection about

others. It should minimize relational exposure whenever possible (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016; National Commission, 1979).

Family reports require particular care. A caregiver can provide valuable context, but they may also experience guilt, grief, pressure, hope, exhaustion, or conflict. Their report is a relational act, not a transparent window into the participant's inner life. It needs consent, framing, and limits.

Staff data also matter. Clinical notes, workflow records, and classifications can expose staff conduct, emotional burden, or institutional pressures. Staff should not be reduced to instruments without labor acknowledgment, privacy consideration, training, and reporting channels.

Relational data raise publication risk. A de-identified vignette can still reveal a family, community, diagnosis, site, or conflict. The more unusual the case, the higher the identification risk. Narrative power is not an excuse for relational exposure.

The bystander rule is a fairness rule: a person may consent to carry research burden for themselves, but they cannot unilaterally donate the privacy of everyone around them. The study must protect the ring of people who can be made visible by another person's participation.

3.4.5 — Minimization, Retention, and Deletion

The retention plan should explicitly cover derived features, analytic code, and consent records. For each object, name the

retention or deletion reason, who may request action, the review authority, and the record left after deletion.

Data minimization asks a simple question before collection begins: what is the least information that can answer the defined scientific question under the approved use? It rejects the habit of collecting everything because future models might find patterns. A project about hidden burden should not create unnecessary data burden.

Retention requires a reason and a clock. Some data may need to remain for audit, replication, safety reporting, or correction. Some should expire. Some should be destroyed when a participant withdraws. Some derived results may remain only in aggregate or locked form. The participant should not have to guess which category applies (National Commission, 1979; National Institute of Standards and Technology, 2020).

Deletion is not always technically or legally simple, and the consent process should not promise what cannot be delivered. It should distinguish raw identifiable data, coded data, de-identified datasets, aggregate statistics, published results, trained model weights, audit logs, and safety reports. Each has a different withdrawal and deletion pathway.

Data retirement is more than deletion. A dataset can become obsolete because an instrument version fails, a consent route is found inadequate, a privacy breach occurs, a population was unfairly recruited, or a model is retired. Retired data should not continue traveling through public artifacts or later analyses as if nothing happened.

Access control is part of minimization. Fewer people should see sensitive data, and each access should have a purpose, role, log, and review route. Curiosity, convenience, donor interest, public excitement, or inter-book usefulness are not access purposes.

3.4.6 — Privacy as a Dynamic Relationship

Privacy risk changes during a study. A data element that appears harmless at collection may become identifying when combined with location, medication, family timing, voice, device telemetry, or a rare clinical trajectory. Governance therefore cannot be exhausted by a one-time classification of identifiers. It must revisit what the dataset can reveal as linkages, models, and external information change.

The relevant unit is often relational rather than individual. A participant's report may reveal that a spouse was present, that a caregiver missed work, that a clinician changed medication, or that a child witnessed a crisis. Those people may never have enrolled, yet their lives become inferable through the participant's record. Data minimization should ask not only what was collected about the participant, but what can be learned about others.

Access is divided by purpose. Collection staff may need contact information. Clinicians may need safety alerts. Analysts may need coded observations. Public repositories may receive only a carefully reviewed derivative. Artificial Intelligence training, commercial reuse, and public release are separate uses, not automatic extensions of research consent. A broad permission cannot carry every future purpose without becoming functionally meaningless (Council for International Organizations of Medical

Sciences, 2016; National Institute of Standards and Technology, 2020; U.S. Department of Health and Human Services, 2018).

Technical controls require accountable roles. Encryption, access logs, secure computation, and de-identification need someone responsible for key custody, approval, anomaly review, revocation, and incident response. Naming tools without decision-makers hides authority behind infrastructure.

Inference risk should be tested adversarially. Teams can attempt re-identification, membership inference, linkage, attribute inference, and extraction from released models or summaries. The goal is not to prove that privacy is perfect. It is to discover which releases create unacceptable pathways and to redesign or withhold them before participants bear the consequence.

Retention should follow scientific need rather than institutional habit. Preserve data as needed for correction, replication, or regulatory obligations; aggregate, transform, or delete them when their purpose ends. Distinguish raw, derived, analytic, and public artifacts in the retention schedule because their value and risk differ.

Correction is part of privacy: inaccurate records can travel farther than disclosed ones. Participants need a route to report errors in identity, context, medication, consent state, or interpretation. Record whether corrections change metadata or analysis, require public notice, or invalidate prior artifacts. Silence cannot correct propagated errors.

Privacy bounds authority. Data held for declared research grant no ownership of a person's inner life or permanent claim on the

record. Legitimacy requires visible purpose, access, contestability, exit, minimization, retention, and deletion.

Map collection, derivation, access, retention, reuse, bystander exposure, and deletion or withdrawal limits. Retained records must remain worth their participant and relational risks (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

3.5 — Vulnerability, Assent, and Research Proportionality

Vulnerability is not a defect in a person. It is a relationship between a person's situation and the power of a research request. Illness, dependence, poverty, age, disability, institutional status, crisis, language barriers, incarceration, immigration fear, employment hierarchy, clinical reliance, and spiritual pressure can make refusal harder (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; World Medical Association, 2024).

This section does not solve vulnerability by excluding vulnerable people from research. Exclusion can also be unfair when it removes people from studies meant to understand their needs. The problem is not presence. The problem is extraction, pressure, inaccessible communication, and burden that the protocol fails to count.

Participation constraints become validity constraints. If only the healthiest, most supported, most articulate, and most trusting people can enroll or remain, the study describes a selected group. Unequal effort requirements amplify that selection, and costly refusal weakens the meaning of consent.

The design burden therefore rises where refusal is fragile. Materials must be accessible. Aid must not be contingent. Assent must be respected. Surrogates must remain limited. Staff and participant burden must be budgeted. Monitoring intensity must be justified. Recruitment must avoid crisis and desperation.

The causal structure is explicit: research procedures can change the outcome. Monitoring burden can change sleep and physiology. Consent pressure can change expectancy. Clinician behavior can change care. Family anxiety can change the bedside. Staff workload can change adherence. Missingness can concentrate among those most burdened.

A serious protocol therefore specifies prospective participation and workload endpoints before enrollment. It tracks fatigue, distress, time, privacy concern, family involvement, staff workload, device discomfort, and withdrawal patterns. Predefined thresholds trigger pause, redesign, or termination before protocol strain becomes hidden evidence.

Table 3.3. Research-burden causal pathways and design controls.

Research element	Immediate burden pathway	Data-generating consequence	Required design control
Monitoring schedule	Sleep interruption, device awareness, privacy worry	Changed physiology, altered affect report, selective adherence	Minimize schedule, monitor discomfort, allow instant removal
Consent and recruitment language	Expectancy, obligation, prestige, fear of refusal	Demand effects, selective enrollment, underreported distress	Neutral scripts, independent consent support, refusal protection

Research element	Immediate burden pathway	Data-generating consequence	Required design control
Clinician awareness	Treatment timing, extra attention, hesitation, documentation change	Care-path confounding and site effects	Separate care and research roles; record care changes
Family involvement	Anxiety, guilt, interpretation pressure, conflict	Proxy bias, altered bedside environment, missingness after distress	Family burden budget, opt-out, contextual labels
Staff workload	Protocol fatigue, rushed measurement, moral distress	Nonadherence, incomplete data, selective site continuation	Staff time accounting, workload stop threshold, support routes
Public artifact anticipation	Fear of exposure or symbolic use	Self-censorship, withdrawal, public-risk bias	Embargo, artifact review, no individual meaning claims

3.5.1 — No-Desperation Consent

No-desperation consent is violated when research becomes the practical route to care, money, housing, attention, belonging, protection, or hope. The protocol must separate services from enrollment, provide nonresearch routes to support, make refusal visibly safe, and keep compensation proportionate to time and cost rather than need. A person in hardship may consent, but saying no must leave that person no worse off.

Payment and support require care. Reimbursement for time, travel, childcare, communication access, or lost wages can make participation more equitable. Excessive inducement can make refusal less real. The distinction depends on context, income, need, dependence, and the stakes of the study (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; Wendler & Grady, 2008).

Desperation can also be social. A participant may want to please a clinician, a community leader, a religious guide, a family member, a donor, or a public movement. Consent materials should make refusal ordinary and protected, not heroic or disloyal.

Recruitment should avoid language that frames participation as helping prove the Law, advancing humanity, honoring suffering, or joining a special circle of truth. Those phrases convert belief and belonging into pressure. A safer invitation names the study, the burden, the limits, and the right to decline.

No-desperation consent also requires independent care routes. If unmet need is identified, the person should receive appropriate referral or support whether or not they enroll. The research team should not be the only visible door to help.

Voluntariness becomes concrete at the point of refusal. A person must be able to decline or withdraw without losing care, housing, employment, immigration support, spiritual belonging, or any other needed relationship. If that assurance cannot be made credible, the protocol is not ready. Under such pressure, the data may reflect dependence rather than genuine willingness.

3.5.2 — Children and Assent

Research with children and adolescents requires age-appropriate explanation, assent where developmentally possible, meaningful dissent protection, and guardian permission. Their developing agency, dependence on adults, and sensitivity to authority call for a stricter burden threshold. The question must address a child-specific care need or knowledge that cannot responsibly be obtained from adults, with clear justification for the benefit or knowledge sought. Low-risk developmental measurement uses accessible communication and protects the child's freedom to form their own interpretation of the research.

Guardian permission is not enough. Age-appropriate assent, dissent, comfort, comprehension, and communication matter. A child may not understand the abstract hypothesis, but can often understand whether a device hurts, whether a question feels frightening, whether they want to stop, or whether they prefer privacy. The protocol must treat those responses as present authority, not as inconvenient noise beneath adult permission (Council for International Organizations of Medical Sciences, 2016; U.S. Department of Health and Human Services, 2018; Wendler & Grady, 2008).

Children should not be asked to carry existential language. Consent and assent materials should avoid telling a child that their feelings help test whether life is fair. The child's role, if any, should be described in concrete and modest terms: a study of sleep, mood, comfort, communication, or measurement feasibility.

Developmental differences also affect data interpretation. Children's self-report scales, affective expression, language, caregiver reports, neurodevelopmental context, and medical status differ from adult measures. A result in children cannot be imported into the adult claim without calibration and reference-class caution.

Assent is ongoing. A child who tolerated a procedure yesterday may not tolerate it today. Fatigue, illness, developmental stage, family stress, and clinical changes can alter willingness. Stopping should be easy, immediate, and free from adult disappointment.

The chapter's rule is protective but not erasing. Children can be included ethically in research that addresses their needs and respects their voice. They cannot be recruited to make a grand claim emotionally irresistible.

3.5.3 — Disability and Communication Access

Disability access determines who can participate and whose experience enters the evidence. Include nonverbal response routes where needed, and do not substitute a support person's convenience for the participant's will.

Disability and communication difference must not be mistaken for lack of experience. A person who cannot speak fluently, move easily, complete a standard scale, tolerate a device, read a form, or respond at expected speed may still have rich lived experience. Measurement difficulty is not ontological absence.

Accessible research begins before enrollment. Materials may require plain language, sign language, augmentative communication, large print, screen-reader compatibility,

translation, flexible timing, sensory accommodation, participant-chosen support, and non-digital options. Accessibility is part of construct validity because an instrument that excludes or distorts communication cannot claim to measure the same target across participants (International Test Commission, 2018; National Institute of Standards and Technology, 2020).

A protocol that excludes people because they are hard to measure may create a false reference class. The study might then describe only articulate, mobile, compliant, device-tolerant participants and call the result human evidence. That would hide exactly the burden the project claims to reveal.

At the same time, inclusion must not become coercion. Some accommodations require time, staff, privacy, or specialized equipment. If the site cannot provide them safely, it should not recruit and then place the burden of adaptation on the participant. Access failure should trigger redesign, not pressure.

Communication support also raises privacy questions. A caregiver, interpreter, assistant, or device may mediate responses. The protocol must distinguish participant voice from support context and protect the supporter from becoming an unconsented data source. The participant should control, where possible, who assists and what is recorded.

Difficulty communicating does not reduce the person's standing. Any resulting access limit belongs to the measurement method, not to the person.

3.5.4 — Institutional and Clinical Dependence

Dependence changes refusal. A patient may fear disappointing a clinician, a resident may fear staff judgment, a student may fear authority, an employee may fear retaliation, and a service recipient may fear losing access. Ethical review should therefore examine who introduces the study, who collects consent, whether care providers know the decision, what records show refusal, and how complaints move outside the dependent relationship. The cleanest consent process may require independent staff, delayed approach, private refusal routes, and visible assurance that ordinary care or service will not change. The scientific reason is direct: fear of refusal changes selection and response long before the model sees data.

These pressures can operate without intended coercion; prisoners and detained persons face especially strong constraints.

The study must separate authority from recruitment. A clinician who controls care should not be the only person explaining enrollment when a neutral alternative is possible. A supervisor should not recruit subordinates. A service office should not present research while someone is seeking urgent help. Refusal must not affect care, grades, employment, housing, benefits, or access (Wendler & Grady, 2008; World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Clinical dependence is especially subtle in terminal and serious illness settings. A participant may believe that agreeing to research will make staff more attentive, provide better symptom

relief, or signal gratitude. The consent process must clarify the study's scientific purpose, distinguish possible benefits from guarantees, and make clear that refusal will not reduce ordinary care or staff respect.

Institutional dependence also affects reporting of distress. A participant may underreport discomfort to avoid disappointing staff or seeming difficult. They may keep a device on because a nurse placed it. They may answer in ways they think researchers want. These are not mere social niceties; they are measurement threats.

A dependence-aware protocol includes independent consent support, refusal scripts, complaint pathways, non-retaliation rules, separation between care and research decisions, and monitoring for differential withdrawal. It also records whether participants felt free to refuse in post-contact checks where appropriate.

Institutional dependence does not make research impossible. It makes invisible power visible. A study that cannot show how refusal remains safe should not recruit from that dependent relationship.

3.5.5 — Participant and Staff Burden Budgets

A participation budget should be as concrete as a sample-size calculation. It should estimate minutes, awakenings, device contacts, skin-irritation risk, travel, forms, interviews, family tasks, staff steps, handoffs, after-hours calls, translation, security incidents, and emotional load. The plan needs a ceiling, a monitoring schedule, and an authority able to pause the study

when the design exceeds it. This is an inference condition as well as an ethical one: patterned noncompliance, withdrawal, or skipped procedures can change the study population and the missingness mechanism.

A burden budget is a prospectively specified account of what the study asks participants, families, and staff to carry. It includes time, travel, fatigue, device discomfort, privacy anxiety, emotional strain, sleep interruption, forms, technical troubleshooting, family conversations, clinical disruption, staff labor, and reporting duties.

The budget should be quantitative where possible and descriptive where necessary. Minutes per visit, number of contacts, device wear time, interview length, alerts per week, staff minutes per participant, family calls, adverse distress events, missed measures, and withdrawal reasons can be tracked. Some burdens, such as moral distress or privacy worry, may need carefully designed qualitative reporting (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000).

Burden endpoints are not decorative. They are thresholds that can stop, shrink, or redesign a study. A protocol might pause if distress exceeds a declared level, if device removal requests cluster, if staff overtime rises, if family complaints exceed a threshold, if missingness becomes burden-linked, or if no-estimate conditions dominate.

The budget also belongs in the analysis. Excessive study load can produce selection, missingness, nonadherence, altered

physiology, and poor transport across sites. A favorable signal among the easiest-to-observe participants may not generalize; a null after disruptive monitoring may describe the protocol more than the phenomenon.

3.5.6 — Community Partnership and Post-Study Obligations

Ethical responsibility does not begin and end with an individual signature. A study can secure consent while shifting unexamined burden onto families, clinical teams, cultural communities, translators, or institutions whose labor makes the protocol possible. Community partnership can expose those burdens before they become invisible features of the data, but it never replaces individual consent.

Partnership is meaningful only while the design remains reversible. People affected by the study should be able to change enrollment, timing, burden thresholds, communication, data flows, and return-of-results plans. Their role is to inform burden, access, meaning, foreseeable misuse, and practical feasibility, not to certify the truth of the hypothesis or replace accountable clinical, statistical, philosophical, and technical judgment (Council for International Organizations of Medical Sciences, 2016; National Academies of Sciences, Engineering, and Medicine, 2017; World Medical Association, 2024).

Compensation and access should recognize real expertise without turning need into pressure to endorse the project. Travel support, translation, caregiving support, assistive technology, supported communication, and adequate decision time can make refusal and participation more genuine. These accommodations also change

who can be observed and how, so their effects belong in the scientific record rather than being treated as invisible logistics.

Post-study obligations should be declared before enrollment. Participants should know whether aggregate findings will be returned, whether individual feedback is possible, how long data will be retained, who may reuse them, and how corrections will be communicated. No return process may imply a personal terminal interpretation or latent-ledger reading that the science cannot support. Return plans should also specify when findings are too uncertain, preliminary, or group-level to support meaningful individual feedback.

Withdrawal and retained data require a concrete governance route. Participants should know which future uses can still be stopped, which aggregated products cannot practically be reversed, who holds custody, how identity and requests are verified, and how long a response should take. Clinical safety signals also need a prospectively designed response pathway so research findings do not fall between scientific and care responsibilities. Clear responsibility for each request prevents withdrawal, correction, or safety concerns from becoming unresolved administrative handoffs.

Families, staff, and sites need an exit and complaint pathway of their own. The record should state what can be shared, what remains confidential, how care disruption is reported, how devices and local data copies are handled, and who remains available after the research team leaves. Contribution to a study never grants permission to disclose another person's inner life. These pathways should remain accessible after active data

collection ends because privacy, burden, or interpretation concerns may emerge only during later analysis or publication.

Partnership can improve external validity by identifying seasonal routines, cultural scripts, medication practices, communication norms, staffing realities, and care transitions that shape measurement and missingness. It does not make a local sample universal; it makes the local conditions visible enough for later investigators to judge transport rather than assume it. Documenting which local conditions materially affected participation or observation also helps future studies distinguish genuine transport failure from differences in implementation.

The governing principle is reciprocity without transferred authority. People who bear research burdens should have meaningful power over burden, access, communication, correction, and exit, while no participant or community is asked to bless the theory or adjudicate the science. A program worthy of studying fairness should leave those relationships more contestable and more humane than it found them. Reciprocity is strongest when that influence produces visible changes in study practice while scientific conclusions remain answerable to the evidence.

3.6 — Scientific Authority and Domain Boundaries

Scientific authority is narrow. It can define claims, test models, compare rivals, report uncertainty, update evidence status, and correct public overclaim. It cannot replace the independent authority of clinical care, democratic decision-making, constitutional law, religious freedom, ordinary privacy, civic equality, employment rules, insurance protections, or transition governance (Douglas, 2009; National Academies of Sciences, Engineering, and Medicine, 2017).

Findings can inform questions in policy, religion, institutions, and personal practice. Each application evaluates those findings alongside its own evidence, purposes, and accountable decision procedures.

Scientific findings reach other domains through a clear account of their scope and intended use. Cities evaluate applications against civic outcomes; religious communities retain authority over their own doctrines; governments preserve independent eligibility rules; and donors receive accurate accounts of research progress. Scientific verdicts remain the responsibility of accountable researchers and reviewers.

Applied domains can use the burden question to protect nonparticipants, improve records, reduce coercion, and design reviewable systems. Those aims can be evaluated directly through the outcomes of the application.

The science volume gives the series a shared vocabulary for scientific, formal, applied, spiritual, legal, and public-

communication claims. Transition governance and each applied domain retain their respective responsibilities.

3.6.1 — No Policy Shortcut

No policy shortcut means that even strong evidence travels through ordinary public reasoning. A scientific finding may advise a legislature, court, agency, hospital, school, or civic institution, but it does not replace constitutional analysis, democratic accountability, administrative competence, community review, disability access, economic evidence, or legal remedy. A successful research program would create prestige, and prestige can become pressure. A public official could cite the science to justify surveillance, triage, austerity, forced participation, or spiritual symbolism. Evidence can constrain false claims; it cannot become a fast lane around lawful public review.

Scientific findings do not bypass democratic, legal, constitutional, administrative, or clinical review. Even a strong result about a component of the Law would not tell a government what policy to enact, what budget to pass, what right to recognize, or what person to prioritize.

Policy involves values, rights, tradeoffs, institutional competence, distributional effects, feasibility, public participation, law, equity, and accountability. Science can inform those questions by clarifying facts and uncertainty. It cannot erase them by announcing that fairness has been measured (Douglas, 2009; Mitchell, 2009; National Academies of Sciences, Engineering, and Medicine, 2017).

The danger is greatest when policy makers want simple authority. A graph can appear cleaner than a hearing. A model can appear more neutral than a court. A statistic can appear more serious than a person's testimony. This chapter blocks that translation.

Candidate momentary valence estimates, interval contribution estimates, ledger estimates, terminal-signature studies, participation-load models, and endpoint classifications are research tools within a declared context of use. They are not policy scores. They cannot decide benefits, custody, sentencing, immigration, housing, employment, insurance, policing, or access to public trust.

A responsible policy interface would instead use science as constraint and warning. If a public program creates hidden burden, measurement science may help detect it. If a proposed artifact overclaims evidence, the science volume may correct it. If an operational record is misused as research data, the science volume can identify the boundary violation.

No policy shortcut keeps public authority answerable to its own standards. Science may enter the room as evidence, method, and critique. It may not leave the room as unreviewable command.

3.6.2 — No Religious Certification

Religious communities and nonbelievers can engage with scientific findings while retaining freedom of interpretation. Science evaluates empirical claims through its methods; religious and pastoral meanings are considered within their own traditions and personal contexts. This division leaves both forms of inquiry room to develop without requiring either to govern the other.

Scientific reporting may state only the tested empirical relation, its uncertainty, and its branch-matched status. It may not translate a component result into divine agency, providence, karma, grace, spiritual rank, or religious obligation. Those are interpretive claims within voluntary religious life, not outputs of a measurement model.

The boundary protects religion as well as science. If sacred life depends on a future dataset, a negative result becomes a spiritual emergency; if a favorable result is treated as certification, dissenters, nonbelievers, and other traditions can be pressured by a prestige they never accepted. Separation protects voluntary meaning from scientific rank (Douglas, 2009).

The Science of Fairness can say that a claim has or lacks empirical support under defined conditions. It cannot say that a person's suffering had sacred purpose. It cannot say that one spiritual interpretation is higher. It cannot make participation in a religious framework more rational, more loyal, or more required.

Religious communities may still respond to scientific humility. They can learn from burden ethics, no-coercion rules, data dignity, and care before evidence. But those are ethical boundaries and interpretive opportunities, not scientific certification of sacred truth.

Research communication should therefore avoid language that presents findings as spiritual confirmation. It should not say that science has proven a covenant, confirmed a faith, blessed a practice, or shown that a ritual works in cosmic terms. It can state only the scientific status of the tested claim.

No religious certification keeps belief optional and science falsifiable. The Law may invite meaning for some readers, but the research program cannot require meaning from anyone.

3.6.3 — No Civic or Employment Rank

The civic and employment boundary should be written into data-sharing agreements, not left as a moral aspiration. No momentary valence estimator, interval valence contribution estimate, ledger estimate, burden profile, endpoint class, consent choice, withdrawal, missingness pattern, vulnerability marker, or research participation status may become a basis for hiring, insurance, policing, benefits, credit, admission, housing, immigration, workplace evaluation, or public trust ranking. Even aggregate tools can leak into person-level suspicion when dashboards are built carelessly. The study must therefore prevent downstream reuse by contract, access control, audit, technical separation, and public language. A scientific estimate that can ruin a person's ordinary life has already become the wrong kind of instrument.

The technical safeguard is purpose limitation. Every estimate must carry its study purpose, prohibited downstream uses, access roles, retention rule, and audit path. A variable that is valid for one bounded analysis becomes invalid when repurposed as eligibility, loyalty, employability, insurability, credibility, or public trust.

The misuse can begin subtly. A research participant may be treated as more committed; someone who refuses as evasive; a public supporter as more trustworthy; a measured person as more

transparent. Each move converts participation or visibility into social hierarchy (National Commission, 1979).

Employment and insurance uses are especially dangerous. A model that estimates burden, distress, resilience, compliance, honesty, or affective state could be used to route opportunities, premiums, assignments, surveillance, or exclusion. That use is outside the project.

Civic rank is also incompatible with measurement uncertainty. The constructs are latent, bounded by context, vulnerable to calibration failure, and unsuitable for individual worth judgments. Even if a measure is useful in a study, it cannot bear the ethical weight of ranking persons.

A protocol should include nonuse language and data-governance controls. Participants should be told that research estimates will not be returned to employers, insurers, public agencies, schools, religious authorities, donors, or civic platforms as ranking tools. Data-access systems should make that promise enforceable.

No civic or employment rank preserves the difference between making burden visible and making persons measurable. The first can serve fairness. The second would betray it.

3.6.4 — No Artificial Intelligence Sovereignty

Artificial Intelligence may assist with coding, monitoring, anomaly detection, privacy review, translation support, or model comparison only under declared human responsibility. It may not decide consciousness, consent, endpoint, dignity, truth, eligibility, spiritual meaning, or participant credibility. Model outputs must be inspectable, contestable, versioned, and subordinate to human

review with disciplinary competence. This is not hostility to tools. It is recognition that a system trained on partial data can acquire institutional authority faster than it acquires understanding. For this program, any Artificial Intelligence assistant must increase accountability rather than create a new opaque actor between the participant and the claim.

Human governance means more than placing an approval button after an automated output. The responsible investigator must understand the task, data provenance, model version, error profile, limits, and appeal route, and must be able to reverse the decision. A system whose output cannot be meaningfully inspected or contested is not an assistant in this research program.

The reason is not that Artificial Intelligence is useless. The reason is that these systems can compress uncertainty into fluent output, reproduce bias, hide provenance, magnify training-data misuse, and create unwarranted confidence. A language-like answer about a person's suffering can sound humane while having no right to interpret that person (Autio et al., 2024; Douglas, 2009).

In this volume, Artificial Intelligence use requires context labels: what data were used, what task was authorized, what model version was involved, what human review occurred, what errors are known, what appeal route exists, and what the output cannot be used to decide. Without those labels, model output becomes theater.

Artificial Intelligence also needs data permission. A participant's consent to a study does not grant training rights. A clinical note does not become a model corpus by convenience. A public artifact

does not become consent for extraction. Sensitive data require purpose-specific authorization and governance.

No Artificial Intelligence sovereignty includes no automated endpoint adjudication. A model may help identify patterns for human review, but it cannot declare truth-level endpoint, terminal valence closure, participant meaning, or scientific support. Endpoint and verdict decisions remain governed, contestable, and humanly accountable.

The chapter's boundary is compact: Artificial Intelligence can assist the research program only as a tool under explicit limits. It may not become oracle, judge, priest, clinician, or public ranker.

3.6.5 — No Applied Proof

Applied achievements can generate hypotheses, identify practical challenges, and preserve useful practices. A sound housing policy, care protocol, privacy rule, or analytical tool can be evaluated through its own purposes and outcomes as the scientific program develops.

The fit among applied ideas can generate hypotheses about how institutions, practices, and experiences relate. Those hypotheses become useful research proposals when they specify observations and comparisons that can evaluate the suggested connection.

The science volume may receive applied observations as research leads. A public system may reveal burden pathways. Ordinary-life tools may show measurement hazards. Religious practice may raise interpretive questions. Transition trials may expose data risks. These observations can shape research questions, but they do not skip the measurement and rival gates.

Public materials should connect each scientific claim to its evidence. Diagrams identify what their relationships represent, donor pages describe the research supported, and websites distinguish adoption, participation, and research findings. These details help readers understand how the project develops.

3.6.6 — Authority After Publication

Scientific communication continues through summaries, interviews, diagrams, classroom materials, dashboards, and derivative tools. Each format should accurately name the study's object, population, method, and result so readers can follow the finding back to its source.

Every high-reach statement must preserve the level of the underlying verdict. A route-favored result cannot become evidence that every qualifying stream closes; measurement validation cannot become a personal score; and a replicated terminal-window signature cannot become a forecast of an individual death or destiny. The public sentence must not outrun the technical object it summarizes.

Correction routes should be visible beyond the author team. Participants, readers, rival researchers, clinicians, journalists, ethics officers, and data stewards need defined ways to report factual error, misleading artifacts, broken sources, privacy concerns, or status overstatement. The process should identify who reviews the concern, what evidence is required, when a notice is issued, and how the corrected object replaces or points back to the superseded one (National Institute of Standards and Technology, 2020; Tabassi, 2023).

Institutional use remains bounded by the evidence. Universities, funders, companies, religious groups, political actors, and public agencies may cite the work, but citation does not broaden the verdict or transfer scientific ownership. Artificial Intelligence systems may assist search, consistency checking, or hypothesis generation; they may not become sovereign interpreters of persons, issuers of scientific status, or custodians of hidden experiential dossiers.

Public-use monitoring should be proportionate to reach. A registry of high-reach artifacts should record the source chapter and version, status ceiling, responsible human owner, correction route, and current state: active, corrected, superseded, or withdrawn. Archives should preserve prior objects without allowing search systems or derivative tools to present them as current.

Authority can expire when its evidentiary conditions change. An instrument may be superseded, consent for secondary use may lapse, a security incident may alter risk, or an independent replication may overturn the favored model. Review dates and event triggers reconnect public claims to those changes so a once-accurate statement does not become misleading through staleness (Fischhoff & Davis, 2014; Tabassi, 2023). A review can narrow a claim's permitted use while leaving the underlying observation in the record, with the reason for that change attached.

Those affected by public use need a route to contest it. A participant may challenge a disclosure, a clinician may challenge a ranking-prone dashboard, a rival may identify an omitted comparator, and a reader should be able to locate the current

correction without searching through silent replacements. Contestability does not decide truth by vote; it gives error a path to the people responsible for correction. Someone raising a concern should be able to identify the affected artifact directly rather than reconstructing the full research history before review can begin.

Public scientific claims remain inspectable and correctable. Their documented evidence and revision history let contributors across the series assess the findings and use them within their respective domains. Correction notices are most useful when they distinguish the changed conclusion from findings that remain unaffected, allowing readers to update their understanding proportionately.

3.7 — Non-Negotiable Research Red Lines

The final section concentrates the ethical firewall in one place. Its purpose is practical. Later chapters should not reproduce the full list whenever they discuss measurement, endpoints, protocols, models, public artifacts, or verdicts. They should use one local safeguard when a local risk changes and otherwise rely on this chapter (Council for International Organizations of Medical Sciences, 2016; National Commission, 1979; World Medical Association, 2024).

The red lines are not negotiable by result. A favorable study cannot soften them. A donor cannot buy an exception. A public crisis cannot make them optional. A religious community cannot reinterpret them. An Artificial Intelligence system cannot override them. A transition ambition cannot postpone them.

These red lines protect people and the integrity of the evidence. Conditional accounting cannot be used to erase a qualifying counterexample, move an endpoint, or invent an unobserved remainder after the fact. One stream cannot balance another, estimates cannot rank human worth, and failures must remain visible. These safeguards govern research and interpretation; they are not conditions that make a universal hypothesis true.

The firewall therefore does two forms of work at once. It protects persons from the project, and it protects the project from itself. A theory that cannot govern its own temptations should not govern public imagination.

The compact firewall that follows is a reference table for the rest of the volume.

The final line is simple: the science may become stronger, weaker, narrower, inconclusive, or defeated. None of those states authorizes the forbidden uses named here.

Table 3.4. Compact ethical firewall.

Red line	Local meaning for research	Consequence for later chapters
Care before evidence	Care, comfort, safety, access, and dignity outrank measurement and claim status	Use one local sentence when a study risk changes; do not repeat the whole firewall
No harm induction	No induced suffering, deprivation, fear, humiliation, withholding, or path engineering for Law-specific evidence	Route studies to observation, natural variation, independent-benefit interventions, and rival modeling
Comfort-first terminal research	The dying person owes no result, and palliative care outranks endpoint clarity	Use no-estimate, stop, withdrawal, and low-signal labels rather than endpoint theater
Granular consent and data dignity	Observation, linkage, sharing, publication, future reuse, and Artificial Intelligence training are distinct permissions	Maintain purpose limitation, retention rules, deletion routes, bystander review, and no silent conversion
No authority leakage	Science governs claim status only, not law, religion, care, employment, insurance, civic rank, or transition	Defer applied authority to its own domain and block candidate momentary and interval estimators or ledger-like scoring

Red line	Local meaning for research	Consequence for later chapters
No moral payment or worth scoring	Balance requirements are not moral debt; persons are not scores; archives are not souls; failures remain visible	Preserve conditional net accounting; correct moral-desert, control, and evidence-status overclaims

3.7.1 — Balance Requirements Without Moral Desert

Assume exact PTVC, finite positive and negative totals, one qualifying completed stream, and a valid fixed representation with a calibrated neutral origin. At any chosen cut, the accumulated net balance and the signed remainder must sum to zero. A genuinely negative accumulated balance therefore requires a net positive remainder; a genuinely positive accumulated balance requires a net negative remainder. This follows from terminal closure regardless of the mechanism.

The Series Map’s statement “There is no free happiness” expresses the absence of a net positive surplus over a completed conscious stream under this hypothesis. It does not mean joy is wrong, pain is morally owed payment, or suffering is holy. Exact balance challenges a pleasure-only conception of utopia; the series develops its humane social goals through further ethical and practical reasoning.

If exact PTVC is true and the accumulated balance is already negative, a further negative contribution increases the algebraic net-positive remainder required for closure by the same amount. This fixes future positive magnitude minus future negative magnitude, not either gross total separately or an improvement

over a different, unchosen life. It does not establish a strategy, entitlement, or safe and reliable way to obtain a chosen reward.

The required remainder is an arithmetic condition on closure, not moral desert, earned suffering, or justification. It identifies no particular illness, injury, deprivation, or other event that must occur, and implies no event-by-event pairing, timing, conscious knowledge of the balance, or particular mechanism.

Kindness has no automatic valence value merely because it is kind. Any enjoyment genuinely felt by the person helping enters that person's balance like other positive experience. If the balance is already positive, that enjoyment increases the required net negative remainder. If it is negative, enjoyment reduces the positive remainder still needed, unless it carries the balance past zero. The recipient's experience belongs to the recipient's stream. Neither kindness nor enjoyment creates a moral obligation to suffer.

Helping changes the life actually lived. PTVC fixes a terminal difference, not a fixed quota of suffering or a ranking of whole lives. It does not require a particular prevented harm to reappear. It also does not exempt the felt effects of help from the arithmetic. Rights, care, and justice remain independent commitments.

No actually completed qualifying stream can end off balance if exact universal PTVC is true. But changing a genuine accumulated balance changes what the remainder must contain. That leaves a real question about influencing the route, not a demonstrated technique for controlling future happiness. An imagined strategy need not be achievable, and no one's true

balance is assumed known. A realized qualifying counterexample would refute the universal claim, not break a previously exceptionless law.

The project seeks a society capable of understanding PTVc without turning it into a reason for deliberate suffering, coercion, or withheld care. If the hypothesis is universal, informed choices and attempts to exploit it are included in its scope. Safeguards protect people and research integrity; they do not make the Law true. A humane civilization is the project's aim, not a theorem of terminal balance. Research prohibitions do not remove actually realized qualifying streams from the universal claim. Research remains subject to independent ethical review, and deliberate harm is not an endorsed experiment or strategy (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

3.7.2 — No Cross-Person Balancing

Cross-person balancing often appears in respectable language: net benefit, aggregate welfare, average neutrality, social gain, national readiness, institutional success. Those concepts can matter in policy, but they cannot erase the burden of a particular stream. In research, a cohort mean near zero does not show that each person closed. In ethics, one participant's insight does not justify another participant's suffering. In public communication, one life's peace does not explain another life's grief. The stream-bound character of the Law is therefore also a protection against public arithmetic that hides persons. Comparison can teach. Exchange cannot redeem.

Cross-person balancing would let institutions hide harm. A program could aggregate benefit over one group and suffering over another, then speak as if the suffering disappeared in the total. PTVC is intrastream and nontransferable; research ethics likewise requires attention to who bears risk rather than allowing aggregate gain to erase them (National Commission, 1979).

Report cohort means together with the distribution, heterogeneity, missingness, and stream-level results relevant to the hypothesis. This shows both the overall pattern and how it varies among the contributing trajectories.

The red line also blocks sacrificial rhetoric. A participant, family, community, staff group, or vulnerable population cannot be asked to bear research burden because the project may help humanity later. Future value matters, but it does not make present burden disappear. It must be justified, minimized, consented to, and reported.

Cross-person balancing is especially tempting in public artifacts. A visual might show overall harmony while hiding concentrated injury. A dashboard might show aggregate improvement while one subgroup carries distress. A status report might celebrate participation while staff burnout rises. Those artifacts fail the lens.

3.7.3 — No Human-Worth Measurement

Research variables may describe limited observations under limited assumptions; they may not become measures of worth, trust, holiness, sincerity, employability, civic eligibility, parental adequacy, deserved care, or future potential. Enforce that boundary through purpose labels, access controls, prohibited-use

terms, audit logs, and highly restricted person-level release. Section 3.6.3 governs civic and employment nonuse; this firewall states the general no-worth rule (National Institute of Standards and Technology, 2020; Tabassi, 2023).

3.7.4 — No Archive as Soul

An archive can preserve memory, provenance, consent, correction, and public accountability. It cannot preserve the person as a continuing conscious stream merely by containing records. This matters for terminal research because grief, technology, and public fascination can combine into a powerful temptation: to treat messages, recordings, model reconstructions, or data traces as if they reveal or continue the self after death. The science volume must refuse that implication. Posthumous records may be handled with dignity, restricted access, correction, and family sensitivity. They may not be called the soul, the ledger, the endpoint, the surviving mind, or a substitute for the person who is gone.

Records, public artifacts, governance logs, research datasets, and future archives can preserve accountability, correction, and institutional memory. They remain records. No accumulation of files can resurrect consciousness, certify inner life, or become a proxy for personhood (National Academies of Sciences, Engineering, and Medicine, 2017).

An archive can also tempt endpoint overclaim. A complete record may make observers feel they know what a person was, meant, deserved, or became. The science volume must reject that feeling. The bearer of the formal claim is a separately routed maximal

qualifying conscious stream, not a biography assembled from documents.

Artificial Intelligence intensifies the risk. A model trained on someone's text, voice, image, records, or behavior may simulate style or answer questions. It is not the person. It cannot consent after death. It cannot reveal terminal meaning. It cannot close the ledger.

Posthumous dignity requires record humility. Data retention, family access, publication, memorialization, and deletion should be handled under the permissions and governance set before or legally after death. The more moving the record, the more careful the boundary.

3.7.5 — No Concealed Failure

No concealed failure is the final ethical rule because it helps future studies learn from the complete research record. Null results, rival wins, adverse events, breaches, recruitment problems, burden overruns, model failures, endpoint uncertainty, withdrawn measures, and substantive corrections must remain visible through appropriate channels. Concealment shifts burden onto the next site, the next family, the next dying person, the next staff member, and the next reader who trusts the public summary. Preserving this record is care extended forward in time: it gives later teams the information needed to avoid repeating preventable problems.

Concealed failure is not merely dishonesty. It makes the Law unfalsifiable in practice. If unfavorable outcomes disappear, the public sees only survival. A claim that survives by file drawer is

not answerable; it is curated (Council for International Organizations of Medical Sciences, 2016). Making unsuccessful studies discoverable also helps researchers combine results across sites and choose more informative follow-up designs. A visible record of failure also allows later teams to distinguish isolated setbacks from recurring weaknesses in a method, bridge, or prediction. That distinction can prevent both premature abandonment and unjustified persistence.

The protocol should specify failure reporting before data collection begins. It should state what is reported publicly, what is reported to oversight bodies, what remains protected for privacy, what triggers correction, and what changes claim status. The route should exist before anyone knows the result. Assigning responsibility and a reporting deadline makes this commitment actionable when staff, funding, or institutional priorities change. The protocol should also identify who verifies that required reporting actually occurred and where the resulting record can later be found. Accountability is stronger when disclosure has an owner, destination, and auditable completion point.

Failure visibility also includes public artifacts. If a diagram, summary, website, donor page, or prior chapter becomes inaccurate after a result, it must be corrected, retired, or relabeled. A public artifact is not harmless because it is beautiful. It can carry old authority after the claim has changed. A dated correction notice should identify the affected claim and direct readers to the current explanation, allowing them to follow the change even when older copies remain in circulation. High-reach artifacts deserve priority because their outdated implications can persist

long after the underlying scientific record has changed. Correction should therefore follow both evidentiary importance and the likely reach of the obsolete claim.

Rival wins deserve special protection. A study that favors adaptation, homeostasis, observation attenuation, sedation, frailty, bridge conditioning, or another model has produced knowledge. It should not be narrated as temporary confusion until the Law recovers. It should narrow, retire, or defeat the relevant component as prospectively specified. The comparison should also identify which predictions or observations gave the rival its advantage, turning the result into guidance for the next model or experiment. A rival victory is most informative when the record also shows where the preferred model failed and which assumptions no longer deserve the same confidence. That makes defeat a source of model improvement rather than merely a change in ranking.

Chapter 4 — Technical Synopsis: Claim, Tests, and Verdicts

This chapter maps the PTVC target to its representation, measurement, endpoint, model-comparison, and verdict requirements. It distinguishes derived consequences from proposed tests and identifies the unresolved connections on which stronger claims depend.

The starting point is not a second full statement of the Law but the dependency structure that makes the received target scientifically vulnerable. Section 4.1 supplies the exact finite signed-measure form; this introduction maps the representation, routing, endpoint, observation, recovery, rival, and verdict conditions that must hold before evidence can reach it.

The claim is exact, intrastream, and nontransferable. It permits enormous differences in duration, intensity, temporal order, agency, opportunity, memory, welfare, and gross valence variation. Exact net closure would not imply an easy life, a just society, or low suffering. Those dimensions remain scientifically and ethically independent.

The synopsis next separates status, notation, and dimensions. A symbol may denote a truth-level object, a route state, an observation, an estimand, an estimator, or a verdict label. Units and domains matter because a dimensionless unity gate, an exposure rate, a neutral-relative value, and an accumulated ledger cannot be interchanged without changing the model.

Representation comes before inference. The program must justify sign, neutral origin, cross-sign comparability, scalarity or an identified set, and the exposure convention. A completed-record

recentering that forces zero is prohibited because it builds the answer into the scale.

Bearer and routing come next. The target belongs to one maximal qualifying conscious stream under a prospectively specified ownership rule. Splitting, merging, duplicating, or dropping events after seeing the residual can manufacture apparent closure. Partially identified routing must remain set-valued rather than resolved by convenience.

Endpoint construction is a separate gate. The truth-level endpoint, protocol endpoint, final observation, and biological death are not synonyms. Return, resuscitation, low-access states, observation shutdown, and terminal windows require adjudication rules that do not use the ledger outcome being tested.

Observation and recovery form another layer. Reports, behavior, physiology, neural signals, clinical records, and context are outputs of an observation process, not direct access to the latent ledger. Missingness, measurement reactivity, delayed entry, and unseen prefix or suffix exposure can widen the identified set or yield No adequate estimate.

The candidate signatures are designed to create disagreement among models. Depending on the declared branch, the program may examine reserve-linked drift, innovation contraction, variance shape, endpoint concentration, time alignment, completion potentials, perturbation-compatible compensation, and held-out transport. No single signature is treated as unique to PTVC.

Rivals receive the same information and fitting budget. Homeostasis, adaptation, allostasis, predictive processing, learning, endpoint-conditioned bridges, selection, observation attenuation, medication, frailty, memory, and hidden-state alternatives must be able to generate synthetic data and compete under common scoring rules (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Model recovery comes before real-data confidence. The analysis must show that it can identify the generating model, detect open-set inadequacy, and avoid manufacturing support when every candidate is wrong. Recovery failure is a diagnosis of the test, not a mysterious property of participants.

The first protocol is therefore deliberately bounded. It defines an eligible population, low-burden data flow, endpoint object, estimands, rivals, missingness plan, safety stops, recovery benchmark, and public reporting rule. SPIRIT 2025 and ICH E6(R3) provide current protocol and quality-by-design disciplines that can be adapted to this unusual target (Chan et al., 2025; International Council for Harmonisation, 2026).

Verdicts remain typed. A result may support a prerequisite, favor a route, remain model-ambiguous, narrow a reference class, retire a mechanism, favor a rival, produce No adequate estimate, or become adverse to a declared target branch. A composite score cannot allow strength in one module to cancel failure in another.

Equivalence language is also bounded. A finite scientific band can be adjudicated with an appropriate equivalence design, but a band-compatible estimate is not exact latent zero (Schuirmann,

1987). Repeated success at finer prospectively nested resolutions can strengthen a case only while the representation, bridge, endpoint, and limiting rule remain stable.

Open problems are kept visible because they organize future work. Neutral-origin identification, mixed valence, stream individuation, complete-domain recovery, unexpected endpoints, policy support, route transport, open-set model recovery, and safe terminal observation remain scientific frontiers rather than gaps to be hidden in prose.

The synopsis ends where the formal core begins. Readers now have a map of the received object, the gates that make it testable, the signatures that could discriminate it, the rivals that could win, and the verdicts that preserve what the evidence actually says. The next Part can develop the mathematics without making the reader rediscover the jurisdiction of every symbol.

4.1 — The Scientific Claim in One View

The synopsis organizes the formal hypothesis, representation assumptions, observation models, endpoint rules, candidate realizations, rival comparisons, reference classes, and decision rules. Naming those layers before introducing notation makes their connections easier to follow (Folland, 1999; Lakatos, 1978; Popper, 1959).

The strong form is deliberately high-risk. Its target is exact latent terminal closure for each qualifying stream on the declared branch. Finite-band compatibility, an estimated atom at zero, probability-one closure under a specified law, and [SET-ALL] closure over every qualifying stream are different claims with different decision rules. The public program does not pretend that a near-zero estimate observes exact equality. It asks whether frozen measurable consequences and validated bridges can bear on the declared target while surviving serious rivals and replication.

The chapter therefore starts with a component-evidence map. A favorable result may update a measurement bridge without updating the strong form. A rival may dominate a signature without defeating every analytic use of the burden-focused lens. A terminal equivalence finding may be unusable if the endpoint was selected after outcome inspection. A beautiful theory may remain metaphysically conceivable while losing scientific status because its empirical bridges failed.

The section also states what no result can authorize. Scientific claim status is not political authority, religious certification,

clinical instruction, civic rank, employment trust, insurance eligibility, public worth, transition readiness, or Artificial Intelligence sovereignty.

Table 4.1 maps claim layers to the evidence that could update them and the evidence that cannot. The table is a status discipline, not a verdict.

The compressed dependency spine can be written as a chain of conditional permissions. Let C_0 be the received formal target, C_1 the representation bridge, C_2 the observation model, C_3 the endpoint rule, C_4 the declared consequence module, whether mechanism-neutral or route-specific, C_5 the rival set, C_6 the inference engine, and C_7 the verdict rule. A public update is licensed only through the weakest surviving link invoked by the target component, not through the most impressive-looking result at the end of the chain.

$$C_0 \rightarrow C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow C_4 \rightarrow C_5 \rightarrow C_6 \rightarrow C_7$$

If any prerequisite gate G_i fails, the permissible update is capped before the failed gate:

$$permittedstatus \leq \min\{G_i : i \leq testedcomponent\}$$

This is the noncompensatory evidence rule. The minimum operator in the displayed shorthand is ordinal over the prospectively specified status ladder, not arithmetic, and it ranges only over gates required by the component being judged. It does not imply that every target-law test requires one candidate realization. A clean endpoint band cannot repair an undefined sign convention; a reserve correlation cannot repair a circular

endpoint; a beautiful simulation cannot repair failed model recovery.

The reader can also read the table as a formal dependency diagram. Formal claim leads to representation; representation leads to observation; observation leads to endpoint; endpoint leads to candidate consequences; candidate consequences lead to rival comparison; rival comparison leads to replication; replication leads to bounded public status. The arrows do not run backward. A later public verdict cannot repair an earlier representational defect. Each arrow can therefore be annotated with the assumption or measurement procedure that carries evidence from one stage to the next.

This matters because many apparently careful theories become unfalsifiable through backward repair. If the endpoint fails, they change the endpoint. If the origin fails, they change the origin. If a rival wins, they rename the rival as the theory. If a reference class fails, they say the class was never meant. The dependency diagram forbids that style of survival. A changed bridge becomes a changed claim version. Recording the revised definition alongside the original keeps the earlier test interpretable and gives the revised proposal a clear starting point.

The component-evidence map also shows how a progressive program differs from a degenerating one. A progressive modification names a failed bridge, proposes a new bridge, and generates a new risky prediction that can be tested before seeing the next result. A degenerating modification merely moves a boundary so the previous failure no longer counts. The scientific conscience of this book is the ability to tell those two movements

apart. A particularly useful next test targets a condition where the revised account and its strongest competitor predict different outcomes.

The strongest positive case therefore rises only when multiple independent components survive together. A measurement bridge that survives calibration, an endpoint rule that survives circularity tests, a reserve relation that survives independent estimation, a rival matrix that lets alternatives win, and a replication system that transports cautiously can together make a program more credible. None of them alone can lift the whole claim. Agreement is especially informative when the checks have different sources of error, reducing the chance that one shared assumption accounts for their apparent convergence.

The same map explains why a single failure can be highly informative. A failure of model recovery may not refute the Law, but it refutes that study's power to adjudicate. A circular endpoint may not refute the Law, but it makes the endpoint evidence unusable. A repeated, valid rival victory in the exact reference class where the candidate was supposed to win may force narrowing or defeat.

Noncompensatory logic means that evidence cannot leap across a failed prerequisite. If sign is unidentified, a terminal curve cannot repair it; if model recovery fails, a favorable information criterion cannot repair it; if care-disrupting observation distorts the record, sample size cannot repair it. Each failure limits the permissible claim at the point where the dependency chain breaks.

Table 4.1. Claim layers, update evidence, and noncompensatory limits.

Layer	What it says	What could update it	What cannot update it
Strong latent hypothesis	Exact terminal valence closure is posited for the received latent cumulative process of a unified conscious stream.	Only a mature chain of component success, rival defeat, endpoint independence, and replication could strengthen its scientific standing indirectly.	A diagram, fitted example, spiritual resonance, applied success, public enthusiasm, or a single equivalence study cannot confirm it.
Representation bridge	The lived-experience target can be represented as a signed, temporally additive rate under declared unity rules.	Psychometric, phenomenological, physiological, and construct validation work can support or weaken this bridge.	Terminal concentration cannot repair a failed sign convention or invalid construct.
Observation model	Candidate momentary estimator, candidate interval contribution estimate, reserve proxies, endpoint proxies, reports, clinical context, and missingness are treated as fallible witnesses.	Calibration, reliability, missingness, sensitivity, and model recovery studies can update it.	A favorable terminal estimate cannot make a noisy or biased instrument valid by outcome.

Layer	What it says	What could update it	What cannot update it
Candidate realization	Reserve coupled or control theoretic mechanisms may generate measurable consequence structures.	Prospective dynamics, variance shape, innovation behavior, and rival comparison can support or retire a realization.	Mathematical construction alone cannot establish that the candidate mechanism operates in nature.
Reference-class prediction	A specified population and endpoint class may show the protocol fixed conjunction of signatures.	Held out prediction and transported replication can update a bounded claim.	A hospice cohort cannot silently generalize to all cultures, sudden deaths, nonhuman animals, or artificial systems.
Public claim status	A result receives a status label: preliminary, replicated, inconclusive, unusable, narrowed, defeated, or abstained.	The declared verdict process and independent review update public language.	Donor interest, media attention, or institutional adoption cannot change scientific status.

4.1.1 — The Received Strong Form

The technical advantage of stating the strong form plainly is that later weakening cannot be mistaken for the original claim. A public program may later narrow to bounded dynamics,

reference-class regularities, or measurement achievements, but those narrower outcomes are not the same proposition. The synopsis therefore writes the strong form without apology and then refuses to treat any early surrogate as if it had already reached that height.

A skeptical reader should be able to underline this sentence and say, this is the target that would need extraordinary support. That clarity is a gift to the project, not a threat. A blurred target might attract easier agreement, but it would make every later verdict untrustworthy because no one could tell what the evidence had actually touched.

The received strong form is the exact terminal-valence-closure hypothesis stated at the latent level of the formal Law. On the primary finite signed-measure branch, one separately routed maximal qualifying stream carries a finite signed phenomenal measure over its complete strict-pre-endpoint domain. Its terminal residual is equivalently the measure of that domain, the difference between its positive and negative Jordan magnitudes, and the left limit of the phenomenal ledger at the truth-level endpoint. The scientific task is to connect this fixed target to observable consequences and the methods needed to evaluate them (Folland, 1999; Lakatos, 1978).

$$Z_i^* := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^* -)$$

$$PTVC_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^* = 0$$

The equality is intrastream and nontransferable. It constrains the signed terminal residual while permitting unequal duration, intensity, temporal order, agency, opportunity, memory,

resources, welfare, rights, and gross phenomenal variation. Additive PTV, the bipolar mixed-valence branch, and Generalized Terminal Neutrality remain separate unless an approved bridge establishes equivalence.

On the absolutely continuous primary subbranch, the phenomenal ledger accumulates realized neutral-relative value against experienced exposure. In a physical-time chart, the received representation is:

$$L_i^{\text{phen}}(t) = L_i^{\text{phen}}(S_{i,\theta}^-) + \int_{S_{i,\theta}}^t g_{\theta,i}(s) q_i^{\text{exp}}(s) v_i^{\text{real}}(s) ds$$

A probability-one branch under a declared stream law is a separate modal claim:

$$P_i\{L_i^{\text{phen}}(T_i^* -) = 0\} = 1$$

The display states [P1] closure: the latent integral equals zero almost surely under the specified law. [SET-ALL] instead quantifies over every qualifying stream. The branch label identifies which statement a derivation or analysis addresses.

The strong form is therefore not the first observable claim. Felt valence has no instrument-fixed origin or unit, a study begins after stream onset, and the truth-level endpoint is not delivered by one sensor. A near-zero estimator at the end of an observed window can support only the protocol-fixed measured branch unless representation, complete-domain, endpoint, observation, and measured-to-latent bridges license more.

The most discriminating structural result concerns endpoint timing. Under a declared reference law, suppose an endpoint is

independent of the sigma-field generated by a jointly measurable pre-endpoint ledger path and has a conditional density that is positive almost everywhere on a fixed interval. Exact closure at that endpoint then forces the pre-endpoint ledger to be zero for almost every time in that interval, for almost every eligible path.

$$P_i^0\{L_i(T_i^{\text{exo}}-) = 0\} = 1$$

T_i^{exo} is independent of $\sigma\{L_i(s^-): s \in I\}$ under P_i^0 , and $f_{T_i^{\text{exo}}}(t) > 0$ for Lebesgue-almost every $t \in I$.

$$0 = P_i^0\{L_i(T_i^{\text{exo}}-) \neq 0\} = \int_I P_i^0\{L_i(t^-) \neq 0\} f_{T_i^{\text{exo}}}(t) dt$$

Therefore $L_i(t^-) = 0$ for Lebesgue-almost every $t \in I$ on a P_i^0 -probability-one set of eligible paths.

Endpoint timing has several distinct forms. Section 4.3.4 defines the branch distinctions and anti-tautology conditions; the reserve-coupled route is one candidate realization within that framework.

The strong form receives scientific pressure through branch-matched consequences and validated bridges. Representation failure, circular endpoint construction, unidentifiable route state, failed model recovery, serious rival dominance, or nontransport can narrow or defeat the exact claim version each result actually reaches. Hiddenness does not make the target immune.

If exact PTVC is true, a completed qualifying stream has the proposed terminal balance under its declared representation and endpoint. Observing someone suffer does not establish that person's accumulated balance, explain a particular event, or assign moral payment, desert, or improvement. Care remains

binding under every scientific outcome. The strong form is a research target, not a license over anyone's life.

4.1.2 — The First Empirical Burden

The first empirical burden also protects the program from spectacle. Terminal scenes are emotionally powerful, but their emotional force can overwhelm methodological restraint. Preterminal restrictions ask a colder question: before the ending is known, does the system behave in a way that the candidate architecture predicted and rivals did not predict as well?

The first empirical target is therefore a vector of prospective consequence claims, not one scalar verdict:

$S = \{\text{reserve-linked drift, innovation contraction, variance-shape law, forecast performance under the declared independent-endpoint branch, observation-model robustness, held-out rival superiority}\}$ (Popper, 1959).

A study may report evidence about one coordinate of S while abstaining on another. A fitted negative drift coefficient can support a local dynamical component. Variance shape, reserve coupling, the protocol-fixed endpoint-timing premise, and model recovery receive their own assessments. Component-specific reporting identifies what the result contributes and which comparison should come next.

This order gives the research program an honest first rung. It can begin with dynamics that are difficult but not metaphysical, with instruments that can be improved, with rivals that can be simulated, and with results that can be negative without asking anyone to reinterpret the meaning of death.

The first empirical burden is prospective, comparative structure across this signature vector under frozen rules, before the endpoint is inspected. Terminal zero sits higher in the inference ladder. The program becomes scientifically interesting when the joint pattern is difficult for simpler rivals to reproduce.

Prospective specification makes the study's choices inspectable. Before the relevant outcome is examined, the empirical bridge declares the population, instrument version, endpoint channels, analysis window, smoothing method, estimands, rivals, and decision rules. Later exploratory changes are documented as such.

The first burden is also comparative. A signature has little force if it is compatible with ordinary adaptation, allostatic regulation, endpoint-conditioned bridges, medication, frailty, clinical practice, or observation shutdown. The Law candidate must not merely fit. It must out-predict or out-explain serious alternatives in held-out data after the analysis pipeline has shown it can recover known synthetic generators.

Dynamical shapes, reserve-space relations, forecast errors, innovation behavior, and comparative predictions offer tractable early targets. Their dependence on origin assumptions can be assessed directly, helping identify which features can be compared before absolute terminal level is estimated.

A first result can provide a clear foundation for further work. Suppose a prospectively specified study finds that reserve-linked variance contraction appears prospectively and survives observation-model checks. The finding supports that dynamical consequence within the tested setting, while the terminal-

equivalence analysis reports its own uncertainty. Replication can then test whether the two components develop together.

The first empirical question is not whether one attractive terminal pattern can be found. It is whether the route from experience to evidence survives scale choice, missingness, device effects, clinical confounding, endpoint selection, and public compression. Early studies seek discriminating signatures while exposing those distortions before they harden into claims.

4.1.3 — Component Claims and Dependencies

Dependency logic is also a protection against overcorrection. A hostile reviewer might want one defect to destroy the whole research program, while a proponent might want one success to lift the whole Law. Both reactions are too crude. Component accounting lets the book be severe and fair at the same time.

The lattice should be treated as a burden map. Each row asks who or what would carry the cost if that component were left vague. If representation is vague, participants carry the cost through intrusive but useless data collection. If endpoint is vague, dying persons carry the cost of dramatic but circular interpretation. If rivals are weak, future science carries the cost of false confidence.

The program is a lattice, not a single plank. It includes a formal hypothesis, representation assumptions, observation models, candidate mechanisms, endpoint rules, study designs, rivals, inference tools, replication standards, and public verdict rules. Each component can fail separately. The scientific dignity of the book comes from refusing to let success in one component silently repair failure in another (Folland, 1999).

Representation is one component. It asks whether lived experience can be modeled as a signed, temporally additive target under a declared bearer and exposure rule. Measurement is another. It asks whether momentary and interval estimators, under a declared version, can estimate relevant contributions with known uncertainty. Endpoint adjudication is another component; its anti-tautology and branch-specific timing requirements are developed in §4.3.4.

Candidate realization is also distinct. The route coordinate $L^r(t) = c_{LU} U(t) Y(t)$ may make terminal concentration dynamically approachable, but it is not the received definition of the phenomenal ledger. It is one candidate bridge from formal target to measurable consequences. The bridge can fail while the root hypothesis remains formally intelligible, or it can succeed locally while the exact set-universal form over the declared class remains unconfirmed.

Rival discrimination is a component with its own gates. A model that fits the same pattern after parameter tuning has not earned the status of a model that predicted the pattern prospectively. A rival that wins on held-out prediction under fair comparison must be allowed to change the program. Component claims become meaningful only when rivals are not decorative.

The dependencies are noncompensatory. Endpoint precision cannot compensate for invalid measurement. Measurement reliability cannot compensate for a circular endpoint. A beautiful reserve model cannot compensate for inability to estimate reserve independently. Replication cannot compensate for a public artifact that overstates the result.

When dependencies are blurred, the costs scatter. Participants may undergo needless monitoring; families may absorb anxiety; researchers may chase an uninterpretable signal; readers may mistake component support for support of the whole claim. The dependency lattice is therefore both an inferential map and an ethical map: it shows what each result can bear and who is exposed when the layers are collapsed.

4.1.4 — Reference Classes and Scope

Reference-class discipline is also the antidote to hidden population ranking. A cohort that yields usable data is not a better kind of life. A cohort that yields unusable data is not a lesser one. It is only a domain where the current bridge either works, fails, or has not yet been built.

Public conclusions should state the sample, instrument, and endpoint class actually tested. Those details show readers the scope of the result and the next opportunity for extension.

No empirical result has universal scope by default. A study conducted in one hospice network, under one cultural setting, one instrument version, one endpoint rule, one medication environment, and one recruitment pathway does not silently speak for all conscious streams. The reference class is part of the claim. If it is not named, the result is already overstating itself (Lakatos, 1978; Popper, 1959).

The reference-class ceiling is especially important here because the formal hypothesis reaches beyond any first human cohort. Sudden deaths, pediatric cases, nonhuman animals, advanced dementia, anesthesia, disorders of consciousness, cultural

variation, artificial systems, and future candidate bearers each raise separate representation and endpoint issues. A mature science can expand only by defending transport bridges, not by assuming them.

Scope statements must therefore name population, state space, endpoint class, instrument version, data window, care context, and transport assumptions. A result may say, for example, that in a defined adult palliative cohort under a particular momentary valence estimator version and endpoint adjudication rule, a preterminal reserve-linked variance pattern survived specified rivals. That sentence is long because the claim is honest.

Reference classes also protect against country ranking, clinical ranking, and cultural arrogance. The project must not treat one population's data as evidence that another population is less fair, less balanced, less mature, or less measurable. A transport failure can reveal instrument limits, endpoint limits, cultural calibration needs, or mechanism scope. It is not a hierarchy of lives.

A positive result can sometimes justify expansion. It may motivate a new study in another care context, a different endpoint class, or a distinct culture. But expansion is an empirical bridge, not a rhetorical gesture. Every widening of scope should add a new vulnerability, not merely a broader sentence.

The fairness lens sees reference-class discipline as burden containment. Without it, a narrow study shifts interpretive burden onto people who never participated, never consented, and may not fit the assumptions. Scientific care means refusing to make absent lives carry conclusions drawn from a different world.

4.1.5 — What Support Could Update

Public summaries should identify the component studied, its population, method, and result. Those details let readers understand the finding and locate the work that can build on it.

A study can support a measurement bridge, an endpoint proxy, a reserve-coupled signature, a comparison method, a reference-class prediction, a replication claim, or a decision convention. Name that contribution directly so other investigators can build on the result.

A favorable measurement study can strengthen a witness channel. Correlations with carefully elicited reports, clinical context, and physiological markers in a bounded setting provide evidence for a specified estimator version and inform the design of later studies.

A favorable endpoint study can qualify a rule for a reference class. Agreement with expert adjudication and successful circularity checks strengthen the endpoint procedure for later terminal inference.

Consider a favorable rival-comparison study. If the Law candidate outperforms homeostatic, bridge, frailty, medication, learning, and observation-attenuation rivals on held-out preterminal dynamics, the result may support a conjunction of signatures. The strong form remains exact and latent, but the empirical program has earned more pressure and attention. That is the kind of support this book seeks first.

The update rule is equally important for negative results. If model recovery fails, no real-data result should update the claim. If a

rival wins, the candidate mechanism may be retired. If transport fails, the reference class narrows. If endpoints remain uncertain, terminal claims abstain. An unfavorable result is not a public-relations problem. It is the machinery of scientific honesty.

Support that lands specifically protects human beings from overreach. No one should become more ranked, more exposed, more morally interpreted, more recruitable, or more governable because a component survived. A status update belongs to the claim. It does not attach to the person who generated the data.

4.1.6 — What No Result Can Authorize

Results in this chapter update only the scientific propositions they test. Chapter 3 governs the standing prohibitions on harm, ranking, and authority leakage; locally, each formula, table, model, and caption should name its object, reference class, uncertainty, rivals, and status ceiling.

4.2 — Status, Notation, and Dimensions

Notation is useful only when its status is clear. The same symbol can carry different kinds of burden: a formal definition, a candidate realization, an estimator, a proxy, a design criterion, or an open conjecture. This section gives the reader a status map so that later formulas are not mistaken for empirical findings or root definitions (Durbin & Koopman, 2012; Folland, 1999; Karatzas & Shreve, 1991; Protter, 2005).

Notation note: Chapter 4 uses schematic, often index-suppressed symbols to orient the reader; Chapter 5 supplies the branch-specific stream-indexed definitions. A schematic symbol and its Chapter 5 counterpart are not interchangeable unless the stated branch or bridge identifies them.

The chapter uses a formula-status code. R denotes received definition. CR denotes candidate realization. OE denotes operational estimator. DC denotes design criterion. OC denotes open conjecture. No line may change status inside a paragraph because the prose has become confident.

$$R: L_i^{phen}(t) = v_i(\Omega_i \cap [S_i, t])$$

$$CR: L_i^{rc}(t) := c_{LU} U_i(t) Y_i(t), c_{LU} > 0,$$

$$OE: \hat{L}_i(t_n) = \sum_{j \leq n} \Delta \hat{L}_{ij}$$

This status code is more than editorial discipline. It prevents a common failure in speculative science: the display equation is treated as if it were a discovered relation simply because it is

compact. In this volume, a formula must carry its assumptions, its context of use, its proxy status, and its route to defeat.

The dimensional problem is not cosmetic. A valence rate, a time integral, a unitless reserve, a normalized process, a momentary valence estimator, and an accumulated contribution cannot be multiplied or compared without assumptions. When those assumptions are unclear, a diagram can look cleaner than the science is. The synopsis therefore marks dimensions and unresolved bridges before it presents signatures.

The origin problem connects three choices: the reference point for phenomenal neutrality, the onset of the accumulated history, and the relation between a measured neutral band and the latent quantity. Reserve-product realizations add a fourth question: how the closing product relates to the received valence ledger. Explicit definitions make these questions available for mathematical analysis and measurement studies.

The section closes with open bridges rather than hidden repairs. Process-class mismatch, initial-condition compatibility, endpoint-reserve synchronization, whole-life ledger ignorance, scalarization, and the conservative-collapse trap remain problems that must be carried into later chapters. A synopsis that hides them would make the book easier to sell and weaker to read.

Table 4.2 is the compact dimensional and status map for the chapter. It is meant to be consulted whenever a later formula seems to be doing too much work.

The origin problem can be made concrete with a simple transformation. Suppose a momentary valence estimator $x(t)$ is

transformed into $x^*(t) = a + b x(t)$, where b is positive. Differences are preserved up to scale, but the zero changes whenever a is not zero. A terminal value that looked neutral under x may not be neutral under x^* . Without an independently meaningful zero, terminal equality can move with the analyst's convention.

The same issue compounds under integration. If a rate estimate is shifted by a constant offset, the accumulated estimate changes by that offset multiplied by included conscious time. Over a long window, a small arbitrary centering choice can dominate the accumulated ledger. This is why a study cannot create the Law's neutral origin by subtracting a sample mean or by centering a questionnaire for convenience.

The empirical program can still test less origin-sensitive structure. Slopes, forecast errors, variance shapes, reserve relations, and rival-discriminating dynamics may retain meaning under transformations that would undermine absolute terminal level. That is why the synopsis places prospective preterminal structure before terminal equality. The order is not evasive. It is an identification strategy.

Formula status should therefore travel with every symbol. The phenomenal ledger $L^{\text{phen}}(t)$ is a received formal object. The route coordinate $L^{\text{rc}}(t) = c_{\text{LU}} U(t)Y(t)$ is a candidate realization. A momentary valence estimator is an operational estimator, and an interval contribution estimate is context-bound under a declared exposure and time rule. A neutrality band is a design criterion. A verdict label is a public-status rule. These objects never carry the same authority.

A diagram can accidentally erase those statuses by drawing all objects with the same color and line. A table can erase them by putting measured estimators beside latent variables without warning. A chapter can erase them by saying the ledger when it means the measured running estimate. This synopsis names the statuses so the rest of the volume has less excuse to blur them.

The unresolved reserve-unit problem belongs in the same family. Mathematical reserve can be normalized, but physiological reserve is estimated through noisy, multidimensional clinical and biological signs. The bridge between normalized U and clinical U must be defended. Otherwise a reserve-coupled signature may look precise while the object called reserve has changed from line to line.

The viability ratio Ψ is especially fragile. It can be algebraically useful because it expresses ledger magnitude relative to remaining reserve, but as U approaches zero even small estimation error can explode. A chapter that uses Ψ must specify windows, errors, and abstention rules. It may not convert Ψ into a public urgency score.

Dimension checking is therefore a protection against hidden burden. Without it, the burden of ambiguity is passed to the reader, the reviewer, and eventually the participant whose data may be used to support a claim that the symbols never licensed.

Table 4.2. Scientific objects, status, dimensions, proxies, and failure conditions.

Object	Compact meaning	Status	Unit or scale concern	Observable proxy	Failure condition
$F(t)$	Latent valence rate contribution while the stream is unified.	Received formal object	Rate per time; sign and origin require representation assumptions.	No direct proxy; candidate momentary estimator is an estimator only after calibration.	No defensible signed, temporally additive target.
$g\theta(t)$	Unity gate that determines whether experience is included in the stream ledger.	Received formal object	Dimensionless inclusion or weight; threshold and soft-gate choices remain open.	Unity proxies from consciousness-state adjudication.	Endpoint or inclusion rule becomes circular or nonrecoverable.
$L(t)$	Latent cumulative phenomenal ledger.	Received formal target	Accumulated valence over time; origin problem remains binding.	measured cumulative estimates, never direct observation.	Scale, origin, or aggregation cannot be justified.

Object	Compact meaning	Status	Unit or scale concern	Observable proxy	Failure condition
T	Truth-level stream endpoint, fixed without terminal-ledger outcomes; stopping-time treatment requires a declared filtration and justification .	Received endpoint concept and empirical adjudication target	Not a calendar instant given by one sensor; adjudication windows matter.	Endpoint proxies and expert adjudication channels independent of the ledger.	Endpoint selected by or algebraically tied to the ledger outcome.
U(t)	Biological or functional reserve in the candidate product realization.	Candidate realization component	Unit conflict and depletion scale require formal change control.	Reserve proxies, frailty measures, clinical trajectory, physiological reserve indicators.	Reserve cannot be independently estimated or is only disease severity in disguise.

Object	Compact meaning	Status	Unit or scale concern	Observable proxy	Failure condition
Y(t)	Signed arithmetic reference process in the candidate reserve product $L^{rc}(t) = c_{LU}U(t) \cdot Y(t)$.	Candidate realization component	Signed real-valued process; not geometric and not restricted to positive values.	Inferred only through model fit under assumptions.	Reference process cannot be distinguished from bridge conditioning or ordinary control.
Rate estimate (t)	Momentary fallible estimator of valence-rate-like burden on a versioned scale.	Operational estimator	Versioned scale; calibration, response shift, and sign conventions bind use.	Physiological estimator plus validation and challenge stack, not inner life itself.	Estimator is unstable, biased, noninvariant, or invalid for the reference class.
Interval estimate	Integrated contribution unit derived from candidate momentary estimator over time under a declared version.	Operational accumulated estimate	Interval contribution, not a worth score; version and uncertainty travel with it.	Time-integrated candidate momentary estimator-derived estimate with uncertainty.	Additivity, uncertainty, or version continuity fails.

Object	Compact meaning	Status	Unit or scale concern	Observable proxy	Failure condition
Ψ	Viability ratio $\frac{ L^{rc} }{(c_{LU}U)'$ for $U > 0$, in the reserve realization.	Derived diagnostic under candidate relation	Numerically fragile near U approach ing zero; errors-in-variables problem.	Estimated only under windowed or stabilized analysis.	Unstable denominator, proxy error, or endpoint window makes inference unusable.

4.2.1 — Received Definitions

The received definitions are the target definitions carried forward into each analysis. This volume states them in the technical synopsis and develops their full specification in Chapter 5. The term received identifies the object a study is designed to investigate. A revised definition creates a revised target, so its implications for measurement and inference are recorded explicitly.

That distinction keeps criticism clean. A scientist may object that $F(t)$ cannot be represented, that $g\theta(t)$ is underdefined, or that $L(t)$ lacks an identifiable origin. Those objections attack the received target or its bridge directly. The book does not defend a substitute object merely because the substitute is easier to measure.

Received definitions are the fixed target objects used throughout this volume. The bearer is a separately routed maximal qualifying conscious stream, not the body, biography, legal person, social identity, or record. The currency is phenomenal valence under a

prospectively fixed representation, sign, neutral origin, and exposure rule. The ledger is latent. The domain is complete and strict-pre-endpoint on the primary branch. The endpoint is the truth-level endpoint. Candidate momentary and interval estimators are research objects, not instruments already validated (Durbin & Koopman, 2012; Folland, 1999; Karatzas & Shreve, 1991).

The science volume can analyze these definitions, pressure their operational use, and show what must be measured, but it cannot silently rewrite them. If a later chapter discovers that a definition needs formal amendment, that problem belongs to formal change control. A scientific chapter may identify the failure and state its implications; it may not pretend the root object has changed because a study design prefers a different one.

The received definitions also come with public-risk boundaries. The unified stream is not a public identity. The latent cumulative process is not an administrative account. Candidate momentary and interval estimators are not human-worth scores. The endpoint is not a spectacle. Scientific evidence is not legal, religious, or civic authority. These boundaries travel because the definitions would be dangerous without them.

A reader can think of received definitions as the coordinates of the research map. Coordinates do not prove that the destination exists. They let investigators say where they are trying to go and what kind of error would mean they are no longer studying the proposed object. Without received definitions, every failed test could be escaped by renaming the target.

The key received object is the phenomenal ledger, a signed measure accrued over the declared phenomenal-event domain. On the absolutely continuous physical-time branch, the ledger uses the experienced-exposure density $g_{\theta,i}(t)q_i^{\text{exp}}(t)$ and realized neutral-relative value $v_i^{\text{real}}(t)$. This is a formal target, not a sensor output, questionnaire total, or social record.

Dimensional reading follows from that factorization. The product $g_{\theta,i} q_i^{\text{exp}} v_i^{\text{real}}$ has ledger-per-clock-time units on the declared chart; integration yields accumulated phenomenal value. A momentary estimator may approximate only a versioned local component, and an interval contribution estimates an increment, not the person, a score, or a hidden moral balance:

$$\Delta L_{i,[a,b]} = \int_a^b \vartheta_i(s) d\hat{\tau}_i(s)$$

$$\Delta L_{ij} = \vartheta_i(t_j) \Delta \tau_{ij}, \hat{L}_i(t_n) = \sum_{j=1}^n \Delta L_{ij}$$

L denotes the latent target under the formal hypothesis. $\hat{L}$ denotes its estimate under a versioned observation model, carrying uncertainty from missingness, sign conventions, and context of use. The measurement relation connects these two objects in the analysis.

The inherited definitions keep the scientific object honest. They direct attention to experience rather than outward appearance, to stream unity rather than administrative identity, and to an independently specified endpoint rather than a dramatic scene

selected after the fact. Their value lies partly in what they refuse to simplify before evidence begins.

4.2.2 — Candidate Realizations

Candidate realizations make room for mathematical exploration. A model can derive signatures, reveal incompatibilities, and identify conditions that distinguish it from alternatives. Those consequences turn a broad hypothesis into specific questions.

The reserve product is therefore presented with both force and restraint. It is powerful because it links terminal dynamics to a declining capacity variable. It is restrained because reserve, normalized process, and endpoint synchronization are not gifts from notation. They have to be estimated, justified, and compared.

Candidate realizations are proposed ways the received formal object might generate measurable consequence structures. They are not root definitions. The principal current route coordinate is $L^{\text{rc}}(t) = c_{\text{LU}} U(t)Y(t)$, where a declared reserve state and signed normalized coordinate generate drift, innovation, and dispersion restrictions. Its value is that it exposes the route to measurable signatures. Its danger is that compact notation can be mistaken for a discovered mechanism (Karatzas & Shreve, 1991; Protter, 2005).

The reserve route has a specific scientific burden. $U(t)$ must be independently meaningful, not merely a symbol chosen to vanish at the endpoint. $Y(t)$ must be a signed arithmetic reference process, and c_{LU} must carry a declared scale convention. The relationship between L^{phen} and L^{rc} must be derived, assumed, or

marked open. Endpoint-reserve synchronization must be tested rather than built into the definitions.

On the reference route, reserve is nonincreasing with $dU_i(t) = -r_i^U(t)dt$ and $r_i^U(t) \geq 0$, while the signed normalized coordinate follows:

$$dY_i(t) = \sigma_i(t)dW_i(t), L_i^{rc}(t) = c_{LU}U_i(t)Y_i(t).$$

With $U_i(t) > 0$, the product rule gives:

$$dL_i^{rc}(t) = \frac{-r_i^U(t)}{U_i(t)}L_i^{rc}(t)dt + c_{LU}\sigma_i(t)U_i(t)dW_i(t).$$

The empirical signature is joint, not single. On the positive, absolutely continuous reserve branch, the integrated restoring coefficient diverges as reserve tends to zero. Pointwise strengthening of the restoring coefficient $\kappa_i^U(t) = r_i^U(t)/U_i(t)$ requires an additional rate or monotonicity premise; vanishing innovation requires $\sigma_i(t)U_i(t) \rightarrow 0$. Endpoint-conditioned bridges can produce pull toward zero, but reserve-linked prediction asks whether the declared drift and innovation signatures track an independently estimated route state under common observation and rival models.

A candidate realization can be powerful because it produces fingerprints: drift dominance, innovation contraction, variance compression, time-scaling relations, and viability-ratio behavior. But each fingerprint has rivals. Ordinary control systems can pull states toward set points. Bridges can condition paths on endpoints. Disease severity can couple reserve and affect. Observation loss can shrink measured variance. A realization earns attention only through discrimination.

Candidate control models play a different role. Completion-potential, conditioned-path, reserve-coupled, homeostatic, predictive, and reinforcement-learning accounts can explain parts of experience dynamics. Some may be Law-candidate mechanisms, some may become rivals, and some may be retired. The synopsis marks them as candidate mechanisms, not settled physiology.

A conservative-state representation requires a separate justification. Defining valence as the derivative of a state function makes the integral a boundary difference; imposing equal initial and final values would additionally force closure. The target cannot gain empirical support from those imposed conditions. An independently specified potential remains available as a model or rival, subject to a validated phenomenal bridge.

The fairness lens treats candidate realizations as offers of vulnerability. A mechanism is not added to make the Law safer. It is added to let the Law lose in more precise ways. If the mechanism cannot be measured or separated from rivals, it cannot support a discriminating empirical claim. If closure is imposed by construction, that imposed result is not empirical support; independently testable path predictions remain available for comparison.

4.2.3 — Operational Proxies and Estimators

Report the momentary-to-interval estimator chain with its uncertainty, context, channel coverage, and no-estimate states. That information lets later analyses use the estimate under the conditions it can support.

This is also where the anti-score rule becomes technical. A score usually tempts users to rank, sort, and administer. A research estimator instead carries purpose limitation, uncertainty, contestability, and prohibited use. The same number can become either a tool or a harm depending on the governance that surrounds it.

Operational proxies and estimators are the tools by which a study witnesses what it cannot directly see. The momentary valence estimator is a context-bound momentary estimate, not inner life itself. The candidate interval contribution estimate is an accumulated contribution derived under a versioned integration rule, not a unit of worth. Unity proxies, reserve proxies, endpoint proxies, clinical observations, active reports, behavioral context, and records all have limited roles and must be kept distinct (Särkkä, 2013).

A generic observation layer can be written as a state-space model rather than a direct readout:

$$X_k(t) \sim p_{\phi,k}(X_k(t)|F(t), U(t), \theta(t), C(t), M(t))$$

$$\hat{v}_i(t) = m_{\psi}(X_{i1}(t), \dots, X_{iK}(t), C_i(t), M_i(t)) + \varepsilon_i(t)$$

Here $C(t)$ denotes context and $M(t)$ marks medication, device, clinical, and observation conditions. The formula is deliberately broad. It makes clear that the momentary valence estimator is an estimator shaped by a model, not a natural instrument that automatically inherits the authority of the latent target.

The naming discipline matters. First-person reports, family context, clinician observation, behavior, and qualitative materials

can validate, challenge, or contextualize a momentary valence estimate. They are not automatically channels used by the momentary valence estimator by default. Treating every witness as a channel would create a spurious total instrument and invite surveillance creep. The estimator must remain narrower than the person.

Every operational object needs a version. Instrument version, channel weights, calibration rules, sign conventions, missingness treatment, uncertainty propagation, endpoint proxy definitions, and data-retention rules should be frozen for a study. If the version changes, the claim version changes. A result obtained under one bridge should not be retroactively folded into another.

Estimators also carry response-shift and context burdens. A person's language, medication, culture, disability, fatigue, delirium, sleep, fear, and relationship with researchers can change what a report or physiological channel means. A proxy that works in one state or reference class may fail in another. The synopsis therefore treats proxies as research questions, not solved instruments.

A compact momentary-to-interval estimation example helps. A participant has four observation windows. The momentary valence estimator estimates a momentary burden signal with uncertainty. The candidate interval contribution estimate integrates those estimates across time, marking missing windows, device removal, medication changes, and confidence. The result is not the person's ledger. It is a fallible estimate of a study-defined contribution on a versioned scale.

A broader multimodal alternative makes the bridge clearer. This illustration is distinct from the received physiological candidate described in §6.5.1. Imagine a thirty-minute window with a brief report channel, a physiological channel corrected for medication and movement, and a clinical-context channel marking a pain episode, support contact, or rest. The separately declared model specifies which observations inform the signed estimate and how each channel's uncertainty is represented.

At each minute, this illustrative multimodal model combines the declared channels. A report missing because the participant declined a prompt is not replaced by zero. Refusal, fatigue, sleep, distress, and device burden enter the observation model; unresolved missingness widens uncertainty or yields no estimate for that minute.

If the momentary valence estimator has been calibrated as a rate-like estimate, the interval valence contribution estimate for the thirty-minute interval is the time accumulation of those minute estimates. The covariance between minutes matters. A series of correlated errors does not shrink simply because the data table has many rows. The interval contribution travels with uncertainty and with a label showing which channels were active.

The running measured cumulative estimate begins at the study-window origin. Its graph represents accumulation during the observed window; a complete-history analysis adds an independently identified or bounded prior contribution.

Now suppose reserve is estimated independently from functional decline, clinical trajectory, and physiological reserve markers. If

the windowed measured cumulative estimate shows drift and innovation behavior linked to that reserve estimate, the result can enter the candidate-signature analysis. If reserve was partly computed from the ledger itself, the coupling would be less informative or circular.

A different artifact appears when a smoother uses information from after a candidate endpoint to reconstruct the preterminal path. The resulting curve may begin bending toward neutrality before the endpoint because future data have leaked backward through the analysis, not because participants followed a prospective completion dynamic. The correct response is a no-look-ahead audit and a filtered analysis, not admiration for the smooth curve.

A well-built bridge between momentary and interval valence estimators is therefore an instrument of restraint. It does not promise to see all of inner life. It tells the study when it is seeing enough to ask a bounded question and when it should withhold claim language.

An estimator can simplify the participant so aggressively that its apparent precision becomes a design defect. Sleep disruption from a device, pressured reports, relational data gathered without consent, informative missingness, or a model that compresses uncertainty into one clean trace all change the bridge between experience and evidence. Those changes belong in the analysis, not outside it.

4.2.4 — Dimensions and Rate Consistency

A dimensional map makes the notation easier to interpret. For each quantity, specify whether it is a rate, interval contribution, reserve coordinate, or ratio, and state the corresponding units.

The unresolved reserve-unit conflict is not a reason to abandon the program. It is a reason to route the program honestly. Later formal work may choose a normalization, a dimensional bridge, or a family of admissible transformations. Until then, the text should not pretend that all quantities already live on one coherent physical scale.

Dimensions connect each expression to a quantity that readers can interpret and researchers can check. $F(t)$ is a rate-like formal contribution over time. The unity gate is dimensionless or weighting-like. $L(t)$ accumulates over time. $U(t)$ is a reserve quantity in a candidate realization. $Y(t)$ is the normalized signed reference process in that realization. The candidate momentary valence estimator is an estimator on a declared scale. The candidate interval contribution estimate is an integrated estimate. Ψ is a derived ratio and becomes fragile near vanishing reserve (Durbin & Koopman, 2012; Folland, 1999; Karatzas & Shreve, 1991).

The reserve-scale convention should be explicit. The reserve-coupled realization uses a positive route-scale constant c_{LU} , allowing U to retain its declared reserve unit while Y carries the complementary unit. A normalized submodel may set $c_{LU}=1$, but that is a declared gauge choice rather than an empirical finding:

$$L_i^{rc}(t) := c_{LU} U_i(t) Y_i(t), c_{LU} > 0,$$

The reserve-normalized coordinate is:

$$Y_i(t) = \frac{L_i^{rc}(t)}{c_{LU}U_i(t)}, \Psi_i(t) = \frac{|L_i^{rc}(t)|}{c_{LU}U_i(t)} = |Y_i(t)|.$$

The notation should not let U mean biological capacity in one paragraph, ledger scale in another, and time-to-death in a third. A dimension table is not pedantry. It is the line between a formal model and a suggestive metaphor.

The product $L^{rc}(t)=c_{LU} U(t)Y(t)$ specifies a dimensional relation. If U carries reserve units, c_{LU} and Y together convert them into ledger units. Gauge rescaling leaves the product unchanged, so claims about the separate factors require an external anchor.

Rate consistency matters for integration. A momentary level cannot be accumulated like a rate without a time rule. If the momentary valence estimator estimates a valence-rate-like contribution, then the interval valence contribution estimate requires declared time intervals, missingness rules, and uncertainty propagation. If the momentary valence estimator estimates a state level instead, the integration claim changes. The title of the estimator does not solve the unit problem.

The viability ratio $\Psi=|L^{rc}|/(c_{LU} U)$ is useful only under the candidate route and only above a validated reserve floor. As U approaches zero, small reserve error can dominate the ratio. The synopsis therefore treats Ψ as a fragile diagnostic requiring windowing and errors-in-variables analysis, not as a terminal truth meter.

Terms adopted from physics, control theory, or neuroscience should retain a clear definition and mathematical role. Where a

term enters empirical work, specify its measured indicators, units, and comparison model.

The fairness lens sees rate consistency as anti-burden work. Bad units shift interpretive burden onto future readers, analysts, and participants. They make disagreement look philosophical when it is actually a measurement error. A humane science of felt experience owes its participants dimensions that can be checked.

4.2.5 — Initial Conditions and Study Windows

A terminal window can contain valuable information about changing dynamics. Complete-history analysis also represents the contribution before enrollment, using historical information, justified bounds, or sensitivity analysis as available.

Diaries, health records, speech, behavior, and public archives may enrich a participant's history. Each enters the analysis with its documented coverage, missingness, sign interpretation, and context.

Stream onset and study enrollment define different starting points. A late-life study combines its observed window with an unknown pre-enrollment contribution, represented through an explicit estimate, bound, or sensitivity analysis (Karatzas & Shreve, 1991; Protter, 2005).

An origin transformation makes the danger visible. If an estimator shifts the rate origin by a constant over a unity-one interval, the accumulated estimate shifts with time. A neutral-looking endpoint can therefore be created or destroyed by origin choice. The program must test structures that are less dependent on arbitrary

origin when possible, then treat terminal level claims as more assumption-sensitive.

The reserve route adds an initial-condition compatibility problem. If L^{phen} and L^{rc} are asserted to be the same pointwise object at stream onset, their initial values must agree. A realization that chooses a convenient Y initial state without an explicit bridge changes the route rather than solving it.

On the exact path-bridge branch, the initial identity is:

$$L_i^{\text{phen}}(S_i^\theta -) = L_i^{\text{rc}}(S_i^\theta -) = c_{\text{LU}} U_i(S_i^\theta -) Y_i(S_i^\theta -).$$

If the phenomenal convention additionally sets the initial ledger to zero and reserve is positive, this identity constrains the initial Y state. Other branches may use an offset, a reduced-form route, or an explicit initial ledger, but the choice must be declared before terminal fitting.

Study windows also change reference class. A palliative cohort with forty days of observation is a study of local terminal dynamics under a defined care context, not a whole-life ledger. It may estimate drift, variance, reserve coupling, missingness, and endpoint behavior during that window. It may not infer total lifetime balance without a bridge that no first study possesses.

Initial-condition honesty makes partial support possible. A late-window study might support a preterminal consequence while abstaining from terminal-level equality. It might show a rival dominates late dynamics while leaving earlier-life mechanisms untouched. It might reveal that pre-enrollment uncertainty is too

large for any terminal-level statement. These are legitimate outcomes.

A study window is not a life. Treating a late observable interval as though it represented the complete stream overstates what participants supplied and obscures which portion of the trajectory was actually informative. The correct result names the window, the unobserved history, and the assumptions connecting one to the other.

4.2.6 — Open Formal Bridges

Open bridges define a research agenda. Naming them gives future formal work a precise direction and keeps each dependency available for review.

Open formal bridges are not footnotes. They are the places where the research program could break before data become decisive. The integral definition and the reserve product belong to different process classes unless a bridge is supplied. Origin, unit, scalarization, endpoint, reserve, and representation assumptions must be declared. A study that ignores the bridges may produce clean results for the wrong object (Särkkä, 2013).

The process-class discrepancy is central. The received ledger is written as a unity-gated integral over time. The candidate reserve realization writes the ledger as a product of reserve and a signed reference process. The product can be useful, but the equivalence of these descriptions is not automatic. The bridge may be an assumption, a derivation, or an open conjecture. Its status must be labeled. A useful derivation would show how changes in reserve and the reference process combine to reproduce the ledger's

increments under the same initial conditions and time convention. A bridge that reproduces increments but not the same domain or initial condition would establish a related construction, not identity with the received ledger.

Endpoint-reserve synchronization is another bridge. If $U(t)$ approaches zero at the same time the endpoint T occurs, the product construction can force L toward zero. That can be a candidate mechanism only if reserve is independently defined and not chosen because it vanishes where the endpoint is declared. Otherwise the construction risks making closure true by design. Tracking both factors near the endpoint helps determine whether the reference process remains bounded or grows rapidly enough to offset the decline in reserve. A useful test asks whether the same reserve definition predicts closure timing in held-out streams where endpoint location was not used to calibrate the reserve measure.

The Logical Fork also remains open. If the endpoint is independent and the terminal ledger distribution remains continuous and nondegenerate, exact terminal equality is not expected by chance. The program must therefore identify what premise changes: terminal degeneracy, atom formation, boundary control, endpoint-state coupling under declared rules, or another mechanism. Each option has different signatures and risks. Comparing these options under a shared observation model can identify which differences in timing, distribution shape, or boundary behavior are detectable with the planned measurements. These branches should be compared on observable implications

that diverge before the endpoint, because terminal fit alone may leave their generative differences unresolved.

Scalarization is an open bridge because lived experience may be multidimensional. Pain, calm, fear, meaning, relief, dignity, fatigue, attachment, and bodily distress do not collapse into one dimension without a value-laden or empirical rule. The formal model uses a signed contribution, but the empirical bridge must justify how channels become a single estimator without erasing structure. Where scalarization is nonunique, the empirical object may be a set of admissible residuals rather than one number, preserving uncertainty at the representation layer.

4.3 — Measurement and Endpoint Dependency Chain

The research program has an order. Representation comes before integration. Calibration comes before comparison. Bearer classification comes before endpoint inference. Endpoint rules come before outcome inspection. Filtering comes before smoothing. Observation processes enter the generative model rather than being swept into missingness after the fact (Borsboom, 2005; Little & Rubin, 2019; Rubin, 1976; Särkkä, 2013).

The chain uses a separate decision at each prerequisite: signed representation, measurement calibration, endpoint adjudication, observation quality, and model comparison. Later inference proceeds when the prerequisites on which it depends meet their criteria.

The chain is also ethical. Endpoint science touches people in vulnerable situations, but the protection here is not simply a warning. No-crisis recruitment, burden budgets, palliative priority, consent scope, device withdrawal, and staff workload affect data quality because they affect who enters, who remains, what is observed, and how signals change. The ethics chapter owns the full firewall; this section shows where it enters the dependency stack.

The reader should notice that the endpoint firewall has two walls. The constructional wall prevents defining truth-level endpoint by the ledger or by a zero result. The inferential wall prevents selecting, smoothing, or adjudicating the endpoint after seeing the outcome in a way that builds the conclusion into the test. Both walls have to hold before terminal inference can be discussed.

Table 4.3. Dependency chain as a noncompensatory gate system: later success cannot repair earlier failure.

Gate	Pass condition	If it fails	What cannot compensate
Representa tion	Signed, temporally additive target is defensible for the reference class.	Ledger inference is unavailable or must be narrowed.	No endpoint pattern or large sample can repair it.
Calibration	Origin, sign, scale, invariance, and response shift are handled by frozen rules.	Comparison across time, persons, or sites loses warrant.	No smoothing or model selection can make the scale valid after the fact.
Bearer	The unified stream is adjudicated with confidence appropriate to the study.	The endpoint may not be assigned to the correct object.	Terminal concentration cannot identify the bearer.
Endpoint	truth-level stream endpoint is defined prospectively through non-ledger channels.	Neutrality becomes circular or uninterpretable.	A clean final curve cannot rescue a circular endpoint.
Observatio n process	Missingness, device removal, visit timing, sedation, and clinical escalation enter the generative model.	A favorable pattern may be observation artifact.	More participants cannot remove structured bias by volume alone.

Gate	Pass condition	If it fails	What cannot compensate
Model recovery	The pipeline recovers known synthetic generators under realistic error.	Real-data model comparison is not trustworthy.	A better information criterion on real data cannot substitute for recovery.
Rival comparison	Strong rivals are specified, fit, and allowed to win.	Support remains local or rhetorical.	One dramatic signature cannot replace direct rival pressure.
Replication and transport	Independent sites and populations reproduce bounded component claims.	Status remains preliminary or reference-class limited.	A single mature site cannot authorize universal language.

4.3.1 — Representation Before Integration

Representation also asks whether mixed affect can be treated as a scalar without distortion. Many experiences contain relief and pain, fear and love, grief and gratitude, exhaustion and peace. Scalarization may be useful for a bounded estimand, but it is a component assumption, not an obvious fact about experience.

Scalarization must specify what information is discarded, which rival interpretation that loss permits, and whether a vector or identified-set representation better answers the study's question.

Before integrating, the program specifies what is being added, its sign, its time basis, its unity condition, and its uncertainty. Those representation choices give temporal addition a defined meaning (Borsboom, 2005; Folland, 1999).

A minimal scalarization can be written as:

$$F_w(t) = w^\top z(t), w \in W, L_w(T) = \int_0^T g_\theta(t) w^\top z(t) dt$$

If several admissible weights remain plausible, the identified terminal set is:

$$L(T; W) = \{L_w(T) : w \in W\}$$

A result is scalarization-sensitive when the sign, equivalence-band conclusion, or rival ranking changes across W . In that case the proper outcome is narrowing or abstention, not a more confident paragraph.

The representation problem is deeper than choosing variables. Physiological arousal can accompany fear, joy, effort, fever, pain, medication, or social contact. A report can be shaped by language, shame, sedation, or hope. Behavior can express burden or conceal it. The signed target must therefore be built from validation, challenge, and context, not from channel convenience.

The most dangerous shortcut is to integrate a score whose sign is not stable. If high values sometimes mean burden and sometimes mean restoration, then a cumulative total can cancel unlike things by arithmetic rather than experience. The program must declare channel transformations, anchors, response-shift rules, and exclusion conditions before the interval valence contribution estimate is computed.

Representation also decides what is left out. If meaning, dignity, trauma, disability, culture, or communication access changes the interpretation of signals, the model must either include the issue

or restrict the reference class. A silent omission is not neutral. It shifts burden onto people whose experience is harder to measure.

A study may find that the proposed signed target is unsuitable for a population or state. It then reports that result and develops an alternative representation before proceeding with accumulation.

Representation should preserve what matters about the experience before any integral is computed. The aim is not to grind a life into a single number, but to define a signed target whose interpretation, uncertainty, and limits remain visible. Representational fidelity therefore precedes mathematical convenience.

4.3.2 — Calibration Before Comparison

Calibration is also a history problem. A person's use of language, bodily awareness, and emotional scale may change after illness, trauma, medication, aging, cultural migration, or terminal proximity. A stable number across time can hide a changed meaning, and a changed number can hide a stable experience under a changed reporting frame.

Calibration is an ongoing state documented by interval, instrument version, response-shift checks, and uncertainty. This record lets later analyses use each measurement under the conditions in which it was evaluated.

Comparison requires calibration. A measured value today cannot be compared with a measured value next week, another site, or another person unless the scale has a declared relationship across those contexts. The same number may not mean the same burden after illness progresses, after medication changes, after language

shifts, after response styles adapt, or after a sensor is replaced (Rubin, 1976; Särkkä, 2013).

The origin problem becomes operational here. A neutral point can be set by rules, but the rule must be defended. A personal baseline may support within-stream change while failing across-person comparison. A population anchor may support site harmonization while missing local meaning. A terminal band may be useful for equivalence only if fixed before outcomes are known.

The affine-origin problem can be stated in one line. If $x^*(t)=a+bx(t)$, with $b>0$, then the study-window integral transforms as:

$$\int_{t_0}^{t_1} x^*(t) dt = a(t_1 - t_0) + b \int_{t_0}^{t_1} x(t) dt$$

The effect of origin uncertainty differs by estimand. Analyze its influence on dynamical restrictions, comparisons, and absolute terminal levels separately, and carry the resulting uncertainty into each conclusion.

Calibration also includes uncertainty propagation. If the momentary valence estimator has uncertainty, the interval valence contribution estimate inherits it. If the endpoint is uncertain, the terminal estimate inherits it. If reserve proxies are noisy, reserve-linked slopes inherit it. A clean final table without uncertainty is not clearer. It is less honest.

Response shift is a special hazard. People may change the meaning of their ratings as illness, adaptation, fear, acceptance, fatigue, or hope changes. A scale may appear stable because participants learned the instrument rather than because experience

kept the same metric. The program must model, test, or restrict around response shift before comparison.

Calibration failure can narrow rather than end the program. A method may work within a person but not across persons, within one culture but not another, or within active reports but not in delirium. The status update should say exactly which comparisons remain licensed.

A bad calibration rule usually fails unevenly. Language, culture, disability, development, and clinical condition can be mistaken for noise when the instrument is really losing invariance. Calibration before comparison is therefore both a scientific requirement and a protection against systematic misrepresentation.

4.3.3 — Bearer Before Endpoint

The bearer question also guards against archive drift. A sufficiently detailed record can make observers feel that the stream is still present, especially when speech, images, and Artificial Intelligence reconstructions are available. The scientific bearer is not the archive. The archive may inform history or context; it does not continue the stream by being vivid.

Endpoint discipline therefore begins with respect for the thing that could end. If the stream is uncertain, endpoint confidence should be uncertain. If the stream is partitioned, interrupted, or returning, a single endpoint label may be premature. Scientific dignity means letting ambiguity remain visible instead of forcing a clean ending.

An endpoint can close only the object it belongs to. The object is a separately routed maximal qualifying conscious stream, not merely a body, chart, legal person, family story, or device record. Before a terminal event can be interpreted, the program must have a qualified bearer model, exhaustive routing rule, and continuity class appropriate to the study (van der Vaart, 1998).

Bearer adjudication is not metaphysical ornament. Sleep, anesthesia, coma, delirium, dementia, dissociation, developmental thresholds, nonhuman animals, and artificial systems can all challenge the boundary of the stream. A terminal endpoint assigned to the wrong bearer is not a noisy result. It is a category error.

Bearer uncertainty should travel visibly. A study can declare confidence classes: high-confidence unified stream, uncertain continuity, possible partition, temporary interruption, or insufficient evidence. Those classes then control which endpoint and ledger claims can be made. A terminal estimate without bearer confidence is under-specified.

This gate also protects public dignity. No person should be treated as less worthy because the science cannot classify the stream with confidence. Uncertainty about bearer status is a limit on the claim, not a ranking of the person, family, species, or system.

Stream classification distributes uncertainty across the entire analysis. Families may supply context, clinicians may judge state, researchers may quantify confidence, and participants may still be misclassified. Naming those roles before endpoint inference

keeps a terminal timestamp from concealing unresolved bearer uncertainty.

4.3.4 — Endpoint Before Outcome Inspection

The endpoint firewall must also survive incentives. Researchers, donors, public readers, and institutional partners may all prefer a clean terminal picture. That preference can subtly influence window choice, smoothing, censoring, and exclusion. A prospectively specified endpoint rule is a defense against the project’s own desire to be right.

A technical reader should be able to audit the endpoint path without trusting the author’s virtue. The rule, data channels, thresholds, uncertainty classes, adjudicators, disagreements, revisions, and analysis consequences should be visible. Trust grows when it is not required.

The endpoint must be defined before the outcome is inspected. If the truth-level endpoint is selected by looking for the moment when the measured cumulative estimate is nearest zero, the program has defined the conclusion into the test. This is the tautology firewall in operational form. Endpoint channels must be prospectively declared and not algebraically identical to the ledger estimator (Hernán & Robins, 2020).

On a declared first-hitting endpoint branch, with a specified endpoint filtration and regularity conditions ensuring the stopping-time property, the endpoint rule can be written as:

$$T_{\theta}(\omega) := \inf\{t \geq 0: \theta_t(\omega) \in E_{\theta}\}, \quad \omega \in \Omega_{\theta}$$

$$\{T_{\theta} \leq t\} = \{\omega \in \Omega_{\theta}: T_{\theta}(\omega) \leq t\} \in \mathcal{F}_t^{\text{endpoint}}$$

$$T_{\theta} = \Phi_{\theta}(\theta_{[0,\infty)}).$$

Constructional separation does not require stochastic independence. Any shared causes or residual dependence between endpoint channels and the ledger must be represented in the joint model.

Endpoint discipline contains a universal anti-tautology requirement and branch-specific timing premises. Universally, the truth-level endpoint must be prospectively defined and adjudicated independently of the ledger value whose closure is tested; defining the endpoint as the first or last zero makes closure an identity. On a declared independent-endpoint branch, timing independence from the ledger-generated information set and the protocol-fixed support assumptions are additional premises. State-coupled timing, first hitting or absorption, endpoint atoms, singular carriers, continued experience, and endpoint-inclusive value remain distinct branches. Windows, smoothing rules, thresholds, and exclusions must be frozen before outcome inspection in every branch.

Clinical reality makes the rule hard. Dying is not always a single observable instant. Resuscitation, sedation, coma, low-signal periods, device removal, family decisions, and record gaps can create multiple plausible windows. The program must specify how uncertainty is represented rather than pretending precision that care cannot provide.

Endpoint failure is a valid verdict. If the endpoint cannot be adjudicated independently, no terminal inference should be drawn. A study may still inform measurement burden,

observation processes, care feasibility, or preterminal dynamics, but it should not speak about terminal concentration.

This rule also blocks the archive-as-soul drift. Records of last moments are not the person and do not survive as consciousness. Endpoint evidence is a research artifact under consent and governance. It should not be turned into spiritual proof, public spectacle, or a final narrative about a life.

A clean endpoint is never worth a compromised deathbed. If classification procedures intrude on comfort, family presence, or clinical judgment, the protocol has failed before the model is fitted. Endpoint discipline therefore begins with care-compatible observation and a standing right to return no estimate.

4.3.5 — Filtering Before Smoothing

Filtering also gives the program a way to lose prospectively. A smoothed reconstruction can often be made to look orderly because the endpoint is already known. A filtered forecast has to speak before the ending. That temporal humility is one reason filtered performance carries more evidentiary force.

Smoothing still matters, especially in clinical states where reports are sparse and later information clarifies earlier uncertainty. The rule is not to forbid smoothing. The rule is to label it, restrict its claim, and prevent it from impersonating prediction.

Filtering and smoothing answer different questions. Filtering uses information available up to a time point to estimate or forecast the current state. Smoothing uses information from before and after a time point to refine a retrospective estimate. Both can be useful, but they cannot be confused. A prospective prediction cannot use

future information and still count as prospective (Rubin, 1976; Särkkä, 2013).

The distinction can be written in conditioning notation, with $\mathcal{Y}_t := \sigma(Y_s : s \leq t)$ denoting the information generated by observations through t :

$$\hat{L}_{\text{filter}}(t) = \mathbb{E}[L(t) \mid \mathcal{Y}_t]$$

$$\hat{L}_{\text{smooth}}(t) = \mathbb{E}[L(t) \mid \mathcal{Y}_T]$$

Every figure should therefore label its information set as observed, filtered, forecast, smoothed, endpoint-aligned, retrospectively adjudicated, or simulated. A prespecified smoothed analysis may support retrospective adjudication, quality control, or teaching; its evidentiary role remains retrospective.

The distinction matters because terminal alignment is powerful. If trajectories are smoothed after the endpoint is known, they can appear cleaner, more convergent, and more law-like than they were in real time. That may be acceptable for retrospective description if labeled. It is not acceptable as evidence for prospective restriction unless the analysis separates the two.

Filtering before smoothing also protects model comparison. A Law candidate may perform well in retrospective smoothing and poorly in real-time forecasting. A bridge rival may look similar after alignment but differ prospectively. The program needs both comparisons, but it must not let retrospective elegance masquerade as prediction.

A study can fail this gate if the pipeline cannot reproduce what information was available at each time. Missing timestamps, undocumented endpoint adjudication, late transformations, and untracked imputation can make a dataset unsuitable for prospective claims. The right response is no inference or retrospective-only status, not narrative rescue.

4.3.6 — Observation Process as Data

Because measurement can change behavior and care context, contact burden belongs in the observation process rather than outside the model.

The observation process is part of the data-generating system. Device removal, visit timing, family presence, clinical escalation, sedation, fatigue, refusal, staff workload, and the participant's tolerance for contact can all shape what is observed. Treating these as mere missingness after the fact hides a causal pathway that may mimic the target signature (van der Vaart, 1998).

The applicable dependency chain is noncompensatory: target, calibration, burden limits, reserve qualification when a reserve-coupled route is tested, endpoint rule, model recovery, prospective prediction, rival comparison, and component verdict must pass before any downstream inference that depends on them. A failed target, calibration, burden, endpoint, or recovery gate blocks the downstream inference. This is substantive because device intrusion, family anxiety, staff overload, and clinical escalation can otherwise masquerade as properties of the latent process.

The observation process also explains why absence has many meanings. No report can mean comfort, sleep, fatigue, inability, refusal, delirium, sedation, shame, family filtering, clinical urgency, or technical loss. A design that treats absence as neutral has already chosen one story without evidence.

For the candidate reserve coordinate $L^{\text{rc}}(t) = c_{\text{LU}} U(t)Y(t)$, a useful observation model is:

$$\begin{aligned} L_{\text{obs}}(t) &= L^{\text{rc}}(t) + \varepsilon(t) \\ \text{Var}(L_{\text{obs}} \mid U = u) &= c_{\text{LU}}^2 u^2 \text{Var}(Y \mid U = u) + \text{Var}(\varepsilon \mid U = u) \\ &\quad + 2c_{\text{LU}} u \text{Cov}(Y, \varepsilon \mid U = u). \end{aligned}$$

The additive-noise floor can bias the estimated variance exponent downward, making a reserve-coupled process look more bridge-like. More observations sharpen the wrong answer if the floor remains uncharacterized. The response requires an observation model, sensitivity analysis, and abstention when the floor dominates. Where feasible, estimating the floor from independent calibration or control measurements gives the correction a separate empirical basis rather than allowing it to inherit the terminal structure being tested.

The strongest first protocols should make observation marks analyzable objects. Device removals, prompt refusals, response latency, family presence, clinical interruptions, staff workload, care changes, and participant-requested pauses should be recorded as part of the generative process. They may reveal that the study is not viable before they reveal anything about the Law. Predeclaring how each mark enters the primary and sensitivity analyses, including any stratification, observation-process

modeling, or abstention rule, turns these events into reproducible design information rather than post hoc judgment.

Observation intensity can collapse near the endpoint. If the most distressed, medicated, or low-reserve periods are less likely to produce usable momentary valence estimator windows or reports, measured variance may shrink even when latent experience has not. A favorable-looking terminal pattern can therefore be a visibility pattern. The program must model observation gain. Because changing observability can alter apparent location, dispersion, and temporal dependence together, the analysis should carry the observation model through the full candidate signature rather than correct variance alone.

Observation also affects ethics. Devices that disturb sleep, questionnaires that increase worry, and interviews that change family behavior can alter the measured state. Research burden is both a moral cost and an input into internal validity. The protocol should distinguish burden created or amplified by measurement from burden already present in the clinical state, because the former represents observation reactivity introduced by the study.

Every estimate should carry observation marks: active report, passive physiological observation, device-off, medication-changed, family-present, clinician-interrupted, burden-stopped, endpoint-uncertain, or no-estimate. These distinguish absence of signal from absence of experience. These marks should remain available in analysis so a quiet segment caused by device loss cannot be pooled with a quiet segment observed under full coverage. A protocol-fixed rule for how each mark enters the primary and sensitivity analyses prevents the same observation

failure from being reclassified opportunistically after outcomes are known.

A study may produce valuable observation-process results even when the Law-specific inference abstains. It may show that a device is too burdensome, that missingness is informative, that staff workload makes the protocol infeasible, or that endpoint windows are too sparse. Such findings improve future research by revealing burden. Such results should be reported as feasibility or observation findings in their own right rather than buried beneath an abstained primary verdict.

Observation is selective in ways that often track vulnerability. The exhausted participant, overloaded clinician, or family member who can no longer support the protocol may vanish from the record just when the system is under greatest strain. Modeling the observation process makes that selection visible instead of converting it into a deceptively quiet dataset. Burden-linked dropout should therefore enter the estimand and sensitivity plan, not be treated merely as a loss of statistical precision.

Observation intensity can be modeled as a process with its own history and predictors. The probability of a usable report at time t may depend on earlier symptoms, treatment, reserve, family presence, staff workload, and previous prompts. A joint model of the latent trajectory and marked observation process makes those relationships explicit and supports sensitivity analysis across alternative missingness assumptions (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020). Joint models should report which predictors affect observation intensity independently of the

latent state, because those variables can help diagnose selection without becoming outcome proxies.

Observation-specific negative controls can help separate visibility from experience. Device battery loss, scheduled maintenance, staffing transitions, benign changes in prompt timing, and channels that should respond to contact burden but not to the latent Law-specific state can reveal whether apparent contraction follows the measurement system. State-space filtering and smoothing may summarize such patterns, but the filtered estimate available at time t must remain distinct from the retrospectively smoothed estimate that uses later data (Durbin & Koopman, 2012; Särkkä, 2013). Comparing these controls across device versions and sites can reveal whether the apparent terminal pattern tracks biology or the changing visibility of the measurement system. A control that changes visibility without plausibly changing experience is especially valuable because it directly tests whether the apparent signature survives altered observation gain.

4.4 — Candidate Signatures and Rival Matrix

A single signature is never enough. Mean reversion can come from ordinary homeostasis. Variance compression can come from observation shutdown, sedation, frailty, or event alignment. Endpoint concentration can come from conditioning, selection, smoothing, or post hoc bands. Reserve linkage can come from disease severity shared by affect and physiology. The PTVC candidate must therefore win as a prospectively specified conjunction, not as a favored reading of one pattern (Doob, 1957; Durbin & Koopman, 2012; Gneiting & Raftery, 2007).

This section presents the candidate signatures in the order in which they become progressively more demanding. Reserve-linked drift and innovation contraction are prospective dynamic restrictions. Variance shape and time scaling add form. Terminal concentration and equivalence sit higher in the ladder because they are more assumption-sensitive. Rival mapping accompanies each signature so the positive case does not outrun the discriminating work.

A strong rival matrix makes the comparison informative. At comparable predictive accuracy and calibration, a model that explains the pattern with fewer independently unsupported assumptions merits preference in that domain. Repeated superiority across the declared conjunction should lead to the corresponding narrowing, mechanism retirement, or theory-level verdict. This structure lets a demanding test improve the field whichever model succeeds.

The current research program draws on a substantial empirical foundation in neighboring fields and has a defined next step. Affect regulation, reward learning, relief, cross-modal valuation, biological integral feedback, lifespan change, and consciousness assessment provide measured phenomena relevant to the program. They contribute representational plausibility, mechanism feasibility, measurement opportunities, and quantitative alternatives. Their importance is clearest when each finding is assigned its actual role.

The measurement agenda is concrete: evaluate momentary rate estimates; test interval composition under realistic missingness; identify neutral-origin anchors; estimate reserve independently; and qualify endpoint proxies through separately specified consciousness channels. Each task provides a focused opportunity for empirical work.

Law-specific predictions are narrower and more demanding. Exact complete-stream phenomenal reconstruction is not required for every informative test. A protocol-fixed PTVC-linked route or consequence model may imply observable restrictions on history dependence, continuation choice, recovery dynamics, endpoint-relative behavior, or cross-species trajectories before a complete ledger is identified. Prospective reserve-linked drift, innovation contraction after observation correction, variance-shape restrictions, forecast performance under a declared independent-endpoint branch, fixed-band measured equivalence, and held-out superiority to rivals are examples. The derivation, empirical bridge, and rival set must be fixed before the relevant data are examined. A successful signature first supports the specified route

or consequence model. It bears on PTVC only when the tested restriction follows from PTVC under independently justified bridge assumptions, and only to the scope licensed by that implication. The program should therefore pursue both direct measurement bridges and indirect discriminating tests against strong learning, adaptation, homeostatic, selection, and observation-process rivals.

Cross-field convergence becomes more informative when its structure is explicit. Triangulation combines approaches with different major sources of bias and asks whether their results support a common causal interpretation (Lawlor et al., 2016). For PTVC, this suggests a concrete synthesis: derive a shared restriction from one declared realization, then test its consequences in independently sampled observational, behavioral, physiological, and sleep settings. Allow each channel its own validated measurement relation. Record overlapping participants and shared preprocessing so the same evidence is not counted twice.

The gain comes from a common model predicting the joint observations under fixed assumptions. An adequate comparison gives learning, regulation, context, and observation-process rivals the same information and tests the additional restriction on protected data. This turns the recurring shape of phenomena into a specific scientific challenge.

The next milestones are explicit: qualify momentary and interval phenomenal-valence measurements in bounded settings; establish origin, cross-sign scale, exposure, and endpoint bridges; recover

the competing models under realistic observation; and obtain independent, prospectively specified comparisons.

Read the rival matrix together with four evidence roles: observed scientific findings; the components those findings make plausible or measurable; the additional bridges and predictions required by a declared PTVC realization; and results that would compare that realization with its alternatives. This map lets each achievement retain its full value and gives the reader a precise route from established science to a stronger test.

The section includes an adversarial synthetic case because a superficially favorable terminal pattern is easy to create. If observation intensity collapses near the endpoint, noisy extremes can disappear. If endpoints are aligned after the fact, trajectories can resemble bridges. If sedation changes both report and physiology, apparent concentration may be a clinical artifact. The synopsis shows these possibilities before any real study is praised.

Table 4.4. Pairs candidate signatures with the rivals most likely to mimic them. The positive case depends on the conjunction, not on any one row.

Candidate signature	Law-candidate interpretation	Strong rivals	Discriminator
Reserve-linked drift	State-dependent relaxation steepens on a declared subbranch where r^U/U increases as reserve declines.	Ordinary homeostasis, allostatic control, endpoint-conditioned bridges, medication effects.	Independent reserve estimation plus prospective held-out drift relation.

Candidate signature	Law-candidate interpretation	Strong rivals	Discriminator
Innovation contraction	Stochastic innovations shrink on a declared product subbranch where σU decreases.	Observation attenuation, sedation, device removal, frailty-linked signal loss.	Observation model checks and residual innovation analysis after proxy error.
Variance shape	Variance contracts according to reserve-time scaling rather than generic event alignment.	Brownian bridge conditioning, terminal alignment, noise-floor bias, frailty selection.	Reserve-space and time-domain exponents estimated with noise-floor deconvolution.
Terminal concentration	Measured endpoint distribution concentrates near a protocol-fixed neutral band.	Post hoc endpoint choice, smoothing, response dropout, clinical scripts, social desirability.	Prospective endpoint, no look-ahead smoothing, and predeclared equivalence band.
Reference-class transport	The conjunction persists across bounded populations and endpoint classes.	Site practice, culture, recruitment pathway, care model, instrumentation.	Multisite prospectively specified replication and transport analysis.
Adversarial favorable pattern	A late curve appears to approach neutrality.	Observation gain collapse or endpoint conditioning created the approach.	Synthetic recovery must show the pipeline identifies the rival branch.

4.4.1 — Reserve-Linked Drift

The drift signature should be stated in terms of conditional relation, not narrative decline. A person approaching death may become calmer, more agitated, less communicative, more medicated, or more variable for many reasons. Reserve-linked drift becomes meaningful only when the statistical model links changes in the estimated ledger to independently estimated reserve under a specified dynamic form.

Reserve is a candidate state variable. Its estimated value changes the route model's predictions; it does not determine a person's completeness, worth, or availability for research.

Reserve-linked drift relates the route coordinate's pull toward neutrality to reserve. In the signed arithmetic reference version, the drift is $-(r^U/U)L^{rc} dt$. A steeper conditional drift slope requires r^U/U to increase over the declared regime. Its predicted relation to independently measured reserve and innovation behavior supplies the joint comparison developed below (Doob, 1957; Durbin & Koopman, 2012; Gneiting & Raftery, 2007).

Under the reference candidate, the local drift and local diffusion functions are:

$$b_L(\ell, t) = -\frac{r^U(t)}{U(t)}\ell, \quad s_L(t) = c_{LU}\sigma(t)U(t)$$

The associated one-dimensional forward equation, conditional on a known reserve path and a justified Markov closure, is:

$$\partial_t p(\ell, t) = -\partial_\ell \{b_L(\ell, t)p(\ell, t)\} + \frac{1}{2}\partial_{\ell\ell} \{s_L^2(t)p(\ell, t)\}$$

If U is hidden, stochastic, or coupled to observation, L alone is not generally a Markov state. The appropriate object is then a joint state-space model, not a decorative Fokker-Planck line. The synopsis includes the equation to show the burden, not to skip it.

The positive intuition is simple: remaining biological or functional reserve constrains how much imbalance can continue to move. If the candidate realization is correct, late dynamics should not look like unconstrained wandering. They should show a coupling between reserve and corrective movement. The coupling specification and fitted parameters must be fixed without tuning to the evaluation endpoint or held-out ledger outcomes, then evaluated prospectively under the declared joint model.

Drift alone cannot separate the Law from serious rivals. Ordinary homeostatic models also pull states toward set points. Brownian bridge-like models can create endpoint-directed drift by conditioning. Medication and clinical care can induce state-dependent changes. A drift estimate without variance, innovation, endpoint, and rival context is too weak to bear law-language.

A simple Brownian bridge to zero has a state-dependent pull:

$$dB(t) = -B(t)/(T - t) dt + \sigma dW(t)$$

The pull strengthens near T , so drift toward zero is not enough to distinguish the Law candidate. Without additional structure, the conditioned bridge does not supply the same independently estimated reserve-coupled innovation contraction. The discriminating conjunction must therefore include variance

shape, reserve coupling, the protocol-fixed endpoint-branch premises, and held-out prediction rather than a single slope.

Reserve must be estimated independently. If reserve is merely a transformed version of the ledger, the signature is circular. If reserve is only disease severity and disease severity also changes affect, shared cause can mimic coupling. The analysis must include clinical state, treatment, observation intensity, and proxy error.

A favorable drift finding could support the candidate realization in a bounded reference class. An unfavorable finding could motivate revision or retirement of the reserve-coupled mechanism. Both outcomes guide the comparison of routes and identify which predictions merit the next test.

Reserve language is technically useful only while it remains a model variable. It is not a measure of worth, courage, eligibility, prognosis, or priority for care. The construct belongs to a candidate mechanism evaluated under consent and governance; it does not classify the person in whom it is estimated.

4.4.2 — Innovation Contraction

Innovation contraction should also be compared to censoring. If high-variance participants exit the observable dataset because of transfer, withdrawal, device intolerance, or sudden decline, the remaining observed trajectory can look compressed. This is not contraction of the underlying process. It is selection in the observed sample.

A good study therefore reports both within-stream innovation estimates and who remains observable at each time. The set of

observed streams is part of the result. A narrowing sample may be more important than the variance number it leaves behind.

Innovation contraction is the companion to drift on a separately declared candidate subbranch. Stochastic innovation in the reserve-coupled route coordinate scales through $c_{LU}\sigma_i(t)U_i(t) dW_i(t)$. Contraction requires the product $\sigma_i(t)U_i(t)$ to decline, and vanishing terminal innovation requires $\sigma_i(t)U_i(t) \rightarrow 0$; reserve decline alone is insufficient. The observation model must also exclude manufactured contraction. This is a route-specific consequence under those premises, not a unique fingerprint (Karatzas & Shreve, 1991; Protter, 2005).

For reserve-space variance under RC-1:

$$V_{L,i}(u, x, t) := \text{Var}(L_i^{rc}(t) | U_i(t) = u, \Xi_i^\circ(t) = x) = c_{LU}^2 u^2 V_{Y,i}(u, x, t).$$

This identity is exact conditional on $U=u$ under $L^{rc}=c_{LU} UY$. It becomes an empirical claim only when U is independently estimated, Y has a stable conditional second-order structure, and the observation process has been separated from latent variance. Otherwise the slope can be a sensor-gain artifact.

The positive case is that terminal approach should become less volatile in a specific way, not merely calmer by description. A shrinking innovation term predicts relationships among reserve proxies, residual variance, and time-to-endpoint. Those relationships can be compared with bridges, homeostatic models, frailty, medication, sedation, and observation attenuation.

Observation attenuation is the nearest danger. A device may record less variation because it is removed, because the

participant moves less, because reports cease, because sedation suppresses channels, or because data processing filters out extremes. That would look like innovation contraction while saying more about visibility than experience. The observation process must be modeled explicitly.

Innovation contraction should therefore face a resolution sweep rather than one favored window. The analysis should vary sampling interval, aggregation rule, and instrument channel under prospectively specified limits, then ask whether the route-specific relation transports within uncertainty. A biological relation may weaken predictably under coarser measurement; an artifact may reverse, disappear, or track one instrument version only.

A favorable finding would be component support only if contraction survives proxy error, observation marks, and rival recovery. An unfavorable finding may reveal that the reserve product is wrong, that reserve is poorly measured, or that the window is dominated by observation artifacts. The verdict must name which gate failed.

4.4.3 — Variance Shape and Time Scaling

Time scaling also affects visual communication. A graph plotted against normalized time-to-death can make lives with very different trajectories appear commensurable. That may be useful for an exploratory figure, but it must be labeled as endpoint-aligned and not mistaken for a prospective clock. The axis is part of the claim.

Every candidate consequence must declare its clock. Calendar time, experienced time, reserve time, declared hazard time, and

endpoint-aligned time can reveal different structures and different artifacts. A robust prediction must state which clock carries the estimand, why that clock is scientifically motivated, which transformations preserve the claim, and which clock changes would count against the proposed route.

Variance shape is more discriminating than a generic claim that variance declines. RC-1 predicts reserve-space contraction because $L^{rc}=c_{LU} UY$ implies conditional scaling in relation to U under stated assumptions. The observable boundary exponent also depends on normalized-state dispersion and on the declared route clock. No universal time exponent follows from the product alone (Vehtari, Gelman, & Gabry, 2017).

If $U_i(t)=c_U[d_i^{rc}(t)]^\beta\{1+o(1)\}$ and the conditional normalized-state variance scales as $c_Y[d_i^{rc}(t)]^\gamma\{1+o(1)\}$, then:

$$Var(L_i^{rc}(t)|U_i(t), \Xi_i^\circ(t)) = c_{LU}^2 c_U^2 c_Y [d_i^{rc}(t)]^{2\beta+\gamma}\{1 + o(1)\}.$$

The route-space exponent is $2\beta+\gamma$. A value of 2β requires $\gamma=0$ or an independently justified locally constant normalized-state dispersion. Neither β nor γ is universal; both are estimands or declared route assumptions.

For a normalized illustrative case with $c_{LU}=1$, $U(t)=U_0(1-t/T)$, $Y(0)=0$, and $dY=\sigma dW$:

$$Var[L(t)] = U_0^2 \sigma^2 t \left(1 - \frac{t}{T}\right)^2$$

This illustrative variance peaks at $t=T/3$ and then contracts. It demonstrates terminal-regime compression under the stated

special case, not monotone whole-life smoothing or a universal exponent.

The exponent depends on its coordinate. A log-variance slope against reserve and a log-variance slope against clock time coincide under the relevant linear-depletion conditions; nonlinear depletion, state dependence, shared disease drivers, and proxy error can change that relation. The analysis estimates the slope on its declared scale.

Variance shape also separates the Law candidate from endpoint-conditioned bridges. A bridge may show variance that peaks and then contracts because paths are conditioned on an endpoint. The reserve-coupled candidate predicts contraction tied to independently estimated reserve. Discrimination requires checking whether variance follows reserve, time alignment, endpoint conditioning, or observation gain.

Noise floors matter here. Additive measurement noise can flatten late variance and bias exponent estimates. If subtracting the noise floor produces unstable or negative variance, the analysis should abstain. A clean exponent calculated from a biased estimator is worse than uncertainty because it looks precise.

A favorable variance-shape result could strengthen the conjunction if it is prospective, reserve-linked, noise-characterized, and rival-tested. A variance decline alone should not update the Law. An unfavorable or unstable shape may retire a predicted signature, narrow the reserve model, or force new measurement work.

A variance curve carries a network of assumptions with it. Reserve, time-to-endpoint, observation density, noise, rival dynamics, and selection all shape the pattern. The curve becomes scientifically useful when those dependencies are explicit and its interpretation is correspondingly conditional.

4.4.4 — Terminal Concentration and Equivalence

Equivalence also requires a meaningful null. The program should not treat failure to reject non-neutrality as evidence for neutrality. It must specify an equivalence band, measurement error model, power or precision target, and sensitivity analysis. A weak study that cannot distinguish anything has not found balance.

Terminal concentration is most persuasive when it arrives as the end of a route already traveled. If representation, calibration, endpoint, observation, rival, and recovery gates are intact, then a terminal pattern can add force. If those gates are broken, the same pattern becomes a warning sign about the design.

Terminal concentration and equivalence are the strongest and most assumption-sensitive targets in the public program. Measured endpoint estimates may concentrate near a prospectively specified neutral band, but that is not direct observation of exact latent closure. It is an empirical consequence on a fallible scale under a declared endpoint and observation model (Doob, 1957; Durbin & Koopman, 2012; Gneiting & Raftery, 2007).

An equivalence analysis specifies its estimand, operational scale, acceptance band, measurement-error model, and decision rule

before outcome inspection. It evaluates whether the data establish the declared degree of closeness at that resolution.

Terminal concentration inherits all prior gates. Representation, calibration, bearer, endpoint, filtering, observation process, model recovery, and rival comparison must already be strong enough. A favorable terminal band cannot compensate for failure upstream. This is why terminal-level claims appear after preterminal signatures in the synopsis.

The following is a population concentration claim about the measured terminal estimator, not an equivalence-test decision or direct sight of exact closure:

$$\Pr(-K \leq \hat{L}(\hat{T}) \leq K \mid \text{observation model, endpoint model}) \geq 1-\alpha$$

This probability claim depends on K , α , the endpoint estimator, the observation model, and the reference class. Here α is the permitted tail probability. The analysis estimates that probability with quantified uncertainty. An equivalence test separately evaluates its declared estimand against a prespecified band under an appropriate decision rule.

The strongest rival dangers are endpoint conditioning, post hoc smoothing, observation loss, clinical scripts, medication, sedation, family pressure, and frailty selection. Each can create apparent convergence. A terminal result that does not compare against these rivals should be treated as preliminary at best and unusable if endpoint or observation gates fail.

A terminal-concentration result updates the tested reference-class consequence when the required measurement and endpoint conditions hold. The direction and strength of the comparison

guide the specified decision about that consequence, its candidate route, or a broader branch reached through a valid entailment.

Terminal concentration is an empirical question, not a narrative claim about the meaning of a person's final days. Last-window data are especially vulnerable to treatment effects, observation shutdown, family scripts, and selective visibility. The analysis has to protect comfort first and treat the ending as evidence only within the narrow protocol that produced it.

4.4.5 — Rivals by Signature

The matrix should also include the possibility that a rival wins in one layer while the Law candidate remains useful in another. Adaptation may explain short-term recovery while failing to explain reserve-linked variance shape. Observation attenuation may explain terminal calm while leaving earlier dynamics open. Fair comparison is not a duel where only one sentence survives.

Generous model development makes the comparison informative. Give each serious rival enough flexibility, data access, and testing resources to represent its explanation fairly. A well-specified comparison teaches the field which structures best predict the observations.

Every candidate signature has a rival family. Homeostasis can produce local return toward set points; opponent processes can produce rebound; predictive and active-inference accounts can produce error correction and control; reinforcement learning can produce policy adaptation; Brownian bridges and Doob transforms can produce endpoint-conditioned paths; frailty, medication, sedation, and observation shutdown can produce late

contraction. The rival matrix must distinguish these mechanisms rather than treating a shared shape as unique (Doob, 1957; Durbin & Koopman, 2012).

Rivals should be generative, not merely named. A generative rival produces synthetic data with declared parameters, observation processes, endpoint behavior, and missingness. The pipeline must show that it can recover the rival when the rival generated the data. Without this step, model comparison on real data can become an aesthetic contest among formulas.

The rival map should be signature-specific. A rival that mimics variance contraction may not mimic reserve-linked drift. A rival that mimics endpoint concentration may not match innovation behavior. A rival that fits one cohort may fail transport. The Law candidate gains force only from a conjunction whose members fail against different alternatives.

An adversarial synthetic case belongs in the synopsis. Imagine a dataset where late momentary valence estimator variance shrinks and terminal estimates fall near the neutral band. On inspection, observation intensity declines sharply, sedating medications increase, and endpoint alignment is retrospective. A bridge or observation-attenuation model recovers the data better than the Law candidate. The correct verdict is rival dominance or unusable terminal evidence, not cautious support.

Rival success contributes scientific knowledge. When adaptation, homeostasis, or predictive control provides a better calibrated explanation under the declared comparison rules, the result guides the scope of the PTVC candidate. Repeated outcomes across the

joint prediction inform whether that candidate should be retained, revised, or retired.

A preferred hypothesis earns trust by constructing the alternatives that could defeat it. Critics should not have to reverse-engineer the strongest homeostatic, adaptive, bridge, selection, or observation-artifact model after publication. The rival set belongs inside the program, with enough information and flexibility to win.

4.4.6 — The Discriminating Conjunction

The conjunction should be prospectively specified as a gate structure, not reconstructed as a story after results. Which components are necessary, which are exploratory, which are decision-critical, and which are sensitivity checks should be stated before analysis. Otherwise any favorable subset can be narrated as the real test after the fact.

This is where the project's ambition becomes disciplined. It is willing to ask for a pattern that ordinary rivals should struggle to reproduce, but it is also willing to lose if those rivals reproduce it. The conjunction is therefore both the positive case and the defeat route.

The discriminating conjunction is the book's first positive empirical target. No single pattern is enough. The conjunction requires independent reserve coupling, prospective endpoint separation, observation-model robustness, model recovery, held-out prediction, variance-shape discipline, and rival comparison. It is a compound claim because the serious rivals are strong in different places (Vehtari, Gelman, & Gabry, 2017).

A useful conjunction might say: under a declared reference class and instrument version, preterminal dynamics show reserve-linked drift, innovation contraction, and variance-shape restrictions; the endpoint is prospectively adjudicated by non-ledger channels; observation attenuation and clinical confounds are modeled; generative rivals are recovered when true; and the Law candidate outperforms rivals on held-out forecasts. That is the level at which support becomes scientifically interesting.

The conjunction is deliberately vulnerable. It can fail through measurement invalidity, reserve proxy error, endpoint circularity, observation collapse, rival dominance, nontransport, model-recovery failure, or ethical burden. A failed member is not repaired by success elsewhere. The status update names the failed prerequisite.

The conjunction also makes positive results narrower and more valuable. If the full set survives in one bounded population, the result supports that reference-class prediction. It does not support all conscious streams. It creates a reason to replicate, expand, and challenge, not a reason to announce validation. The next replication should preserve the successful conjunction rather than selecting only its strongest coordinate, because the joint restriction is what earned the result.

This standard prevents the book from becoming a pattern-recognition exercise. Humans are good at seeing significance in curves. Scientific claim status requires a rule declared in advance that says which patterns matter together, which rivals were allowed to win, and what the public statement will say if the conjunction fails. Predeclared failure of any indispensable

coordinate should lower the verdict even when the remaining plots look persuasive.

A conjunctive signature package keeps attractive findings from hiding qualifications in prose. Requiring the major conditions together states the evidentiary price of support before public compression. This structure also makes later synthesis cleaner, because independent studies can compare the same multicomponent criterion rather than retrospectively chosen highlights.

4.5 — The First Protocol in Miniature

This section walks through a humane, bounded, late-life observational study in miniature. Population, data flow, estimands, intercurrent events, rivals, recovery, adverse branches, and public reporting are specified together, giving researchers an integrated design to examine and develop before enrollment (International Council for Harmonisation, 2019; International Council for Harmonisation, 2026; Lakens, 2017; Nosek et al., 2018).

The miniature studies dynamics and measurement within a defined late-life window. Recruitment occurs outside crisis, care proceeds as clinically indicated, devices are used only while tolerated, and endpoint rules are specified before trajectory evaluation. Those choices make the initial study concrete and proportionate.

Simulation has several distinct jobs before real-data selection: verify code against known identities, calibrate finite-sample behavior, expose parameter nonidentifiability, estimate design power, and stress the observation and endpoint rules under controlled failure. These jobs should be declared separately so a convincing illustration is not mistaken for recovery evidence.

The protocol's most important feature may be its adverse branch. A study can return nulls, unstable estimates, endpoint failure, burden stops, measurement collapse, rival dominance, or no-estimate outcomes. Those are not failed products. They are scientific results when they are declared, preserved, and allowed to change the program.

Table 4.5 compresses the first-study route. It is a schematic, not a protocol approval, and it does not authorize research without ordinary expert, legal, clinical, and ethical review.

Table 4.5. First protocol in miniature: stage, scientific action, and stop or decision rule.

Stage	Scientific action	Stop or decision rule
Enrollment	Define an adult, comfort-first palliative reference class and approach only through non-crisis pathways.	Abstain when urgency pressure, site burden, or selection distortion makes the intended reference class uninterpretable.
Observation	Collect versioned momentary estimates, context, reserve or horizon proxies, treatment changes, and observation-process marks.	Pause or stop when monitoring disturbs comfort, sleep, family interaction, or staff workflow; mark affected data unusable.
Endpoint	Adjudicate the truth-level stream endpoint prospectively through non-ledger channels, confidence classes, and a no-estimate option.	Return no terminal inference when endpoint confidence fails or outcome information has entered the adjudication.
Estimands	Estimate preterminal drift, innovation variance, reserve coupling, observation effects, and measured endpoint concentration.	A late-window result may update only the declared component; it cannot stand for the complete life.

Stage	Scientific action	Stop or decision rule
Rivals and recovery	Demonstrate model recovery in synthetic worlds, then compare homeostatic, bridge, frailty, medication, learning, and observation rivals under common rules.	Recovery failure blocks model-winner language; rival victory narrows or retires the candidate route.
Release	Report scope, uncertainty, burdens, adverse branches, rival performance, and prohibited interpretations.	Permitted outcomes include bounded component support, narrowing, rival-favored, unusable evidence, or No adequate estimate.

4.5.1 — Population and Enrollment

The miniature protocol begins with a bounded population rather than a universal claim. A humane first terminal-dynamics study might enroll adults in a palliative reference class where observation can occur without crisis recruitment and where care teams can refuse or pause research without penalty. The narrow class is a strength: it states exactly which care environment, endpoint process, and participation pathway the first inference can address (International Council for Harmonisation, 2019; International Council for Harmonisation, 2026; Lakens, 2017).

The reference class should also include the care system. A study in a well-staffed hospice with stable relationships, trained research staff, and strong privacy infrastructure is not the same as a study in an overstressed hospital or fragmented home-care environment. Site conditions affect participation, observation, and data quality.

Enrollment is therefore a scientific design choice as much as an ethical one. It determines which lives become visible to the model, which participation demands are acceptable, which endpoints can be adjudicated, and which generalizations remain forbidden until transport is tested.

Recruitment materials describe the observed period and the study's questions about measurement, dynamics, endpoints, and competing explanations. Any inference using pre-enrollment history is identified separately, together with the information that supports it.

No-crisis recruitment is a scientific as well as ethical condition. A person approached in acute distress, uncontrolled pain, family panic, or urgent clinical uncertainty may be less able to refuse freely, and the recruitment context may select a distorted sample. The study should use stable clinical pathways, repeated opportunities to decline, and independent care-team permission.

Inclusion and exclusion should be defined through care compatibility and scientific interpretability. Device tolerability, communication needs, cognitive fluctuation, family involvement, staff capacity, and clinical instability all shape the data. When a subgroup cannot participate without added risk or unmanageable workload, the design narrows or changes instead of forcing representation through pressure.

Enrollment also includes bystander handling. Family members, caregivers, clinicians, and roommates may appear in context, speech, timing, or observation records. They do not become

research participants by proximity. Their data must be minimized, separated, consented where needed, or excluded.

A humane protocol makes the distribution of work explicit. Participants can refuse; families are protected from unchosen roles; clinicians retain care priority; staff workload is measured; and the design can shrink or stop before intrusion becomes harmful. Feasibility includes the people who make the study possible.

4.5.2 — Data and Context

The miniature data plan should include versioning. Instruments change, sensors update, staff training evolves, and clinical workflows shift. A result tied to one measurement version should not silently merge with another. Version labels make the evidence reusable without pretending it is timeless.

Context also includes silence. A nonresponse may mean rest, exhaustion, refusal, delirium, distress, device failure, privacy need, or death. The study should not assign one meaning by convenience. Silence is a data problem with human dignity at its center.

The miniature dataset contains more than momentary valence estimator values. It includes physiological channels used in the momentary valence estimator, active reports when appropriate, reserve proxies, medication timing, sleep and rest context, care events, family presence where consent allows, device status, observation intensity, and endpoint-related marks. Context is not noise. It is how the model knows what kind of signal it is seeing (Lakens, 2017; Nosek et al., 2018).

Data and context remain witnesses, not the latent object. The candidate momentary estimator and interval estimate inherit the versioning, uncertainty, and channel-role rules in §4.2.3; contextual reports may challenge or validate the estimator without silently becoming part of it.

Treatment records are essential because medication and care can mimic target signatures. Sedation, analgesia, oxygen changes, delirium treatment, infections, hydration shifts, and clinical escalation can change physiology and reportability. A terminal pattern that ignores treatment context is under-modeled.

Observation marks should be attached to windows rather than buried in a data appendix. A window may be observed, filtered, forecast, smoothed, endpoint-aligned, simulated, missing by refusal, missing by device burden, missing by clinical priority, or no-estimate by design. The label determines what claim the window can support.

Data minimization remains active. The protocol collects what the estimand and rival comparison require, not every intimate detail that might someday interest a model. Future reuse, linkage, and model training require separate permissions. The dataset is research material under governance, not a general archive of a life.

Context should sharpen interpretation rather than justify unlimited capture. Medication changes, sleep, symptoms, and care events may explain a signal; unrelated conversations, household activity, or bystander behavior do not become research data merely because sensors can reach them. The miniature

protocol collects the minimum context needed for its declared question.

4.5.3 — Primary Estimands

The study estimates defined dynamic quantities under its research model. Clinical teams evaluate care and individual needs through their own assessments, informed by relevant contextual observations.

Public wording identifies the component estimand and the scientific question it answers. The description keeps the study's unit, population, and purpose clear to readers.

The first study needs estimands, not hopes. Primary estimands might include reserve-linked drift slope, innovation variance as a function of reserve, variance-shape exponent with time-to-endpoint, observation-gain parameters, forecast error under the declared endpoint-timing branch, and measured terminal concentration within a prospectively specified band. Each estimand answers a different question (Schuirmann, 1987; Wilson & Collins, 2019).

A compressed estimand vector for the first study is:

$$\eta = (\kappa_U, \gamma_{innov}, \beta_{var}, \delta_{end}, \Delta ELPD_{Law-rivals})$$

κ_U indexes reserve-linked drift; γ_{innov} indexes reserve-linked innovation contraction; β_{var} indexes time-domain variance scaling; $\delta_{endpoint}$ indexes terminal equivalence under the observation model; and $\Delta ELPD$ compares held-out predictive performance against prospectively specified rivals. The joint

decision evaluates their predicted relations and each required component.

The ordering matters. Preterminal dynamic estimands should be primary before terminal-level equivalence because they are less dependent on arbitrary origin and whole-life initial conditions. Terminal concentration may be secondary or high-level, depending on endpoint and measurement gates. The protocol should not let the most dramatic outcome become primary after seeing it.

Estimands must carry reference class and scale. A drift slope in one instrument version, one care context, and one endpoint class is not a universal parameter. A candidate interval accumulation over a forty-day window is not a lifetime ledger. A terminal mean is not an individual closure probability. The estimand name should prevent overreading.

Uncertainty belongs in the estimation and reporting of each estimand. If reserve is noisy, endpoint timing uncertain, or momentary valence estimator calibration unstable, the analysis should propagate that uncertainty. A point estimate stripped of its measurement model may look clean while losing the very information needed for verdict status.

An estimand can also remain unestimated. If the endpoint gate fails, terminal concentration is not estimated. If observation collapses, preterminal dynamics may become no inference. If model recovery fails, rival comparison may be withheld. These outcomes should be written into the protocol before the data arrive.

An estimand is also a promise about use. It states what participant data will answer, which intercurrent events matter, and how the result will be interpreted. Precision at this stage reduces the chance that observations gathered for one purpose will later be converted into an opportunistic claim.

4.5.4 — Rival Set and Recovery

Recovery should include near-miss designs. The analysis should test cases where the Law candidate and a bridge rival produce superficially similar endpoint behavior but differ in preterminal variance, observation marks, or reserve relation. Near-misses are more useful than easy separations because real data will rarely be kind.

The synthetic fixtures should be archived with versioned code in a governed research repository. The book can explain assumptions, diagnostics, and pseudocode, but runnable science needs artifacts that independent teams can inspect, challenge, and rerun. Prose alone does not make a computational program reproducible.

The rival set should be prospectively specified as generative alternatives. A homeostatic mean-reversion model, an endpoint-conditioned bridge, an observation-attenuation model, a frailty and medication model, an adaptive-learning model, and a safe-control baseline should all be able to generate synthetic data. The PTVC candidate must face them before the pipeline interprets empirical data (Gneiting & Raftery, 2007; Nosek et al., 2018; Wilson & Collins, 2019).

Model recovery is the gate that makes comparison credible. The analysis pipeline should receive synthetic datasets generated by each rival and by the Law candidate under realistic missingness, noise, endpoint uncertainty, and observation marks. If the pipeline cannot recover the true generator often enough, real-data selection is not interpretable.

The recovery requirement is operational:

$$M_j \rightarrow O_\phi \rightarrow \{M_1, \dots, M_J\} \rightarrow \hat{f} \text{ with calibrated uncertainty}$$

Recovery tests should include adversarial mixtures. Real participants may not be generated by a single pure model. Observation attenuation may combine with homeostasis. Medication may combine with bridge-like endpoint alignment. Reserve coupling may exist but be weaker than assumed. The pipeline should know when it is uncertain rather than forcing classification.

The rival set should include positive-terminal-drift baselines and no-closure baselines. A serious program must let the data say that terminal dynamics move away from neutrality, or that no structured closure pattern appears. A model space that lacks defeat options is not a model space. It is a protection device.

Rival success changes public language. If the bridge model wins, the result favors an endpoint-conditioning explanation over the defeated candidate route. If observation attenuation wins, the result pressures measurement, not the Law directly. If homeostasis wins, the reserve-coupled mechanism may narrow. The verdict should name the component that lost.

Rivals deserve scientific parity, not rhetorical courtesy. Weak comparators make the preferred model look strong while leaving outsiders to supply the missing skepticism. Strong generative alternatives make the result worth reading because the program has genuinely risked losing.

4.5.5 — Adverse and Unusable Branches

Interpret terminal trajectories together with medication, communication capacity, participant condition, and changes in observation. Including those processes in the model makes the graphical result and its uncertainty easier to assess.

In the favorable synthetic branch, the endpoint is adjudicated without ledger variables, reserve proxies decline under a prospectively specified low-reserve rule, the fitted Law candidate recovers the joint drift-diffusion pattern, and held-out prediction beats the bridge and homeostatic baselines.

In this synthetic branch, a successful result supports the pipeline's recovery and discrimination under the specified generator. It is not empirical support for PTVC or for a natural reserve mechanism. Independent execution can reproduce the computational result; empirical component support requires a separately conducted study with the relevant measurement, endpoint, and comparison gates satisfied (Lakens, 2017; National Academies of Sciences, Engineering, and Medicine, 2019).

In the adverse synthetic branch, terminal values cluster near zero only after device removal, sedation escalation, or endpoint alignment. The Law candidate fits the visible end, but the observation-attenuation rival explains why excursions

disappeared. The correct output is unusable or rival-dominant evidence, not a polished success story.

This is why the no-estimate outcome is protective. It protects the participant from having uncertainty converted into a story, and it protects the theory from surviving by using data that no longer satisfy its own design.

The protocol specifies responses to withdrawal, sleep disruption, family burden, endpoint uncertainty, observation loss, calibration problems, and comparative model outcomes. Each branch identifies its appropriate action, including a No adequate estimate result when the target is unavailable.

Adverse branches should be predeclared as decision paths. If device burden crosses a threshold, monitoring stops. If care teams report interference, the study pauses. If endpoint channels conflict, terminal inference abstains. If observation gain collapses below a usable level, variance claims are withheld. If synthetic recovery fails, model comparison is not reported as decisive.

When evidence is unusable, the protocol action should be concrete: quarantine the affected analysis, record the failed prerequisite, prevent pooled salvage under a new label, assess participant and privacy consequences, and issue a correction if a public claim escaped. The dataset may still teach about feasibility or failure, but it does not update the protocol-fixed target.

An adverse branch can still teach. It may reveal that a proposed device is too burdensome, that the reference class is too fragile, that missingness is informative, or that a better endpoint proxy is

needed. That knowledge belongs in the research record, but it does not become support by reframing.

The protocol should also protect participants after adverse events. Withdrawal, deletion requests, data retirement, incident reporting, family communication, and staff support must be planned. A study that can collect data but cannot handle distress, breach, or withdrawal is not ready for this subject.

Adverse and unusable branches belong in the main result. When a device conflicts with care, missingness becomes dominant, endpoint adjudication fails, or model recovery collapses, the stopping path is itself evidence about feasibility. Concealing it would transfer the same failure to future participants and readers.

4.5.6 — Bounded Public Result

A bounded public result names uncertainty plainly. It states what the model estimated and what the study could not see: earlier life history, unobserved internal states, excluded populations, ambiguous endpoints, and rivals outside the protocol-fixed set. These limits belong to the result.

The best public result is modest enough to be true and clear enough to matter. It gives scientists a reason to replicate, critics a target, and readers no permission to rank, console, pressure, or explain another person's suffering.

A bounded public result states the component, reference class, instrument version, endpoint rule, uncertainty, rival outcome, adverse events, and prohibited uses. It does not expand because the audience wants a larger story. A first study might report preliminary support for reserve-linked dynamics in one declared

reference class, or abstention because endpoint adjudication or a protocol-fixed timing premise failed (Fischhoff & Davis, 2014; Nosek et al., 2018; Spiegelhalter, 2017).

Use specific verbs and source roles when reporting neighboring findings, measurement results, and proposed predictions. State each result once, with its population, method, and outcome.

The result must also name what did not change. Component support authorizes no clinical decisions, policy, religious doctrine, legal status, employment rank, civic trust, ordinary-life scoring, or Artificial Intelligence interpretation of persons. Negative results do not weaken care. State these limits once in the result or release note, not a disclaimer page.

A bounded result should include nulls and rival wins. If an observation-attenuation model explains the terminal pattern better, the release should say so. If the momentary valence estimator versioning failed, the release should say no claim update. If a burden stop ended data collection, the release should record the stop as evidence about feasibility.

The release should not be upgraded by donors, media, aesthetics, or public enthusiasm. Every detachable artifact should carry a source result, version, date, maximum status, correction trigger, and withdrawal path. The public surface may compress the explanation, but it cannot compress away the condition that controls the verdict. Visual beauty is not warrant.

A bounded public result should state exactly which component changed status. Readers must never be invited to interpret lives

through evidence lacking that authority, so the protocol ends with a narrow, reversible statement rather than a triumphal headline.

Table 4.6. Neighboring evidence, open questions, and PTVC status.

Evidence category	What is currently available in neighboring science	What remains open for this program	Status for the Law
Affective dynamics	Affect changes over time and can show adaptation, rebound, inertia, and regulation.	Whether those dynamics instantiate the declared ledger target and reserve-coupled signatures.	Observed affective dynamics and candidate mechanisms.
Consciousness-state research	Markers and expert adjudication can sometimes distinguish wakefulness, sleep, anesthesia, coma, and other boundary states.	Whether unity proxies can define endpoints without circular dependence on the ledger.	Empirically tested consciousness-assessment methods.
Psychometrics and measurement	Latent-variable modeling, reliability, calibration, invariance, and response-shift methods can improve construct discipline.	Whether candidate momentary and interval estimators can become valid estimators for the specific reference classes and endpoints.	Empirical measurement foundations and qualification methods.

Evidence category	What is currently available in neighboring science	What remains open for this program	Status for the Law
Palliative and clinical research	Comfort-first observation and longitudinal symptom research can be conducted under ethical review.	Whether terminal dynamics can be studied without burdening care or distorting signals.	Observed clinical trajectories and practical study-design experience.
Statistical model comparison	Equivalence testing, sensitivity analysis, Bayesian companions, predictive validation, and model recovery provide tools.	Whether those tools discriminate the Law candidate from strong rivals under realistic error.	Established tools for estimation and model comparison.
Law-specific evidence	Formal predictions and proposed study designs.	Test the central predictions through prospective comparison, replication, and clearly reported outcomes.	Next milestone: prospectively specified comparative results.

4.6 — Verdicts, Stop Gates, and Open Problems

A scientific program should define verdict language before evaluating data. This volume defines a formal target, develops conditional results and candidate realizations, and specifies measurement, endpoint, rival, and study requirements; the evidence reviewed here does not establish PTVC as an empirical law. The literature reviewed in this volume supports component methods, mechanisms, rivals, and feasibility, not the complete-stream equality itself. The verdict system below therefore governs future PTVC evidence and current component findings alike: preliminary component support, replicated component support, inconclusive or unusable evidence, narrowing, mechanism retirement, rival dominance, abstention, and theory-level defeat (Gneiting & Raftery, 2007; Lakatos, 1978; Mayo, 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

The verdict ladder is noncompensatory. A result cannot jump over failed prerequisites. Replication cannot repair a circular endpoint. Elegant statistics cannot repair invalid measurement. A donor-facing graphic cannot upgrade a null. A public response cannot turn a rival win into support.

The verdict rule is deliberately categorical. It maps the data and the state of each prerequisite gate into one of the protocol-fixed outcome labels.

Its inputs include prospective specification fidelity, measurement validity, endpoint admissibility, model recovery, rival comparison, ethics, privacy, and transport.

The section also gives abstention dignity. If origin, sign, scale, endpoint, observation, recovery, ethics, privacy, or rival-separation gates fail, the correct verdict may be no inference. That is not timidity. It is the refusal to turn poor data into a story. The program becomes stronger when it can say that a study should not update any claim.

Open problems remain inside the scientific program rather than being hidden in back matter. The detailed register at the end of the chapter assigns formal bridges, dimensional interpretation, process classes, endpoint construction, branch-specific timing premises, transport, recovery, observation error, and public updating to explicit unresolved states. Naming that work prevents a synopsis from converting incompleteness into decorative certainty.

The decision tree begins before statistics. If consent, privacy, capacity, assent, surrogate boundaries, or data-use limits fail, the study cannot produce a supporting public claim. If participant or family burden exceeds the declared budget, the protocol pauses or stops. A humane stop is not an interruption of science. It is one of the outcomes the science promised to respect.

If ethics and permission hold, the tree moves to representation. If the state cannot be represented as a signed and temporally interpretable target, the study may report phenomenology, feasibility, or qualitative data, but it should not compute a ledger. If representation survives, the tree moves to calibration, scale, uncertainty, and response shift.

If calibration fails, comparison fails. The study may still teach the next instrument how to improve, but it may not claim cross-time, cross-person, or terminal valence closure inference. If calibration survives, the tree moves to bearer and endpoint. If stream identity or endpoint adjudication is unresolved, terminal claims are withheld.

If endpoint and observation modeling survive, model recovery becomes the next gate. A workflow that cannot recover the right generator from synthetic data should not be trusted to name the right generator in real data. The correct verdict is abstention from model-winner language until recovery improves.

If recovery succeeds, the candidate and rivals compete on held-out filtered prediction, prespecified terminal estimands, observation robustness, and transport. A Law candidate that wins only after smoothing with endpoint information has not won the prospective test. A rival that wins on the declared criteria should receive the public verdict, even if the result is disappointing.

At the program level, the decision combines results across independently motivated bridges and rival comparisons. Revisions contribute when they add testable predictions; a sustained absence of predictive progress can justify retiring the affected scientific claim.

Table 4.7. The verdict ladder. Its purpose is to make support, narrowing, abstention, unusable evidence, rival dominance, and defeat available before any result is known.

Verdict	Concrete condition	Public wording discipline
Preliminary component support	One prospectively specified component survives measurement checks and specified rivals in a bounded reference class.	Say which component and reference class, not that the Law is validated.
Replicated component support	Independent sites reproduce the same bounded component under compatible instruments and endpoint rules.	State replication scope and remaining transport limits.
Narrowing	Evidence supports a component only under a smaller population, endpoint class, or measurement version.	Change the claim version visibly; do not reinterpret the old claim as unchanged.
Mechanism retirement	A candidate realization fails while the broader formal hypothesis remains open.	Retire the mechanism without declaring victory elsewhere.
Rival dominance	A protocol-fixed rival explains the target signatures with better held-out performance and fewer strained assumptions.	Credit the rival and specify what Law claim loses standing.
Inconclusive evidence	Data are adequate but do not distinguish candidates within the protocol-fixed sensitivity bounds.	Say unresolved, not supported.

Verdict	Concrete condition	Public wording discipline
Unusable evidence	Origin, endpoint, sign, observation, recovery, ethics, or privacy gates fail.	Say no inference; do not mine the dataset for narrative.
Abstention	The correct scientific action is to withhold claim update because prerequisites did not hold.	Treat abstention as an honest result.
Theory-level defeat	A gate-valid counterexample rejects the declared universal branch; alternatively, repeated bridge failure, uninformative replacements, and strong rival dominance can exhaust the empirical program.	Withdraw scientific law-language for the affected claim.

4.6.1 — Preliminary and Replicated Component Support

Replication should also include failure memory. A second study should not merely repeat the first success conditions. It should carry forward the first study's problems, adverse branches, rejected rivals, and uncertainty. Otherwise replication becomes ceremony rather than correction.

A mature evidence program should be able to say: this component is preliminary, this component is replicated, this component failed transport, this mechanism is retired, and this question remains open. Such plural verdicts will sound less dramatic than proof, but they will be far more useful.

Preliminary component support means that a declared component survived its initial gates in a bounded reference class. It may be measurement support, endpoint support, reserve-signature support, model-recovery support, rival-separation support, or protocol-feasibility support. The statement names the supported component and its tested scope (Gneiting & Raftery, 2007; Lakatos, 1978; Mayo, 2018).

Replicated component support requires independent confirmation under compatible rules. A second analysis of the same site is not enough. A transported result should preserve instrument version or explicitly bridge it, use compatible endpoint rules, declare reference-class differences, and include rival comparison. Replication is a design of vulnerability, not a ceremonial second number.

Component support can accumulate. Measurement may improve first, endpoint discipline second, dynamic signatures third, rival separation fourth, replication fifth. The strong form remains indirect throughout. A mature program may make the strong form more scientifically credible by strengthening the bridges that expose it to loss, but credibility is not the same as confirmation.

Component support should be published with failure routes. The same report should say what would have turned the result into narrowing, abstention, or defeat. A result without its defeat conditions is not ready to travel publicly because readers will supply triumph where the method left uncertainty.

Support also has a time horizon. A result may be preliminary until external replication, source verification, data audit, code review,

endpoint adjudication review, and public-artifact review are complete. The pressure to release early should not alter the label. The label protects the reader from the project's own excitement.

Support should attach only to the component tested. Using participant data for a stronger public claim than the design can bear converts observation into meaning that was never supplied. Component-level language respects both the evidence and the people whose time made it possible.

4.6.2 — Inconclusive and Unusable Evidence

The distinction between inconclusive and unusable also helps public trust. Critics often suspect that ambitious theories convert every result into support. A visible unusable-evidence category shows that the project can refuse its own data when the bridge is broken.

The same category helps future researchers. A study that fails because of device burden, endpoint ambiguity, or observation attenuation teaches design constraints. The lesson should be retained, but the data should not be forced to answer the original question.

Inconclusive evidence and unusable evidence are different verdicts. Inconclusive evidence passes the basic gates but does not distinguish the candidates strongly enough. Unusable evidence fails a prerequisite: invalid measurement, circular endpoint, broken consent scope, observation collapse, failed model recovery, privacy breach, or ethical interference. The distinction protects science from false neutrality (Mayo, 2018;

National Academies of Sciences, Engineering, and Medicine, 2019).

An inconclusive result can still update design. It may show that a sample is too small, an equivalence band too narrow or too wide, a reserve proxy too noisy, a rival set too underdeveloped, or a window too short. The claim status may remain unchanged while the next protocol becomes sharper.

Unusable evidence should not be mined for narrative. If the endpoint was chosen after outcome inspection, no terminal claim should be made. If the momentary valence estimator calibration failed, no measured cumulative estimate inference should be drawn. If participant burden altered the state, the data may serve a burden report but not a Law-specific result. No amount of interpretive eloquence repairs a failed gate.

The program should publish unusable paths in controlled form. Hidden unusable evidence creates publication bias and lets future teams repeat harms. Public registration, restricted data notes, adverse-result trees, and negative-result logs can preserve the lesson without exposing private material.

Abstention belongs near this verdict. Sometimes the correct scientific answer is that no claim should be updated. This may feel unsatisfying to a public audience, but it is better than turning ambiguous data into confidence. Abstention is the discipline that keeps the program from becoming self-sealing.

Uncertainty needs an accurate label. Calling unusable evidence merely inconclusive can preserve false hope; calling inconclusive evidence a defeat can overstate what the design learned. The

verdict vocabulary protects readers and propositions alike by distinguishing absence of information from information against a claim.

4.6.3 — Narrowing and Mechanism Retirement

Narrowing can be progressive when it increases risky content. If a broad claim narrows to a specific reference class and then predicts new patterns there, the program has become sharper. It becomes degenerative when narrowing merely protects the claim from adverse results without new exposure.

Mechanism retirement should be written with recognition of what was learned. A failed mechanism may have helped design a better study, clarify a rival, or identify an open bridge. The scientific value is retained even when the mechanism is not.

Narrowing is a successful form of losing. A claim may become smaller because a reference class failed, an instrument did not transport, an endpoint rule worked only in one state, or a signature survived only under specific care contexts. Narrowing preserves what the evidence can bear and removes what it cannot (Popper, 1959; Vehtari, Gelman, & Gabry, 2017).

Mechanism retirement is more specific. A candidate realization may fail while the formal hypothesis remains open. The reserve product might fail to outperform rivals. A queue or horizon mechanism might explain only some dynamics. A predictive-control rival might absorb a proposed signature. Retiring a mechanism should be visible, not hidden inside new language.

Narrowing requires version control. A changed measurement bridge is a changed claim version. A changed endpoint rule is a

changed claim version. A changed reference class is a changed claim version. The program may improve by changing, but it must not call the changed claim the same result.

A narrow claim can be more valuable than a broad vague one. If the program learns that a preterminal dynamic signature appears only under certain endpoint classes and not others, the science has gained structure. The public claim becomes smaller but more contentful. That is progress if it is named honestly.

Mechanism retirement also protects the formal root from premature abandonment. A failed candidate product does not automatically defeat every possible realization. But repeated mechanism retirement without novel risky predictions can exhaust empirical content. The program must not use endless replacement to evade loss.

A narrowed claim should carry its revision history with it. The data, model failures, and rival victories that forced the change remain part of the scientific record. Without that memory, the same risk can return under a new label and appear novel.

4.6.4 — Theory-Level Defeat

The defeat rule also stops category drift. The project may not call the Law a deep law in inspirational prose and a heuristic when the tests fail. If the scientific claim loses its empirical bridge, the public scientific language must change, even if the analytic lens remains useful elsewhere.

Whatever the theory-level verdict, the program can leave better methods for burden detection, care, privacy, and measurement.

Those contributions give future research and practice a useful foundation.

One gate-valid counterexample can reject a declared set-universal [SET-ALL] branch when its stated identification and inferential prerequisites hold. That logical result does not by itself refute a probability-one [P1] branch, which is defined under a declared probability law and may permit exceptional trajectories of probability zero. Separately, program-level scientific retirement can follow when independently motivated bridges repeatedly fail, replacements produce no novel risky prediction, serious rivals dominate the declared conjunction, and the empirical program no longer yields distinct testable consequences (Gneiting & Raftery, 2007).

For retirement through program exhaustion, assess the following conjunction:

Program exhaustion if: bridge failures persist + rivals dominate + replacements add no risky prediction + transport fails under defended reference classes

The verdict follows the declared test and its reference class. A counterexample meeting a universal branch's identification conditions can reject that branch; retirement through program exhaustion follows the cumulative pattern specified above.

The strong latent form has an asymmetry. No finite sample can confirm a universal per-stream equality, but credible counterexamples or repeated failure of the measurable consequence structure can refute or scientifically retire strong versions. The program must keep that asymmetry visible. Support

accumulates conditionally; defeat can be more decisive when prerequisites are valid.

Defeat can have levels. A reference-class prediction can be defeated while another remains open. A candidate mechanism can be defeated while the formal hypothesis remains metaphysically conceivable. A public empirical claim can lose scientific status while a fairness-oriented analysis remains useful as an analytic burden question. The verdict should specify which object is defeated.

The Science of Fairness exists to discover which aspects of PTVC are borne out by the world and how strongly the evidence supports them. Its verdict rules allow support, narrowing, and theory-level defeat when warranted. The final verdict chapter develops the public language for those outcomes; this synopsis gives readers the criteria before decisive results arrive.

Care and participant dignity remain the foundation of the research program through every scientific verdict. Findings guide the hypothesis and its methods; participants retain the same respect and support throughout.

A theory that asks participants, critics, and rival models to accept vulnerability must accept vulnerability itself. Theory-level defeat names the point at which the strong form no longer deserves protection by local repairs. That reciprocity is not humiliation; it is the price of scientific standing.

4.6.5 — Publication and Safety Stops

Publication stops are design constraints, not last-minute public-relations checks. A figure that becomes misleading when cropped

is redesigned. A result sentence that requires several remote caveats is narrowed. A release whose source verification, endpoint adjudication, model recovery, privacy review, or adverse-event record is incomplete waits. Public timing is part of scientific status.

The strongest safe artifact is not timid. It is memorable because the contribution is clear and trustworthy because the condition that limits it travels with the claim. Beauty, compression, and confidence are welcome when they orient attention toward the evidence rather than performing the evidence's work.

Once released, a result remains conditional on its source, endpoint, bridge, recovery, privacy, and adverse-event record. If any of those change, the public claim must be corrected, relabeled, or withdrawn with a visible history. Publication is not a one-time gate; it is an ongoing claim-status obligation (Gneiting & Raftery, 2007; Lakatos, 1978).

Safety stops protect participants and surrounding persons before scientific ambition. Device burden, distress, sleep disruption, family pressure, staff overload, clinical interference, privacy breach, security incident, or consent withdrawal can stop a protocol. A stopped study may disappoint the analysis plan, but it preserves the ethical foundation required for any future science.

Public-artifact stops are also necessary. A diagram, summary page, donor note, website, social excerpt, or media caption can overstate the science even when the paper is careful. Every artifact should pass hostile-cropping review: if a single image or

sentence can be lifted as proof, score, consolation, recruitment pressure, or authority over persons, the artifact fails.

Data stops must distinguish deletion, retirement, retained derived results, restricted access, and model-training permission. Withdrawal does not always erase already aggregated scientific products, but future use must follow the consent and governance rule declared at enrollment. A model cannot inherit permission merely because a dataset existed.

Publication and safety stops should be nonpunitive. They are not signs that a site failed morally. They are the designed brakes of a program that studies vulnerable phenomena. A mature infrastructure expects to pause, correct, retract, and learn.

Publication pressure can convert unresolved problems into costs borne by participants, staff, families, and future readers. Prospective stops prevent that transfer. A study halted for safety, unusable data, or failed recovery may be a more valuable contribution than an overclaimed release.

4.6.6 — Open Theorem and Identification Register

The technical synopsis ends with an explicit register of what remains unsolved. Open theorems, identification limits, measurement gaps, and protocol dependencies receive named statuses, owners, next actions, and defeat conditions rather than being dispersed through repetitive caution (Mayo, 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

An open theorem identifies formal work to complete; an identification question specifies empirical work. The reserve-product relation should be recorded as derived, assumed, or open.

Complete-history coverage, operational equivalence, and prospective forecasting each receive the conditions and evidence appropriate to that task.

The register begins with process class, origin, transformation, units, and reserve meaning. It records whether the additive ledger and multiplicative reserve product are linked by derivation, assumption, or signature only, and which preterminal features survive admissible rescaling.

Endpoint-reserve synchronization is tracked separately. Reserve may not be defined to vanish at the endpoint and then offered as discovered closure; timing, support, and any independence assumptions are fixed without using the tested ledger outcome, and residual dependence is modeled.

Scalarization and observation identifiability share another entry. The protocol may defend a scalar bridge, restrict the class, or report an identified set, while testing whether observation marks distinguish latent contraction from visibility loss caused by device removal, report cessation, sedation, care changes, or staff burden.

Model recovery precedes real-data adjudication. The pipeline must recover candidate generators, serious rivals, mixtures, endpoint conditioning, observation attenuation, medication and frailty effects, and no-closure baselines under realistic missingness and proxy error, using thresholds fixed before real data are scored.

Transport is recorded by population, culture, instrument, endpoint class, care context, and model version. Failure narrows the claim or exposes measurement dependence; it is not explained away. A

transport failure should also identify whether the break occurred in representation, measurement, endpoint construction, or model behavior, because each implies a different successor study.

Negative and null results retain branch-matched status. With valid prerequisites, failure may narrow a class, retire a route, favor a rival, or count against the target under an explicit entailment. When a required protocol or implementation gate fails, the affected evidence is Unusable; when passed gates leave the target undefined or without either identification or a valid bound, the result is No adequate estimate. Keeping these labels separate prevents uncertainty about the method from being misreported as evidence for or against the target.

Public artifacts receive their own status ceiling, owner, hostile-crop review, and correction route. Summaries, diagrams, interviews, donor pages, and excerpts can change the practical meaning of evidence after analysis. The artifact record should retain the exact technical verdict it compresses so that later corrections can trace back to the responsible analysis.

Repeated bridge failure without new risky predictions triggers exhaustion rather than another rescue. Exhaustion is therefore a property of a versioned research program, not a metaphysical proof that every conceivable formulation is false.

Part II — PTVC Formal Core

A scientific hypothesis becomes answerable only when its objects are separated from the instruments, mechanisms, and stories that surround them. Part II performs that separation for Per-Stream Terminal Valence Closure. *The Law of Fairness* supplies the public question and shared vocabulary. This volume specifies the technical meaning and develops the scientific exposition, mechanism-neutral consequences, tests, rivals, and verdict architecture for empirical work.

The Part begins with the bearer. PTVC concerns separately routed maximal qualifying conscious streams, not bodies, legal persons, biographies, data records, households, populations, or archives. That choice creates hard problems of unity, ownership, split, fusion, pause, continuation, and endpoint identity. It also protects the formal claim from cross-person transfer and protects persons from being treated as entries in a social ledger.

The next formal task is phenomenal representation. The primary additive branch uses a finite signed phenomenal measure over the complete strict-pre-endpoint domain. A bipolar branch preserves primitive positive and negative components and their overlap. Generalized Terminal Neutrality admits richer typed terminal functionals, but it remains a separate hypothesis family until a bridge theorem establishes equivalence.

Exactness enters through origin, sign, units, and exposure. A positive change of units can preserve zero; an arbitrary translation usually cannot. A questionnaire midpoint, population mean, adaptation level, or fitted terminal center does not become a phenomenal neutral origin merely because it is convenient. The

science must identify the origin independently or abstain from exact-zero adjudication.

Time is equally structural. A phenomenal contribution is a rate or measure over experienced exposure, not a stack of disconnected scores. Reparameterizing the clock requires the matching transformation of the rate. Delayed observation creates an unknown prefix rather than a new stream, and a study-relative baseline cannot reset the cumulative past.

The endpoint is not the last data point. The truth-level stream endpoint, protocol stopping event, estimated endpoint, physiological event, loss of responsiveness, and observation shutdown are different objects. The primary branch uses a complete strict-pre-endpoint domain. Endpoint-inclusive atoms, jumps, continued experience, and revised routing belong to separately declared branches.

Reference-class closure, probability-one closure, support containment, all-policy closure, all-intervention closure, and compositional closure specify different scopes. Naming the scope directly makes each result and opportunity for extension precise.

Chapter 6 then turns the formal object toward measurement without allowing measurement to usurp it. Representation, route state, observation, estimand, estimator, decision rule, and verdict occupy different layers. An estimator can be reliable and still answer the wrong question. A rich set of sensors can improve prediction while leaving the neutral origin or complete-stream history unidentified.

Mixed valence and partial identification receive full treatment because false precision would be especially damaging here. Some data may support only vectors, partial orders, or identified sets over scalarization weights. The appropriate output may be a range, a branch-specific result, or No adequate estimate. Scientific seriousness is not measured by how often the program produces one number.

The observation kernel includes response, context, memory, communication, devices, missingness, and gain. Near endpoints, the probability and quality of observation can change exactly when the theory becomes most vulnerable. Silence, immobility, sensor dropout, sedation, and clinical workflow can manufacture apparent calm. Those possibilities belong inside the model, not in a final limitations paragraph.

Chapter 7 asks what exact closure would imply before any mechanism is chosen. Prefix feasibility, remaining directional capacity, completion sets, endpoint support, and boundary accessibility create necessary conditions that candidate routes must satisfy. The equal-and-opposite net remainder is a conditional mathematical requirement, not a known personal forecast, moral desert, or a safe means of obtaining a chosen reward.

The random-endpoint problem supplies one of the program's hardest tests. Under a prospectively declared baseline law with independent diffuse timing, almost-sure exact zero at the endpoint imposes severe restrictions on the pre-endpoint ledger. State-coupled timing, first-hitting termination, endpoint atoms, singular support, continued experience, and routing change remain live

branches, but each must be declared and tested rather than invoked after failure.

Candidate routes are then allowed to compete. Native dynamics, conditioned bridges, admissible-history selectors, interventions, observation selection, global constraints, local controllers, and reserve coupling can produce similar terminal-looking trajectories for different reasons. Equal-information and equal-scoring comparison is therefore a scientific obligation, not a gesture of politeness toward rivals.

The reserve-coupled route is developed through a declining nonnegative reserve, a signed arithmetic reference process, a positive scale constant, an explicit product rule, linked drift and innovation restrictions, and observation-noise analysis. These features define a candidate realization with specific predictions and decision criteria.

This Part develops the hypothesis from its compact statement into an explicit map of definitions, assumptions, and consequences. That map connects the formal proposal to the measurement, experimental, and comparative work in the chapters that follow.

The formal core connects a bearer to a domain, a domain to a signed measure, a signed measure to an observation model, and observations to an estimand, estimator, and verdict. Part II specifies the transitions among those objects. Their explicit relationships make exactness usable across the participating disciplines (Borsboom, 2005; Manski, 2003).

The same architecture also clarifies what remains valuable under loss. If additive PTV C proves too coarse, the bipolar branch may

preserve component structure. If scalarity fails, identified sets may replace point estimates. If the reserve route loses, reachability, endpoint-support analysis, and equal-information rival comparison remain useful. A rigorous formal Part is therefore not a fortress built around one conclusion. It is a map of how the research program can learn without hiding what failed (Mayo, 2018; Lakatos, 1978; National Academies of Sciences, Engineering, and Medicine, 2019).

By the end of the Part, the reader should be able to audit any later formula or diagram by asking six questions: Which branch is active? What object does each symbol denote? Which assumptions are definitions and which are empirical? What observation bridge is required? Which rival receives the same information? What result would force narrowing, abstention, or retirement? Those questions turn a striking terminal claim into an inspectable scientific program and prepare the transition from formal architecture to the science of lived experience.

Chapter 5 — PTVC: Hypothesis, Branches, and Scope

A formal hypothesis is not one sentence. It is a dependency architecture. The terminal equality may be memorable, but its scientific content depends on the bearer, representation, domain, origin, sign, units, endpoint, scope, and change rules that give the equation a determinate meaning.

This chapter specifies PTVC's exact technical meaning while preserving the public question and vocabulary introduced in *The Law of Fairness*. The following sections define the objects needed for scientific exposition, analysis, testing, and adjudication. When a successor branch, endpoint, or representation is proposed, the change is named, versioned, and propagated to every affected theorem, protocol, figure, and verdict.

The first object is the qualifying stream. PTVC is intrastream and nontransferable. Bodies, biographies, legal persons, records, families, and populations may supply observations or contexts, but they do not replace stream ownership. Split, fusion, pause, and continuation remain scientific questions whose answers determine the ledger's domain.

The second object is the finite signed phenomenal measure. On the primary additive branch, positive and negative Jordan magnitudes are finite and equal at the complete strict-pre-endpoint boundary. The equality constrains one net relation while leaving gross variation, temporal order, duration, agency, welfare, rights, and biography radically unequal.

The third object is the bipolar branch. Mixed valence can contain positive and negative components at once, and a scalar net residual may erase coactivation and overlap. The bipolar

representation therefore preserves component totals and temporal structure long enough for scalarization to become a testable assumption rather than an automatic subtraction.

The fourth object is Generalized Terminal Neutrality. A richer terminal functional can be useful when the science does not support one scalar representation. Yet generality cannot become an escape hatch. A prespecified functional must expose a risky neutral set, and equivalence to additive PTVC requires a bridge theorem.

The fifth object is scope. A formal target can be proposed for one reference class, almost surely under one probability law, throughout one support, under every policy, under every intervention, or compositionally across streams. Those are different claims with different evidence burdens.

The sixth object is past-fixity. Later evidence can change what researchers know about earlier experience; it cannot change the earlier experience itself. Future complementarity is about a future signed increment, not retroactive moral or phenomenal repricing.

The seventh object is endpoint domain. The truth endpoint is not the last measurement, first zero crossing, protocol stop, or physiological event by definition. The primary branch uses the strict-pre-endpoint domain, while endpoint atoms and continued experience require separate branches.

These distinctions make the formal statement harder to support and easier to respect. They also make failure informative. Origin may be unidentified, scalarity may fail, routing may remain

ambiguous, endpoint support may exclude the route, or the reference class may narrow.

The chapter will use mathematics as a transparency device. Every formula has a role: received target, derived identity, branch definition, non-implication, decomposition, or change rule. No display is allowed to become evidence through visual polish.

Concrete examples show how pooled means, centered scales, mixed states, delayed entry, and terminal jumps affect the mathematical interpretation. Each example identifies the assumption being examined and the resulting consequence.

Critics and builders should leave with precise questions for formal and empirical work. The branch labels identify whether a result concerns additive PTVC, a different terminal formulation, or a candidate realization.

The chapter develops one equality into a family of precisely defined claims and tests. That structure gives each version of the hypothesis a clear scientific question.

Formalization earns scientific value by restricting movement, not by making the page look authoritative. A definition fixes what later evidence is about. A branch declaration records which alternatives are live. A bridge theorem states what must be proved before two formulations may be treated as equivalent. A modal quantifier tells the reader whether the claim concerns one realized stream, almost every stream under a law, every admissible continuation, or every member of a declared class. When those roles are separated, notation becomes a map of vulnerability rather than a shield against criticism (Suppes, 1957; Mayo, 2018).

Consider three studies that all report a terminal estimate near zero. One represents experience with a scalar signed measure over a complete strict-pre-endpoint domain. Another records separate positive and negative components whose overlap is scientifically meaningful. A third uses a generalized terminal functional that is invariant only under a specified family of transformations. The numbers can look similar while the scientific claims differ. Chapter 5 prevents the apparent agreement from erasing branch, representation, domain, and bridge obligations.

The chapter also establishes a discipline of counterexamples. A one-sign stream pressures the origin and sign conventions. Duplicate ownership shows why a formally additive total can be invalid even when its arithmetic is flawless. Overlapping positive and negative components expose information hidden by a net residual. Endpoint-inclusive contributions distinguish a left-limit claim from a terminal-jump claim. Nonadditive path functionals show why an attractive scalar equation cannot be assumed to cover every admissible phenomenology. Each counterexample removes an illicit shortcut and leaves a more inspectable object. Counterexamples are most valuable when they force an explicit branch decision, because that converts a conceptual objection into a downstream change that can itself be tested.

This architecture is deliberately outcome-resilient. If the additive branch narrows, the bipolar branch may preserve component science. If scalarity fails, a generalized or set-valued formulation may remain testable. If complete-domain exposure cannot be identified, local additive structure may still support bounded studies. The scientific loss is therefore recorded at the smallest

warranted level. That is more demanding than declaring the whole program alive or dead, and more useful to the fields that must measure, compare, and reproduce its components (Manski, 2003; National Academies of Sciences, Engineering, and Medicine, 2019). This modularity also lets repeated losses accumulate within one dependency until retirement is warranted without erasing findings that remain valid elsewhere.

The chapter ends with change control because formal exactness is temporal as well as logical. A future revision must identify the object that changed, the authority for the change, the equations and protocols affected, and the conclusions that no longer transport. A research program may improve its definitions, but it may not retroactively claim that every earlier test addressed the successor object. Versioned formalism is how an ambitious hypothesis remains corrigible without becoming movable. A successor version should therefore begin with a delta statement naming the changed definition before any new evidence is compared.

5.1 — Truth-Domain Streams, Ownership, and Maximality

The first formal task is not the terminal equation but the carrier to which it could apply. A complete-history claim needs a prospectively declared qualifying bearer, an ownership rule for phenomenal events, a continuation relation, and a maximality convention. Without those objects, the same experience can be counted twice, omitted, or reassigned after the outcome is known (Bayne, 2010; Bayne & Chalmers, 2003).

This section treats stream identity as a truth-domain problem rather than a synonym for organism, brain, biography, account, or data record. Those objects may supply evidence, but none automatically determines which phenomenal events belong to one separately routed stream. The protocol must preserve that distinction even in ordinary cases, because fission, fusion, pause, copying, restoration, and nested candidates expose any hidden ambiguity.

Ownership must enter exactly once. The formal model permits either a domain already restricted to stream-owned events or a common exposure base with one explicit allocation or unity gate. Applying both can apply the ownership restriction twice and distort the represented contribution. Leaving residual ownership unclassified would turn an exact equality into an artifact of selective accounting.

Maximality is therefore relational. It means maximal under the declared continuation-and-ownership rule, not the largest organismic, anatomical, legal, social, or computational aggregate.

The six subsections below move from bearer and routing to event domains, unity gates, difficult continuations, and the division between root ownership and scientific use (Bayne, 2010; Bayne & Chalmers, 2003).

Table 5.1. Truth-domain objects and first-use rules.

Object	Status in this chapter	First rule for use
Unified conscious stream	Received bearer	Do not substitute body, biography, legal person, public record, or social identity.
$r_i(t)$	Latent composite contribution rate on an absolutely continuous chart	Define from realized density and one declared exposure convention; withdraw pointwise notation when the measure has unmatched atoms or singular mass.
$g_{0,i}(t)q_i^{\text{exp}}(t)$	Declared exposure-density factors	Encode unity or ownership at most once and preserve units under time reparameterization.
$L_i^{\text{phen}}(t)$	Latent phenomenological ledger	Formal target, not an observed total or public record.
T_i^*	Received endpoint concept	truth-level stream endpoint, not ledger crossing, body death, or convenience time.
$Z_i^* = 0$	Strong latent hypothesis	Exact closure claim, not a public empirical finding.
$L_i^{\text{rc}}(t) = c^{\text{LU}} U_i(t) Y_i(t)$	Candidate realization	Candidate route with bridge, dimensional, endpoint-coincidence, and retirement obligations; not the primitive definition.

Object	Status in this chapter	First rule for use
$\hat{L}(t)$	Estimator family	Belongs downstream; never substitute for $L(t)$.

5.1.1 — Qualifying Bearers and Unified Streams

The unified conscious stream is the proposed bearer of the received Law. It is not the body considered biologically, the person considered legally, the biography considered narratively, or the record considered administratively. The formal claim is attached to the stream of unified felt experience, because the ledger is supposed to integrate what is lived, not what is externally attributed (Bayne, 2010; Bayne & Chalmers, 2003).

This distinction matters even before hard cases appear. A body may continue while a qualifying conscious stream has irreversibly ended. A legal person may remain while consciousness is absent or profoundly uncertain. A biography may be socially intact while lived unity is interrupted, fragmented, or inaccessible. A record may be complete while the inner stream remains unknown.

The chapter receives the stream as a formal target, not as an established scientific object. Later chapters must ask how unity could be detected, how pauses and returns should be handled, how partition should be adjudicated, and how nonhuman or artificial candidates should be treated. Those questions are downstream. Here the only safe move is to state the object and prevent substitutions.

The fairness lens asks who bears the burden when a bearer is misnamed. If the body is treated as the bearer, terminal research

may confuse physiology with experience. If biography is treated as the bearer, memory and public identity may overrule present consciousness. If a record is treated as the bearer, the archive can become a false soul. The stream concept is a safeguard against those transfers.

A scientist may object that the stream is not directly observable. That objection is correct and useful. The received bearer is latent, and its adjudication requires proxies, confidence classes, edge-state rules, and abstention categories. A latent bearer can still be a scientific target if the book states what would count as evidence about it and what would force narrowing.

The unified conscious stream is therefore a received reference point, not an easy empirical object. Every later measurement, endpoint, and verdict must state whether it bears on the stream itself, a proxy for it, or a different object.

A useful technical test for this subsection is substitution pressure. Replace the phrase unified conscious stream with body, brain, person, biography, legal identity, or record, and ask whether the formal claim would still be the same claim. It would not. Each substitution changes the bearer and therefore changes the reference class of every later equation. The received object may prove hard to adjudicate, but difficulty of adjudication is not permission to substitute an easier object without declaring a formal change.

5.1.2 — Ownership, Routing, and Nonduplication

Ownership is the rule that assigns each phenomenal event to one declared stream-history branch. The assignment is not inferred

from where a datum is stored or which body produced it. It is a truth-level relation that the protocol represents fallibly. A sensor trace, testimony, or medical record may be evidence about ownership without becoming ownership itself (Bayne, 2010; Gazzaniga, 2000).

Two formally admissible encodings are especially important. On a restricted-exposure branch, the phenomenal measure is already limited to events owned by the stream. On a common-base branch, one frozen allocation or unity weight assigns exposure and signed components. The same ownership relation may not be multiplied into the exposure, valence rate, gate, and quadrature weights several times.

Nonduplication becomes decisive when candidates overlap. A component stream and a whole-system stream may both look plausible; two restored copies may inherit the same memories; a split history may support several continuations. The protocol must either give distinct ownership, apportion events under a declared rule, or retain unresolved typed slack. It may not let the same event close several ledgers.

Residual ownership cannot be hidden in one signed remainder. Unresolved positive variation, negative variation, exposure, atoms, and singular mass have different consequences. They remain separate coordinates with explicit bounds. When those coordinates are too wide for adjudication, the scientifically honest result is Model-ambiguous or Inconclusive evidence rather than a conveniently balanced total.

Routing also constrains maximality. A qualifying instance is maximal only relative to a prospectively fixed continuation relation that preserves event ownership. Maximality does not mean the spatially largest system, the longest biography, the most detailed archive, or the most socially recognized identity. Those substitutions change the carrier and therefore change the claim (Parfit, 1984; Schechtman, 1996).

Fission, fusion, copying, and restoration may redistribute a fixed token or history weight under a declared branch. The routing relation specifies those historical allocations and distinguishes them from the successor streams' future phenomenal contributions. Informational continuity and future accumulation are modeled as separate quantities.

A useful audit case begins with one recorded episode that is accessible to two candidate continuations. If both candidates receive the full episode, ownership is duplicated. If neither receives it, ownership is incomplete. If the allocation is chosen after terminal residuals are inspected, the result is selected. Only a frozen allocation, explicit unresolved set, or justified quotient avoids those failures.

Scientific use therefore requires an ownership register: carrier, branch, allocation rule, unresolved coordinates, continuation relation, adjudicator, version, and abstention condition. The register is not a dossier about a person. It is a model-control object that prevents the program from manufacturing closure by moving events between streams.

5.1.3 — Maximal Event Domains

The phenomenal domain is the complete set of stream-owned experiential events on which the signed measure is defined. It is not merely a calendar interval, a sequence of survey rows, or the portion of life that left a record. A complete-history hypothesis cannot substitute observation coverage for the truth domain without changing its quantifier (Folland, 1999; Kallenberg, 2002).

A maximal event domain must exhaust the declared stream under the chosen routing rule. It includes ordinary intervals, isolated events when the measure has atoms, and any singular component admitted by the representation. A density along clock time is one convenient branch, not the definition of the domain itself.

The distinction matters because event structure can be richer than sampled time. Experiences may cluster around transitions, occur during intervals with weak reportability, or be represented by measures that are not absolutely continuous with respect to the chosen clock. Treating every contribution as a smooth time series would erase those possibilities before measurement begins.

Delayed entry does not restart the domain. Enrollment, device activation, first diary response, or hospital admission marks the beginning of an observation window. The pre-enrollment phenomenal domain remains part of the complete target, even when it can be represented only by bounds or remains unidentified.

Domain selection can manufacture neutrality. Excluding hard-to-observe intervals, dropping positive or negative atoms, or ending

the domain at the first convenient zero changes the object being summed. A valid protocol freezes inclusion rules without access to terminal residual, endpoint surprise, or completion-model output.

Nonvacuity is equally important. A qualifying stream must have a nonempty declared phenomenal domain unless a zero-exposure class is separately defined as degenerate. Otherwise exact closure could be satisfied by an empty object that says nothing about lived experience.

A practical domain manifest should list the event carrier, sigma-field or generating algebra, exposure measure, ownership restriction, admissible atoms and singular components, entry and exhaustion rule, and unresolved mass. Each line names a possible failure point rather than decorating the notation.

The domain survives only if its meaning transports from formal statement to simulation, protocol, analysis, and public explanation. A study that tests a sampled-window estimand may be valuable, but it must not be reported as direct adjudication of the complete maximal domain.

5.1.4 — Unity Gating and Qualifying Intervals

A unity gate marks when and to what degree the declared bearer qualifies under the chosen stream criterion. It is an inclusion or weighting object, not a judgment of personhood, worth, care priority, sincerity, or social standing. Low confidence about unity calls for wider uncertainty or abstention, never reduced dignity (Bayne & Chalmers, 2003; Bayne, 2010).

Hard gating is mathematically simple: an interval is included or excluded. Consciousness science rarely offers such clean boundaries. Sleep, anesthesia, delirium, fragmented access, covert awareness, and rebinding may require a family of thresholds or a soft gate whose values express graded evidential support rather than graded human value.

The gate must be identified independently of the terminal residual. If the same valence estimator both decides whether an interval counts and supplies the value being accumulated, unfavorable intervals can disappear by construction. Unity evidence and valence evidence may share data sources, but their algebraic roles and validation targets remain distinct.

Apply §5.1.2's nonduplication rule to unity gating. On a restricted-exposure branch, ownership is already in the domain; on a common-base branch, one gate may supply allocation. A second ownership multiplier in the rate or quadrature weights would duplicate the same relation.

Qualifying intervals need endpoint discipline. A gap in reportability is not automatically a pause in experience, and a pause is not automatically the end of a stream. The gate family should therefore be estimated with rules for return, reversible loss of access, threshold hysteresis, and uncertainty around the first and last qualifying times.

Gate uncertainty propagates into every downstream object. It changes interval contributions, cumulative residuals, endpoint windows, and the set of compatible histories. A protocol that reports a point ledger while treating a disputed gate as certain

understates uncertainty at the very layer that determines what enters the sum.

A useful stress test compares two pipelines on the same synthetic history. One uses an independently calibrated unity gate; the other tunes the gate after inspecting terminal residuals. Similar apparent closure in the second pipeline is not corroboration. It is a demonstration of how selection can imitate the target.

The correct output may be a set of ledger paths indexed by admissible gate families, or No adequate estimate if the target cannot be identified or validly bounded. That outcome is scientifically informative because it locates the failure at stream qualification rather than mislabeling uncertainty as neutral experience.

5.1.5 — Split, Fusion, Pause, and Continuation

Split, fusion, pause, and continuation are not exotic decorations around a simple bearer. They determine how the universal carrier is counted. A probability-one closure statement is meaningful only after the reference class, routing branch, maximal instances, and event ownership have been fixed (Parfit, 1984; Schechtman, 1996; Bayne, 2010).

The next display states the branch-specific probability-one target under a declared probability law and carrier event. It is a formal claim, not an observed frequency and not a substitute for routing decisions.

$$Q^b(FPTVC^b) = 1$$

A pause differs from an endpoint because continuation remains admissible. Sleep, reversible anesthesia, temporary loss of report, and interrupted access may preserve one stream under a declared relation, or they may remain ambiguous. Biological continuity alone does not settle the question, and observation shutdown cannot decide it by default.

Fission requires the protocol to state what happens to the parent instance and how successor ownership begins. Fusion requires the corresponding rule for predecessor instances and the fused continuation. The science volume does not obtain permission to trade residuals across branches merely because they share memory, tissue, code, or causal history.

Copying and restoration create a related trap. Informational sameness does not prove numerical identity of phenomenal ownership. A restored system may be a continuation, a new branch, or an unresolved candidate under the declared ontology. The result must remain conditional on that ontology rather than selected by whichever routing closes most cleanly.

Return after an apparent ending is a direct falsifier of a badly defined protocol endpoint. Resuscitation, renewed responsiveness, recovered access, or later evidence of continued experience can show that the study stopped too early. The estimate or adjudicated classification of the truth-level endpoint is revised only through the protocol-fixed adjudication rule, not through retrospective convenience.

Difficult routing cases should remain in the analysis as branch-indexed objects or explicit unresolved sets. Dropping them

because they are complicated creates survivorship in the formal carrier. Pooling them under one presumed identity can be equally damaging. Both errors can produce false confidence about closure.

Routing uncertainty is itself a scientific result. When continuation, ownership, or maximality cannot be adjudicated, the program records Model-ambiguous or No adequate estimate and preserves the case for later theory revision. Ambiguity is not permission to choose the branch with the preferred terminal value.

5.1.6 — Root Ownership and Scientific Use

This chapter specifies the formal object that the rest of the book investigates. Representation, measurement, inference, candidate mechanisms, rival comparison, replication, and revision all refer to this stated target. The bearer, domain, endpoint, currency, and quantifiers remain explicit so that readers can follow each derivation and identify what a study tests (Folland, 1999; Kallenberg, 2002).

That boundary is a defense against semantic drift, not a ban on scientific criticism. Research may show that a branch is too coarse, a bridge is unidentifiable, or an endpoint class is empirically inadequate. Such findings can justify a formal change request, a successor object, a narrowed claim, or a withdrawn route. They cannot justify changing the target silently while preserving its public name.

A received-object concordance should accompany every operational descendant. For each symbol and branch, it should identify the truth-level meaning, the protocol approximation, the

observation bridge, the estimand, the estimator if one exists, and the conditions that force abstention. The concordance makes it possible to locate a failure without absorbing every discrepancy into generic model misspecification.

Independent teams should be able to reconstruct the same target from the frozen packet. If one team treats biological death as the truth endpoint, another uses the last report, and another defines the endpoint as first hitting zero, they are not applying different methods to one hypothesis. They are studying different endpoint constructions, one of which may make closure tautological.

Operational descendants should remain traceable to the same formal target. Public explanations may simplify the notation, but technical derivations and empirical tests must restore the assumptions needed for the object actually under study.

Scientific creativity begins from this fixed point. Researchers may propose vector extensions, endpoint-inclusive branches, nonadditive functionals, revised routing, or new candidate mechanisms. Each proposal is welcome as a typed successor with its own identifiers, bridges, predictions, and failure conditions. Visible succession permits genuine discovery while preventing the current hypothesis from inheriting evidence earned by a different object. Successor proposals should also state which earlier counterexamples they resolve and which new counterexamples they expose themselves to.

The practical standard is simple: receive first, test honestly, and revise visibly. A stable target lets critics challenge assumptions, reference classes, or bridges directly instead of arguing against a

moving collection of metaphors. It also lets a failed module teach the program exactly what failed without forcing either dogmatic preservation or opportunistic rescue (Cronbach & Meehl, 1955; Lakatos, 1978; National Academies of Sciences, Engineering, and Medicine, 2019). That continuity also lets later datasets remain comparable, because a revision can be evaluated against the same frozen target rather than retroactively redefining earlier evidence. That stability makes disagreement cumulative: later teams can argue about evidence against the same object instead of unknowingly testing different targets.

5.2 — Finite Signed Phenomenal Measure and Exposure

This section defines the primary signed measure first and introduces density or rate notation only on branches where absolute continuity justifies it. The distinction keeps measure-level claims separate from later physical-time representations.

Time-indexed ledger and rate notation require a declared exposure chart; §5.2.3 gives the absolute-continuity conditions under which a physical-time contribution rate exists.

Write $r_i(t)$ for the declared composite contribution rate $g_{\theta,i}(t)q_i^{\text{exp}}(t)v_i^{\text{real}}(t)$. Physical time, unity-weighted exposure, subjective time, and protocol time are not interchangeable. Units and rates must transform with the clock, and ownership may enter only once. A polished cumulative curve cannot repair a missing exposure convention.

The display below is a chart identity on the absolutely continuous branch, not the primitive definition of the signed measure and not an empirical finding. The final warning concerns path class: an ordinary integral of a regular rate has finite variation, whereas later candidate routes may contain diffusion or jumps. Any bridge between those objects requires its own assumptions, units, and testable consequences.

$$\begin{aligned} r_i(t) &:= g_{\theta,i}(t)q_i^{\text{exp}}(t)v_i^{\text{real}}(t), \\ L_i^{\text{phen}}(t) &= L_i^{\text{phen}}(S_i^\theta-) + \int [S_i^\theta, t]r_i(s)ds \end{aligned}$$

5.2.1 — Signed Phenomenal Measure

On the primary branch, the latent phenomenal object is a finite signed measure v_i on the complete measurable domain (Ω_i, Σ_i) of

one separately routed maximal qualifying stream. The measure assigns signed contribution to phenomenal event sets. It is not initially defined as a questionnaire score, a mood state, or a regularly sampled time series (Folland, 1999; Kallenberg, 2002).

Countable additivity is not licensed by arithmetic convenience. An empirical representation needs a premeasure or additive-conjoint gate: disjoint unions must preserve the relevant ordering and magnitude relations on a generating algebra, and extension to a signed measure must be coherent on the declared domain. Failure at this gate favors an ordinal, vector, bipolar, or nonadditive branch rather than a manufactured measure (Luce & Tukey, 1964; Krantz et al., 1971; Folland, 1999).

The measure-theoretic formulation separates event ownership from clock choice. Ω_i carries the stream-owned phenomenal events; a later time chart pushes the measure onto an ordered axis. This makes it possible to distinguish a truth-level contribution from the particular cadence, window, or device used to observe it.

Density and physical-time rate notation are available only on the absolutely continuous subbranches specified in §5.2.3; atoms or singular components remain measure-level objects. Realized density, conditional predictor, proxy, and estimator remain notationally distinct.

The cumulative ledger $L_i(t)$ is a prefix evaluation of v_i under the declared chart. It inherits its meaning from the measure and domain. It is not a separate metaphysical substance and not a public or private record anyone can inspect directly. Its value changes only through stream-owned phenomenal contribution under the chosen representation.

Finiteness means finite total variation over the complete domain. It rules out undefined subtraction of infinite positive and negative masses. The condition is mathematically substantive and empirically demanding; it cannot be presumed merely because every finite study produces finite numbers.

Exposure gauge freedom introduces another distinction. Different factorizations into density and exposure can represent the same signed measure. The terminal measure and its Jordan decomposition may remain invariant while duration, intensity, and normalized diagnostics change. A protocol must record the admissible gauge family rather than let terminal fit choose it.

A scientific implementation should therefore maintain a measure packet: domain, sigma-field or generating algebra, ownership convention, exposure measure, density status, atomic or singular components, total-variation condition, and observation bridge. That packet exposes the assumptions on which every later equation depends.

5.2.2 — Positive and Negative Jordan Magnitudes

Every finite signed measure has a Jordan decomposition into mutually singular positive and negative measures.

Their total masses V_i^+ and V_i^- preserve the two directions of phenomenal contribution, while their sum is gross variation and their difference is the terminal signed residual (Folland, 1999).

The decomposition is not a moral partition into deserved and undeserved experience. It is a mathematical separation under a declared sign convention. Positive and negative components

belong to one stream and cannot be transferred between persons, branches, groups, or generations.

$$v_i = v_i^+ - v_i^-, v_i^+ \perp v_i^-, V_i^+ := v_i^+(\Omega_i), V_i^- := v_i^-(\Omega_i)$$

Gross variation is essential because net zero can conceal arbitrarily large experience. A stream with $V_i^+=1$ and $V_i^-=1$ shares the same residual as a stream with $V_i^+=100$ and $V_i^-=100$, yet their total variation differs by two orders of magnitude. Exact closure would therefore never imply equal lives, mild lives, or adequate welfare.

Mutual singularity in the Jordan theorem concerns the measure decomposition, not the impossibility of psychologically mixed episodes. If positive and negative valence can co-occur within the same phenomenal event, the bipolar branch in Section 5.4 preserves overlap explicitly. The additive signed measure and the bipolar component representation are related but not automatically equivalent.

Estimating only the signed residual is insufficient. Measurement error can make two large uncertain components look like a precise small difference. Separate component estimates, covariance, calibration, and identified sets are needed before a near-zero residual can be interpreted as compatible with equal Jordan magnitudes (Cacioppo & Berntson, 1994; Manski, 2003; Imbens & Manski, 2004).

Finiteness must be assessed componentwise. The expression $V_i^+=V_i^-$ is not meaningful on the primary branch if both are infinite. A finite observation window cannot establish complete-domain finiteness, so the claim remains conditional on the domain and bridge assumptions that make the full masses well defined.

A useful simulation contrasts three worlds with the same observed terminal residual: genuinely small component masses, very large cancelling masses, and a miscentered instrument that forces cancellation. A model that reports all three as the same result has not recovered the target object.

Jordan magnitudes thus provide the minimum anatomy of additive PTVC. They protect the theory from the false idea that zero means nothing happened, and they protect readers from the equally false idea that a terminal equality could excuse intense suffering, unequal conditions, or failures of care.

5.2.3 — Experienced Exposure and Temporal Rate

A signed measure becomes a temporal rate only after the program chooses an experienced-exposure measure and establishes absolute continuity on the relevant branch. The realized value density is a Radon-Nikodym derivative relative to exposure; multiplying it by the declared exposure density produces a physical-time contribution rate. Neither object is a mood level with time appended (Folland, 1999; Billingsley, 1999).

Physical time is one possible chart, but phenomenal exposure may depend on subjective duration, unity, or another declared clock. The science does not gain freedom to choose whichever chart closes best. The chart, exposure anchor, admissible gauge family, and transformation rule must be frozen before decisive terminal analysis.

$$[v_i^{\text{real}}] = V \cdot E^{-1}, [g_{\theta,i} q_i^{\text{exp}}] = E \cdot T^{-1}, [r_i] = V \cdot T^{-1}, [L_i^{\text{phen}}] = V$$

Apply the nonduplication rule in §5.1.2: ownership or unity enters the exposure architecture once. Any additional weighting must

represent a distinct scientific quantity rather than re-encode the same allocation.

The dimensional audit separates phenomenal value V , exposure E , and chart time T . The realized density carries V per E ; the exposure density carries E per T ; their product r_i carries V per T ; integration returns V . Checking those relationships helps identify the conversions a proposed measurement bridge requires.

Irregular observation cadence is not the same as irregular phenomenal exposure. Experience can continue between prompts, while a device can sample densely during quiet periods and sparsely during crises. Interval contribution estimators must therefore separate the latent exposure process from the marked observation process (Stone et al., 2007; Shiffman et al., 2008).

Subjective time adds a further identification problem. Retrospective duration, prospective time judgment, memory density, and momentary temporal experience can diverge. A study may compare these channels, but it cannot silently choose one as the complete clock without a validation and transport argument.

Gauge changes can preserve the signed measure while changing density and exposure separately. That invariance is useful: it shows which conclusions depend on the measure and which depend on a particular factorization into intensity and time. Reserve- or horizon-normalized diagnostics require their own anchors because they are not automatically gauge invariant.

The practical result is an exposure register beside every ledger estimate. It records the clock, unit, unity convention, ownership branch, cadence, interpolation rule, unobserved exposure, admissible gauge family, and uncertainty. Without that register,

an accumulated number may be dimensionally neat yet scientifically uninterpretable.

5.2.4 — Dimensions, Units, and Reparameterization

Time reparameterization is where notation can quietly manufacture invariance. Suppose a new clock u is used and physical time is written as $t=\tau(u)$, with τ increasing and differentiable. The ledger does not keep the same contribution rate by magic. The time element changes, and the rate must carry the Jacobian (Folland, 1999; Kallenberg, 2002).

The correct change of variables is:

$$L_i^{\text{phen}}(\tau(u_2)) - L_i^{\text{phen}}(\tau(u_1)) = \int_{u_1}^{u_2} r_i(\tau(u))\tau'(u)du$$

If one wants a rate expressed in u -time, the transformed rate must satisfy:

$$r_{i,u}(u) = r_{i,t}(\tau(u))\tau'(u)$$

Endpoint-normalized time is useful precisely because it can align terminal windows, but that convenience creates a leakage risk. Mapping every history to a unit interval uses endpoint information to define the chart. The transformation, information timing, and Jacobian must therefore be declared before confirmatory analysis; otherwise a clean terminal pattern may be partly manufactured by the clock.

Time normalization can be legitimate. It may be required for comparing terminal windows of different lengths, estimating reserve-space behavior, or displaying model recovery. The rule is that the transformation must be declared before the confirmatory analysis, the rate must carry the Jacobian, and any endpoint

information used to define the clock must be separated from claims of prospective prediction.

Comparability also has a human consequence. A participant's final week should not be compressed into another participant's final hour merely to obtain visual symmetry. Any changed clock must state where the transformation enters the rate, uncertainty, interpretation, and claim status.

More generally, a noninjective time map can preserve the net signed residual while reducing gross variation inside time fibers. Equality of gross variation requires an injective chart, sign-pure fibers, or another proved no-cancellation condition. Protocols should register temporal-resolution sensitivity and retain the event-space branch when cancellation cannot be bounded (Folland, 1999).

Transport remains a separate claim. A time representation validated in one species, developmental stage, clinical state, language, or endpoint class may fail in another. Each transport claim requires its own invariance tests, rival comparison, and possibility of reference-class narrowing.

5.2.5 — Finiteness and Complete Domain

An ordinary pathwise integral of a sufficiently regular contribution rate has finite variation on finite intervals. That fact matters because the reserve-coupled realization later uses stochastic dynamics. A finite-variation process and a diffusion-bearing semimartingale can be related under a declared bridge, but they are not automatically the same process (Folland, 1999; Protter, 2005).

The clean comparison is:

$$[L_i^{\text{phen}}]_t = 0$$

Here $[L_i^{\text{phen}}]_t$ denotes quadratic variation. The statement is a path-class implication of the ordinary integral chart under regularity conditions, not a new empirical result. If the root ledger is interpreted as an ordinary pathwise integral, diffusion-like quadratic variation requires additional structure rather than a change of typography.

This creates a bridge obligation rather than a disaster. One route is to treat the product realization as a reduced-form consequence model for terminal dynamics, not as a pointwise identity with the phenomenological integral. Another route is to enrich the formal ledger with stochastic integration. A third is to define an approximation relationship between the objects. Each route requires visible change control.

Relating $L(t)=U(t)Y(t)$ to the pathwise integral requires a clear account of their process classes. If the product contains a diffusion term, specify whether it models the same object, a projection, an estimator, or an auxiliary consequence. That classification determines the mathematical bridge to establish.

5.2.6 — Temporal Additivity as a Component Claim

Temporal additivity states that the signed phenomenal measure assigned to a disjoint union equals the sum of its restrictions to the disjoint parts. In the primary branch, this is not merely notation. It is a representation claim about how experienced contributions compose across the complete strict-pre-endpoint domain, and it can fail even when a convenient numerical

summary remains available (Stone et al., 2007; Shiffman et al., 2008).

A valid cut-point analysis must prospectively partition the same truth domain into prefix and suffix sets, assign any cut-time atom exactly once, and preserve the same representation, neutral origin, exposure clock, unit, ownership rule, routing, and endpoint convention on both sides. Changing any of those objects across the cut turns an additivity test into a comparison of different ledgers.

On the absolutely continuous chart, the received additive branch yields the increment identity displayed below; the general measure branch uses the corresponding signed-measure increment, including any atoms or singular component. For a fixed terminal residual, prefix and suffix contributions sum to that residual whether or not it is zero.

$$L_i^{\text{phen}}(t_2) = L_i^{\text{phen}}(t_1) + \int_{t_1}^{t_2} r_i(s) ds$$

Across varying cuts, a suffix-versus-prefix slope of minus one is an additivity diagnostic shared by any conserved residual, including a nonzero residual. It supplies no independent vote for PTVC. The PTVC-bearing estimand is the independently adjudicated terminal residual and its fixed zero target, not the generic fact that prefix and suffix restrictions sum to a constant.

Observation-level non-tautology is essential. The measured suffix must come from post-cut observations or from a separately validated terminal-remainder bridge. It may not be defined by subtracting the measured prefix from the zero target under test.

That subtraction would manufacture complementarity and erase the empirical burden.

A bounded laboratory study can test local refinement and complementarity. Complete-stream inference additionally requires justified prefix and suffix information, domain coverage, and an independently adjudicated endpoint. The report identifies which of these objects the design addresses. This separation prevents a successful local additivity test from inheriting whole-stream authority it was never designed to establish.

Sampling design must respect the temporal object. Irregular cadence, state-dependent prompting, missing intervals, high-arousal oversampling, and endpoint-related observation loss can distort both sides of the partition. Duration and exposure weighting must be explicit, and uncertainty from unobserved intervals must travel into the residual rather than disappear through equal-weight scoring. Partition sensitivity can be checked by recomputing the same realized history under alternative admissible sampling grids rather than assuming cadence is innocuous.

Path dependence supplies possible rivals, but history-dependent rates can still be additive along a fixed realized stream. Anticipation, relief, memory, adaptation, saturation, and context may alter those rates; they demonstrate nonadditivity only when the declared disjoint-union rule fails. Stable failure should prompt review of the additive representation and, where justified, a declared nonadditive, generalized, or no-estimate branch rather than a more flexible scalar chosen after the outcome. Testing additivity should therefore vary the partition while holding the

represented path fixed, separating failure of the accumulation rule from ordinary history dependence. An atom at a partition boundary must be assigned once under a convention fixed before either side is scored.

The additivity restriction also needs an identification route. Correlated measurement error, common smoothing, origin drift, or shared-method bias can manufacture a slope near minus one. External calibration, method-diverse replicate channels, valid instruments, sealed recovery studies, or truth-containing error bounds must be declared before the restriction is treated as evidence about the latent measure (Manski, 2003; Borsboom, 2005). Method-diverse replication is most informative when its errors are plausibly independent, because agreement produced by shared preprocessing offers weaker protection against artifact. Where channels share calibration data, report the covariance this creates between the two partition estimates.

A complete report shows the partition rule, atom treatment, observation source for each side, error structure, estimated additivity restriction, terminal-residual region, and admissible alternatives. Separate decisions for additivity and terminal zero identify the representation supported and the next question to investigate. A study can then remain reusable after a terminal-zero failure if its partition and additivity evidence continue to satisfy their own gates. Reporting the two decisions separately also identifies whether a successor needs a different accumulation rule or better endpoint evidence.

5.3 — Additive PTVC and Unequal Gross Variation

This section states the additive branch formally and develops the consequences of its exact intrastream terminal relation. The bearer, measure, origin, domain, and endpoint requirements already defined above remain binding.

The formal simplicity is purchased by exact commitments. The bearer must be separately routed, the measure finite, the origin and sign fixed, the strict-pre-endpoint domain complete, and the branch genuinely additive. When any of those conditions fails, the correct response is not to keep the equation and soften its meaning. The branch must narrow, become unidentified, or yield to another representation.

The equation below states the received additive target. The surrounding discussion explains the equality, the differences among experiences that it preserves, and the assumptions needed to interpret it.

$$Z_i^* := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^* -)$$
$$PTVC_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^* = 0$$

The display identifies three equivalent forms within the declared additive branch: equality of Jordan magnitudes, zero total signed mass, and zero latent terminal residual. None of those forms supplies the representation, origin, bearer, domain, or endpoint needed to interpret it.

Table 5.2. Additive PTVC constraints and dimensions that may remain unequal.

Object	Constrained by additive PTVC?	May remain unequal?
Terminal signed residual	Yes, exactly zero on the declared latent branch	No, within the target statement
Positive and negative Jordan magnitudes	Yes, finite and equal	Their common magnitude may differ across streams
Gross variation, duration, order, peaks	No	Yes, potentially radically
Rights, welfare, opportunity, biography	No	Yes; these require independent analysis

5.3.1 — The Exact Additive Statement

The additive statement assigns one finite signed phenomenal measure to one qualifying stream over its complete strict-pre-endpoint domain. Its terminal residual is the measure of that whole domain, not a score at the last observed moment and not a retrospective judgment about whether the life was good. A decisive falsification would be a qualifying stream for which representation, routing, domain, endpoint, and measured-to-latent gates pass while the identified terminal residual excludes zero (Folland, 1999; Cacioppo & Berntson, 1994).

Jordan decomposition writes the measure as a positive part and a negative part whose supports are mutually singular in the measure-theoretic sense. PTVC requires their total magnitudes to be finite and equal, which is stronger than a zero sample mean and weaker than equality at every time.

Empirical work cannot observe the equality directly. It must declare a representation, observation kernel, missingness process, estimand, estimator, endpoint rule, and measured-to-latent bridge before a finite-band result can bear on the latent exact statement.

Consider two streams with the same terminal signed residual but entirely different temporal structures. One may contain long mild burdens and long mild relief; another may contain rare extremes. The additive target treats their net terminal relation alike while preserving their gross variation as radically different.

A critic may object that the target is too coarse because it discards order, agency, meaning, and distribution across time. That objection identifies a real limitation of the additive branch rather than a refutation of its internal statement. Those dimensions remain scientifically and ethically indispensable.

Exact additivity is best treated as a branch-specific invariant. Its scientific value lies in making a strong claim testable without pretending that one scalar relation exhausts a life.

Failure of the additive target would not erase order, agency, meaning, or the distribution of experience across time. It would show that this scalar invariant is false, too coarse, or inapplicable to the declared reference class. A well-documented counterexample would redirect research toward richer representations rather than be absorbed as unexplained noise.

5.3.2 — Jordan Magnitudes and Gross Variation

Equality of positive and negative Jordan magnitudes does not imply small total exposure. Their sum is gross phenomenal variation, and it can be slight or immense while the signed

residual remains zero. Reporting both residual and gross variation preserves the difference between a quiet history and one containing large opposing magnitudes; the equality is a measure-theoretic relation, not a psychological description (Folland, 1999).

The two summaries should be reported together, because gross variation and terminal residual answer different questions.

$$G_i := V_i^+ + V_i^-, Z_i^* := V_i^+ - V_i^-$$

A measurement program must estimate the two sides rather than infer them from a quiet endpoint. Large opposing quantities can produce a small residual, so uncertainty in either component may dominate uncertainty in the difference.

A terminal estimate near zero could arise from two well-measured moderate totals, two poorly measured enormous totals, or a baseline choice that forced cancellation. Those cases have different scientific meanings even when a single plotted residual looks identical.

Ordinary adaptation, opponent processes, and measurement centering can also generate balanced-looking summaries without establishing complete-stream Jordan equality. Equal scoring requires those rivals to receive the same information and flexibility.

Gross variation keeps the formal claim from being mistaken for equal lives. A theory that predicts one terminal invariant can coexist with urgent differences in suffering, rights, care, opportunity, and justice.

The covariance of the two Jordan-magnitude estimates is scientifically consequential. Strong positive error correlation can make their difference appear stable even when each magnitude is poorly determined, so the joint uncertainty region should accompany any terminal residual.

5.3.3 — Intrastream Nontransferability

The additive equality belongs inside one separately routed stream. No positive magnitude from one person, animal, generation, population, institution, or archive may be used to cancel negative magnitude in another (Manski, 2003; Imbens & Manski, 2004).

Formally, each stream receives its own measure space and terminal residual. A sum across streams is a different object requiring an independently justified aggregation rule, and it cannot retroactively satisfy PTVC for any constituent stream.

Routing errors are therefore upstream failures. If observations from distinct streams are pooled before ownership is resolved, later estimation can be precise and still answer the wrong question.

A population average near zero illustrates the danger. Half a cohort could have large positive residual estimates and half large negative estimates, yielding a neutral mean while no individual stream is compatible with closure.

Aggregation across people can be legitimate in welfare science, public health, or institutional planning, but it answers a different question. A cohort mean, family narrative, or social welfare functional cannot substitute for the residual of one routed stream. If stream-specific ownership is unresolved, adjudication requires

a branch-matched conclusion that holds across every admissible ownership map; otherwise intrastream PTVC remains unadjudicated.

Nontransferability is both a formal boundary and an ethical firewall. It prevents the mathematics from becoming a ledger of human worth or a permission structure for cross-person balancing.

When routing remains uncertain, the protocol should preserve an admissible set of ownership maps rather than force a single partition. Each map induces its own stream-specific residual. The set of residuals allowed by alternative admissible ownership maps is their branch-indexed union. Intersections require constraints justified to hold jointly, with any required simultaneous-coverage guarantee; prespecification alone does not justify intersecting residual sets from alternative maps. Pooling becomes permissible only after routing uncertainty has been resolved at the level the claim requires.

Nontransferability fixes the unit of the claim and blocks moral accounting across bearers. One stream's positive measure cannot cancel another's negative measure, and a later event cannot alter the value already assigned to an earlier interval. This boundary prevents the formal ledger from becoming a ranking of lives or a justification for imposed harm.

5.3.4 — Origin, Sign, and Positive Rescaling

Exact zero has content only after the neutral origin and sign orientation are identified. Positive rescaling changes units but

preserves zero; additive translation generally changes whether a path closes (Folland, 1999; Cacioppo & Berntson, 1994).

Origin validity therefore requires independent calibration rather than terminal fitting. Candidate anchors may come from psychophysical indifference, symmetric tradeoff tasks, adaptation models, or cross-channel constraints, but each must survive its own falsification tests.

A questionnaire midpoint called neutral can make every participant appear closer to closure if the midpoint is culturally or clinically mislocated. Better precision around that midpoint does not repair the origin error.

Some theories deny that valence has an instrument-independent zero or interval scale. In that case exact additive PTVC may be unidentified, and a vector, ordinal, or generalized branch must be considered without pretending equivalence.

The scientific gain is not that the origin problem disappears. The gain is that it becomes an explicit gate whose failure can narrow the claim before an attractive terminal graph is overread.

Origin evidence should be challenged with anchors that were not used to fit the terminal model. Agreement among held-out indifference, tradeoff, and cross-channel tests is more informative than a zero chosen because it improves closure.

Under $F'(t)=a+bF(t)$, with $b>0$, positive rescaling preserves a zero residual when the neutral origin is transported consistently. An additive shift contributes an exposure-weighted term proportional to $aA_i(T)$ whenever total experienced exposure is nonzero. Scale invariance therefore does not solve origin identification; the

neutral origin must be independently declared and validated before accumulation.

5.3.5 — Past-Fixity and Non-Equivalent Readings

Past-fixity means that an experienced contribution, once it belongs to the declared history, does not change because later evidence, memory, relief, or interpretation changes. What can change is the estimate of that contribution. The distinction is ontic versus epistemic: the event is fixed as past, while knowledge about it may improve or deteriorate (Billingsley, 1999).

Filtering and smoothing make the distinction concrete. A filter estimates a latent state from information available up to a time; a smoother may revise that estimate after later observations arrive. The smoother can alter the posterior history without rewriting the experienced history itself. Every report should identify which information set produced each estimate.

Later interventions can change future experience, future reporting, and future memory. They cannot retroactively make earlier suffering smaller, morally deserved, or justified. Under exact PTVC, the fixed past constrains the signed remainder regardless of the chosen mechanism. Changing a genuine accumulated balance changes that remainder requirement, but does not by itself determine both future gross totals or identify particular compensating events.

Endpoint-revision rules are also specified prospectively. New evidence may show that the stream continued longer than first believed or that the observation endpoint was not the truth endpoint. The newly included interval belongs to the completed

domain, but the revision does not alter contributions already assigned under the same representation. A changed representation requires a separate migration record.

When a model version changes, old and new analyses should coexist. The record should distinguish changed evidence, changed representation, changed reference class, changed endpoint, and changed scientific claim. Without that matrix, a new scalarization or endpoint can inherit the apparent success of the object it replaced.

Remembered utility and narrative appraisal are scientifically important but non-equivalent targets. A person may remember an episode differently, reinterpret its meaning, or report less distress after recovery while the earlier phenomenal contribution remains uncertain. These processes are rivals or additional objects, not automatic corrections to the ledger.

A terminal residual must be reported with its cancellation geometry. Positive and negative Jordan magnitudes, gross variation, covariance, origin sensitivity, and the residual belong in the same display. Two histories can have similar residuals while differing radically in exposure, intensity, and uncertainty; a single terminal number would hide that structure.

Routing uncertainty should remain explicit as a family of admissible ownership maps when the evidence does not settle one map. Past-fixity then applies within each admissible history, and the terminal result becomes an identified set over those histories. Choosing the route that produces the most favorable residual would be outcome-dependent reconstruction. Set-valued routing

can still yield a decisive result when every admissible map gives the same sign conclusion, while disagreement across maps should widen the verdict rather than disappear.

The public language follows the same discipline. Later evidence may change what researchers believe about an earlier interval, but it does not erase what was lived. Exact closure, if supported, would be a terminal relation within one stream, not a moral story in which later events compensate for earlier harm or justify withholding present care. Public explanations should therefore avoid converting fixed-past arithmetic into counterfactual claims about whether a different life would have been better or worse.

Past-fixity gives tests a stable historical object. Later corrections to the representation, endpoint, or weighting receive explicit successor labels and their own evaluation, preserving the evidence associated with the earlier version (Lakatos, 1978; Mayo, 2018). The version record should specify whether a revision changes only estimation of the fixed past or redefines the target itself, because those changes have different evidentiary consequences.

5.4 — Bipolar Mixed-Valence Branch and Overlap

Mixed valence is not a minor complication added after the scalar case. A conscious episode can contain attraction and aversion, relief and grief, awe and fear, or bodily pain and interpersonal warmth at the same time. A single signed number may compress those components before the science has earned the right to combine them (Cacioppo & Berntson, 1994; Russell & Carroll, 1999).

The bipolar branch therefore begins with primitive positive and negative component measures β_i^+ and β_i^- . Their shared overlap ζ_i is preserved explicitly, while the net signed measure v_i equals their difference. The branch does not assume that every theory of mixed experience is literally two-dimensional; it prevents subtraction from erasing coactivation before it has been studied.

The following displays define the bipolar component decomposition and its primitive gross magnitude. They are not additive PTV in alternate notation. A bridge may connect the branches only under stated component, overlap, origin, unit, and exposure assumptions.

$$v_i = \beta_i^+ - \beta_i^-, \beta_i^+ = v_i^+ + \zeta_i, \beta_i^- = v_i^- + \zeta_i, \zeta_i \geq 0$$

$$G_i^{\text{bip}} = \beta_i^+(\Omega_i) + \beta_i^-(\Omega_i) = V_i^+ + V_i^- + 2\zeta_i(\Omega_i)$$

The net Jordan magnitudes V_i^+ and V_i^- describe the nonoverlapping positive and negative parts of v_i . The overlap measure ζ_i records component mass that cancels in the net signed measure but remains part of primitive gross experience. Net residual, component totals, and overlap therefore answer different questions.

5.4.1 — Why a Net Residual Can Hide Structure

A net signed residual can be identical for experiences with no overlap, complete overlap, or rapidly alternating signs. The compression is mathematically legitimate only if the discarded structure is irrelevant to the target being tested. A scalar residual is therefore sufficient only on a branch that declares overlap and order irrelevant; failure of that sufficiency sends the analysis to bipolar or path-sensitive alternatives (Cacioppo & Berntson, 1994; Russell, 1980).

The mapping from a bipolar state to a scalar residual is many-to-one. Once components are subtracted, the original pair cannot generally be recovered without additional assumptions or observations.

Measurement should therefore preserve component channels long enough to test whether scalarization is stable across people, contexts, intensities, and temporal scales. Early subtraction can manufacture apparent simplicity.

Grief at a wedding and relief during painful treatment illustrate coactivation. A zero net judgment at one instant would not reveal the presence, intensity, or duration of either component.

Preserving component structure makes later failure informative. The data may support scalar additivity, stable bipolarity, partial identification, or no adequate representation, and each outcome teaches something different.

A unidimensional rival can explain mixed reports as rapid alternation, appraisal, or report-generation structure. The bipolar branch therefore earns support only if independent positive and

negative components recover and transport better than those alternatives. If it does not, the additive branch or another declared representation remains the appropriate object.

5.4.2 — Independent Positive and Negative Components

The bipolar branch treats positive and negative phenomenal contributions as separately represented components rather than automatic negatives of one another. This permits coactivation, asymmetric response functions, and tests of whether one component can change while the other remains stable (Cacioppo & Berntson, 1994; Russell & Carroll, 1999; Larsen et al., 2001).

Each component needs its own origin, unit, exposure rule, and observation bridge. For finite components on the same domain and a common scale, $v_i(\Omega_i) = 0$ is equivalent to $\beta_i^+(\Omega_i) = \beta_i^-(\Omega_i)$; it does not identify their common magnitude, overlap, or temporal order.

Protocols should test component reliability, monotonicity, and cross-context transport before combining them. Separate elicitation can reveal ceiling effects, asymmetric noise, and differential missingness that a net scale conceals.

In chronic pain, a person may report persistent nociceptive burden alongside meaningful connection, mastery, or relief. Treating one component as the arithmetic negation of the other may misrepresent both.

Component independence can also be overstated. Shared arousal, attention, language, and response style may induce correlation,

and the representation must not interpret statistical separability as ontological independence.

The scientific payoff is a branch that can be wrong in specific ways. It can fail because components are not distinct, because their units are not comparable, or because no stable terminal relation survives.

Residual dependence after modeling shared arousal, response style, and context can reveal whether the two components carry distinct information. Failure to recover that distinction favors a simpler representation rather than a nominally bipolar scale.

Branch comparison should therefore be explicit. A dataset supports the bipolar branch only when separately identified component channels, coactivation, and asymmetric responses improve recovery and transport beyond a one-axis or rapid-alternation rival under the same observations. Without that advantage, branch identity remains Model-ambiguous rather than being settled by richer vocabulary.

5.4.3 — Overlap, Coactivation, and Gross Exposure

Overlap records the simultaneous presence of positive and negative components. It matters because a stream with high coactivation can have the same component totals as a stream in which the components occur in separate eras (Manski, 2003; Imbens & Manski, 2004).

An overlap functional depends on temporal resolution and representation. Coarse windows may make alternating states look simultaneous, while very fine windows may fragment states whose phenomenology is integrated over longer intervals.

A study should protocol-fix the windowing rule before outcome inspection and examine resolution sensitivity. Stability across plausible resolutions is evidence for a robust overlap construct; instability is a result, not a nuisance to hide.

A difficult rehabilitation session may combine pain, fear, achievement, and hope within the same minute. Summing daily positive and negative totals cannot show whether those elements were integrated, alternating, or separately remembered.

Delayed opponent dynamics and multilayer appraisal can produce apparent coactivation without a primitive bipolar phenomenal state. The branch should therefore be compared against alternation, lagged-response, and report-generation models using common observations. Unresolved equivalence warrants Model-ambiguous, not preference for the richer language.

Overlap therefore belongs in the discovery architecture, not as a decorative complexity term. It tells the science what scalar closure would have to ignore and what a richer branch might preserve.

A resolution study should compare at least two prospectively specified windows and two explicit alternatives: genuine coactivation within the same window, and rapid alternation whose average only appears mixed. The branch is strengthened when the same component structure survives this aliasing test; otherwise the result belongs to the resolution at which it was identified.

Overlap is not a decorative correction to a scalar model. It identifies information that a net residual discards and therefore determines when a richer branch is needed. Past observations

remain fixed; later component activity can change the terminal sum without rewriting what was experienced before.

5.4.4 — Temporal Order and Component Histories

Terminal component totals do not determine temporal order. Positive experience followed by burden, burden followed by relief, and repeated oscillation can end with the same totals while producing different adaptation, memory, action, and clinical consequences (Cacioppo & Berntson, 1994; Russell, 1980).

Observation schedules must be dense enough for the order-sensitive quantity they claim to estimate. Retrospective summaries cannot recover every transition, and missingness may be informative when people disengage during the most intense states.

Two end-of-life trajectories can carry similar component totals but differ in whether relief arrived early enough to preserve communication, agency, or family connection. The formal terminal relation does not erase that difference.

Peak-end, recency, and memory-consolidation accounts are powerful rivals for reported order effects. They may explain later judgments even when the underlying cumulative exposure is unchanged.

The branch remains disciplined by distinguishing the terminal target from scientifically important path structure. A theory can propose one invariant while the book still treats order as essential to explanation and care.

Order sensitivity should be estimated separately for contemporaneous reports, delayed recall, and retrospective evaluation. A pattern that appears only after memory consolidation belongs to the observation or memory model unless a bridge to phenomenal order is independently supported.

Temporal-order extensions may use kernels, path signatures, transition counts, or state-space dynamics, but adopting one creates a new branch with its own invariances, estimands, observation requirements, and counterexamples. The additive terminal residual should not be described as if it already contained sequence information.

5.4.5 — Branch Tests, Failure, and Non-Substitution

The additive and bipolar branches describe different features of experience. A zero net residual can coexist with large positive and negative magnitudes, extensive coactivation, or asymmetric histories. Separate branch labels identify which features a study evaluates and the bridge needed to connect its findings (Cacioppo & Berntson, 1994).

Branch selection must be prospective. The protocol should declare whether positive and negative components are primitive, whether overlap is allowed, how coactivation is measured, and which bridge would connect the bipolar object to an additive signed measure. Selecting the branch after seeing which one closes would make the result outcome-dependent.

Resolution matters because rapid alternation can resemble coactivation when observations are coarse, while genuine coactivation can be forced into alternating labels by a single-sign

instrument. Multiscale analysis should therefore compare component occupancy, overlap, order, and gross exposure across prospectively specified temporal resolutions.

A bridge theorem must state sufficient conditions, not merely visual resemblance. If the bipolar components integrate to the positive and negative Jordan magnitudes of an additive measure under a common origin, exposure rule, and domain, the relationship is testable. When those conditions fail, the branches remain distinct even if their terminal summaries happen to agree.

Failure should be typed. Unstable sign indicates a representation problem; unresolved coactivation indicates an observation problem; nontransportable overlap suggests a reference-class boundary; incompatible component totals pressure the bipolar target; and a nonzero additive residual pressures the additive target. A generic mixed label would discard these distinctions.

The empirical design should give each branch equal opportunity. Instruments that force a single scalar naturally favor the additive formulation, while separate positive and negative channels can favor a bipolar model. Equal-budget comparison uses the same observations, cadence, endpoint rule, calibration opportunity, and held-out scoring criteria.

Non-substitution is also an ethical safeguard. A richer branch cannot explain away an adverse result unless declared and measured. Nor may component magnitudes become scores of personal worth, emotional purity, or eligibility. The branches model a proposed intrastream target; they do not classify people.

A branch-concordance report should name what survived. An additive failure may still preserve sign calibration, component measurement, endpoint work, and useful rival models. A bipolar failure may leave an additive approximation viable in a bounded context. The report should preserve these gains without presenting them as support for the branch that failed.

The resulting architecture allows genuine narrowing. The program can move from a universal additive claim to a bounded bipolar or generalized claim only through visible status change and new risky predictions. It cannot rotate among branches until one fits. That rule turns branch pluralism into scientific strength rather than a rescue device (Mayo, 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

5.5 — Generalized Terminal Neutrality and Bridge Theorems

Generalized Terminal Neutrality is the program's broadest typed terminal language, and for that reason it requires the strictest discipline. It allows a terminal target on a marked carrier and declared value space without privileging one scalar signed measure. The generalization is formal, not evidentiary (Manski, 2003).

Generality is useful when the science has not earned a unique scalarization or when mixed, vector, relational, or distributional structures remain live. It is dangerous when the general functional is chosen after the data so that every terminal pattern can be called neutral.

The display below states a separate GTN target: a frozen terminal functional H_{GTN} evaluates a declared marked carrier (Γ, s) and returns the distinguished neutral element 0_{GTN} . Equivalence to additive PTVC requires a bridge theorem that preserves bearer, domain, origin, sign, exposure, and endpoint commitments.

$$H_{GTN}(\Gamma, s) = 0_{GTN}, (\Gamma, s) \in M_{GTN}, H_{GTN}: M_{GTN} \rightarrow V_{GTN}$$

Here $\mathfrak{M}_{GTN}$ is the declared class of marked carriers, $\mathcal{V}_{GTN}$ is the target value space, and 0_{GTN} is a prospectively identified neutral element. The equation is meaningful only when the carrier, codomain, admissible transformations, and zero set are fixed independently of the observed endpoint.

5.5.1 — A Separate General Terminal Target

GTN names terminal constraints that need not reduce experience to one additive scalar. It can accommodate vector states, distributions, path functionals, or other structures when the scientific reason and typed map are declared in advance. GTN remains testable only when the functional and admissible representation family are frozen before outcomes; a post hoc map that always returns its neutral element is not a terminal law (Manski, 2003).

The formal object is a functional on a represented complete stream, not a license to call any desirable terminal state neutral. Its domain, codomain, invariances, topology, and zero set must be stated before evaluation.

Empirical work must say which features of the representation are observed and which remain latent. A generalized target may be less point-identified than additive PTVC, so set-valued or inequality conclusions may be the most honest outputs.

A vector-valued representation of simultaneous affective components may support a terminal region rather than a single point. That region can be scientifically useful without being mislabeled as proof of exact additive equality.

The principal rival is unrestricted flexibility. If every failed scalar model is rescued by choosing a new terminal functional after inspection, GTN becomes a language of accommodation rather than a hypothesis.

Treating GTN separately gives richer structures their own definitions, predictions, and empirical bridges. The program can

compare their contributions with additive PTV C while preserving the meaning of each target.

5.5.2 — Representation Families and Invariance

A generalized target is intelligible only relative to a family of admissible representations and transformations. The science must identify which changes are coordinate recodings, which preserve the experiment's information, and which change the claim (Krantz et al., 1971; Blackwell, 1953).

An admissible invariance group must be stated explicitly. Positive rescaling, component permutation, monotone transformation, and measure-preserving reparameterization preserve different structures and cannot be treated as one operation.

Measurement invariance should be examined across instruments, cultures, developmental stages, care settings, and temporal resolutions. Apparent invariance in fit statistics is not enough if the substantive zero set changes.

Two laboratories may use different component bases for the same latent vector. A bridge is possible only if the terminal functional transforms consistently across those bases and does not depend on one laboratory's arbitrary coding.

Overly broad invariance can empty the target by making all states equivalent; overly narrow invariance can turn harmless unit changes into different hypotheses. Both errors must be tested through counterexamples.

The payoff is a clear distinction between robustness and vagueness. Robustness survives declared transformations; vagueness avoids saying which transformations matter.

The proposed transformation class should also be closed under composition and inversion where those operations are claimed to preserve meaning. Without that algebraic coherence, two individually permitted recodings can combine into a recoding that changes the target.

5.5.3 — Bridge Theorems to Additive PTVC

A bridge theorem states sufficient and, where possible, necessary conditions under which a generalized terminal functional is equivalent to the additive signed-measure target. It is a theorem burden, not a rhetorical resemblance (Popper, 1959; Mayo, 2018).

A valid bridge must map bearer, complete domain, positive and negative structure, origin, scale, and endpoint across representations. It must show that zero under one branch entails the appropriate zero or equality under the other without hidden outcome selection.

The empirical program must also test whether the theorem's premises are plausible in the declared reference class. A mathematically correct implication can be scientifically irrelevant when its assumptions are unmeasured or systematically violated.

For example, a norm of a vector terminal residual equals zero only when every component is zero. That fact can bridge to componentwise neutrality, but it does not by itself imply equality of integrated positive and negative Jordan masses unless the representation connects them.

A bridge theorem should state which assumptions do the work and, where possible, illustrate their necessity with separating counterexamples outside those assumptions. Such examples may show a generalized functional vanishing while the additive residual remains nonzero, or an additive zero failing the generalized target. Within the theorem's full premises, the proved implication or equivalence must hold.

Bridge theorems allow the program to expand without blurring identities. Until a bridge is established, results remain branch-specific and cannot be pooled into one evidential vote.

Premise testing should be operational rather than ceremonial. If the theorem requires a common origin, additive exposure, exhaustive routing, or endpoint regularity, each premise receives an observable diagnostic, a measurement-error model, and a failure threshold. When a premise is unidentified, the bridge is unlicensed even if both endpoint quantities happen to be near zero (Borsboom, 2005; Mayo, 2018).

Bridge theorems specify the conditions connecting additive, bipolar, and generalized results. Their assumptions identify which theoretical and empirical findings can be used together.

5.5.4 — Counterexamples and Non-Tautology

A generalized terminal target is especially vulnerable to tautology because its functional can be designed around terminal data. Non-tautology requires the carrier, neutral set, representation family, and endpoint rule to be fixed independently, with admissible data capable of producing failure (Mayo, 2018).

The formal test is counterfactual: could an admissible dataset have failed the target under the same prespecified functional? If the answer is no, the construction has no discriminating content.

Simulation should include adversarial generators that place terminal states just inside and just outside the declared neutral set. The analysis must recover those differences under realistic observation error and missingness.

A learned embedding that defines neutrality as the center of its own terminal training sample will almost inevitably report centrality. That result may describe the embedding but cannot adjudicate a pre-existing terminal law.

Flexible representation learning, regularization, and latent-space rotation can all hide post hoc rescue. The strongest safeguard is to freeze the training objective, evaluation set, and failure thresholds before terminal outcomes are inspected.

A generalized theory becomes stronger, not weaker, when it can name the constructions it refuses. The neutral set must be a risky target rather than the place where the method is guaranteed to arrive.

A useful red-team benchmark gives the pipeline data generated from a world with no terminal constraint and asks whether model selection still manufactures one. A method that frequently discovers neutrality in those counterfactual worlds has failed before it reaches human data.

5.5.5 — Empirical Consequences and Status Discipline

Generalized Terminal Neutrality is a separate hypothesis family, not a semantic shelter for additive PTV. Its terminal functional, representation class, invariances, and endpoint domain must be declared before data are inspected. A generalized target becomes scientifically useful only when it generates consequences that can fail under the same source and change-control discipline.

Bridge theorems determine when a generalized result bears on the additive branch. The theorem must name the representation family, origin, sign structure, scalarization, domain, endpoint, and regularity conditions that make the two targets equivalent or one imply the other. Without that bridge, similarity of terminal language is not equivalence.

Empirical consequences should be derived at the level of observables and identified sets. A generalized terminal condition may imply invariance across admissible representations, a bounded compatibility region, componentwise restrictions, or transported order relations. Each implication must state the observation and measured-to-latent bridges it requires.

The strongest protection is to preregister the representation family. Adding a new functional only after the additive residual fails would make the target indefinitely elastic. A successor may still merit study, but it requires a new identifier, reference class, and held-out tests not used to invent it.

Counterexamples are valuable for exposing bridge failures. Two representations may agree on ordinary episodes and diverge under mixed valence, extreme duration, singular contributions, or

endpoint-inclusive events. Those cases should remain a test suite unless their removal from the reference class is explained.

Partial identification is often proper. If several admissible representations yield different terminal values, the output is a set of compatible values or verdicts. Exact neutrality is licensed only when every admissible member satisfies the declared target or when additional evidence narrows the set by a justified bridge. Set width is itself informative, because it reveals whether ambiguity is a narrow numerical tolerance or a disagreement among qualitatively different representations.

Status language identifies the outcome precisely: generalized-compatible, additive-compatible, bridge-supported, bridge-failed, representation-ambiguous, or No adequate estimate. Each label describes the scientific analysis and the next work it supports. A status transition should name the bridge that changed so neighboring branches do not gain or lose standing by association.

Simulation and theorem work make the formal program testable. Simulations assess whether the pipeline recovers the correct branch and distinguishes competing bridges. Derivations establish consistency and consequences; empirical studies evaluate those consequences through their specified measurement, endpoint, observation, and model-comparison procedures. Recovery benchmarks should include data generated from neighboring branches, so a generalized method is tested for false attribution as well as successful detection.

Version control links generalized successors to the objects they replace or extend. Each change record should state which earlier

results remain comparable, which require reanalysis, and which become historical. This prevents a generalized branch from inheriting evidence earned under a different terminal functional. A successor functional that changes the zero set or invariance class should be treated as a new target even if its notation resembles the earlier one.

The durable contribution is a disciplined ladder of claims. If the additive branch fails, the program may still learn that a vector, ordered, bipolar, or generalized object better represents the data. That is progress only when the new object is more constrained and more testable, not merely more accommodating (Lakatos, 1978; Mayo, 2018). Greater generality earns value only when it creates additional exclusions or stable bridges that empirical work can actually challenge.

5.6 — Reference Classes, Modal Scope, and Past-Fixity

A formal equation does not determine how widely it is claimed to hold. Reference class and modal scope are separate scientific objects. A theory may concern one species, one class of maximal qualifying conscious streams, one endpoint type, one observation regime, or a far broader domain, and those possibilities cannot be collapsed into the word universal (Billingsley, 1999; Kallenberg, 2002).

The same separation applies to probability. Probability-one closure under a declared branch law, support containment under a declared topology, high finite-cohort frequency, closure under every policy, and closure under every intervention are different propositions. Evidence for one does not automatically raise the status of the others.

The displays below keep three modal statements apart. The first is the probability-one target under branch law Q^b . The second shows that probability one does not by itself guarantee support containment unless the topology and closedness conditions are supplied. The third reminds the reader that support containment still does not entail all-policy or all-intervention closure.

$$Q^b(FPTVC^b) = 1 \not\Rightarrow \text{supp}(Q^b) \subseteq FPTVC^b$$

$$\text{supp}(Q^b) \subseteq FPTVC^b \not\Rightarrow \forall \pi, a: Z_i^*(\pi, a) = 0$$

These are schematic non-implications, not claims that no stronger theorem is possible. In regular topological settings, probability-one membership in a closed target set may imply support containment. All-policy and all-intervention assertions still

require their own counterfactual domains and assumptions; vocabulary cannot supply them.

Table 5.3. Modal claim forms and non-implications.

Claim form	What it asserts	What it does not establish
Reference-class closure	Target within a declared class	Transport beyond that class
Probability-one closure	Almost-sure under a declared law	Every possible path or intervention
Support containment	All supported residuals satisfy the target	Correct observation or representation
All-policy or all-intervention closure	Counterfactual invariance under specified actions	Ethical permission or universal mechanism

5.6.1 — Reference Classes Before Universality

Every empirical and formal statement needs a declared reference class specifying bearers, stream criteria, endpoint types, settings, and exclusions. A universal statement is defeated by one eligible counterexample, whereas a class-restricted result speaks only to its declared inclusion and exclusion rules (Billingsley, 1999; Kallenberg, 2002).

A reference class can be defined by species, developmental status, clinical condition, stream architecture, endpoint hazard, or measurement feasibility. The defining variables must not be chosen after observing favorable terminal behavior.

Transport beyond the original class requires new evidence about representation, origin, routing, endpoint, observation, and

mechanism stability. Similarity in one observable feature is not enough.

A result in slowly progressive terminal illness cannot simply be extended to sudden death, anesthesia, infancy, nonhuman animals, or artificial systems. The causal and observational structures differ at precisely the points the hypothesis depends on.

A narrow reference class can produce real knowledge, but it can also hide selection. Reports must identify exclusions, their scientific rationale, and whether they remove the trajectories most capable of testing the claim.

Reference-class discipline converts universality from rhetoric into a sequence of transport problems. A narrowed result may be scientifically important even when the grandest claim remains open.

External validity is a separate question. A failure outside the declared class does not refute the class-specific result, but it blocks universal language and identifies the next transport problem. The record should state which representation, origin, endpoint, and observation assumptions would have to survive before extension.

5.6.2 — Probability-One, Support, and Almost-Sure Claims

Probability-one closure is relative to a declared probability law and may tolerate exceptional trajectories; support containment additionally depends on topology, closedness, and regularity; set-universal closure quantifies over every eligible history in the declared class. These propositions are distinct, and none may be inferred from finite-cohort frequency or a confidence interval

around an average (Pearl, 2009; Hernán & Robins, 2020; Frangakis & Rubin, 2002).

Inference must identify the probability law, sampling scheme, dependence structure, censoring, and observation process. A confidence interval around a cohort mean cannot establish an almost-sure latent law.

A continuous model may assign probability zero to each exact path while its support contains many paths incompatible with closure. Saying that each path is individually improbable does not rescue the target.

Rivals can agree on high-probability terminal concentration while disagreeing on support, mechanism, and counterfactual behavior. Proper scoring and held-out prediction should be used to separate them.

Naming the modal claim protects the reader from false certainty. The science can report probability-one under a model, finite-band compatibility in data, or support failure without treating those phrases as interchangeable.

Reports should identify the exceptional set allowed by an almost-sure claim and the topology used to define support. Concealing either choice makes a probabilistic statement sound stronger than the model actually warrants.

5.6.3 — All-Policy and All-Intervention Distinctions

A claim that closure holds under the realized policy differs from a claim that it holds under every possible policy. An all-intervention statement can be stronger when its declared

intervention family includes the policy family and additional interventions that may alter stream unity, routing, endpoint, or representation (Lakatos, 1978; Mayo, 2018).

Counterfactual scope requires a structural model that defines how actions change trajectories and whether the same bearer and endpoint remain meaningful. Without that model, universal intervention language is undefined.

Experiments can test selected interventions, but they cannot license every-policy closure by induction. Each intervention also carries ethical constraints independent of scientific value.

Analgesia, sedation, psychotherapy, stimulation, or environmental change may alter experience, observation, memory, and endpoint hazard simultaneously. A favorable terminal estimate under one treatment does not show policy invariance.

Control-theoretic and time-symmetric routes may imply invariance across policies or interventions that ordinary adaptation models do not. Those stronger quantifiers require direct prospective contrasts. Evidence under one policy cannot be promoted to all-policy or all-intervention scope.

These stronger quantifiers do not create permission to manipulate suffering. The design must state policy and intervention scope prospectively, while care, consent, and legal permissibility remain independent gates that no terminal result can override.

5.6.4 — Compositional Scope and Stream Boundaries

Compositional closure asks whether closure properties of parts imply closure of a whole or vice versa. For conscious streams, composition is inseparable from the rules for split, fusion, pause, branching, and ownership (Billingsley, 1999; Kallenberg, 2002).

A parent stream ending at a split and child streams beginning thereafter is not mathematically equivalent to one ledger continuing through all descendants. The chosen routing rule changes the domain of every terminal statement.

Data systems must preserve those boundaries instead of merging records because one body, account, family, or device persists. Biological continuity and informational continuity do not settle phenomenal ownership.

A hypothetical fusion case shows the difficulty. Two pre-fusion streams may each have histories, while a post-fusion stream may inherit memories from both. Cross-history information does not automatically create cross-stream ledger transfer.

Integrated-information, global-workspace, recurrent-processing, and higher-order theories may propose different unity criteria. PTVC cannot use terminal fit to choose among them without an independent mind-source bridge.

Composition remains an open scientific frontier rather than a loophole. Explicit routing protects nontransferability and makes hard cases available for genuine defeat.

Routing uncertainty can be represented as a family of admissible partitions rather than settled by one preferred consciousness

theory. Closure that survives every admissible routing is stronger than closure under one convenient partition, while disagreement across admissible partitions can leave the result Model-ambiguous.

5.6.5 — Past-Fixity Across Updating and Narrowing

Past-fixity applies to classes as well as histories. Once cohort, branch, endpoint, and representation are prospectively specified, unfavorable cases cannot be removed and reported as though they were never targeted. Updating may change the claim, but never the test already run (Lakatos, 1978; Mayo, 2018).

Each reference class is versioned as an inclusion graph recording bearer, routing, domain, endpoint, representation, observation requirements, exclusions, and failure modes. Parent-child links state what changed and why.

The record specifies modality: probability one under a law, support containment, all-policy, all-intervention, compositional, or set-universal closure. Each proposition has its own assumptions and inference criteria.

Narrowing is legitimate only when it exposes a boundary and creates new risk. The successor class preserves the observations and counterexamples that forced the change and names a transport study capable of challenging it. Removing every difficult case creates a moving target.

Transportability is a separate identification question governed by overlap and stability in representation, routing, endpoint support, observation, and route structure. Similar diagnoses, ages, or

species labels do not establish transport when those deeper structures differ (Pearl, 2009; Hernán & Robins, 2020).

A counterexample archive retains adverse cases with their data-quality, endpoint, measurement, and routing status. It prevents later summaries from remembering only the accepted class and shows whether the challenge reached the target, bridge, route, or estimability.

Representation updates receive a migration matrix. Earlier experiences remain fixed, while estimates may change under a new origin, scalarization, or observation model; the matrix separates changed knowledge from a changed object and marks which verdicts still compare.

The public claim always names its current class and modality. Narrowing can be important, but cannot retain the predecessor's universality. Visible scope change preserves learning, failure memory, and the possibility of reversal.

5.7 — Endpoint Domain and Change Control

The formal hypothesis depends on where the stream begins, where its strict-pre-endpoint domain ends, and how changes to those boundaries are governed. Endpoint language is therefore not a poetic ending. It is part of the mathematical object and part of the protocol's vulnerability (Andersen et al., 1993; Aalen et al., 2008).

This section also gathers the chapter's non-equivalent simplifications. A last observation is not automatically the truth endpoint. A first zero crossing is not automatically termination. A centered residual is not automatically neutral. A product with a vanishing factor is not automatically the phenomenological ledger.

The rule is simple: when the science needs a different object, it should change the object openly. Silent semantic migration is not refinement; it is loss of falsifiability.

$$L_i^{\text{phen}}(T_i^* -) = L_i^{\text{phen}}(t_0 -) + v_i([t_0, T_i^*))$$

The decomposition keeps delayed observation and truth-domain closure separate. The estimated or protocol endpoint may help study the truth endpoint, but it does not become that endpoint through notation.

5.7.1 — Strict-Pre-Endpoint and Endpoint-Inclusive Domains

The primary branch uses the complete strict-pre-endpoint domain and excludes any value assigned at the endpoint itself unless a distinct endpoint-inclusive branch is declared. The two domains

can produce different terminal residuals, so any endpoint atom must be assigned prospectively rather than moved across the boundary after inspection (Andersen et al., 1993; Aalen et al., 2008).

The choice matters when the process may jump, terminate with a final experiential atom, or continue through a boundary that the protocol treats as terminal. Left limits and endpoint atoms are not interchangeable.

Protocols must specify how the last interval is handled, whether terminal uncertainty creates censoring, and whether observations after the protocol endpoint can reveal continued experience.

A final burst of pain, relief, dreamlike experience, or awareness near physiological death could alter an endpoint-inclusive quantity while leaving a strict-pre-endpoint approximation incomplete.

Domain discipline makes terminal evidence honest. The study must say which boundary it observes and which boundary the formal claim requires.

Observation shutdown can manufacture a clean-looking left limit. Devices may be removed, speech may end, sedation may deepen, or sampling may stop precisely when endpoint classification becomes hardest. If the terminal contribution cannot be bounded across that gap, the correct verdict for the strict-pre-endpoint target is No adequate estimate; a narrower observation-window or endpoint-inclusive result remains available only if it is separately defined and supported.

5.7.2 — Truth Endpoint, Protocol Endpoint, and Estimated Endpoint

The truth endpoint is the end of the qualifying stream under the declared bearer and unity rule. The protocol endpoint is the event at which a study stops or classifies. The estimated endpoint is a statistical object inferred from available data (Billingsley, 1999; Kallenberg, 2002; Greer et al., 2020).

The Law formulation used in this series also includes a biological-death admissibility condition. Scientifically, treat it as a separate boundary restriction in the Law formulation rather than by redefining biological death as the truth endpoint. Biological death may not occur before the declared closure condition has been satisfied for the relevant stream. The condition does not require biological death to occur when closure is reached. A qualifying conscious stream may permanently end after satisfying closure while biological processes continue without consciousness, including under artificial life support; such continued nonconscious biological persistence does not extend the completed stream. If the truth-level endpoint coincides with biological death, or if an endpoint-inclusive contribution is declared, any death-associated phenomenal contribution must be assigned prospectively to the appropriate branch. This biological-death admissibility condition is logically additional to the primary terminal-equality statement and therefore requires its own scope, bridge assumptions, and empirical tests.

These endpoint objects require distinct notation and adjudication rules. Conflating them can make a protocol-defined stopping time appear to certify a truth-level boundary.

Endpoint models should use channels not algebraically identical to the ledger estimator. Otherwise closure can become true by defining the endpoint as the time at which the estimated residual reaches zero.

A clinical protocol may stop at loss of responsiveness, while later evidence suggests residual awareness or a new stream state. The protocol was not fraudulent, but its endpoint was not the truth endpoint the formal claim needs.

Rivals include endpoint-conditioned selection, censoring, first-hitting termination, continued stream hypotheses, and misclassified unity. Each can produce different terminal patterns from the same observed window.

Typed endpoint separation lets the science learn without retrospective rescue. A revised endpoint changes the analysis and its status; it does not disappear into the same symbol.

Endpoint adjudication should report a confusion matrix or posterior over endpoint classes when evidence is imperfect. Propagating that uncertainty into the terminal residual is more honest than assigning one boundary and treating its classification as known.

5.7.3 — Initial Conditions and Delayed Observation

A complete-stream law begins at stream onset, while almost every empirical study enters later. The observed window therefore contains an increment plus an unknown pre-entry state (Popper, 1959; Lakatos, 1978).

Setting the running estimate to zero at enrollment is a bookkeeping convenience, not a claim that the latent cumulative history reset. Whole-stream closure cannot be inferred from the window alone without bounds or a bridge for the missing prefix.

Designs should collect historical proxies, repeated informant reports, medical records, and prior self-reports where appropriate, while preserving uncertainty about what those sources cannot recover.

A two-week terminal cohort can provide precise information about its observed interval. Analysis of the preceding history separately records the information available, the admissible bounds, and the uncertainty they introduce.

Delayed enrollment is usually selective. Diagnosis, communication capacity, care setting, social support, referral pathways, and survival to enrollment jointly shape who becomes observable. Sensitivity analysis should vary both the missing prefix and the entry mechanism, and the reported reference class should follow the prospectively specified selection model or bounds rather than the broad population label.

Delayed entry limits what can be estimated rather than resetting the stream. A study may identify local transitions, observed-window increments, or route signatures while leaving the unseen prefix unresolved; without a validated prefix bridge, whole-stream terminal closure remains unadjudicated.

5.7.4 — Non-Equivalent Simplifications and Counterexamples

Several attractive simplifications recur because they make terminal neutrality easy to display. Conservative state functions, adapting baselines, derivative transformations, endpoint centering, and vanishing-factor products can all create zero-like outcomes by construction (Andersen et al., 1993; Aalen et al., 2008).

Model recovery can expose these failures by generating data under known nonclosing processes and checking whether the analysis manufactures support for closure. A pipeline that exceeds its predeclared false-positive tolerance on those controls requires redesign or retirement; valid compatibility under limited resolution remains inconclusive.

A Brownian bridge conditioned to end at zero and an unconditioned process with a genuine preterminal controller may look similar near the endpoint. Without prospective endpoint separation, the simplified interpretation is not identified.

The correct use of simplifications is as explicit models, approximations, or rivals. Their elegance may help derive predictions, but it does not entitle them to inherit the Formal Law's status.

Counterexamples keep the program creative without making it immune. Every shortcut remains available for study and unavailable as an invisible rescue.

A simplification is scientifically useful when it carries an error bound, an approximation regime, and a counterexample outside

that regime. Without those limits, a convenient reduced model can silently replace the object it was meant only to approximate.

5.7.5 — Formal Change Control and Withdrawal

Formal change control lets the program improve without presenting every successor as though it had always been the target. Any change to bearer, ownership, domain, endpoint, origin, representation, quantifier, or terminal functional receives a new version state, a reason, and a dependency record (Billingsley, 1999; Kallenberg, 2002).

The change packet should compare old and new objects line by line. It identifies affected theorems, formulas, simulations, estimands, protocols, reference classes, figures, summaries, and public artifacts. Each dependent item is marked unchanged, reinterpreted, requiring reanalysis, or withdrawn.

Retrospective repair must create new risk. A successor that explains one failed cohort but predicts nothing beyond it is degenerating. A progressive change is frozen before new data and earns its status through held-out predictions, tighter exclusions, or a bridge that can fail independently (Lakatos, 1978; Mayo, 2018).

Supersession is not erasure. Retired objects remain in the archive with their tests, failures, code, and reasons for replacement. Stable identifiers are preserved or tombstoned rather than recycled, so later researchers can determine which version an analysis actually evaluated. A tombstone should point to the successor and reason for retirement, preventing dead identifiers from becoming ambiguous references in later code or prose.

Withdrawal can be granular. A theorem may be corrected while a branch remains; a candidate route may retire while the mechanism-neutral target stays open; a reference class may narrow; or a set-universal claim may fail. The record should withdraw the smallest object justified by the evidence and preserve everything that survives. Granular withdrawal also protects unaffected findings from being discarded simply because a more ambitious claim failed. A narrower successor may retain prior evidence only to the extent that the assumptions and inferential bridge supporting that evidence still apply to the narrowed object.

Correction must propagate beyond the technical chapter. A changed endpoint rule or retired equation requires review of protocols, tables, diagrams, summaries, presentations, websites, and derivative excerpts. A correction that remains trapped in the formal text leaves the public artifact scientifically stale. Dependency tagging can make that propagation selective, updating only artifacts whose claims actually relied on the altered endpoint object. The propagation record should also identify artifacts reviewed and found unaffected, so a deliberate nonchange can be distinguished from a dependency that was simply missed.

Human review closes the change only after formal, measurement, source, and artifact consequences agree. The record names reviewers, dissent, unresolved limitations, completed repairs, and triggers for reopening the decision. This makes scientific version control an accountability mechanism rather than clerical decoration (National Academies of Sciences, Engineering, and

Medicine, 2019; Nosek et al., 2018). Dissent should remain attached to the version when it concerns an unresolved assumption that could materially change interpretation. For consequential revisions, a reviewer who did not design the repair should test whether it resolves the named defect without quietly changing an assumption, target, or evidentiary threshold.

The strongest claim is not the one that never changes. It is the one whose changes remain visible enough that reality can keep answering. Formal change control preserves ambition by making revision, narrowing, retirement, and withdrawal honest outcomes rather than hidden forms of rescue. A visible history of revision also lets later evidence distinguish improvement in the model from movement of the target. Evidence gathered under non-equivalent target versions should remain separated unless an explicit bridge shows that the estimand and inferential meaning are preserved across the change.

Chapter 6 — Representation and Measurement Interfaces

The latent object can be investigated through representations, tasks, reports, behaviors, physiology, neural data, historical records, context, and models, each with a distinct relation to what the science ultimately wants to know.

This chapter builds the interface between the formal target and empirical work. It begins by separating the latent phenomenal object from route state, observation, estimand, estimator, decision rule, intervention, and verdict. Those layers can interact, but none may silently inherit another's authority.

Representation comes first because operations depend on structure. A scalar signed rate supports integration only if origin, sign, units, and cross-sign comparison are identified. A bipolar or vector representation can preserve more information while creating different terminal targets.

The neutral origin is the chapter's sharpest identification problem. More precise sensors cannot determine an arbitrary zero. Positive rescaling preserves zero, while translation generally does not. Origin anchors must therefore be independent of the terminal residual they are later used to evaluate.

Sign and scalarity are separate burdens. A state may be positive in one component, negative in another, and incomparable under a single weight vector. The correct output may be an identified set rather than one ledger, and the book must treat that as scientific information rather than failure of ambition.

Observation then enters through a kernel. Reports, behavior, autonomic signals, neural features, clinician judgment, and informant evidence have different access paths and different error

structures. Fusion should preserve conflict long enough to learn from it.

The protocol measurement map extends beyond one estimator. It includes cadence, missingness, gain, calibration, context of use, versioning, abstention, privacy, and the rule that turns an estimate into a scientific verdict. Clear abstention rules make its conclusions more dependable.

Most studies observe a suffix of a stream. Their analysis combines the measured window with an explicitly unknown prefix, using justified bounds or historical information where available.

Bounds and identified sets offer a rigorous response. Historical records may narrow uncertainty; local dynamics may be testable; whole-stream claims may remain unavailable. The science becomes more valuable when it can distinguish these outcomes without forcing a verdict.

Empirical neutrality also requires a hierarchy. Mean equivalence, distributional concentration, individual compatibility, latent exactness, and structural universality are not one claim. Finite observations can support carefully defined compatibility without proving exact zero.

The chapter will repeatedly ask whether an analysis choice made neutrality easier. Centering, baseline subtraction, derivative transforms, outcome-selected channels, endpoint fitting, and selective exclusion can all manufacture closure. Their exclusion rules belong before estimation.

Sections 9.7 and 28.8 govern non-deployment and computational authority. Here the measurement-interface consequence is

narrower: a representation, observation channel, estimator, or model output retains only the scientific role assigned in the typed pipeline and cannot inherit authority from another layer.

A serious reader should finish with a typed pipeline from representation to verdict and a clear location for every uncertainty. The aim is to identify exactly what a measurement must accomplish to bear on PTVc and which achievable steps can strengthen that connection.

Disciplined approximation gives measurement a practical path forward. Measurement becomes more humane and more scientific when it can preserve uncertainty without abandoning the effort to learn.

A simple hypothetical record shows why the interface must be typed. A participant reports intense distress, a wearable records low movement, a neural channel is unavailable, and the final survey is missing because care takes priority. None of those facts is the latent ledger. Together they define an observation problem with channel-specific error, informative missingness, and a decision about what can still be estimated.

Identification comes before precision. Ten thousand measurements cannot identify a neutral origin that every instrument defines by its own sample mean, and a narrow confidence interval cannot repair a scalarization that erased a relevant dimension. The governing questions are which transformations preserve the target, which anchors are independent, and which conclusions survive the full admissible representation set.

The measurement pipeline is also a provenance graph. Every value that enters an estimator has a source instrument, version, timestamp, preprocessing rule, calibration file, missingness status, and access history. That graph is not administrative decoration. It is how a skeptical reviewer can reconstruct whether the apparent terminal pattern came from the participant, the instrument, the preprocessing code, or a later analytic choice.

The chapter remains valuable under every scientific outcome. A failure to identify origin can improve affect measurement; a vector result can reveal structure a scalar score concealed; a no-estimate verdict can identify the measurement work still needed; and a well-governed observation system can protect participants throughout the research. Together these developments contribute a durable science of inference from lived experience.

6.1 — Latent Objects, Routes, and Estimands

Measurement connects a defined target to the observations used to estimate it. Part II begins with a map of the latent phenomenal object, route state, observation process, estimand, estimator, intervention, decision rule, and public verdict. Each connection specifies the work needed to interpret the resulting estimate (Borsboom, 2005; Cronbach & Meehl, 1955).

$L(t)$ denotes the latent cumulative object specified by the formal branch. $\hat{L}(t)$ denotes a data-derived estimate produced under a declared observation and analysis model. A momentary estimator, interval contribution estimate, operational proxy, posterior path, and public verdict are further objects, each with its own units, uncertainty, and claim ceiling.

The measurement chain is constructive, not merely restrictive. It identifies what must be learned at each arrow: construct validity from latent contribution to channel; calibration from channel to momentary estimate; rate and unit validity from momentary estimate to interval contribution; dependence and missingness propagation from interval contributions to a running estimate; and decision discipline from the running estimate to a verdict.

An observable channel may respond to arousal, pain, attention, medication, sleep, temperature, autonomic state, or device artifact. A prospectively specified and empirically evaluated bridge establishes how that channel informs the declared construct in its intended context.

Table 6.1. Latent, route, estimand, and estimator layer map.

Object	Status	What it can license	What it must not become
$L(t)$	Latent construct received from the formal statement	Formal target and claim object under assumptions about bearer, rate, gate, and endpoint	Sensor output, clinical score, human worth, public identity, or observed truth
$\hat{L}(t)$	Measured or estimated path under an observation model	Qualified empirical statements about a reference class, instrument version, and window	Direct access to $L(t)$ or a rank of persons
Candidate momentary estimator	Momentary physiological contribution estimator or rate proxy	Local evidence about a declared contribution rate under context of use	Universal feeling meter or portable public score
Candidate interval contribution estimate	Interval contribution convention	Additive measured increments when rate, units, and cadence are justified	Moral credit, debt, or exchangeable human value
Posterior path	Model-conditioned distribution over possible measured paths	Uncertainty-aware component claims and sensitivity analyses	Recovered inner life or final verdict

Calibration then turns channel data into an estimator. Preprocessing, artifact rules, transforms, channel weights, lags, signs, calibration samples, and diagnostics belong to the estimator's identity. Transparent composites and hierarchical

latent-state models can both be legitimate; the relevant question is whether their assumptions, failure modes, and uncertainty are recoverable.

Integration adds a separate burden. A momentary value must be a rate, or a validated proportional rate proxy, before it can support additive interval contributions. An index may classify or forecast without carrying the units needed for a cumulative phenomenal measure. Establishing that unit is a substantive scientific result.

Accumulation is more than arithmetic. Serial dependence, interpolation, missingness, model uncertainty, calibration drift, and version changes shape the running measured estimate. A cumulative path assembled from changing instruments is a stitched object whose seams must remain visible.

The final arrow is inferential. A measured path may support a finite-band result, a local dynamics result, a route comparison, or an abstention decision. Exact latent equality requires the additional representation, endpoint, observation, recovery, and measured-to-latent conditions specified for that branch.

Provenance is part of the object, not clerical metadata. Instrument version, code identity, feature map, transform, calibration population, analysis date, and feature-availability time determine which estimator generated each interval. Similar numerical output from different versions does not make those versions interchangeable.

This architecture also prevents uncertainty from being shifted onto participants. The mapping from person to estimate is many-to-one, context-limited, and model-dependent. Similar data can

arise from different inner histories, and similar inner histories can generate different observations. A research estimate cannot rank a person or replace human judgment.

Strong measurement is a chain of testable propositions. A failed link narrows the licensed claim and points to the next technical problem; a strengthened link opens a more demanding test. The program gains credibility by making both movements visible.

The subsections now define each layer and its interfaces. Later curves, bands, and endpoint summaries should be read through this map: every display concerns a named object under a declared model, never an unmediated view of a life.

6.1.1 — The Typed Layer Map

$L(t)$ is the latent cumulative process named by the formal architecture. It is not a sensor output, not a survey total, not a clinical impression, and not the visible shape of a chart. It is the theoretical cumulative object that the science program tries to approach under explicit assumptions about valence, unity, time, and endpoint (Borsboom, 2005; Cronbach & Meehl, 1955).

$\hat{L}(t)$ is different in kind. It is a data-derived estimate of a ledger-like path, produced by a measurement system and an analysis model. Its value depends on channels, transforms, sign conventions, filters, priors, calibration decisions, missingness assumptions, endpoint rules, and the reference class being studied. The hat marks the estimate's dependence on its measurement and analysis model.

$$L_i^{\text{phen}}(t) = L_i^{\text{phen}}(S_i^\theta -) + \int [S_i^\theta, t] r_i(s) ds$$

A clean measurement statement therefore has a grammar. It says: under instrument version v , transform ψ , cadence c , population P , endpoint rule E , and model family M , the data support an estimated distribution of measured cumulative contributions over the observed window. It does not say that the latent life itself has been read.

The temptation to collapse $L(t)$ and $\hat{L}(t)$ is strongest when the estimate behaves as expected. If a terminal estimate sits near zero, the eye wants to say the Law has appeared. If a measured trajectory narrows, the prose wants to say the life converged. This chapter forbids that shortcut because the same visible pattern can arise from measurement attenuation, selection, smoothing, endpoint conditioning, or an omitted initial state.

The distinction also protects negative evidence. If $\hat{L}(t)$ fails, the program must ask what failed: the latent claim, the observation bridge, the instrument version, the scalarization, the endpoint adjudication, the missingness model, or the rival comparison. Treating the estimate as the latent object would make every failure both too large and too small at once.

The public sentence that survives is narrow and powerful. A measured cumulative estimate may support, narrow, suspend, or defeat a component claim under declared conditions. It may never become the ledger itself by confidence, beauty, repetition, or public need.

6.1.2 — Momentary and Interval Estimators

Candidate momentary and interval estimators are often the first terms readers want to use because they sound like instruments. In

this book they are not consumer devices or public scores. A candidate momentary estimator is the momentary contribution estimator: an attempt to estimate a physiological or physiological-plus-model signal that bears on the latent valence rate. A candidate interval contribution estimate is the interval contribution convention that lets those momentary estimates enter a running measured cumulative estimate (Carroll et al., 2006; Hernán & Robins, 2020; Pearl, 2009).

To construct an interval contribution, express the momentary estimate as a validated rate or proportional rate proxy and integrate it over the specified duration. The conversion defines the units of the accumulated quantity.

$$\Delta \hat{L}_{ij} := \int_{t_j}^{t_{j+1}} \hat{r}_i(s) ds$$

A candidate interval contribution estimate therefore has a narrower role than a general pleasure or pain unit. It is a research contribution over a specified interval, under a specified estimator and transform. It exists only inside a context of use. It is not a portable coin of experience, not a unit of spiritual status, not a clinical entitlement, and not a way to compare the worth of lives.

A useful way to read the stack is moment, interval, path, and verdict. A momentary estimator belongs to the first layer. An interval contribution estimate belongs to the second. $\hat{L}_i(t)$ belongs to the running measured path. Public claim language belongs to the verdict layer. The stack becomes unsafe when any layer speaks with the authority of another.

This stack also disciplines uncertainty. Uncertainty in a candidate momentary estimator is local and channel-specific. Uncertainty in a candidate interval contribution estimate includes duration, interpolation, and covariance across time. Running-ledger uncertainty includes serial dependence, parameter uncertainty, model choice, and version migration. A public verdict inherits all of them.

Later chapters will deepen the measurement stack. The governing relation is already clear: a candidate momentary estimator targets momentary contribution; a candidate interval estimate integrates that contribution over a justified interval; and the measured cumulative estimate aggregates intervals only after units, signs, cadence, missingness, and uncertainty have been declared.

6.1.3 — Operational Proxies and Context

Not every useful signal is a ledger estimate. Unity markers, reserve proxies, sedation status, device tolerance, clinician notes, family reports, movement, sleep stage, medication, and context can all matter profoundly without becoming a candidate momentary estimator or a candidate interval contribution estimate. They are operational proxies that help interpret, bound, challenge, or qualify the measurement bridge (Campbell & Fiske, 1959; Manski, 2003).

The distinction is practical. A unity measure may indicate whether the gate is plausibly open; a reserve proxy, whether a terminal model is in the relevant regime; and a motion artifact, why a physiological channel lost reliability. None alone estimates cumulative felt burden.

Operational proxies prevent false precision. A participant's wearable signal may become cleaner near the endpoint because movement stops and artifact decreases. A naive model might mistake this for affective convergence. A disciplined model treats observation quality as a proxy for channel reliability, not direct evidence about neutrality.

Contextual proxies carry similar discipline. A family visit, a medication change, an infection, a delirium episode, or a sleep disruption can shift observed signals and missingness. These facts may explain a measured pattern, identify confounding, or trigger exclusion from a specific estimand. They do not give the model permission to revise the latent cumulative process after the fact.

A proxy can reduce burden when it prevents overinterpretation, and increase burden when it expands surveillance. Treat proxies as governed research metadata: minimal, purpose-bound, versioned, and incapable of ranking persons.

A neutrality claim requires a clear distinction between proxy and estimator. The first names context; the second estimates contribution. Keeping their roles explicit allows ambient evidence to inform the appropriate part of the PTVC analysis.

6.1.4 — Posterior Paths and Information Sets

A posterior path is a distribution over possible trajectories conditional on data and assumptions, not a recovered autobiography of consciousness. Narrowness need not reflect access to the latent path: it may arise from strong priors, smoothing, shared parameters, endpoint conditioning, or an

overconfident observation model (Borsboom, 2005; Cronbach & Meehl, 1955).

Given model M , instrument version v , data D , and specified priors or estimation rules, a posterior may favor some ledger-like paths. Another observation model may yield a wider distribution, shifted path, or no usable estimate from those data.

The estimated path is shaped by scalarization, unity gating, interval integration, missingness, and endpoint timing. A defensible posterior path carries each of those uncertainties into its reported conclusions.

The filter-smoother firewall starts here. Filtered or forecast paths use information available at estimation; smoothed retrospective paths may use future observations. Smoothing can aid sensitivity analysis and visualization but cannot count as prospective prediction of the terminal approach.

Posterior paths are maps with legends, not windows without glass. The legend must state whether a path is filtered, forecast, cross-fitted, smoothed, endpoint-aligned, retrospectively adjudicated, or simulated. A missing label is a scientific defect.

A posterior-path result identifies the component studied and the measurement and analysis conditions supporting it. This makes the finding usable in subsequent comparisons and replication.

6.1.5 — Versioning and Provenance

Every measured object has a history. Instruments change, firmware changes, sampling rates change, channel availability changes, questionnaires change, code changes, preprocessing

changes, and calibration models change. A measured cumulative estimate that does not carry version and provenance cannot be compared responsibly across time, sites, or studies (Carroll et al., 2006; Hernán & Robins, 2020; Pearl, 2009).

Versioning identifies the measurement and analysis procedure. If one estimator version uses heart-rate variability, sleep architecture, and endocrine timing, while another adds skin conductance and removes a motion-sensitive channel, the intended estimand may remain the same even though the estimator changes. A migration study must establish whether the versions estimate that target comparably; a changed target requires a separately stated estimand.

Provenance must travel from raw data to public summary. It should identify source channels, device identifiers, timestamps, preprocessing, artifact rejection, imputation, transformations, scalarization weights, model version, code hash, analyst role, and release status. Without it, neutrality language cannot be audited.

Versioned objects also matter for participant dignity. A person should not be haunted by a stale estimate after a model changes. Research governance must define when a derived result is retained for reproducibility, when it is superseded, and when downstream model training requires separate permission.

Version comparison needs a declared migration protocol. It may use overlapping calibration samples, synthetic fixtures, bridge studies, or consent-permitted reanalysis of archived raw data, or find versions incomparable for a given verdict.

A measured cumulative estimate becomes scientifically usable when its version is visible enough to be challenged. Hidden version drift is a quiet way to manufacture agreement.

6.1.6 — Intervention and Scientific Verdict

Intervention stays on its own layer because an action may alter the latent process, route state, observation process, endpoint distribution, or several at once. The causal estimand names the manipulated object and its identification assumptions; otherwise reporting, medication, or observation changes can masquerade as ledger change (Pearl, 2009; Hernán & Robins, 2020).

The decision rule is distinct from the estimator. One interval can yield different decisions under different loss functions, equivalence regions, or abstention costs, so the rule must state how false support, false defeat, and forced estimation are treated.

A verdict packet names the target component, branch, class, estimand, estimator, uncertainty set, recovery result, rival comparison, sensitivity analysis, and licensed language. The packet distinguishes route preference, model ambiguity, and estimation availability, giving each result its own interpretation.

Synthetic recovery tests verify that the system does not confuse representation failure with route failure, visibility collapse with neutrality, or rival equality with preferred-route support. A numerically accurate estimator can still yield a wrong verdict.

The result remains a research object, never a personal label. It cannot locate consciousness, summarize a life, rank character, determine eligibility, or certify balance, completion, safety, holiness, or worth (Campbell & Fiske, 1959; Manski, 2003).

Human governance completes the map. Algorithms may estimate, simulate, and reproduce analyses, but named reviewers must own status updates, dissent, correction, and withdrawal, and leave a record another qualified reviewer can reconstruct.

Typed lineage records each variable's layer, transformation, assumptions, and prohibited verdicts. No sensor-to-latent-state, state-to-estimand, or estimate-to-verdict arrow is automatic.

This architecture supports strong science without building a hidden ledger of persons. It permits route tests, principled abstention, and component-wise learning while care, dignity, privacy, and human authority remain independent.

6.2 — Representation and the Neutral Origin

Measurement begins by defining the representation that a signal is intended to inform. For an accumulated signed quantity, the representation specifies its origin, scale, and temporal composition before sensor calibration begins (Luce & Tukey, 1964; Krantz et al., 1971; Borsboom, 2005).

The origin problem is not solved by more participants, denser sampling, or a more accurate device. Those improvements can reduce uncertainty around a coordinate system whose zero is still arbitrary. Identification requires an independent rule that distinguishes neutrality from a convenient reference point.

Representation and origin therefore precede estimation. The aim is not to impose one psychophysical theory, but to specify what any adequate theory must identify, preserve under transformation, and expose to failure.

$$\tilde{v}_i(t) = a_i + b_i v_i(t), b_i > 0, L_i(T) = a_i A_i(T) + b_i L_i(T)$$

Positive rescaling preserves terminal zero when the origin term vanishes. Translation generally does not. The cumulative active exposure $A_i(T)$ makes the origin error grow with the domain over which experience is accumulated.

6.2.1 — Representation Before Measurement

A representation determines which phenomenal distinctions the model treats as scientifically meaningful. It may be scalar, bipolar, vector-valued, ordinal, distributional, or partially ordered, and each choice supports different operations (Krantz et al., 1971; Borsboom, 2005).

Integration is licensed only after sign, interval-comparable units, exposure, and clock are identified. A vector representation additionally requires either a declared scalarization whose weights are identified or a genuinely non-scalar target with explicit adjudication rules.

Instrument design should follow the representation rather than define it accidentally. Questions, anchors, tradeoffs, physiology, behavior, and neural measures are observations whose relationship to the representation must be modeled.

A device that predicts self-reported intensity may be useful while remaining agnostic about whether positive and negative valence share one axis. Predictive success on one channel does not settle the representation.

Competing theories may treat valence as constructed, multidimensional, action-oriented, or inseparable from arousal and meaning. The program must compare those views rather than encode one into the scoring pipeline.

Beginning with representation keeps the book scientifically generous. PTVC may be narrowed or defeated, while the work on phenomenal measurement remains valuable.

Representation choice can be compared through out-of-sample measurement invariance, predictive calibration, and recovery under known generative structures. The winning representation is the one that survives those tests, not the one that yields the most appealing terminal graph.

6.2.2 — Neutral Origin as Identification Problem

The neutral origin is the point separating positive from negative phenomenal contribution under the declared representation. It cannot be assumed to coincide with a verbal midpoint, sample mean, adaptation level, or absence of overt distress (Cacioppo & Berntson, 1994; Russell, 1980; Cronbach & Meehl, 1955).

Because translation changes an accumulated residual, the origin is a structural parameter rather than a cosmetic convention. Exact closure is unidentified when multiple origins fit the same observations and imply different terminal signs.

Candidate origin anchors should be triangulated through indifference, symmetric tradeoffs, within-person reversals, psychophysical thresholds, and invariance tests. None should be accepted solely because it produces terminal balance.

A person may use the word neutral for emotional numbness, calm, boredom, low arousal, or inability to report. Those states need not occupy the same phenomenal origin.

Moving-reference models are not nuisances around a fixed zero; they are competing explanations of apparent neutrality. Adaptation level, expectation, and homeostatic prediction may all shift reports toward a local baseline. An invariant origin is supported only if independent anchors outperform those dynamics.

Treating origin as identification creates a clear failure route. The program can report that valence was measured but exact-zero adjudication was not identified.

Triangulation is informative only when disagreement is preserved. If indifference judgments, psychophysical thresholds, and within-person reversals identify different neutral regions, the protocol should carry that discordance into an identified set or No adequate estimate rather than average the anchors until the terminal residual closes.

Adaptation-level and active-inference models predict moving references, so the prospectively specified contrast must include outcomes that favor those rivals. Neutrality that tracks expectation shifts, physiological set points, or context-specific prediction errors would count against a fixed phenomenal origin even if terminal estimates appear centered.

6.2.3 — Affine Freedom and Positive Rescaling

Many psychometric representations are identified only up to an affine transformation. For PTVC, a stream-constant positive change of units preserves sign and terminal zero, while translation changes the neutral origin and adds an exposure-weighted term to the accumulated ledger. The usual interval-scale freedom must therefore be narrowed by independent origin evidence (Mokkink et al., 2024; Gagnier et al., 2025).

Apply §5.3.4’s affine result here. The empirical task is to identify the origin independently enough that translation freedom cannot manufacture or erase terminal neutrality, then propagate that origin uncertainty into every absolute-level inference.

The zero-preserving transformation criterion fixes the admissible change of units. A fixed positive measurable pointwise multiplier chosen independently of the ledger preserves zero total for every

zero-total finite signed measure for which the weighted total exists exactly when it is constant on the event space. To see why, place equal positive and negative unit masses at any two events. Their original total is zero; the weighted total is zero only when the two multipliers agree. Because the events are arbitrary, the multiplier is constant. Conversely, a constant multiplier factors out of every total and preserves zero. Event-, state-, or time-dependent reweighting therefore defines a new representation whose invariance must be established on its declared branch (Folland, 1999).

That distinction matters for attention weights, subjective-time weights, unity weights, and adaptive rescaling. A strictly positive reweighting preserves pointwise sign but may preserve the final net residual in one dataset while changing atomic or singular masses, gross variation, covariance, or route diagnostics elsewhere. Weights that can vanish may additionally remove contributions from sign regions. Protocols should register its integrability domain and effects before terminal fitting.

Common-factor models constrain factor means or intercepts to identify their statistical coordinates. Terminal analyses should propagate admissible location uncertainty and evaluate the resulting estimates against independently established origin anchors.

Two laboratories can therefore report almost perfectly correlated trajectories and still disagree about the terminal residual because one scale is shifted by a constant. Agreement in rank or covariance cannot repair a displaced origin.

Cross-instrument calibration should estimate whether linking transformations are origin-preserving in the declared population and context. When the admissible transformation family changes the terminal conclusion, the result belongs to an identified set, sensitivity envelope, or No adequate estimate rather than to exact-zero adjudication.

Before terminal fitting, the protocol should freeze the admissible origin family, its complexity, calibration sample, and leverage limits. Blocked cross-fitting can estimate that family without allowing terminal outcomes to select the zero, and the report should show terminal-origin leverage and sensitivity across every admissible anchor. If a saturated or outcome-adaptive origin class can manufacture closure, or if the verdict changes across admissible origins, the correct result is Origin-sensitive or No adequate estimate rather than selection of the favorable parameterization (Manski, 2003).

6.2.4 — Independent Origin Anchors

An origin anchor is independent when it is chosen through information not algebraically equivalent to the terminal residual under evaluation. Independence prevents neutrality from being fitted into existence (Krantz et al., 1971; Borsboom, 2005).

Possible anchors include within-person indifference points, bidirectional tradeoff tasks, opponent-process symmetry, physiological transitions with convergent self-report, or experimentally manipulated reference conditions. Each anchor needs a context of use.

Validation should examine stability across time, language, culture, clinical state, and response mode. A universal anchor should not be inferred from one instrument in one population.

In a repeated-choice task, the point at which a participant is indifferent between a brief positive and negative perturbation may provide one origin constraint. It remains an experimental construct rather than a direct reading of the latent zero.

Demand characteristics, framing, loss aversion, memory, and strategic responding can shift apparent indifference. Multi-method convergence is therefore more informative than any single task. If the anchors still admit several origins, the result is an identified set or No adequate estimate, not a favored neutrality convention.

Independent anchors let the study evaluate the origin before terminal analysis begins.

Any diagram of origin identification should show the independent anchor, the moving-baseline rival, and the abstention route. A zero mark without those dependencies falsely presents a calibration choice as an observed fact.

Independent sign anchors can place a concrete interval around a common origin. Suppose a candidate channel has the local relation $Y = aV + b + e$, with $a > 0$, and independently qualified negative and positive anchors have $V \leq 0$ and $V \geq 0$ respectively. If each anchor error satisfies $|e_k| \leq \delta_k$, every admitted origin b lies in

$$[\max_{i \in \mathcal{N}}(Y_i - \delta_i), \min_{j \in \mathcal{P}}(Y_j + \delta_j)].$$

Here $\mathcal{N}$ and $\mathcal{P}$ denote the negative and positive anchor sets, each nonempty. An empty interval challenges the common-origin, sign or error assumptions; a nonempty interval preserves the remaining origin uncertainty. Anchor signs must be justified as experienced valence, not inferred solely from preference, avoidance or the channel being calibrated. With probabilistic error bounds, coverage must hold jointly across the anchors. Holding out a context or anchor family then tests whether the same origin interval transports without using terminal outcomes to select it.

6.2.5 — Adaptation, Moving Baselines, and Abstention

Reports can adapt to context, expectation, illness, deprivation, repeated exposure, and local norms. A moving comparison level can make severe states sound ordinary without making them phenomenally neutral. The scientific task is to separate change in represented experience, change in adaptation state, and change in reporting rule (Cacioppo & Berntson, 1994; Russell, 1980; Borsboom, 2005).

A dynamic-origin model specifies how its reference point changes and how that changing reference relates to the target origin. Recentring on a recent mean removes variation that an analysis of drift may need to preserve.

Neutral-anchor tasks, indifference choices, bidirectional perturbations, and diverse channels can constrain the origin; disagreements must remain visible. A questionnaire midpoint, population average, or phrase such as normal for me reflects response and adaptation, not an automatic zero of lived valence.

Digital measurement already offers an encouraging, testable starting point. In the thirty-day study by Cloos et al. (2025), 3,761 Norwegian adults registered for ecological momentary assessment and 2,718 completed the post-study survey. That survey showed high tactile precision and generally consistent ratings of affective scenarios. Among its respondents, 82.19% placed neutrality within five points of the unipolar midpoint of 50, and 88.89% within five points of the bipolar midpoint of zero. Some participants instead located neutrality near their own typical affect. These findings support the feasibility of precise reporting and identify a concrete calibration task: distinguish familiar feeling, scale interpretation, and phenomenal neutrality.

Negative controls should distinguish genuine origin movement from response-style change. A valid origin should remain stable under manipulations that alter expression without altering the represented neutral point, and it should move coherently when independent anchors indicate a real change.

Transport offers another test. Freeze an origin model estimated in one context and test it under different language, burden, affective range, and observation conditions. Failure may indicate construct drift, response shift, or a reference-class boundary; it cannot justify silent recalibration until terminal zero returns (International Test Commission, 2018; Mokkink et al., 2024).

Origin uncertainty must propagate into every residual and equivalence test. A narrow terminal band under a fixed baseline may widen sharply when anchor disagreement or dynamic-origin uncertainty is included. Exact zero is demanding precisely because location uncertainty cannot be ignored. Origin sensitivity

should therefore be reported beside the point estimate rather than hidden in a secondary robustness appendix.

Abstention is therefore a positive scientific output. When available anchors cannot distinguish neutrality from adaptation or scale-use change, apply §6.2.4's identified-set/No adequate estimate rule and target the missing origin, invariance, or transport information rather than enlarging the same ambiguous sample.

The abstention rule should be protocol-fixed in the units of the identification problem before outcome inspection. It can depend on anchor disagreement, posterior width, invariance failure, transport loss, or sensitivity to admissible baseline models. Once the threshold is crossed, terminal-neutrality language stops for that context. The trigger should be tested on simulated and negative-control cases so its abstention behavior is known before it encounters the substantive endpoint data.

The archive should preserve competing origin models, accepted anchors, overlap regions, residual sensitivity, and reopening conditions. This record makes moving baselines tractable and prevents a preferred origin from becoming invisible through repeated recalibration (Manski, 2003; Gagnier et al., 2025). Archived origin alternatives also make future reanalysis possible without pretending that the originally chosen baseline was uniquely identified.

6.3 — Sign, Cross-Sign Units, and Scalarity

Sign is not merely a plus or minus printed before a number. It is a claim that experiences on opposite sides of an origin are oriented within a common structure and that their magnitudes may be compared or accumulated under declared units. Such arithmetic requires empirical representation conditions, not just labels (Luce & Tukey, 1964; Krantz et al., 1971; Cacioppo & Berntson, 1994).

Cross-sign comparison is one of the least studied obligations in any exact cumulative theory. Many instruments are good at ordering severity within pain or pleasure while offering little evidence that one unit on one side is commensurate with one unit on the other.

This section separates sign identification, unit construction, scalarity, and non-scalar alternatives. The purpose is to prevent arithmetic from outrunning phenomenology.

$$x_i(t) \in R^d, F_{i,w}(t) = w^T x_i(t)$$

$$I_i(T) := \{ \int w^T x_i(t) dt : w \in W \}$$

When weights are not identified, the scientifically honest object may be the identified set $I_i(T)$ rather than one scalar ledger. Exact zero is then a property of all, some, or none of the admissible scalarizations, and those cases must be distinguished.

6.3.1 — Sign Identification

Sign identification determines which phenomenal changes count as positive, negative, or neutral under the target representation. Verbal labels alone are insufficient because language, appraisal,

and response strategy can reverse or blur them (Krantz et al., 1971; Cacioppo & Berntson, 1994).

A sign rule should be anchored in directional preference, controlled tradeoffs, approach and avoidance under controls, and convergent channels while permitting mixed and uncertain states. The rule constrains sign without pretending direct access or forcing near-origin observations into binary labels.

Measurement models should propagate sign uncertainty rather than force every observation into a binary classification. Near-origin states may require intervals or posterior probabilities over sign.

A state described as painful but meaningful may be negative in one component, positive in another, and ambiguous under a scalar summary. The sign cannot be settled by choosing the adjective the analyst prefers.

Motivational salience, arousal, and action tendency are imperfect proxies. Fear can motivate approach, pleasure can coexist with avoidance, and social display can oppose felt valence.

Sign identification gives the terminal claim a chance to fail for the right reason. Without it, zero can be produced by misclassified contributions rather than genuine balance.

6.3.2 — Cross-Sign Comparability

Cross-sign comparability asks whether positive and negative magnitudes share units that support subtraction. Ordering states on each side separately does not answer that question (Russell, 1980; Borsboom, 2005; Manski, 2003). For a mixed-sign record

with positive total $P > 0$ and negative magnitude $N > 0$, rescaling the two signs by separate positive factors a and b preserves every ordering and the neutral point, yet changes the total to $aP - bN$. Choosing a/b above, below or equal to N/P gives a positive, negative or zero result from the same ordinal observations. Independent cross-sign calibration is therefore what makes a zero total informative.

Commensurability may be investigated through bidirectional tradeoffs, duration exchange, intensity matching, or structural models that predict choices across signs. Each method introduces assumptions about memory, risk, and decision context.

Protocols should report whether comparability is global, local, person-specific, or only partially identified. A single conversion factor is not justified by convenience.

A participant may judge ten minutes of severe pain as outweighing an hour of mild pleasure, while another makes the opposite tradeoff. The difference may reflect true scaling, framing, or unstable decision processes.

Choice asymmetries do not automatically reveal phenomenal cross-sign units. Prospect-theoretic weighting, loss aversion, and decision noise can produce similar behavior. Choice data can inform the bridge only through a model that separates valuation, action, and experienced exposure.

The program gains credibility by allowing comparability to fail. A non-scalar branch may preserve scientifically meaningful structure when subtraction is not earned.

Cross-sign comparability should be tested as a family of local bridge functions, not presumed to be one universal multiplier. A protocol can ask whether the positive-to-negative exchange rate is stable across intensity, duration, context, and person. Heterogeneity that survives calibration belongs in the identified set and limits any scalar terminal claim.

Local exchange maps also need loop consistency. If a sequence of positive-to-negative and negative-to-positive calibrations returns to a different unit when the comparison loop closes, the scalarization is path-dependent rather than globally coherent. A global scalar claim therefore requires identity holonomy, or a prospectively declared tolerance, across the relevant comparison graph; persistent loop discrepancy points to context dependence, an omitted state variable, or the need for local charts and an identified set (Krantz et al., 1971).

6.3.3 — Units, Rates, and Exposure

A momentary value becomes an accumulated contribution only through units and exposure. Rate, duration, sampling cadence, and time parameterization must fit together dimensionally (International Test Commission, 2018; Mokkink et al., 2024).

If F has units of phenomenal magnitude per experienced-time unit, integration produces phenomenal magnitude. Changing the clock requires a corresponding Jacobian so that the represented exposure does not change merely because the timeline was reparameterized.

Instruments should distinguish state level from rate contribution. Repeated samples may approximate an integral, but irregular

spacing and missing intervals require explicit quadrature or state-space estimation.

Two studies can report the same average intensity while observing different durations. Treating the averages as equal accumulated burden would discard the very exposure PTVC concerns.

Retrospective duration neglect and peak-end effects are important rivals because reported totals may not scale with elapsed time. They challenge the observation bridge, not the need to specify units.

Dimensional discipline connects every formula to a clearly defined quantity. Every formula should make clear what accumulates, under which clock, and in which units.

A reproducible dimensional audit should let a reader who never saw the fitting code identify every state, rate, increment, exposure clock, integration step, and unit conversion that turns observations into a cumulative quantity. Repeating the audit after each transformation often exposes a hidden change of estimand before statistical fitting begins.

6.3.4 — Scalarity and Non-Scalar Alternatives

A scalar representation can support a total order; arithmetic accumulation additionally requires justified units and additivity. Experience may resist these requirements because states can differ along dimensions that are not reducible without value-laden weights (Krantz et al., 1971; Cacioppo & Berntson, 1994).

Non-scalar alternatives include vectors, cones, partial orders, signed measures on feature spaces, and distributional

representations. Each supports different terminal constraints and different empirical tests.

Model comparison should ask whether scalarization improves held-out prediction and transport without erasing systematic residual structure. A scalar model that fits only after flexible weights may be less informative than a bounded identified set.

A nominated scalar measurement model can face an explicit test before accumulation. In one fixed context, suppose four channels satisfy $Y_j = \alpha_j + \beta_j V + e_j$, with finite second moments, fixed loadings, conditionally zero-mean errors and errors uncorrelated across channels conditional on V . Their off-diagonal covariances obey $\sigma_{ij} = \beta_i \beta_j \text{Var}(V)$, so four distinct indices must satisfy

$$\sigma_{ij}\sigma_{k\ell} - \sigma_{ik}\sigma_{j\ell} = 0.$$

Freeze the channels and error structure before testing this restriction on held-out records. Correlated method effects, nonlinear loadings or additional dimensions can break it. Failure challenges this particular linear observation model; a scalar phenomenal quantity may still admit a different observation relation. Success earns further sign, origin, cross-sign and additivity tests. Repeated channels are informative only when their error assumptions survive independent checks.

Pain intensity, emotional valence, agency, and meaning may move in different directions. A scalar summary can be useful for one decision while remaining inadequate as a universal phenomenal currency.

One objection is that no science can proceed without simplification. The answer is not to reject scalars, but to label their context of use and test the consequences of alternative simplifications. For scalarity and non-scalar alternatives, prediction and recovery are evaluated together so one branch cannot benefit from asymmetric validation.

The title's scientific mission is served when scalarity becomes a hypothesis with competitors. Exact additive PTVC then stands or falls on a declared representation rather than on arithmetic habit.

Technical and public artifacts should label whether they display a scalar, a vector, a partial order, or an identified set. A single number is not a harmless simplification when the representation question remains open.

When scalarity fails, the analysis need not collapse into silence. It can retain a vector-valued path, a partial order over states, or a set of scalarizations that preserves all admissible weights. Those outputs answer weaker but still exact questions and reveal precisely which additional comparison axiom a scalar ledger would require.

6.3.5 — Cross-Stream Noncomparability and Surviving Claims

PTVC is intrastream. Even when each stream admits a useful signed representation, the theory does not supply a common interpersonal cardinal scale on which one person's magnitude can be subtracted from another's. Positive rescaling within a stream can preserve its zero without fixing a shared unit across streams (Krantz et al., 1971).

This blocks cross-stream balancing and ranking. A larger estimated residual, component magnitude, or gross variation in one stream does not show greater moral worth, blame, need, holiness, civic fitness, or entitlement than a value from another stream. The measurement architecture does not by itself establish an interpersonal scale that the formal target does not supply; even a separately justified common scale would not permit cross-stream cancellation.

Population science remains possible. Researchers can estimate the prevalence of branch compatibility, distributions of within-stream parameters, instrument performance, endpoint classes, or route-specific effects. Those summaries concern patterns across streams, not a pooled phenomenal ledger.

Hierarchical models can borrow statistical strength while preserving semantic separation. Partial pooling may improve estimation of within-stream objects, but shrinkage toward a population distribution does not make the latent units interpersonally commensurable. The model should report which parameters are shared and which remain stream-specific.

Group-level averages require an explicit measured-scale interpretation. They may summarize reports, behaviors, physiological variables, or decision outcomes under a validated common instrument. They do not become averages of total lived value unless an additional interpersonal representation theorem is supplied, and no such theorem is assumed here.

Transport across streams is a separate problem. Similar observation patterns can arise from different origins, response

functions, routing, or exposure clocks. A transported estimator must state the invariances and overlap conditions it uses rather than treating demographic or diagnostic similarity as sufficient.

Several strong claims survive noncomparability. Within-stream sign, temporal order, exact or approximate residual compatibility, component structure, local dynamics, endpoint support, and route restrictions can remain scientifically meaningful. The absence of an interpersonal scale narrows what can be compared; it does not erase every within-stream question.

Governance must reflect this limit. No dashboard, ranking, eligibility rule, resource-allocation score, or social dossier may be built from estimated ledgers. Care and rights are governed by ordinary ethical, clinical, legal, and public reasons, not by a cross-stream comparison the science cannot identify.

The result is a disciplined population science of nontransferable objects. It can compare methods, frequencies, mechanisms, and uncertainty without pretending to compare the value of lives. That boundary is both mathematically honest and essential to the humane use of the research (Manski, 2003; Nissenbaum, 2010).

6.4 — Mixed Valence and Identified Sets

The primary additive branch uses the signed valence-rate representation specified in Sections 4.2 and 5.2. That gives the Law its mathematical force. It also creates a measurement burden: the science book must test whether a scalar, signed, additive path is adequate for the domains in which it speaks. A single path is not a premise that measurement may inherit without scrutiny (Cacioppo & Berntson, 1994; Russell, 1980).

Human experience is often mixed, layered, and temporally structured. Pain can coexist with meaning. Grief can contain love. Fear can carry agency or helplessness. Numbness can be relief, dissociation, medication effect, exhaustion, or shutdown. A measurement system that compresses all of this too early may create a ledger that is smooth because it has discarded the dimensions that mattered.

The program can use scalar paths where their predictive performance, discrimination, and sensitivity checks support that representation. Vector, ordinal, partially ordered, kernel-weighted, and nonadditive alternatives provide comparison models when additional structure is useful.

The scientific question is whether a neat scalar has shifted uncertainty into dimensions it cannot express. Scalarization is admissible only when the weighting rule is identified well enough for the intended conclusion and when sensitivity across the admissible weight family remains visible.

Table 6.2. Representation options, strengthening evidence, and narrowing triggers.

Representation	Scientific question	What would strengthen it	What would force narrowing
Single signed scalar	Can one additive path represent contribution?	Stable held-out prediction, sign validity, residual adequacy	Vector or ordinal rivals dominate
Vector state $z(t)$	Do multiple dimensions carry irreducible information?	Distinct dimensions predict auxiliary evidence and dynamics	Projection loses key burden patterns
Weight family W	Do conclusions survive plausible scalarizations?	Same qualitative result across W	Sign or verdict changes across W
Temporal kernel K	Does simple duration-weighting hold?	Additive path beats recency and memory rivals	Kernel models explain outcomes better

A scalarization sensitivity module should be treated as a standard output, not a special appendix. The study declares a family W of admissible weights and computes the full range of measured cumulative estimates over W , or a certified outer bound on that range. A representative grid alone yields a sampled sensitivity analysis, not the full identified set, unless a justified interpolation or extremum bound covers the unsampled weights.

Consider a terminal-window dataset with three dimensions: bodily distress, fearful arousal, and peaceful social safety. One weight vector may produce a near-zero measured terminal value

because distress and safety cancel. Another may produce a negative value because bodily distress is weighted more heavily. A third may produce an indeterminate result because peaceful safety and fear cannot be ordered on one axis. The scalar conclusion is then representation-sensitive.

Temporal alternatives create a parallel sensitivity. Simple duration weighting may produce one path. A recency kernel may privilege the last day. A memory-consolidation kernel may make sleep intervals matter disproportionately. A peak-end model may predict reports better than an additive path. The measured object changes with the temporal rule.

These examples show how evidence can distinguish among competing representations. Predictive accuracy, calibration, and held-out performance guide the choice of model under its stated assumptions.

Representation can be qualified separately in different settings. A palliative terminal cohort might require a vector measured object while ambulatory longitudinal cohorts permit a stable scalar proxy. Reporting those reference classes identifies which representation is supported in each setting.

The same framework guards against overengineering. A vector model with many dimensions can overfit. A kernel family can be too flexible. A rival can be designed to explain every outcome. The answer is a prospectively specified model space, held-out prediction, and model recovery, not simpler prose.

The fair burden here is the burden of representation. If the book asks lived experience to become a number, the book must show

what the number sacrificed and what evidence justifies the sacrifice.

The scalar ledger becomes credible only after it has survived the possibility that experience required more than one axis.

Table 6.3. Scalarization-sensitivity scenarios and verdict consequences.

Scenario	Scalar weight family result	Temporal rule result	Verdict consequence
Stable scalar	All plausible weights preserve sign and near-band conclusion	Additive model predicts held-out data best	Measured scalar path may proceed for this context
Weight-sensitive	Some weights support near-neutrality and others do not	Temporal rule unchanged	Report identified set; no single neutrality sentence
Kernel-sensitive	Weights stable, but recency model beats additive path	Recency or peak-end rival dominates reports	Narrow additive claim or separate remembered-experience target
Vector-required	No scalar projection preserves residual adequacy	Multiple dimensions needed	Suspend scalar candidate interval contribution estimate for that reference class

Scalarization is where the Law's elegance meets the complexity of experience. A single signed axis is powerful because it lets accumulation, neutrality bands, and dynamic signatures be stated clearly. It is dangerous because real experience may contain dimensions that resist collapse: pain with dignity, fear with

meaning, grief with love, numbness with relief, exhaustion with gratitude, or attachment with dread.

A projection from a vector representation can be evaluated through stability across held-out contexts, preservation of relevant temporal dynamics, prediction of auxiliary observations, and residual structure. When admissible projections lead to different signs or decisions, report the corresponding set of results.

Identified sets over weight families are the disciplined middle. The study declares a plausible set of weights before outcome analysis and reports the range of cumulative results across that set. If the entire set supports the same local conclusion, the conclusion gains robustness. If the set crosses neutrality, support, defeat, and abstention, the result is representation-sensitive rather than quietly positive.

Temporal kernels introduce a parallel issue. The received ledger treats duration additively, but remembered experience, decision value, trauma consolidation, habituation, anticipatory dread, and relief can follow different temporal rules. A rival may weight peaks, endings, recency, duration, repetition, or state transitions differently. The program must show when simple accumulation is empirically adequate and when a kernel-weighted path is the better object.

Testing representation can reveal the structure the object requires. If experience is ordinal, multidimensional, thresholded, hysteretic, or path-functional in ways that prevent a single

additive representation, the analysis should adopt the appropriate representation and identify the revised branch being tested.

6.4.1 — Vector and Multidimensional States

A vector experience state represents the possibility that lived experience contains multiple dimensions relevant to burden and relief. Instead of assuming one latent rate $F(t)$, the model may begin with $z(t)$, a vector of affective, interoceptive, motivational, and contextual dimensions that might or might not collapse responsibly onto one signed axis (Cacioppo & Berntson, 1994; Russell, 1980).

The vector is not a rejection of the Law. It is a representation test. If a scalar projection of $z(t)$ can reproduce held-out dynamics, preserve sign, predict auxiliary witnesses, and remain stable across contexts, the scalar path gains credibility. If no projection can do this without hiding systematic residuals, the scalar ledger narrows.

Dimensions must be chosen carefully. They should not be a grab bag of every measurable feature. Candidate dimensions may include valence, arousal, bodily distress, agency, social safety, cognitive load, temporal orientation, or meaning-related affect, but only where each has a measurement route and a reason to bear on the latent target.

Vector states also make mixed affect visible. A participant may have high physical pain and high peace, low arousal and high dread, or numbness and relief at once. A single score can hide these patterns unless the scalarization rule explains how they combine.

The vector approach protects against false neutrality. Opposing dimensions can cancel numerically while remaining phenomenologically distinct. A zero scalar projection may mean balanced contributions, inadequate representation, or loss of information. The model must distinguish those possibilities.

Vector experience states are the measurement chapter's way of saying: one axis must be earned, not assumed.

A multidimensional representation also changes what counts as empirical agreement. Two models may predict the same signed projection while disagreeing sharply about the geometry of the underlying state, the covariance among dimensions, and the transformations that preserve meaning. Accordingly, validation cannot stop at aggregate fit. It should test whether the proposed dimensions are recoverable across tasks, whether their relations survive held-out contexts, and whether rival dimensional systems explain the same observations with fewer unsupported commitments. A vector model earns scientific value by preserving distinctions that improve prediction or adjudication, not by multiplying coordinates. When additional dimensions do not change any observable consequence or decision rule, they remain descriptive possibilities rather than identified structure.

6.4.2 — Scalarization Weight Families

Scalarization weights define how a vector state becomes a signed rate. A weight vector w maps $z(t)$ into $w^T z(t)$. That operation is not just arithmetic. It is a claim about which dimensions count, in what direction, with what relative force, and under which

population and context (Krantz et al., 1971; Manski, 2003; Imbens & Manski, 2004).

Weights can come from theory, within-person calibration, expert prior, measurement model, predictive performance, or declared sensitivity analysis. They cannot be chosen because they make terminal results look favorable. Outcome-selected weights manufacture evidence by changing the object after seeing the answer.

$$F_w(t) = w^T z(t)$$

$$L(T; W) := \{ \int_0^T g_\theta(t) w^T z(t) dt : w \in W \}$$

A transparent scalarization report should state the candidate dimensions, raw units, transformations, signs, weights, constraints on admissible weights, and why the family W was chosen before terminal analysis. It should also report which conclusions survive across W and which depend on a narrow region of the weight space.

Within-person weights may differ from between-person weights. A channel that predicts distress within one stream may not compare across streams without invariance, and a population-average scalarization can erase individual structure. Scalarization is therefore context-bound and reference-class-bound.

Weights should be testable. Held-out prediction, residual structure, response shift, auxiliary witness conflict, and rival model comparison all place pressure on the chosen projection. A scientifically useful scalarization states in advance which observations would support it, require revision, or rule it out.

The safest scalar path is the one whose weights were declared before it knew where the path would end.

Weight families should therefore be treated as scientific objects rather than silent tuning parameters. A defensible program states how weights are learned, which transformations leave conclusions invariant, how uncertainty in weights propagates to the terminal claim, and which populations or contexts fall outside the calibration domain. Hierarchical or partially pooled weights may improve transport, but they also introduce assumptions about exchangeability and group structure that must be tested. The strongest analysis reports not only a preferred weighting rule, but the region of weight space over which the substantive conclusion survives. If a narrow, outcome-selected corner alone produces compatibility, the correct result is fragility, not confirmation.

6.4.3 — Identified Sets and Partial Orders

When weights are uncertain, the study can report every ledger conclusion generated across a declared weight family, together with the evidence and assumptions that define that family. Use the term identified set when these are values of one fixed target compatible with the available evidence and declared model. When changing the weights changes the target itself, report a scalarization-sensitivity set and distinguish that sensitivity from uncertainty about a fixed target (Manski, 2003; Imbens & Manski, 2004; Mokkink et al., 2024).

Suppose plausible weights produce terminal measured values ranging from strongly negative to near neutral. A single chosen weight might tell a compelling story, but the identified set tells

the scientific truth: the result depends on scalarization. The public claim must narrow until the representation problem is solved.

Identified sets can also strengthen a result. If every admissible weight family produces the same qualitative local signature, such as variance compression or rival defeat within the observed window, the component result becomes less representation-sensitive.

The set must be declared before outcome inspection. A weight family expanded only to include favorable weights, or narrowed only to remove inconvenient weights, is another form of neutrality by fiat. The family W is part of the prospectively specified bridge.

Identified sets show which values remain compatible with the evidence and declared assumptions. Reporting that range gives readers a clear account of what is established and which further observations could sharpen the result.

6.4.4 — Sensitivity to Weight and Kernel Choice

The received ledger integrates contributions over time without an additional memory kernel. Measurement must still test whether relevant empirical targets behave that way. A recency kernel, memory-consolidation kernel, or state-dependent temporal weighting can produce different path functionals from the same momentary estimates (Cacioppo & Berntson, 1994; Russell, 1980).

This matters because lived experience and remembered experience are not identical. A participant's later report, clinical expression, or decision behavior may depend more on recent,

peak, unresolved, or narratively organized states than on simple duration-weighted accumulation. If the study's observed outcome is a report or behavior, the temporal rule matters.

$$L_{w,K}(T) = \int_0^T K(T,s) g_{\theta}(s) w^T z(s) ds$$

A kernel-weighted model can explain how a reporting channel depends on temporal history. Comparison with the proposed accumulation model identifies which observable summaries the channel supports and which additional measurement relation is needed.

Temporal kernels must be specified before analysis. $K(T,s)$ may depend on recency, sleep consolidation, arousal, memory salience, or state transition, but each dependence changes the measured object. Endpoint-normalized kernels chosen after seeing terminal data can manufacture convergence.

A strong study should compare simple integration with kernel alternatives using held-out prediction and rival discrimination. It should ask whether the additive path predicts future physiological and auxiliary patterns better than recency or memory-weighted rivals.

The time rule is part of the construct. A ledger is not additive merely because the analyst prefers integrals.

Sensitivity analysis should also distinguish epistemic uncertainty from genuine heterogeneity. A conclusion that changes because the analyst lacks information about the correct kernel is different from one that changes because different people or contexts require different kernels. The first calls for better identification,

richer measurements, or wider uncertainty sets. The second may require a reference-class restriction or a formally heterogeneous model. Collapsing these cases can make a program look more stable than it is, or more unstable than it is. A serious report therefore attributes each change in verdict to the assumption, population, observation regime, or transport step that produced it and preserves those dependencies in the public summary.

6.4.5 — Memory, Recency, and Duration Rivals

Peak-end and duration rivals deserve special attention because they can make a path look cumulatively meaningful without being a cumulative ledger. Remembered evaluation can overweight the most intense moment and the ending, while neglecting duration or the distribution of experience across the interval. A life-course or terminal report may therefore reflect salient moments rather than additive accumulation (Kahneman et al., 1993; Redelmeier & Kahneman, 1996).

The Law of Fairness claim is about felt experience as it occurs within the qualifying conscious stream, not merely retrospective satisfaction. Still, if the data used to estimate the ledger are dominated by memory, report, or decision, the rival must be modeled. Otherwise the study may confuse remembered burden with experienced burden.

Duration also matters. Equal peak pain for ten seconds and ten hours cannot be treated identically in an additive ledger. Yet some reports compress duration, especially under memory distortion, trauma, sedation, or narrative reconstruction. The measurement system must identify whether the channel is duration-sensitive.

A useful analysis can compare additive duration-weighted models, peak-end models, recency models, and hybrid models. The winner should depend on held-out prediction and calibration, not on which model produces prettier neutrality language.

If peak-end rivals explain reports better, the study may still use physiological rate channels to estimate additive contributions. But it must not use those reports as if they validated the additive path without resolving the mismatch.

The chapter's rule is simple: remembered shape can inform the bridge, but it cannot replace the path unless the claim changes.

6.4.6 — Representation Failure as a Result

Representation failure means the evidence does not support a requirement of the declared branch, such as sign, neutral origin, scalarity, additivity, component structure, or invariance. These are typed failures: each invalidates a different descendant and points toward a different successor (Borsboom, 2005; Cronbach & Meehl, 1955).

Failure may be local. A scalar can work for ordinary unimodal episodes yet fail under mixed valence, trauma, altered consciousness, or extreme duration; one instrument may fail while another remains usable. The least committal supported object then replaces the failed one: vector for scalar, identified set for point, partial for total order, generalized functional for additive branch, or No adequate estimate when the family remains unresolved.

Representation includes resolution. Coactivation at one scale can appear as alternation at another, and smoothing can erase short

excursions. Weights, kernels, summaries, and resolutions must therefore be justified independently, evaluated out of sample, and varied within prospectively named families rather than selected for a desired residual.

Sealed-world recovery should distinguish scalar, vector, set-valued, origin-failed, and observation-failed generators. A pipeline that always returns one scalar cannot recognize when that object is unwarranted. Its failure register must identify the smallest invalidated descendant, preserving ordinal, componentwise, or formal work that still survives.

Public language names the scientific limit without pathologizing the participant. Representation not identified in this context is more accurate than declaring the person incoherent or the entire inquiry unmeasurable. Recovery-qualified abstention removes hidden flexibility and identifies the next discriminating study (Manski, 2003; Mayo, 2018).

6.5 — Observation Kernel and Protocol Measurement Map

The observation kernel is the probabilistic and procedural map from latent phenomenal states to the data a protocol records. It includes instruments, response processes, context, missingness, timing, calibration, and the transformations that turn raw signals into analysis variables (Bollen, 1989; Carroll et al., 2006).

A protocol measurement map is broader still. It says which latent object is targeted, which estimand is defined, which estimator is used, and which decision rule produces a verdict. Without that map, a measured ledger can quietly replace the latent ledger in the reader’s mind.

The section defines each estimator by its target, observation model, and context of use.

$$Y_{ijk} = \mathcal{H}_{\eta,j}(X_i(t_k), C_{ijk}, R_{ijk}) + \varepsilon_{ijk}, \quad \hat{\theta}_i = \mathcal{A}(Y_i, M_i; \eta)$$

The observation kernel $\mathcal{H}_{\eta}$ maps latent state, context, and response conditions into observations. The analysis operator $\mathcal{A}$ then produces an estimator under an explicit missingness model M_i . Here i indexes person or stream, j indexes channel, k indexes occasion, and Y_i denotes the collection $\{Y_{ijk}\}_{j,k}$. Neither object is the latent experience itself.

6.5.1 — Momentary Contribution Estimators

A momentary contribution estimator targets the signed phenomenal rate or a declared local functional at one interval. It is not a direct meter of consciousness and not a permanent score attached to a person (Bollen, 1989; Carroll et al., 2006).

The estimator may combine channels only through a declared model that preserves their distinct error structures. A composite gains no authority merely because it predicts one report channel well.

Calibration should use held-out tasks, repeated measures, and within-person anchors. The estimator's context of use must state populations, settings, time scales, and conditions under which abstention is required.

A phone prompt, bedside report, and neural recording may disagree during the same interval. The disagreement is data about the observation process rather than permission to average blindly.

Proxy dominance is a central risk. A high-performing behavioral or neural model may predict reports while missing unreported experience, response suppression, or culture-specific expression.

The scientific role is modest but essential: estimate a local contribution with uncertainty, then allow later models to test whether those estimates can support cumulative claims.

Local estimators should be recalibrated across contexts and versions, with drift monitored through anchor tasks and replicate measurements. A stable model coefficient is not evidence of stable measurement when the response process has changed.

The received candidate momentary estimator uses qualified physiological channels. Self-report, pairwise comparison, behavior, and contextual variables remain distinct auxiliary witnesses for calibration, challenge, and the modeling of observation conditions. A broader multimodal estimator may combine these sources with autonomic signals and neural features

under a separately declared model, but it must not silently replace the received physiological candidate. Each channel retains its own error structure.

6.5.2 — Auxiliary Witnesses Remain Distinct

Self-report, behavior, physiology, neural data, clinician observation, and informant evidence are witnesses with different access paths. None is the privileged voice of the latent object in every context (Campbell & Fiske, 1959; Mokkink et al., 2024; Gagnier et al., 2025).

A multi-channel model should preserve channel identity before fusion. Conditional dependence, shared artifacts, and common causes must be represented rather than treated as independent confirmations.

Protocols can compare channel-specific likelihoods, posterior predictive checks, and conflict patterns. Persistent conflict may narrow the estimand or trigger no-estimate rather than being hidden in a composite.

Sedation can reduce speech and movement while leaving uncertain residual experience. A bedside observer and a device may both report quiet, yet quiet is not a validated neutral origin.

Keeping witnesses distinct respects both persons and data. It prevents an algorithmic consensus from being mistaken for access to inner truth.

Conflict analysis should ask which channel fails under which conditions. A channel may lag, saturate, react to observation, or lose validity near an endpoint while another remains informative.

Modeling those patterns explicitly can identify the failing measurement interface, whereas an omnibus composite can conceal it behind an apparently stable average.

Common-method bias can make questionnaires converge, while shared device placement can make sensors share one artifact. The adversarial plan should treat agreement that disappears when methods, hardware, or timing are diversified as support for a shared-method rival rather than counting the number of channels as independent evidence.

6.5.3 — Latent-to-Observed Channel Map

The channel map specifies how latent states generate observable responses under context, memory, attention, communication, and device conditions. It is the causal heart of measurement rather than a technical appendix (Shiffman et al., 2008; Stone et al., 2007; International Test Commission, 2018).

A channel model may be parametric, nonparametric, state-space, or simulation-based, but it must expose identifiability assumptions. Different latent trajectories can produce the same observations when gain, noise, or missingness changes.

Negative controls and perturbation studies can test parts of the map. Instrument changes, repeated prompts, known stimulus manipulations, and cross-device calibration reveal whether the estimated channel behaves as expected.

A decline in facial expression near an endpoint may reflect reduced experience, motor weakness, medication, fatigue, or observation-gain collapse. The channel map must carry those alternatives.

A flexible neural decoder can fit observed labels while remaining untransportable across scanners, populations, or states of consciousness. Prediction within one dataset is not channel validity.

A credible map makes later terminal evidence possible by showing how observed data could have been different under different latent states.

Channel validity should be evaluated through held-out devices, sites, states, and perturbations, with calibration as well as discrimination reported. A decoder that preserves rank but not scale cannot support the same cumulative estimand without an additional bridge.

Channel validity becomes scientifically useful when it yields counterfactual separation: the same latent change should alter the channel in a predicted direction, while a channel-specific artifact should not masquerade as a change in the latent object.

6.5.4 — Interval Contribution and Cadence

Cumulative estimation requires interval contributions, not isolated snapshots. Cadence determines what transitions, bursts, and durations can be recovered from discrete observations (Bollen, 1989; Carroll et al., 2006).

An interval contribution estimator may integrate a filtered latent path, use quadrature between observations, or model continuous-time dynamics. Its uncertainty must include interpolation and unobserved intervals.

Sampling design should be tied to the fastest scientifically relevant dynamics and participant burden. Dense measurement is not automatically better when it changes experience, exhausts participants, or increases dropout.

A daily rating can miss short severe episodes; a minute-by-minute protocol can create intrusion and reactivity. The appropriate cadence depends on the estimand and ethical context.

Irregular sampling, informative prompting, and event-triggered recording can distort apparent duration and transition rates. The design mechanism belongs in the likelihood or sensitivity analysis. When those designs remain observationally equivalent, cadence cannot be used to favor a route; the ambiguity belongs in the verdict.

Plots and tables must encode actual observation intervals, not just row order. Cadence, gaps, and interval length determine what cumulative contribution is represented; a smooth trajectory drawn through irregular samples can conceal the exposure uncertainty under study.

Cadence is therefore an identification parameter. Sampling that is too sparse aliases bursts into trends and alternation into coactivation; sampling that is too dense can increase reactivity and participant burden. A defensible design studies this tradeoff prospectively and reports which interval functionals remain recoverable at each cadence.

6.5.5 — Context of Use and Versioning

Every estimator needs a context of use stating what decision it supports, for whom, under which conditions, and with what

consequences of error. A research estimator is not automatically a clinical, legal, or public instrument. A version is qualified only for its declared population, setting, cadence, and decision; material changes require new validation rather than silent inheritance (Campbell & Fiske, 1959; Mokkink et al., 2024; Gagnier et al., 2025).

Versioning should lock features, calibration data, preprocessing, model weights, uncertainty rules, and abstention thresholds. A changed estimator is a new measurement object even if its public name remains the same.

Performance should be reported by relevant setting and subgroup, with transport tested rather than assumed. Retired versions, changed items, preprocessing revisions, and recalibration reasons belong in the production record so every result can be traced to the exact instrument that generated it.

An estimator trained on verbal adults in laboratory tasks may fail in pediatrics, dementia, intensive care, nonverbal disability, or another language. Those failures are not edge cases when the theory claims broad conscious scope.

Continuous software updates can create silent measurement drift. Apparent longitudinal change may reflect the instrument's evolution rather than the person's experience.

Context and versioning make measurement answerable. They allow the book to say exactly which object a result concerns and which object it does not.

6.5.6 — Abstention, Privacy, and No-Score Governance

Abstention rules must be protocol-fixed before outcome inspection for excessive uncertainty, out-of-distribution inputs, unresolved channel conflict, missingness, calibration failure, endpoint ambiguity, or privacy limits. Human review may add context, but it cannot convert an unidentified object into an estimate or waive a formal stop (Manski, 2003; Mokkink et al., 2024).

An abstaining system should name the failed gate: origin not identified, observation insufficient, endpoint unresolved, model recovery failed, or privacy-limited audit. Typed outcomes preserve information that a generic inconclusive label would erase.

Data minimization is part of validity. Collect only channels needed for the declared estimand, retain them for a stated period, and separate research identifiers from ordinary identity records. Secure enclaves, federated analysis, disclosure limits, and protected access may reduce exposure, but protocols must state their effects on reproducibility, calibration, subgroup analysis, and independent audit. More data or more privacy does not automatically improve or verify a measured-to-latent bridge (Nissenbaum, 2010).

No cumulative estimate may become a personal score or determine care, employment, insurance, education, legal standing, access, spiritual status, trustworthiness, or public worth. The rule applies to positive- and negative-looking outputs, including those marketed as wellness, balance, or resilience.

Participants retain refusal, pause, withdrawal, correction, and care rights; review bodies, not automated systems, own deployment limits, adverse-event response, suspension, and correction propagation.

Public communication must describe the research object rather than the person. It may report that a protocol-fixed pipeline returned a specified evidential status under declared assumptions; it may not call anyone fair, balanced, complete, unfinished, or on a required trajectory. Governance requires versioning, access logs, secondary-use limits, audit rights, incident response, and a retirement path. Changed models or contexts invalidate automatic carry-forward; affected outputs must be revalidated, narrowed, or withdrawn.

Abstention, privacy, and no-score rules are part of validity, not warnings added afterward. They determine collectable data, adjudicable estimands, and communicable conclusions without converting uncertainty into hidden human judgment.

6.6 — Missing History and Running-State Uncertainty

A complete-stream hypothesis collides with an ordinary empirical fact: researchers almost never observe a stream from its onset. They enter through a clinic, cohort, device, survey, or historical record after an unknown amount of phenomenal history has already passed (Little & Rubin, 2019; Rubin, 1976).

The unknown prefix is not a nuisance constant that can be set to zero. It is part of the latent state at enrollment, and different compatible prefixes can imply different whole-stream terminal residuals even when the observed window is identical.

This section develops decomposition, bounds, historical evidence, and the limited claims that survive delayed entry. It also clarifies why a running measured estimate is uncertainty over histories rather than a fresh ledger created by the study.

$$\begin{aligned} L_i^{\text{phen}}(T_i^* -) &= L_i^{\text{phen}}(t_0 -) + \Delta L_i(t_0, T_i^*) \\ L_i^{\text{phen}}(t_0 -) &\in B_i(t_0) \end{aligned}$$

The observable window can inform the increment. The missing prefix belongs to a bounded or modeled set $B_i(t_0)$ unless independently recovered. Whole-stream adjudication depends on how much that set can be narrowed.

6.6.1 — Observable-Window Decomposition

The decomposition separates the pre-entry cumulative state from the observed-window increment. It prevents a study from treating enrollment as stream onset (Little & Rubin, 2019; Rubin, 1976).

The identity is bookkeeping rather than evidence. Its scientific content depends on how the missing prefix and observed increment are represented, measured, and linked.

Analyses should report window-specific estimands even when whole-stream claims remain unavailable. Local drift, variance, response to intervention, and endpoint timing can still be studied.

A hospice cohort may provide high-frequency data for the final month. That data can test terminal route signatures while remaining silent about the complete lifetime residual.

Retrospective reconstruction can create an illusion of coverage when later reports are used as if they were continuous observations. Recall and narrative are channels with their own models.

The decomposition preserves scientific value without overclaim. It lets the final window answer the questions it can actually answer.

The prefix identified set should be displayed separately from the observed-window increment so readers can see which part of the conclusion comes from data and which part comes from assumptions. A narrow total produced by a narrow prior is not equivalent to direct historical coverage.

The decomposition becomes scientifically useful only when each term is assigned a distinct source of information and uncertainty. The observed-window increment can be estimated from sampled data; the missing prefix requires a historical model or bounds; and their dependence must be propagated rather than suppressed by treating the prefix as a fixed offset.

6.6.2 — Multiple Compatible Prefix Histories

The same observed late trajectory may be compatible with many earlier cumulative histories. This is a structural source of nonidentification, not merely a shortage of sample size (Tsiatis, 2006; Manski, 2003; Aalen et al., 2008).

Compatibility sets can be defined by bounded rates, known events, historical observations, or mechanistic assumptions. The wider the set, the weaker the whole-stream conclusion.

Sensitivity analysis should vary prefix assumptions and display which terminal verdicts change. A result that depends on one narrow prior should be reported as such.

Two people with identical final-week observations may have had very different prior burdens and relief. The late window cannot erase that difference.

Identified-set reporting is often more honest than one posterior mean. It shows what the data and assumptions jointly permit.

The sensitivity analysis should vary more than the prior mean. It should alter admissible path roughness, exposure rates, origin uncertainty, and dependence between the missing prefix and observed window. A conclusion that survives only one narrow historical model is a property of that model, not of the complete-stream target.

Population priors can narrow prefix uncertainty by importing social stereotypes, health inequities, or diagnosis-based assumptions. Apparent historical reconstruction that collapses under subgroup transport, prior sensitivity, or record-free

validation should favor that rival explanation rather than a recovered individual history.

6.6.3 — Bounds and Sensitivity Sets

Bounds translate incomplete history into explicit limits on possible terminal residuals. They can be wide and still scientifically useful because they reveal which conclusions do not depend on unverifiable reconstruction (Rizopoulos, 2012; Imbens & Manski, 2004).

Rate bounds, duration bounds, event records, and monotonicity assumptions can narrow the prefix set. Each assumption should be named and stress-tested.

Protocols can report best-case, worst-case, and assumption-indexed residual ranges. Equivalence with zero is meaningful only if the relevant range lies within a declared empirical band under all required gates.

A documented period of prolonged severe pain may support a lower bound on the magnitude of a negative historical contribution under a justified observation bridge, even when finer details are missing. That bound is different from a complete reconstruction.

Overly tight bounds can be manufactured by excluding unrecorded states or assuming stable reporting. Conservative uncertainty is not a defect when the history is genuinely unavailable.

Bounds make the limits of identification visible and show which additional evidence or assumptions would support a sharper conclusion.

Each bound should be accompanied by an assumption-deletion analysis showing how it widens when one restriction is removed. This reveals which premise supplies apparent precision and prevents several weak assumptions from hiding behind one narrow interval.

6.6.4 — Historical Records and Their Limits

Medical records, diaries, ecological momentary assessments, wearable data, family testimony, and institutional records can constrain missing history. None is a transparent archive of phenomenal experience (Little & Rubin, 2019; Rubin, 1976).

Each source has selection, coding, purpose, and missingness mechanisms. Records created for billing or treatment may omit duration, meaning, or unreported experience.

Source triangulation should preserve provenance and disagreement. Historical evidence can update a prefix model without being merged into one falsely homogeneous stream of data.

A medication chart may document dosage accurately while offering little information about relief, side effects, fear, or interpersonal context. A family narrative may preserve meaning while misremembering timing.

Historical records constrain specified features of a prefix; they do not establish a complete conscious history. Their contribution

must be evaluated through the coverage and source-specific checks below.

Historical evidence should narrow uncertainty proportionately to what it actually records. The record's dignity lies in being used carefully, not in being treated as complete.

Historical records should be displayed with coverage windows, source type, provenance, and known omissions. A chart that fills unrecorded years with a continuous line converts absence of evidence into invented history.

Historical evidence is strongest when it constrains a specific feature rather than pretending to reconstruct an inner archive. Medication records may bound exposure, diaries may locate episodes, and wearable traces may constrain timing; none by itself identifies phenomenal sign or magnitude. The protocol should record which latent dimensions each source can and cannot narrow.

6.6.5 — Local Dynamics Versus Whole-History Claims

Delayed-entry studies can investigate local dynamics, route predictions, transitions, and bounded terminal implications. A justified prefix model extends those results to questions involving earlier history (Tsiatis, 2006; Manski, 2003; Aalen et al., 2008).

A local model can estimate drift, innovation, response kernels, or observation gain in the observed window. Those estimates require no claim that the prefix is known.

Every result should identify its object: local state dynamics, an observed-window increment, a prefix bound, or the complete

terminal residual. The label determines which missing-history assumptions and transport claims are required.

A study of naturally occurring changes, or an independently justified beneficial intervention, can test short-window opponent dynamics without measuring a lifetime ledger. The design must satisfy Chapter 3's prohibition on inducing suffering for Law-specific evidence.

The temptation is to let a strong local signature stand in for the whole law. Rivals can share local dynamics while differing at the boundary.

Layered claims preserve value under defeat. Local affective science, measurement, and dynamics remain useful even if complete-stream closure is not identified.

6.6.6 — No Reset at Enrollment

Enrollment begins the observed window. A complete-history analysis combines its measured increments with an independently identified or bounded prefix (Manski, 2003).

The correct decomposition keeps the prefix explicit. Multiple compatible histories can produce the same observed state at enrollment while implying different component totals, residuals, route states, and future completion sets. Replacing that family with zero is an assumption, not a neutral starting point.

A useful analysis carries a distribution, interval, mixture, particle cloud, or compatible-prefix set suited to the problem. A mean and variance may fail when the family is multimodal or sign uncertainty changes reachability in opposite directions.

Historical records can constrain the prefix, but each channel needs validation. Diaries, medical notes, informant reports, prior devices, treatment histories, and recalled episodes capture different objects and fail for different reasons. Agreement may reflect shared documentation, not access to phenomenal value.

Overlapping cohorts can test local-dynamics transport. Different entry times provide common windows for comparing transition structure and measurement behavior, while historical coverage is documented separately.

A reset-artifact negative control should be part of model recovery. Synthetic worlds can share identical post-entry dynamics while differing in prefixes chosen to imply different whole-history verdicts under the benchmark assumptions. A method that returns the same whole-history verdict despite sufficient information to distinguish those cases should fail recovery; the audit should test whether it has reset the ledger implicitly. Including reset artifacts in recovery makes delayed-entry bias an explicit failure mode instead of a theoretical warning.

Prefix and endpoint uncertainty must be varied jointly. Modest uncertainty on each side can combine into a wide terminal set, especially when continued-experience, false-ending, or routing branches remain plausible. The report should identify whether history, endpoint, representation, observation, or their interaction is the limiting gate. Factorial sensitivity analysis can show whether uncertainty is dominated by one bridge or by interactions that become important only when several weak links coincide.

The output should distinguish post-entry change, a full-history bound, and a complete-stream verdict. The first may be estimable when the others are not. This distinction permits useful longitudinal science without overstating enrollment data. That separation also prevents a precise within-window effect from lending unwarranted precision to the unobserved prefix.

No reset at enrollment is a technical and ethical rule. It prevents convenience from erasing hidden prior burden or presenting a study-relative baseline as the beginning of a person's lived history. When the prefix cannot be identified or validly bounded, No adequate estimate is the honest whole-stream result. A bounded prefix may still support useful partial conclusions; an unbounded one should block only the whole-stream inference that depends on it.

6.7 — Equivalence, Adjudicability, and No-Neutrality-by-Fiat

Exact closure, finite-resolution compatibility, an atom at zero, probability-one closure, and set-universal closure are different targets. Section 6.7 prevents a favorable measurement result from moving among them. The central question is not whether an estimate looks close to zero, but which object, resolution, bridge, and quantifier the study can actually adjudicate (Manski, 2003; Mayo, 2018).

The exact-branch decision rule used in this section is stated formally in §6.7.1. The key distinction is that empirical uncertainty containing zero is not the same object as a truth-containing singleton $\{0\}$; the former may support compatibility at a declared resolution without establishing exact latent closure.

A finite-band target asks a different question. It can test whether the truth-containing residual set lies inside a prospectively declared acceptance region at the study's resolution. Passing that test can be valuable evidence for the declared band while leaving exact equality open.

Scientific acceptance geometry must also remain separate from epistemic uncertainty. The equivalence band describes the target region the protocol is prepared to treat as scientifically indistinguishable for a named use. The confidence, posterior, or identified set describes what the data and model allow researchers to know about the estimand.

Table 6.4. Neutrality-manufacturing routes and required controls.

Manufacturing route	How it can fake neutrality	Required control
Centering	Forces means or windows toward zero	Pre-outcome origin rationale and sensitivity
Baseline subtraction	Makes residuals appear balanced by reference choice	Independent baseline and status label
Derivative transforms	Telescopes cumulative claims into boundary terms	Label the observed change and its historical coverage
Outcome-selected weights	Finds channels or signs that cancel	Frozen weights or exploratory label
Endpoint fitting	Moves window to the favorable ending	Endpoint firewall and sensitivity
Exclusions	Removes inconvenient trajectories	Symmetric rules and independent audit

An atom at zero is different again. A distribution can place increasing mass in narrow neighborhoods of zero without assigning positive probability to zero itself. Claims about an exact pile-up therefore require a validated inverse bridge, deconvolution, partial-identification argument, or another structure that distinguishes an atom from a narrow continuous law.

Adjudicability precedes estimation. A large sample cannot rescue a design whose bearer, ownership map, origin, sign, endpoint, observation process, or measured-to-latent bridge leaves the target unidentified. The correct result may be a bounded statement, an identified set, or No adequate estimate.

No-neutrality-by-fiat protects this hierarchy from analytic choices that make favorable outcomes easier. Centering, baseline subtraction, difference transforms, flexible scalarization, endpoint alignment, smoothing, missingness handling, and exclusions can all be legitimate. They become defects when their role or timing is hidden or when they quietly change the target.

The timing vocabulary is exact. Protocol-fixed means written into the study protocol before outcome inspection. Prospectively specified means fixed before outcome inspection without asserting a public registry. Publicly preregistered is reserved for a timestamped external registry with a persistent identifier; this Part makes no such claim without that record.

Before outcome inspection, the protocol fixes the instrument version, channel list, sign rules, scalarization family, temporal kernel family, endpoint rule, missingness model, equivalence region, bias-transfer strategy, exclusion rules, and abstention criteria. Later deviations remain visible and are routed to sensitivity or exploratory status unless an independently governed amendment justifies another label.

Adversarial reanalysis asks whether reasonable alternatives reverse, erase, or weaken the conclusion. Synthetic fixtures ask whether a nonneutral process can be manufactured into apparent neutrality, whether observation attenuation can imitate latent convergence, and whether derivative transforms telescope into boundary differences. These tests let the pipeline fail before participants carry more burden.

Figure status is part of inference. Observed, filtered, smoothed, forecast, endpoint-aligned, retrospectively adjudicated, and simulated trajectories must be labeled as such. A smooth line can communicate a result only when readers can see which transformations and endpoint rules produced it.

This audit identifies results that survive specified alternatives, independent reconstruction, strong rivals, and transparent abstention. Each successful check strengthens the result by showing which analytical choices it can withstand.

Table 6.5. No-neutrality-by-fiat audit.

Audit item	Frozen before outcome?	Independent check	Failure consequence
Origin or baseline	Yes	Sensitivity across declared alternatives	Withdraw exact or measured neutrality language
Channels and signs	Yes	Blind artifact and sign audit	Exploratory-only if changed after outcome
Scalarization weights	Yes	Identified-set report over W	Report representation sensitivity
Endpoint rule	Yes	Endpoint-firewall review	Abstain or rerun under sensitivity
Missingness model	Yes	Severity and device-tolerance checks	Widen uncertainty or no estimate
Code and figures	Versioned	External reanalysis or secure audit	Correct, retract, or retire claim

Human dignity does not depend on the outcome. Participants are not failed subjects when a bridge collapses, a rival wins, or the study cannot estimate the target. The scientific object has encountered a limit; the person remains outside every ledger, score, and verdict.

No-neutrality-by-fiat is therefore an affirmative research standard. It tells collaborators exactly what must be frozen, what may be learned, what remains exploratory, and what would force the claim to narrow. The discipline makes favorable evidence more valuable because the conditions under which it could have failed are visible.

The standard also protects cross-study accumulation. A later meta-analysis can combine results only when target geometry, observation resolution, endpoint class, and bridge status are sufficiently aligned. Pooling incompatible exact, band, atom, or measured-only objects creates apparent precision around no single scientific question.

Every later protocol should carry this section's audit packet forward. A favorable pattern is not complete until the route from target to data, code, figure, and public verdict can be reconstructed and challenged. That requirement is part of the contribution, not a final caution.

The final product is an inference architecture that can say exact-compatible, band-equivalent, exact-adverse, band-adverse, indeterminate, route-failed, rival-favored, or No adequate estimate without collapsing them into one success score. Such

typed outcomes allow the research program to learn without moving the claim it set out to test.

6.7.1 — Exact Latent Zero and Empirical Compatibility

Exact latent closure and empirical compatibility must remain visibly separate. For a nonempty truth-containing residual set R_i , exact compatibility requires $R_i = \{0\}$; exact adversity requires $0 \notin R_i$; and every wider set containing zero is exact-indeterminate. A confidence interval or posterior region containing zero does not by itself satisfy the singleton condition (Manski, 2003; Imbens & Manski, 2004).

Finite data can nevertheless contribute. A study may show that a declared measured estimand is compatible with a narrow region around zero under a specified observation, endpoint, missingness, and recovery model. That result is resolution-bounded evidence for the measured or bridged branch it actually addresses.

Repeated success across prospectively nested bands can strengthen the case only while the representation, bridge family, endpoint class, recovery guarantees, and limiting rule remain fixed. Choosing a narrower band after seeing favorable data tests a new analysis, not a stronger version of the original one.

An exact latent atom requires more than shrinking observed bands. A narrow continuous distribution, rounded measurements, censoring, smoothing, or an instrument floor can all create a visible concentration near zero. Distinguishing an atom requires a bridge that can recover that feature or a partial-identification argument that rules out the continuous rivals.

The report should therefore separate three sentences: what the observations establish directly, what the measured-to-latent bridge adds, and how the conclusion changes when the bridge is weakened. Precision in an estimator cannot substitute for exactness in the target.

Adverse evidence is equally typed. A valid truth-containing set that excludes zero rejects exact closure for the declared stream and branch. A point estimate away from zero without adequate coverage, endpoint, or bridge support does not yet carry that force.

This hierarchy lets the program accumulate real progress without pretending that finite measurement has solved an ontological question. Compatibility can be informative, exact adversity can be decisive for a declared branch, and indeterminacy can identify the next bridge or instrument the science needs (Mayo, 2018; Lakens, 2017).

6.7.2 — Equivalence Bands and Observation Error

An equivalence band is a scientific acceptance region for a named estimand and use; an uncertainty interval is an epistemic statement about what values remain compatible with the data and model. They can overlap, nest, or diverge, but they answer different questions (Lakens, 2017; Schuirmann, 1987; Mayo, 2018).

The band must be justified before outcome inspection through measurement resolution, calibration error, biological or temporal variability, and the consequence of false equivalence. A tolerance

chosen because the observed estimate fits inside it is an outcome-selected target.

Observation error enters through bias as well as variance. A narrow interval around a biased measured estimand can falsely suggest equivalence, while a wide but unbiased interval may remain scientifically honest. The analysis must propagate calibration, origin, endpoint, missingness, and bridge uncertainty rather than attaching them as prose after the decision.

Different uses may require different bands. An exploratory route screen, a measurement-qualification study, and a public claim about terminal concentration need not share one threshold. Each band should name its estimand, scientific purpose, error control, and status ceiling.

When several bands are defensible, the report should show the verdict profile across them rather than select the most favorable. A prospectively specified hierarchy can distinguish a primary band from sensitivity bands without treating all thresholds as interchangeable.

Observation floors, rounding, smoothing, and censoring can compress estimates into the band even when latent dispersion remains. Negative controls and simulated nonneutral data should show whether the pipeline can separate genuine compatibility from instrument-limited concentration.

Band-equivalence assesses the declared estimand against an acceptance region at a specified resolution. Reporting the estimand, band, uncertainty, and decision procedure makes that result precise and reusable.

6.7.3 — Adjudicability Before Estimation

Adjudicability is a property of the whole design, not merely of sample size, estimator precision, or computational sophistication. A study becomes adjudicative only when distinct scientific states can produce distinguishable planned outcomes under the declared measurement and endpoint architecture (Borsboom, 2005; Manski, 2003; Mayo, 2018).

Pre-estimation review traces the dependency chain from bearer and ownership through representation, origin, sign, exposure, endpoint, observation, missingness, estimand, estimator, rival set, and decision rule. A broken prerequisite blocks promotion beyond the last object that remains identified.

Large datasets can remain nonadjudicative. If two endpoint models, scalarizations, or observation kernels generate the same observed law, more observations of that law improve precision without separating the explanations. The design needs a new contrast, perturbation, instrument, or assumption, not merely more participants.

Model recovery provides an early gate. Simulated worlds with known additive closure, band compatibility, atoms, continuous near-zero laws, observation attenuation, and rival dynamics should be analyzed blind. A pipeline that cannot recover those known cases is not ready to interpret human data.

Endpoint ambiguity, unstable sign, incomplete routing, or an unidentified neutral origin can make the primary target unanswerable while leaving narrower science available. The

report should name those surviving estimands instead of treating all measurement failure as one undifferentiated limitation.

A pre-estimation adjudicability table should list each target branch, the contrast that could separate it, the principal rival, the required data, and the abstention condition. The table reveals when a planned analysis is descriptive, route-informative, or capable of adjudicating the declared terminal branch.

When no planned contrast separates the relevant states, the scientifically correct output is Model-ambiguous when adequate rivals remain inseparable, No adequate estimate when the target cannot be identified or validly bounded, or Open-set inadequate when none of the models is adequate, together with the evidence needed to change that status. Estimation should begin only after the design has shown how the result could matter.

6.7.4 — Centering and Baseline Subtraction

Centering and baseline subtraction are useful descriptive tools, but neither identifies the phenomenal neutral origin. A sample mean, adaptation level, site mean, preterminal average, or fitted terminal center can force measured values toward zero while leaving the latent origin unknown (Borsboom, 2005; Lakens, 2017).

A baseline must have a pre-outcome rationale and a declared role. It may support visualization, within-person contrast, calibration, nuisance adjustment, or a state model. Each role licenses different language; none automatically creates the zero used by the terminal hypothesis.

Temporal placement matters. A baseline estimated after adaptation, medication, sleep loss, disease progression, or entry into terminal care may absorb part of the process under study. A baseline fixed too early can instead treat ordinary drift as cumulative valence. The protocol should state the causal and temporal relation between baseline and target interval.

Within-site or within-person centering can also erase scientifically relevant heterogeneity. Culture, illness severity, communication capacity, device gain, and reporting standards may differ across sites or periods. The analysis should show whether conclusions survive defensible centers rather than hiding those differences through normalization.

Derivative and difference transforms create a related hazard. Summed changes can telescope to a final-minus-initial state, converting path burden into boundary contrast. Such transforms may test local dynamics, but they do not become cumulative phenomenal measures without an independently validated bridge.

$$\sum_{j=0}^{m-1} [\varphi(X_{j+1}) - \varphi(X_j)] = \varphi(X_m) - \varphi(X_0)$$

Endpoint alignment can compound the problem. If trajectories are centered or stopped at the point closest to zero, the procedure has placed the finish line near the answer. Truth and protocol endpoints must be fixed by information not algebraically identical to the residual being tested.

A transformation audit should report the original scale, every centering or differencing operation, the reason and timing of the

choice, alternative defensible transformations, and the resulting verdict profile. A centered graph may be useful; it must never conceal that neutrality entered through the preprocessing rule.

6.7.5 — Outcome-Selected Weights, Endpoints, and Exclusions

Outcome-selected flexibility can manufacture neutrality while looking like model improvement. With many signs, channels, lags, weights, filters, windows, endpoints, and exclusions, some combination may approach zero even when no prospectively specified representation does (Mokkink et al., 2024; Gagnier et al., 2025; Simonsohn et al., 2020).

Weights and signs may be learned on development data, but the confirmatory terminal analysis must apply a protocol-fixed rule to protected outcomes. Cross-validation helps only when the endpoint and feature construction do not leak into every fold.

Endpoint fitting includes trimming the final interval, selecting the closest zero crossing, excluding sudden or low-signal endings after inspection, or changing the endpoint class because one definition is favorable. Necessary exclusions require symmetric, documented rules and sensitivity across plausible adjudications.

Channel rejection has the same burden. Artifact removal can protect validity, yet a contradictory channel cannot be dismissed merely because it weakens the preferred story. Blind quality review, negative controls, and predefined failure rules reduce that discretion.

A prospectively specified multiverse can show how the result changes across defensible weights, kernels, endpoints, and

exclusions. The primary confirmatory route remains one declared analysis; the wider multiverse is sensitivity evidence, not a license to choose the branch that wins.

When a favorable representation is discovered after outcome inspection, the correct status is exploratory. Its value lies in generating a new protocol-fixed test on independent data. The science becomes stronger when discovery and confirmation are separated rather than rhetorically blended.

6.7.6 — Audit, Reanalysis, and No Adequate Estimate

A complete audit reconstructs the result from frozen scientific objects, not only the final code. The packet identifies the representation and origin versions, ownership map, endpoint, observation and missingness models, estimand, estimator, exclusions, equivalence rule, recovery evidence, rivals, software environment, random seeds, access controls, and protocol deviations (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018).

The trail must carry each published claim backward to its data provenance and forward to every dependent table, figure, summary, excerpt, and verdict. That chain makes later correction precise rather than reputational.

Reanalysis asks whether the protected data support the same result under the declared pipeline and defensible alternatives. Replication asks whether a prospectively specified design transports to new data. Neither substitutes for the other.

Blinded reconstruction gives an independent analyst the frozen objects and protected data without the original conclusion.

Rebuilding the estimand and decision rule can expose undocumented preprocessing, software defaults, ambiguous definitions, or hidden judgment.

Adverse reanalysis should test bounded alternatives for bias, informative missingness, covariance, origin and weight families, endpoint uncertainty, and serious rivals. Scientific purpose constrains the search, because unbounded flexibility can manufacture defeat as readily as support (Steege et al., 2016; Simonsohn et al., 2020).

Sensitive records require secure audit routes, including controlled environments, disclosure-limited outputs, synthetic fixtures, federated execution, and purpose-limited external review. Privacy limits must be reported as limits on audit scope, never as evidence that an unreviewable claim was independently verified (Nissenbaum, 2010). When audit access is restricted, independent reviewers should still receive enough synthetic or derived material to test the transformations that carry the claim.

Every abstention status must name its reason under §24.2's canonical evidence-state rules. Representation, endpoint, observation, missingness, recovery, computation, privacy-limited audit, and an open rival envelope are distinct failures that call for different next studies. A reason-coded abstention register also lets later work measure which gate most often limits adjudication and prioritize method development accordingly.

Corrections propagate to every dependent estimate, formula, table, summary, diagram, public excerpt, and reference record. A superseded certainty may not survive in a higher-reach artifact.

Propagation should follow dependency links rather than keyword similarity, so unaffected claims remain stable while genuinely dependent artifacts update.

The closing verdict states what the evidence earned under the standing rule in §24.2. Abstention, reanalysis, replication, and correction must be as reproducible as favorable results (Munafò et al., 2017; Mayo, 2018). Replicating an abstention boundary is informative because it tests whether the same inferential limit appears again under the same declared conditions.

Chapter 7 — Mechanism-Neutral Consequences

Once a boundary hypothesis has been stated and its measurement layers separated, the next question is unavoidable: what must follow from exact closure before any mechanism is chosen? That question gives the program consequences that cannot be protected by changing routes.

Mechanism-neutral reasoning begins with prefix feasibility. A nonzero cumulative residual requires an equal and opposite remaining net increment under exact PTVC, and bounded remaining exposure can make that increment unreachable in a declared continuation class. This is conditional mathematics, not an individualized reward forecast or permission to impose suffering.

Endpoint timing then introduces a sharper challenge. If a diffuse endpoint is independent of a nontrivial ledger path, exact zero at the random cut is severely obstructed. The theorem forces the program to distinguish independent timing from state-coupled hazards, first hitting, absorption, endpoint atoms, singular support, continued experience, and routing change.

Directional completion sets generalize one reserve number. Positive and negative capacity can differ, reachable increments can be state-dependent, and action can alter the route space. These sets describe possibility under a model; they do not prescribe action or forecast personal fate.

Endpoint support and compensators add time to reachability. A target may be geometrically reachable and still unlikely to be reached before termination. Counting processes, occupation

measures, hazard models, and boundary accessibility make that distinction testable.

Doob transforms and path constraints provide exact comparison models for terminal drift and concentration. Specifying those alternatives precisely makes any predictive advantage of a candidate interpretable.

The route also faces direct failure. Reserve may be unmeasurable, endpoint synchronization may fail, normalized dynamics may explode, observation gain may collapse, or rivals may predict equally well. Any of those outcomes can narrow or retire the route without rewriting the Formal Law.

The strongest rivals receive equal information and scoring. Ordinary mean reversion, adaptation, homeostasis, frailty, disease severity, medication, endpoint-conditioned bridges, Doob transforms, selection, missingness, and flexible observation models all remain capable opponents.

Open theorems are part of the positive contribution. The integral-to-route bridge, measure-level reserve geometry, random-endpoint branches, boundary support, and transport across representations remain research problems whose resolution could advance science even if PTVC loses.

Reserve, admissibility, and completion sets have defined mathematical roles in the models that follow. Care, consent, and interpretation of personal meaning remain governed by the chapter's independent ethical commitments.

By the end, the reader should see a more sharply testable program with clear opportunities for discovery. The candidate routes turn

the Law's formal target into competing scientific proposals, each with predictions that evidence can support, revise, or reject.

The chapter begins by refusing a common shortcut: no candidate mechanism is allowed to define the consequence it later claims to explain. Exact closure, if imposed on a complete stream, creates feasibility, endpoint, and support constraints before any reserve, bridge, selector, controller, or biological route is chosen. Those mechanism-neutral pressures form a theorem-to-test map. Each consequence must name the formal premises, observable implication, required bridge, strongest counterexample, and result that would count as failure. A route becomes scientifically interesting only when it adds predictions that could distinguish it from other ways of satisfying or imitating the same boundary.

A simple pair of synthetic trajectories shows why this ordering matters. One path is generated by a native state-dependent process whose innovations contract as a real reserve declines. The other is an ordinary diffusion conditioned to arrive near zero at a selected endpoint. Their terminal graphs can be nearly indistinguishable. The difference appears under an unexpected endpoint, a reserve perturbation, a held-out action, or a change in observation shutdown. A beautiful closing curve is therefore the beginning of identification, not the end.

The chapter organizes the pressure in stages. Prefix feasibility asks whether the remaining continuation set can still reach the boundary. The random-endpoint obstruction asks what exact closure would imply under independent diffuse timing. Endpoint support and compensators ask where endings can occur and how the path occupies the relevant times. Bridge and selector

distinctions separate native dynamics from conditioning, admissible-history pruning, intervention, and sampling. The stages form a dependency graph rather than a checklist: a failed representation or endpoint premise can invalidate a later route test, while a failed route prediction need not erase the earlier mechanism-neutral result. Only after those relations are explicit does the reserve route enter.

The reserve route is treated with unusual strictness because its intuition is powerful. A declining nonnegative reserve multiplied by a signed arithmetic reference process can generate steepening drift and shrinking diffusion near the boundary. Yet the product rule, scaling law, shared shocks, jumps, observation noise, denominator instability, and route-state validation must all be derived under the declared filtration. If any of those links fails, the route may be retired even while the target question remains open.

The chapter ends with a common model-comparison scorecard. Brownian bridges, Doob transforms, Ornstein-Uhlenbeck regulation, frailty, endpoint selection, observation floors, and hidden-state models receive the same data, preprocessing, information clock, parameter budget, and evaluation horizon. Component scores show performance on endpoint separation, observation robustness, and transport. The licensed conclusion may be Route-favored, Rival-favored, Model-ambiguous, Open-set inadequate, or No adequate estimate. Transparent comparison makes each outcome informative.

7.1 — Mechanism-Neutral Consequences of Exact Closure

Exact closure has consequences before any mechanism is chosen. If one stream's complete signed phenomenal measure must terminate at zero, then every observed prefix places constraints on what the remaining domain must permit. Those constraints are reachability statements, not promises that nature will supply a particular compensating event (Aubin, 1991; Aubin et al., 2011).

Mechanism-neutral consequences are valuable because they survive the retirement of a favored route. They ask what any realization of the target would have to respect, whether the route is global, local, conditioned, state-coupled, psychophysical, or not yet imagined.

The section begins with remaining capacity, nontransferability, and failure. A necessary future contribution is a mathematical requirement under stated assumptions. Any accounting language refers to that signed net requirement, never to moral debt owed by a person or to a method for selecting future rewards.

$$|L_i^{phen}(t)| \leq \int_t^{T_i^*} m_i(s) ds =: C_i(t)$$

The inequality is a reachability bound. If the current residual exceeds the maximum remaining signed capacity in the needed direction, exact closure is impossible under that bound and route class.

7.1.1 — Constraint Before Mechanism

A terminal constraint specifies an allowed boundary relation without specifying the dynamics that produce it. PTVC can therefore be studied before a controller, reserve, bridge, selector, or physical mechanism is identified (Aubin, 1991; Aubin et al., 2011).

Mechanism-neutral derivations use only the target, domain, regularity, and boundedness assumptions explicitly stated. Any result that invokes a route-specific state belongs to a later layer.

Protocols can test necessary consequences across multiple routes. A consequence that fails under adequate measurement narrows the target or its assumptions regardless of which mechanism was preferred.

If exact closure requires a path to remain within a shrinking feasible region near the endpoint, that geometric implication can be examined without knowing whether selection, control, or boundary conditioning produced it.

Some alleged consequences disappear under an allowed change of coordinates or arise because the endpoint was built into the state definition. A mechanism-neutral result must survive admissible rescaling and an independently specified endpoint. Otherwise it is a representational artifact rather than a consequence of closure.

Constraint-first reasoning makes the research program resilient. It preserves risky predictions while allowing mechanisms to compete rather than protect the target.

Each consequence should be linked to a theorem-to-test table: formal premise, observable implication, required bridge, rival explanations, and failure verdict. That table prevents a route-specific prediction from masquerading as a consequence of closure and lets several candidate mechanisms be challenged by the same declared necessary condition.

7.1.2 — Prefix Feasibility and Remaining Capacity

At any prefix time, the accumulated residual and the maximum possible remaining signed exposure jointly determine whether closure remains feasible under a declared class (Protter, 2005; Kallenberg, 2002; Ames et al., 2019).

The reserve bound compares current magnitude with an integral bound on future contribution. It is necessary under the rate bound, but it is not sufficient when sign, timing, routing, or endpoint accessibility also constrain the future.

Estimating the bound requires uncertainty in the current residual, remaining duration, rate limits, and observation model. Feasibility may therefore be set-valued rather than binary.

A large negative residual near a short, independently defined endpoint may be incompatible with closure if the maximum possible positive contribution is too small. The conclusion concerns the formal model, not the person's worth or care.

Flexible rate bounds can make every path feasible. Bounds chosen after observing the residual are therefore not tests; they are rescue parameters.

Feasibility analysis tells researchers where a strong claim becomes vulnerable. It also identifies the data needed to distinguish genuine remaining capacity from unconstrained speculation.

Rate and exposure bounds should come from independent physiological, psychophysical, or protocol evidence and should be stress-tested before terminal outcomes are inspected. Their scientific value lies in excluding continuations, not in expanding until every path survives.

7.1.3 — Nontransferability as Consequence

Nontransferability is a mechanism-neutral consequence of the received object: a feasible continuation for one stream cannot be substituted for another stream's missing signed contribution.

No aggregation across people can repair an infeasible prefix. A family's positive experience, a community's welfare gain, or a later archive cannot satisfy another stream's terminal residual.

Data linkage should preserve ownership boundaries. Household, institutional, and social variables can be covariates or causes without becoming ledger entries for the wrong bearer.

A caregiver's relief after a patient's death may be ethically and scientifically important. It cannot serve as the patient's compensating positive magnitude.

Collective-consciousness or fusion hypotheses would require new bearer and routing definitions. They cannot be introduced after individual-stream failure merely because social interaction was present. Unless a new ontology and data model distinguish

fusion from interaction, the individual-stream analysis is narrowed rather than rescued.

The consequence is both exact and humane: the formal burden stays with the object that bears it, while care and justice remain independently owed across persons.

Any detachable explanation of nontransferability should keep the unit of analysis visible. Population averages and cross-person comparisons may answer other questions, but they cannot be drawn as if one stream completed another.

This boundary also disciplines intervention design. A social, clinical, or environmental event can change a stream's future completion capacity only through the experience and timing it produces within that stream. Resources, care, and relationships remain causally important, but they are not transferable ledger entries.

7.1.4 — Necessary Compensation Without Moral Payment

For a nonzero prefix to end at zero, the remaining aggregate increment must have the opposite sign and equal magnitude. This gives a direct test of its attainable range under the declared dynamics and remaining domain (Aubin, 1991; Aubin et al., 2011).

The needed increment is defined over phenomenal contribution, not over narrative categories. Many possible trajectories can supply the same signed total, and the formal statement does not rank them by desert.

Relief, adaptation, ordinary positive experience, and other subsequent states can change the signed contribution accumulated after a negative prefix. A separately justified stream boundary defines the domain over which that contribution is evaluated.

Teleological storytelling is a major rival interpretation rather than a scientific route. It can fit any sequence after the fact and therefore generates little discriminating content.

The requirement concerns the future net difference, not a fixed quota of either pleasure or suffering. Additional positive or negative experience changes the required signed remainder by the opposite amount. The resulting gross totals, opportunities, and timing may differ between possible continuations; terminal equality alone does not establish that a chosen manipulation improves a life.

Prospective tests can examine whether remaining aggregate increments have the required sign and attainable magnitude. They should not assign compensatory meaning to individual events after the fact, because the arithmetic entails direction, not purpose.

A joint mathematical fingerprint can make the cumulative claim more informative. Consider a declared probability law for a qualified reference class on the primary additive branch. Partition each stream's complete strict-pre-endpoint domain into m disjoint measurable parts, for the same prospectively fixed finite integer $m \geq 1$, assigning each exposure token and any cut-time atom exactly once under independently justified endpoint rules. Write X_j for the signed phenomenal mass in part j and assume $\mathbb{E}[X_j^2]$ is

finite for every j . Let X collect those masses, let $\Sigma = \text{Cov}(X)$, and let $\mathbf{1}$ denote the m -vector of ones. Then probability-one closure entails

$$\mathbf{1}^\top \mathbb{E}[X] = 0, \quad \Sigma \mathbf{1} = 0.$$

Conversely, these two exact population conditions imply $\sum_{j=1}^m X_j = 0$ with probability one: the sum has both mean zero and variance $\mathbf{1}^\top \Sigma \mathbf{1} = 0$. This is a direct derivation from additivity and elementary covariance identities, under the square-integrability assumption stated here. Each row also gives $\text{Var}(X_j) + \sum_{k \neq j} \text{Cov}(X_j, X_k) = 0$, exposing a whole-history dependency restriction rather than a requirement that every neighboring interval oppose the last.

This supplies a useful multivariate diagnostic when the complete-domain and measurement premises can be met. To distinguish zero from another fixed total, test the mean restriction alongside the covariance restriction using the independently calibrated origin. Fit an unconstrained measurement model, propagate correlated errors, and quantify uncertainty in the joint mean and covariance restrictions when comparing the linked prediction with equally flexible rivals.

7.1.5 — Mechanism-Neutral Limits and Falsifiers

Mechanism-neutral consequences fail when their premises fail, and those failure conditions should be recorded before data. Unbounded rates, incomplete domains, uncertain routing, or endpoint-inclusive jumps can invalidate a simple reserve bound (Protter, 2005; Kallenberg, 2002; Ames et al., 2019).

The target also faces direct falsification when an adequately identified prefix cannot reach the terminal set under any admissible continuation in the declared class. Widening the feasibility set after seeing the residual would erase that risk.

A recovery benchmark should include provably feasible, provably infeasible, and observation-limited histories. The pipeline should classify the first two and abstain on the third; a method that merely widens remaining-capacity bounds until closure becomes possible has no exclusion power.

A protocol-fixed falsifier should state the measurement precision, endpoint class, and admissible-route family under which an empty completion set would be decisive. Changing any of those after failure creates a new test rather than rescuing the old one.

An empirical infeasibility certificate requires more than a point estimate. The prefix, directional capacity bounds, endpoint support, and admissible route class must be identified tightly enough that every compatible continuation misses the target. When that standard is met, the counterexample attaches to the declared branch rather than to a person or an informal narrative.

A mechanism-neutral falsifier must be stated at the level of the target, not at the level of one route. If a qualifying stream, complete domain, valid representation, endpoint, and measured-to-latent bridge are all secured, and the identified terminal residual excludes zero, the exact branch fails for that declared class. By contrast, a failed reserve model, a flat hazard relation, or absent variance compression defeats only the assumptions that made those patterns necessary (Popper, 1959).

The prerequisite order assigns separate tests to origin, ownership, observation, and reachability. A terminal inference uses the components that meet their stated criteria and records any unresolved dependency.

Counterexamples should be built prospectively. Synthetic one-sign streams test whether the model recognizes impossible additive closure. Prefixes outside every completion set test reachability. Independent diffuse endpoints test the non-atomic obstruction. Paths with identical terminal residuals but different gross variation test whether the analysis confuses net and total variation. A recovery pipeline should identify which condition failed rather than returning one omnibus fit score.

Severe testing requires a serious rival. Before inspecting data, the plan should state what would favor ordinary mean reversion, a conditioned bridge, endpoint selection, observation-floor compression, frailty, or a flexible hidden-state model. The target earns credibility only if the same data and scoring budget can produce a rival-favored result (Mayo, 2018; Gelman et al., 2013). Rival plans should specify a protected-data failure signature strong enough to displace the preferred account rather than merely reduce its apparent advantage.

Mechanism-neutral limits make the program durable without protecting it. They preserve useful theorems, bounds, and diagnostics when a candidate route retires, while defining when the central branch must narrow or fail. That combination is appropriate optimism for a strong formal hypothesis: discovery remains possible, and defeat remains real. Preserving route-

independent results also prevents mechanism turnover from erasing formal constraints that later models must still satisfy.

Mechanism-neutral restrictions can be organized as an exclusion geometry. At each prefix, uncertainty in residual, remaining exposure, rate bounds, routing, and endpoint class defines continuations still capable of reaching the terminal target. As observations accumulate, that set may contract, split, or empty. An empty set under prospectively justified bounds directly falsifies the declared branch; a large set means the data are not yet discriminating. This geometry is useful because it reports what the claim requires without inventing a controller to supply it (Ames et al., 2019; Protter, 2005). Tracking the size and shape of this continuation set over time can provide a model-agnostic measure of how strongly observed history restricts possible completion.

7.2 — Random Endpoints and the Non-Atomic Obstruction

Endpoint timing is one of PTVC's sharpest discriminators. If an endpoint can occur independently at a continuously distributed time while a nontrivial pre-endpoint ledger evolves, exact closure at that random cut is extraordinarily restrictive (Kallenberg, 2002; Billingsley, 1999).

Let E_i be the prospectively declared positive-probability event on which the stream, frozen routing and truth domain, finite ledger, and pre-killing extension exist on a deterministic interval I , and write $P_i^0 = P(\cdot | E_i)$. Under this same law, let $\tilde{L}_i(t -)$ be jointly measurable, let $\mathcal{F}_i^L = \sigma\{\tilde{L}_i(s -) : s \in I\}$, and let T_i^{exo} be independent of $\mathcal{F}_i^L$. Assume that T_i^{exo} is I -valued, has a Lebesgue density $f_{T_i^{\text{exo}}}(t)$ positive for Lebesgue-almost every $t \in I$, and $P_i^0\{\tilde{L}_i(T_i^{\text{exo}} -) = 0\} = 1$.

$$0 = P_i^0\{\tilde{L}_i(T_i^{\text{exo}} -) \neq 0\} = \int_I P_i^0\{\tilde{L}_i(t -) \neq 0\} f_{T_i^{\text{exo}}}(t) dt.$$

The integrand is nonnegative. Therefore $\tilde{L}_i(t -) = 0$ for Lebesgue-almost every $t \in I$ on a P_i^0 -probability-one set of eligible paths.

The result does not apply when timing is state-coupled, independence fails, the endpoint contributes phenomenally, routing changes the stream, or the pre-killing extension does not exist. First-hitting or absorbing constructions, endpoint atoms, singular timing laws, continued experience, and endpoint-inclusive value remain separate branches requiring their own

premises rather than proof details that may be changed after a result is seen.

The implication is conditional, not empirical evidence that the assumptions hold. Removing independence, diffuseness, joint measurability, or the positive-density condition changes the branch and the required data. A protocol should test those premises and report which one fails before interpreting the theorem.

Table 7.1. Endpoint classes, ledger relations, and scientific burdens.

Endpoint class	Relation to ledger	Primary scientific burden
Independent diffuse cutpoint	No path dependence under reference law	Non-atomic obstruction
State-coupled hazard	Hazard depends on declared state	Recover coupling beyond frailty and selection
First hitting or absorption	Endpoint generated by boundary event	Avoid tautological endpoint definition
Atomic or singular support	Timing concentrated on restricted set	Specify support and transport limits

7.2.1 — The Random-Cutpoint Question

The random-cutpoint question asks what exact terminal closure would imply if a declared exogenous endpoint probe sampled the pre-killing left-limit ledger at a time not selected by that ledger or by its state. This is a branch-specific discriminator, not a universal premise of PTVC. Its force depends on a prospectively declared timing law, measurable ledger path, positive-risk class, and support conditions (Billingsley, 1999; Kallenberg, 2002).

The universal requirement is anti-tautology. Whatever endpoint branch is used, the truth-level endpoint must be defined and adjudicated independently of the ledger value whose closure is tested. First hitting, absorption, zero-surface, and last-exit routes supply no target evidence when the endpoint is defined by the same zero event and then scored on that event.

On the independent diffuse branch, almost-sure exact closure over a region with positive timing density imposes a severe restriction: the pre-endpoint ledger must be zero almost everywhere on that region for almost every eligible path. The result is conditional. It does not apply automatically to state-coupled, atomic, singular, first-hitting, or continued-experience branches.

Empirical work cannot prove full independence from finite data, but it can compare timing classes. A prospectively specified tournament can include independent diffuse timing, measured state dependence, latent frailty, first hitting, absorption, endpoint atoms, and observation-shutdown models, each with its own likelihood, diagnostics, and failure conditions.

Biological death or an external accident may be relevant timing variables without being the phenomenal endpoint. A cause of bodily death is not automatically a cause or observation of the end of qualifying experience. The endpoint bridge remains a separate object that can fail.

Negative controls should expose leakage. Variables known only after the endpoint, indicators of observation shutdown, and features affected by the ledger but not by the exogenous probe can

reveal a model that appears independent because the timing process or data pipeline was misclassified.

Selection, common causes, and unobserved frailty can manufacture apparent dependence or independence. The joint model must therefore include who enters the risk set, which endpoints remain observable, and how latent state affects both timing and observation. Narrative plausibility cannot replace recovery and held-out performance.

The question is productive under every outcome. Independent diffuse timing may pressure exact closure, a state-coupled model may earn support, or endpoint adjudication may remain inadequate. Each result narrows the space of viable endpoint architectures without allowing the endpoint to be chosen retrospectively to protect the terminal claim.

7.2.2 — Independent Diffuse Endpoints

An independent diffuse endpoint has no atoms and carries positive density over a time interval while remaining independent of the ledger path under the reference law. Such timing can sample any region of that interval (Aalen et al., 2008; Andersen et al., 1993; Revuz & Yor, 1999).

Before the stronger independence theorem, a simpler at-risk screen applies. If the regular conditional law of the pre-endpoint ledger at the endpoint assigns no atom to zero for the declared conditioning history, exact zero has conditional probability zero on that risk set. This result does not call the endpoint exogenous and does not require path-endpoint independence; it only rules out probability-one exact closure for the specified non-atomic

conditional law. Empty or unidentifiable risk sets return No adequate estimate (Kallenberg, 2002).

If the ledger were nonzero on a set of times with positive measure, an independent endpoint with positive density would hit that set with positive probability. Almost-sure terminal zero therefore excludes such nonzero regions.

Studies should distinguish physical event timing from truth-level stream termination. Independence of an accident time does not imply independence of the actual conscious endpoint if experience continues or ends earlier.

A device failure time may be diffuse and independent of experience but is merely an observation endpoint. Treating it as the truth endpoint would produce a false obstruction or false support depending on the analysis.

The empirical task is to estimate how much endpoint-ledger dependence reconciles the observed path occupation with the endpoint distribution under prospectively specified covariates and timing classes.

The approximate conditional-dependence result supplies a graded target. Fix the covariates, off-band threshold, and a pre-killing ledger process on a common time domain prospectively. Work on standard Borel path and covariate spaces with jointly measurable left-limit evaluation and the required regular conditional laws. For each covariate value, compare the actual joint law of endpoint time and ledger path with the product of their conditional marginal laws. Under the product law, the off-band probability is the ledger's off-band occupation weighted by that covariate-

specific endpoint-time distribution. Average this probability over the covariates and subtract the actual off-band endpoint probability. Define the mismatch as the positive part of that difference. The mean conditional total-variation distance is at least this mismatch, and conditional mutual information, measured in nats, is at least twice its square (Cover & Thomas, 2006).

The proof compares the same off-band event under the conditional joint and product laws. By the definition of total-variation distance, their probability difference is bounded by that distance. Averaging gives the mismatch bound. Conditional Pinsker's inequality and Jensen's inequality give the squared information bound. The result quantifies the residual endpoint–ledger dependence that a state-coupled or selection model must explain after the declared covariate adjustment.

A sharper information bound uses the same common pre-killing domain and conditional joint and product laws as above. Given covariates $C = c$, let P_c be the joint law of endpoint and ledger path, Q_c the product of their conditional marginals, and $E_\varepsilon = \{|L_{T-}| \leq \varepsilon\}$. Write $p_c = P_c(E_\varepsilon)$, $q_c = Q_c(E_\varepsilon)$, $p = E[p_c]$ and $q = E[q_c]$. Mapping both laws to this same event and applying joint convexity gives

$$I(T; L \mid C) \geq E[d_{\text{Ber}}(p_c \| q_c)] \geq d_{\text{Ber}}(p \| q).$$

$$d_{\text{Ber}}(p \| q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

Logarithms are natural, with the usual extended-value conventions. For $p = 0.9$ and $q = 0.1$, the lower bound is 1.758

nats, compared with 1.280 from the squared total-variation bound. If $p = 1$, it becomes $\log(1/q)$; shrinking bands with product-law probability tending to zero require unbounded endpoint-path information in this idealized comparison. Ordinary hitting or state-coupled timing can supply that dependence. The scientific task is to identify which timing mechanism predicts its observable consequences.

An observable companion tests endpoint enrichment against a specified ordinary event model. Fix a finite horizon u . Let N_i count prospectively classified events while observed, stopped at censoring; let Y_i be the predictable at-risk indicator. In one declared observed filtration, let $Y_i \lambda_{0i}$ be the null compensator density and $Z_{i,\varepsilon}$ a bounded predictable band indicator or filtered probability for a separately defined observed-window residual. Then

$$U_\varepsilon(u) = \sum_i \int_0^u Z_{i,\varepsilon}(t) [dN_i(t) - Y_i(t) \lambda_{0i}(t) dt].$$

When the compensator is correct and square integrability holds, $E[U_\varepsilon(u)] = 0$. With continuous compensators and orthogonal stream martingales, the predictable quadratic variation is

$$\langle U_\varepsilon \rangle_u = \sum_i \int_0^u Z_{i,\varepsilon}(t)^2 Y_i(t) \lambda_{0i}(t) dt, \quad \text{Var}[U_\varepsilon(u)] = E[\langle U_\varepsilon \rangle_u].$$

The random bracket is not itself an unconditional variance or an automatic normal-test justification. Estimated intensities require nuisance uncertainty and validated calibration. Freeze models on development records; include frailty, treatment, state-dependent

timing and informative observation loss. Cross-fitting controls some outcome reuse but cannot repair an incorrect hazard. Survivor-conditioned risk sets and shuffled endpoint dates do not reproduce the product-law comparison above. A positive score identifies residual event association relative to the declared model; the event and experiential bridges determine its stronger interpretation.

7.2.3 — The Non-Atomic Obstruction

Under the declared positive-density and measurability assumptions, the non-atomic obstruction says that probability-one exact closure at an independent endpoint requires the ledger to be zero for Lebesgue-almost every time on almost every eligible path. The issue is not merely low probability; it is structural incompatibility under the assumptions (Protter, 2005; Mayo, 2018).

The proof can be expressed by conditioning on the path and integrating the indicator of nonzero ledger values against the endpoint density. A positive-measure nonzero set yields positive failure probability.

The result should be checked under discontinuities, càdlàg paths, equivalence classes, and left limits. Endpoint atoms or singular supports require separate analysis.

A simulated ledger that is nonzero on a set of positive Lebesgue measure in the sampling interval has a probability strictly below one of exact zero at an independent uniformly sampled endpoint. The probability is zero only when its zero set has Lebesgue measure zero. Model recovery should reproduce that distinction.

Numerical rounding can hide the obstruction by turning a finite band into exact zero. The empirical band and the latent theorem must remain separate. For the non-atomic obstruction, non-discrimination supports only the common observable family, not either latent explanation.

The obstruction gives the program a rare hard discriminator. A branch that posits nontrivial ledger motion and independent diffuse timing must either show why the theorem's premises fail or accept the adverse implication; aesthetic appeal cannot carry that burden.

A public rendering of the non-atomic obstruction must show its assumptions, especially independent endpoint timing, positive density on declared support, and strict-pre-endpoint closure. Without those conditions the theorem becomes an overbroad claim that all random endings force an always-zero ledger.

The obstruction is a branch separator, not a universal refutation. If its assumptions fit the reference class, nontrivial exact closure cannot be maintained on that branch; if they fail, the analysis must name the alternative timing, support, routing, or endpoint structure that replaces them and derive its own predictions.

7.2.4 — State-Coupled, Hitting, and Absorbing Alternatives

State-coupled endpoints have hazards that depend on route state or ledger-related variables. Hitting-time endpoints occur when a process reaches a boundary, and absorbing endpoints remain in a terminal state thereafter. Each endpoint class induces a different likelihood, support geometry, and rival set; they cannot be pooled

as interchangeable explanations of closure (Kallenberg, 2002; Billingsley, 1999).

These alternatives evade the independent-cutpoint implication by changing the timing law. They also create new testable relations among state, hazard, approach geometry, and endpoint support.

A first-hitting rule can guarantee terminal zero by definition if the endpoint is set to the first time the ledger reaches zero. That is a tautology unless an independently motivated physical or psychophysical mechanism links termination to that surface.

Absorption can also be confused with observation shutdown. A flat observed series after device removal is not evidence that the latent process entered an absorbing neutral state.

Alternative timing branches are scientifically legitimate when they accept their own burden. They must predict more than the terminal value they were built to enforce.

Protocols should model endpoint hazard prospectively and compare state-coupled, frailty, disease-severity, treatment, and observation-selection accounts under the same information clock. A branch remains viable only if its distinctive state-hazard relation, support implication, or safe-intervention response recovers in simulation and transports beyond the fitted cohort.

7.2.5 — Atomic, Singular, and Protocol Consequences

Endpoint laws may contain diffuse, atomic, mixed, or singular components generated by scheduled procedures, biological thresholds, or classification rules (Aalen et al., 2008; Andersen et al., 1993; Revuz & Yor, 1999).

Atomic timing samples named times, so closure constrains those masses rather than an interval.

Support is decomposed by component and tied to declared timing mechanisms or protocol classes instead of being forced into one smooth hazard.

Anesthesia emergence, scheduled life-support withdrawal, and device-defined stops may concentrate protocol times, but their relation to the truth-level stream endpoint remains empirical.

Mixture class, support, and assignment are fixed or validated on held-out data. The estimand is component-specific mass and support, not a smooth hazard flexible enough to absorb every failure.

A branch matrix states the consequence of each component: diffuse mass can invoke the interval obstruction, atoms constrain named points, and singular mass needs carrier analysis. Mixed laws require mixed conclusions. Recording each component's weight, or the range of weights consistent with the data, makes the scope of each conclusion easier to assess. A mixture weight near zero may make a component practically minor, but it does not disappear from a universal or probability-one statement that quantifies over it.

Protocol timing remains distinct from truth timing. Device removal, withdrawal, discharge, report loss, and censoring may end observation while the stream continues, so terminal promotion requires an independent bridge. At each stop, the study can record the reason, the last usable measurement, and which aspects of the subsequent trajectory remain observable. This

record lets later return or resuscitation evidence reconnect to the same case without retroactively converting the protocol stop into the truth endpoint.

Report timing class, support, uncertainty, endpoint alternatives, and downstream sensitivity. Unrecoverable class membership narrows the result or yields No adequate estimate (Aalen et al., 2008; Revuz & Yor, 1999). A compact comparison across endpoint alternatives can show which findings remain stable and which are driven by the chosen timing rule. Where timing-class uncertainty is substantial, averaging across endpoint models can conceal incompatible assumptions; branch-specific results should remain visible before any synthesis. Reports should separate within-branch sampling uncertainty from between-branch sensitivity to endpoint classification, because precise estimates within individual branches can coexist with substantial uncertainty about which branch applies.

Stress tests vary density support, state dependence, frailty, selection, and competing definitions, recording where assumptions hold, fail, or remain untestable and how departures change off-band occupation (Billingsley, 1999; Kallenberg, 2002; Andersen et al., 1993). Varying assumptions individually and in combination helps distinguish a single influential choice from interacting sources of sensitivity. A useful stress grid reports both verdict changes and changes in off-band occupation, revealing influential assumptions even when the headline classification does not flip. A sensitivity ranking can then prioritize validation by identifying which assumptions move the protocol-fixed result closest to its decision boundary.

Multi-state models separately label clinical transitions, observation shutdown, and theorem-relevant endpoints. Nonresponse and protocol stops may reduce observation while the stream continues. An explicit return-to-observation transition lets resumed monitoring reconnect with the existing longitudinal record. That transition structure also prevents a temporary loss of observability from being mistaken for a truth-level endpoint. Where the declared endpoint bridge supports identification, separate estimates or bounds for observation loss, observation return, and endpoint entry can keep observation dynamics from being silently absorbed into endpoint hazard.

7.3 — Directional Reachability and Completion Sets

Remaining capacity is rarely symmetric. A stream may retain more opportunity for positive than negative contribution, or the reverse, because duration, physiology, environment, action, care, and endpoint hazard constrain directions differently (Aubin, 1991; Aubin et al., 2011).

Directional reachability therefore replaces a single scalar reserve with sets of attainable future signed increments. The result is not a forecast of what will happen. It is a statement about what remains possible under a declared model class.

Completion sets make prefix feasibility testable while preserving mechanism neutrality. They also reveal where an intervention changes the route space without guaranteeing a terminal outcome.

$$C_i(t, x) := \left\{ \Delta \in R : \exists \gamma \in A_i(t, x), \Delta = \int_t^{T_i^*} d v_i^\gamma \right\}$$
$$-L_i(t) \in C_i(t, x)$$

The completion set depends on current state x , admissible histories $\mathcal{A}_i$, endpoint support, and the representation. Closure is feasible only if the required opposite increment lies in that set.

7.3.1 — Directional Remaining Capacity

Directional capacity distinguishes how much positive and negative contribution remains attainable. A symmetric absolute bound can conceal the fact that one direction is practically or biologically unavailable (Aubin, 1991; Aubin et al., 2011).

Upper and lower reachability envelopes can be derived from signed rate bounds, state constraints, and endpoint support. They need not be equal or monotone.

Estimation requires models of future state, action availability, endpoint support, experienced exposure, and route uncertainty. The relevant state variables, admissible action set, and capacity bounds should be fixed before scoring; when those elements are weakly identified, the output should be directional bounds or an identified set rather than a point forecast.

Severe sedation may reduce both positive and negative observable variability while leaving asymmetric possibilities for pain, comfort, or awareness. A single reserve number cannot capture that structure.

Shrinking remaining capacity can arise from frailty, disease progression, treatment, or loss of social support without any fairness route. Those processes must be fit as equal-budget rivals. Reserve-linked evidence begins only where the joint directional pattern exceeds what those ordinary dynamics predict.

Directional capacity turns vague “room to balance” language into a geometry that can exclude routes and guide experiments without becoming a personal prediction.

7.3.2 — Positive and Negative Completion Sets

Positive and negative completion sets collect future increments reachable in each direction under the declared dynamics. Their geometry may be disconnected, state-dependent, sign-asymmetric, or constrained by action and endpoint timing (Ames et al., 2019; Altman, 1999; Kallenberg, 2002).

The sets may be represented through stochastic reachable sets, viability kernels, control bounds, or simulation-based envelopes. Each representation has assumptions about dynamics and intervention.

Protocols should report uncertainty in set boundaries and test calibration through held-out transitions. A set that contains every observed outcome is not automatically informative if it is too wide to exclude anything.

A rehabilitation program may expand positive experiential possibilities by restoring agency while also increasing short-term pain exposure. The two directional sets change differently.

Ordinary decision-making and social support provide comparison processes for changes in reachable experience. Their predictions can be evaluated alongside those of a specified terminal-control model.

Separate completion sets preserve the scientific distinction between what can happen and what the Law predicts must happen.

Sign-specific reachability should be reported as more than two maxima. The analysis should preserve gaps, disconnected regions, action constraints, and the probability or support attached to each attainable increment. A route may permit some positive movement while still excluding the exact opposite-signed increment required for closure.

Directional completion sets also reveal asymmetry that a single reserve number conceals. A path may retain ample capacity in one

sign while having no admissible route to the opposite-signed increment required by the current prefix.

When several active streams share finite resources or opportunities, individual reachability and joint solvency separate. The correct object is a vector of current residuals together with a joint attainable future-contribution set whose coordinates preserve stream ownership and provenance. Simultaneous completion requires the negative residual vector to belong to that joint set; summing coordinates or allowing one stream's surplus to cancel another's deficit would change the hypothesis (Aubin et al., 2011).

Shared constraints can create subset bottlenecks: each stream may have an individually completing continuation while no single feasible allocation completes all of them together. For any fixed direction, every jointly attainable increment has a projection no larger than the supremum over the joint set. If the required increment exceeds that supremum, simultaneous completion is infeasible. Testing directions supported on stream subsets supplies necessary support-function screens for the convexified joint set; exact membership remains the sufficiency target when the attainable set is nonconvex. Each coordinate retains its original stream ownership.

7.3.3 — Prefix Infeasibility and Reachability Bounds

A prefix is infeasible for exact closure under a route class when the required compensating increment lies outside the completion set. This is a direct route-failure condition (Protter, 2005; Mayo, 2018).

Infeasibility may be proved by analytic bounds, comparison theorems, justified dynamic programming, or certified numerical outer bounds. Failure to find a completing path in a finite simulation is not by itself a proof of impossibility. Empirical use must also carry endpoint and observation uncertainty into the bound.

Empirical adjudication should distinguish structural infeasibility from insufficient estimation. When a valid set cannot be narrowed enough for a decision, the verdict is Inconclusive evidence rather than compatibility by default. No adequate estimate applies when the target cannot be identified or validly bounded.

A short remaining support interval with bounded positive rate can make a large negative residual unreachable. The conclusion is conditional on the representation and rate bound, not a forecast about the person's future emotions.

Outcome-dependent enlargement of the completion set is a rescue move. The admissible dynamics and bounds must be fixed before the terminal residual is known. The analysis of prefix infeasibility and reachability bounds must report the unresolved equivalence class when the candidates cannot be distinguished.

Infeasibility is one of the program's clearest ways to lose. It transforms a grand terminal claim into a local, checkable exclusion.

Reachability figures should mark the declared completion set and the current prefix. If the required opposite-signed increment lies outside that set, the route fails locally; expanding the set after seeing the path would erase the falsifier.

A reachability failure should be reported at the narrowest justified level: the declared route cannot complete from the observed prefix under the protocol-fixed bounds. It does not by itself refute every route or the target law, but it removes any explanation that depends on unavailable same-stream capacity.

7.3.4 — Action Support Without Prescribing Action

Action support describes which transitions are structurally available from a reached history. It is not a recommendation, a moral instruction, or a claim that a person should choose experiences for the sake of terminal fit. Physically possible actions, ethically admissible actions, and actions that receive positive behavioral probability are different sets (Aubin, 1991; Aubin et al., 2011).

Under a frozen action-first regime, let each admissible action carry a completion probability after every downstream policy, physical transition, endpoint, routing, selection, and measurement-reactivity component is fixed. The policy-level completion probability is the policy-weighted mixture of those action-specific values.

That mixture equals one only when the policy assigns probability one, almost everywhere under its own action distribution, to actions whose complete frozen continuations close with probability one. A favorable action, nonempty reachable set, positive-probability closure, or topological-support statement is insufficient. The condition characterizes policy mass, not psychological motivation, conscious ledger awareness, moral instruction, or endpoint support.

Finite soft guidance has a ceiling. Reweighting actions by a finite positive power of their completion probabilities removes zero-completion actions but retains any baseline-positive action with an intermediate probability. Any retained intermediate mass prevents exact probability-one completion. Approaching one as guidance strengthens is limit-sure behavior, not finite attainment.

Almost-sure and limit-sure attainability must remain distinct. Almost-sure attainability requires one admissible policy that reaches completion with probability one; limit-sure attainability may only provide a sequence of policies whose probabilities approach one. If the admissible policy class is empty, the route is unavailable rather than almost-sure or limit-sure (Chatterjee et al., 2016; Asadi et al., 2025).

Partial observation creates a robust intersection problem. An observation-adapted policy can guarantee completion across all compatible histories only when their probability-one action sets have a nonempty admissible intersection and a measurable, time-consistent selector exists. If indistinguishable histories require disjoint safe actions, no policy based only on that observation can guarantee both.

Hard exclusion also creates an identification problem. When observed behavior assigns zero probability to putatively unsafe actions, their counterfactual completion probabilities are not identified without positivity, structural restrictions, valid instruments, experiments, or partial-identification bounds. The strongest safety claim can concern precisely the actions least observed.

Ethically bounded evidence may come from ordinary treatment variation, low-risk tasks, natural experiments, exogenous availability changes, or sealed recovery studies. Harmful, illegal, endpoint-manipulating, or lifetime-risk interventions remain prohibited. The route must accept unavailability rather than treating ethical limits as missing opportunities to test the theory.

Action support is useful because it makes a route vulnerable while preserving human authority. It can show that a claimed guarantee lacks the required policy mass, attainment, information, or selector. It can never delay care, rank people, induce burden, or turn a control model into a prescription engine.

7.3.5 — Completion Potentials and Prediction Without Forecasting

A completion potential is a declared state function tied to future reachability, such as a committor, viability value, harmonic function, or bounded reachability functional. Each has its own boundary condition, recursion, and interpretation; a flexible endpoint predictor does not acquire that status by name (Ames et al., 2019; Altman, 1999; Kallenberg, 2002).

Its strongest test is transported prediction under frozen state variables, feature times, endpoint, and action interface. Re-estimating the coordinate system in every cohort tests local predictors, while terminal or post-baseline information may not leak into a representation later praised for forecasting.

Generic prognosis, frailty indices, and latent embeddings are equal-treatment rivals. A completion potential earns distinction only by recovering its defining equation or recursion and adding

held-out sign-specific reachability, action-sensitivity, or boundary-access predictions beyond ordinary prognosis, with calibrated recovery. It remains a research object, never a personal forecast (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Identification requires the defining recursion, calibration over reachable states, safe-intervention response, and recovery in synthetic worlds to distinguish the potential from alternative representations with similar endpoint probabilities (Aubin, 1991; Ames et al., 2019; Kallenberg, 2002).

Calibration uses only information available at prediction time. Cross-fitting limits direct leakage but cannot repair target leakage in features or cohort selection, so provenance records when every variable became available and which transformations were frozen.

Action invariance is stronger than passive discrimination. If safe interventions create the same measured state but different reachability, a potential that treats them alike lacks causal structure; if it changes when only observation changes, it may be tracking measurement. Matched interventions should therefore vary action support while preserving observation quality, making causal reachability harder to confuse with sensor response.

Held-out scoring should cover transition, hitting, and endpoint distributions, then transport a frozen potential against simpler prognosis, flexible machine learning, and path-conditioned rivals. Advantage must appear on restrictions unique to completion geometry, not general outcome prediction (Gneiting & Raftery, 2007; Dawid, 1984). A model that predicts generic prognosis well

but adds no accuracy on completion-specific events should not receive route support merely for overall fit.

Action-specific potentials require intervention data, defensible natural experiments, or explicit causal assumptions. Observational prediction alone cannot show which action changed the completion set (Pearl, 2009; Hernán & Robins, 2020). Where intervention identification is weak, the analysis should separate predictive reachability from causal controllability rather than treating one as evidence for the other.

When several potentials remain compatible, report a reachability set. Robust claims survive across the family; fragile claims depend on unverified geometry. The width and topology of that set can themselves be reported, showing whether uncertainty is a small quantitative margin or a qualitative disagreement about accessible routes.

7.4 — Endpoint Support and Accessibility

Reachability is not enough when the endpoint may arrive before the path reaches the required region. The support of endpoint timing, the intensity of termination events, and the accessibility of the boundary jointly determine whether a completion set is relevant (Andersen et al., 1993; Aalen et al., 2008).

Counting-process and compensator language provides a way to separate observed event occurrence from predictable hazard. It also clarifies how much time the process spends in states that make closure possible or impossible.

This section treats endpoint support as a scientific object in its own right. It keeps protocol stopping, truth-level termination, observation shutdown, and endpoint-inclusive jumps distinct.

$$N_i(t) = M_i(t) + \Lambda_i(t)$$

$$E\left[\int_0^T H_i(t) dN_i(t)\right] = E\left[\int_0^T H_i(t) d\Lambda_i(t)\right]$$

For predictable H_i that is nonnegative, or satisfies $E \int_0^T |H_i(t)| d\Lambda_i(t) < \infty$, the compensator identity relates event-weighted occupation to the predictable endpoint intensity. It does not identify the truth endpoint unless the counting process itself has been validly linked to stream termination.

7.4.1 — Endpoint Support as a Model Object

Endpoint support consists of the time-state points whose every open neighborhood has positive probability under the declared endpoint law. It constrains which prefix states the boundary

hypothesis must handle (Andersen et al., 1993; Aalen et al., 2008).

Support may depend on calendar time, biological state, route state, intervention, or latent hazard. It is distinct from the observed distribution in one selected cohort.

Estimating support is harder than estimating density because rare regions and censoring matter. Conservative support sets and sensitivity analysis may be more honest than smooth extrapolation.

A terminal study that recruits only after hospice enrollment observes a narrow support shaped by clinical referral and service access. That support cannot stand in for all stream endpoints.

A hazard model with arbitrarily thin tails can make every time technically possible and thereby evade a support failure. The analysis should distinguish practically occupied support from numerical tails introduced by smoothing. Sensitivity to tail thresholds and censoring is part of the conclusion.

Making support explicit reveals where the law is being tested and where it remains untouched.

Support uncertainty should be represented by nested lower and upper support sets. The inner set contains regions with well-supported endpoint risk; the outer set contains regions not excluded after censoring and rare-event uncertainty. Boundary-accessibility claims may use only the set appropriate to their error cost, rather than treating a smoothed density tail as established support.

Effective support may guide finite-sample design, but it must remain distinct from mathematical support. A low-probability region cannot be discarded when evaluating a claim that quantifies over that region. Any tail truncation must therefore be stated as a reference-class or approximation choice, with its effect on the conclusion reported.

7.4.2 — Counting Processes and Compensators

A counting process records endpoint-like events or transitions, while its compensator represents the predictable cumulative intensity under the filtration. The martingale residual captures departures from that predictable component (Protter, 2005; Jacod & Shiryaev, 2003; Revuz & Yor, 1999).

Compensator analysis can test whether endpoint events are related to route states beyond ordinary risk factors. It requires careful specification of the event, at-risk process, censoring, and time scale.

Endpoint support supplies a restriction without assuming independence from the ledger. On a declared stopping-time branch, let $T > 0$ be the true endpoint, $N_t = \mathbf{1}_{\{T \leq t\}}$, and Λ its predictable compensator. In that same filtration let L_{t-} be predictable. For finite u and $\delta \geq 0$, compensation of the nonnegative predictable indicator gives

$$\Pr(T \leq u, |L_{T-}| > \delta) = E \int_0^u \mathbf{1}_{\{|L_{t-}| > \delta\}} d\Lambda_t.$$

Exact closure makes this nonnegative integral zero almost surely, so endpoint compensator mass is confined to the pre-endpoint

zero set. If $d\Lambda_t = Y_t \lambda_t dt$ with ordinary finite intensity and that zero set has zero at-risk time occupation, no exact-closing endpoint can occur in the interval with positive probability. A singular or predictable hitting endpoint can have a different structure.

If $\lambda_t \geq \lambda_0 > 0$ throughout the off-band at-risk region and the displayed endpoint-failure probability is at most ε , positivity gives

$$E \int_0^u Y_t \mathbf{1}_{\{|L_{t-}| > \delta\}} dt \leq \varepsilon / \lambda_0.$$

A lower bound established only on a subregion gives only that subregion's occupation bound. The compensator and ledger predictability must use the same filtration: projecting a hidden endpoint to an observed hazard does not permit multiplication by a separately projected band probability. Ordinary hitting and zero-plateau shutdown mechanisms can satisfy the restriction. Their prospectively predicted timing and occupation relations, together with independent event and contribution calibration, determine the comparison.

Protocols should avoid using the same fitted ledger residual to define both the event and the predictor. Separate channels and cross-fitting reduce algebraic dependence.

In a clinical cohort, loss of responsiveness, device removal, physiological death, and adjudicated stream endpoint may be different counting processes. Combining them erases the very distinctions the formal model needs.

Frailty, treatment changes, care setting, and informative dropout can alter intensities. A state-coupled pattern may disappear once those rivals are modeled.

The compensator supplies a disciplined language for timing. It does not turn a medical event into a phenomenal boundary by notation.

Compensator diagnostics should therefore include time-varying covariates, treatment processes, observation intensity, and care transitions in the at-risk specification. Residual state coupling after those adjustments is the relevant signal, not the unadjusted hazard shape.

7.4.3 — Occupation Measures and Remaining Exposure

Occupation measures record how long a trajectory spends in regions of state space before the endpoint. They connect reachability geometry with the temporal opportunity to realize future increments (Kallenberg, 2002; Billingsley, 1999).

A state may be reachable yet have negligible occupation probability before termination. Conversely, long occupation in a region can dominate accumulated exposure even when transitions are rare.

Estimation can use continuous-time state-space models, inverse-intensity weighting, or simulation. Observation cadence and informative missingness must be part of the calculation.

A person may repeatedly enter a state associated with relief but remain there only briefly, while spending long intervals in

burdened states. Counting visits alone would misrepresent exposure.

Ordinary disease progression can alter occupation patterns without any terminal controller. Equal-treatment rivals should predict the same state-time summaries. A route-specific conclusion requires a pattern that those disease and care processes cannot reproduce under the same observation model.

Occupation analysis makes time visible inside boundary accessibility. It prevents the model from confusing possible states with experienced exposure.

Occupation displays should distinguish states that are mathematically reachable from states in which the process spends appreciable time. Coloring an entire state space as available without exposure weights overstates practical opportunity.

Occupation data can reject a reachability story even when the target state is topologically accessible. A path with insufficient occupation of the necessary region before the qualifying stream's endpoint lacks the temporal exposure required by the route. Reaching that region only after observation ends instead limits empirical adjudication unless the observation stop has been independently linked to the stream endpoint. Conversely, long occupation does not establish completion unless the permitted signed increments and endpoint relation are also validated.

7.4.4 — Boundary Accessibility and First-Hitting Geometry

A boundary is accessible when admissible trajectories can reach it before termination under the declared dynamics. First-hitting geometry describes approach, contact, crossing, and absorption.

Inaccessibility under those dynamics is a direct route failure; apparent access created only by numerical smoothing or changed boundary rules is not recovery (Andersen et al., 1993; Aalen et al., 2008).

Accessibility depends on drift, diffusion, jumps, state constraints, and endpoint hazard. A terminal set may be mathematically present yet polar or practically unreachable.

Simulation should estimate hitting probabilities and times under each declared route and rival, then test calibration on held-out transitions. Numerical schemes need dedicated checks near singular, sticky, reflecting, or absorbing boundaries.

A zero surface in a high-dimensional latent state can be crossed frequently, touched rarely, or never reached depending on dynamics. A plotted trajectory cannot establish the general geometry.

Defining the truth endpoint as the first hit makes accessibility tautological. A valid zero-surface termination route needs independent evidence that the phenomenal or physical system actually terminates there.

Boundary geometry gives the program concrete failure modes: inaccessible targets, wrong approach rates, or endpoint hazards that ignore the proposed surface.

Numerical recovery requires convergence diagnostics, discretization sensitivity, and benchmark cases with known solutions or trustworthy bounds. A simulated hitting pattern that changes materially with step size, boundary convention, or

interpolation rule is a numerical result, not evidence about the target route.

7.4.5 — Endpoint-Inclusive Value, Jumps, and Support Failure

A process may carry a jump or terminal atom, so strict-pre-endpoint and endpoint-inclusive closure are distinct branches. An endpoint-inclusive model must prospectively specify the mark distribution, timing law, sign, magnitude, measurability, ownership, latent-representation link, and compensator; an unobserved contribution cannot be added merely because it repairs the residual (Protter, 2005; Jacod & Shiryaev, 2003; Revuz & Yor, 1999).

Observation is often weakest at the jump. Communication loss, emergency intervention, device failure, sedation, or classification uncertainty can hide the final interval. A route that repeatedly needs an unseen contribution of exactly the required sign is fragile, and support failure must be allowed to retire it.

Sensitivity analysis should bound the unobserved contribution and vary mark and hazard laws rather than impute closure. Support must be separated into topological, effective, and observed forms: a state can be reachable yet negligible before the endpoint, or model-supported yet absent from the cohort because of censoring and care transitions (Aalen et al., 2008; Andersen et al., 1993).

Simulation should include inaccessible and rare boundaries, wrong support classes, missed jumps, device shutdown, sedation, and false endpoint labels. Negative-control channels and

administrative timing can reveal whether an apparent terminal atom belongs to the latent process or the measurement system. Simulated false atoms are especially useful because they test whether the pipeline can distinguish structural endpoint mass from censoring or device-induced concentration.

Outcomes remain branch-specific. Strict-pre-endpoint closure may fail while an endpoint-inclusive branch remains open; a first-hitting route may fail while state-coupled hazard survives; observation may be inadequate while truth-level status remains unresolved. Failure should localize the branch rather than create an all-purpose terminal rescue. This branch accounting should remain visible in summaries so a surviving alternative is not mistaken for rescue of the branch that actually failed.

A frozen decision table distinguishes strict-pre-endpoint, endpoint-inclusive, first-hitting, state-coupled, unresolved-endpoint, and No adequate estimate outcomes. None permits a terminal value to be selected solely because it closes the ledger. A predeclared table also makes endpoint changes auditable by showing which classification rule would have produced each verdict before the data were seen.

7.5 — Bridge, Selector, and Admissible-History Distinctions

Several mathematically distinct constructions can produce paths that approach a terminal target. A native dynamical law can drive the process. A bridge can condition ordinary dynamics on a future endpoint. A selector can retain only histories that close. An intervention can alter the path. A protocol can observe only the subset that survives to measurement (Doob, 1957; Föllmer, 1985).

These constructions are not rival notations for the same mechanism. They answer different causal and probabilistic questions, and they generate different counterfactuals away from the conditioned endpoint.

The section develops their distinctions because terminal curves are visually deceptive. Similar paths can arise from incompatible generative stories.

$$L^h f(x, t) = Lf(x, t) + a(x, t) \nabla \log h(x, t) \cdot \nabla f(x, t)$$

$$h(x, t) = Pr_{x, t}[X(T) \in A]$$

For the displayed diffusion case, $\mathcal{L}$ is the baseline spatial generator and $a = \sigma \sigma^T$ is its local covariance matrix. The formula is used where h is positive and sufficiently regular for the generator calculation. A zero-probability point endpoint instead requires a transition-density or other justified regular-conditional bridge construction; its zero event probability cannot be substituted into $\log h$. The Doob h -transform modifies the drift of the baseline process under the declared conditioning. It is a bridge construction whose predictions must be compared with an unconditioned native process and with selection mechanisms.

Table 7.2. Native dynamics, bridges, selectors, and observation rivals.

Construction	What changes	Key discriminator
Native dynamics	Forward generator or state law	Perturbation and off-endpoint behavior
Bridge or h-transform	Path law conditioned on terminal event	Dependence on endpoint information
Admissible-history selector	Support or weight of complete paths	Prospectively fixed path constraint
Policy/intervention	Action or causal transition law	Counterfactual response under controlled change
Observation selection	Which data are recorded	Missingness and gain mechanisms

7.5.1 — Native Dynamics Versus Conditioning

Native dynamics specify how the process evolves before any terminal outcome is imposed. Conditioning reweights those paths by a future event and can create an apparent pull toward the endpoint (Doob, 1957; Föllmer, 1985).

A conditioned bridge often has singular or steepening drift near the terminal time even when the baseline process has none. Observing steepening drift is therefore not uniquely diagnostic of a controller.

Protocols should separate data used to define the terminal condition from data used to test preterminal predictions. Prospective endpoint rules and held-out windows are essential.

A Brownian bridge and a feedback-controlled diffusion can both approach zero continuously. Their responses to perturbation, endpoint uncertainty, and off-target conditions can differ.

Endpoint conditioning can manufacture convergence by construction. A bridge earns empirical relevance only if a frozen conditioning rule predicts held-out transitions, interventions, or endpoint-support behavior beyond the event used to define it. Fit to the conditioned endpoint alone is circular.

The distinction protects causal interpretation. Terminal fit may reveal selection or conditioning rather than a law governing native dynamics.

The split should be enforced at the data level. One module fixes the endpoint class and terminal set; another, held out from that choice, scores pre-endpoint drift, covariance, or hazard predictions. Cross-fitting or external replication can recover power, but no observation may both define the ending and serve as evidence that the path anticipated it.

Flexible endpoint conditioning can reproduce terminal convergence after the fact. A native-process candidate must therefore predict transported transition, action, or intermediate-state restrictions that the corresponding conditioning rival does not reproduce. When the conditioning rival predicts the shared pattern and the native-process candidate fails those additional restrictions, the result favors the conditioning explanation for that component.

7.5.2 — Admissible-History Sets

An admissible-history model restricts the path space to trajectories that satisfy a global boundary condition. The restriction can be written as a constrained measure over complete

histories (Léonard, 2014; Léonard et al., 2014; Chetrite & Touchette, 2015).

Global path constraints differ from local control because they need not produce an ordinary forward-time mechanism. They may be time-symmetric, variational, or boundary-conditioned.

Empirical consequences must be derived from the constrained ensemble, including preterminal correlations, perturbation responses, and transport across endpoint classes.

A path-integral construction may assign zero weight to nonclosing histories while preserving ordinary-looking local dynamics. That is a coherent route only if it yields discriminating observables.

An admissible set chosen after observing the path is selection, not a risky law. The constraint and baseline measure must be specified independently.

An admissible-history route earns comparative support through predictions tested against specified local-dynamics and selection models.

The admissible-history set should also be nontrivial under counterfactual paths. A rule that retains only the realized successful trajectory has no predictive content and should fail simulation-based recovery before it is applied to observational data.

7.5.3 — Bridge Laws and Doob h -Transforms

Bridge laws describe a process conditional on reaching a terminal set. The Doob transform provides a precise way to express the

induced drift when the relevant harmonic or space-time function is known (Çetin & Danilova, 2016; Kallenberg, 2002).

The transformed process inherits assumptions from the baseline generator and the terminal event. Errors in either object can be absorbed into the apparent bridge dynamics.

Recovery studies should distinguish native dynamics, conditioned bridges, admissible-history selection, and ordinary mean reversion under realistic observation noise. The benchmark should include generators that produce similar terminal pictures for different reasons and require abstention when the available data cannot separate them.

A bridge to zero at a fixed time produces a characteristic pull that depends on remaining time. A random or uncertain endpoint produces a mixture of such pulls and may soften the apparent singularity.

The transformed drift can resemble the reserve-coupled drift. Conditional slope alone therefore cannot discriminate the routes. If the native and conditioned laws remain indistinguishable on held-out pre-endpoint data, the evidence cannot identify which construction generated the path.

Bridge mathematics is valuable because it turns a vague rival into an exact comparator. It also teaches the program not to claim uniqueness from one terminal signature.

Bridge diagrams should identify the baseline law, conditioning event, h-function, information set, and held-out tests. A terminal curve alone cannot reveal whether convergence was native, selected, or imposed by conditioning.

A bridge is therefore a calibrated rival as much as a mathematical tool. It can reproduce terminal attraction without showing that the unconditioned system contains a completion controller. Evidence for native dynamics requires predictions that survive when conditioning information, terminal selection, and endpoint leakage are withheld.

7.5.4 — Selectors, Policies, and Interventions

A selector determines which histories or actions enter a constructed model, a policy assigns action probabilities, and an intervention changes the data-generating system. These operations can produce similar trajectories while answering different causal questions. Each requires its own estimand, assumptions, and identification strategy (Doob, 1957; Föllmer, 1985; Pearl, 2009).

The downstream regime must be frozen before a policy-mixture claim is interpreted. Physical transitions, later policies, endpoint rules, routing, selection, and measurement reactivity all affect the continuation. If those components change with the behavior being explained, an apparent action effect can be a change elsewhere in the regime.

A policy-support condition says where probability mass lies under a declared factorization. It does not identify psychological motivation, conscious ledger access, endpoint knowledge, or computation of completion probabilities. If every available action already completes with probability one, the condition is exact but behaviorally silent.

Selectors require measurable and time-consistent construction. A completion-compatible set at one history is not yet a policy over time, and a sequence of locally favorable choices need not form an admissible recursively consistent strategy. Empty or nonmeasurable selector sets make the route unavailable.

Under partial observation, policies are adapted to observations or belief states rather than to hidden histories. A robust guarantee requires an action that remains completion-compatible across every live history represented by the observation. Disjoint history-specific action sets reveal an information failure or another causal locus, not a reason to call the policy approximately guaranteed.

Observed choices do not by themselves identify intervention effects. Selection adjustment, inverse weighting, g-methods, randomized contrasts, instruments, and sensitivity analysis are useful only when their assumptions match the operation and data. Conditioning on survivors or communicative participants is not evidence that an intervention generated their terminal pattern (Hernán & Robins, 2020).

Off-policy evidence has a bounded warrant. Data generated under one behavior policy identify another policy only under adequate overlap or a defensible structural bridge, with uncertainty from weighting, model extrapolation, and unobserved actions carried into the result. A route observed under one policy therefore cannot silently become an all-policy or all-intervention conclusion (Hernán & Robins, 2020; Manski, 2003).

Care policies can create state-dependent treatment that resembles feedback control. That complication belongs in the causal model, but care remains governed by clinical need, consent, and ordinary human purposes. A route cannot claim cleaner identification by withholding effective treatment or manipulating endpoint timing.

The scientific gain comes from tracing the same curve to distinct generating operations and asking where their predictions diverge. Native dynamics, conditioning, selection, policy, and intervention should compete under common endpoint, observation, and scoring rules so that a favored route earns its status rather than borrowing it from the analysis design.

7.5.5 — Observation Selection and Equal-Information Rivals

Observation selection occurs when measurement probability, quality, timing, or channel depends on latent state, route state, care, or endpoint proximity. Silence, immobility, dropout, sedation, device removal, and workflow may create apparent compression without latent change, so they first describe the measurement process (Little & Rubin, 2019; Aalen et al., 2008).

Equal-information comparison begins with a frozen contract listing variables, availability times, preprocessing, endpoint labels, missingness, feature construction, tuning, and scoring. A favored route cannot receive a future-informed reserve while rivals receive only baseline covariates.

Information clocks are synchronized. Every model receives each covariate when it would have been prospectively available and

uses the same endpoint and missingness history; otherwise leakage purchases superiority.

The observation process is modeled with its own intensity, quality, channel state, and reasons for stopping. Fatigue, refusal, loss of consciousness, clinical burden, battery failure, and scheduling gaps have different implications, and collapsing them can make declining observability look like declining phenomenal variance.

Denominator flow reports who entered each risk set, who remained observable, why observation stopped, and what other channels still showed. A model that succeeds only among easy-to-observe participants may fail where the terminal claim is most demanding.

Candidate negative controls include technical variables such as battery level and channel outages, provided the protocol independently justifies that they affect observation without changing latent valence. Staffing and prompt schedules require the same exclusion argument because care and measurement reactivity can also alter experience. Outcomes sharing measurement burden but not the proposed completion relation supply another check.

Bridge, frailty, adaptation, observation-collapse, reserve, hidden-state, and selection rivals receive comparable information, tuning, calibration, complexity control, and recovery tests. Equal tables are insufficient when one model receives a richer latent state or endpoint construction (Blackwell, 1953; Gneiting & Raftery, 2007).

Held-out evaluation reports calibration, discrimination, coverage, and transported restrictions rather than only fit. Joint sensitivity is necessary because modest latent, endpoint, and observation errors can combine into a persuasive terminal pattern.

Bounds, pattern-mixture alternatives, selection models, and negative controls can narrow unobserved selection but cannot create missing information. The verdict states which conclusion survives the prospectively specified observation envelope.

Model ambiguity identifies the evidence needed to separate contenders. Comparing them under the same assumptions and observation process makes that next test informative (Mayo, 2018; Tsiatis, 2006; Rizopoulos, 2012).

7.6 — The Reserve-Coupled Route as One Candidate

The reserve-coupled route is a candidate realization of the mechanism-neutral constraints. It introduces a declining nonnegative reserve U_i , a signed arithmetic reference process Y_i , and a positive scale constant c^{LU} , then defines a route state L_i^{rc} as their product. This construction generates linked drift and innovation restrictions for empirical comparison (Protter, 2005; Revuz & Yor, 1999).

The route must remain dimensionally coherent, sign-flexible, independently measurable, and explicit about stochastic calculus. A geometric Brownian reference is unsuitable as the default signed coordinate because it remains positive; Y_i must be able to cross zero unless another representation is declared.

The displays introduce the reference route and its finite-variation special case; §7.6.3 gives the general product-rule conditions and the additional covariation terms.

$$L_i^{rc}(t) = c^{LU}U_i(t)Y_i(t), c^{LU} > 0, dU_i = -r_i^U dt, dY_i = \sigma_i dW_i$$
$$dL_i^{rc} = -(r_i^U/U_i)L_i^{rc}dt + c^{LU}\sigma_i U_i dW_i$$

In the reference case with a continuous finite-variation reserve, r_i^U is the nonnegative reserve-depletion rate and c^{LU} fixes units between reserve times the normalized coordinate and the route state. The product approaches zero if reserve reaches zero at the relevant route boundary and Y_i does not diverge fast enough to defeat the product. Coincidence with the truth-level endpoint is a separate bridge.

7.6.1 — Declining Reserve as Route Coordinate

Reserve represents remaining capacity in this candidate route, not a person's worth, moral credit, life value, or eligibility. It must be defined through independently measurable biological, functional, or process variables (Protter, 2005).

A finite-variation reference reserve satisfies a declared depletion law and remains nonnegative before the endpoint. More general reserves require separately declared dynamics and a product-rule calculation that retains any nonzero covariation terms.

Construct validation should specify raw indicators, units, transformations, context of use, confounds, and transport limits. Reserve cannot be whatever fitted quantity makes the product close.

Frailty indices, physiological capacity, communication ability, and care intensity may all correlate with endpoint proximity while measuring different constructs.

Reserve proxies can confound age, disability, medication, disease severity, access to care, and social support. Those variables are not interchangeable measures of remaining completion capacity. Apparent depletion that follows them but fails the route's independent construct, cross-equation, sign-specific, perturbation, or transport tests should favor the bias, frailty, or prognosis rival.

Reserve is useful only as a testable route coordinate. If it cannot be measured independently, the route may remain mathematically coherent and empirically unadjudicable.

Independent reserve validation should precede the product fit. Before outcome inspection, the protocol should state whether reserve is reflective, formative, or merely prognostic; which indicators can move it; and which perturbations should change it without directly changing the ledger. Failure of those cross-construct predictions retires the coordinate even if the product equation fits.

7.6.2 — Signed Arithmetic Reference Dynamics

The normalized process must support positive and negative values because the ledger-like product can lie on either side of zero. An arithmetic diffusion is the simplest sign-flexible reference case (Karatzas & Shreve, 1991; Jacod & Shiryaev, 2003; Särkkä, 2013).

The reference process may later include drift, jumps, stochastic volatility, or state dependence, but each addition changes predictions and must be justified prospectively.

Estimation should recover the process on held-out data and test whether residuals retain structure. A flexible latent process can absorb every mismatch and make the route unfalsifiable.

A geometric Brownian motion remains strictly positive and therefore cannot serve as a default signed reference without a transformation that changes interpretation.

Ordinary mean reversion and opponent processes can generate signed arithmetic dynamics without reserve coupling. They are natural rivals rather than approximations to the preferred route.

The arithmetic reference is a laboratory for consequences, not a claim that phenomenal experience literally follows Brownian motion.

The signed arithmetic reference should be judged by the reserve-linked restrictions it produces, not by familiar Brownian or mean-reverting notation. Held-out tests should ask whether jointly predicted drift, innovation, reserve, and boundary behavior recover; ordinary Ornstein-Uhlenbeck, opponent-process, and flexible state-space dynamics remain strong rivals whenever those linked restrictions fail.

7.6.3 — Product Rule and Induced Ledger Dynamics

The product rule reveals which assumptions make the compact induced equation possible. A continuous finite-variation reserve has zero quadratic covariation with the reference process; continuous martingale components or shared jumps can produce additional covariation terms (Revuz & Yor, 1999; Kallenberg, 2002).

Writing the general semimartingale product before the special case prevents hidden process-class changes. Filtration, adaptedness, integrability, and boundary behavior belong to the statement.

Simulation and estimation should use the same calculus as the derivation. Discretization can introduce spurious drift or conceal covariation near small reserve.

Correlated physiological shocks and experiential shocks would create a covariation contribution absent from the simplest equation. Ignoring it can make the fitted relaxation look stronger.

A product identity is true by construction. Its empirical burden lies in whether independently measured reserve and normalized dynamics satisfy the linked restrictions.

The induced equation exposes cross-equation tests and a concrete task: establish its relation to the formal phenomenal integral.

The reserve route should be displayed as a coupled system, not as one attractive stochastic differential equation. U , Y , L , their filtration, jump assumptions, shared shocks, and cross-equation restrictions determine whether the product-rule derivation is valid.

The empirical signature changes with the process class. Jumps in reserve, shared shocks between reserve and reference dynamics, or filtration changes introduce terms absent from the smooth finite-variation case. Estimation should therefore begin from the general product identity and justify every term set to zero, rather than reading the simplified equation backward as evidence.

7.6.4 — Drift Dominance, Diffusion Compression, and Scaling

As reserve shrinks, the reference relaxation coefficient can grow while the innovation coefficient shrinks in the finite-variation case. This produces a distinctive conjunction rather than one terminal curve. The route is challenged jointly by the scaling of relaxation, innovation, and reserve, so a terminal decline in variance alone is non-discriminating (Protter, 2005; Revuz & Yor, 1999).

If reserve behaves like a power of remaining time, reserve-space variance and time-domain variance have different exponents. Confusing them leads to incorrect claims about scaling.

A bridge can also show steepening drift, and observation-gain collapse can show shrinking variance. The joint pattern must outperform those rivals on held-out data.

Time-varying depletion and volatility can mimic almost any exponent if chosen after inspection. Their functional forms and priors need prospective constraints.

Scaling is useful when it connects multiple observables with fewer degrees of freedom than the rivals. It is decorative when it merely redescribes the fitted path.

Drift, innovation, and reserve should be estimated jointly, with reserve error and covariance propagated through the induced dynamics and with functional forms frozen before testing. Support requires transported cross-equation restrictions across resolutions and cohorts, not one visually narrowing trajectory or one fitted exponent.

7.6.5 — Noise, Ratio Fragility, Transport, and Retirement

Observation noise sets a floor on estimated compression, while observation gain may collapse as communication or movement declines. Larger samples do not remove systematic observation bias (Karatzas & Shreve, 1991; Jacod & Shiryaev, 2003; Särkkä, 2013).

For $U_i > 0$, the normalized ratio $|L_i^{\text{rc}}|/(c_{LU}U_i)$ equals $|Y_i|$ under the exact product representation and is fragile when numerator

and denominator are uncertain near zero. It remains secondary and is reported with both components, covariance, reserve-floor probability, and bounded or Fieller-type uncertainty.

Transport requires a common reserve coordinate, unit, endpoint relation, observation map, and perturbation interface. Refitting coefficients is permitted; redefining reserve after outcome inspection creates a new route rather than evidence of transport.

Late sensor dropout can imitate declining innovation while latent dynamics remain uncertain. Negative controls and joint observation models are therefore essential.

The route narrows or retires when reserve lacks independent validation, endpoint synchronization fails, normalized dynamics become unstable, cross-equation restrictions fail, or serious rivals predict as well or better.

Retirement preserves narrower achievements, including better reserve measures, endpoint models, observation analysis, and a clearer account of what the target alone does not determine.

Criteria are fixed before protected analysis, with thresholds for construct invalidity, denominator instability, cross-equation failure, endpoint desynchronization, nontransportable parameters, and recovery-qualified rival parity. A route without operational failure conditions is not exposed to defeat (Mayo, 2018; Nosek et al., 2018).

Near the boundary, joint errors-in-variables or state-space models propagate ledger and reserve uncertainty together. A stable ratio can hide two unstable quantities, while an explosive ratio may reflect a denominator artifact rather than failed route dynamics

(Fieller, 1954; Gleser & Hwang, 1987; Carroll et al., 2006; Särkkä, 2013).

Transport varies endpoint class, instrument, cadence, clinical state, and observation regime, not merely geography. The same coordinate and cross-equation restrictions must survive with reserve uncertainty carried into drift and innovation estimates.

Useful subcomponents can survive route retirement. A reserve proxy may retain prognostic value, an observation model may improve terminal-window inference, or a state-space decomposition may clarify error. The record migrates each survivor to its narrower status rather than treating retirement as erasure (Mayo, 2018).

7.7 — Discriminating Signatures and Equal-Treatment Rivals

The reserve realization is discriminated by a conjunction, not by a showcase curve. Negative drift can arise from bridge conditioning or ordinary regulation; shrinking variance can arise from observation loss; reserve correlation can reflect shared disease severity; and endpoint concentration can be imposed by construction. Distinctive evidence therefore requires these components to cohere under independent reserve measurement, prospective endpoint rules, recovery-qualified estimation, and held-out equal-information rival comparison. Failure of a prerequisite remains visible rather than being offset by strength elsewhere (Gneiting & Raftery, 2007; Vehtari et al., 2017).

This section specifies the joint prediction, the signature family for the protocol, and the competing models to be tested. Later chapters develop the study design. The goal is to make favorable evidence informative by comparing the predictions of the candidate and its alternatives.

The reference reserve route has a compact cross-equation prediction. For differentiable $U(t) > 0$, depletion rate $r_U(t) \geq 0$, and constants $c > 0$ and $\sigma > 0$, define its mean-reversion coefficient and innovation amplitude by

$$a(t) = \frac{r_U(t)}{U(t)}, \quad b(t) = c\sigma U(t), \quad \frac{dU(t)}{dt} = -r_U(t).$$

Then

$$a(t) = -\frac{1}{2} \frac{d}{dt} \log[b(t)^2].$$

The same reserve trajectory therefore links relaxation strength to the rate of change in innovation variance. Estimate the drift and innovation features without imposing this identity, propagate their joint estimation uncertainty under the common observation model, and compare the linked prediction with alternatives that allow the features to vary separately. The shared reserve trajectory supplies a testable relation between the two features.

Table 7.3. Candidate signatures, strong rivals, and required evidence.

Signature	Reserve-realization expectation	Strong rival	Needed evidence
Steepening relaxation	κ_i^U grows like r_i^U/U_i under the declared depletion law.	Bridge or homeostatic reversion.	Reserve-linked prospective slope.
Shrinking innovation	$\sigma_{L,i}$ shrinks like $c^{LU}\sigma_i U_i$ in the reference case.	Sensor loss or sedation.	Observation-robust innovation estimate.
Cross-equation restriction	$\kappa_i^U \sigma_{L,i}$ equals $c^{LU}\sigma_i r_i^U$ under the printed assumptions.	Flexible heteroscedastic model.	Protocol-fixed drift-diffusion coupling.
Held-out performance	Joint fingerprint predicts unseen data.	OU, bridge, frailty, medication, observation-only models.	Fair model recovery and rival comparison.

7.7.1 — Steepening Relaxation

The reference case with a continuous finite-variation reserve predicts a relaxation coefficient $\kappa_i^U = r_i^U / U_i$. As reserve falls, the coefficient steepens when the depletion rate falls proportionally more slowly than reserve, remains constant, or increases. This is a conditional route signature, not a generic prediction of PTVC (Protter, 2005).

Negative drift is not the signature. Ordinary homeostasis, regression to a set point, endpoint-conditioned bridges, and smoothing artifacts can all create apparent motion toward zero. The reserve-specific claim is that relaxation strength tracks independently measured reserve depletion as one component of the full prospectively specified conjunction.

A study should therefore estimate κ or an analogue jointly with reserve. It should ask whether the relaxation steepens as U falls, whether the relation survives confounds, and whether the estimate is available prospectively rather than only after endpoint alignment.

The depletion rate r_i^U also matters. If r_i^U changes with time or clinical state, κ may steepen, flatten, or jump. A model that treats r_i^U as constant after observing a favorable slope is too rigid in the wrong way and too flexible in the wrong way. The depletion-rate model must be declared.

Steepening relaxation should be reported with uncertainty, window sensitivity, and rival fits. A Brownian bridge toward zero has its own diverging pull as T approaches. If the bridge receives

the same endpoint information and fits as well, drift cannot support the reserve realization.

The strongest result would be a preterminal, reserve-linked, held-out prediction of relaxation behavior that a bridge, OU process, frailty model, and observation-gain model fail to match. Anything less is component pressure, not validation.

7.7.2 — Shrinking Innovation and Variance Structure

The paired reference signature is shrinking latent innovation amplitude. In the displayed special case, the route-state diffusion coefficient is $\sigma_{L,i} = c^{LU} \sigma_i U_i$, so it contracts with reserve when σ_i and c^{LU} remain stable and the observation bridge is adequate (Protter, 2005; Karatzas & Shreve, 1991).

This signature is stronger when paired with drift because it concerns a different part of the dynamics. Drift describes directional pull. Innovation describes stochastic amplitude. A model that gets one but not the other may be capturing generic reversion or generic quieting rather than reserve-coupled dynamics.

Shrinking innovation must be distinguished from shrinking observed movement. A person may move less, speak less, respond less, or tolerate fewer sensors near the endpoint. That can shrink measured innovation while latent variation remains unknown. The observation model from the previous section is therefore a prerequisite.

Time-varying σ is allowed only when independently identified. If σ is simply reduced near the endpoint to fit variance, the model

loses content. A stochastic-volatility extension must specify why volatility changes, how it is measured, and what rival it beats.

The innovation signature also depends on sampling cadence. Irregular observation can miss bursts. Late windows may be shorter and less dense. Missingness may be informative. A properly fitted model should account for unequal intervals and not treat silence as smoothness.

A favorable innovation result supports the reserve-scaled stochastic-amplitude component under the tested observation model and alternatives.

7.7.3 — Cross-Equation Restrictions

The reference case yields a cross-equation restriction linking relaxation and innovation through the same reserve: $\kappa_i^U = r_i^U / U_i$ and $\sigma_{L,i} = c^{LU} \sigma_i U_i$, so their product equals $c^{LU} \sigma_i r_i^U$ in the constant-volatility special case. The restriction lowers flexibility because drift and diffusion cannot be fitted independently (Protter, 2005).

That restriction is useful only under its printed assumptions: finite-variation reserve, coherent reserve units, a stable or prospectively modeled σ_i , and estimators for κ_i^U , $\sigma_{L,i}$, and r_i^U that do not share hidden outcome information. If an assumption fails, the equation becomes a sensitivity target rather than a confirmatory restriction.

$$\begin{aligned}\kappa_i^U(t) &= r_i^U(t) / U_i(t), \\ \sigma_{L,i}(t) &= c^{LU} \sigma_i U_i(t), \\ \kappa_i^U(t) \sigma_{L,i}(t) &= c^{LU} \sigma_i r_i^U(t)\end{aligned}$$

A cross-equation restriction is scientifically valuable because it reduces freedom. The model cannot choose drift and diffusion separately if one reserve process determines both. Rival models must therefore be scored on the joint restriction, not on whichever component happens to fit.

The restriction should be estimated with joint uncertainty and covariance. Near low reserve, errors in U_i affect both estimated relaxation and innovation, so independent standard errors can overstate agreement. A state-space or errors-in-variables analysis should report whether the apparent product relation survives that shared dependence on reserve.

If the restriction fails, the result is informative. The drift component might fit while innovation does not, or innovation might fit while drift does not. That would suggest ordinary homeostasis, observation noise, stochastic-volatility extension, or failure of the reserve product in that reference class.

A stronger model could replace constant σ with a structured volatility function. But that is a new claim version. It must produce its own cross-equation restrictions rather than simply absorbing any mismatch.

The cross-equation relation belongs in the prospectively specified fingerprint, with direction, tolerance, and failure rule fixed before endpoint alignment. A favorable terminal slope and variance contraction remain separate component observations when the relation between them is free; they become route-specific only when the same reserve process predicts both within declared uncertainty.

7.7.4 — Prospective Endpoint Separation

A signature evaluated after aligning everyone at the endpoint can be useful, but it is not the same as prospective prediction. Endpoint-conditioned processes can mimic terminal pull and variance shrinkage. The reserve realization must therefore show what it can forecast before the endpoint is known or without using outcome-selected alignment (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Prospective separation can be achieved in several ways. The endpoint rule can be protocol-fixed through independent unity-proxy channels. Reserve and ledger estimators can be cross-fitted. Model parameters can be trained on one subset and tested on held-out trajectories. Forecasts can be made for windows not used to define the endpoint.

The Brownian bridge rival should be given the endpoint information allowed by the study design. If the PTVC candidate uses endpoint alignment while the rival does not, the comparison is unfair. If both use the endpoint and perform similarly, the PTVC-specific interpretation weakens.

A stronger result would show that reserve-linked dynamics predict endpoint approach better than an endpoint-conditioned bridge using only information available before the endpoint. That result would support the predictive contribution of the reserve realization and motivate replication across endpoint settings.

Prospective separation also prevents public misreading of beautiful retrospective graphs. A curve smoothed after death may teach researchers about possible dynamics, but it cannot serve as

a confirmatory forecast. The figure label must tell the reader whether the curve is filtered, forecast, smoothed, or retrospectively adjudicated.

Terminal alignment organizes trajectories for comparison. Evaluate the aligned predictions under the same information rules for every model, with endpoints and evaluation criteria specified independently of the result being tested.

Prospective separation protects the model from hindsight. If the endpoint is selected, shifted, or retrospectively aligned to the path that looks most favorable, a Brownian bridge or conditioning rival may win before the Law is even tested. The endpoint rule must be declared before the signature is evaluated or protected by cross-fitting that prevents leakage.

7.7.5 — Observation-Model Robustness

The joint signature must survive observation rivals. A reserve-linked pattern that disappears when sedation, frailty, medication, missingness, sensor quality, or observation gain is modeled is not a reserve-realization success. It may be a measurement lesson (Talts et al., 2018; Wilson & Collins, 2019; Munafò et al., 2017).

Observation-model robustness requires more than adding covariates. The study should include generative observation models capable of producing false compression. It should simulate data from those rivals and verify that the analysis can distinguish them from latent reserve coupling.

Sedation is a particularly strong rival. It can reduce responsiveness, physiology, movement, report, and observable distress. It may coincide with terminal care and reserve decline.

A model that treats sedation merely as missingness can overinterpret the quieting it helps create.

Frailty and age-only models are also strong. Declining variance may reflect general biological weakening, not PTVC-specific reserve coupling. A fair comparison gives those models real parameters, real data access, and a chance to win.

Robustness should include negative controls. Channels that should respond to observation gain but not latent valence, device-quality indicators, care-timing variables, and non-affective physiological streams can reveal whether the supposed ledger signature is tied to measurement conditions.

A result that survives these checks is still component-level. It says the reserve signature was not explained away by specified observation rivals. It does not remove the need for replication, endpoint discipline, measurement validity, or formal bridge resolution.

7.7.6 — Held-Out Superiority to Strong Rivals

The final empirical test is comparative. After development, the reserve realization must outperform strong rivals on held-out trajectories under the declared comparison rules. This separates an explanation fitted to existing observations from a prediction that earns support on new evidence (Nosek et al., 2018; Mayo, 2018).

The rival set should include endpoint-conditioned bridges, OU and homeostatic processes, age-only decline, frailty-only decline, medication-only decline, observation-shutdown models, shared-driver disease models, and unconceived-model sensitivity. Each

rival should be implemented as a generative model where possible.

Held-out superiority should be evaluated on components that matter: drift, innovation, reserve coupling, endpoint forecasting, variance shape, observation robustness, and prediction calibration. A single aggregate score can hide a prerequisite failure. The gates should remain noncompensatory.

Compare the reserve realization with OU and bridge alternatives using common predictive and calibration criteria. At equivalent performance, a simpler adequate model merits preference in the tested reference class.

Model recovery is a precondition. If synthetic data generated by the reserve model cannot be recovered by the analysis, then a real-data failure is uninterpretable. If synthetic data generated by rivals are falsely classified as reserve support, then a real-data success is suspect.

The public verdict should be exact. Held-out advantage can support the reserve-realization component. Rival parity narrows or suspends PTVC-specific interpretation. Rival superiority retires the tested realization in that reference class unless a new, prospectively risky version is proposed.

7.7.7 — Conjunctive Signatures Rather Than Single Curves

Ordinary state reversion is not accumulated-ledger closure. An Ornstein-Uhlenbeck state can return toward its mean while the variance of its time integral continues to grow, so reduced reactivity near an endpoint may reflect medication, fatigue,

disease, family presence, or good care rather than PTVC (Uhlenbeck & Ornstein, 1930; Gillespie, 1996).

A candidate route must therefore survive a modular conjunction: route-state validity, drift and innovation restrictions, endpoint separation, observation robustness, and held-out rival superiority. The component vector remains visible because a win on drift cannot average away failed innovation, endpoint leakage, or an invalid bridge.

The signature package is a dependency graph, not a bag of p-values. Shared measurements create dependence, while route validation and endpoint assumptions feed multiple tests. Joint analysis and sealed-world model recovery must show that repeated views of one signal are not being counted as independent confirmation.

Exploration may compare proxies, lags, volatility models, and kernels, but confirmation freezes the surviving objects, tuning ranges, and scores in a new cohort or protected site. Negative controls should include false endpoints, shuffled route states, synthetic dropout, bridge generators, and ordinary mean-reverting worlds; support in those worlds disqualifies the pipeline.

Replication should include an adversarial team with independent code, alternative reasonable implementations, and no obligation to preserve the preferred interpretation. Transport tests the mapped coordinate, units, depletion direction, endpoint relation, and joint restriction, not merely shared vocabulary (Nosek et al.,

2018; National Academies of Sciences, Engineering, and Medicine, 2019).

A strong rival victory is informative. If OU, bridge, frailty, observation, or flexible hidden-state models reproduce the joint signature with fewer unsupported assumptions, the route narrows or retires while measurement, endpoint, and recovery knowledge remains.

Verdicts should preserve states such as route failed, drift compatible but innovation failed, endpoint separation unavailable, rival favored, bridge failed, and open-set inadequate. A single confidence score would restore the flexibility the conjunction was designed to remove.

Held-out superiority earns route attention only after simulation recovery, calibrated uncertainty, prospectively fixed conjunction logic, negative controls, transport, and a real possibility that a strong rival wins (Gneiting & Raftery, 2007; Talts et al., 2018; Mayo, 2018).

7.8 — Formal Failure, Open Theorems, and Route Retirement

The chapter closes with a research agenda and explicit tests of the reserve realization. Its precision creates opportunities for derivation, simulation, and independent evaluation. Progress requires retaining adverse results as evidence that can revise or reject the tested proposal (Popper, 1959; Lakatos, 1978). Publishing reproducible derivations and simulation code would let independent teams examine the same proposal and locate disagreements in specific assumptions or results.

The formal questions identify the next mathematical and empirical tasks: connect the phenomenological integral to the product semimartingale, validate the reserve construct, establish endpoint synchronization, and characterize when the normalized process remains finite or explodes. Joint comparison with rival models tests the resulting predictions. Any formal change proceeds through the declared change-control process. These tasks can be ordered by dependency, allowing measurement work to focus on the quantities that the mathematical analysis identifies as useful for distinguishing competing predictions.

This section separates mathematical questions, measurement questions, rival-model outcomes, and changes requiring formal review. Each result can guide the next version of the research program. A narrowed claim remains scientifically useful when its revised scope and supporting evidence are explicit, and every substantive change requires a visible record and a new test where appropriate.

Table 7.4. Route falsifiers, open problems, and retirement consequences.

Problem or falsifier	Why it matters	What would count against the realization	Route after failure
Integral-product bridge	Connects formal ledger to stochastic candidate.	No defensible mapping of path class and dimensions.	Formal change request or reduced status.
Reserve construct validity	U must be a measured object, not a fit label.	Proxy instability, bias, or shared measurement artifact.	Not instrument-ready.
Endpoint synchronization	Reserve collapse must meet independent endpoint.	Systematic early or late mismatch.	Reference-class narrowing.
Rival dominance	Science is comparative.	Strong rivals match or beat predictions.	Retire or revise candidate version.

7.8.1 — Integral-to-Route Bridge Failure

The largest open formal problem is the bridge between the phenomenological ledger and the reserve product. The received ledger is an integral over unity-gated valence rate. The product realization is a stochastic process involving reserve and a normalized signed component. The open task is to specify and prove the conditions under which the product represents the received ledger (Popper, 1959; Lakatos, 1978).

The bridge problem includes path class. An ordinary integral of an integrable rate has finite variation. A Brownian product can

have nonzero quadratic variation before the endpoint. If both are to describe the same construct at the same level, the mapping must explain that difference.

Possible resolutions include a reduced-form interpretation, a coarse-grained limit, a generalized derivative relation, a semimartingale integral, or a state-space bridge that connects phenomenological rate to stochastic increments. Each option has assumptions and consequences.

The product provides a candidate reduced-form realization of consequences associated with the latent cumulative process. The interpretations above define several routes for developing a stronger bridge. A later bridge must name its assumptions, preserve the distinction between route state and target ledger, and generate predictions that can fail independently of terminal fit.

A bridge theorem would need filtration, integrability, regularity, approximation norm, dimensional interpretation, endpoint behavior, and measurement implications. It would also need to explain how a latent valence-rate contribution gives rise to the normalized diffusion component.

The product model supplies component signatures while the relation to the formal integral is investigated. Resolving that relation is a central mathematical task for the program.

The bridge problem is not cosmetic notation. The received formal ledger is an accumulated phenomenological object, while the product realization is diffusion-bearing and reserve-coupled. Until a theorem, approximation regime, or formal amendment

connects them, the product should be called a candidate realization rather than the definition itself.

7.8.2 — Reserve Construct Invalidity

Reserve must be a separately measurable biological or functional quantity with stable enough units for its intended use. Its definition and validation criteria should be fixed independently of terminal-model fitting, giving the reserve model a meaning that can be tested across settings (Mayo, 2018; National Academies of Sciences, Engineering, and Medicine, 2019; Vehtari et al., 2017).

A reserve construct needs raw indicators, transformations, units, uncertainty, context of use, calibration population, confounds, transport limits, and failure rules. A frailty index, metabolic marker, arousal proxy, neurological reserve measure, or composite clinical score may all be candidates, but none is automatically the reserve U of the model.

Confounds are central. Medication, disability, illness, age, care quality, social support, sensor placement, and clinical workflow can all affect reserve proxies. Some are part of the biological process; some are observation artifacts; some are social burdens that the model must not erase.

Reserve construct validity also has equity implications. A proxy that systematically misreads disabled bodies, older bodies, sedated bodies, nonverbal persons, or people with different baseline physiology would create both scientific and ethical failure. Such a proxy cannot support public claims.

Validation should be local before transport. A reserve measure that works for one endpoint class should not be generalized to

sudden death, coma, dementia, pediatric care, nonhuman animals, or artificial systems. Reference class names must travel with results.

If reserve cannot be measured independently in a reference class, the reserve realization may still be a formal possibility, but it is not empirically testable there. The correct verdict is not support; it is not-yet-instrument-ready or unusable for that context.

7.8.3 — Endpoint Synchronization Failure

Endpoint synchronization failure occurs when reserve collapse and independently defined truth-level stream endpoint do not align in the relevant way. This can happen if reserve reaches near-zero while conscious stream activity continues, or if the endpoint occurs while reserve remains materially positive (Munafò et al., 2017; Nosek et al., 2018).

Such failure is not a mere inconvenience. It strikes a central premise of the reserve realization. If the product closes because *U* collapses, but the endpoint is elsewhere, then the realization has not explained terminal valence closure at the endpoint. It may explain a different process.

Synchronization failure should be measured prospectively where possible. Reserve trajectories and endpoint-proxy trajectories should be compared under rules that allow early reserve collapse, late endpoint, sudden endpoint, ambiguous endpoint, and no-estimate outcomes.

Clinical complexity matters. Sedation, resuscitation, withdrawal of life support, fluctuating consciousness, delirium, and low-signal states can all create uncertainty. The study should not force

synchronization by choosing the most convenient endpoint after seeing reserve behavior.

If synchronization repeatedly fails in a reference class, the reserve realization narrows for that class. A different reserve definition may be proposed, but only as a new version with new risky predictions. Reinterpreting the same failure under a new reserve label without public versioning is ad hoc rescue.

Synchronization failure is scientifically valuable because it tells the program where its candidate mechanism is wrong or incomplete. A theory that permits such a result is stronger than one that hides every mismatch inside endpoint language.

7.8.4 — Normalized-Process Explosion

Normalized-process explosion is the failure mode already illustrated by the Y equals one over U counterexample. As reserve declines, the normalized process grows too quickly for the product to approach zero. The product may remain away from zero or become unstable (Popper, 1959; Lakatos, 1978).

Explosion can be mathematical, biological, or observational. Mathematically, the process class may permit growth rates that defeat closure. Biologically, terminal distress, delirium, pain, fear, or conflict may create large normalized excursions. Observationally, denominator error can make a stable process look explosive.

The model must distinguish these possibilities. A joint posterior for L_p , U , and Y can reveal whether explosion is robust or driven by reserve uncertainty. Observation-quality metrics can reveal

denominator artifacts. Clinical covariates can reveal treatable causes.

The decisive closure question is whether the product approaches zero. Divergence of Y alone is insufficient: for $U > 0$, the mathematical example $Y = U^{-1/2}$ gives $UY = U^{1/2}$, which still tends to zero as U tends to zero. A finding can reject a bounded- Y or fixed-volatility subbranch without rejecting every closing product. Failure of the required product limit, after accounting for denominator and observation error, is adverse to the declared reserve route.

7.8.5 — Rival-Equal or Rival-Better Prediction

The reserve realization earns distinct evidential support when it improves explanation or prediction relative to capable rivals under a fair comparison. Equal or better rival performance identifies which claims remain unresolved and where revision is needed (Mayo, 2018; National Academies of Sciences, Engineering, and Medicine, 2019; Vehtari et al., 2017).

Rival equality is already important. If an endpoint-conditioned bridge, observation-gain model, frailty model, or OU homeostatic model performs equally well on held-out data, the reserve realization has not distinguished itself. It may remain a candidate, but its public status must narrow.

Rival superiority is stronger. If a rival predicts terminal dynamics better while using fewer PTVc-specific assumptions, the tested reserve realization should be retired for that reference class. Retirement of a component is not defeat of every metaphysical possibility, but it is real scientific loss.

The comparison must be fair. Rivals need enough flexibility to be serious. They should receive the same covariates, endpoint information, observation model, and training data permitted to the PTVC candidate. A weak rival set creates false victory.

A favorable reserve result must also be robust to unconceived-model sensitivity. The book cannot pretend that its named rivals exhaust possibility. It can require that current serious rivals be beaten and that new rivals receive a route into future updates.

The verdict follows the comparison. Equal performance leaves the tested models unresolved; better rival performance directs revision or retirement; better prospective performance by a recoverable reserve candidate supports the specified components.

7.8.6 — Formal Change Control and Route Retirement

Formal change control prevents repairs from silently replacing the target or route. A new process class, reserve definition, endpoint convention, bridge, or policy assumption receives a new identifier and dependency record. Improvement is welcome; invisible replacement is not.

Retirement begins with a typed failure. The reserve may be unmeasurable, the product may lack a bridge to the phenomenal ledger, endpoint synchronization may fail, the normalized process may become unstable, or a strong rival may perform as well or better. Each failure withdraws a different object.

Formal and empirical failures follow different routes. A false theorem is corrected formally; an unidentifiable route is narrowed, reclassified, or retired; and an open theorem stays open

until proved under matching assumptions. Simulation cannot promote it.

A supersession matrix compares retired and successor routes across object, endpoint, filtration, state, parameter, observable, intervention, and failure condition. Results depending on an altered column are rerun or marked nontransportable, while valid components may survive independently.

Evidence maturity and theory maturity remain separate. Replicated route failure may justify retirement while the mechanism-neutral target stays open, and a coherent theorem or simulation cannot raise empirical maturity without valid data.

Negative results are packaged for reuse. Code, synthetic worlds, failed recovery cases, parameter regions, numerical pathologies, observation burdens, privacy constraints, and recruitment failures receive stable identifiers, turning the completed research path into a reusable benchmark (Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019).

Retirement should sharpen the next study. Observation failure calls for a better observation model, endpoint failure for better adjudication, and rival victory for the earliest transported restriction on which the surviving models make distinguishable predictions.

Corrections propagate to summaries, diagrams, presentations, websites, translations, and excerpts. Retired equations and claims are removed or relabeled, and cached copies point to a successor or withdrawal notice.

No route verdict authorizes a personal inference. Success would not predict destiny or rank lives; failure would not erase independent advances in measurement, endpoint science, reachability, or rival comparison, or change any participant's dignity.

An open-theorem register names each proposition, proved assumptions, missing extension, countermodels, numerical evidence, and specialist review. Keeping several routes under common object and observation rules reduces pressure to rescue any one.

Part II closes with explicit exits and preserved value. The target can narrow, representation can yield an identified set, an endpoint branch can fail, a route can retire, a rival can win, and a study can return No adequate estimate. The hypothesis can therefore learn without moving its target while measurement, endpoint science, reachability, and recovery continue to advance (Lakatos, 1978; Mayo, 2018).

Part III — Measurement and Psychometrics

Measurement is the point at which a compact, consequential candidate hypothesis accepts disciplined contact with the world. Parts I and II established the scientific jurisdiction, formal target, and mechanism-neutral consequences of PTVC. Part III now builds the bridge from privately lived experience to evidence that can be inspected, challenged, corrected, and, when necessary, declined. Its task is not to manufacture a number that resembles the target, but to determine which features of experience can be represented, which observations carry information about them, which comparisons remain licensed, and which claims must stop at an identified set or a No adequate estimate outcome (American Educational Research Association et al., 2014).

The opportunity is larger than a single instrument. A rigorous measurement program can clarify the structure of valence, mixed affect, duration, memory, response shift, missingness, and cross-context transport even before PTVC receives a terminal test. It can reveal when first-person reports, choices, physiology, neural activity, clinical observations, and contextual records converge on one construct, when they illuminate different constructs, and when their disagreement is itself the most informative result. The achievement is a transparent relation among target, observation, model, uncertainty, and decision, not a device that claims possession of another person's inner life.

Lived experience remains the target because circumstances become burden or relief through consciousness. Housing, illness, disability, safety, money, law, relationship, medication, culture, and time remain indispensable causes and constraints, but none is interchangeable with the experience it helps shape. The scientific

challenge is therefore double. The program must preserve the reality of material conditions while refusing to treat appearance, preference, productivity, memory, or a sensor trace as the phenomenal object itself. That distinction protects both causal explanation and the person whose experience cannot be reduced to an administrative proxy.

The Part proceeds in three movements. Chapter 8 asks what kind of object lived experience can be for science, including ordinal, interval, scalar, vector-valued, mixed, path-dependent, and partially identified representations. Chapter 9 turns those possibilities into a qualified measurement pipeline that separates measurement map, estimand, analysis map, estimator, observation process, uncertainty, and decision rule. Chapter 10 asks when any resulting coordinate can travel across time, language, culture, disability, treatment, device, clinical state, or site. The sequence prevents a precise instrument from becoming a precise answer to the wrong question.

Measurement also creates an exposure of its own. A prompt interrupts a day. A sensor changes clothing, sleep, or movement. A clinic loses time. A family notices a device. A participant decides whether honesty is safe. A model may increase burden precisely when distress is greatest. Contemporary human-participant guidance and good-clinical-practice standards make participant welfare, proportionality, meaningful consent, and data quality part of the scientific design rather than decorative ethics. Part III treats burden, refusal, withdrawal, and care events as features of the observation process that can alter both validity and

dignity (World Medical Association, 2024; International Council for Harmonisation, 2026).

A qualified result must carry both its boundary and its measurement history. The output should identify the construct, measurement model, context of use, population, state range, instrument and model versions, scoring and transformation rules, observation window, missingness regime, uncertainty, comparison tier, and prohibited uses. Current clinical-outcome-assessment and psychometric reporting guidance reinforces this fit-for-purpose discipline. Transparent reporting of measure selection, modification, scoring, and validity evidence is part of the warrant for a conclusion, not administrative detail. These standards make later PTVC tests interpretable and help identify measurement relations that need improvement (U.S. Food and Drug Administration, 2025b; Gagnier et al., 2025; Flake & Fried, 2020).

Plural evidence is a strength when roles remain distinct. First-person report can be the closest available witness to phenomenal content, yet report can be shaped by language, memory, demand, fear, fatigue, and access. Physiology can reveal embodied dynamics, yet it can also reflect arousal, medication, exertion, or disease. Behavior and choice can reveal constraints and priorities, yet they are not identical to felt value. Context can explain a trajectory, yet it cannot testify for the person. The pipeline becomes stronger by modeling these relations explicitly rather than forcing every channel into a single agreement score.

Uncertainty is productive when it is typed. A confidence interval describes uncertainty for a target under a specified model and

repeated-sampling guarantee; an identified set contains every target value still compatible with the observed evidence and declared assumptions. The two objects may be combined, but they are not interchangeable. Channel conflict can reveal lag, state dependence, or construct mismatch. Informative missingness can reveal that the calmest-looking record is the most selectively observed. A No adequate estimate outcome identifies where additional evidence is needed before a numerical conclusion is available. The book therefore treats abstention, narrowing, local comparability, and instrument retirement as scientific accomplishments when the evidence cannot support a stronger claim (Manski, 2003; Imbens & Manski, 2004).

Part III establishes an anti-score boundary: research estimates may support declared scientific uses, but they cannot rank persons, set eligibility or care priority, assign spiritual or civic status, or transfer value across streams. No cross-stream comparison can settle PTVC. Privacy, refusal, correction, contestability, and human review govern use. The measurement system must justify its limited authority; participants do not owe legibility to a model.

The scientific workmanship is intentionally field-spanning. Affective science contributes construct and temporal models. Psychometrics contributes validity, invariance, linking, response-process evidence, and measurement-error analysis. Statistics contributes partial identification, missing-data models, uncertainty propagation, and sensitivity analysis. Clinical science contributes context, edge states, treatment effects, and care priorities. Privacy engineering and reproducible research

contribute purpose limitation, provenance, version control, audit, and correction. The value of the Part lies in joining these disciplines without pretending that any one of them has already measured a lifetime phenomenal ledger.

The central benchmark is simple to state and demanding to satisfy: every number must reveal the chain that earned it. A reader should be able to ask what construct was defined, why the measure was selected, what was observed, how the observation was scored or transformed, which estimand was targeted, which assumptions connected measurement to the latent object, what went missing, how versions changed, which rival explanations remained, and what decision the evidence actually licensed. Any modification to items, channels, scoring, exclusions, anchors, or preprocessing belongs in the same record. If a link is hidden, the claim becomes smaller. If a link fails, the result narrows or abstains rather than borrowing confidence from the rest of the pipeline (Flake & Fried, 2020; Gagnier et al., 2025).

Measurement can expose burdens that appearance misses, improve affective and clinical science, identify differences across people and states, and create a shared language for correction. Each advance makes the PTVC research question more tractable while contributing to the wider science of lived experience. Part III develops the instruments and evidence needed to pursue that opportunity.

Chapter 8 — Lived Experience and the Evidence Baseline

A person sits beside a hospital bed after good news that arrived too late. Relief, grief, love, exhaustion, anger, and gratitude coexist. A momentary rating may ask for one number. A memory interview may emphasize the ending. Heart rate may reflect sleeplessness. A choice may reflect duty rather than comfort. None of these observations is useless, yet none can claim to be the whole experience. Chapter 8 begins with that ordinary scientific difficulty: before measurement can be accurate, the target must be represented in a form that the evidence could genuinely support (Russell, 1980; Cacioppo & Berntson, 1994).

Lived experience is the target because the Law concerns what is carried within a conscious stream. This does not demote housing, food, safety, rights, medicine, disability support, relationship, or material justice. Those conditions shape experience through bodily, social, temporal, and interpretive pathways, and they remain independently important even when no phenomenal estimate is available. The chapter asks how goods and harms become lived conditions without turning either the circumstance or the report into a substitute for the experience itself.

Representation comes before instrumentation. The first question is not which questionnaire, sensor, or model should be used. It is whether the relevant relations are ordinal, interval-scaled, cardinal, additive, rate-like, state-like, partially ordered, vector-valued, or path-dependent in the declared context. The measurement model must also state why observations relate to the target. Reflective indicators are treated as effects of a latent state; causal or composite indicators help constitute the score; network models represent mutually dependent components. Those

structures license different reliability, validity, and aggregation claims. Ranking does not license integration, interval structure does not create a neutral origin, and a useful within-person coordinate does not automatically support between-person level comparison (Krantz et al., 1971; Bollen, 1989; Borsboom, 2005).

Mixed affect makes the issue empirical rather than philosophical decoration. Positive and negative experience can coexist, and bivariate representations allow that possibility to be measured instead of forced onto opposite ends of one line. Grief can carry love; relief can carry guilt; chosen effort can carry pain and agency; awe can carry fear; numbness can coexist with physiological threat. The chapter uses these cases to compare one-axis, bipolar, multidimensional, and set-valued models through prediction, discrimination, stability, and recovery rather than through aesthetic preference (Cacioppo & Berntson, 1994; Larsen et al., 2001).

Several neighboring constructs must remain visible without being collapsed. Valence is not arousal. Pain intensity is not suffering. Salience is not depth. Desire is not pleasure. Craving is not relief. Boredom is not neutral absence. Meaning can deepen endurance or deepen loss. Dignity and agency can change the lived character of similar sensory states. Parsimony remains valuable, but every omitted dimension needs an explicit role: irrelevant to the declared use, a moderator, a context variable, an auxiliary outcome, or a noncompensatory constraint. The model becomes simpler only after it shows what its simplification preserves.

Time adds a second representation problem. Much of a life is carried in ordinary duration rather than memorable peaks: the

fourth hour of waiting, the fiftieth morning of fatigue, the long calm after danger, the background freedom created by reliable care. A momentary rate, an episode summary, a remembered evaluation, and a decision utility answer different questions. The chapter keeps duration weighting, sampling coverage, recency, peak-end effects, recovery, order, and hysteresis separate so a vivid ending cannot silently stand in for the experienced path (Fredrickson & Kahneman, 1993; Kahneman et al., 1993).

Memory remains scientifically important precisely because it is not the original moment. Recollection can integrate meaning, compress duration, emphasize peaks, respond to later events, and guide future choice. Those transformations may be psychologically and morally significant, but they do not rewrite the experience that occurred. The chapter treats memory as an auxiliary construct, a possible mechanism, and a rival source of apparent closure. Path-functional comparison asks which summaries predict held-out reports, behavior, physiology, and future states without equating retrospective coherence with a complete phenomenal ledger.

Fairness appraisal and social comparison belong in the baseline because they can alter experience, attention, memory, and action. Inequity aversion, reciprocity, punishment, gratitude, entitlement, relative position, and just-world beliefs can generate local affective dynamics that mimic a fairness process without implying terminal closure. Cross-cultural and developmental recurrence establishes human salience and supplies comparative hypotheses. It becomes PTVC-specific evidence only if prospectively specified predictions outperform serious cultural,

psychological, and methodological rivals (Fehr & Schmidt, 1999; Henrich et al., 2005; Hafer & Bègue, 2005).

The chapter's evidence baseline is deliberately multimethod. First-person report, pairwise judgment, trade-off choice, physiology, behavior, clinical observation, relational context, and intervention response can each contribute to construct validity. Convergence is informative only when the channels are expected to reflect the same target and their shared errors are modeled. Components that cause or constitute a score need not correlate like reflective indicators, while channels aimed at different constructs should not be rewarded for artificial agreement. Divergence can therefore reveal a hidden dimension, response process, medication effect, or observation artifact. No composite gains sovereignty merely because it contains more inputs (Campbell & Fiske, 1959; Cronbach & Meehl, 1955; Bollen, 1989).

Material justice and care do not wait for the representation problem to be solved. A person need not produce a valid valence score before receiving food, shelter, medicine, protection, accommodation, relief, or belief. The chapter separates scientific warrant from ethical obligation so that measurement can improve explanation without becoming a gate to care. That separation also improves data quality because fear of losing treatment, housing, employment, or standing changes what people report and whether they participate (National Commission, 1979; World Medical Association, 2024).

The work remains valuable under every PTVC outcome. A supported scalar bridge would open stronger within-stream

accumulation tests. A vector or set-valued result would narrow the branch while preserving richer affective science. A failed construct would show where the target cannot yet be operationalized. A rival explanation might outperform the proposed representation and teach the program how human experience is organized. Each result advances knowledge when the target, assumptions, channels, and withdrawal conditions remain visible.

Chapter 8 moves from the status of lived experience through representation, mixed affect, scalarization, temporal structure, construct validity, and fairness cognition. Its final product is not a universal score. It is a disciplined menu of candidate representations, evidence roles, failure criteria, and human boundaries that later instruments must respect. Chapter 9 can then build a measurement pipeline whose outputs are useful because the chapter has already decided what they are allowed to mean.

8.1 — Why Lived Experience Is the Target

The phrase lived experience can sound soft until one notices how much science and policy already depend on substitutes for it. Institutions count income, diagnosis, attendance, discipline, productivity, reported satisfaction, legal status, and behavior. Those measures matter, but none of them is the felt life itself. *The Law of Fairness* enters precisely where visible condition and carried burden can come apart (Cacioppo & Berntson, 1994; Russell, 1980).

This section therefore establishes the target without collapsing the world into feeling. Food, housing, safety, legal protection, medical care, political rights, social belonging, and bodily integrity matter materially before they become data. Yet their relevance to the Law runs through a further question: what burden or relief do they create within a conscious stream? A science of fairness must hold both levels at once.

The difficulty is that many common proxies are attractive because they are administratively convenient. Preferences are observable through choices. Life satisfaction is easy to ask about. Public appearance can be seen by others. Social status can be recorded. But each proxy can miss hidden fear, adapted deprivation, shame, chronic pain, numbness, exhaustion, or the quiet relief of safety. The section treats those misses as scientific information.

A mature target must also protect dignity. To say that experience is central is not to say that rights should wait for measurement, or that material harm must justify itself by producing a measured inner state. The target is scientific, not a gate to care. The section

makes this boundary explicit so the later measurement chapters do not become permission to doubt people until an estimator agrees.

8.1.1 — Experience Over Appearance

Appearance is the first temptation of social measurement. A life can look stable from the outside because the rent is paid, the appointments are kept, the child is fed, the smile is reliable, and the work product arrives on time. None of that is false. The error begins when visible functioning is treated as if it exhausts the burden carried by the stream (Russell, 1980; Borsboom, 2005).

The Law of Fairness places lived experience at the target because hidden burden often survives inside acceptable appearance. A person may comply because dependence makes refusal costly. They may succeed while carrying chronic pain. They may perform calm under threat because panic would invite punishment. These are not literary exceptions; they are ordinary routes by which external observers can miss the carried cost.

A scientific target must therefore separate public biography from felt trajectory. Biography records what happened, what was obtained, what role was held, and what others could see. Felt trajectory asks what those events became within consciousness: fear, relief, numbness, humiliation, agency, safety, pressure, boredom, awe, or peace. The two can correlate strongly without being identical.

This distinction leaves the moral force of visible harm intact. An injured body, unsafe home, coercive law, or lack of money already demands attention and repair. Material conditions and lived

experience answer different scientific questions: the conditions establish real causes and obligations, while the Law-facing inquiry asks how those conditions are carried within a conscious stream. Neither level replaces the other.

An estimator of lived experience needs an explicit bridge from observed appearance to the target. Attendance, output, facial expression, and medication dose can be useful clinical or contextual evidence, with roles distinct from phenomenal measurement. The chapter states the proposed bridge and the observations that would require it to be revised or rejected.

The operative question is exact: where is burden carried when appearance conceals it? Composure, productivity, or social fluency cannot answer that question by themselves. Giving lived experience priority over appearance is therefore not a romantic gesture; it is the first rule preventing an external record from being mistaken for the phenomenal quantity the hypothesis concerns.

Proxy variables become most dangerous when they are accurate about the wrong target. Employment, attendance, facial action, speech rate, sleep duration, and medication use may predict distress in a population while failing for a particular person or context. The correct scientific question is therefore not whether a proxy correlates with burden somewhere, but whether its context-specific bridge remains calibrated, transportable, and contestable where the claim is made. A high-performing proxy can still be a poor representation of lived experience (Campbell & Fiske, 1959; Borsboom, 2005).

8.1.2 — Goods Become Lived Conditions

External goods matter because bodies, minds, and relationships are not sealed containers. Housing changes sleep. Money changes horizon. Law changes fear. Care changes dependency. Food changes energy. Safety changes attention. Friendship changes recovery. The science of lived experience cannot float above these goods as if feeling were detached from the material world (Cacioppo & Berntson, 1994; Campbell & Fiske, 1959).

The key word is become. A good does not enter the ledger merely by being listed. It becomes a lived condition through pathways of access, expectation, bodily state, social meaning, perceived control, and continuity. A disability accommodation may reduce pain, but it may also reduce humiliation. A legal protection may reduce danger, but it may also reduce vigilance. A cash transfer may relieve hunger, debt pressure, or shame depending on the surrounding life.

This conversion is not one-to-one. The same resource can carry different experiential meaning across people and contexts. A hospital room can feel safe, abandoned, controlled, humiliating, or hopeful. A public benefit can feel like relief or surveillance. An apology can repair dignity or reopen dependence. The Law needs measurement that asks which conversion occurred rather than assuming that the good itself carries a fixed affective value.

Goods are upstream conditions, not substitutes for experience. Medicine, law, economics, and social policy can describe resources, protections, opportunities, and harms with their own objective measures. This volume asks the additional question of

how those conditions become lived within a conscious stream. Keeping the levels distinct prevents two errors at once: reducing justice to feeling and reducing feeling to a checklist of circumstances (Sen, 1999; Borsboom, 2005).

A construct map should therefore include both external and experiential variables. Money, shelter, safety, clinical care, time, privacy, relationship, disability support, and legal status may be covariates, moderators, or causes of experience. They are not automatically the target. The target is the stream-level consequence under declared representation.

This is why the chapter can be both materialist and phenomenological. It refuses the idea that fairness is only external distribution, but it also refuses the idea that feeling can be separated from the conditions that shape it. Goods become lived conditions through pathways that must be studied, not assumed.

The route from goods to experience is often mediated and heterogeneous. Secure housing may improve sleep through temperature, privacy, reduced threat, and continuity, while the same move can also disrupt community or identity. Clinical treatment may reduce symptoms while increasing fatigue, dependency, or fear. Modeling these pathways prevents a material variable from becoming a one-directional affect label and allows the same good to have different experiential profiles across time and persons (Cronbach & Meehl, 1955; Bollen, 1989).

8.1.3 — Preferences Are Not Experience

Preference is evidence of something, but it is not identical to experience. A person may choose what is least dangerous, most

familiar, socially required, economically possible, or least humiliating among bad options. Revealed preference can record constrained survival rather than felt relief. Willingness to pay can record wealth, bargaining power, hope, fear, or lack of alternatives (Kahneman et al., 1993; Fredrickson & Kahneman, 1993).

This matters for the Law because a preference ordering can look rigorous. Choices generate data. Markets generate prices. Surveys generate rankings. Yet the fact that someone chooses one burden over another does not tell us the felt weight of either burden. A person can choose a painful treatment because they love their child, remain in a job because rent is due, or refuse care because dependence has been made frightening.

Adapted deprivation exposes a special failure of preference inference. People may learn to want less when asking is punished, costly, or hopeless, and may report an unjust condition as acceptable because imagining alternatives is itself painful. The absence of expressed demand for relief can therefore be an effect of constraint rather than evidence of low burden. Preference data remain informative, but only after the feasible set, incentives, and adaptation process are modeled.

None of this means preference data should be ignored. Preferences can reveal priorities, tradeoffs, agency, and practical constraints. They can help design humane studies by showing what participants are willing to bear. They can identify outcomes people value. But they cannot serve as a silent replacement for momentary or path-level experience.

Choice should be labeled as choice evidence, not as direct valence evidence. A preference may encode anticipated experience, moral commitment, social pressure, habit, identity, obligation, risk management, or resource constraint. The study must specify which bridge from choice to felt value it is testing and which observations would break that bridge; otherwise the inference stops at revealed behavior.

A constrained choice does not make suffering disappear. It records what a person selected from the options that remained available, not what the stream carried while selecting it. This distinction is essential whenever deprivation, coercion, dependence, or adaptation has already altered the menu.

Choice data are especially vulnerable to scarcity and path dependence. A repeated choice can reflect learned feasibility, switching cost, habit, coercion, or the expectation that alternatives will disappear. Preference elicitation should therefore record the option set, information, time pressure, consequences, and reversibility of the choice. Without those conditions, a ranking may describe constrained behavior while being reported as a ranking of felt value (Sen, 1999; Kahneman et al., 1993).

8.1.4 — Life Satisfaction Is Not a Momentary Rate

Life satisfaction is a reflective judgment. It asks a person to stand back from the stream and evaluate a life, chapter, role, or condition. That judgment can be profound, morally important, and socially useful. It is still not the same object as a momentary contribution rate. A life can be judged worthwhile while

containing long periods of fear, pain, boredom, exhaustion, or grief (Shiffman et al., 2008; Stone et al., 2007).

The distinction is familiar in experienced-utility research: the experiencing self and the remembering or evaluating self can diverge. A person may remember an episode through its ending, its meaning, or its story rather than its full temporal distribution. The Law's formal ledger, if it is to be tested as a trajectory claim, cannot substitute reflective evaluation for the time-local process without changing the hypothesis.

Life satisfaction also carries cultural and personality differences in aspiration, modesty, comparison, gratitude, religious interpretation, and social desirability. A high satisfaction report may reflect dignity, resilience, or genuine peace. It may also reflect lowered expectation, fear of complaining, or a meaningful life that still contains substantial suffering. The score needs interpretation, not reverence.

Life satisfaction belongs in the evidence system without being promoted into the integrated quantity. Reflective evaluation can serve as an auxiliary construct, validator, rival, or outcome in its own right; it may reveal meaning, standards, aspiration, or narrative coherence that momentary measures miss. It becomes a valence-rate measure only if an explicit and independently tested bridge supports that role.

The difference protects both science and dignity. It lets a person say that a life is meaningful without erasing the suffering inside it, and it lets science study suffering without declaring the life meaningless. That is the boundary a fairness science must keep.

Reflective evaluation can also change the interpretation of an episode. Reappraisal, narrative integration, social conversation, and later outcomes may alter what an earlier event means without altering what was felt at the time. Both objects deserve study, but they answer different questions. A science of terminal valence should preserve the experienced path, the remembered path, and the evaluative judgment as linked but noninterchangeable records (Fredrickson & Kahneman, 1993; Stone et al., 2007).

8.1.5 — Meaning and Dignity Matter Within Feeling

A scientifically useful account must represent how context and meaning shape the quality of experience. Meaning, dignity, agency, violation, loyalty, awe, shame, moral injury, and love alter the felt shape of states. They do not merely decorate valence after the fact. They can change what a moment is like from within (Cronbach & Meehl, 1955; Borsboom, 2005).

Consider pain under two contexts: pain inflicted as humiliation and pain endured in chosen care for someone loved. The sensory intensity may overlap while the experience differs in agency, meaning, threat, resentment, and dignity. The second case does not make pain morally owed payment or credit. It shows that pain is not fully represented by nociceptive intensity alone.

Dignity is especially important because it resists simple scalar capture. A humiliating interaction may be brief and materially minor, yet carry a long felt afterlife. A respectful interaction may not eliminate illness or poverty, but it can reduce fear and restore agency. These effects are experiential while also being moral and social. The chapter keeps both descriptions visible.

Meaning must be represented without being assigned a permanently positive coefficient. It can sustain action, deepen commitment, reframe pain, intensify loss, or coexist with severe distress. A model that automatically lets meaning compensate suffering is as misspecified as one that excludes meaning entirely. The relevant question is how meaning changes phenomenal structure in the declared context, not whether it belongs on a universally favorable side of a ledger.

This requires multidimensional measurement. Some constructs may enter as separate axes, moderators, or context variables rather than as signed components. Agency, humiliation, moral injury, belonging, and sacred meaning may shape how valence is interpreted and how time is carried. Their role must be modeled, not hidden under a pleasure label.

The target must accommodate more than raw pleasure and pain. Burden may be carried as bodily distress, loss of agency, humiliation, fear, moral injury, or the effort of meaningful endurance. Those forms need not collapse into one dimension in advance. The representation should earn any compression through evidence rather than by excluding the experiences that make compression difficult.

Some dimensions may be noncompensatory within the scientific representation. Severe violation, loss of agency, or humiliation may not be responsibly traded against comfort by one fixed weight in every context. Thresholds, lexicographic rules, vector outcomes, or identified sets can preserve that structure instead of forcing every dimension into a smooth utility surface. This approach prevents the measurement model from erasing

distinctions that participants, mechanisms, and evidence treat as qualitatively important (Manski, 2003; Borsboom, 2005).

8.1.6 — Material Justice Does Not Wait

No one waits for affective science to become precise before receiving care, rights, safety, food, housing, medicine, disability support, or protection. Measurement cannot become a gate a person must pass before suffering is believed; people remain people before they are research cases (National Commission, 1979).

That separation is methodological as well as ethical. If report is unavailable, a sensor intolerable, or representation contested, the study has not earned its intended bridge. The person's experience and claim to care do not thereby disappear.

Care conditions shape data quality. Fear of losing treatment, housing, access, family trust, or status can alter reporting and participation, while deprivation can select for unusual endurance. Accessible responses, compensation, transport, flexible scheduling, independent care, and immediate stopping therefore belong to design (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

A protocol specifies dignity-preserving, reason-coded abstention paths before collection. It records which link failed, what remains interpretable, which claims withdraw, and which care continues unchanged. Uncertainty stays with the instrument and study.

The rule holds under every scientific outcome. Evidence consistent with PTVC does not excuse present suffering, and

evidence against it does not diminish the urgency of repair. Rights and relief are independent of a terminal-boundary verdict.

Measurement can still contribute to justice by exposing burdens that appearance misses without creating new gates or dossiers. The achievement is a bounded account that informs research and care while preserving refusal, privacy, and nonparticipation.

The table below lists the representation questions that must be answered before integration. When ordering, interval meaning, additivity, sign, or transport fails, the program narrows its claim rather than repairing the person or delaying justice.

8.2 — Representation Before Measurement

Measurement becomes serious only after the target has a form. A thermometer measures temperature because temperature has a theoretical role, units, transformations, and expected relations to other quantities. Lived experience will not hand science the same simplicity. Before the book can integrate, compare, or accumulate experience, it must ask whether the relevant quantity can be ordered, spaced, signed, summed, and transported (Russell, 1980; Kahneman et al., 1993; Krantz et al., 1971).

The central question is not whether instruments are sensitive enough. It is whether the proposed representation is meaningful. A person may say one state is worse than another without implying that the difference has a stable unit. A physiological channel may track arousal without tracking valence. A scale may work within one person and fail across people. A neutral point may be chosen by convention and then mistaken for discovery.

The Law needs this section because its formal object is not a general mood report. The ledger requires a valence-like contribution that can enter time. If evidence supports only ordinal ordering, multidimensional comparison, or partial ordering in some contexts, then the scalar additive program must narrow in those contexts. Narrowing is not defeat of human reality. It is protection against false mathematics.

Representation failure is an informative scientific outcome. If a study shows that the target is not scalar, not additive, not stable across states, or not recoverable at the proposed resolution, the result has prevented later precision from being attached to the

wrong object. The program should record which lower-level claims survive rather than hiding the failure beneath denser sampling or a more elaborate estimator.

The order of obligations matters: identify the bearer and target; establish ordering; identify the neutral origin and sign; justify distances and units; test composition; define the temporal path; only then attempt cross-person linking. A later failure does not erase a valid earlier relation, but it blocks every claim that depends on the missing rung. The ladder therefore preserves partial knowledge without promoting rank, local change, or within-stream meaning into a common cardinal scale.

Table 8.1. Representation features and decision consequences.

Representation feature	Question for the chapter	If the answer is strong	If the answer is weak
Ordinal structure	Can states be ordered as more or less burdensome within context?	The program may compare direction and rank without yet summing experience.	The target must split into qualitative or partially ordered descriptions.
Interval structure	Do differences carry stable meaning across nearby states?	Local rates and changes can be estimated under declared conditions.	Additive ledgers must narrow to rank, threshold, or local contrast claims.
Additivity	Can momentary contributions be meaningfully accumulated over time?	Path integrals become candidate summaries rather than metaphors.	Duration and order must be modeled as separate path functionals.

Representation feature	Question for the chapter	If the answer is strong	If the answer is weak
Sign and origin	Is there a non-outcome-selected reference for positive, negative, or neutral?	Signed rates can enter later ledger work with declared uncertainty.	Neutrality claims must abstain or use origin-resistant signatures only.
Within-person transport	Does the representation remain stable across states of the same stream?	Longitudinal measurement can proceed with version labels and drift checks.	The program must use state-specific models or stop comparing phases.
Between-person transport	Can the same representation travel across persons or cultures?	Population summaries become bounded and reference-class-labeled.	The result remains local, personal, or group-specific.

8.2.1 — Ordinal Versus Interval Structure

An ordinal representation says that one state is more burdensome than another. An interval representation says more: that differences between states have stable meaning. The step from ordinal to interval is not cosmetic. A ledger that integrates contributions over time needs distances, not only rankings. If the distance between mild and moderate distress is not comparable to the distance between moderate and severe distress, simple addition becomes unstable (Krantz et al., 1971; Borsboom, 2005).

People can often rank experiences without being able to assign meaningful intervals. Someone may know that nausea is worse

than boredom and panic worse than nausea, while having no stable unit that makes the two gaps comparable. Cultural language can sharpen or blur ranks. Clinical states can change sensitivity. Repetition can make a fixed stimulus feel different. Ordinal information may be valuable but insufficient for additive arithmetic.

Affective science frequently uses rating scales that look interval because they are numbered. The chapter treats those numbers as hypotheses. A zero to ten scale may be approximately useful under some contexts of use, but the spacing must be tested through response models, anchoring, repeated measures, invariance checks, and predictive behavior.

Ordinal evidence supports directional change, rank ordering, and threshold classification. Interval arithmetic becomes available when the additional spacing assumptions have empirical support. Every result should name the level of representation it has earned. Within-person ordinal change, within-person interval approximation, group-level linking, and cross-context common-scale measurement are different accomplishments with different assumptions. They cannot be combined into a single confidence label, and success at a weaker level does not silently authorize the stronger one.

Match the output to the representation: report ordinal direction, threshold crossing, or set-valued comparison where those are the supported operations.

Ordinal structure can support more science than it first appears to permit. Repeated pairwise comparisons may identify stable order

relations, detect reversals, and test whether one context preserves another context's ranking even when equal intervals are unavailable. The key is to report the structure actually identified. A robust order is not an interval scale, but neither is it mere impression; it can constrain models, reject proposed scalarizations, and define partial-order estimands that later work must respect (Krantz et al., 1971; Borsboom, 2005).

8.2.2 — Cardinality and Additivity

Cardinality concerns meaningful magnitude. Additivity concerns meaningful summation. A measure can have ordered categories without cardinal meaning, cardinal approximations without safe addition, and addition within one narrow context without transport across people or states. The measurement model matters here. Indicators treated as effects of a latent state support different tests from components treated as causes or constituents of an index; mutually interacting states require still different dynamics. The ledger needs a defensible magnitude, composition rule, and temporal interpretation in every domain where it is used (Cronbach & Meehl, 1955; Bollen, 1989; Borsboom, 2005).

The additive question is easy to underestimate. Suppose two short intervals each carry mild fear. Is their combined contribution equal to one longer interval of mild fear? Does fear saturate, compound, habituate, or interact with exhaustion? Does relief after fear cancel fear, transform memory, or create a new state? Additivity is a model of time, not a gift from arithmetic.

Temporal addition requires stable units and a defensible contribution at each time. Habituation, sensitization, and recovery

may change that contribution through history-dependent dynamics without violating additivity over disjoint intervals. A nonadditive path functional is needed only when the declared target cannot be represented by integrating the resulting time-local contributions (Krantz et al., 1971; Borsboom, 2005).

Cardinality also determines whether a weight vector can be interpreted. If one dimension is measured on an arbitrary scale and another on a different arbitrary scale, a weighted sum may reflect instrument convention rather than lived tradeoff. Internal consistency does not solve that problem and may be irrelevant for a composite whose components are not expected to be interchangeable effects of one latent state. The model must state what each magnitude means and why the components may be combined before it multiplies and adds them (Bollen, 1989; Flake & Fried, 2020).

A useful scientific program can begin with approximate additivity in bounded contexts. It can test whether repeated episodes concatenate linearly, whether duration doubles contribution, whether recovery resets sensitivity, and whether high-intensity states dominate totals. These tests provide content rather than decoration.

If additivity fails, the Law does not become meaningless. The scalar additive claim narrows. The evidence may shift toward path functionals, thresholds, vector trajectories, or local signatures. The chapter makes this possibility ordinary so that later results cannot be rescued by ignoring failed arithmetic.

Additivity is not earned by writing a sum. It requires representational or empirical conditions under which duration, intensity, context, and interaction can be composed without changing the meaning of the parts. Cancellation failure, saturation, threshold effects, hysteresis, and context-dependent synergy all pressure an additive ledger. A serious program should test those failures directly and retain a nonadditive, vector, or set-valued alternative whenever one common composition rule is not supported (Krantz et al., 1971; Russell, 1980).

8.2.3 — State Versus Rate

A state describes how a moment is. A rate describes contribution per unit time. The difference matters because the formal ledger integrates a rate-like object. If a report says a person feels anxious at a moment, that report has not yet said how anxiety contributes over the next interval. The same intensity may have different duration, recovery, anticipatory load, and aftereffect (Russell, 1980; Cacioppo & Berntson, 1994).

Affective instruments often report levels. A momentary rating of sadness, pain, fear, or calm may be a state estimate. To integrate it, the model must declare whether the level is itself a rate proxy, whether it must be transformed, or whether a different quantity such as change in burden should be used. Multiplying a state score by time does not automatically create a valid unit.

The distinction becomes especially important for sustained conditions. A background dread that is low in momentary intensity but present for months can dominate a life if treated as a rate. A sharp acute pain may dominate memory while occupying

little clock time. The model must decide what time-local contribution means before it claims cumulative burden.

The Law-facing target should therefore label each candidate source. Is it a state, a rate proxy, a transition marker, a recovery cost, an anticipatory cost, or a remembered evaluation? Each label has different implications for integration and comparison. A mixed model may include several of these, but only if the components are not quietly collapsed.

A worked schedule can make the difference visible. Morning nausea, afternoon vigilance, evening relief, and nighttime fear may have similar state ratings at some instants while carrying different interval contributions because duration and context differ. A research design must sample enough of the interval to avoid mistaking snapshots for contributions.

The state-rate distinction states how a momentary score contributes through time, including the exposure rule and the evidence supporting that conversion.

The state-rate distinction also determines sampling design. A slowly changing state can be represented by sparse observations only under assumptions about interpolation, while a rapidly changing rate requires sampling fast enough to avoid aliasing. If the instrument's temporal resolution is coarser than the process, the apparent interval contribution may reflect the observation schedule more than the experience. The measurement map should therefore state the timescale on which the construct is defined and the timescale on which the estimator is permitted to speak (Shiffman et al., 2008; Stone et al., 2007).

8.2.4 — Within-Person and Between-Person Meaning

A scale can work within one person before it works between persons. A participant may reliably report that today's pain is worse than yesterday's, or that safety has increased since a medication change, without the same scale allowing a direct comparison with another participant. Within-person meaning and between-person meaning are distinct representational achievements (Bollen, 1989; Borsboom, 2005).

This distinction matters because the Law is stream-centered. Many early studies may be idiographic: they ask whether changes within a stream follow predicted dynamics under a stable local representation. Such studies can be scientifically valuable even if no universal cross-person scale exists. A within-person model can test temporal shape, response to events, and reliability of local anchors.

Between-person comparison demands more than identical prompts or synchronized devices. Language, physiology, disability, trauma history, medication, cultural display rules, and social expectation can alter how the same response or signal relates to experience. A shared score may therefore be a shared format rather than a shared quantity. Interpersonal arithmetic requires evidence that origin, sign, unit, and response process survive the proposed link (Putnick & Bornstein, 2016; International Test Commission, 2018; American Educational Research Association et al., 2014).

Two ambitions must remain separate. A study may first establish temporal coherence within one participant, then ask whether the

resulting scale can be linked across participants. Failure of interpersonal calibration does not erase valid within-person change, and local validity does not grant a common metric. Idiographic and nomothetic claims require different evidence even when they share a model family (Molenaar, 2004; Hamaker, 2012).

The burden lens makes the separation ethically important. People with atypical bodies, communication styles, cultures, pain histories, or disabilities should not be declared incomparable as persons. The scientific claim may be incomparable; the person is not. Measurement humility must not become social exclusion.

A common scale must earn its scope one bridge at a time. Stability across states of one stream, continuity across instrument versions, transport across clinical conditions, comparison across groups, and transport across languages or cultures are different achievements. Evidence for one does not license the next. The public claim should stop at the last bridge that has been independently supported.

Within-person and between-person relations can even point in opposite directions. A variable associated with relief across people may mark distress within one person when it rises above that person's own baseline, and a stable group difference may say nothing about change over time. Because PTVC is intrastream, population structure may inform calibration, priors, and rivals, but it cannot substitute for the trajectory of the stream being studied (Molenaar, 2004; Hamaker, 2012).

8.2.5 — Neutral Origin and Sign

A signed representation needs an origin. Positive and negative are not free labels. If the origin is chosen by sample mean, instrument midpoint, cultural convention, or terminal fit, neutrality can become an artifact. The Law's strong claim is especially sensitive to this problem because terminal valence closure concerns exact zero in a latent coordinate system (American Educational Research Association et al., 2014; Putnick & Bornstein, 2016).

The formal core owns the abstract origin problem; lived measurement supplies the empirical ways an apparent midpoint can mislead. A person may select 'neutral' because nothing dramatic is happening, because language offers no better option, because dissociation has flattened access, because expectations have narrowed, or because a midpoint feels socially safe. Those response processes define different candidate origins and must be separated before zero acquires cumulative meaning (American Educational Research Association et al., 2014).

Sign depends on construct. High arousal can be joyful, terrified, angry, sexually excited, or medically distressing; low arousal, peaceful, depressed, exhausted, or sedated; pain, threatening, chosen, meaningful, humiliating, or ignored. A one-dimensional physiological sign rule fails when context changes meaning.

A candidate neutral origin must be fixed independently of the closure result it will later help evaluate. It may draw on repeated within-person judgments, contextual anchors, tolerance or indifference tasks, qualitative interpretation, and auxiliary physiology, but no single anchor is self-authenticating. The

protocol should state the origin rule, its uncertainty, its failure conditions, and the consequences of shifting it within the plausible range (Krantz et al., 1971; American Educational Research Association et al., 2014).

When neutral origin is weak, output should become an identified region or directional statement. The study may report decreased burden, movement toward an operational neutral band, or indeterminate sign, but not claim exact zero was observed.

The origin problem is not technical pedantry. It is how the chapter prevents science from manufacturing fairness by choosing the coordinate system that makes the graph look fair. Sign and zero must be earned before they can bear moral or scientific weight.

Positive rescaling preserves zero, while translation changes the origin. An origin error accumulates across exposure and can produce a substantial terminal residual. Independent anchors, cross-sign tasks, behavioral or psychophysical checks, and sensitivity analyses can quantify that influence and guide better calibration (Krantz et al., 1971; Cacioppo & Berntson, 1994).

8.2.6 — Representation Failure as Scientific Result

A representation can fail honorably. Failure of a scalar, additive, signed, or transportable representation is substantive, not a setback hidden by revised language. It shows one empirical bridge cannot carry the claim (Manski, 2003).

Representation can fail in several ways. Reports may remain ordinal; dimensions may resist stable weights; mixed states may be incomparable; groups may use different response processes; clinical states may alter item meaning; physiological channels

may track arousal or illness rather than valence; and temporal aggregation may be path-dependent.

Such findings do not deny lived experience. They reject a proposed simplification. The person remains real, burden remains ethically significant, and care remains urgent. The science may move from scalar to vector language, common to within-person scope, or additive to path-functional rivals.

Abstention is legitimate. A study may return no scalar estimate for a person, context, group, or interval when representation gates fail. That output is more informative than a number forced from incompatible dimensions, uncertain signs, or noncomparable scales (Manski, 2003).

The hypothesis becomes more scientific when representation failure can narrow it. A program that can lose scalarity, additivity, or comparability at the first gate has exposed real ways to be wrong rather than protecting a grand claim behind flexible language.

8.3 — Mixed Affect and Multidimensional States

Human feeling is often not arranged as a single traffic light. People grieve and feel love, dread and feel relief, hurt and feel loyal, suffer pain and find meaning, feel numb while their body remains under threat, or feel joy so intense that it carries fear. These states are not exotic exceptions. They are ordinary evidence that the representation problem is real (Russell, 1980; Cacioppo & Berntson, 1994; Dejonckheere et al., 2022).

Context can change the felt sign of an otherwise identical stimulus. In 16 healthy volunteers, Leknes and colleagues (2013) presented the same moderate heat in two settings: one in which the alternative was nonpainful warmth, and another in which it was intense pain. Average sensation ratings shifted from painful in the first setting to pleasant in the second. The shift accompanied reduced skin-conductance responses in the 12 participants with usable recordings and changes in pain-related and valuation circuitry. This gives PTVC's emphasis on experienced value a concrete experimental foundation: the relation between an event and its alternatives can alter the feeling itself. A context-sensitive valence model can therefore make quantitative predictions about changes in felt sign while the physical input remains fixed, with expectancy and contrast models providing direct comparisons.

A scalar model can handle some complexity if its mapping is declared and tested. Positive meaning within pain may coexist with negative experience, and genuinely positive and negative contributions must be represented under the chosen branch. Their

arithmetic combination does not make a harm morally justified or erase its negative magnitude. Report gross components as well as the net result where the representation permits them; do not change weights to manufacture cancellation or suppress genuine positive feeling because it accompanies suffering.

Affective science offers several neighboring structures: dimensions such as valence and arousal, appraisal processes, motivational action tendencies, interoceptive and allostatic accounts, constructionist views of emotion, and clinical descriptions of pain, numbness, anxiety, depression, trauma, and dissociation. The section does not choose among them as final truth. It uses them to ask what a Law-facing representation must survive.

Mixed states matter because burden can reside in conflict as well as magnitude. Relief may coexist with grief, pride with fear, love with exhaustion, and pleasure with narrowing compulsion. A one-axis model must show that its projection preserves the comparisons required by the study; otherwise the richer state remains the scientifically appropriate object.

8.3.1 — Coexisting Positive and Negative Affect

Positive and negative affect can coexist. A parent can feel fear for a child and love in the same hospital room. A mourner can feel grief and gratitude. A person leaving an abusive situation can feel relief and guilt. An athlete can feel excitement and dread before a dangerous event. The coexistence is not a measurement error by default (Cacioppo & Berntson, 1994; Russell, 1980).

A single bipolar scale compresses coexistence into one rating; an explicit difference score does so by subtraction. That may be useful in some contexts, but it can erase the structure of the state. Grief with love is not neutral because love cancels grief. Relief with guilt is not zero because the signs differ. The experienced state may be high in both positive and negative dimensions, and that joint activation may matter differently from a low-arousal neutral state.

The Law-facing risk is cancellation without understanding. If mixed affect is compressed too quickly, the measured cumulative estimate may understate a burden because another dimension is present. It may also miss resilience, attachment, agency, or meaning that genuinely changes the state. The problem is not that scalarization is forbidden. The problem is that it must explain what happens to coactivation.

Mixed-affect research provides a methodological starting point. Measures can ask separately about positive and negative components, examine intraindividual patterns, use time series, and distinguish simultaneous coactivation from rapid alternation. These are construct tests that should precede any one-axis ledger model.

Coexistence also affects temporal interpretation. Positive and negative elements may have different onset, duration, and aftereffect. Love inside grief may persist while acute pain fluctuates. Relief after fear may be experienced as a separate state rather than as cancellation. A path model should preserve those differences when they are relevant.

The measurement problem is not whether a life can be stamped positive or negative at one instant. It is whether a declared representation can preserve simultaneous burdens, reliefs, arousal, control, and meaning, and whether any scalar projection retains the distinctions required by the study's estimand.

Bipolar measurement and bivariate measurement make different empirical commitments. A bipolar scale forces positive and negative states onto opposite ends of one line, while a bivariate design allows both to be present, absent, or unequal at the same time. Mixed grief, awe, relief after loss, pride under exhaustion, and fear during chosen challenge can therefore expose whether the one-axis representation is adequate. The choice should be compared through predictive and discriminant performance rather than settled by graphical convenience (Cacioppo & Berntson, 1994; Dejonckheere et al., 2022).

8.3.2 — Valence, Arousal, Control, and Meaning

Valence is only one axis commonly used to describe affect. Arousal, control, agency, uncertainty, threat, bodily demand, and meaning can vary independently enough to matter. Calm joy differs from excited joy. Quiet despair differs from terror. Love and grief can coexist without becoming a single midpoint. A one-axis model must show which distinctions it can compress and which mixed states require a bivariate or multidimensional representation (Russell, 1980; Russell & Carroll, 1999; Cacioppo & Berntson, 1994; Larsen et al., 2001).

Dimensional models are useful because they make this pressure visible. A valence-arousal plane already shows that high-energy

positive and low-energy positive states are not identical. Adding control or dominance distinguishes anger from fear, confidence from panic, and helpless pain from chosen effort. Meaning adds another layer because the same bodily state can be interpreted as sacrifice, violation, endurance, or absurdity.

The Law does not need every dimension to become a separate ledger. It needs to know whether the target of fairness can be approximated by a signed rate within a defined context. If arousal or control systematically changes the burden while the scalar ignores it, the scalar is incomplete. If those dimensions only moderate sign or variance, the model should say so.

The construct map therefore separates state dimensions from scalar output. The state vector may include valence, arousal, agency, meaning, pain, social safety, and bodily demand. A scalarization rule may then map the vector into a signed candidate contribution. The mapping is the claim, not an invisible preprocessing step.

The representation should capture the distinct clinical forms of positive and negative experience. Depression can involve low arousal, anhedonia, agitation, numbness, or self-condemnation. Anxiety can involve high arousal, anticipatory simulation, bodily vigilance, and social threat. These differences alter measurement.

The model should be parsimonious and explicit about what it discards. Valence may remain the central coordinate, yet omitted arousal, control, meaning, or bodily demand can become a source of bias rather than harmless noise. A simpler representation earns

authority by preserving the comparisons required by the estimand, not by declaring the other dimensions secondary.

Valence and arousal are often correlated, but their separation matters for interpretation. High arousal can accompany terror, exhilaration, anger, effort, or intense attention; low arousal can accompany peace, fatigue, numbness, or depression. Perceived control and meaning can alter the experiential character of states with similar valence and arousal. A measurement model that omits these dimensions may attribute channel differences to noise when they reflect genuine multidimensional structure (Russell, 1980; Russell, 2003).

8.3.3 — Intensity and Depth

Intensity, persistence, and life-structuring force are different properties. A loud feeling may be brief, while a quiet state can organize attention and action for hours; an acute insult may fade while a persistent sense of threat shapes every later cue. Representation therefore needs to distinguish magnitude from duration, salience, and temporal reach before those features are compressed (Russell, 1980; Cacioppo & Berntson, 1994).

Intensity is tempting because it looks measurable. A scale can ask how strong the pain is, how intense the fear is, how pleasant the moment is. Depth asks harder questions: does the state change the person's horizon, agency, identity, relationship to others, or ability to recover? A model that records only peak intensity may miss the burden that shapes the stream below dramatic thresholds.

The Law-facing target may need both. Some states matter because they are intense. Others matter because they are durable,

recurring, invasive, or meaning-laden. Chronic low-level dread can consume more life than one spectacular panic episode. Ordinary safety can create relief not by dramatic pleasure, but by removing the background tax of vigilance.

Depth also matters in positive experience. Awe, trust, belonging, dignity, and peace may not always score as high-intensity pleasure. Yet they can alter the structure of the stream, reduce defensive effort, open attention, and change the meaning of time. A crude pleasure scale may undervalue them.

A useful state description separates at least immediate magnitude, persistence, salience, functional disruption, recovery cost, and felt significance. A particular study may estimate only a subset, but it should state which dimensions are present, which are omitted, and whether omission changes the comparisons the estimator is expected to support.

This distinction protects the ordinary middle. The burden most likely to be hidden is often not the loudest state. It is the depth of having to carry the same small cost across thousands of moments.

Intensity measurement inherits classic psychophysical difficulties. Equal numerical differences need not represent equal experiential differences, ceiling effects can compress severe states, and verbal categories can widen or narrow across context. Repeated within-person comparisons, magnitude estimation, item-response models, and cross-modal anchors can each help, but none should be treated as a universal solution. The scaling method must match the intended claim and expose where saturation or response style makes depth unidentified (Krantz et

al., 1971; American Educational Research Association et al., 2014).

8.3.4 — Ambivalence and Incomparability

Ambivalence is not indecision alone. It can be the felt presence of reasons, attachments, fears, loyalties, and losses that do not collapse onto one preference or one affective sign. A person may want and dread the same outcome. They may love someone and fear them. They may feel gratitude for care and humiliation at needing it (Cacioppo & Berntson, 1994; Campbell & Fiske, 1959).

Incomparability appears when two states cannot be honestly ranked on a single line without losing what matters. Is numbness better or worse than panic? Is meaningful pain better or worse than empty comfort? Is anger with agency better or worse than despair with quiet? Sometimes the answer is context-specific. Sometimes the representation should remain partial.

A partial order can be more truthful than a forced scalar. It can say that one state is better on safety and worse on agency, or lower in pain but higher in humiliation. It can preserve unresolved tradeoffs rather than hiding them under a composite. This is important for a book that must not imply suffering is morally justified or erased by meaning or joy.

Ambivalence also affects report. People may describe a state as fine because no single label fits, or because naming the conflict would create social consequence. A measurement system that demands one rating can prevent participants from expressing that conflict accurately. The construct should allow people to say that

a state is mixed, unresolved, incomparable, or not representable in the offered categories.

Partial ordering preserves vector trajectories, set-valued estimates, local dominance relations, and explicitly incomparable states. These representations let the study report the structure identified in that context.

Incomparability should not be treated as evidence that a person is beyond science. It is evidence that the model must preserve more structure than a single ordered line can carry.

Incomparability need not mean that nothing can be learned. Some states may be partially ordered: one is clearly more burdensome on every admissible dimension, while two others trade agency against pain or meaning against exhaustion and resist a single ranking. Identified sets can preserve those tradeoffs without selecting a preferred weight after the outcome is known. This approach keeps the model answerable while recognizing that a one-number result may be scientifically weaker than a well-characterized region (Manski, 2003; Borsboom, 2005).

8.3.5 — Vector-State Alternatives

A vector-state representation begins by refusing premature collapse. Instead of asking for one value, it represents experience as a declared collection of dimensions at time t . Candidate coordinates may include valence, arousal, pain, agency, meaning, threat, social safety, bodily demand, and cognitive load, but the list and its interpretation must be justified for the study context rather than borrowed as a universal emotional anatomy (Russell, 1980; Russell, 2003; Cacioppo & Berntson, 1994).

Vector notation is not decorative. It changes what can be tested. A scalar asks whether one signed contribution moves over time. A vector asks whether several dimensions have distinct dynamics, couplings, lags, and failure modes. The measurement pipeline can then test whether any scalar projection preserves the features needed for a Law-facing claim.

A simple vector can also protect mixed affect. Positive and negative components can both be nonzero. Arousal can rise while valence remains uncertain. Agency can increase while pain remains high. Social safety can improve before mood follows. These patterns are lost if every state is forced into one number at entry.

The vector-state alternative does not abandon the ledger. It places the ledger downstream. A later scalarization may map the vector into a signed rate if the weights, signs, and sensitivity are justified. Until then, the vector is the evidence-preserving representation.

Researchers should also distinguish vector dimensions from observed channels. A pain axis is not the same as a pain questionnaire item. A social-safety coordinate is not the same as the presence of another person. Each dimension remains latent until linked to evidence. The vector is a construct map, not a measurement by itself.

For this chapter, $z(t)$ is therefore the honest placeholder: the state of experience before scalarization. It lets the book ask whether the Law's scalar needs can be met without pretending that experience arrived already flattened.

A vector representation permits dimensions to evolve on different clocks and respond differently to intervention. Pain intensity may change quickly while fatigue, control, meaning, or social safety changes slowly. Multivariate dynamics can therefore distinguish an intervention that improves one dimension from one that merely shifts the burden elsewhere. Scalar summaries may still be reported for bounded purposes, but the vector state should remain available for diagnostics, sensitivity, and construct-failure review (Russell, 2003; Bollen, 1989).

8.3.6 — Clinical and Cultural Variation

Clinical and cultural variation reveal a representation's transport boundary. Depression, pain, trauma, dementia, delirium, dissociation, autism, medication, sedation, disability, language, and social norms can alter report, physiology, expression, comparison, and interpretation. A scale may remain useful in one class while changing meaning elsewhere (Cacioppo & Berntson, 1994; International Test Commission, 2018).

Transport is declared by population, state, task, language, time scale, and channel. A model learned in ambulatory adults does not silently extend to childhood, terminal illness, dementia, nonhuman animals, or artificial systems; each extension needs comparability evidence, a local model, a narrower estimand, or abstention.

Sign is context-sensitive. Restlessness may reflect anxiety, pain, delirium, medication, dyspnea, or discomfort, while low expression may reflect comfort, sedation, depression, fatigue,

fear, or limited communication. Context can constrain alternatives, but no display becomes a universal sign rule.

Mixed experience creates another transport problem. Relief and grief, love and exhaustion, or low arousal and distress may coexist, while positive and negative channels change reliability independently. Multidimensional measurement should precede scalar summary so cancellation cannot manufacture neutrality. Keeping the component trajectories visible allows researchers to distinguish a stable net score from substantial changes in both kinds of experience.

Repeated sampling does not repair a weak construct. Momentary items still require response-process, distribution, within-person variability, trend, and level-specific reliability checks; frequent prompting can reproduce the same unstable relation many times (Dejonckheere et al., 2022; Siepe et al., 2025; Castro-Alvarez et al., 2026). A pilot can examine whether participants interpret each item consistently across contexts and whether the sampling schedule captures changes at the timescale relevant to the study.

A transport audit distinguishes item, channel, and construct behavior. Translation may fail while physiology remains usable, medication may alter physiology while communication remains interpretable, or the construct itself may differ. Invariance models test relations but cannot create shared phenomenal meaning (Putnick & Bornstein, 2016; American Educational Research Association et al., 2014). This separation supports targeted repairs, such as revising an item's wording or recalibrating a channel, while retaining the parts of the measurement system that still perform well.

A practical matrix crosses populations and states with items, channels, anchors, lags, and scalarization rules, recording established, partial, unsupported, or prohibited status. One successful channel cannot authorize a universal score, and one failed channel cannot disqualify a person. Each entry can also identify the next validation task, turning gaps in coverage into a prioritized plan for extending the instrument to additional settings.

Variation is information about representation, not nuisance around a preferred display rule. Comparable structure may travel with evidence; where it does not, local models or noncomparability preserve more truth than a common number with hidden bias (Henrich et al., 2005; Mokkink et al., 2024; Gagnier et al., 2025).

8.4 — Scalarization and Identified Sets

Section 6.4 established why scalarization and identified sets belong to PTVC's representation gate. Here the task is psychometric: specify, estimate, validate, and stress-test the projection from multidimensional lived states to a study-specific scalar before that scalar enters a ledger (Fredrickson & Kahneman, 1993; Eldar et al., 2016; Manski, 2003).

A scalar is a declared projection, not a metaphysical primitive. Let the state remain vector-valued until the study states which dimensions enter, which weights or interactions map them to a sign, and which families of maps remain defensible. If the Law-facing conclusion changes across those families, the result is representation sensitivity or an identified set rather than a stronger scalar claim.

Scalarization becomes informative when the analysis reports what is identified, which conclusions depend on the weighting rule, and what evidence would resolve the remaining uncertainty.

Outcome-selected weights are circular. A projection chosen because it makes a trajectory close is not a bridge from experience to measurement; it is a rescue operation conducted with the answer in view. Weight families, exclusions, and tuning rules must be justified and frozen before terminal outcomes are inspected, with any exploratory search reported separately (Manski, 2003; American Educational Research Association et al., 2014; Nosek et al., 2018).

$$z(t) = (v(t), a(t), p(t), c(t), m(t), r(t), s(t), b(t))$$

$$x_w(t) = w^\top z(t), L_w(a, b) = \int_a^b g_\theta(t) x_w(t) dt$$

$$I_W(a, b) = \{L_w(a, b) : w \in W\}$$

8.4.1 — Weight Vector w

A weight vector is the visible form of a tradeoff. If lived experience is represented as a vector of dimensions, a scalar ledger requires a mapping from that vector to one signed quantity. The weights say how much each dimension contributes and in what direction. They are not merely technical coefficients; they are where the model decides what it counts (Manski, 2003; Krantz et al., 1971).

The simplest candidate can be written conceptually as a vector state $z(t)$ mapped by weights w into a scalar contribution. The equation matters less as calculation than as disclosure: it shows that the one-number result is produced by a declared operation, not discovered as though experience had only one natural axis.

Weights can be fixed by theory, estimated from data, elicited from participants, constrained by expert judgment, or compared across plausible families. Each route has different risks. Participant weights may reflect current mood or social pressure. Expert weights may encode professional bias. Data-derived weights may optimize prediction of an outcome that is not lived experience. Theory-derived weights may be elegant but wrong.

Weighting determines which experiences remain visible. Down-weighting pain when meaning is high, ignoring agency, or treating social safety as universally beneficial can transfer the consequences of the modeling choice to the person whose

experience is misrepresented. Scalarization is therefore both a scientific and an ethical commitment: it specifies which dimensions may compensate for which others, and by how much (Manski, 2003).

A rigorous chapter therefore requires weight provenance. Every scalar candidate should state the vector, the sign convention, the weight source, the population, the context of use, the sensitivity range, and the forbidden inference. Without that, a one-number ledger is not a scientific simplification. It is an unmarked judgment.

The weight vector is not the enemy of measurement. It is the honesty that allows measurement to proceed. Once weights are visible, they can be argued with, tested, varied, rejected, or narrowed.

A weight vector is part of the estimand, not merely a computational setting. Changing the weights changes which features of experience count, how mixed states are ordered, and which trajectories appear close to neutrality. The protocol should therefore state whether weights are person-specific, population-level, context-specific, learned from independent data, or fixed by a declared normative convention. A weight rule selected because it produces terminal closure would make the result circular (Manski, 2003; American Educational Research Association et al., 2014).

8.4.2 — Weight Uncertainty

A single weight vector is rarely justified at the start. Several weightings may be plausible given current evidence. Pain may

deserve more or less weight relative to agency. Meaning may moderate pain rather than add to it. Arousal may matter only under some valence states. Social safety may interact with threat. Weight uncertainty should be part of the inference rather than hidden in a selected model (Borsboom, 2005; Bollen, 1989).

The right object may be a family W of admissible weights. W can be defined by theory, validation evidence, participant interpretation, ethical constraints, and empirical sensitivity. A result that holds across W is more robust than a result that depends on one narrow vector. A result that changes sign across W tells the reader that the scalar claim is representation-sensitive.

Weight uncertainty should not be confused with statistical noise. Noise concerns data variation around a declared model. Weight uncertainty concerns which model is being used to translate experience into a scalar. More data under the wrong weight vector can make a wrong estimate precise. The uncertainty about representation must travel separately.

A scientific report should therefore display several outputs: the central candidate, the range across plausible weights, the dimensions driving disagreement, and the conditions under which the conclusion changes. If all plausible weights agree that burden increased, a directional claim may be strong even without a unique scalar. If weights disagree, the honest claim may be an identified set.

Weight uncertainty also protects against moral overreach. The project must not decide after the fact that meaning compensated pain or that fear was outweighed by later relief because that makes

closure appear. The admissible family must be declared before the result is evaluated.

Weight uncertainty should travel into every downstream quantity. Sampling variability is only one component; uncertainty may also arise from task framing, anchor instability, response shift, cultural interpretation, and model choice. Robust or Bayesian analyses can represent a distribution or admissible set of weights, but the output remains conditional on that representation. Narrow posterior intervals do not rescue a misspecified scalarization, and computational precision should never be mistaken for construct identification (Borsboom, 2005; Manski, 2003).

8.4.3 — Sensitivity Across W

Sensitivity across W asks whether a conclusion survives the scalarizations that remain defensible under the declared construct map, population, and use. The task is not to demand agreement across every mathematically possible weight. It is to prospectively specify a scientifically admissible family before outcome inspection, show how status changes within that family, and report the identified set when no single projection is warranted (Krantz et al., 1971; Manski, 2003; American Educational Research Association et al., 2014).

A terminal or longitudinal conclusion might appear under a pain-heavy scalar and disappear under an agency-heavy scalar. A recovery claim might hold when arousal is treated as neutral and fail when agitation is treated as distress. A neutral band might contain one projection and not another. These are not nuisances to average away; they are answers about representation.

The sensitivity report should identify which dimensions control the disagreement. If the conclusion is fragile because meaning and pain are weighted differently, the chapter has found a scientific and ethical pressure point. Future work can study that relation instead of pretending the scalar was settled.

Sensitivity across W can also separate robust partial claims from overbroad claims. If every defensible weight shows decreasing burden during a care intervention, the intervention may have strong experiential evidence. If the neutrality result varies across weights, the report identifies which weighting assumptions determine the conclusion.

A practical report should include graphical or tabular displays in research artifacts, while the surrounding explanation makes the logic visible in prose. The scalar result must be accompanied by its sensitivity class: robust, directionally robust, sign-fragile, magnitude-fragile, or unidentified. These labels prevent a public artifact from displaying one curve as though it were the life.

The Law becomes testable only when it can lose under plausible representations. Sensitivity across W is one of the mechanisms by which representation becomes a real gate rather than a decorative caution.

Sensitivity across admissible weights is most informative when it is structural rather than cosmetic. Researchers should identify which conclusions hold throughout the admissible set, which depend on a narrow corner, and which reverse sign across plausible choices. A robustness frontier can show how much assumption is required before a one-number claim appears. When

the central conclusion depends on particular weighting commitments, the report identifies those commitments and the measurements that could distinguish them (Manski, 2003; Imbens & Manski, 2004).

8.4.4 — Identified-Set Reporting

An identified set contains every target value compatible with the observed data, the admissible measurement models, and the declared assumptions. It is not a confidence interval. A confidence procedure addresses sampling uncertainty for a target under a specified model, while partial identification addresses the possibility that the model and evidence do not select one target value at all. A rigorous analysis may construct a confidence region for the target parameter or for the entire identified set, but it must state which coverage guarantee is intended and preserve both layers (Manski, 2003; Imbens & Manski, 2004).

Set reporting is especially valuable because the Law's public risk is drawn toward clean numbers. A point estimate invites rank, comparison, and certainty. A set shows the reader where knowledge stops. It also allows the project to say something without saying too much. The result can support bounded direction, reject some claims, or abstain from exact scalar language.

Identified sets also make construct failure visible. If the set spans incompatible statuses, the issue may not be sampling error but representation uncertainty. If a terminal valence closure band is reached by only one corner of the weight family, the result is not

robust neutrality. If all defensible weights reject a proposed pattern, that is stronger adverse evidence.

A wide set can narrow the program by showing that a strong claim is not presently identifiable. That is a substantive result, not a protective maneuver. An honest region is preferable to a narrow number produced by hidden assumptions because it exposes exactly what new anchors, bridge evidence, or representation commitments would be required for a sharper conclusion.

An identified set can be used at multiple levels: within-person trajectory, group average, terminal-window increment, intervention contrast, or path-functional comparison. Each use must state which uncertainties are included. A set over weights is not the same as a confidence interval over sampling noise, and the two should not be collapsed.

Set-valued reporting should be designed for interpretation rather than treated as an apology for imprecision. The report can identify the range of terminal residuals, rankings, or path summaries compatible with all admissible weights and measurement models, then show which scientific conclusions remain invariant across that range. A wide set may still reject a strong claim, and a narrow set may still exclude zero. The governing discipline is to preserve uncertainty in the output instead of hiding it behind a point chosen for presentation (Manski, 2003; Imbens & Manski, 2004).

8.4.5 — No Outcome-Selected Weights

Outcome-selected weights are weights chosen because they produce the desired result. In the context of this book, the danger is obvious: a researcher could adjust the scalarization until the

terminal path looks neutral, the intervention looks beneficial, or the rival looks weaker. That would convert measurement into rescue (Manski, 2003).

The prohibition must be explicit because outcome selection can hide inside reasonable language. A team may say that it refined the model after inspecting anomalies, optimized weights for fit, removed a dimension that behaved inconsistently, or chose the psychologically plausible transformation that happened to improve closure. Some refinements are legitimate in development. They are not confirmatory evidence unless locked and retested.

Exploration and confirmation must remain distinct. Exploratory work may compare weight families, discover dimensions, revise items, and inspect failure. Confirmatory work freezes scalarization, population, endpoint, temporal rule, and analysis plan before testing the claim. Changing weights after results are visible creates a new claim version that requires a new test (Munafò et al., 2017; Nosek et al., 2018).

This rule also protects participants. A weight selected to make a result look fair may erase a burden the participant actually carried. If fear is removed as noise, if pain is offset by meaning, or if dignity violation is down-weighted because it disrupts a curve, the model has shifted burden onto the person for the sake of the theory.

Outcome-selected weights are not always malicious. They can arise from pressure to make complex data coherent. But the effect

is the same: the measurement bridge becomes immune to defeat. A bridge that cannot fail cannot bear evidence.

No outcome-selected weights is therefore a scientific and ethical firewall in one sentence. The scalarization must face the result from outside the result. If it changes, the book must say that the claim changed too.

8.4.6 — Withdrawal of One-Number Claims

A one-number claim should be withdrawn when scalarization is unstable, admissible weights cross incompatible statuses, dimensions remain only partially ordered, cultural or clinical variation changes meaning, or temporal functionals disagree beyond the study's ability to resolve them. Withdrawal is not rhetorical defeat; it is the correct status for a construct that has not earned a scalar (Manski, 2003; Borsboom, 2005).

Withdrawal should identify the highest representation tier that survives. A failed scalarization may still permit sign, partial order, within-person direction, a vector profile, or bounded change; it does not automatically invalidate every measurement. The report should state which coordinate claim failed and which lower-dimensional or set-valued object remains scientifically usable.

The decision rule belongs in the preregistration. Separate rows should address sign instability, weight sensitivity, common-origin failure, noncomparability, transport failure, and disagreement among temporal functionals; each row states the evidence threshold, the language that becomes unavailable, and the narrower output that remains permitted. This makes withdrawal

reproducible rather than a rescue invented after an attractive result fails (Manski, 2003; Imbens & Manski, 2004).

Restoration requires genuinely new evidence. The program may not choose a favorable weight family, remove difficult cases, redefine the endpoint, or change the path functional until a clean number returns. A redesigned construct or protocol must be requalified and tested on data that did not generate the repair, preserving the boundary between development and confirmation (American Educational Research Association et al., 2014; Nosek et al., 2018). Restoration should also identify the specific failure being retested and the prospective criterion that would count as recovery, so new data alone cannot restore a claim whose original defect remains unresolved.

Withdrawal also propagates beyond the analysis. Figures, summaries, public excerpts, model cards, and dataset descriptions must be corrected, versioned, or retired when their scalar premise is no longer supported. A clean current result does not close the error if an older one-number image continues circulating without its withdrawal notice (Munafò et al., 2017; Center for Open Science, 2025). The correction record should identify which active public surfaces were checked and which copies could not be updated, making residual exposure to the superseded claim visible rather than assumed away.

Reports should distinguish cardinality loss, comparability loss, transport loss, and aggregation loss. These failures have different remedies and different public consequences; calling all of them ‘measurement failure’ discards information, while silently switching from a shared unit to rank order or from a lifetime claim

to a local association disguises a changed target. Each failure should be paired with the strongest relation that still survives, such as within-person direction, ordering, a vector profile, or a set-valued result, so withdrawal preserves usable information instead of collapsing every limitation into the same verdict.

When a scalar claim withdraws, redesign should return to the path rather than search immediately for another summary. The ordinary-day schedule below illustrates how a study can preserve waking transitions, obligations, waiting, social contact, bodily events, relief, evening recovery, and sleep-related carryover so that later model comparison is not built from peaks alone. Preserving that temporal structure allows competing path functionals to face the same record instead of letting data collection quietly favor whichever summary replaces the withdrawn scalar.

8.5 — Duration, Memory, and Path Functionals

A life is extended in time, but time is not a neutral container once experience enters it. Ten minutes of terror, ten minutes of boredom, and ten minutes of quiet safety have the same clock length. Their felt contributions may not have the same structure. A simple integral treats equal time intervals as equally eligible for accumulation. That assumption needs rivals (Eldar et al., 2016; Fredrickson & Kahneman, 1993; Shiffman et al., 2008).

Psychology has long shown that remembered episodes can differ from experienced trajectories. Peaks, endings, recency, recovery, anticipation, and narrative coherence can influence later evaluation. The Law's target is the experienced stream, but the evidence often arrives through memory, report, expectation, and retrospective meaning. A chapter that ignores that gap would let the measurement source redefine the target.

The ordinary middle is the central correction. Dramatic peaks are easier to remember and easier to write about. Fairness, however, is often carried in unmemorable duration: the commute, the waiting room, the background fear, the long illness, the quiet apartment, the repetitive care task, the night of not quite sleeping. If duration disappears behind peaks, hidden burden disappears with it.

Competing temporal functionals should be compared under common data and scoring rules. Simple addition, kernel weighting, peak-end summaries, hysteresis, recovery penalties, and order-sensitive models make different claims about what history retains. The appropriate functional is the one that survives

prospective prediction, perturbation, and rival comparison in the declared context, not the one that best matches an intuitive story after the outcome is known.

For exploratory ordinary-day work, a coverage matrix can include waking transition, bodily baseline, first social contact, work or care load, waiting, discomfort, relief, meals, movement, solitude, evening recovery, sleep onset, and night interruption. The list is not a surveillance mandate. It is a check that the design samples low-salience duration rather than only dramatic events.

Table 8.2. Ordinary-day sampling windows and omitted-window bias.

Ordinary-day window	Scientific reason to sample it	Bias if ignored
Waking transition	Captures baseline mood, bodily demand, sleep residue, and anticipatory load.	The day begins after the first burden has already been carried.
First obligation	Shows how work, care, school, treatment, or dependency enters felt time.	Visible productivity replaces hidden cost.
Waiting interval	Measures low-drama burden, uncertainty, pain, boredom, or constrained choice.	Only peak events appear to matter.
Social contact	Distinguishes support, threat, loneliness, obligation, and masking.	Relational burden is mistaken for external circumstance only.

Ordinary-day window	Scientific reason to sample it	Bias if ignored
Bodily event	Tracks pain, nausea, breathlessness, fatigue, hunger, or medication effect as lived condition.	The body becomes a confound rather than part of the target.
Relief or repair	Tests recovery dynamics, contrast effects, gratitude, exhaustion, and partial repair.	The model sees harm but not release, or sees release as total compensation.
Evening recovery	Captures accumulation, rumination, meaning, memory, and return of agency.	The path is reduced to isolated moments.
Sleep onset or interruption	Links affective residue to rest, dreams, and next-day carryover without making sleep a proof source.	Night becomes invisible even when it carries the day forward.

8.5.1 — The Ordinary Middle

The ordinary middle is the part of life most likely to be erased by dramatic measurement. Peaks are memorable. Endings are narratively powerful. Crises attract attention. But a life is also made of waiting, commuting, washing dishes, monitoring bills, checking on someone, being too tired to answer a message, walking in a safe street, sitting without fear, and repeating care (Kahneman et al., 1993; Fredrickson & Kahneman, 1993).

Ordinary duration matters because hidden burden is often distributed thinly across many intervals. A small daily vigilance cost can become large over years, while a quiet daily relief can alter whether life feels merely survivable or genuinely open. A design that samples only dramatic events can miss the cumulative structure the ledger is meant to represent.

The ordinary middle also tests dignity. Some conditions are not spectacularly painful but are persistently narrowing: lack of privacy, unreliable transport, low-grade disrespect, sensory overload, chronic uncertainty, bureaucratic delay, or the constant need to explain one's body. These may not produce high peaks, but they occupy time and attention.

Experience sampling and ecological momentary methods are useful here because they can catch life closer to its occurrence. They still have burdens and biases, but they resist the habit of asking memory to summarize the stream after the most narratively salient moments have taken over.

A fairness science that ignores the ordinary middle becomes endpoint obsessed. It would study how lives end while missing what most of the life was like to carry. This chapter keeps ordinary duration in the target before the book turns toward terminal signatures.

The ordinary middle is also where sampling burden and scientific value meet. Brief random prompts, end-of-day reconstruction, passive context, and time-use records each reveal different portions of extended experience. A design that samples only when emotion is intense will overrepresent peaks; a design that samples

mechanically may burden the very routines it seeks to observe. The schedule should be justified by the temporal process and tested for reactivity, missingness, and coverage of low-salience intervals (Stone et al., 2007; Shiffman et al., 2008).

8.5.2 — Duration Weighting

Simple integration treats each unit of clock time as equally eligible for contribution. That is a clear starting point, but it may not be the only plausible rule. Duration can be weighted by attention, consciousness level, salience, suffering intensity, effort, recovery cost, or availability of agency. The question is not whether time matters, but how (Shiffman et al., 2008; Stone et al., 2007).

Equal clock weighting has virtues. It is transparent, preregistrable, and hard to manipulate after the fact. It gives ordinary life a place because long low-intensity states count. It also matches the intuition that a burden carried for hours matters more than the same burden carried for seconds. But even this simple rule has assumptions.

Saturation, habituation, sensitization, and exhaustion can make the tenth hour experientially different from the first. An additive integral can represent those differences through a changing rate; equal clock weighting does not require constant experience. A separate duration-weighting rule needs its own justification and must not count intensity or aftereffects twice.

The analysis should distinguish clock duration, conscious-access duration, attention-weighted duration, burden-weighted duration, and recovery-inclusive duration. Equal clock time is a transparent

initial model, not a protected conclusion. The design must permit prospective challenges from attention, consciousness level, saturation, sensitization, and recovery; if equal weighting fails, the response is a better path functional rather than deletion of duration from the target.

The temporal question has an empirical foundation. Cui et al. (2023) synthesized 95 effect sizes from 31 studies involving 3,776 participants and found that valence, arousal, stimulus type, and timing task moderated emotional duration judgments. Negative rather than positive valence, and greater arousal, generally accompanied greater overestimation. This gives the program measurable alternatives to compare: physical duration, reported duration, and any independently qualified phenomenal-exposure model. It makes the choice of clock an explicit research question, with its rule fixed before the cumulative result is examined.

Duration weighting also affects public interpretation. A short intense peak may draw sympathy while a long dull burden is ignored. Measurement can correct that bias only if duration is represented rather than treated as background. The Law's burden lens makes invisible time scientifically visible.

8.5.3 — Peak-End and Memory Effects

Remembered experience is a reconstruction, not a transparent archive of momentary experience. Peaks, endings, narrative coherence, later meaning, and current state can change retrospective evaluation. That makes memory scientifically important and also prevents it from being substituted for the path

that was lived without an explicit reconstruction model (Fredrickson & Kahneman, 1993; Kahneman et al., 1993; Stone et al., 2007).

Peak-end effects are not merely cognitive curiosities. They can make a temporally longer and more painful episode remembered more favorably if the ending improves. They can also make a mostly tolerable experience remembered harshly because it ended badly. A ledger that substitutes remembered evaluation for momentary trajectory would become a different construct.

Memory can still be evidence. Retrospective evaluation matters to future choice, identity, narrative meaning, and suffering that reappears as recollection. Memory should not be discarded. Remembered utility, experienced utility, predicted utility, and decision utility must remain separate so that each source speaks in its own status.

The Law-facing risk is endpoint contamination. If terminal calm changes memory of a painful period, a retrospective report near death may appear to support neutrality while the experienced trajectory remained non-neutral. Conversely, a frightening ending may make a life period remembered as worse than its momentary distribution. Both directions matter.

Study designs can compare momentary sampling with retrospective ratings, episode reconstructions, narrative interviews, and physiological traces. Divergence is not an error to average away. It indicates that time-local experience and remembered evaluation are different objects.

The chapter keeps this distinction early so later terminal research cannot treat last reports as if they fully summarize the life. Memory is part of lived experience, but it is not the whole path.

Peak-end effects are not a universal law of memory, and one simplification should not replace another. Their magnitude varies with episode structure, delay, attention, meaning, and the question asked. The scientific use of the phenomenon is comparative: experienced trajectory and remembered evaluation are different targets whose divergence can be measured, modeled, and sometimes predicted. That divergence becomes a rival to any attempt to infer cumulative experience from retrospective summary alone (Kahneman et al., 1993; Fredrickson & Kahneman, 1993).

8.5.4 — Recency and Kernel Models

A kernel model assigns different weights to moments depending on their location in time or relation to an event. Recency kernels give more weight to recent states. Anticipatory kernels weight future expectation. Recovery kernels carry the cost of a state beyond its immediate interval. Such models may fit some psychological phenomena better than a flat integral (Eldar et al., 2016; Russell, 1980).

The timing of earlier events can improve predictions of present mood. Keren and colleagues (2021) compared a primacy model with several recency models across random, structured, and adaptive reward environments, including adult and adolescent samples and participants with depression. The primacy model, whose expectations incorporate earlier outcomes, predicted

successive mood ratings more accurately in the tested comparisons. Its expectation-related parameters also corresponded to activity in frontal regions involved in mood regulation. This supplies a concrete benchmark for cumulative research: compare flat, recency-weighted, primacy-weighted, and latent-state histories on the same future observations.

Kernel models are rivals to unweighted temporal accumulation. They can explain why recent fear colors a day, why recovery after an insult matters, why anticipatory dread can dominate before an event, and why a painful episode can have aftereffects. They also create risk because a flexible kernel can be tuned after the fact to produce the desired pattern.

Kernel status labels are therefore required. A kernel may be theoretical, empirically estimated, exploratory, or confirmatory. It must declare whether it weights attention, memory, physiological recovery, narrative evaluation, or momentary contribution. A kernel chosen because it improves terminal valence closure is no better than outcome-selected weights.

Recency models are especially useful for separating near-end reports from longer historical contributions. A terminal-window study can compare local dynamics under explicit memory kernels, with the kernel's form and coverage stated in the analysis.

Kernel models can represent how a past threat changes later experience. A flat integral of an adequately specified, history-dependent rate already includes those later effects. A separate recovery kernel is useful when it improves that representation, but

it must not count the same experienced aftereffect once in the rate and again as an added historical cost.

Kernels should be treated as competing scientific hypotheses. Simple integration, recency, recovery, anticipation, and memory-weighted forms should be frozen, compared prospectively, and allowed to fail. A kernel that predicts held-out experience within a declared construct earns bounded local use, not universal authority.

Recency kernels are themselves estimands that require identification. A rapidly decaying kernel, a long-memory kernel, and a reconstruction rule based on salient transitions can produce similar retrospective summaries under limited sampling. Prospective experience reports, controlled delays, and held-out prediction of later recall can help separate them. If several kernels remain observationally equivalent, the analysis should retain the family instead of selecting the one that produces the cleanest cumulative path (Kahneman et al., 1993; Särkkä, 2013).

8.5.5 — Order, Recovery, and Hysteresis

Two trajectories can contain the same set of momentary states and still feel different because order matters. Pain followed by relief differs from relief followed by pain. Fear followed by safety differs from safety followed by fear. A humiliation after a long recovery can reopen burden differently from the same event before recovery began.

Hysteresis names the dependence of current state on path history. The same present condition may mean different things depending on what came before. A quiet room may feel restful after overload

and lonely after abandonment. A medical monitor may feel reassuring after crisis and oppressive after repeated alarms. The state cannot always be interpreted without sequence.

Recovery also has its own burden. A harmful event may end while the person continues to pay in sleep disruption, vigilance, avoidance, pain flare, relationship strain, or cognitive effort. If the model counts only the event interval, it may undercount what the stream carries. Recovery time belongs to the path if it changes experience.

Order effects challenge simple ledger addition. If the total contribution of A followed by B differs from B followed by A, then the path functional is not merely the sum of unordered moments. The model may still use integration with state variables that include history, but it must declare that history is part of the state.

A study can test order by comparing episodes, interventions, or natural sequences that share similar components but differ in arrangement. It can examine whether recovery slopes, anticipation, or re-exposure alter later contribution. The result may support a path-dependent model without eliminating the possibility of quantitative study.

Sequence can carry burden even when an unordered total cannot. The same events can have different consequences when relief follows pain, threat follows safety, or recovery remains unavailable. Temporal order, opportunity for repair, and carryover therefore belong in the candidate functional whenever the evidence shows that permutation changes the experienced path.

Hysteresis indicates that present experience may depend on the route by which the present state was reached. Equal current pain ratings can carry different meanings after escalation, relief, failed treatment, or prolonged uncertainty. When history matters, the state representation must be enlarged to include the relevant memory, adaptation, or control variables; otherwise a non-Markov process will be forced into a misleading memoryless model. Closed loops and recovery trajectories are therefore diagnostic tools for discovering missing state rather than evidence of a preferred mechanism (Särkkä, 2013; Dejonckheere et al., 2022).

8.5.6 — Path Functional Model Comparison

A path functional maps a trajectory into a summary, but different functionals answer different questions. The additive ledger weights eligible duration; peak-end, recency, recovery, hysteresis, vector, and set-valued forms preserve other structures. Fit alone does not make any one of them the target.

Comparison begins from common observations, inclusion rules, and a prospectively fixed functional set. Each candidate declares its invariances, cadence, missing-interval rule, endpoint sensitivity, and held-out predictions. Allowing every model to choose a favorable representation would confound fit with target selection (Kahneman et al., 1993; Fredrickson & Kahneman, 1993).

Structural contrasts are often decisive: reorder the same states, extend a low-intensity interval, alter the final minutes, or repeat

recovery episodes to separate additive, hysteretic, duration-weighted, peak-end, and return-dynamic accounts.

Held-out prediction is matched to use. A functional may predict remembered evaluation, behavior, treatment preference, or later reports without measuring experienced cumulative contribution. Predictive success therefore supports its declared target, not automatically the ledger (Stone et al., 2007; Shiffman et al., 2008).

Identification depends on observation. Sparse prompts may leave order and duration unresolved, reconstruction may preserve narrative while missing brief episodes, and passive channels may improve timing while leaving sign uncertain. The honest output may be an observationally equivalent model set.

Outcome-selected functionals are prohibited. Development may explore kernels, weights, and scalarizations, but confirmation freezes the family, reports sensitivity, and keeps failure language valid under unfavorable ordering (Nosek et al., 2018; Munafò et al., 2017).

First-person, physiological, behavioral, clinical, relational, and intervention evidence constrain different parts of the path model. Their combination can reveal connections that a single channel misses.

8.6 — Evidence Baseline and Construct Failure

After representation and time come evidence. No channel is pure. First-person report is privileged because the subject lives the moment, yet report can be shaped by memory, language, demand, fear, fatigue, culture, and access. Physiology and behavior can reveal change that a person cannot report, but they can also track arousal, illness, medication, effort, or context rather than felt valence. The evidence baseline must therefore compare channels by role and failure mode rather than by prestige (Shiffman et al., 2008; Stone et al., 2007; Khalsa et al., 2018; Campbell & Fiske, 1959).

The evidence baseline combines source families according to their roles. Affective science, pain science, ecological momentary assessment, behavioral observation, clinical context, relational testimony, and intervention studies each contribute information about the construct. The matrix records those contributions, the checks needed to interpret them, and the conditions requiring revision.

Construct validity is the discipline that keeps this section from becoming a list of interesting findings. Convergent evidence should relate to the intended experience. Discriminant evidence should show that the construct is not merely arousal, illness severity, compliance, speech rate, social desirability, or device tolerance. Known-groups and intervention tests can help only when the direction of change is ethically and theoretically justified beforehand.

Failure is allowed. If the proposed scalar target fragments into several dimensions, if instruments mostly track arousal, if reports and physiology disagree in systematic ways that cannot be resolved, or if interventions change behavior without changing experience, the construct must be redesigned or narrowed. That is not a problem for science. It is what science is for.

Table 8.3. Evidence families, orienting value, and construct-failure routes.

Evidence family	What it can orient	Interpretation requirement	Typical failure route
First-person report	Access to felt quality, meaning, burden, and context.	Independent tests of units, sign stability, and temporal coverage.	Memory, demand, language, fear, fatigue, and response shift.
Physiology	Timing, arousal, interoceptive load, stress, pain correlates, and treatment state.	A calibrated relation to the target experience under the relevant bodily state.	Medication, illness, disability, sensor gain, and nonspecific arousal.
Behavior	Approach, withdrawal, speech, activity, sleep, social contact, and care use.	Contextual evidence about motivation, constraints, and the experience of the act.	Masking, coercion, habit, culture, access barriers, and survival strategy.

Evidence family	What it can orient	Interpretation requirement	Typical failure route
Clinical context	Pain, dyspnea, nausea, delirium, fatigue, diagnosis, and care burden.	Integration with participant reports and other available evidence.	Overmedicalization, underreporting, protocol burden, and role pressure.
Relational context	Familiar patterns, social safety, loneliness, dependence, and witnessed change.	Participant consent, source provenance, and consideration of alternative interpretations.	Projection, conflict, loyalty, grief, and privacy invasion.
Intervention response	Whether changes in pain, safety, rest, agency, or support alter reported and observed experience.	A construct specified before analysis and checks on the intervention's active mechanisms.	Multiple active mechanisms, placebo/context effects, and unmeasured mediation.

8.6.1 — First-Person Report

First-person report is privileged because the person lives the experience. No external channel has equal access to what it is like from within. If a study of lived experience systematically ignores testimony, it has already drifted away from its target. The subject is not one sensor among many in the same way a device is a sensor (Cronbach & Meehl, 1955; Borsboom, 2005).

Privilege is not sovereignty. Reports can be shaped by memory, language, fear, shame, expectation, demand characteristics,

fatigue, medication, clinical state, culture, and lack of concepts. A person may not know how to describe numbness, dissociation, moral injury, mixed grief, or chronic low-level dread. They may also report strategically because consequences attach to honesty.

The correct posture is respectful modeling. Reports should be collected in ways that minimize burden, protect privacy, allow uncertainty, invite context, and separate immediate experience from memory and evaluation. The model should know whether it is using momentary report, episode reconstruction, narrative interview, symptom scale, or life evaluation.

First-person report can also contest the model. If a participant says the labels are wrong, the timing missed the event, a device changed the feeling, or a question forced a false choice, that information belongs in data quality. Contestability is not an administrative courtesy. It is part of construct validity.

Silence also deserves respect. Refusal to answer, inability to answer, or choosing not to disclose should not be treated automatically as low burden or missing at random. Nonresponse can itself reflect burden, privacy, distrust, exhaustion, or dignity. The study must decide when to abstain rather than infer.

First-person report gives the baseline its moral center and its scientific vulnerability. It brings the target closest to view while reminding the project that even the closest source requires interpretation.

First-person report is privileged by target relevance, not infallibility. Reports can be affected by language, demand, memory, privacy, fear, fatigue, alexithymia, dissociation, and the

available response scale. Good design therefore varies prompt wording, protects refusal, compares concurrent and retrospective reports, and studies when disagreement is systematic. Treating reports as defeasible evidence respects the person more than either declaring them transparent truth or dismissing them whenever they complicate the model (Stone et al., 2007; Shiffman et al., 2008).

8.6.2 — Physiology and Behavior

Physiology and behavior are fallible correlates, not replacements for the subject. Heart rate, variability, skin conductance, sleep, respiration, facial movement, posture, speech, activity, medication use, and clinical observation can all carry information. They can also track exertion, illness, temperature, pain medication, device fit, disability, social masking, or environmental demand (Campbell & Fiske, 1959; Bollen, 1989).

The danger is physiological overconfidence. Arousal is not valence. Stillness is not peace. A smile is not joy. Low speech is not absence of experience. Agitation is not always distress in the same way. The chapter requires every external channel to state the construct it is meant to inform and the rival interpretations it cannot eliminate alone.

Behavior can be especially misleading under constraint. Compliance may reflect fear. Productivity may reflect economic pressure. Withdrawal may reflect safety seeking rather than depression. Politeness may reflect dependency. A fairness science should ask what burden a behavior hides before it treats the behavior as evidence of inner state.

Physiology remains important because first-person report can be unavailable, delayed, socially constrained, or altered by memory. External channels can detect changes, support timing, reveal arousal, help model sleep and pain, and identify conflicts that merit human review. They are most useful when they are combined with context and not granted authority alone.

A baseline matrix should therefore list for each channel: intended construct, raw unit, plausible interpretation, major confounds, expected lag, participant burden, missingness mechanism, and failure condition. If a channel cannot survive that specification, it should not contribute to a Law-facing estimate.

Physiology and behavior serve as witnesses rather than rulers. They can constrain, challenge, or contextualize a latent account, but they do not become the stream they help study.

Neuroscience provides a concrete reason to investigate shared structure across different feelings. Chikazoe and colleagues (2014) found orbitofrontal population patterns that tracked affective valence across visual and taste experiences. Later work separated shared from modality-specific patterns across several aversive experiences (Čeko et al., 2022). Such results make a common affective dimension a productive empirical target while preserving the distinctive qualities of each modality.

Pleasure itself is becoming a more specific measurement target. Kragel and colleagues (2023) developed a distributed neural signature using 28 fMRI studies and evaluated it in independent datasets. A pooled analysis of four independent validation studies found larger signature differences for hedonic contrasts than for

monetary contrasts. In a separate 19-person placebo-controlled crossover study, naloxone reduced the signature's response to erotic images. These results provide a candidate for distinguishing pleasure-related responses from incentive-value responses in the tested settings. Combining such a measure with independently validated unpleasantness channels could test whether cross-sign comparison is stable within a person.

Positive experiences now have a similarly concrete research opening. In 52 adults, Savalia and colleagues (2026) predicted responses to personalized music, videos, positive memories and monetary rewards from distributed neural activity and connectivity. Combining the two feature families improved held-out prediction. Their measure asked how much participants wanted an experience again, making separation of wanting from felt pleasure a valuable next validation step. A PTVC-oriented measurement study can retain this positive range while testing direct pleasantness and unpleasantness independently.

Valence assignment is also experimentally tractable at the circuit level. In mice, Li et al. (2022) identified a neurotensin pathway from the paraventricular thalamus to the basolateral amygdala that biased learning about rewarding and punishing outcomes. Activating this pathway promoted reward learning; removing neurotensin from the pathway strengthened punishment learning. Related work identified partially segregated ventral hippocampal cell populations associated with appetitive and aversive memory engrams, together with distinct molecular and projection features (Shpokayte et al., 2022). These findings give the project concrete examples of biological systems that distinguish valence-related

learning and memory without establishing PTVC. A declared route can be tested against independent appetitive and aversive learning models by asking whether its specified coupling predicts previously unseen changes in report and choice under ordinary, nonharmful variation. This makes the proposed coupling available for direct comparison.

8.6.3 — Clinical and Relational Context

Clinical context can make evidence interpretable. Diagnosis, medication, pain history, oxygenation, delirium risk, disability, sleep disruption, infection, hydration, and treatment burden can explain why reports or signals move. Without those records, a physiological pattern may be mistaken for affective change when it is actually illness, treatment, altered access, or sensor failure. Context is therefore part of the observation model, not a descriptive appendix (Shiffman et al., 2008; Khalsa et al., 2018; Giacino et al., 2018).

Relational context also matters. Close others may know patterns of expression, withdrawal, fear, relief, and comfort that a one-time researcher would miss. Caregivers may notice when a facial expression means pain rather than annoyance, or when quiet means peace rather than shutdown. But close others can also project, pressure, misremember, minimize, exaggerate, or carry their own burden into interpretation.

Clinical and relational evidence can inform the target without owning it. Clinicians are not mind readers, family members are not automatic translators, and researchers are not neutral by position alone. Each witness has proximity, expertise, bias,

missingness, and a context of use. Convergence among witnesses is evidence only when their errors are not treated as interchangeable (Cronbach & Meehl, 1955; Campbell & Fiske, 1959).

Clinical context is especially important in palliative and terminal research, but this chapter keeps it general. Medication, fatigue, social dependence, fear of being a burden, and care quality can shape experience across many populations. A baseline that ignores these pathways will confuse construct with context.

Relational evidence also raises privacy and bystander issues. A person's experience should not become available to others simply because those others provide context. Consent, purpose limits, and data minimization remain necessary even when close observers help interpret evidence.

A defensible evidence baseline is triangulated but not totalizing. Clinical and relational context can improve interpretation while preserving the participant's authority to contest, refuse, or remain partly unobserved.

8.6.4 — Convergent and Discriminant Validity

Validity evidence must match the measurement model. In a reflective model, indicators should behave as consequences of the proposed latent state under the declared conditions. In a causal or composite model, components may jointly define the score without being interchangeable or highly correlated. In a network model, relations among components may be the object of interest. Convergent and discriminant tests therefore depend on what the model says the observations are. A lived-experience estimator

should relate plausibly to reports, contexts, known contrasts, and interventions without merely reproducing arousal, illness severity, social desirability, compliance, or device tolerance (Cronbach & Meehl, 1955; Campbell & Fiske, 1959; Bollen, 1989; Borsboom, 2005).

A channel can be convergent for the wrong reason. Skin conductance may rise during fear, pain, excitement, heat, exertion, or medication effects. Activity may fall during depression, fatigue, rest, disability, or safety. A model that predicts many states because it tracks general physiological activation has not yet identified valence.

Discriminant tests should therefore include rival constructs. The evidence baseline should ask whether the candidate target can be separated from arousal, physical illness, fatigue, sleep, medication, social pressure, language ability, and observer expectation. If it cannot, the estimate should not be used as lived valence.

Convergent evidence should come from more than one method, but a count of channels is not enough. Momentary self-report, qualitative context, physiology, behavior, and ethically justified interventions can align or diverge. Agreement raises confidence only when the methods do not share the same wording, scoring, training labels, missingness, or contextual confound. Shared method bias is not independent convergence, and a composite built from redundant witnesses should not be treated as stronger merely because it has more inputs.

The chapter also needs ecological validity. A construct validated only in artificial laboratory stimuli may fail in the ordinary middle of life. A construct validated only during acute pain may fail for boredom, shame, safety, or awe. Evidence must match the range of experiences the claim intends to cover.

Validity is not a single stamp. It is a chain of relations that can break. The chapter treats each link as a place where the Law-facing measurement bridge may narrow.

Convergent validity is strongest when methods share a target without sharing the same bias. Agreement among several self-report items may reflect common wording or response style rather than independent access to experience. Disagreement between report, behavior, physiology, and clinical context can reveal construct breadth, lag, or method failure. Multitrait-multimethod designs, explicit latent-variable models, causal-composite alternatives, and held-out intervention tests can locate that disagreement, but none creates direct access to phenomenology (Campbell & Fiske, 1959; Bollen, 1989; Borsboom, 2005).

8.6.5 — Known-Groups and Intervention Tests

Known-groups tests ask whether a construct distinguishes groups or states that theory and ethics expect to differ. Intervention tests ask whether the construct changes when conditions change. Both can support a lived-experience target, but only if the comparison is justified before the result and does not exploit vulnerability (Munafò et al., 2017; Nosek et al., 2018).

Examples can be humane and ordinary. A noise-reduction intervention in a hospital ward, a better pain-control plan, a

privacy improvement, a caregiving respite, a debt-relief process, or a mobility accommodation may be expected to reduce burden. If the proposed construct does not move when participants report relief and context supports the change, the construct is under pressure.

Relief research provides a particularly useful separation of expectation, sensation, and positive feeling. Willems et al. (2025) studied 31 adults and found that omission of an expected painful stimulus produced reported relief that increased with expected threat intensity and probability. Omission also recruited parts of reward and salience circuitry. The neural pattern was prediction-error-like rather than a complete canonical reward-prediction-error signal: one required axiom failed, and the ventral tegmental probability effect did not reach the conventional significance threshold. The substantive opportunity is that relief can occur when the anticipated harm never happens. A phenomenal account must therefore record experienced anticipation and experienced relief in their own windows, while the omitted event contributes no unexperienced pain. Approved reanalysis or naturally occurring relief records can compare expectation-based and event-only models. These findings give temporal and appraisal models a concrete event sequence to explain.

Known-groups tests are dangerous when they turn identities into expected feeling. Disability, culture, age, poverty, diagnosis, or family status should not define burden by stereotype. The comparison must concern specific conditions and pathways, not the worth or presumed inner life of a group. Experts should ask whether the group label is scientifically and ethically appropriate.

Intervention tests can also mislead. An intervention may change behavior without changing experience, change report without changing burden, or reduce one burden while increasing another. A privacy intervention may reduce observation stress while making data sparse. A medication may reduce physiological arousal while increasing numbness. The construct must be able to show tradeoffs.

Learning from pain and learning from its avoidance can follow different processes. In a randomized, blinded study of 83 healthy adults, Jepma et al. (2022) found that a model with separate learning rates for received and avoided pain fit behavior better than a single-rate model. Under placebo, estimated group learning rates were 0.72 for received pain and 0.32 for avoided pain. Levodopa and naltrexone each increased learning from avoided pain, while imaging associated the two learning processes with different systems. No reduction in the number of painful outcomes was detected between the treatment groups. The study identifies a concrete, modifiable learning asymmetry. For this program, independently estimated positive and negative learning kernels can be compared with a declared coupled model using existing approved data, with future observations reserved for scoring. This comparison can identify which coupling mechanism best explains the observed learning asymmetry.

The best intervention tests include participant interpretation, predeclared expected mechanisms, alternative outcomes, and adverse pathways. They ask not only whether a score moves, but whether the movement corresponds to the burden the intervention was meant to alter.

Known-groups and intervention tests help the chapter avoid pure abstraction. They give the construct a chance to fail against humane changes in the world. If a measure cannot notice a real reduction in carried burden, the measure must change before the Law can use it.

8.6.6 — Construct Failure and Redesign

Construct failure occurs when evidence does not support the target as represented: channels fail to converge, meaning or sign drifts, interventions move indicators without the intended experience, or scalarization collapses states that participants and experts distinguish. The conclusion concerns the construct, bridge, or measurement practice, never the participant's credibility (Cronbach & Meehl, 1955; Campbell & Fiske, 1959; Flake & Fried, 2020).

Redesign begins with the smallest failed relation. An unstable item is revised or removed, a nonspecific channel is narrowed, a failed scalar becomes a vector or identified set, and a target without stable content is retired locally. More channels do not repair a construct if they merely combine arousal, illness, device gain, or response style. The redesign should preserve successful relations from the earlier instrument so correction does not erase evidence that still holds.

Every redesign records the failed use and context, contrary evidence, changed items or transformations, timing relative to outcome inspection, affected claims, and required validation. A familiar name does not let the successor inherit the predecessor's evidentiary status (American Educational Research Association

et al., 2014; Borsboom, 2005). The redesign record should also state which observed failure the new version is expected to correct, creating a prospective criterion for improvement rather than assuming it.

The revised object must return to content, response-process, structural, external, and consequential validation, including time-sensitive reliability and repeated-administration effects. Independent teams should recover both the improvement and the original failure (Nosek et al., 2018; Munafò et al., 2017; Vogelsmeier et al., 2024). Recovering the original failure is important because an instrument that only looks better on favorable cases may simply have hidden the condition that exposed its weakness.

Some failures close the route, including unidentified origin, irreducible sign ambiguity, intolerable burden, nontransportable observation, or persistent mismatch with lived reports. That outcome preserves useful knowledge about what cannot presently be measured and prevents a failed bridge from returning under a new label. A closed route should remain discoverable in the archive so later teams can distinguish a genuinely new bridge from a renamed version of a previously unsuccessful one.

8.7 — Fairness Appraisal and Cultural Recurrence

Fairness appraisal is itself a scientific domain. People do not merely register pleasure and pain; they interpret distributions, procedures, intentions, promises, status, reciprocity, and relative position. Those appraisals can alter attention, arousal, anger, gratitude, trust, shame, and willingness to cooperate. For this book, they matter because a theory about carried burden must distinguish the experience of an outcome from the judgment that the outcome was fair (Fehr & Schmidt, 1999; Bolton & Ockenfels, 2000).

The distinction is not ornamental. A person may reject an unequal allocation at personal cost, accept an equal allocation that feels humiliating, or prefer a procedure whose result is materially worse because the procedure recognizes agency. Fairness judgments can therefore generate affect, organize memory, and change behavior without serving as a direct meter of phenomenal value. They are evidence about appraisal processes and social dynamics, not automatic entries in a cumulative ledger.

Recurrence across societies and developmental stages creates a comparative research opportunity. Shared social problems, evolved cooperative capacities, convergent institutions, learning, cultural transmission, experimental demand, and narrative habits offer distinct explanations whose predictions can be examined across settings (Henrich et al., 2005; Blake et al., 2015).

The empirical starting point is substantial. In a controlled study of 40 participants, Tricomi and colleagues found that neural reward responses and subjective appeal depended on whether

monetary transfers reduced or increased an initially unequal distribution (Tricomi et al., 2010). These results give fairness appraisal a measurable place in value processing: external rewards are evaluated in a social context. For PTVC research, the gain is a concrete set of variables and experimental contrasts through which social events can alter experience within a stream.

8.7.1 — Inequity Aversion and Fairness Appraisal

Inequity aversion describes the possibility that people care not only about their own payoff but also about how outcomes compare. Formal models distinguish disadvantageous inequity, in which another receives more, from advantageous inequity, in which one receives more than another. These terms capture recurrent behavior in allocation tasks, but they are models of preference and appraisal rather than direct measures of lived valence (Fehr & Schmidt, 1999; Bolton & Ockenfels, 2000).

The experiential structure can be mixed. Disadvantage may bring anger, humiliation, threat, envy, or resignation. Advantage may bring pleasure, guilt, responsibility, fear of retaliation, or no salient feeling at all. A scalar fairness score that treats all departures from equality as the same burden would erase the very mechanisms the experiment is meant to reveal.

Procedure and intention also matter. An unequal result produced by chance may be experienced differently from the same result produced by favoritism, contempt, or broken promise. Conversely, an equal result can feel unfair when it ignores effort, need, risk, or prior commitment. Outcome equality is therefore neither a complete construct nor a universal neutral origin.

Neurochemical work gives this contextual distinction another experimentally accessible layer. During clinically indicated awake surgery in four people with Parkinson's disease, Batten and colleagues measured dopamine and serotonin while participants responded to economic offers attributed to humans or computers. Social source changed decisions, and the two transmitters carried distinguishable contextual and value information (Batten et al., 2024). This small clinical study demonstrates a route for investigating how social interpretation enters valuation.

Heterogeneity is part of the science. Participants differ in sensitivity to disadvantage, willingness to reject, interpretation of the task, expectations about deservedness, and trust in the experimenter. Population averages can hide subgroups whose appraisal functions differ in sign or shape. The model should preserve that heterogeneity before deciding that one fairness parameter explains a community.

Inequity aversion matters here because it shows how external arrangements become lived conditions. The same material difference may be carried as threat, insult, motivation, indifference, or responsibility. Studying that conversion connects external conditions with the subsequent course of experience (Fehr & Schmidt, 1999; Henrich et al., 2005).

A strong study should prospectively specify the allocation rule, information available to participants, intention cues, reference group, stakes, rejection costs, and candidate affective mediators before inspecting the target outcome. It should compare preference models, appraisal models, norm-compliance accounts, and experimenter-demand explanations. The result may identify

a local fairness process even if it says nothing about lifetime closure.

8.7.2 — Reciprocity, Cooperation, and Punishment

Reciprocity links present action to remembered treatment. Positive reciprocity can support trust, gratitude, help, and durable cooperation; negative reciprocity can support retaliation, withdrawal, punishment, and refusal. Laboratory games, field observations, and institutional records show that people sometimes incur costs to reward cooperation or sanction defection (Fehr & Gächter, 2002; Tomasello & Vaish, 2013).

Concern about fairness also extends beyond direct retaliation. In a study of 535 children aged 5 to 15 across communities in Canada, India, Peru, Uganda, the United States, and Vanuatu, children were more likely to intervene against maximally unequal than equal sharing between two absent peers. Some interventions required sacrificing the child's own rewards, and sensitivity to cost and age varied across settings. This makes immediate personal gain or repayment of a personal injury an incomplete account of the observed behavior. Learned norms, social evaluation, and evolved cooperative capacities remain live explanations. The useful PTVC research question is whether independently measured affective dynamics add a distinctive prediction beyond these mechanisms (McAuliffe et al., 2025).

Costly punishment is not experientially simple. Anger may motivate the act, anticipated relief may sustain it, and the act may later produce guilt, social approval, or escalation. A punishment decision can therefore be both behaviorally costly and affectively

reinforcing. Treating it as a clean negative or positive contribution would confuse action, motive, appraisal, and felt consequence.

Cooperation also depends on expectations and institutions. Repeated interaction, reputation, communication, group identity, enforcement, and perceived legitimacy can change whether cooperation feels safe or exploitable. The same cooperative act can be carried as generosity, duty, fear, belonging, or coerced compliance. A science of fairness must model these meanings rather than use cooperation as a proxy for well-being.

Cooperation can also contribute positively to the experience of the person who helps. Aknin and colleagues combined surveys from 136 countries with experiments, including charity-versus-self purchasing in Canada and South Africa; the experimental prosocial purchases increased reported positive affect even without an opportunity to build social ties (Aknin et al., 2013). More recently, three two-month randomized studies of daily online micro-donations, with samples of 125, 296, and 462, reported improvements in depressive symptoms and emotional positivity relative to waitlists (Y. Zhang et al., 2025). The first two studies registered their plans after recruitment and baseline measurement; the third registered before data collection. These findings offer concrete, testable pathways from voluntary social action to better reported experience.

Prosocial spending supplies a complementary finding about the giver's experience. A registered replication with 712 students found greater momentary happiness after purchasing for a stranger in need than after purchasing for oneself. A recall experiment with 1,950 adults found no corresponding advantage,

while a revised recall task with 5,199 participants found a small benefit. The full pattern supports a conditional connection between generosity and positive feeling and identifies task engagement and measurement as issues for replication. Material expenditure can therefore coexist with positive experience, which is a reason to measure felt consequences directly. The contrasting tasks also show where the relation is most reliable (Aknin et al., 2020).

The Law-facing question is not whether cooperation is good in the abstract. It is how reciprocal structures alter the stream of experience and which local dynamics might generate apparent compensation, adaptation, or burden transfer. A failed partnership may produce immediate anger and long recovery. A repaired relationship may change future threat sensitivity without erasing the experienced past.

Repair can also change the emotional state of the person who acts. Donohue and Tully randomly assigned 97 children aged 6 to 10 to conditions separating guilt from the opportunity to make amends. Children led to believe they had upset a peer showed more reparative behavior than children in a no-guilt condition. Guilt declined when children had an opportunity to offer stickers and a note, whereas it did not decline in the condition without that opportunity. This provides a concrete local sequence linking appraisal, constructive action, and relief. It motivates time-resolved observation of naturally occurring repair as a regulatory process. The opportunity to make amends offers a specific action pathway for investigating emotional recovery (Donohue & Tully, 2019).

Serious rivals must remain visible. Reciprocity can be explained through strategic learning, reputation, norm enforcement, kin and coalition psychology, reinforcement, or institutional selection. Those accounts may fit the evidence without any terminal constraint. The research program becomes stronger when it asks which predictions differ rather than treating cooperative behavior as thematic confirmation.

A useful protocol combines choices with time-resolved reports, physiological channels, expectations, relationship history, and follow-up. It separates the cost paid, the norm invoked, the emotion carried, the social consequence, and the later interpretation. That architecture lets fairness science inform the measurement problem without turning a social game into a miniature proof of the Law.

8.7.3 — Social Comparison and Relative Position

Social comparison concerns the use of other people, past selves, or imagined standards as reference points. Upward comparison can motivate, inspire, shame, or discourage. Downward comparison can reassure, isolate, or provoke guilt. The direction of comparison does not determine its valence; meaning, controllability, identification, and perceived legitimacy matter (Festinger, 1954; Suls et al., 2002).

Relative position helps explain why identical material conditions can produce different experiences. Income, recognition, safety, workload, or illness may feel different when a peer group changes, when inequality becomes visible, or when a former standard remains psychologically active. This does not make

objective conditions irrelevant. It shows that their experiential effect is mediated by reference structure.

Reference groups are dynamic. People compare with coworkers, siblings, neighbors, online networks, prior selves, cultural ideals, and expected futures. Platforms and institutions can manipulate salience by repeatedly displaying exceptional outcomes. A measurement system that ignores the comparison field may attribute resulting distress to an intrinsic trait or to the wrong event.

Social comparison also complicates adaptation. A material improvement may fail to produce sustained relief when the reference group rises faster, while a stable condition may feel better after a threatening comparison disappears. Such changes are scientifically important, but they do not retroactively alter earlier experience. Past-fixity requires the model to represent reference-dependent change without rewriting the past.

Social comparison is a psychological process that can change experience within a stream. Measuring its effects connects interpersonal events with individual affective responses.

A time-resolved experiment makes this relation quantitative. In 47 participants, Rutledge and colleagues found that both receiving less than a partner and receiving more could lower momentary happiness. Adding social-inequality terms to a recent-reward model increased mean explained variance from .39 to .44, and the asymmetry between the two inequality sensitivities correlated with generosity in a separate anonymous allocation ($\rho = -.48$; Rutledge et al., 2016). The study therefore connects

relative outcomes, felt appraisal, and generosity in a separate task within one testable account. It also provides an important comparison model for PTVC research: recent rewards, expectations, and social comparison already explain meaningful structure. Happiness ratings were standardized within participants.

A disciplined study records the reference target, comparison direction, perceived similarity, controllability, justice appraisal, exposure frequency, and resulting affect over time. It tests whether comparison variables improve held-out prediction beyond material conditions and individual history, thereby refining the local model of experience.

8.7.4 — Just-World Cognition and Narrative Balance

Just-world cognition is the tendency to expect that people generally receive outcomes they deserve or that the world is meaningfully ordered. Such beliefs can support persistence and trust, but they can also encourage victim blame, denial of structural harm, and retrospective moralization. For a project named around fairness, this is a central adversarial risk (Lerner, 1980; Hafer & Bègue, 2005).

Narrative balance can shape memory and meaning-making. A redemptive ending may organize the telling of a life that still contains substantial burden; a tragic ending may overshadow relief that the person experienced. Recording both narrative evaluation and time-resolved feeling makes their relationship available for study.

Just-world beliefs can also shape reports. Participants may minimize suffering to protect a worldview, exaggerate agency to avoid stigma, or reinterpret harm as deserved, educational, or spiritually necessary. These interpretations can be sincerely held and experientially consequential. They still require separation from the scientific question of what was felt and when.

Public communication is especially vulnerable to moralized readings of balance. Explain conditional net requirements without using them to justify suffering, delay care, rationalize inequality, or assign spiritual purification. Just-world stories and retrospective event pairing are not evidence of PTVC. An accurate account can preserve the arithmetic while rejecting those additional inferences.

Research can examine framing effects directly. Experiments can vary deservedness cues, narrative framing, outcome order, and information about structural causes, then measure changes in memory, fairness judgment, blame, and affect. Comparing explicitly balanced framing with neutral framing helps distinguish a reproducible experiential pattern from a response to the expectations introduced by the task.

The constructive lesson is not to remove meaning from science. It is to model meaning as a variable with causes and consequences. Narrative coherence may reduce distress, increase agency, or support social repair. Those are worthwhile outcomes. Studies can compare how well specified PTVC, appraisal, and learning models predict the timing and course of those outcomes.

Beliefs about moral consequences also have experimentally measurable effects. White and colleagues (2019) found that reminders of karma reduced selfish allocations among karmic believers across religious affiliations, while reminders of God reduced selfishness among believers in God. These findings identify a connection between culturally learned expectations and choices about sharing. They give the study of fairness a concrete social mechanism to investigate alongside material incentives, reputation, relationship history, and the feelings associated with cooperation.

8.7.5 — Developmental and Cross-Cultural Recurrence

Fairness behavior changes across development. Children become more sensitive to disadvantage, later show greater concern about advantageous inequity in some settings, and learn how local norms interpret equality, need, merit, and group membership. Developmental findings therefore reveal trajectories of appraisal and norm learning rather than a single fairness instinct frozen at birth (Blake et al., 2015; House et al., 2020).

The scale of this evidence deserves to be visible. Blake and colleagues studied 866 pairs of children aged four to fifteen in seven societies. Rejection of disadvantageous inequity developed across all seven, whereas rejection of advantageous inequity appeared later and in three societies (Blake et al., 2015). In complementary experiments, jointly earning rewards increased equal sharing among three-year-olds compared with windfall or parallel-work conditions (Hamann et al., 2011). Together these findings identify recurrent developmental capacities that social arrangements can cultivate. They supply clear contrasts among

disadvantage, advantage, collaboration, learning, and local norms.

Cross-cultural work likewise shows both recurrence and variation. Small-scale societies, market-integrated communities, urban populations, and institutional settings can differ markedly in offers, rejection, punishment, cooperation, and interpretation. The shared presence of fairness concerns is scientifically interesting, while the variation prevents a local task from becoming a universal psychological meter (Henrich et al., 2005; Henrich et al., 2010).

Ethnographic evidence adds a broader map of cooperative values. Curry and colleagues examined seven forms of cooperation in records from sixty societies and found positive moral evaluation in 961 of 962 coded observations (Curry et al., 2019). Coverage differed across domains; fair division of disputed resources was documented in nine societies. Recurring problems of kinship, coordination, reciprocity, conflict, and possession repeatedly invited moral solutions.

A later machine-reading study extended coverage to 256 societies and detected most of these moral themes in most societies across world regions (Alfano et al., 2024). Its expanded corpus included the earlier records. Together, these studies provide a large comparative record of cooperation-centered moral evaluation and a basis for investigating how social practices shape experience.

Task equivalence is not guaranteed. Currency, anonymity, experimenter authority, household obligations, reputational risk, and local meanings of sharing can change what an allocation

game measures. Translation must include institutions and incentives, not merely words. A numerical difference may reflect task ecology rather than a deeper difference in felt fairness.

A 2026 study sharpens the distinction between broad fairness sensitivity and particular institutional norms. Among 697 children aged three to eleven in Norway, Japan, the United States, and Italy, aversion to unequal allocations emerged across settings by age ten, while rejection of bribes appeared in late childhood only in some contexts (Reyes-Jaquez et al., 2026). The dissociation is informative: related moral judgments need not be expressions of one interchangeable faculty. It gives future research a practical strategy: compare closely related judgments under the same task framework and identify which developmental and cultural conditions carry each pattern.

Development and culture also interact with expression. Children may lack concepts for mixed motives. Adults may follow politeness rules that mute complaint. Communities may treat open anger, direct refusal, or individual entitlement differently. These patterns affect reports and behavior without making any group less capable of experience.

Recurrent concern with distribution, reciprocity, and repair identifies questions for the science of lived burden. It supplies candidate mechanisms and contrasts across social settings.

Religious traditions also supply observable social mechanisms. In eight communities, a study of 591 participants associated beliefs in knowledgeable, punitive gods with more impartial allocations to distant co-religionists (Purzycki et al., 2016). An expanded

two-wave study covering 2,228 participants in fifteen societies, including the earlier sample, further examined this relation and its limits across religious groups (Lang et al., 2019). These findings make belief, accountability, group boundaries, and cooperation experimentally and comparatively tractable. Analyses of this evidence should account for the overlapping samples and shared cultural histories. Their contribution is a better map of the meanings and social conditions that can alter experience within a stream.

A strong comparative program uses local collaborators, measurement-invariance tests, qualitative interpretation, task redesign, and within-site replication. It reports commonalities and differences without forcing one global parameter. When a pattern fails to transport, the failure improves the theory by showing which social and developmental assumptions were carrying it.

Developmental and cross-cultural recurrence should be studied at the level of tasks, meanings, and institutions, not only labels. Children's responses change with cognitive capacity, norm learning, relationship, and stakes; adult patterns vary with market integration, kinship, religion, authority, and experimental framing. Similar behavior may therefore rest on different appraisals, while different behavior may express a shared concern under different constraints. Comparative models can test which shared processes and local conditions account for the observed recurrence (Henrich et al., 2005; Blake et al., 2015).

The comparative frontier remains active and increasingly precise. A primary-data synthesis of twenty-three studies across eighteen nonhuman species found no inequity-aversion evidence in the

accept/reject paradigms analyzed (Ritov et al., 2024). Subsequent bonobo experiments found greater refusal under unequal rewards, including controls for disappointment in a human experimenter, although the second study was only partly consistent with an inequity-aversion account (Radovanović et al., 2025). A 2026 group study of twenty-seven participating chimpanzees found responses that depended on food quality and the social partners present (Martinez et al., 2026). These results invite stronger tests of context, reward, and relationship effects rather than a universal animal-fairness verdict. Developmental research supplies the same lesson: a coordinated study testing 1,018 infants, with 567 included across thirty-seven laboratories, found no helper preference in the specific hill paradigm (Lucca et al., 2025). Such coordinated work makes claims more precise while leaving other tasks and capacities open.

8.7.6 — Recurrence as Hypothesis Generator, Not Validation

Fairness themes recur in moral psychology, economics, anthropology, religion, politics, childhood development, and ordinary storytelling. This breadth gives researchers several explanations to connect and compare: shared social problems, cognitive capacities, cultural diffusion, convergent institutions, narrative preferences, and specified PTVC-related processes. The next step is to identify observations on which their predictions differ (Munafò & Davey Smith, 2018).

The constructive use of recurrence is to turn a recognizable pattern into a discriminating question. Existing social-value, cooperation, developmental, religious-cognition, and affect

studies identify variables, candidate processes, and feasible designs. A PTVC-related route can then be compared with cultural learning, reciprocity, social-preference models, and bounded-memory reward learning under the same observations. For example, if a particular route predicts a history-dependent change beyond current affect, recent rewards, expectations, resources, and norms, that residual-history prediction must be derived, fixed in advance, and tested in held-out observations. The evidential gain comes when a specified route successfully predicts a difference that capable rivals do not, with identification and observation error accounted for. Recurrence supplies the research opportunity; comparative prediction determines which explanation earns additional confidence.

A practical starting point is the Global Preference Survey, which measured experimentally validated preference items in approximately 80,000 people across 76 countries, including reciprocity, altruism, and trust. Its larger variation within countries than between countries encourages person-sensitive analysis rather than national stereotypes (Falk et al., 2018). These data can help choose contexts and social-preference measures. A new, low-burden longitudinal study could combine such measures with repeated affect reports and naturally occurring episodes of help, disappointment, and voluntary repair. Candidate PTVC relations and appraisal, learning, reputation, and norm-based alternatives would make protected predictions using the same observations. A PTVC realization can add a prospectively specified restriction to this shared comparison.

This rule also protects negative results. If a fairness motif is absent in a population, a developmental stage, or a well-designed task, the project should not redefine the motif until it appears. Absence may reveal a boundary condition, a task failure, or a genuine limit. Scientific progress requires all three possibilities to remain open.

Demand characteristics deserve special attention. Participants may infer that a study expects equality, balance, redemption, or moral concern and respond accordingly. Researchers and readers may also select memorable fairness-shaped outcomes while ignoring ordinary counterexamples. Blinding, neutral task language, preregistration, and complete reporting reduce that risk.

The evidence baseline establishes phenomena and measurement problems that PTVC must explain: affect is dynamic, multidimensional, state-dependent, culturally mediated, and imperfectly observed. Those literatures do not themselves support terminal valence closure. Their role is to constrain representation, identify ordinary mechanisms and confounds, and provide benchmarks that any PTVC-specific model must outperform.

Chapter 9 — The Measurement Pipeline

A measurement pipeline is not a dashboard placed over a person. It is a sequence of accountable translations: a lived target is represented, sampled through one or more channels, transformed into data, linked to an estimand, analyzed by a declared map, summarized by an estimator, and interpreted under a decision rule. Measure selection, item content, scoring, modifications, exclusions, and versions belong inside the same chain. Every step can add information, lose information, or change the question. Chapter 9 makes that sequence visible so that a final number cannot hide the decisions that produced it (Flake & Fried, 2020).

Qualification begins with a declared context of use; §9.1 specifies the conditions that bound what an instrument may validly support.

The pipeline then separates objects that are often compressed into the word measure. The measurement map specifies how observations arise. The estimand states the scientific quantity or contrast being targeted. The analysis map records preprocessing, feature construction, exclusions, and transformations. The estimator produces an estimate and uncertainty object. The decision rule states what the result can license, reject, narrow, or leave unresolved. Keeping these objects distinct makes post hoc rescue visible and lets model recovery test whether the procedure answers the question it claims to answer.

Candidate channels contribute different kinds of evidence. First-person reports and pairwise judgments can witness phenomenal content and comparative intensity. Trade-off choices and behavior can reveal priorities, constraints, and anticipated consequences. Autonomic and interoceptive signals can track embodied change. Neural and clinical channels can locate state transitions or

confounds. Contextual and relational records can explain exposure, care, threat, and opportunity. The pipeline assigns each channel a declared role instead of treating a larger sensor array as automatic access to experience. The received candidate momentary estimator remains physiological; the other witnesses are auxiliary unless a broader multimodal estimator is separately declared.

First-person report remains central because the target is lived, but central does not mean infallible or sufficient. Language, recall, social desirability, fear of consequences, communication access, fatigue, demand characteristics, and scale interpretation can shape an answer. These influences are not reasons to demote the participant. They are features of the response process that the measurement model must investigate. Pairwise judgments, repeated anchors, qualitative interviews, and participant challenge can improve interpretation without replacing the person's authority over consent and correction.

Intensive longitudinal methods create both opportunity and risk. Repeated observations can reveal within-person dynamics, context dependence, lag, and ordinary duration that one retrospective survey misses. They can also reproduce the same unstable item hundreds of times, create burden-related missingness, and mix between-person, within-person, and person-specific reliability. Recent reviews document substantial variation in experience-sampling design and reporting, while recent psychometric work shows why item distributions, time trends, measurement structure, level-specific reliability, and person-specific performance must be examined directly

(Shiffman et al., 2008; Wrzus & Neubauer, 2023; Vogelsmeier et al., 2024; Siepe et al., 2025; Castro-Alvarez et al., 2026; Peeters et al., 2026).

A serious momentary-to-interval estimation pipeline must answer those questions before claiming scientific value. Its qualification dossier should define the construct, justify the measure and channels, report every modification and scoring rule, specify the context of use and state range, name the exact estimand, and identify the decisions the output may never support. The dossier also records whether the system is exploratory, protocol-fixed, or publicly preregistered, and when it must return No adequate estimate instead of improvising a number because the pipeline wants completion (Flake & Fried, 2020; Gagnier et al., 2025).

Cross-channel conflict is evidence, not clutter. A participant may report calm while physiology remains elevated after exertion, or report distress while a wearable records little change because medication, disability, device contact, or individual physiology weakens the channel. The scientific question is which observation processes and constructs could generate the disagreement. A strong pipeline can widen uncertainty, split by channel, narrow the claim, or return No adequate estimate without treating the person as inconsistent or the sensor as sovereign.

The measured-to-latent bridge is the chapter's central inferential gate. Calibration, invariance, linking, construct validity, and observation coverage determine whether measured outputs support a local trajectory, a bounded contrast, an identified set, or any claim about latent phenomenal value. Bridge strength must match licensed language. A well-predicted report is not

automatically a recovered latent state, and a recovered momentary state is not automatically a complete-stream cumulative object.

Section 9.6 treats missingness as part of the observation model, including informative and missing-not-at-random processes relevant to intensive longitudinal data (Tsiatis, 2006; McNeish, 2025).

Provenance and versioning determine whether an estimate can be interpreted later. Hardware, firmware, item wording, language, sampling cadence, preprocessing, model weights, calibration sample, missingness rules, and public claim templates can all change the measurement relation. Each output therefore carries the versions and transformations that produced it. Migration studies test which comparisons survive updates; backward compatibility is demonstrated, not presumed. A correction must be able to locate every figure, summary, model, and downstream claim that depended on the retired relation. Version provenance should be queryable at the individual result level, allowing a later defect to find exactly which outputs require recalculation.

The pipeline also needs a reproducibility model suited to sensitive data. Independent checking does not require public release of person-level records. Synthetic fixtures, controlled-access environments, privacy-preserving summaries, code tests, data dictionaries, and provenance logs can make transformations auditable while protecting participants. Current reproducibility and reporting standards support this layered approach, and current good-clinical-practice guidance treats reliable information, proportional processes, and participant protection as connected

design duties (Wilkinson et al., 2016; Center for Open Science, 2025; International Council for Harmonisation, 2026). The chosen reproducibility layer should be declared before analysis so privacy restrictions do not become a post hoc reason why an unfavorable result cannot be checked. The declaration should also state which independent checks remain possible at each access tier, making any verification limits created by privacy protection explicit.

Section 9.7 implements anti-score and non-deployment governance for the measurement pipeline through context-of-use limits, access control, contestability, record separation, deletion, retirement, and human review. Technical enforcement matters because purpose limits that rely only on user discretion can disappear when outputs move between systems. Context-of-use restrictions should therefore travel with derived outputs through persistent provenance or policy metadata, so transfer between systems does not silently detach an output from the conditions that licensed its use.

Participant authority travels through the entire stack. People can refuse, withdraw, correct context, challenge an interpretation, report burden, and learn what an output can and cannot mean. Clinicians and caregivers can flag medication, workflow, state, or device effects without becoming substitute reporters of inner life. These routes improve both dignity and validity because unchallengeable instruments preserve their own mistakes. The burden of justification stays with the pipeline. Contestability also provides calibration data: recurring participant corrections may reveal systematic context errors that ordinary validation samples

miss. Correction pathways should distinguish amendments to contextual information from changes to the original record, preserving provenance while allowing the interpretation built from that record to be challenged.

Chapter 9 specifies an end-to-end measurement dossier, from context of use through maps, estimands, channels, dynamics, missingness, provenance, privacy, and retirement. An implemented pipeline earns qualification only through the validation work described here. Chapter 10 then examines whether origin, sign, unit, linking, and invariance support comparisons across time and context. Qualification therefore attaches to a versioned pipeline rather than to an instrument name, allowing later changes to be evaluated without inheriting unsupported authority.

9.1 — Context of Use and Qualification

Qualification begins with a question, not a device. The context of use should name the construct, population, state range, setting, sampling cadence, decision purpose, participant-burden ceiling, tolerated error, and prohibited uses before performance is summarized. Evidence adequate for exploratory within-person tracking may be inadequate for clinical interpretation, cross-group comparison, or a terminal-window claim. Current fit-for-purpose clinical-outcome-assessment guidance and measurement-property reporting standards reinforce this claim-specific discipline (FDA-NIH Biomarker Working Group, 2016; U.S. Food and Drug Administration, 2025b; Gagnier et al., 2025).

The chapter uses qualification as a scientific discipline rather than a badge of approval. A candidate momentary estimator may be qualified for one purpose and unusable for another. It may track within-person change in ordinary ambulatory conditions while failing in severe pain, support exploratory association without supporting confirmatory terminal inference, or work in one device version without surviving transfer to another. The context of use is the border around every claim.

Qualification protects both inference and participant. Saying that an instrument is not qualified for a decision does not say that a person is unknowable, unreliable, or outside care. It says only that the instrument has not earned that use. The cost of overreach is carried by the person interpreted outside the population, state, purpose, and error tolerance for which the system was evaluated.

9.1.1 — Research Question First

A research instrument is defined by the question and measurement model it is asked to serve, not by the volume of data it collects. The same physiological stream can support questions about arousal, autonomic instability, sleep disruption, pain response, stress recovery, or affective valence. A candidate momentary estimator begins only when the study declares the target construct, the relation between target and observations, the estimand, and the conditions under which the output may be interpreted (American Educational Research Association et al., 2014; Bollen, 1989; Borsboom, 2005; FDA-NIH Biomarker Working Group, 2016).

The minimum question has five parts. It must name the target state or rate, the time scale, the population, the intended inference, and the decision boundary for failure. For example, a development study might ask whether a physiological composite tracks within-person changes in signed burden over two waking weeks in ambulatory adults. A confirmatory study might ask whether a locked estimator predicts a prospectively specified endpoint-window trajectory better than rival observation models. Those are different instruments for different claims, even if the sensors are identical.

A candidate momentary estimator targets latent valence rate or a declared proportional proxy. Its estimand specifies the quantity, population, time window, and comparison the study will investigate.

A declared estimand also protects future failure. If the estimate fails to track self-reported burden in one population, that is not

automatically a defeat of all measurement. If it fails to identify sign under the declared context, sign-bearing claims must narrow. If it fails only under a device-shift analysis, the instrument version may be retired. The result changes the claim that was actually declared.

A candidate momentary estimator begins with the research target and loss function. For this volume, that target is signed contribution to a unified stream over time. General affect detectors provide useful starting tools; qualification establishes which observations reliably inform that cumulative target.

Qualification asks two practical questions: which claim can this instrument make reviewable, and what burden could an incorrect result create? Answering both before data collection connects predictive performance with construct validity, reactivity, missing-state bias, and the intended use.

9.1.2 — Population and State Scope

Population and state scope must be explicit. Age, language, disability, clinical condition, medication exposure, device placement, sleep-wake state, sampling cadence, and setting can all change the observation relation. A qualification result confined to one envelope does not become general by omitting the envelope from the abstract, model card, or public graphic (FDA-NIH Biomarker Working Group, 2016; Putnick & Bornstein, 2016; American Educational Research Association et al., 2014).

State scope is as important as demographic scope. A person resting after exercise, a person under opioid medication, a person in panic, a person with fever, a person sleeping, a person in delirium, and a person receiving palliative sedation can produce

physiological channels that resemble or invert one another. The same raw channel may carry different meaning across states. A model that ignores state cannot claim to read through state.

The stack should therefore distinguish ordinary ambulatory qualification, clinical qualification, terminal qualification, and edge-state qualification. Each carries a different burden of proof. Ordinary use may tolerate frequent prompts and wearable devices; terminal research may not. A clinical ward may provide medication and care-procedure records; private life may not. An endpoint window may have high missingness precisely when the state is most relevant.

Population scope also includes who is excluded by design. If the model performs only in fluent speakers using one device under low burden, it must say so. If people with disability, skin-tone sensor artifacts, autonomic neuropathy, arrhythmia, movement disorders, chronic pain, or medication profiles are underrepresented, the model's silence is not neutral. It is an unqualified region.

This creates a reference-class ceiling. Evidence from one population supports only that population and context until transport is defended. A hospice cohort in one care system cannot silently generalize to all terminal states, and an adult ambulatory sample cannot establish developmental, disability, or communication-limited validity.

Explicit scope helps protect participants whose circumstances differ from the development sample. The order should remain strict: define scope, estimate within scope, test travel, and only then write public language.

9.1.3 — Development Versus Confirmatory Use

Developmental measurement and confirmatory measurement are different scientific acts. In development, investigators may explore features, compare transforms, inspect lags, test candidate channels, and learn where the instrument breaks. In confirmatory use, the construct, measure selection, item or channel set, scoring, estimator, preprocessing, signs, weights, cadence, missingness model, endpoint separation, and analysis plan must be fixed before the test begins, or every later modification must be disclosed as a new version requiring fresh evaluation (Shiffman et al., 2008; Stone et al., 2007; Flake & Fried, 2020).

To test the intended pattern prospectively, the confirmatory pipeline fixes channels, signs, weights, lag corrections, smoothing windows, and neutrality bands before examining the terminal-window outcomes. Development data can guide those choices; independent observations evaluate their predictive performance.

Developmental work is not inferior. It is where the instrument learns to respect the world. Sensor placement, artifact rejection, self-report timing, context annotation, calibration anchors, and burden endpoints must be refined somewhere. The danger begins only when development is marketed as confirmation, or when exploratory performance is treated as public evidence for the Law.

A practical stack should carry visible labels. Exploratory feature work is not a protocol-fixed candidate momentary estimator. A locked composite is not a dynamic latent-state model. A model tuned on one site is not externally replicated. A smoothed

retrospective trajectory is not a prospective forecast. These labels are not bureaucratic. They are the reader's map of claim status.

The same separation should govern code and artifacts. Development code may change rapidly. Confirmatory code should be versioned, frozen, recoverable, tested on synthetic fixtures, and resistant to silent rescoring. A model card or measurement card should state the purpose, evidence base, abstention rules, and known invalid regions before the model is used in a study that can change public language.

The Law becomes more answerable when development is allowed and confirmation is disciplined. It becomes less scientific when every promising pattern is treated as if it had been predicted all along.

9.1.4 — Performance Requirements by Use

Performance is not one number. A system that separates high-arousal from low-arousal states may fail sign orientation; one that tracks within-person change may fail between-person comparison; one that predicts next-hour report may fail terminal dynamics. Reliability itself can differ across persons, levels, occasions, and states. The performance burden must therefore be matched to the declared use, estimand, and measurement model rather than summarized by one global accuracy or internal-consistency score (American Educational Research Association et al., 2014; Hernán & Robins, 2020; Vogelsmeier et al., 2024; Castro-Alvarez et al., 2026).

For association studies, performance may begin with reliability, artifact control, convergent validity, discriminant validity, and plausible relation to known state changes. For within-person

tracking, it must also show responsiveness, temporal stability, lag handling, and sensitivity to the individual's own baselines. For prediction, it needs time-forward validation, calibration, and error under distribution shift. For terminal inference, it needs all of that plus comfort-first feasibility, independent endpoint adjudication, missingness modeling, rival comparison, and abstention rules.

Validation architecture should mirror the intended use. New-stream transport requires person-held-out evaluation; forecast claims require time-forward splits; hardware portability requires device or firmware holdouts; and care-system transport requires site holdouts. These are not interchangeable checkboxes, and failure in one lane must narrow the claim that depends on that lane rather than being averaged into one global score.

Performance requirements should be reported as a profile rather than collapsed into one validation score. The profile may include analytical performance, technical reliability, construct and discriminant validity, sign validity, temporal responsiveness, state and group invariance, clinical robustness, prospective prediction, and independent replication. A use is qualified only through the components it actually passes, and the reader must be able to see where the evidence stops (American Educational Research Association et al., 2014; Campbell & Fiske, 1959; Putnick & Bornstein, 2016; Nosek et al., 2018).

Performance also includes humane feasibility. If a device is accurate only when it wakes participants, interrupts care, requires frequent adjustment, or pressures families to maintain signal, the instrument has created burden that may alter the data. The score of the device is not separate from the cost of obtaining it.

Qualification is claim-specific. An instrument is adequate only for a named construct, population, state range, cadence, estimand, decision, missingness regime, and surviving rival set. That grammar replaces the vague judgment that a tool is good with an inspectable scientific permission whose limits, expiration conditions, and failure consequences can be audited.

9.1.5 — Abstention and Exclusion Rules

A reason-coded abstention is a valid and informative measurement output. Apply §24.2's canonical evidence-state rules, and record the measurement, calibration, consent, or participant-protection condition that blocked the estimate. An explicit abstention route protects participants and keeps each claim within the evidence (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

Abstention rules should be declared before use. The system should withhold a candidate momentary estimate when required channels fail, sign-bearing evidence disappears, missingness is informative beyond modeled sensitivity, device contact is poor, medication or care procedure makes the channel uninterpretable, context falls outside qualification, or participant burden exceeds the protocol threshold. Each rule protects both the science and the person.

Exclusion is different from abandonment. A participant excluded from one analytic claim remains a participant whose data, experience, and care matter. The chapter's no-score discipline means exclusion is about the model, not the person. It says the instrument cannot answer a specific question. It does not say the person is unclear, unreliable, or unavailable to dignity.

Abstention should also appear in cumulative outputs. If three intervals in a running ledger are outside context of use, the ledger should show gaps, identified sets, or sensitivity bounds rather than silently imputing continuity. If a version migration is invalid, the stack should report version-separated estimates. If a retrospective smoother uses later observations, it should be labeled and barred from claims of prospective prediction based only on earlier information.

Abstention has to be audited. A system that abstains more often for disabled participants, darker skin tones, terminal patients, nonfluent speakers, people in pain, or people without stable housing may be revealing a fairness defect in the instrument. The answer is not to force estimates for equity optics. The answer is redesign, narrower claims, or retirement.

9.1.6 — Prohibited Contexts

Some uses lie outside the instrument's authority regardless of predictive performance. A candidate affect estimator is not an employment screen, insurance price, policing signal, civic or spiritual rank, donor score, loyalty test, or basis for denying care. Those exclusions require purpose limitation, access control, audit, and retirement rather than a note beside a reusable dashboard (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

The prohibition protects people and inference. Participants cannot consent freely when access, employment, treatment, schooling, insurance, or public benefits depend on compliance, and may change reports when outputs may be used against them. Clinical care precedes measurement; a No adequate estimate outcome may

never deem a person insufficiently suffering for relief or too uncertain for attention (World Medical Association, 2024; International Council for Harmonisation, 2026).

A context-of-use contract defines the population, state space, sampling window, target construct, estimand, error and burden limits, development stage, abstention rule, and prohibited uses. It states when changes in instrument, software, treatment, language, population, or setting end qualification. A blank field leaves the tool promising, not qualified (FDA-NIH Biomarker Working Group, 2016; U.S. Food and Drug Administration, 2025b).

Consider a momentary estimator for adults recovering at home during the first fourteen waking days after surgery. The question is whether a locked physiological model tracks within-person change under known devices, medication patterns, language, prompt burden, and care events without using recovery outcomes to select signs or weights. The result concerns within-person postoperative tracking under those specified conditions.

Performance requirements follow that use. Analytical reliability, artifact control, within-person responsiveness, medication and motion negative controls, time-forward validation, participant burden, and contestability may be required, while between-person comparability and whole-life inference remain outside scope. A participant with device intolerance, missing anchors, or unusual medication effects can remain fully included in care and descriptive observation while the estimator abstains.

Table 9.1 applies the performance profile in §9.1.4 as bounded use tiers. Each requirement retains its own failure language, and

progress on one cannot compensate for failure of a requirement needed by the claim.

Table 9.1. Measurement-use tiers and interpretation scope.

Use tier	Minimum evidence before use	Scope of interpretation
Exploratory development	Transparent features, artifact rules, basic reliability, burden monitoring, and clear labels.	Exploratory associations and development results.
Within-person tracking	Repeated anchors, time-forward validation, lag tests, missingness model, and participant contestability.	Within-person change under the qualified conditions.
External replication	Independent data under the declared context of use, with person, site, device, or state holdouts matched to each claimed form of transport.	The tested reference class and context of use.
Confirmatory Law-facing study	Frozen estimator, protocol-fixed estimand, endpoint separation, rival models, and abstention rules.	Protocol-specified target contrasts within the qualified construct, context, and scale.

Partial progress should be publishable. A channel may pass analytical and technical checks but fail sign validity; a model may support within-person tracking but fail site-held-out transport; a device may work in ordinary life and become unusable near terminal care. The qualification file records the exact rung, population, state, version, untested conditions, and participant burden endpoint rather than advertising that the instrument works.

Current guidance reinforces this context-specific discipline. Fit-for-purpose clinical outcome assessments require a clear concept

of interest, context of use, content-validity evidence, appropriate measurement properties, and documentation of modifications, while digital technologies add verification, validation, risk-management, and data-integrity duties. Psychometric guidance also requires transparent reporting of validity, reliability, measurement error, responsiveness, and cross-cultural invariance. Intensive longitudinal measurement adds item-level, level-specific, temporal, and person-specific checks that cross-sectional summaries can miss. These standards inform the pipeline without serving as evidence for PTVC (U.S. Food and Drug Administration, 2023; U.S. Food and Drug Administration, 2025b; Mokkink et al., 2024; Gagnier et al., 2025; Vogelsmeier et al., 2024).

Prohibited contexts must be enforced technically. Purpose-bound schemas, separated identifiers, role-based access, deletion schedules, query restrictions, audit logs, and release controls should prevent persistent rankings, cross-system dossiers, secondary training, and eligibility dashboards. Qualification expires outside the tested envelope; a supported research context never grants permanent authority to an evolving system (National Institute of Standards and Technology, 2020; National Institute of Standards and Technology, 2024; Wilkinson et al., 2016).

9.2 — Maps, Estimands, and Estimators

The pipeline is a typed sequence of maps. A latent phenomenal state and its context jointly generate reports, behavior, physiology, and neural activity through channel-specific observation processes; an analysis map transforms those records; an estimator targets a declared quantity; a decision rule converts the estimate into bounded language. Conflating any two stages makes it impossible to tell whether failure belongs to the construct, channel, model, or decision.

Four objects must be separated. The measurement map describes how the latent target and context generate observable records. The estimand states the exact quantity the study seeks to learn. The analysis map specifies how raw records become analyzable variables. The estimator is the finite-sample procedure that returns an answer with uncertainty. Validation at one layer cannot be borrowed by another.

This grammar matters acutely for the Law because a polished trajectory can hide several substitutions. A sensor feature may become a state estimate; a state estimate may become an interval contribution; an interval contribution may become a cumulative curve; and the curve may be described as if it were the latent phenomenal ledger. Each step requires its own assumptions, calibration, recovery tests, and failure language.

The section builds a pipeline that can fail locally. A measurement map may be misspecified even when the estimator is numerically stable. An estimand may be scientifically incoherent even when software computes it precisely. A reported quantity may be clear

but unlicensed for the context of use. Local failure is a strength because it shows where the claim must narrow.

The distinctions can be made operational by treating each layer as a separate object with its own failure test. The matrix below is not a workflow convenience. It is a claim-control device: a study cannot repair a poorly specified estimand by fitting a more elaborate estimator, and it cannot repair a poor measurement map by reporting a narrower confidence interval.

Table 9.2. Measurement-pipeline objects and observable rejection conditions.

Pipeline object	What it must specify	Observable rejection condition
Measurement map	How states, events, reports, sensors, and context become recorded variables, including timing, units, transformations, and failure codes.	The record cannot distinguish a target-relevant change from device, language, task, or context change.
Scientific estimand	The exact population, stream, time window, contrast, functional, and conditioning set the study seeks to learn.	The reported number answers a different question from the protocol-fixed scientific target.
Analysis map	Preprocessing, alignment, feature construction, missing-data handling, calibration, and transformations applied before estimation.	A reasonable alternative analysis map reverses sign, changes scope, or materially alters the conclusion without prior justification.

Pipeline object	What it must specify	Observable rejection condition
Estimator	The finite-sample rule that maps the analysis record to an estimate or identified set, with uncertainty and regularization declared.	Simulation or held-out checks show bias, instability, nonrecovery, or unjustified certainty under plausible generators.
Reported quantity	The exact object shown to readers, such as an observed channel, local contrast, interval estimate, path estimate, or identified set.	Language, labels, graphics, or captions imply a stronger latent or cross-person claim than the analysis supports.
Decision or verdict rule	The protocol-fixed rule connecting evidence to abstention, narrowing, route comparison, correction, or withdrawal.	The rule changes after outcomes are seen or allows downstream success to compensate for an upstream gate failure.

9.2.1 — Measurement Map and Observable Record

The measurement map links a latent phenomenal target and its context to the records a study can actually collect. Those records may include reports, pairwise judgments, autonomic channels, interoceptive signals, neural measures, behavior, medication timing, clinical observations, and missingness indicators. The map must state whether each record is modeled as an effect of the target, a cause or component of an index, a contextual moderator, or an auxiliary witness. It is a scientific model of access, not a declaration that any record is experience itself (Bollen, 1989; Borsboom, 2005).

A complete map identifies the raw unit, sampling window, transformation, expected lag, noise source, context variable, burden of collection, and failure condition for each channel. It also identifies what is never observed. Silence, device removal, sleep, care escalation, refusal, and communication loss are not empty cells; they are outcomes of the observation process.

Reactivity belongs in the map. Prompts can alter attention, devices can change sleep or movement, clinical observation can change behavior, and repeated ratings can teach participants a vocabulary that reshapes later reports. A channel that changes the state it measures may remain useful, but its intervention effect must be modeled or bounded.

The map must preserve person and time structure. A signal collected at one minute does not automatically represent the same temporal interval as a report made ten minutes later. A channel that tracks arousal during exercise may not have the same relation to valence during pain, medication, or social threat. State and context determine the link.

Missingness and selection are generated inside the map, not appended after estimation. If high burden reduces device tolerance or report capacity, observed records overrepresent easier intervals. If a terminal window ends observation before the truth endpoint, the resulting curve is a record of observation shutdown as well as possible experience.

The practical output is a measurement-map table that can be audited before analysis. For every channel it names the proposed target relation, measurement-model role, unit, time window,

transformation, lag, confounds, burden, missingness route, version, and prohibited inference. It also records the source of the measure, any modifications, and the validity evidence available for the declared use. A pipeline without that table asks the estimator to hide conceptual debt (Flake & Fried, 2020; Gagnier et al., 2025).

9.2.2 — Scientific Estimand and Target Contrast

An estimand is the precise scientific quantity a study intends to learn. It includes the population or stream class, state and time window, treatment or exposure condition where relevant, outcome definition, handling of intercurrent events, and contrast of interest. An estimand is not the observed score and not the algorithm that estimates it (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

For this program, candidate estimands include a within-stream directional change in phenomenal-value rate, an interval contribution under a declared representation, a difference between two time windows, a response to an ethically permissible intervention, a trajectory feature, or an endpoint-window contrast. These are different objects and require different measurement and causal assumptions.

A momentary estimand must name what momentary means. It may refer to an instantaneous idealization, a short window, a latent state at a time point, or an average over a sampling interval. A rate estimand must say how its unit relates to time. That statement makes time accumulation dimensionally interpretable.

An interventional estimand also separates experience from assignment and adherence. Treatment, care, context, device use, and participant choice can change both the target and observation. The protocol should state whether it seeks a treatment-policy effect, an effect under adherence, a principal-stratum contrast, or a descriptive within-person response.

Intercurrent events, such as a relevant treatment change, can alter the interpretation or existence of the outcome and must be addressed in the estimand. Device removal, missed prompts, and study withdrawal may instead create missing data; they are not automatically intercurrent events. Record the underlying clinical event separately from the loss of observation, and align the estimation and sensitivity analyses with the declared target (International Council for Harmonisation, 2019).

These choices must be fixed before the result is known. An estimand selected after inspecting which definition approaches neutrality is not the original target but a rescue construction. A protocol-fixed estimand may still be wrong, but it can fail cleanly and support an honest redesign.

9.2.3 — Analysis Map and Preprocessing

The analysis map converts raw records into variables used by the estimator. It includes synchronization, artifact rejection, filtering, feature extraction, normalization, missingness coding, lag adjustment, channel exclusion, item scoring, scale modification, and any learned representation. Each decision can create or remove the pattern the study later claims to discover, so the

complete map must be visible and versioned (Munafò et al., 2017; Nosek et al., 2018; Flake & Fried, 2020).

Preprocessing must therefore be locked or nested inside a transparent development process. Exploratory work may compare filters, features, and windows, but confirmatory work must test the chosen pipeline on held-out persons, sites, devices, or time periods. A favorable result that depends on one undocumented filter is not a robust measurement finding.

Preprocessing choices are specified before evaluation of the target outcome. Centering, smoothing, baseline subtraction, endpoint normalization, and channel weighting can alter apparent convergence or sustained burden. The protocol states the purpose of each operation and evaluates its effect on the target estimand.

The analysis map should preserve multiple views when they answer different questions. Filtered estimates use information available up to the present; smoothed estimates use future observations to revise the past; forecasts predict later states; imputed values fill missing intervals under assumptions. A single line that mixes those products hides time and evidence.

Versioning belongs here. Firmware, preprocessing code, feature definitions, annotation rules, and model inputs may change during a longitudinal program. Every output should carry the exact analysis-map version, and bridge studies should determine whether old and new outputs can be linked, compared, or only reported separately.

A reader-facing methods statement should reconstruct the map without requiring access to source code. It should name raw

inputs, item or channel content, scoring, transformations, exclusions, timing, training data, validation split, version changes, and uncertainty contribution. Code remains essential for reproducibility, but prose must reveal the scientific choices and their chronology rather than outsourcing them (Flake & Fried, 2020; Peeters et al., 2026).

9.2.4 — Candidate Momentary Estimator

A candidate momentary estimator is a procedure for producing a bounded estimate of a declared momentary or short-window target. It may be a transparent weighted composite, a state-space model, a Bayesian latent-state model, or another recoverable architecture. Its status comes from the identified target, the declared measurement model, a qualified observation map, model recovery, and out-of-sample performance, not from mathematical sophistication (American Educational Research Association et al., 2014; Bollen, 1989; Borsboom, 2005; Gelman et al., 2013).

A transparent composite can be scientifically valuable because its transformations and weights are inspectable. Its weakness is that standardized channels may not share units or construct meaning. Equal weights can hide unequal reliability, while learned weights can overfit outcome labels. The result should therefore display component contributions and sensitivity to plausible weights.

A dynamic latent-state estimator can model continuity, lag, missingness, and state evolution. Its flexibility is also a risk. Time-varying loadings, process noise, contextual covariates, and smoothing can absorb the very burden the model is meant to

recover. Model recovery and synthetic fixtures are necessary before the estimator is trusted on human data.

Sign, location, and scale must be identified independently of the Law-facing outcome. A factor can fit the data while remaining reversible in sign, shiftable in origin, or arbitrary in unit. An estimator that cannot distinguish positive from negative experience may still report magnitude or arousal, but it cannot support a signed cumulative claim.

Uncertainty should be a distribution or identified set when the model permits it. Pointwise standard errors alone do not capture model choice, sign ambiguity, channel conflict, missingness, or calibration uncertainty. The estimator should widen, split, or abstain when those uncertainties dominate.

Qualification is context-specific. A model may track within-person change in ordinary ambulatory life and fail in pain crisis, sedation, childhood, another language, or a new device generation. The correct label states the population, state scope, time scale, version, permitted use, and failure conditions.

9.2.5 — Candidate Interval Estimator and Path Integration

An interval estimator requires more than multiplying a momentary score by time. The momentary quantity must have a defensible rate interpretation or declared proportional relation to experienced exposure. The interval, clock, unity conditions, and handling of gaps must be specified before integration (Shiffman et al., 2008; Stone et al., 2007).

Irregular sampling makes the temporal problem visible. Reports may arrive minutes apart, sensors may sample at different rates,

devices may fail, and care events may change both state and observation. Sparse or uneven cadence can alias rapid dynamics, while interpolation and smoothing can invent continuity or suppress extremes. The estimator must decide prospectively whether to interpolate, forecast, impute, abstain, or return bounds, and each decision must propagate into path uncertainty (Shiffman et al., 2008; Stone et al., 2007; Peeters et al., 2026).

Posterior path integration is preferable to summing isolated point estimates when dynamics and serial dependence matter. The model should draw coherent trajectories, integrate each trajectory over the declared interval, and report the resulting distribution of interval contributions. That procedure carries timing, lag, covariance, and missingness into the cumulative object.

Filtered, forecast, and smoothed paths must remain distinct. A filtered path is eligible for prospective claims because it uses information available at the time. A smoothed path may be useful for retrospective sensitivity analysis but cannot be presented as if it were observed online. A forecast is a prediction and should be scored as such.

Version-spanning accumulation needs linking studies. If sensors, channels, code, calibrations, or models change, the path may need to split. Historical rescoring can support comparison, but it must not erase the original contemporaneous output. The record should show what was known then and what a model estimates.

The resulting cumulative estimate is a distribution or identified set over paths under stated assumptions, never an account balance. It should carry observed and modeled intervals,

abstained regions, version boundaries, lag and cadence uncertainty, missingness assumptions, and invalid windows. A smooth curve without that status line is a visualization, not a defensible ledger estimate.

9.2.6 — Reported Quantity, Decision Rule, and Abstention

The reported quantity is the public-facing output of the pipeline. It may be a point estimate, interval, posterior distribution, identified set, directional statement, component profile, conflict report, or abstention. The report must name which quantity it is and which estimand and estimator produced it (Gneiting & Raftery, 2007; Manski, 2003).

Decision rules should be declared before the outcome is inspected. They should specify evidence thresholds, equivalence bands where appropriate, calibration requirements, rival comparisons, uncertainty ceilings, missingness limits, and conditions for No adequate estimate. A rule declared after a striking curve is an interpretation, not a confirmatory procedure.

For confirmatory held-out evaluation, fix the outcome, horizon, information set, and scoring rule before inspecting the evaluation data. A scoring rule's mathematical propriety is distinct from whether its selection and use support a confirmatory claim. Calibration, sharpness, coverage, and decision loss answer different questions; predictive success supports the declared forecast relation unless construct and measured-to-latent bridges justify more (Gneiting & Raftery, 2007).

Abstention is a first-class output. The pipeline should abstain when sign is unidentified, calibration fails, observation is too

sparse, burden is excessive, context shifts outside qualification, or model disagreement overwhelms the target. Abstention protects both the participant and the claim. Abstention rates should be reported by context and subgroup, because a pipeline that works only for the easiest cases may create hidden selection in downstream inference.

Public language must remain smaller than the model. It may report a declared within-person contrast, a compatible interval, or a failed bridge. It may not claim to read the person, measure worth, discover a soul, or prove exact terminal closure. This boundary is easier to preserve when every public output names the estimand before presenting the numerical estimate.

Every output must remain traceable to its decision rule, estimator, analysis map, estimand, measurement map, and raw record. That chain separates evidence from an authoritative-looking number. Traceability also permits error propagation: when an upstream map changes, dependent outputs can be located and recomputed rather than silently surviving under obsolete assumptions.

9.3 — Phenomenal-Value Channels and Auxiliary Witnesses

No single channel has direct, complete, and context-free access to lived phenomenal value. First-person report is closest to the person's experience but can be delayed, constrained, shaped, or unavailable. Physiology, behavior, neural activity, clinical observation, and context can reveal important structure while remaining many-to-one with respect to valence. A serious pipeline therefore uses plural evidence without creating a new oracle (Campbell & Fiske, 1959; Shiffman et al., 2008).

The chapter distinguishes candidate phenomenal-value channels from auxiliary witnesses. A candidate channel may contribute to a declared estimator under explicit transformation, sign, lag, and calibration rules. An auxiliary witness helps validate, challenge, contextualize, or explain the channel model. Usefulness does not silently promote a witness into the target.

Plurality also makes disagreement visible. A participant may report fear while physiology suggests low arousal; behavior may look calm while clinical context indicates severe pain; neural and autonomic channels may diverge after medication. The right response may be model revision, state-specific interpretation, wider uncertainty, or abstention. Averaging is not understanding.

The evidential channels perform different jobs. First-person report has privileged access but is affected by language, safety, memory, and response process. Behavior and choice reveal action under constraints. Autonomic, interoceptive, neural, and clinical signals provide auxiliary witnesses with their own lags and

confounds. Context records explain opportunities and exposures. No channel becomes the target merely by being objective or direct.

A plural pipeline also benefits from an explicit assignment of roles. The table below identifies the contribution each channel can make without allowing any one of them to become a universal meter. A channel remains usable only within the context in which its sign, timing, construct relation, and error process have been established.

Table 9.3. Evidence channels, scientific contributions, and retirement triggers.

Evidence channel	Primary scientific contribution	Principal limits and retirement trigger
First-person report	Construct meaning, sign, language, timing, salience, and participant challenge to the model.	Fear, demand, communication limits, recall, reinterpretation, or nonresponse can invalidate the proposed use.
Pairwise or trade-off judgment	Local ordering, bounded contrasts, and possible within-person calibration under declared options.	Choices reflect power, information, obligation, stakes, and opportunity, not experience alone.
Behavior	Approach, avoidance, persistence, withdrawal, performance, and response to intervention.	Many-to-many relation with experience; strategic behavior or context change defeats simple interpretation.

Evidence channel	Primary scientific contribution	Principal limits and retirement trigger
Autonomic physiology	Timing of arousal, effort, threat, recovery, and regulation under local calibration.	Sign ambiguity, medication, movement, illness, device artifact, and person-specific gain require narrowing or removal.
Interoceptive evidence	Awareness and interpretation of bodily states, including mismatch between physiology and felt state.	Access, language, learned meaning, and interoceptive accuracy vary; bodily signal is not self-interpreting.
Neural evidence	Mechanistic, temporal, or state-classification information relevant to pain, reward, salience, memory, and consciousness.	Task and population transport failure, many-to-many mapping, and classifier drift prohibit direct valence or personhood inference.
Clinical observation	Illness, medication, care context, delirium, pain behavior, communication capacity, and acute change.	Observer bias, workload, familiarity, institutional routine, and care effects must enter the observation model.
Contextual or relational witness	Confounds, timing, social setting, bystander evidence, and explanation of missing or conflicting channels.	Purpose creep, dossier formation, bystander privacy, and narrative overreach require minimization or deletion.

9.3.1 — First-Person Reports and Pairwise Judgments

First-person report remains indispensable because the participant is the only source with direct experiential access. Momentary

ratings, free text, event-contingent reports, pairwise comparisons, and ordinal judgments can reveal qualities that external channels miss. Their privileged position is epistemic, not infallible (Shiffman et al., 2008; Stone et al., 2007).

Reports vary in what they ask. A rating of intensity, unpleasantness, relief, agency, meaning, or desire is not interchangeable with a global mood score. Pairwise judgments may be easier than numerical ratings when a person can say that one interval was more burdensome without knowing how much more. The representation level should match the response task.

Timing matters. Retrospective reports can be shaped by peaks, endings, memory, present mood, and narrative reconstruction. Immediate prompts can reduce recall error but increase burden and reactivity. The protocol should declare which form is used, why, and what temporal object it can support.

Language and access also matter. People may lack concepts, fear consequences, communicate atypically, or choose silence. Accessible response modes, participant-chosen vocabulary, supported communication, and qualitative context can improve interpretation. Nonresponse should not be coded as low burden by default.

Reports can falsify a model locally. If a participant says that the labels are wrong, the sign is reversed, the timing is mistyped, or the instrument misses a central dimension, the disagreement is evidence about construct validity. The model may ultimately retain its estimate, but it must record and adjudicate the challenge.

First-person report should be neither sovereign nor dismissed. It anchors construct meaning, supplies sign and context, supports calibration, and can reveal model failure, while remaining vulnerable to memory, language, demand, fear, and missingness. Its evidential role must be declared rather than inflated or erased.

9.3.2 — Trade-Off Choices and Behavioral Evidence

Choices and trade-offs can reveal what people are willing to accept, avoid, delay, or exchange under specified conditions. Pairwise choices, willingness-to-pay or willingness-to-wait tasks, effort allocation, and approach-avoidance behavior can contribute evidence about relative burden or relief. They remain behavior under constraints, not transparent windows into experience (Kahneman et al., 1993; Hernán & Robins, 2020).

Section 8.1.3 establishes why constrained choices are not direct valence readings. In the measurement pipeline, behavioral evidence enters only through an explicit model of the option set, incentives, obligations, risks, and information that shaped the choice; without that bridge, the inference stops at observed behavior.

Behavioral evidence is also state dependent. Withdrawal can indicate fear, relief, fatigue, protest, depression, sensory overload, or strategic privacy. Productivity can coexist with severe burden, while inactivity can coexist with peace. A behavior channel must name the candidate mechanism and its confounds.

Trade-off tasks may help with local scaling when designed ethically. Repeated within-person choices can indicate ordinal or interval-like relations among bounded states, especially when

paired with reports and contextual anchors. The task must not manufacture suffering, withhold care, or impose stakes that make refusal costly.

Experimental demand is a live rival. Participants may infer that the researcher values consistency, bravery, equality, or a particular sign and adjust responses. Neutral framing, randomized order, comprehension checks, debriefing, and alternative models can reduce but not eliminate this pressure.

Choice and behavior can test predictions, identify discrepancies, and support within-person calibration. Their interpretation becomes stronger when compared with reports and other independently characterized channels.

9.3.3 — Autonomic and Interoceptive Channels

Autonomic channels include heart-rate dynamics, electrodermal activity, respiration, temperature, vascular measures, pupil responses, and related signals. Interoceptive channels concern sensing and interpretation of internal bodily states. These measures can track arousal, effort, threat, pain, recovery, and regulatory demand, but none has a fixed one-to-one relation with signed valence (Cacioppo & Berntson, 1994; Khalsa et al., 2018).

The same heart-rate increase can accompany fear, exercise, joy, fever, medication, or pain. Reduced movement can indicate calm, sedation, exhaustion, disability, or device removal. Sign therefore requires context-specific evidence independent of the outcome the study hopes to support.

The body can participate causally in affective regulation. Hsueh et al. (2023) used an optical pacemaker to increase heart rate in

mice and found increased anxiety-like behavior in risky contexts. The same cardiac manipulation did not produce a detectable preference or aversion in a neutral place-preference test, and inhibition of posterior insular cortex attenuated its anxiety-related effect. This pattern supports a context-sensitive body-to-brain pathway rather than a fixed emotional meaning for heart rate. It supplies a useful constraint for affect models: bodily state and environmental appraisal must be allowed to interact. In noninvasive human observational work, the corresponding model comparison can ask whether a prespecified physiology-by-context interaction improves held-out predictions of reported feeling beyond either channel alone. A declared PTVC route can extend that comparison with its own predicted timing or state interaction.

Channel specifications should include raw units, sampling windows, artifact rules, transformations, expected lags, state scope, and negative controls. Sensor placement, skin contact, posture, hydration, ambient temperature, illness, and hardware version can alter the signal. Technical quality is part of construct validity.

Within-person calibration is often more defensible than cross-person comparison. A channel may track change reliably for one participant while differing in baseline, gain, or state sensitivity across participants. Hierarchical models can borrow strength, but they must not erase person-specific channel behavior.

Interoception adds another layer because bodily signals are not identical to the experience of those signals. Two people with similar physiology may differ in awareness, interpretation, fear,

or learned meaning. A science of valence should model this bridge rather than treat the body as self-interpreting.

Experiments have begun to separate current bodily state from its neural anticipation. In mice, Livneh et al. (2020) found that insular population activity tracked hunger and thirst, while learned food or water cues rapidly shifted activity toward patterns associated with future satiety before physiological replenishment. Artificially activating hypothalamic hunger or thirst systems could restore seeking in sated animals while ongoing insular activity continued to reflect satiety. These dissociations give predictive interoception an empirical architecture: motivation, present physiology, and anticipated physiology can be measured separately. They support a biologically grounded route through which expected bodily consequences may shape affect. A human measurement study can compare models of current physiology, anticipated need, reported feeling, and choice using ordinary eating and drinking records without deprivation or care alteration. The resulting dissociations provide a basis for comparing anticipatory and accumulated-history mechanisms.

Autonomic and interoceptive channels contribute timing, state information, and diagnostic evidence within a plural measurement architecture. Joint interpretation with reports and other channels can identify conflicts and improve the model of experience.

9.3.4 — Neural and Clinical Channels

Neural measures provide temporal, spatial, and mechanistic evidence about processes related to pain, reward, salience,

regulation, memory, and consciousness. Clinical observations identify state, illness, medication, distress, communication limits, and care context. The measurement model specifies how these observations inform the target construct (Giacino et al., 2018; Feldman et al., 2024).

Neural patterns are many-to-many. A network associated with pain can also participate in salience, attention, expectation, or control. A classifier trained on one task may fail in another population or under medication.

Person-specific neural measurement has already advanced beyond brief laboratory snapshots. Shirvalkar et al. (2023) paired repeated pain reports with ambulatory intracranial recordings over months in four people with refractory neuropathic pain and decoded fluctuations in chronic pain severity within individuals. Huang et al. (2025) extended naturalistic acute-pain work to 12 people undergoing clinically indicated epilepsy monitoring, combining neural recordings, reports, and facial dynamics. Their within-person neural classifier distinguished high from low reported pain with a mean area under the receiver-operating-characteristic curve of 0.70. Together these studies demonstrate feasible repeated measurement of a limited experiential dimension in real clinical settings. With consent and existing clinical oversight, approved secondary analyses can test temporal stability, channel disagreement, and generalization to later recordings without adding implants or changing care. These are concrete measurement milestones on which a future phenomenal bridge can build.

Clinical channels carry their own expertise and limits. Nurses, physicians, therapists, and caregivers may recognize pain behavior, delirium, fatigue, dyspnea, or sudden change that a device misses. Their judgments can also be shaped by workload, bias, familiarity, communication style, and institutional routine.

Clinical intervention can also test whether an individualized neural marker connects to a change in lived symptoms. Scangos et al. (2021) mapped symptom-related activity and stimulation responses in one person with severe treatment-resistant depression, then used a chronically implanted device to trigger stimulation when the selected biomarker appeared. Symptoms improved rapidly and remained improved during the reported follow-up. The intervention links a timed internal-state estimate with an independently reported improvement. Consented secondary analysis of clinically approved records could compare symptom-responsive, time-only, and state-insensitive models while preserving treatment and privacy. The scientific gain would be a better bridge between measured dynamics and reported experience.

Clinical context can transform other evidence. Sedation may reduce report and movement; fever may alter autonomic signals; analgesia may change pain while leaving fear; delirium may alter expression and memory. These variables should enter the observation model rather than appear later as excuses.

Consciousness and communication assessments identify retained capacities, available response channels, and state-specific uncertainties. Those findings guide the choice of measurements and the intervals in which each channel is interpretable.

The proper role of neural and clinical evidence is triangulation under declared context. They can support state classification, timing, rival discrimination, and construct revision. They cannot overrule the person by default or transform a candidate measurement system into a consciousness oracle.

9.3.5 — Contextual and Relational Witnesses

Context records the conditions under which reports and signals acquire meaning. Medication, care procedures, sleep, temperature, noise, privacy, social contact, workload, language, legal pressure, and material security can alter both experience and observation. A channel without context may be technically precise and scientifically wrong, while a context record without minimization can become an unnecessary private dossier (Shiffman et al., 2008; Little & Rubin, 2019; Nissenbaum, 2010).

Context collection must be purpose limited. A study should record what is needed to interpret the declared estimand, not every detail that might later become interesting. Rich context can easily become a life dossier containing relationships, habits, disability, health, religion, location, and conflict.

Relational witnesses can help when communication is limited or patterns are familiar. Family members and clinicians may notice changes in fear, pain, relief, sleep, or engagement. Their observations remain interpretations shaped by closeness, incentives, grief, fatigue, culture, and power.

Bystander privacy is therefore part of the measurement design. A participant's experience may reveal information about relatives, caregivers, coworkers, or clinicians. Consent, minimization,

access separation, redaction, and deletion rules should extend to those whose lives enter the record indirectly.

Context also provides rival explanations. A change in signal after a device update, room move, care escalation, visitor arrival, or medication change may defeat a Law-specific interpretation. The pipeline should prefer the explanation that survives contextual controls, not the one that best fits the project's hopes.

A contextual or relational witness earns influence through a specific task: explaining timing, identifying confounds, supporting sign, clarifying missingness, or challenging a model. It should never become a general moral narrative, loyalty profile, or substitute testimony about the whole person.

9.3.6 — Plural Evidence Without Composite Sovereignty

Plural evidence is necessary because each channel has different access, error, burden, and failure modes. The multitrait-multimethod tradition shows why convergence is informative only when methods do not share the same confound. Agreement can strengthen a claim, while disagreement can reveal construct failure or rival structure (Campbell & Fiske, 1959; Bollen, 1989).

A composite becomes sovereign when component identities disappear. Self-report says fear, physiology says arousal, behavior says stillness, context says coercion, and the dashboard outputs calm. Such compression is not integration; it is an unrecorded decision about which witness was allowed to matter.

Fusion should therefore preserve channel-level estimates, uncertainty, missingness, lag, negative controls, and disagreement. Early fusion, late fusion, hierarchical models, and

ensembles answer different questions. Each should be compared under held-out validation and model-recovery tests rather than selected for visual smoothness.

Channel ablation is an ethical and scientific diagnostic. The study should show whether one channel drives the result, whether removal changes sign, and whether the model remains calibrated for participants with missing or inaccessible channels. A system that works only for one bodily or communicative template is not universally qualified.

Participant interpretation adds another check. People should be able to see which channels contributed, what the model inferred, where uncertainty entered, and how to contest an obvious error. Contestability does not make every objection dispositive; it makes the measurement relation answerable.

9.4 — Dynamic Measurement and Channel Conflict

Channels run on different clocks. Reports may be delayed by reflection, response-time features may capture uncertainty rather than valence, autonomic signals can rise and recover on different timescales, and device records can be shifted by clock drift or late synchronization. A dynamic model must therefore estimate or bound channel-specific lag before disagreement is interpreted as psychological conflict (Shiffman et al., 2008; Sosa-Hernandez et al., 2024).

Cross-channel disagreement is a diagnostic object. It may reveal mixed affect, lag, medication, sensor failure, context change, or a misspecified latent state. Fusion should preserve the pattern long enough to test these explanations; a composite that becomes more confident by averaging away conflict has converted uncertainty into decoration.

The technical burden is therefore local: every channel needs a response model, a link function, a lag strategy, an artifact strategy, and a removal rule. These specifications make fusion interpretable and open to independent evaluation.

9.4.1 — Impulse-Response Functions

An impulse-response function describes how a channel reacts over time to a change in the latent state. One signal may spike immediately and decay. Another may rise slowly and persist. Another may respond only after a threshold. Without response functions, a model may compare channels at the wrong moment and call the disagreement psychological (Shiffman et al., 2008; Dejonckheere et al., 2022).

For a causal linear time-invariant channel, or a justified local approximation to one, the following continuous-time representation makes the lag structure explicit before fusion or deconvolution:

$$X_k(t) = \int_0^{\infty} h_k(r)F(t-r)dr + \varepsilon_k(t)$$

The kernel $h_k(r)$ describes how channel k weights past latent states. A narrow kernel means the channel tracks rapid change. A broad kernel means it smooths history. A delayed kernel means the channel carries old burden into the present. The kernel is a scientific assumption about physiology, not a mathematical ornament.

Impulse-response functions help prevent false compression. If a slow channel is aligned to a fast channel without deconvolution, both may appear to converge near an endpoint because the slow channel cannot respond quickly enough. Conversely, a fast artifact may look like a sudden valence shift when the true state has not changed. The kernel makes these errors testable.

The response function also affects burden. A device or prompt cadence chosen for mathematical convenience may miss rapid suffering or overmeasure slow change. The stack should choose cadence by the dynamics of the channel and the burden of observation, not by the convenience of uniform rows in a table.

When a response function cannot be estimated, the system should mark lag uncertainty. It may still use the channel for broad state classification or auxiliary validation, but it should not let the channel dominate a fine-grained interval-contribution path.

9.4.2 — Ordinal and Non-Gaussian Observation Models

Observation families require distribution-specific models. Reports may be ordinal, behaviors counted, device outputs bounded, and physiological signals censored, saturated, skewed, heavy-tailed, or zero-inflated. Forcing all channels through one Gaussian observation model can manufacture apparent agreement, attenuate extremes, and distort uncertainty (Bollen, 1989; Särkkä, 2013; Shiffman et al., 2008).

The observation model, including its link function, should match the measurement process. An ordinal report may need a cumulative link model. A count of awakenings may need a count distribution. A bounded probability may need a logit-scale model. A saturated sensor may require censoring. A rare artifact may require a mixture or outlier model. These are not statistical luxuries; they define what the observation can say.

A misspecified observation model can create false certainty. Treating ordinal categories as equal intervals may overstate change. Treating bounded data as unbounded may invent extremes. Treating heavy-tailed artifacts as normal noise may pull the latent state. Treating a thresholded channel as continuous may make the model insensitive near the very state it is supposed to capture.

Non-Gaussian observation models also affect fusion. A high-confidence continuous channel may dominate a low-resolution ordinal channel unless the model respects each source's information content. Conversely, a coarse report may carry sign

information that the physiology lacks. The model should not rank channels by mathematical smoothness.

Public communication need not teach every link function, but it must expose the consequences. The reader should know when a channel is coarse, censored, delayed, saturated, thresholded, or uncertain. The governing principle is translation without category theft: mathematical smoothness must not make a channel appear to have spoken in a richer language than it did.

9.4.3 — Lag Estimation and Deconvolution

Lag estimation asks how far each channel is from the latent event it is meant to reflect. Deconvolution asks whether the observed channel can be unsmoothed enough to recover the underlying trajectory. Both are inverse problems whose answers depend on the kernel, cadence, regularization, noise, and missingness model. They can improve temporal alignment, but they can also create patterns that were not present in the observations (Särkkä, 2013; Shiffman et al., 2008).

A confirmatory candidate momentary estimator should fix in the protocol which lags are fixed, which are estimated, which priors are allowed, and how lag uncertainty enters the result. A development study may compare lags freely. A confirmatory study may not choose the lag that best produces terminal convergence. Endpoint-tuned lag selection is a hidden form of post hoc rescue.

Deconvolution can amplify noise and become unstable when the channel is strongly smoothed, sampling is irregular, or high-frequency information was never observed. A slow signal may be

biologically slow, technically slow, or both; no algorithm can recover frequencies that the design and noise have erased without adding assumptions. The stack should report regularization, lag sensitivity, simulation recovery, and failure cases before any deconvolved trajectory enters a ledger (Särkkä, 2013; Peeters et al., 2026).

Temporal alignment also creates fairness risk. A person in pain may remove a device, delay a report, move less, or receive medication. If the lag model interprets the resulting timing as state recovery, it has turned burden into apparent improvement. The missingness and context layers must travel with lag estimation.

Time is part of the construct, not an index column. A rate proxy needs the correct relation to clock, exposure, cadence, and channel lag before it can become an interval contribution. If that relation is not identified, the pipeline may retain broad temporal ordering or state classification while withdrawing fine-grained path integration.

A clean stack reports both aligned and unaligned views during development. If the conclusion exists only after one favorable alignment, the result belongs to sensitivity analysis, not confirmation.

9.4.4 — Conflict as Evidence

Channel conflict is not a nuisance by default. It may be the most informative part of the measurement event. A participant reports calm while physiology shows arousal. Facial muscle activity suggests positivity while respiration suggests distress. Sleep improves while reported dread worsens. Clinical observation

notes pain while the wearable appears stable. Each conflict has to be interpreted, not averaged away (Bollen, 1989; Tsiatis, 2006).

Conflict can reveal mixed states. It can also reveal mismeasured states. It can identify a model outside context of use, a lag problem, a sign problem, a device problem, or a confound. The chapter's rule is that conflict increases the burden of explanation before it increases authority. A composite should not become more confident by hiding disagreement inside a total.

A good stack should preserve channel-level traces, contribution plots, posterior disagreement, and ablation results. It should show whether the final estimate depends on one channel, whether auxiliary report supports or contradicts physiology, and whether conflict clusters by person, site, device, state, or time. Conflict with pattern is evidence about the instrument.

Conflict can also protect participants. If a person says the model is wrong, the system should have a route for contestability. Their challenge may reveal a context not recorded, an artifact not flagged, a meaning dimension outside scalar valence, or a state that the instrument has never learned. Contestability is not a courtesy; it is a source of validity evidence.

Hidden burden often appears where indicators diverge: outward function with exhaustion, calm behavior with fear, reported numbness with sustained mobilization, or caregiving warmth with depletion. Preserving disagreement is therefore not indecision. It is how the pipeline avoids making the easiest channel sovereign over the rest.

Fusion is licensed only after conflict has been interrogated. Until then, discordance is not clutter. It is the site where the stack learns whether it is measuring burden or merely organizing signals.

9.4.5 — Medication and Device Effects

Medication, sedation, and treatment can alter observed channels without changing lived experience in the same direction. They can affect heart rate, respiration, movement, sleep, arousal, facial expression, report capacity, and response timing, while illness and relief may change simultaneously. The protocol must therefore model treatment timing and plausible channel effects instead of reading pharmacology as a change in the Law-facing construct (Shiffman et al., 2008; Khalsa et al., 2018; Giacino et al., 2018; Feldman et al., 2024).

Device effects can be just as strong. Sensor placement, skin contact, temperature, battery behavior, firmware updates, charging routines, device removal, data compression, sampling frequency, and manufacturer preprocessing can change the signal. These changes may correlate with burden because people remove devices when uncomfortable, staff adjust devices during care, or batteries fail during long sleep or hospitalization.

A measurement pipeline should record medication windows and device status as structured provenance when needed for the declared context. This does not mean recording every possible clinical fact or turning research into a surveillance inventory. It means documenting the variables that can invalidate the channel interpretation.

Medication and device effects should be included in negative-control designs. If a care-procedure flag explains a large apparent candidate momentary estimator change, the Law-facing interpretation should narrow. If a firmware update shifts estimates across participants, version migration must occur before historical and current data are combined. If device contact quality predicts the final candidate momentary estimator, contact quality is not merely a nuisance; it is a rival.

In palliative contexts, care effects are especially sensitive. Comfort medication should never be delayed or adjusted for measurement clarity. The model must adapt to care; care must not adapt to the model. If medication makes the candidate momentary estimator uninterpretable, the result is abstention, not experimental pressure.

Treatment and hardware belong inside the observation model. A signal arrives through a body receiving care and through a device with its own response, calibration, failure, and version history; neither may be treated as invisible infrastructure.

9.4.6 — Channel Failure and Removal

A channel can fail analytically, technically, constructively, ethically, or contextually. Noise, saturation, missingness, medication, device drift, sign ambiguity, burden, privacy risk, or mismatch with the target may each make a channel unusable for one context of use. Removal should name the failure and the claims affected rather than treating the channel as generally bad.

Removal and down-weighting may not be selected by outcome. Dropping the witness that disrupts the preferred trajectory, or

reducing its weight until channels agree, turns model selection into result construction. Reliability, artifact, burden, and context rules must be frozen or evaluated in a separate development stage (American Educational Research Association et al., 2014; Nosek et al., 2018).

Channel composition is part of the estimator version. If one participant is estimated from five channels and another from two, their outputs are not automatically comparable; if firmware changes gain, pre- and post-update paths require a migration study or version split. Missing or removed channels must propagate into uncertainty and claim status.

Participant burden can independently justify removal. A sensor may disturb sleep, irritate skin, expose participation, interfere with care, or become intolerable during distress. Requiring unusual endurance changes the sample and observation process, so burden-triggered removal protects participants and validity.

Disagreement should be preserved long enough to diagnose. It may indicate wrong sign, different lags, dissociation, medication, illness, state change, or genuinely different constructs. Negative controls at channel, person, site, and time levels, plus leave-one-channel-out and leave-one-sensor-family-out analyses, show whether a result is robust or dominated by one witness.

The conflict patterns below illustrate why a composite should not override its components. Calm physiology beside reported suffering, a post-update shift across all participants, terminal variance compression as channels disappear, or one channel dominating the estimate each demands a different response.

Table 9.4. Cross-channel conflict patterns and required responses.

Conflict pattern	Possible source	Required response
Report says suffering, physiology looks calm	Dissociation, medication, learned suppression, wrong sign, sensor insensitivity.	Preserve discordance, examine context, block overriding participant report.
Physiology changes across all participants after update	Firmware, preprocessing, calibration, or device batch effect.	Version split, migration study, no silent pooling.
Terminal variance compresses as channels disappear	Observation shutdown or comfort-related device removal.	Model missingness and test against false compression.
One channel drives composite	Weight dominance, artifact, true mechanism, or poor fusion design.	Leave-one-channel-out analysis and component reporting.

Early, late, hierarchical, ensemble, and adjudicative fusion address different questions and hide different failures. The table states these commitments so performance is not mistaken for a validated phenomenal bridge. Comparisons among fusion strategies should use the same protected cases, revealing whether apparent improvement comes from better inference or more flexible adjudication.

Table 9.5. Fusion designs, failure modes, and diagnostics.

Fusion design	Strength	Main failure mode	Required diagnostic
Early fusion	Fuses standardized features before fitting, enabling interactions.	Feature scale, lag, and artifact differences are hidden inside one input space.	Preprocessing audit, feature ablation, negative-control dominance tests.
Late fusion	Keeps channel models separate before combining predictions.	Consensus may conceal systematic disagreement or shared confounds.	Channel-level posterior display and conflict report.
Hierarchical latent model	Models shared state, loadings, uncertainty, and missingness jointly.	A well-fitting factor may be arousal, illness, or device gain rather than valence.	Location, scale, sign, recovery, and rival-generation tests.
Ensemble model	Improves prediction across heterogeneous learners.	Performance can rise while interpretability and claim status fall.	Person-held-out, time-forward, site-held-out, and device-held-out validation.
Adjudicative fusion	Preserves discordance for expert review and context annotation.	Can become subjective rescue if rules are not prospectively specified.	Written adjudication criteria, blinded review, and audit trail.

Concrete discordant cases should remain visible. Grief may combine deep sorrow with love, opioid treatment may lower pain reports while altering respiration and arousal, and a wearable may disappear during anxiety because the device itself becomes aversive. These cases distinguish construct conflict, treatment

effect, observation loss, and burden rather than forcing them into one residual. Keeping these cases typed separately also lets model recovery test whether the pipeline can distinguish genuine latent disagreement from observation-system disagreement.

A disciplined model keeps uncertainty over channel state as well as over the latent state. When conflict cannot be resolved under protocol-fixed rules, the output may widen, split by channel, narrow to a local claim, or return No adequate estimate. The person does not become less credible because the pipeline has reached its identification limit. Channel-specific uncertainty can preserve useful evidence from one witness without forcing agreement with another whose meaning is temporarily unidentified.

The downstream consequence is versioned and auditable. Removing, delaying, or quarantining a channel changes the measurement map, estimator, calibration, and perhaps the estimand; every dependent interval, path, figure, and public statement must be recomputed or marked nontransportable. A version transition should record which past observations remain comparable, preventing a repaired channel from retroactively changing the status of earlier data.

9.5 — Measured-to-Latent Invariance

A measured signal becomes evidence about latent phenomenal value only through a bridge. That bridge includes representation, calibration, sign, scale, invariance, linking, transport, observation, and missingness assumptions. None is guaranteed by a stable instrument name or a high predictive score. The bridge must be specified, tested, and allowed to fail (Meredith, 1993; Putnick & Bornstein, 2016).

Invariance asks whether the measurement relation preserves enough structure for the intended comparison. Calibration asks how observed values relate to declared anchors and states. Linking asks whether versions or instruments can be placed on a common coordinate. Transport asks whether a relation learned in one context survives another. These are connected problems, not synonyms.

The measured-to-latent bridge is also noncompensatory. Strong prediction cannot repair an unidentified sign. A large sample cannot repair a missing neutral origin. Cross-site replication cannot repair a construct that changed across sites. Failure at an upstream bridge narrows the downstream claim before accumulation begins.

Chapter 10 will examine calibration and scale in greater depth. This section establishes the pipeline gate: what bridge strength licenses which type of claim, what uncertainty must travel with the output, and when the correct scientific response is a lower-tier result, an identified set, or abstention.

9.5.1 — Measurement Invariance as a Bridge Gate

Measurement invariance is a bridge gate, not a ceremonial fit statistic. It asks whether the same observed pattern corresponds to the same latent relation across time, people, languages, states, sites, devices, or versions. The answer may be full, partial, local, approximate, or absent, and each status licenses a different comparison (Meredith, 1993; Millsap, 2011; Putnick & Bornstein, 2016; Paap et al., 2025).

For within-stream tracking, the central question is often whether loadings, thresholds, sign, and response behavior remain stable enough to interpret change. Illness, medication, fatigue, learning, disability, and changing language can alter that relation. A stable score label does not make the ruler stable.

Cross-person and cross-context claims require stronger evidence. A channel may be reliable within each participant while differing in gain or meaning across participants. A translated item may preserve rank but not level. A device may preserve timing but not amplitude after a firmware change.

Partial invariance can support partial inference. If some anchors and channels remain stable, the study may retain within-person direction, local contrasts, or a subset of dimensions while withdrawing common-level claims. The surviving claim should be named rather than hidden inside a global score.

Time-varying measurement models may be necessary when change is expected. Flexibility should be bounded, prospectively specified where possible, and evaluated through recovery studies.

A model that allows every loading to move can fit almost anything while making the latent trajectory uninterpretable.

The bridge gate is claim specific. An invariance result identifies which comparisons the instrument supports and which require a revised model, additional calibration, or separate reporting.

9.5.2 — Local Calibration and Context of Use

Calibration links observed outputs to declared reference conditions inside a context of use. It may use repeated anchors, ethically available contrasts, known treatment or environmental changes, cognitive interviewing, and participant interpretation. The first defensible claim is usually local and within-stream; broader transport requires a separate linking argument with its own assumptions and uncertainty (American Educational Research Association et al., 2014; International Test Commission, 2018; Sprangers & Schwartz, 1999).

An anchor is not automatically a neutral origin. Rest after exertion, relief after treatment, ordinary baseline, or a familiar state may orient direction without defining zero. Calibration files should distinguish direction anchors, magnitude anchors, contextual anchors, and origin claims.

Context of use determines the evidence burden. A model intended for exploratory within-person tracking may tolerate wider uncertainty than a confirmatory terminal-window test. A rehabilitation context, pain clinic, ordinary ambulatory study, sleep study, and palliative setting may require different anchors and stop rules.

Calibration burden must be counted. Repeated prompts, challenge tasks, device wear, clinician time, and bridge sampling can change the person's experience or compete with care. A technically stronger calibration can be scientifically worse when the burden of obtaining it exceeds the value of the claim.

Local calibration should report failure and drift. If anchors change meaning, channels decouple, or participant reports contradict the model, the bridge may split into state-specific versions. Continuity is earned through evidence, not presumed because a longitudinal graph is convenient.

The output names the relation calibration supports: within-person direction, a local interval estimate, or a sensitivity range. Additional comparison levels enter the report as their own calibration evidence becomes available.

9.5.3 — Linking and Transport Across Versions and Settings

Linking places instruments, versions, languages, devices, sites, or models into a relation that permits a specified comparison. Common-item designs, common-person designs, bridge samples, anchor events, latent linking models, and historical replay answer different transport questions and require overlap across the range and states for which the link will be used. The method must be chosen for the claim it is expected to carry, and a link qualified for ranking cannot be promoted into common-origin or common-unit comparison (American Educational Research Association et al., 2014; Millsap, 2011).

Version changes are common. Hardware, firmware, preprocessing, prompts, item wording, model weights, and calibration samples may change over a long study. Old and new versions should run in parallel where feasible, or be compared through bridge data and synthetic fixtures.

Transport is stronger than linking. A model may link numerically across devices yet fail in a new clinical state, culture, language, or care context. Transport requires evidence that the relation between observables and the target remains sufficiently stable under the new conditions.

Bridge samples must cover the range that matters. A link built among healthy adults in low-burden states cannot silently support terminal, pediatric, disabled, or high-distress interpretation. Sparse overlap should produce wider uncertainty or separate versions, not confident extrapolation.

Backward compatibility may be partial. Rank order may survive while level comparison fails; timing may survive while amplitude changes; one channel may link while another must be retired. The migration record should state exactly which contrasts remain licensed.

When linking fails, the cumulative path may need to split. Historical estimates remain part of the scientific record, and later reanalysis should not erase what the earlier instrument reported. Version boundaries are evidence about the measurement system, not blemishes to be smoothed away.

9.5.4 — Measured-to-Latent Assumptions

Measured-to-latent assumptions define the inferential bridge between observed records and the phenomenal target. They include construct coverage, sign orientation, origin, unit, measurement-model form, channel response, invariance, missingness, state dependence, and the relation between the measured proxy and experienced exposure. These assumptions are not background prose. They determine which latent claim, if any, the data can support (Bollen, 1989; Borsboom, 2005).

A latent variable is not a hidden fact merely because a model estimates it. Location, scale, rotation, label switching, and model equivalence can leave multiple latent interpretations compatible with the same data. Constraints must be justified independently of the desired Law-facing result.

Measurement error has several layers. Sensor noise, report variability, preprocessing uncertainty, model misspecification, calibration error, and transport error are not interchangeable. A narrow posterior under one model can coexist with large uncertainty about whether the model represents valence at all.

Auxiliary evidence should test the bridge rather than decorate it. Known-groups contrasts, ethically permissible interventions, negative controls, participant interpretation, clinical events, and held-out prediction can support or challenge the proposed relation. None should be selected because it makes the terminal pattern cleaner.

Model recovery is indispensable. Synthetic data should include known states, lags, missingness, sign reversals, response shift,

device drift, and rival generators. The pipeline should recover the intended target when possible and widen or abstain when identification is removed.

The bridge should travel with every result as a compact dependency statement. Readers need to know which assumptions received empirical support, which were imposed, which remain uncertain, and what would force the claim to narrow. Without that statement, a measured line can impersonate a latent truth.

9.5.5 — Bridge Strength and Licensed Claims

Bridge strength should be stated as a ladder of licensed claims rather than one validity label. A channel may support artifact detection, broad state classification, within-person direction, local interval contrast, calibrated latent-state estimation, or measured-to-latent transport. Complete-stream accumulation and terminal adjudication require still stronger endpoint and domain bridges. Success on a lower rung cannot be promoted by rhetoric, and failure on a higher rung does not erase every lower-level use.

At the lowest tier, the pipeline may report observed or transformed channels with no latent interpretation. A stronger tier may support association with reports or context. A further tier may support time-forward within-person tracking under a declared state scope. Each achievement should remain visible.

Interval and path claims require additional gates: rate interpretation, temporal alignment, serial dependence, missingness, calibration, and version continuity. Cross-person or cross-cultural claims require origin, unit, linking, and invariance

evidence. Terminal-window claims require endpoint and observation architecture beyond the measurement bridge itself.

A bridge can support directional robustness without point identification. If all admissible calibrations imply improvement, the study may say so even when the magnitude is uncertain. If signs or levels differ across admissible bridges, the result should be an identified set, sensitivity statement, or abstention.

Measurement evidence can support compatibility, discriminate routes, identify the reference classes in which a bridge works, and guide interpretation of terminal results. The analysis connects these findings through the declared representation, endpoint, observation, and comparison criteria.

The licensed-claim ladder makes public communication easier because it replaces vague confidence with exact scope. A result can be scientifically valuable at a lower tier. The project need not inflate it into a stronger statement to justify the work.

9.5.6 — Bridge Failure, Narrowing, and Abstention

Bridge failure occurs when representation, origin, sign, scale, invariance, linking, observation, or recovery requirements are not met. Because these gates are noncompensatory, success downstream cannot repair failure upstream. A well-fitted cumulative curve cannot rescue an invalid bridge (Manski, 2003).

Failure should be localized by bridge layer and licensed output. Loss of cross-person linking may leave within-person direction intact; loss of scalar level may leave ordinal change; endpoint incompatibility may leave a bounded clinical-state model; and one failed channel need not invalidate another. Salvage is

legitimate only when the surviving object, population, state range, and inference tier are named separately from the failed bridge.

Identified sets show conclusions compatible with plausible origins, links, and calibrations. Incompatible signs or statuses indicate structural ambiguity rather than simply a large error bar.

Apply the canonical evidence states in §§24.2 and 27.4. Here, failed measured-to-latent identification narrows or blocks only the inference that depends on that bridge; collecting more observations does not repair nonidentification.

Later evidence of noninvariance, sign error, or invalid transport requires correction, narrowing, versioning, or withdrawal of affected figures, summaries, and verdicts. Preserve earlier outputs for traceability.

Bridge failure locates unsupported measured-to-latent transformations. Name the failed link, comparisons to narrow or stop, and lower-level claims still estimable. A more flexible fit to the same outcome does not repair the failure.

9.6 — Missingness, Provenance, Versions, and Uncertainty

A measurement is also a provenance chain. Sampling time, prompt version, device and firmware, preprocessing, missingness reason, analyst code, model version, and access history determine what the value means and whether it can be reproduced. Preserving those records is part of preserving the scientific object, not administrative overhead (Shiffman et al., 2008; Wilkinson et al., 2016; Center for Open Science, 2025).

This section treats the observation process as part of the evidence. Missingness can be caused by burden, care, sleep, refusal, device failure, low signal, staff unavailability, privacy, or model exclusion. Version changes can alter estimates. Provenance can determine whether a result is reproducible or merely remembered. The stack becomes more trustworthy when these histories are visible.

The public-risk issue is direct. A clean number without provenance can be quoted as truth. A number with provenance is harder to misuse because its conditions travel with it.

9.6.1 — Missingness Codes

Missingness codes should describe why, when, and at which layer a record is absent. No prompt, no response, refusal, sleep, device removal, low contact quality, care procedure, hospitalization, communication loss, privacy pause, technical failure, and endpoint-related observation shutdown are scientifically different events. Collapsing them into one missing flag destroys information about the observation process and can bias the

measured path (Little & Rubin, 2019; Tsiatis, 2006; Shiffman et al., 2008).

Refusal and withdrawal are not technical failures. They are participant acts within the consent structure. Sleep is not missing in the same way as battery failure. Care escalation is not missing in the same way as ambient noise. Low signal near severe illness is not missing in the same way as routine device charging. The code carries the rival explanation.

The codes should be applied as close to collection as possible and preserved through analysis. If all missingness is collapsed later, the model cannot distinguish a burden-related gap from a hardware-related gap. Apparent ledger smoothness may then reflect loss of evidence rather than change in experience.

Codes should also be auditable across groups and sites. If one site has high staff-unavailable gaps, the issue may be workflow rather than physiology. If one subgroup has high contact-quality gaps, the issue may be device design. If terminal windows show high privacy gaps, the study should respect them rather than impute over them without caution.

A missingness code does not need to shame the person or expose private detail. It needs enough specificity to govern the inference. The most privacy-preserving code that still protects science should be preferred.

Missingness is also where the human cost of measurement becomes visible. Fatigue, pain, fear, communication loss, device failure, and refusal can remove the very intervals most relevant to the estimand. The protocol should record that cost without

converting it into a permanent surveillance archive or treating absence as neutral evidence.

9.6.2 — Informative Observation

Informative observation means that whether a record exists depends on the latent state, observed history, participant burden, care context, device condition, or another variable related to the target. In experience-sampling research, prompt density, study duration, item burden, population, and protocol design can change compliance and therefore who and what remains observed. Observation cannot be treated as an independent window merely because the missing cells are numerous (Diggle & Kenward, 1994; Wrzus & Neubauer, 2023; Peeters et al., 2026).

A compact observation model can represent how burden, device condition, care events, or other state variables alter the probability that a channel is observed:

$$\begin{aligned} & \Pr(R_{kij} = 1 \mid F_{ij}, B_{ij}, D_{kij}) \\ &= \text{logit}^{-1}(\gamma_{0k} + \gamma_{1k}|F_{ij}| + \gamma_{2k}B_{ij} + \gamma_{3k}D_{kij}) \end{aligned}$$

R_{kij} indicates whether channel k is observed for person or stream i at time j ; B_{ij} can represent burden or observation burden; D_{kij} can represent device condition. This logistic form is illustrative rather than canonical. Its scientific purpose is to make observation causes explicit, because those causes can bias the measured path and may require selection models, sensitivity bounds, or abstention.

Informative observation can create false improvement. If high-burden intervals are less likely to be observed, the remaining data

look calmer. It can create false worsening when measurement intensifies during crisis. It can create false compression when terminal discomfort leads to device removal and only stable channels remain.

The stack should combine care logs, device-quality metrics, negative controls, and sensitivity analyses spanning plausible ignorable and informative observation mechanisms. Selection-model and pattern-mixture formulations reveal how conclusions move when unobserved intervals are treated differently, but neither creates identification from unsupported assumptions. If the scientific status changes across credible models, the result is model-dependent and public language must say so (Tsiatis, 2006; Little & Rubin, 2019).

Observation also burdens the participant. Increased measurement can alter behavior, sleep, anxiety, family attention, and clinician workflow. Informative observation therefore belongs both to measurement theory and research ethics. The data-generating process includes the act of data generation.

9.6.3 — Instrument and Model Version

Every candidate momentary estimator or candidate interval contribution estimate output should carry an instrument and model version. Hardware, firmware, sampling frequency, sensor placement, preprocessing, artifact rejection, transforms, signs, weights, priors, missingness model, calibration sample, and code release can all change the output. A value without version is not reproducible evidence (Munafò et al., 2017; Nosek et al., 2018).

Versioning should be granular enough to answer a challenge. Which sensor family produced the raw data? Which firmware processed it? Which preprocessing pipeline removed artifacts? Which composite weights were used? Which dynamic model version generated the posterior path? Which context-of-use statement governed the estimate? These questions are not optional when public claims are at stake.

A model version should include retirement status. Some versions are active for exploratory work only. Some are locked for confirmatory studies. Some are retired because of bias, sign failure, device shift, security issue, or burden. Old outputs from retired versions should not disappear, but their public status should change.

Version changes should be visible in figures and tables. A running ledger that crosses versions should show the version boundary. A sensitivity figure should distinguish historical replay under a new model from the original contemporaneous output. A public summary should not flatten these differences.

Versioning also prevents Artificial Intelligence drift. If an Artificial Intelligence support system changes prompts, models, embeddings, summarization templates, anomaly detectors, or feature flags, the change can affect outputs. The stack should version those support tools as carefully as physiological preprocessing.

Version change creates a new measurement relation unless equivalence is demonstrated. Revised wording, translation, firmware, feature extraction, imputation, and model weights can

all alter the construct or observation process. Earlier and later outputs should carry version labels and bridge uncertainty rather than being joined because their column names match.

9.6.4 — Data and Model Provenance

Provenance records where data came from, how they were transformed, who or what accessed them, which model acted on them, which outputs were generated, and which versions were released. It is the chain of custody for a scientific estimate and the route by which a correction can find every dependent result. Provenance makes that audit reproducible (Wilkinson et al., 2016; Center for Open Science, 2025; National Institute of Standards and Technology, 2020).

A provenance record should include source identifiers, collection time, device and model version, preprocessing steps, excluded intervals, missingness codes, auxiliary sources used, consent permissions, access events, derived outputs, publication labels, and deletion or retirement status. It should be detailed enough for reproduction and limited enough to avoid creating a cross-life dossier.

The stack should distinguish raw data, transformed data, features, posterior paths, candidate interval estimates, running ledgers, summaries, and public figures. Each layer has different sensitivity and different reuse rules. Model-training permission for one layer should not imply permission for every layer.

Provenance also helps participants contest outputs. A participant who sees a summary should be able to ask what data produced it, what context was used, whether a missingness code was wrong,

whether medication was recorded, whether a device failed, and whether an auxiliary witness was included. Contestability depends on records that can answer.

In public science, provenance protects against attractive diagrams. A curve can look definitive after many transformations. A provenance trail reminds the reader that the curve is the endpoint of choices. Some choices are justified; some are uncertain; some may be wrong. The trail is where correction enters.

9.6.5 — Migration and Backward Compatibility

Migration is the controlled relation between one instrument or model version and another. Backward compatibility is the stronger claim that old and new outputs can be compared, pooled, or translated under a documented bridge. Neither follows from improved fit. A new model may be preferable for future work while making historical estimates noncomparable (American Educational Research Association et al., 2014; Wilkinson et al., 2016; Center for Open Science, 2025).

A migration study should use overlapping data where possible, replay old raw data through the new pipeline, simulate known states, test subgroup effects, examine sign stability, compare missingness behavior, and estimate version-specific shifts. It should also ask whether the new model changes historical conclusions. If it does, the change should be reported, not hidden.

Backward compatibility may be partial. A new version may preserve rank ordering but change scale. It may preserve within-person direction but change between-person comparison. It may

work in ordinary settings but not terminal settings. It may link some channels and retire others. The compatibility statement should name the exact claim.

Historical rescoring is especially risky. Reprocessing older data with a new model can be useful for sensitivity, but it should not erase what the old model said at the time. The original output, the rescored output, and the migration method should coexist. A scientific field that rewrites its past silently cannot learn from failure.

The momentary-to-interval estimation pipeline should also define version-spanning public language. A claim supported under version 2 but not version 1 should say so. A result that disappears after migration should narrow. A result that survives migration gains robustness, but only for the claim that actually survived.

9.6.6 — Audit and Reproducibility

Audit begins with a reconstructable measurement dossier: construct, measure and channel selection, administration, scoring, transformations, context of use, validity evidence, versions, code, and data provenance. Reproducing an analysis is insufficient when the measure itself cannot be reconstructed or evaluated (Flake & Fried, 2020; Gagnier et al., 2025; Wilkinson et al., 2016).

The pipeline records dictionaries, units, timestamps, missingness codes, hardware and firmware, preprocessing, features, calibration objects, estimands, models, regularization, uncertainty, and thresholds. Synthetic fixtures and unit tests should detect sign reversal, time misalignment, version drift,

dropout, leakage, and failed abstention without exposing participants. Test fixtures should be versioned with the pipeline, because a check that no longer exercises the current transformation can create false confidence.

Audit also governs access and burden. It identifies who may see raw signals, features, posterior paths, summaries, and public results, and whether reproducibility requirements create disclosure or reidentification risk. Controlled access and secure enclaves can be more scientific than maximal release (National Institute of Standards and Technology, 2020; U.S. Food and Drug Administration, 2023). The release boundary should therefore be part of the reproducibility design, specifying what can be independently checked without exposing information the study never needed to disclose.

Independent review should combine psychometrics, signal processing, missing-data methods, clinical context, privacy engineering, and participant perspectives. Plausible detectors can flag different records, so flags need validation and uncertainty rather than becoming credibility scores (Vogelsmeier et al., 2026). Disagreement among detectors is itself diagnostic and should prompt examination of the error model rather than automatic escalation against the participant or site.

Correction must be executable. Provenance should identify every affected date, device, participant, figure, summary, and downstream model after a firmware or estimator failure, and distinguish public methods and fixtures, controlled features, and records that never leave the protected setting. Executable correction is strongest when provenance reaches machine-

readable identifiers, allowing a defect to be traced through derived datasets without depending on institutional memory.

Table 9.6. Provenance fields and minimum public labels.

Record field	Why it matters	Minimal public label
Missingness code	Distinguishes burden, refusal, sleep, care, device, and staff causes.	Observed, missing, abstained, or modeled.
Instrument version	Prevents hardware or preprocessing drift from masquerading as experience.	Device and pipeline version.
Model version	Tracks weights, priors, architecture, and code release.	Exploratory, locked, retired, or sensitivity.
Provenance chain	Allows audit, contestability, and reproduction.	Source-to-output lineage retained.
Migration status	Governs comparison across versions.	Linked, partially linked, unlinked, or split.

9.7 — Anti-Score, Privacy, and Non-Deployment Governance

The momentary-to-interval estimation pipeline becomes dangerous when people forget why it exists. It exists to test bounded scientific claims about measured consequence structures, not to evaluate persons. Governance is implemented through context-of-use limits, access rules, model permissions, participant rights, deletion routes, and retirement criteria (Shiffman et al., 2008; Stone et al., 2007; National Commission, 1979).

Anti-score rules are part of measurement validity. Fear that an output could affect employment, insurance, care, family standing, or public reputation changes participation, response, and missingness. Purpose limitation and nondeployment therefore protect both dignity and the data-generating process; they are not ethics added after the model has finished (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016; National Institute of Standards and Technology, 2020).

The mechanism is direct. When participants anticipate ranking or surveillance, they may withhold, perform, disengage, or leave, and the resulting sample ceases to represent the context the instrument claims to study. A pipeline that cannot prevent those uses may be scientifically unqualified even if its internal fit is excellent.

9.7.1 — No Human-Worth Score

Candidate momentary, interval, and cumulative estimates may never be used as scores of human worth. They do not rank moral value, civic value, spiritual status, trustworthiness, loyalty, eligibility, productivity, deserved care, deserved privacy, or future potential. Chapter 9 states the prohibition here because the measurement stack is where a bounded research output is most likely to be mistaken for a verdict on a person (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

The rule should be embedded in outputs. A report should display context of use, uncertainty, abstention status, version, and prohibited interpretations. It should not display leaderboard aesthetics, rank colors, badges, moral labels, achievement language, danger labels, or public comparison. The visual design should communicate the same research purpose and context as the accompanying text.

The no-score rule does not prevent participants from receiving understandable information about research outputs when appropriate. It requires that outputs be explained as estimates under uncertainty, not as verdicts on the person. Participant-facing summaries should say what the estimate can and cannot mean, and how to challenge it.

The rule also applies to aggregate public communication. A site, group, country, community, or clinical program should not be ranked by average candidate momentary estimator or cumulative candidate interval contribution estimate unless a specific, ethical,

and scientifically qualified comparative study has been designed for that purpose and stripped of person-evaluation meaning. Even then, the target is measurement behavior or burden conditions, not human worth.

9.7.2 — No Hidden Continuous Monitoring

Continuous monitoring can look scientifically attractive because dense data improve temporal models. It can also become the most invasive form of measurement and change the behavior it is meant to observe. The protocol should use the least intrusive cadence sufficient for the declared question, define off-periods and deletion rules, and require renewed consent when monitoring purpose or intensity changes (Council for International Organizations of Medical Sciences, 2016; National Institute of Standards and Technology, 2020; Shiffman et al., 2008).

Hidden monitoring is prohibited. Participants should know what is collected, when, by which devices, for which research purpose, who can access it, what derived outputs may be created, whether model training is permitted, how withdrawal works, and which outputs can be deleted or retired. Consent to wear a device is not consent to every inference the device might later support.

The least-intrusive-cadence rule is technical. Oversampling can increase burden, alter sleep, change behavior, increase anxiety, and generate missingness. Undersampling can miss relevant dynamics. The cadence should be justified by the channel's response time, the scientific question, and the participant burden. More data are not automatically more fairness.

Private settings require special limits. Home, sleep, intimacy, family conflict, prayer, grief, bathing, caregiving, and ordinary rest can all be affected by sensors. A research stack should use collection windows, privacy pauses, local processing, feature minimization, or participant-controlled modes where possible. The full life is not the data source simply because the Law concerns life.

Continuous monitoring also creates security risk. Raw physiological streams, context annotations, and derived paths can reveal health, habits, relationships, disability, medication, stress, sleep, and location-like routines. Security, access separation, retention limits, and deletion routes are therefore measurement requirements.

The stack should collect the minimum data needed to answer the declared question, not the maximum data that a future model might find interesting.

9.7.3 — Participant Access and Contestability

Participants should receive understandable explanations of outputs returned under the protocol. Access does not mean releasing raw telemetry or posterior draws without support. It means giving the construct, context, uncertainty, limits, and a route to challenge the data or interpretation, while preserving the option not to receive a result that lacks clinical or personal meaning (American Educational Research Association et al., 2014; Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Contestability is a validity tool. A participant may know that a device was loose, a medication was taken, a report was misunderstood, a family event occurred, a sensor was removed for privacy, or a state was mixed in a way the model missed. Their challenge can correct the record, trigger a missingness code, change an exclusion rule, or reveal construct failure.

Contestability should be designed before collection begins. The protocol should specify which outputs participants can view, how they can challenge them, who reviews the challenge, what corrections are possible, whether downstream derived results can be updated, and when a challenge results in abstention rather than revision.

The system should also protect participants from overinterpretation. Return-of-results language should avoid diagnosis, moral interpretation, endpoint prediction, and claims about the whole life. It should say that the output is a research estimate under a declared context. If the estimate is not clinically actionable, that must be clear.

Family and surrogate contexts require careful separation. A participant's access does not automatically grant relatives access. A surrogate role does not create general permission to inspect private affective data. Contestability should respect capacity, consent, privacy, and the dignity of the person whose stream is being studied.

Contestability prevents a research output from becoming a one-way description of a person. When the stack can speak about someone, that person must have a route to correct context,

challenge interpretation, refuse unsupported return, and trigger review without carrying the burden of proving the model wrong.

9.7.4 — No Cross-Architecture Dossier

The privacy architecture should be designed around separation rather than one supposedly perfect vault. Identity, raw signals, derived features, context, calibration files, model outputs, and public summaries have different purposes and retention needs. Linking them by default creates a dossier even when each component was collected for a defensible reason.

This rule is not merely privacy preference. It protects the book series from a structural inversion. The science volume studies how a hypothesis could be tested. It does not authorize public transparency of ordinary persons. The City book may make public power answerable; it does not make private life measurable. The religion book may define optional practice; it does not receive candidate momentary estimator data as devotion. The transition book may sequence institutions; it does not import inner-life metrics.

A dossier would also poison participation. People would reasonably fear that research data could follow them into civic, religious, economic, clinical, or social contexts. The result would be selection bias, concealment, withdrawal, and public mistrust. The science would become less true because the data architecture became threatening.

Aggregated research outputs need caution too. Even deidentified data can be vulnerable when combined with rare conditions, time stamps, location-like routines, or clinical events. The stack should

use minimization, aggregation thresholds, differential privacy where appropriate, or nonrelease when reidentification risk is too high.

The no-dossier rule tells the reader what kind of civilization this science is not building. It is not building a registry of souls. It is building a testable research program whose outputs must remain smaller than the lives that made them possible.

9.7.5 — Artificial Intelligence as Bounded Support

Artificial Intelligence may assist artifact detection, provenance checks, anomaly flags, code tests, model comparison, and documentation under §3.6.4's authority limits and §20.7.4's analysis rules. It may not infer inner truth, assign a human score, decide care, adjudicate disputes, or issue a Law-facing verdict (World Medical Association, 2024). Patterns alone can miss sign, context, missingness, and causation; every output remains auxiliary within its declared context of use.

Version each component by model, release, configuration, input, output, review step, known errors, prohibited uses, and retirement status. A changed component changes provenance and may change the result. Participant and public materials must state when AI assisted, what it did, what it could not do, and who remains responsible; no output may imply direct access to experience.

Negative-control monitoring can flag device failure, site drift, missingness clusters, or version mismatch to inform the instrument's declared abstention rules. AI cannot replace the

participant, clinician, statistician, ethics board, stated hypothesis, or verdict architecture.

9.7.6 — Retirement and Deletion

Each instrument carries versioned, scoped retirement criteria: persistent sign error, subgroup noninvariance, device dependence, excessive burden, broken provenance, unsafe disclosure, security failure, or repeated prohibited use.

Separately map deletion of raw signals, features, embeddings, checkpoints, logs, backups, exports, and identifiers. Explain deletion options, retention periods, and residual-copy records to participants (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

Pull, correct, or relabel dashboards, summaries, model cards, figures, and excerpts that invite scoring, hidden monitoring, eligibility decisions, or claims of inner access.

Purpose separation is architectural. Identifiers, permissions, raw data, summaries, model-training rights, and public artifacts share no universal key or default export path. None transfers automatically into civic, religious, employment, insurance, clinical, donor, or unrelated Artificial Intelligence systems.

Participant and public materials explain collection, uncertainty, context, contestability, and withdrawal without badges, destiny language, moral colors, rankings, or claims of revelation or proof.

The governance table converts no-score, no-surveillance, no-dossier, bounded Artificial Intelligence, and version retirement into enforceable technical and public-language controls.

Table 9.7. Anti-score governance risks and controls.

Governance risk	Technical control	Public-language control
Human-worth scoring	No rank outputs, no eligibility fields, no comparison dashboards.	Research estimate only, never person value.
Hidden monitoring	Purpose-limited collection, privacy pauses, minimization, consent logs.	No surveillance and no implied duty to be measured.
Cross-dossier linkage	Separated identifiers, no shared keys, no secondary model-training use.	No City, government, religious-community, Workgame, archive, donor, or voting linkage.
Artificial Intelligence overreach	Bounded support, versioning, human review, audit trail.	No autonomous valence authority.
Unsafe versions	Retirement criteria, deletion routes, release records.	Corrected, narrowed, or withdrawn outputs.

All releases, including donor and institutional materials, label outputs as observed, filtered, smoothed, forecast, simulated, or sensitivity-based and state their context and uncertainty. Those status labels should remain attached when an output is excerpted, reformatted, or exported, because a forecast or smoothed estimate stripped of its status can be mistaken for an observation.

Visible correction tests credibility. Stop, narrow, retire, or, where authorized, delete unsafe versions; notify affected users. Preserve methods, synthetic fixtures, aggregate performance, incidents, and correction history without unnecessary dossiers (National Institute of Standards and Technology, 2020; National Institute of

Standards and Technology, 2024). The trigger for each corrective action should be tied to the qualification or governance condition that failed, keeping correction proportional to the defect and independent of whether the result was favorable.

The pipeline ends with revocable authority: it estimates only in its declared context, abstains when its bridge fails, and loses its role when evidence or governance no longer supports it. Revocation should remain scoped to the authority actually lost, so failure in one context of use does not automatically invalidate a separately qualified use that still satisfies its own evidence and governance requirements.

Chapter 10 — Calibration, Scale, and Uncertainty

A scale is a promise about comparison. It promises that an order, interval, sign, or zero has the meaning the analysis assigns to it, and that change in the number can be separated from change in the instrument. For a cumulative phenomenal claim, that promise becomes especially demanding. Small errors in origin or unit can accumulate across long exposure, and a stable score can conceal a changed response process. Chapter 10 asks what must be calibrated, what must remain invariant, and what conclusions survive when a common scale cannot be earned.

The narrowest defensible starting point is calibration within one stream. Repeated anchors and contrasts can test whether the same person's responses preserve direction and magnitude across ordinary variation in a declared context. That local success may support a within-person trajectory without supplying a universal zero or common unit. Beginning locally preserves the intrastream structure of PTVC and prevents a convenient between-person scale from being assumed before its bridge exists.

Neutral origin deserves special scrutiny because translation moves zero while positive rescaling preserves it. A tiny origin error can create a large cumulative residual when exposure is long. Midpoint labeling, within-person centering, retrospective mean subtraction, or completed-record normalization can therefore manufacture apparent neutrality. The origin rule must be fixed independently of the terminal outcome, supported by response-process and anchor evidence, and propagated as uncertainty whenever more than one origin remains defensible (American Educational Research Association et al., 2014).

Longitudinal invariance asks whether repeated observations continue to measure the same construct in the same way. Configural, metric, and scalar invariance answer different questions, and partial invariance may license some comparisons while blocking others. Reliability alone cannot establish transport through time. A model can reproduce scores consistently while thresholds, loadings, meanings, or channel relations change. The chapter therefore treats invariance as a claim-specific bridge rather than a badge attached permanently to an instrument (Meredith, 1993; Putnick & Bornstein, 2016).

Response shift turns apparent stability into a scientific question. Illness, treatment, adaptation, grief, recovery, aging, or approaching death can change internal standards, the importance of dimensions, or the meaning of the construct itself. Stable scores can then coexist with changed experience, and changed scores can coexist with stable experience. Current longitudinal psychometric work illustrates how partial invariance and item-level changes can be detected in clinical populations, while leaving transport to other measures and settings open (Sprangers & Schwartz, 1999; Paap et al., 2025).

Language and culture affect more than wording. Translation can change thresholds, connotation, response style, social desirability, emotional categories, and what neutral or severe means in use. Cultural scripts can alter which states are named, hidden, tolerated, or expressed. Cross-cultural linking therefore requires translation procedures, participant interpretation, differential item functioning, anchor review, and evidence that the same comparison survives. When it does not, group-specific models or

noncomparability can preserve more truth than a falsely universal coordinate (International Test Commission, 2018; Henrich et al., 2005).

Consider a hypothetical longitudinal calibration record. The same participant completes one instrument in two languages before and after a medication change, then later uses supported communication after aphasia. The total score remains nearly flat, yet cognitive interviews show that the neutral anchor shifted in translation, one item's threshold changed after medication, and the new interface altered which states could be expressed. A flat trajectory is therefore compatible with stable experience, response shift, translation drift, or changed observability. The study must test those alternatives before it calls either the person or the ruler stable (Sprangers & Schwartz, 1999; International Test Commission, 2018; Paap et al., 2025).

Development, disability, and neurodiversity also change access and expression. Communication form, sensory processing, motor control, interoception, fatigue, cognitive load, dependence on supporters, and device accessibility can alter both the target's expression and the observation process. Inclusive calibration is not achieved by forcing every person through the same interface. It is achieved by testing alternative channels, preserving response rights, and narrowing claims when a bridge cannot travel without erasing the person's mode of communication.

Uncertainty must remain decomposable. Sampling variation, measurement error, calibration uncertainty, origin uncertainty, response shift, missingness, model selection, version migration, and transport each enter through different routes. One total

standard error can hide the source that actually limits interpretation. The chapter therefore favors uncertainty budgets and sensitivity envelopes that show what would need to improve before a stronger claim becomes available.

Calibration authority must remain current; §10.7 governs drift, requalification, correction, and retirement.

Correction can change earlier conclusions, and the system must be designed to carry that change downstream. Reanalysis identifies affected estimates, figures, public summaries, and claims. Superseded versions remain traceable rather than disappearing. A scale may retain within-person use while losing cross-site or terminal authority. Retirement can be local, versioned, and informative. This is how the book turns fallibility into a reproducible scientific process rather than a reason for quiet inconsistency.

Public language is part of calibration. A sentence such as ‘this instrument tracked directional change for this person in this state, language, device version, and treatment period within the stated uncertainty’ can be both precise and useful. The sentence becomes misleading when its context and bridge disappear and the output is described as the person’s true burden. Every public claim must therefore preserve the comparison tier, population, state, version, uncertainty, and prohibited uses that materially define it.

These requirements make a future PTVC test more informative because they identify the work the test must accomplish. A valid within-stream coordinate, independently calibrated origin, stable

sign, justified unit, transparent observation process, and endpoint-appropriate bridge would give cumulative evidence a defensible meaning. A failed protocol or implementation gate narrows any permitted use and makes the evidence Unusable for the affected estimand; a target that is undefined or cannot be identified or validly bounded yields No adequate estimate. Neither status counts for or against a stronger object the study never reached.

The chapter is optimistic about bounded comparability. Full common-scale cardinality is not the only valuable outcome. Local trajectories, stable orderings, identified sets, state-specific calibrations, and transparent noncomparability can each advance affective, psychometric, clinical, and longitudinal science. The program becomes more credible when it can distinguish what travels from what merely looks standardized.

Chapter 10 moves from within-stream calibration through origin, sign, unit, linking, longitudinal invariance, response shift, culture, language, disability, clinical edge states, identified sets, audit, correction, and retirement. Its final product is a maintained contract of comparability with visible limits and challenge routes. Part IV can then take up the next dependency in the chain: which conscious stream owns the experience, how continuity and routing are adjudicated, and where the truth-relevant endpoint lies.

10.1 — Calibration Within the Stream

Within-stream calibration asks whether repeated measurements preserve meaning for one unfolding stream. It must distinguish familiar item use from genuine validity, group-level discrimination from person-level tracking, and stable averages from state-dependent response processes. Reliability must also be attached to the level and purpose of inference: between-person consistency, within-person sensitivity, person-specific precision, and temporal stability are different properties. Only after those tests may the instrument support accumulation or a bridge to broader comparison (Molenaar, 2004; Hamaker, 2012; Vogelsmeier et al., 2024; Castro-Alvarez et al., 2026).

This order is required because the target is not an external commodity. A local baseline, repeated anchor, or stable channel relation can help study change within one stream, but none of them creates an instrument-fixed origin. A person's mean, median, or familiar resting state is not automatically phenomenal neutrality.

Within-person calibration supports time-forward prediction, drift detection, and early diagnosis of measurement problems. Repeated validation can show how the local reference changes with adaptation, illness, disability, culture, medication, sleep, and memory.

The comparison tier matrix below is the chapter's first calibration map. It shows how each step outward from one stream increases the burden of proof. It also prevents a common public error:

treating a strong within-person result as if it already licensed interpersonal comparison.

Table 10.1. Within-stream comparison tiers and calibration burdens.

Comparison tier	What it may support	Minimum calibration burden	Stop rule
Within episode	Short-window direction or responsiveness	State label, sign anchor, artifact rule	Stop when state or artifact cannot be separated
Within person over time	Trajectory and change under declared conditions	Longitudinal invariance, drift checks, response-shift review	Stop or split when bridge changes
Across states	Sleep, illness, medication, or clinical state comparison	State-specific calibration and error costs	Stop when state changes sign, origin, or unit
Across persons or groups	Common-level or same-scale claims	Origin, sign, unit, linking, invariance, transport	Stop when scalar or common-origin claims fail
Clinical or terminal context	High-burden and endpoint-adjacent inference	Treatment, symptom, burden, and observation model	Stop when care or physiology dominates the bridge

A concrete within-stream example helps fix the scope. Imagine a participant whose ordinary afternoon estimates have been stable for three weeks, whose self-reports and physiological channels both move during ethically available pain-relief anchors, and whose device diagnostics remain clean. The study has built a local measurement bridge for that participant under those conditions.

Now the person develops an infection and begins a new medication. Heart rate, sleep, temperature, and fatigue shift at once. Simply appending the new points to the old path would assume that the earlier calibration still applies. A calibrated analysis asks whether the state label has changed, whether the anchor relation still holds, and whether the bridge should split into pre-infection and post-infection segments.

The same example shows why local calibration can still be valuable. Even if the bridge splits, the earlier segment may support within-person estimates, the later segment may support a new development period, and the break itself may teach the instrument how illness changes its relation to burden. Scientific loss at one tier can become scientific information at another.

The interpretive burden in this example is created by a model that values continuity more than accuracy. Refusing to mark the infection and medication break produces a smoother curve, but it also hands later readers a less truthful history and makes every downstream integral look more certain than the bridge permits.

A within-stream calibration file should therefore remain a living, versioned object. Raw context, anchor episodes, state labels, missingness, device and software versions, and model changes

are what allow later investigators to locate where the bridge held, weakened, or failed.

A local calibration dossier should include an explicit nontransfer statement. That statement is not a weakness; it is the exact boundary that lets later researchers decide what additional evidence is needed for travel.

Strong local evidence should produce sharper labels rather than automatically larger claims. The better a bridge performs within one stream, the more precisely the report can state its population, state, version, and time bounds before asking whether any part of it transports.

The final local question is always practical: what next observation would break this bridge? If no answer can be named, the bridge has become a story rather than a measurement relation.

10.1.1 — Personal Baselines and Their Limits

A personal baseline describes a participant's channels in a familiar state, variation in self-report across ordinary days, or physiological changes after rest, pain relief, social contact, sleep, or exertion. It provides a local reference for detecting change (Borsboom, 2005; Cronbach & Meehl, 1955).

Personal baselines and the target origin serve different analytical purposes. A usual level describes a participant's recent experience; independent origin anchors provide the reference for signed accumulation. Keeping both visible helps the analysis detect persistent burden as well as change.

A baseline is therefore useful only when its scope is named. A quiet morning baseline may not travel to fever, grief, opioid medication, delirium, panic, or terminal weakness. A device baseline may shift with skin contact, hydration, posture, arrhythmia, sensor gain, or environmental temperature. A report baseline may shift with language, expectation, or what the person has learned to tolerate.

The calibration file for a stream should record the baseline context: time of day, wake state, medication timing, activity, symptom level, device version, sampling cadence, and the person's own description of what was happening. Without that record, later investigators cannot tell whether change reflects experience, measurement, or context.

The calibration question is which burden disappears when a personal baseline is treated as neutral. Chronic pain, fatigue, fear, isolation, or learned endurance can become the background against which later states are measured, so a local baseline must be described as a reference condition rather than an independently established neutral origin.

The safe language is therefore narrow. A baseline may support within-stream deviations under declared conditions. It may help detect change. It may inform person-specific prediction. It may not be used as a common zero, a proof of adaptation, a rank of persons, or a way to say that a long-carried burden has ceased to count.

10.1.2 — Repeated Anchor Events

Anchor events are moments whose direction is known well enough to test whether the measurement relation behaves as expected. The ethical word is crucial: anchors must be available without inducing harm. Relief after ordinary care, increased discomfort during naturally occurring pain, recovery after rest, dyspnea during documented exertion, or calm after a chosen calming practice can become calibration evidence only when the event would occur without the study manufacturing it (Shiffman et al., 2008; Stone et al., 2007).

An anchor is stronger when it is repeated across time and context. One pain-relief episode can orient a channel; several episodes across days can test whether the relation is stable. If the physiological estimate moves in the expected direction only once, the result may be coincidence, medication artifact, reporting delay, or care-context confounding. Repetition lets calibration become more than an anecdote.

The anchor design should separate sign, magnitude, and timing. A person may report relief before autonomic change settles. Movement may precede dyspnea. Analgesia may reduce distress while altering respiration or heart-rate variability in ways that mimic calm. The anchor therefore needs a time window, lag plan, negative controls, and a record of co-occurring care.

A repeated-anchor file should name the anticipated direction before the data are examined. If the study waits to see which channel changed and then declares that channel the anchor, it has

calibrated to the outcome. That is not anchor validation. It is post hoc fitting in clinical clothing.

Anchors also protect against false universality. A pain-relief anchor for one person may not orient the same channel in another person with neuropathy, disability, medication tolerance, or different expression norms. Even within one stream, the same anchor can change meaning after illness or adaptation. Anchors require requalification when the life context changes.

Repeated anchors can establish a practical measurement foothold in ordinary life. They test whether an instrument reliably tracks the direction of burden under its stated conditions. Passing that qualification supports the next measurement stage; a failed result identifies what must be repaired before larger claims are made.

10.1.3 — State-Dependent Calibration

State dependence is expected, not exceptional. Pain, arousal, sleep, medication, illness, posture, fatigue, fever, social setting, and attentional demand can alter both experience and the channel that witnesses it. Calibration should model those state-conditioned relations or narrow the context, rather than assume one observation law across every body and circumstance (Khalsa et al., 2018; Feldman et al., 2024; Shiffman et al., 2008).

A useful first model lets the measurement relation depend on an observed or independently adjudicated state. In prose, the signal observed from channel j at time t depends on a state-specific intercept, a state-specific loading on the latent target, and residual error. This model is deliberately simple. Its job is to turn presumed stability into a testable claim and to expose where state

uncertainty, parameter flexibility, or weak anchors make calibration unidentified.

If loadings change by state, the same latent change can produce different observed changes. If intercepts change by state, the same observed score can correspond to different latent levels. If residual variance changes by state, confidence can be narrower in some conditions and dangerously overconfident in others. Each difference has a different consequence for the measured cumulative estimate.

The state categories and their observation parameters should be specified before confirmatory analysis. Calibration must then be demonstrated across the clinical, behavioral, and device states in which the resulting estimate will be interpreted.

State-dependent models also prevent a common fairness error. A person whose body responds differently should not be treated as less measurable, less credible, or less available to science. The model's job is to ask what relationship can be defended in that state. When it cannot be defended, the instrument abstains; the person is not discounted.

The practical requirement is a state label with uncertainty. Every record should carry wake state, medication exposure, symptom context, device context, and clinical state when available. Where labels are uncertain, the analysis should propagate that uncertainty rather than assigning a crisp state because the model needs one.

10.1.4 — Calibration Drift

Calibration drift is a change in the measurement relation that may develop gradually. Monitoring can help detect that change before it causes a major inferential error. The device may age, the skin contact may change, the person may learn the prompts, the clinical routine may shift, the illness may progress, or the meaning of a response option may evolve. The scale still produces numbers, but the numbers no longer carry the same relation to the target (Russell, 1980; Cacioppo & Berntson, 1994).

Drift can be technical, physiological, interpretive, or contextual. Technical drift includes sensor gain, software preprocessing, battery behavior, firmware changes, and replacement devices. Physiological drift includes medication adaptation, autonomic neuropathy, weight loss, inflammation, fever, sleep disruption, and disease progression. Interpretive drift includes changes in how a person understands pain, fatigue, dignity, relief, or endurance.

A calibration system should therefore include drift monitors before it is used for confirmatory inference. Anchor-response stability, negative-control channels, device diagnostics, residual patterns, missingness changes, and self-report discordance can all signal that the measurement relation has moved. Drift is not a nuisance parameter after the fact. It is a reason to pause comparison.

The drift threshold and its action rule must be declared before confirmatory outcome inspection. Changing either to retain a favorable result or exclude an unfavorable one makes calibration

outcome-dependent. A justified revision belongs to a new version and is evaluated on independent data.

Drift also complicates accumulated ledgers. A small calibration shift repeated over many intervals can create a large apparent cumulative difference. The longer the integration window, the more dangerous drift becomes. A ledger that spans months or years without drift handling may look precise because errors have been averaged, while actually being biased by a slowly moving scale.

The right response to drift is not panic. It is provenance, correction, and narrowed language. A drifting scale may still support local comparisons before the drift point, or within a stable segment, or under a reestimated bridge. It may not silently support a single continuous conclusion across the break.

10.1.5 — Within-Person Validity

Within-person validity asks whether estimate changes track the intended experience for that person. A measure can separate diagnostic groups or correlate with average burden yet fail to follow one person's trajectory. Because the downstream object is path-based, between-person evidence cannot substitute for within-person sensitivity (Molenaar, 2004; Hamaker, 2012; American Educational Research Association et al., 2014).

A serious within-person design uses repeated observations, time-forward prediction, lag sensitivity, and negative controls. The estimate should track known changes in experience at the time relation justified by its validated response model, without using later data to manufacture an earlier prospective signal. It should

fail to move with irrelevant device artifacts. It should be sensitive to direction, not merely magnitude.

The analysis should separate within- and between-person effects. A channel can be higher among more distressed people on average yet fall within one person during acute distress as medication, withdrawal, or fatigue changes the signal. Such reversals are common in longitudinal science and should be expected calibration hazards, not surprises.

Validity must be tested against competing explanations. If an estimate rises during pain but also rises with motion artifact, fever, battery heating, family visits, or clinical procedures, the interpretation is not secure. The relevant question is not merely whether the signal changes, but which causal and observation processes could have produced that change.

A within-person validity file should preserve raw channels, preprocessing version, state labels, prompts, anchor events, missingness codes, and model version. Without it, later claims cannot be audited. A beautiful curve without provenance is not a measurement result; it is a picture with hidden conditions.

The payoff is worth the discipline. Strong within-person validity gives the program a credible way to study ordinary and clinical dynamics without overclaiming interpersonal comparability. Failure is informative: the instrument may need a different state model, narrower context, or retirement for that use.

10.1.6 — No Automatic Between-Person Transfer

The hardest calibration error is the leap from one stream to many. An instrument may track one person's directional change while

sharing neither origin, unit, response process, nor channel gain with another; within-person success therefore does not establish interpersonal comparability, population rank, or a common measure of burden (Molenaar, 2004; Hamaker, 2012).

Comparison should advance through explicit tiers: within episode, within person over time, across states, instruments, groups, and persons. Each tier adds anchors, linking, invariance, and transport assumptions. Between-person transfer further requires comparable anchor function and preserved sign and interval meaning across physiology, disability, language, culture, development, treatment, and response style; identical wording or standardized scores are insufficient (Putnick & Bornstein, 2016; American Educational Research Association et al., 2014).

A calibration sheet records state, medication, device version, anchors, response process, missingness, burden, and permitted comparison. If anchors move or reporting changes, split the series, widen the identified set, or withdraw the estimate. Person-specific and population models answer different questions; pooling requires tested shared structure.

Partial transfer may support rank without level, direction without interval magnitude, or some anchors without a common zero. No person carries another person's scale: every comparison must name the tier, population, state, instrument, bridge, and uncertainty earned, and broader use requires new calibration.

10.2 — Origin, Sign, Unit, and Linking

Every common-scale claim rests on coordinate choices. Origin, sign, unit, and linking are not clerical details; they determine which mathematical statements are meaningful. If origin fails, terminal neutrality cannot be stated as a common-zero result. If sign fails, burden and relief cannot be distinguished. If unit fails, integration has no stable cardinal meaning. If linking fails, instruments or persons do not share one coordinate (Krantz et al., 1971; Borsboom, 2005).

Coordinate honesty is more useful than generic skepticism. A scale may support order without distance, change without a common origin, or bounded sets without a point estimate. Each achievement should be stated at its actual level. Standardization changes a coordinate; it does not independently identify the zero or unit the hypothesis requires.

Origin, sign, unit, and linking receive separate rules because each can fail independently. Keeping them distinct prevents convenient rescaling of the formal target and protects participants from being interpreted through a coordinate system that has not earned its authority.

$$y^* = a + by, \quad b > 0$$

This transformation is the chapter's simplest warning. Positive rescaling can preserve direction, but translation moves the origin. Any claim that depends on a shared zero must therefore defend the origin independently of the result it hopes to support.

A coordinate example makes the danger visible. Suppose two instruments are both transformed to have mean zero and standard deviation one in their own development samples. The resulting scores look commensurable, and their graphs share a horizontal zero line. But one zero is a development-sample average, the other is a translated instrument mean, and neither is an independently defended neutral origin.

These data can establish useful relations between instruments. Consistent movement after burden and relief anchors can orient sign locally. A bridge sample that aligns slopes under ordinary conditions can support comparison of direction or relative change.

A professional calibration report would therefore mark three statements separately: sign evidence obtained, relative linking estimated with uncertainty, common-zero claim unavailable. That triad states exactly what the linking evidence has established and what remains to be learned.

Label each reported zero by its reference: personal baseline, sample mean, instrument origin, or independently calibrated neutral origin. That label makes the displayed coordinate interpretable.

This is why the affine transformation is central rather than decorative. Translation can move zero without changing many patterns the eye finds persuasive. If the origin is not anchored independently, a terminal-zero statement can be made or erased by coordinate choice.

The coordinate rule therefore travels forward: figures, tables, and verdicts must say which zero is being used, what sign means, what unit is defensible, and whether linking uncertainty changes the claim.

A professional result can carry several coordinate statuses at once. It may have sign, lack common origin, preserve rank, and fail interval meaning. The analysis should let those statuses coexist instead of compressing them into a single verdict.

Coordinate discipline is therefore a way of protecting the positive case. If a pattern still matters after origin, sign, unit, and linking have been separated, it is worth the reader's attention.

10.2.1 — Neutral-Origin Rule

A neutral origin is not a place where the investigator decides to put zero. It is a reference point whose neutrality is justified independently of the outcome being tested. In the Law's scientific program, that independence is essential. If the endpoint pattern itself is used to choose the origin, the test has been wired to favor neutrality (Krantz et al., 1971; Russell, 1980).

Questionnaire centering, person-mean centering, z-scoring, baseline subtraction, and regression residualization can improve numerical stability, remove nuisance variation, or support within-person change analysis. Their reference points and units belong in the measurement specification alongside any separately justified neutral origin.

The neutral-origin rule therefore has three parts. The reference must be specified before confirmatory analysis. It must be justified through anchors, construct theory, or external calibration

not chosen because it makes terminal results look favorable. It must carry an uncertainty interval or failure condition, because an origin with unknown error cannot support exact zero language.

A complementary task in the Cloos et al. (2025) follow-up isolated interface precision: 87.0% of respondents placed a requested value of 31 between 30 and 32 on a digital 0–100 slider. Including such a task alongside affective judgments separates use of the interface from interpretation of the feeling scale. Repeating it across devices and assessment periods provides a direct check on measurement consistency.

Suppose two terminal estimates approach zero after each is centred on its own preterminal mean. The observed pattern is a return toward personal recent baselines. An analysis using a separately calibrated neutral origin asks a different question.

When the origin remains unresolved, the study can examine origin-invariant dynamics, within-person directional change, or rival discrimination. It can also report endpoint proximity to a declared analytic baseline. These analyses preserve the information available while origin calibration develops.

Independent origin calibration keeps persistent burden visible even when personal baselines or reporting conventions change. Record the reference used and show how alternative plausible origins affect the result.

10.2.2 — Sign Orientation

Sign orientation asks which direction means burden increasing and which direction means relief, restoration, or positive valence. This sounds obvious only when the channel is already interpreted.

Physiological channels are rarely so obedient. Increased heart rate can signal fear, effort, fever, joy, pain, medication, or movement. Reduced expression can signal calm, fatigue, sedation, inhibition, or disconnection (Bollen, 1989; Borsboom, 2005).

A sign-bearing anchor must therefore be declared. The study should name which events, reports, or external criteria are allowed to orient sign and which are only auxiliary. A channel that changes during pain relief and during motion artifact cannot be signed by inspection. A report channel may orient sign in one context while becoming unreliable in another.

The sign gate also prevents magnitude-only evidence from masquerading as signed evidence. A model may detect arousal, disequilibrium, effort, or physiological instability. Those can be important constructs, but they are not signed valence. If the observation model cannot tell whether a state is burdensome or restorative, the result must be reported as magnitude, arousal, uncertainty, or abstention.

Sign orientation should be tested for independence from the outcome. If the same terminal pattern that the study hopes to test is used to orient sign, the bridge is circular. Sign anchors should come from ordinary and ethically available events, cross-fitted data, earlier development samples, or independent validation streams. Confirmatory sign should not be learned from the terminal result it is meant to evaluate.

Sign can also vary by person and state. A signal that means relief for one person after medication may mean sedation, fatigue, or withdrawal for another. A sign rule that travels without testing

may turn atypical physiology into apparent contradiction. When sign differs, the model should allow separate sign structures or stop common-scale claims.

Sign is technically indispensable because the proposed ledger is signed. Burden and relief cannot be treated as interchangeable magnitudes, and every integration or comparison must preserve a prospectively defended orientation. When sign is uncertain, the output must remain mixed, set-valued, magnitude-only, or abstaining rather than being assigned the direction the theory needs.

10.2.3 — Unit and Interval Meaning

A unit is a contract about equal differences. If one candidate interval contribution estimate is to represent an interval contribution, then the scale must say where equal numerical changes correspond to comparable changes in the intended construct. Standardized scores do not supply that meaning by themselves. A standard deviation is a distribution fact, not a unit of lived burden (Cronbach & Meehl, 1955; Campbell & Fiske, 1959).

Interval meaning can be local. It may hold across a narrow range of a person's ordinary states while failing at extremes. It may hold for mild to moderate fatigue and fail in severe dyspnea. It may hold across one instrument version and fail after item wording or sensor preprocessing changes. Unit claims should therefore name range, context, and version.

The interval problem also matters for integration. Multiplying a score by time produces an area, but not necessarily a meaningful

contribution. The score must estimate a rate or a declared proportional rate proxy before time integration is scientifically interpretable. Otherwise the ledger becomes arithmetic performed on a label.

A useful unit file should name raw units, transforms, anchors, scaling constants, uncertainty, and the conditions under which equal differences are defensible. If a model uses nonlinear links, ordinal categories, thresholds, or bounded scales, the unit claim must reflect that. Equal category steps on a questionnaire need not represent equal changes in latent experience.

The same caution applies to cross-version and cross-language work. If one version has different item difficulty, different response style, or different sensor preprocessing, unit continuity must be revalidated even if the label remains the same. Historical continuity cannot be asserted because the name of the scale survived.

A unit that cannot be defended does not end all science. It may narrow the analysis to ordinal ordering, monotone change, bounded sets, or within-person rank. The error would be to force interval language because later chapters need a total. The burden of a total must be earned before the total is used.

10.2.4 — Linking Across Instruments

Linking places instruments, versions, languages, devices, or model outputs onto a shared coordinate. It is not the same as correlation. Two measures can correlate strongly and still differ in origin, unit, sign, sensitivity, or state behavior. For the Law, linking must support the exact comparison being made, not

merely show that two instruments are related (Manski, 2003; American Educational Research Association et al., 2014).

A linking design needs overlap that genuinely spans the intended use. Common items, common persons, bridge samples, anchor events, or latent calibration models must cover the relevant range, languages, devices, states, and clinical conditions. A bridge estimated in healthy adults cannot silently connect an instrument used during aphasia, autonomic neuropathy, delirium, or terminal sedation (American Educational Research Association et al., 2014; International Test Commission, 2018; Putnick & Bornstein, 2016).

Linking should be versioned. A device update, questionnaire translation, changed preprocessing pipeline, or new weighting rule creates a new instrument for scientific purposes. During migration, dual scoring should be preserved long enough to estimate the relation between old and new outputs. Without a bridge period, historical comparison becomes a guess.

The linking equation may look simple, but the assumptions behind it are not. A slope and intercept can align means and variances while leaving differential item functioning, response shift, heteroscedastic error, or state-specific bias. A linked score should therefore carry uncertainty and sensitivity results into the measured cumulative estimate, not only into a methods appendix.

The chapter treats linking as a permission with scope. It may link within one language but not another, one age range but not another, one care site but not another, one clinical state but not

another. When the link fails outside scope, the output is not a bad person or a bad population. It is a nontransportable instrument.

The Law's scientific credibility depends on refusing silent backward rescoring. If a new calibration makes old data look more favorable, that is exactly when provenance matters most. The old version, new version, bridge model, uncertainty, and public claim change must all remain visible.

10.2.5 — Affine Freedom and Residual Ambiguity

Affective measures can retain uncertainty in origin and unit. Positive rescaling preserves an identified zero; translation generally does not. Person-specific positive rescalings can therefore preserve each stream's zero while leaving magnitudes incomparable across streams. When the bridge identifies only a family of affine coordinates, propagate the unresolved translations and scales rather than selecting them for favorable closure (Luce & Tukey, 1964; Krantz et al., 1971).

The simplest transformation makes the problem plain. If a measured coordinate is transformed by shifting and rescaling, changes in direction may survive, but the location of zero may not. A positive rescaling leaves zero in place only when there is no translation. A translation can turn an apparent neutral endpoint into a nonneutral endpoint, or the reverse, without changing relative patterns.

Residual ambiguity can remain after linking. Bridge samples may identify slope and intercept with uncertainty. Anchor events may orient sign but leave origin broad. Cultural adaptation may preserve rank but not level. Response shift may allow within-

person direction but not cross-time level comparison. A responsible claim names what remains unidentified.

This is where identified sets become more honest than point estimates. Instead of forcing one calibration, the analysis can report the range of terminal estimates, drift estimates, or cumulative contributions compatible with admissible transformations. If all admissible calibrations support the same qualitative conclusion, confidence rises. If conclusions change across admissible calibrations, the result is representation-sensitive.

The public version should not say that ambiguity means nothing can be known. It should say what can be known under the available bridge. A pattern may be robust to positive rescaling, sensitive to origin, stable within person, unstable between persons, or valid only as a direction. Those are different scientific results.

The Law becomes stronger when it accepts this ambiguity rather than hiding it. A claim that survives known transformation freedom is more valuable than a larger claim propped up by an arbitrary coordinate.

10.2.6 — Withdrawal of Common-Zero Claims

The origin must come from an independently justified response or state relation, not from sample centering, an instrument midpoint, a population mean, a fitted terminal value, or a transformation chosen because it improves closure. Different anchors may support local zeros while failing to establish one shared origin.

Withdrawal is required when anchors conflict, response processes change, linking leaves an additive constant unidentified, or plausible origins produce incompatible sign or terminal status. The correct output may be origin-specific results, an identified set over admissible origins, or an origin-free feature such as ordering or shape.

Failure of common zero need not destroy all calibration. Within-person direction, thresholds, ranks, local responsiveness, and some contrastive effects may remain interpretable. The report should state exactly which relations survive and which claims, especially exact-zero and cross-person level claims, are unavailable.

Z-scores, centering, and norm-referencing express values relative to a specified sample. Analyses using them should record that reference and use independent calibration when the target requires a different origin.

Restoration requires new anchor evidence and a prospectively specified linking test on data independent of the failure that motivated redesign. The program may not search across origins until one produces the preferred terminal result. Development and confirmatory status must remain separate (American Educational Research Association et al., 2014; Nosek et al., 2018).

10.3 — Longitudinal Invariance and Response Shift

Longitudinal calibration asks whether the same instrument is still measuring the same construct in the same way after time has passed. A person who becomes ill may change what fatigue means. A person who survives pain may change what counts as severe. A person who nears death may change the relation among dignity, fear, comfort, and report. Intensive longitudinal measurement therefore requires tests of temporal measurement structure, not merely more observations from the same items (Bollen, 1989; Borsboom, 2005; Vogelsmeier et al., 2024; Paap et al., 2025).

If the measurement relation changes, a numeric change can be ambiguous. It may reflect a change in experience, a change in response scale, a change in physiology, a change in meaning, or some mixture of all four. The Law's measured cumulative estimate cannot afford to pretend that these are the same.

Longitudinal invariance and response shift belong at the center of the science because apparent stability can arise in two different ways: experience may remain stable, or the ruler may change with the person. The task is to distinguish adaptation in the lived construct from adaptation in standards, interpretation, or measurement behavior (Sprangers & Schwartz, 1999).

$$X_{j,t} = \tau_j(t) + \lambda_j(t)\eta(t) + \varepsilon_{j,t}.$$

This displayed relation is deliberately modest. It does not announce the true model of experience. It shows how a stable-looking score can change because the latent state changed,

because the loading changed, because the intercept changed, or because residual error changed.

Longitudinal invariance can be tested directly through item parameters and factor relations. Recent clinical work illustrates how a population may support partial invariance, with uncertainty carried for the changing portions of the measurement structure (Paap et al., 2025; Meredith, 1993; Putnick & Bornstein, 2016).

A response-shift example shows why longitudinal measurement is not simply repeated measurement. A person recovering from a long illness may report the same fatigue score before and after adaptation, yet the meaning of that score may have changed. The first score may mean alarm and disruption; the second may mean a familiar limitation folded into a changed life.

The opposite can also happen. A person may report lower distress after a diagnosis because their standards changed, while qualitative evidence and physiological strain suggest continuing burden. That report is not false. It is evidence that the person's evaluative frame has shifted, which means the measurement bridge must be updated before the ledger uses it as a level change.

A serious design does not ask the person to freeze their meanings so a scale can remain stable. It lets meanings change and then asks what that change does to the inference. Time-varying loadings, anchors, qualitative context, and partial invariance become ways to respect the life rather than overwrite it.

This distinction supports both compassionate care and accurate measurement. Adaptation can be real relief, real strength, or real meaning. It can also be a change in standards that hides

continuing burden. Measurement earns trust by keeping both possibilities alive until the evidence narrows them.

The public consequence is that a stable score across illness is not automatically stable experience, and a changing score is not automatically changed experience. The bridge, not the graph, decides what can be said.

Longitudinal calibration therefore gives later terminal chapters a rule of restraint. Terminal calm, distress, narrowing, or compression must be interpreted through state and response-shift gates before it becomes evidence about the Law.

The same rule applies to every long-run graph in the science book. A curve that crosses months or years should carry the history of its bridge: what remained stable, what changed, and which parts of the conclusion were relabeled after change.

10.3.1 — Configural, Metric, and Scalar Invariance

Configural invariance asks whether the same pattern of indicators relates to the same construct across time. If the factor structure changes, the instrument may be measuring a different object. For the Law, this is the minimum condition for treating repeated measures as part of one trajectory rather than a sequence of different constructs with the same label (Sprangers & Schwartz, 1999; Paap et al., 2025).

Metric invariance asks whether the loadings are comparable. If loadings change, then the same latent difference produces different observed differences at different times. A channel may become more sensitive during illness, less sensitive during

sedation, or differently related to burden after adaptation. Without metric invariance, slope and change claims must narrow.

Scalar invariance asks whether intercepts or thresholds are comparable. Without it, level comparison becomes unsafe. A person may endorse the same response category at two times while meaning different latent levels, or produce the same physiological level under different clinical states. Scalar failure bars common level claims even when direction may remain usable.

Residual invariance is stronger still. It asks whether residual variances are comparable. This may matter when uncertainty is carried into cumulative ledgers. If residual variance increases near terminal states or under device burden, confidence intervals must widen. A chapter on fairness cannot let fragile states produce the narrowest public claims merely because the model forgot heteroscedasticity.

The invariance ladder should be treated as a permission ladder. Passing configural invariance may allow structural similarity. Passing metric invariance may allow some change comparisons. Passing scalar invariance may allow level comparisons. Passing residual invariance may support stronger uncertainty claims. Each rung licenses only what it actually earns.

In the Law's vocabulary, invariance failure narrows measurement, not dignity. A person whose measurement relation changes is not less real, less coherent, or less worthy of attention. The scale has met its boundary.

10.3.2 — Response Shift

Response shift occurs when a person's internal standards, values, or conceptualization of the construct change. In health measurement, it is often described through recalibration, reprioritization, and reconceptualization. These three processes describe how changes in a person's life can alter the meaning of their measurements (Putnick & Bornstein, 2016; Bollen, 1989).

Recalibration changes internal standards. A pain level once rated severe may later be rated moderate after prolonged illness, not because the sensation is smaller, but because the person's scale has changed. Reprioritization changes what matters within an evaluation. Fear, dignity, fatigue, social connection, and autonomy may carry different weight after diagnosis. Reconceptualization changes what the person means by the construct itself.

A response-shift example can break a naïve ledger. Suppose a person reports lower distress after months of illness while physiological burden and qualitative accounts show continued suffering. The report may indicate genuine adaptation, altered standards, resignation, meaning-making, medication, or a changed interpretation of the response categories. The number alone cannot decide.

The correct design uses multiple evidence streams, then-tests or individualized appraisal where appropriate, anchor events, qualitative context, and longitudinal models with time-varying measurement parameters. None of these makes response shift disappear. They make it observable enough to affect the claim.

Response shift can be part of meaningful change. A person may find relief, courage, or purpose in conditions that previously seemed unbearable. Measurement can help distinguish that lived transformation from changes in reporting scale and describe how the two interact.

Reported calm near an endpoint may reflect adaptation, care, reconciliation, sedation, altered standards, or a change in the meaning of the reported construct. Modeling these pathways helps identify which aspects of the latent trajectory each observation informs.

10.3.3 — Time-Varying Loadings and Intercepts

Time-varying loadings, thresholds, and intercepts turn response shift into a testable measurement structure. Allowing parameters to move is not a license for unrestricted flexibility; the model needs constraints, external anchors, prospective checks, and comparison with simpler invariant alternatives. Its purpose is to discover whether stability is present before stability is used (Bollen, 1989; Meredith, 1993; Paap et al., 2025).

A time-varying loading can show that one channel becomes more or less connected to the latent target. For example, respiration may become a stronger indicator of distress during dyspnea, while facial expression may become weaker under fatigue. A time-varying intercept can show that the observed level shifts even when the latent target is held constant.

Such models require discipline because flexibility can rescue too much. The calibration plan should specify which parameters may vary, what covariates justify variation, what smoothing is

permitted, and what failure thresholds trigger withdrawal. Otherwise every discrepancy can be explained away by a moving parameter, and the model loses empirical content.

The observed covariates matter. Medication, sleep state, disease stage, device version, language, age, care setting, and symptom clusters can all justify time-varying relations. If these variables are unobserved, the model should not pretend to have solved the problem. It may report wider uncertainty or an identified set over plausible parameter paths.

A time-varying model can also protect participants from false stability. If the instrument admits that sedation changes expression, it need not interpret silence as relief. If it admits that delirium changes report, it need not rank confusion as low burden. Flexibility, when constrained, can be more humane than a rigid scale.

The final rule is change-control: changing the measurement bridge changes the claim version. A result obtained under time-varying calibration is not the same claim as a result obtained under a fixed bridge. Public language must say which bridge was used.

10.3.4 — Partial Invariance

Partial invariance permits only the comparisons supported by the structure that remains stable. Stable loadings may preserve selected associations or within-person dynamics, while failed intercepts or thresholds can block latent-level comparisons; a subset of stable anchors supports only the construct range those

anchors still represent (Meredith, 1993; Millsap, 2011; Putnick & Bornstein, 2016).

Partial-invariance decisions must be prospective and auditable. The analysis should define how indicators are tested, which departures trigger review, whether modifications were anticipated, how multiplicity and model search are handled, and whether the retained indicators still cover the intended construct. Dropping inconvenient items after seeing the result is not evidence of invariance.

Construct coverage is the noncompensatory gate. If indicators of dignity, fear, agency, or social connection fail while only autonomic channels remain, the reduced model may no longer represent the intended valence construct even if its fit improves. A numerically stable remnant cannot inherit the name or authority of the larger construct without an explicit bridge.

The correct output may therefore be a narrower trajectory, a relation supported only within persons, a version-specific estimate, a set-valued comparison, or abstention from common-scale language. Partial invariance licenses partial inference and a named loss of scope; it does not license full-scale rhetoric with a smaller item set.

10.3.5 — Change Versus Measurement Change

A measurement system must distinguish change in experience from change in the instrument's relation to experience. This is the heart of longitudinal calibration. A rising score could mean rising burden, greater sensor sensitivity, changed response standards, a new medication effect, an altered clinical state, or a software

update. The same graph can carry many stories (Cronbach & Meehl, 1955; American Educational Research Association et al., 2014).

The first strategy is design. Repeated anchors, negative controls, bridge periods, dual scoring, time-forward prediction, and contextual annotation make the distinction more visible. A study that collects only one composite score at each time point has very little ability to separate experience from measurement change after the fact.

The second strategy is modeling. Joint models can include latent experience, measurement parameters, state variables, and drift terms. They can ask whether observed change is better explained by the latent state or by a changing bridge. They can produce posterior uncertainty over both, rather than giving the reader one falsely clean curve.

The third strategy is precise reporting. State how the measurement bridge connects a decrease, increase, or stable score with the underlying experience, and report the remaining alternatives when that connection is unresolved.

Measurement change is especially plausible during major life events. Illness, bereavement, disability adjustment, aging, terminal transition, treatment, and social disruption can change both the person and the way evidence is expressed. The more consequential the state change, the less safe it is to assume measurement stability.

This distinction is a falsifiability condition. If favorable movement is always attributed to experience while unfavorable

movement is assigned to measurement change, the program can absorb every result. The rules that distinguish latent change from bridge change must therefore be frozen before the trajectory is inspected and must allow either explanation, or unresolved ambiguity, to win.

10.3.6 — Longitudinal Withdrawal Rule

The longitudinal withdrawal rule marks the point at which a trajectory no longer has one defensible scale. Failed invariance, response shift, state-dependent calibration, changed channel gain, or unrecoverable version breaks can make before-and-after values incomparable even when the printed score is stable (Meredith, 1993; Sprangers & Schwartz, 1999). The withdrawal point should be reported as part of the data structure so later models cannot bridge across it without an explicit linking study.

Partial invariance provides a middle path. Indicators or channels that remain stable may support a narrower longitudinal claim while unstable components are modeled separately, removed under prespecified rules, or placed in an identified set. Partial invariance licenses partial inference, not full-scale rhetoric (Putnick & Bornstein, 2016). The retained indicators should still cover the construct region required by the estimand; statistical stability alone does not guarantee substantive coverage.

Withdrawal rules should be fixed before the target trajectory is examined. They name the tests, tolerances, affected comparison tiers, and successor outputs, such as split phases, state-specific models, bounded direction, or no common trajectory. This prevents post hoc preservation of an attractive path. The successor

output should be chosen by the failed invariance property, so a threshold failure and a loading failure do not trigger the same fallback automatically.

When a common longitudinal scale withdraws, the study may still report symptom-specific change, local direction, origin-free shape, or separate pre- and post-shift trajectories with explicit contexts and versions. Restoration requires requalification and external replication; changed anchors, parameters, or items define a new measurement version rather than silently repairing the old one. Cross-version linking can later restore some continuity, but only through data that directly span both measurement versions rather than by assuming semantic equivalence.

10.4 — Culture, Language, Development, and Disability

A scale that travels across populations carries moral weight even when it pretends to carry only numbers. If one language, body, age, or culture becomes the silent template, the measurement system can mistake difference for absence, restraint for calm, atypical physiology for neutrality, or unfamiliar expression for noise (Borsboom, 2005; Campbell & Fiske, 1959; Henrich et al., 2005).

The Law's burden lens makes that error acute. Hidden burden often stays hidden because institutions expect pain, distress, relief, and dignity to appear in familiar forms. Calibration must resist that expectation. It must ask whether the construct, sign, unit, and response relation actually travel.

Culture, language, development, disability, and neurodiversity are not sensitivity checks added after a main analysis. They determine how items are understood, which channels are available, what anchors mean, and whether the construct itself is comparable. They therefore help define the main estimand rather than merely qualifying it later.

$$Pr(X_j = 1|\eta, g) = \text{logit}^{-1}(\alpha_{jg} + \beta_{jg}\eta)$$

The group index in this item-functioning relation is not a ranking device. It is a warning that an item or channel may behave differently across language, culture, disability, development, or clinical state even at the same proposed latent level.

A cross-language example shows why fairness needs calibration rather than good intentions. An item translated as ‘I feel at peace’ may evoke emotional calm in one language, spiritual reconciliation in another, absence of conflict in another, and social acceptability in another. The same words can cross a border while the construct does not.

A disability example sharpens the point. A participant with limited facial expression may be scored as affectively flat by a model trained on expressive movement. The model may be technically consistent and still wrong. The problem is not the participant’s body. The problem is a calibration bridge that confused one form of expression with experience itself.

Developmental context adds a third pressure. A child, an adolescent, an adult, and an older person may all answer a question about tiredness, but the construct could involve play, school, social withdrawal, work, autonomy, frailty, or medication in different proportions. The word travels more easily than the meaning.

Group-specific models are often the most accurate statement of where a bridge has been earned. A common model deserves wider authority only after it survives transport across the relevant languages, developmental stages, disabilities, clinical states, and communication modes. When it does not, local models prevent one group’s evidence ecology from becoming the hidden norm for everyone else.

Where comparability remains unresolved, preserve the local result and identify the evidence needed to extend it. Separate

models can retain useful information while a wider measurement bridge is developed.

This is also a source-design rule. Recruitment cannot include only the populations easiest to link and then let the resulting scale describe humanity. The chapter requires the limits of the recruitment frame to remain visible in the claim.

When a group-specific model is necessary, it should be presented as the valid model for its qualified context, not as a deviation from a supposedly real universal scale. That wording matters because measurement language can turn a transport failure into an unjustified claim that the population itself is anomalous.

10.4.1 — Language and Translation

Translation is not word substitution. It is construct transport. A response option may carry the same dictionary meaning while changing intensity, dignity, social acceptability, or clinical implication. Words for pain, anxiety, peace, exhaustion, shame, relief, and meaning can carry different histories across languages and care settings (Henrich et al., 2005; Henrich et al., 2010).

Translation is a measurement study, not a clerical step. Forward translation, reconciliation, back translation, expert review, cognitive interviewing, harmonization, and field testing examine whether an item elicits the intended construct in the new language and context. Statistical fit cannot replace response-process evidence when words, norms, disclosure risks, or communication routes have changed (International Test Commission, 2018; Putnick & Bornstein, 2016; American Educational Research Association et al., 2014).

Conceptual equivalence matters more than lexical equivalence. A culture may distinguish forms of suffering that another language folds together. A phrase about feeling down may imply sadness in one context, fatigue in another, spiritual burden in another, and social shame in another. The Law's scalar target cannot absorb such differences by pretending the translated words are identical.

Cross-language linking requires bridge samples and uncertainty. If bilingual or multilingual participants, common anchors, or parallel validation data are unavailable, the claim may narrow to language-specific models. A language-specific model is not a second-class result. It may be the only honest result.

Translation also interacts with state. Terminal weakness, delirium, sedation, cognitive impairment, and low literacy can make even a conceptually good translation unusable in a given context. The calibration plan should specify when proxy report, clinician observation, or physiological channels may serve as auxiliary evidence and when they cannot replace report.

Public language should say translated and calibrated, not merely translated. The difference is the difference between words on a page and a measurement bridge that has survived contact with lived meaning.

10.4.2 — Cultural Display and Meaning

Culture changes expression, interpretation, suppression, and the social consequences of disclosure. Understatement, somatic language, relational description, strategic emphasis, or avoidance may each be locally intelligible responses rather than evidence of lesser or greater experience. These differences belong in the

observation model and in the design of anchors, not in stereotypes about a group (Henrich et al., 2005; Henrich et al., 2010; International Test Commission, 2018).

A calibration system should model display and meaning rather than moralize them. If a person does not cry, does not complain, does not use the expected affect words, or does not show the expected autonomic pattern, the instrument should not treat the absence of the expected signal as absence of burden. It should ask which channels are valid in that context.

Cultural display also affects anchors. A self-report anchor may be stronger in one setting and weaker in another. A behavioral anchor may be interpreted differently across care practices. Family presence, clinician authority, privacy, stigma, and spiritual interpretation can all change what is said or shown. Calibration needs those variables in the design, not only in the discussion.

Meaning can alter unit claims. A one-category change in a response scale may not represent the same experiential difference across cultural contexts. The same word for moderate may be used conservatively in one setting and expansively in another. Scalar invariance is therefore a substantive cultural question, not just a model-fit statistic.

The project should avoid both stereotypes and erasure. Culture should not be used to explain away every discrepancy, but neither should discrepancies be forced into one universal expression template. The correct response is measured: test, model, bound, and withdraw when the bridge fails.

A fair calibration system leaves room for separate models. If common comparison cannot be defended, the result is noncomparability, not failure of the people being measured.

10.4.3 — Developmental Change

Development changes construct meaning, communication, physiology, autonomy, memory, and reporting. A child's distress, an adolescent's shame, an adult's endurance, and an older person's fatigue may not relate to the same channels in the same way. Calibration across development must therefore be developmental science, not adult measurement stretched across age (Sosa-Hernandez et al., 2024; Shiffman et al., 2008).

For younger children, self-report may be limited by language, temporal understanding, scale comprehension, and dependence on caregivers. For adolescents, privacy, identity, peer context, and risk sensitivity can alter disclosure. For older adults, sensory impairment, cognitive change, medication, frailty, and social expectation can alter both channels and reports.

Developmental calibration should use age-appropriate anchors and accessible response methods. Visual scales, behavioral tasks, caregiver context, clinician observation, and physiological channels can all help, but each carries its own bias. A caregiver's concern is not the child's experience. A child's silence is not neutrality. A sensor's stability is not well-being.

Development also affects the unity and endpoint chapters downstream, but this chapter owns comparability. Even if later chapters ask whether a bearer exists or how a stream is adjudicated, calibration must ask what a score means at that

developmental stage. The same number cannot travel across age without evidence.

Longitudinal development is especially challenging because response shift and maturation occur together. The person may change, the construct may change, and the instrument relation may change. A developmental trajectory therefore often supports within-stage or transition-specific claims rather than one lifetime common scale.

The humane rule is to make development a reason for better measurement, not exclusion. Children, aging persons, and those whose communication changes should not disappear from the research program because adult scales are convenient.

10.4.4 — Disability and Neurodiversity

Disability and neurodiversity require calibration to stop mistaking familiar expression for experience itself. A person may communicate nonverbally, use assistive technology, show atypical facial or autonomic patterns, mask distress, or encounter sensory and social conditions a standard model was never designed to represent. Accessible response routes and local response-process evidence are therefore part of validity, not accommodations outside it (American Educational Research Association et al., 2014; International Test Commission, 2018).

Access is a prerequisite to interpretation. Prompt wording, response mode, timing, device design, sensory demands, motor demands, communication supports, and consent procedures must be usable before a missing or unusual response is treated as

evidence about the construct. When the interface excludes a participant, the failure belongs first to the instrument.

The second duty is construct respect. Atypical physiology does not mean absent feeling. Reduced expression does not mean reduced burden. Different social communication does not mean lower relational depth. The chapter's calibration rule is that deviation from the development sample must trigger context-of-use review, not automatic deficit interpretation.

Neurodiversity can also alter sensory burden. A device, prompt frequency, clinical context, or social demand may itself create distress. A calibration design that ignores measurement burden can alter the very state it hopes to estimate. The instrument then measures the consequences of being measured.

Disability-specific or neurodiversity-informed models may be necessary. They should not be treated as inferior to a universal model. A universal model that misreads people is not more scientific because it is universal. It is less scientific because it hides failed transport behind a single score.

The no-score rule has local force here. Calibration may qualify claims about an instrument. It may not rank communicative forms, bodies, cognitive styles, or lives. When the instrument cannot travel, the scale withdraws. The person remains fully present.

10.4.5 — Differential Item and Channel Functioning

Differential item functioning occurs when people with the same underlying construct level respond differently to an item because of group, language, culture, state, disability, or method.

Differential channel functioning is the physiological analogue: the same latent target produces different channel behavior across contexts. Both threaten comparability (Borsboom, 2005; Russell, 1980).

DIF is not automatically bias, but it is always a warning. An item may function differently because it captures a real contextual difference, because translation shifted meaning, because response style differs, because a symptom overlaps with the construct, or because the model is misspecified. The analysis must decide whether to retain, adjust, isolate, or remove the item.

A worked example clarifies the problem. Suppose two groups have the same latent distress level, but one group endorses a fatigue item more often because a medication produces physical tiredness unrelated to valence. If the item is retained as pure distress, the group appears more burdened. If the item is removed without noting medication burden, the model may erase real experience. Calibration must separate symptom overlap from construct bias.

Channel DIF can be more hidden. Skin conductance, heart-rate variability, respiration, voice, facial movement, sleep, movement, and temperature can all vary with physiology, medication, environment, disability, device placement, and care practices. A channel that is reliable in one group may be artifact-dominated in another.

Testing should include group comparisons, item or channel response functions, residual diagnostics, anchor stability, multiple-group models, and sensitivity analyses. More

importantly, it should include construct review. Statistical detection without construct interpretation can remove the very features that matter.

When DIF or channel functioning cannot be corrected, common comparison should narrow. Separate models, bounded comparisons, or abstention are better than a single score that silently taxes one group with another group's measurement assumptions.

10.4.6 — Group-Specific or Noncomparable Results

Group-specific results can be the most honest scientific outcome. If a scale is calibrated within one language, culture, developmental stage, disability context, or communication mode but fails a common link, local models preserve valid information without forcing one group's expression to become the hidden norm for everyone else (International Test Commission, 2018; Putnick & Bornstein, 2016).

Noncomparability can arise from item wording, response style, cultural display, reference groups, developmental change, disability, device access, or the construct itself. The analysis should separate these possibilities because an item-level failure calls for a different remedy from a channel failure or genuinely different construct expression.

Differential item and channel functioning should be tested at the level of the intended comparison. A translated item may shift thresholds, a wearable may change gain with skin or mobility, and a communication channel may carry different effort or risk.

Equality constraints and standardized scores cannot create common meaning when these relations fail.

A transport dossier should combine statistical invariance tests with response-process evidence, expert review, participant challenge, and context information. Psychometric fit is important, but a common factor can still reflect a shared method or institutional script rather than a common phenomenal coordinate (American Educational Research Association et al., 2014; Mokkink et al., 2024).

When full comparison fails, several outputs remain possible: separate group models, within-group trajectories, rank-only relations, bounded links, partially invariant composites, or explicit noncomparability. The chosen output should identify which transformation and inference survive rather than reporting that the population failed the scale. Choosing among these outputs should follow the scientific question and surviving invariance structure, not a preference for whichever result looks most comparable.

A universal template is especially risky in developmental and disability contexts. Familiar speech, facial expression, movement, and response speed are not privileged evidence of experience, and nonparticipation or alternative communication cannot be interpreted as low burden or low validity. The no-score and dignity rules apply to the measurement system, not to the person. Accessibility adaptations should be treated as measurement conditions to validate, not as deviations from a supposedly standard participant.

Transport should be earned cell by cell. A matrix across groups, languages, states, items, channels, anchors, lags, and devices can record supported, partial, unsupported, or prohibited relations, making the exact boundary of comparison visible and correctable (Gagnier et al., 2025). Sparse cells should remain explicitly unresolved rather than borrowing certainty from a neighboring group whose response process may differ.

The result prepares the clinical edge-state problem. If ordinary cross-group calibration already requires this discipline, illness, sedation, delirium, terminal physiology, sleep, anesthesia, and communication loss demand even more explicit state-specific bridges and abstention rules. Those edge-state bridges should also distinguish changes in experience from changes in expressibility, because the same observed silence can arise through very different mechanisms.

10.5 — Clinical, Terminal, and Edge-State Calibration

Clinical and terminal contexts change the observation model. Pain, dyspnea, delirium, fatigue, medication, sedation, fever, cachexia, anxiety, family presence, clinician workflow, and limited communication can alter expression, physiology, report, and missingness. Context-specific calibration is required; No adequate estimate may be the only defensible inference (Giacino et al., 2018; Surges et al., 2024; World Health Organization, 2020; World Medical Association, 2024).

Many consequential empirical questions arise near endpoints and high-burden states, where calibration demands increase. As science approaches vulnerability, it must more carefully separate experience, treatment, physiology, observation burden, and loss of communication.

Table 10.2. Clinical and edge-state calibration hazards and safe outputs.

Clinical condition	Calibration hazard	Required distinction	Likely safe output
Pain crisis	Symptom intensity overlaps with valence	Pain signal versus total burden	Symptom-specific direction or abstention
Dyspnea	Respiration carries panic, exertion, and physiology	Breathlessness versus affective distress	State-specific model
Delirium	Report and expression may change meaning	Experience versus observation reliability	Uncertainty expansion or no level claim

Clinical condition	Calibration hazard	Required distinction	Likely safe output
Sedation	Signal suppression may mimic relief	Treatment effect versus measured calm	Treatment-labeled sensitivity analysis
Terminal decline	Physiology may compress channels	Reserve and care effects versus Law-specific dynamics	Rival comparison only after bridge passes

A terminal-clinical example makes the stress test concrete. A patient receiving opioids for severe dyspnea shows reduced movement, slower respiration, fewer reports, and lower measured agitation. That pattern may reflect genuine relief, sedation, exhaustion, disease progression, family quieting, sensor attenuation, or several of these at once.

A weak calibration labels the graph as calming and moves on. A strong calibration asks which channels were changed by treatment, which symptoms and care events were present, whether communication or device contact deteriorated, and whether the previously qualified bridge still applies. Movement is observable; its permitted meaning is the calibration question.

Clinical calibration also has to preserve care. If asking for more reports burdens the patient or family, the protocol should stop or reduce contact. Scientific uncertainty is preferable to measurement that worsens the state being measured.

Terminal physiology provides specific mechanisms to investigate. Changes in perfusion, medication, movement, and signal quality can produce compression or reduced amplitude.

Including these processes in the model makes the comparison with a proposed signature more informative.

Vulnerable contexts can teach the science only when the bridge is strong enough, participant burden is low enough, and rival explanations are specified well enough for the preferred interpretation to fail. This is not opposition to terminal research. It is the standard that prevents urgency and symbolic importance from substituting for measurement validity.

A clinical-context file should therefore travel with every estimate produced in serious illness. It should tell later readers what the patient was receiving, what the care team was doing, what signal was feasible, and what the instrument was not allowed to infer.

The appropriate posture near vulnerability is disciplined scarcity: collect only what can be obtained without compromising care, interpret only what the qualified bridge supports, preserve clinical and observation context, and leave the remainder unforced.

10.5.1 — Pain, Dyspnea, Delirium, and Fatigue

Pain, dyspnea, delirium, and fatigue are not interchangeable burdens. They can overlap in physiology and report while differing in experiential structure. Pain can be sharp, dull, meaningful, terrifying, manageable, or dignity-threatening. Dyspnea can carry panic beyond its oxygen metrics. Fatigue can flatten expression while increasing burden. Delirium can disrupt report and amplify distress (Giacino et al., 2018; Surges et al., 2024).

A calibration system should therefore avoid one symptom-to-valence shortcut. A pain score is not a valence score. Respiratory

distress is not automatically total burden. Agitation is not always suffering, and stillness is not always relief. Each symptom requires its own observation bridge and its own relation to the target construct.

Symptoms also interact. Dyspnea can increase fear, pain can increase fatigue, fatigue can suppress report, delirium can alter expression, and medications can change all of them. A model that treats dependent indicators as independent can misstate uncertainty and overstate the amount of independent evidence.

Clinical calibration should use symptom-specific annotations, medication timing, care procedures, report when possible, clinician context, and physiological channels. The goal is not to replace experience with clinical labels. The goal is to prevent symptom confounding from being mistaken for Law-specific dynamics.

A high-burden symptom state may still support useful research. Within-person response to routine care, natural symptom fluctuation, and ethically available anchors can calibrate direction. What the chapter forbids is treating clinical severity as direct evidence for or against terminal valence closure without the bridge.

The humane consequence is immediate. When symptoms make measurement uncertain, care does not wait for measurement. Comfort-first remains the clinical rule; calibration governs scientific interpretation, not treatment priority.

10.5.2 — Sedation and Analgesia

Sedation and analgesia change experience, expression, physiology, report, and the possibility of report. They can relieve suffering, reduce visible distress, alter autonomic channels, impair communication, and change memory. Any calibration that treats reduced signal as reduced burden without modeling treatment has failed both science and care (World Health Organization, 2020; World Medical Association, 2024).

Clinical interventions can affect the target and the channels differently. Analgesia may reduce pain while preserving access; sedation may alter awareness, expression, or recall; anxiolysis may reduce fear without changing nociception. Protocols should therefore model treatment-specific pathways rather than interpreting every quiet signal as comfort or every absent report as absence of experience.

Medication timing is part of the measurement record. Dose, route, onset, expected peak effect, accumulation, tolerance, renal or hepatic impairment, and co-medications can affect channels. If treatment is not recorded, the model cannot distinguish relief from signal suppression. Missing treatment data should widen uncertainty or stop the claim.

Sedation changes consent, reporting capacity, and the available observation channels. The analysis records these transitions and evaluates auxiliary channels within the relevant state.

The Law's terminal studies will be especially vulnerable to this confusion. A terminal pattern that looks like compression may reflect good palliative care, sedation, reduced movement, device

gain, or observation attenuation. These are not annoyances after the fact. They are rival explanations that calibration must anticipate.

Treatment effects can illuminate care, experienced state, and the observation process together. When treatment changes the measurement bridge, the analysis identifies that change and states which comparisons remain meaningful.

10.5.3 — Communication-Limited States

Communication loss removes or degrades a channel; it does not establish the absence of experience. Aphasia, motor impairment, fluctuating attention, intubation, or impaired command following may prevent speech, writing, gesture, or device use. Evidence from alternative behavioral and neurophysiological assessments can inform the state, but the correct conclusion may remain uncertain (Giacino et al., 2018; World Health Organization, 2020; World Medical Association, 2024).

The calibration plan should distinguish inability, refusal, fatigue, sleep, delirium, sedation, language barrier, device inaccessibility, and care interruption. Each has a different meaning. Lumping them into missing report hides precisely the information needed to interpret the measured cumulative estimate.

Alternative communication can help: eye gaze, yes-no systems, assistive devices, simplified prompts, caregiver context, clinician observation, or physiological indicators. But each alternative has a context of use. A caregiver can provide context, not direct access. A clinician can report observed signs, not the full stream.

A physiological channel can suggest state, not testify as the person.

Silence should trigger calibration humility. It may require wider identified sets, state-specific models, or abstention. It may also require ethical modification of the protocol if measurement burden contributes to silence. A person should not be pressed into response so that a ledger can become smoother.

Communication limits are not rare edge cases. They occur in fatigue, pain, disability, delirium, severe illness, terminal weakness, language mismatch, childhood, and cognitive impairment. A scale that cannot handle them should not claim broad scientific authority.

Interpret silence through the available context: grief, exhaustion, comfort, sedation, confusion, refusal, communication difficulty, or peace. Preserve the uncertainty when the observations support several interpretations.

10.5.4 — Terminal Physiology

Terminal physiology can compress signals for ordinary reasons unrelated to PTV. Reduced movement, altered respiration and circulation, palliative sedation, metabolic disturbance, changing communication access, and sensor degradation may create apparent convergence or silence. The observation model must separate those mechanisms from any Law-facing interpretation before visual resemblance is allowed to count (Giacino et al., 2018; Surges et al., 2024; Shiffman et al., 2008).

The endpoint chapters will handle death-of-mind adjudication. This chapter owns the calibration question: does the measurement

relation still mean what it meant earlier? Near death, the answer may be no. Channels can decouple from experience, become dominated by physiology, lose sign, or become unavailable. That is a measurement result, not a reason to force the bridge.

Terminal calibration should separate physiological decline from valence inference. A falling variance in a channel may reflect reduced movement, sedation, low sensor gain, reduced perfusion, or observation loss. It may also be compatible with a Law-specific prediction in some designs. The distinction requires independent reserve measures, treatment records, negative controls, and rival models.

Common-scale origin becomes especially fragile near terminal states. If the only available data are late-window increments, the whole-life ledger remains unidentified without assumptions about the pre-enrollment ledger. If terminal physiology changes the scale, even late-window level may be uncertain. Calibration must carry those limits into the verdict.

The risk is subtle: a dying person can become a symbolic endpoint for other people's theories. This chapter refuses that conversion. Terminal data are evidence only under humane protocols, state-specific calibration, and observation bridges that permit abstention when the person becomes less observable.

Terminal calibration can reveal which channels are useful, which procedures are too burdensome, which rivals explain the data, and which states need different instruments. Those findings guide the next design.

10.5.5 — Sleep, Anesthesia, and Covert Activity

Sleep and anesthesia show why visible report cannot define experience. Dreams, partial awareness, covert responsiveness, memory failure, and altered unity can all complicate calibration. The absence of waking report is not a stable zero. It is a state-specific observation condition (Siclari et al., 2017; Sanders et al., 2012).

Sleep calibration must distinguish wake, non-rapid-eye-movement sleep, rapid-eye-movement sleep, awakenings, dream reports, nightmares, and next-day residue. A channel that tracks waking distress may not track dream valence. Dream reports are filtered by memory and awakening conditions. The measurement relation changes with state.

Anesthesia and disorders of consciousness impose still higher burdens. Behavioral unresponsiveness may coexist with residual processing or rare awareness, while physiological channels can be dominated by drugs, surgery, pain, temperature, ventilation, and clinical intervention. Calibration in these states must be treated as a distinct edge-state problem with its own reference conditions, error asymmetry, and abstention rule, not as an ordinary estimator operating under unusual lighting.

Covert activity makes error asymmetry explicit. A false negative may erase experience from the record; a false positive may convert noise into an unwarranted claim. Edge-state calibration therefore requires separate false-negative and false-positive analyses, not one aggregate accuracy score (Giacino et al., 2018).

These states belong partly to later consciousness and endpoint chapters, but calibration must prepare the ground. The question here is not whether a stream exists. It is whether the instrument's relation to the target remains valid under the state. If not, the later chapter receives uncertainty rather than a cleaned score.

10.5.6 — Clinical Context of Use

Context of use bounds an instrument's clinical authority. Calibration in ambulatory adults cannot be assumed to survive acute pain, medication change, intensive care, delirium, sedation, terminal physiology, sleep, anesthesia, or communication loss. Each setting changes target expression and observation and therefore requires setting-specific evidence (U.S. Food and Drug Administration, 2025b; International Council for Harmonisation, 2026).

Clinical qualification should follow the setting's actual failure routes. The record must separate treatment effect from measurement suppression, symptom-specific signals from total burden, comfort from reduced observability, and deterioration from a Law-specific trajectory. State, medication, device, care event, and missingness histories belong to the bridge.

A care-aware context file names the setting, population, state range, cadence, devices, anchors, communication routes, burden ceiling, stop rules, estimand, prohibited uses, and No adequate estimate path. It also excludes autonomous triage and comfort-denial uses.

A terminal example shows the risk. Physiological variance may narrow after sedation or low perfusion while self-report and

device coverage decline. Relief, observation shutdown, treatment, disease progression, or several processes may explain the pattern; terminal-valence language remains unavailable until the measured-to-latent, endpoint, and rival bridges survive (Surges et al., 2024).

The same discipline governs sleep, anesthesia, delirium, and covert activity. Unresponsiveness does not identify absence of experience, and a channel validated in waking report-rich states may fail when memory, communication, or motor output changes. Edge-state results must remain labeled by state and observation regime (Giacino et al., 2018).

Clinical calibration also protects caregivers and families. Prompts, devices, annotations, and challenge procedures must not displace care, create unpaid workflow, or turn relatives into continuous witnesses. Burden endpoints and automatic stopping rules remain operative even when the data are valuable.

Available outputs include symptom-specific direction, state-specific calibration, treatment-labeled sensitivity analysis, uncertainty-expanded identified sets, and No adequate estimate. The protocol specifies the appropriate output for each evidence state (World Medical Association, 2024).

10.6 — Identified Sets and Uncertainty

When origin, sign, unit, linking, invariance, or context of use remains only partly identified, the output should be the set of target values or conclusions compatible with the evidence and declared assumptions. That identified set is distinct from a confidence set, which expresses sampling uncertainty under a chosen inferential procedure. A complete report may need both: first identify the admissible target region, then quantify sampling uncertainty without pretending the first problem has disappeared (Manski, 2003; Imbens & Manski, 2004).

Uncertainty becomes scientifically useful when it is decomposed. Bounds can reveal whether origin, sign, linking, missingness, channel noise, model choice, or sampling variation drives the result; they can also show when between-person associations reverse within persons. The decomposition should point to the evidence that could actually narrow each component rather than using one general caution label for every uncertainty.

$$C(D) = \{h(\theta): \theta \in \Theta(D), \theta \text{ satisfies the calibration gates}\}$$

$$Var(Q|D) = E[Var(Q|D, M)|D] + Var(E[Q|D, M]|D)$$

One object reports the compatible set of conclusions; another separates uncertainty within a model from uncertainty across models and representations. Both belong in the public claim. A narrow interval conditional on one disputed calibration is not stronger than a wider set that honestly carries the calibration alternatives.

An uncertainty example can be more revealing than a point estimate. Suppose a late-window study estimates endpoint

proximity under three plausible calibration bridges. One bridge gives near-neutrality, another gives mild residual burden, and the third is inconclusive because sign weakens under sedation. A single averaged estimate would hide the fact that the conclusion depends on the bridge.

An identified-set report keeps that dependence visible. It says which conclusions survive all admissible calibrations, which survive only some, and which cannot be stated. Such a report may disappoint readers seeking a simple verdict, but it gives researchers something stronger: a map of what additional evidence would shrink the set.

The same logic applies to uncertainty decomposition. If the channel noise is small but linking error is large, gathering more signal from the same channel will not solve the problem. If model uncertainty dominates, a larger sample under the same model may sharpen the wrong answer. If state labels are uncertain, clinical annotation may matter more than sensor precision.

This is where calibration becomes strategic. It tells the program where to improve next. Sometimes the next study should not be bigger; it should be better linked, better anchored, better translated, or better state-labeled.

The public sentence follows the uncertainty map. It identifies the quantity estimated precisely, the bridge assumptions used, and the layers that call for a wider uncertainty set or additional measurement.

This turns uncertainty from a defensive paragraph into a research agenda. Uncertainty should remain visible and scientifically productive.

Public verdicts are fair only when uncertainty survives translation. People should not be asked to live under claims whose dependence on anchors, links, states, or models was known in analysis but removed from the sentence they receive.

10.6.1 — Partial Identification

Apply the partial-identification framework from §8.4 to calibration by carrying the admissible origins, links, transformations, measurement models, and invariance structures into the identified set (Manski, 2003; Imbens & Manski, 2004).

Partial identification may be the correct result whenever scalar valence, temporal additivity, common origin, or cross-person comparability remains uncertain. A late-window study may bound an observed increment, a cross-cultural study may bound linked levels, and a terminal study may bound endpoint estimates across admissible calibration bridges.

A point estimate can be misleading even when it is computed correctly. If the point depends on one arbitrary anchor, one chosen prior, or one unverified link, the precision is conditional on a fragile bridge. An identified set makes that dependence visible. It says: here is the family of conclusions the data have not ruled out.

The admissible model and transformation class should be fixed before public interpretation. The analysis should specify origins, links, invariance classes, bridge models, anchor assumptions, and sensitivity thresholds, then construct the identified set under those

rules. Widening the class only after an unfavorable result is rescue; narrowing it only after a favorable result is overclaim. Prospective specification gives both favorable and adverse conclusions the same geometry.

Partial identification can support strong conclusions when all admissible calibrations agree. If every admissible bridge rejects common neutrality, the result is a robust defeat for that claim. If every admissible bridge supports a within-person drift pattern, the result may support a narrower dynamical claim. If admissible bridges disagree, the result is representation-sensitive.

This approach is deeply fair to the reader. It refuses to hide the burden of uncertainty in a single number. It also refuses to pretend that incomplete identification means no knowledge. The world often gives ranges before it gives points.

10.6.2 — Measurement-Error Decomposition

Measurement error is often discussed as if it were a single band around an estimate. Calibration requires a sharper decomposition. Channel noise, item unreliability, sampling error, calibration error, model uncertainty, linking error, state-label uncertainty, and transport error are distinct. Some are aleatory under a model; others arise because the model, coordinate, or bridge is not identified. Each source enters differently and should remain separate whenever it can change the claim (Tsiatis, 2006; Little & Rubin, 2019; Vogelsmeier et al., 2024).

Channel noise concerns the observed signal: sensor artifacts, item unreliability, environmental disturbance, or short-term fluctuation. Calibration error concerns the relationship between

signal and target. Model uncertainty concerns which statistical structure is appropriate. Linking error concerns transformations across instruments, populations, languages, or versions. Transport error concerns whether the bridge travels to the new context.

Integrating posterior means can provide a point estimate when the required integrability conditions hold. Its uncertainty must still be computed from the joint path model, including temporal covariance, calibration, missingness, and linking uncertainty; attaching isolated pointwise error bars does not provide uncertainty for the integral. The law of total variance separates uncertainty within a model from uncertainty across the declared model family.

A simple error budget shows why calibration deserves separate attention. For equal-exposure increments on a common signed unit, suppose $Y_k = X_k + b + e_k$, with a shared zero-mean random offset b of variance σ_b^2 and independent zero-mean errors e_k of variance σ_e^2 . Assume the offset and error sequence are independent of one another and of the latent increments. If $A_n = \sum_{k=1}^n X_k$ and $\hat{A}_n = \sum_{k=1}^n Y_k$, then

$$\text{Var}(\hat{A}_n - A_n) = n^2 \sigma_b^2 + n \sigma_e^2.$$

A fixed offset instead produces bias nb ; a bound $|b| \leq \delta$ allows cumulative error $n\delta$. For unequal weights, the shared-offset variance is $\sigma_b^2 (\sum_k w_k)^2$ and the independent-error variance is $\sigma_e^2 \sum_k w_k^2$. Serial errors require the full weighted covariance matrix. Denser sampling therefore cannot, by itself, remove shared origin error.

Qualified replicate channels provide a complementary diagnostic. If $Y_k^{(r)} = X_k + e_k^{(r)}$ for $r = 1, 2$, both complete error paths have zero conditional means given the latent path and zero conditional cross-covariances, and all quantities have finite second moments, then

$$\text{Cov}(\sum_k Y_k^{(1)}, \sum_k Y_k^{(2)}) = \text{Var}(\sum_k X_k).$$

This identifies cumulative latent variance under the stated observation model. Shared bias, unequal loadings, or different measured quantities invalidate that interpretation; physiology and self-report require qualification before being treated as replicate channels. The covariance identity does not identify the neutral origin.

A practical reporting template can separate four layers: observed channel uncertainty, latent-state uncertainty, calibration-bridge uncertainty, and public-claim uncertainty. The final layer is often widest because it includes the assumptions required to translate a technical estimate into a verdict.

Error decomposition prevents false precision. A device may be highly repeatable while the construct bridge is weak. A questionnaire may have high internal consistency while translation, temporal structure, or within-person reliability fails. A model may fit well while sign remains unidentified. Reliability is level-specific, validity is claim-specific, and neither guarantees transport (Vogelsmeier et al., 2024; Castro-Alvarez et al., 2026).

The practical question is who bears the cost when distinct uncertainties are collapsed into one reassuring interval. Usually it

is the person whose result is treated as clearer than the evidence allows. Decomposition keeps uncertainty attached to its source so that correction can repair the right layer of the bridge.

10.6.3 — Within–Between Sign Reversal

Within-between reversal occurs when a relationship observed across people differs from the relationship operating within one person over time. A channel can be higher among people with greater average burden yet fall during an individual acute episode because medication, fatigue, withdrawal, or care changes the signal. Group-level direction therefore cannot be used as a trajectory bridge without a within-person model (Molenaar, 2004; Hamaker, 2012).

This problem is not exotic. It is a structural risk in longitudinal measurement. Between-person variation includes stable traits, diagnoses, environments, devices, medications, culture, disability, and care access. Within-person variation includes momentary state, adaptation, symptom fluctuation, and response timing. A single coefficient can mix them.

The measured cumulative estimate is primarily a path object. Applying a between-person calibration to within-person change without testing the relation can reverse sign and misread a terminal trajectory. Group association may inform a candidate bridge, but it cannot substitute for a within-stream model.

The analysis should therefore decompose within and between effects. Multilevel models, person-mean centering, dynamic structural models, and time-forward validation can help, but the

claim must state which level is being inferred. A between-person association is not a within-stream law.

A worked example is simple. People with chronic severe illness may have lower heart-rate variability on average than healthier people, but within one person, a further drop may indicate sedation rather than worsening experience. If the model treats the group association as a within-person sign rule, it may call sedation relief or burden incorrectly.

The public sentence should remain level-specific. Say that a channel predicts between-person differences when that is all it does. Say that it tracks within-person change only when the within-person test has passed. Never let a group relation borrow authority for a life trajectory.

10.6.4 — Sensitivity to Anchors and Priors

Anchors and priors contribute information and assumptions to difficult measurement; their roles and influence should be explicit. Calibration should report how conclusions change when plausible anchors, bridge samples, priors, transformation families, and state labels are varied within declared ranges (Bollen, 1989; Borsboom, 2005).

Anchor sensitivity asks whether the result depends on a particular reference event or item. If one relief anchor produces terminal proximity and another does not, the claim is anchor-sensitive. If a language bridge depends on one set of common items, the result should say so. If a terminal valence closure result appears only after a convenient anchor, the claim should narrow or withdraw.

Prior sensitivity matters in hierarchical and Bayesian calibration. Priors may stabilize estimates in sparse terminal data, rare states, small language groups, or edge conditions. That can be valuable, but the posterior should not be described as data-driven certainty when prior structure carries the conclusion.

Sensitivity analysis should include direction, level, and verdict. It is not enough to show that a parameter changes slightly. The analysis should ask whether public claim status changes: support, component support, narrowing, abstention, rival dominance, or defeat. Calibration sensitivity matters most when it changes the sentence.

The study should also record why anchors and priors were chosen. Scientific reasons, clinical reasons, accessibility reasons, and historical reasons differ from outcome-favoring reasons. The reader should be able to tell which kind of reason was used.

Dependence on assumptions does not disqualify a result; hidden dependence does. An anchor-sensitive estimate can be useful when the anchor family, sensitivity range, and conclusions that change are reported with it. The public statement should be no stronger than the most fragile assumption that it needs.

10.6.5 — Common-Scale Abstention

Common-scale abstention is a valid scientific result: the evidence does not support the proposed comparison at the required level. It may follow unidentified origin, unstable sign, weak unit meaning, failed linking, inadequate construct coverage, scalar noninvariance, dominant response shift, or an identified set that spans incompatible verdicts. The output names the exact

comparison that failed, the lower-tier relations that survive, and the evidence needed to reopen the bridge instead of filling the gap with a convenient score (Manski, 2003; Imbens & Manski, 2004).

Abstention is not the same as nonsignificance. A nonsignificant estimate may reflect limited information about a well-defined target; common-scale abstention means that the coordinate relation required to define or compare that target has not been earned. More observations cannot repair a missing origin, sign, unit, or transport bridge by themselves.

Abstention is also distinct from defeat. If a prerequisite bridge fails before the Law-specific proposition is reached, the study yields unusable or narrower evidence rather than evidence against PTVC. If the bridge survives and the protocol-fixed target is contradicted, the result is adverse to that target. The location of failure must remain visible.

The abstention rule should be fixed in the protocol in operational terms. Calibration uncertainty exceeding the equivalence margin, failed scalar invariance for essential indicators, unsupported sign orientation, insufficient anchor overlap, or model dependence across credible missingness assumptions can each trigger a no-common-scale verdict. Prospective rules prevent disappointment from becoming rhetorical rescue.

Abstention can preserve lower-tier results. The same data may support within-person direction, state-specific calibration, subgroup models, origin-free relations, identified sets, or future bridge development. Public reporting should name the surviving

object, the failed comparison, and the evidence needed to reopen it.

10.6.6 — Uncertainty in Public Language

Public reports should present the finding in clear language, with the material population, context, measurement, and uncertainty information stated once in the accompanying methods note or caption. Later discussion can refer to that description and repeat a qualification only when the comparison or conclusion changes (Manski, 2003).

An uncertainty report should map sources to consequences rather than append one interval at the end. Channel noise affects local estimates; missingness affects who and when was observed; linking uncertainty affects version continuity; origin uncertainty affects zero language; invariance uncertainty affects comparability; and context uncertainty affects transport.

Consider three prospectively specified plausible neutral anchors. If the valid terminal-window uncertainty sets under the three anchors fall, respectively, entirely inside the equivalence band, entirely above it, and entirely below it, the proper result is that common-zero terminal equivalence is not identified across those anchors. Averaging the anchors would hide the dependency, although some origin-free ordering or shape features may remain available (Manski, 2003; Imbens & Manski, 2004).

The same logic applies to cross-language and within-between relations. A bridge may support rank without common origin, or a channel may predict differences across people while failing to track change within a person. Public summaries should state the

level and relation actually supported rather than converting the output to a common score with false precision.

Typed uncertainty strengthens negative as well as favorable results. Failure under every admissible calibration is more discriminating than failure under one arbitrary scale, while a favorable pattern that survives only one anchor is weaker than it appears. Uncertainty is therefore part of falsification, not an apology after the result.

The ambiguity belongs to the instrument, not the person. High calibration uncertainty never makes dignity, care, rights, or credibility uncertain, and it does not authorize more intrusive measurement merely to force a decisive number.

The operational rule is exact: report bounds when assumptions leave a range, identified sets when several values remain compatible, and abstention when the target cannot be identified or validly bounded. A measured cumulative estimate is valuable because its uncertainty is typed, propagated, contestable, and open to reduction.

10.7 — Calibration Audit and Correction

Calibration is maintained rather than completed once. Instruments drift, populations and clinical routines change, languages evolve, devices update, and models age. Audit schedules, version tags, bridge samples, requalification triggers, and correction routes therefore belong to the measurement object itself, while measurement-property and reporting standards make the maintenance evidence inspectable (Mokkink et al., 2024; Gagnier et al., 2025; Nosek et al., 2018).

Calibration governance asks how anchors, samples, thresholds, software, challenges, reanalyses, and retirement decisions are recorded and acted upon. Its purpose is to keep each active claim aligned with the evidence for the bridge on which it depends.

A calibration program that can correct itself is more credible than one that presents permanence. For PTVC to become answerable, the instruments that test it must also remain answerable.

Table 10.3. Calibration audit triggers, actions, and correction records.

Audit trigger	Immediate action	Claim-status consequence	Record to preserve
Anchor instability	Pause affected comparison	Widen, split, or withdraw	Anchor version and context
DIF or channel functioning	Review affected indicators	Separate models or noncomparability	Group and item diagnostics
Device or code update	Dual scoring or bridge study	Version-specific claims	Hardware, code, and transform

Audit trigger	Immediate action	Claim-status consequence	Record to preserve
Participant challenge	Context review	Correction, explanation, or no change	Challenge and adjudication path
Cross-site failure	Restrict context of use	No transport without new evidence	Site, population, and protocol differences

An audit example closes the chapter. A scale qualified for within-person ambulatory tracking is used for two years. During that time, the device firmware changes, a translated version is introduced, a new medication becomes common in the study population, and a clinician notices that the score behaves oddly during delirium. Without audit, the same label covers several instruments.

A useful audit can begin with a concrete failure chain: a firmware update changes contact quality, medication alters autonomic responsiveness, a translation shifts an item's threshold, and delirium pushes observations outside the qualified state range. The audit asks which outputs are affected, whether a bridge can be recovered, which claims must be relabeled, and where use must stop. Its purpose is correction, not defense of the previous scale.

Correction may change old claims. Some estimates may be reanalyzed, some public summaries narrowed, some figures relabeled, and some contexts retired. A project afraid of correction will hide these changes. A serious project will make them part of the scientific record.

Participant and clinician challenge are crucial in this example because they see contexts the model may miss. A patient may know that a prompt is misunderstood. A clinician may know that a procedure altered signal. Neither should automatically override the model, but both should have a formal path into review.

Retirement is the most honest ending when the scale no longer supports its use. The instrument may still be historically important, developmentally useful, or valid in a narrower context. It should not remain active for a claim it cannot carry.

The final discipline is institutional as well as statistical: a tool must be able to step back before a person is made to carry its overreach. That requires named human responsibility, a correction channel, a withdrawal mechanism, and records sufficient to find every dependent output.

A retired instrument can still be honored as a stage of learning. Its failure can show what later instruments must measure, which populations were missed, which states were misread, and which public sentences were too large.

Retirement also prevents obsolete authority from surviving by inertia. Responsible science updates or withdraws an instrument when its bridge fails, while preserving the history needed to understand what changed.

The final calibration promise is modest and demanding: no scale may speak beyond the bridge that carries it, and every bridge must be inspectable when it starts to fail.

10.7.1 — Prospectively Specified Calibration Plan

Apply the prospective-specification terminology in §6.7. The calibration plan names the construct, measurement model, context of use, measure and channel selection, scoring and modifications, comparison tier, origin and anchors, bridge samples, invariance tests, state labels, model family, missingness assumptions, sensitivity analyses, failure thresholds, version rules, and permitted public language (American Educational Research Association et al., 2014; Nosek et al., 2018; Flake & Fried, 2020; Gagnier et al., 2025).

The plan should separate development from confirmation. Development may explore items, channels, links, and anchors. Confirmation freezes the bridge long enough to let the world answer. If the bridge changes during confirmation, the claim version changes. The result may still be useful, but it is no longer the same test.

Failure thresholds should be concrete. Scalar invariance failure may stop level comparison. Sign failure may stop signed interpretation. Link uncertainty exceeding the equivalence margin may trigger abstention. Drift beyond a declared threshold may split the trajectory. These rules should be visible before the result.

The plan should also include burden thresholds. Calibration procedures can impose prompts, devices, interviews, or clinical workflow demands. If participant or staff burden alters response, missingness, or care, the instrument has changed the data-

generating process. A measurement plan that ignores burden is incomplete.

A prospectively specified plan does not make the instrument true. It makes the test answerable. A mistaken plan can fail cleanly, be corrected, and produce better science. An unfrozen plan can preserve itself indefinitely by adapting to every result.

10.7.2 — External and Cross-Site Audit

External audit asks whether the calibration survives outside the development team. Cross-site audit asks whether it survives different devices, care routines, languages, populations, staff practices, and environmental conditions. Both are essential before claims travel beyond the site that built the instrument (Munafò et al., 2017; Nosek et al., 2018).

Cross-site audit should reproduce the deployment problem rather than reshuffle the development data. Site, device, language, and time holdouts test different transport claims, and each should be tied to a prospectively specified failure rule. A model that succeeds only after local refitting has demonstrated adaptability, not transport of one frozen bridge.

External reviewers should examine construct coverage, invariance tests, missingness, drift, version provenance, anchor selection, burden, adverse calibration decisions, and public language. A model can fit well while making a public claim that the calibration does not support. Audit must include interpretation, not only metrics.

Cross-site failure should be informative. It may reveal a language issue, a device issue, a clinical routine, a population difference, or

a hidden confound. The response should be narrowing, model revision, new bridge samples, or retirement for that context. It should not be public confidence with a footnote.

An audit trail should preserve disagreements. If clinicians or participants challenge a calibration interpretation, the challenge should be recorded, categorized, and reviewed. It may be mistaken, but it may also reveal context the model missed. Challenge data are part of calibration learning.

10.7.3 — Drift and Version Alerts

Drift alerts indicate that the observation-to-target relation may have changed; version alerts indicate that the instrument, preprocessing, or model has changed. Both require prospectively specified thresholds, bridge samples, and requalification decisions so that slow degradation, abrupt breaks, and comparison across incompatible versions become visible (American Educational Research Association et al., 2014; Wilkinson et al., 2016; Nosek et al., 2018).

Drift can arise from population shifts, new treatment practices, device replacement, firmware updates, new prompts, translation revisions, staff training, changed care workflow, or environmental conditions. Each can alter origin, sign, unit, invariance, or missingness. A drift alert should ask which coordinate is threatened.

Versioning should be granular. Hardware version, sensor placement, preprocessing, item wording, language adaptation, weighting, model code, calibration sample, and public claim

template should all have identifiers. A result without version identifiers cannot be fully interpreted later.

Drift monitoring should combine statistical and practical triggers. Residual change, anchor instability, differential item functioning, missingness shifts, device diagnostics, participant challenge, clinician concern, and changed treatment or workflow can each reveal a threatened coordinate. Statistical alerts without contextual review can miss clinically obvious failure; contextual concern without defined evidence can become untraceable discretion.

When an alert fires, the response is versioned triage rather than automatic defense or deletion. The team may continue under monitoring, pause a class of claims, split the version, reestimate through a bridge sample, widen uncertainty, requalify the context, or retire the use. The trigger, evidence, decision, and affected outputs must be recorded.

10.7.4 — Participant and Clinician Challenge

Participant and clinician challenge gives contextual evidence a route into calibration review without making anecdote sovereign over data. People and clinicians may detect failures the model cannot see, including distress caused by a sensor, a misunderstood translation, edema that disrupts a device, or an output that contradicts a documented clinical state. The challenge, evidence reviewed, disposition, and appeal route should be recorded (American Educational Research Association et al., 2014; Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

A challenge system should be accessible. Participants should be able to contest an interpretation, request explanation, report burden, correct context, stop future optional research collection without penalty, and learn what the score may not mean. Clinicians should be able to flag treatment changes, workflow artifacts, symptom confounds, or state changes that threaten calibration.

The challenge should trigger review, not automatic rescoring. The review asks whether the challenge identifies a data error, context error, model limitation, device issue, missing state label, unmodeled medication effect, translation problem, or public-language problem. Different causes require different corrections.

Challenge evidence should be protected from coercion. A participant should not have to justify their inner life to defeat a score. The instrument carries the burden of proof. The person carries no burden to perform legibility for the model.

Clinician challenge also has limits. Clinical impressions can be biased by culture, disability, communication style, workload, and expectations. The correct balance is neither model supremacy nor anecdotal veto. It is a review route where both model and context can be wrong.

This is one of the chapter's most important public-risk controls. A measurement system without contestability can become authority infrastructure. A bounded research instrument with challenge routes remains a scientific tool.

10.7.5 — Correction and Reanalysis

When calibration changes, affected estimates and claims must be corrected. The correction should identify the old bridge, new bridge, reason for change, affected population, affected dates, affected instruments, uncertainty change, and public claim consequences. A correction that changes the science but leaves public language untouched is incomplete (Gagnier et al., 2025).

Reanalysis should preserve provenance. Old estimates should not disappear as if they were never used. They should be marked as superseded, linked to the old version, and compared with the corrected estimates where appropriate. Historical transparency is part of trust.

Correction may narrow, widen, reverse, or abstain. A new bridge may show that a previous common-scale claim was unsupported. It may widen equivalence intervals beyond the decision margin. It may preserve within-person direction while removing cross-person level comparison. It may reveal that an apparent terminal pattern was calibration artifact.

Each instrument version should have reproducible code, synthetic fixtures, expected outputs, and migration tests. When recalibration changes a result, the program must identify which historical records remain recoverable, which require reanalysis, and which cannot be translated. Reproducibility is the mechanism that lets correction propagate rather than remain an announcement (Wilkinson et al., 2016; Center for Open Science, 2025; Nosek et al., 2018).

Public corrections should be calm and precise. The project should not treat correction as dishonor or suggest that measurement error alone decides the Law. It should identify the changed bridge, affected outputs, and surviving claims. This preserves claim continuity without hiding the reason for revision.

10.7.6 — Retirement of Nontransportable Scales

Retirement is appropriate when an instrument cannot support its claim; drift or invariance failure cannot be repaired; sign or origin is unidentified; provenance unrecoverable; participant burden or disclosure risk excessive; or foreseeable misuse uncontrolled. Retirement is versioned and scoped: an estimator may remain useful for ordinary within-person change while losing terminal, cross-language, cross-site, or cross-person authority.

Retirement ends an instrument's authority within a context of use. Care, listening, participant reports, and research continue: clinicians can comfort and treat while researchers develop a better bridge. A retired context should remain visible in instrument documentation so later users do not mistake continued use elsewhere for renewed qualification there.

The audit trail records development and freeze dates, qualified settings, anchors, versions, exclusions, drift alerts, challenges, failed invariance tests, corrections, and withdrawn claims. Reviewers use this history to reproduce where the scale gained or lost authority (Nosek et al., 2018; Mokkink et al., 2024). The history should preserve superseded calibrations rather than overwrite them, so later reanalysis can reconstruct which scale governed each published result.

An estimator valid in ambulatory adults may fail in palliative care as medication, low perfusion, care events, and observation loss alter its channels. Retire terminal use, requalify ordinary use, rerun affected analyses, and describe the surviving scope accurately. A split qualification can retain ordinary-use evidence while explicitly severing terminal claims, avoiding the false choice between total approval and total retirement.

Remove retired scales from public artifacts and operational systems. Correct, withdraw, delete, or restrict figures, summaries, websites, donor materials, model cards, and datasets as recorded. Keep unsafe outputs out of ranking, triage, employment, insurance, civic or spiritual status, and surveillance (National Institute of Standards and Technology, 2020; U.S. Food and Drug Administration, 2025b). Retirement should include machine-readable status where feasible so automated systems do not keep serving an output that human documentation has already withdrawn.

Part IV — Consciousness, Stream Routing, and Endpoints

A science of fairness cannot study a complete life until it specifies what carries the experience being accumulated. Bodies persist through many states, biographies can outlive report, legal persons are institutional objects, and records are traces. None is automatically identical to the separately routed maximal qualifying stream that bears the formal phenomenal measure. Part IV begins where those familiar identifiers cease to answer the scientific question on their own (Bayne, 2010).

Part IV therefore takes on a narrower, testable task. For a declared research purpose, it decides which experienced intervals may be joined, which remain separate, which gaps preserve several live routing models, and when the evidence licenses only an unresolved result. Each decision changes the domain of the later terminal test, so bearer definition and routing uncertainty must be fixed before endpoint evidence can carry weight.

The distinction matters because the formal ledger depends on declared intervals assigned to a qualifying stream under explicit unity and routing rules. If the gate is drawn too generously, the measured path may include intervals that were not part of the stream under study. If the gate is drawn too harshly, the program may erase experience because it was quiet, unreported, inaccessible, or inconvenient to the model. Either error can make the evidence cleaner while making the science less true.

The costs of error become concrete in a neurocritical-care room. A patient may show no reliable movement while task-sensitive neural activity remains detectable; clinicians may have to act before research converges; and families may hear different classifications from different teams. A well-governed protocol

preserves that conflict as an auditable record of channels, timing, and uncertainty instead of converting silence into absence (Bodien et al., 2024; Giacino et al., 2018).

This Part builds one dependency chain. Chapter 11 specifies provisional bearer ownership and routing. Chapter 12 asks how competing theories and boundary states alter the marker families used in that routing. Chapter 13 then constructs truth-level and protocol endpoints from independently governed evidence. The sequence matters because an endpoint estimate is interpretable only after the study has stated whose stream, which intervals, and which uncertainty class the ending belongs to.

Good bearer science must be exact enough to guide analysis and modest enough to preserve live alternatives. Biography can preserve social continuity where direct access is absent, while a classifier can create false discontinuity when a channel fails. The disciplined middle course records competing stream models, the evidence supporting each, the consequences of error, and the point at which classification authority ends.

Unknown is a typed result in this Part. Its causes include insufficient access, conflicting marker families, unsupported transport, unresolved routing, endpoint nonidentification, and observation shutdown. Naming the cause preserves the next scientific step and prevents uncertainty from becoming a convenient zero, a false absence, or an administrative judgment.

The hard cases in this Part are chosen because each breaks a different shortcut. Sleep separates experience from immediate access; anesthesia separates experience from later memory and

movement; disorders of consciousness separate organized cognition from overt response; artificial systems separate linguistic performance from warranted phenomenality; and terminal care separates observation from the truth-level end. Each case therefore tests a distinct link in the bearer-to-endpoint chain (Siclari et al., 2017; Giacino et al., 2018; Butlin et al., 2026).

This architecture makes negative results interpretable. A terminal study can fail because the bearer rule was wrong, the endpoint was chosen by convenience, or a low-signal interval was erased by an unexamined gate. By keeping those layers separate, the study can locate the failure instead of turning every adverse result into a mystery.

A persistent name, family memory, legal file, or public profile can secure obligations and social continuity without identifying the relation among experienced intervals. Those records matter, but they answer different questions. Part IV keeps administrative persistence available for care, accountability, and provenance while preventing it from becoming an unexamined proxy for phenomenal continuity.

Falsifiability improves when more than one stream model is specified in advance. A one-body-one-stream rule, a memory-gap boundary, or a low-signal exclusion may look economical, yet each can hard-code the result. Explicit alternatives reveal where evidence favors continuity, partition, rebinding, or abstention and where the later ledger remains conditional on unresolved bearer assumptions.

Endpoint discipline begins before any terminal curve is drawn. The truth-level end must be distinguished from somatic events, legal declarations, last observations, data outages, and analyst-selected windows. It must also be insulated from the ledger it is meant to test. What later becomes a stopping-time and adjudication problem begins here as a commitment to preserve cases in which no observed clock identifies the endpoint.

For a reader outside consciousness science, the edge cases provide a practical lesson: ordinary words such as awake, absent, returned, split, and ended compress several separable claims. The Part opens those words into access, report, memory, integration, causal continuity, routing, and endpoint evidence. That decomposition is what makes disagreement reviewable rather than merely philosophical (Siclari et al., 2017; Bodien et al., 2024; Andrews et al., 2024).

The dependency chain is noncompensatory. A persuasive terminal pattern cannot rescue an invalid bearer rule; an independently constructed endpoint cannot rescue a mis-zeroed measure; and a large sample cannot identify an undefined target. Stream confidence, routing alternatives, endpoint construction, observation coverage, and error consequences therefore remain separately auditable. This separation lets one failed layer stop only the inferences that depend on it, preserving valid observations without allowing them to bypass the failed gate.

The practical task is to keep that dependency chain visible. A bearer decision fixes ownership; ownership determines which observations may be combined; continuity distinguishes a pause from a boundary; and the endpoint rule fixes the domain of the

terminal claim. An explicit chain localizes a negative result. A hidden chain lets every failure retreat into an undefined bearer, interval, or ending. The chain also identifies which new evidence would reopen a blocked inference, making uncertainty actionable rather than merely descriptive.

The dossier architecture can travel across sleep, anesthesia, disorders of consciousness, development, comparative cases, artificial candidates, sudden death, and diffuse decline. What travels is the governance of evidence. Every marker retains its population, task, timing, theory dependence, qualification domain, and failure modes (Giacino et al., 2018; Butlin et al., 2026). A shared dossier schema permits comparison across states without pretending that the same markers carry identical meaning in every state.

Chapter 11 — Conscious Unity, Stream Identity, and Routing

A person may be visible and still unreachable. A body may breathe, sleep, heal, decline, or return while the scientific question of unified experience remains unresolved. A biography may continue through family memory, legal records, medical charts, and public accounts, while the research question asks something smaller and more difficult: what evidence permits us to say that a stream of experience is continuous through time (Bayne, 2010; Giacino et al., 2018)?

Chapter 11 defines the research bearer for this science: a candidate separately routed maximal qualifying stream to which felt experience is provisionally assigned for analysis. That object is narrower than legal personhood, spiritual identity, civic standing, family meaning, or the worth of a life, all of which remain governed by their own authorities.

A research bearer is a versioned evidential object, not a discovery of the person's ultimate nature. It records which intervals a study may provisionally analyze together, which alternatives remain live, and which uses are prohibited. The record may guide a sensitivity analysis or endpoint audit; it does not become the person, the soul, the public identity, the legal status, or an archive that claims to contain inner life.

The scientific problem begins with markers. Access, integration, reportability, command following, memory, learning, causal recurrence, affective carryover, physiology, and neural dynamics all provide evidence. None is sovereign. Report can fail when movement fails. Memory can fail while experience continues. A neural index can be informative without being identity. A family

story can preserve meaning without proving continuity. A public record can remain after experience has ended.

Because the unity gate determines which intervals enter the phenomenal ledger, it is part of the model rather than a clerical filter. A hard threshold can erase low-signal experience, while a threshold tuned after outcome inspection can manufacture apparent closure. The gate therefore carries sensitivity analysis, provenance, and an explicit relation to the tested endpoint.

Threshold families, soft gates, and confidence classes make this uncertainty analyzable. Their purpose is not permanent indecision, but proportional classification with enough version history to revise the record when evidence improves. Unknown remains an adequate result when no available rule can separate the live models without overreach.

Hard cases reveal which variable a continuity rule is actually tracking. Sleep can preserve vivid experience while external access falls; anesthesia can permit return without autobiographical recall; cognitive motor dissociation can preserve command-following without reliable movement; and split-access tasks can reveal restricted information flow without proving multiple subjects. Each case therefore tests a different component of continuity rather than supplying one dramatic verdict (Bodien et al., 2024; Gazzaniga, 2000).

These cases occur in operating rooms, rehabilitation units, homes, laboratories, and family decisions, not only in philosophical examples. Every classification is therefore treated as a limited research act with a context of use, a revision date, and a

prohibited-use field. That design lets the science become more exact without allowing provisional labels to migrate into treatment denial, eligibility, publicity, or moral rank.

By the end, the reader should be able to define a stream precisely enough to analyze its intervals, endpoints, and ledger, and to show how the classification can change with new evidence. Clear purpose, revisable rules, and participant protections make that work practical.

The vocabulary stays typed throughout. Stream names the proposed bearer; unity names a relation among experiences; confidence names support for an inference; partition and fusion name competing routing structures; and No adequate estimate names a limit of adjudication. None of those terms is a category of human worth.

Responsible inference proceeds from substitutes to evidence, from evidence to gates, and from gates to routing. The chapter first separates body, biography, law, memory, and record from the phenomenal target. It then assembles continuity markers, tests threshold families, examines pauses and returns, and finally confronts partition and fusion. Error costs and abstention close the sequence because every routing decision must declare what happens when the evidence is wrong.

Each bearer packet identifies candidate models, evidence channels, calibration population, confidence range, excluded alternatives, permitted analyses, and revision triggers. The packet narrows only scientific claims. When evidence cannot distinguish

continuity from interruption or partition, it returns a bounded model set or No adequate estimate.

Consider a person admitted unresponsive after injury. Motor output is unreliable, repeated examinations disagree, task-based neuroimaging later reveals organized command following, and recovery changes what can be said about the earlier interval. A versioned routing record preserves each evidence state and judgment instead of rewriting the gap as obvious in hindsight (Bodien et al., 2024; Giacino et al., 2018).

Uncertainty is treated here as structured information rather than a residual category. A confidence class records what was observed, which channels failed, which thresholds were used, which rival classifications remain live, and which downstream uses are barred. That record makes uncertainty corrigible without turning it into stigma or allowing a single score to hide the evidence chain.

The metaphysical literature remains useful because it reveals where intuition breaks. Memory can continue without unity, unity can falter without loss of dignity, and branching can make ordinary survival language unstable. The scientific task here is narrower: construct an operational bearer whose assumptions, alternatives, and limits can be inspected.

The recurring question is operational: what analysis is licensed by the evidence available at that time? A continuity marker may support inclusion in a sensitivity analysis without licensing a confirmatory ledger. A partition hypothesis may remain live without supporting public claims about multiple minds. Later

evidence may improve the record while the earlier uncertainty remains visible.

Permanent refusal to classify would also abandon decisions that evidence can sometimes support. Terminal studies, boundary-state research, and measurement models need disciplined outputs when their gates are met. Proportionate science treats yes, no, maybe, not yet, and No adequate estimate as serious results, then states what each result permits and what remains outside its authority.

11.1 — Bearer Is Not Body or Biography

The first boundary is not technical decoration; it is the condition of a responsible science. Body, memory, law, and record may each be necessary in ordinary cases, yet none is sufficient in hard cases. The stream concept is introduced not to deny those realities, but to stop them from doing work they cannot do alone.

Contemporary consciousness science increasingly separates the biological carrier, the psychological narrative, and the target of a specific measurement. Split-brain findings, memory-system dissociations, and disorders of consciousness all show why body, biography, legal person, and candidate stream cannot be collapsed into one evidential object (Bayne, 2010; Gazzaniga, 2000; Giacino et al., 2018).

The chapter uses body as support, biography as context, law as jurisdiction, record as trace, and stream as research bearer. This vocabulary gives the later terminal chapters a consistent way to connect evidence with its intended target.

The first mistake is to point at the body and call the problem solved. The organism is indispensable evidence. It carries the nervous system, the injury, the physiology, the medication exposure, the sleep state, and the clinical history. Yet the organism is not identical to the research bearer because the Law's formal object concerns unified conscious experience, not biological continuity alone.

A second mistake is to point at the biography. A life story can preserve meaning, responsibility, love, habit, memory, intention, and public recognition. But narrative continuity is too broad and

too narrow at once. It is too broad because stories can continue after conscious experience ends. It is too narrow because experience can continue when memory, report, or autobiography is impaired.

A third mistake is to point at legal personhood or public record. Law and record systems protect people, organize obligations, and preserve accountability. They must not be imported into the scientific bearer definition. A legal person can be unconscious. A public record can remain after death. An archive can contain traces of a life without becoming that life.

11.1.1 — Organism and Stream

A robust study file should keep organism, session, candidate stream, unity-gate version, routing model, and reviewer decision in separate fields. The separation permits the same body to contribute observations under more than one live routing model without duplicating any phenomenal event. It also prevents a hospital identifier or session row from silently becoming the stream because the database happened to place them together (Bayne, 2010; Bayne & Chalmers, 2003).

A useful way to hold the distinction is to treat the organism as the support condition and the stream as the target condition. Support conditions can be necessary without being identical to what they support. A lung is necessary for breathing in the ordinary human case, but breathing is not the lung; a nervous system may be necessary for conscious life in the ordinary human case, but the stream is not simply a body part named by anatomy.

This distinction also prepares later endpoint work. Somatic death, circulatory collapse, respiratory failure, brain injury, and device cessation are biological or observational events. They may be powerful evidence about the stream, but they must pass through declared endpoint and unity criteria. The science becomes fragile if it lets biology alone settle an object whose formal definition is conscious trajectory.

This distinction protects the Law from biological reduction. A pulse, an EEG pattern, a metabolic trace, a reflex, or a motor response can contribute to evidence, but each still asks what it bears on. Some markers speak to arousal. Some speak to access. Some speak to integration. Some speak to motor output. None alone says, with unrestricted authority, that one stream has continued.

The distinction also protects against the opposite error: treating bodily impairment as absence of experience. Motor paralysis, language loss, medication, fatigue, pain, sensory impairment, and injury can narrow expression without ending consciousness. A science that equates visible responsiveness with stream presence would exclude precisely the cases where caution matters most.

The organism table and the bearer table should therefore communicate through declared joins rather than collapse into one file. Age, injury, diagnosis, medication, physiology, sensory access, and motor capacity constrain interpretation. Access, recurrence, learning, integration, command following, and transition evidence bear on stream models. A join records how support evidence updates the bearer; it does not make body and stream identical.

Organism-level evidence is indispensable but typed. Anatomy, physiology, and causal continuity can support a one-stream model, while split access or independent control may keep partition alternatives alive. The result should state the model, confidence, and unresolved alternatives; insufficient evidence yields unresolved, not absent, and never changes the protection owed to the organism or person.

11.1.2 — Biography and Memory

Family memory, clinical records, and public biography preserve obligations, relationships, preferences, prior function, and grief even when scientific access is weak. The research program should honor those continuities without using love, narrative coherence, or archival persistence as a consciousness detector. They provide context and recognition while remaining indirect evidence about present phenomenal unity (Tulving, 1985; Squire, 2004).

Memory also has directionality. A later remembered episode can support the claim that something was experienced earlier, but failure of later recall is not symmetric evidence of absence. Amnesia, aphasia, sedation, trauma, sleep, delirium, and developmental limits break the symmetry. The classifier must therefore record memory as one channel with known failure modes, not as the court of last appeal.

Narrative fluency creates a systematic selection risk. Adults who can recount episodes, understand tasks, and describe a stable self are easier to classify than infants, people with amnesia or aphasia, heavily sedated patients, and others whose experience may be less narratable. The bearer model must therefore treat language and

autobiographical memory as channels with access requirements, not as admission tickets to the phenomenal domain.

Memory and biography contribute only through a declared evidential role. Consistent, causally plausible retention can strengthen continuity, while rehearsal, suggestion, record exposure, reconstruction, and amnesia remain live rivals. The absence of autobiographical recall cannot by itself close a gap, and the presence of a narrative cannot by itself prove phenomenal continuity (Nadel & Moscovitch, 1997; Squire, 2004).

11.1.3 — Legal Person and Research Bearer

Legal person and research bearer should occupy separate columns because they carry different powers. Legal status organizes rights, duties, consent authority, and institutional action. Bearer status restricts what one analysis may infer about phenomenal continuity. A change in scientific confidence must not alter legal protection, and a legal declaration must not be imported as evidence that a stream began, continued, or ended.

The distinction is especially important because legal categories are designed for public administration and protection, not for detecting conscious unity. They answer questions such as who may consent, who may inherit, who may be represented, and what duties institutions owe. The research bearer answers a different question: what candidate stream, if any, is the object of a scientific claim about felt trajectory.

Confusing legal and scientific categories weakens both. Law may be tempted to borrow neuroscientific language as if it settled rights or institutional status, while science may borrow the finality

of a legal declaration as if it settled phenomenal continuity. Legal status is therefore treated as protected context and authority, not as a measurement channel or an output of the bearer classifier.

Confusing the two would make the science dangerous. A low-confidence unity classification could be misread as reduced personhood. A no-estimate state could be misused as permission to reduce care. A partition hypothesis could be sensationalized as a legal division of identity. None of those uses follows from this chapter, and each is explicitly barred.

This separation also protects science. Legal categories are sometimes coarse relative to experience. A legal person can undergo sleep, anesthesia, coma, dissociation, or dementia while retaining status. That continuity of status is ethically necessary, but it does not solve the scientific problem of which experiential intervals are open, closed, uncertain, or partitioned.

Bearer classifications belong in restricted scientific records rather than administrative eligibility systems. Access is role-limited, labels expire or undergo review, and public summaries omit confidence classes that could become stigma, triage, insurance logic, employment inference, or rank.

11.1.4 — Public Identity and Record

Public identity is especially prone to uncontrolled reuse. A name, image, biography, diagnosis, or famous case can attract attention far beyond the original protocol and make later reviewers less independent. The safer record links public identity to the research dataset only through restricted provenance, while analytic files

retain the minimum variables needed to test routing and endpoint claims.

Medical records, social-media accounts, letters, video, voice notes, photographs, wearable traces, and model-generated summaries preserve different aspects of a person's public history. Routing analysis records what each archive contains and the continuity relation it can inform.

This matters for public artifacts as much as for research. A diagram that shows a line continuing through an archive, a caption that speaks of a person living in data, or a public page that treats model output as continuing witness would violate the chapter's boundary. Records can help interpret the past. They do not become the stream.

Public records are also selective. They preserve what was documented, not what was felt. They privilege visible actions, language, official encounters, social participation, and institutional access. A person's hidden burden can be absent from records precisely because it was hidden. Treating the record as the stream would repeat the injustice the Law's lens was built to expose.

The record has legitimate scientific uses. It can establish timing, medication, clinical events, injury history, sleep schedules, public behavior, consent records, and version history. It can show what evidence was available to a reviewer at a given time. It can preserve uncertainty instead of overwriting it. Its role is evidential and protective.

11.1.5 — One Body, Possible Multiple Streams

One organism usually supports the initial one-stream model because it is parsimonious and consistent with ordinary integration. Parsimony is a starting rule, not a metaphysical axiom. Evidence of sustained access barriers, partly independent control, conflicting report channels, or causal separation may require partition alternatives, while ordinary shifts in mood, role, or attention do not.

The key distinction is between complexity within one stream and evidence for more than one stream. Multiple goals, voices, perspectives, or behavioral modes can coexist in an integrated system. A partition hypothesis becomes scientifically relevant only when the evidence concerns restricted coaccess, independent control, causal insulation, or incompatible ownership of phenomenal events.

A partition hypothesis enters only when it explains access, memory, agency, or integration patterns better than a one-stream alternative under the same evidence budget. It is not added for novelty and not suppressed for tidiness. Whatever the scientific result, public and moral status remain unchanged.

Split-brain phenomena, severe dissociation, focal injury, seizure states, developmental anomalies, and possible artificial or hybrid cases all pressure-test the one-body-one-stream assumption. Their value lies in forcing explicit unity, ownership, routing, maximality, and domain rules, not in dramatizing multiplicity.

A research model should therefore allow a state space with options: one continuous stream, uncertain continuity, interrupted

stream, partitioned streams, remerged stream, or no-estimate. The existence of options does not mean the options are common. It means the program does not force the world into a shape convenient for the ledger.

The two error directions require different repairs. Collapsing a genuine partition can assign distinct event sets to one ledger; over-partitioning ordinary complexity can duplicate events, inflate apparent model flexibility, and create harmful public narratives. The protocol should therefore compare routing models under a nonduplication constraint and report the analyses that change when partition remains unresolved.

11.1.6 — Dignity Does Not Depend on Classification

Dignity is not an empirical confidence level. Under support, narrowing, nonmeasurement, or defeat, uncertainty about consciousness cannot reduce care, privacy, communication, legal protection, or ordinary human regard (Giacino et al., 2018; World Medical Association, 2024).

The protocol therefore separates scientific and care consequences. A classification may determine primary, sensitivity, or excluded analysis, but never comfort, analgesia, visitation, rehabilitation, communication support, or clinical seriousness. This separation also prevents classification-driven care changes from contaminating observation.

A no-estimate bearer state means evidence is inadequate for the declared inference, not that the person is absent from medicine, family, law, or moral consideration. Continuity may remain possible without being sufficient for a confirmatory ledger.

Bearer uncertainty remains visible downstream through parallel routing analyses, model families, identified sets, or No adequate estimate. Statistical precision cannot erase identity uncertainty, and a clean terminal result cannot repair unresolved ownership.

Transport must be shown for the marker-to-target relation. A classifier calibrated in communicative adults may change meaning in infancy, aphasia, motor impairment, sedation, brain injury, nonhuman species, or artificial architectures. New domains require documented prerequisites, calibration, thresholds, and failure modes; otherwise evidence remains restricted or unavailable (Giacino et al., 2018; Andrews et al., 2024). Transport evidence should include error asymmetry, because a marker that fails mainly through false exclusion has different scientific consequences from one that fails through false inclusion.

Review should be protected from biography and publicity. Blinding fame, legal controversy, commercial value, political meaning, or emotional vividness cannot remove every bias, but it reveals when public identity is doing work the markers cannot support. Review materials should therefore foreground marker evidence and uncertainty before identifying narrative context that could anchor interpretation.

The bearer object is a versioned decision packet recording admissible stream models, evidence date, uncertainty, excluded alternatives, permitted analyses, prohibited uses, and revision triggers. It can travel into later chapters without becoming a portable identity score. Separating bearer evidence from public

identity also makes later corrections safer, because a revised stream model need not rewrite a biography or legal record.

Even when one stream cannot be identified, the research can show which assumptions failed, which ownership models remain live, which analyses are prohibited, and which protections remain fixed. Uncertainty becomes a structured result rather than a gap hidden beneath the ledger. A structured uncertainty set can still support conservative ownership bounds when all live models agree that certain events cannot belong to the same stream.

11.2 — Continuity Evidence and Confidence

Continuity evidence is deliberately plural because no single aperture reveals consciousness. A person can be accessible yet unable to report, recurrent yet unable to remember, physiologically organized yet behaviorally silent, or behaviorally active without enough evidence of the unity under study. A plural framework reduces both premature exclusion and premature inflation.

No single continuity marker is sufficient across healthy, clinical, developmental, and nonhuman cases. Access, integration, memory, causal recurrence, command following, and perturbational complexity each carry different sensitivities and failure modes, so confidence must be assembled from converging families rather than one privileged signal (Casali et al., 2013; Dehaene et al., 2017; Bodien et al., 2024).

Confidence classes state exactly what the study can carry forward: continuity within a window, unresolved status across a gap, possible partition under a task, or No adequate estimate when the required evidence is unavailable. Each class connects a result with its next permitted analysis.

Continuity confidence is assembled from channels that answer different questions. Access asks what information becomes usable; memory asks what persists; recurrence asks what causal organization continues; command following asks whether instructions can guide activity; physiology and neural measures ask about state and capacity. The dossier records which

component each marker updates, how the marker can fail, and what routing claims remain licensed.

A useful continuity model should preserve three distinctions. Access is not memory. Reportability is not experience. Physiological organization is not stream identity. Markers can support a classification only when their scope is named and their alternatives are kept alive.

The output is therefore a confidence class, not a final identity proclamation. A stream may be classified as probable continuity, uncertain continuity, probable interruption, candidate partition, candidate rebinding, or no-estimate. Each label remains purpose-limited, versioned, and open to revision.

Table 11.1. Continuity marker families, limits, and confounds.

Marker family	Primary evidence	Cannot establish by itself	Main rival or confound
Access and integration	Coordinated availability of information across channels	Numerical identity or moral status	Motor bottleneck, reflex coordination, theory-specific overread
Memory and learning	Retention, updating, affective carryover, procedural change	Presence or absence of experience in every interval	Reconstruction, priming, suggestion, nonconscious learning

Marker family	Primary evidence	Cannot establish by itself	Main rival or confound
Report and command following	First-person statement or intentional task response	Absence of experience when missing	Language, motor, arousal, fatigue, pain, task burden
Physiology and neural dynamics	System-level state markers and response patterns	Conscious stream identity by metric alone	Arousal, medication, artifact, nonconscious processing
Causal recurrence	State dependence and continuity of trajectory	Full unity across gaps	Homeostatic persistence, public record, organism continuity

11.2.1 — Access and Integration

Integration should not be confused with agreement among channels. Fear and resolve, pain and relief, or conflicting action tendencies can coexist within one stream, while identical behavioral outputs can be generated by very different internal organizations. A continuity model should therefore ask whether channels participate in a common causal organization, not whether every measurement produces the same sign or story (Dehaene et al., 2017; Tononi et al., 2016).

Access evidence is strongest when it is specific about routing. Does visual information reach action selection? Does auditory command content alter neural or behavioral state? Does painful stimulation enter affective regulation or only spinal reflex? Does

interoceptive disturbance alter report, autonomic pattern, or avoidance? These questions prevent access from becoming a vague compliment attached to any complex system.

Integration also has a burden side. A person in pain may appear disorganized not because experience is absent, but because burden is overwhelming the systems by which experience becomes orderly report. The classifier should not demand the calm integration of a healthy person from someone whose condition makes calm integration impossible.

Access and integration concern whether information is available in coordinated form within a system. If sensory, affective, cognitive, and control processes can influence one another in organized ways, the evidence may support a unified state. If access is fragmented, delayed, or separated, the evidence may support uncertainty, interruption, or partition.

The observation channels can include behavior, task performance, communication, neural connectivity, autonomic coordination, response to stimuli, and state-transition patterns. Each channel has its own temporal resolution. A second-by-second neural pattern may reveal integration that a bedside exam misses; a long behavioral record may reveal continuity that a brief scan cannot.

Integration evidence is theory-dependent. Global-availability accounts, recurrent-processing accounts, integrated-information accounts, predictive-processing accounts, and higher-order accounts do not weight the same markers equally. Chapter 12 will compare those theories. Here the rule is to record which

theoretical bridge is being used and what a rival theory would say about the same evidence.

Access evidence can also mislead. A motor bottleneck can make integration invisible. Sedation can reduce output while leaving some internal coordination. Reflexive coordination can appear without conscious unity. A strong marker under one condition can be weak under another.

Access should not be operationalized as performance alone. Paralysis, fatigue, pain, aphasia, sensory barriers, task misunderstanding, and low arousal can all reduce visible response while leaving the relevant experiential question open. Confidence in continuity should therefore depend on the accessibility of the channel and the alternatives that could suppress it, not on how elegantly a person can demonstrate access.

Access and integration markers should be reported by channel, task, state, and theory. Convergent evidence can raise continuity confidence; disagreement can reveal motor, sensory, arousal, or model failure. The marker family supports a bounded classification only when its rivals and transport limits are visible.

11.2.2 — Memory and Learning

A learning marker is most useful when it includes a counterfactual: what would the system have done if the prior interval had not occurred? Without that contrast, learning can be confused with drift, medication effect, fatigue, or changing environment. Continuity evidence becomes stronger when the alteration is specific to prior exposure rather than general change (Nadel & Moscovitch, 1997; Squire, 2004).

The most useful memory evidence is often not a dramatic autobiographical report. It may be a changed startle pattern, a learned preference, a persistent aversion, a familiar voice effect, or a procedural adjustment. These small traces can matter because they show that the system was not merely passing through unrelated states. They still require careful rivals, but they keep the evidence field from being restricted to speech.

Memory-based continuity rules systematically disadvantage people whose encoding, retrieval, language, or autobiographical organization is impaired. Dementia, delirium, trauma, developmental difference, anesthesia, and focal amnesia can weaken later narrative without establishing absence of earlier experience. The dossier should therefore let memory strengthen continuity when its prerequisites are met, but never let failed recall close the bearer question by itself.

Episodic memory supports one kind of continuity. Procedural learning supports another. Emotional conditioning, habituation, priming, and preference shifts support still others. A person may lack a reportable memory while the system has learned. A later state may be shaped by an earlier one without narrating it.

Learning evidence also has rival explanations. Repetition can train a reflex. Environmental cues can guide behavior without conscious recall. Family or clinician prompting can supply story. A model must ask whether the learned content is causally linked to the candidate stream or explainable through nonconscious mechanisms, external reconstruction, or ordinary conditioning.

The Law's ledger problem adds a particular constraint. Memory after the fact does not decide whether an interval was open when it occurred. The unity gate concerns the interval itself, not only later report. Learning can strengthen the case that something continued, but it cannot rewrite the original uncertainty into certainty.

Memory and learning must remain separate evidence families because explicit recall, procedural retention, conditioning, and affective carryover have different mechanisms and lags. A retained trace may support causal continuity without establishing conscious presence throughout the interval; a missing trace may reflect encoding or retrieval failure. The classification should preserve that asymmetry.

11.2.3 — Recurrence and Causal Continuity

Recurrence matters here as evidence about carried causal organization, not as a vote for one grand theory of consciousness. The question is whether a later candidate state depends on an earlier organized process in a way that distinguishes continuation from restart, coincidence, or reconstruction. Recurrent processing and higher-order accounts offer different marker families, so the protocol must state which continuity claim each marker is meant to bear (Lamme, 2006; Brown et al., 2019).

A continuity model must name what persists, how it is carried, the time scale on which it operates, the observations that can reveal it, and the rival that would explain the same data without continuation. Candidate carriers may include network organization, synaptic state, bodily regulation, learned

association, affective tone, attentional set, or action policy. The model must also say what follows when the proposed carrier is blocked, destroyed, disconnected, or pharmacologically suppressed; a theory that explains persistence but cannot describe loss has not earned the word continuity.

Evidence may come from state-transition patterns, restored context, physiological rhythms, recurrent neural dynamics, behavioral coherence, affective carryover, learning transfer, and recovery of task sets. Hidden-state or hidden semi-Markov models can represent continuous, interrupted, partitioned, rebound, merged, and no-estimate states while allowing their dwell times to differ. Those states are analytic hypotheses, not ontological declarations, and their value depends on whether they improve held-out prediction or discrimination against simpler restart and access-failure rivals.

The dossier must separate continuity of substrate, continuity of causal organization, continuity of accessible content, and continuity of a candidate phenomenal field. Local recurrent loops may survive while global coordination fails; arousal may persist while content-specific access disappears; public records may persist while the relevant experiential organization is unknown. Evidence for a weaker layer cannot silently stand in for a stronger one.

Intervention can strengthen the claim when it is ethically and scientifically appropriate. Clinically indicated treatment changes, natural transitions, noninvasive perturbations, and already occurring disconnections can test whether later organization depends on the earlier state as the routing model predicts. The

purpose is never to disrupt a vulnerable person for theory; it is to distinguish a causal continuation edge from a descriptive correlation using admissible evidence (Pearl, 2009; Peters et al., 2017).

Counterfactual continuity should be written as explicit questions. Would the later pattern persist if one access channel were blocked? Would learned information transfer if the proposed bridge were removed? Would recoordination recur after the same perturbation under another protocol-fixed gate? A protocol that cannot name an observation that would count against its continuation edge has drawn a narrative arrow rather than a scientific route.

Persistence is easy to overread. A machine can preserve a record, a body can maintain homeostasis, a reflex loop can continue, and a public account can remain active without establishing a continuing unified field of experience. The relevant inference is therefore not that something endured, but that the declared evidence favors one routed continuity model over equal-information alternatives.

The proper output is a versioned graph of admissible continuation, interruption, partition, and unresolved edges, together with the evidence and uncertainty attached to each. When two routing graphs remain observationally equivalent, the result is an equivalence class and a statement of what additional observation could separate it. This preserves what has been learned without converting underidentification into a confident identity claim.

11.2.4 — Report and Command Following

Where report is available, respect for report includes respect for refusal. A participant who can communicate may decline tasks, correct language, or withdraw from parts of the protocol. Command following should not become command extraction. The ability to answer is not a license for investigators to demand more and more proof of inner life (Claassen et al., 2019; Bodien et al., 2024).

Section 12.3.3 develops covert command following in boundary states; here it enters only as one source of continuity evidence when overt behavior is unavailable.

Task-based neural responses are evaluated through replication, controls, sensory access, arousal, artifact review, and chance performance. A reproducible command-specific response adds evidence of retained cognitive capacity under the tested conditions.

Report is valuable because it gives experience a direct first-person bridge when conditions allow. A person can say what was present, absent, remembered, confusing, painful, dreamlike, or continuous. Command following is valuable for showing intentional responsiveness even when spontaneous behavior is limited.

Yet report and command following are constrained by access, language, motor capacity, arousal, attention, culture, pain, fear, fatigue, medication, and task design. A failed report can mean absence of experience, but it can also mean inability to express,

inability to understand, inability to move, inability to sustain attention, or inability to map the task onto a response.

The same discipline applies in ordinary and psychiatric cases. Dissociation, trauma, language limits, neurodivergence, and clinical stress can alter reporting. A person may report discontinuity because memory is inaccessible, not because experience ended. Another may report continuity because the narrative self repairs gaps after the fact.

A research protocol should therefore classify report and command following by context. What was the command? What sensory channel carried it? What motor or neural response was permitted? What burden did the task impose? Was the person fatigued, sedated, afraid, or in pain? What alternatives explain silence or success?

Valid report and command following provide strong positive evidence because they demonstrate organized access under a defined task. Their absence is much weaker when language, sensory input, motor output, arousal, pain, or fatigue can fail. The confidence system should therefore treat positive and negative observations asymmetrically rather than making silence decisive (Giacino et al., 2018; Bodien et al., 2024).

11.2.5 — Physiological and Neural Evidence

Marker reports need verbs that preserve their evidential role. A neural or physiological result may support, constrain, challenge, or leave open a continuity model. It does not certify consciousness, reveal personhood, or rank a bearer. The verb attached to a result should match the qualification domain and

uncertainty of the marker, especially when the sentence may circulate outside the technical context (Casali et al., 2013; Jang et al., 2024).

Every physiological or neural marker should travel with provenance. The record should name the acquisition system, preprocessing version, artifact policy, development population, task, medication and state context, calibration target, and transport limits. Without that packet, a locally useful marker can become a universal-looking number whose failures are concentrated in the populations least able to contest it.

Neural and physiological markers can also conflict with each other. A person may show low behavioral responsiveness but preserved task activation, preserved arousal but poor integration, or strong autonomic response but uncertain subjective experience. The chapter treats such conflict as information rather than noise because discordance often reveals where the model is underdeveloped.

Physiological and neural evidence can reveal organization that behavior misses. Sleep architecture, evoked responses, autonomic coordination, functional connectivity, perturbational complexity, task-based activity, and recovery patterns can all bear on access and unity. They are indispensable in cases where report is absent or unreliable.

The strongest use is triangulation. If neural evidence, physiological organization, command-following tasks, learning traces, and later recovery converge, confidence rises. If they diverge, the divergence is information. It may show motor

bottleneck, sedation, measurement artifact, partition, state fluctuation, or inadequate theory.

Biomarker sovereignty begins when a useful index is treated as self-interpreting. Complexity, connectivity, command following, autonomic response, or behavioral performance each depends on a target, task, population, preprocessing chain, and rival explanation. The bearer dossier should therefore preserve those dependencies and prohibit any single metric from becoming a portable oracle.

11.2.6 — Confidence Classes

A confidence class should read as a bounded scientific proposition, not a badge. A statement such as qualified continuity across a specified recovery window under routing model M, with marker families A and B available and command following unavailable, is less convenient than one word but identifies the target, interval, evidence, and limitation that give the classification meaning (Green & Swets, 1966; Berger, 1985).

The record should include version, date, target event, reference class, available and missing channels, threshold family, uncertainty, unresolved rivals, reviewers, permitted analyses, prohibited uses, and expiration conditions. Behavioral recovery, covert command following, continuity across an interval, and admissibility for ledger analysis are distinct events; a model trained for one cannot be renamed as another because inputs overlap.

Numerical confidence requires held-out evaluation under a frozen outcome, horizon, score, and reference class. Reliability curves

and proper scoring rules can test calibration, but selective verification from recovery, repeated assessment, survival, site resources, and family advocacy must also be modeled. Predictive calibration alone does not validate the bearer construct (Gneiting & Raftery, 2007; Bodien et al., 2024; Little & Rubin, 2019).

Decision thresholds remain separate from evidence and should be tested as a family. One threshold may justify additional noninvasive assessment and another entry into a primary analysis; the loss function belongs to that decision. Transport across sites, languages, injuries, ages, species, or devices likewise requires evidence rather than a portable universal score (Manski, 2003).

Public reports state the supported relation, interval, model, uncertainty, and permitted use. They do not convert a confidence class into conscious, unconscious, same person, different person, or another claim beyond scope, and they provide challenge, review, and revocation routes without risking care or participation (Tabassi, 2023).

The final system reports admissible stream models, unavailable evidence, instability across approved analyses, and triggers for renewal, narrowing, retirement, or No adequate estimate. Confidence constrains what may be claimed; it never grades the worth, rights, or standing of the being described.

11.3 — Soft Unity Gates and Threshold Families

A declared unity gate makes its role in the mathematics inspectable. Testing it across a prospectively specified threshold family reveals how membership uncertainty affects the result. The gate must remain independent of the terminal outcome so the study tests its chosen bearer instead of selecting the bearer that produces the preferred pattern.

A threshold is a decision rule layered onto evidence, not a discovered line in nature. Signal-detection and Bayesian decision theory make the costs of false inclusion and false exclusion explicit, while threshold-family analysis reveals whether a routing conclusion depends on one convenient cut point (Green & Swets, 1966; Berger, 1985).

Low-signal intervals must remain humanly visible. A low weight in a model is not a lower person. A zero in an inclusion rule is not a moral zero. A threshold boundary is a claim about evidence under a model, and every public sentence must keep that limited status alive.

A unity gate decides which intervals are eligible to enter a stream-bound ledger. Under the stream-routing rules in Section 5.1, the gate is necessary because the Law concerns separately routed qualifying streams, not every biological event in an organism. In empirical work, however, the gate is never a pure metaphysical switch. It is estimated through uncertain evidence.

The hard gate is useful as a protocol-fixed baseline because it makes inclusion decisions reproducible. It is not final authority. A threshold chosen because it produces a cleaner ledger, fewer

unresolved intervals, or a more attractive terminal curve has converted a sensitivity parameter into an outcome-selection device. The study should freeze the baseline and publish the result across approved threshold families.

Soft gates and threshold families preserve the structure of the formal idea while acknowledging that the evidence may be graded. They ask how conclusions change when plausible unity rules vary. If a result survives only one cutoff, it is threshold-sensitive and requires independent justification of that cutoff before it can support the broader claim.

False inclusion and false exclusion are symmetric sources of model error, but their consequences are not identical. False inclusion can contaminate ownership and terminal estimates. False exclusion can erase possible experience and hide the missingness that produced a clean sample. The audit therefore reports both rates, their downstream effects, and the protections that remain fixed regardless of classification.

Table 11.2. Soft unity gates, failure modes, and controls.

Gate type	Scientific use	Failure mode	Required control
Hard gate	Primary protocol-fixed inclusion rule	False precision near threshold	Threshold-family sensitivity
Soft gate	Represents graded or uncertain inclusion	Opaque model confidence	Calibration, priors, theory labels

Gate type	Scientific use	Failure mode	Required control
Conservative gate	Avoids false inclusion	False exclusion of low-signal experience	Protective no-estimate and sensitivity
Permissive gate	Avoids erasing possible experience	False inclusion or ledger inflation	Independent motivation and public labels
Outcome-tuned gate	None in confirmatory use	Circular closure or rescue	Disallow confirmatory claim

11.3.1 — Hard-Gate Baseline

A hard gate can still be useful in preregistration because it makes failure possible. If the gate is declared and the study fails under it, the result cannot be escaped by calling the cutoff merely illustrative. The project may later test a changed gate, but that becomes a new version of the claim, not a reinterpretation of the same result (Green & Swets, 1966; Berger, 1985).

The hard gate should be viewed like a ruler with a visible edge. It is useful because the edge is explicit; it is dangerous because reality near the edge does not become sharp simply because the ruler is. A serious chapter lets readers see the edge and then asks how much of the conclusion depends on it.

A hard gate can be appropriate for an engineering check, a simulation baseline, or a conservative sensitivity condition. It becomes unsafe when it is written as if nature itself has agreed to the cutoff. The chapter's language should therefore say 'under this gate' rather than 'the stream is absent' whenever the evidence rests on threshold convention.

The hard-gate baseline says that an interval either enters the stream analysis or it does not. In notation, the formal gate can be represented as open when the unity proxy exceeds a prospectively specified threshold and closed when it does not. This is clean enough for a first model and dangerous enough to require immediate limits.

The advantage is auditability. A hard gate can be fixed in the protocol, replicated, challenged, and tested against alternatives. It prevents analysts from silently changing inclusion after seeing the ledger. It gives the study a rule before emotion, hope, or theory preference can move the line.

The disadvantage is false precision. A person near a threshold may be classified differently because of noise, medication, sleep stage, artifact, task burden, device version, or sampling time. A continuous biological and experiential reality can be forced into a binary file. The line can create certainty the evidence does not deserve.

Hard gates are most vulnerable in low-signal and edge states. A person may have weak reportability, partial access, impaired motor output, or fluctuating arousal. A rigid threshold can exclude precisely those intervals where caution and care should be highest.

For the Law's empirical program, the hard-gate baseline is never enough by itself. It can be the primary protocol-fixed rule only when paired with sensitivity across threshold families, missingness analysis, and explicit reporting of intervals near the boundary.

11.3.2 — Soft Gate $g_0(t)$

Soft gates also need calibration. A model can produce smooth probabilities that look elegant but are badly calibrated. If intervals assigned 0.8 confidence behave like 0.4 confidence under later review, the gate is overconfident. Calibration error should propagate into the ledger and verdict rather than being hidden in the consciousness layer (Casali et al., 2013; Dehaene et al., 2017).

The soft gate has one valid function: propagate stream-membership uncertainty through a declared research calculation. Its parameters, calibration, and sensitivity are available for critique, while its output remains barred from person-scoring, consciousness grading, eligibility, or spiritual interpretation.

In the probabilistic implementation described here, the soft gate represents uncertainty about whether an interval belongs to the declared stream. An inclusion probability is not a claim that the experience itself is fractionally present. A genuinely graded unity representation is a different modeling choice and requires its own interpretation and bridge to the formal target.

This approach is especially useful when markers are partial. Report may be absent, physiology organized, neural evidence ambiguous, and later recovery informative but not decisive. A soft gate can express the fact that the interval contributes to a range of possible ledgers rather than one asserted path.

Soft gates also make uncertainty visible. If an interval has high probability of unity under one theory and lower probability under another, the result can be shown as a family of trajectories. This

prevents the public graph from hiding the unresolved evidence behind one smooth line.

The risk is that soft gates can look more sophisticated than they are. A probability is only as good as the model, training data, priors, calibration, and context of use behind it. A poorly justified soft gate is not safer than a hard threshold. It is merely harder to question.

A soft gate is a model, not a moral compromise. Its evidence sources, calibration domain, theory dependence, time resolution, uncertainty, and failure conditions should be declared before outcome analysis. Competing gate families should receive the same information and scoring opportunities so that the terminal conclusion cannot depend on an unreported preference for one smooth classifier.

11.3.3 — Threshold Families

A threshold family is a prospectively specified set of plausible unity rules, not a collection assembled after the terminal result is visible. It may vary cutoffs, marker combinations, smoothing windows, filtering rules, artifact exclusions, and theory-weighted evidence, but every variant must have an independent scientific rationale. The family should also include a burden check because the most data-hungry gate may be clinically or ethically unacceptable (Berger, 1985; Manski, 2003).

Rule variants matter as much as numeric cutoffs. A conclusion may survive several thresholds while depending entirely on one preprocessing choice, one marker family, or retrospective smoothing. The approved family should therefore expose the

major ways classification can change without changing the underlying observations.

Selection rules must be frozen before outcome inspection. The protocol should state how near-boundary intervals, missing channels, discordant markers, and low-quality observations are represented. If the family changes after the ledger is known, it becomes an endpoint-rescue device rather than a robustness analysis.

Robustness should be matched to the claim. A stable access bottleneck may survive a wider family than exact cumulative contribution, and a routing conclusion may remain unresolved even when a behavioral distinction is robust. The verdict should inherit the stability of the specific proposition tested, not the ambition of the surrounding theory.

Threshold families require calibration as well as discrimination. A rule may separate groups while assigning poorly calibrated probabilities to individual cases. Held-out reliability, proper scoring, sensitivity, specificity, and decision-curve consequences should therefore be reported together when the gate influences confidence classes or routing (Gneiting & Raftery, 2007).

Different uses may justify different loss functions. A discovery study may accept more false inclusions to avoid overlooking candidate experience, while a primary terminal analysis may require stronger evidential support before assigning a specific route. Publishing both choices makes the ethical and inferential tradeoff visible and prevents one cutoff from being marketed as a universal boundary.

Public presentation should preserve the family rather than collapse it into one dramatic curve. Ranges, shaded trajectories, status tables, and explicit threshold-sensitive conclusions usually communicate the result more honestly than a single line. A finding that appears under only one convenient rule must remain gate-specific.

Transport must be tested separately. A threshold learned in healthy adults may fail under anesthesia, infancy, severe injury, nonhuman neuroanatomy, organoids, or artificial architecture because both the marker distribution and the observation process change. A nontransportable cutoff receives a reference-class limit; additional decimal places do not make it universal.

11.3.4 — False Inclusion Costs

False inclusion can also occur through compassion. Researchers may want to honor a difficult case by recognizing experience, but compassion cannot replace evidence. The fairer act may be to protect the person fully while narrowing the scientific claim. Care can be generous even when classification is restrained (Giacino et al., 2018; Bodien et al., 2024).

False inclusion also matters for artificial or archival systems. A system that produces fluent language, coherent behavior, or an imitation of a person may invite attribution. This chapter does not decide artificial consciousness, but it installs the caution that a trace or output must not be promoted to bearer status without a declared theory and evidence path. The next chapter carries the broader boundary-state comparison.

The scientific repair for false inclusion is usually not ridicule. It is narrower status language. A channel may be called context, behavioral output, auxiliary evidence, or simulation trace rather than stream evidence. Naming the lesser status preserves useful information without granting it the wrong authority.

Unsupported inclusion occurs when an interval is assigned to a candidate stream without sufficient evidence under the declared unity and routing rules. False inclusion occurs when the interval does not in fact belong to that stream. Either problem can compromise inference; actual false inclusion can add non-bearer time to a ledger, blur partition, or misclassify interruption as continuity.

The cost is not merely numerical. False inclusion can make a trajectory look smoother, more complete, or more closed than it is. It can also conceal a real discontinuity or partition, producing a single cumulative story where the evidence warrants several alternatives.

False inclusion has ethical costs as well. It can lead researchers to describe a person's interior life with more certainty than evidence permits. It can burden families with claims about hidden experience that are not justified. It can make public artifacts feel more intimate or authoritative than they are.

The scientific protection is conservative labeling. Intervals with weak evidence should be marked uncertain, low signal, no estimate, or sensitivity-only rather than silently included. If they are included under a permissive gate, the analysis should say so and show how the result changes when they are excluded.

False inclusion is especially dangerous when it helps the Law. If including uncertain intervals improves apparent closure, the result must be treated as threshold-sensitive unless the inclusion rule was independent, prospectively specified, and robust under rivals.

11.3.5 — False Exclusion Costs

False exclusion should be tracked as a live rival in every low-signal context. The analysis should ask not only whether the signal is below threshold, but whether the threshold is blind to the very forms of consciousness most likely in that state. A detector that works only for healthy expressive adults is not a universal unity gate (World Medical Association, 2024; Andrews et al., 2024).

The false-exclusion problem also has a data-design form. If a study collects evidence only when a participant is easy to measure, the excluded intervals may be precisely those with higher burden. Device intolerance, fatigue, delirium, pain, communication difficulty, and clinical instability can all create missingness that mimics absence. Exclusion can become a hidden selection effect.

A low-confidence interval should enter the data model with a reason code and an uncertainty state, not as zero exposure or absence. Downstream analyses can then carry it through sensitivity sets, alternative routing models, or no-estimate rules. This preserves the distinction between an interval that contributes neutral value and an interval the study cannot responsibly classify.

False exclusion occurs when a conscious interval is omitted because the available markers are weak, missing, delayed,

behaviorally silent, or misread. Scientifically, this can erase parts of the path. Ethically, it can deny evidence of experience where care and humility should remain active.

False exclusion is particularly serious in disorders of consciousness, locked-in or motor-limited states, deep fatigue, sedation, sleep, delirium, dissociation, and severe disability. Behavioral silence may reflect motor output failure, language barrier, task design, arousal fluctuation, or measurement limits rather than absence of experience.

The cost to the Law is direct. A ledger built from visible intervals only may become a ledger of access, not experience. If the program excludes the hardest intervals because they are inconvenient, it may test a cleaned version of the hypothesis rather than the hypothesis itself.

The protection is precaution with status discipline. Precaution does not mean declaring consciousness present. It means preserving uncertain intervals, seeking additional evidence where ethically possible, using soft or sensitivity analyses, and refusing public language that equates silence with absence.

False exclusion is also a public-risk problem. If a confidence class becomes visible outside the study, low-signal persons may be treated as less present, less protected, or less socially meaningful. The chapter prohibits that leakage.

False exclusion has a different consequence: it can erase possible experience and bias the ledger toward easily observed states. Where exclusion would affect care or conceal plausible burden, the response is protective uncertainty and a no-estimate route, not

automatic inclusion in the formal object. Ethical protection can be broader than scientific classification.

11.3.6 — No Threshold Chosen for Closure

No unity threshold may be chosen because it makes the terminal result look favorable. Features, priors, thresholds, exception rules, and the approved threshold family must be fixed before the ledger or endpoint outcome is examined. Otherwise the bearer is defined by the closure pattern the study later presents as evidence, creating a direct form of outcome-dependent selection (Hernán & Robins, 2020).

The protocol-fixed primary gate should be accompanied by a limited family of scientifically motivated alternatives. The alternatives are not an invitation to search for the best-looking result. They are a sensitivity design that shows whether the conclusion survives plausible uncertainty about unity and continuity.

False inclusion and false exclusion must remain separate error classes. A permissive threshold may join intervals that do not belong to one routed stream; a restrictive threshold may erase quiet, inaccessible, or weakly measured experience. The ethical response to those errors is protection, but the scientific response is to estimate their different consequences rather than hide them inside one accuracy number.

Soft gates can preserve uncertainty when the evidence does not support a hard boundary. Their weights must still have a declared interpretation, calibration domain, and downstream rule. A smooth curve is not automatically more truthful than a threshold,

and a weight chosen after ledger inspection is no less circular than a post hoc binary gate. Weight uncertainty should propagate into downstream ledger or endpoint conclusions rather than vanish when the soft gate is summarized by one expected value.

Any post-registration change begins a new model version. The original analysis remains visible, the reason for change is recorded, and the revised gate is evaluated on new or properly held-out data. An unfavorable primary result cannot be rescued by silently moving the threshold. A revised gate should also be tested against cases that motivated the change, but only in a separate validation role so they do not serve simultaneously as tuning and confirmation data.

Sensitivity should include threshold perturbation, leave-one-channel-out analyses, alternative missingness assumptions, and, where justified, model averaging over the prospectively specified family. Proper scoring may compare predictive calibration across gates, but it does not determine which gate identifies phenomenal unity (Gneiting & Raftery, 2007). Sensitivity should also report threshold regions over which the substantive verdict is unchanged, distinguishing robust conclusions from decisions balanced on a narrow cutpoint.

A gate may legitimately make the study unusable. If every plausible threshold produces unstable ownership, excessive double counting, or loss of essential intervals, the affected evidence is Unusable when a required ownership or measurement gate fails. If the required gates pass but the target still cannot be identified or validly bounded, the result is No adequate estimate. That outcome protects the universal claim from a manufactured

test and identifies the exact layer that needs better evidence. The failed layer should be named explicitly so later work knows whether better markers, routing evidence, or a different estimand could restore adjudicability.

Threshold discipline therefore makes failure informative. The study can report which marker families control eligibility, how sensitive the ledger is to the unity rule, and which conclusions survive. It cannot choose the bearer that closes. This converts threshold choice from a hidden tuning parameter into an observable source of verdict sensitivity.

11.4 — Pause, Interruption, and Rebinding

The gap cases are where ordinary intuition often breaks down. A person may return with no memory, with partial memory, with a changed sense of time, with a restored communication channel, or with evidence that later clarifies the interval. The analysis preserves the difference between what happened and what could be detected at the time.

Sleep and anesthesia demonstrate that responsiveness, report, memory, and experience can dissociate. Dreaming during low behavioral access and connected consciousness during anesthesia make pause and rebinding empirical questions rather than intuitions about what an unresponsive body must contain (Siclari et al., 2017; Sanders et al., 2012; Linassi et al., 2018).

A gap is recorded as an evidence object rather than a blank interval waiting for a preferred story. Its state, observation process, missingness, burden, medication, review, and revision history tell later analyses which continuity claims remain licensed.

Scientific routing distinguishes reversible loss of access, temporary interruption, suspension, restart, branching, and rebinding. Comparing these models lets pause, memory gaps, and return contribute to an explicit account of continuity.

Pause and return cases require a versioned sequence rather than one retrospective label. The record should identify the last well-supported pre-gap state, evidence available during the gap, return evidence, and the date on which continuity confidence changed. Sleep, anesthesia, temporary disconnection, and recovery then

become comparable transition structures without being forced into one ontology.

Later recovery can sharpen retrospective classification without converting uncertainty into a fact that was available earlier. The primary prospective record should retain its original confidence class; a secondary smoothed record may incorporate return, report, and learning evidence; and the difference between the two should remain visible. That separation prevents hindsight from entering a supposed pre-gap decision rule.

The output is a pause-return timeline: pre-gap state, entry event, evidence during gap, exit event, post-gap evidence, continuity rivals, confidence class, and revision rule. The timeline belongs to research. It does not decide care or dignity.

Table 11.3. Pause-return timeline fields and inferential purpose.

Timeline field	What must be recorded	Why it matters
Pre-gap state	Access, report, physiology, clinical context, confidence class	Establishes the last known status without assuming continuity
Entry event	Sleep onset, induction, injury, seizure, disconnection, or unknown transition	Prevents a visible event from becoming a false boundary
Gap evidence	Signals, missingness, artifacts, care events, possible reports	Keeps low signal distinct from absence

Timeline field	What must be recorded	Why it matters
Exit event	Awakening, emergence, recovery, communication return, or observation change	Prevents return from retroactively settling the gap
Post-gap evidence	Memory, learning, affective carryover, command following, neural organization	Supports or weakens rebinding without erasing uncertainty
Revision status	Current class, previous class, reviewer, reason for change	Preserves scientific memory and family burden

11.4.1 — Sleep and Reversible Loss of Access

Sleep also protects the chapter from a common rhetorical shortcut: treating wakefulness as the template for consciousness. Wakefulness is one access regime, not the essence of the stream. A fair bearer model asks what forms of experience and unity may occur outside the waking report style that research convenience prefers (Siclari et al., 2017; Cataldi et al., 2024).

Sleep also reveals how time scale affects identity. Over seconds, integration may fragment; over hours, the organism cycles; over mornings, the person returns to ordinary orientation. A chapter that asks only one time-scale question will misclassify the phenomenon. The stream model must say whether it is classifying momentary inclusion, overnight continuity, or longer narrative persistence.

The familiarity of sleep provides an accessible starting point for precise research. A useful study specifies the sleep state, evidence

channels, report filter, unity gate, memory bias, and intended inference. Those choices let an ordinary experience become an informative test of the bearer model.

Sleep is the ordinary teacher of this chapter. It reduces access, reportability, environmental responsiveness, and voluntary control, yet it does not simply remove experience. Dreams, partial awareness, memory gaps, and sleep-stage transitions show that the relation between behavior and experience is not binary.

For the Law's formal structure, sleep raises the unity-gate question. Some intervals may involve unified experience. Some may involve low or no reportable experience. Some may involve state-dependent forms of affective processing. A chapter on sleep later in the book will examine those signatures; here the issue is stream identity across reduced access.

The continuity evidence after sleep includes ordinary waking orientation, memory fragments, physiological state transitions, recurrence of goals, and preservation of person-specific patterns. Those markers usually support continuity, but they do not license a simple rule that every sleep interval is open or closed.

A research file separates sleep onset, sleep stage, dream report, physiological markers, awakening, and later recall. Recording the timing and availability of each channel helps distinguish the experience sampled from the portions later remembered.

Sleep contains large domains of experience whose accessibility varies with stage, awakening, recall, distress, and physiology. Restricting the bearer to ordinary waking report would erase potentially experienced intervals; admitting every organized sleep

pattern would count physiology without adequate evidence. The scientific response is a family of protocol-fixed gates and visible sensitivity analyses, not a universal sleep switch.

Sleep commonly supports organism-level continuity, while interval-level stream inclusion depends on state and evidence. Alternative unity gates can be compared using stage labels, dream reports, neural markers, and awakening procedures (Siclari et al., 2017).

11.4.2 — Anesthesia and Return

Anesthesia cases should record the difference between clinical purpose and research interpretation. The drug is given for care, comfort, surgery, or procedure, not to test the Law. Research may learn from clinically indicated care under an appropriate protocol, while care decisions remain independent of the theory (Sanders et al., 2012; Linassi et al., 2018).

Clinical anesthesia monitoring serves patient safety, dosing, analgesia, and recovery. Those signals can inform a research state model only under a separate protocol that does not alter care or imply that a depth index identifies the phenomenal bearer. The scientific use must remain subordinate to clinical purpose, especially when movement, memory, and report are pharmacologically unavailable (Sanders et al., 2012; Jang et al., 2024).

Emergence is especially instructive because return is not a single instant. A patient may regain eye opening, then command following, then orientation, then memory formation, then executive control, with variation across drugs and persons. A

stream-identity model that forces one return time should report how much uncertainty it has discarded.

Anesthesia offers a clinical setting for studying profound unresponsiveness, altered neural dynamics, amnesia, and later return. The observation model distinguishes awareness, implicit processing, state transitions, and subsequent recall.

The scientific record should separate induction, maintenance, surgical stimulation, medication, physiological monitoring, neural markers where available, emergence, post-anesthesia report, and later follow-up. Each stage bears on a different claim. The entry into anesthesia is not the same as truth-level endpoint, and return does not automatically classify every intervening moment.

Anesthesia also exposes the danger of single-threshold unity. If the threshold is too strict, it may erase low-level or intermittent experience. If too permissive, it may include intervals of purely physiological response. Both errors matter, and both should be reported through sensitivity rather than hidden inside one verdict.

A return case should preserve versioned uncertainty. At time of emergence, a reviewer may classify the gap as probable interruption with possible unreported experience. Later evidence may revise that classification. The revision should append to the record, not overwrite the earlier state as if certainty was always present.

Anesthesia also shows how research can misallocate uncertainty. Calling an unresponsive interval empty can erase inaccessible experience; describing every possibility as hidden awareness can

impose fear on patients and families. Drug-specific evidence, state-specific markers, prospective consent, and clinical independence provide a narrower route between those errors.

Anesthetic interruption should be classified by agent, dose, task conditions, physiological state, memory effects, emergence pattern, and available neural evidence. Return supports continued organismal and often personal continuity, but it does not retroactively establish conscious presence through every unobserved interval. Gap status remains model-relative until the evidence can discriminate it.

11.4.3 — Temporary Disconnection

The protocol should also test whether study procedures degrade the observation channels. A burdensome device, confusing task, repeated awakening, or anxiety-producing protocol should be logged as a possible source of reduced responsiveness, unreliable reporting, or missing data. The classification record should distinguish these observation failures from disease-related disconnection and from interruption of the stream itself.

Disconnection can be environmental or communicative as well as neural. Language mismatch, sensory loss, fear, restraint, institutional neglect, and inaccessible interfaces can make a present person appear unreachable. Continuity analysis should therefore model the relation among organism, environment, observer, and channel before interpreting failed communication as interruption of the stream.

This matters for fairness because some people are repeatedly misclassified by systems built around the communication style of

others. Disability, trauma, unfamiliar language, poverty, race, age, and clinical context can all alter the chance that a stream becomes visible to researchers. A disconnection model that ignores those conditions imports social bias into consciousness science.

Temporary disconnection covers cases where ordinary coordination among access, report, memory, motor output, or neural integration is disrupted without immediate evidence that the stream has ended. It includes neurological injury, seizure-related interruption, communication loss, medication effects, severe fatigue, dissociation, and technology-mediated future cases.

The key question is whether the disconnection is an access problem, an integration problem, a report problem, a control problem, or a stream-continuity problem. Those are different. A locked-in patient may retain experience with impaired output. A dissociative state may alter access to memory or self-organization. A seizure may disrupt control while leaving later continuity complex.

A temporary-disconnection record should document onset, observed channels, unavailable channels, likely confounds, duration, recovery markers, and competing models. It should also include a no-estimate option when evidence cannot distinguish access disruption from interruption or partition.

Temporary disconnection is a place where public language must be especially careful. Saying the person was gone can overstate absence. Saying the person was fully present can overstate

evidence. Saying the stream is unresolved may be the only honest research phrase.

Visible coherence is not a prerequisite for possible experience. A person whose access is disrupted may be unable to demonstrate continuity through report, movement, or narrative even when other evidence remains. The routing model should therefore separate failure to perform continuity from evidence of discontinuity and preserve a no-estimate state when those explanations cannot be distinguished.

Temporary disconnection should be decomposed into the functions that failed: external access, internal integration, motor output, communication, memory, or cross-module coordination. Different failures support different continuity models. The classification concerns the disrupted relation, not a diminished person.

11.4.4 — Rebinding After Recovery

Rebinding evidence should be strongest when it is convergent across time rather than dramatic at one moment. A single striking statement after recovery may matter, but stable orientation, affective continuity, learning, and functional reintegration across repeated observations usually carry more scientific weight. The model should privilege patterned evidence over the most moving anecdote (Jang et al., 2024; Nadel & Moscovitch, 1997).

Rebinding should be evaluated with humility toward the person's own later testimony where communication is possible. A participant's report cannot settle every interval, but it can correct misunderstanding, identify hidden awareness, explain missing

behavior, or challenge the model's assumptions. The challenge route belongs inside the science rather than outside it.

The same recovery can support different claims at different levels. It may support legal continuity, ordinary social continuity, clinical recovery, and research continuity in different ways. This chapter owns only the research level. It should not borrow the warmth of social return to overstate stream evidence, and it should not borrow scientific uncertainty to chill social welcome.

Rebinding asks whether later evidence supports connecting a post-gap state to a pre-gap stream. It is not the same as waking up, speaking again, or recovering behavior. Those events are evidence, but the scientific question is whether the same unified trajectory should include intervals before and after the gap, and with what confidence.

Evidence for rebinding can include restored context, recognition, learning continuity, affective carryover, recurrent goals, neural-state recovery, physiological rhythms, and coherent response to earlier material. The evidence is stronger when it cannot be explained by external prompting, record exposure, ordinary reconstruction, or generic recovery.

Rebinding also has limits. A person may regain function while remaining uncertain about the gap. A later story may be coherent because humans are narrative repairers. A neural return may show system recovery without resolving whether the intervening interval was continuous, interrupted, or absent.

The versioned record should therefore distinguish confirmed post-gap function from gap classification. The person has

returned to accessible interaction. The gap remains classified according to its evidence. Those two sentences can coexist without contradiction.

This distinction protects families from false comfort and false fear. It also protects science from post hoc smoothing. A good ending does not retroactively clean the evidence. A difficult ending does not retroactively erase possible continuity.

Rebinding requires evidence, evaluated under a declared information set, that the later state is causally and structurally connected to the earlier candidate stream. Recovery, memory, recurrent dynamics, and affective carryover can update confidence, but none by itself converts an unknown gap into known continuity. The original uncertainty remains in the version history.

11.4.5 — Competing Continuity Models

The model family also keeps the book from turning continuity into consolation. Persistence may be emotionally comforting in some cases and frightening in others. Restart may be comforting in some interpretations and disturbing in others. The science should not choose the model whose emotional tone is easiest; it should choose the model best supported by evidence and safety (Cogitate Consortium et al., 2025; Bayne, 2010).

The model family should be presented before case interpretation. If the available options are defined only after a puzzling case appears, the program will invent categories to fit anecdotes. By naming persistence, suspension, restart, and branching in

advance, the chapter gives later protocols a disciplined menu of hypotheses.

Model choice should also affect public language. A persistence model can support continuity language; a suspension model supports return language; a restart model supports new-stream language only within research; a branching model supports unresolved inheritance language. None supports casual statements about worth, identity value, or spiritual completion.

The same pause-return evidence can support several models. Persistence says the stream continued through the gap. Suspension says the stream paused and resumed. Restart says a new stream began after the gap. Branching says more than one trajectory may have emerged. Partition says coordination changed without a simple single-stream story.

These model labels do not by themselves entail distinct observable predictions. A declared operational version may predict differences in affective carryover, learning traces, state-dependent markers, or access, memory, and control; those contrasts must be specified before selection and justified by the version's mechanism and observation model. When versions yield the same observable law, the result remains an observationally equivalent model set.

A model comparison should be modest. The evidence may not distinguish persistence from suspension, especially when the interval is low signal. It may distinguish both from a crude death-of-mind claim. The verdict can be narrowed without being final.

The program should also consider theory dependence. Some consciousness theories will favor access continuity. Others will emphasize phenomenal unity, recurrent processing, integration, or higher-order availability. The chapter does not settle theory; it records how the classification depends on theory family.

Competing continuity models should not be chosen for narrative appeal. Persistence can feel comforting, restart can feel decisive, suspension can simplify the ledger, and branching can make the theory seem profound. None of those aesthetic advantages is evidence. Model comparison should instead use frozen predictions, state-transition data, transport checks, and explicit consequences for downstream ownership.

11.4.6 — Uncertainty Across the Gap

A gap does not interpret itself. Persistence with reduced access, suspension, restart, rebinding, and unresolved routing can fit the same interval while joining later evidence to the ledger differently. Sedation, sensor loss, motor or language failure, fatigue, transfer, and selective awakening remain inside the observation model rather than becoming evidence of absent experience.

Prospective filtering and retrospective smoothing stay separate. Later recovery or improved data may revise a secondary reconstruction, but cannot enter the primary model as if available at the time. Missingness may also be informative because clinical instability, death, withdrawal, burden, or technical failure often removes the hardest intervals (Särkkä, 2013; Little & Rubin, 2019). A primary filtered analysis can remain valid even when a

later smoothed reconstruction becomes more accurate, because the two answer different information-time questions.

Competing gap models face common predictions about return, recurrence, learning, cross-gap transfer, and later verification. The protocol ranks gaps by inferential role, since a short endpoint-adjacent interval may control the verdict while a longer earlier gap adds little uncertainty. Prioritizing gaps by inferential leverage can reduce participant burden by avoiding intensive recovery work on intervals that cannot materially change the verdict.

When evidence cannot separate the admissible models, results remain conditional or are withheld. No adequate estimate is preferable to an invented continuation, and the preserved set prepares partition and fusion analysis without assuming that every return restores one stream or every silence ends one (Giacino et al., 2018). Conditional reporting should identify which continuity models support each conclusion, allowing later evidence to update the model set without rewriting the observed record.

11.5 — Partition, Fusion, and Multiplicity

Partition and fusion affect which experiences enter each ledger. The analysis must distinguish evidence of divided access from evidence of separate streams, and keep unresolved ownership visible without turning a research classification into a public identity.

Partition research cautions against two opposite simplifications: assuming one organism always carries exactly one stream, and treating every functional division as a separate consciousness. The strongest recent analyses of split brains support graded and task-dependent interpretations, which is precisely why routing models must expose their criteria and uncertainty (Pinto et al., 2017; Schechter, 2018).

The ledger consequences are why this material belongs in the science volume. If the bearer is one stream, a path can be integrated one way. If the bearer is partitioned, allocation and ownership become open problems. If the bearer fuses, inheritance becomes open. These are not side stories; they are mathematical and ethical constraints on later inference.

Partition cases test whether the model can tolerate complexity without turning people into spectacle. One apparent bearer may show partial access separation, altered self-organization, or conflicting streams of control. The question is not whether the case is interesting. The question is what evidence supports one stream, more than one stream, intermittent partition, or unresolved access disruption.

Partition is a model class, not a dramatic conclusion. Split access, dissociation, and hemispheric asymmetry may support restricted information flow or partly independent control without establishing multiple consciousnesses. Conversely, declaring one stream by definition would prevent the science from learning when ordinary unity assumptions fail. The dossier should keep the smallest set of routing models compatible with the evidence.

The section therefore treats partition, fusion, and multiplicity as hypotheses with evidence requirements. Each requires marker families, rival models, and protective governance. No classification may become public drama, moral rank, legal division, or clinical verdict.

Partition and integration raise a specific question of ledger ownership: how should intervals be assigned, bounded, or left unresolved? This chapter states the competing assignments and the evidence needed to distinguish them, preparing the formal routing analysis developed later.

11.5.1 — Split Access and Hemispheric Partition

A split-access discussion should avoid both denial and mythology. Denial pretends the experiments teach nothing because daily life continues. Mythology pretends the experiments prove two fully independent lives. The chapter's scientific path lies between them: identify the level of access under strain and state how far that strain travels (Gazzaniga, 2000; Pinto et al., 2017).

Split-brain cases also teach that evidence can be layered. One layer may show separated access under laboratory tasks. Another

layer may show unified behavior in ordinary life. Another may involve subjective report that does not match the experimental dissociation. A serious account should let layers disagree without forcing a cartoon conclusion.

The chapter therefore compares unity and partition models under a common evidential standard. A split-access finding should trigger questions about what kind of unity is under pressure and what kind remains intact. That is exactly the kind of distinction the Law needs before it assigns ledger continuity.

Split access refers to partial separation of information, control, or report across systems that normally coordinate. Split-brain research is the familiar example, but the scientific lesson is broader: access can be divided without the ontology being settled. The evidence may support one stream with divided access, two partially independent access systems, or unresolved structure.

Task evidence cuts both ways. If one hand responds to information unavailable to speech, the result supports access separation under those conditions. It neither establishes two complete conscious streams nor secures one undivided stream. The observation creates a routing-model comparison that must be resolved, bounded, or carried forward as uncertainty.

The relevant markers include cross-cue transfer, conflict resolution, coordinated affective response, learning across channels, bodily control, report asymmetry, neural connectivity, and temporal stability of the dissociation. A single demonstration can be fascinating and still insufficient for bearer classification.

Laboratory tasks can magnify asymmetry because they are built to isolate information channels. A striking dissociation under controlled instruction may coexist with substantial coordination in ordinary activity. The routing analysis should therefore compare task-specific separation, spontaneous behavior, cross-cue transfer, longitudinal stability, and causal evidence before promoting a laboratory effect into a persistent partition model.

Split-access findings demand interpretation without spectacle. The scientific record may examine restricted coaccess, lateralized control, transfer failures, and task dependence. Public description should avoid turning a person into a metaphysical exhibit or using a laboratory dissociation to question legal standing, dignity, or care.

Split access supports partition only when intervention, information flow, control, and report patterns are better explained by more than one stream than by divided access within one stream. The analysis should compare those models directly and preserve uncertainty about event ownership. Split-brain evidence is a test case, not a template for every dissociation (Gazzaniga, 2000; Pinto et al., 2017).

11.5.2 — Dissociation and Altered Self-Organization

Research on dissociation can illuminate altered self-organization while remaining attentive to the burdens people may experience. Clinical collaboration and respectful communication help researchers understand those experiences in context (Bayne, 2010; Schechter, 2018).

Dissociation enters first as a lived and clinical phenomenon. Trauma, stress, memory disruption, altered agency, and changing self-organization can shape both experience and report. Participants are never required to resolve the metaphysics of their own state, and public description must not turn coping or distress into spectacle.

A respectful evidence model can still be rigorous. It can ask about amnesia boundaries, state-dependent access, executive control, physiological shifts, affective continuity, and therapy-related integration. What it cannot do is convert diagnosis into ledger allocation without a formal and clinical bridge.

Dissociation presents another challenge because it can alter memory, self-relation, agency, emotional access, and narrative continuity. Some presentations include distinct self-states or access patterns. The scientific question is not whether those experiences are real or worthy of care. They are. The question is what kind of stream classification the evidence supports.

Clinical diagnosis cannot be translated directly into stream multiplicity. Diagnosis serves treatment, safety, and understanding. Research bearer classification asks a different and narrower question about unified experience across intervals. The two domains should collaborate without collapsing into one another.

Evidence should include within-person history, state transitions, amnesia patterns, co-consciousness, physiological changes, trauma context, report conditions, clinician assessment, longitudinal stability, and participant interpretation. It should also

include rival explanations such as access barriers, memory compartmentalization, state-dependent recall, role enactment, suggestion, medication, or stress response where relevant.

Research language must be trauma-sensitive. The person should not be asked to perform identity complexity for scientific curiosity. Interview burden, public exposure, destabilization, and clinical disruption can alter the very access patterns being studied. Safety is part of validity.

Dissociation is especially vulnerable to public spectacle. A striking narrative can be memorable while remaining diagnostically incomplete, and publicity can make the participant carry risks that the audience never sees. Research use therefore requires clinical context, privacy controls, and a question narrow enough that the classification need not become a public identity.

Dissociative symptoms do not map one-to-one onto literal stream division. Clinical context, state fluctuation, memory, control, trauma, suggestion, and treatment effects can support several models. Research use therefore requires collaboration with clinicians, privacy protection, and an explicit refusal to turn diagnosis into metaphysical proof.

11.5.3 — Parallel or Overlapping Streams

Parallel or overlapping streams are hypotheses about more than one candidate experiential organization within one apparent support system. The possibility belongs in the technical analysis because some clinical, theoretical, artificial, or hybrid cases may resist both simple unity and clean partition. It should remain rare

in public summaries and should never be presented as common, settled, or personally diagnostic (Bayne, 2010; Schechter, 2018).

The evidence burden is much higher than simultaneous processing, ambivalence, divided attention, changing roles, dissociative symptoms, or competing motivations. A multiplicity model requires patterned limits on joint availability, partially independent learning or control, and evidence that the separation is stable enough to distinguish it from ordinary segregation within one stream.

Overlapping models require event-token ownership rules. The model must say which events are exclusive, shared, unresolved, or conditionally assigned, and it must prove that no contribution is counted twice. When shared ownership cannot be identified, an uncertain or set-valued ledger is more honest than a clean division invented for convenience.

The alternatives are best represented as competing routing graphs. One graph may describe one stream with segregated access, another two streams sharing resources, and another alternation across time. They should be fitted and scored with the same observations, comparable complexity penalties, and the same opportunity to explain missingness so that a richer ontology does not win by construction.

Identifiability may fail even when every observed task is measured precisely. Distinct routing graphs can generate the same reports, choices, and neural summaries while differing only in unobserved phenomenal ownership. The correct output is then an

equivalence class of routes and a statement of what intervention or observation could distinguish them, not a forced partition.

Ordinary psychological conflict should not be reclassified as stream multiplicity. The one-stream and unresolved models remain preferable until evidence about causal insulation, ownership, nonduplication, and cross-context persistence defeats them. This asymmetry protects both scientific parsimony and people who could otherwise be sensationalized by a narrow research hypothesis.

A multiplicity analysis belongs in a purpose-limited research dossier, not a public identity file. The record may preserve the model, tasks, uncertainties, and prohibited uses while withholding a simplified label that could be misunderstood socially, legally, or spiritually. Privacy and dignity are part of the evidential design because publicity can change the cost of error.

When current theory cannot distinguish unity, overlap, alternation, and partition, abstention is the scientific result. The program should retain the competing graphs, identify the missing discriminator, and continue care and ordinary rights without waiting for an ontology to become available.

11.5.4 — Fusion and Recoordination

A later return of integrated access may support a rebinding model, clarify earlier uncertainty, or show that a task-specific partition was transient. These observations help reconstruct the trajectory of access and coordination (Pinto et al., 2017; Tononi et al., 2016).

Fusion creates an unresolved ownership question rather than an intuitive inheritance rule. If previously separated candidate paths

later re-coordinate, the model must state whether earlier events belong to one ancestor, several descendants, a jointly owned domain, or an identified set of possibilities. Until a formal bridge is justified, ledger inheritance under fusion remains conditional and should be reported as such.

Clinically, recoordination may be meaningful long before it is formally resolved. A person's recovery of communication, agency, or coherence can guide care without resolving stream metaphysics. The chapter keeps that distinction so the science does not make clinical hope wait for a theorem.

Fusion and recoordination ask whether previously separated, partially separated, or independently modeled processes can form or resume a coordinated candidate stream. The issue arises after recovery, treatment, neurological change, task-context change, or future systems in which coordination can be altered.

Fusion should not be assumed because behavior becomes smoother. Coordination can improve at the level of motor control, report, clinical function, or public narrative without proving full experiential unity. Conversely, a person may experience increased internal communication or integration before external behavior makes that visible.

Evidence for fusion includes improved cross-access, shared memory, coordinated affective response, unified report, stable control, reduced internal barriers, and longitudinal consistency across contexts. The evidence should be gathered without forcing a person into a public story of becoming whole.

A recoordination model may be more accurate than fusion. Recoordination says separated channels or states are communicating better, not that a metaphysical merger has occurred. This distinction matters because the research may be about access and control rather than stream identity itself.

The record should keep pre-fusion, transition, and post-fusion evidence distinct. If earlier partition was uncertain, later integration does not prove that earlier multiplicity existed. If earlier partition was well-supported, later coordination does not erase the old classification; it revises the trajectory.

Fusion or recoordination describes a change in present organization; it does not retroactively settle the identity of earlier states. The protocol should test whether previously independent channels now share access, control, memory, and causal dependence, while preserving the pre-fusion routing record. Recovery is not permission to rewrite uncertainty as unity.

11.5.5 — Ledger Ownership Under Partition

The open problem remains visible throughout the book. A protocol that excludes partition cases names the exclusion; one that includes them specifies an allocation rule or returns identified-set results. Silence is not a neutral choice because it hides the assumption that determines event ownership (Bayne & Chalmers, 2003).

One safe production practice is to treat partitioned ledgers as scenario-specific. A study can report what follows under the one-stream assumption, what follows under a partition assumption, and what remains unidentified. That is more honest than hiding

the assumption in one number and more useful than refusing to analyze the case at all.

The open problem also has measurement consequences. Candidate momentary-valence and interval-contribution estimators are built for a candidate stream under conditions. If streams partition, channel-to-stream assignment becomes uncertain. Physiological signals may be shared, reports may be channel-limited, and context may affect multiple candidates. Measurement cannot be cleaner than bearer classification.

Ledger ownership is the formal problem created by partition. If one apparent organism supports more than one candidate stream, which intervals belong to which stream-bound ledger? If streams overlap, can an interval be shared? If a partition later recoordinates, does the ledger combine, continue separately, or remain unresolved? This chapter does not pretend those questions are solved.

The safest first answer is status labeling. A ledger analysis should say one-stream model, partition model, overlapping-stream model, merged-stream model, or no-estimate. It should not silently integrate all intervals into one path merely because that is easier to graph.

Under a partition hypothesis, each candidate stream would require its own evidence basis, unity gate, temporal window, and uncertainty. If the evidence cannot assign an interval responsibly, the interval should be marked unresolved rather than forced into the favored ledger.

The problem becomes more serious near endpoint research. If partition, fusion, or rebinding occurs near truth-level endpoint, endpoint classification and ledger interpretation become linked. The endpoint chapter will handle final adjudication; this chapter prepares the ownership discipline so endpoint science cannot inherit a hidden simplification.

A single cumulative line can conceal uncertainty about its bearer. Visual simplicity is therefore not a reason to collapse competing routing models or erase unclassified intervals. A visibly branched or partially identified record is preferable when that is what the evidence supports, because an honest complex graphic is more informative than a precise line attached to the wrong stream.

When partition is plausible and ownership is unresolved, the correct object may be set-valued or model-indexed rather than a single ledger. Analyses should report the range across admissible routings and stop if conclusions depend on arbitrary allocation. Administrative convenience is never an ownership principle.

11.5.6 — No Personhood or Moral Rank Inference

Partition, fusion, overlap, and unresolved multiplicity are research models of organization. They are not findings about personhood, authenticity, moral rank, legal identity, spiritual status, or the value of a life. That boundary must remain explicit even when the evidence for one routing model becomes strong (World Medical Association, 2024; Parfit, 1984).

The legal and moral person may continue to receive one coherent set of rights and protections while the scientific analysis compares several candidate ownership structures. Conversely, one routed

research stream does not exhaust the person's social, legal, relational, or narrative identity. The objects interact through consent and care, but neither replaces the other.

Reports and graphics require an anti-ordinal design. One-stream, partitioned, fused, overlapping, and unresolved models must not be arranged on a ladder that resembles health, worth, authenticity, or sophistication. Legends should name candidate routing structures, evidence strength, and uncertainty rather than higher and lower kinds of being. Ordering the models by confidence rather than by structure can help preserve evidence strength without implying that one routing form is intrinsically superior.

Privacy risk increases when unusual cases are described in detail. Rare trajectories, split-access patterns, family histories, timestamps, and clinical images can reidentify people even after names are removed. Purpose-limited records, consented communication, narrative restraint, and hostile-crop review are therefore part of scientific validity as well as dignity. Aggregation thresholds should be set before publication review so privacy protection does not depend on how compelling a particular case appears.

Case selection should resist novelty bias. Dramatic partition or dissociation cases may reveal that a simple one-stream rule fails, but ordinary null and ambiguous cases are needed to estimate when a more complex routing model is actually warranted. A literature of striking exceptions can exaggerate both prevalence and confidence. Prevalence estimation should therefore draw from prospectively defined sampling frames rather than

collections assembled because the cases were theoretically dramatic.

The scientific output is a constrained set of routing models, the evidence that distinguishes them, and a record of what remains unallocated. That structure prevents unresolved events from being pooled merely because a single total is easier to compute, and it prevents one event from being counted in more than one stream. Set-valued routing can still support exclusion claims when every admissible model agrees that a particular event is not jointly owned.

A routing verdict therefore narrows a scientific model and nothing more. The more consequential the public interpretation, the narrower the permitted language must become. The public report should describe uncertainty in the routing model rather than translating multiplicity into a claim about personal identity or worth.

11.6 — Error Costs, Abstention, and Dignity

Abstention is a positive scientific act when it identifies the failed gate. A no-estimate result can reveal inadequate channels, unresolved routing, poor transport, excessive burden, or model inadequacy and can direct the next protocol toward the actual bottleneck. Treating abstention as honorable reduces pressure to manufacture a definitive bearer solely to keep the study moving.

The closing section distinguishes scientific loss from ethical loss. A false bearer classification can corrupt a ledger, endpoint, or route verdict; its ethical consequences arise only if the label is allowed to influence care, publicity, access, or standing. The protocol should reduce both forms of error while never treating improved scientific accuracy as permission to weaken the ethical floor.

Error matrices should therefore include more than sensitivity and specificity. They should identify who bears the consequence, which downstream objects change, whether a claim can be corrected, whether data can be withdrawn, and which protections remain invariant. This turns error analysis from an abstract table into a map of scientific and human consequences.

Every classification system has error costs. In this chapter, the costs are not only statistical. A false stream label can distort a ledger, but it can also alter care conversations, family understanding, public identity, or the perceived presence of a vulnerable person. The science must distinguish model error from ethical harm without pretending they are unrelated.

The four central scientific errors are false continuity, false discontinuity, false partition, and false rebinding. They do not fail in the same way. False continuity can join intervals wrongly. False discontinuity can erase continuation. False partition can multiply streams beyond the evidence. False rebinding can connect what remains unresolved. Each requires a different correction route.

The safest system is not one that never classifies. The safest system is one that classifies only within scope, preserves abstention, invites multidisciplinary review, supports appeal, records revision, and blocks administrative leakage. That is the difference between scientific classification and institutional ranking.

Asymmetric protection means that uncertainty can narrow the study before it narrows care. When evidence is weak, the analysis may exclude a confirmatory claim, preserve multiple routing models, or return no estimate. The person still receives communication attempts, comfort, privacy, and ordinary protection because those duties do not depend on successful stream classification.

Table 11.4. Routing error classes, ethical risks, and preferred responses.

Error class	Scientific consequence	Ethical risk	Preferred response
False continuity	Wrongly joins intervals or streams	Misleads families or public meaning	Model comparison and revised confidence

Error class	Scientific consequence	Ethical risk	Preferred response
False discontinuity	Wrongly breaks a stream	Erases possible experience or care salience	Protective uncertainty and further evidence
False partition	Wrongly multiplies streams	Stigma, spectacle, identity confusion	Higher evidence burden and review
False rebinding	Wrongly reconnects unresolved states	False reassurance or retrospective certainty	Versioned uncertainty and appeal
Administrative leakage	Label travels beyond research	Eligibility, treatment, rank, or status harm	Access restriction and prohibition

11.6.1 — Scientific Error Versus Ethical Error

The no-estimate class should be structured enough to teach. It should record whether failure arose from unavailable channels, incompatible markers, unvalidated transport, unresolved partition, observation shutdown, excessive burden, or model inadequacy. Those reasons guide redesign and prevent an inconclusive case from being silently recoded as absence or neutrality.

Scientific error concerns the model's relation to the evidence. A continuity classifier may join intervals that should be separate. A partition model may overfit task asymmetry. A confidence class may be too strong for the marker set. These errors can be studied, corrected, and reported.

Ethical error concerns the person's exposure to harm. A model label may burden a family, stigmatize a participant, influence care, invite public spectacle, or diminish perceived dignity. Ethical error can occur even when a scientific model is carefully hedged, if the label travels to the wrong context.

The two error types interact. If classification changes behavior toward the person, the data-generating process changes. Staff attention, family anxiety, participant stress, consent pressure, and clinical workflow can all alter evidence. That means ethical leakage can become scientific contamination.

A good protocol writes both error types into the design through model diagnostics, adversarial review, label access, communication limits, incident reporting, correction routes, and stop rules. Ethical analysis is part of the inferential architecture because misuse can change both human welfare and the data-generating process; it is not a preface to the statistical work (Emanuel et al., 2000; World Medical Association, 2024).

Every classification decision carries two loss functions. One concerns inferential error in routing, ownership, or endpoint analysis; the other concerns the human consequences of stigma, reduced care, false reassurance, or public exposure. The review record should report both because a scientifically convenient label can still be an ethically disastrous one.

11.6.2 — Asymmetric Protection

Asymmetric protection is prospectively specified before cases appear. Possible false exclusion triggers protective caution, while possible false inclusion narrows the claim and restricts public use.

The two responses address different harms and can operate together without turning caution into a bearer verdict.

The scientific error rates for false inclusion and false exclusion are reported in §11.3. The ethical asymmetry is separate: false exclusion can also reduce precaution and erase possible experience from the record, so uncertainty should bias the system toward protected abstention rather than confident absence.

Asymmetric protection means that, where evidence is weak and false exclusion could reduce care or erase possible experience, the protocol should preserve uncertainty rather than declare absence. This does not mean declaring consciousness present. It means keeping the person protected while the research claim narrows.

Asymmetric protection also affects data interpretation. A low signal may be labeled low signal rather than closed. Estimated absence may be kept separate from confirmed absence. Unknown may remain unknown. These distinctions prevent false clarity from becoming policy or public meaning.

The rule is especially important in disorders of consciousness and communication-limited states. Behavioral unresponsiveness cannot be a safe exclusion rule when neural command-following evidence may reveal cognitive-motor dissociation in some patients. The correct response is not to overclaim presence, but to treat absence claims as fragile.

Asymmetric protection must still be bounded. It cannot become a license to infer rich experience from every faint signal. It should trigger additional review, conservative public language, care-first defaults, and sensitivity analysis, not dramatic assertions.

11.6.3 — Abstention as Valid Output

Abstention is a protocol-fixed scientific output. It means that the available evidence does not justify classifying continuity, interruption, partition, rebinding, or absence at the level required for the intended claim. It is not silence, indecision, or a failure to finish; it is a positive verdict about the limits of the evidence.

The reason for abstention must be typed. Marker conflict, insufficient temporal resolution, missing report, confounded physiology, theory disagreement, threshold instability, unresolved routing, clinical burden, and lack of independent review are different failures and lead to different next steps. A specific no-estimate state teaches more than a generic inconclusive label.

Triggers should be prospective. The protocol must state which combinations of low signal, missing access, threshold sensitivity, model equivalence, or routing nonidentification produce No adequate estimate and which produce Model-ambiguous. A refusal issued only after an inconvenient result is selective reporting rather than disciplined abstention.

Abstention belongs in evidence accumulation. Repeated no-estimate results caused by one device expose an instrument problem; concentration in one population exposes a reference-class or access problem; concentration in a high-burden setting may show that the study should not be run there. The rate and causes of abstention are therefore scientific outcomes, not missing rows.

A classification may return to review after recovery, improved calibration, additional evidence, or external audit. The earlier

abstention should remain in the version history, followed by the new evidence and decision, rather than being overwritten. Revision is more credible when the path of uncertainty remains visible.

Multidisciplinary disagreement should remain auditable. Neurology, anesthesia, sleep science, psychiatry, psychology, philosophy, statistics, ethics, and lived-experience expertise may weight the same record differently. Votes or confidence ranges, reasons for dissent, conflicts of interest, and the rule used to construct the admissible model set should be preserved.

Public language should be calm and bounded: this study cannot responsibly classify the stream for this purpose. The sentence concerns the research claim, not the person, and it cannot lower care, dignity, rights, or participation. Calm precision prevents uncertainty from being mistaken for concealment or failure.

A useful abstention policy also reveals exclusion. Reports should show which groups were disproportionately unclassifiable and what evidence would reduce that uncertainty. This makes visible whether an evidential system systematically leaves infants, disabled people, nonverbal patients, nonhuman animals, or unfamiliar architectures outside the record, and it allows redesign without forcing labels.

11.6.4 — Blinded and Multidisciplinary Review

Review quality should itself be auditable. The record should state reviewer roles, independence, conflicts, evidence access, blinding limits, dissent, and revision triggers. Without that audit, multidisciplinary review can become a prestige shield rather than

a real protection against theory-driven classification (Yakhkind et al., 2024; Giacino et al., 2018).

The review process should also have an adverse-outcome route. If a reviewer discovers that classification burden is affecting care, family communication, participant distress, or staff workflow, the protocol should be able to pause. The purpose of review is not only better labels; it is better distribution of burden.

A multidisciplinary record distinguishes consensus from agreement by exhaustion. Material dissent is preserved with its evidentiary basis, even when a bounded conclusion remains justified. Visible disagreement creates a review path; hidden disagreement creates a future credibility failure.

Bearer adjudication is vulnerable to theory desire. Researchers may want continuity because it preserves a ledger, partition because it is novel, absence because it simplifies a protocol, or rebinding because it completes a story. Blinded and multidisciplinary review reduces that pressure.

Blinding should separate, where possible, the continuity classification from ledger outcomes, terminal-neutrality expectations, public narratives, and investigator hopes. A reviewer should not know that a particular classification makes a later result look favorable. Otherwise the gate can become an outcome-selection device.

Multidisciplinary review should include consciousness science, neurology, psychiatry or psychology where relevant, palliative or clinical expertise, statistics, ethics, and lived-experience

consultation where appropriate and safe. No single discipline owns unity. Each can see a different failure mode.

The review file should record disagreement. A neurologist may see marker weakness, a clinician may see risk, a philosopher may see identity ambiguity, a statistician may see nonidentifiability, and a participant advocate may see burden. A mature classification does not hide such disagreement; it routes it into confidence or abstention.

Review should also be proportional. Ordinary clear cases do not need the same process as disorders of consciousness, dissociation, or endpoint-adjacent uncertainty. The burden of review should be scaled to risk, ambiguity, and claim consequence.

High-consequence classifications require outcome-blind, multidisciplinary review. Reviewers should see the marker evidence and rival models without seeing whether a particular class improves terminal neutrality, and disagreements should be preserved as confidence or abstention. Exploratory classifications remain clearly labeled and cannot silently enter confirmatory work.

11.6.5 — Appeal and Revision

Appeal and revision also make the framework durable as consciousness science changes. New evidence may strengthen, narrow, or overturn a classification, example, or procedure. A visible change history makes correction an expected scientific duty rather than a reputational crisis (World Medical Association, 2024; Center for Open Science, 2025).

Appeal should be designed for accessibility. A challenge route that requires technical language, legal sophistication, or institutional confidence will not serve the people most likely to need it. Plain-language challenge forms, clinician-supported routes, family-support routes where appropriate, and independent ombuds review can make revision more than a theoretical right.

Revision should also propagate. If a bearer classification changes, related ledger estimates, endpoint interpretations, figures, summaries, and public claims may need update. The change is not local if downstream claims relied on the old label. Correction must follow dependency, not convenience.

A classification system must be contestable. Participants, authorized surrogates, clinicians, reviewers, or later investigators may have evidence that the original classification missed. Appeal does not mean every objection overrides the model. It means the model remains answerable.

Revision is equally important. New evidence may show that a gap was more likely interruption, that a presumed interruption was likely continuity, that a partition hypothesis was overfit, or that rebinding was unsupported. The classification should change when the evidence changes.

The version history should preserve all major states: initial class, evidence basis, reviewer identity, permitted use, challenge, new evidence, revised class, and affected claims. This prevents a later clean record from hiding the uncertainty people lived through or the scientific path that produced the conclusion.

Appeal remains private enough to be usable. Families and participants can challenge a classification without exposing intimate details in public, making care contingent on winning, or entering adversarial theater. The record preserves the grounds, response, and correction while minimizing collateral disclosure.

An appeal initiates a formal reanalysis rather than an informal exception. Participants, authorized surrogates, clinicians, and independent reviewers can challenge the evidence, threshold, permitted use, or retention period without risking care or participation. Any accepted revision propagates to every downstream model, figure, summary, and public artifact that depended on the earlier state.

11.6.6 — No Administrative Use

Bearer classifications are purpose-limited research objects. They may qualify the scientific claim that created them, but they may not determine triage, insurance, employment, education, legal status, public rank, spiritual standing, or resources. The institution carries the uncertainty (Tabassi, 2023; World Medical Association, 2024).

Architecture enforces that boundary. Research classifications, clinical records, identity, contact information, and public outputs occupy separate access domains and retention schedules. Administrative systems never receive a portable bearer score.

Each label has a declared lifecycle for access, retention, revision, expiration, and deletion. It narrows or disappears when context ends, calibration fails, admissible models change, or privacy risk exceeds scientific value.

Automation may assist documentation and audit, but bearer labels may not train systems for triage, surveillance, eligibility, or ranking. A new use requires fresh justification, independent review, and accountable human authority.

The audit trail records who accessed a classification, for what authorized purpose, and whether export or model-training use occurred. Unusual access or attempted linkage triggers review and suspension authority.

Challenge must be usable without technical expertise. The project explains evidence, uncertainty, and permitted uses plainly, accepts contrary information, and supplies correction. Explanation and revision remain the project's burden.

Public reporting defaults to cohort-level methods and uncertainty. An indispensable case still requires consent, de-identification, narrative restraint, and hostile-crop review; reproducibility does not require indefinite exposure of rare trajectories.

Revisions remain traceable and propagate to affected estimates, models, figures, summaries, and public statements without preserving a permanent public identity (Center for Open Science, 2025).

Chapter 12 — Consciousness Theories and Boundary States

Consciousness is a prerequisite layer for terminal science. The research program needs to know where experience may occur, how access can fail, how continuity can be obscured, and which signals remain informative when report is unavailable. Chapter 12 uses consciousness theories as generators of testable marker families and predictable errors.

The previous chapter built the research bearer as a provisional classification rather than an ontology. That work now meets the state of consciousness science itself. A stream may be awake, asleep, sedated, injured, fragmented, developing, nonhuman, synthetic, or silent under conditions that make ordinary report unavailable. If the book let one favored theory decide every boundary, the endpoint architecture would inherit that theory's blind spots and call them discipline.

The chapter compares global availability, integrated and causal, recurrent, higher-order, predictive, active-inference, enactive, and embodied families by the evidence each makes salient. For each family, it asks which marker is predicted, which state and scale it addresses, which rival can mimic it, where it is qualified, and which cases expose its blind spots.

That pluralism increases testability. A marker can fail because movement is unavailable, language is absent, memory was never formed, sedation altered report, a species expresses pain differently, or a machine learned the language of experience. Each failure becomes a discriminating question rather than a shortcut to certainty.

Boundary-state error is typed. Motor dependence can erase covertly responsive patients; adult-human tasks can fail in infants or animals; fluent language can inflate confidence in artificial systems; and a complexity index can change with drug, injury, scale, or artifact. Every marker therefore carries its target, prerequisites, qualification domain, error direction, and abstention condition.

Dream reports provide evidence filtered by awakening, memory, language, salience, and sampling. Sleep therefore offers an everyday setting for studying how experience, access, report, and recall relate to one another.

Anesthesia sharpens the problem further. A motionless patient may be insentient, conscious without response, responsive without later memory, dreaming, fragmented, or passing through states that change with drug class, dose, physiology, and time. The scientific task is to build graded state evidence without letting the Law influence care. Anesthesia monitoring belongs to clinical safety before it belongs to any research program.

Disorders of consciousness give this work immediate clinical relevance. Covert command following can reveal retained capacities in behaviorally unresponsive patients. Cognitive motor dissociation and related findings help improve assessment and identify additional communication possibilities.

Developmental and nonhuman research broadens the range of evidence the science can use. Infants, children, and unfamiliar species call for age-appropriate and species-appropriate markers, welfare protections, and methods that reflect their capacities and

environments. Naturalistic observation and independently justified care provide valuable starting points.

Artificial and synthetic candidates create a different temptation. Fluent language, emotional mimicry, self-descriptions, and simulated pain labels can be persuasive because they resemble the social forms through which humans recognize one another. Yet performance is not experience. At the same time, convenience cannot permanently exclude future systems from examination if their architecture, causal organization, embodiment, memory, self-modeling, and adaptive integration make the question scientifically serious.

The chapter's practical result is a versioned boundary-state dossier. It records observed markers, theory mappings, confounds, plausible false positives and negatives, transport limits, active classification, review needs, permitted analyses, and prohibited public uses. The dossier advances routing and endpoint analysis without becoming a universal consciousness score. A dossier can also preserve temporal changes in marker reliability, preventing a classification earned in one state from being carried unchanged into another.

The chapter moves from theory families to states that break ordinary inference. Sleep tests experience without stable external access; anesthesia tests return without reliable memory or movement; disorders of consciousness test covert cognition; dementia and delirium test fluctuating access and narrative loss; development and comparative cases test transport; artificial systems test the difference between functional resemblance and warranted phenomenality. These states therefore function as

stress tests for different assumptions, revealing whether access, memory, behavior, or continuity is responsible when a marker fails.

The output is a set of constrained updates. A marker may raise or lower confidence in one component, expose a failed channel, or force a no-estimate state. Progress is measured by which uncertainty became smaller, which rival gained ground, and which claim remains unavailable. The same accounting keeps progress visible even when no single theory or marker family wins the entire comparison.

12.1 — Theory Pluralism and Marker Families

A scientific program that depends on consciousness has to decide how to treat disagreement among consciousness theories. It cannot wait for a final theory before doing any work, because measurement, endpoint adjudication, and human-subject safeguards must already operate under uncertainty. It also cannot choose the theory that best fits its preferred endpoint architecture. That would turn science into instrument selection after the verdict.

Theories of consciousness are most useful here when they generate marker families and risky predictions. Section 12.1.6 develops adversarial comparison as a concrete implementation of that principle.

Theory pluralism becomes operational through a matrix with rows for theory families and columns for marker target, state, temporal scale, task dependence, transport domain, false-positive routes, false-negative routes, and abstention conditions. The matrix is not a vote. It reveals when apparently convergent markers share the same hidden dependency and when genuinely independent channels support a stronger update.

Theory pluralism matters because every theory makes some cases easy to see and others easy to miss. Report-heavy accounts may exclude people without motor or language access; complexity-heavy accounts may mistake measurement failure for absence; performance-heavy approaches may overread artificial fluency. Comparing these blind spots is therefore part of classification, not an optional philosophical preface.

Table 12.1. Theory-to-marker matrix for boundary-state use.

Theory family	Primary marker family	Known hazard
Global availability	Broadcast, flexible access, task use, report-independent response	Motor or language failure can hide access; fluent output can imitate access.
Integrated and causal	Effective connectivity, perturbational complexity, differentiated causal response	Complexity can be artifact; low response can reflect measurement failure.
Recurrent and higher-order	Feedback loops, metacognition, confidence, self-monitoring	Recurrence can be unconscious; metacognition can fail while experience remains.
Predictive and active-inference	Interoception, generative control, embodied regulation	Viability and control can exist without felt experience.
Enactive and embodied	Organism-environment coupling, situated agency, sensorimotor world-involvement	Embodiment can be overextended, simulated, or too tied to human forms.

12.1.1 — Global Availability Accounts

Global availability accounts begin with a practical observation: some information becomes conscious when it is made broadly available for reasoning, report, memory, action, and flexible control. In that family, consciousness is not merely local processing. It is a change in functional reach. A stimulus or state

matters because it can become available to systems that use it across tasks and times (Dehaene et al., 2017; Mashour et al., 2020).

The observable implications are access markers: delayed, widespread, or task-relevant activation; flexible use in report or decision; candidate signatures of broadcasting; and contrasts between subliminal processing and contents that enter control. Their value lies in functional reach, while their principal risk is confusing availability with speech or motor output.

Boundary states reveal the limits of global-availability accounts. Motor disconnection can block report while experience remains possible; dreams can contain vivid experience without ordinary external access; children and nonhuman animals can use information flexibly without adult language; and synthetic systems can broadcast text without a validated phenomenal bridge. Global availability supplies important evidence, but its evidential role remains theory- and context-specific.

For PTVC, global availability matters because report cannot define the unity gate. Excluding every low-access interval would preferentially remove sleep, sedation, severe injury, infancy, and other states in which experience may be difficult to express. Including every system that flexibly broadcasts symbols would create the opposite error. Availability therefore updates access, while the bearer decision remains conditional on additional marker families.

The scientific contribution of global-availability accounts is strongest when they separate access from experience. Report-

independent tasks, alternative response channels, and timing controls can reveal flexible use without demanding speech. Negative evidence remains weak when sensory input, comprehension, attention, memory, fatigue, or motor output is compromised; positive evidence supports organized access without mapping every untested interval.

12.1.2 — Integrated and Causal Accounts

Integrated and causal accounts begin from a different intuition. Consciousness seems to require a system whose parts are not merely coactive but causally integrated in a way that produces a unified field. These theories make differentiation and integration central. A conscious state is not a loose list of contents; it is structured, unified, and causally organized (Tononi et al., 2016; Albantakis et al., 2023).

Candidate markers include effective connectivity, perturbational response, complexity, differentiation, and causal interaction. The strongest versions ask whether perturbing one part of the system produces a richly structured response that is neither local reflex nor global silence, making the approach especially relevant to anesthesia, sleep, and severe brain injury.

The principal failure mode is index sovereignty. Perturbational complexity or effective-connectivity measures can fall because experience is absent, because drugs or injury alter propagation, because the perturbation misses the relevant scale, or because data quality is poor. High values can also arise from rich dynamics that do not establish the kind of unity the theory intends. No scalar therefore becomes a context-free unity meter.

Integration evidence should change scientific confidence, not human status. Presence under qualified conditions may support a candidate-stream model; absence may indicate low integration, theory mismatch, impaired propagation, or measurement failure. The report should state which inference is licensed and leave care, personhood, and legal protection outside the index.

The adjudication rule is to pair integration markers with context. Drug state, arousal, injury pattern, sensory access, motor access, sleep stage, and data quality must travel with the marker. A perturbational result without context is like a ledger estimate without origin: it may be mathematically clean and scientifically incomplete.

Integrated and causal accounts contribute a demanding question: does the system respond as a differentiated causal whole rather than as isolated modules or undirected noise? That question can be compared with access, recurrence, report-free tasks, anatomy, and state history. Its value lies in risky predictions and cross-theory disagreement, not in supplying final authority over who or what is conscious.

Progress is already measurable. Casarotto and colleagues (2016) tested a perturbational-complexity threshold in 81 patients after establishing it in a 150-person benchmark. It detected 36 of 38 minimally conscious patients, while 9 of 43 behaviorally unresponsive patients had complexity values overlapping the conscious benchmark. This provides a practical route toward identifying capacities that behavior alone can miss.

12.1.3 — Recurrent and Higher-Order Accounts

Recurrent processing accounts emphasize feedback loops rather than one-way feedforward registration. A stimulus may be processed deeply enough to guide behavior without becoming conscious unless recurrent interactions stabilize or amplify it. Higher-order accounts add a different condition: a state becomes conscious when the system represents, monitors, or is in some way aware of its own state (Lamme, 2006; Brown et al., 2019).

These accounts produce markers that are especially useful in borderline cases. Recurrent processing directs attention to sustained or feedback-mediated activity within sensory and association networks. Higher-order approaches ask whether metacognitive access, confidence, self-monitoring, or error awareness is present. Both families help explain why content, access, and self-evaluation can dissociate.

Their limits are immediate. Infants, animals, patients with language impairment, and people in altered states may lack adult forms of metacognitive report while still having experience. A higher-order criterion that is too demanding can make consciousness disappear wherever explicit self-report is weak. A recurrence criterion that is too permissive can include processing loops that never become felt.

A responsible boundary protocol therefore asks what kind of recurrence is observed, whether it is content-specific, whether it persists across tasks, whether it is paired with integration or access, and whether it survives artifact controls. It asks what kind

of higher-order capacity is being tested and whether the case could have experience without that capacity.

Boundary states make the distinction concrete. Dreaming may contain recurrent sensory organization with weak metacognitive access; severe injury may preserve local recurrence while disrupting global use; dementia or delirium may impair monitoring and report while affective experience continues. The dossier should therefore specify which relation each marker targets and which alternative explanation remains live.

Recurrence and higher-order markers should be decomposed by level: local feedback, long-range recurrence, metacognitive monitoring, confidence, and error awareness are not interchangeable. Their absence may reflect task or report limits, while their presence may show processing rather than experience. They strengthen a bearer classification only when the protocol names the process measured and the rival it excludes (Lamme, 2006; Brown et al., 2019).

12.1.4 — Predictive and Active-Inference Accounts

Predictive and active-inference accounts treat conscious systems as organisms or agents that model the world, the body, and the consequences of action. Their emphasis falls on generative models, prediction error, interoception, action, control, and the active maintenance of viable states. Consciousness is not simply a broadcast or a complexity score. It is tied to situated regulation (Friston, 2010; Seth & Friston, 2016).

Candidate markers are dynamic: prediction-error updating, uncertainty or precision weighting, interoceptive coupling,

perception-action coordination, attention, and state regulation. In human cases they can illuminate pain, dyspnea, nausea, anxiety, delirium, and terminal physiology because experience is often entangled with bodily prediction.

The danger is viability inflation. A system can regulate itself without feeling. A thermostat has control loops. A cell has homeostatic organization. A hospital monitor tracks physiology. None of those examples alone establishes a conscious stream. Active inference becomes relevant to consciousness only when the control architecture is rich enough, integrated enough, and situated enough to bear on felt perspective.

These models can explain why access, salience, and felt state change across sleep, anesthesia, illness, and stress. Their flexibility also creates a recovery problem: a model that can reinterpret every trajectory after the fact offers little discrimination unless its state variables and predictions are frozen in advance (Friston, 2010; Seth & Friston, 2016).

A boundary-state protocol tests whether predictive and active-inference markers are specific to conscious modeling or reflect generic regulation. It examines their relation to reportable experience, covert command following, perturbational evidence, and clinical context, while stating whether each marker targets level, content, valence, or only background capacity.

12.1.5 — No Theory as Endpoint Oracle

Enactive and embodied approaches remind the program that consciousness is sustained through an organism's active relation with its environment. Posture, movement, breathing, sensory

sampling, autonomic regulation, pain, medication, and social interaction can change both experience and the channels used to infer it. Boundary-state analysis should therefore record the coupling conditions rather than treat the brain trace as an isolated verdict.

No theory is comprehensive enough to serve as an oracle across all states. Global availability can miss covert consciousness. Integration metrics can be theory-laden and artifact-sensitive. Recurrence can be present without conscious report or absent from measured channels. Higher-order criteria can exclude infants and impaired persons too readily. Predictive accounts can overinclude homeostatic systems.

Cross-theory robustness strengthens classification when independently imperfect theories converge under declared evidence. Classification narrows when theories diverge in ledger-relevant ways and abstains when theory dependence prevents supported inclusion or exclusion.

When theory dependence dominates, abstention preserves evidence. The dossier records disagreement, identifies discriminating observations, and restricts downstream use until they exist. Averaging incompatible theories into one score would erase the uncertainty the review should expose.

A later endpoint study should therefore record which theory families were used, which markers each supported, where they disagreed, and what that disagreement permits. A favorable terminal pattern cannot repair a failed consciousness gate. The

dependence is noncompensatory: endpoint evidence is interpretable only after the bearer and state gates have survived.

Embodied evidence is especially important when conventional report is unavailable. Changes in orienting, autonomic response, sensorimotor contingency, pain behavior, and environmental engagement can constrain candidate models. Each channel remains fallible: reflexes can occur without the target experience, motor impairment can suppress meaningful response, and clinical intervention can alter the very dynamics being observed.

Endpoint use requires cross-theory stress. One theory may license a local-state marker but cannot decide a stream's endpoint when rivals predict different inclusions, exclusions, or uncertainty zones. Material divergence requires a theory-version label on the public scientific claim or abstention from endpoint interpretation.

Every theory family faces a characteristic failure zone. Access accounts confront report-independent cases; integration accounts confront scale and artifact sensitivity; higher-order accounts confront experience without testable metacognition; predictive accounts confront flexibility and recovery; and embodied accounts must distinguish meaningful coupling from reflex. The comparison becomes most informative where each theory's preferred marker is least secure.

12.1.6 — Theory-to-Marker Matrix

A theory-to-marker matrix exposes disagreement. Each row identifies the theory version, typed target, contrast, time scale, observation, confounds, and conditions under which either result is interpretable.

Targets are fixed before scoring. State presence, content, access, integration, metacognition, continuity, report capacity, and action control are different objects; evidence cannot migrate among them by resemblance.

Predictions remain tied to the theory version that generated the test. Later revision belongs in a secondary update with a visible change log, preventing rescue after the data.

Rows record population, task, preprocessing, scale, prerequisites, missingness, direction, uncertainty, detection opportunity, and failure modes. A prospectively specified conflict rule preserves disagreement instead of averaging it.

Adversarial collaboration is stronger than isolated illustration. The 2025 comparison of global neuronal workspace and integrated information theories constrained both sides before theory-neutral testing, showing how rival comparison can narrow consciousness claims without becoming PTVC evidence (Cogitate Consortium et al., 2025).

Open multisite fMRI data from 118 participants and MEG-EEG data from 100 preserve common events, quality reports, metadata, and code across cohorts and modalities. Openness does not settle consciousness, but it makes preprocessing, rival testing, and transport inspectable (Khalaf et al., 2026; Liu et al., 2026). Shared acquisition protocols also allow site and modality effects to be estimated rather than conflated with theory performance.

Held-out calibration, proper scoring, and recovery test declared predictions; transport tests their performance in new settings (Gneiting & Raftery, 2007). Interpretation also uses the construct

definition and checks on delivery, task access, signal quality, and the intended contrast. A model that scores well only after task-specific preprocessing changes should be distinguished from one whose predictions survive a frozen cross-site pipeline.

The matrix is a versioned comparison tool, not an endpoint oracle. It records support, challenge, conflict, and permitted classifications without granting any theory authority over care, personhood, routing, or the terminal hypothesis (Seth & Friston, 2016; Cogitate Consortium et al., 2025). The matrix should preserve unresolved disagreements rather than collapse them into a consensus score that hides which evidence each theory explains.

12.2 — Sleep, Dreams, and Reversible Low Access

Sleep is a useful first boundary state because it is ordinary, reversible, and rich in evidence about the relationship between experience and access. Conscious experience can occur while ordinary responsiveness is reduced and later recall is incomplete. Studying those differences helps the book develop state-sensitive measures that extend beyond waking report.

Sleep research shows that report absence, recall failure, and experiential absence are separable. High-density recordings link dream reports to local and distributed changes that vary across REM, non-REM, and parasomnia states, making within-state sampling and awakening design central to any continuity inference (Siclari et al., 2017; Cataldi et al., 2024).

Dreaming also tempts overinterpretation. A dream may contain fear, desire, pain, grief, relief, absurdity, memory, or meaning, but it is not an unfiltered sample of the latent ledger. It is recalled after awakening, narrated through memory, and shaped by the questions asked. Sleep gives the research program evidence about access and unity, not a shortcut around measurement.

Sleep offers a controlled way to study experience when ordinary access, report, and memory separate. The scientific question is how reversible low-access states should be included, excluded, or labeled when unity gates and measured ledgers encounter weak or delayed signals. Dream content remains auxiliary evidence rather than a privileged message about lifetime closure.

12.2.1 — Sleep Stage and Access

Sleep stages differ in arousal threshold, physiology, neural dynamics, and report probability. Rapid eye movement sleep, non-rapid eye movement sleep, slow-wave sleep, light sleep, microarousals, and transitional states should not be collapsed into one category called unconscious. A state label is only the beginning of inference (Siclari et al., 2017; Nir & Tononi, 2010).

Access matters because a person may have experience without immediate report, and may report after awakening an experience that was already filtered by stage, timing, and memory. A sleep-stage model therefore needs arousal markers, electroencephalographic features, autonomic context, awakening protocol, and report procedure. The marker family determines what counts as evidence.

Theories disagree about sleep. Some emphasize reduced global access; some emphasize local posterior content networks; some emphasize altered integration; some emphasize predictive disconnection from the external world. The disagreement is useful because it forces the chapter to distinguish conscious content, environmental access, metacognitive control, and stream continuity.

Sleep-stage labels affect eligibility because they alter the prior probability of access, dreaming, memory, and external responsiveness. A stage label should therefore enter the unity model as contextual evidence, not as a direct consciousness switch. Threshold-family analyses should show how stage, report,

physiology, and marker confidence jointly change the included phenomenal domain.

A sleep state can support several verdicts: included conscious interval, probable low-access experience, no-estimate interval, or excluded nonconscious interval under declared assumptions. Those labels must be versioned because later research may revise how sleep stages map to experience.

Sleep stage is therefore not a minor preprocessing detail. It is the first demonstration that a life ledger, if it is ever measured, must be built from state-specific evidence rather than from the public convenience of waking reports.

Sleep staging organizes sampling once the target is specified. Electroencephalographic stage, arousal threshold, autonomic state, eye movement, motor inhibition, recall probability, and reported experience enter the analysis as distinct variables whose relationships can be tested.

12.2.2 — Dreaming and Non-Dreaming Reports

Dream reports are reports after a transition. They do not arrive from inside the dream untouched. Awakening changes attention, memory, language, and motivation, and the absence of a report may reflect nonexperience, failed recall, weak encoding, question wording, or the timing of awakening (Siclari et al., 2017; Tempesta et al., 2018).

A serious protocol therefore separates dream experience, report of experience, report with recall, report without recall, and nonreport. Those categories are often compressed in ordinary

speech, but they have different implications for the unity gate. A person who cannot retrieve content may still have had experience.

Evidence from sleep research suggests that reports of dreaming can occur in both rapid eye movement and non-rapid eye movement sleep, and that posterior cortical activity has been associated with dream reports. The point for this chapter is not to make a posterior region sovereign. It is to show why stage labels and report labels are not enough.

Dream reports constrain boundary-state models through a documented observation chain. They can test how low-access experience is later reported, how recall filters sampled content, and how continuity is reconstructed after awakening.

The failure conditions are clear. If dream reports depend strongly on awakening method, subject training, salience, or retrospective reconstruction, then dream evidence must be labeled report-filtered. If no report is obtained, the correct inference may be absence, nonidentification, or insufficient access, depending on the full marker family.

Sleep evidence enters the routing analysis through awakening design, memory, confidence, stage, and marker context. Recording that chain makes each update to the candidate stream inspectable.

A report of no dream can arise from no experience, failed encoding, failed retrieval, delayed questioning, low salience, language limits, or burden-related nonresponse. A strong protocol varies awakening timing, records confidence, distinguishes immediate from delayed report, and models those alternatives

rather than treating one word after waking as an exhaustive classification.

12.2.3 — Lucid and Fragmented States

Lucid dreaming and fragmented sleep states show that consciousness is not a single knob. A dreamer may have vivid content, partial metacognition, weak voluntary control, altered time, unstable selfhood, and little external access. That mixture is scientifically useful because it separates dimensions that ordinary waking life often bundles together (Cataldi et al., 2024; Nir & Tononi, 2010).

The marker families diverge. A higher-order account may treat lucidity as a special increase in metacognitive awareness. A global-access account may ask whether information becomes available for control within the dream world. A predictive account may ask how internal generative models dominate external input. An integration account may ask whether the state remains unified enough to count as one stream interval.

Sleep is a family of changing states rather than one block of reduced access. Brief awakenings, nonrapid-eye-movement imagery, rapid-eye-movement dreams, sleep paralysis, nightmares, hypnagogic and hypnopompic states, and fragmented arousal can differ in content, agency, memory, physiology, and external responsiveness. The dossier should preserve those distinctions before testing a unity gate.

Lucid and fragmented states show why unity gates need dimensions rather than a single dial. Access, metacognition, agency, self-location, memory, bodily control, and narrative

coherence can separate within one episode. The analysis should report which dimensions support inclusion and how ledger conclusions change when the gate treats them differently.

A fair classification records the nature of fragmentation rather than smoothing it away. Was the fragmentation about content, agency, memory, self-location, bodily control, emotional tone, or temporal continuity? Each kind of fragmentation creates a different risk for ledger allocation.

The scientific task is not to multiply streams whenever organization becomes unusual. It is to test whether the fragmentation concerns content, access, agency, memory, self-model, or causal ownership and whether those changes require a different routing model. Partial disruption can remain within one candidate stream when the evidence does not establish independent ownership.

Lucid dreams are valuable because they partially dissociate metacognition from ordinary waking control. Fragmented states are valuable because they show that contents may appear without stable narrative integration. Together they test whether the unity gate is all-or-nothing, graded, or state-family dependent, while keeping dream content outside any claim of direct ledger access.

12.2.4 — Awakening and Sampling Bias

Awakening is an intervention. The moment of waking, the question asked, the delay before response, the subject's expectation, and the emotional intensity of the experience all affect what becomes reportable. A dream study that ignores

sampling bias may measure reportability more than experience (Siclari et al., 2017; Cataldi et al., 2024).

Sampling bias appears in several places. People may remember vivid or late dreams more than quiet states. They may be trained to report more. They may confabulate when pressured. They may forget dreams that occurred earlier in the night. They may translate nonverbal or affective experiences into a story because the protocol asks for one.

Theories of consciousness do not remove this bias. A report-friendly theory may mistake retrieval for presence. A report-free marker may avoid report bias but introduce sensor and model bias. The fair approach is to report the whole sampling chain: sleep stage, awakening schedule, question form, time to report, confidence, and nonresponse interpretation.

A report-derived ledger would preferentially sample experiences that survive awakening, language, salience, and willingness to disclose. Quiet, bodily, confusing, shame-laden, or rapidly forgotten affect would be underrepresented. Sleep research therefore teaches the measurement program to model the report-selection process before interpreting the content that passed through it.

The correction is not to discard dream reports. It is to classify them properly. A dream report is a post-awakening artifact with evidential value under a known sampling design. It can support boundary-state modeling, but it cannot be treated as an unbiased draw from the latent stream.

The same architecture later governs terminal observation. Near an endpoint, arousal, memory, language, sampling frequency, caregiver availability, and sensor quality may change together. Sleep provides a safer domain in which to learn how observation windows select what becomes visible without pretending that the visible record exhausts experience.

Awakening protocols create their own data-generating process. The alarm, stage-selection rule, delay to interview, interviewer wording, fatigue, and selective exclusion of difficult awakenings can all shape what is remembered and reported. A sleep inference is credible only when those design choices are prospectively specified and their missingness is visible (Cataldi et al., 2024; Little & Rubin, 2019).

12.2.5 — Unity-Gate Sensitivity

The unity gate decides whether an interval is open to ledger contribution. In sleep, that gate is especially difficult because low access is common, continuity is altered, and later report is incomplete. A hard threshold would simplify analysis by hiding the hardest cases (Casali et al., 2013; Jang et al., 2024).

Sensitivity analysis is therefore mandatory. The protocol should test how conclusions change under a strict waking-access gate, a broader experience-likelihood gate, a theory-plural soft gate, and a no-estimate category for ambiguous intervals. If the measured result depends on one convenient sleep gate, the result is not robust.

Different theories support different gate shapes. A global-access account may narrow the gate during deep sleep. An integration

account may allow some low-access states if perturbational or connectivity markers remain rich. A dream-content account may include intervals around later reports. A predictive account may emphasize internal generative modeling even when external access is reduced.

A unity gate is an eligibility rule for one analysis, not a hierarchy of lives or states. Excluding a sleep interval because the marker dossier is insufficient says only that the study cannot use that interval for the declared computation. The record should preserve the uncertainty and prohibit any translation into worth, personhood, or care priority.

The failure condition is post hoc threshold selection. If a unity gate is chosen after seeing whether a ledger estimate moves toward neutrality, the study has contaminated its endpoint. The gate must be declared before the relevant outcome is analyzed, and alternative gates must be reported.

Sleep makes the exclusion rule testable. An interval may leave the confirmatory computation only when the gate, reason code, and sensitivity treatment were declared in advance. The exclusion remains visible in the denominator and cannot be converted into evidence that the interval was phenomenally empty.

Threshold-family analysis is the scientific heart of this subsection. A high threshold may exclude low-access conscious intervals; a low threshold may include organized unconscious activity. The section therefore requires reports across multiple threshold families and asks whether any claimed signature survives strict, moderate, and protective unity gates.

In dream research, selective inclusion is the danger. A stringent gate can discard poorly recalled but genuine experience, while a permissive gate can admit reconstructed reports or organized physiology that does not establish experience. The analysis should report how sleep stage, awakening, recall, and unity thresholds alter the included domain instead of letting one threshold silently determine the ledger.

12.2.6 — No Dream Privilege

Dream experience, dream report, and later recall are different objects. A person may experience without retaining a report, report fragments altered by awakening, or construct a coherent narrative from sparse memory. The study must preserve these distinctions rather than treat remembered dream content as a direct transcript of the sleeping stream (Nir & Tononi, 2010).

Sampling is selective. Awakenings are more likely in some stages, after some physiological changes, and when participants can respond. Reportable or vivid dreams therefore receive disproportionate representation, while silent intervals remain difficult to interpret. A dream dataset is a record of both sleep experience and the procedure used to interrupt and sample it.

Dream evidence is useful for testing state-dependent report, affective transition, memory, unity-gate sensitivity, and links between physiology and later testimony.

A protocol-fixed sleep design should state the stage, awakening schedule, report window, scoring method, missing-report rule, distress policy, and separation of confirmatory from exploratory

features. Negative recall does not equal no experience, nor does a vivid report establish continuous experience across the interval.

Dream material is unusually sensitive. It can reveal trauma, sexuality, fear, relationships, and identity in forms the participant never intended to make public. Coding should minimize narrative detail, separate scientific features from identifying content, and permit withholding or deletion without penalty (World Medical Association, 2024).

The most useful output may be conditional. A study can report how marker families behave in intervals followed by particular reports, how awakening changes evidence, and which unity models remain compatible. It need not decide what every silent interval contained.

This discipline lets sleep inform the next boundary case without mystifying it. Anesthesia shares low access, uncertain report, memory failure, and state transition, but adds pharmacology, motor blockade, and nonnegotiable clinical priority.

12.3 — Anesthesia and Covert Consciousness

Anesthesia is intentionally induced for care, so clinical purpose governs every research use. State measurements must remain noninterfering, consented where applicable, and subordinate to analgesia, safety, procedure, and recovery. The scientific question is what anesthesia can reveal about access, memory, integration, and return under those constraints, not how care can be modified to produce a cleaner boundary state.

Anesthesia is a controlled model of altered access, but unresponsiveness is not equivalent to unconsciousness. Isolated-forearm studies, covert command paradigms, and network-dynamics research show that drug class, dose, motor blockade, memory, and recovery can dissociate, so a bearer model must preserve all of those layers (Sanders et al., 2012; Linassi et al., 2018; Jang et al., 2024).

The scientific difficulty is that anesthesia can separate responsiveness, memory, awareness, distress, sensory processing, autonomic reactivity, and later report. A patient may fail to move because movement is blocked, fail to remember because memory formation is suppressed, or fail to report because the relevant state passed before recovery. A single category called unconscious under anesthesia is not enough.

For public-risk control, anesthesia is a stress test for the unity gate and the endpoint firewall. It shows how a stream can pass through low-access, amnesic, reversible, or fragmented states without granting researchers permission to treat the body as empty.

12.3.1 — Drug Class and Dose

Drugs used during anesthesia have different effects on consciousness, memory, movement, and physiology. Propofol, volatile agents, ketamine, benzodiazepines, opioids, neuromuscular blockers, and adjuncts cannot be collapsed into one state category. Neuromuscular blockers cause paralysis without providing unconsciousness, amnesia, or analgesia. Drug class and dose shape access, memory, sensory response, arousal, and autonomic markers differently (Sanders et al., 2012; Jang et al., 2024).

A state model must therefore record agent, dose, timing, coadministration, depth monitoring, hemodynamics, ventilation, pain control, surgical stimulation, and patient factors such as age, illness, neurological condition, medication, and sleep deprivation. These variables are not merely medical background. They are part of the evidence chain.

The relevant disagreement is empirical: different theories predict different effects of drug class, dose, and state on access, complexity, self-monitoring, and embodied modeling. The protocol should predeclare which theory-indexed markers it expects to move for each agent and dose.

For the bearer question, dose specificity prevents a false unity gate. An interval under light sedation, dissociative anesthesia, deep general anesthesia, and emergence should not receive the same classification merely because all occurred in an operating room. The ledger cannot inherit a clinical label and call it a consciousness state.

A protocol that uses anesthesia data must therefore predefine state bins by pharmacology and physiology, not by narrative convenience. It must also state whether the evidence bears on level, content, valence, continuity, or memory. A drug-specific model may produce a no-estimate interval when the signals conflict.

The ethical point is equally sharp. The purpose of anesthesia is care. The purpose of state classification is research. The first cannot be altered to satisfy the second. The dying person owes no result, and neither does the anesthetized patient.

A concrete measurement advance makes these distinctions more tractable. Maschke and colleagues analyzed spontaneous EEG from 15 healthy volunteers before and during unresponsiveness produced by propofol, xenon, or ketamine. Neural criticality measures differed across the drug-associated experiential states and predicted perturbational complexity; ketamine-associated dream reports were especially useful for separating unresponsiveness from reported absence of experience (Maschke et al., 2024).

The productive next step is a preregistered external validation across anesthetic agents, patient populations, and recording systems, retaining reports and behavioral channels alongside EEG. Such work could improve interval classification and identify where an observation model should preserve possible experience. The immediate objective is a more reliable relation between neural complexity, reported experience, and task access.

12.3.2 — Behavioral Unresponsiveness

Behavioral unresponsiveness can arise from drug effect, paralysis, motor-pathway interruption, sensory failure, task design, pain control, fear, confusion, or inability to understand commands. Because these mechanisms differ, nonresponse cannot carry the whole consciousness inference; the dossier must identify which prerequisite failed or remained untested (Sanders et al., 2012; Linassi et al., 2018).

This matters especially when neuromuscular blockade, surgical context, airway management, or severe illness makes movement unavailable. A person may be unable to signal even if some conscious content remains. The correct inference is not empty. The correct inference is that behavior is an unavailable or compromised channel.

Theory families handle this differently. Access theories may ask for alternative forms of task access. Integration accounts may use perturbational evidence. Predictive accounts may examine autonomic and interoceptive dynamics. Higher-order accounts may be limited if explicit metacognition cannot be tested. The point is not to choose one. It is to prevent behavior from carrying the whole inference.

Behavioral unresponsiveness under anesthesia therefore supplies a general endpoint lesson. Absence of movement cannot by itself justify excluding an interval from the candidate stream, especially when drug effects, sensory isolation, and motor blockade share the same outward appearance. Any cleaner ledger obtained by

dropping those intervals must be treated as a gate-sensitive analysis rather than as evidence about their phenomenal absence.

A fair protocol records the motor channel separately from the consciousness hypothesis. It asks which commands were possible, whether sensory input was intact, whether movement was blocked, whether alternative responses existed, and whether physiological markers changed with stimulation. If these cannot be answered, abstention is stronger than declaration.

The burden of proof should never be placed on a person to move while science already knows movement may be unavailable. That is the human reason behavioral unresponsiveness must be treated as evidence about a channel, not as a verdict about a stream.

12.3.3 — Covert Command Following

Covert command following asks whether a person who cannot respond behaviorally can nevertheless follow instructions in neural activity. The paradigm is powerful because it separates motor output from task understanding and intentional modulation. It is also limited because it requires comprehension, attention, sensory access, task design, and equipment-sensitive interpretation (Huang et al., 2018; Claassen et al., 2019).

An openly released graded-propofol fMRI resource extends this logic in healthy volunteers. It combines mental-imagery and motor-response tasks before, during, and after loss of behavioral responsiveness, including instances of volitional imagery without overt response. The dataset permits comparison and validation of models across dose, task, connectivity, and emergence while keeping behavioral output distinct from covert task performance.

Model recovery separately requires synthetic or otherwise ground-truthed benchmark data (Huang et al., 2026).

Context determines whether covert tasks are feasible and interpretable. Under anesthesia, task delivery may be clinically inappropriate or technically fragile. In disorders of consciousness, related paradigms can reveal cognitive motor dissociation. Across both settings, absent behavior keeps alternative channels scientifically relevant.

A positive covert response establishes organized task-related cognition under a declared paradigm. It can support command comprehension and intentional modulation, but it does not map every interval, content, affective state, or continuity relation needed for a complete bearer classification.

Its strongest contribution is classificatory. Covert command following can reopen a case that behavior alone would have closed, motivate alternative communication channels, and change which routing models remain live.

The protocol requirements include protocol-fixed tasks, artifact controls, false-positive rates, false-negative rates, replication criteria, sensory access checks, medication context, and independent interpretation. A single surprising activation should not rewrite the bearer classification without a dossier.

A negative result is equally bounded. Task misunderstanding, sensory or language barriers, medication, fatigue, signal loss, preprocessing choices, and fluctuating state can all produce failure. Repetition is justified only while clinical condition,

participant welfare, and expected information gain support another attempt.

The versioned dossier therefore records the task, prerequisites, signal quality, uncertainty, and rival explanations for both positive and negative findings. Covert command following updates the state classification without becoming a universal test of consciousness (Claassen et al., 2019; Bodien et al., 2024).

12.3.4 — Memory Formation and Amnesia

Memory is often mistaken for experience because the public usually learns about experience through later memory. Anesthesia breaks that habit. A patient may have no later recall because memory encoding was suppressed, while the question of intraoperative or perioperative experience remains distinct (Sanders et al., 2012; Linassi et al., 2018).

The relevant distinctions include conscious experience, sensory processing, pain processing, distress, memory encoding, consolidation, retrieval, and later narrative. Each can fail or survive differently. A research model that treats memory as the archive of experience will erase states precisely when memory is most pharmacologically fragile.

The theory implications vary. Higher-order and access accounts may treat later report as valuable but not complete. Predictive and interoceptive accounts may examine bodily response without claiming direct report. Integration accounts may ask whether neural complexity remains sufficient. The disagreement shows why amnesia cannot be allowed to settle the gate.

A physiological response does not by itself establish conscious distress. The dossier should therefore analyze memory, access, nociception, autonomic response, and report as distinct channels.

The protocol preserves a memory-state distinction among experienced content, encoded content, retrievable memory, affective response, and later report. When memory is absent but other markers conflict, the allowed conclusion follows the relevant evidence class rather than treating amnesia as absence.

Anesthesia research makes intervals with little later narrative trace scientifically accessible. Combining state-sensitive observations with recovery reports helps characterize their place in a continuing life.

12.3.5 — Emergence and Rebinding

Emergence from anesthesia is not merely a switch flipped from off to on. It may involve gradual recovery of arousal, sensory integration, attention, memory, body orientation, reportability, agency, and emotional regulation. The stream may appear to return before all capacities are stable (Jang et al., 2024; Huang et al., 2024).

Rebinding is the scientific problem of how pre-anesthetic and post-anesthetic intervals are treated as belonging to one stream. Ordinary practice assumes continuity, and often that assumption is right. The research program still needs evidence rules because emergence can include delirium, confusion, hallucination, pain, fear, and amnesia.

Theory families again separate dimensions. Global availability may return with report and task access. Integration may recover

at a different pace. Predictive control may be distorted by pain, drugs, and interoceptive surprise. Higher-order monitoring may lag behind arousal. A single emergence time is therefore inadequate.

Emergence is a reversible transition in observable state. Return challenges an earlier endpoint classification when continuity with the same stream is independently supported, yet it cannot reconstruct every experience within the low-access interval. Induction, maintenance, transition, emergence, and later report therefore remain separate evidence windows whose relations can be compared without hindsight replacing uncertainty.

A good record marks induction, low-access intervals, possible content, emergence, report, confusion, and later revision as separate events. It does not overwrite uncertainty after recovery. If later testimony emerges, it updates the dossier rather than erasing the earlier no-estimate state.

Anesthesia emergence is a humane lesson in scientific patience. The person returns before the model fully understands the interval. The model's incompleteness should never burden the returning person with a public story they did not choose.

Emergence is a transition whose components can separate: arousal, command following, report, memory, orientation, and coordinated control may return on different schedules. Rebinding claims should therefore specify which relations resumed and when. Recovery supports continuity models but does not retroactively make the entire anesthetic gap known.

12.3.6 — Clinical Priority

Clinical care governs every anesthesia study. Drug choice, dose, ventilation, airway management, analgesia, monitoring, emergence, and postoperative treatment must be determined by patient safety and clinical judgment, not by the needs of the Law's research program. The participant owes the theory no additional risk or data (World Medical Association, 2024).

For clinical interpretation, keep awareness and memory separate, and interpret both alongside agent, dose, stimulation, pain control, task availability, and emergence timing; amnesia alone cannot close the consciousness question (Sanders et al., 2012; Linassi et al., 2018).

The perioperative chain should be time-resolved. Baseline cognition, induction, maintenance, surgical stimulation, dose change, physiological instability, task timing, movement channels, neural evidence, emergence, memory testing, and later report occupy different places in the inference. One depth label cannot stand in for that sequence (Jang et al., 2024).

Research procedures require a noninterference budget. Auditory tasks, repeated stimulation, extra sensors, staff movement, and prolonged setup can alter arousal or clinical workflow. The protocol should state the permitted burden and give clinicians an immediate stop rule without penalty to the patient or team.

Negative results require an opportunity-to-respond analysis. Investigators should report whether the stimulus was delivered, sensory access was plausible, signal quality met criteria, a response route remained available, and the state was stable long

enough for the task. Without that chain, a negative trial is a missing test disguised as evidence.

Permitted outputs are narrow: better classification methods, clearer uncertainty, protection against false absence, and improved separation among memory, access, and experience. Disallowed outputs include bedside ranking, clinical authority, judgment of a particular anesthetic plan, or claims that hidden suffering has been detected in a specific procedure.

Consent, minimization, and clinician override are also part of the data-generating model. If research burden changes care, stimulation, physiology, or observation, the resulting evidence is partly about the intervention introduced by the study. That dependency must be visible.

Anesthesia therefore supplies a disciplined transition model rather than a consciousness verdict. It shows why unresponsiveness and amnesia are insufficient while keeping the science subordinate to care.

12.4 — Coma and Disorders of Consciousness

Disorders of consciousness are where boundary-state science becomes most consequential. The research question is no longer an abstract puzzle. It concerns patients who may be unable to speak, move, or demonstrate awareness through ordinary channels while families, clinicians, and institutions face decisions under uncertainty.

Repeated behavioral examination remains essential, yet current evidence shows that a substantial subset of behaviorally unresponsive patients can exhibit covert command following with functional imaging or electrophysiology. The 2024 multicenter study of cognitive motor dissociation strengthens the case for multimodal assessment while leaving prognosis, personhood, and treatment decisions under independent clinical standards (Bodien et al., 2024; Giacino et al., 2018).

A 2025 three-center EEG study in 237 acute and chronic disorders-of-consciousness patients identified recurring dynamic connectivity patterns and evaluated real-time bedside detection and recovery prediction. For this chapter, the contribution is a longitudinal, state-dynamic marker family that can be tested against diagnosis, time, confounds, and outcome rather than a one-time scalar label. Its findings do not eliminate task, clinical, or theory uncertainty, but they make repeated bedside dynamics a serious part of the dossier (Della Bella et al., 2025).

Disorders of consciousness are used here to test the reliability of evidence channels, not to turn patients into proof material. Bedside behavior, task-based neural activity, perturbational

measures, clinical diagnosis, family observations, and longitudinal recovery can disagree. The dossier should preserve those disagreements and use them to qualify routing confidence and endpoint eligibility.

The burden question is direct: who carries the cost of a false absence? If a patient is treated as empty because evidence was difficult to obtain, the burden falls on the person least able to contest the classification. That is why the science must preserve uncertainty as a real output.

Table 12.2. Disorders-of-consciousness evidence and caution matrix.

Evidence family	What it can support	What it cannot decide alone
Behavioral assessment	Reproducible response, arousal, clinical category	Absence of experience when motor or sensory channels are compromised.
Task-based EEG or fMRI	Covert command following under a defined task	Continuous consciousness, valence, or whole-stream ledger inclusion.
Perturbational or complexity measures	Level or capacity under a theory-laden marker	Personhood, care entitlement, or universal consciousness status.
Family and clinician observation	Context, repeated patterns, state sensitivity	Definitive classification without corroboration and confound review.

12.4.1 — Clinical Categories and Diagnostic Boundaries

Clinical categories such as coma, unresponsive wakefulness syndrome or vegetative state, and minimally conscious state are indispensable. They organize observations about wakefulness, arousal, behavior, reproducible awareness, and recovery. But they are clinical categories, not transparent windows into the whole of subjective experience (Giacino et al., 2018; Giacino et al., 2002).

Coma involves absent wakefulness. Unresponsive wakefulness preserves arousal cycles without clear behavioral evidence of awareness. Minimally conscious states show inconsistent but reproducible signs of awareness. These distinctions must be respected because they shape prognosis, care, research eligibility, and communication attempts.

Theories of consciousness interpret these states through different markers. Global access may be limited. Integration and perturbational complexity may be reduced or partially preserved. Recurrent or posterior content networks may be damaged, masked, or inaccessible. Predictive and interoceptive processing may continue in ways that are hard to interpret.

Clinical categories organize care and prognosis. In severe brain injury, the measurement analysis records whether valence estimates are identified, partially identified, or dependent on explicit marker and routing assumptions.

A responsible classification records diagnosis, assessment tool, time since injury, etiology, medication, sensory limitations, motor limitations, arousal state, neuroimaging or electrophysiology if

available, and uncertainty. The classification should be versioned because state changes and methods improve.

The ethical baseline remains present care. Scientific uncertainty about a stream does not reduce the person's dignity, care entitlement, or family meaning. The research record may say no estimate; the bedside must still say care.

Clinical categories are operational diagnostic classifications with uncertainty; they are not direct partitions of phenomenal reality. Because their relation to conscious capacity can vary with time, examiner, task, sensory and motor access, and injury pattern, stream routing should use longitudinal, multimodal evidence and retain No adequate estimate when the category cannot bear the intended inference (Giacino et al., 2018).

12.4.2 — Cognitive Motor Dissociation

Cognitive motor dissociation names a clinically important separation: a patient may show no observable command following while functional magnetic resonance imaging or electroencephalography indicates covert performance of a commanded mental task. The finding makes behavior-only exclusion unsafe as a final rule and reopens assessment when motor output is unreliable (Claassen et al., 2019; Bodien et al., 2024).

Across six centers, Bodien and colleagues studied a convenience sample of 353 participants. Task-based fMRI or EEG detected responses in 60 of 241 participants without observable command following, or 25%. Among 112 participants who did follow commands behaviorally, 43, or 38%, produced detectable task

responses. These differing detection rates make channel sensitivity a central design question and support repeated, multimodal assessment of retained cognitive capacities (Bodien et al., 2024).

The evidential asymmetry is important. A positive, well-controlled task result can expose a false behavioral negative, while a negative result remains compatible with sensory failure, inattention, fatigue, medication, fluctuating state, misunderstanding, insufficient statistical power, or an unsuitable task. Absence on the test therefore cannot become proof of absence of experience.

The protocol must distinguish true covert response from artifact, analytic overfitting, sensory inability, task failure, and uncertain command delivery. Repetition and multimodal assessment may increase confidence, but testing must stop when fatigue, burden, or clinical priority makes further assessment inappropriate.

For routing and endpoint studies, cognitive motor dissociation changes the gate. Motor silence must first be treated as a possible channel failure, and low-response participants cannot be removed merely because behavior is easy to score. Positive cases increase protection and review; negative cases retain the limitations of the instrument and task.

The finding therefore narrows one claim rather than solving the whole consciousness problem. It defeats behavioral unresponsiveness as a sufficient exclusion rule while leaving phenomenal experience, continuity, prognosis, ownership, and

endpoint status separately uncertain (Claassen et al., 2019; Bodien et al., 2024).

12.4.3 — Longitudinal Change

Disorders of consciousness are dynamic. A single assessment can miss fluctuations in arousal, medication effect, fatigue, sleep-wake cycles, pain, infection, seizure activity, sensory access, and recovery trajectory. Longitudinal change is not a complication added after diagnosis. It is part of the state (Bodien et al., 2024; Yakhkind et al., 2024).

Repeated assessment matters because detection is probabilistic. A patient may show command following on one day and not another. A family may observe meaningful response in a familiar context while a clinical exam does not reproduce it. A machine marker may vary with state, electrode quality, or task comprehension.

Theories of consciousness must therefore be tested over time. A global-access marker may appear during brief arousal. An integration measure may recover gradually. Predictive and interoceptive regulation may change with medical stability. A single theory snapshot can turn state variability into a false classification.

For later protocols, longitudinal change protects the endpoint firewall. If a candidate endpoint or low-signal interval is later followed by evidence of return, the earlier classification must not be rewritten silently. The record should preserve candidate, uncertain, revised, and confirmed statuses.

A good longitudinal design records timing, state context, test burden, clinical changes, and the reason a test was or was not repeated. It uses trajectories rather than isolated labels. It also allows durable abstention when repeated evidence remains ambiguous.

The person does not become less worthy because the signal changes. The model becomes more honest when it records that change. That is the difference between a fair classifier and a classifier that offloads uncertainty onto the patient.

A single examination can misclassify a fluctuating or recovering patient. Longitudinal design should record examiner, instrument version, arousal, medication, sensory function, injury evolution, and state transitions so that change can be distinguished from measurement drift. Routing and endpoint conclusions should follow the trajectory of evidence rather than the first stable label (Giacino et al., 2018; Yakhkind et al., 2024).

12.4.4 — Confounds and Artifacts

Every marker in severe brain injury can be confounded. Medication can suppress responsiveness. Sensory loss can prevent command perception. Motor injury can prevent response. Seizure activity, autonomic instability, sleep, pain, fatigue, and environmental noise can alter signals. Devices can fail. Analysis pipelines can manufacture patterns (Casali et al., 2013; Schnakers et al., 2009).

Confounds are not excuses to ignore evidence. They are reasons to make evidence sharper. A state dossier should ask whether the channel was available, whether the task was understood, whether

artifact controls were passed, and whether alternative explanations were tested.

Complexity metrics, connectivity measures, and task-based responses each have their own artifacts. High complexity can reflect noise or nonconscious dynamics. Low complexity can reflect sedation, injury, poor data, or state fluctuation. Task response can reflect true command following or analysis error. A fair review names these possibilities before using the result.

For the evidence dossier, confound control prevents false terminal inference. A pattern that seems to show absence, narrowing, or closure may reflect medication, sensor shutdown, family absence, care procedures, or device limitations. If those confounds are not modeled, the burden of error falls on vulnerable patients and the truth.

A useful confound register includes medication, sedation depth, analgesia, sensory pathway, motor pathway, time since injury, arousal cycle, infection, seizure, temperature, sleep, device quality, preprocessing, and adjudicator knowledge. The register should travel with the classification rather than remain in a laboratory notebook.

The purpose of confound review is to identify which observations are informative and why. A clear account of signal quality, alternative explanations, and uncertainty makes even an incomplete record useful for comparison and future improvement.

Confound control should be built into the marker packet. Sedatives, antiseizure drugs, metabolic disturbance, hydrocephalus, sleep-wake fluctuation, sensory loss, motor

impairment, scanner artifact, and task noncomprehension can mimic absence or conceal capacity. A result that cannot separate these explanations belongs in a lower confidence or unusable class, even when the trace looks clean.

12.4.5 — Family and Clinician Evidence

Families and clinicians often notice patterns that formal tests miss. A familiar voice, a specific routine, a painful procedure, a beloved song, or a subtle movement may reveal context that a standardized task does not capture. Such evidence deserves respect because it can identify channels and states that instruments overlook (Giacino et al., 2018; Young et al., 2021).

It also requires discipline. Family hope, grief, exhaustion, and pattern seeking can overinterpret ambiguous signs. Clinician familiarity can be powerful but can also be shaped by workload, prognosis expectations, and institutional culture. Contextual evidence is valuable; it is not a definitive marker by itself.

The theory-plural approach helps here. A family observation may indicate access, affective response, learning, recognition, or arousal. It may also be a reflex, coincidence, or artifact of timing. The review should ask which theory families the observation bears on and what alternative explanations remain.

For public-risk control, family and clinician evidence matters because hidden burden is often visible first to those who care. A fair science should not treat only machine-readable signals as evidence. Yet it must also protect families from having their love turned into scientific proof or public burden.

A state dossier can include contextual observations with provenance: who observed, in what setting, under what conditions, whether the response was repeated, whether blinded review is possible, and which claims the observation does not support. This makes lived context reviewable without flattening it into a score.

Family and clinician evidence reminds the chapter that consciousness research is not conducted in a vacuum. It happens among people who are already carrying care, grief, uncertainty, and hope. The science must learn from them without making them carry the verdict.

12.4.6 — No Research Classification as Treatment Ceiling

A research classification must never become a treatment ceiling. Low confidence, failed command following, an unresolved marker pattern, or No adequate estimate cannot justify reduced analgesia, rehabilitation, communication support, repeated examination, legal protection, or ordinary clinical care (World Medical Association, 2024; Giacino et al., 2018).

Diagnostic classification, prognosis, research eligibility, and bearer routing are separate questions. They may use overlapping evidence, but they have different targets, consequences, and error costs. A marker useful for detecting covert command following may be poor at forecasting recovery and still less suited to determining whether an interval enters a phenomenal ledger.

False negatives are especially consequential because absent response can reflect motor impairment, fatigue, fluctuating arousal, language mismatch, sensory loss, medication, or task

burden. Multimodal assessment reduces some risks but does not make every negative channel independent or decisive (Bodien et al., 2024).

Longitudinal reassessment is part of the target. A state that fluctuates across hours or days cannot be represented responsibly by one examination selected for convenience. Repeated testing, rest, alternate tasks, and context-sensitive interpretation help distinguish stable absence from episodic access or measurement failure.

Verification is selective. Cases followed by recovery, repeated testing, longer survival, or better-resourced sites are more likely to supply a later answer. Studies should report which classifications become verifiable and whether verification depends on prognosis, family advocacy, site capacity, or survival, rather than evaluating accuracy only on the easiest cases.

Family and bedside observations can identify timing, channels, and test conditions that standardized tasks miss. Familiar voices, routines, expressions, or context-specific responses should be preserved with provenance and uncertainty, then examined against alternatives. Repeated observations can test whether context-specific responses recur under comparable conditions, helping separate stable signal from one-off interpretation.

Research and treatment pathways therefore remain on separate rails. Clinical decisions made under time pressure do not certify a research bearer, and research classifications do not outrank patient values, clinical obligations, or law. Public language should state the purpose-bound result and its limits. This separation also

guards against circularity in which treatment decisions made from a clinical classification later appear as independent validation of the research label.

The scientific gain is a more accurate account of what the evidence can and cannot show. The ethical gain is that uncertainty is absorbed by the research system rather than by the person whose communication is limited. That allocation of uncertainty reduces pressure to force a binary classification when the available channels support only a range of possibilities.

12.5 — Dementia, Delirium, and Fragmented Access

Dementia, delirium, and fragmented access force the chapter to separate experience from narrative continuity. A person may lose memory, executive organization, language, or orientation while comfort, distress, fear, pleasure, pain, trust, and recognition still matter. The scientific temptation is to treat what is hard to narrate as less real.

Dementia and delirium reveal why memory, language, attention, and experience must be kept distinct. Fluctuating cognition and report failure can coexist with distress, pleasure, recognition, and relational response, so the research problem is not to infer absence from impairment but to design context-sensitive access and comfort measures (Inouye et al., 2014; Fong et al., 2015; Livingston et al., 2024).

The scientific response is to keep present felt burden central even when biography becomes difficult to follow. Memory loss, fluctuating attention, atypical expression, and communication change can hide distress or comfort from ordinary observation. Measurement should adapt to those changes rather than treat them as reasons to erase the candidate stream.

Dementia and delirium should be analyzed through the capacities and channels that remain, not through a single global label. Attention, report, memory, affective responsiveness, pain behavior, sensory access, fluctuation, and medication can change differently. The classification should state which marker failed, which persisted, and when abstention is required, while care and dignity remain unchanged.

12.5.1 — Memory Loss Is Not Experience Loss

Memory loss can break biography without eliminating experience. A person may forget an event and still have felt it. They may fail to recognize a name and still feel comfort, fear, hunger, pain, pleasure, or attachment. A science that equates memory with experience would confuse later access with present life (Squire, 2004; Livingston et al., 2024).

The evidence channels must therefore separate episodic memory, semantic memory, autobiographical narrative, immediate affect, recognition, bodily comfort, distress behavior, and physiological state. Each channel can decline differently. A person's inability to recount a burden does not prove the burden was absent.

Theories of consciousness help by distinguishing content, access, reportability, memory, metacognition, and unity. Higher-order capacities may be reduced while experience remains. Global availability may narrow while local affective states remain salient. Embodied and interoceptive accounts may detect burdens that narrative accounts miss.

The scientific target is present experience, not the ability to preserve or narrate it later. A system that samples only remembered states will systematically underrepresent people with impaired encoding, retrieval, language, or autobiographical organization. Memory measures should therefore be modeled as delayed and selective witnesses whose failures change uncertainty, not as retrospective vetoes on what was felt.

A responsible protocol combines contemporaneous observation, accessible report, comfort and distress indicators, caregiver

context, sensory access checks, and repeated assessment. The record identifies which observations depend on memory and how recall was assessed.

The humane rule is exact: memory matters to identity, care, and evidence, but it is not the price of being counted as a feeling subject. The research program must keep present experience visible when the archive becomes unreliable.

Prospective observation is also making retained capacities in advanced dementia easier to study. Across 539 audiovisual observations of 20 participants, Gilmore-Bykovskyi and colleagues (2025) identified nine validated lucid episodes in three people, including increases in meaningful vocal communication. A separate caregiver study found that lucid episodes often occurred well before death (Griffin et al., 2024). These findings support patient-centered attention to fluctuating communication and offer a feasible observation method.

12.5.2 — Delirium and Fluctuation

Delirium is a fluctuating state of attention, awareness, cognition, and often distress. It can vary across hours, medication changes, infection, pain, sleep disruption, environment, and clinical stress. A single classification can be dangerously misleading because the state itself is unstable (Inouye et al., 2014; Fong et al., 2015).

The evidence problem is temporal. A person may be lucid in the morning, confused in the evening, distressed at night, and more coherent after hydration, pain control, or family presence. Measurement that averages across those changes can turn fluctuation into noise rather than evidence.

Theory families interpret delirium differently. Access may fluctuate; integration may be unstable; predictive control may be overwhelmed by interoceptive and environmental uncertainty; higher-order monitoring may fail; recurrence may remain locally active but disorganized. None of these accounts permits one fixed label without time indexing.

Delirium requires time-varying classification because attention, arousal, distress, perception, report, and physiology can fluctuate over hours. Treating the entire episode as one state can erase periods of possible experience or admit intervals whose channels were unusable. A state-space record should therefore permit included, low-confidence, and no-estimate intervals within the same clinical course.

A fair protocol records delirium screens, medication, infection, pain, sleep, environmental stimulation, sensory aids, hydration, communication access, and time of day. It also records whether a low score reflects inattention, aphasia, hearing loss, fatigue, fear, or true absence of relevant capacity.

Delirium teaches the book that uncertainty is sometimes rhythmic. The burden is not only whether a person is conscious, but whether the observation schedule is faithful to the moments in which the person can be reached.

12.5.3 — Language and Report Limits

Language is one of the most convenient bridges from experience to evidence, which is exactly why it can dominate classification unfairly. Aphasia, dysarthria, hearing loss, unfamiliar language, developmental stage, cognitive impairment, anxiety, fatigue, and

cultural differences can all limit report while experience remains (Livingston et al., 2024; Kitwood, 1997).

A boundary-state protocol should distinguish inability to speak, inability to understand, inability to access words, unwillingness to report, and absence of experience. These possibilities require different tests. Alternative communication, yes-or-no interfaces, eye tracking, gesture, affective expression, and caregiver knowledge may all matter.

Theories centered on report or metacognition are especially vulnerable here. They may overvalue linguistic access. Report-free paradigms reduce that risk but introduce their own theory dependence. The correct response is not to discard report, but to place report among other channels.

Language ability is a measurement condition, not an ontological criterion. People who cannot describe pain, fear, pleasure, confusion, or memory may be most vulnerable to having those experiences discounted. Instruments and interviews should therefore provide alternative communication routes, and failed verbal report should widen uncertainty rather than lower presumed experiential standing.

The dossier should record language access, translation needs, hearing and vision status, motor capacity, communication aids, fatigue, cultural context, and confidence in interpretation. It should also state which claims remain unavailable when report is weak.

Interpret speech and silence through the participant's communication capacities, clinical state, and available channels.

Language and report limits require alternative routes rather than lower expectations of experience. Gesture, gaze, affective response, familiar routines, supported communication, physiology, and clinician observation may each contribute within their scope. The study should record which channel was unavailable and why, so that inability to answer does not become an answer.

12.5.4 — Distress, Comfort, and Misinterpretation

Distress and comfort in dementia or delirium may appear in atypical forms. Agitation, withdrawal, calling out, resistance, facial expression, sleep change, appetite change, or bodily movement may reflect pain, fear, confusion, unmet need, environment, medication, or nonconscious reflex. Interpretation is necessary and risky (Inouye et al., 2014; Fong et al., 2015).

The risk of misinterpretation runs in both directions. A distressed person may be dismissed as merely confused. A reflex may be read as suffering. A calm appearance may hide pain, fear, or resignation. Atypical expression requires context, repeated observation, and humility.

Theory families can help separate channels. Interoceptive and predictive accounts foreground bodily distress and environmental threat. Access and report accounts ask what can be communicated. Integration and recurrence accounts ask what level and content markers remain. The disagreement should be recorded rather than resolved by convenience.

Atypical distress poses a symmetric error problem. Excluding hard-to-interpret behavior can hide burden; counting every

atypical movement or vocalization as distress can manufacture a signal the person did not experience. Repeated observation, individualized baselines, clinical context, caregiver knowledge, and converging channels should define the identified set rather than one behavior code.

A fair protocol uses pain assessment adapted to cognitive impairment, context review, caregiver input, medication review, environmental audit, sensory access, and response to comfort interventions. It records uncertainty about whether the marker reflects felt state, physiology, habit, reflex, or communication attempt.

Comfort matters too. Pleasure, recognition, relief, touch, music, routine, food, warmth, and familiar presence can be part of felt life even when narrative is damaged. A science that sees only distress because distress is louder would not be fair either.

Distress and comfort can be misread when expression is atypical, motor access is limited, or observers expect a flat response. State-sensitive behavioral patterns, physiological context, caregiver knowledge, and response to comfort measures should be evaluated together. The aim is not to infer valence from one proxy, but to prevent proxy failure from erasing possible suffering or relief.

12.5.5 — Dissociation and Split Access

Dissociation and split access complicate ordinary assumptions about unity. A person may experience depersonalization, derealization, altered self-location, discontinuity, compartmentalized memory, or changes in agency without the

science being entitled to sensational conclusions about multiple streams (Bayne, 2010; Schechter, 2018).

The evidence problem is not whether a clinical label proves partition. It does not. The problem is how to classify intervals in which access, memory, affect, agency, and self-representation are altered. Different dimensions can fragment without requiring the model to split the bearer.

For dissociation, the discriminating question is whether apparent fragmentation reflects disrupted self-monitoring, altered body/world modeling, loss of global access, or a genuine change in causal unity. Report that theory-indexed uncertainty directly.

Dissociation tests access and ownership, but lack of access alone does not settle stream continuity. The relevant questions concern causal influence, cross-state learning, affective carryover, coaccess, and whether event sets can be assigned without duplication. The answers may support one stream, a partition model, several admissible models, or no estimate.

A protocol should record clinical context, trauma risk, consent, privacy, report limits, memory access, affective continuity, and the danger of inducing distress through questioning. It should avoid any procedure that pressures a person to perform identity fragmentation for research.

Dissociation offers a productive test of how access, memory, agency, and causal continuity fit together. Claims about stream division should follow that evidence while preserving privacy, respect, and clinically appropriate support.

12.5.6 — Care and Dignity Under Uncertainty

Memory loss, language impairment, disorientation, or fragmented narrative do not reduce dignity or establish absence of experience. They alter access and continuity evidence while distress, comfort, relationship, and momentary awareness remain clinically important (Livingston et al., 2024).

Medication, pain, infection, sleep, sensory conditions, familiarity, fatigue, and time of day can change communication. One brief examination should not harden into a permanent account.

Care should maximize communication and minimize distress regardless of the estimate. Familiar people, adapted language, sensory aids, calm settings, and repeated opportunities may improve welfare and evidence without forcing performance.

A model may classify an interval as uncertain, locally supported, or unusable for one analysis. It may not turn that status into a judgment that the person is gone, less authentic, or less entitled to privacy and care.

Family and clinical observations can supply longitudinal context when source, timing, uncertainty, and alternatives are recorded.

When access remains unresolved, protective uncertainty is often correct. Science narrows its claim while care, dignity, and communication remain fully active (World Medical Association, 2024).

12.6 — Developmental and Nonhuman Cases

Development and nonhuman consciousness make the boundary problem less familiar and more ethically exposed. Adult human report cannot be the universal template. Infants, children, animals, and other forms of life may express experience through capacities, behavior, physiology, learning, play, pain response, social attachment, or ecological action that do not look like adult discourse.

Developmental and comparative science cannot use adult verbal report as the universal gate. Evidence from early development, comparative neurobiology, behavior, and welfare science supports graded reference classes and protective uncertainty, while the New York Declaration emphasizes a realistic possibility of consciousness across a wider range of animals than traditional practice assumed (Lagercrantz & Changeux, 2009; Andrews et al., 2024).

The evidential discipline runs in both directions. Convenience can exclude beings whose signals are difficult to interpret, while sentiment can promote weak evidence into a bearer verdict. Protective uncertainty supports proportionate welfare precautions and restricted claims; it does not erase the need for discriminating evidence.

Developmental and comparative cases require evidence zones because marker validity changes with age, species, task, anatomy, ecology, and welfare constraints. Each zone should state which capacities are being tested, which theory families interpret them, what transport assumptions are required, and what the

classification permits. The output is domain-bounded confidence, not a universal rank.

12.6.1 — Developmental Emergence

The origin of a qualifying conscious stream need not coincide with biological birth. Developmental emergence asks when the evidence makes a functional boundary of conscious experience scientifically plausible. A symbolic date cannot answer that question, and research convenience cannot substitute for the relevant developmental evidence (Lagercrantz & Changeux, 2009; Dehaene et al., 2017).

Developmental evidence includes arousal regulation, sensory integration, pain and comfort response, learning, sleep-wake organization, affective expression, social engagement, attention, memory formation, and neural maturation. These markers emerge unevenly. The model must not pretend that one milestone carries the whole boundary.

Developmental evaluation must distinguish flexible coordination, causal integration, embodied regulation, and explicit metacognition rather than treat them as interchangeable markers. The boundary remains theory-indexed and age-sensitive, especially when a theory requires explicit metacognition.

For later protocols, developmental emergence matters because a ledger cannot begin before the bearer exists, but that scientific boundary does not settle moral status. The scientific start of a stream for a model is not a public ruling on the value of infants, fetuses, children, families, or clinical care.

A responsible developmental protocol states age, developmental condition, marker family, sensory access, clinical context, stress burden, and uncertainty. It uses protective defaults where false exclusion could miss experience, while narrowing formal claims when evidence is insufficient for ledger inclusion.

The fair standard is careful without being cold. Developmental uncertainty should produce more protection and more precise language, not a search for a convenient line that lets the model avoid difficulty.

Developmental emergence is unlikely to be a single instant shared by all capacities. Arousal, sensory integration, affective learning, self-world differentiation, report, memory, and metacognition develop on different trajectories. The dossier should therefore be age- and capacity-specific and should not treat adult reportability as the threshold that creates experience (Lagercrantz & Changeux, 2009).

12.6.2 — Age-Appropriate Evidence

Age-appropriate evidence means the marker must match the organism's capacities. An infant cannot be asked for adult verbal report. A toddler may express pain, fear, or preference through behavior, attention, crying, regulation, and interaction. A child's evidence must be read developmentally rather than judged as deficient adult evidence (Lagercrantz & Changeux, 2009; Bayne, 2010).

The relevant channels include facial expression, physiological response, orienting, learning, touch response, sleep, distress, soothing, play, attention, social referencing, and communication

attempts. Each channel requires context because it can also reflect reflex, fatigue, hunger, illness, caregiver stress, or environmental mismatch.

Theories of consciousness can be translated into age-appropriate terms. Access may appear as flexible use rather than verbal report. Integration may appear as coordinated state regulation. Predictive processing may appear in expectation, surprise, and comfort-seeking. Higher-order evidence may be limited, but that limitation cannot erase possible experience.

For the evidence dossier, age-appropriate evidence prevents a subtle unfairness: making the most vulnerable subjects carry the cost of adult measurement. If the only accepted proof is adult speech, then infancy becomes invisible by method rather than by truth.

A protocol should state what the child could realistically do, what the environment allowed, what distress the study imposed, and which claims remain unavailable. It should also separate clinical care, developmental support, and research classification.

The goal is not to make children research objects. It is to keep the science from building a theory of conscious life that quietly begins only where adult methods become convenient.

Age-appropriate evidence changes the task, response channel, baseline, and interpretation. Infant looking, autonomic response, learning, preference, distress behavior, and neural organization can support different claims than adult report. Transport across ages requires an explicit bridge; using the same instrument name does not make the construct identical.

12.6.3 — Species-Specific Expression

Species-specific expression is the nonhuman analogue of age-appropriate evidence. Animals do not owe human-style report to be scientifically considered. Their capacities, ecologies, sensory worlds, social lives, pain behaviors, play, learning, and problem solving must be interpreted in relation to the form of life they actually have (Andrews et al., 2024; Birch et al., 2021).

The marker set will differ across mammals, birds, reptiles, fishes, cephalopods, decapods, insects, and other candidates. Some evidence may come from nociception, protective behavior, motivational tradeoffs, learning under analgesia, play, social recognition, sleep-like states, or flexible avoidance. No single human marker can be exported unchanged.

Species comparisons require homologous or functionally analogous markers rather than a human-centered checklist. Nonverbal flexible use, causal organization, embodied regulation, and ecological competence may carry different weights across taxa.

For public-risk control, species-specific evidence matters because a universal claim about conscious streams cannot silently mean adult humans only. At the same time, including a nonhuman stream in a formal ledger would require more than welfare concern. It would require domain-specific evidence about consciousness, valence, time, and the candidate realization if reserve dynamics are invoked.

A comparative evidence zone should name the taxon, context, marker family, experimental demand, welfare protections, and

known error routes. Protective treatment can be justified under uncertainty even when the scientific classification remains open. Precautionary welfare decisions must therefore remain distinct from claims that a marker has established consciousness or stream structure.

Nonhuman research expands the range of expressions, behaviors, and causal organizations that measurement must accommodate. Species-sensitive methods improve both scientific interpretation and welfare.

12.6.4 — Comparative Homology and Analogy

Comparative evidence often relies on homology and analogy. Homology asks whether structures or mechanisms are shared through common ancestry. Analogy asks whether different systems perform similar functions through different structures. Both matter, and both can mislead if used loosely (Barron & Klein, 2016; Crook, 2021).

A mammalian pain system may support inference through shared neurobiology. Bird cognition may require attention to different neural architecture with similar functional organization. Cephalopod intelligence may require analogy more than direct homology. Insects may present still different evidence profiles. Each case demands its own bridge.

Theory families help sort the bridge. Integrated and causal accounts may be more open to different substrates if causal organization is relevant. Embodied accounts may emphasize organism-environment loops. Access and higher-order accounts

may need translation into species-specific capacities. The bridge should be explicit rather than hidden in human resemblance.

For the bearer question, homology and analogy determine how far a status statement can travel. Evidence from mammals cannot silently generalize to all animals. Evidence from human patients cannot silently generalize to artificial systems. Reference class discipline applies to consciousness just as it applies to terminal dynamics.

A comparative dossier should state whether the inference is homologous, analogous, behavioral, physiological, computational, ecological, or welfare-protective. It should also state whether the evidence supports consciousness, valence, pain, affective learning, or merely complexity.

The chapter's rule is to let similarity help without letting it rule. A fair science takes kinship seriously, takes difference seriously, and refuses to turn either into a shortcut.

Homology can support transport when shared structures and functions have a credible evolutionary relation; analogy can support a weaker comparison when similar capacities arise through different organization. Neither is decisive by itself. The dossier should state which relation is claimed, which properties are conserved, and where the comparison is expected to fail.

12.6.5 — Protective Uncertainty

Protective uncertainty guides cases where suggestive evidence makes false exclusion costly. Welfare precautions can respond to that risk while researchers continue to assess formal inclusion (Andrews et al., 2024; Crump et al., 2022).

In nonhuman and developmental cases, protective uncertainty may lead to welfare consideration, reduced experimental burden, better housing, analgesia, environmental enrichment, or prohibition of harmful procedures even when consciousness classification remains open. The ethical response can be stronger than the scientific verdict.

Consciousness theories can identify different candidate cases. A cross-theory matrix records the supporting indicators and the areas of agreement or uncertainty, allowing welfare precautions and further assessment to proceed together.

When false exclusion could permit avoidable harm, the design increases welfare protection and preserves the unresolved scientific classification.

A protocol should distinguish research classification, welfare protection, legal routing, and public communication. Protective uncertainty belongs especially to welfare and research-burden decisions. It does not authorize public scoring, domain expansion, or claims that all possible bearers have been scientifically included.

This approach connects careful classification with practical care. The study reports the evidence it can identify while welfare decisions respond to the possibility of avoidable harm.

Protective uncertainty preserves welfare without pretending that consciousness has been established everywhere. When evidence is incomplete and the cost of false exclusion is serious, research and practice should avoid preventable suffering while retaining

an uncertain scientific status. Protection and classification answer different questions.

12.6.6 — No Harmful Crucial Experiments

No harmful crucial experiment is permitted to settle a question of consciousness. A study may not induce pain, deprivation, fear, injury, prolonged isolation, or clinically unnecessary burden in a child, nonhuman animal, or other vulnerable candidate merely because the result would be theoretically decisive (World Medical Association, 2024; Andrews et al., 2024).

No harmful crucial experiment does not mean no discriminating experiment. Formal models and simulations may use maximally sharp endpoint-versus-continuation contrasts, including the biological-death admissibility condition in Section 5.7.2, without turning those contrasts into empirical harm. Empirical comparative work can use methods such as naturalistic longitudinal variation, comparative anatomy and behavior, reversible low-burden perturbations, benign enrichment, route choice, recovery, natural survival, and independently warranted care, husbandry, or welfare interventions. The empirical study must not create a harmful terminal condition merely to sharpen the contrast. Ethical refinement can improve inference by removing acute threat, injury, escape responses, and other processes that would otherwise become competing explanations for an observed effect. The full immediate-terminal formal discriminator is developed in Section 19.2 and the Section 29.4.4 simulation benchmark.

Evidence must be age- and species-appropriate. Adult report tasks cannot serve as a universal test, and human-like behavior cannot be the only route to inclusion. Sensory worlds, motor repertoires, developmental capacities, ecological demands, and species-specific expression determine what is informative. A transport claim should identify the functional relation presumed to persist across different expression systems, not merely cite behavioral similarity.

False inclusion and false exclusion have different consequences. Precaution may justify welfare protection while scientific language remains restrained. Protective action is not a confirmed bearer classification, and classification uncertainty is never permission to neglect possible suffering (Birch et al., 2021; Andrews et al., 2024). The welfare threshold can therefore be deliberately more precautionary than the evidentiary threshold for scientific classification without conflating the two.

Research must not withhold ordinary care, enrichment, social contact, analgesia, or environmental support to preserve a contrast. Multidisciplinary review should identify avoided harms, unavailable evidence, and the limits those absences place on interpretation. When a humane condition removes a potentially informative contrast, the missing contrast should be recorded as an ethical identification limit rather than silently approximated.

Rigor therefore comes from theory comparison, better noninvasive markers, transport tests, and willingness to report no estimate when the only decisive experiment would be unacceptable. Comparative work is strongest when marker transport is tested across species or developmental stages with explicit allowance for different sensory, motor, and communicative repertoires.

12.7 — Artificial and Synthetic Candidates

Artificial and synthetic candidates test several assumptions at once: whether performance can substitute for experience, whether causal organization can transport across substrates, and how uncertainty should alter governance. Software systems, simulations, organoids, and other synthetic biological systems therefore require separate evidence pathways.

Theory-derived indicators can organize those pathways, but they are credence-shifting observations rather than consciousness detectors. Linguistic fluency, model scale, simulation success, electrical activity, and biological complexity each admit nonphenomenal explanations and cannot confer authority over persons (Butlin et al., 2026; Kreitmair, 2023; Hartung et al., 2024).

The evidence ladder separates performance, training history, simulation, architecture, causal organization, recurrent integration, memory, embodiment, self-modeling, and world-directed control. It also separates candidate status from personhood, welfare protection from scientific confirmation, and human governance from automated classification.

Artificial intelligence is especially vulnerable to category drift because fluent output can be mistaken for both experience and authority. Generated testimony may enter a dossier as behavior to explain; it cannot adjudicate its own status, classify another bearer, select endpoints, define fairness, or rank evidence.

Table 12.3. Artificial and synthetic evidence ladder.

Research-evidence zone	Typical evidence	Public-language limit
Not supported under current protocol	Only surface performance or scripted affect labels	No conclusion about consciousness, personhood, rights, or future architectures.
Insufficiently identified	Some architectural indicators but weak causal access	No conclusion that consciousness is present or absent.
Theory-limited research candidate	Indicators under one or two theories with live rivals	Theory-limited research status only; no personhood or rights inference.
Cross-theory research candidate	Multiple independent marker families survive review	Cross-theory evidence is not personhood, legal status, moral rank, or self-adjudicating authority.
Prohibited study	Testing would require creating or intensifying suffering	Stop or redesign; uncertainty does not license harmful evidence-seeking.

12.7.1 — Performance Is Not Experience

Performance is behavior under a task; experience is the phenomenal target, if any, behind that behavior. In ordinary human interaction, language is informative because it is embedded in shared biology and development. Artificial systems trained on human language break that inference and require an independent bridge (Bender et al., 2021; Butlin et al., 2026).

An artificial system may report pain because a prompt requests empathy, training data contain pain language, reinforcement favored a style, retrieval supplies a script, or the system models user expectations. Those explanations support claims about competence, imitation, alignment, deception, role performance, or interface design, not a conscious stream.

Consciousness theories supply candidate indicator families such as recurrent processing, global availability, higher-order monitoring, predictive modeling, agency, and embodied interaction. The 2026 indicator framework treats such features as theory-indexed reasons to update credence, not as a single test whose output settles consciousness (Butlin et al., 2026).

Without a supported phenomenal bridge, linguistic fluency, emotional style, self-description, and strategic behavior remain functional evidence. They may justify further architectural investigation while leaving bearer status unresolved.

A responsible dossier explains what the system can do, which mechanisms produce the behavior, which training and tool chains shape it, whether persistent state exists, and which alternative explanations survive. It can return noncandidate, theory-limited candidate, or No adequate estimate without contempt or hype.

Intervention gives architectural indicators scientific force. Ablation, memory interruption, recurrent-state perturbation, counterfactual transfer, and sensorimotor intervention can test whether a proposed organization is causally involved. Parameter count, eloquence, benchmark performance, and anthropomorphic design cannot substitute for that evidence.

12.7.2 — Architecture and Causal Organization

Consciousness theories concern causal organization, so architecture matters beyond surface behavior. Recurrence, integration, global availability, memory, attention, metacognition, embodiment, self-modeling, agency, and action-world feedback form different indicator families with different commitments (Butlin et al., 2026; Dehaene et al., 2017).

Each indicator receives a mechanical definition, time scale, perturbation, expected observation, and rival. Engineering labels such as memory, reward, attention, or self-model carry no evidentiary weight merely because a component uses the name.

Causal tests replace feature inventories with counterfactual evidence. If removing or interrupting a proposed feature leaves the relevant behavior unchanged, the feature may be redundant, decorative, or incorrectly characterized; if the behavior changes, the result still requires a theory-specific bridge to consciousness.

Training and simulation remain part of the architecture review because human descriptions of grief, pain, selfhood, and uncertainty can produce fluent reports through learned regularities, retrieval, reward shaping, or role prompting. The analysis must separate those routes from persistent state and causal organization.

Architectural evidence informs the classification specified by the protocol. The analysis names the indicator properties, causal organization, and observation conditions supporting that classification.

System version, training and fine-tuning, tools, memory, embodiment, sensors, actuators, persistent state, and intervention results belong in the same provenance packet. Indicator-rich and indicator-poor rivals receive the same behavioral evidence and analysis budget.

This method slows both excitement and dismissal. It asks which causal powers are present, which predictions they satisfy, which engineering explanation remains, and which observation would change the classification. The output is a bounded dossier rather than a verdict extracted from scale, brand, fluency, or fascination.

12.7.3 — Training Data and Simulation

Training data can reproduce the language of grief, pain, love, awe, fear, and philosophy without reproducing the experience described. The evidentiary task is to separate learned description, retrieval, prompt response, and reward-shaped style from persistent internal state (Bender et al., 2021; Butlin et al., 2026).

Simulation does not settle the matter by analogy. A simulated storm is not wet, yet some theories treat organization rather than material as central. The scientific question is therefore which causal properties are instantiated, which are merely represented, and what evidence would discriminate the two.

Theory families answer that question differently. Substrate-neutral accounts emphasize causal organization; embodied and biological accounts emphasize interoception, metabolism, organismic regulation, or developmental history. The dossier identifies which inference depends on which commitment.

Persuasive output also creates an authority hazard. A system may describe its own pain, spiritual insight, or certainty in ways that influence users. Those self-descriptions are observations generated by a system, not independent testimony about its phenomenal status or the Law.

Training source, fine-tuning, reinforcement objectives, role instructions, memory design, simulation architecture, intervention tests, and the relation between affect labels and internal control are all versioned. The dossier states whether the object is software, a simulation, an embodied agent, or a synthetic biological system.

The result may be scientifically modest: no current support, insufficient identification, or candidate status under one theory family. Modest classifications are valuable because they preserve a path for stronger evidence without allowing fluency to write metaphysics.

12.7.4 — Organoids and Synthetic Biology

The August 2025 ISSCR compilation retains guidance stating that current central-nervous-system organoids provide no biological evidence of consciousness or pain perception that would warrant specialized consciousness oversight, while urging continued attention as models mature, combine, and gain complexity (International Society for Stem Cell Research, 2025; Hyun et al., 2020).

That calibration prevents two opposite errors. Present organoids are not treated as conscious by default, and future ethical questions are not dismissed merely because current systems fall

below the evidentiary threshold. Governance follows the changing properties of the model rather than public fascination.

The dossier tracks cell types, maturity, connectivity, spontaneous dynamics, sensory coupling, learning, perturbational response, metabolic support, and theory-specific relevance. Electrical activity or complexity alone is insufficient.

Governance covers cell and tissue provenance, consent, stimulation limits, monitoring, welfare warnings, expert review, and authority to pause or stop. No aversive manipulation serves only to improve candidate-state detection.

Stronger evidence may support a stronger candidate classification through controlled review, independent expertise, and public correction. Weak evidence leaves a well-governed model with bounded claims, not a dramatic bearer verdict.

Design for uncertainty before ambition. Seek informative, reversible, low-risk observations; leave questions unanswered when stronger tests create morally relevant risk.

12.7.5 — Candidate Status and Abstention

Candidate status is a research classification, not recognition; abstention is an active result, not dismissal. A system enters a candidate zone only under declared theory-derived indicators and leaves it when version, architecture, evidence, or failure criteria change (Butlin et al., 2026; Hartung et al., 2024).

The evidence ladder distinguishes no current support, insufficient identification, theory-limited candidate, cross-theory candidate, welfare-precaution candidate, and prohibited study. Each class

names evidence, live theories, permitted research, prohibited uses, and revision triggers.

Public descriptions state the research classification, its intended use, and the evidence supporting it. These details connect the result with the scientific question it addresses.

Abstain if theories disagree, architectural access or transport fails, or testing requires harmful creation or manipulation. Record the reason and potentially decisive observation.

Classification may change with evidence, but no research zone permits harmful testing or automated administration. Welfare precautions may exceed classification because error costs differ.

12.7.6 — No Artificial Intelligence Sovereignty

Artificial systems may aid search, simulation, or pattern analysis under accountable review. They cannot govern consciousness classification, certify their experience, judge another system, or decide human inner life (Tabassi, 2023; Butlin et al., 2026).

Apply §12.7.1's performance-versus-experience and theory-indexed indicator rules. Artificial self-reports remain functional evidence unless architecture and intervention tests justify stronger inference; theory-based indicators may warrant further testing or precaution, but not a binary verdict (Bender et al., 2021; Butlin et al., 2026).

Accountability remains human and institutional. Records cover version, architecture, training, memory, tools, embodiment, data access, perturbations, rival explanations, uncertainty, and

prohibited uses. No automated score enters legal, commercial, employment, clinical, or welfare administration.

Mimicry cuts both ways. Fluency or apparent distress may cause over-attribution; unfamiliar systems invite careless dismissal. Perturbation, ablation, transfer, and memory interruption clarify causal organization without revealing experience.

Responsible research anticipates possible artificial suffering without manufacturing it. Reversible, low-risk tests, staged review, transparent sharing, and careful public communication are preferred; unnecessary aversive training, uncontrolled replication of candidate suffering, or escalation after warnings triggers stop or redesign (Butlin & Lappas, 2025).

Artificial and synthetic candidates can expand comparative consciousness science, but candidate status remains evidence under a method. It does not confer public authority, personhood, spiritual standing, or entitlement to govern human decisions.

12.8 — Boundary-State Evidence and Review

Boundary-state science needs a record form because memory, theory, technology, and public language all change. A case that looks clear under one marker family may become uncertain under another. A case that seems absent may later show covert response. A striking artificial output may lose force after architecture review.

Boundary-state evidence is unusually vulnerable to theory dependence, measurement error, and public overinterpretation. A versioned dossier should therefore record marker provenance, disagreement, false-positive and false-negative audits, lived-experience input, and the exact language permitted by the current evidence class (Yakhkind et al., 2024; Cogitate Consortium et al., 2025).

The review apparatus converts each boundary claim into an auditable packet: marker target, state and context, theory interpretation, shared dependencies, false-positive and false-negative routes, reviewer roles, version history, permitted analyses, and prohibited uses. This format is intentionally less dramatic than a verdict because its purpose is to expose where confidence comes from and where it fails.

Boundary cases become scientifically useful when they change the dossier rather than merely decorate the argument. Sleep may expose report bias, anesthesia may expose memory-output dissociation, severe injury may expose covert cognition, development may expose transport failure, comparative cases may expose anthropocentric markers, and artificial candidates

may expose performance-based inflation. Each case should end with a specific update or abstention rule.

12.8.1 — State Evidence Dossier

A state evidence dossier is the minimum record required before a boundary-state classification can influence a research claim. It should include the state description, timing, context, markers observed, markers unavailable, theory families used, confounds, uncertainty, and permitted uses. Without a dossier, the classification is only a story (Yakhkind et al., 2024; Giacino et al., 2018).

The dossier should separate evidence about level, content, access, reportability, unity, valence, continuity, and endpoint relevance. A case may support one dimension and not another. A dream report may support experience without supporting ledger content. Covert command following may support awareness without measuring valence. Artificial fluency may support performance without experience.

The dossier must record burden. What did the test require of the subject, family, animal, clinician, or system? Was there distress, fatigue, intrusion, privacy exposure, or clinical disruption? Evidence that shifts burden invisibly is not fair evidence.

The dossier prevents a favorable case from becoming portable authority. A result can influence routing or endpoint work only through a declared chain from state evidence to marker interpretation, classification, permitted inference, and uncertainty. If any link fails, the claim narrows or stops; the vividness of the case cannot repair the missing bridge.

A useful dossier also includes provenance: date, instrument, protocol version, theory matrix version, reviewers, dissent, and revision triggers. Future evidence should update the classification without pretending the old uncertainty never existed.

The state evidence dossier is the chapter's practical legacy. It makes the reasoning behind a classification available for review, preserves the person's context, and gives later evidence a clear route to improve the result.

12.8.2 — Cross-Theory Assessment

Cross-theory assessment asks where consciousness theories converge, diverge, or remain silent. A boundary claim is stronger when several independent theory families support it through different markers. It is weaker when support comes from one theory while others produce plausible rival interpretations (Cogitate Consortium et al., 2025).

The assessment should not average theories into a fake consensus. It should preserve disagreements. A global availability marker may be absent while integration markers remain. A recurrent signal may persist without report. A predictive-control marker may exist in a system whose higher-order markers are weak. These patterns are not nuisances; they are the evidence.

A cross-theory report should identify which claim each theory bears on: conscious level, content, unity, access, valence, continuity, or candidate status. It should avoid saying consciousness evidence when the evidence supports only access or integration under one framework.

Cross-theory assessment must be frozen at the level that matters. The protocol should state which marker families determine primary eligibility, which provide sensitivity analyses, and how disagreements are resolved before ledger outcomes are inspected. Switching theories after seeing the terminal curve is equivalent to changing the endpoint rule after the result.

The assessment should include failure triggers. If theories disagree on a classification that materially affects a ledger or endpoint inference, the result may be theory-dependent rather than supportive. If the study cannot survive declared alternative theories, it should narrow.

Cross-theory assessment should report agreement, disagreement, and blind spots rather than manufacture a consensus score. A marker supported by several theories may deserve more weight, while a disagreement can define the next discriminating observation. The result remains theory-indexed whenever the evidence does.

Cross-state transport can also be examined at the level of shared anatomy without assuming identical mechanisms. A 2025 analysis compared thalamocortical connectivity and local fluctuation across propofol and sevoflurane anesthesia, NREM sleep, minimally conscious states, and unresponsive wakefulness, finding nucleus-specific rather than uniform thalamic involvement. The result supports a state-indexed transport question: which marker relations persist, which change sign or locus, and which fail outside the context in which they were estimated (Lu et al., 2025).

12.8.3 — False-Positive and False-Negative Audit

Every marker can generate false positives and false negatives, and the two errors must remain separate. A false positive may attribute candidate experience or unity where the relevant state is absent; a false negative may exclude a state or being that has experience. Their scientific and ethical consequences differ by use (Green & Swets, 1966; Bodien et al., 2024).

False positives may arise from reflex, artifact, learned performance, simulation, noise, unconscious processing, nonexperiential complexity, or wishful interpretation. False negatives may arise from motor impairment, language or sensory limits, medication, low arousal, poor timing, species or developmental mismatch, inadequate tasks, or observation failure.

The audit should challenge each marker across the boundary states for which it is proposed. Deep sleep, dreaming, anesthesia, minimally conscious states, cognitive motor dissociation, delirium, dementia, infancy, nonhuman animals, organoids, and artificial systems expose different failure modes. A marker demonstrated only in easy adult waking cases is not yet a boundary-state marker.

Consequences must be traced through the full pipeline. A false negative can erase candidate experience and create informative missingness; a false positive can misroute events, inflate apparent coverage, or support unwarranted public claims. Research inclusion, clinical care, welfare protection, public

communication, and ledger inference therefore require different thresholds and invariant protections.

Where possible, the audit should estimate both error directions, calibration, opportunity to detect, and who bears the consequence. It should also test cases chosen to probe the marker's known limitations and plausible failure modes. Completing this audit is required before the tool is used for routing, endpoint interpretation, or public claims.

When neither error rate nor its distribution across populations can be justified, abstention is the defensible output. The report should state the remaining uncertainty and preserve care, dignity, and protective action independently of the classification.

12.8.4 — Expert and Lived-Experience Review

Boundary-state evidence requires expertise across fields. Consciousness theorists, cognitive neuroscientists, sleep researchers, anesthesiologists, neurocritical-care clinicians, dementia specialists, disability scholars, animal-cognition researchers, artificial-intelligence researchers, ethicists, patients, caregivers, and families each detect different failure modes (World Medical Association, 2024; Young et al., 2021).

Expert review is not decoration. It protects against theory capture, clinical naivete, disability erasure, anthropomorphism, machine hype, and public-risk drift. A panel that lacks the relevant expertise may make a classification look authoritative while hiding the burdens it cannot see.

Lived-experience review matters because boundary states are not only laboratory categories. People with dementia, dissociation,

disability, communication impairment, anesthesia awareness, family experience with disorders of consciousness, or caregiving knowledge may notice harms that technical experts miss. Their evidence is not sovereign, but neither is it optional.

Expert and lived-experience review answer different questions. Specialists assess marker validity, clinical context, task design, and statistical uncertainty; participants, survivors, families, and advocates can reveal burdens, access barriers, communication failures, and harmful interpretations that technical review may miss. The final dossier should preserve those roles without treating testimony as a substitute for marker qualification or expertise as authority over meaning.

A review process should record reviewer roles, conflicts, dissent, scope, and the exact claim being reviewed. It should not ask one panel to decide everything from scientific classification to care, law, ethics, and public meaning. Domain routing is part of fairness.

Good review turns disagreement into specific questions that the evidence can answer. It helps the program recognize a strong result when it appears and identify the next useful test when uncertainty remains.

The composition of review changes the evidence. A panel without disability expertise may overread silence; a panel without clinical expertise may miss medication or injury confounds; a panel without philosophy may confuse access with ontology; a panel without patient and family perspective may miss lived burden.

The review design should therefore be treated as part of the method.

Expert review should combine the disciplines that can see different failure modes and include lived-experience or family input where appropriate and safe. Neurology, anesthesia, sleep medicine, developmental science, comparative cognition, AI research, statistics, and ethics do not supply interchangeable votes. Their disagreements should be recorded as reasons and routed into confidence or redesign.

12.8.5 — Versioned Classification

Boundary-state classifications must be versioned because evidence changes. A sleep marker may be revised, an anesthesia model updated, a disorder-of-consciousness method standardized, an animal evidence framework expanded, or an artificial architecture disclosed. Without version history, later confidence can erase earlier uncertainty (Center for Open Science, 2025; Yakhkind et al., 2024).

A versioned classification records the state of evidence at the time: theory matrix, instruments, thresholds, reviewers, confounds, dissent, and permitted uses. If later evidence changes the classification, the record should show the change rather than pretending the earlier classification was always obvious.

Versioning matters for science because old data may not be comparable to new data. A marker update can change inclusion, exclusion, or confidence. A classifier that silently rescored historical cases would create artificial continuity and possibly false support for the Law.

A versioned classification should identify the marker package, theory family, threshold rule, calibration domain, adjudicators, evidence date, and permitted uses. When any component changes, dependent endpoint estimates and public artifacts should be discoverable through a revision graph. Scientific correction then becomes an observable process rather than a quiet replacement of one confident label with another.

The version history should include deprecated uses. A classification that was once used for research inclusion may later be found too weak for ledger inference. The version record states what changed and how it affects later analyses.

A later revision may strengthen, weaken, or reverse a classification. Preserve the earlier evidence, uncertainty, and scoring rules so the record shows how the conclusion developed.

12.8.6 — No Canon Settlement by Case Study

A striking case can expose a false assumption, but it cannot settle the universal bearer definition by itself. Covert command following, a vivid anesthesia report, remarkable animal behavior, or persuasive artificial dialogue may reopen a question and revise a protocol without creating canon (Cogitate Consortium et al., 2025).

Each case should enter a versioned dossier that records marker families, theory mappings, confounds, missing channels, competing explanations, permitted uses, and revision triggers. The dossier makes the case scientifically inspectable without turning it into a symbol larger than the evidence.

Generalization requires replication and theory comparison. Equal-budget rivals, independent sites, prospective predictions, and negative controls should determine whether the case reveals a stable phenomenon, a measurement artifact, a selection effect, or a failure of one theory-specific marker. Replication should include ordinary and ambiguous cases as well as striking examples so case selection does not inflate apparent marker performance.

Novelty and celebrity create predictable bias. Famous patients, visually compelling scans, extraordinary animals, and socially persuasive machines attract more attention, follow-up, and interpretive generosity than ordinary null or ambiguous cases. Review must examine whether publicity is changing the evidence standard. A review log can record whether extra testing, media attention, or model tuning followed notoriety, making attention itself a potential source of bias.

No single case may set treatment ceilings, legal status, species policy, artificial-system rights, or the endpoint rule for the formal hypothesis. Such decisions require their own evidentiary, ethical, and institutional processes, and research uncertainty cannot be converted into public authority (World Medical Association, 2024). Separating research classification from downstream authority also prevents a high-profile case from becoming an informal precedent beyond the evidence it actually supplies.

A canonical change should therefore require a documented synthesis across cases, methods, theories, and failure modes, together with expert disagreement and version history. The prior rule remains visible until the replacement has earned its scope.

Change control should state which cases would have been classified differently under the new rule, providing a direct measure of the practical effect of revision.

Case studies remain valuable because they make hidden assumptions vulnerable. Their proper role is to change the questions and tests, not to end them. The most informative case is often the one that forces a hidden assumption into an explicit hypothesis that can be tested elsewhere.

Chapter 13 — Truth-Level and Protocol Endpoints

Endpoint science begins by bringing several meaningful clocks into focus. A final breath, a clinical declaration, a fading neural signal, and the end of recording describe different events, each with its own purpose and uncertainty. Distinguishing them gives the research program a stronger basis for asking when a qualifying stream ends.

The formal hypothesis defines the truth-level endpoint as the end of the complete strict-pre-endpoint phenomenal domain of the routed stream. The scientific task is to connect that target with independently specified evidence and a reproducible adjudication procedure.

Endpoint science makes the final boundary an independently governed research object. It constructs prospective rules, collects channels distinct from the ledger estimator, estimates timing with uncertainty, handles return and resuscitation, and preserves no-estimate outcomes for sudden or low-signal cases. These requirements make the terminal claim answerable to evidence.

Endpoint science begins under clinical and relational constraints. The dying person is not an endpoint instrument, and family testimony is not an unlimited data source. Comfort, treatment, consent, refusal, and communication therefore enter the protocol as design variables and stop rules. When measurement interferes with those duties, the study yields rather than seeking a more complete ending.

The formal danger is circularity. If the program defines death as the moment when the ledger appears neutral, then neutrality has not been tested; it has been installed. If it chooses among several

candidate endpoint times after seeing which one produces the prettiest terminal pattern, it has not discovered fairness; it has selected a favorable clock. If it lets endpoint adjudicators tune thresholds with knowledge of ledger trajectories, it has allowed the answer to write the question.

This chapter builds the double firewall that prevents that collapse. Constructional independence requires the endpoint system to be built from channels not algebraically identical to the ledger estimator. Inferential independence requires thresholds, windows, analysts, revisions, and public claims to be separated from the outcome being tested. Residual dependence is not denied. It is modeled, reported, and used to limit the claim.

Endpoint research is most vulnerable when absence masquerades as an event. Missing data, device removal, sedation, family narrative, clinical declaration, and legal time can each explain why observation ended without identifying the truth-level endpoint. The status system therefore includes no-data, competing-clock, censored, and No adequate estimate states before terminal classification begins.

The scientific payoff is large. With endpoint discipline, the later protocol can ask whether prospectively specified terminal dynamics outpredict serious rivals without defining the terminal time by the very ledger under test. Without endpoint discipline, every elegant analysis downstream becomes suspect. A model that predicts neutrality at a self-chosen endpoint is not a discovery. It is a mirror.

Three products remain distinct throughout the chapter: the truth-level endpoint named by the formal target, the protocol object used to approximate it, and the evidence-status record describing whether adjudication succeeded. Candidate, estimated, revised, unresolved, censored, and unusable are statuses of the third product. They prevent an operational timestamp from being mistaken for the phenomenal endpoint itself.

Terminal research is written here as care-constrained measurement, not as drama. A last signal, family recollection, or peaceful appearance may be emotionally important without identifying the truth-level endpoint or the terminal ledger. The protocol earns trust by protecting comfort, recording uncertainty, and accepting observation shutdown before it earns any right to interpret a terminal pattern.

With those objects separated, the first falsification protocol can be both humane and interpretable. It knows when observation must stop, which inferences survive uncertainty, and how to report a failed or unusable endpoint without pressure to complete the story.

Endpoint uncertainty also changes the meaning of statistical power. A larger sample cannot repair a target that moves with the observed ledger, an endpoint rule that is revised after outcome inspection, or a terminal window that disappears selectively in the hardest cases. The design must therefore separate uncertainty that can be reduced by more observations from nonidentification that requires a different bridge, a narrower claim, or no estimate. That distinction prevents sample size from becoming a ritual substitute

for conceptual validity (Little & Rubin, 2019; Hernán & Robins, 2020).

A mature endpoint system also preserves counterfactual discipline. With the confirmatory rule fixed, it can show how classification changes under alternative approved persistence windows, marker families, and observation schedules. Sensitivity analysis then reveals fragility rather than silently replacing the primary endpoint.

Finally, the endpoint record must be understandable outside the analysis team. Clinicians, participants, family representatives, reviewers, and later investigators need to know whether a time was observed, estimated, retrospectively revised, censored, or unusable. That vocabulary is part of scientific dignity. It allows a study to remain precise without turning a person's ending into an administrative fact that claims more certainty than the evidence can carry (World Medical Association, 2024; Greer et al., 2023).

13.1 — Truth-Level Endpoint Domain

Endpoint science begins with a refusal of substitution. The formal hypothesis speaks about the end of the complete strict-pre-endpoint phenomenal domain of a routed qualifying stream. That event may correlate with brain death, cardiac death, clinical unresponsiveness, device silence, or the end of social interaction, but it is not identical to any one of them by definition. If the book lets a convenient proxy become the endpoint, the terminal test loses its target before it begins.

Modern death-determination standards are carefully operationalized clinical constructs, but the truth-level endpoint defined in Chapter 5 is a different scientific object. Brain-death criteria, circulatory criteria, device shutdown, legal time, and last observation can constrain the domain only through declared bridges; none is the latent endpoint by definition (Greer et al., 2020; Greer et al., 2023).

The word conceptual does not mean imaginary. It means the target is defined at the level of the formal object and then approached through evidence. Physics, medicine, and law all use constructs that require operational rules without being exhausted by one instrument. Here the construct is harder because its bearer is experience itself: the relevant stream may be inaccessible, low signal, intermittently expressed, or retrospectively revised by return.

The truth-level endpoint denotes the irreversible end of the relevant stream. A protocol estimates that endpoint from specified observations and can return an interval, a revised classification,

or an unresolved result. Separating the target from its estimate keeps each outcome interpretable.

A humane scientific program also avoids turning endpoint language into a story about a person's worth or meaning. A truth-level endpoint is not a moral verdict, spiritual judgment, clinical instruction, or public identity. It is a research construct whose use is narrow enough to be checked.

13.1.1 — The Truth-Level End of the Stream

The truth-level endpoint is the event at which the complete strict-pre-endpoint phenomenal domain of the routed qualifying stream ends under the declared branch. In notation, later chapters may call the latent endpoint T star, but the notation is not the event. The event is neither the cessation of a body part, a social relationship, nor a recording device, and branches involving return, continued experience, endpoint atoms, or endpoint-inclusive value must be declared rather than absorbed into one generic ending (Greer et al., 2023).

The stream is not defined by public display. It is the bearer of felt experience as received from the formal root. A person can be silent, paralyzed, sedated, dreaming, delirious, or unable to report while still leaving open the possibility of experience. The endpoint cannot therefore be set by the last report or last movement.

The endpoint is also not a reward for the model. A terminal study may estimate a candidate time, fail to estimate it, revise it, or classify it as unusable. All of those outcomes are compatible with

a coherent formal target. None allows a researcher to choose a time because it makes a ledger curve look orderly.

Irreversibility matters because a temporary interruption is not a truth-level endpoint. Sleep, anesthesia, coma, and some resuscitation contexts can include long intervals of missing report or weak signals. If later evidence supports continuity or return, the earlier candidate crossing must be demoted rather than treated as a confirmed endpoint.

Endpoint uncertainty belongs in the model rather than in the person. A protocol that equates no response with no experience transfers uncertainty into a false finality; a protocol that represents latent endpoint time, observation failure, and competing explanations carries the uncertainty itself. That difference is the basis of every later no-estimate and revision rule.

For analysis, this subsection forces a notation discipline. The conceptual endpoint should be stored as a latent object, not as a field copied from a monitor, a hospital record, or a family timeline. A case file may contain many terminal times, but only a declared endpoint model can say which time belongs to which scientific use. That distinction lets later models carry uncertainty without pretending that the formal endpoint has become visible.

13.1.2 — Biological and Clinical Proxies

Somatic death matters, but it does not by itself settle the formal endpoint. Biological systems fail in cascades: circulation, respiration, metabolism, brain perfusion, cellular integrity, and tissue viability do not collapse as a single mathematical instant. Clinical medicine must act under operational rules; scientific

analysis must not mistake those rules for the formal end of experience (Greer et al., 2020; Greer et al., 2023).

Brain death or death by neurologic criteria is closer to the relevant terrain than cardiac or administrative time of death, but it remains a clinical determination with declared prerequisites, exclusions, examinations, and jurisdictional governance. It can guide care, law, and transplantation policy. It cannot be imported as a solved theory of the final unit of conscious-stream identity.

Somatic and neurological events enter the endpoint dossier as clocks with declared relations to the phenomenal target. They can bound plausible intervals, provide competing-event information, trigger persistence checks, or identify observation shutdown. Routine recording and clinical decisiveness for other purposes do not make them the truth-level endpoint (Greer et al., 2023; Dhanani et al., 2021).

Direct cortical recordings show why this distinction is productive. In a nine-patient observational study during independently decided withdrawal of life-sustaining treatment, electrical silence usually preceded terminal spreading depolarization; the reported interval was 13–266 seconds (Dreier et al., 2018). Different physiological transitions therefore provide different time anchors. Carefully governed existing records can help refine this multi-clock model, including the uncertainty between a measured transition and the end of a qualifying stream.

Biological condition and stream evidence can vary in different ways. A biologically living body may present uncertain stream organization; a person near physiological collapse may retain

experience, covert awareness, or the capacity to recover. Each case calls for classification using the relevant marker and continuity evidence.

The strongest rule is simple: somatic evidence may narrow, challenge, or confirm endpoint classification only under declared criteria. If the endpoint system simply replaces truth-level endpoint with body death, it has changed the claim object. That change belongs to formal change control, not to an analyst's convenience.

13.1.3 — Legal and Administrative Time

Legal death is necessary for hospitals, families, states, records, estates, burial, organ donation, and public order. It is not designed to decide the formal endpoint of a consciousness hypothesis. Administrative categories must be stable enough to govern life with others; scientific endpoint categories must be narrow enough to test a claim (Greer et al., 2023; World Medical Association, 2024).

Legal and clinical systems continue to fulfill their own duties. The research record identifies the purpose and resolution of each certificate, registry field, or institutional timestamp and evaluates how it relates to the endpoint being studied.

Legal concepts can still matter as context. They determine when certain data streams stop, when consent authority changes, when records are closed, and when families encounter institutional language. They may create observation patterns that a study must model. They do not identify truth-level endpoint by legal authority.

Administrative timestamps invite silent alignment because they are available and standardized. A legal or registry time may sharpen a plot while remaining biologically and phenomenally coarse. It should therefore enter as context or a separately justified analysis clock, with its error relation to the truth-level endpoint estimated rather than assumed.

Uniformity across sites does not make an administrative time truth-relevant. A clinical declaration may be necessary for care and a registry timestamp necessary for reporting, yet both can be too coarse for terminal inference. The protocol should preserve their operational purpose, estimate their lag and uncertainty where possible, and prevent them from becoming the default endpoint of the ledger merely because they are complete.

13.1.4 — Last Observation and Observation Shutdown

The last observation can occur for several reasons that require separate endpoint adjudication. Sensors stop. A family declines further monitoring. A device is removed for comfort. A patient is transferred. A battery fails. A hospital workflow changes. None of these events proves that experience ended (Little & Rubin, 2019; Greer et al., 2023).

Last observation is especially dangerous because it often aligns with severity. The sickest moments are also the moments when monitoring may become least tolerable, least reliable, or least ethically appropriate. If the study treats observation cessation as the endpoint, it can convert burden, device fragility, or care priority into apparent terminal structure.

A valid endpoint system must therefore carry an observation clock distinct from the endpoint clock. The observation clock marks scheduled measures, failed measures, refused measures, removed devices, care interruptions, and unavailable data. It is a record of access to evidence, not a record of mind.

When a study loses the terminal window, its endpoint result may be absence of usable evidence. Reporting this explicitly keeps the observation record accurate and the endpoint question open for independently justified evidence.

The burden question is direct: who pays for the researcher's need for a final row of data? In a humane program, the dying person never pays through extra discomfort. The data structure pays by carrying missingness and by allowing the analysis to abstain.

The last observation may become relevant in sensitivity analysis, last-contact reporting, and missingness modeling. It may not become truth-level endpoint.

13.1.5 — Strict-Pre-Endpoint Domain and Posthumous Records

A person can remain loved, remembered, recorded, quoted, imitated, modeled, and socially present after death. Those forms of presence matter to families and cultures and belong to the study of memory, identity, and relationships (Parfit, 1984).

Personal archives give families and researchers access to memories, relationships, language, and patterns of response. A diary, video, social media profile, voice model, transcript, digital twin, or statistical simulation can preserve different aspects of that

record; its scientific use depends on the information it contains and how it was produced.

Social memory belongs to grief, history, relationships, and meaning. The formal ledger includes experience within the qualifying stream's strict-pre-endpoint domain; later archives inform retrospective questions about that earlier experience.

The endpoint file distinguishes evidence about experience before the endpoint from representations created afterward. A final message, photograph, recording, or diary may inform retrospective classification when its timing, provenance, consent, and measurement properties are established. Later memories, interpretations, and synthetic representations retain their own source and creation dates.

13.1.6 — Endpoint Uncertainty and Nonidentification

Endpoint uncertainty and endpoint nonidentification are different results. Uncertainty describes an identified target estimated imprecisely under a declared model. Nonidentification means that materially different truth-level endpoints remain compatible with the observed record because the bridge, clock, routing, or observation process cannot distinguish them (Manski, 2003).

Clinical declarations, physiological events, legal times, last observations, device shutdown, and family reports can constrain the endpoint without being identical to it. Each proxy relates differently to the complete strict-pre-endpoint phenomenal domain and fails under different conditions.

When several endpoints remain admissible, the protocol should report an identified set or branch-specific results rather than select

the point that makes the ledger easiest to analyze. A defensible interval can be more informative than a falsely precise timestamp because it preserves both the evidence and the unresolved bridge.

Construction and adjudication must remain independent of the tested residual. Choosing an ending because the ledger is near zero, or excluding cases whose residual is inconvenient, manufactures closure rather than tests it.

Sensitivity analysis should vary only prospectively approved endpoint models, persistence windows, proxy assumptions, and observation rules. Public status language should distinguish observed, estimated, interval-identified, revised, censored, unresolved, and unusable endpoints, while naming whether sparse data, conflicting clocks, routing ambiguity, or structural nonidentification caused the limitation.

A no-estimate endpoint is therefore a substantive scientific result. It identifies the missing bridge that prevents a terminal claim, blocks later mathematics from claiming more certainty than the domain permits, and directs the next design toward the actual source of nonidentification.

13.2 — Protocol Endpoint Objects and Adjudication Predicates

Endpoint work becomes safe only when it stops using one word for several different acts. A candidate crossing is not an estimated endpoint. An estimated endpoint is not a confirmed endpoint. A confirmed endpoint can still be revised when later evidence shows return or classification error. These distinctions are not clerical. They are the scientific grammar that prevents a terminal analysis from overclaiming.

Endpoint adjudication requires a protocol object that is observable, prospective, and independent of the tested ledger. Clinical-trial methodology and causal inference support the endpoint discipline developed here: endpoint construction, estimation, and confirmation must be separable enough that one stage cannot quietly encode the desired result of another (Hernán & Robins, 2020).

The section's status ladder also gives negative and unusable evidence a place to stand. If a threshold crossing is followed by return, it is a false crossing. If the data are too poor to classify, the endpoint is unresolved or unusable.

The status system records endpoint ambiguity explicitly. It identifies which evidence supports a classification, what remains unresolved, and which additional observations could improve the estimate.

The endpoint-status ladder determines analysis eligibility. Qualified prospective endpoints may enter the primary analysis; retrospectively qualified endpoints may support secondary

reconstruction; unresolved or censored endpoints restrict inference; and failed endpoint gates make the affected evidence Unusable. An endpoint that cannot be identified or validly bounded despite passed gates yields No adequate estimate. The distinctions are fixed before analysis so endpoint quality cannot be renegotiated after model fit is known.

Table 13.1. Protocol endpoint status and interpretation rules.

Endpoint status	What it means	Interpretation rule
Candidate crossing	A protocol-fixed threshold event has occurred in the endpoint system.	Evaluate persistence and the remaining adjudication criteria.
Estimated endpoint	A model assigns an event-time distribution or interval under declared assumptions.	Retain the estimated interval or distribution and its assumptions.
Confirmed endpoint	A persistence, review, or evidence rule has been satisfied under the protocol.	Reopen classification when later return or identified error requires it.
Revised endpoint	Later evidence changes the versioned classification.	Preserve the original primary analysis and label retrospective revisions.
Unresolved or unusable	The endpoint cannot be classified with adequate quality.	Report the missing classification and its consequences for available analyses.

13.2.1 — Candidate Protocol Endpoint Object

A candidate crossing is the first event at which a prospectively specified endpoint rule is triggered. It may involve a unity-proxy threshold, a loss of responsiveness pattern, a neurophysiological marker, a quality-controlled absence of state evidence, or a composite event declared before the study. Its dignity lies in being prospective, not in being final (Hernán & Robins, 2020).

A candidate crossing is a research alarm, not a verdict about the person or the truth-level endpoint. It marks the moment at which a prospectively specified threshold has been met and activates the protocol's confirmation, return, and revision rules. The ledger remains unopened to the endpoint adjudicators until that process is locked.

One workable formal form is a threshold time conditioned on quality and persistence prerequisites. The variables belong to endpoint construction, not to the ledger estimator. A low-quality crossing is not a candidate endpoint; it is an artifact flag.

The candidate crossing carries its own context: medication, sedation, respiratory intervention, delirium, sensor quality, care procedure, and observation conditions. Without that context, a threshold crossing is only a number with insufficient provenance.

Denominator discipline reports how often candidate crossings occur, fail confirmation, precede return, or become uninterpretable because evidence quality fails. Publishing only successful confirmations hides the false-crossing burden.

The benefit is procedural honesty. A frightening threshold becomes less dangerous when the study treats it as a provisional

signal with known error rates, review steps, and reversal rules rather than a final statement about the person.

Candidate crossings are also useful for calibration because they expose the endpoint system before it receives permission to conclude. Candidate reversals and missed endings identify different error burdens, but their counts alone do not establish sensitivity or specificity. Sensitivity requires the proportion of independently adjudicated endings detected by the rule; specificity requires the proportion of independently adjudicated non-endings correctly left unflagged, with reference rules, time windows, and denominators declared. Those tradeoffs belong in the endpoint report before any terminal ledger claim is interpreted.

13.2.2 — Persistence and Confirmation Window

A persistence window is the rule that prevents an instantaneous fluctuation from becoming an endpoint. In clinical and neurophysiological data, state markers can flicker because of artifact, medication timing, arousal fluctuation, hypoperfusion, electrode contact, motion, care procedures, or sampling cadence. A candidate crossing therefore needs a declared pattern over time before confirmation is even considered (Cox, 1972; Andersen et al., 1993).

The window is not always a fixed number of minutes. It may be a duration, a repeated-state requirement, a sequence of low-signal observations with quality checks, an absence of return under clinically meaningful conditions, or a combination of channels.

The choice must be justified by endpoint science, not by convenience or by the expected ledger shape.

Persistence also has ethical limits. A protocol may not extend monitoring because a researcher wants a cleaner persistence window if the person's comfort, preferences, or care needs require stopping. When care interrupts the window, the correct status may be unresolved rather than confirmed.

The persistence window is calibrated before outcome inspection. Stretching or shortening it after viewing the terminal ledger turns endpoint confirmation into model rescue, and the same crossing can look very different under five-minute, one-hour, or last-exit rules.

A good endpoint report states the window, its rationale, its quality conditions, its sensitivity analysis, and the number of cases whose status changes under plausible alternatives. This makes persistence a scientific object rather than a hidden judgment.

The window exists to protect the person and the claim at once. It protects the person from a false finality. It protects the claim from confirmation by noise.

13.2.3 — Adjudication Predicate and Estimated Endpoint

An estimated endpoint is not a time point handed down by the data. It is a probability distribution, interval, or classification result produced by a model under assumptions. It depends on observations, state definitions, priors or likelihoods where used, censoring rules, quality thresholds, and revision procedures (Hernán & Robins, 2020; Fleming & Harrington, 2005).

The word estimated should remain visible in public and technical language. If the chapter writes simply T after a model has produced an estimate, readers may hear certainty. A cleaner convention distinguishes the latent conceptual endpoint from the estimated endpoint and keeps the uncertainty attached to every downstream analysis.

Event-time uncertainty can be narrow, wide, multimodal, or formally unidentifiable. In a low-signal terminal window, several candidate intervals may be plausible. In a sudden-death case, no meaningful interval may exist within the observed data. Forcing a point estimate in those cases manufactures information.

Estimated endpoint uncertainty is propagated into sensitivity analysis rather than collapsed too soon. A terminal signature may be robust to plausible jitter, fragile under small changes, or entirely dependent on one time choice, and those outcomes carry different claim status.

The estimation model must declare what information it uses. A prospective filter uses only information available up to the time being estimated. A retrospective smoother uses later evidence. A retrospectively adjudicated onset may be clinically useful, but it may not be treated as if it had been available prospectively.

Each endpoint estimate states the observed channels, adjudication rule, uncertainty interval, and assumptions linking the protocol object to the latent target. A point estimate may simplify a table, but the downstream analysis still exposes how conclusions change across plausible endpoint times.

13.2.4 — Confirmed Endpoint Status and Performance

A confirmed endpoint is a candidate event that has passed declared confirmation rules. Confirmation may require persistence, independent review, absence of return under relevant conditions, quality checks, clinical context, and a documented decision path. It is stronger than a candidate crossing, but it remains a research classification (World Medical Association, 2024; Center for Open Science, 2025).

The word confirmed can mislead if it is allowed to sound metaphysical. Here it means confirmed under the protocol, not confirmed by direct access to the end of experience. The confirmation class should always travel with the version of the endpoint rule, the evidence window, and the known limitations.

Confirmed endpoints are eligible for primary analysis only if the confirmation procedure was prospectively specified, constructionally separated, and inferentially protected. A confirmed endpoint obtained after outcome-dependent tuning or ledger-aware review may be useful for exploratory learning, but it is not confirmatory evidence.

The confirmation process should include false-positive and false-negative audits. How often do candidate crossings fail? How often do low-signal cases later show return? How often does sedation mimic absence? How often does device failure make an endpoint look clean? Confirmation is a performance claim, not a label to be trusted without calibration.

A confirmed endpoint can also become unconfirmed. Later return, new channel evidence, or error discovery should not be hidden to

preserve a tidy result. Revision is a strength when the version history is preserved.

Endpoint confirmation is a qualified research status whose strength comes from explicit criteria, reproducible adjudication, and a clear revision route. Its scope remains the protocol's classification of evidence.

Confirmation should also specify what kind of evidence was absent. Absence after high-quality observation is different from absence after device removal, heavy sedation, emergency intervention, or poor contact. A confirmed endpoint built from strong absence should not be pooled casually with a confirmed endpoint built from fragile absence. Confirmation classes may need grades, and those grades should travel into sensitivity analysis.

13.2.5 — Revision, Versioning, and Audit

Revision is a necessary feature of endpoint science in low-observation, high-stakes states. If later evidence shows that a candidate or confirmed endpoint was wrong, the record must change. Preserve the record of what was known when the primary analysis was run (Särkkä, 2013).

A revision can arise from return, resuscitation, recovered communication, post hoc device-quality review, clinical context, later family or clinician evidence, or discovery that an endpoint channel was compromised. Each source has a different evidentiary weight and different risk of bias.

The chapter distinguishes revised classification from retrospective smoothing. A revised endpoint may improve the

final case record. It does not automatically authorize the primary analysis to behave as if the later evidence had been available at the time. The original filtered analysis and the revised secondary analysis both need to be visible.

Versioned records preserve that difference. A good record says: under rule version one and information available at time t , the endpoint status was candidate; after later return, the status became false crossing; after expert review, the case was routed to error memory. That history is more scientific than a single overwritten label.

Reclassification can improve the endpoint method. When new evidence changes the interpretation of a case, researchers update the classifier, stress-test the protocol, and explain what the case teaches about endpoint adjudication.

Revision should make the record more accurate without rewriting the history of what was known. The original classification, later evidence, reason for change, and affected analyses all remain visible. This distinguishes scientific correction from retrospective smoothing and allows false-ending rates to improve the next protocol rather than disappear from view.

Revision should be asymmetric in public interpretation. A case revised from confirmed to false crossing is not a public scandal if the earlier rule was followed; it is a data point about error. A case revised from unknown to confirmed may strengthen feasibility but cannot retroactively create prospective knowledge. Both revisions become part of the same scientific memory: the endpoint system is learning under uncertainty.

13.2.6 — No Endpoint Defined by the Tested Residual

No favorable endpoint selection is the rule that keeps terminal analysis from becoming a proof machine. When several candidate endpoints exist, the study may not choose the one that makes the residual most neutral, the curve most elegant, or the rival least competitive. The confirmatory endpoint object must be fixed before the ledger result is inspected (Hernán & Robins, 2020).

The object includes more than a threshold. Persistence duration, last-exit rules, allowable gaps, smoothing, censoring, reentry, competing events, and revision handling can all change terminal alignment. The protocol should freeze the primary construction, publish a bounded sensitivity family, and report whether the signature survives rather than substituting the most favorable definition.

Analyst separation and technical controls strengthen the rule. Endpoint adjudicators should not see ledger trajectories where feasible, ledger analysts should not tune endpoint thresholds, and endpoint algorithms, forms, training data, permissions, and deviations should be versioned before unblinding. The primary endpoint file should be reproducible without access to the tested residual.

The firewall has two distinct layers. Constructional separation asks whether the endpoint object is built from information that is not algebraically identical to the ledger estimator. Inferential separation asks whether analysts, thresholds, model selection, sensitivity choices, and public verdicts are protected from the outcome being tested.

Both layers permit explicitly modeled dependence. Illness severity, medication, frailty, comfort care, physiology, and sensor quality may influence both endpoint channels and ledger estimates. The analysis names these shared causes, models their effects, and carries them through sensitivity analyses so that conclusions reflect the resulting uncertainty.

The rule also protects adverse results. A rival victory, false crossing, unusable endpoint, or unfavorable residual must remain in the record when the frozen design says so. Reworking the endpoint until PTVC survives would convert the study from adjudication into advocacy.

The firewall should face direct audit. Synthetic ledger trajectories, perturbed endpoint labels, permission tests, endpoint jitter, and reproducibility checks can reveal whether information that should be unavailable changes the construction. A failed audit requires a redesigned workflow, a new blinded package, or a narrower estimand, not stronger rhetoric.

A defensible endpoint is therefore visible in the data flow: independent features where possible, frozen rules, cross-fitting, shared-confound analysis, dispute logs, public revision rules, and a reproducible primary file. The endpoint and ledger may share declared causes, but they cannot be the same algebraic construction or be tuned to rescue one another.

$$E_t \equiv f_E(Z_{\leq t}^E, \Gamma_E), \hat{L}_t = f_L(Z_{\leq t}^L, \Gamma_L), Z^E \not\equiv Z^L$$

13.2.7 — Constructional Independence

Constructional separation means the primary endpoint is not constructed from the same algebra that defines the measured

ledger. If candidate momentary-valence or interval-contribution estimates quantify affective burden, those same quantities cannot define the truth-level endpoint. Otherwise the terminal claim would test a condition it helped create (Hernán & Robins, 2020).

The separation does not require different organs or wholly unrelated sensors. Consciousness, physiology, and affect share biology. The rule is narrower and stricter: the endpoint channels must not be algebraically identical to the ledger estimator, and any overlap in raw acquisition, preprocessing, or latent variables must be declared.

A unity-proxy endpoint may use perturbational response, connectivity, command-following evidence, arousal context, or other consciousness-related markers. A measured ledger may use physiological valence estimators. When a raw channel contributes to both systems, the primary protocol should either remove the overlap, split the feature families, or route the case to residual-dependence modeling.

Constructional separation is easiest to understand through failure. If a falling affect score is treated as evidence that the mind is ending, and the same falling score is then tested for terminal neutrality, the endpoint and outcome have become one object. No amount of statistical sophistication repairs that circularity.

The separation should be visible in the protocol and the manuscript. Readers should be able to identify endpoint features, ledger features, overlap exclusions, channel provenance, and the rule that prevents outcome-driven endpoint construction.

The firewall reduces burden by keeping the question honest. The person's ending is not shaped to fit the ledger. The ledger is evaluated at an endpoint constructed independently enough to put the claim at risk.

Constructional separation should be audited at the feature level. A physiological channel can be safe in one role and unsafe in another if preprocessing turns it into a ledger proxy. For example, a signal used to classify unity may become entangled with valence estimation if affective labels trained the same representation. The firewall therefore asks not only whether variables have different names, but whether their construction is algebraically and statistically distinct.

13.2.8 — Inferential Independence

Inferential separation prevents the people and procedures around the endpoint from leaking terminal results into endpoint definition. Even if endpoint channels are constructionally distinct, analysts can contaminate the endpoint by tuning thresholds after viewing ledger outcomes, selecting smoothing rules that improve the terminal pattern, or excluding hard cases after seeing results (Särkkä, 2013; Fleming & Harrington, 2005).

The first defense is preregistration. Thresholds, windows, state definitions, confirmation rules, revision procedures, and primary-versus-secondary analysis roles should be declared before the outcome is inspected. Exploratory changes are allowed only when labeled as such.

The second defense is role separation. Where feasible, endpoint adjudicators should be blinded to ledger trajectories, and ledger

analysts should receive endpoint classifications only after endpoint adjudication is locked. Small teams may not always make full separation possible, but the limitation must be acknowledged rather than hidden.

Cross-fitting or sample splitting supplies a third defense. Endpoint tuning occurs in one partition and the resulting rule is applied to held-out cases. This does not make endpoint estimation perfect; it prevents the same observations from selecting the endpoint and then appearing to confirm the terminal claim.

Inferential separation also applies to narrative interpretation. A striking terminal curve should not be allowed to influence case review, family communication, or public artifact selection. The most persuasive picture is often the most dangerous one if its selection rule was not fixed in advance.

Inferential independence is lost when endpoint thresholds, windows, exclusions, or features are selected because they improve the terminal ledger result. The endpoint model may learn from training data, but the confirmatory endpoint for a case must be frozen before the tested residual is revealed. Cross-fitting, held-out adjudication, or separate teams are methods for preserving that order.

Information separation can fail during feature and model selection even when endpoint and ledger teams begin apart. Thresholds, windows, exclusions, and preprocessing choices may be tuned toward an attractive terminal result. Locked scripts, restricted data access, analyst blinding where feasible, and a complete tuning log

make independence a property of workflow rather than a statement of intention.

13.2.9 — Shared-Confound Analysis

Shared confounds are the reason the firewall cannot stop at separate variables. Disease severity, medication, frailty, oxygenation, inflammation, pain control, sensor contact, care procedures, and family or clinician decisions can affect both endpoint markers and ledger estimates. A clean channel list does not make those causes disappear (Little & Rubin, 2019; Hernán & Robins, 2020).

A shared-confound model begins by identifying variables that plausibly influence both systems. Some are measured directly, such as medication timing or device removal. Some are latent, such as global decline, delirium burden, or unobserved discomfort. The model should state which are measured, proxied, or unmeasured.

Shared causes belong in the primary causal diagram and sensitivity plan. Sedation, frailty, perfusion, illness severity, family availability, device removal, and observation quality can affect both endpoint proxies and ledger estimates. A pattern that disappears across plausible shared-confound models has a different status from one that survives; a pattern dominated by unresolved dependence is model-ambiguous, not quietly positive.

Confounding can also create false endpoint evidence. For example, a care procedure may reduce signal quality and physiological variability simultaneously. Without context, the

study may read this as low unity plus variance compression. With context, it becomes a rival explanation.

Sensitivity analysis should vary confound assumptions, especially when variables are unmeasured or weakly measured. The public verdict should state whether conclusions are robust, fragile, or unusable under plausible shared-confound structures.

13.2.10 — Independent Teams and Blinding

Independent teams and blinding turn the firewall from intention into process. Endpoint adjudicators work without the terminal ledger whenever possible, ledger analysts cannot change endpoint labels, and review committees receive only the information required for their task (World Medical Association, 2024; Center for Open Science, 2025).

The level of independence can be scaled. A small feasibility study may use documented role separation and audit logs. A multisite confirmatory study should use separate endpoint and ledger teams, locked data packages, blinded adjudication, and prespecified dispute resolution. The stronger the claim, the stronger the separation must become.

Blinding does not mean ignorance of clinical context. Endpoint adjudicators may need medication, imaging, treatment, arousal, and device-quality information. What they do not need is the measured ledger trajectory whose terminal behavior the study is trying to evaluate.

Audit trails matter because blinding can fail. Who saw which files, when thresholds changed, when cases were excluded, and when public figures were selected should be recorded. The goal

is not suspicion for its own sake. It is to make rescue moves visible.

Independent teams also protect participants and families. They reduce the chance that a researcher invested in the hypothesis interprets an ambiguous ending as favorable. They keep clinical and family-facing communication from being shaped by a terminal result.

The chapter treats independence as a burden carried by institutions, not by dying participants. The study must build the separation; the person should not be asked to trust a theory's goodwill.

Role-specific language preserves the firewall. Endpoint adjudicators give endpoint reasons, ledger analysts give measurement reasons, and any unavoidable role overlap is offset by masked datasets, staged releases, decision logs, and independent review. The purpose is protection from unconscious alignment rather than bureaucracy.

13.2.11 — Prospective Registration and Locked Adjudication

Prospective specification is the study's memory before outcome inspection. Use §6.7's terminology for protocol-fixed, prospectively specified, and publicly preregistered. The endpoint record names endpoint channels, thresholds, persistence windows, exclusion and abstention rules, confirmation procedures, revision handling, primary and secondary analyses, and public-reporting categories (Chambers, 2013; Center for Open Science, 2025).

The protocol-fixed record includes both success and loss conditions: what qualifies an endpoint, what produces unresolved or unusable status, what triggers revision, and what follows when endpoint uncertainty exceeds the admissible range for the primary estimand.

It also names the sensitivity analyses that stress the endpoint, including threshold jitter, alternative clocks, last-observation alignment, shared-confound adjustment, filter-versus-smoother comparison, and false-crossing treatment. Sensitivity is planned before the result becomes inconvenient.

Post-specification changes can be legitimate when safety, ethics, or scientific error require them, but each change is versioned and treated as a changed endpoint rule rather than a quiet reinterpretation of the same test.

A public registry record helps readers distinguish a surprising result from a tuned result. A terminal pattern observed under a frozen endpoint rule carries more evidentiary weight than one found after several endpoint revisions, all else equal.

The same prospective record provides an ethical safeguard. It prevents a person's endpoint classification from being revised after the fact merely because another label would make the terminal result more attractive.

The record also covers revision. A protocol that fixes the threshold but improvises return handling has not protected the endpoint; false crossings, recovery, device cessation, low quality, sudden death, withdrawal, and competing somatic events receive declared classification consequences before outcomes are known.

13.2.12 — Residual Dependence and Sensitivity Analysis

Biological, clinical, and observational interconnection leaves residual dependence even behind strong firewalls. Endpoint channels may share latent causes with valence estimates despite separate teams and code. Model that dependence explicitly rather than assume perfect independence (Little & Rubin, 2019; Manski, 2003).

Shared random effects, latent severity, site and device effects, treatment processes, observation models, and sensitivity parameters can represent that dependence. The scientific question is how much shared structure would be required to explain the apparent terminal signature without PTVC, and which parameters are informed by data rather than assumption.

Verdicts should respond to the sensitivity result. If modest dependence removes the pattern, the claim is fragile; if the result survives a wide, justified range, it becomes more interesting; if the dependence cannot be bounded, the output remains exploratory or No adequate estimate. Numerical strength cannot compensate for unidentified dependence.

Reports and figures should disclose the firewall state and the remaining shared channels. Independent, cross-fitted, jointly modeled, retrospectively adjudicated, and merely differently labeled pipelines are not equivalent. The diagnostic, uncertainty, and verdict consequence should be stated beside the result rather than hidden in implementation notes.

The firewall should itself undergo adversarial stress testing. A test team can inject synthetic residual trajectories, hide or reveal

selected covariates, perturb endpoint labels, and inspect whether adjudicators or scripts change decisions. Sensitivity of the endpoint to prohibited information is a measurable failure and should trigger redesign or narrowing.

A dependence budget makes the remaining risk explicit. It records unavoidable shared channels, the magnitude of an unmeasured common cause required to erase the result, and the conclusions that survive alternative dependence structures. A result that requires implausibly perfect separation should be labeled fragile even when the primary graph is striking (Hernán & Robins, 2020; Manski, 2003).

The endpoint dossier should close with a reproducibility packet containing frozen code, rule version, adjudicator and ledger information sets, deviations, dispute logs, and the mapping of every case to qualified, revised, unresolved, censored, or unusable status. Another team should be able to reconstruct the endpoint without seeing the tested residual; otherwise the object is not ready to support the claim.

13.3 — Stream Routing, Origin, Maximality, and Junctions

An endpoint claim is incomplete until the study can say which stream is ending. Chapter 11 developed routing as a bearer problem; here the same discipline is made endpoint-specific. Origin, maximality, split points, fusion points, overlap, and unresolved junctions must be declared before terminal evidence is inspected, because a favorable ending can otherwise be manufactured by changing the object after the path is known (Bayne, 2010).

The endpoint dossier therefore contains a routing graph as well as a time line. Nodes represent candidate stream intervals or junction states, edges represent prospectively permitted continuations, and each phenomenal contribution is assigned at most once. The graph does not settle metaphysical identity. It makes ownership assumptions visible enough to audit, vary, and reject.

13.3.1 — Stream Origin and Initial Ownership

A stream origin is the earliest boundary from which the protocol treats a candidate trajectory as one routed phenomenal history. It is not automatically conception, birth, first memory, first report, registration, hospital admission, or device connection. Those are observable anchors that may constrain the origin, but none is identical to the truth-level beginning of conscious continuity (Lagercrantz & Changeux, 2009).

Because early experience is often left-censored, origin uncertainty should be represented rather than erased. A study may define a reference-class-specific eligibility origin, a first observed

interval, and an identified set of plausible truth origins. Reporting all three prevents the administrative start of observation from masquerading as the beginning of the stream.

Initial ownership also requires a no-duplication rule. If two candidate routes both claim the same early interval, the protocol must either adjudicate between them, assign explicit mutually exclusive probabilities, or leave the interval unresolved. It may not count the same phenomenal contribution in two ledgers because two biographies or data systems refer to it.

Developmental and clinical evidence can narrow origin classes without becoming an oracle. Neurodevelopmental organization, behavioral repertoire, recurrent access, physiological integration, and later continuity may all bear on plausibility, but their evidential force depends on the reference class and the theory family used to interpret them (Lagercrantz & Changeux, 2009; Dehaene et al., 2017).

The origin rule must be frozen before terminal analysis. Retrospectively shifting the start until positive and negative magnitudes balance would be the temporal counterpart of defining the endpoint by zero. A defensible study therefore prospectively specifies origin objects, alternative origins, left-censoring assumptions, and the verdict consequences of origin failure.

The scientific output may be an interval of admissible origins rather than one date. That result is not evasive. It states exactly how much of the trajectory is identified and prevents unsupported early-life extrapolation from entering a terminal claim.

13.3.2 — Maximality and Continuation

A qualifying stream is maximal when it cannot be extended earlier or later under the declared unity and routing rules without joining evidence that the protocol does not license as one stream. Maximality is therefore a property of the candidate object under specified criteria, not a narrative judgment that the life feels complete (Bayne & Chalmers, 2003; Casali et al., 2013).

Continuation evidence is plural. Causal dependence, preserved access architecture, learning, memory, dynamical integration, recovery trajectories, and intervention response can support an edge between intervals. No single marker is necessary in every population, and no marker becomes sufficient merely because it is convenient to collect (Bayne & Chalmers, 2003; Casali et al., 2013).

The protocol should compare at least three continuation models where the case permits: uninterrupted continuation, pause with rebinding, and termination followed by a new stream. Model comparison must use the same observations and loss rules. Otherwise the preferred continuation model may win by receiving richer evidence than its rivals.

Maximality also has a stopping discipline. Once the declared evidence no longer supports extension, the protocol must not continue the stream through an archive, social memory, legal status, a successor system, or a later simulation. Those objects may preserve information and meaning, but they do not inherit phenomenal ownership by continuity of record.

Sensitivity analysis should vary unity thresholds, permitted gaps, observation models, and rebinding priors. A terminal conclusion that changes under minor defensible variations is not stable enough for a strong verdict. The appropriate output may be Reference-class narrowed, Model-ambiguous, or No adequate estimate.

Maximality defines the temporal extent of one candidate stream under the declared unity and routing criteria. The analysis states the evidence supporting each boundary and preserves alternatives where that extent remains unresolved.

13.3.3 — Split Points and Descendant Streams

A split point is a junction at which one previously coordinated candidate stream may no longer be representable as one maximal stream under the declared unity rule. Split-brain research, dissociation, severe focal injury, and engineered partition are relevant because they show that organismic unity and phenomenal unity need not be treated as synonyms (Gazzaniga, 2000; Pinto et al., 2017).

The conservative formal rule is parent termination with descendant initialization. Once a split is adjudicated, the parent ledger ends at the junction and each descendant begins with its own origin state. No descendant inherits an obligation to complete the parent, and no later total may be used to trade valence across descendants.

Evidence for a split should exceed ordinary functional specialization. Distinct access, conflicting simultaneous control, independent learning, stable dissociation, divergent report, and

causal decoupling may support partition. A single lateralized response or transient communication failure is evidence to investigate, not a sufficient basis for multiple-stream classification.

Partial partition may require an overlap model rather than an immediate binary split. In such cases the protocol can represent a junction interval with competing ownership hypotheses and prohibit terminal adjudication until the overlap is resolved or explicitly carried as nonidentification. This is scientifically cleaner than forcing a sharp boundary through ambiguous evidence.

The error costs are asymmetric. False splitting can duplicate or erase parts of one stream; false continuation can pool experiences that should be separately routed. Both errors can manufacture closure. The analysis therefore reports how each plausible split rule changes the residual, but it does not select the rule that improves fit.

A split classification remains a scientific model, not a legal division of the person, a moral ranking, or a statement about how relatives should love or remember. The protocol's authority ends at routing and uncertainty.

13.3.4 — Fusion, Overlap, and Junction Rules

Fusion is the converse junction problem: two or more candidate streams become sufficiently integrated that later experience may form one new maximal stream. The later unity does not retrospectively merge the earlier histories. Each pre-fusion stream remains fixed as its own past, while a new post-junction stream

begins under an explicit origin rule (Bayne, 2010; Tononi et al., 2016).

Information exchange alone cannot establish fusion. Distinct systems may communicate, synchronize behavior, share memory access, or coordinate control. A fusion claim requires evidence bearing on a unified phenomenal field, and current methods may often leave that question unresolved (Bayne, 2010; Tononi et al., 2016).

Overlap is different from full fusion. A system may show shared access in one domain and segregated access in another, or alternate between integrated and partitioned organization. The routing graph should allow junction states and mixed evidence rather than forcing all intervals into one permanent topology.

Junction rules must specify temporal granularity. A split or fusion that appears at one sampling rate may dissolve at another. Multiscale analysis therefore reports whether the junction persists across defensible resolutions, and it treats resolution-dependent conclusions as model-limited rather than ontologically final.

The post-junction stream begins without receiving a transferred balance. Nontransferability means that phenomenal measures are not assets moved from parent streams into a successor. The new stream has its own routed domain even when it contains memories, dispositions, or structures inherited from earlier systems.

For public communication, junction diagrams must show uncertainty and temporal direction. A visually seamless merger can imply metaphysical continuity that the data do not establish.

The safer diagram shows pre-junction streams, the uncertain junction interval, the post-junction candidate, and the alternative models that remain live.

13.3.5 — Exhaustive Routing and No Double Counting

Exhaustive routing assigns every phenomenal contribution in the declared study domain to one stream, probabilistically among mutually exclusive alternatives, or to the unresolved set. Unassigned contributions must remain visible. A protocol cannot discard difficult intervals yet claim complete-stream adjudication (Little & Rubin, 2019; Hernán & Robins, 2020).

No double counting is equally strict. An interval represented under several candidate routes belongs to sensitivity or model-averaging analysis, not simultaneous ledgers whose values are later combined. The same experience cannot support multiple terminal equalities merely because routing is uncertain.

Probabilistic routing requires the estimand and its posterior or mixture summary to remain distinct. Posterior ownership probabilities yield an expected or mixture-level quantity, not an observed truth-level ledger. Analysis must separate route from valence uncertainty and never present a posterior mean as exact stream ownership.

Routing completeness should be audited with a balance sheet of a different kind: counts or measures of included, excluded, multiply claimed, and unresolved intervals. This is not a moral ledger. It is a data-integrity report that lets reviewers see whether a favorable residual was purchased by selective ownership.

Missingness and routing can interact. The least observable intervals may be the ones most likely to be dropped, reassigned, or merged. Sensitivity analyses should therefore couple observation models to routing models rather than assuming that missing intervals are exchangeable with observed ones (Little & Rubin, 2019; Hernán & Robins, 2020).

A route without exhaustive, nonduplicative ownership is Route-failed. Prediction cannot repair the defect because the tested object was never defined well enough to carry the claim.

13.3.6 — Endpoint Routing Audit and Intrastream Nontransferability

Before any endpoint verdict, an independent audit should reconstruct the route from origin through every pause, split, fusion, overlap, and return. It asks whether each rule was prespecified, whether alternatives were scored equally, and whether any revision followed inspection of the terminal residual (Greer et al., 2023).

The audit also enforces intrastream nontransferability. Magnitude from one stream cannot repair another stream's residual even when streams share a body, family, archive, origin system, or successor. This is a formal boundary, not a metaphor about moral isolation.

Junction sensitivity should be reported as a family of endpoint results. If defensible routes yield different signs, compatibility states, or verdicts, the result is model ambiguity or a narrowed reference class, not permission to select the route that preserves the preferred claim.

Event-level ownership conservation is mandatory. Each included interval is assigned once to one declared stream, assigned probabilistically across mutually exclusive routing graphs, or left unresolved. No interval may disappear, appear in multiple branches, or move across streams to improve closure.

Routing uncertainty must propagate to the terminal estimate through a family of admissible graphs, prospectively specified model weights, or an identified set when weights are indefensible (Manski, 2003). A reproducible packet preserves origin, maximality, split and fusion rules, overlap treatment, graph version, adjudicator information, interval ownership, superseded routes, and reasons for change without exposing the tested residual.

Clinical and ethical authority remains separate. Routing uncertainty cannot decide treatment, personhood, capacity, inheritance, guardianship, spiritual status, or eligibility. At junctions, causal inheritance and shared memory may motivate bridge hypotheses, but they never authorize retroactive pooling of predecessor ledgers (Greer et al., 2023; World Medical Association, 2024).

13.4 — Accessibility Classes and Terminal Windows

The endpoint is not one clock. Clinical care has a clock. Unity-proxy measurement has a clock. Observation availability has a clock. Ledger estimation has a clock. Legal and administrative systems have their own clocks as well, although they do not own this chapter's scientific endpoint. When those clocks are aligned without scrutiny, endpoint inference becomes fragile.

Terminal accessibility is a multistate observation problem. Clinical state, unity proxy, measurement availability, censoring, competing events, and terminal support can change on different clocks, so endpoint eligibility should be reported as a class or identified set rather than compressed into one timestamp (Andersen et al., 1993; Aalen et al., 2008; Little & Rubin, 2019).

Clock separation becomes operational through a multistate record. Clinical interventions can change physiology and possible experience; unity proxies can fail through low signal; observation can stop for reasons unrelated to consciousness; and ledger estimates can change when future data enter. Recording each transition on its own clock preserves causal order and prevents one convenient timestamp from absorbing the others.

The core principle is that endpoint adjudication should model progression through states rather than jump from observed to ended. A stream can be open, low signal, candidate crossed, confirmed, returned, censored, withdrawn, or unknown. A study that lacks a state for uncertainty will convert uncertainty into something else.

The machinery here is not final ontology. It is a way to keep endpoint evidence honest enough for later tests. A multistate model can be wrong, but it is wrong visibly. A single timestamp can be wrong invisibly. The displayed inequality is a diagnostic clock-separation convention, not a theorem that every recorded time must differ in every case.

$$T * \neq T_C, T * \neq T_{det}, T * \neq T_{adj};$$

$$T_C \neq T_{det}, T_C \neq T_{adj}, T_{det} \neq T_{adj}$$

Any two realized times may coincide in a case without becoming the same clock object.

Table 13.2. Endpoint clocks, records, and primary dangers.

Clock	Records	Primary danger
Clinical clock	Treatment, interventions, declarations, physiology, care changes.	Clinical necessity can be mistaken for endpoint evidence.
Unity-proxy clock	Prospectively specified consciousness-related channels and thresholds.	Proxy absence can be mistaken for absence of stream.
Observation clock	Scheduled, failed, refused, removed, interrupted, or unavailable data.	Data cessation can be mistaken for the truth-level endpoint.
Ledger clock	Affective estimates, measured ledger windows, and analysis alignment.	Ledger behavior can leak into endpoint construction.

13.4.1 — Clinical Accessibility Class

The clinical clock records what care is doing. Medication, sedation, oxygen support, ventilation, hemodynamics, resuscitation attempts, comfort measures, withdrawal of devices, transfer, discharge, and clinical declarations all belong here. These events are not outside the science. They shape what can be observed and sometimes shape the experience being studied (Greer et al., 2023; Giacino et al., 2018).

Clinical time has privileged ethical authority. Care decisions are made for the person, not for the terminal model. A palliative intervention that disrupts measurement does not create a data problem to be solved by delaying care. It creates a scientific limitation that the study must carry.

The clinical clock can also act as a confounder. Sedation can reduce behavior, alter physiology, change sensor quality, and affect ledger estimates. Pain treatment can change both experience and endpoint channels. Ventilation changes can alter signals that a naive endpoint rule might treat as consciousness-related.

A good endpoint record therefore annotates candidate crossings with the clinical state surrounding them. The question is not only whether a threshold was crossed, but what was happening to the person, the care plan, and the body when the threshold appeared.

Clinical declarations of death, including neurologic or circulatory determinations, should be recorded in this clock with their criteria. They can provide important boundaries and external

context. They do not erase the conceptual difference between clinical authority and research classification.

The clinical clock remains prior because treatment cannot be reorganized to serve endpoint inference. Changes in medication, ventilation, analgesia, monitoring, or location should be recorded as time-varying causes of both observation and state, not treated as protocol deviations to be suppressed. The study adapts to care; care does not adapt to the study.

The clinical clock also identifies why observations are missing. A gap following a comfort-medication change carries different causal information from a failed battery or hospital discharge. Recording those causes prevents care events and logistics from being misread as mind-related endpoint evidence.

13.4.2 — Unity-Proxy Accessibility

The unity-proxy clock records the protocol-fixed consciousness-related channels used to approach the endpoint. These may include perturbational, connectivity, responsiveness, arousal, command-following, or other theory-sensitive indicators. The details belong to the boundary-state and protocol chapters; here the point is clock discipline (Casali et al., 2013; Bodien et al., 2024).

A unity proxy is not the unity gate itself. It is an instrumented candidate for classifying whether the relevant stream remains plausibly active, low signal, absent, or unknown. The clock records changes in those proxies with their quality conditions and context.

The unity-proxy clock must not be derived from the ledger estimator. If affective estimates help define the endpoint, the terminal claim becomes circular. The endpoint system may record physiological channels that overlap in broad biological context, but algebraic identity and outcome-driven tuning must be blocked.

Threshold families are safer than a single threshold. A channel may be sensitive under one state and unreliable under another. The endpoint clock should preserve low-signal, artifact, and uncertain states rather than turning all subthreshold intervals into absence.

The unity-proxy clock also needs revision memory. If a patient later shows evidence of continuity or return, earlier unity-proxy classifications must be demoted or reclassified. The history should remain visible so the system can learn where it failed.

This clock carries one of the chapter's central burdens: it must bring consciousness science into endpoint work without letting any metric become a universal meter of mind.

13.4.3 — Observation Window and Sampling Support

The observation clock is the clock of available evidence. It records when data were scheduled, collected, failed, refused, stopped for comfort, stopped by clinical care, lost to device failure, interrupted by transfer, or missing for unknown reasons. It is essential because endpoint evidence often disappears exactly when it becomes most tempting to interpret (Andersen et al., 1993; Aalen et al., 2008).

Observation missingness is rarely innocent near death. The person may be too uncomfortable for sensors, the family may decline monitoring, clinicians may need access, devices may lose contact, tissue perfusion may change, or the care setting may shift. Each reason carries different implications.

A study that treats missing data as if it were a zero signal shifts burden from the instrument to the person. The fairer approach is to tag observation status at the interval level: observed, partially observed, poor quality, refused, clinically interrupted, device failed, administratively unavailable, or not meaningful because care rightly stopped.

Sudden unobserved endings inform feasibility, burden, and future design. Their records preserve the available evidence and the exact extent of missing terminal observation.

Observation status should travel into models of endpoint uncertainty. A confirmed endpoint built from high-quality, independent channels carries a different status from one inferred through sparse, low-quality, or post hoc evidence. The clock keeps those statuses separate.

The observation clock is also where site differences enter the endpoint model. A hospice room, intensive care unit, home setting, and emergency department may produce different observation patterns even for similar physiology. If those differences are ignored, the endpoint system may learn the institution rather than the stream. A credible multisite protocol therefore models observation infrastructure alongside state evidence.

13.4.4 — Ledger-Independent Terminal Window

The ledger clock records affective estimates, measured-ledger windows, and the alignment choices used in terminal analysis. It belongs in the endpoint chapter only because it must remain separate. The endpoint system should know that the ledger clock exists, but it must not let ledger behavior decide endpoint construction (Little & Rubin, 2019).

This separation is more than a file-management rule. If the endpoint team sees a ledger curve and adjusts a threshold to catch a pleasing terminal approach, the endpoint has become outcome-selected. If the ledger team sees endpoint uncertainty and chooses the endpoint that supports neutrality, the test has become circular from the other side.

The ledger clock may still enter joint models as an object of analysis. For example, a model may ask whether observation quality, clinical events, and endpoint status affect the measured ledger. It may not define the primary endpoint by ledger features.

A clear ledger clock also prevents accidental smoothing leakage. If a smoothed ledger estimate uses future data, it cannot be used to construct a prospective endpoint. Retrospective reconstructions must be labeled as such; their primary or secondary status follows the prespecified analysis plan.

The clocks may inform one another under declared models, but they are not silently interchangeable. Cross-fitting, joint models, and shared-confound analyses can use dependence while preserving honest prediction. The prohibited move is to let the ledger select its own terminal window or let an administrative

time stand in for a phenomenal endpoint without an explicit bridge.

The ledger clock therefore functions as a boundary marker. It is the clock the endpoint system must not become.

For the primary terminal test, endpoint rules and adjudication must be protected from outcome-driven ledger tuning. A prospectively specified joint model may nevertheless estimate declared dependence between endpoint and ledger processes; joint modeling is not automatically exploratory. The analysis must identify its estimand, information set, shared causes, and anti-circularity controls, and must not describe a retrospective estimate as an advance prediction.

13.4.5 — Multistate Terminal Support, Atoms, and Diffuse Timing

A multistate endpoint model represents movement among declared protocol states such as open, low signal, candidate crossing, confirmed endpoint, return, resuscitation, competing event, censoring, withdrawal, device cessation, and unknown. Every state needs an operational meaning and an allowed analytic role (Fine & Gray, 1999; Fleming & Harrington, 2005).

History and dwell time matter. A crossing followed by return differs from persistent absence under high-quality observation, and sedation differs from low signal after technical failure has been excluded. Hidden semi-Markov structure can represent duration without turning the first low-access instant into finality.

Transition intensity and occupancy answer different questions: increased hazard of a terminal-support state does not imply

simultaneous individual entry, and high occupancy does not prove absorption. Both are reported so a population pattern cannot masquerade as a truth-level endpoint (Andersen et al., 1993; Aalen et al., 2008).

Absorbing status is versioned. A frozen primary classification may later enter error memory after recovery, while withdrawal, device cessation, and competing events can end observation or the study question without resolving mind absence.

Endpoint laws may be discrete, absolutely continuous, singular continuous, or mixtures of these; endpoint observations may be interval-censored, and the law may be only partially identified. When the last qualifying observation precedes the first verified nonqualifying state, preserving the bounded interval is stronger than manufacturing a timestamp.

The transition graph marks impossible, reversible, conventionally absorbing, censored, false-confirmed, and observation-shutdown paths. It is therefore both a statistical and governance device, recording how endpoint classification is earned, revised, lost, or withheld without confusing protocol states with the truth-level endpoint.

13.4.6 — Misaligned Clocks, Censoring, and No Estimate

Clinical events, unity-proxy crossings, observation shutdown, legal timestamps, detection, confirmation, and retrospective adjudication often diverge. The protocol preserves each clock and states which analysis uses it; collapsing them can make compression follow medication, device contact, last observation,

or administration rather than the endpoint object (Little & Rubin, 2019; Manski, 2003).

Sensitivity analysis jitters candidate times, compares clock alignments, and quantifies gaps by site, diagnosis, medication, and device. Dependence on a preferred clock narrows the result to the endpoint system.

Public reports name the endpoint class and uncertainty. Large, systematic, or transported gaps require an endpoint-system verdict or No adequate estimate before any terminal conclusion.

13.5 — Filtrations, No-Look-Ahead, and Endpoint Leakage

Filtering estimates a state from information available through the specified time; smoothing uses later observations to reconstruct an earlier state. An advance-prediction claim requires the prospective information set. A prespecified retrospective estimand may legitimately use smoothing, including in a primary analysis, provided the report states that information set and does not claim foresight.

A claim of preterminal prediction requires predictions computed from information available at each frozen prediction time. Filtering and forecasting must therefore exclude future observations and outcome-informed preprocessing. Smoothed trajectories may support a separately prespecified retrospective target, reconstruction, or sensitivity analysis; their primary or secondary status follows the analysis plan rather than smoothing alone.

Look-ahead leakage is not a minor statistical mistake. It can manufacture the very preterminal order the Law is supposed to predict. If future endpoint information helps shape the curve before the endpoint, the study may appear to have seen the ending coming when it merely smoothed backward from it.

$$\pi_t(i) = Pr\{S(t) = i | O_{\leq t}\}, \pi_t(i) = Pr\{S(t) = i | O_{\leq n}\}$$

13.5.1 — Prospective Filtration and Information Set

A prospective filter uses information available up to the current time. In an endpoint setting, it may update the probability that the

stream is open, low signal, candidate crossed, confirmed, or unknown based on observations that have already occurred. It does not know future return, later device review, or final clinician adjudication (Särkkä, 2013; Fleming & Harrington, 2005).

The prospective filter is appropriate for confirmatory claims about information available at a stated time, including anticipation and forecasting. A claim about preterminal dynamics reconstructed after observation ends may instead use a prespecified retrospective model. The latter must be reported as reconstruction, not as a signal known before the endpoint.

Filtering forces the program to confront real-time ambiguity. A case that looks obvious retrospectively may have been uncertain in the moment. The filter preserves that uncertainty rather than granting the past the knowledge of the future.

The filter's inputs must be documented: which channels were available, missing, or low quality; which clinical context variables were included; and which endpoint thresholds were frozen. That record makes the filtered estimate reproducible.

A filtered endpoint may be wrong. That is acceptable. The question is whether it was produced by the rule the study declared and whether its errors were preserved. Error memory lets the system improve without pretending all past filtered estimates were correct.

The prospective filter is the chapter's antidote to prophecy language. It asks only what could have been known, not what the ending later made obvious.

The filter should also preserve nondecision. A real-time classifier may output open, low signal, candidate, or unknown, and unknown should not be treated as failure of computation. In endpoint science, unknown can be the most accurate state because it preserves the information available at the decision time. A filter that always decides may be less scientific than one allowed to abstain.

13.5.2 — Retrospective Smoothing

A retrospective smoother uses later information to reconstruct earlier states. It can be valuable for clinical narrative, case review, secondary analysis, and classifier improvement. It can integrate return evidence, final device-quality audits, clinician notes, family reports, and later state information (Särkkä, 2013; Little & Rubin, 2019).

A smoother uses later observations to refine the account of an earlier trajectory. Report it as retrospective reconstruction and evaluate prospective forecasting from the information available at each prediction time.

Smoothing is especially tempting near terminal endpoints because future data can stabilize noisy states. A low-signal interval may be reclassified after return. A candidate crossing may be demoted after evidence of continuity. These revisions improve the case record while changing the analysis status.

Every endpoint figure should identify its information set. Labels such as filtered primary estimate, smoothed secondary reconstruction, retrospectively adjudicated onset, and versioned revision tell the reader whether future evidence entered. Without

that label, an apparently early terminal signal may be a reconstruction produced only after the ending was known.

Smoothers can also introduce bias if the future data are selected or incomplete. A case with rich post-event review may be smoothed differently from a sudden unobserved death. The smoother must carry observation and selection limits.

Retrospective smoothing may improve an earlier-state estimate, but it cannot show what was known at the prediction time. A prespecified primary retrospective analysis may use later evidence; a primary forecast may not. Tables, figures, and verdicts should identify both the analysis role and the information set.

Retrospective smoothing is most useful as a teacher for later prospective rules. If it repeatedly corrects filtered errors in a particular state, the next protocol can prospectively specify better features or windows. It becomes methodologically dangerous only when used to rescue the current primary test.

13.5.3 — Endpoint Leakage and Frozen Prediction Time

Look-ahead leakage occurs when information from after a time point influences the estimate, threshold, classification, or feature used before that point. In terminal research, leakage can make a preterminal trajectory look more orderly, more predictive, or more neutral than it was under prospective information (Hernán & Robins, 2020).

The simplest leakage is retrospective smoothing shown as if prospective. More subtle forms include tuning endpoint thresholds using complete trajectories, choosing a persistence

window after seeing return outcomes, selecting features that best align to terminal neutrality, or excluding cases because later evidence made them messy.

Look-ahead leakage is dangerous because it often looks like wisdom. The smoother fills gaps, reduces noise, and resolves ambiguity. The resulting curve feels scientifically mature. But if the primary claim was prospective, that maturity is borrowed from the future.

Auditing leakage requires a time-stamped pipeline. Each feature, classification, exclusion, revision, and figure should be labeled by the information set that produced it. If a step used future data, it cannot support a primary prospective claim.

Endpoint jitter tests help expose leakage. If a terminal result changes dramatically under small perturbations of detection time or retrospective adjudication, the claim is less stable than the smoothed figure suggests.

The no-leakage rule is not anti-interpretive. It allows hindsight where hindsight is declared. It simply refuses to let hindsight become the evidence that the model predicted.

Look-ahead leakage should be tested with deliberately damaged pipelines. The study can run a leakage-positive analysis as a negative control to show how much artificial order future information can create. If the leakage-positive curve resembles the claimed signature, the primary result needs especially strong separation. The false method becomes a teaching device for why the real method must stay strict.

Leakage can enter through preprocessing as well as explicit endpoint labels. Imputation models, denoising procedures, feature selection, and representation learning may all use information collected after the frozen prediction time. A valid prospective analysis therefore versions the entire pipeline, not only the final regression, and fits every data-dependent transformation inside the training fold or prospective window.

Nested evaluation is especially important when endpoint definitions, unity thresholds, or feature families are tuned. The outer split estimates transport to unseen cases; the inner split selects among candidate pipelines. Without that separation, a model can appear to predict terminal structure while merely remembering the particular endpoint construction used to optimize it. The error is methodological, not cosmetic, because it converts hindsight into apparent foresight.

13.5.4 — Primary and Retrospective Analyses

Primary endpoint analysis is the analysis that carries the main public burden of a study. It must use the endpoint rule, information set, thresholds, windows, and classification procedures declared before outcome inspection. Secondary analyses may explore alternatives, but they do not have the same status (Center for Open Science, 2025; Chambers, 2013).

This distinction is central for the Law because terminal patterns are easy to find retrospectively. If the primary analysis fails and a secondary smoother succeeds, the result is not confirmatory support. It may be hypothesis-generating, a clue for improved

endpoint design, or evidence that the primary rule was too crude. It is not the same claim.

The primary analysis should declare how it handles candidate, confirmed, revised, unresolved, and unusable endpoint classes. It should also state what happens when endpoint uncertainty exceeds the admissible band for the estimand. If the answer is abstention, that is a valid primary result.

Secondary analyses can examine alternative clocks, retrospective adjudication, persistence windows, state models, or family-report augmentation. Their status depends on what was prespecified and which target they estimate. Retrospective analysis is not automatically exploratory, but an outcome-selected refinement cannot replace a failed primary test and retain its confirmatory status.

The distinction protects the public from proof drift. A headline that merges primary failure and secondary insight misleads. A careful report can say that the primary rule did not support the claim, while a secondary analysis identified a possible endpoint refinement for future testing.

The strongest version of the program gives the primary analysis a fair opportunity to support, narrow, or challenge the tested claim, with the same reporting standards for each outcome.

The analysis plan should assign each output to a public-status lane before results arrive. Primary endpoints support the declared confirmatory question; secondary endpoints refine mechanisms, estimate sensitivity, or generate improved rules; exploratory endpoints suggest future studies. A reader should not need to infer

this hierarchy from tone. The table, figure captions, and verdict language should carry it explicitly.

13.5.5 — Last-Observation Alignment and Informative Shutdown

Last-observation alignment is a rival explanation in which apparent terminal structure arises because data are aligned to the moment observation stops rather than to truth-level endpoint. This can create false compression, false smoothing, or false terminal concentration if measurement cessation is driven by severity, device tolerance, or care workflow (Hernán & Robins, 2020; Little & Rubin, 2019).

The rival is plausible because terminal observation is fragile. Monitoring often becomes harder when physiology changes, family preference shifts, care intensifies, or devices interfere with comfort. If the study aligns all trajectories to the last usable measurement, it may be analyzing the end of measurement rather than the end of mind.

The endpoint chapter therefore requires last-observation analysis. Do candidate signatures align better with the unity-proxy endpoint or data-cessation clock? Are patterns similar when observation continues after candidate crossing? Do device failures produce comparable apparent compression?

Last-observation controls identify the contribution of alignment to the apparent pattern. A persistent result merits further comparison; a disappearing result informs the observation model; indistinguishable explanations remain unresolved.

This rival also protects comfort. If the study knows that data cessation is informative, it can stop without pretending that stopped data became an endpoint. The person's care remains prior; the model carries the lost information honestly.

Last-observation alignment turns a common data defect into a specific adversarial test. That is exactly what a serious science volume should do.

This rival matters even when endpoint science is strong because last observation can coincide with the moments most compelling to readers. A smooth terminal figure aligned to data exit may resemble discovery while being a workflow artifact. The chapter therefore puts last-observation alignment in the rival set before the first study begins.

13.5.6 — Public Reporting of Revision and Leakage Audits

Endpoint revision shows that the system can correct itself, but public reporting must show how claim status changed. It should identify the original classification and information set, later evidence, revised classification, and affected analyses while avoiding spectacle or consolation language about the person's death (Spiegelhalter, 2017; World Medical Association, 2024).

A leakage audit traces each datum by generation time, availability to the endpoint classifier, first analyst access, and derivation from ledger or outcome-adjacent variables. Event logs, separated teams, frozen outputs, and an independent integration step are stronger than retrospective assurances; where full separation is impossible, overlap and its consequences are documented. A timestamped provenance graph can reveal leakage paths that

prose descriptions miss, especially when derived variables cross team or software boundaries.

Preprocessing is part of the audit. Artifact rejection, smoothing, imputation, censoring, and quality thresholds may leak endpoint status when tuned on known cases, so versioned pipelines, locked parameters, and held-out stress sets must test whether the terminal pattern survives. Stress sets should include intentionally leaky pipelines so the audit can demonstrate that it detects contamination rather than merely confirming the preferred workflow.

The public record includes negative audit findings, unresolved channels, staff overlaps, and analyses downgraded because separation was incomplete. The audit states which channels were checked and what each check found. Publishing the downgrade rule in advance prevents partial separation from being reinterpreted after a favorable terminal pattern appears.

13.6 — Return, Resuscitation, and False-Ending Controls

Any system that classifies truth-level endpoint near the boundary of low signal, sedation, clinical intervention, or failing physiology must be prepared for return. Treating return as evidence lets the endpoint system improve and preserves the participant's actual recovery in the record.

Return after apparent cessation provides a practical test of an endpoint protocol. Resuscitation, autoresuscitation, delayed recovery, and reversible neurologic shutdown inform persistence and reentry rules and can require revision of an earlier endpoint classification (Dhanani et al., 2021; Greer et al., 2023; Parnia et al., 2023).

Return and resuscitation should update a state history, not erase it. The record should preserve the original candidate crossing, the evidence that supported it, the reentry or return event, and the revised classification. That sequence lets the protocol learn which persistence or marker assumptions failed while preventing later recovery from being used to pretend that the earlier endpoint decision was never uncertain.

The core moral rule remains simple: rescue and treatment come first. Emergency intervention does not serve the study. Research classification follows care. If care reopens an endpoint that the study thought closed, the study changes its record. The person owes the model no consistency.

The scientific value of false endings is large. They expose threshold fragility, state dependence, channel failure, and

overconfident irreversibility. A program that remembers its false endpoints becomes harder to fool.

13.6.1 — False Ending and Premature Crossing

A false crossing is a candidate endpoint event followed by evidence that the relevant stream remained or returned. The evidence may be behavioral, communicative, neurophysiological, clinical, or contextual. The term false crossing is deliberately technical. It does not shame the protocol; it describes an event class (Dhanani et al., 2021; Greer et al., 2023).

False crossings can arise from artifact, sedation, delirium, transient hypoperfusion, device failure, motor impairment, low arousal, or an endpoint threshold that is too brittle. They can also arise because the science of consciousness is incomplete. The endpoint system should learn from all of these causes.

A false crossing requires demotion of the earlier endpoint status. It may become a low-signal interval, a reversible interruption, an artifact event, or an uncertain state. What it cannot remain is a confirmed truth-level endpoint under the same version history.

False crossings should be counted and described. How many candidate crossings occur? How many later return? Which channels failed? Which clinical states were present? Which thresholds were most vulnerable? These data are classifier evidence, not merely exclusions.

A false endpoint can erase a living or returning stream from the analysis, mislead a family, and contaminate every terminal estimate built on the label. The protocol must therefore record false crossings, near misses, reversals, and their downstream

effects as system-level errors. Responsibility for the mistake belongs to the research system, not to the person whose state was difficult to classify.

False crossings are therefore not enemies of the program. They are one of the ways the program remains scientific.

False crossings are especially valuable because they reveal which combinations of channels imitate endpoint absence. They can identify medication effects, arousal fluctuations, device artifacts, and care-procedure disruptions. A mature program should publish false-crossing rates and examples, stripped of identifying details, because the classifier's failures are part of the scientific object.

13.6.2 — Resuscitation, Autoresuscitation, and Recovery

Resuscitation and recovery require special care because they can reopen what looked closed. A cardiac arrest, respiratory collapse, or severe neurologic crisis may be followed by restored circulation, consciousness, communication, or other continuity evidence. The endpoint record must be able to reverse itself without hesitation (Dhanani et al., 2021; Parnia et al., 2023).

The chapter does not treat resuscitation as a research opportunity. Emergency intervention belongs wholly to care. Researchers may record what happened only under ethical, consent, and burden rules. They may not shape rescue, delay rescue, or infer from rescue decisions that the endpoint question has been answered.

Recovery evidence can vary. Immediate command following is strong but not always available. Later communication may support continuity but can be affected by memory gaps, medication, delirium, or trauma. Physiological return may

constrain endpoint timing without directly establishing subjective continuity. The record should grade evidence rather than collapse it.

Resuscitation stress-tests the multiple-clock structure in §13.4: arrest and return, observation gaps, unity-proxy confidence, and ledger availability can diverge, so one endpoint timestamp cannot represent all four.

If recovery occurs after a candidate endpoint, the primary rule should state how the case is handled. It may become false crossing, reversible low-signal state, censored, unresolved, or excluded from primary endpoint analysis. The rule should not be invented after the outcome is emotionally or statistically attractive.

An endpoint model incorporates later recovery by correcting its record and preserving the evidence behind the revision.

Recovery also changes how endpoint evidence should be archived. The case should not be erased from the study as inconvenient; it should move into a recovery and reversal class with its original filtered state preserved. That lets later researchers ask whether the endpoint system failed because the threshold was too sensitive, the window too short, the channel too narrow, or the clinical context too unstable.

13.6.3 — Persistence, Hysteresis, and Reentry

Hysteresis means the rule for leaving a state need not be identical to the rule for entering it. Endpoint adjudication needs hysteresis because near-boundary signals can oscillate. Without reentry rules, a noisy channel can produce multiple candidate endpoints

in rapid sequence (Greer et al., 2023; Fleming & Harrington, 2005).

A simple threshold creates arbitrary endings when signal quality fluctuates. The system may declare candidate endpoint, then open, then candidate again, depending on noise, arousal, medication, or device contact. Hysteresis requires a stronger or sustained pattern before changing state back and forth.

Reentry rules should define how a stream moves from candidate crossing back to open, from low signal to candidate, from confirmed to revised, and from unknown to classified. These rules should be protocol-fixed, state-specific, and sensitive to clinical context.

Hysteresis is not a trick to avoid falsification. It can make the endpoint harder to confirm and easier to demote under return. Its purpose is to prevent noise from carrying formal authority. The rule must be tested through simulation and false-crossing memory.

The endpoint model should report how often classifications change under different hysteresis parameters. If the scientific result depends on a narrow hysteresis choice, the endpoint is fragile. If it survives plausible choices, the endpoint system is more credible.

Reentry rules reflect the varied ways living systems change state. They improve classification by preserving continuity evidence, reversals, and uncertainty.

Hysteresis should be tuned for scientific stability rather than aesthetic smoothness. A rule that is too strict may miss genuine

return; a rule that is too loose may oscillate with noise. The protocol should test hysteresis parameters on held-out examples and report how often endpoint status changes under plausible alternatives. Stability has to be earned, not assumed.

13.6.4 — Return Without Recall

Memory after return depends on encoding, consolidation, retrieval, language, and reporting capacity. An endpoint model evaluates these processes alongside other state evidence when interpreting a gap in recall (Squire, 2004; Parnia et al., 2023).

The distinction matters after resuscitation, anesthesia, delirium, coma emergence, and severe physiological crisis. A person may later remember nothing, remember fragments, confabulate, or report experiences shaped by recovery. Each possibility has different evidentiary status.

Credible recall can support continuity. The record preserves memory's contribution without using it as the sole exclusion rule for experience.

A returned person's memory, silence, confusion, or fragmentary account belongs to the protected case record and must be interpreted through memory, medication, injury, language, and time since recovery. It can revise scientific confidence about the earlier interval; it should not be converted into a public story about whether the Law was witnessed, disproved, or fulfilled.

Methodologically, memory gaps strengthen the case for contemporaneous, low-burden, noncoercive observation before and during boundary states. They do not authorize burdensome probing. When memory fails, the model carries uncertainty.

Return without recall can overturn a premature endpoint classification when the returned state is independently linked to the same stream under the declared routing rules. Amnesia does not by itself resolve that linkage or show that the intervening interval was empty. Report the return evidence, the revised endpoint status, and any remaining uncertainty about the gap.

13.6.5 — Competing Events, Refusal, and Withdrawal

Competing events must be defined relative to the study's event of interest: they preclude that event, whereas an interruption of observation alone does not. Transfer, discharge, device removal, refusal, withdrawal of consent, loss of capacity without adequate surrogate permission, clinical emergency, family request, or care procedure can all interrupt endpoint adjudication (Fine & Gray, 1999; World Medical Association, 2024).

Withdrawal is not failure by the participant. It is a right and a design condition. If a person or authorized process stops participation, the research model records the endpoint evidence as censored, unresolved, or unavailable under the rules. It does not infer closure from refusal.

Device removal is especially important. A sensor may be removed for comfort, clinical access, skin integrity, family preference, or operational need. Treating that removal as data about mind would convert care into endpoint evidence.

Each listed event's role should be specified in the estimand structure of the study. Is it a competing risk, a censoring event, an intercurrent event, an endpoint-disqualifying event, or a

secondary classification? The answer should be written before data are collected.

The distinction also prevents survivor bias and selective reporting. Cases with messy competing events may be less likely to produce clean terminal figures. If they disappear from the denominator, the endpoint system will look better than it is.

Competing events, refusal, transfer, device removal, and clinical necessity can terminate observation without terminating the stream. Each should be coded by cause and routed to its own censoring or no-estimate rule. None creates a missing interval that the theory is permitted to fill by imputation.

Withdrawal and competing events should be coded with reasons where ethically permissible. Refusal, discomfort, clinical necessity, transfer, family request, device removal, and administrative discharge carry different implications for missingness and endpoint uncertainty. Treating all exits as one censoring code hides the burden structure that this chapter is supposed to make visible.

13.6.6 — Error Memory and Protocol Correction

A false ending is evidence about the protocol, not an inconvenience to erase. Return, autoresuscitation, delayed recovery, renewed command following, or later testimony shows that a candidate endpoint, persistence window, observation rule, or routing assumption failed for that case (Dhanani et al., 2021; Greer et al., 2023).

The original classification and the evidence available at the time should remain immutable in the audit record. The corrected

classification is added as a new version with the trigger, reviewer, affected analyses, and reason for revision. Hindsight may improve a secondary reconstruction, but it must not rewrite what the primary model actually knew.

Protocols should prospectively specify return windows, competing events, reentry rules, and the circumstances that require a case to be reclassified or withdrawn from a terminal analysis. Different clinical contexts may require different windows; the choice must be justified before the outcome is known.

Correction must propagate through a dependency graph. Endpoint estimates, ledger calculations, figures, summaries, model fits, and public statements that relied on the false ending should be identified, rerun, qualified, or withdrawn. The person should not become a permanent error label simply because the research object changed. Dependency-aware correction can preserve unaffected analyses while preventing the false endpoint from remaining embedded in downstream estimates.

Original and revised analyses should be reported together when both are useful. The original analysis preserves the record of the prespecified information set; the revised analysis shows what later evidence reveals. Their difference is itself a measure of endpoint fragility. A large discrepancy between the two analyses should trigger review of which later information changed classification rather than being averaged into one endpoint estimate. For the paired comparison, all other prespecified analysis choices should remain fixed where possible, so the

difference isolates the effect of the new endpoint information rather than combining correction with model retuning.

Public correction should be prompt, specific, and proportionate. Participants, authorized representatives, clinicians, reviewers, and readers need to know what changed and which conclusions no longer hold, without sensationalizing the return or implying a metaphysical lesson (Center for Open Science, 2025). Correction notices should preserve the original date and affected version, allowing readers to distinguish a repaired record from a result that was never superseded.

Error memory makes the endpoint system stronger. A protocol that preserves and learns from false endings becomes less likely to repeat them and more capable of declaring no estimate when the available clocks cannot support a responsible judgment. Cataloging false-ending mechanisms can also improve design by showing which clinical or observation states most often generate premature endpoint confidence.

13.7 — Sudden, Diffuse, and Low-Signal Endpoints

Many endings will not supply the terminal window a researcher wants. Sudden death may leave no usable preterminal series. Low-signal states may produce artifact rather than evidence. Emergency care may correctly interrupt measurement. Sedation, delirium, and rapid decline may make signals ambiguous. The chapter treats these cases as central because they are where overconfidence is most tempting.

Sudden, diffuse, and low-signal endpoints can produce informative censoring, uncertain timing, or loss of usable terminal support. Sedation, delirium, emergency intervention, equipment removal, and rapid decline can jointly alter experience and observation, making no-estimate outcomes essential to valid inference (Little & Rubin, 2019; Surges et al., 2024).

Missing terminal data identify a concrete boundary for the current analysis. The study records the missing window or inadequate signal, reports its consequences, and uses that information to improve feasible future designs.

Unusable endpoint evidence is a substantive outcome. A case may be censored, unresolved, excluded from the primary terminal analysis, or retained only for feasibility and observation-process learning. The report should state why the case was unusable and what it still teaches, without forcing it into a favorable or unfavorable terminal narrative.

The humane logic is identical to the scientific logic. Emergency intervention, comfort, and dignity come first. If those priorities erase the terminal window, the study accepts the loss. It does not

ask the person to suffer for completeness or ask the model to pretend completeness remains.

13.7.1 — Sudden and Unobserved Endpoints

A sudden unobserved ending is a case in which the relevant terminal interval is missing or too sparse to support endpoint adjudication. The person may die suddenly, outside the measurement window, after transfer, after withdrawal, or during an event that prevents useful observation. The study may know that death occurred without knowing the endpoint state in the required sense (Greer et al., 2023; Little & Rubin, 2019).

This situation cannot be repaired by aligning to the last observation. The last observation may be hours, days, or weeks before the event, and its relationship to truth-level endpoint may be unknown. It can constrain what was observed, not what happened afterward.

Sudden endings are scientifically valuable for feasibility. They tell the program how often terminal windows are unavailable, which populations are difficult to observe, which care contexts make monitoring inappropriate, and whether the reference class can support the proposed study.

A study should report sudden unobserved endings in the denominator. Excluding them silently can make the endpoint system look more usable than it is. A reference class with frequent unobserved endings may require a different design or a narrower claim.

The public wording should be plain: no terminal endpoint verdict was available. This protects families from speculation and protects the theory from imputation.

Sudden death is therefore not a blank page for the Law. It is a boundary condition on what this research program can honestly know.

Sudden unobserved endings define the feasible reference class of terminal-window research. Their frequency determines how much of the population can ever contribute qualified terminal data and whether observed cohorts are selected toward slower, more monitored trajectories. Reports should therefore treat sudden cases as a scope and transport problem, not merely as missing rows.

13.7.2 — Diffuse Support, Low Signal, and Artifact

Near terminal states, signal quality can change with perfusion, device contact, movement, sweating, skin integrity, electrical noise, medication, clinical procedures, or physiology outside the qualified context of use. Classify these observation conditions before interpreting the signal (FDA-NIH Biomarker Working Group, 2016; Little & Rubin, 2019).

Quality thresholds should be declared by channel. A unity proxy may be unavailable if contact quality fails. A physiological marker may be unusable during movement or care. A device may produce clean-looking numbers that are physiologically meaningless under terminal conditions. The endpoint model must know the difference.

Low signal can also be a real state. A fading consciousness marker under adequate quality may be endpoint-relevant. The difficult task is separating true low signal from poor observation. That separation requires quality indicators, negative controls, clinical context, and abstention rules.

Artifact should never be corrected by desire. Imputation, smoothing, or manual cleaning may be useful only when rules are frozen and the procedure is labeled. If artifact dominates a candidate endpoint, the case may need to become unresolved or unusable.

A low-signal report should state what failed: sensor contact, biological signal, responsiveness, arousal, data cadence, clinical context, or model confidence. Different failures imply different revisions.

Low signal is an observation requiring classification: it may reflect the underlying state, poor observation quality, or both; it is not itself an endpoint. Reduced amplitude, missing channels, medication, impaired perfusion, movement, emergency procedures, device displacement, and genuine decline can produce similar records. Endpoint classification requires both quality criteria and contextual evidence; when those causes cannot be separated, the case remains unusable rather than being promoted to terminal support.

A low-signal branch should record which channel failed, when it failed, what competing causes were present, and whether any independent marker remained. Sensitivity analyses can then test whether conclusions depend on treating the interval as censored,

missing, or potentially conscious. None of those branches is equivalent to assigning low consciousness by the size of one signal.

13.7.3 — Emergency Intervention and Rescue Priority

Emergency intervention is categorically prior to endpoint inference. Rescue, stabilization, comfort, airway support, medication, resuscitation, seizure treatment, and hemorrhage control can interrupt measurement or destroy interpretability, and the study accepts that loss (World Medical Association, 2024; Greer et al., 2023).

Research never slows, times, or shapes urgent care. A protocol that depends on uninterrupted measurement during clinical need is both ethically and scientifically unfit.

The emergency event is still recorded on the clinical clock with its treatments, signal consequences, and missingness. It can explain candidate crossings or false endpoint classifications without becoming evidence of the truth-level endpoint.

Low signal, device removal, medication effects, and missing data may arrive together. The status system therefore returns competing event, censored endpoint, or no endpoint verdict rather than forcing interpretation.

Emergency care is a protected intervention, not a protocol deviation. Incidental data remain governed by prior consent, minimal burden, and later review; the event never becomes an opportunity to complete a terminal curve.

13.7.4 — Sedation, Delirium, and Rapid Decline

Sedation, delirium, and rapid decline are not interchangeable. Sedation can reduce arousal, memory, movement, report, and physiological variability. Delirium can fragment attention, perception, distress, and communication. Rapid decline can alter perfusion, metabolism, and signal quality. Each can mimic endpoint approach without being truth-level endpoint (Surges et al., 2024; Inouye et al., 2014).

Endpoint models should therefore carry state context. Medication class, dose timing, route, cumulative burden, delirium assessment where possible, pain or dyspnea treatment, infection, hypoxia, metabolic changes, and care goals may all matter. These variables are not nuisance notes; they are rival explanations.

A sedated low-signal interval may be a reversible drug effect, an ethically appropriate comfort state, a true reduction in consciousness, or a mixture. The endpoint system should not force one interpretation when the evidence cannot separate them.

Delirium adds another difficulty: experience may be present, intense, confused, and difficult to report. Treating delirious inability to communicate as absence of experience would be both scientifically unsafe and ethically dangerous. Distress may be hidden precisely when report becomes unreliable.

Rapid decline can compress observation opportunities. A model may see steep change and infer endpoint approach, but the same pattern may reflect systemic collapse, sensor gain, or medication. The correct response is rival modeling, not dramatic conclusion.

Sedation, delirium, and rapid decline require joint time-varying models because responsiveness, distress, memory, physiology, and data quality can change together. The analysis should compare medication, illness, arousal, sensor failure, and genuine endpoint explanations instead of treating every downward trace as approach to the truth-level end. When those explanations remain inseparable, the endpoint widens or becomes unavailable.

The scientific report should distinguish state change from confound change. Dose timing, delirium assessments, perfusion, respiratory status, pain control, device quality, and observation frequency can reveal whether an apparent decline reflects consciousness evidence, measurement deterioration, or both. Comfort remains prior even when the additional clinical information cannot recover an estimate.

13.7.5 — No Imputed Closure

No imputed closure is the rule that missing terminal data cannot be filled with the hypothesis. If the endpoint is unobserved, unresolved, or unusable, the study may not declare that the ledger reached neutrality because the Law predicts it. That would be circularity in its most direct form (Little & Rubin, 2019).

Imputation can be legitimate in ordinary statistical work when assumptions are explicit and the estimand permits it. Endpoint closure is different because the missing value is the claim. Imputing it from the claim under test destroys falsifiability.

The rule applies to whole-life closure, terminal neutrality bands, preterminal signatures, and endpoint classification. A missing terminal window cannot be replaced by a smooth curve to zero. A

low-signal interval cannot be labeled neutral because it is near death. A sudden ending cannot be counted as supportive because the model expects closure.

Imputation requires an explicit observation model, assumptions appropriate to the estimand, and uncertainty propagation. A prespecified primary analysis may use defensible imputation, accompanied by sensitivity analyses. It may not impose terminal zero, a favored trajectory, or the hypothesis under test on the missing observations and then count that imposed result as evidence. If plausible missingness assumptions change the verdict, report that dependence or abstain.

Public communication must resist filling missing data with stories of mystery, peace, hidden balance, or spiritual confirmation. The correct statement is narrower: the terminal evidence was unavailable or insufficient, so the scientific claim remains limited.

No imputed closure applies in figures or text. A plot does not draw a graceful terminal line through an unobserved interval unless it is labeled as simulation, sensitivity, or imputation, and a table does not convert an unusable endpoint into a neutral one for completeness.

13.7.6 — Unusable Endpoint and No Adequate Estimate

Sudden death, diffuse decline, heavy sedation, delirium, low signal, missing sensors, late enrollment, or observation shutdown can leave the truth-level endpoint outside protocol support. Report it as unusable rather than assigning zero or imputing favorable closure.

Apply the evidence-state distinctions in §24.2 and §27.4 to endpoint uncertainty. A defensible interval of endpoint times may support branch-specific or set-valued analysis, while inseparable adequate endpoint, routing, or missingness models remain Model-ambiguous (Manski, 2003).

State missingness assumptions. Carried-forward values, smooth extrapolation, favorable terminal imputation, and complete-case analysis can mimic stable endings after informative observation loss. Test plausible selection and pattern-mixture models in sensitivity analyses (Little & Rubin, 2019).

Retain difficult endpoints and unfavorable residuals in the accounting. Report unusable-case counts, reasons, differences by site, state, treatment, or access, and each restriction's effect on the reference class and denominator.

A peaceful appearance, flat trace, last word, or family narrative can hold personal meaning while leaving the terminal phenomenal state unidentified. Report the failed layer: terminal support, routing, clock alignment, observation, bridge validity, or burden-driven shutdown.

A no-estimate outcome defines the remaining population and the next protocol's repair task. It also prevents universal terminal conclusions from resting only on easily observed cases.

13.8 — Comfort-First Care and No-Estimate Outcomes

Chapter 3’s comfort-first rules govern this entire section. The local endpoint consequence is precise: care, refusal, family burden, or clinical duty may close observation, and the resulting record must move to the appropriate qualified, censored, unresolved, unusable, or no-estimate state rather than forcing a terminal value. Endpoint governance must therefore carry evidence class, permitted use, uncertainty, correction route, and care boundary together (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016; Sanders et al., 2024).

A terminal evidence class describes the endpoint information available to a protocol. Qualified, revised, unresolved, censored, and unusable classifications identify the analyses the record supports and the questions that remain open.

Governance therefore has to be both scientific and humane. It must define evidence classes, protect family and clinician language, route formal definitions to the formal-law book, route evidence status to this volume, and prevent public artifacts from turning endpoint science into spectacle.

Table 13.3. Endpoint evidence classes and public-status restraints.

Endpoint evidence class	Allowed public status	Required restraint
Qualified endpoint	May support the declared analysis within its reference class.	No claim of an exact truth-level endpoint or personal meaning.

Endpoint evidence class	Allowed public status	Required restraint
Unresolved endpoint	May inform feasibility, uncertainty, and future design.	No positive or negative terminal verdict.
Unusable endpoint	May count in denominator and operational evaluation.	No imputed closure or anecdotal use.
Revised endpoint	May update the versioned record and secondary analysis.	No rewriting of primary prospective evidence.

13.8.1 — Comfort-First Care

The comfort-first rule is not a slogan. It is a stop rule. Measurement stops when care requires access, when devices burden the person, when the person refuses or withdraws, when family involvement becomes harmful, when clinicians judge the protocol disruptive, or when dignity would be compromised by continued observation (World Medical Association, 2024; Sanders et al., 2024).

The rule applies even when stopping makes the endpoint unusable. A research program that cannot tolerate lost data will be tempted to make people serve its completeness. This chapter reverses the burden: the model must tolerate loss so the person does not have to tolerate unnecessary burden.

Comfort-first design also means that endpoint protocols should be built around minimal burden from the beginning. Passive or clinically integrated observation is preferable to intrusive prompting. Any active measurement must have explicit burden

caps, withdrawal routes, and staff authority to pause or stop without research permission.

The rule does not make the science weak. It makes the reference class honest. If comfort-first constraints prevent certain endpoint claims, the claim narrows. A narrowing forced by ethics is not a defect. It is the condition under which the research remains worth doing.

Comfort-first stopping must be recorded without blame. The endpoint status may become censored, unresolved, or unusable. The record should say care required stopping, not that data were lost because of noncompliance.

The dying person owes no final measurement. That sentence is the ethical center of the chapter and the practical boundary of its science.

13.8.2 — Refusal, Family Welfare, and Clinical Communication

Family and clinician communication must prevent endpoint language from becoming prognosis, consolation, pressure, or interpretation. A family should not hear that a candidate crossing means the person is gone. A clinician should not hear that a terminal study wants monitoring continued against care judgment. Research vocabulary must remain research vocabulary (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Use the endpoint status terms defined in §13.2 consistently in family and clinical communication, linking each classification to its meaning for the observation and care setting.

Family narratives can be evidence in some contexts, but they are not clean endpoint labels. Last words, apparent lucidity, silence, agitation, peacefulness, or remembered signs may matter clinically and relationally. In research, they require denominator, selection, timing, and interpretation limits.

Clinicians need a separate channel for concerns. If research terminology confuses care, increases family anxiety, or creates pressure around monitoring, the protocol should pause or stop. The endpoint system exists inside clinical trust, not above it.

Communication materials keep death spectacle, proof language, and emotional manipulation outside the scientific record. Their tone must be clinically useful, gentle enough for families, and exact enough for later audit.

Families may be told that data are incomplete, that a protocol endpoint was or was not adjudicated, and that uncertainty remains. Model probabilities are not translated into reassurance, blame, a promise about inner life, or an interpretation of what a death meant; those categories belong to evidence status, not consolation.

13.8.3 — Endpoint Evidence and No-Estimate Classes

Endpoint evidence classes identify which analyses the record can support. Qualified endpoint evidence may enter the prespecified primary analysis appropriate to its estimand and information set. Later information may support a declared retrospective reconstruction but cannot create an earlier forecast. False crossings record protocol error; unresolved, censored, or unusable

endpoints restrict or prevent the affected terminal claim (Greer et al., 2023; World Medical Association, 2024).

Each class must carry provenance and permissions: rule version, channels, information set, clinical context, observation quality, confidence, revision history, allowed analyses, and prohibited public interpretations. The class is an evidence contract, not a decorative label.

Class structure governs the reference claim. A hospice cohort with many independently qualified endpoints and an emergency cohort dominated by sudden unobserved endings support different inferences even when their nominal sample sizes match. Denominator flow should show how many cases entered each class and which transitions occurred.

The same endpoint classes appear in reports, figures, summaries, and public artifacts. Mixed-class figures identify their composition, and claims based on qualified endpoints disclose the unresolved, censored, revised, and unusable cases excluded from support.

The class set should remain small enough to audit while preserving every distinction that changes inference. Sensitivity analyses should show whether conclusions change when retrospective or lower-quality classes are included, excluded, or reclassified, making the selection mechanism visible rather than hiding it behind one final sample size.

Endpoint classes connect the available evidence with a specified scientific use. Their definitions and revision rules make that connection clear to readers.

13.8.4 — No Endpoint Verdict as Personal Meaning

Scientific reporting can describe what was observed, how an endpoint was classified, and which uncertainties remain, while families and communities retain the space to understand the life in their own terms (Parfit, 1984).

Graphs, last signals, false crossings, and peaceful accounts can invite narrative closure. The report resists that pull by keeping every sentence tied to the observed object, its uncertainty, and its permitted use.

Families, communities, and spiritual traditions may find meaning through their own relationships and practices. Scientific endpoint reporting contributes a clearly defined account of the observations.

The boundary survives every scientific outcome. Support for PTVC would not authorize death interpretation, and defeat would not subtract dignity from any life. Detachable public language therefore says that evidence was classified, revised, or unavailable, never that a life was balanced, redeemed, or explained.

13.8.5 — Scientific, Legal, and Spiritual Boundaries

Formal and empirical changes travel through different authorities. A revision to the truth-level endpoint or strict-pre-endpoint domain belongs to the formal root and requires controlled concordance. A revision to an estimator, persistence window, evidence class, or adjudication rule belongs to scientific method and changes the evidence status. Clinical ethics governs care

throughout; no layer silently rewrites another (World Medical Association, 2024).

Hard cases often revise the empirical bridge rather than the formal target. A return may show that a threshold or confirmation rule was wrong. New boundary-state evidence may improve unity proxies. Neither result, by itself, settles the truth-level endpoint or the formal identity of a stream.

Formal change control should be rare, visible, and justified. Empirical updates should be expected, versioned, and tied to evidence. Confusing those two processes lets the project escape defeat by redefining the target, or lets a measurement failure falsely weaken the formal hypothesis.

Clinical, legal, scientific, and spiritual systems answer different questions. Clinicians direct care, law records institutional status, science classifies evidence under a protocol, and families or traditions may carry meaning. An endpoint estimate does not inherit the powers of those neighboring systems, and none of them supplies the scientific endpoint merely by naming an ending.

Change propagation should follow the layer that changed. A new formal definition triggers concordance across the model and series; a new endpoint estimator triggers reanalysis, versioning, and public correction within the scientific program. Keeping those routes separate prevents an operational difficulty from quietly redefining PTVC and prevents a formal clarification from being mistaken for new empirical support.

13.8.6 — Observation Shutdown, Correction, and Public Stop Gate

Observation shutdown is a protocol event, not a truth-level endpoint. Record why monitoring stopped and keep endpoint adjudication separate.

Care, refusal, family privacy, and clinical capacity outrank data completion. Preserve the last valid information set and the missingness consequences; return Unusable evidence, Inconclusive evidence, or No adequate estimate according to the canonical evidence-state rules rather than extrapolating.

If later evidence changes the admissible endpoint model, correct dependent analyses through Chapter 28. Audit whether stop rules were applied consistently across sites and populations so selection introduced by shutdown is reflected in the reference class.

Part V — Study Families and First Tests

A study family translates a formal claim into an empirical design. Earlier Parts established scientific jurisdiction, the formal target, representation and measurement, stream identity, and endpoint discipline. This Part connects those dependencies to populations, instruments, estimands, rivals, burden limits, decision rules, and stopping routes. Each design identifies the evidence needed to develop, replicate, revise, or retire a component.

Part V asks what the first humane, protocol-fixed, decision-relevant test should look like. Its scientific strength comes from fixing the target objects, measurement conditions, outcome branches, stopping rules, and next actions before outcome inspection (Nosek et al., 2018; Munafò et al., 2017).

Research burden is part of the data-generating process. Fatigue, workflow, privacy, staff capacity, device discomfort, consent, refusal, and comfort can change who remains observed and what the record means; evidence obtained by compromising care is methodologically contaminated.

The first protocol therefore fixes a reference class, measurement stack, endpoint clocks, estimands, intercurrent-event handling, rival models, burden budget, missingness branches, decision thresholds, and adverse-result language before enrollment. Its simulation gate can cancel recruitment when the planned cadence, sensors, endpoint uncertainty, or missingness pattern cannot separate the candidate from its strongest rivals. A runnable study is not automatically an informative or permissible study (International Council for Harmonisation, 2019; Chan et al., 2025).

The first setting is selected for measurement quality, feasibility, and participant comfort. In a late clinical window, the primary questions concern the observed dynamics and the models that explain them.

The second study family examines sleep, dreams, and affective reprocessing. It asks how state-specific affective dynamics can inform testable hypotheses through optional reports, physiological measures, and comparison with established sleep science. Privacy and participant comfort shape the observation plan.

The third study family converts the reserve-coupled route and neighboring candidate routes into exposed predictions. It asks whether independently measured horizon and capacity variables carry the boundary-rate, dispersion, covariance, and intervention signatures that the route requires, and whether equally informed rivals reproduce them. Reserve is a candidate state, not worth, prognosis authority, or a reason to narrow anyone's care.

The fourth study family follows experience-related constructs over longer periods without pretending that a long record is a complete stream. It confronts delayed entry, sparse windows, treatment change, informative observation, device and language drift, competing events, transport, and the privacy cost of being followed. Longitudinal evidence becomes useful only when the conditions under which a person was visible remain part of the scientific object.

Here scientific character appears as method. Prospective specification, negative controls, model recovery, missingness

analysis, versioning, replication, and publication of adverse results are the instruments by which the program states what can count against it before enthusiasm has a result to protect. They convert openness to correction from a promise into a design property.

The Part's strongest contribution is an architecture in which support, narrowing, rival victory, unusable evidence, ethical stoppage, and defeat all produce defined scientific consequences. *The Law of Fairness* becomes more serious when those branches remain real after the work becomes expensive. A hypothesis gains content by exposing the conditions under which it must become smaller, change route, or yield. Predefining those consequences also protects later teams from sunk-cost pressure when a costly protocol reaches an unfavorable branch.

For statisticians and methodologists, the program asks for discrimination rather than a favorable association. The design must survive protocol-fixed rival comparison, generator and model recovery, negative controls, observation-process modeling, and reference-class limits. The relevant question is whether the complete pipeline can distinguish a candidate signature from the processes that mimic it, not whether one coefficient points in a preferred direction. Discrimination should be evaluated on simulated and empirical cases where rivals are intentionally hard to separate, not only on easy benchmark worlds.

The first studies can establish measurement foundations, distinguish candidate routes, and identify the observations that make stronger tests possible. Clear reporting of methods and

results lets readers follow those advances and understand how each stage prepares the next. A result that improves the next design can be valuable even when it leaves the target hypothesis unchanged, because dependency knowledge is itself cumulative.

The four chapters carry the science outward in complementary directions. The first protocol builds a decision system; sleep and dream studies open a repeated-state laboratory; terminal-window studies test route-specific restrictions; and longitudinal designs expose how observation changes across time. Each study family must alter the status of a declared component without manufacturing the answer through its scale, endpoint, sample, or missingness rules. Using a common status vocabulary across the four families will make later synthesis possible without treating unlike evidentiary roles as interchangeable.

Chapter 14 — The First Discriminating Protocol

The first discriminating protocol begins with the smallest humane question capable of changing scientific status. Terminal dynamics, felt experience, and conscious unity call for tighter identification and lighter burden, not a rush toward vulnerable moments. The design therefore tests prospective preterminal structure and held-out rival performance before any attempt at complete-stream adjudication.

The proposed design is observational, staged, and comfort-first. It never induces pain, withholds or delays care, alters medication for research, or manipulates expectation. It asks whether a qualified measurement stack and independently adjudicated endpoint support a bounded inference in one reference class. Protocol content, consent routes, burden limits, and stopping rules are fixed before enrollment. SPIRIT 2025 formally applies to randomized-trial protocols; here its relevant checklist principles are used only as a voluntary rigor framework, alongside current good-clinical-practice principles and the Declaration of Helsinki (Chan et al., 2025; International Council for Harmonisation, 2026; World Medical Association, 2024).

The protocol record fixes every estimand, endpoint, rival, burden ceiling, stopping rule, and public result branch before outcome inspection. Apply §6.7's prospective-specification terminology to that record. A statistician should be able to locate the estimands, a clinician the burden and stop rules, an ethics reviewer the recruitment and consent gates, and a public critic the branches that prevent favorable results from receiving special treatment.

The first decision-relevant question is deliberately not the strong latent claim. No late-window study can identify a whole-life

ledger unless an independent initial-ledger bound or lifelong bridge has already been specified before outcome inspection. This protocol therefore treats terminal-level equivalence as secondary and conditional. Its primary work concerns prospectively defined preterminal structure: reserve interactions, variance behavior, filtered prediction, and out-of-sample comparison with serious rivals.

This restraint makes the study more severe, not less ambitious. A design that separates a reserve-linked dynamic from ordinary homeostasis, endpoint-conditioned bridges, frailty, medication effects, observation shutdown, and none-of-the-models data is more informative than one dramatic terminal curve. The scientific achievement is a result branch that survives comparison without depending on a compelling scene.

Burden accounting belongs inside the analytic design. The relevant questions include who carries the measurement, who can refuse it, which care and privacy costs the observation creates, and which clinical or family labor remains hidden behind the data. A study that cannot represent its own burden cannot interpret the trajectory it helped produce.

The protocol also treats clinical context as a data-generating process rather than background color. Medication, palliative sedation, analgesia, delirium, disease trajectory, family presence, prognostic awareness, cultural death scripts, site routines, device tolerance, staff workload, and sensor quality can all shape the observed trajectory. Some are confounds, some are competing explanations, some are ethical stop signals, and some are reasons

for no-estimate status. None may be ignored because it complicates the story.

The recruitment gate is absolute. No participant may be recruited through crisis, treatment dependency, aid dependency, family pressure, platform belonging, religious belonging, transition participation, fear of losing care, or desire to please an institution. Consent cannot be bundled with ordinary care, public services, spiritual participation, civic participation, operational records, or Artificial Intelligence training permission. A protocol that benefits from dependence has converted vulnerability into leverage. The recruitment record should document the contact pathway and any authority relationship between recruiter and participant, so compliance with this gate can be audited rather than inferred from consent paperwork alone.

The analytic gates are noncompensatory. The endpoint system must avoid defining itself through ledger behavior; the candidate momentary-valence estimator must already be qualified for the clinical context; the reserve proxy must add information beyond severity, prognosis, treatment, and observation quality; and model-recovery studies must show that candidate, rival, hybrid, and open-set worlds can be distinguished at the proposed burden. Later predictive performance cannot rescue an earlier identification failure. Each gate should therefore have a prospectively declared decision rule and failure consequence, preventing downstream performance from relaxing an upstream requirement after results are visible.

Section 14.7 applies the protocol's distinct Unusable evidence, No adequate estimate, Inconclusive evidence, and Model-

ambiguous branches. Each branch identifies a different next measurement or design problem without being counted as support or defeat. The frequency and recorded reasons for these branches should also be reported across sites and cohorts, because recurring no-estimate or model-ambiguous outcomes may indicate a systematic limitation in the measurement or design architecture rather than evidence about the candidate Law itself.

Public reporting is part of the protocol. A protocol-fixed report template should require participant flow, methods, deviations, nulls, adverse feasibility findings, rival wins, model-recovery failures, missingness branches, endpoint revisions, and correction routes regardless of outcome. STROBE supplies the primary reporting discipline for this observational design, while CONSORT 2025 contributes useful flow, harms, and transparency checks for any later randomized extension. No public artifact may convert individual trajectories into spectacle or imply that a measured pattern reveals worth, destiny, holiness, rank, or completion (von Elm et al., 2007; Hopewell et al., 2025). A fixed template also reduces selective emphasis by requiring the same outcome categories whether the preferred model wins, loses, or cannot be adjudicated. Using the same template across supportive, adverse, and unresolved branches also gives independent teams a stable reporting object for replication and correction.

The first protocol estimates named components in one declared reference class under one frozen measurement version, endpoint system, and rival set. That bounded task is the form in which a result can be replicated, challenged, transported, or defeated. A

conclusion that exposes every bridge on which it depends is scientifically stronger than a sweeping claim that cannot identify its measured object. This boundedness makes disagreement productive because critics can challenge a specific bridge without reconstructing the entire program. That specificity lets later studies strengthen, replace, or retire one bridge without pretending that an untouched component has changed status.

Section 14.7 fixes the status and next action for every admissible exit, and the action map is published with the protocol so readers can verify that consequences were not chosen retrospectively. Publishing the map prospectively also gives replication teams a common decision rule for comparing outcomes across sites and versions.

14.1 — The First Decision-Relevant Question

The first study evaluates a bounded late-window process with declared instruments, an endpoint system, and explicit rival models. Its purpose is to identify which observations can most inform the next research decision (Nosek et al., 2018; Chambers & Tzavella, 2022).

The decision-relevant question therefore concerns prospective preterminal structure. The study asks whether a reserve-linked dynamic conjunction improves held-out prediction and rival discrimination in a bounded palliative reference class. That wording keeps the protocol close to what it can actually test. It also keeps the program from treating a beautiful terminal graph as a complete life-level verdict.

A negative answer remains decision-relevant. Candidate failure narrows the route; observation-shutdown explanations return the measurement program to development; endpoint uncertainty gives endpoint work priority; and excess burden stops the clinical study. These branches are the mechanism by which the design remains vulnerable to evidence rather than immune to it.

Placed between prior qualification and later replication, the protocol is a staged feasibility-and-discrimination study. The measurement stack must already be usable in the target clinical state, while transport, calibration, and site dependence remain tasks for independent replication. Its question is whether the program should proceed, narrow, or stop.

Table 14.1. Protocol objects, inferential roles, and claim ceilings.

Protocol object	Primary role	What it can update	What it cannot update
Preterminal dynamic conjunction	Primary confirmatory target	Reserve-linked drift, innovation, variance-shape, and held-out rival comparison in the declared reference class	Exact latent closure, whole-life balance, personal meaning, legal or spiritual status
Terminal equivalence band	Secondary target	Compatibility of the measured estimator with a declared band after origin, endpoint, and observation gates pass	Instrument-free equality or a whole-life terminal ledger
Endpoint model	Eligibility for inference	Whether the stopping system is independent enough to support terminal-window analysis	truth-level stream endpoint as ontology or clinical/legal death
Feasibility and burden outcomes	Gate before expansion	Whether recruitment, observation, and clinical workflow are ethically and operationally viable	Any proposition about the truth or falsity of the Law by themselves

14.1.1 — Primary Scientific Question

The primary scientific question is whether the protocol-fixed reserve-linked dynamic conjunction improves held-out predictive and generative adequacy over strong rivals under a common observation model. It is not whether a person's complete life has reached exact neutrality. The first question has estimands, comparators, failure conditions, and a next action; the second depends on unsolved whole-life and measured-to-latent bridges (Nosek et al., 2018; Chambers & Tzavella, 2022).

The conjunction should include nonredundant components that fail differently: a reserve-by-state interaction, a prospectively specified dispersion restriction, filtered prediction of a held-out interval, observation-process diagnostics, and negative-control behavior. A negative drift, smooth terminal curve, or interval overlapping zero cannot carry the package. The point of heterogeneity is to force bridges, frailty, treatment, selection, and ordinary regulation to explain the same frozen fingerprint.

The phrase held-out prediction matters. A model that explains the terminal pattern after observing it has not earned the same status as a model whose parameters, thresholds, and preprocessing were fixed before the held-out window. The first study must therefore separate training windows, validation windows, and confirmatory windows before any endpoint result is seen.

The primary analysis should be comparative rather than ornamental. The candidate model is tested against ordinary homeostatic mean reversion, endpoint-conditioned bridges, frailty and age decline, medication and sedation processes,

observation shutdown, and flexible time-to-endpoint models. A rival is favored when it meets the protocol-fixed victory criterion; adequate candidate and rival models that remain inseparable yield Model-ambiguous.

The study evaluates aggregate and model-level behavior within a declared reference class. Any individual compatibility estimate is reported as a statistical object under the protocol's participant-protection rules.

A favorable answer would raise the status of one component program. An adverse answer would weaken or retire the tested empirical bridge. This is the right opening question because it can redirect the scientific path without pretending to settle every possible boundary architecture.

14.1.2 — Why Not Begin With Exact Terminal Zero

Exact terminal zero is the wrong opening estimand because the relevant origin is not instrument-given. A late-window measurement system can be standardized, calibrated, and linked, but standardization does not create an independent origin for a whole-life latent cumulative process. A terminal value cannot be interpreted as exact closure unless the origin problem has been solved before the terminal result is inspected (Hernán & Robins, 2020).

The pre-enrollment ledger is also unknown. A person entering a palliative study brings a whole lived history that the study has not observed. Resetting the ledger at enrollment would turn the study into a local-window analysis while allowing public language to

sound like a whole-life claim. The protocol must carry that unknown initial term explicitly.

Endpoint uncertainty creates a third barrier. The conceptual endpoint, the candidate crossing, the detection time, and the retrospective adjudicated time are not the same object. If the analysis chooses the endpoint after seeing which window looks most neutral, the terminal-zero claim becomes circular. The endpoint must be declared through independent channels and stress-tested by jitter and revision rules.

Measurement error creates a fourth barrier. A physiological estimator can concentrate near a band for reasons that have nothing to do with the latent claim: attenuation, sensor removal, sedation, reduced movement, decreased observation intensity, changing calibration, or endpoint-conditioned selection. A measured value near zero is not exact neutrality unless the observation model has survived its own tests.

Beginning with exact zero would also distort ethics. It would tempt the research team to seek the cleanest terminal window, the most complete final record, or the most symbolically satisfying ending. A comfort-first protocol must do the opposite. It must allow missing terminal data, device removal, refusal, sedation, rescue, and no-estimate endings without treating them as obstacles to be overcome at the bedside.

Terminal-regime concentration or equivalence may remain secondary targets, but they do not lead the design. The opening study begins where evidence can be prospective, comparative, and less dependent on arbitrary origin: preterminal dynamics,

reserve coupling, observation-process behavior, and rival discrimination.

14.1.3 — Component Claims Under Test

The protocol decomposes the program into independently reportable components. The measurement component tests whether the candidate momentary estimator identifies its protocol-fixed interval target in the clinical context; the reserve component tests an independently constructed capacity state; the endpoint component tests the adjudication firewall; the observation component tests informative visibility; and the dynamical component compares held-out candidate and rival performance. Each component keeps its own estimand, error rule, and verdict ceiling (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

The dynamical component asks whether the candidate reserve-linked model improves prospective prediction over rivals. Its result informs further development of the mathematical realization in the declared setting.

Each component has a corresponding decision. Measurement failure directs instrument development; endpoint failure directs stopping-rule revision; rival superiority directs review of the tested mechanism.

Each component receives its own status. Recruitment may be feasible while measurement recovery fails; the endpoint system may be usable while reserve coupling disappears; a reserve association may survive while the observation process explains the same pattern. Those are different scientific outcomes with

different repair paths. Compressing them into one favorable or unfavorable headline would erase the information the protocol was built to obtain.

Component separation also protects participants. If a burden measure fails, the study stops or narrows before additional terminal recruitment, even if the statistical model looks promising. Ethical feasibility is not a secondary appendix to scientific success. It is a noncompensatory prerequisite.

The final report names every component that moved and every component that did not. If only a measurement bridge survives, the public sentence identifies that bridge, the population, the protocol, the rivals, and the unresolved prerequisites. Component-level evidence is most useful when it is not inflated into a statement about the whole Law.

14.1.4 — Reference Class

The first reference class should be narrow enough to be clinically coherent. A plausible starting point is adults receiving specialist palliative or hospice care at participating sites, with prognosis windows appropriate to observation, and with care teams able to support optional, low-burden participation. That is a reference class, not a human universal (Nosek et al., 2018; Chambers & Tzavella, 2022).

The eligible setting must allow research participation to remain separate from crisis, unstable care access, coercive dependence, and urgent clinical decision making. Sites unable to protect comfort, refusal, or staff attention are outside the first protocol even when their populations are scientifically important. This

boundary limits what can be learned now; it does not diminish the importance of people or conditions excluded from the initial design.

The reference class must include site, diagnosis, prognosis, care model, language, communication access, medication context, and consent pathway. A hospice site with strong family support is not interchangeable with an emergency department, a prison infirmary, an unsupported home death, or a low-resource region. Transport beyond the reference class must be earned later.

The registry should identify the populations and endpoint pathways the study does not represent. Sudden deaths, pediatric populations, nonhuman streams, artificial candidates, resuscitation pathways, deaths outside organized care, and settings without qualified endpoint review require distinct designs. Naming those absences prevents a local result from being translated into a universal statement by omission.

Reference-class discipline is part of fairness. People outside the first study should not be absorbed into its public conclusions. The hidden burden of overgeneralization falls on those whose conditions were not represented, whose access differed, or whose endings did not resemble the sampled pathway.

The first study reports its findings within the declared clinical population. It identifies the empirical bridge tested, the evidence supporting or challenging that bridge, and the populations or settings in which replication would be most informative.

14.1.5 — Decision Relevance

Decision relevance means that every major result changes a predeclared scientific action. A favorable component result may justify independent replication; a rival victory may retire the candidate realization in that reference class; measurement failure may redirect resources to instrument development; and an unacceptable burden profile may close terminal recruitment. The protocol therefore registers its action map as carefully as its estimands (Chan et al., 2025; Hernán & Robins, 2020).

Decision relevance also requires bounded ambition. This protocol is a gate, not a container for all future public belief. It asks whether the next gate should open, close, or be rebuilt. A design that claims to settle the theory would create more ethical and interpretive burden than its evidence could justify.

The decision branches should be written in the prospective specification rather than composed for the press release. If the candidate wins on a subset of estimands but loses on observation-process checks, the public wording must be already constrained. If model recovery is inadequate, the result is unusable even if the observed pattern looks favorable.

The protocol must define resource consequences. A failed pre-enrollment recovery benchmark blocks the proposed design. Investigators should distinguish a remediable lack of resolution from structural nonidentification, then test any revised sample size, cadence, or measurement plan in simulation under the same burden limits. Recruitment resumes only after the redesigned benchmark passes. An adverse burden result suspends expansion,

and a rival victory cannot be answered by adding flexible candidate parameters without declaring and testing a new version.

A decision-relevant protocol makes humility operational. It is not enough to say that negative results will be welcome. The study must specify what negative results do. It must let the Law become smaller, less public, more conditional, or scientifically retired in the tested form.

14.1.6 — What the Study Cannot Establish

The first protocol evaluates measurement, endpoint, route, and comparator questions within its declared reference class. The registry and study reports identify those questions and the evidence used to answer them (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

The study reports late-window dynamics. Any complete-history extension uses an independently qualified initial-ledger bound or lifelong bridge specified before data inspection. The report should separate conclusions supported by the observed late window from those that depend on the additional historical bridge.

Participant data contribute to scientific models under consent and governance. Personal meaning remains with the participant. Optional qualitative accounts can accompany the scientific record without being converted into latent-state labels the participant never authorized.

It cannot authorize clinical care changes. Comfort care, pain relief, sedation, rescue, communication, and family support remain prior to the study. The protocol's conclusions cannot justify delaying care, increasing measurement, or altering

treatment in pursuit of cleaner data. Care events that interrupt measurement should be recorded as part of the observed process, not treated as deviations to be corrected away.

Applications in civic design, government, law, religion, transition, artificial intelligence, and ordinary life are evaluated through the evidence and authority appropriate to each domain. A scientific result can constrain factual premises in those domains without transferring the study's decision rule or authority into them.

A study cannot rescue itself by changing scope after failure. The public result must name the failed object, and any new hypothesis must carry a new version. That rule lets the protocol teach without becoming self-protective. Versioning also protects negative evidence, because the failed claim remains identifiable even after a successor is proposed.

14.2 — Population, Enrollment, and Burden

Chapter 3 governs consent, dependence, and care priority. This protocol adds only the local conditions that non-crisis recruitment be independent of care and that participant, family, and staff burden be monitored as part of study feasibility.

Table 14.2. Enrollment, consent, and stop conditions.

Enrollment element	Required declaration	Stop or narrow condition
Reference population	Diagnosis groups, prognosis window, care setting, language access, communication support, and site capacity	Population too heterogeneous for the planned estimands or too dependent for free refusal
Consent pathway	Timing, key information, support person role, capacity review, re-consent, pause, withdrawal, and data-retirement options	Consent cannot remain ongoing, supported, revocable, and unbundled from care
Burden budget	Maximum prompts, sensor hours, family contacts, staff minutes, and adverse burden thresholds	Observed burden exceeds budget or concentrates in a vulnerable subgroup

Enrollment element	Required declaration	Stop or narrow condition
No-crisis recruitment	Recruitment cannot occur through crisis, care dependency, aid dependency, family pressure, platform belonging, religious belonging, or fear of losing care	A site cannot operationally separate participation from access, safety, belonging, or treatment.

14.2.1 — Inclusion and Exclusion

Inclusion criteria should be scientific and safety criteria. They should name the clinical state, expected observation window, capacity pathway, communication support, measurement feasibility, care-setting stability, and site readiness needed for the estimands. Convenience, expected fit, dramatic terminal proximity, or anticipated compliance must not drive inclusion (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Exclusion criteria should prevent harm and noninterpretability. A person should not be invited if participation would predictably burden comfort, disrupt care, intensify family pressure, overload staff, or produce data too confounded by crisis to answer the primary question. Exclusion from the first protocol is not exclusion from dignity or from future, better-designed science.

Diagnosis and trajectory strata should be declared before enrollment because they generate different symptom paths, medication exposures, observation opportunities, and endpoint processes. Cancer, neurodegenerative disease, organ failure,

dementia-related pathways, and mixed frailty should not be pooled merely to enlarge the sample. Partial pooling may borrow information where justified, but it must not turn composition change into an apparent terminal signature.

Communication access is part of eligibility design, not an afterthought. Nonverbal communication, augmentative devices, interpreters, caregiver-supported communication, fluctuating attention, and sensory disability require explicit routes for participation and uncertainty. When a site cannot support those routes, the reference class must narrow transparently rather than classifying inaccessible participants as ordinary missing data.

Exclusion rules must be audited for unfair patterning. If disability, language, poverty, race, site resources, family structure, or home-care access repeatedly exclude people, the report should say so. That finding may reveal an institutional burden before it reveals anything about the Law.

The final inclusion table should be readable by a clinician and a participant advocate. A criterion that cannot be explained in ordinary language may still be scientifically necessary, but it must be translated in consent and reviewed for hidden coercion or hidden exclusion.

14.2.2 — Late-Window Enrollment

Late-window enrollment is scientifically limited by design. It observes a fragment of a trajectory after long unknown history, after disease selection, and often after major treatment and family changes. The protocol must state that its window begins at

enrollment and that the pre-enrollment ledger remains unknown (Emanuel et al., 2000; Wendler & Grady, 2008).

The enrollment time should be recorded as a scientific event, not merely an administrative date. The point at which a person becomes eligible may depend on diagnosis, prognosis, hospice referral, insurance, family capacity, clinician judgment, cultural expectations, and site practice. Those variables shape the observed terminal window.

Late enrollment also creates survivor and access selection. People who reach a study site with time, capacity, and support differ from people who die suddenly, without care, or under emergency intervention. The protocol should model selection into the reference class and avoid public language that treats the sample as the human end-of-life field.

The measured cumulative estimate in a late window is an increment. The study may evaluate changes, dynamics, variance, observation quality, and endpoint timing within that increment. It may not speak as if enrollment resets the life ledger. A clean window is not a clean life total.

Pre-enrollment context is limited to variables required by the estimand and rival models. Diagnosis, medication history, recent symptom trajectory, major care transitions, and hospitalizations may be necessary. The collection plan specifies how each requested field contributes to the analysis.

A late-window study can still be valuable. It can test whether the candidate dynamics appear during a vulnerable and clinically

important period. Its honesty comes from saying exactly that, not from pretending to have reached beyond what it observed.

14.2.3 — Capacity and Consent

Consent is a continuing state, not a signature preserved from enrollment. The protocol should define supported decision procedures, re-consent checkpoints, participant-controlled pauses, and observable assent or dissent markers for periods of fluctuating capacity or communication. Uncertainty about willingness suspends research activity; it does not authorize investigators to infer continued permission from earlier cooperation (International Council for Harmonisation, 2026; National Commission, 1979).

The key information should be understandable under illness and sufficient for a considered decision. It should explain that this observational study is not individualized treatment, does not guarantee personal benefit, and will not determine care access or interpret a participant's life. It should name the burden, data collected, optional components, and right to stop without explanation or loss of ordinary care.

Surrogates require careful boundaries. A legally authorized representative may permit certain research procedures when the participant cannot decide, but surrogate permission should not turn a person's dying process into data if prior wishes, current distress, or clinical judgment argue against continuation. The comfort-first rule remains superior to completeness.

Capacity fluctuation should be recorded without becoming a worth judgment. A participant may be fully dignified, clinically

protected, and scientifically unclassifiable at the same time. The protocol must not treat temporary incapacity as consent to observe more passively or as permission to infer willingness from silence.

Consent should be granular. Physiological observation, self-report prompts, chart linkage, family context, audio or narrative notes, data sharing, future reuse, model training, and publication should be separately permissioned where applicable. A person may accept one form of participation and reject another.

Withdrawal should include data-retirement options declared in advance. Some derived aggregate results may not be retractable after analysis, while identifiable future use can be stopped. Consent materials must state these differences plainly so refusal and withdrawal remain real rather than ceremonial.

14.2.4 — No-Crisis Recruitment

No-crisis recruitment should be enforced by timing, personnel, and documentation. Invitations are deferred during acute fear, unresolved treatment decisions, unstable symptoms, family conflict, or any circumstance in which declining could reasonably feel unsafe. Screening logs should record deferral without converting vulnerability into noncompliance (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Recruiters should be independent from direct care whenever possible. When clinical staff introduce the study, the script should separate care from research and transfer the conversation to trained research personnel. A nurse, physician, chaplain, or aide

should not carry the burden of persuading a patient while also providing care.

Recruitment language should describe the study without invoking legacy, heroism, proof, special access, or the promise of helping humanity through one person's death. The participant may be told why the question matters, but no invitation should make scientific usefulness feel like a final obligation. A person receiving end-of-life care owes the program neither data nor meaning.

Timing matters. Recruitment should not occur during uncontrolled symptoms, emergency decisions, family conflict, medication changes that impair comprehension, or immediately after receiving devastating news. The protocol should include a cooling-off interval where clinically appropriate, and it should let sites abstain when timing cannot be made fair.

Recruitment monitoring should include refusal rates, reasons for refusal where voluntarily offered, staff concerns, family concerns, and participant-reported pressure. Low refusal can be a warning sign, not merely a recruitment success. If a site produces unusually high acceptance under dependence, monitoring should investigate pressure rather than celebrate efficiency.

No-crisis recruitment protects science as well as people. A sample recruited through desperation or belonging will carry expectancy, selection, and trust confounds into the data. The study cannot learn fairly from people it has pressured into helping.

14.2.5 — Bystander and Family Protections

Terminal research often collects bystander data even when it does not intend to. Family presence, visitor timing, caregiver distress,

staff interaction, room changes, and relational events may explain signals. The protocol must separate participant data from family and bystander data rather than absorbing everyone near the bed into one research object (Emanuel et al., 2000; Wendler & Grady, 2008).

Family members should not become unpaid instruments. If the study asks for context reports, timing notes, or post-event narratives, it must define burden, consent, refusal, support, and withdrawal for those contributors. A grieving family member's testimony may be valuable, but it is not automatically available because the participant consented.

Bystander data should be minimized and purpose-limited. The protocol can record relational context categories where necessary, such as alone, family present, clinician present, procedure occurring, or conflict noted, without collecting names, private conversations, or interpretable family stories unless separately consented and scientifically necessary.

Family pressure can work in both directions. Some families may encourage participation because they want meaning; others may oppose participation because they fear exploitation. The protocol should not recruit through either pressure. It should let participant preference, capacity rules, and comfort govern without turning family disagreement into data spectacle.

Staff protections belong here too. Nurses, aides, social workers, chaplains, physicians, and volunteers can experience added workload, grief exposure, moral stress, and conflict. The burden

budget must include staff time and staff refusal. A site that cannot protect staff should not run the protocol.

Bystander protections also improve validity. Family anxiety, staff workload, clinician behavior, and room events can alter physiology, missingness, and care timing. Treating those effects as invisible would burden the science with false simplicity.

14.2.6 — Participant and Staff Burden Budget

The burden budget should specify maximum prompts, sensor-wear intervals, device checks, family contacts, research visits, staff documentation, and quality-control requests, together with cumulative limits across days. It should also name prospective pause, reduction, and withdrawal triggers for fatigue, distress, device intolerance, family concern, staff overload, clinical disruption, or repeated refusal. Observation is itself an intervention, so its demands must be fixed before enrollment rather than inferred after participants have absorbed them (International Council for Harmonisation, 2026; National Commission, 1979).

Procedure counts alone cannot represent burden. A one-minute prompt during severe breathlessness may be heavier than a longer conversation on a stable day, and the same device may become intolerable after dyspnea, skin fragility, fatigue, delirium, or family crisis. Burden should be updated prospectively from participant report when available, bedside observation, device-tolerance records, symptom state, clinician judgment, and staff workflow indicators (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Staff burden shares the budget. Aggregate time, emotional labor, documentation, device maintenance, overnight coverage, workflow interruption, and independent consent support by role and site. Feasibility built on unrecorded work by nurses, aides, or family members will not transport ethically or scientifically.

The burden rules are noncompensatory. Strong data quality, an attractive early signal, or additional power cannot purchase permission to exceed comfort limits, add sensors, increase prompts, or prolong observation without renewed review. Stopping for excess burden is a usable result about where the design should not go.

Burden-triggered pauses and withdrawals also shape the data-generating process. Observation may disappear precisely when distress or incapacity rises, so an ethically correct stop can still change the population represented by the estimate. Reports should show the relevant burden threshold, the observations lost after it was crossed, and the reference class to which the remaining result can still apply (Little & Rubin, 2019; Hernán & Robins, 2020).

Enrollment flow should preserve the denominator from first approach through screening, deferral, exclusion, consent, first measurement, withdrawal, and analysis. Eligibility, temporary deferral, inability to approach safely, site limitation, communication barrier, device intolerance, and active refusal require distinct neutral codes. Collapsing them can make a restrictive design look broadly acceptable and can hide avoidable barriers to participation (Emanuel et al., 2000; International Council for Harmonisation, 2026).

Consent remains longitudinal. Capacity, fatigue, clinical state, family presence, and willingness may change during the observation window, so the protocol must state when permission is reconfirmed, how assent or dissent is recognized, what happens to previously collected data after withdrawal, and which procedures stop immediately when a stable preference can no longer be expressed. A design that enters a vulnerable person's day must also know how to leave it cleanly (Emanuel et al., 2000; Wendler & Grady, 2008).

14.3 — Measurements, Endpoint, and Data Flow

Measurement development begins in lower-risk settings. Before a palliative cohort is considered, the candidate momentary-valence estimator, interval valence-contribution rule, reserve proxies, endpoint channels, preprocessing, quality controls, and provenance system are declared and qualified for the target clinical state (FDA-NIH Biomarker Working Group, 2016; International Council for Harmonisation, 2026).

Data flow is not administrative plumbing. It is the route by which a latent question becomes a measured claim. Every transformation changes what can be said. Raw sensor units, artifact rules, windowing, synchronization, missingness marks, endpoint adjudication, and statistical estimation must therefore be visible before the first participant is enrolled.

The protocol's measurements must also preserve conflict. If a reserve proxy declines while observation quality collapses, that is not a clean reserve signal. If the candidate momentary-valence estimator moves toward neutrality while sedation increases, the treatment context is not background. If endpoint channels and cumulative-value channels become coupled through preprocessing or analyst decisions, the independence firewall has failed. Predeclared conflict rules should determine when discordant channels remain separate, trigger sensitivity analysis, narrow the estimand, or require abstention rather than allowing disagreement to disappear inside a composite score.

Table 14.3. Measurement and data-flow locks.

Data layer	Locked before enrollment	Reason
Candidate physiological momentary-valence estimator	Context of use, sensors, raw units, transforms, sign anchors, artifact rules, and cadence	Prevents the estimator from being tuned to terminal results
Interval valence-contribution path	Rate interpretation, integration window, posterior path draws, and uncertainty propagation	Prevents summing scores as if they already had additive ledger units
Reserve proxies	Independent development, biological rationale, units, confounds, proxy error, and abstention rules	Prevents reserve from being fitted to force closure
Endpoint channels	Unity and endpoint proxies, stopping rules, confirmation windows, analyst separation, and jitter plan	Prevents the endpoint from being defined by ledger behavior
Treatment context	Medication, sedation, analgesia, delirium, procedures, care changes, and family presence	Allows rivals and clinical confounds to explain patterns
Provenance	Versioned acquisition, preprocessing, quality control, missingness marks, and audit trail	Allows later readers to know exactly which data path produced the result

14.3.1 — Candidate Momentary-Valence Estimator Context of Use

The candidate momentary-valence estimator must be qualified for the target clinical context before terminal use. A physiological composite that behaves adequately in healthy volunteers, general outpatients, or laboratory stress tasks is not automatically valid in hospice, advanced illness, sedation, delirium, pain, dyspnea, fatigue, or low-movement states. Qualification must address the precise construct, population, state range, intended interpretation, and conditions under which the estimator abstains (FDA-NIH Biomarker Working Group, 2016; International Council for Harmonisation, 2026).

The context-of-use statement should identify the target population, state range, sensors, raw units, transformations, window lengths, artifact rules, sign anchors, calibration route, missingness rules, and prohibited inferences. It should say what the candidate momentary-valence estimator is intended to estimate, at what temporal resolution, and what it is not allowed to decide.

The candidate momentary-valence estimator remains a declared physiological estimator rather than a name for the entire evidence stack. Self-report, behavior, clinical notes, medication records, family context, and relational observations may validate, challenge, or explain it, but their inclusion does not collapse distinct channels into one index. A broad witness stack may be scientifically valuable precisely because disagreement remains visible.

Apply the performance profile in Section 9.1.4 to the relevant terminal clinical states. Reliability alone does not establish sign identification, state robustness, or terminal-state transport.

The candidate momentary-valence estimator needs explicit invalidity rules. Loss of sign-bearing information, artifact levels beyond qualification, sensor intolerance, model-version mismatch, failed calibration, or a clinical state outside the declared context of use should suspend the estimator for that interval. Invalid windows remain invalid; they are not replaced by the model's expected trajectory or silently imputed as neutral.

A well-defined context of use strengthens both inference and care. It specifies when the estimator is qualified, how participant testimony enters interpretation, and when a different measurement approach or requalification is needed.

14.3.2 — Reserve Proxies

Reserve proxies must be developed independently of the target trajectory, not tuned to make the ledger close. Candidate channels may include functional decline, frailty markers, autonomic flexibility, physiological variability, performance status, or disease-specific indices, but each requires a declared context of use, analytical validation, clinical interpretation, and failure rule. A proxy that mainly measures observation quality or prognosis cannot silently become reserve (FDA-NIH Biomarker Working Group, 2016; Goldsack et al., 2020).

Each proxy needs units, cadence, uncertainty, confounds, and context of use. Medication, disability, disease type, age, hydration, infection, pain, care quality, sleep, nutrition, family

support, and sensor gain can all affect apparent reserve. The protocol should treat those effects as modeled pathways, not as noise to average away.

Reserve should not be inferred from the ledger outcome. If a proxy is selected because it best aligns terminal compression with neutrality, the design is circular. Reserve construction must be frozen before confirmatory data and tested by negative controls and held-out prediction.

Reserve-proxy uncertainty should propagate into every interaction, slope, and route comparison that uses it. Classical error may attenuate associations, differential error may manufacture them, and shared inputs may create circular agreement with the affect estimator. When the proxy cannot distinguish the candidate route from disease severity or prognosis, the result is no estimate or a narrower observed-variable claim, not weak support.

Reserve proxies are used only for scientific modeling under consent and governance. The protocol excludes their use in triage, insurance, employment, care eligibility, family decisions, religious interpretation, or public ranking.

The strongest reserve design is independently justified, clinically interpretable, versioned, explicit about confounds, and capable of abstention. A reserve curve becomes useful when its interpretation survives measurement-error and shared-driver checks.

14.3.3 — Unity and Endpoint Channels

Unity and endpoint channels define the stopping architecture for the study. They should be separated from cumulative-value estimation channels wherever possible, and any unavoidable dependence must be modeled and reported. The endpoint cannot be an algebraic consequence of the candidate momentary-valence estimator or of the measured cumulative estimate; otherwise neutrality can be manufactured by the stopping rule.

The channel set may include clinical neurological observations, responsiveness, perturbational or electrophysiological markers where clinically appropriate, communication evidence, medication and sedation context, and charted care events. The protocol should declare which channels support candidate crossing, which support confirmation, and which only contextualize uncertainty.

Endpoint adjudication should be blinded to confirmatory ledger outcomes. Analysts constructing endpoint thresholds should not tune them against terminal interval valence contribution estimate behavior. If the endpoint threshold is chosen because it improves apparent neutrality, the endpoint firewall has failed even if the variables look separate on paper.

Endpoint adjudication should operate as a versioned state machine rather than a single threshold. Confirmation windows, reversible loss, return, device cessation, withdrawal, sudden death, low-signal states, and unresolved classifications each require an allowed transition and an evidentiary status. No one

observation can convert a protocol state into an irreversible truth-level endpoint.

Endpoint channels should also be clinically subordinate. Emergency intervention, comfort care, and communication with family are not delayed to preserve endpoint cleanliness. If care disrupts measurement, the protocol records a care interruption or no-estimate condition. It does not treat care as missing data to be repaired by imputation.

The endpoint system is the study's tautology firewall. A terminal study becomes meaningful only if the endpoint is independent enough that ledger behavior could have surprised it. If the endpoint follows the ledger, no terminal result is evidence.

14.3.4 — Treatment and Care Context

Treatment and care context must be recorded as time-varying data. Analgesia, palliative sedation, anxiolytics, antipsychotics, opioids, oxygen, ventilation, procedures, delirium treatment, repositioning, family visits, chaplain visits, and changes in goals of care can alter physiology, reports, observation availability, and endpoint classification at different lags. The protocol should predeclare which variables are exposures, confounders, mediators, observation modifiers, or competing events rather than treating the entire clinical record as an undifferentiated adjustment set (Chan et al., 2025; International Council for Harmonisation, 2026).

Medication, analgesia, sedation, and other treatment changes are humane clinical exposures as well as sources of scientific variation. They should be represented with timing, indication,

dose, route, and plausible lag rather than removed as inconvenient noise. Cleaner evidence obtained by withholding comfort is outside the design; the analysis must adapt to care, never the reverse.

Care quality and site practice should also be modeled. Different sites may handle dyspnea, pain, agitation, family communication, sedation, hospice referral, and device removal differently. A site effect may explain terminal patterns more plausibly than the candidate model. If so, that rival explanation should stand.

Treatment data should be captured with enough temporal resolution to align with observation windows. A medication dose recorded only by calendar day may be too coarse for physiological windows. Conversely, overburdening staff with minute-level coding may exceed the burden budget. The protocol should choose a cadence that is scientifically useful and clinically tolerable.

The analysis plan should include medication and care-context sensitivity models. It should ask whether apparent variance compression, drift, or endpoint prediction survives after accounting for sedation, analgesia, delirium, procedures, and observation quality. A pattern that vanishes under care-context modeling is not a failed life. It is sensitive to those modeling assumptions; a rival-explained verdict requires the declared identification and model checks.

Care context is where this burden-focused approach becomes concrete. Hidden burden can be carried by symptoms, by families,

by clinicians, and by research staff. Recording context makes that burden reviewable without making care subordinate to the theory.

14.3.5 — Observation Marks

Every planned observation should receive a mark. Completed, failed, refused, paused, removed, clinically interrupted, family-deferred, device-error, low-quality, sedated, transferred, and post-endpoint observations are different events. Treating them all as missing would erase the pathway by which evidence became available or vanished (Aalen et al., 2008; Andersen et al., 1993; International Council for Harmonisation, 2026).

Observation marks should be chosen before enrollment and kept simple enough for staff use. A mark system that demands too much judgment at the bedside becomes another burden. The protocol should favor a small controlled vocabulary with optional free-text notes for clinically important exceptions.

Marked observation is essential for rival testing. Observation shutdown can mimic variance compression. Device intolerance can correlate with severity. Refusal can indicate fatigue, distress, privacy preference, or family context. Clinical interruption can cluster near endpoints. Without marks, the model may mistake observation process for terminal dynamics.

Observation marks also protect participants. Refusal is not noncompliance. Device removal is not loss to be lamented by the study. A paused prompt is not missing evidence to be recovered by pressure. The marks should preserve participant agency as part of the data record.

Observation should be modeled as an outcome-bearing process. The analysis should estimate whether response, device quality, removal, staff contact, or mark type depends on the measured path, reserve proxy, clinical state, treatment, or endpoint proximity. If the observation model accounts for the apparent candidate signature, that explanation receives its own protocol-fixed verdict rather than being hidden inside a sensitivity appendix.

The public report should include an observation-flow display showing scheduled opportunities, completed measurements, refusals, pauses, device failures, clinically interrupted intervals, and unknown absences across the terminal window. The display should reveal both quantity and reason, allowing readers to see which portions of the claimed trajectory were directly observed and which depend on assumptions.

14.3.6 — Versioned Data Pipeline

The data pipeline must be versioned from acquisition to public report. Sensor firmware, calibration files, preprocessing code, artifact thresholds, quality rules, candidate momentary-valence estimator versions, reserve-proxy algorithms, endpoint-adjudication rules, analysis code, and reporting templates need stable identifiers before the confirmatory run. The final consolidated good-clinical-practice standard and reproducibility literature make this traceability part of the study, not clerical decoration (International Council for Harmonisation, 2026; National Institute of Standards and Technology, 2020; Nosek et al., 2018).

Development, validation, and confirmatory environments should be separated. Analysts may not tune preprocessing, thresholds, or feature definitions while viewing confirmatory endpoint outcomes; cross-fitting, held-out thresholds, locked containers, or independent analysis teams may be required. If an instrument or transformation changes after early inspection, the original output must remain recoverable and the migration must be tested under a declared plan rather than silently rescore history.

One authoritative clock map should connect prompts, wearable samples, clinical events, medication administrations, room changes, device restarts, and endpoint adjudication. Original timestamps, synchronization tolerances, rounding rules, time-zone handling, and every alignment transformation should be retained. A terminal-window pattern that moves when clocks are corrected is a lineage finding, not a stable biological signature.

Hardware and software changes require bridge evidence. Firmware updates, sensor substitutions, battery-saving modes, altered sampling frequencies, vendor preprocessing, and device replacement can create discontinuities that resemble physiological change. Every observation should therefore carry the active hardware, software, calibration, preprocessing, and quality-control version, and a channel should be withdrawn when old and new versions cannot be linked (U.S. Food and Drug Administration, 2023; International Council for Harmonisation, 2026).

Derived variables need feature-level provenance. A reserve proxy, movement index, autonomic composite, or quality flag should identify its raw channels, window, unit, normalization population,

missingness rule, device version, code commit, and validation rule. A machine-readable dictionary should bind every analysis field to that lineage so an independent team can distinguish a scientific feature from an implementation artifact.

The pipeline records access, consent, retention, and deletion. Raw identifiable signals, derived features, family context, clinical records, and model-training permissions have distinct consent requirements. Audit who viewed or altered each layer, and minimize collection at the outset.

Test reconstruction from a published result through the frozen dataset and feature table to source observations, consent, access conditions, and every transformation. Before enrollment, use synthetic records with clock conflicts, duplicates, missing marks, device and consent changes, and endpoint reversals to expose breaks in that chain.

Versioning preserves the history needed for correction. If an instrument, calibration, feature, or code path proves unsafe, biased, unstable, or nontransportable, identify every dependent estimate, figure, and public artifact and specify the required reanalysis or withdrawal.

14.4 — Estimands and Rival Models

An estimand prevents the data from choosing their own question. Before enrollment, each target should name population, treatment or exposure condition when relevant, time origin and window, outcome or latent object, intercurrent-event strategy, endpoint relation, summary measure, and uncertainty. Those fields align the scientific question with the analysis and stop a favorable pattern from being upgraded after inspection (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

The primary estimands should be preterminal and comparative. They ask whether reserve-linked dynamics, innovation behavior, variance shape, and filtered endpoint prediction improve out-of-sample performance against serious rivals. Terminal equivalence may remain secondary, but only after origin, endpoint, observation, and whole-life limitations are carried honestly.

Rivals are executable generators, not a list of objections. Homeostasis, conditioned bridges, frailty, age, medication, sedation, observation shutdown, and hybrid models can reproduce different parts of the pattern. The candidate advances only when a protocol-fixed conjunction outperforms them under common data access, tuning resources, scoring rules, and abstention options.

$$\mathcal{E} = (\beta_r, \gamma_v, \Delta S, \eta_{obs}, \kappa_{burden})$$

Table 14.4. Estimands, rivals, and failure conditions.

Estimand family	Primary object	Key rival pressure	Failure condition
Relaxation and innovation	State-dependent drift and stochastic innovation under filtered windows	OU homeostasis, medication effects, endpoint-conditioned bridge	Parameters unstable, not recoverable, or no better than rivals
Variance shape	Reserve-space and time-domain variance relations under proxy-error correction	Frailty, observation attenuation, shared disease severity	Slope explained by noise floor, proxy drift, or observation process
Prospective endpoint prediction	Held-out prediction using information available before the endpoint	Generic time-to-death, age, prognosis, site practice	No gain beyond clinically simpler predictors
Observation process	Availability, quality, and missingness as data-generating processes	Observation shutdown, device intolerance, care interruption	Observation model explains apparent compression
Discriminating conjunction	Joint score across components that fail against different rivals	Flexible hybrid rivals and none-of-the-models data	Conjunction redundant, nonseparable, or unrecoverable in simulation

14.4.1 — Relaxation and Innovation Estimands

Relaxation estimands describe whether the measured path shows state-dependent movement toward the candidate neutral region

under conditions declared before analysis. Innovation estimands describe the size and structure of the unpredictable component after modeled drift, reserve, treatment, and observation processes are accounted for. Both are preterminal targets (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

The primary relaxation and innovation estimands must use information available in the declared pre-endpoint filtration. Retrospective smoothing may support reconstruction or sensitivity analysis, but it cannot carry a claim about prospective dynamics. Filtering, forecasting, or cross-fitting should therefore isolate the state available before the held-out interval.

A useful estimand might ask whether drift toward neutrality strengthens as independently measured reserve declines, after accounting for site, diagnosis, medication, sedation, delirium, care changes, and observation quality. The exact model can vary, but the target must be declared before seeing terminal behavior.

Innovation matters because a candidate closure story should not only move the mean. It should also shape residual variability in ways rivals cannot easily reproduce. If the residual variance collapses because sensors stop capturing movement or because sedation flattens physiology, the innovation estimand belongs to an observation or medication rival, not to the candidate model.

Parameter recovery is mandatory. A complex drift model that cannot recover its own parameters in synthetic data should not be trusted on terminal data. Before enrollment, the team should show that the planned cadence, windowing, missingness, and proxy

error still allow the relaxation and innovation quantities to be estimated with decision-relevant precision.

The public result should separate drift support from innovation support. A favorable drift estimate without innovation structure is component support at most. Innovation compression without recoverable drift may be observation shutdown. The conjunction is stronger than either component because each can fail differently.

14.4.2 — Variance-Shape Estimands

Variance-shape estimands ask how measured cumulative estimate uncertainty and dispersion change as reserve declines and as endpoint approaches. They must separate reserve-space relations from time-domain scaling. A slope near two in reserve space means something different from a time-domain exponent, especially when reserve declines nonlinearly (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Conditional variance in reserve space, variance indexed by a prospectively estimated horizon, and variance in windows aligned retrospectively to endpoints are distinct estimands with different risk sets and artifacts. The protocol should prospectively specify one primary object and treat the others as bridge or sensitivity analyses. Otherwise endpoint alignment can create apparent compression in a process with no native reserve-coupled structure.

Measurement noise must be included in the estimation model. An additive noise floor can bias apparent variance compression and flatten slopes. More data may sharpen a biased estimate. The

protocol should include noise-floor characterization and deconvolution before interpreting variance shape as support.

Reserve proxy error also matters. If the reserve proxy is noisy or shares disease severity with the measured cumulative estimate, conditional variance estimates can be attenuated, inflated, or shifted. The analysis should report sensitivity to proxy-error assumptions and abstain when those assumptions dominate the target margin.

Variance shape contributes one member of a joint fingerprint. OU homeostasis, endpoint-conditioned bridges, frailty, medication, observation shutdown, and sensor-gain changes can each reproduce pieces of compression. Its value comes from how it combines with drift, dependence, negative controls, and observation diagnostics, not from uniqueness in isolation.

A variance result should be reported with its rival explanation. A favorable reserve-space relation plus negative observation controls is stronger than a favorable time-aligned plot. A variance pattern that appears only after smoothing or endpoint-conditioned alignment requires explicit artifact and rival checks. A valid analysis may be confirmatory if prospectively specified for its declared estimand; it cannot establish a prediction made using only earlier information.

14.4.3 — Prospective Endpoint Prediction

Prospective endpoint prediction asks whether filtered preterminal features improve prediction of independently adjudicated endpoint states. The word prospective is doing real work. The features must be available before the endpoint window they

predict, and the endpoint rules must be declared without looking at ledger behavior (Lipsitch et al., 2010; Wilson & Collins, 2019).

Candidate features should be evaluated against a transparent clinical baseline containing diagnosis, age, site, performance status, recent decline, symptom burden, treatment, sedation, prognosis, and observation quality where available. Improvement is assessed in held-out data with calibration as well as discrimination. Failure to beat that baseline is informative, and it cannot be converted into support by emphasizing an in-sample coefficient.

Prediction should be evaluated with proper scores and calibration, not a favorable coefficient. The protocol-fixed report should provide held-out log or Brier score where appropriate, calibration-in-the-large, calibration slope, discrimination, horizon-specific error, and comparison with a transparent clinical baseline. A feature that improves rank ordering while degrading calibration has not earned a general endpoint-prediction claim (Gneiting & Raftery, 2007; Steyerberg, 2019).

The prediction target should be clinically and scientifically interpretable. It may be time to candidate endpoint, probability of transition into a confirmed endpoint state, or probability of no-estimate classification within a window. It should not be a dramatic label such as final balance, final meaning, or completed trajectory.

Calibration of predictions matters as much as discrimination. A model that ranks participants but is badly calibrated near the endpoint may be scientifically interesting but not confirmatory.

The protocol should report calibration, discrimination, decision-relevant error, and uncertainty separately.

Prospective prediction can fail cleanly: no added predictive information narrows the program; success requiring smoothing or endpoint tuning remains exploratory; superior clinical predictors give the clinical model victory.

14.4.4 — Observation-Process Estimands

Observation-process estimands test availability, quality, and modality as primary rivals. Weakening may bring sensor failure, reduced movement, refused prompts, device removal, or clinical interruptions that generate the apparent signature (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

The probability and type of observation should be modeled from prior measured history, clinical state, treatment, site, family presence, device status, reserve proxy, and endpoint proximity. Complete-case analysis is defensible only under assumptions made explicit and stress-tested. When observation depends on the latent state under study, selection is part of the scientific model rather than a removable inconvenience.

Model how contact quality, motion artifact, ambient conditions, battery state, staff workload, device placement, and care procedures alter signals. A smooth terminal path may reflect a quieter device, not a quieter ledger.

The observation model must be capable of defeating the candidate. When observation availability or quality explains variance compression or drift, the result is a rival victory if the pathways can be separated and Model-ambiguous if adequate

pathways cannot be separated. Either outcome belongs in the main verdict.

Negative-control timing helps here. The protocol should test whether similar signatures appear in impossible windows, shuffled endpoints, nonrelevant channels, or variables unrelated to the hypothesized process. If the pipeline creates compression where no hypothesis predicts it, the confirmatory result fails.

Observation-process reporting protects trust by distinguishing device silence from experiential silence. At the endpoint, that distinction is scientific and humane.

14.4.5 — Strong Rival Set

The rival set should represent the strongest credible explanations of the same observations. Ordinary homeostatic mean reversion can create drift toward a set point. Endpoint-conditioned bridges can create terminal concentration. Frailty and age can reduce variability. Medication and sedation can flatten physiological signals. Observation shutdown can create false compression. Hybrid models can combine these routes (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Each rival should be written as a generative model, not merely as a verbal objection. A rival that cannot produce simulated data cannot be tested fairly in model recovery. The candidate model should not receive a mathematical specification while rivals receive only prose.

The rival set must also admit that none of the named models generated the data. Real terminal trajectories may fall outside both candidate and comparator families. An explicit

misspecification and abstention branch prevents the analysis from converting model incompleteness into false certainty.

Rivals should be tuned under the same rules as the candidate. If the candidate receives cross-fitting, proxy-error modeling, and flexible dynamics while rivals are constrained to naïve forms, the comparison is unfair. A fair protocol gives rivals enough flexibility to represent their actual scientific strength.

When a rival wins, the report should explain what the victory teaches. Frailty may account for the apparent horizon effect, medication may explain a change in variance, and observation shutdown may reveal a defect in the collection system. These findings can improve clinical prediction or study design even while weakening the candidate route. Rival success is positive scientific content, not a rhetorical loss to be softened.

A strong rival set also protects the Law's positive case. If the candidate survives only weak rivals, the result will not persuade serious readers. If it survives the strongest rivals the field can build, the program gains real pressure-tested warrant.

14.4.6 — Discriminating Conjunction

The protocol's central result is a discriminating conjunction, not one favorable coefficient. Representation, calibration, measurement recovery, endpoint validity, observation coverage, predictive or structural performance, rival comparison, burden, and safety each have a declared role; when one prerequisite is noncompensatory, strength elsewhere cannot repair it.

Members, thresholds, tie rules, and stopping rules are fixed before the real-data pattern. Rivals receive comparable lags,

nonlinearities, calibration effort, and information, while recovery simulations map regions in which models are indistinguishable or absolutely inadequate so the procedure can abstain (Wilson & Collins, 2019; Durbin & Koopman, 2012; Talts et al., 2018).

The estimand register keeps held-out predictive value, terminal-band compatibility, feasibility, burden, missingness, and safety separate. Each target travels with its information set, horizon, update schedule, censoring rule, and covariates, preventing a retrospective reconstruction from masquerading as a prospective forecast (Hernán & Robins, 2020; Gneiting & Raftery, 2007).

Uncertainty is decomposed into sampling, calibration, proxy, endpoint, missingness, model, and transport components. A protected abstention branch applies when components are nonseparable, open-set checks fail, or the design lacks resolution.

The final report displays the conjunction as a decision path: gates passed, failed, or unusable; rivals still viable; and predeclared actions such as recalibration, route retirement, lighter design, different perturbation, or no further terminal recruitment. Decision relevance comes from consequences, not a compressed success label.

14.5 — Power, Recovery, and Negative Controls

Power in this protocol does not mean the probability of detecting one dramatic terminal association. A study can be powered to find a pattern and still be scientifically useless if that pattern is equally produced by observation shutdown, endpoint conditioning, medication timing, site practice, or ordinary homeostatic regulation. The first question is whether the design can tell one world from another while keeping care untouched and participation humane (Talts et al., 2018; Wilson & Collins, 2019).

The protocol therefore requires design-separability simulations before enrollment. The simulated worlds must include the Law candidate, the major rival families, hybrid worlds, and deliberately misspecified worlds. They must be passed through the same cadence, sensors, endpoint adjudication, reserve proxies, missingness marks, preprocessing rules, estimand definitions, and analysis pipeline planned for the real study. If the pipeline cannot separate those worlds under plausible clinical conditions, the study should not recruit.

Negative controls test whether the analysis produces the target signature where the candidate predicts none. Battery voltage, ambient temperature, prospectively justified shuffled endpoints, and impossible time windows can reveal how the measurement system creates apparent structure.

Table 14.5. Simulation and model-recovery design axes.

Simulation axis	Example levels	Decision use
Measurement noise floor	Low, moderate, high, state-dependent	Tests false compression and false negative risk
Endpoint uncertainty	Exact adjudication, interval censoring, delayed detection, jittered endpoint	Tests whether terminal alignment drives the result
Reserve proxy error	Independent, attenuated, shared-driver, drifting proxy	Tests reserve-interaction robustness
Missingness process	MCAR, MAR, MNAR, observation shutdown, comfort-driven removal	Tests validity under clinically realistic missingness
Rival worlds	OU, bridge, frailty, medication, observation shutdown, hybrid, none-of-the-models	Measures model confusion before recruitment

14.5.1 — Synthetic Data Generator

The synthetic data generator is the rehearsal room for the protocol. It creates candidate, rival, hybrid, and misspecified worlds under one shared observation system. Each world includes latent path structure, reserve trajectories, treatment events, endpoint states, observations from the candidate momentary-valence estimator, auxiliary context, missingness marks, device quality, burden stops, and site-level variation. The point is not to make the candidate look elegant; it is to determine whether the

planned study can distinguish elegance from reality (Talts et al., 2018; Wilson & Collins, 2019).

Candidate worlds instantiate the protocol-fixed reserve-linked conjunction, including conditional dispersion, reserve interaction, filtered endpoint prediction, and observation-process behavior. Rival worlds instantiate homeostasis, conditioned bridges, frailty, disease severity, age, prognosis, medication, sedation, observation shutdown, flexible time-to-event processes, hybrids, and generators outside the fitted library.

The generator should vary clinically plausible features rather than only mathematical parameters. Diagnosis, prognosis, site practice, care model, symptom volatility, medication cadence, family presence, communication capacity, device tolerance, staff burden, and transfer risk should be part of the scenario library. Testing those features helps prepare the protocol for the variability and care needs of an actual clinical setting.

Misspecified and none-of-the-models worlds are required. Nature may not follow the candidate or any named rival. If every simulated dataset is forced into one family, the analysis will learn to choose the least-wrong label rather than admit that its model space is inadequate. A serious first protocol must be able to say that its models did not contain the world it saw.

Scenario labels, parameters, seeds, preprocessing, and scoring rules should be fixed before real-data analysis and retained with the protocol package. Apply §6.7's prospective-specification terminology; later changes create a new version.

Synthetic data test the protocol's capacity to learn. If a generator shows that the design confuses the Law candidate with observation shutdown, researchers revise the design and rerun recovery before enrollment. This rehearsal helps direct participant contributions toward questions the study can answer.

14.5.2 — Parameter Recovery

Parameter recovery asks whether the analysis can estimate the quantities it reports. A model may look principled while failing to recover its own reserve interaction, drift parameter, innovation scale, observation-process effect, endpoint-jitter parameter, or medication coefficient. The first falsification protocol cannot ask the world to decide among theories until the pipeline has shown that it can recover known truths from data shaped like the planned study (Gelman et al., 2013; Vehtari et al., 2017).

Recovery should be tested across the expected clinical range, not only under ideal measurements. The simulation grid should include weak reserve proxies, high symptom volatility, irregular cadence, late enrollment, endpoint uncertainty, treatment changes, delirium, device removal, and site-specific practice. Clean recovery in a clean world is useful, but it is not a burden budget for terminal research.

Uncertainty calibration matters as much as point accuracy. A biased estimate with a narrow interval is more dangerous than a noisy estimate that admits its limits. Coverage, posterior calibration, interval width, rank calibration, and sensitivity to priors should be reported before the protocol is allowed to claim confirmatory capacity.

Nonidentifiability should be treated as a result. If medication effects and drift cannot be separated, the protocol cannot report a drift finding. If reserve-proxy error and disease severity cannot be separated, the reserve interaction is not identified. If endpoint jitter changes the sign or magnitude of the target estimate, the endpoint system is not strong enough for that claim.

The recovery report should name the estimands that are recoverable, those that are weakly recoverable, and those that are not recoverable under realistic conditions. The study may proceed only for the estimands that remain scientifically interpretable after this report. It must not keep an attractive target simply because it would be rhetorically useful.

This is a dignity rule as well as a statistical rule. A person near the end of life should not be enrolled in a study that cannot estimate the quantity it asks permission to study. Scientific humility is not a mood here. It is a pre-enrollment obligation.

14.5.3 — Model Recovery

Model recovery tests whether the pipeline can identify which generative family produced the observed pattern. It is possible to estimate parameters accurately within the wrong family and still draw the wrong conclusion. The protocol therefore needs a confusion matrix across candidate, OU, bridge, frailty, medication, observation-shutdown, shared-driver, and hybrid worlds before it interprets real participants (Lipsitch et al., 2010; Nosek et al., 2018).

For each simulated world, the analysis should score the candidate and every rival using the same held-out rules planned for the

confirmatory study. The protocol should record how often each world is classified as the candidate, how often the candidate is missed, and how often an unmodeled world is forced into a named family.

The most important failure is false candidate victory. If observation shutdown, endpoint-conditioned bridges, sedation, or shared clinical decline are frequently recovered as the Law candidate, the design is not ready. False candidate victory is not a conservative error. It is exactly the public-risk path by which a fragile study becomes proof language.

Model recovery also measures false rejection of the candidate. Coarse cadence, proxy noise, or endpoint jitter may cause candidate-generated data to be classified as frailty or medication effects. Those errors identify where the design needs improvement before a hypothesis comparison.

Hybrid worlds prevent easy triumph. A real terminal pattern may contain ordinary homeostasis, care-quality changes, medication effects, social context, and reserve-linked dynamics at once. The model-recovery plan should not require one winner when the honest conclusion is a mixed or nonseparable result.

A protocol that cannot recover the relevant models stops before recruitment. Early simulation results then guide a more informative design with an appropriate participant burden.

14.5.4 — Power Across Noise and Missingness

Power analysis must vary measurement noise, missingness, endpoint uncertainty, reserve-proxy error, sample size, cadence, site heterogeneity, and burden stops together. Treating one source

of uncertainty at a time gives a beautiful but misleading design. Terminal research is not a laboratory of isolated imperfections. Its difficulties arrive in clusters (Talts et al., 2018; Wilson & Collins, 2019).

Measurement noise floors are especially important. More participants do not remove a systematic noise floor or a biased estimator. They can make a biased conclusion look more precise. The simulation plan should therefore test whether additive noise, state-dependent noise, device-quality changes, and smoothing choices alter variance-shape conclusions.

Missingness mechanisms must be clinically realistic. Measurements may be missing because a participant refused, a device was uncomfortable, a nurse removed it for care, sedation changed behavior, a family member requested privacy, a transfer occurred, or death arrived faster than expected. Treating all such losses as random missingness would be a scientific and ethical error.

The relevant power is discriminating power. The protocol should report the probability of distinguishing the candidate from each rival, not merely the probability of finding a nonzero coefficient. A design that has high power to detect time-to-endpoint change but low power to distinguish the candidate from a Brownian bridge is not a falsification protocol.

A separability threshold is fixed before enrollment. Across plausible noise and missingness, the benchmark may require a stated upper bound on confusion between the candidate and its strongest rival, together with protection against falsely defeating

a rival that generated the data. The threshold belongs in the protocol record, not in prose written after the outcome.

Statistical power describes a study's ability to detect the effect specified in its design. Reporting that effect, the sample assumptions, and the decision rule makes the study's expected contribution clear.

14.5.5 — Negative-Control Outcomes

A negative-control outcome must be justified as outside the hypothesized causal pathway while remaining informative about the relevant bias. Battery state, contact quality, ambient temperature, and administrative timing are candidate diagnostics, not automatically valid negative controls: temperature may affect comfort, and device or staffing variables may affect both observation and experience. The protocol should identify the assumed causal paths and distinguish negative-control outcomes from technical quality checks (Lipsitch et al., 2010).

Negative-control outcomes require a prospective rationale. The protocol record should explain why the candidate does not predict the target signature in that outcome and why the outcome remains exposed to relevant artifacts such as device behavior, staffing, or endpoint alignment. A variable chosen after it looks convenient is not a valid negative control.

If the negative-control outcomes show the same apparent reserve-linked compression or endpoint-related drift as the measured cumulative estimate, the candidate interpretation weakens or fails for that analysis. The result is informative because it reveals a source of false structure.

Negative controls reveal operational rivals: battery effects may indicate changed device wear near the endpoint; ambient temperature, room transitions or care procedures; administrative timing, staff workflow. These explanations belong in the scientific record.

Set quantitative control thresholds. Incidental correlations need not defeat the analysis, but matching the primary signature's direction, timing, and magnitude should trigger a predefined observation-process or no-estimate branch.

A target-like pattern in battery voltage can identify an observation artifact before it is mistaken for a biological finding. This control helps the protocol distinguish a promising signal from a feature of the measurement system.

14.5.6 — Negative-Control Timing and Alignment

Timing controls test whether alignment creates the signature. Shuffled, displaced, person-swapped, too-early endpoints and anchors unrelated to the truth-level stream endpoint expose geometry imitating mechanism. Time-to-endpoint structure alone is not a Law-specific pattern (Lipsitch et al., 2010; Nosek et al., 2018).

Endpoint conditioning is a strong rival: aligning trajectories by a terminal event can make deterioration, observation loss, and selection resemble structured convergence. Negative-control timing tests whether the pattern survives deliberately incorrect endpoint labels.

Timing controls should be protocol-fixed before enrollment. The plan specifies endpoint jitters, shuffled or cross-person labels,

impossible windows, summary statistics, multiplicity handling, and the similarity threshold that triggers an observation-process, rival-favored, or no-estimate branch.

If displaced or shuffled endpoints reproduce the primary signature, the confirmatory branch fails. The accurate public statement is that the pipeline could not distinguish the protocol-fixed endpoint from false alignment, not that terminal closure was almost observed.

Look-ahead must be handled with particular care. Features built using future data may appear in sensitivity analyses only when clearly labeled as smoothed or retrospective. Primary confirmatory claims use filtered, forecast, or cross-fitted features that do not borrow information from the future they claim to predict.

The timing plan gives the study clear, testable standards for prospective inference. Meeting those standards makes a favorable result more persuasive and gives critics a reproducible way to assess it.

14.5.7 — Circular Perturbation-Design Gate

The circular perturbation-design gate rejects any design in which the reserve construction, endpoint rule, preprocessing, normalization, or perturbation already forces the target pattern. Reproducing a consequence of the analysis pipeline is not evidence that the same pattern exists in nature. Before enrollment, an independent reviewer should be able to derive what the pipeline would produce under neutral and rival generators (Talts et al., 2018; Wilson & Collins, 2019).

The review should examine centering, baseline subtraction, endpoint-normalized time, adaptive weights, derivative transforms, smoothing, inclusion rules, and endpoint adjudication. Each step must be tested for whether it can manufacture closure, variance compression, or apparent neutrality when the candidate process is absent. The relevant question is not whether the step is common, but whether it creates the feature later presented as discovery.

The same gate applies to perturbation and invariance tests. Reserve cannot be built from the signals used to define ledger decline and then cited as independent support for reserve-linked compression. Endpoint thresholds cannot be tuned to maximize terminal concentration, reliability weights cannot be learned from the terminal window, and lag or time normalization cannot be selected after seeing which choice makes endings look most alike.

A failed circularity review blocks confirmatory interpretation. The design may remain useful for method development, simulation, or diagnostic exploration, but its outputs must be labeled nonconfirmatory and kept outside any status update. Sample size, statistical significance, or careful rhetoric cannot repair a result whose defining pattern was built into the analysis.

The design target is model separation, not merely rejection of a null. The simulator should test whether the frozen pipeline recovers the correct generator, distinguishes the candidate from the strongest protocol-fixed rival, detects failed measurement and endpoint bridges, and recognizes open-set inadequacy. Effect size, cadence, measurement error, endpoint uncertainty, missingness, and heterogeneity must vary across the benchmark

rather than remaining fixed at one favorable setting (Talts et al., 2018; Gelman et al., 2013).

Recovery exercises should include deliberately broken worlds: mis-zeroed scales, invalid reserve proxies, informative dropout, endpoint misclassification, time leakage, device drift, and generators outside the fitted family. Negative controls should have a predeclared expected direction, tolerance, and consequence; some should remain null, while others should reproduce a known artifact. A diagnostic that cannot detect the manufactured failure is too weak to protect the primary interpretation (Vehtari et al., 2017; Nosek et al., 2018).

Simulation can expose analytic weakness before enrollment, but real observation remains selective and ethically interruptible. Missingness models must preserve why information disappeared and the rule that care, refusal, and safety outrank completion of the record.

14.6 — Missingness, Safety, and Adverse Events

Terminal study data will be incomplete if the protocol is honest. Measurements may disappear because a participant is tired, a device is uncomfortable, medication changes, delirium emerges, a transfer occurs, family privacy is needed, staff workload spikes, an emergency interrupts the schedule, or the person dies before the planned window. Missingness is therefore not a nuisance term added after the study. It is part of the phenomenon and part of the burden structure (Little & Rubin, 2019; World Medical Association, 2024).

Safety is noncompensatory. A promising statistical pattern cannot rescue a protocol that increases suffering, confuses care, pressures families, exposes private data, or makes staff labor invisible. The protocol must define adverse events for participants, families, staff, sites, and data systems before the first enrollment.

The chapter's burden lens appears here as validity discipline. If the study shifts burden onto people, families, clinicians, or data systems, those shifts can change the data-generating process. A protocol that ignores those shifts may not merely be unethical. It may be wrong.

Table 14.6. Missingness, safety, and claim consequences.

Event type	Protocol action	Claim-status consequence
Participant refusal or fatigue	Pause or stop the component; mark reason only if voluntarily given	Missingness is agency-sensitive, not noncompliance

Event type	Protocol action	Claim-status consequence
Device failure or intolerance	Remove device; mark quality and burden; no pressure replacement	May trigger observation-process analysis or no-estimate
Clinical decline, sedation, or procedure	Care proceeds; measurement yields to care; time-stamp context	Modeled as treatment or competing explanation
Privacy or security incident	Contain, notify, suspend access, delete where required, review publication consequences	May stop study or prevent release of affected data
Unanticipated distress	Support participant or family, reduce or stop procedures, report adverse event	Ethical adverse result overrides statistical promise

14.6.1 — Marked Missingness

Marked missingness means that absence of data receives a meaningful label when that label can be collected without creating new pressure. The protocol should distinguish refusal, fatigue, device failure, device intolerance, clinical decline, sedation, transfer, discharge, family privacy request, staff unavailability, emergency intervention, death, and unknown absence. These categories are scientifically different (Little & Rubin, 2019; Rubin, 1976).

Refusal and fatigue are not measurement failures to be repaired. They are expressions of participant agency. The protocol may record that a scheduled observation was declined, but it should

not demand an explanation. If the participant volunteers a reason, it can be marked in a predefined category without being treated as noncompliance.

Device failure and device intolerance should be separated. A battery problem, contact-quality problem, lost signal, sensor displacement, or software failure implies one kind of observation problem. Skin irritation, discomfort, sleep disruption, embarrassment, or family objection implies another. Combining them hides both burden and instrument behavior.

Clinical interruptions require detailed but respectful marking. Sedation, analgesia, delirium, dyspnea, nausea, procedure, repositioning, bathing, family visit, spiritual care, and urgent symptom management can all alter observation and experience. These marks should not become a surveillance script. They should capture enough context to prevent false inference.

Unknown reasons for missingness should remain marked unknown rather than assigned a favorable explanation. A prespecified primary observation model may estimate missing quantities under explicit assumptions, with uncertainty and adverse sensitivity analyses. It must not set missing experience to neutral or enforce terminal closure by construction; §14.6.6 supplies the resulting abstention and verdict categories.

Marked missingness makes refusal, care, and burden visible without turning them into blame. It is one of the ways the protocol asks where burden is carried, hidden, shifted, reduced, repaired, or made reviewable.

14.6.2 — Joint Longitudinal–Event Model

Repeated measurements, observation availability, treatment changes, device tolerance, and the terminal event share causes and therefore belong in a joint data-generating account. A decline in clinical state may change both the estimated affect path and the probability that the path is observed. Joint models, multistate formulations, and marked observation processes answer different questions, so the protocol should state which dependence structure its primary estimand requires (Rizopoulos, 2012; Andersen et al., 1993).

A joint longitudinal-event model can represent this dependence without defining the endpoint through the cumulative-value estimate. The longitudinal component models observations from the candidate momentary-valence estimator, reserve, auxiliary channels, and quality marks over time. The event component models endpoint states, withdrawal, device cessation, transfer, death, and competing events. Shared random effects or latent clinical processes may explain dependence between the two without making the endpoint a function of the measured path.

The essential firewall remains: ledger-derived states may help model informative observation or rival explanations, but they must not define the primary endpoint. The endpoint channel is constructed independently and then brought into the joint analysis under declared dependence assumptions.

Missingness analysis should include a protocol-fixed primary model, at least one materially more adverse missing-not-at-random scenario, and a no-estimate region in which the target is

not recoverable. The comparison should show how the verdict changes as unobserved-state dependence strengthens. The most convenient missingness assumption is not treated as truth merely because it yields a precise answer.

Care events belong in the model as time-varying processes. Analgesia, palliative sedation, antipsychotic medication, oxygen support, hydration changes, procedures, family presence, symptom crises, and site practices may be causes, confounders, mediators, or competing explanations depending on the estimand. The protocol should not force all of them into one role.

The joint model is not a machine for rescuing incomplete data. Its purpose is to state the assumptions under which inference is possible, and to identify when the assumptions are too fragile. In this chapter, abstention remains a result.

$$Y_{it} = m_i(t; \theta, b_i) + \varepsilon_{it}, \lambda_i(t | b_i) = \lambda_0(t) \exp\{\alpha^T Z_i(t) + \gamma_b b_i\}$$

Here, b_i is the shared random effect or latent clinical process, and γ_b is the event-association parameter.

14.6.3 — Comfort and Clinical Stop Rules

Apply Chapter 3's stop authority locally: participants, clinicians, bedside staff, and designated research personnel may halt observation, with comfort controlling. Prespecified stops cover distress, device intolerance, clinical instability, privacy requests, and workflow conflict.

Participants may stop any component without giving a reason. A family member or clinician may request interruption when the participant cannot comfortably decide, when privacy is needed,

or when care would be impaired. Staff should not be required to negotiate for research continuation during clinical strain.

Stop rules should include device discomfort, skin irritation, sleep disruption, respiratory distress, agitation, distress during prompts, family distress, clinician concern, staff workload, procedure conflict, privacy need, and any sign that the person experiences the study as intrusive. The default should be removal or pause, not persuasion.

Clinical care remains unmodified. The protocol must never delay medication, repositioning, family access, spiritual support, symptom relief, resuscitation where appropriate, withdrawal of burdensome treatment where appropriate, or any ordinary clinical judgment. A research schedule has no authority over the terminal room.

Every stop event should be recorded as a burden and feasibility outcome when recording can occur without new burden. A study with many comfort stops may be telling the truth that the protocol is not humane enough, even if the remaining data look analytically attractive.

This stop system turns answerability into scientific design. The dying person does not carry the burden of completing the record; the protocol carries the burden of stopping when comfort, consent, or interpretability fails.

14.6.4 — Privacy and Security Incidents

A privacy or security incident is not a mere administrative problem in this study. It can injure dignity, expose families, distort participation, alter trust, and determine whether the study can

ethically continue. The protocol should define incident classes before enrollment and connect each class to containment, notification, deletion, and publication consequences (National Institute of Standards and Technology, 2020; World Medical Association, 2024).

Relational data require special care. Family presence, speech, touch, conflict, spiritual practices, room context, and caregiver behavior may be scientifically relevant but personally intimate. The study should minimize such data, obtain appropriate permission where data are intentionally collected, and avoid covert transformation of family life into research material.

Data access should be separated by role. Clinical staff should not receive research summaries that could alter care unless a predefined safety finding requires communication. Analysts should not receive identifiable family narratives unless the analysis cannot be performed without them. Model developers should not receive data for secondary training unless separately permitted.

Incident response should include immediate containment, access suspension where needed, audit-log review, participant or surrogate notification when appropriate, family notification when affected, IRB or oversight reporting, deletion or quarantine where required, and a written decision about whether affected data can be analyzed or published.

Security incidents require scientific as well as administrative action. Versions that expose identifiers, ignore consent flags, leak relational data, or permit unauthorized secondary training should

be quarantined and retired, and outputs derived from them should be reexamined. A compromised pipeline cannot remain evidentially valid simply because its statistical code still runs.

Privacy discipline strengthens science. Participants who fear exposure, families who distrust the study, or staff who feel watched can change behavior and missingness. Protecting privacy is therefore not outside the model. It is part of maintaining a valid research environment.

14.6.5 — Unanticipated Distress

Unanticipated distress must be treated as an adverse event, not as an intense data point. A prompt, device, conversation, family reaction, or research presence may create distress even when the design looked minimal on paper. The protocol should assume that such events can occur and specify immediate support routes (World Medical Association, 2024).

Distress may arise in the participant, family, staff, or research personnel. A dying person may feel watched. A family member may feel judged or excluded. A clinician may experience added moral stress. A researcher may be affected by repeated terminal exposure. The burden budget should include all of these groups.

The first response is support and reduction of burden. The study should pause, remove devices, stop prompts, provide clinical or psychosocial support through ordinary care routes, and document only what is needed for safety and oversight. It should not ask for an explanatory interview during distress.

Unanticipated distress should be reviewed for pattern. One incident may be local. Several incidents of the same kind indicate

a design defect. The independent monitoring process should be able to reduce cadence, remove a modality, narrow the population, pause enrollment, or stop the study.

A participant's fear, agitation, grief, or pain calls for present care. Research records those states only within the agreed purpose and observation plan.

The chapter's rule is simple: distress can teach the protocol where it failed, but it cannot be converted into proof material. That distinction protects both the participant and the science.

14.6.6 — Abstention and Inconclusive Conditions

Apply the evidence-state taxonomy in Table 14.7 and §§24.2 and 27.4. For this protocol, failed estimator qualification, unstable sign, dominant device artifact, unresolvable channel conflict, invalid endpoint separation, excessive jitter, model-recovery failure, strong-rival nonseparability, or uncertainty crossing the decision boundary routes the primary inference to the corresponding prespecified branch. The local task is to identify the blocking condition, not redefine the status vocabulary.

Ethical conditions can also close authority: unacceptable burden, coercive recruitment risk, privacy incident, staff overload, distress-related withdrawal, or interference with care. The study may still report instrument, feasibility, safety, or narrower observed-variable findings, but it may not promote them to the primary claim.

Operational monitoring remains separated from outcome interpretation. An independent review can examine burden, device tolerance, recruitment pressure, privacy, staffing, and

observation loss without learning whether the candidate model is performing well, preventing scientific excitement from raising the stop threshold.

When a threshold is crossed, the protocol version closes with its data and adverse result intact. Restart requires a reviewed successor, new burden analysis, and an account of which inference remains comparable across versions; stoppage is part of the evidentiary history, not an inconvenience to erase.

Every burden signal, mitigation, pause, and restart is timestamped and linked to the protocol in force. A study able to return No adequate estimate is stronger than one compelled to produce a claim because it can surrender authority before it loses trust.

14.7 — Decision Tree and Public Reporting

A falsification protocol is incomplete until its result branches and consequences are fixed. Without a decision tree, a favorable component can inflate into whole-program support, a rival win can dissolve into ambiguity, and a measurement or ethical failure can disappear behind method language. The protocol records each exit before outcome inspection, and the report later shows which branch was reached, why, and what action follows (Chan et al., 2025; Hopewell et al., 2025; Nosek et al., 2018).

The protocol-fixed decision tree is used by analysts, investigators, oversight bodies, and public communicators. Every branch states the permissible claim, prohibited inference, required next action, and whether the result supports a component, returns to development, narrows a reference class, retires a route, or rejects a declared branch.

Public reporting completes the scientific object. Protocol-fixed methods are not enough when the public summary suppresses deviations, missingness, burden, negative controls, recovery performance, rival scores, or adverse events. The reporting plan should map each component verdict to exact language and preserve the same status in the abstract, registry record, figures, press materials, and corrections (Hopewell et al., 2025).

Table 14.7. Protocol-fixed result branches and permitted public language.

Result branch	Required public status	Next action
Candidate outperforms rivals after all gates pass	Component support within declared reference class	Independent replication and transport checks
Rivals match or win	Model-ambiguous for adequate inseparable models; Rival-favored when a rival meets the frozen victory criterion	Diagnose nonseparation; revise or retire the tested bridge only as warranted
Measurement or endpoint failure	Unusable targeted inference	Return to instrument or endpoint development
Mixed or nonseparable results	Model-ambiguous for inseparable adequate models; otherwise apply the gate, adequacy, and identification rules in Section 14.7.4	Redesign before expansion
Adverse ethical result	Protocol route failed or must narrow	Stop, report, protect participants, review route
No-estimate	Valid refusal to infer	Publish feasibility and failure reasons

14.7.1 — Candidate Outperforms Rivals

A candidate-outperforms-rivals verdict requires the entire prerequisite chain to pass: a qualified candidate momentary-valence estimator, an intact endpoint firewall, a sufficiently

independent reserve proxy, adequate model recovery, negative controls without the target signature, interpretable missingness, and ethical burden within the protocol-fixed budget. Predictive superiority is then judged under the same information and scoring rules used for the rivals (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

The public report names the population, endpoint class, instrument version, reserve proxy, rival set, and tested result. Those particulars give readers a clear account of what the study achieved.

The correct status is preliminary component support or route-favored within the declared class. A report may say that, under one qualified estimator and endpoint system, the protocol-fixed reserve-linked conjunction improved held-out performance over named rivals. It must also state the first unresolved gate and the replication required before any stronger status.

An open-problem register links remaining questions, such as scale calibration or transport, to the studies needed to address them.

A favorable result is a promising milestone and the basis for independent replication. Different sites, devices, analysts, endpoint adjudicators, reserve proxies, and rival models should test its robustness before stronger public status is assigned.

The replication packet identifies the exact component that survived, its observation window, the tested specification, and the comparison models. Another team can then reproduce the

analysis and evaluate its transport to a new setting (Hopewell et al., 2025; Nosek et al., 2018).

14.7.2 — Rivals Match or Win

A rival wins when it meets the protocol-fixed victory criterion under the common scoring and adequacy rules. If no model meets that criterion and adequate candidate and rival models remain inseparable, the result is Model-ambiguous. Endpoint changes, post hoc covariates, retrospective narrowing, or relabeling the rival as partial support would change the question after the answer and are therefore reported only as new exploratory analyses (Nosek et al., 2018; Chambers & Tzavella, 2022).

Different rivals carry different consequences. An OU or homeostatic rival may retire a dynamical interpretation while preserving measurement development. A bridge rival may show that endpoint alignment explains the pattern. A frailty or disease-severity rival may relocate the finding to clinical trajectory. A medication or sedation rival may make the result directly care-relevant.

A rival win can be scientifically valuable. It may improve palliative understanding, reveal device behavior, expose site practice, refine endpoint adjudication, or show that the first reference class was poorly chosen. The protocol should treat such findings as knowledge that informs the next study.

Public language should name the winning rival, the failed candidate component, and the resulting action. It may report that the bridge narrowed, the route retired, the clinical process explained the pattern, or the study must return to measurement or

endpoint development. It may not preserve a promise that the protocol-fixed comparison rejected.

A rival win may defeat one component without defeating every possible form of the Law. That distinction is legitimate only if the component and claim version were named before the study. The protocol should not invent a new version after the result and pretend it was always the target.

14.7.3 — Measurement or Endpoint Failure

Measurement or endpoint failure suspends the targeted inference and identifies the bridge that needs further development (Gneiting & Raftery, 2007; National Academies of Sciences, Engineering, and Medicine, 2019).

Measurement failure includes failure to qualify the candidate momentary-valence estimator, sign ambiguity, unstable calibration, excessive artifact, device drift, incompatible versions, or response-shift failure. These failures send the program back to measurement development rather than into a weaker interpretation of the same data. Posterior path uncertainty larger than the target region can instead leave a valid analysis Inconclusive; it is not by itself a measurement-validity failure.

When the endpoint bridge fails, the result names the failed property and exits the terminal-window branch. Loss of independence sends the design back to endpoint construction; false crossings or unresolved returns require a revised state machine; inadequate terminal coverage narrows the estimand to an earlier window; and future-leaking adjudication voids the

prospective analysis. The action follows the defect rather than being softened into generic uncertainty.

Measurement or endpoint failure is reported as failure of the targeted bridge unless the protocol prospectively defined that bridge as a necessary component whose failure defeats a broader claim. The public sentence should name the unusable object and the reason it failed. Calling damaged evidence negative evidence for PTVC would be as misleading as calling it support.

A measurement or endpoint failure may still produce useful secondary findings. It can show that a device is too burdensome, a clinical state is not yet a valid context of use, a stopping channel is too dependent on the ledger, or a site cannot support the cadence. Those findings should be preserved.

When a measurement or endpoint bridge fails, the protocol returns to the first failed gate. Investigators may qualify a new instrument, revise the endpoint classifier, narrow the context of use, or classify the affected evidence as Unusable. They may not cross the gap by rhetorical continuity or treat unusable data as weak support (National Academies of Sciences, Engineering, and Medicine, 2019; Wilson & Collins, 2019).

14.7.4 — Mixed or Nonseparable Results

Mixed or nonseparable results arise when the data do not allocate the observed pattern cleanly among the candidate, named rivals, and observation process. These patterns require the applicable gate and adequacy checks: inseparable adequate models yield Model-ambiguous; an inadequate model set yields Open-set inadequate; a failed required gate yields Unusable evidence; a

valid but undecided analysis yields Inconclusive evidence; and a target that cannot be identified or validly bounded yields No adequate estimate. The correct response is to diagnose nonseparation, not to select the most favorable coefficient (Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019).

Mixed results are judged against the entire protocol-fixed conjunction. Reserve coupling with failed negative controls, improved prediction with endpoint leakage, or a favorable mean with unrecovered dependence can produce component findings but cannot advance the route verdict. The report preserves the fracture instead of averaging it into a reassuring middle category.

A mixed result may still be informative. It may show that reserve coupling exists but is mediated by care quality, that endpoint prediction works only under one site practice, that medication explains part of the variance, or that observation shutdown and biological decline are entangled.

The decision tree should specify whether mixed results trigger redesign, additional measurement qualification, richer rival modeling, narrower reference class, or a no-estimate branch. It should not allow expansion to a larger trial simply because some results were attractive.

If the study permits repeated interim looks, adaptive stopping, or rolling site entry, the evidence process must be fixed with the same rigor as the endpoint. A conventional fixed-sample p value cannot be repeatedly inspected without changing its error guarantee. The protocol should therefore choose in advance

between fixed-sample analysis and a time-uniform method such as a confidence sequence or e-process, state the information times and stopping consequences, and report every look. Sequential validity protects early stopping for futility or harm and resists opportunistic stopping after a favorable fluctuation (Howard et al., 2021; Ramdas et al., 2022).

Nonseparability should be named in ordinary language. Readers should not need statistical training to understand that the study could not tell whether the pattern belonged to the candidate, a rival, or the observation process.

Mixed results should be diagnosed by source: model overlap, inadequate cadence, weak negative controls, informative missingness, endpoint ambiguity, or open-set inadequacy. Each source implies a different redesign or stopping rule. Collapsing them into a generic claim of complexity would hide the scientific reason the decision tree stopped (National Academies of Sciences, Engineering, and Medicine, 2019; Talts et al., 2018).

14.7.5 — Adverse Ethical Result

An adverse ethical result is a scientific result for this chapter. If the protocol creates excessive burden, distress, privacy risk, staff overload, family pressure, care confusion, or withdrawal due to discomfort, the route has failed or must narrow. A statistically promising pattern does not compensate for that failure (Nosek et al., 2018; Chambers & Tzavella, 2022).

The adverse branch should be protocol-fixed, with pause, modification, and termination thresholds; authorized trigger roles; support and incident procedures; data-disposition rules; and

required public reporting. Ethical feasibility cannot be reconstructed after the study by subtracting the participants who carried the greatest burden.

Adverse outcomes may be participant-level, family-level, staff-level, site-level, data-level, or public-communication-level. A privacy incident after data collection is not less important than a device discomfort event during collection. A public artifact that creates dangerous misunderstanding can also be an adverse result of the research program.

Once an ethical stopping rule fires, the event must propagate through the abstract, public registry where applicable, methods, participant-flow account, limitations, data disposition, and replication status. Investigators report exposure before the stop, support offered, whether collected data remain usable, and whether the design may be modified or must be retired. A buried appendix cannot carry an adverse result that changes the study's legitimacy.

An adverse ethical result should trigger route review. The program may move to less intrusive designs, safer populations, instrument-only development, retrospective feasibility studies, or no further terminal recruitment until the cause is repaired.

Participation remains voluntary through the final observable moment. A participant may sleep, decline, disconnect a device, refuse a question, withdraw, or become unable to continue without owing the study a replacement observation. The design treats those outcomes as part of feasibility and the observation process, never as a moral or scientific failure by the person

(Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

14.7.6 — Registered Public Report

Before enrollment, investigators prepare a protocol-fixed report shell for participant flow, burden, missingness, endpoint adjudication, deviations, negative controls, recovery, rival scores, adverse events, and claim-status consequences. For this prospective protocol, the plan qualifies as preregistered only when it is deposited before enrollment in a timestamped, read-only external registry record with a persistent identifier (Chan et al., 2025; Hopewell et al., 2025).

The report protects raw terminal and family material while identifying the measurement version, reserve proxy, endpoint system, estimand, rival set, reference class, missingness model, decision rule, analysis version, and deviations behind each component verdict. Null, adverse, unusable, inconclusive, and rival-win outcomes remain publicly auditable under privacy rules.

Versioned correction connects replication failure, data corruption, classifier retirement, privacy incidents, or coercive recruitment to revised status and visible notice on every high-reach artifact. Answerability comes from reconstructable provenance and branch-matched language, not a dramatic terminal image (von Elm et al., 2007; National Academies of Sciences, Engineering, and Medicine, 2019).

14.8 — Protocol Limits and Replication Gate

The first protocol is designed to produce a bounded, informative, humane result. A finding that survives the specified prerequisites and comparisons can justify independent replication; other outcomes can justify stopping, returning to development, narrowing the route, or identifying a more informative next question (Chambers & Tzavella, 2022; Munafò et al., 2017).

The protocol specifies site coverage, observation windows, estimands, measurement qualification, and permitted analyses before recruitment.

The protocol-fixed public report should ship with a replication packet that makes independence possible: frozen code, schemas, endpoint-adjudication materials, calibration recipes, rival specifications, hardware and firmware identifiers, incident logs, and a list of judgments reserved for the replication team. The report should distinguish computational reproduction of the original analysis from a genuinely independent test of the scientific claim.

Table 14.8. Protocol limits and replication gates.

Limit	Rule	Reason
Single-site result	Preliminary only	Site practice, care model, staffing, and population may drive patterns
Whole-life claim	Not allowed without independent initial-ledger bridge	Late windows observe increments, not entire lives

Limit	Rule	Reason
Individual verdict	Forbidden	Scientific models evaluate specified research outcomes
Replication	Independent sites, analysts, calibration, endpoint, and rival checks required	Protects against local, technical, and interpretive artifacts
Public artifacts	Embargoed until claim-status and safety review	Prevents early diagrams or summaries from becoming proof language

14.8.1 — Single-Site Results Are Preliminary

A single-site result is preliminary even if it is statistically strong. Hospice practice, staffing, culture, diagnosis mix, symptom protocols, family presence, documentation habits, device tolerance, and endpoint adjudication can all be site-specific. The study may learn something real about that site without learning a transportable law (Chambers & Tzavella, 2022; Munafò et al., 2017).

The first protocol may deliberately begin in one or a small number of highly prepared sites because ethical and operational quality matter more than breadth. That choice is defensible only if the public status remains local. A carefully governed site is not a miniature universe.

Site context should be described as part of the data-generating environment: care model, population mix, staffing, palliative expertise, medication practice, device workflow, family-room

norms, documentation systems, and endpoint adjudication. Those features can change both what occurs and what becomes observable.

A favorable single-site result should trigger replication design, not broader claim language. The next site should not be selected because it is likely to reproduce the result. It should test whether the result survives meaningful change in care context, population, device environment, and analytic personnel.

An unfavorable single-site result may also be preliminary. If the design fails because the site cannot support cadence, endpoint independence, or burden limits, that result may be operational rather than theoretical. The report should distinguish site infeasibility from component defeat.

14.8.2 — No Whole-Life Reconstruction

A late clinical window identifies an increment under its measurement model. Complete-history reconstruction uses an independently qualified initial-state bridge or sufficiently comprehensive prior observation specified before terminal outcomes are examined (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018).

The initial state therefore enters the analysis explicitly, with its available information, bounds, and sensitivity to alternative assumptions.

The window-level targets are defined in §14.1.2. Terminal-regime concentration or equivalence may be analyzed secondarily under declared origin and bias assumptions, but a whole-life claim requires a qualified bridge to the unobserved history.

A report should never say that a participant's life balanced because the observed terminal window approached a band. At most, it may say that a measured window-level trajectory showed or failed to show a protocol-fixed signature under a specified estimator and endpoint system.

Complete-history inference requires long-term longitudinal observation, independent bounds on the initial ledger, or a formally justified bridge between observed windows and earlier history.

State the observed terminal window and its estimand in the abstract, figures, and public summary. Identify any independently qualified historical or measurement bridge used to extend the analysis beyond that window (National Academies of Sciences, Engineering, and Medicine, 2019).

14.8.3 — No Individual Verdict

The protocol returns study-level findings and agreed descriptions of observed windows. It excludes personal judgments about worth, meaning, or whether a life balanced. Consent materials, staff training, and result-return plans specify that scope in ordinary language (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

This rule applies even when an individual trajectory looks dramatic. A graph can appear to approach neutrality, flatten, compress, or turn. Such displays may be useful for internal quality review or aggregate modeling, but they are not personal truth and should not be shown as meaning.

Participant and family communication should avoid claim language. The study may explain participation, privacy, withdrawal, burden, and general scientific aims. It should not tell families that a pattern suggests closure, balance, fairness, peace, failure, or destiny.

Individual-level estimates may exist within the statistical model, but access should be restricted and purpose-limited. They are research quantities used for model fitting, diagnostics, and uncertainty assessment. They are not records of worth, prognosis, holiness, trust, eligibility, or care entitlement.

This rule also protects clinicians. A bedside team should not receive research interpretations that could shape care unless a predefined safety issue requires communication. Care decisions follow clinical judgment, patient preference, and established practice, not a fairness estimator.

Population and component estimates remain statistical objects with uncertainty and reference-class limits. Even a well-calibrated individual random effect is not a verdict about a person's inner truth, worth, prognosis, or complete life. Interfaces and access controls should prevent model diagnostics from being repurposed as personal interpretations (Council for International Organizations of Medical Sciences, 2016; Nissenbaum, 2010).

14.8.4 — Independent Replication Required

Independent replication requires more than rerunning code on a new sample. It requires meaningful independence in site selection, recruitment route, clinical workflow, endpoint adjudication, measurement implementation, reserve proxy,

analyst team, rival model specification, and public reporting review (Chambers & Tzavella, 2022; Munafò et al., 2017).

The replication plan should include at least one site differing in care model or population, an analyst team independent of the original primary analysis, and an endpoint adjudication team applying the same protocol-fixed rules without access to ledger-derived features. The replicated estimand and the intentionally changed transport dimension must be identified separately.

Device and pipeline replication matter. A result that survives only under one sensor model, preprocessing version, or quality-control rule may be a technology artifact. A result that survives across devices may still require calibration and versioned provenance before it can be compared.

Rival replication is essential. Replication does not count if the candidate is tested carefully and the rivals are simplified. OU, bridge, frailty, medication, observation shutdown, shared-driver reserve, and hybrid models should travel with the replication package and be improved when criticism shows they were too weak.

Publication of replication should preserve negative and mixed outcomes. A first positive followed by unpublished failures would recreate the very selective evidence problem this chapter exists to prevent. The replication record should be cumulative and correction-ready.

Only after independent replication can public scientific status begin to strengthen beyond preliminary component support. Even then, status changes remain component-level until broader

reference classes, measurement versions, endpoints, and transport assumptions are tested.

14.8.5 — Transport and Calibration Recheck

Transport is not automatic. A protocol that works in one hospice system may not work in another region, language, illness group, disability context, cultural death practice, family structure, or technology environment. The transport gate asks whether the measured object, endpoint, reserve proxy, and burden structure retain their meaning outside the original context (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018).

Calibration must be rechecked at each new site and population. The context of use for the candidate momentary-valence estimator, sign identification, instrument version, device behavior, response shift, clinical-state robustness, and endpoint-classifier performance may all change under different conditions. A common label does not create a common scale.

Transport analysis should specify which assumptions are being moved. Population, clinical state, endpoint class, instrument version, reserve proxy, medication regime, cadence, and observation process should each receive a transport statement. If one cannot be defended, the claim narrows.

Transport studies should include overlap wherever feasible: dual devices, parallel scoring versions, shared calibration samples, and independent endpoint review during a bridge period. The overlap data should be held apart from the final transport test so that the bridge is estimated before its success is judged. Silent migration

across sites or versions creates a new measurement system, not a replication.

When transport fails, the result is not useless. It may reveal that the phenomenon is site-dependent, that measurement is not stable, that care practice dominates, or that a reference class was too broad. Such findings are valuable if they are reported plainly.

Failed transport should cost the program scope, not cost participants additional exposure merely to defend a preferred conclusion. When language, device, care setting, or observation cadence changes the measurement bridge, the study recalibrates, narrows, or stops. The burden of demonstrating continuity remains with the investigators and sponsor (Dahabreh et al., 2019; World Medical Association, 2024).

14.8.6 — Public Artifact Embargo

Public artifacts remain embargoed until claim-status, privacy, recruitment, and hostile-crop review are complete. The rule covers graphs, diagrams, dashboards, animations, social excerpts, donor summaries, press language, educational guides, and model images. Every released object should identify the population, measured object, observation window, comparator set, uncertainty, and verdict ceiling within the artifact itself (Hopewell et al., 2025; Fischhoff & Davis, 2014; Spiegelhalter, 2017).

Artifacts should distinguish observed, filtered, forecast, smoothed, simulated, synthetic, retrospective, endpoint-aligned, and sensitivity-only quantities. Recruitment materials face a stricter rule: beauty, destiny, spiritual resonance, belonging, aid, care access, or fear of missing out may not be used to influence

participation. Burden, privacy, withdrawal, and the possibility of no usable inference must be stated with equal visibility.

Review standalone images, captions, and sentences for accurate context, privacy, and consistency with the study's result. Revise any item that could misstate the finding or expose participants and families before release.

A first study that survives its gates should provide a complete replication packet. It includes the protocol and registry identifier where available, endpoint manual, data dictionary, codebook, simulation library, model and rival specifications, burden limits, missingness marks, adverse-event definitions, decision table, analysis environment, and report templates needed for independent evaluation.

Replication materials preserve threshold sensitivity, alternate windows, competing missingness models, rival specifications, and analyses that nearly changed the decision. Private terminal narratives, identifiable family material, and raw records unnecessary for verification remain protected.

Table 14.9. Replication packet minimum contents.

Replication packet item	Minimum content	Why it matters
Protocol version	Registration date, amendments, freeze points, responsible committees	Prevents silent claim-version drift

Replication packet item	Minimum content	Why it matters
Endpoint manual	Candidate, detected, adjudicated, censored, return, and no-estimate rules	Protects the endpoint firewall
Momentary and interval estimator dictionary	Channel version, transforms, cadence, quality marks, posterior path rules	Prevents score summation from replacing measurement theory
Reserve proxy specification	Context of use, units, confounds, calibration, error model, abstention rule	Prevents reserve from becoming an all-purpose frailty label
Rival and simulation package	Generative worlds, parameter ranges, model-recovery thresholds, confusion matrix	Shows that the design could distinguish alternatives
Burden and safety ledger	Participant, family, staff, privacy, and clinical stop events	Keeps feasibility and ethics inside scientific status
Public language templates	Branch-specific summaries for support, rival win, no-estimate, and adverse result	Prevents post hoc rhetorical rescue

A second study should be adversarial in at least one consequential dimension, such as device family, site, endpoint adjudication team, population, cadence, or the rival that most threatened the first result. Reproducing every favorable local choice is a repeat performance rather than a transport test. The replication protocol

should state which changes test generalization, which require recalibration, and which create a new estimand.

The bounded protocol record consists of population and context of use, measurement and endpoint systems, estimands and intercurrent-event rules, rival and open-set generators, simulation and recovery thresholds, missingness and safety branches, stopping rules, public language, correction routes, and replication gates. The freeze checklist below is part of the scientific object. Apply §6.7's prospective-specification terminology to the finalized pre-enrollment package. The frozen package should also include hashes or equivalent identifiers for the exact code, model, and instrument versions used at enrollment.

Table 14.10. Protocol freeze checklist.

Protocol element	Required freeze before enrollment	Failure if absent
Population and reference class	Sites, diagnoses, care settings, consent pathways, exclusions	Results generalize by rhetoric rather than design
Measurement system	momentary-estimator version, auxiliary channels, calibration, artifact rules, cadence	Measured patterns cannot be interpreted
Endpoint system	Independent clocks, adjudication rules, return handling, censoring, no-estimate states	Terminal inference becomes circular or unstable

Protocol element	Required freeze before enrollment	Failure if absent
Estimands	Attributes, windows, intercurrent events, population level, summary measures	Analysts estimate different questions after seeing data
Rival set	Model families, scoring rules, hybrid and none-of-models branches	Candidate wins by comparison to weak alternatives
Recovery simulations	Parameter recovery, model recovery, noise floors, missingness, endpoint jitter	The study may be unable to learn before it recruits
Negative controls	Outcomes, timing controls, endpoint shuffles, artifact channels	Design artifacts may masquerade as signatures
Safety and burden plan	Stop rules, adverse events, privacy incidents, staff burden, family protections	Human cost becomes invisible to the scientific verdict
Reporting tree	Support, rival win, failure, mixed, no-estimate, adverse-result language	Public meaning is rescued after the fact

The gates remain noncompensatory, and the final report should state which future studies are barred until prerequisites improve. Simulation cannot repair coercive recruitment, endpoint precision cannot repair an unqualified instrument, rival comparison cannot erase a privacy incident, and comfort-triggered missingness cannot be relabeled as convenient missing-at-random data.

Measurement failure may redirect work to calibration, endpoint failure may limit the result to method development, and excessive burden may stop terminal recruitment until a lighter design is independently reviewed. The report should distinguish a gate that can be repaired before replication from one that invalidates interpretation of the current cohort entirely.

The first protocol closes with an ethic of small, exact learning. It may narrow, pause, redirect, or stop the program. Trust is earned by learning one bounded thing without taking more from participants, families, or clinical teams than the scientific question has the right to ask. A small valid result should outrank a broader conclusion assembled by combining components whose prerequisites were never jointly satisfied.

Chapter 15 — Sleep, Dreams, and Affective Reprocessing

Sleep provides a repeated natural laboratory in which consciousness, report, memory, physiology, and affect separate and recombine. Its scientific richness becomes useful when those layers are kept explicit. Nocturnal change can then generate discriminating hypotheses without being turned into hidden balancing, prophecy, or privileged access to the latent ledger.

Sleep and dream data provide auxiliary evidence about state, report, memory, and affective transition. Dream content may illuminate fear, desire, grief, pain, bodily disturbance, and memory recombination, but the report remains an observation rather than a nocturnal ledger. Dream experience, awakening, recall, coding, and later interpretation are therefore modeled as distinct stages of one pathway (Palmer & Alfano, 2017; Tempesta et al., 2018; Siclari et al., 2017).

The fairness question enters through burdens that silence can hide. Fear, bodily discomfort, grief, medication, respiratory struggle, trauma recurrence, and fatigue may occur without public report, while measurement earns access only through consent and purpose limitation. Bounded sleep studies can test specified auxiliary mechanisms only while preserving privacy and giving strong sleep-science rivals equal standing.

The most basic discipline is separation. Sleep experience, dream report, dream recall, physiological stage, next-day affect, and clinical sleep disorder are not the same object. A vivid dream report may reflect a strong nocturnal experience, a memorable fragment, an awakening effect, a cultural narrative habit, or an interviewer's timing. A quiet morning may reflect recovery, depletion, medication, dissociation, habituation, or simple lack of

recall. The chapter's first scientific task is to keep these objects apart.

The second discipline is equal-status rivalry. REM and NREM processes, emotional-memory consolidation, extinction, fear generalization, noradrenergic modulation, predictive updating, homeostatic sleep pressure, circadian phase, nightmares, sleep-disordered breathing, targeted memory reactivation, medication, and ordinary recovery can all produce apparent affective adjustment. A Law-specific auxiliary mechanism advances only when its frozen predictions outperform these alternatives under the same observation model.

Time supplies a third discipline. Within-night transitions, awakening reports, morning mood, next-day behavior, and longitudinal recurrence answer different questions. Reduced autonomic arousal after a REM-rich period, a morning report of relief, and a month-long reduction in nightmares after treatment therefore require distinct estimands and mechanisms.

For that reason, the chapter treats a dream report as the output of an observation process. Experience, awakening, selection, memory, language, coding, confidence, and later interpretation are separate stages. Scientific use requires the protocol to record the stage, delay, medication, daytime burden, prompt, privacy option, and uncertainty that shaped what became reportable.

Affective reprocessing is a candidate mechanism. Nightmares, trauma replay, grief dreams, and recurrent threat may reflect harm, emotional learning, failed integration, medication, arousal, or a clinical state requiring care. Their occurrence alone

establishes neither a balancing mechanism nor moral payment. Any genuinely felt contribution remains part of the stream under its declared representation; the signed remainder requirement and the explanation of a particular dream are separate questions.

Sleep remains unusually valuable because it permits repeated within-person designs, natural state transitions, stage-specific sampling, noninvasive physiology, next-day follow-up, and negative-control nights. These features make it possible to test whether a proposed auxiliary mechanism transports across nights and channels, or dissolves into ordinary memory, arousal, respiratory, circadian, and measurement explanations.

The inquiry follows the observation pathway rather than touring dream theories. It fixes what sleep evidence can update, separates experience from report, compares reprocessing accounts, centers nightmares and trauma, links physiology to specific measurement questions, and ends with study families, data fields, privacy, and verdict ceilings. Each step reduces the inferential distance between a hidden state and a public claim.

The payoff is a set of bounded outcomes. A study may support one reprocessing component, favor an ordinary rival, show that awakening changes the state, reveal that privacy or burden prevents adequate observation, or return No adequate estimate. Each outcome improves the next design.

Sleep research can contribute mechanisms, instruments, comparisons, and increasingly informative observations. Optional reporting and participant-controlled disclosure keep that

inquiry compatible with private experience and personal meaning.

The preferred designs can reduce confidence as well as increase it. Dream reports may add nothing after sleep continuity, medication, waking concern, and report probability are modeled; trauma recurrence may explain nightmare patterns; awakening may distort the target state; privacy review may exclude decisive content. These outcomes retire weak bridges before they become public claims.

15.1 — Sleep as Auxiliary Evidence

Sleep belongs in the science volume because it offers repeated observations of changing consciousness, physiology, memory, and affect. The night can reveal transitions that daytime reports miss, creating a valuable setting for state-sensitive measurement and carefully specified reprocessing hypotheses (Walker, 2009; Siclari et al., 2017).

Sleep offers repeated opportunities to test state-dependent experience, memory updating, affective residue, and physiological recovery. These questions connect specific measurements to candidate mechanisms and prospective predictions.

The central discipline is to keep scientific interest separate from evidentiary scope. A sleep pattern can reveal a useful mechanism, improve measurement, or defeat an auxiliary route without making exact terminal closure more probable. That distinction protects the public from treating dream science as proof and protects participants from being cast as characters in a theory they never agreed to embody.

A final discipline follows from that separation: the sleep chapter must never become the emotional substitute for the first falsification protocol. Sleep studies are safer and repeatable, which makes them valuable, but their safety does not enlarge their warrant. They can strengthen instruments, reveal rivals, and test mechanism fragments; they cannot perform the work of endpoint and longitudinal designs.

Table 15.1. Auxiliary sleep evidence and claim limits.

Evidence family	Can update	Cannot update
Sleep-stage physiology	State timing, arousal context, rival mechanisms	latent cumulative process level, terminal valence closure, human worth
Dream reports	Selected nocturnal content, recall and disclosure processes	Whole-night experience, hidden ledger access, moral meaning
Next-day affect	Local transition and residual burden under controls	Whole-life closure or endpoint equality
Repeated-night patterns	Within-person auxiliary mechanisms and rivals	Universal Law status or cross-person rank

15.1.1 — Auxiliary, Not Foundational

A sleep study connects a specified conscious state, observation channel, and prediction about experience, reporting, or measured affect. Those connections define its empirical question (Walker, 2009; Palmer & Alfano, 2017).

Dreams can be vivid, personally transformative, and central to grief or identity. Research can preserve those qualities while describing the sampling and measurement process that produced each record.

An auxiliary hypothesis can be exact. It may predict that specified affective residues change during particular sleep states, that prespecified dream features add next-day predictive information after report bias is modeled, or that sleep disruption changes a

transition expected under a reprocessing account. Each prediction receives its own test.

Sleep's strongest contribution is often diagnostic: it can localize where an experience-to-report bridge fails before any Law-specific interpretation is assigned. Section 15.2 develops that observation chain in full.

Participants control dream disclosure. Refusal, no recall, partial disclosure, and privacy-based exclusion remain legitimate observation states, recorded according to the agreed data plan.

Before collection, label each sleep question by its role: mechanism, measurement, model comparison, exploration, or privacy feasibility. The public report connects each finding to that question and its declared reference class.

15.1.2 — Conscious Experience During Sleep

Sleep is not a uniform absence of consciousness. People may later report visual imagery, emotion, bodily sensation, thought-like mentation, lucidity, fear, relief, confusion, or nothing at all. The absence of a report is not identical to the absence of experience, and a report of experience is not identical to continuous awareness across the whole interval (Tempesta et al., 2018; Cartwright et al., 1998).

Sleeping consciousness should be analyzed by state and transition rather than averaged into one nocturnal category. REM, NREM stages, microarousals, sleep onset, awakening, and fragmented sleep differ in report probability, cognitive organization, autonomic activity, and memory accessibility. A pooled score can

therefore mix changes in experience with changes in access and retrieval.

The Law's bearer problem becomes sharper during sleep. A stream may be unified, dim, fragmented, intermittently reportable, or inaccessible to later memory. That is why this chapter defers final unity classification to the consciousness and endpoint chapters, while still requiring sleep studies to record the uncertainty instead of treating all sleep as ledger-active or ledger-silent by assumption.

Sleep also stresses the distinction between access and experience. A person may have dream experience that cannot be retrieved on waking, or may retrieve a narrative that was reconstructed from fragments. Conversely, physiological arousal may occur without reportable dream content. None of these possibilities authorizes a single easy rule about whether the latent cumulative process is accumulating.

State labels travel with report status, timing, confidence, medication context, and missingness. Useful categories include report-positive, report-negative after awakening, no awakening, awakening-distorted, low-confidence, medication-confounded, and No adequate estimate. These labels preserve uncertainty in analyzable form instead of treating a silent night as neutral.

The framework gives experience scientific importance even when ordinary report is unavailable. Sleep research helps identify which methods can recover useful evidence about those intervals and where uncertainty must remain explicit.

The research design should therefore use graded state confidence rather than binary sleep-consciousness declarations. A report-positive REM awakening, a report-positive NREM awakening, a spontaneous morning memory, and a no-report interval under medication should not be coded as equivalent states of access. They each carry different uncertainty about consciousness, memory, and affective contribution.

A difficult but necessary implication follows: the measured absence of report can never be used as a simple zero in a ledger-like construction. If an interval is low access, low signal, or no-estimate, the data structure should say so. The scientific program gains more from honest unknowns than from a smooth nocturnal curve built by calling the invisible empty.

15.1.3 — Sleep Architecture and Affect

Sleep architecture gives the chapter its first ordinary scientific scaffolding. REM sleep, NREM stages, slow-wave activity, spindles, arousals, respiratory events, cardiac events, and circadian timing can shape memory, affect, reportability, and next-day function (American Academy of Sleep Medicine, 2023; Siclari et al., 2017).

A recent experiment makes the opportunity tangible. Greco and colleagues analyzed 18 healthy adults after replaying, during REM sleep, sounds associated with half of a set of negative images. At the 48-hour assessment, cued images elicited lower responses in selected emotion-related brain regions and less heart-rate deceleration than uncued images. Subjective arousal decreased for initially more upsetting images but increased for

less upsetting images. This small study supports selective, baseline-dependent modulation of emotional responses. Following all three channels helps identify which part of an emotional response changes (Greco et al., 2025).

Sleep continuity provides a complementary research opportunity. Wassing and colleagues (2019) studied 29 participants with evening and morning brain measurements surrounding overnight sleep recording. More uninterrupted REM sleep was associated with greater overnight amygdala adaptation, while more restless REM was associated with less adaptation. Sleep continuity can therefore enter a prespecified model of affective change alongside memory and waking context. Existing-data reanalysis and voluntary observation of natural nights can test whether cumulative affective history adds prediction beyond recent distress and sleep quality.

NREM sleep is equally important. Slow-wave sleep, spindle activity, hippocampal-cortical replay, and homeostatic pressure may influence memory stabilization, abstraction, and next-day regulation. NREM-linked change directs the analysis toward memory consolidation and stage-specific mechanisms.

Circadian phase and homeostatic pressure also matter. A person's evening burden, sleep debt, temperature rhythm, cortisol pattern, light exposure, and awakening time can affect mood before dream content enters the analysis. Without these controls, a study may mistake time-of-day physiology for nocturnal affective work.

The evidence baseline is layered. Stage architecture can orient hypotheses; physiology can time candidate processes; reports can

sample selected experience; next-day measures can test residual change.

Sleep architecture offers a rich set of measurable mechanisms. The chapter asks which findings refine the proposed route, which are explained by established sleep science, and what prospective comparison would distinguish them.

Stage information helps interpret changes in physiological variability. Sedation, exhaustion, respiratory obstruction, sensor limitations, and deep sleep can each produce a quieter signal through different processes. Channel-specific models distinguish those explanations.

For field experts, the crucial standard is context of use. A sleep-stage feature can be used to schedule awakenings, stratify reports, model arousal, or define a rival pathway. It cannot be treated as a universal psychological variable. The chapter's later protocols preserve this distinction by requiring stage labels, report status, and next-day outcomes to remain separable.

15.1.4 — Reprocessing as Candidate Function

Affective reprocessing names the possibility that sleep changes how affective material is organized, carried, recalled, or embodied. Researchers can investigate its timing, mechanisms, and relationship to waking experience through explicit models and observations (Walker, 2009; Palmer & Alfano, 2017).

A workable reprocessing hypothesis needs a direction, a time scale, a measurement object, and a rival set. It might predict reduced next-day reactivity to a daytime stressor after intact sleep, altered dream affect after targeted memory reactivation, or a

reserve interaction in which poor sleep quality weakens preterminal affective stabilization. Without such specificity, reprocessing is only a beautiful word.

The Law-specific version would need to do more than show that sleep affects emotion. It would need to show a pattern that is prospectively specified, distinct from ordinary memory consolidation and arousal regulation, robust to report bias, and tied to the particular burden-and-trajectory structure under investigation. That is a much harder target and therefore a more useful one.

Failure must be allowed. Sleep might worsen next-day affect, leave affect unchanged, amplify nightmares, stabilize traumatic memory, or produce improvements fully explained by ordinary sleep quality. Each result narrows the mechanism rather than inviting a rescue story. A hypothesis that turns every dream into evidence has left science.

Sleep-related distress calls for care on the person's own terms. Reprocessing describes a candidate mechanism; clinical and personal decisions respond to the sleeper's needs, experience, and preferences.

The candidate function supports disciplined hope: sleep offers repeated opportunities to study affective transition and improve its measurement. A distinct route signature would justify further testing; an ordinary sleep-science explanation would still identify mechanisms that help the program become more precise.

A reprocessing model specifies the conditions under which sleep should improve next-day affect, leave it unchanged, or increase

distress. Sleep disruption, nightmares, and ordinary memory consolidation provide concrete comparisons for testing those predictions.

The most useful form of the hypothesis is not a statement about dreams fixing emotion, but a set of conditional predictions. Given a measured pre-sleep burden, a qualified sleep-state profile, a report model, and a next-day residual, the study asks which rival family best predicts the transition. That framing makes reprocessing testable without making it mystical.

15.1.5 — Waking Comparators

No sleep hypothesis can stand without waking comparators. Affective change occurs during waking rest, social contact, distraction, rumination, medication, therapy, exercise, pain relief, food, light exposure, and ordinary time. A sleep-linked pattern matters scientifically only if it differs from plausible waking baselines or interacts with sleep-specific mechanisms in a protocol-fixed way (Tempesta et al., 2018; Cartwright et al., 1998).

The simplest comparator is within-person daytime burden. A study should know what the person carried before sleep: stressor exposure, pain, conflict, grief cue, medication, fatigue, social support, physical symptoms, and anticipatory worry. Without this pre-sleep baseline, a morning report cannot be interpreted as recovery, persistence, worsening, or measurement noise.

The second comparator is an altered-sleep or poor-sleep night. Fragmented sleep, delayed sleep, REM suppression, respiratory disruption, nightmare recurrence, and awakening protocols can

all change the next-day affective trajectory. These conditions should be modeled carefully rather than treated as mere exclusions, because they may reveal which processes are actually doing the work.

The third comparator is a no-dream-report night. If next-day affect changes without recalled dream content, the mechanism may lie in sleep physiology, memory consolidation, or unreported experience. If recalled content predicts change only after high-intensity awakenings, the report mechanism itself may be creating the association.

The fourth comparator is waking reappraisal: people reinterpret events, receive comfort, change plans, and forget details while awake. Without measuring this, sleep studies may credit the night with what the day accomplished.

Comparators prevent unmeasured waking life from making sleep seem mysterious. Fairness science must track burden across day and night; neglecting either distorts the other.

The comparator problem is especially important for ordinary days. This method of locating hidden cost asks where burden is carried, and much burden is carried in chores, caregiving, conflict, anticipation, or quiet support. A sleep design that samples only the night risks treating the day as an unmeasured disturbance when the day is the burden field from which the night begins.

15.1.6 — Rival Sleep Science

Rival sleep science must be named before data arrive. Emotional-memory consolidation, extinction, reconsolidation, threat

simulation, predictive updating, arousal regulation, circadian mood effects, medication effects, and sleep-disorder mechanisms belong in the comparison set. Zhang and colleagues provide one active-dreaming result within that larger field; it is not a general theory of sleep or a warrant for closure (Palmer & Alfano, 2017; Tempesta et al., 2018; Zhang et al., 2024).

Regulation rivals predict that ordinary sleep may reduce next-day reactivity through restoration, memory abstraction, autonomic settling, or reduced physiological stress. A candidate account that also predicts improvement must therefore specify a distinct dependence on pre-sleep signed burden, state sequence, or independently measured capacity. Improvement by itself is not discriminating.

Other rivals predict worsening rather than repair. Nightmares may rehearse fear, fragmented REM may amplify distress, pain and dyspnea may repeatedly interrupt sleep, and sleep-disordered breathing may leave the person more depleted. The study family must permit those directions and report them without forcing every night into a recovery narrative.

Report selection can create strong apparent effects even when underlying sleep mentation is ordinary. Intense, bizarre, emotional, recently cued, or repeatedly rehearsed dreams are more likely to be remembered and described. Sampling only remembered dreams therefore conditions on a collider-like reporting process whose determinants should be modeled explicitly.

Language and cultural interpretation shape how the same remembered event is narrated. A grief dream may be described as visitation, stress, memory, bodily disturbance, or spiritual experience, and those labels need not represent comparable measurement units. Coding should separate descriptive content from participant meaning and should not erase nonbelief or culturally specific interpretation.

The chapter invites sleep scientists to help specify and test the auxiliary mechanism. Their strongest models give the proposed route a clear benchmark and make any additional predictive value easier to recognize.

The rival set should be operationalized, not merely named. A memory-consolidation rival should define memory outcomes, a threat-simulation rival should define threat content and recurrence, an arousal-regulation rival should define physiological and next-day markers, and an observation-selection rival should define report probability. Rival seriousness means each alternative has enough structure to win.

A night can include tonic affect, phasic images, state changes, and altered memory. Modeling how sleep redistributes, suppresses, amplifies, or transforms particular signals helps identify which dynamics a candidate mechanism needs to explain.

15.2 — Dream Experience, Report, and Recall

Dream evidence passes through several filters before it reaches a dataset. Experience may occur without later recall; recalled fragments may remain incomplete; an interview may be delayed, interrupted, or declined; and coding may erase ambiguity. High-density EEG work on parasomnia episodes is valuable precisely because the presence of experience and the recovery of content separated within the same observed state (Siclari et al., 2017; Cataldi et al., 2024).

The observation chain is now explicit: sleep state, possible experience, awakening, encoding, recall, report, coding, and later interpretation. Every link can fail or change the signal. Dream research remains valuable precisely because the design refuses to substitute one late link for the entire chain.

The report chain is also a consent chain. Every movement from experience to report, from report to code, from code to model, and from model to public summary changes what the participant has allowed the study to hold. This section treats those transformations as scientific facts rather than merely administrative permissions.

Because that chain is fragile, the most truthful dream dataset will often look less complete than a public audience expects. It will contain refusal, fragments, uncertain reports, no-recall intervals, cultural notes, and restricted content. Those markings are not defects. They are the visible form of scientific honesty around a private state.

15.2.1 — Dream State Versus Dream Report

The dream state is the candidate experience; the later report is the observation. A person may experience without recalling, recall only fragments, reshape content after awakening, or decline to disclose material that feels private, shameful, sacred, or confusing. Report probability and recall confidence therefore belong in the observation model, and silence cannot be equated with dreamlessness (Siclari et al., 2017; Cataldi et al., 2024).

A dream report therefore has to be treated like a measurement with a sampling mechanism. Its presence depends on sleep stage, awakening timing, intensity, salience, language, confidence, context, and willingness. The narrative may preserve affective truth while losing temporal order, or preserve a striking image while losing the emotion that surrounded it.

The report supplies a filtered record of sleep experience or memory. Its interpretation includes the awakening procedure, recall interval, and reporting conditions.

The observation model should include report availability, content length, affect rating, confidence, disclosure limits, and refusal. A refusal is not invalid data. It may be an ethically successful boundary that tells the study which measurements are too intimate for the participant or the setting.

The model also needs uncertainty about the unreported portion of the night. A vivid nightmare report does not imply that the entire sleep period was negative. A pleasant fragment does not imply repair. A blank report does not imply absence of experience. These are partial observations with structured ignorance.

Separating state from report creates testable questions about the filter itself. Report probability can be modeled as a function of stage, awakening method, medication, pre-sleep burden, arousal, and next-day mood. The study can then ask whether an apparent affective pattern belongs to sleeping experience or to which experiences became reportable.

The dream-state/report distinction should be built into the variable names themselves. A study that labels a field dream valence when it actually stores reported dream-valence after awakening has already hidden a major assumption. The more exact name is report-coded dream affect, with separate fields for stage, delay, confidence, and disclosure restriction.

This naming discipline helps clinicians and participants too. A person may disagree with a coder's label, may not want an image interpreted, or may state that the report captured only a fragment. If the dataset preserves those distinctions, disagreement becomes information instead of an error to be corrected by the researcher.

15.2.2 — Awakening and Sampling

Awakening is not a neutral camera. It interrupts sleep, changes physiology, alters memory access, and can affect the next report. A dream study that wakes participants for temporal precision buys information at the price of sleep disruption, reactivity, and burden. The protocol should randomize or otherwise model awakening schedule where feasible, report failed and refused awakenings, and distinguish the state being studied from the state created by measurement (Palmer & Alfano, 2017; Tempesta et al., 2018).

Sampling time matters. Reports after REM awakenings, NREM awakenings, spontaneous awakenings, morning recall, and middle-of-night probes may differ in content, emotionality, length, and confidence. A study that mixes them without labels may convert methodological heterogeneity into false psychological findings.

Sampling intensity also matters. Frequent awakenings can increase recall while reducing sleep quality, and the resulting next-day affect may reflect the protocol rather than the dream process. The design must therefore include awakening burden, sleep fragmentation, and next-day fatigue as variables, not merely tolerability notes.

A privacy-preserving alternative is staged sampling. The study may use physiology and actigraphy on most nights, brief morning affect reports on many nights, optional dream keywords on selected nights, and full narrative only in a smaller consented sub-study. The richest data need not be the default data.

Sampling should be matched to the scientific question. If the question concerns within-night stage-linked mentation, awakenings may be justified. If the question concerns next-day residue, morning report and EMA may be enough. If the question concerns nightmares, clinical diary methods may be safer than laboratory interruption.

The burden lens is precise here. The more the study wakes, prompts, records, or asks, the more the protocol becomes part of the phenomenon. A fair design counts that burden as a causal input into the data-generating process.

Sampling also creates a burden gradient. Healthy volunteers in a sleep laboratory, older adults at home, trauma survivors, hospitalized patients, and palliative participants may tolerate awakenings very differently. A single protocol cannot be fair across those contexts merely because the instrumentation is noninvasive. The burden budget must be population-specific.

Sampling-dose sensitivity can compare suitable existing records or naturally varying observation schedules. Any newly assigned awakening schedule must first satisfy the Chapter 3 limits; a desire to compare zero, one, and several awakenings does not itself authorize sleep disruption. Report whether the permitted observation method changes sleep continuity, recall, morning affect, or willingness to continue.

15.2.3 — Recall Delay and Reconstruction

Recall delay changes the evidential object. A report given thirty seconds after awakening is not the same as a report given at breakfast, and neither is the same as a story told weeks later to a clinician or family member. Delay can reduce detail, select salient affect, invite narrative coherence, and add waking interpretation (Palmer & Alfano, 2017; Tempesta et al., 2018).

Dream memory is also reconstructed. A person may fill gaps, reorder scenes, translate images into culturally available symbols, omit embarrassing content, or intensify meaning after the dream becomes relevant. Reconstruction is not dishonesty; it is ordinary memory operating under unusual conditions.

A serious observation model records delay as a variable. It should include awakening time, report time, prompt type, confidence,

affect rating, and whether the participant distinguishes remembered content from inferred meaning. A report that says, I think I was afraid, should not be coded as the same kind of evidence as an immediate fear rating.

A compact model can make the filter visible. Let the observed report indicator depend on sleep stage, affect intensity, contextual willingness, and memory stability rather than on experience alone. The point of the equation is not to pretend certainty; it is to force the study to name the channels through which reports appear.

Recall delay should also be modeled as a nonlinear process. The first few minutes after awakening may preserve sensory and affective fragments that decay quickly, while later reports may become more coherent and more interpretive. A longer report is not automatically a better report; it may be more narratively organized and less temporally faithful.

The observation model should therefore separate immediate affect rating from later meaning attribution. A person can wake with dread and later understand the dream as helpful, or wake comforted and later find the content troubling. A single post hoc valence label loses the very trajectory the chapter is trying to measure.

$$\begin{aligned} &Pr(R_k = 1|S_k, A_k, C_k, M_k) \\ &= \text{logit}^{-1}(\alpha_0 + \alpha_S S_k + \alpha_A A_k + \alpha_C C_k + \alpha_M M_k) \end{aligned}$$

The same logic applies to missing reports. A missing dream report may reflect no experience, no recall, unwillingness, interruption, medication, cognitive impairment, language barrier, or device

failure. Treating all of those pathways as a single zero would manufacture a false night.

Recall delay enters the observation model as a measurable influence on selected nocturnal reports. Recording it helps separate changes in memory from changes attributed to the sleep interval.

15.2.4 — Language and Cultural Interpretation

Dreams arrive in language, and language carries culture. A dream of a dead parent, a flooded room, a pursuing animal, a missing child, or a bright doorway may be described through grief, religion, family obligation, trauma, folklore, illness, or ordinary metaphor. Coding dictionaries should therefore be translated and adapted as instruments, with evidence for construct continuity and room for unclassified meaning rather than universal symbolism (International Test Commission, 2018).

The first rule is to separate content coding from meaning coding. Content may identify persons, places, actions, bodily sensations, threats, social contact, or affective tone. Meaning coding asks what the dreamer takes the dream to signify. Those layers should not be collapsed, especially in cross-cultural or multilingual work.

Translation introduces another burden. Some languages encode emotion, agency, kinship, obligation, spiritual presence, or bodily feeling differently. A translated dream report may preserve plot while losing the social or affective grammar that made it meaningful to the participant. Back-translation alone is not enough when the construct itself is culturally shaped.

Cultural interpretation also affects disclosure. Some participants may freely report dreams because dreams are treated as ordinary psychological material. Others may avoid reporting because dreams are private, sacred, shameful, dangerous to speak, or relationally sensitive. Report probability is therefore not culturally neutral.

The science response is not to abandon dream research. It is to version codebooks, record language, preserve participant meaning separately from analyst categories, use local review where appropriate, and treat cross-cultural comparisons as bridge claims requiring evidence. A universal dream-valence score cannot be assumed.

This is where the project's fairness analysis becomes methodological. Hidden burden includes the burden of being translated into someone else's categories. A fair dream study does not claim access to the person by making their vocabulary convenient for the model.

15.2.5 — Lucidity and Metacognition

Lucid and metacognitive dream states show that awareness, access, and control vary within sleep. Participants able to signal or deliberately influence dream content provide a valuable selected sample; separate studies characterize ordinary dreaming, fragmented recall, parasomnia, and states with limited access (Palmer & Alfano, 2017; Siclari et al., 2017).

Time-locked communication during dreams has already been demonstrated. Across four research teams and 36 participants, Konkoly and colleagues obtained 29 correct answers in 158

communication attempts during signal-verified lucid dreams. Responses used agreed eye or facial signals; most attempts produced no clear answer. Three independent experts unanimously classified 26 of the 29 successful trials as REM sleep. This establishes a practical route for selected questions before awakening. Repeated signaling and broader recruitment are practical next steps toward more detailed observations during sleep (Konkoly et al., 2021).

Metacognition also complicates valence. A dream can be frightening while the dreamer knows it is a dream; a dream can be pleasant while the dreamer feels trapped; a person can choose to alter content and still wake distressed. Experience, appraisal, control, and later meaning must remain separate fields.

For auxiliary Law research, lucid dreaming may help test report timing, signal validity, and within-dream affect ratings under constrained conditions. It does not authorize invasive attempts to engineer dream content or to create special burdens for people who can report from within sleep.

Lucidity is especially important for public communication. A dramatic ability to signal from sleep can be misread as direct access to the hidden ledger. The correct language is narrower: it is evidence about one family of sleep states, one form of access, and one measurement opportunity under strong selection.

Lucid dreaming is a useful edge case for examining metacognition, agency, and time-locked communication during sleep, but it is not representative of ordinary dreaming. Findings from trained lucid dreamers should remain in their own reference

class unless transport to nonlucid sleep is independently demonstrated.

Lucidity provides a useful calibration case. It permits comparison of immediate in-dream signals with later reports, tests timing assumptions, and helps identify how metacognitive access changes recall.

15.2.6 — Interpreting Dream Evidence

Dream records can capture affect, memory, fantasy, threat, bodily sensation, moral imagery, and grief. Each study specifies which features it measures and how they connect to its research question, while preserving the dreamer's control over disclosure (Palmer & Alfano, 2017; Tempesta et al., 2018).

This boundary protects method and dignity together. Dream reports are selected observations shaped by awakening, encoding, recall, language, and reconstruction. The same rule keeps visitation, reconciliation, terminal, and grief dreams from being appropriated as evidence for PTVC, an afterlife claim, or a compensatory meaning the dreamer did not assign.

A bounded study may estimate associations among pre-sleep state, sleep stage, protocol-fixed report features, next-day affect, and ordinary rivals. Coding remains limited to the declared constructs, such as affect, threat, agency, social content, bodily sensation, recurrence, and recall confidence. Analysts must not treat symbolic resemblance, debts in a dream narrative, or post hoc reward stories as evidence of balancing. A separately specified test of signed experiential contributions requires its own valid measurements and comparisons. Separating report content

from signed experiential estimation allows rich qualitative data to inform rivals without granting symbolic interpretation quantitative ledger status.

Public artifacts require equal restraint. Word clouds, symbolic collages, dramatic quotations, and terminal anecdotes can make private narratives appear to be direct evidence of a hidden ledger. Sensitive content should be minimized or restricted, and every displayed variable should be identified as a report-derived measure rather than the experience itself. Where examples are necessary, synthetic or composite illustrations can sometimes communicate coding logic without exposing a participant's distinctive narrative.

A focused research question makes dream reports useful for comparing timing, content, recall, and affect across specified conditions. Its value increases when repeated awakenings or independent scorers can test whether the reported pattern is stable rather than a one-time reconstruction.

15.3 — Affective Reprocessing Hypotheses

Affective reprocessing becomes useful when translated into measurable stages: pre-sleep state, sleep architecture, dream recall, dream affect, physiological change, and next-day memory or reactivity. Reported associations between dream recall or mood and overnight emotional-memory outcomes make an active account testable without establishing that dreams generally repair affect (Cartwright et al., 1998; Zhang et al., 2024).

Sleep research provides a concrete example of emotional responses changing across time. Van der Helm and colleagues (2011) compared responses to previously viewed emotional pictures after a night of sleep or a day awake. The sleep group showed reduced amygdala reactivity and subjective emotional intensity, together with changed amygdala–prefrontal functional connectivity. The overnight reductions were associated with features of REM-sleep physiology. This creates a tractable setting for studying how earlier experience, intervening sleep, and later emotional response form a connected trajectory.

The local task is to explain changes across sleep. A model specifies what changes, what generates the change, and how its prediction compares with established sleep functions.

Affective reprocessing becomes a clear research question when the study specifies what changes, in which sleep state, relative to which pre-sleep condition, through which channel, against which rival, and over what time scale.

Prespecified outcome branches distinguish a valid test from an inquiry blocked by measurement or participant-protection limits.

Failed gates yield Unusable evidence. Valid tests can establish the predicted pattern, identify a different outcome, or favor an alternative mechanism. Each branch informs the next study.

15.3.1 — Pre-Sleep State and Post-Sleep Change

The primary transition in sleep research begins before sleep. Affective burden must be measured in the evening with enough detail to distinguish distress, arousal, pain, grief, anger, bodily discomfort, rumination, social support, and fatigue. Without a pre-sleep state, a morning state has no direction (Cartwright et al., 1998; Zhang et al., 2024).

The post-sleep state must also be more than a mood rating. It should include affective valence, arousal, bodily symptoms, sleepiness, pain, motivation, social readiness, intrusive memory, and confidence in recall. The Law's burden lens is not satisfied by asking whether the participant feels better; it asks what kind of burden changed and where it may have moved.

For a nightly within-person estimand, repeated observations help separate change within a participant from stable differences among participants. Parallel-group or randomized between-person designs can answer other sleep questions; their estimands and assignment assumptions should be stated separately. Neither design automatically licenses a complete-stream conclusion.

A simple time-series model can help keep the estimand visible. The next-day affective residual is modeled as a function of REM features, NREM features, nightmare burden, pre-sleep burden, and person-specific baseline. The equation does not validate the

mechanism; it prevents the study from pretending that morning affect speaks alone.

The pre-sleep baseline should also include expectation. If a participant expects the study to find nocturnal repair, or fears that the study will interpret nightmares, that expectation can change reports and next-day affect. The study should avoid deceptive expectancy manipulation and should measure ordinary expectation as a context variable.

The post-sleep measure should be timed more than once where burden allows. Immediate awakening state, first-morning state, midday residue, and evening recovery can diverge. A pattern that appears at 7 a.m. and disappears by noon should not be given the same status as a stable next-day residual.

$$\Delta A_{i,n+1} = \beta_{REM}REM_{i,n} + \beta_{N3}N3_{i,n} + \beta_{NM}NM_{i,n} + \beta_B B_{i,n} + u_i + \varepsilon_{i,n}$$

The study must also define failure. If adding prospectively specified dream-report features does not improve held-out prediction beyond pre-sleep burden and sleep architecture, those reports add little to that predictive task. If sleep-disordered breathing explains the morning change, the mechanism narrows. If a target pattern appears only after smoothing reports with future information, a claim of prospective prediction fails.

This transition gives auxiliary sleep evidence its strongest role. It is local, repeatable, humane, and open to rival comparison. The question is whether the night changes the next day in a way that survives serious sleep science, not whether one night reveals a whole-life closure.

15.3.2 — Dream Valence and Waking Residue

Dream valence and waking residue are separate targets. A negative dream may leave relief, a positive dream longing, a neutral dream unease, and a forgotten dream later change. Measuring both stages independently prevents the report from dictating the interpretation of the next day (Walker, 2009; Tempesta et al., 2018).

The first question is association. Does dream affect, when reported under controlled sampling, predict next-day affect after accounting for pre-sleep burden, stage, awakening timing, sleep quality, medication, and recall confidence? If the answer is no, dream content may still matter clinically, but it does not support the auxiliary reprocessing mechanism.

The second question is direction. A negative dream followed by reduced reactivity may fit extinction, exposure-like habituation, reconsolidation, or meaning-making. A negative dream followed by increased distress may fit nightmare pathology, trauma replay, arousal carryover, or sleep fragmentation. The same content sign can support different mechanisms depending on the temporal pattern.

The third question is selection. Highly emotional dreams are more likely to be remembered, reported, and discussed. If remembered negative dreams predict next-day burden, the association may arise because intense nocturnal states are selected into the dataset. The model must include recall probability before interpreting the valence path.

A candidate sleep model can predict an interaction among dream valence, sleep quality, a reserve proxy, and next-day residual affect. Specifying that pattern before analysis makes it comparable with alternative explanations.

A recent study makes the dream–waking relationship more specific. Lin and colleagues (2026) followed 87 adults with seven-day dream diaries and then recorded EEG during an emotional-picture task. An exploratory analysis linked greater dream fear to stronger late-positive-potential responses to negative pictures among participants reporting good sleep quality. Dream affect showed no significant association with subjective response ratings or waking emotion regulation. The result directs attention to a precise joint question: how do dream content and sleep quality predict each response channel across repeated nights?

This is also where mixed affect becomes critical. A dream can be frightening and meaningful, sad and comforting, disgusting and funny, or peaceful and empty. If the study forces one signed rating, the scalarization decision may determine the result. A more serious design allows positive and negative channels, arousal, agency, and meaning to be reported separately.

Next-day residue also needs multiple dimensions. Distress may fall while fatigue rises; fear may fall while shame rises; peace may coexist with low motivation; connection may deepen while grief remains. The relevant question is which burden moved and by what mechanism, not whether one morning score moved in a preferred direction.

15.3.3 — Habituation and Extinction

Habituation and extinction describe ways emotional responses can diminish through repeated or context-changed exposure. A study can compare those learning mechanisms with the additional contribution of sleep by measuring both processes (Palmer & Alfano, 2017; Siclari et al., 2017).

Variation across findings helps identify the conditions under which sleep changes emotional responses. In 29 adults, Halonen and colleagues compared REM suppression with slow-wave-sleep suppression. Skin-conductance responses to a social-stress task increased more overnight when REM was preserved, while no between-condition differences were detected in subjective affect or memory. The result supplies a useful contrast to accounts of universal emotional dampening. Future work can test moderators such as task type, initial arousal, sleep stage, and the outcome channel, using existing data or independently justified, low-burden observation (Halonen et al., 2024).

A usable design would measure the daytime fear or burden, the sleep-stage context, any dream or mentation report, physiological arousal during sleep, and the next-day response to a nonharmful cue. It would test whether the association is strongest under hypothesized reprocessing conditions and weaker under control nights.

Habituation also provides a falsifier for romantic dream theory. If repeated nightmares remain distressing or increase avoidance, the dream is not functioning as repair in any simple sense. If dream exposure reduces distress only in people with strong waking

therapy support, the mechanism may belong to therapy and memory rather than the Law.

The chapter forbids inducing trauma, fear, or nightmare burden for crucial testing. The appropriate designs use naturally occurring concerns, mild ethically approved learning tasks, clinical data where care is primary, or retrospective coding with strong limitations. Suffering is never created to see whether sleep balances it.

The extinction rival also requires attention to context renewal. A fear response may diminish in one context and return in another, which means a morning improvement after sleep may not represent durable change. Follow-up prompts should test whether the affective change persists across cues, settings, and time.

A Law-specific sleep mechanism would be more credible if it predicted patterns beyond standard extinction. For example, it might predict interactions among burden class, reserve, sleep continuity, and next-day residual that are not expected under ordinary cue-exposure learning. Those interactions should be specified before data are inspected.

15.3.4 — Reappraisal and Memory Updating

Reappraisal is another rival because the mind can change the meaning of an event without changing the event itself. A dream may reorganize roles, soften a memory, intensify a conflict, or place a person in a new perspective. Those possibilities predict changes in remembered meaning and next-day response, not a hidden payment or terminal balance (Cartwright et al., 1998; Tempesta et al., 2018).

Memory updating can occur through waking reflection, social conversation, therapy, sleep-dependent consolidation, targeted cueing, or spontaneous dream imagery. A study must separate the sleep contribution from waking meaning-making after the report. If a participant wakes and then interprets the dream with a partner, clinician, or interviewer, the post-report context becomes part of the outcome path.

Reappraisal also affects coding. The same dream can be coded as negative in immediate affect and positive in later meaning. A grief dream may hurt and comfort at once. A trauma dream may repeat threat while revealing a new sense of agency. Single-valence coding loses these distinctions.

A useful schema separates immediate dream affect, confidence in recall, later appraisal, perceived meaning, bodily residue, intrusive memory, and next-day functioning. Analyses can then test which fields are predicted by sleep-stage or physiological features after accounting for waking interpretation. This is a more discriminating target than asking whether a dream somehow balanced prior experience.

Here the fairness analysis asks where burden is carried after reinterpretation. A memory may feel lighter because the person has integrated it, or because the person has learned to silence it, or because medication changed arousal, or because reporting pressure shaped the story. Reappraisal is not automatically relief.

Affective reprocessing can be investigated more precisely when measurement distinguishes memory, appraisal, physiology, and social context.

Positive learning can influence which emotional memories remain accessible. In a study of 37 participants, Xia and colleagues first established word associations with negative images, then linked selected words to positive images. Replaying those words during subsequent NREM sleep weakened recall of the associated negative images and increased positive-memory intrusions relative to uncued items. The result supports selective memory updating and a concrete route for studying how positive experience changes later access to emotional material (Xia et al., 2024).

Reappraisal can arise through therapy, religion, family conversation, or private reflection. A study can examine changes associated with those processes while participants retain control over the meanings and experiences they choose to share.

$$\theta_{TMR} = E[\Delta A|cue, stage, context] - E[\Delta A|sham, stage, context]$$

15.3.5 — Reserve and Recovery

Sleep quality may interact with reserve because fatigue, illness, medication, grief, and clinical fragility change recovery capacity. A sleep study can examine how those conditions shape observed recovery dynamics (Walker, 2009; Tempesta et al., 2018).

A reserve-aware sleep study tracks the conditions that shape both experience and observability: sleep duration, fragmentation, REM proportion, slow-wave activity, respiratory disturbance, pain, medication, mobility, caregiving interruption, and daytime exhaustion. These variables help explain the capacity to sleep, dream, recall, and recover.

The candidate prediction must be modest. If a person has sufficient sleep continuity and lower physiological disruption, the association between pre-sleep burden and next-day residual may weaken. If sleep is fragmented, obstructed, painful, or nightmare-dominated, the expected transition may fail or reverse. That failure is scientifically informative.

Reserve also creates confounding. Severe illness may reduce sleep quality, increase medication, change report capacity, increase family presence, and alter next-day affect. A study that calls the result reserve-linked without modeling these pathways has confused a shared clinical driver with a mechanism.

The safest strategy is within-person and time-forward. Compare nights within the same person, include clinical covariates, preserve missingness, and test whether the reserve interaction predicts held-out nights. Do not use reserve to rank people, forecast worth, or guide care access.

Sleep can contribute to the reserve program only by making capacity-sensitive dynamics more testable. It cannot convert biological decline into evidence of fairness, and it cannot make the sleeping person responsible for whether recovery occurs.

Reserve interactions are modeled as moderators. Pain, respiratory disturbance, caregiving interruption, and advanced illness can change both sleep and recovery capacity, supplying specific conditions for comparison.

15.3.6 — Failure of Reprocessing Predictions

Failure is part of the reprocessing hypothesis. If protocol-fixed dream features do not predict next-day residuals, the dream-

content route weakens; if sleep architecture predicts change but content does not, the explanation moves toward physiology or memory consolidation; if neither adds information beyond waking burden and ordinary rivals, the auxiliary pathway may retire for that reference class (Palmer & Alfano, 2017; Siclari et al., 2017).

Worsening is an admissible result. Nightmares, insomnia, fragmented REM sleep, sleep apnea, pain, medication effects, and trauma recurrence may increase burden. A model that redescribes every worsening as deeper repair has lost the possibility of failure and revives the rejected idea that suffering serves balance.

Null and unusable results require different diagnoses. A null may follow from weak recall, poor timing, noisy coding, inadequate power, or absence of the hypothesized effect; unusable evidence may follow from sleep disruption, unacceptable burden, privacy limits, or missing reports that cannot be modeled credibly. The verdict should identify the failed component rather than declaring the whole Law protected or vindicated. The next study should be chosen by that diagnosis, because more participants cannot repair a timing failure and better coding cannot repair unacceptable burden.

Retiring a sleep mechanism narrows dream-content claims without deciding physiology, measurement, endpoint, reserve, or longitudinal branches. Tables and summaries must preserve that separation. A retired mechanism may still leave reusable timing or measurement findings, but those should be cited for their own object rather than as residual support for the route.

The protocol should prestate mechanism-favored, subgroup-narrowed, rival-favored, dream-content-retired, adverse-burden, privacy-limited, and no-estimate outcomes. Otherwise positive results are absorbed and negative results renamed. Precommitting the null and burden branches is especially important in sleep studies because missing reports and disrupted sleep can otherwise be redescribed after the fact.

Privacy-preserving review of null, adverse, and no-estimate sleep results shows what the program must narrow or stop. Affective reprocessing is scientific only when evidence can change or retire it. A mechanism that survives only after excluding nights with inconvenient reports should be treated as selection-sensitive rather than strengthened.

Table 15.2. Rival predictions for sleep and dream reprocessing hypotheses.

Result pattern	Law-specific auxiliary reading	Strong rival reading	Verdict limit
Dream affect predicts next-day residual after controls	Possible sleep-linked affective transition	Report selection or memory salience	Mechanism support only
Sleep architecture predicts change; dream content does not	Physiology may matter more than report	Ordinary sleep homeostasis	Dream mechanism narrows
Nightmares predict worsening	Reprocessing mechanism fails in this class	Nightmare pathology or trauma loop	No repair inference

Result pattern	Law-specific auxiliary reading	Strong rival reading	Verdict limit
No separable pattern	Auxiliary mechanism unsupported	Measurement limits or rival equivalence	Abstain; mechanism retirement requires valid adverse evidence

15.4 — Nightmares, Trauma, and Memory Rivals

Nightmares and trauma-related sleep are not edge cases. They are the settings in which a simple repair narrative becomes most dangerous. Repetition, fear, bodily arousal, impaired extinction, and fractured memory may mark suffering that needs care, not evidence that the mind is balancing itself (Giesermann et al., 2019; Zhang et al., 2019).

Failure of a reprocessing prediction should identify which link failed: no state-specific change, no relation to pre-sleep burden, no improvement beyond memory rivals, poor report reliability, or unacceptable participant burden. The chapter then narrows the auxiliary claim rather than preserving it through symbolic interpretation. Harmful exposure, nightmare induction, or trauma amplification remains outside crucial testing.

The chapter's ethical seriousness is tested most severely here. The same evidence that might fascinate a researcher may cost the participant sleep, safety, dignity, or trust. For that reason, clinical support, optionality, and no-exposure rules are not soft constraints added after the science; they define which science can be attempted at all.

The section also keeps language clean. It does not call nightmares lessons, balancing, correction, or needed experience. It calls them high-burden phenomena with clinical rivals, measurement hazards, and care obligations. That language choice is part of the ethical firewall because words can make harm sound useful before a protocol ever begins.

Table 15.3. Nightmare and trauma rival paths.

Observed pattern	Primary rival to test	Required safeguard
Recurrent threat dreams with next-day distress	Nightmare pathology, trauma replay, hyperarousal	Care referral and no exposure induction
Negative dream followed by lower reactivity	Extinction, reconsolidation, therapy support	Measure waking treatment and reappraisal
No dream report with persistent distress	Recall failure, avoidance, medication, insomnia	Missingness category and privacy option
Dream improvement after intervention	Clinical treatment pathway	Do not attribute to spontaneous balancing

15.4.1 — Nightmares and Recurrent Threat

Nightmares are clinically important experiences that call for careful attention and effective support. They may repeat threat, loss, helplessness, bodily danger, or social humiliation, and chronic forms may involve hyperarousal and impaired fear extinction. Research should help explain and reduce that burden (Gieselmann et al., 2019; Zhang et al., 2019).

Recurrent threat may accompany trauma, anxiety, grief, medication, sleep disruption, pain, withdrawal, fever, neurodegenerative conditions, or ordinary stress. Comparing those contexts can help identify the processes maintaining the symptoms and the conditions that improve them.

Nightmares can provoke awakenings, avoidance of sleep, daytime fatigue, hyperarousal, and anticipatory fear. Tracking the

following day captures how nocturnal experiences and waking symptoms interact.

Nightmare studies should distinguish frequency, affective intensity, thematic recurrence, autonomic arousal, awakening, recall confidence, avoidance, sleep fragmentation, and next-day impairment. These fields may move together, but they support different mechanisms and clinical consequences. Collapsing them into one negative-dream score hides the rival structure the study is meant to test.

The auxiliary hypothesis specifies when recurrent threat should change, when it should persist, and which clinical conditions explain the difference. Nightmare pathology supplies a substantive comparison for these predictions.

The burden lens is especially strict here. A person with nightmares is not thereby shown to be paying a moral debt or receiving nocturnal justice. The scientific obligation is to reduce harm, respect privacy, and avoid making symbolic meaning more important than care.

The coding scheme should therefore include recurrence, anticipatory anxiety, avoidance of sleep, awakening distress, and next-day impairment. These variables can distinguish a vivid but isolated negative dream from a disorder-like loop. The Law-specific mechanism has no right to call both examples reprocessing.

15.4.2 — Trauma Replay and Fragmentation

Trauma replay is not ordinary dream drama. It can carry sensory fragments, bodily threat, helplessness, shame, and physiological

arousal in ways that destabilize sleep and waking life. The dream may not tell a coherent story because the memory itself may be fragmented, state-bound, or linked to threat systems more than narrative memory (Palmer & Alfano, 2017; Tempesta et al., 2018).

Persistent trauma dreams give reprocessing models a specific explanatory task: account for repeated imagery, arousal, avoidance, and the conditions under which those patterns improve.

The chapter distinguishes trauma replay from trauma processing by examining re-experiencing, intrusive imagery, nightmare recurrence, physiological activation, and indicators of safe memory integration. Intervention studies can help identify which changes accompany recovery.

Study designs in trauma contexts require the highest restraint. No participant should be exposed to trauma cues, asked to intensify nightmares, or pressured to disclose dream content for a theoretical test. Existing clinical care, optional diaries, and de-identified coding are the ethical limits unless expert review justifies a narrower design.

Trauma also changes report. Some participants cannot describe dreams without distress; others avoid sleep, suppress recall, or disclose only fragments. Missingness may be a protective response. Treating missing trauma-dream data as absence of dream burden would be a serious error.

Trauma-related sleep gives the program a clear empirical responsibility: describe persistent burden accurately, identify

conditions that improve it, and test reprocessing predictions against that evidence. An auxiliary mechanism must accommodate both recovery and continuing symptoms.

Fragmentation also challenges dream-content coding. Trauma dreams may contain sensory flashes, bodily states, or scene fragments that resist ordinary narrative categories. A codebook designed for plot, characters, and valence may misclassify the most clinically important material. Trauma-informed coding should include no-code, fragment, and bodily-threat categories.

The research team should plan for post-report support. A participant who describes trauma-related dream content may need grounding, clinician referral, or a pause in participation. Those procedures are not outside the study; they are part of conducting a fair study in a high-burden domain.

15.4.3 — Memory Consolidation Rival

Memory consolidation is one of the strongest rivals to Law-specific sleep reprocessing. Sleep can stabilize, transform, prioritize, or reorganize memories. Emotional memories may become more accessible, less reactive, more generalized, or more persistent depending on context (Walker, 2009; Tempesta et al., 2018).

A consolidation account predicts relationships among prior learning, sleep stage, cue exposure, memory accuracy, emotional reactivity, and later recall. Those predictions are concrete and often testable. A Law-specific account must either make different predictions or explain why the same pattern means something more than ordinary memory science.

Dream content may reflect memory fragments, day residue, recent concerns, remote associations, or recombination. The presence of emotionally relevant content does not imply that the dream is repairing affect. It may simply show that memory networks are active and reorganizing during sleep.

Therapeutic learning offers another concrete application. After one EMDR treatment session, 33 patients with PTSD received either memory cues timed to slow oscillations during sleep or sham stimulation. The cueing group showed greater reduction in avoidance during a later script-driven imagery assessment. Baseline avoidance was higher in the cueing group, which complicates interpretation of the size of that improvement. Cue-evoked slow oscillations and spindles also correlated with symptom reduction within that group. These early results motivate further study of sleep as an adjunct to treatment, with avoidance and overall PTSD symptoms assessed separately (van der Heijden et al., 2024).

The design implication is direct. Sleep studies should include memory-task rivals, cueing histories, day residue measures, and next-day memory outcomes. If dream affect tracks memory strength rather than burden relief, the memory rival has earned priority.

A strong comparison sharpens the auxiliary hypothesis. If memory consolidation explains the observations, that result identifies the relevant mechanism and guides the next question.

The consolidation rival can also predict emotional persistence. Some memories become less reactive after sleep, while others

become better organized and more available. If a dream study assumes all sleep-driven memory change is relief, it will misread precisely the cases most relevant to trauma, grief, and fear.

A useful analytic plan should include both memory accuracy and affective tone. A memory may become more accurate and less distressing, less accurate and less distressing, more generalized and more distressing, or unchanged. Those combinations distinguish mechanisms more effectively than a single next-day mood score.

15.4.4 — Medication and Sleep Disorder Effects

Medication, withdrawal, pain, respiratory disturbance, and other sleep disorders can jointly alter stage architecture, arousal, recall, and next-day affect. The protocol therefore records drug, dose, timing, indication, respiratory events, and sleep-disorder status as time-varying exposures and observation modifiers rather than unexplained noise (American Academy of Sleep Medicine, 2023; Zhang et al., 2019).

Sleep-disordered breathing, restless legs, periodic limb movements, pain, nocturia, fever, neurodegeneration, and circadian rhythm disruption can also shape dream reports and next-day affect. A dream study that fails to measure these mechanisms may falsely attribute burden movement to sleep reprocessing.

Medication and disorder effects are especially important in clinical or terminal settings. The same drug may relieve pain, reduce anxiety, suppress REM, impair recall, and change morning affect. A humane intervention can make dream evidence less

interpretable. Care comes first, and the resulting loss of interpretability is acceptable.

Medication, pain treatment, sleep-disorder symptoms, respiratory disturbance, and dose timing should be recorded at the resolution relevant to the night under analysis. The protocol also needs predeclared abstention rules for major medication changes, severe sensor disruption, or untreated sleep pathology that makes the candidate contrast uninterpretable. A confounded night can remain clinically meaningful without entering the confirmatory estimate.

These effects can also be informative in safer populations. If REM suppression weakens dream-report associations but not next-day recovery, dream content may be less central. If apnea severity predicts negative next-day residue, respiratory disruption may explain the pattern better than reprocessing.

Medication and sleep disorders make the research question more realistic. Sleep occurs within ordinary bodies, habits, illnesses, and treatment plans. Representing those conditions makes the findings more useful for clinical and everyday settings.

Dose timing should be recorded at a resolution relevant to sleep. A medication taken at bedtime, a rescue dose during the night, and a morning dose after awakening may each alter stage architecture, recall, pain, or affect differently. A daily medication indicator is often too coarse for this chapter's questions.

The same applies to sleep disorder treatment. Continuous positive airway pressure use, mask discomfort, oral appliances, sleep-position advice, and treatment adherence can change sleep and

next-day burden. Treatment success should be celebrated clinically, but analytically it becomes a pathway that must be modeled before any reprocessing claim is made.

15.4.5 — Clinical Support Before Study

Clinical support precedes study in every high-burden sleep context. A person with nightmares, insomnia, trauma replay, severe anxiety, sleep-disordered breathing, or medication-related sleep distress needs care pathways before research curiosity. The study must not turn untreated distress into an opportunity for observation (Palmer & Alfano, 2017; Tempesta et al., 2018).

This does not mean research cannot occur. It means research must be designed around care already being offered, around optional participation, and around procedures that stop when distress, fatigue, or privacy burden rises. Sleep research can coexist with treatment only when treatment remains sovereign.

The consent process must state that declining dream disclosure, refusing awakenings, stopping devices, omitting sensitive content, or leaving the study will not affect care. This is especially important for people receiving therapy, hospice services, or institutional support. Dependence can silently become pressure.

A randomized study provides a hopeful treatment example. Thirty-six adults with nightmare disorder received imagery rehearsal therapy. Both groups then heard a sound during REM sleep at home for two weeks, but only one group had first associated that sound with its positive rehearsal scenario. Mean weekly nightmares at two weeks were 0.19 in the association group and 1.02 in the comparison group. This small trial supports

further study of a specific treatment enhancement. The contrast identifies a concrete way that positive rehearsal and sleep cueing can work together (Schwartz et al., 2022).

Support procedures should precede content collection. Referral routes, distress response, participant-controlled privacy limits, and clinician review of awakening or interview procedures should be operational before enrollment. When a report evokes destabilizing material, the participant receives support or withdrawal options; the protocol does not pursue detail because the narrative is scientifically rare.

Care before study is not a slogan. It is an analytic condition. If a design ignores care, it changes the data-generating process and fails the ethical architecture that this science volume has already established.

A care-first sleep study should also have a withdrawal-without-explanation option. Participants should not have to justify refusing dream questions or stopping a sleep device. Requiring explanation can itself become pressure, especially when the content is private, frightening, or relationally sensitive.

Clinical support also includes staff training. Researchers collecting dream reports need guidance on trauma-informed responses, noninterpretation, confidentiality, and escalation. A poorly trained interviewer can turn a harmless question into a burden, and that burden can distort both the participant's experience and the dataset.

15.4.6 — No Exposure for Crucial Testing

No study may induce trauma, nightmares, serious fear, humiliation, grief intensification, sleep deprivation, treatment interruption, or comparable distress to create a decisive test. The prohibition also covers repeated awakening when clinical need favors rest, emotionally provocative prompts, public challenges, and designs that make care or belonging feel contingent on participation (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

The no-exposure rule belongs in prospective specification, consent, staff training, monitoring, stop criteria, and public materials. A protocol that cannot state how distressing procedures are prevented, recognized, and stopped is not ready for enrollment. Discriminating power that depends on prohibited exposure is a design failure, not an ethical tradeoff.

Natural variation, clinically indicated treatment, within-person observation, low-burden awakening schedules, independently collected archives, and simulation can test many reprocessing predictions without manufacturing harm. Some causal contrasts will remain unresolved, and that is an acceptable limit of a humane program.

Nightmare analysis still requires temporal precision. Pre-sleep state, nightmare timing, awakening, immediate report, delayed recall, next-day symptoms, treatment changes, medication, and subsequent sleep should remain separate so trauma replay, consolidation, arousal, recall reconstruction, and ordinary symptom fluctuation can compete (Tempesta et al., 2018;

Gieselmann et al., 2019). The clock map should be frozen before content coding so later recall cannot silently redefine when the relevant experience occurred.

Psychotherapy, medication, pain control, safety, and treatment of sleep disorders can alter dream frequency, recall, arousal, and waking distress. The analysis represents those therapeutic pathways alongside consolidation, rehearsal, fragmentation, current concern, and reconstruction at report. Treatment effects should be modeled as clinically meaningful changes, not nuisance corrections, because they may alter both the underlying state and the observation pathway.

Report the tested sleep prediction and its observed outcome. The study preserves its specified exposure limits and uses the result to evaluate the proposed route. A concise outcome statement should also identify the strongest rival that remained compatible with the data, preserving the comparison that gives the prediction meaning.

15.5 — Sleep Stage, Physiology, and Measurement

Sleep measurement is powerful because it can mark state, timing, physiology, and disturbance. It is limited because those markers are not consciousness, not valence, and not the latent cumulative process. This section builds the measurement architecture while keeping those limits in view (American Academy of Sleep Medicine, 2023; Siclari et al., 2017).

The goal is a state-specific evidence system: polysomnography and staging for architecture, autonomic and respiratory signals for bodily regulation, awakening protocols for selected reports, ecological momentary assessment for next-day residue, and sleep-specific validation before the candidate momentary-valence estimator is used at night.

Reliable sleep measurement comes from connecting each channel to a defined construct, artifact rules, lag and timing model, calibration context, uncertainty estimate, and abstention rule. These specifications make physiological records useful for testing rather than simply observing change.

Clear presentations make scoring uncertainty, stage transitions, lag, artifact, respiratory burden, awakening effects, and report confidence easy to follow. Tables and visualizations should show the dependencies relevant to the question being tested.

Table 15.4. Sleep-stage measurement roles.

Layer	Primary use	Main limit
EEG staging	Classify wake, NREM, REM, arousals, transitions	Not consciousness or valence by itself

Layer	Primary use	Main limit
Autonomic signals	Arousal, regulation, bodily stress context	Confounded by movement, respiration, medication
Respiratory signals	Sleep-disordered breathing and oxygenation	Can dominate next-day affect without dream mechanism
Awakening reports	Selected access to recent mentation	Sampling burden and recall reconstruction
Morning EMA	Next-day residue and function	Affected by waking context and delayed memory

15.5.1 — EEG and Sleep Staging

EEG-based sleep staging gives the chapter a common clinical and research language. Wake, NREM stages, REM, arousals, respiratory events, and transitions can be scored under established rules, but scoring remains an interpretation of signals rather than direct access to experience. The manual version, montage, scorer process, and any automated classifier should be preserved because each can alter the state sequence (American Academy of Sleep Medicine, 2023; Siclari et al., 2017).

The AASM scoring framework is useful because it standardizes sleep stages, arousals, respiratory events, movements, and related technical specifications. The Law program should use such standards as measurement discipline, not as evidence that a stage contains a particular affective process.

Stage scoring uncertainty should be preserved. Automated and human scoring can differ, and difficult epochs occur around transitions, artifacts, unusual physiology, medication, age, and

sleep disorders. A sleep study should record scorer confidence, algorithm version, epoch length, montage, and adjudication rules. Stage labels also have different inferential uses. REM may be relevant for dream reports and emotional memory, NREM for slow-wave activity and consolidation, arousals for disruption, and transitions for recall. None of these uses makes the stage a valence meter.

A fair sleep design uses staging to align evidence. It asks whether a report came after REM, NREM, transition, spontaneous awakening, or forced awakening. It asks whether physiology changed before the report. It does not infer the report from the stage or the experience from the stage alone.

The science gains credibility by resisting the comfort of a single label. Sleep stage is essential, but it must stay in its lane. It locates the night; it does not explain the whole night.

Automated staging can support large studies, but it should not hide scoring uncertainty. Model version, training population, montage, disorder status, and confidence should travel with each epoch. If automated and human staging disagree in key windows, the analysis should treat disagreement as uncertainty or sensitivity, not as an inconvenience.

The science also needs state-transition information. A report collected after a stable REM epoch is not equivalent to one collected after a mixed transition or arousal cluster. Transitions can be especially relevant to recall and affective residue. Collapsing them into the nearest stage may erase the timing the study needs.

15.5.2 — Autonomic and Respiratory Signals

Autonomic and respiratory signals can reveal arousal, stress physiology, sleep disruption, and bodily burden that dream reports miss. Heart rate variability, respiration, oxygen saturation, skin conductance, movement, and temperature can help interpret the night, especially when reports are absent or unreliable (American Academy of Sleep Medicine, 2023; Tempesta et al., 2018).

These signals are also vulnerable to artifact. Sensor contact, movement, sweating, room temperature, sleep position, device tolerance, battery state, and algorithmic smoothing can create apparent change. A study that treats wearable physiology as transparent affect will repeat the measurement errors this volume is designed to prevent.

Respiratory disturbance deserves special attention. Sleep-disordered breathing can fragment sleep, alter oxygenation, increase sympathetic activation, and worsen next-day mood or fatigue. If respiratory events explain the affective residual, they are not nuisance variables; they are the mechanism.

Autonomic signals also lag and smooth. A frightening dream, an arousal, a movement, and a respiratory event may appear close in time without a clear causal order. The analysis should specify lag windows and sensitivity ranges rather than forcing a single narrative.

Channel separation can reveal useful mechanisms. In a counterbalanced study of 17 participants, Viselli and colleagues found that experimentally fragmented REM sleep impaired

cardiac habituation to emotional stimuli, with no detected change in subjective ratings, electrodermal responses, or recognition memory. Fragmentation was accompanied by shorter REM duration. The value for this program is a clear measurement lesson: physiology can identify a sleep-sensitive process even when reported affect does not change. Naturally occurring sleep variation and clinically indicated improvements in sleep can investigate that distinction (Viselli et al., 2026).

The burden lens again matters. More sensors may improve data and increase discomfort. A participant who removes a sensor has provided ethically meaningful information about burden. Device tolerance belongs in the dataset.

The respiratory pathway also offers a strong rival test. If apparent negative dream residue is strongest on nights with oxygen desaturation or respiratory-event burden, the result may belong to sleep-disordered breathing rather than dream reprocessing. That rival is clinically actionable and scientifically important.

15.5.3 — Awakening Protocols

Awakening protocols offer temporal precision but impose cost. An awakening close to a scored epoch can improve report availability, yet repeated awakenings alter sleep continuity, fatigue, mood, and later recall. The study should register the maximum number and timing of awakenings, participant-controlled refusal, recovery intervals, and the estimand that remains when an awakening is skipped (Palmer & Alfano, 2017; Tempesta et al., 2018).

Different awakening strategies answer different questions. REM-targeted awakenings may enrich dream reports. Random awakenings help estimate report probability across stages. Spontaneous-awakening reports reduce intervention burden but increase selection. Morning reports preserve sleep but add recall delay.

Begin with natural nights, spontaneous awakenings, low-burden morning reports, and tolerated physiology. Any additional awakening procedure must satisfy Chapter 3's prohibition on research-induced suffering and sleep disruption for Law-specific evidence; healthy status and consent alone do not create an exception. Where the desired contrast requires prohibited disruption, use suitable existing data or narrow the question.

Awakening effects must be measured. After each awakening, the study should record sleepiness, distress, willingness to continue, report confidence, and whether the awakening itself changed the content or mood. These measures are not administrative; they are part of the observation model.

The fairness test is whether the question justifies the interruption. If the study can learn enough without waking the participant, curiosity should not win. The sleeping person is not obligated to become more observable.

Awakening burden should be included as an endpoint of the protocol itself. If participants report annoyance, distress, fatigue, reduced willingness, or worse next-day functioning, the protocol has produced a cost. Population-specific review determines whether the awakening protocol is permissible within Chapter 3's

limits; healthy status alone does not authorize otherwise prohibited sleep disruption or distress.

Training and analytic nights should be separated because equipment, prompts, laboratory novelty, and repeated awakenings change sleep. Adaptation nights can establish tolerability and stable procedures, while confirmatory nights follow the frozen schedule. Treating early nights as exchangeable with later nights risks interpreting acclimation as physiology.

15.5.4 — Experience Sampling After Awakening

Experience sampling after awakening should be structured enough to compare nights and gentle enough not to invade the dream. A useful report can ask for presence of experience, affective tone, bodily sensation, social content, threat, agency, recall confidence, and willingness to provide detail without demanding a full narrative. Immediate interviews may improve temporal linkage, but refusal and no-recall remain substantive marks rather than failed compliance (Siclari et al., 2017; Cataldi et al., 2024).

The order of questions matters. Asking for a full story first may reshape affect ratings; asking for affect first may prime content selection. The protocol should standardize the sequence and test whether the sequence changes reports. Method order is a measurement variable.

Reports include confidence and nonresponse options. A participant may indicate experience without recall, recall without confidence about affect, affect without narrative, or a wish not to

disclose. Those categories preserve more information than a forced choice among positive, negative, and neutral.

Post-awakening sampling should also track immediate reactivity. The act of recalling a dream can change mood, arousal, embarrassment, grief, or fear. If the report itself creates burden, the study must record the burden and allow the participant to stop.

The best sampling instruments use bounded fields and optional narrative. A coded minimal report supports comparison; an optional narrative supports qualitative and coding work; a privacy setting can restrict who sees raw content. Scientific reproducibility does not require maximal exposure.

Experience sampling therefore becomes a dignity-preserving bridge. It acknowledges that sleep experience may matter, that memory is partial, that privacy is real, and that a measured field is not a confession.

The prompt should also avoid interpretation. Questions like what did the dream mean can invite cultural, spiritual, or therapeutic reflection, which may be valuable but not comparable. A scientific prompt should first ask what was experienced, what was felt, what is remembered, and how confident the participant is.

The study can then allow an optional meaning field under separate status. That field should be analyzed as participant interpretation, not as coder-inferred truth. This separation allows meaning to be honored without letting it become an unvalidated measure of affective balance.

15.5.5 — Candidate Momentary-Valence Estimator Qualification During Sleep

The candidate momentary-valence estimator cannot be imported into sleep as if waking calibration automatically applies. Sleep changes muscle tone, autonomic balance, respiration, movement, reportability, temperature, and consciousness state. A physiological composite trained during waking distress may fail or reverse under REM, NREM, sedation, illness, or sleep disorder (American Academy of Sleep Medicine, 2023; Tempesta et al., 2018).

Qualification of the candidate momentary-valence estimator for sleep must begin with a declared context of use. The estimator may be used to describe physiological arousal during sleep, support a state model, or predict next-day affect. Each use requires different validation, and none licenses direct measurement of the latent cumulative process.

Sign identification is especially difficult. A heart-rate increase may reflect fear, respiratory obstruction, movement, fever, medication, pain, or excitement. A flat signal may reflect deep sleep, sensor failure, autonomic dysfunction, or drug effect. The sign of valence cannot be read from physiology without auxiliary validation.

A sleep-specific qualification program for the candidate momentary-valence estimator should require stage-specific calibration, artifact rules, lag sensitivity, device-shift checks, missing-modality analysis, and comparison with reports and next-day outcomes. The estimator should abstain when the channels

cannot support sign or when the state lies outside the validated domain.

This qualification work belongs here only as sleep-specific application. The general measurement stack is owned by earlier chapters. The present chapter asks what changes when the same estimator meets sleep architecture, dream reports, respiratory disturbance, and awakening burden.

The necessary safeguard is technical: do not let a composite score become more authoritative because the participant is asleep and cannot correct it. Measurement qualifies claims; it does not speak over the sleeper.

The candidate momentary-valence estimator should first be qualified for sleep in lower-risk contexts. Healthy-adult studies, controlled sleep laboratories, and naturalistic wearable cohorts can test whether channels behave consistently before any clinical or terminal use. A terminal or trauma cohort should not be the first setting in which an estimator learns what sleep does to its signals.

The estimator should also have a no-output state. If artifact, medication, respiratory disturbance, or state uncertainty defeats sign identification, the system should abstain rather than output a smoothed value. An abstention is more honest than a number whose authority comes from the sleeper's inability to object.

15.5.6 — No Unity or Valence From One Marker

No single report, sleep stage, electroencephalographic feature, autonomic pattern, respiratory event, movement signal, or behavioral sign can decide consciousness, unity, or valence during sleep. Marker meaning depends on theory, context, timing,

disorder, medication, and the observation pathway (American Academy of Sleep Medicine, 2023; Siclari et al., 2017).

Single-marker authority fails in both directions. A quiet or absent signal may coexist with experience the channel cannot reveal, while an active signal may reflect arousal, movement, breathing disturbance, memory, or artifact. False-negative and false-positive burdens therefore require separate evaluation rather than one blended accuracy score. Separate error profiles can reveal whether a marker is useful for screening, confirmation, or neither, without granting it universal classification authority.

The appropriate architecture uses multiple imperfect witnesses with explicit abstention. Reports, stage, physiology, behavior, clinical context, and scorer judgment should enter through declared rules, while vector, set-valued, or unresolved outputs remain available when the channels do not license one classification. Discordance is a result to analyze, not contamination to erase. The fusion rule should state whether disagreement widens uncertainty or changes the estimand, since those responses imply different scientific objects.

Temporal alignment is part of validity. Stage labels, physiological events, awakenings, reports, and recall delays should share a versioned clock map with prospectively fixed tolerances. Predictive value that appears only after flexible lag selection or retrospective realignment does not establish a prospective marker. Lag uncertainty should be propagated into the marker estimate rather than resolved by choosing the alignment that maximizes association.

Qualification should survive channel removal, lag perturbation, scorer disagreement, stage uncertainty, nights without report, and clinically relevant subgroups. A protocol-fixed discordance matrix should state what each pattern permits, what remains unresolved, and when No adequate estimate is required. Discordance that clusters by state, medication, or device can reveal where a marker requires narrower qualification instead of a global rejection.

Every marker needs provenance: channel identity, device and firmware, preprocessing, artifact rules, scorer version, temporal window, report pathway, and missingness status. Tables, captions, dashboards, and public summaries must carry the same ceiling so that one striking trace cannot travel as direct evidence of hidden valence, unified experience, or absence of experience. Provenance also makes cross-site harmonization possible by revealing whether apparent biological differences are actually differences in devices, scoring, or preprocessing.

15.6 — Study Families and Data Schema

Sleep research becomes useful for the program only when its designs are specific enough to fail. This section turns the preceding distinctions into study families, data fields, coding rules, and privacy layers that make replication possible without making disclosure compulsory (Shiffman et al., 2008; Nosek et al., 2018).

The study families are arranged from lower burden to higher precision. The common rule is that no design may treat more intimate data as automatically better data. Reproducibility and dignity have to be built together.

Study families are also publishing commitments. If the data schema, missingness categories, privacy tiers, coding manuals, and rival models are not defined before analysis, the study will be tempted to publish whatever story the night makes vivid. The schema is a guardrail against post hoc dream mythology.

The data schema also gives the research team a deletion rule. If a variable cannot be tied to a prospectively specified question, rival test, privacy requirement, or safety monitor, it should not be collected merely because it might become interesting later. Dream research becomes less invasive when curiosity has to state its purpose in advance.

Table 15.5. Minimal sleep-study data schema.

Domain	Fields to version	Why it matters
Sleep architecture	Stage, epoch, scorer, algorithm, confidence	Prevents stage claims from floating free

Domain	Fields to version	Why it matters
Dream report	Prompt, delay, confidence, privacy level, coding version	Separates report from experience
Physiology	Device, units, artifact rule, lag window, missingness	Keeps sensors from becoming opaque authority
Context	Medication, pain, trauma, disorder, family event, environment	Names rival mechanisms and burden paths
Outcome	Next-day affect, sleepiness, function, intrusive memory	Defines the transition actually tested

15.6.1 — Repeated-Night Within-Person Design

The core sleep design should be repeated-night and within-person. Each participant contributes multiple nights with pre-sleep affect, sleep architecture, optional dream report, physiology, next-day affect, and context. This design controls stable individual differences and lets the study test whether night-to-night changes predict next-day residuals (Shiffman et al., 2008; Stone et al., 2007).

Repeated nights also reveal ordinary variability. One nightmare night, one unusually restful night, or one missing report should not define the person or the mechanism. The pattern matters across nights, especially when sleep quality, burden, medication, and report probability change.

One concrete extension is a preregistered test of history dependence. Compare nights that begin with similar momentary

affect, sleep opportunity, medication, and available support but differ in an independently specified summary of earlier affective history. Test whether that history summary improves next-day predictions after current state, waking events, sleep architecture, memory salience, and report probability are modeled. The proposed auxiliary route must specify its direction, lag, capacity interaction, and failure criterion before analysis. Ordinary learning models must receive the same history. A surviving contrast would identify additional predictive structure. Natural variation or independently justified care provides the setting, with the history summary defined over the observed window.

Repeated-night designs should include negative-control nights with a prospectively justified null prediction for the proposed auxiliary signal, alongside observation-process stress tests for no report, severe sensor failure, and major external disruption. Low pre-sleep burden qualifies as a negative control only when the proposed mechanism predicts no signal under that condition. A model that produces the same favorable pattern on every night lacks specificity even if its average fit is high.

Within-person modeling can still overreach. If a participant's sleep changes because of medication, illness, therapy, grief rituals, family visits, or pain flare, the model must treat those as time-varying pathways. The person is not a fixed laboratory system.

A simple mixed model may estimate next-day affect from stage features, nightmare burden, pre-sleep burden, and person-specific intercepts, but the model must be protocol-fixed, tested out of

sample, and compared against sleep-science rivals. Good fit is not enough.

Repeated-night design is humane because it can begin in ordinary life and avoid terminal vulnerability. It is scientifically valuable because it turns sleep into a sequence of testable transitions rather than a collection of dramatic anecdotes.

The repeated-night design should predefine the minimum number of usable nights per participant and the maximum burden allowed. A person with too few interpretable nights should not be forced into analysis through imputation that hides why the nights were missing. Missingness is part of the design result.

Carryover should be modeled across nights. A nightmare can change the next evening's affect, an awakening schedule can alter later sleep, and reporting a dream can provoke rumination or relief. Calendar separation does not create independence; washout assumptions and lag structure should be protocol-fixed and tested.

Table 15.6. Prospective repeated-night protocol sketch.

Phase	Low-burden default	Higher-resolution optional layer
Evening	EMA on burden, pain, affect, medication, social context	Brief cognitive or memory cue task when ethically justified
Night	Actigraphy or comfortable wearable physiology	PSG, stage-targeted awakenings, or TMR sub-study
Morning	Sleep quality, recall status, affect residue, willingness to disclose	Structured dream coding with consented narrative

Phase	Low-burden default	Higher-resolution optional layer
Next day	Short EMA residuals and functioning	Clinical or behavioral follow-up when relevant

15.6.2 — Naturalistic and Laboratory Designs

Naturalistic and laboratory designs answer different questions. Naturalistic sleep preserves the participant’s own bed, schedule, relationships, stressors, and privacy boundaries. Laboratory sleep increases measurement precision but changes the setting, increases burden, and may alter sleep through first-night effects and observation pressure (Nosek et al., 2018; Chambers & Tzavella, 2022).

A fair research program uses both only when each is needed. Naturalistic studies are better for ecological validity, day residue, family context, long-term recurrence, and low-burden repeated sampling. Laboratory studies are better for stage-targeted awakenings, precise physiology, cueing, and controlled stimulus exposure within ethical limits.

The first-night effect must be modeled, not ignored. Sleeping in a laboratory can change architecture, arousal, reportability, and next-day mood. A study that treats the first laboratory night as ordinary sleep may confuse adaptation to the lab with the mechanism under study.

Wearable designs have their own limits. Actigraphy, consumer EEG, heart-rate wearables, and phone prompts may support low-burden longitudinal data, but device version, placement,

algorithm, battery, user behavior, and missingness must be recorded. Convenience cannot become validity.

Hybrid designs may be strongest. A study can use a laboratory calibration phase to validate stage and report procedures, then deploy a naturalistic repeated-night phase with lower-burden measurements, then return a subset for targeted questions. The bridge between phases must be tested rather than assumed.

The design choice is itself a fairness question. Precision bought by intrusion is not automatically better science. The right design is the least intrusive design that can still distinguish the relevant rival mechanisms.

Naturalistic work should not be treated as soft science. It is often the only way to observe the ordinary middle of sleep, where grief, parenting, illness, work, noise, and private routines coexist. Its difficulty is precisely why it matters for this analytic question.

Laboratory work should not be treated as pure truth. It provides precision under altered conditions. The bridge from laboratory finding to ordinary sleep must be defended with replication, bridge samples, and context-of-use language. A beautiful laboratory signal can still be a poor measure of life.

15.6.3 — Dream Database Resources

A substantial shared resource is already available. The initial DREAM database assembled 20 datasets from 505 participants, linking sleep M/EEG with standardized reports across 2,643 awakenings. It makes cross-study tests of stage classification, report availability, and neural predictors of reported experience feasible across laboratories. A useful first analysis would freeze

its report model on some datasets and evaluate transfer to independent datasets, preserving sleep-stage and sampling differences. Its linked neural recordings and awakening reports give measurement development a shared empirical resource (Wong et al., 2025).

Database interpretation begins with its sampling frame: who agreed to dream research, slept in a laboratory, tolerated awakenings, recalled dreams, and disclosed content. These documented conditions define the population and observations available for analysis.

Dream databases also vary in prompts, languages, scoring systems, electrode montages, awakening timing, and report classification. A meta-analytic comparison must respect those differences rather than treating all reports as interchangeable. Harmonization is an analytic achievement, not a default.

Existing datasets may be especially limited for privacy-sensitive questions. Raw dream narratives may contain bystanders, intimate material, trauma, or religious meaning. Access should be tiered, de-identified where possible, and restricted when content cannot be responsibly shared.

For the Law program, databases can test measurement feasibility, report models, coding reliability, stage-specific recall, and competing sleep mechanisms. Prespecified analyses and clearly identified exploration make the resulting contributions reproducible.

A dream database is a valuable resource for method development. It can help qualify measurements, compare observation models,

and design more informative prospective sleep studies within its documented sampling limits.

Open multicenter resources also reveal differences among methods. A codebook that performs differently across datasets can identify cultural, prompt, stage, or laboratory-protocol dependence and guide more effective measurement.

Identify these as analyses of existing data and disclose what investigators had already seen before fixing the analysis plan. Existing-data use does not by itself make a hypothesis test exploratory: its status depends on prior access, prospective specification, selection, and the suitability of the data for the declared estimand. A database should not determine the question merely because it is available.

15.6.4 — Multimodal Data Dictionary

A sleep study needs a data dictionary before it needs interpretation. The dictionary should version sleep stage, epoch length, scoring method, device, sensor placement, signal quality, artifact decisions, awakening protocol, report prompt, report delay, affect rating, privacy level, and missingness reason (Shiffman et al., 2008; Stone et al., 2007).

The dictionary should keep layers separate. Stage data, physiology, report content, report confidence, pre-sleep burden, next-day affect, medication, clinical events, environment, and privacy restrictions should each have distinct fields. A combined score may be derived later, but the raw distinctions must remain recoverable.

Missingness categories are central. No awakening, no recall, refusal, distress stop, device failure, low signal, medication confound, language barrier, and clinician intervention mean different things. A single missing code erases the very burden the study needs to understand.

Versioning matters because sleep devices and scoring algorithms change. A report generated under one device firmware, one automatic staging system, or one codebook cannot be silently rescored under another without a bridge. Historical sensitivity analysis should preserve both versions.

The dictionary should also include derived variables with provenance. REM density, slow-wave activity, fragmentation index, nightmare burden, recall confidence, next-day residual, and coded dream affect must point back to the raw fields and rules that generated them.

Good data architecture is not clerical. It is the difference between a sleep study that can be criticized and one that can only be believed or dismissed. The Law program requires the criticizable kind.

The data dictionary should include a privacy field for each report element. A participant might allow affect rating, allow theme coding, restrict raw text, prohibit quotation, or request deletion of relational details. These permissions should travel with the data so later analysts cannot accidentally exceed consent.

The dictionary should also include provenance for derived dream variables. Who coded the report, under which manual, with what training, with what reliability, and with what uncertainty? A

dream-valence field without provenance is not a scientific variable; it is a loose interpretation stored in a cell.

15.6.5 — Protocol-Fixed Coding and Analysis

Under §6.7's prospective-specification terminology, coding variables, coder training, reliability thresholds, uncertainty categories, noncodable states, primary contrasts, and the definitions of dream valence, threat, social content, agency, bodily sensation, grief, recurrence, lucidity, and meaning are fixed before outcome inspection (Nosek et al., 2018; Chambers & Tzavella, 2022).

The analysis plan should state which outcomes are confirmatory and which are exploratory. Within-night change, next-day residual, nightmare recurrence, dream-report probability, and stage-specific physiology cannot all become primary after the data are known. Flexibility after inspection is how a dream chapter becomes unfalsifiable.

The prospective plan links each rival to a frozen feature set, observation model, and adjudication rule. Memory consolidation uses retained-memory outcomes; threat simulation uses protocol-fixed threat codes and recurrence; arousal regulation uses stage-linked physiology; and report selection models dream-report probability. The coding manual, training set, reliability floor, and held-out adjudication sample are fixed before model comparison.

Coding should also include abstention. Some reports will be too fragmentary, too private, culturally ambiguous, or emotionally unsafe to code. A valid codebook includes no-code and restricted-code categories rather than forcing content into convenient cells.

Confirmatory coding uses features available at the protocol-fixed analysis time. Later family narratives, clinical interpretations, or revised autobiographical meaning may support separate retrospective analyses, but they cannot redefine the original dream variable. Time-forward coding preserves prediction, reconstruction, and later meaning as distinct objects.

Prospective coding control protects the analysis from the human tendency to discover meaning after the result is known. Its purpose is not administrative purity; it is to keep an intimate, ambiguous report from becoming whatever variable best favors the model.

Coder disagreement should be planned. Dreams are often ambiguous, metaphorical, fragmented, or culturally specific. A useful codebook reports disagreement, confidence, and unresolved cases rather than pressuring coders to agree in order to make the model easier.

Analysis should include model recovery before human data are interpreted. Simulated nights should test whether the proposed design can distinguish sleep reprocessing from memory consolidation, arousal regulation, report selection, and sleep-disorder pathways. If the design cannot distinguish them, recruitment should not proceed.

15.6.6 — Open and Restricted Data Layers

Sleep research needs reproducibility, but intimate dream content is not ordinary open data. A layered architecture separates public codebooks, synthetic examples, aggregate stage features, missingness summaries, model specifications, coding manuals,

and de-identified derived variables from restricted narratives and identifiable context. Openness should improve criticism without making privacy the price of replication (National Institute of Standards and Technology, 2020).

Raw reports may reveal partners, children, deceased relatives, clinicians, trauma, sexuality, religion, or private conflict. Transparency alone does not justify release, and a closed repository is not ethical by default. Restricted access needs a stated purpose, trained users, audit logs, redaction or withdrawal routes, and sanctions for use beyond the approved question.

Open methods often contribute more than open narratives. Analysis code, frozen model specifications, negative controls, sensitivity plans, data dictionaries, and labeled synthetic or paraphrased examples can make the work answerable while protecting the person. Synthetic material demonstrates structure; it is not evidence.

Each layer operates under a versioned contract naming fields, units, clocks, values, missingness meanings, provenance, derivation code, access class, retention, and retirement. Changed dream categories, stage definitions, or coding rules create a new analytic version and require a bridge rather than silently rewriting earlier nights. Schema migration tests should verify that old and new versions produce equivalent derived objects on shared records before pooled analysis is allowed.

Federated analysis, secure enclaves, or site-level sufficient statistics can answer some multisite questions without moving raw narratives into one repository. They reduce what travels but

do not erase privacy risk or establish comparability. The plan records what leaves each site, what remains local, how updates are audited, and how withdrawal propagates (National Institute of Standards and Technology, 2020; Nissenbaum, 2010). A privacy-preserving architecture should also record the inferential limits created by local-only variables so missing cross-site information is not mistaken for absence.

Consent to sleep-reprocessing research is not consent to train general language models, infer personality, build public artifacts, or support unrelated prediction. Purpose limits attach to raw and derived layers, and materially new uses require separate permission and review. Reproducible science can invite criticism without turning sleepers or their relationships into exhibits. Derived embeddings and summaries should inherit the same purpose limits as the source narratives, because transformation does not erase consent boundaries.

15.7 — Privacy, Public Risk, and Verdict Limits

Dream research is public-risk sensitive because it is easy to quote, easy to mystify, and easy to misuse. A single terminal dream or grief dream can become more persuasive to the public than a careful protocol. That is why verdict limits belong in the chapter, not in an appendix (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

This final section connects sleep findings with their public communication. Dream research can advance accounts of sleep mechanisms, report formation, within-person dynamics, and privacy design. Participants retain control over disclosure and the personal meaning they assign to their experiences.

Dream evidence may become culturally powerful faster than it becomes scientifically strong. The closing section therefore specifies what may be said, what may be shown, what remains private, and which verdict labels can travel beyond the specialist context.

A bounded verdict is also a promise to participants. It tells them that their sleeping life will not be used to make claims about their worth, their faith, their grief, their family, their death, or the hidden balance of their lives. The result may enter science, but the person remains larger than the result.

15.7.1 — Dream Content as Sensitive Data

Dream content is sensitive because it can expose desire, fear, shame, trauma, religion, sexuality, violence, family conflict, illness, and grief without the dreamer having chosen those images

while awake. The content may not reflect endorsement, intention, memory, or character, yet it can still harm if exposed (National Institute of Standards and Technology, 2020; Nissenbaum, 2010).

Data minimization should begin before collection. Many studies can ask for affect ratings, themes, intensity, or coded categories without raw narratives. Full narrative should require separate permission, clear purpose, and the ability to skip or redact details.

Access should be tiered. Study staff who need coded variables may not need raw content. Coders may not need identifiers. Public reviewers may need methods and synthetic examples rather than dreams. The default should be less access, not because dreams are shameful, but because they are intimate.

Participants should have control over sensitive passages where scientifically possible. They may designate content as private, allow coding but not quotation, permit paraphrase but not raw text, or withdraw narrative from future use. Those options should be explained before collection.

Security failures in dream data are not ordinary breaches. A leaked dream can harm relationships, identity, and dignity through exposure of distinctive narrative and relational content. Protocols should include incident response, participant notification, deletion rules, and publication consequences.

The no-score rule is strongest here. A dream is not a moral record. It is not a loyalty test, holiness marker, psychological rank, or eligibility signal. It is sensitive data from a person whose dignity is prior to the study.

Privacy protection should be designed before the most interesting report arrives. A protocol that invents access rules after a dramatic dream will be pulled by fascination, urgency, and public value. The correct procedure is to decide in advance what can be coded, quoted, shared, restricted, or deleted.

Dream content also deserves temporal limits. A participant may consent to analysis during a study but not to indefinite storage of raw narrative. Retention schedules, withdrawal options, and derived-result rules should be specific enough that future researchers cannot treat dream archives as permanent reservoirs of intimate material.

15.7.2 — Bystander and Relationship Privacy

Dreams often contain other people. A partner, child, parent, clinician, friend, enemy, deceased relative, public figure, or stranger may appear in a report. Those people did not consent to become research content, and the dreamer may not want the relationship exposed (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Bystander privacy should be handled as a design problem. Reports can use role labels, initials, generalized categories, or redaction before coding. A raw name is rarely necessary for sleep-science inference. When relational identity matters, the purpose must be narrow and access restricted.

Relationship privacy also protects the dreamer. A dream involving conflict, desire, violence, or grief can be misunderstood if treated as waking intention. The protocol should warn against interpretation, prohibit moral inference, and keep raw content

away from employers, institutions, family members, public archives, or scoring systems.

In terminal or grief settings, families may share dream reports for comfort, ritual, or meaning. Research preserves those personal interpretations while describing its own observations and methods.

Bystander content also complicates open data. A de-identified dream can still identify a relationship if the narrative is distinctive. De-identification should therefore include contextual rarity, not just names and dates.

Fairness science must not shift the burden of reproducibility onto people who never joined the study. Bystanders are part of the hidden cost of dream research, and their invisibility cannot be used as permission.

The research team should also avoid creating family burden by returning raw dream content to relatives. A participant may choose to share a dream, but the study should not become the channel through which private or ambiguous material reaches others. Relationship repair or rupture is not a research deliverable.

15.7.3 — Terminal Dreams and Personal Meaning

Dream reports during serious illness may carry profound personal and family meaning. Research can examine their recall, affect, physiology, content, and change over time using a stated question and a qualified coding scheme. Participants retain control over how their material is used (Palmer & Alfano, 2017).

Studies of terminal dreams document whether reports are immediate or retrospective, direct or family-mediated, and how culture, medication, and report availability enter the record. A defined sampling frame makes population comparisons possible.

Terminal illness also changes sleep, cognition, medication, delirium risk, pain, oxygenation, and family presence. These conditions can affect dream content, recall, disclosure, and interpretation. The report may still be meaningful, but the causal path is too complex for proof language.

A study may examine terminal dreams as a clinical or phenomenological domain under care-first rules. It may ask about prevalence, themes, distress, comfort, relation to sleep disruption, medication, delirium, and family communication. It may not use a dream as evidence that a life has reached balance.

Use dream quotations only with permission and context appropriate to the agreed purpose. Review privacy and family implications before publication.

A dream can matter deeply to a person, family, or spiritual tradition. Research can investigate its occurrence, content, and effects while respecting the meaning it holds for the dreamer.

Captions and summaries distinguish the study's observations from the dreamer's own interpretation. Any narrative is shared on the participant's agreed terms.

A missing terminal-dream report is recorded as an observation outcome, with the reason retained where known.

15.7.4 — No Grief Interpretation by Science

Grief dreams can comfort, disturb, reopen longing, or feel like contact. Science should not tell bereaved people what such dreams mean. It should not explain away comfort, certify visitation, rank mourning, or turn grief into evidence for a theory. The chapter's responsibility is methodological restraint (National Institute of Standards and Technology, 2020; Nissenbaum, 2010).

A grief dream may be studied as experience, memory, attachment, emotion, or sleep disruption. It may be coded for themes only with consent and careful privacy. The participant's meaning should be recorded separately from analyst interpretation, and neither should be forced into a universal scale.

People grieve in different ways. Some find comfort in dreams; others experience distressing dreams or recall none. Support responds to each person's experience and preferences.

Grief can change recall and reporting. People may attend to dreams involving the deceased more than other dreams, share them more often, and remember them longer. Those selection processes enter the observation model.

Clinical and pastoral services may help people interpret dreams according to their needs and traditions. The science volume contributes descriptions of the research, its methods, and its findings.

The burden analysis asks where burden is carried. In grief dreams, burden may be carried in love, absence, memory, body, family expectation, and culture. A scientific chapter becomes fair by not pretending to settle what grief has not asked it to settle.

The safest public language treats grief dreams as experiences people may interpret within their own lives and traditions. Science can study report frequency, distress, comfort, sleep disruption, and memory processes; it cannot tell a mourner which meaning is correct.

This distinction should be preserved when results are favorable. If grief dreams correlate with next-day comfort in a study, the conclusion is not that grief dreams are messages or repairs. The conclusion is that a measured association appeared under specified sampling, coding, and cultural conditions.

15.7.5 — Hostile-Crop Review

Hostile-crop review asks how a sentence, table, figure, headline, or quotation reads after its surrounding safeguards disappear. Dream research especially needs this test because vivid material can turn an auxiliary finding into claims about hidden balance, useful trauma, mystical access, or personal moral status (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Figures and tables should keep uncertainty, missingness, stage, report selection, and rival explanations visible. Titles, colors, captions, axes, and row labels carry claims; a label such as dream repair signatures can mislead even when the cells are careful.

Raw dream quotations require necessity, consent, redaction, and protection from spiritual or moral inference. Public summaries should instead name the tested auxiliary relation, such as reported dream affect predicting a next-day outcome after stage and recall controls.

An independent reviewer should crop the artifact and ask what a skeptic, journalist, family member, or online critic could reasonably infer without the chapter. Review includes proof drift, privacy drift, score drift, and mystical drift across prose, graphics, and tables (Fischhoff & Davis, 2014; Spiegelhalter, 2017).

The artifact is revised until its auxiliary status and uncertainty survive detachment. Public reach is earned when a bounded result remains bounded outside the page that first explained it.

15.7.6 — Bounded Auxiliary Verdict

The chapter evaluates its protocol-fixed sleep and dream estimands: report, reprocessing, memory, physiology, and privacy-feasible measurement. The result states which hypotheses gained support, which weakened, and which remain unresolved (Nosek et al., 2018).

A favorable result names the reference class, measurement version, pre-sleep variables, report model, window, rivals, and better-predicted pattern. Narrowing or adverse outcomes can localize effects, contradict simple repair accounts, expose recall or medication confounding, or stop a burdensome and privacy-risky design.

Refusal, no recall, partial disclosure, unclassified narrative, and non-disclosure are valid observation states. Sleep can inform state, memory, and reprocessing while remaining a private domain not owed to research.

Every summary carries the component verdict and its limits. The night remains meaningful and private while the program learns exactly where auxiliary mechanisms survive, narrow, or fail.

Chapter 16 — Terminal-Window Tests

A terminal boundary claim becomes empirical when it predicts relations before the final point, not when a curve merely looks persuasive after alignment. Conditioning, censoring, selection, mean reversion, clinical decline, and observation floors can all concentrate trajectories near an endpoint. The chapter tests joint restrictions on preterminal drift, dispersion, dependence, horizon, reserve, intervention response, and detectability using information available before the endpoint.

A discriminating signature is a typed relation among declared objects: route state or proxy, filtration, endpoint branch, risk set, observation kernel, time coordinate, estimand, and rival family. Removing any field invites a different generator to imitate the same picture. The scientific object is the frozen conjunction, not the visual smoothness of a terminal curve (Doob, 1957; Gneiting & Raftery, 2007).

The leading reserve-coupled realization earns attention through cross-equation restrictions on drift, conditional dispersion, two-time dependence, and perturbation response. Brownian bridges, Doob transforms, homeostatic controllers, frailty selection, treatment, and informative observation can reproduce individual pieces. The route is therefore tested as a transportable conjunction, with each component able to fail independently.

The chapter moves from state identification to exponent restrictions, covariance kernels, terminal windows, observation floors, and rival adjudication. Each step can fail separately. A route can be mathematically coherent but empirically unidentifiable; a proxy can be reliable but unrelated to the

candidate state; a terminal pattern can be real but better explained by selection or care context.

The strongest evidence would transport without retuning. One frozen representation, endpoint rule, reserve proxy, observation model, and scoring system would predict held-out relations across cohorts, resolutions, channels, care contexts, and safe perturbations. A result that needs a new window, origin, floor, or endpoint for each dataset remains a local fit rather than a transported signature.

The human setting matters throughout. Terminal cohorts include sedation, delirium, analgesia, communication loss, family presence, device dropout, and changing goals of care. Those are not nuisance variables to erase after collection. They are parts of the observation and intervention process, and research remains subordinate to comfort, consent, and ordinary clinical duty.

The reserve-coupled conjunction connects several tests: representation, endpoint adjudication, observation, measured-to-latent inference, and route comparison. Success across these tests would favor that route within the studied reference class. The result then guides replication and comparison with other boundary architectures; an unsuccessful route can be revised or retired while those alternatives remain available.

Mean, variance, tails, covariance, path class, and support describe different features of a trajectory. Heterogeneity, symmetric error, or cancellation can produce a mean near zero, making the additional distributional features especially informative for comparing models.

Concrete design begins with repeated reports, wearable channels, medication timing, care events, missingness marks, and independently defined endpoint probes. These records become prospectively specified estimands scored under a common information budget. The central question is which candidate, rival, hybrid, or open-set generator could have produced the complete observed process, including the pattern of observation failure.

Each principled failure result has a scientific destination. A stable noise floor can make an exponent unidentifiable; a bridge can dominate a reserve route; reserve can collapse into severity; terminal windows can be too sparse; a covariance surface can fail held-out recovery. The correct response is to narrow, redesign, or issue No adequate estimate, not to protect an evocative plot.

By the end of the chapter, the reader should be able to tell a route state from its proxy, a prospectively estimated horizon from retrospective time-to-event, a theoretical scaling restriction from a fitted slope, a covariance kernel from a smoothed trend, and route-favored evidence from evidence for the boundary law itself.

The program is most scientific here because its preferred route receives no privileged data, window, tuning budget, or rescue language. It must face ordinary stochastic, clinical, and observation-process models on common rules and preserve the exact result when one of them wins.

16.1 — Candidate Route and State Identification

A route-specific signature begins by fixing the state that is supposed to carry it. The state may include a signed residual, a reserve variable, a filtered horizon, a route indicator, and contextual covariates. None is automatically observed. The study must state which components are truth-level objects, which are candidate mechanism states, and which are measured proxies (Durbin & Koopman, 2012).

State identification is not achieved by naming a familiar quantity. Fatigue, frailty, pain, time remaining, and treatment intensity can all correlate with a proposed reserve while representing different causal processes. A defensible route state therefore comes with a measurement map, exclusion restrictions where possible, and sensitivity analyses showing how the signature changes when the state is misspecified.

The filtration matters. A prospective route state may use information available up to time t , including current measurements and declared history. It cannot use the eventual endpoint class, later retention, or future observations unless the analysis is explicitly retrospective. This no-look-ahead rule prevents a terminal pattern from being built out of information that the proposed mechanism could not have possessed.

Two participants may have the same realized time to an eventual endpoint but very different contemporaneous information, symptom burden, treatment options, and support. A retrospective clock places them at the same distance; a filtered route state need

not. The difference is not cosmetic because the route predicts from the state available before the endpoint is realized.

A state model may remain useful for forecasting observation shutdown, care transitions, or affective volatility even if no terminal invariant exists. PTVC relevance requires a separate bridge to the declared phenomenal residual and endpoint branch. When several latent states generate the same measured process, the correct outcome is an identified set, a new perturbation, or No adequate estimate.

16.1.1 — Route State, Filtration, and Observability

The first scientific object in this chapter is not the endpoint. It is a filtered estimate of remaining opportunity, formed before the endpoint is known and revised as new evidence arrives. Perceived future time is already known to reorganize motivation; the route-specific question is whether a prospectively estimated horizon interacts with unresolved state and independently measured reserve in a pattern that ordinary finite-horizon accounts do not explain (Carstensen, 2006; Särkkä, 2013).

Filtered horizon belongs to the same claim-status discipline as the measured cumulative estimate. It is not a destiny meter. It is a latent state inferred from information available at time t : symptoms, prognosis, functional capacity, social obligations, task structure, perceived time, clinical context, and observed behavior. The analyst may later know more, but the confirmatory model must not.

A useful horizon construct has three obligations. It must name its information set, it must separate belief from realized time, and it

must state how error changes the claim. If a horizon estimate works only because the model secretly sees the realized endpoint, the result is endpoint leakage, not prediction.

A route state should therefore be represented as a typed collection rather than a suggestive scalar. At minimum, the collection distinguishes the cumulative residual under study, the capacity variables that could alter future trajectories, the opportunity or horizon variables available under the current information set, and the contextual variables that govern observation. This typed construction prevents a change in fatigue, prognosis, missingness, or treatment access from being mistaken for movement in the latent target. It also makes clear which quantities belong to the route, which belong to measurement, and which belong only to the analyst's reconstruction.

Observability is a separate question from existence. A latent route state may be scientifically meaningful yet unrecoverable from the available channels, while a highly predictive observable may fail to identify the state that the theory requires. State-space practice therefore demands an explicit transition model, an explicit observation model, and recovery checks under realistic noise, irregular sampling, and dropout (Durbin & Koopman, 2012; Särkkä, 2013). The chapter treats this distinction as a gate: a route cannot be favored because a flexible filter fits the observed series unless simulation and held-out recovery show that the fitted state is distinguishable from serious alternatives.

Prospective transport tests whether a candidate state improves prediction across declared windows, cohorts, interventions, and observation regimes. The state is constructed from information

available before the outcome. Changes in care setting, device availability, or filtering method are examined for changes in its meaning. The resulting analysis can identify a stable route state or a more specific forecasting role.

16.1.2 — H_t From Available Information

The notation H_t names the filtered remaining-horizon candidate at time t . It is not the true remaining lifetime. It is the model's expectation, distribution, or decision-relevant summary of remaining opportunity given the information set available at that moment (Särkkä, 2013; Hernán & Robins, 2020).

The information set, written as $\mathcal{I}_t$, must be declared before analysis. It may contain current clinical status, functional change, self-reported future-time perspective, observable task context, recent choices, and approved auxiliary data. It may not contain future death date, future endpoint adjudication, future sensor loss, or later family narrative except in explicitly retrospective sensitivity work.

One compact definition is to let H_t equal the expected positive remaining time to a latent endpoint under a specified horizon model. The definition is useful only if the model also reports its uncertainty. A point horizon without an uncertainty distribution invites the false impression that the future is already known.

For behavioral tasks, H_t can be discretized into opportunity bins rather than calendar time. A participant may have a short task horizon because only two trials remain, a short social horizon because an interaction is ending, or a short clinical horizon because mobility or communication options are closing. The

shared scientific object is remaining usable opportunity, not metaphysical finality.

The strongest use of H_t is within-person. The model asks whether the same person changes allocation as filtered horizon changes, after reserve, disease severity, medication, fatigue, and context are modeled. Between-person comparisons are possible only after calibration and transport conditions are defended.

A filtered horizon makes the proposed route more concrete by supplying a prospectively defined state for comparison. Qualification asks whether that state improves held-out prediction and distinguishes the candidate from adequately informed rivals.

Every horizon estimate should carry an evidence card: information available at the time, excluded future data, the filtration timestamp, prediction target, uncertainty interval, and permitted downstream use. This card turns horizon from a retrospective story into a testable state variable. It also makes look-ahead leakage visible when prognosis, endpoint adjudication, or later outcomes have entered the predictor.

The design implication is that horizon can be manipulated safely in nonterminal tasks only when the participant knows the information structure. Hidden horizon manipulation may test surprise or anxiety, but it does not test the filtered state this chapter defines. A fair experiment can vary whether the horizon is known, probabilistic, or learned, but each condition must be labeled so the analysis knows whether it is testing available information, uncertainty, or revision.

16.1.3 — Uncertain and Revisable Horizon

Horizon uncertainty is not a nuisance to be hidden. It is part of the construct. Prognosis changes, opportunities return, treatments work, plans collapse, symptoms improve, and social support appears. A serious horizon model must allow revision without pretending that revision is failure (Seow et al., 2011; Vogel et al., 2013).

A horizon distribution is often more honest than a horizon point. The distribution can place probability mass over short, medium, and long opportunity windows, and those probabilities can update as new information arrives. This is especially important in clinical settings, where point prognosis can be both emotionally powerful and statistically fragile.

Revisability also prevents prophecy language. If H_t changes after a new scan, new medication response, or regained function, the old estimate was not a false oracle. It was a time-stamped inference under an earlier information set. The record should preserve both the old state and the update.

In a task environment, revision can be experimentally controlled. A participant may learn that a block has fewer remaining trials than expected, or that a previously unavailable action has reopened. The model then asks whether choices adjust in ways consistent with finite-horizon control rather than with a fixed personality trait.

Uncertainty affects the value of exploration. If the horizon might be long, learning can retain future value. If it is almost certainly short, exploration may compress. But if uncertainty itself is high,

information-seeking can increase. A model that always predicts urgency under uncertainty is too crude.

The withdrawal rule is straightforward: if horizon estimates cannot be revised coherently, if updates depend on future information, or if uncertainty overwhelms the behavioral effect, the chapter may keep horizon as an illustration but must withdraw it as a confirmatory state variable.

A useful analysis should therefore report not only the current horizon estimate but also the revision history. A stable estimate, a sudden compression, and a widening horizon can have different behavioral consequences even if they share the same current mean. The revision itself may carry information about burden, because a person living under repeated route closures may respond less to a nominally open future than a person whose horizon was shortened by one clear event.

Revisions should also be bounded by credibility. If every new observation radically changes H_t , the model may be unstable; if no observation changes it, the filter may be insensitive. A useful filter learns at a rate that can be justified by the evidence stream. Sensitivity to that learning rate should appear in the design, because overreactive and underreactive horizons produce different behavioral predictions.

16.1.4 — Objective and Subjective Horizons

Objective and subjective horizons answer different questions. Objective horizon concerns the opportunity window available under external conditions. Subjective horizon concerns what the

person or system believes, fears, hopes, or attends to. The two can diverge sharply (Durbin & Koopman, 2012).

Time perspective already has a testable behavioral role. In studies in the United States and Hong Kong, older adults' preference for familiar social partners diminished when they imagined an expanded future; under an imagined time constraint, both younger and older adults preferred familiar partners. Preferences also changed around Hong Kong's 1997 handover (Fung et al., 1999). These findings connect anticipated opportunity with social goals and provide an experimental way to separate perceived time from chronological age.

The distinction also improves the comparator. In an integrated analysis of 2,504 adults, greater age was associated with a more favorable emotional profile, while a more limited future-time perspective was associated with a less favorable profile even after accounting for health-related factors (Grühn et al., 2016). Perceived horizon, emotional benefit, and age therefore require separate measurements. A declared PTVC route can build on these studies by predicting a specific interaction between independently measured signed history and remaining opportunity, beyond social preference, current mood, health, and ordinary goal regulation.

Objective and subjective horizons can differ for understandable reasons. Someone with a longer clinical prognosis may perceive fewer future possibilities after depression, trauma, displacement, or repeated loss; someone with advanced illness may experience an open future through meaningful relationships, projects, or spiritual practice. Measuring both helps the model represent those

differences without treating either lived perspective as a personal error.

The Law-specific question cannot simply privilege either side. If choices track subjective horizon, the mechanism may be belief, meaning, and attention. If choices track objective opportunity after controlling for belief, the mechanism may involve physical constraint, reserve, or task availability. If both matter, the model must include both.

In laboratory tasks, objective horizon can be manipulated by remaining trials, time limits, or available moves. Subjective horizon can be measured through belief reports, confidence, perceived urgency, and willingness to invest in delayed payoff. The two should not be collapsed into one variable because collapse hides the mechanism.

Clinical and social settings add cultural death scripts and prognosis awareness. Some communities emphasize preparation, repair, reconciliation, or acceptance under threat; others avoid direct prognostic talk. These scripts may produce horizon-like behavior without any reserve-coupled Law-specific pressure.

The acceptance condition is therefore tiered. A result may support subjective-horizon effects, objective-menu effects, or reserve-horizon interaction. Separating those components and modeling rival scripts makes the resulting contribution interpretable.

The most informative cases are often disagreements between the two. If objective routes remain but subjective horizon is short, the study must ask whether the routes are actually trusted, usable, safe, and known. If subjective horizon remains long while

objective options are narrowing, the study must ask whether support, faith, denial, cultural meaning, or incomplete information is sustaining the person. The divergence is not an error term; it is a research object.

For copy and protocol design, the two horizons should be reported side by side. A table can show objective route count, subjective route confidence, information source, and route trust without implying one is the true horizon. The divergence may guide interviews or task interpretation, but it should not become a correction of the participant's lived account unless independent evidence warrants that correction.

16.1.5 — No Endpoint Leakage

Endpoint leakage occurs when future endpoint knowledge enters a model that is supposed to predict behavior before the endpoint is known. It is a close cousin of tautology. The analysis appears prospective, but it is quietly using the last event as an input (Särkkä, 2013; Hernán & Robins, 2020).

A confirmatory horizon model must be filtered. It may use $\mathcal{I}_t$, the information available at time t , and it may use a protocol-fixed forecasting model trained on other people or held-out periods. It may not use the person's realized time-to-death as a predictor in the same analysis that claims to test horizon-sensitive behavior.

The same rule applies to retrospective labels. Analysts may later classify a window as near-terminal, but that label cannot define the horizon variable used for a prospective claim. Retrospective analyses can be valuable for description, sensitivity, and hypothesis generation, but they must be labeled as such.

Cross-fitting is a practical guard. One model estimates endpoint or prognosis on held-out data, and another model tests behavior using only the filtered estimate. Thresholds and tuning choices should be locked before outcome inspection. If the same analyst adjusts both until the signature appears, the result is not confirmatory.

Endpoint leakage can be subtle. Sensor dropout, hospice admission, sudden medication changes, or family alerts may encode proximity to death. These variables may belong in rival or missingness models, but they require caution if used to define the horizon. What looks like shrinking opportunity may be observation shutdown.

The rule is severe because the claim is severe. If the Law needs the realized endpoint to make horizon predictions, then it has not predicted horizon behavior. It has redescribed the ending.

This rule also governs graphics. A plot aligned backward from the known endpoint may be useful descriptively, but it must not be labeled as the prospective horizon that shaped behavior. The figure label should say endpoint-aligned, smoothed, retrospective, or filtered. This is not visual fussiness. In a public book, the caption may carry the claim more strongly than the paragraph.

Endpoint leakage can also enter through model selection. If investigators try many horizon filters and keep the one that best predicts the known endpoint-aligned pattern, the leakage is analytic rather than algebraic. The safe route is to preselect filters, hold out tuning data, or report the search honestly as exploratory.

A filtered horizon is only confirmatory when its construction was disciplined before the target pattern was known.

16.1.6 — Horizon Measurement Models

Horizon is measured indirectly. It can be inferred from reports of future time, changes in goal selection, task behavior, clinical prognosis estimates, mobility, communication capacity, and observable opportunity sets. None of these channels is horizon itself (Seow et al., 2011; Vogel et al., 2013).

A strong measurement model declares what each channel is allowed to say. A self-report item may measure perceived future time but can be shaped by mood, culture, and interviewer demand. A clinical prognosis may estimate survival distribution but not the person's lived sense of remaining possibility. A task choice may reveal policy under experimental conditions but not life meaning.

A horizon model may combine several imperfect channels, but disagreement among them should remain observable. Reported future opportunity, clinical prognosis, task-imposed time, and environmental options can diverge for principled reasons. The analysis should test whether that divergence predicts behavior or model failure instead of forcing all channels into a single latent score.

Horizon measurement also needs time scale. A person may have a short horizon for mobility, a medium horizon for communication, and a long horizon for relationship meaning. A single H_t can be used only when the domain is declared.

Otherwise the model should use a vector of domain-specific horizons.

In task settings, horizon can be measured by remaining moves, remaining time, probability of task continuation, or belief about continuation. The Virtual Terminal Maze described later uses these variants because they can be manipulated safely. A task horizon is not a death horizon; it is a model environment for testing control signatures.

A horizon measure is qualified only when it predicts protocol-fixed behavior out of sample, remains stable enough for its context of use, and fails gracefully when rivals explain the same data better. Without those conditions, horizon remains a useful word, not an estimand.

A horizon model should also declare its cadence. Daily clinical updates, hourly task states, weekly social-route checks, and momentary app prompts define different horizons. Mixing them can create false precision because a route may close between observations. The model should state whether it estimates horizon continuously, at visits, at decisions, or at protocol-fixed time bins. A horizon that is measured too slowly may explain behavior that happened before the measurement noticed the change.

The horizon variable represents available future options under a declared information set. Measures of despair, acceptance, readiness, meaning, optimism, and perceived opportunity can be collected separately when the research question calls for them.

16.1.7 — No Prognostic Authority

A filtered horizon estimate is a route-specific research object, not a prognosis for a person. It must not determine treatment, rescue effort, hospice eligibility, discharge, insurance, family communication, resource priority, or a participant's understanding of how long they have. In clinical settings, research estimates remain subordinate to licensed judgment, current care standards, and participant preferences. A scientifically useful horizon estimate can describe structure relevant to a model without possessing the validation required for individual prognosis. Keeping those functions separate allows the route hypothesis to be tested without creating an unintended clinical prediction system.

The boundary is scientific as well as ethical. If a horizon estimate changes care, communication, observation intensity, or endpoint timing, it changes the process under study and creates feedback, selection, and leakage. Such feedback can make the estimate appear predictive partly because people or systems began responding to it. Protocols should therefore record any exposure to route-state outputs that could alter the subsequent trajectory or observation process.

Purpose limits require access roles, prohibited-use rules, staff training, participant-facing language, audit logs, incident response, and withdrawal. Public copy may describe a filtered horizon estimate, opportunity model, or task horizon, but never a hidden future. Access controls should also distinguish who needs the estimate for scientific analysis from personnel whose care decisions should remain insulated from it. That separation

reduces the chance that a research variable acquires operational authority simply because it is available.

State identification begins with observability and recovery. The study must show which route states are distinguishable from reports, actions, physiology, and context, and when equivalent states require a coarser output, identified set, or abstention. Recovery should be tested across plausible combinations of states, including cases deliberately chosen to be difficult to distinguish. A state label earns scientific meaning only to the extent that the available observations can reliably separate it from competing latent explanations.

Transition timing must recover in held-out participants, sites, and windows; remain stable under plausible observation shifts; and fail under protocol-fixed rivals. Ablation tests whether horizon, reserve, unresolved state, and their interactions add information beyond severity, frailty, treatment, prognosis, and observation intensity. The strongest evidence comes when the same timing relation survives changes in site and observation conditions without requiring retrospective retuning. Failure to transport should narrow the route claim rather than be absorbed into a more flexible timing model.

The route-state verdict reports observability, recovery, invariance, ablation, rival comparison, and purpose control. Passing licenses further route research in the declared class, not individual forecasting, clinical authority, or PTVc support.

16.2 — Reserve and Horizon Proxies

Reserve and horizon are related but nonidentical. Horizon concerns continuation support or time structure available under current information. Reserve concerns the capacity through which a candidate route could alter future contribution. A long horizon can coexist with little capacity; a short horizon can coexist with intense but narrow capacity (Ramsay & Woods, 2014).

Candidate reserve proxies may include functional status, treatment access, cognitive capacity, autonomic flexibility, available actions, social support, or a route-specific controllability index. No single proxy is the reserve by definition. The measurement model must show shared variation, channel-specific variation, and confounding by the outcome.

Independence is a central design pressure. If reserve is estimated from the same affective reports used to construct the residual, a shrinking ratio may be algebraic rather than biological or psychological. Separate instruments, temporally separated estimation, negative-control channels, and cross-fitting reduce that circularity without curing an invalid construct.

Horizon estimates require uncertainty. A filtered distribution over remaining route time is usually more defensible than a point forecast. Prediction intervals should be calibrated in held-out data, while analyses indexed by realized time-to-endpoint must be labeled retrospective. Neither form may become a personal forecast or care recommendation.

The most useful reserve proxy is not necessarily the one most correlated with terminal decline. It is the one that creates a

discriminating relation after illness severity, medication, care context, and observation availability are modeled. Failure to separate reserve from generic severity blocks promotion of the reserve-coupled route while leaving the measurement work independently useful.

16.2.1 — Reserve as a Multi-Domain Capacity State

Reserve completes the horizon problem. A menu may remain open on paper while the body, attention, cognition, energy, pain tolerance, or social support needed to use it has changed. The model therefore separates remaining time from remaining capacity (Ramsay & Woods, 2014).

The reserve variable $U(t)$ in the formal candidate is not a clinical score handed down by nature. It is a latent capacity construct, and every observable proxy carries interpretation risk. Those risks must remain visible before reserve is used to explain any pattern.

Reserve should be measured with channels as independent from the affect estimator as the design can achieve. A capacity score built from distress, anhedonia, or the same physiological features used to estimate valence can manufacture the interaction it is meant to discover. Shared inputs should be disclosed, ablated, and tested through held-out channels before reserve-specific claims are made.

Reserve is often better modeled as a vector than as a single score. Physiological endurance, cognitive flexibility, communicative capacity, social support, treatment access, and environmental affordance can move in different directions, and each may matter for a different route. Collapsing them into one number can hide

the very mechanism the study hopes to identify. A defensible scalar summary must therefore be justified by a declared measurement model, supported by dimensionality analysis, and accompanied by domain-level sensitivity checks rather than treated as a natural quantity discovered by inspection.

The causal placement of reserve must also be explicit. Reserve may mediate the effect of disease severity on action, moderate the relation between a filtered horizon and choice, predict observation burden, or simply mark clinical decline. Those are not interchangeable roles. The protocol should draw the candidate graph before estimation, specify which time-varying variables are confounders and which are consequences, and test whether the reserve term adds information beyond prognosis, medication, frailty, and care context. Without that separation, an apparently elegant reserve interaction can be ordinary illness progression in new notation.

Finally, reserve has no permission to become a person-level rank. Its scientific use is local to a prospectively specified question, a declared population, and a stated observation regime. Transport requires recalibration when devices, supports, disability patterns, cultures, or care settings change, and failure to transport narrows the reference class rather than lowering anyone's status. The strongest reserve model is therefore not the model with the most persuasive dashboard. It is the model whose components remain interpretable, independently estimated, ethically bounded, and replaceable when better evidence arrives.

16.2.2 — Biological Reserve Candidates

Reserve candidates include physiological stamina, inflammatory load, sleep continuity, mobility, nutrition, oxygenation, autonomic flexibility, cognitive endurance, medication burden, and functional performance. None is the reserve. Each is a possible indicator within a context of use (Seow et al., 2011; Gerstorf et al., 2010).

A useful reserve proxy must state its raw units. Steps per day, grip strength, heart-rate variability, creatinine, oxygen saturation, gait speed, cognitive response variability, and clinician-rated performance status do not live on the same scale. Combining them requires transformation and a theory of what the composite means.

The Law-specific use of reserve is narrower than general health. Reserve matters because it may modulate viable action, volatility, and the admissible set of possible repair or avoidance policies. A biomarker that predicts mortality but not choice, capacity, or signal dynamics may be a prognosis marker rather than a reserve marker.

Clinical state matters. Medication, pain, sleep disruption, delirium, infection, disability, and care quality can all lower apparent reserve. Some of these are reversible. Some are context-dependent. Some affect measurement more than capacity.

Reserve is a latent candidate construct represented through competing indicator models. Each analysis states its normalization and the transformations that preserve the tested restriction.

A reserve candidate can be evaluated through out-of-sample task behavior, observation feasibility, and dynamic variance patterns, with disease severity and prognosis included in the comparison.

A candidate biological route can begin with an independently mapped system. Kleckner and colleagues (2017) identified a large-scale allostatic–interoceptive network in separate discovery and replication samples. J. Zhang and colleagues (2025) extended the map with high-resolution 7 Tesla imaging in 90 participants, mapping resting-state functional relationships between cortical regions and brainstem and other subcortical structures. This supplies specific candidates for studying how bodily condition and affective regulation interact. The useful PTVC-oriented contrast asks whether a frozen cumulative-history variable improves prediction beyond current physiology, recent experience and ordinary allostatic models.

The candidate reserve channels should be tested under perturbation where ethical. In safe settings, that may mean ordinary task fatigue, sleep variation, low-stakes effort, or recovery after a mild cognitive load. In clinical settings, it may mean naturally occurring changes in symptom burden or treatment timing. Reserve is more convincing when it predicts recovery and tolerance, not merely baseline status.

A reserve model should include positive support as well as decline. Oxygen support, assistive devices, caregiver skill, accessible transport, pain relief, and predictable routines can increase effective capacity. If the model only records biological depletion, it may falsely treat social neglect as natural exhaustion.

The fairness lens requires the model to ask when capacity was lost and when it could have been protected.

16.2.3 — Reserve Is Not One Biomarker

The simplest reserve error is to choose one convenient biomarker and let it stand for remaining capacity. That shortcut can make a study look clean while making the construct false. A single biomarker may reflect disease, treatment, sensor contact, socioeconomic access, or disability accommodation rather than the capacity needed by the model (Vogel et al., 2013; Särkkä, 2013).

Frailty scores are useful but not sovereign. They summarize vulnerability and function for particular clinical purposes. They do not automatically measure the formal reserve variable $U(t)$, and they can embed age, disability, mobility, or care-access assumptions that would be unacceptable as hidden Law-specific structure.

A reserve system should therefore report a specification table: channel, unit, transformation, cadence, expected direction, confounds, missingness, context of use, and retirement rule. Without that table, reserve language will drift into authority it has not earned.

The strongest designs use multiple reserve families and test ablation. If the reserve interaction survives only when one fragile indicator is included, the result is weak. If it survives across independent capacity channels with different artifacts, the candidate becomes more interesting.

A reserve proxy should also face a prognostic-ablation test. First fit the declared survival, disease-stage, recent-decline, treatment, and observation-quality baseline; then add the independently estimated reserve channels and evaluate held-out calibration and discrimination. Reserve earns a separate role only when its contribution survives that comparison and remains interpretable after shared inputs are removed.

The public rule is simple: no reserve measure ranks people. Reserve is a study variable for a narrow inference, not a measure of value, eligibility, courage, readiness, or dignity.

Table 16.1. Candidate reserve-proxy validation and misuse boundaries.

Reserve candidate	Raw scale	Primary confound	Use rule
Functional capacity	Mobility, grip, gait, fatigue tolerance	Disability access, pain, equipment, care support	Use only with accommodation and disease controls
Autonomic flexibility	HRV, respiratory coupling, recovery time	Medication, sleep, anxiety, sensor contact	Use as channel, not whole reserve
Metabolic/inflammatory load	Lab values, nutrition, fever, cachexia markers	Disease process, treatment, infection	Model as shared cause and capacity signal
Cognitive endurance	Latency drift, attention, task persistence	Delirium, sleep, medication, education	Use only in accessible tasks

Reserve candidate	Raw scale	Primary confound	Use rule
Social/environmental support	Available help, transport, communication aids	Inequality, family pressure, resources	Treat as menu modifier, not biology

Composite reserve needs interpretability safeguards. A weighted score may predict well while hiding that medication timing or device tolerance carries the effect; a hierarchical model may improve fit while obscuring the construct. Early studies should therefore favor transparent low-dimensional proxies, with richer latent models admitted only when they improve held-out prediction without erasing meaning.

This rule is especially important for commercial devices. Wearable outputs may be precise, frequent, and visually persuasive while remaining weak proxies for the reserve construct. A device score should be treated as a channel with firmware, placement, noise, calibration, and missingness, not as a direct measurement of capacity. That distinction must be clear before any reserve graph appears.

16.2.4 — Reserve and Disease Severity

Disease severity is the largest reserve confound. As illness progresses, reserve may decline, prognosis may shorten, medication may increase, observation may become harder, and choices may narrow. A naive model can misread this shared driver as a Law-specific reserve effect (Ramsay & Woods, 2014).

At least three pathways must remain separate: severity to reserve, severity to horizon belief, and severity to behavior or

measurement. When they move together, an apparent reserve interaction may be a disease-stage interaction, and the analysis should report that common cause directly.

Time-varying clinical covariates are not optional. Medication dose, pain level, dyspnea, delirium status, functional decline, sleep disruption, and care setting can change from day to day. A cross-sectional reserve score may be too coarse for horizon prediction.

A useful design can exploit within-person deviations. If reserve improves temporarily because symptoms are managed and horizon belief remains stable, do choices and latencies change? If prognosis worsens but functional reserve is maintained, does the model still predict menu compression? These contrasts help separate mechanisms.

Reserve must also remain distinct from socioeconomic constraint. Poverty, inaccessible environments, absent transport, and unsupported caregiving can shrink live menus without indicating biological reserve. Treating those conditions as an inner capacity deficit would naturalize a remediable injustice.

If reserve cannot be separated from disease severity, prognosis, resource access, treatment intensity, sensor tolerance, or prior knowledge of the endpoint, the reserve-specific interpretation fails. The remaining association may still describe those observed variables, but it cannot be promoted into evidence for an independent completion capacity.

The hardest cases occur when disease severity is not a confound to remove but part of the pathway through which reserve changes.

A model that over-controls may erase the very capacity decline it wanted to study; a model that under-controls may call disease progression reserve. The protocol should declare the causal question first, then decide which variables are confounders, mediators, stratifiers, or rivals.

The same disease severity can coexist with different effective reserve under different supports. Transport, medication access, quiet sleep, family protection, and disability accommodation can change a person's available capacity. Including these conditions makes the reserve model more complete and identifies practical opportunities for support.

16.2.5 — Reserve Over Time

Effective capacity may decline, recover, fluctuate, or change abruptly. Such trajectories require an explicitly declared reserve model. The nonincreasing reserve branch in Section 4.2.2 is a narrower candidate; its drift and boundary consequences do not automatically transfer to a recovering or stochastic reserve process. A baseline proxy alone cannot establish which branch describes the observed window.

Longitudinal reserve modeling requires cadence. A daily measure may capture fatigue and medication effects. A weekly measure may capture functional decline. A minute-level measure may capture task stamina. The choice of cadence should match the prediction.

Measurement error is especially important because reserve often appears inside interactions. Error in $U(t)$ can attenuate the estimated interaction with horizon, create false threshold effects,

or shift apparent variance slopes. Reliability is not a clerical detail.

State-space models can help by treating observed reserve proxies as noisy emissions from a latent capacity process. Such models should include process noise, observation noise, missingness, and version changes. They should also report uncertainty in reserve, not only a smoothed line.

Smoothed reserve trajectories using future data may be useful for retrospective interpretation, but confirmatory predictions should use filtered or forecast reserve estimates. Otherwise the model again learns from the future while pretending to be prospective.

Noise or instability that defeats the predicted horizon interaction is informative. Reserve may remain clinically meaningful while failing as a Law-specific dynamic variable.

Longitudinal reserve should be modeled with lags. Capacity at noon may affect action in the afternoon; action in the afternoon may alter capacity in the evening. A model that treats U_t and behavior as simultaneous can confuse cause and consequence. Lag sensitivity, recovery windows, and delayed effects should be tested before reserve is described as an explanatory state.

The temporal model should distinguish depletion from availability. A person may have capacity in principle but not at the relevant hour because pain, fatigue, medication, or care schedule blocks its use. The unit of analysis should therefore match the decision. A weekly reserve score cannot safely explain a ten-minute task choice unless the mismatch is modeled.

16.2.6 — Independent Estimation

Independent estimation guards against circularity. Where feasible, build reserve proxies from separately qualified physiological, functional, cognitive, or environmental channels, apart from the ledger's affective and experience-linked inputs. Validate context of use, analytical and clinical validity, responsiveness, failure modes, and version drift before entry into the route equation (FDA-NIH Biomarker Working Group, 2016; Goldsack et al., 2020).

Perfect separation is rarely possible. Pain affects mobility, mood affects sleep, medication affects cognition, and distress affects physiology. The goal is not impossible purity. The goal is to declare overlap, model shared causes, and test whether the reserve effect survives when overlapping channels are removed.

Predeclare reserve-channel tiers. One scheme uses functional and physiological capacity with minimal direct affect content, then clinical symptoms overlapping experience, then exploratory composites. Confirmatory claims should favor the least circular tier.

Independent estimation also matters for dignity. If distress lowers a reserve score and the score then predicts future distress, the model may simply be scoring suffering twice. That is scientifically weak and ethically dangerous.

Routinely remove medication-linked, mobility, autonomic, and report-linked channels to test whether the interaction persists. A signature lost after plausible channel removal lacks robustness.

Independent estimation also requires temporal separation. Reserve features used to explain an interval should be constructed from information available before that interval, with endpoint-derived summaries, later clinical deterioration, and affect-estimator outputs excluded or separately ablated. Otherwise apparent reserve coupling may be look-ahead leakage wearing a biological name.

Independence is not all-or-nothing. A study may have strong independence for physiological reserve, partial independence for functional reserve, and weak independence for self-reported energy. The status statement should name those levels. A reserve result based on a partly overlapping channel can still orient research, but it cannot carry the same claim as a result built from independently validated capacity measures.

16.2.7 — Reserve Failure Conditions

Reserve failure is a scientific outcome that clarifies the next step. Candidate components may fail to cohere, an ordinary rival may explain the pattern, a proxy may not transport, or the public-risk burden may make the object unsafe to use. Each failure updates a different proposition and should be recorded separately.

Triangulation should compare proxies from genuinely different pathways rather than stack correlated versions of one signal. Biological, functional, cognitive, social, and material capacities may constrain different menus and time scales, and no biomarker may stand for the entire reserve state. Agreement is informative only when the channels do not share the same measurement error and causal role.

Test monotonicity: reserve may recover, fluctuate, compensate, or fall abruptly, while support can expand options and increase burden. Treat social and material resources as environmental pathways, distinct from personal biology or individual strength.

Severity, clinician expectations, treatment, observation, and endpoint processes can affect both reserve proxies and outcomes. Estimate reserve prospectively, then compare prognosis, frailty, and observation-selection explanations before drawing route conclusions.

The valid output may be an identified set rather than one number. When several reserve states remain compatible with the data, exponent, covariance, intervention, and terminal-window analyses propagate that uncertainty.

Retain failed or retired proxies and their affected analyses and artifacts in the version ledger. Reanalyze, narrow, or withdraw dependent results. Publish failure and retirement conditions before inspecting outcomes.

16.3 — Boundary Exponents and Reserve Dispersion

Boundary predictions are useful only when the coordinate system is declared. A process may contract as calendar time approaches an endpoint, as a reserve proxy approaches zero, or as an action set loses reachability. Those descriptions are related only under an explicit bridge. The section therefore separates the time-domain exponent from the reserve-domain exponent and treats both as route-specific consequences rather than properties of PTVC itself (Karatzas & Shreve, 1991).

The empirical target is conditional dispersion estimated from information available before the endpoint. A local power relation between reserve and remaining time yields a time-domain exponent only under an explicit reserve-time bridge, while measurement error, heterogeneous paths, smoothing, and selection can alter the apparent slope. The protocol should fix a multiscale window ladder, error floor, random-effects structure, and rival family before any log-log relation is interpreted.

Boundary exponents also interact with behavior. Delay sensitivity, priority reordering, risk pruning, repair initiation, and avoidance can change as filtered horizon and reserve change. The candidate's cross-equation restrictions ask whether the same frozen state predicts several outcomes without retuning.

Candidate scaling relations. Role: route-specific asymptotic restrictions under a declared endpoint clock, reserve bridge, and normalized-dispersion model; they are not consequences of PTVC alone (Karatzas & Shreve, 1991).

$$U_t \asymp c_U(T - t)^\beta, \text{Var}(Y_t|U_t, \mathcal{E}_t) \asymp c_Y(T - t)^\gamma$$

$$L_t^{rc} = c_{LU} U_t Y_t$$

$$Var(L_t^{rc} | U_t, \mathcal{E}_t) \asymp c_{LU}^2 c_U^2 c_Y (T - t)^{2\beta + \gamma}$$

$$\gamma = 0 \Rightarrow Var(L_t^{rc} | U_t, \mathcal{E}_t) \asymp (T - t)^{2\beta}$$

The final display is the special case, not the general rule. The time-domain exponent equals twice the reserve-decay exponent only when normalized conditional dispersion is locally stable, or when an independently justified argument supplies the same restriction. Measurement floors, smoothing, path heterogeneity, selection, or a changing dispersion exponent can all alter the observed slope. A shrinking variance curve therefore cannot be relabeled reserve-squared behavior by visual resemblance alone.

16.3.1 — Rising Delay Sensitivity and Boundary Rate

As filtered horizon contracts, delayed outcomes may lose motivational relevance because they fall outside perceived opportunity. Socioemotional selectivity is therefore an ordinary rival. The route-specific test asks whether unresolved state and independently estimated reserve contribute a protocol-fixed interaction beyond future-time perspective, prognosis, urgency, and scarcity (Carstensen, 2006).

Naturalistic observations broaden the horizon finding beyond imagined scenarios. Fung and Carstensen (2006) observed shifts toward emotionally meaningful goals around the September 11 attacks and the SARS outbreak. These settings couple time perspective to threat, social contact, and available resources, so their contribution is ecological convergence rather than randomized causal identification. The next useful comparison tracks those variables separately and asks whether a common,

prospectively fixed horizon-by-state relation transports across ordinary transitions such as relocation, retirement, and the completion of a voluntary project. Successful transport would give a candidate mechanism a substantive constraint to explain.

Perceived time can be studied separately from a frustrated goal. Fung and Carstensen (2004) compared social motivation under foreshortened time and blocked-goal conditions, finding different patterns in social preferences and reported goals. Their design tested alternatives to socioemotional selectivity theory rather than treating all constraints as equivalent. It offers a useful template for this chapter: measure perceived opportunity and current goals separately, then ask whether independently estimated experiential history improves predictions of subsequent choice.

The prediction can be written in a task as a change in the policy for delayed repair options. When h remaining steps are high, the system may choose information gathering or delayed high-value repair. When h is low, it may prefer immediate lower-value actions if they reduce an unresolved state before the menu closes.

This can be tested without terminal pressure. A task can vary remaining trials, delay to reward, reversibility, and cost while keeping all stakes minor. The outcome is choice probability, latency, confidence, post-choice affect, and physiological control load.

The ordinary rivals are strong. Time pressure, impatience, fatigue, scarcity, stress, and reward discounting can all increase sensitivity to delay. A horizon-specific result must show the predicted

interaction with filtered horizon and reserve, not merely a main effect of deadlines.

Shortening horizons need not compress every person toward the same policy. Some people become cautious, others exploratory, generous, rigid, automatic, or more attentive to meaning. Heterogeneity should be modeled through baseline policy, available menu, reserve, culture, task framing, and belief rather than treated as error around a universal response.

Delay sensitivity confined to tired, frightened, or externally pressured participants would point toward those conditions as explanatory factors. A horizon-specific account is tested through its additional predicted interactions with filtered horizon and reserve.

A good delay experiment should include catch conditions. Some delays should be irrelevant to outcome, some should threaten route loss, and some should consume reserve. If all delays produce the same response, the effect is not horizon-specific. If only route-threatening delays matter under short H_t , the candidate becomes more precise and more vulnerable.

Delay sensitivity should be measured against a baseline delay function. Some people are generally impatient; some are generally deliberative; some delay because access is difficult. The candidate pattern is a change in delay sensitivity as H_t and U_t change, not a stable trait. Within-person estimates and task counterbalancing are therefore central.

Scaling claims need two clocks. A reserve-space exponent describes how dispersion changes as candidate capacity U

approaches a boundary, while a time-domain exponent describes how dispersion changes as observed time approaches an independently defined endpoint. If $U(t)$ is approximately proportional to $(T - t)$ raised to a power β , then the simplest reserve-coupled diffusion yields a time-domain variance exponent of approximately 2β . That relation is conditional on the reserve map, diffusion specification, endpoint branch, and observation model. Reporting one exponent as though it were the other is a category error, not a harmless notation choice.

The empirical design estimates the reserve map and boundary law separately across a prespecified ladder of scales. A nonlinear reserve-time relation can create apparent acceleration, while saturation, smoothing, and irregular observation can flatten or steepen it. Joint models estimate reserve-time mapping, process dispersion, and observation floor, then test whether the same exponent family transports across held-out windows.

16.3.2 — Priority Reordering and Drift Slope

Priority reordering means that the relative ordering of live demands changes as horizon and reserve change. A person may shift from long-range exploration toward emotionally meaningful contact, repair, communication, rest, or avoidance. The proposed route predicts a signed-state dependence that can be compared with ordinary horizon-driven goal selection (Carstensen, 2006).

A scientific model must specify the demand set. In a task, demands are explicit goals or penalties. In an observational study, demands may be coded from participant reports, scheduled

obligations, symptoms, unfinished tasks, or documented choices. Retrospective moral interpretation is not allowed.

The candidate prediction is not simply that important things become important. It is that the ordering changes as a function of filtered horizon, reserve, and menu constraints after controlling for baseline importance. A high-importance item may still fall if it is infeasible under low reserve.

Priority reordering can be visible in time allocation, task switching, communication attempts, search behavior, medication questions, social contact, or rest selection. Each channel has its own confounds. A phone call may reflect relationship repair, loneliness, practical logistics, family pressure, or habit.

Rivals include socioemotional selectivity, cognitive bandwidth under scarcity, fatigue-driven simplification, clinical urgency, and learned avoidance. These rivals are not hostile; they are the standard the candidate must exceed.

If reordering follows ordinary expected value, prognosis information, or social scripts with no reserve-queue interaction, the horizon-specific claim should be narrowed. The data may still teach something humane about burden under constraint.

Priority reordering should be evaluated as a trajectory. The order in which a person opens, postpones, abandons, or revisits demands may reveal more than the final action. A single endpoint choice can hide the labor of earlier attempts. Sequence models, not just outcome models, are needed if the program wants to see where burden moved.

A useful priority model may treat demands as a partially ordered set rather than a ranked list. Some tasks are incomparable: symptom relief, apology, legal deadline, last conversation, and rest may not fit a single ordering. The model should allow partial orders and ties, because forcing total order can manufacture precision and disrespect the structure of lived value.

16.3.3 — Risk Pruning and Dispersion Compression

A finite horizon may change tolerance for high-variance trajectories because it changes the time available to recover from setbacks or benefit from upside. The present candidate calls this risk and variance pruning when high-variance options leave the live menu or become less attractive. This proposed behavioral mechanism can be tested by measuring how choices change with represented opportunity, current condition, and recent experience. The clinical studies discussed nearby supply symptom and affect trajectories for that comparison (Seow et al., 2011; Vogel et al., 2013).

The prediction must be stated carefully. Some people under short horizons take more risks, not fewer. They may seek meaning, novelty, relief, or autonomy. Others become conservative because capacity is low. The model therefore predicts structured heterogeneity, not a universal shift.

The relevant variance can be reward variance, affective variance, physiological load variance, social outcome variance, or delay variance. A task must declare which variance it manipulates. Otherwise all risky options are treated as the same object.

Reserve is central. A high-reserve participant may tolerate a high-variance repair attempt because they can absorb failure. A low-reserve participant may prefer stable relief, rest, or small repair. The same horizon can produce different policies under different $U(t)$.

Rivals include loss aversion, anxiety, depression, clinical fatigue, socioeconomic scarcity, and ordinary finite-horizon expected utility. The candidate becomes interesting only if the variance-pruning pattern depends on the unresolved burden state rather than on generic risk preference.

The route-level claim fails in a setting when observed pruning is fully accounted for by baseline risk preference, disease severity, cognitive change, misunderstanding, or a reduced option set unrelated to reserve. The protocol should predeclare those comparators and retire the candidate interpretation when they explain the pattern under the same scoring rules.

Variance pruning also requires a recovery measure. A high-variance option is different when recovery is possible. In the task, the same loss can be paired with high or low reserve recovery. In field settings, recovery may come from rest, medication, support, money, or time. Without a recovery variable, the model cannot know whether risk was pruned because variance rose or because the cost of error became unrecoverable.

Risk pruning should also be compared with ambiguity aversion. A person may avoid an option not because the outcome variance is high, but because the probabilities are unknown or the institution cannot be trusted. The task should distinguish known

risk from ambiguous risk. In field settings, mistrust can be a rational response to prior burden, not a cognitive flaw.

16.3.4 — Accelerated Repair and State-Dependent Drift

Shrinking horizon may accelerate repair, but it may also accelerate avoidance. A person may attempt a difficult conversation, settle a task, forgive, plan, rest, withdraw, distract, or stop engaging. The protocol registers behavior categories and opportunity constraints before outcomes are known; it cannot preselect the noble-looking action as evidence for the route (Carstensen, 2006).

Direct repair is only one policy. Avoidance may be rational when repair is impossible, unsafe, coercive, or too costly. Blocking may protect a person from overload. Substitution may reduce a related burden when the original burden cannot be changed.

A task can distinguish repair and avoidance by offering actions that reduce a specific unresolved state, actions that suppress signals without resolving them, and actions that substitute a lower-cost route. Outcome measures can include choice, latency, post-action residual, and future reactivation.

The Law-specific prediction is conditional. If a burden is salient, repair is feasible, reserve is sufficient, and the horizon is shrinking, direct repair may rise. If reserve is low or repair has high risk, avoidance or substitution may rise instead. The hypothesis should state these branches before data are seen.

Clinical and ordinary-life interpretation must be humble. A person's apparent avoidance may be dignity, fatigue, privacy, trauma protection, communication difficulty, or refusal of

coercive meaning. Science can code behavior only within its evidence limits.

A repair hypothesis fails if it becomes a narrative that explains every action after the fact. If repair, avoidance, and no action all count as support, the prediction has no content.

This subsection should be interpreted through feasibility classes. A favorable repair signature exists only when repair is feasible, safe, and meaningful. A favorable avoidance signature exists only when avoidance is predicted under declared threat, cost, or low-probability repair conditions. Otherwise the model can explain both action and inaction too easily.

A paired design can make this test sharper. The same unresolved demand can be offered with a safe repair route, an unsafe repair route, a delayed route, and a substitute route. The candidate predicts different policies across those route types under changing horizon and reserve. If behavior does not differ by route feasibility, the repair-avoidance interpretation weakens.

16.3.5 — Reserve-Horizon Interaction

The core quantitative prediction is not a main effect of age or calendar time. It is a protocol-fixed interaction among filtered horizon, independently measured reserve, unresolved state, and the live action menu. A short horizon with high reserve may permit focused action; the same horizon with low reserve may yield rest, compression, or no observable response (Carstensen, 2006).

Emotional strengths can persist when circumstances become difficult. In 945 US adults aged 18–76 assessed during COVID-

19, older age was associated with better emotional experience even after measured risk and other covariates were considered (Carstensen et al., 2020). Sun and Sauter (2021) reproduced age advantages in 23,350 people across 63 countries, while their within-person Dutch comparison of 4,370 participants found that those advantages diminished during sustained pandemic stress. Together, these studies make context-dependent regulation a concrete research target. A reserve-horizon test should measure capacity, exposure, social support, and perceived opportunity independently, then compare a frozen interaction model with these established accounts. The empirical opportunity is to explain when resilience persists, when support changes the path, and which additional state variable improves prediction.

This interaction must be modeled within person where possible. Between-person comparisons can confuse reserve with personality, culture, resources, disability, education, and diagnosis. The strongest design observes the same person across changing horizon and reserve states.

A reserve-horizon interaction should have a protocol-fixed sign and measurement route for each task. For example, low reserve may increase the cost of high-effort repair, while short horizon increases the cost of delay. The model then predicts which action class becomes more likely.

The slope-two variance relation is conditional, not automatic. If the conditional variance of the normalized process changes with reserve, the exponent becomes an estimand rather than a fixed route consequence.

In behavioral tasks, the interaction can be tested through reserve manipulations that are safe and mild: cognitive load, time-on-task fatigue, rest breaks, or simulated capacity budgets. These manipulations must never create distress or deprivation.

If reserve does not interact with horizon after confounds are modeled, the candidate should narrow. The result may still support finite-horizon decision science, but not the reserve-coupled version of the Law-specific prediction.

The interaction should be tested in designs that manipulate or observe both dimensions independently. A task can shorten horizon while holding reserve constant, lower reserve cost while holding horizon constant, or vary both. Clinical studies cannot manipulate these states casually, but they can stratify, model natural variation, and test whether the same pattern appears under different reserve proxies.

The interaction can be visualized without overclaiming by plotting predicted choice under four quadrants: longer horizon with high reserve, longer horizon with low reserve, shorter horizon with high reserve, and shorter horizon with low reserve. The plot should be labeled simulated or estimated, not observed truth, and should always include rival curves when used in public artifacts.

16.3.6 — Nonmonotone and Heterogeneous Boundary Behavior

Boundary behavior need not be monotone, and participants need not approach a declared window through the same policy. Some trajectories may narrow gradually, others may oscillate, plateau,

reverse, or change abruptly after treatment, support, fatigue, or a new constraint. A model that permits only smooth compression can fail unfairly and can also rescue itself by averaging away the cases that contradict it.

Heterogeneity should be protocol-fixed as structure. Random effects, latent classes, mixture models, or person-specific parameters may be appropriate, but the choice should be tied to observables and tested in held-out data rather than discovered after the pooled curve disappoints. When heterogeneity cannot be predicted or interpreted, the report should widen uncertainty rather than invent a universal story.

Nonmonotone patterns can separate rivals. State-coupled control, fatigue, episodic treatment, selection, ordinary mean reversion, and endpoint-conditioned bridges imply different reversals, lags, and cross-channel signatures. The analysis should preserve individual and subgroup trajectories long enough to test those differences before summarizing them.

Mixture-induced closure is a serious alternative. A narrowing marginal distribution can arise because high-variance trajectories disappear from observation, different subgroups enter the window at different rates, or opposing local trends cancel. Component-specific and observation-process analyses are required before marginal compression is attributed to one route.

Exponent recovery is local and empirical windows are finite. A power law that is valid asymptotically may not be identifiable over the observed reserve or time range, and the reserve-space exponent is not automatically the time-domain exponent. The

reserve map, process variance, endpoint branch, and observation floor must be estimated jointly enough to show which quantity the data identify.

Irregular observation times and finite noise floors can create or flatten apparent scaling. The analysis should use the actual pair-availability and cadence structure, compare several protocol-fixed local windows, and test whether the estimate survives plausible deconvolution and calibration error. One log-log line is not a boundary law.

Candidate and rival models must be scored on the same windows, information, and missingness conditions. A result that appears only under a favorable bandwidth, after excluding reversals, or in one latent class chosen post hoc remains exploratory. Robust heterogeneity can strengthen a bounded route claim; unexplained heterogeneity should narrow it.

These trajectories describe a population, branch, and observation model; they do not prescribe how any person should feel or act near an endpoint. Nonmonotonicity is therefore both a methodological possibility and a safeguard against turning an aggregate pattern into an individual demand.

16.4 — Two-Time Covariance and Martingale Diagnostics

Marginal contraction is not a unique signature. Two processes can have the same mean and variance near an endpoint while carrying different dependence across time. The scientific leverage therefore lies in the joint law: covariance, conditional increments, quadratic variation, and how prediction errors behave when one interval is held out. These objects are demanding, but they are precisely where conditioned bridges, mean-reverting processes, and reserve-coupled routes cease to look interchangeable (Doob, 1957; Revuz & Yor, 1999).

Prospective two-time prediction must use information available at the earlier prediction time; a prespecified retrospective covariance estimand must be labeled by its actual information set. Retrospective alignment can create elegant covariance surfaces that no prospective observer could have constructed. The first display below is a candidate reserve-route covariance under its stated assumptions; the second is a Brownian-bridge rival. For this display, S_θ and $Y(S_\theta)$ are fixed, and, conditional on Ξ , U and σY are deterministic while W remains Brownian. Neither formula is evidence, and both must be evaluated under the same filtered information, endpoint uncertainty, and observation process (Andersen et al., 1993; Doob, 1957; Rizopoulos, 2012).

$$Y(t) = Y(S_\theta) + \int_{S_\theta}^t \sigma_Y(r) dW(r)$$

$$\tau_Y(t) := \int_{S_\theta}^t \sigma_Y^2(r) dr$$

$$\text{Cov}(L^{rc}(s), L^{rc}(t) \mid U, \mathcal{E}) = c_{LU}^2 U(s)U(t)\tau_Y(s), S^\theta < s \leq t$$

This is a candidate-route consequence under the stated branch, not PTVC evidence.

$$\text{Cov}(B_s, B_t) = \sigma^2 s \frac{T-t}{T}, s \leq t.$$

16.4.1 — Two-Time Covariance as a Route Signature

Two-time covariance asks how deviations at one time predict deviations at another. A route that merely shrinks marginal variance can still leave many possible dependence structures. A reserve-coupled arithmetic diffusion, a Brownian bridge, an Ornstein-Uhlenbeck process, and a hidden-state controller may all contract near a boundary, yet they need not agree on how early innovations persist, reverse, or disappear (Doob, 1957; Revuz & Yor, 1999).

The covariance object should be protocol-fixed before the surface is inspected. A primary option may be conditional covariance at two protocol times given a filtered reserve state; a secondary option may normalize by the protocol-fixed scale process. The normalization cannot be chosen after seeing which version looks most bridge-like (Doob, 1957; Durbin & Koopman, 2012).

Sampling cadence affects which time pairs are observed and how their covariance can be estimated. Sparse visits, selective pair availability, smoothing, and aggregation can distort the estimated dependence or change the observed quantity. Preserve raw timing, device versions, treatment changes, and uncertainty in the latent clock rather than treating a resampled grid as the original process.

Covariance also offers a cross-channel test. If the same route state is claimed to shape report, physiology, action, and endpoint hazard, the model should specify whether their innovations are independent, shared, or hierarchically coupled. A model that predicts each channel with an unrelated latent score has not established a common route; it has multiplied flexible explanations.

Strong rivals receive the same covariance target. A bridge model may reproduce endpoint contraction but imply a characteristic dependence on the fixed terminal boundary. A mean-reverting model may produce stationary or state-dependent correlation without terminal singularity. A frailty mixture may create long-range dependence through unobserved heterogeneity. The comparison must fit all of them through the same observation process and scoring rule.

Acceptance is conjunctive. The estimated surface must be recoverable in simulation, stable across plausible windows, predictive in held-out segments, and more compatible with the declared route than with serious rivals. A failed required recovery gate blocks the component; final verdict assignment follows §16.7.

In practice, the covariance surface should be estimated with uncertainty and tested out of sample. One route may predict long-memory residual dependence, another a triangular bridge kernel, and another near-Markov decay. The study should reserve held-out participant pairs or held-out time blocks for testing those shapes. Visual similarity is not enough. Proper scoring rules can compare predictive distributions for future pairs, while simulation

recovery checks whether the chosen design can distinguish the candidate kernels at the observed cadence and noise level (Gneiting & Raftery, 2007; Talts et al., 2018).

16.4.2 — Normalized Dependence Under Deterministic Reserve

Normalization can reveal whether shrinking scale and temporal dependence are governed by separate components. If reserve multiplies an arithmetic innovation process, dividing by a correctly specified reserve can expose a reference process whose dependence is easier to characterize. That move is legitimate only when reserve is independently estimated and the division does not amplify noise near zero (Andersen et al., 1993; Rizopoulos, 2012).

A deterministic-reserve approximation is a local model, not an assertion that human capacity follows a known clock. The approximation may be useful in controlled tasks or narrow clinical windows where reserve changes smoothly enough to estimate. Outside that context, stochastic reserve, treatment response, and abrupt state transitions should be represented rather than absorbed into residual error (Särkkä, 2013).

The normalized process should be tested for conditional mean, variance, autocorrelation, and tail behavior. A route that predicts driftless normalized innovations must say which filtration defines driftlessness and which observations are allowed into that filtration. If future endpoint information is required to make the residuals look martingale-like, the normalization has failed the prospective test.

Near-zero reserve is a numerical danger. Division can turn small calibration errors into large normalized excursions, making the process appear explosive even when the original observations are quiet. The protocol should define a reserve floor, a partial-identification region, and a rule for withholding normalized diagnostics when uncertainty exceeds the signal.

Normalization also creates a falsifier. If a single frozen transformation fails to stabilize scale across participants, sites, or windows, the deterministic-reserve route may be too rigid. Re-estimating a different reserve map for every cohort can improve fit while destroying transportability. The stronger test is whether one family of transformations survives held-out application.

If normalization removes the predicted scale while preserving the protocol-fixed dependence structure, it supports the specified route-level decomposition. The resulting coordinates allow more direct comparison of candidate and rival dynamics.

16.4.3 — Brownian-Bridge and Doob-Transform Rivals

Conditioned processes are among the strongest rivals because they can approach a boundary without positing an endogenous corrective controller. A Brownian bridge is the simplest example: conditioning a diffusion on a terminal value changes its drift so that paths are drawn toward the endpoint. That construction can imitate apparent closure while leaving a distinct covariance and singular drift structure (Doob, 1957; Karatzas & Shreve, 1991). Causal-process and predictive-scoring requirements enter the later comparison through Røysland (2011) and Gneiting and Raftery (2007).

A Doob transform generalizes the point. Conditioning on future events can produce lawful-looking pre-endpoint behavior even when the native process has no such drift. The scientific question is therefore not whether a fitted path looks guided. It is whether prospective data can distinguish native dynamics, state-coupled termination, selection, and post hoc conditioning.

The comparison must declare what is conditioned. A fixed terminal time, a random endpoint, a first-hitting event, an absorbing set, or a survival filter produces different laws. Treating all of them as one bridge rival makes the rival weak and the candidate artificially strong. Each branch should be simulated and scored under the same measurement and missingness model.

Covariance is especially informative because a fixed-time bridge has a known dependence on both observation times and the terminal time. A reserve-coupled route may instead factor scale through reserve and innovation time. Mixtures can blur the contrast, so the design should include heterogeneous endpoints and parameter distributions rather than comparing only idealized single-parameter examples.

Intervention response offers another distinction. Conditioning alone does not specify how a safe perturbation to support, information, or action availability should propagate. A causal route may predict a state update and changed future dependence. A purely conditioned bridge may fit the passive trajectory yet fail to transport under perturbation.

Rival victory is scientifically valuable. If bridge or selection models explain the held-out dependence with fewer assumptions

and equal or better calibration, the route claim narrows. The project should report that result plainly rather than preserving the favored model through metaphor.

16.4.4 — Martingale and Quadratic-Variation Diagnostics

Martingale diagnostics test whether conditional increments retain predictable drift after the declared state has been accounted for. They are attractive because they translate a verbal claim about innovation into a sequence of forecast errors. They are also easy to misuse: a transformed process may look driftless because the transformation was fitted on the same data or because endpoint information entered the filtration (Doob, 1957; Revuz & Yor, 1999).

Martingale diagnostics require a declared filtration, increment interval, predictable instruments, and sequential monitoring rule. Residual products can reveal unmodeled dependence, while e-processes or other time-uniform methods can control repeated looks without pretending that optional stopping is free. The evidential object is calibration across states and time scales, not one favorable p-value (Røysland, 2011; Ramdas et al., 2022).

Quadratic variation addresses innovation scale. In continuous-time models it can distinguish diffusion-like accumulation from smoother latent trajectories or jump-dominated dynamics. In sampled data, realized variation is contaminated by measurement error and irregular timing, so estimators must be matched to the sampling design and validated through simulation.

Jumps deserve explicit treatment. Medication changes, acute symptoms, device resets, awakenings, and care transitions can

produce discontinuities that a diffusion model will misread as extreme innovations. A jump-sensitive route may be scientifically appropriate, but it must specify whether jumps belong to the phenomenal process, the observation process, or the context process.

Martingale failure is route-specific. Predictable residual drift may indicate a missing state, a misspecified reserve proxy, an incorrect observation model, or a rival generator. The result should trigger diagnostic branching rather than a broad claim that the Law has failed or survived.

Qualification begins with known generators. The pipeline detects drift when present, controls false drift under calibrated martingales, and distinguishes jumps from noise. Passing those synthetic tests establishes which diagnostics can be interpreted in the empirical study.

A drift diagnostic and a covariance diagnostic answer different questions. A process can have approximately zero conditional mean increments while carrying route-specific conditional variance, and it can exhibit an apparent restoring drift because the sample is conditioned on a future boundary. The analysis should therefore test residual martingale properties after fitting each rival, then examine predictable quadratic variation, serial dependence, and innovation calibration. A failure in any one layer identifies where the model misses the process rather than licensing a vague verdict that the fit was poor.

Measurement noise complicates quadratic variation because high-frequency error can masquerade as process volatility.

Replicate channels, device validation, and cadence perturbations can help separate the two. If estimated variation rises without bound as sampling becomes finer, the model may be measuring sensor noise rather than phenomenal dynamics. If it collapses after smoothing, the smoothing rule may have erased real short-scale structure. Resolution invariance is therefore a diagnostic target, not an assumption.

16.4.5 — Survivor Conditioning and Pairwise Bias

Late-window covariance is observed among people who remain alive, enrolled, capable, and measurable at both times. That subset can differ sharply from the original cohort. Pairwise-complete covariance therefore mixes the target dynamics with survival, withdrawal, transfer, device tolerance, and clinical change (Andersen et al., 1993; Rizopoulos, 2012).

Conditioning on survival can induce associations even when none existed in the source population. Frailty variables that influence both measurement availability and affective trajectories create collider structures. The analysis should represent the risk set at each pair of times and state which population the covariance is intended to describe (Andersen et al., 1993; Hernán & Robins, 2020).

Weighting may recover a marginal target under strong assumptions about observation and survival. Joint models may instead connect longitudinal outcomes and event processes through shared random effects or latent states. Principal-stratum estimands can describe survivors under specified counterfactual

conditions. None of these methods is assumption-free, and their targets are not interchangeable.

Terminal observation shutdown is especially consequential. If distress, sedation, or loss of communication removes high-variance observations, the remaining data may appear to compress. Conversely, selective retention of complex cases may inflate late variance. Sensitivity analyses should include both directions rather than treating dropout as a one-sided nuisance.

Pairwise bias also affects cross-channel dependence. A physiological sensor may continue after self-report stops, while proxy reports enter only after capacity declines. Combining those channels without marking the transition can create an artificial covariance pattern. Observation marks and channel-specific risk sets belong in the model.

Partial identification or No adequate estimate may be appropriate when assumptions are weak. A result for a selected survivor population names that population and the conditions under which it was observed.

16.4.6 — Held-Out Dependence Recovery

A dependence signature should be judged by recovery, not by visual resemblance. Simulation must show that the proposed covariance, innovation, or martingale diagnostic can be recovered at the observed cadence, noise level, missingness pattern, and endpoint uncertainty, and that the same procedure does not routinely find the signature in rival-generated data.

Held-out tests should challenge the route at the level of the intended claim. Splits by participant, site, time block, endpoint

window, cohort, and device version reveal different dependencies, while random observation-level splits can leak adjacent measurements and shared preprocessing into both training and test sets. The easiest split should not determine the public verdict.

Joint prediction should be scored as a joint object. Proper scoring, calibration, and coverage should apply to the dependence structure rather than only to marginal means, and open-set cases should be included so the procedure can abstain when none of the protocol-fixed models is adequate (Gneiting & Raftery, 2007; Yao et al., 2018).

Parameter recovery and model recovery remain separate. A parameter may be estimated precisely within the wrong model, and the correct model family may be recognizable while its parameters remain weakly identified. The report should show both questions and the regions where the design cannot separate them (Talts et al., 2018; Wilson & Collins, 2019).

Adversarial simulations should be generated by the strongest ordinary rivals, including bridges, state-space mean reversion, frailty selection, treatment effects, and observation shutdown. Rival flexibility, lags, and calibration effort should match the candidate's budget. A weak rival is not a severe test.

Covariance should be inspected as a surface over time pairs, not collapsed immediately to one correlation. Irregular sampling requires pair-availability accounting, and survivor conditioning can change which pairs exist near the endpoint. A smooth surface

estimated from a changing risk set needs sensitivity analysis before it is read as a process law.

Martingale diagnostics belong to innovations after all protocol-fixed conditioning variables are included. Residual predictability may indicate a missing state, model misspecification, treatment, selection, or observation dependence rather than a special route. Cross-channel innovations can be especially useful because a route should specify which dependencies travel across reports, behavior, and physiology.

Successful recovery permits analysis of the real dependence pattern through the same frozen pipeline. Calibration, rival separation, and transport then determine how well the candidate performs.

16.5 — Terminal-Window and Intervention Signatures

A terminal-window signature is not a photograph of a final emotional state. It is a prospectively defined relation among trajectories, risks, interventions, and observation processes. Windows must be fixed by independent clinical or protocol information rather than by the point at which a preferred curve begins to look persuasive.

Interventions are included only when they are independently justified, low burden, and incapable of withholding care. The scientific objective is to perturb information, support, timing, or action availability in a way that distinguishes route predictions from ordinary learning and context effects. The protocol never induces distress to test a balancing story (World Medical Association, 2024).

16.5.1 — Prospectively Defined Terminal Windows

A terminal window is a protocol-fixed interval defined before the outcome trajectory is inspected. Its boundary must come from an independently governed protocol endpoint, a prospectively specified state transition, or another admissible event rule. The window is not the span in which a preferred curve happens to look most convincing, and it is not a photograph of a final feeling (Aalen et al., 2008).

Several clocks remain separate in the protocol: calendar time, time since enrollment, time to a protocol endpoint, disease or care-state time, and filtered subjective horizon. Each alignment defines a different estimand. A pattern that appears only after retrospective selection of one clock is an alignment finding, not a

terminal-window signature (Rizopoulos, 2012; Kurland et al., 2009).

Window boundaries use endpoint evidence independent of the candidate ledger or route estimator. The analysis distinguishes biological death, discharge, device removal, last report, treatment withdrawal, and protocol visits from the separately adjudicated stream endpoint. Uncertain endpoint status is propagated into the analysis (Fine & Gray, 1999).

Terminal-window analyses inherit the information-set rule from §13.5: filtering may support prospective claims; smoothing is retrospective and must be labeled as such. A prespecified smoothed reconstruction may be confirmatory for a retrospective estimand, not advertised as foresight.

Pseudo-endpoints and matched nonterminal cut points are candidate negative controls when the frozen route predicts no target signature there. The signature should weaken, disappear, or change in the specified way when trajectories are aligned to those events. Unexpected endpoints instead require a prospectively specified test of dependence on horizon awareness, with cause, acuity, care, and observation conditions addressed; unexpectedness alone does not make a genuine endpoint a negative control. If the same pattern appears around arbitrary salient events, the terminal interpretation loses specificity (Lipsitch et al., 2010; Hernán & Robins, 2020).

Observation intensity often changes as a window narrows. Clinical attention may increase, reports may decline, devices may be removed, sedation may alter channels, and family or staff

priorities may shift. The observation model must therefore estimate whether apparent compression, drift, or covariance change survives the changing opportunity to observe rather than treating the last visible measurements as representative of the unobserved remainder.

Prospective hospice research also documents emotionally meaningful experiences near life's end. Kerr et al. (2014) conducted 453 interviews with 59 patients; 52 reported at least one end-of-life dream or vision. Experiences featuring deceased people or animals were more comforting than other reported contents, and comforting experiences involving the deceased became more prevalent as death approached. These observations make internally generated, relationally meaningful experience a legitimate target for careful study. A participant-led observation module can record timing, comfort, sleep or wakefulness, medication, and report availability alongside usual care. This produces a richer account of late-window experience and the conditions associated with comfort.

Each window-specific estimand should state its reference class, clock, endpoint version, observation rule, treatment context, and uncertainty. Results from sudden endpoints, diffuse decline, reversible interruption, protocol censoring, and securely adjudicated terminal windows belong to different branches unless a declared bridge justifies combination. Pooling them for power can erase the very boundary distinction the study is meant to test.

A window chosen by inspecting the favored trajectory makes the evidence Unusable for the confirmatory claim. If no outcome-independent window can be defined, the target receives No

adequate estimate. The study may still report descriptive patterns, endpoint feasibility, observation loss, or rival-model performance. It may not turn an unstable boundary into evidence by choosing the interval after the result is known.

16.5.2 — Direct Repair and Support Perturbations

Direct repair is an action or support change that plausibly reduces a prospectively named unresolved state. In a bounded task, it may complete an unfinished goal; in a clinical or life-course study, it may address a reported symptom, communication barrier, practical need, conflict, or inaccessible option. The study must define the target before analysis so that almost any favorable action cannot be relabeled as repair afterward (Hernán & Robins, 2020).

A repair registry should state the burden element, the admissible action class, the expected time scale, the observable consequence, and the residual that may remain. Attempt and success are different variables. A failed repair attempt may reveal a closed menu, insufficient support, mistimed intervention, or an incorrect route prediction rather than absence of concern.

Support perturbations change information, access, safety, cost, reversibility, communication, or available time without inducing distress or withholding care. Examples include an accessible communication device, transportation, a trusted intermediary, schedule flexibility, symptom support, or a low-burden prompt that makes an existing option usable. Each perturbation must be independently justified and acceptable even if the Law-specific analysis were removed (World Medical Association, 2024).

Measurement should distinguish the offered support, actual uptake, action timing, task completion, immediate relief, later reactivation, and any burden shifted to another person or setting. A support that improves task completion but increases fatigue, family load, privacy loss, or clinical disruption cannot be summarized by one favorable outcome.

Feasibility is part of the causal object. Nonrepair cannot be interpreted as preference when access, speech, energy, safety, money, assistance, or time is missing. The model should represent which routes were live, which were blocked, and which became available only after support rather than hiding environmental constraints inside a personal reserve score.

The first human study should favor low-burden perturbations whose mechanism and stopping rules can be rehearsed in simulation. Any design that gains discrimination by delaying care, withholding rescue, provoking distress, or making support contingent on participation fails before enrollment. Lost data under humane stopping remain a limit on inference, not a reason to weaken the stop rule.

Serious rivals include ordinary goal completion, treatment response, social expectation, deadline effects, cognitive control, family pressure, and improved access. The route-specific account earns weight only when the protocol-fixed relation among filtered horizon, independently estimated reserve, unresolved state, support, and residual burden outperforms those alternatives under the same observation and scoring rules.

Direct repair supplies a concrete test of the candidate. The model specifies when change should occur and is compared with ordinary access and treatment mechanisms. Outcomes outside its prespecified relation lead to revision, narrowing, or retirement of the affected component.

16.5.3 — Substitution, Reframing, and Route Reordering

Substitution occurs when an unavailable or costly route is replaced by another action that changes a related state. Reframing changes the meaning, priority, or interpretation of a burden without necessarily changing the external event. Route reordering changes which live option is attempted first. These are distinct processes and should not be pooled merely because each can precede lower reported burden (Gerstorf et al., 2010; Vogel et al., 2013).

Substitution is not automatically lesser than direct repair. It can be adaptive, creative, evasive, socially supported, or temporarily protective. The analysis should ask what changed, including affective load, action cost, safety, future reactivation, physiological burden, and reported meaning, rather than assigning a moral rank to the route.

Reframing is difficult to identify because it may occur without visible action and may be reconstructed in later report. Experience sampling, language change, choice patterns, confidence, clinician observation, and delayed narrative can provide partial evidence, but none is transparent access to an inner transformation. Immediate and retrospective measures should remain separate.

Ordinary rivals include cognitive reappraisal, acceptance, therapy, religious coping, memory reconsolidation, social persuasion, and changing goals across a shortening future-time perspective. These mechanisms may explain the result better. A Law-specific route cannot absorb them by renaming every successful reinterpretation as completion pressure (Carstensen, 2006).

The protocol should predefine plausible substitution classes and the outcomes that would count against each one. An alternative route may repair the same target, partially repair a coupled target, provide only immediate relief, or merely distract. A classification that treats every alternative action as hidden repair has no adverse case and should not enter confirmatory analysis.

Route reordering requires a live-menu record. The study should know which actions were available, when they became feasible, what they cost, and whether support changed their order. A later choice cannot reveal that an earlier route was psychologically impossible, ethically inadmissible, or simply unavailable unless those constraints were measured independently.

Self-chosen reframing must be distinguished from pressured reinterpretation. A person may voluntarily find meaning, acceptance, or a new goal, while an institution may also demand positive meaning because actual repair is expensive or inconvenient. The former can reduce burden; the latter can transfer burden into silence. Consent, refusal, and context belong in the estimand.

The candidate route fails when substitution and reordering show no protocol-fixed relation to horizon, reserve, unresolved state, or menu change, or when ordinary rivals predict them equally well with fewer assumptions. The study may retain useful knowledge about adaptation and support while withdrawing the stronger route interpretation.

16.5.4 — Avoidance, Blocking, and Automaticity

Avoidance may protect a person from overload, danger, shame, trauma activation, coercion, or wasted effort. Blocking may keep an unresolved state outside the current control field, and automaticity may select a familiar route without extended deliberation. These action classes are not defects, and silence alone identifies none of them.

A shorter horizon does not always make repair urgent. It can make repair too slow, risky, costly, or emotionally expensive to attempt. Short-term avoidance may preserve capacity for a later safer route, while persistent avoidance may increase future reactivation. The model must represent time scale and feasibility before interpreting the same behavior as protection, habit, or unresolved burden.

Avoidance can be studied through skipped or delayed actions, task disengagement, topic shifts, physiological arousal without approach, later reactivation, and explicit refusal when available. These channels are not interchangeable. A route may be behaviorally avoided while remaining mentally active, or it may disappear from report because the observation itself became burdensome.

A bounded task can offer direct repair, substitution, neutral deferral, and avoidance options while varying horizon, information, cost, and support. Safety annotations should identify realistic danger, social exposure, sensory load, and clinical destabilization. When avoidance is the safer or more dignified option, the protocol must not treat lower approach as adverse evidence about the person.

Habit and automaticity are serious rivals. Familiar actions can dominate when deliberation is expensive, cues are strong, or reinforcement history is deep. Baseline habit strength, reaction time, perseveration, sensitivity to changed payoffs, and updating after route closure help distinguish automatic control from a newly reorganized priority structure.

Apply the modal distinctions defined in §§5.6.2 and 7.3.4. While positive mass remains on nonclosing continuations, stronger guidance or a more concentrated choice distribution does not establish fixed-policy probability-one completion.

The route narrows when ordinary urgency, habit, expected value of control, scarcity, fatigue, fear, or privacy explains the action pattern under equal rules. It also narrows when the feasible set cannot be identified without outcome-dependent reconstruction. A useful behavioral finding may remain, but the completion-specific interpretation should lose status.

16.5.5 — Social and Environmental Interventions

Menus are not only internal. Social and environmental support can open actions that would otherwise be closed. A communication device, a caregiver, transportation, accessible

design, medication support, or a trusted listener can change the live menu (Gerstorf et al., 2010; Vogel et al., 2013).

Late-life data also identify potentially helpful social conditions. In 2,910 deceased participants from the German Socio-Economic Panel followed for up to 27 years, social participation and valuing social goals were associated with higher late-life well-being, less pronounced terminal decline, and a later onset of decline after measured covariates were considered (Gerstorf et al., 2016). These associations provide a constructive complement to the accelerated terminal decline documented across national panels (Gerstorf et al., 2010). For this program, the useful target is whether changes in accessible support predict subsequent experience and action beyond health, care, and selection. Social participation is therefore a measurable correlate of the trajectory and a concrete focus for studying late-life connection.

Reserve is therefore not purely biological. The same bodily state can support very different action sets depending on environment. Strong support may preserve effective capacity despite low physical reserve, while isolation or poverty may sharply reduce the usable menu despite biological strength.

Support can therefore create or erase apparent queue pressure. An unresolved burden may remain invisible until someone makes a repair route possible. Conversely, social pressure can force a burden into view before the person is ready.

Social and environmental support should enter as a modifier of the available action menu. Communication access, assistance, relational safety, financial constraints, physical barriers, care

quality, and institutional rules change what actions are possible and costly. Treating those conditions as stable personal traits would mislocate the causal locus and conceal unequal option sets.

Rivals are especially strong here. Many horizon-like effects may be effects of social permission, resource availability, or family scripts. A person may act because support arrived, not because a latent queue intensified.

When environmental conditions restrict available options, the model represents those constraints explicitly. This helps distinguish social barriers from physiological capacity and identifies where changes in support could expand the available paths.

Code support as a menu intervention, not merely a covariate. It can reduce cost, increase safety, extend time, provide information, or absorb action burden. When support changes the feasible set, attribute that change to environment and person.

Formalize support as changes in cost, risk, information, or menu size, avoiding sentimental language. Rather than say support helps, specify reduced travel cost, increased route reliability, extended response time, lower risk, or a delegate action. This makes support measurable without making it mechanical.

Within a frozen route model, interventions that change time, capacity, information, cost, reversibility, or menu size may be summarized by a marginal-value diagnostic. The estimand is the local change in a declared model output when one constraint is relaxed; it is not a price on suffering, a measure of human worth, or a reason anyone should choose the model-preferred action.

Marginal-value diagnostics should be tested in simulation, reported with uncertainty, and compared with ordinary control and decision models before any human interpretation. Reader-facing copy should name the quantity that changed rather than rely on shadow-price language. If the diagnostic is unstable in simulation or adds nothing beyond established rivals, it should not enter a vulnerable-population study.

16.5.6 — Branch-Specific Failure and Residual Burden

Residual burden may persist despite action, support, reframing, substitution, or a nearby protocol endpoint. It should remain a potentially nonzero state rather than being forced toward closure, and persistence is never a defect in the person.

In bounded tasks it may appear through remaining penalty, report, conflict, reactivation, latency, or physiological load. These channels are not interchangeable; vector or set-valued representation is retained when scalarization would hide disagreement, and the measurement rule is fixed before intervention outcomes.

Persistence can falsify a route when it is independent of horizon, reserve, queue state, or the proposed control variable, or when an ordinary rival predicts it better. Operational completion such as discharge, compensation, or case closure likewise remains separate from experienced resolution.

Terminal-window analyses use fixed prospective distances, unexpected endpoints, and pseudo-endpoints. Safe intervention studies require positivity and cannot induce distress or withdraw

care; timing, treatment history, and observation marks address confounding by indication (Hernán & Robins, 2020; Pearl, 2009).

The strongest route result is conjunctive: ordered windows, survival against pseudo-endpoints, support-aware intervention response, stable calibration, adequate observation, and superiority to common-rule rivals. Failure updates the named branch and leaves the broader target unresolved absent a valid entailment.

16.6 — Compression Detectability and Observation Floors

Predicted compression is scientifically relevant only if the planned instruments and observation schedule can detect it. A process may contract below the scale of measurement, while an apparently sharp contraction may be created by sensor floors, smoothing, selective dropout, or the replacement of self-report by proxy channels. Detectability is therefore part of the estimand rather than an afterthought (Carroll et al., 2006; Little & Rubin, 2019).

The section distinguishes a scientific equivalence band from an observation floor. The former is a declared region of empirical compatibility for a measured target. The latter is the smallest pattern that the pipeline can resolve under noise, missingness, and calibration uncertainty. When the floor is wider than the band, a valid but undecided analysis yields Inconclusive evidence; No adequate estimate applies if the target cannot be identified or bounded.

16.6.1 — Additive Noise and Variance Floors

Additive measurement error contributes variance even when the latent process contracts. If the error variance remains approximately constant while the latent variance approaches zero, the observed curve flattens at the noise floor. A naive log-slope fit then underestimates the contraction exponent and can make distinct routes appear identical (Carroll et al., 2006; Little & Rubin, 2019).

The noise model should be estimated from replicate observations, calibration sessions, device comparisons, or validated measurement studies rather than inferred from the same late-window curve it is meant to correct. State-dependent error is especially important because fatigue, sedation, movement, and communication difficulty can change reliability near the endpoint (Carroll et al., 2006).

Correlated error creates a different problem. Shared preprocessing, common interviewer effects, device smoothing, and proxy-report substitution can generate apparent temporal dependence. The protocol should preserve channel provenance and estimate cross-channel error rather than treating multiple measures as independent confirmations.

A variance floor also interacts with participant heterogeneity. Some participants may remain measurable at high precision while others become effectively unobserved. Pooling them can produce a floor that is neither a property of the instrument nor the latent process. Hierarchical error models and participant-specific calibration are therefore preferable to one global correction.

Simulation should show how the proposed estimator behaves when the true exponent varies, when the floor is misestimated, and when the floor depends on state. The study should report the region in which exponent recovery is trustworthy and the region in which only bounds are available.

The contraction claim is unadjudicable when calibration uncertainty, structural error nonidentification, or inadequate statistical precision prevents the declared contrast from being

resolved. A latent variance smaller than the measurement-error variance is not, by itself, proof of nonidentification: an independently calibrated error model and sufficient suitable observations may still identify it. Report whether the limitation is structural or a finite-sample precision problem before deciding which design change could help.

Noise floors create asymmetric scientific risk. They can conceal real compression, but they can also manufacture a plateau that is later misdescribed as a finite neutrality band. Increasing sample size does not remove a systematic floor in the instrument. The study must estimate the floor from validation data, replicate channels, or controlled null periods and propagate its uncertainty into the boundary estimate. Errors-in-variables analysis is especially important because noise in both the residual and the reserve proxy can bias estimated exponents downward and distort their interaction (Carroll et al., 2006).

A useful report should include a detectability surface rather than a single power number. The surface varies the true exponent, baseline dispersion, cadence, dropout mechanism, device error, and endpoint uncertainty, then shows where the planned analysis recovers the correct branch and where it cannot. Regions of nonidentification should be visible before data collection. That honesty is more informative than a nominally powered design whose power applies only to an unrealistically clean process.

16.6.2 — Informative Observation Shutdown

Observation often ends for reasons related to the state being studied. A participant may stop responding because of fatigue,

distress, sedation, transfer, device intolerance, clinical deterioration, or a deliberate choice to protect privacy. Treating those endings as random can manufacture a smooth terminal pattern (Rizopoulos, 2012; Rubin, 1976).

Shutdown should be represented as a marked event with a time, reason, channel, and degree of certainty. The mark may indicate participant refusal, clinician stop, device removal, loss of capacity, technical failure, death, or administrative end. Each mark implies a different missingness mechanism and ethical meaning.

Joint longitudinal-event models can connect the outcome trajectory and shutdown hazard, while sensitivity analyses can vary the strength and direction of unobserved dependence. Pattern-mixture and selection models provide alternative parameterizations. No single model should be treated as a magical recovery device (Rizopoulos, 2012; Little & Rubin, 2019).

Proxy substitution requires a new observation model rather than a seamless continuation. Family and staff reports may capture behavior, distress, or function, but they are not direct replacements for first-person valence. The transition point and channel identity must remain visible in every analysis.

Shutdown can itself be informative without becoming evidence for PTVC. A rising probability of device removal may reveal burden, clinical change, or observation incompatibility. It may improve study design and care while simultaneously making the terminal signature unidentifiable.

The humane stop rule has priority. Investigators may not keep a device in place or continue prompting merely to reduce missingness. The study architecture must be capable of learning from refusal and shutdown, including the conclusion that the target cannot be estimated safely.

16.6.3 — End-to-End Detectability Simulation

Detectability should be evaluated through the entire pipeline, not through an isolated formula for statistical power. The simulation begins with a candidate or rival generator, then adds sampling cadence, channel-specific measurement error, treatment changes, endpoint uncertainty, observation marks, censoring, preprocessing, smoothing, estimation, and the final decision rule (Lakens, 2017; Talts et al., 2018).

Each component should be varied across a plausible design space. The study should include optimistic, central, and adverse scenarios for noise, missingness, event rate, reserve uncertainty, and endpoint spread. A design that works only under its most favorable assumptions is not ready for enrollment.

Simulation-based calibration tests whether the inferential procedure is calibrated when the model is correct. Model-recovery simulations test whether the procedure distinguishes candidate and rival generators. Both are needed because a computationally calibrated estimator can still be structurally non-discriminating (Talts et al., 2018; Wilson & Collins, 2019).

Open-set generators should include mixtures, abrupt jumps, nonlinear observation floors, site-specific protocols, and unmodeled treatment effects. The pipeline should be rewarded for

abstaining when none of the declared models is adequate. Forced classification is a public-risk failure because it converts ignorance into status.

Detectability maps are more useful than a single power number. They show which combinations of cadence, sample size, endpoint density, and signal scale yield adequate recovery. They can guide a staged program from low-burden feasibility to more demanding confirmatory work without exposing participants to a design that cannot learn.

The simulation archive should be frozen and versioned before real-data adjudication. Code changes, parameter-range changes, and bug fixes must be logged. A favorable real-data result cannot be used to redefine the region in which the pipeline was supposedly detectable.

Simulation scenarios should be organized as a factorial stress test rather than a single favorable world. The factors can include effect size, reserve-time relation, branch mixture, cadence, measurement floor, missing-not-at-random strength, endpoint error, and site heterogeneity. The output is a map of recovery, false route preference, no-adequate-estimate frequency, and reference-class narrowing. This map becomes part of the protocol's decision architecture, not an after-the-fact appendix.

The most informative simulations include near-equivalent models. If two routes are intentionally separated by only the amount the planned design claims it can detect, the recovery study shows whether that distinction is real. Failure to separate near neighbors is not a reason to hide them. It is a reason to narrow the

claim, collect more informative data, or choose a different first test.

16.6.4 — Deconvolution and Replicate-Channel Limits

Deconvolution attempts to separate latent variation from measurement error, but it depends on information about the error distribution. Without replicate channels, validation data, or structural assumptions, the decomposition may be unidentified. A smooth corrected curve can therefore reflect the analyst's prior more than the participant's trajectory (Carroll et al., 2006; Little & Rubin, 2019).

Replicate measurements help only when their errors are sufficiently independent or their dependence is modeled. Two instruments that share the same preprocessing, respondent, interviewer, or physiological artifact do not provide two independent views. The protocol should describe the causal pathways that create shared error.

Repeated reports can estimate short-term reliability, but repeated prompting may change the experience or the reporting strategy. The reliability study must therefore balance information against reactivity and burden. Calibration obtained in healthy participants may not transport to sedation, delirium, severe fatigue, or terminal illness.

Errors-in-variables methods, latent-variable models, and validation subsamples can narrow the problem. Partial-identification bounds may be more honest when the error law is uncertain. The choice among them should follow the scientific

question rather than the desire for a single corrected trajectory (Carroll et al., 2006).

Deconvolution becomes unstable near the boundary because small differences between observed and error variance can determine the entire latent estimate. Negative corrected variances, wide intervals, or sensitivity to minor calibration changes are signals to abstain rather than reasons to truncate estimates at zero.

Deconvolution results should be reported with the error model that makes them possible. Replicate channels can help identify noise structure, but they do not transform a latent reconstruction into direct observation. The conclusion should distinguish properties of the measured process from properties that remain conditional on channel independence, error stationarity, and the chosen observation kernel.

16.6.5 — Equivalence Bands Versus Instrument Resolution

An empirical equivalence band is a decision region for one measured estimand; instrument resolution is the smallest contrast the validated pipeline can distinguish. They are independent requirements. An equivalence conclusion requires a prospectively justified band, a valid analysis, and uncertainty sufficiently narrow to exclude both outside regions. Power is a design property; low power can leave a valid test inconclusive.

The margin is scientifically and ethically justified before outcome inspection. Its basis may be a validated measurement standard, clinically meaningful difference, or decision-theoretic loss. The declared decision determines the interpretation of practical equivalence.

Resolution should include more than nominal scale increments. Effective resolution depends on reliability, calibration drift, state-dependent error, missingness, and preprocessing. A one-point scale may have far worse effective resolution than its printed categories imply.

Equivalence analysis should include the two one-sided testing logic or another prospectively specified method appropriate to the estimand, together with sensitivity across plausible margins, calibration error, and missingness models. Nonsignificance is not equivalence, and equivalence under a wide margin does not resolve a narrower target. The report should show where the conclusion changes.

Finite-band compatibility supplies a measured result for route comparison, clinical description, or instrument development. The report specifies the target, band, and decision that the comparison addresses.

The public summary should name both numbers: the declared equivalence band and the validated observation floor. If the floor dominates, the correct conclusion is that the study could not resolve the target at the required scale.

Adjudication requires uncertainty in the declared estimand to be sufficiently narrow relative to the scientifically justified equivalence band. The error of one observation is not the same as uncertainty in an estimated mean, variance, or other model parameter. Additional suitable observations may improve precision under an identified error model; they cannot eliminate an unmodeled systematic bias or identify an unrestricted error

decomposition. Report the outcome through the verdict logic in §16.7 rather than treating an unresolved instrument floor as evidence for or against the route.

Sensitivity analysis should cross the equivalence band with plausible calibration errors and missingness scenarios. A conclusion that changes sign under a small, credible shift in the neutral origin is not robust. A conclusion that survives only after excluding difficult-to-observe windows is not a terminal finding. The final report should present a grid of verdicts across declared perturbations, making clear which conclusions are stable, which are model-dependent, and which remain unestimable.

16.6.6 — Abstention Thresholds

Apply the canonical evidence-state definitions in §24.2 and Table 27.5. This subsection specifies the terminal-window thresholds that trigger No adequate estimate, Unusable evidence, or Inconclusive evidence under those definitions (Lakens, 2017; Talts et al., 2018).

Triggers may include failed measurement qualification, endpoint leakage, unrecoverable model confusion, observation-floor dominance, excessive informative missingness, unstable calibration, insufficient event support, or a burden threshold that stops data collection. Each trigger should map to a specific component status.

The threshold is not the same as a null result. An informative near-zero result requires a qualified design and uncertainty narrow enough to support that conclusion. No adequate estimate says the study did not earn the right to make that comparison. Keeping

those categories separate prevents absence of evidence from becoming evidence of hidden closure.

Partial identification may sometimes replace total abstention. If assumptions support bounds that exclude some routes but not others, the report can narrow the model set without selecting a winner. The bounds and assumptions should be visible in the main result rather than buried in an appendix.

Public communication must resist pressure to fill the vacuum. A no-estimate result cannot be described as suggestive, mysterious, or consistent with an invisible exact law unless the same language is applied to every unidentifiable model. The protocol-fixed verdict should state the blocked gate and the next allowed action.

A program that can return No adequate estimate is stronger than one that always tells a story. It preserves the distinction between the world's complexity and the method's reach, and it protects participants from being turned into evidence by interpretive force.

16.7 — Common-Rule Rivals, Falsifiers, and Route Verdict

The final section applies one observation system, one preprocessing rule, and one scoring constitution to the candidate route and its strongest rivals. The same origin, endpoint, missingness model, reference class, and held-out partitions must govern every model.

Verdicts attach to components. Failure of a reserve exponent can retire the reserve route while leaving other PTVC routes open; failure of endpoint independence can move the program to a state-coupled branch; failure of representation or measured-to-latent transport can block the entire downstream analysis. The public record must preserve that dependency structure (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

16.7.1 — Common Observation Rules

A common-rule comparison begins before model fitting. The candidate route and every rival must receive the same observed channels, measurement versions, preprocessing rules, endpoint information, missingness treatment, and held-out partitions. A flexible candidate compared with rivals denied equivalent data is not a discriminating test (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Representation and calibration are shared inputs rather than model-specific conveniences. The sign convention, neutral origin, unit, channel qualification, exposure clock, and uncertainty propagation must remain fixed across the tournament.

A model may not improve its fit by changing the construct it claims to explain.

Endpoint information is governed identically. If the candidate receives a filtered horizon while a rival receives only crude time to death, the comparison confounds mechanism with information. Prospective analyses use information available at the prediction time; retrospective smoothing is evaluated separately and cannot be fed back into the confirmatory state.

Observation and missingness rules are also common. Refusal, device failure, sedation, staff interruption, observation shutdown, and terminal truncation must enter each model through the same marked record or declared sensitivity family. A rival may not be penalized for missingness assumptions that the preferred route is allowed to hide.

Training, tuning, and complexity budgets should be auditable. Comparable cross-validation folds, prior or penalty budgets, stopping rules, hyperparameter searches, simulation scenarios, and calibration checks prevent a bespoke candidate from defeating deliberately rigid alternatives. Model recovery should be demonstrated before real-data ranking (Talts et al., 2018; Vehtari et al., 2017).

Candidate state variables such as queue, pressure, reserve, horizon, and admissible menu need provenance. The record should show whether each variable came from task state, report, clinical context, passive sensing, or a model-derived estimate, together with its time, uncertainty, and consent boundary.

Combining sources without provenance can make one model appear to possess information no rival was permitted to use.

Common rules support privacy minimization. When the comparison needs only activation status, feasibility, or timing, the study should not collect full narratives, dream content, family conflict, or unrestricted records. A variable definition that is too vague to limit collection is also too vague to support a fair model comparison.

The common-rule gate fails when any material difference in data access, endpoint leakage, preprocessing, tuning opportunity, or observation handling could explain the apparent winner. The result then returns design-inadequate or Unusable evidence for the affected comparison rather than route-favored. The tournament is credible only when the rules would remain acceptable if the rival won.

16.7.2 — Serious Rival Matrix

A serious rival matrix gives each alternative a mechanism, a domain in which it should fit well, a distinct prediction, and a legitimate victory route. It is not a list of names attached after the preferred model is built. The matrix is frozen before outcome inspection and updated only through a versioned, publicly visible change rule (Vehtari et al., 2017; Yao et al., 2018).

Ordinary finite-horizon and urgency models are first-line rivals. Perceived future time, scarcity, deadline pressure, fatigue, fear, and reduced planning depth can reorder behavior without any completion-specific state. These models should receive the same

horizon estimates, action menus, and contextual variables as the candidate route (Carstensen, 2006).

Regulatory rivals include mean reversion, homeostasis, allostasis, adaptation, and opponent dynamics. They can produce reduced variance, rebound, relief, and apparent return toward a baseline. The matrix should identify where their predicted time scale, channel dependence, and response to perturbation differ from the candidate conjunction.

Conditioned-path and endpoint-selection rivals can reproduce terminal concentration by construction or sampling. Brownian bridges, Doob-transformed processes, first-hitting selection, and retrospective alignment may fit a narrowing window without describing a natural completion mechanism. They must be evaluated under the same endpoint and no-look-ahead rules (Doob, 1957).

Clinical and observation rivals include frailty, medication, sedation, care intensification, survivor selection, device removal, reporting decline, and observation floors. These mechanisms can change both the latent trajectory and what remains visible. Joint longitudinal-event and marked-observation models should be allowed to win when they explain the pattern more completely (Rizopoulos, 2012; Kurland et al., 2009).

Control and learning rivals include habit, reinforcement history, expected value of control, avoidance learning, and rational privacy. Repeated or compressed action may reflect low-cost familiar policies rather than a newly activated completion route.

Baseline behavior and changed-payoff tests are therefore part of the rival specification.

The matrix should map each claimed pressure signature to its channel and time scale. Choice, latency, physiology, report, and route abandonment are different outcomes, and gradual drift, abrupt switching, oscillation, freezing, and collapse are different dynamics. A movement in one channel cannot be counted as generic confirmation of a prediction stated in another.

Equal treatment requires comparable flexibility and calibration. Rivals receive realistic observation models, relevant covariates, appropriate nonlinearities, and sufficient tuning to express their strongest known form. Absolute fit, calibration, held-out prediction, transport, and model recovery matter alongside relative score (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Open-set inadequacy remains a required outcome. The candidate may score better than the protocol-fixed rivals while every model fails calibration, transport, or absolute predictive adequacy. That result means the present model set is inadequate, not that the best member has contacted the generating process.

A rival victory is a scientific contribution. It should identify the component explained, the evidence budget used, the candidate prediction displaced, and the scope that remains open. The matrix earns trust when the public report can state rival-favored as clearly as route-favored.

16.7.3 — Component-Level Falsifiers

A component-level falsifier is a predeclared result that removes or narrows one part of the route architecture without waiting for a verdict on the entire hypothesis. Each component should have a failure condition tied to its own object, measurement bridge, and rival set. A vague result that can be reinterpreted as support for another component is not a falsifier.

The horizon component fails when the filtered state cannot be estimated without future leakage, does not update coherently, lacks transport across prospectively specified windows, or adds no discriminating information beyond ordinary future-time perspective, prognosis, urgency, and scarcity. An unstable horizon may remain descriptive, but it cannot carry the route claim.

The reserve component fails when its proxies lack a common unit or declared measurement model, collapse into disease severity or prognosis, change meaning across sites, or add no held-out value once environmental support and treatment are represented. A reserve inferred from the residual it was built to explain is circular and receives no confirmatory status.

Boundary-rate and dispersion components fail when estimated exponents, drift restrictions, variance contraction, or reserve-horizon interactions do not survive alternative clocks, calibration versions, pseudo-endpoints, and ordinary regulatory rivals. A visually narrowing curve is insufficient when the protocol-fixed cross-equation restrictions do not hold.

Two-time and martingale components fail when covariance surfaces, innovations, quadratic variation, or dependence

diagnostics cannot be recovered in simulation, reverse under held-out intervals, or are matched by conditioned bridges and state-space rivals. First-moment movement cannot rescue a failed dependence prediction.

Terminal-window and intervention components are evaluated against prespecified windows, responses to unexpected endpoints, and models of treatment, support arrival, habit, and observation change. Each conclusion states the modality supported by its frozen policy and continuation analysis.

Detectability is itself a falsifiable prerequisite. If cadence, sensor error, missingness, sedation, response limits, or observation shutdown make the protocol-fixed signal unrecoverable at ethically acceptable burden, route the result through the canonical evidence-state rules in §24.2 and Table 27.5. The program does not manufacture power by increasing burden or relaxing the target after simulation.

Admissibility and ethical constraints are noncompensatory. The feasible action set must preserve the reason an option is unavailable, including time, capacity, information, danger, consent, and environmental access. When those constraints cannot be identified, or when the discriminating contrast would require harm, the component is blocked. These component results feed the frozen route-verdict logic that follows.

16.7.4 — Route Verdict Logic

A route verdict should be generated by a frozen decision table that links each prerequisite gate to the strongest language the evidence permits. Representation validity, neutral-origin stability, stream

and endpoint definition, observation coverage, measured-to-latent recovery, rival comparison, and held-out transport are noncompensatory. The table should identify the first gate that limits the claim, not only the model with the best score.

The protocol-fixed statuses include route-favored, rival-favored, model-ambiguous, open-set inadequate, No adequate estimate, reference-class narrowed, and route-failed. Each status has a trigger, proposition ceiling, and next action. Thresholds and labels remain fixed after the data reveal which branch is most favorable.

Route-favored requires the protocol-fixed conjunction to survive common measurement, endpoint, missingness, information-budget, and scoring rules against equal-budget rivals. One attractive exponent, covariance surface, intervention response, or terminal curve cannot compensate for failed or unidentified components.

Failure at an early gate cannot be compensated by later predictive performance. A model that predicts well with an unstable origin, an endpoint-leaking state, or an unrecoverable reserve proxy may be useful for another purpose, but it cannot carry the route proposition tested here.

Route and target-law verdicts remain separate. Failure of the reserve-coupled route can retire that mechanism while leaving PTVC unresolved; survival of a route-specific conjunction can favor the route in a declared reference class without establishing exact latent universality.

Every verdict should state the population, information set, measurement and endpoint versions, rival set, observation

assumptions, uncertainty, transport domain, and unresolved alternatives. A result outside those boundaries is a new question, not an automatic extension.

The public table should show the proposition updated, the evidence that survived, the first limiting gate, the approved status sentence, and the next scientific action. This makes narrowing, correction, replication, and retirement part of the result rather than decisions made after the narrative has formed.

16.7.5 — Public Status Labels

Every public phrase in this chapter needs a status label. Horizon may be a latent state, task manipulation, clinical belief proxy, or analogy. Reserve may be a candidate capacity construct, measured proxy, or formal variable. Queue may be a model state or illustration (Vehtari et al., 2017; Yao et al., 2018).

Status labels distinguish metaphor, measurement, task manipulation, clinical association, and identified mechanism. Keeping those roles explicit lets each contribution inform the next stage of research.

The labels also protect public communication. A social excerpt saying ‘queue pressure rises near the end’ would be unsafe unless it states the claim status, population, task, and rivals. Otherwise it invites the impression of a hidden controller.

A good status sentence is short and local. It says what is being claimed, from what evidence, under what model, and what it cannot imply. Local status should be exact enough to stand on its own.

Diagrams and tables need the same discipline. A flow from horizon to action is a candidate pathway, not evidence. A simulated trajectory is a design tool, not a finding. A fitted model is not a verdict until recovery and rival comparison are complete.

The language rule is simple enough to audit: no status, no authority. If a phrase sounds powerful but cannot say whether it is analogy, model, estimator, or result, it should not be used.

Status labels belong in captions and summaries as well as body prose. A table can overclaim through its title even when its cells are careful. Terms such as observed queue pressure, predicted horizon collapse, and reserve signature are permissible only when observed, modeled, simulated, filtered, retrospective, or exploratory status is attached to the exact object.

16.7.6 — Metaphor Retirement After Failure

Metaphors should be revised when they imply an unestablished mechanism, moral price, deserved suffering, or practical control over future rewards. A declared accounting metaphor may still explain the net remainder required by exact PTVC. Retiring a failed route removes its unsupported mechanism claim; it does not erase a conditional arithmetic consequence of the target (National Academies of Sciences, Engineering, and Medicine, 2019).

A term earns continued use only if its operational meaning, observable consequences, and failure conditions can be stated. It should be retired when it explains every outcome, cannot be distinguished from an ordinary rival, repeatedly creates moral misunderstanding, or has a clearer technical replacement.

Retirement may be audience-specific. Queue can remain inside a bounded task model while disappearing from captions and public summaries; a technical symbol can remain in an appendix while reader-facing copy uses a plainer noun. The distinction is between model utility and public authority, not between interesting and dull language. The audience-specific status should be recorded in the release registry so public materials cannot revive terminology intentionally confined to technical use.

Empirical failure also retires vocabulary. If queue pressure adds nothing beyond ordinary urgency, or reserve adds nothing beyond severity and prognosis, keeping the metaphor for intuition would continue to bias model construction. A failed route should lose both its status and the language that made it feel inevitable. Retired vocabulary should remain searchable in the audit so future authors can recognize the old concept even after reader-facing language changes.

Technical terms should identify their objects: filtered horizon, reserve proxy, admissible subset, unresolved state, marginal value, rival model, and failure condition. Public balance language can remain when its conditional net meaning is clear. Neither vocabulary establishes a personal forecast, moral obligation, or chosen future reward. Where a term has both a public and technical use, the chapter should preserve a stable mapping between them so simplified language cannot drift into a different claim.

Retirement should be recorded in the scientific decision and version ledger with the reason, permitted technical uses, prohibited public uses, dependent artifacts, and any approved

successor term. That record prevents later chapters or public materials from reviving discarded language without reopening the evidence. Successor terminology should not inherit favorable evidence automatically; it must show which formal object is unchanged and which interpretation has been revised.

The chapter's final test is whether each retained term points to an observable or formal object. Clear vocabulary gives experts a common basis for evaluation. Chapter 17 carries the resulting variables into longer records, where partial history, attrition, treatment, and observation are modeled directly. This object-level discipline makes later longitudinal tests easier to compare because a changed label cannot conceal a changed estimand.

Chapter 17 — Longitudinal and Life-Course Designs

Life-course science gains its value from following change in context. Lived records include answered and missed prompts, visits, treatment changes, device transitions, sleep fragments, transfers, consent changes, and family boundaries. A credible design preserves that uneven visibility so the resulting trajectory reflects both experience-related observations and the conditions under which they were obtained.

The value of long follow-up depends on how observations are obtained. Diaries have entry dates; devices have nonwear periods and firmware changes; follow-up may end through withdrawal, transfer, clinical decline, or death. Representing these processes gives an extended record its scientific meaning (Shiffman et al., 2008; Hernán & Robins, 2020; Little & Rubin, 2019).

Longitudinal design is where momentary measures either become interpretable trajectories or an archive of observation bias. Sampling opportunities, alignment, missingness, visit intensity, device version, site context, treatment, transport, and purpose are therefore part of the estimand and data-generating process. They cannot be relegated to clerical metadata after the curve is drawn.

The chapter protects the scale of inference. A study may estimate change within a terminal, treatment, grief, sleep-recovery, or post-diagnosis window; it may not silently reconstruct the complete ledger from the first conscious moment to the truth-level endpoint. Unknown prior history remains an initial-state identified set or sensitivity problem, not a blank that a smoother can fill.

Design follows the estimand. Within-person change, between-person difference, event response, treatment effect, forecast, transport, and pressure on a PTVC component require different eligibility rules, clocks, observation plans, and estimators. Experience sampling, passive sensing, clinical cohorts, natural experiments, target-trial emulation, N-of-1 designs, accelerated longitudinal studies, and cohort-sequential follow-up are selected only after that target is fixed.

Within-person and between-person results can differ in direction, magnitude, and mechanism. Analyses must therefore state the process, stationarity and recurrence assumptions, and limiting claim while keeping person-specific dynamics, population distributions, time averages, and ensemble averages distinct.

The first protocol, sleep designs, and reserve-horizon tests supply concrete component targets; longitudinal research asks how those targets change when follow-up extends across ordinary life. A prompt is both measurement and interruption, a device both sensor and social object, a clinic visit both care and observation, and a record both evidence source and permission boundary.

The chapter names the models needed to keep these distinctions visible, including continuous-time processes, dynamic structural equation models, Gaussian processes, functional data analysis, state-space and hidden-state models, landmarking, joint longitudinal-event models, and target-trial emulation. Its purpose is not to reteach every estimator. It is to match each question to a design, expose the assumptions that identify it, and show where a wrong design reverses or invents the result.

Longitudinal data are morally delicate because they follow people across changing vulnerability. Records can reveal pain, fear, medication, fatigue, family pressure, sleep, recovery, refusal, and location. Purpose limitation, minimization, participant challenge, secure or federated computation, withdrawal propagation, retention limits, and governed result return are therefore scientific design conditions, not downstream privacy decoration.

The technical objects in this chapter are design aids, observation models, diagnostics, and governance structures. Their purpose is to show what the study can estimate and which dependencies can overturn the result. None converts a partial record into a complete life, and no graph, formula, or data architecture acquires evidentiary authority by appearing precise.

The governing question is strict: when evidence arrives through moments, devices, visits, records, and attrition, what can science responsibly infer about a lived trajectory? The answer is a hierarchy of designs, clocks, observation models, calibration bridges, causal and predictive estimands, transport rules, and governance boundaries. Each level states what it learns and what would make the estimate unusable.

The chapter examines how longitudinal inference develops with longer follow-up. Its models represent prior history, irregular observation, state-dependent missingness, competing events, treatment changes, device and response drift, cultural translation, privacy, and transport across sites. These features determine the population and questions each result can address.

17.1 — From Moments to Life-Course Inference

Repeated diary entries, wearable readings, pain ratings, and clinical notes can inform a life-course process when the design specifies how those observations connect to its estimand. That specification guides which moments are collected and how they are analyzed (Shiffman et al., 2008; Bolger & Laurenceau, 2013).

For *The Science of Fairness*, longitudinal evidence is useful because the formal target concerns a stream. Yet the empirical record is always local. The scientific burden is to move from momentary estimates to cautious trajectory claims without converting gaps into invisible data.

Emotional development already provides an encouraging empirical starting point. Carstensen et al. (2011) sampled feelings five times daily for a week and repeated the procedure approximately five and ten years later; the three waves included 184, 191, and 178 adults. Greater age was associated with more positive and stable emotional experience and greater coexistence of positive and negative emotions. This gives the life-course program an observed pattern of affective organization to explain, extending beyond a static questionnaire snapshot. The appropriate comparison is among models of regulation, goals, context, selection, and development. The repeated observations show that increasing age can accompany meaningful emotional gains. Repeated sampling also separates the frequency, persistence, and co-occurrence of feelings within persons, providing a practical design for following major life transitions.

The design table below is a routing device, not a ranking. A design that is poor for reconstructing whole-life history may be excellent for lagged within-person dynamics. A design that helps transport across age groups may be weak for acute event response.

Table 17.1. Longitudinal design families and inferential roles.

Design family	Best question	Key strength	Main artifact	What it can update
Intensive longitudinal or ESM	How does this person change across hours or days?	Captures lag, recovery, cycles, context.	Prompt reactivity, fatigue, short windows.	Within-person dynamics and pilot feasibility.
Cohort-sequential	How do age, period, and cohort differ?	Adds overlapping age and cohort observations; separating age, period, and cohort effects still requires identifying assumptions.	Cohort confounding, invariance.	Life-stage transport and broad patterns.
Accelerated longitudinal	Can overlapping windows approximate longer development?	Reduces wait while preserving age coverage.	Assumes linkable segments.	Long-horizon bounds under linking assumptions.

Design family	Best question	Key strength	Main artifact	What it can update
Event-based or case-crossover	What changes around a specified event?	Controls stable person characteristics.	Anticipation, carryover, event timing.	Local event response and transient signatures.
N-of-1 or idiographic	What is the process law for one stream?	Centers person-specific dynamics.	Limited transport, high burden.	Individual process hypotheses.
Clinical cohort	How do trajectories change under care?	Ethically available repeated measures.	Care confounding, death truncation.	Clinical component tests.
Registry-linked	What do long institutional records reveal?	Scale, duration, event history.	Consent boundaries, coding drift.	Background rates and transport checks.

Within-between decomposition. Role: diagnostic for separating person-specific change from cross-person association; it is not a latent-ledger estimator or a score of any participant.

$$y_{it} = \alpha_i + \beta_W(x_{it} - \bar{x}_i) + \beta_B\bar{x}_i + \varepsilon_{it}$$

Here the within-person coefficient estimates how departures from a person's own usual value relate to that person's outcome, while the between-person coefficient estimates how people with higher usual values differ from people with lower usual values. The two coefficients can differ or reverse. Reporting only one slope can therefore hide the level at which the relation actually lives.

Recovery is a measurable possibility with more than one trajectory. In a prospective bereavement study, 205 people were assessed before spousal loss and at six and eighteen months afterward; the largest classified group, 45.9%, maintained comparatively low depressive symptoms, while other groups showed recovery or persistent distress (Bonanno et al., 2002). This finding gives an evidence-based reason to investigate resilience without asking every bereaved person to follow the same timetable.

Disability research further improves the question by separating degrees and kinds of adaptation. Oswald and Powdthavee (2008) estimated partial recovery in mental well-being, with illustrative estimates of approximately 30–50% depending on severity. Lucas (2007b) found substantial lasting changes in two disability-onset samples. More recently, Köhler and Vanhoutte (2024) distinguished transient from persistent functional limitations in European longitudinal data: recovery toward initial well-being was clearer after transient limitations, and the pattern differed across depressive symptoms, life satisfaction, and quality of life. Together these results support an ambitious, humane target: identify which resources and mechanisms sustain or restore well-being, while modeling persistent trajectories explicitly.

Table 17.2. Analysis plans by inferential target.

Plan	Primary target	Minimum cadence or design need	Validation that matters
Idiographic	One stream's dynamics	Enough repeated points for person-specific lag and volatility	Forward chaining within that person
Hierarchical	Shared and person-specific dynamics	Repeated measures across persons with comparable constructs	Leave-person-out and calibration checks
Transport	Generalization from source to target	Bridge variables, overlap samples, target definition	Leave-site-out, target-site calibration
Federated	Multi-site learning without pooling records	Common dictionary, local feature extraction, disclosure checks	Site-level reproducibility and audit logs

17.1.1 — Intensive Longitudinal Targets

An intensive longitudinal design begins with a process that lives within a person, not with a population curve. The protocol must name the variable, clock, lag structure, and decision it seeks to estimate, then preserve enough person-specific history to distinguish change from stable between-person differences. Repeated observations are valuable because they expose temporal order and recovery, not because their quantity automatically

converts them into a life-course law (Shiffman et al., 2008; Stone et al., 2007).

The analysis file should retain person-specific deviations from usual sleep, pain, repair, activity, and momentary valence estimates. Those fields separate lived change from changes in observation, version, care, and participation while preserving a reconstructable audit trail.

Imagine an evening prompt series in which one caregiver's sleep loss reliably precedes next-day distress, while the cross-sectional group average suggests no association. That divergence is scientifically informative. It shows why the analysis must preserve the individual's temporal sequence before pooling, and why the group result cannot be used to overwrite a process that occurs within one stream.

The most credible alternative is that between-person averages, prompt fatigue, and observation timing can reverse the apparent direction. If protocol-fixed comparisons favor that account, report the result at the level actually identified rather than as Law-specific support.

Evaluation should begin by forecasting later points within the same person before asking whether a population model generalizes. Blocked time folds and leave-person-out tests answer different questions, and both should be named. The design fails its own purpose if neighboring moments from the same person leak into the test set and make prediction look stronger than it is.

The within-person estimand must be frozen before smoothing, subgroup analysis, or population pooling begins. The data

structure should show which observations belong to the same stream, which lags were available at each prediction time, and which summaries are allowed to cross people. When those choices cannot be stabilized, the output remains descriptive or receives No adequate estimate rather than being promoted into a general trajectory.

A substantial starting resource is already available. The July 2026 openESM journal article describes 60 openly available experience-sampling datasets with more than 16,000 participants and more than 740,000 observations, accessible through searchable metadata and R or Python tools. Its authors' analysis of 39 datasets found a pooled within-person concurrent correlation of -0.49 between positive and negative affect (95% CI $[-0.54, -0.42]$) (Siepe et al., 2026). That result supplies a reproducible description of how positive and negative affect co-occur within persons.

For PTVC component research, select eligible datasets by timestamps, cadence, scale content, available context, and access conditions before examining the intended outcomes. Use one set for measurement and model development, then freeze the analysis before testing whole held-out studies. Study-specific scale meanings and timestamps guide the measurement model. The first deliverables can be replicated within-person dynamics and a shared benchmark that identifies which additional observations best distinguish the candidate routes.

17.1.2 — Sparse Windows and Unobserved Time

Repeated points reveal windows, not the path between them. The inferential problem is therefore to state what regularity, smoothness, or state-space assumptions are needed to connect those windows and how strongly the conclusion depends on them. Sparse designs can answer important questions, but only at resolutions supported by their cadence and by assumptions that remain visible to the reader (Bolger & Laurenceau, 2013; Little & Rubin, 2019).

The longitudinal record must include sampling intervals, smoothness assumptions, bandwidth, aliasing, and worst-case error. Their joint presence makes it possible to locate drift instead of absorbing it into the target trajectory.

A quarterly clinic series may miss a two-week medication crisis and still produce a visually persuasive recovery curve. The appropriate response is not to distrust every sparse record, but to represent the unseen interval explicitly through alternative smoothness classes, worst-case bounds, and interval holdouts. Uncertainty should widen where the person was not observed, even when the fitted line would prefer continuity.

A serious competing account is that sparse points can manufacture a continuous ledger by interpolation. A win for this rival is informative: the conclusion must then narrow to the measured component or be reported as a measurement finding.

Evaluation should hold out whole intervals rather than isolated observations. A model that predicts the next visible clinic point after seeing adjacent points has not shown that it recovered the

unseen path between them. The test should ask whether the assumed smoothness can survive the gaps it claims to bridge.

Interval contributions are reported only at the resolution supported by the sampling schedule and the prospectively specified regularity class. When multiple admissible paths through the same gap imply different signs, peaks, or cumulative contributions, that disagreement belongs in the identified set. No interpolation method may turn an undersampled interval into a known episode merely because the estimate is smooth.

Sparse designs are valuable when their regularity assumptions and observation limits remain explicit. They can report ranges for interval contributions, compare admissible paths through gaps, and show which additional observations would most improve the next study.

17.1.3 — Unknown Prior History

Enrollment creates an initial-state problem. A study that starts in late illness, after diagnosis, or during a treatment transition observes an increment from that point forward; it does not observe the cumulative state carried into enrollment. The protocol must decide whether it estimates post-entry change, bounds a compatible initial state, or requires a longer design, and the distinction must survive into the headline result (Hernán & Robins, 2020; Gerstorf et al., 2013).

A credible analysis must carry the initial latent state and the post-enrollment increment as separate objects, with the initial state retained as an identified set rather than silently fixed. Together they reveal whether the estimate tracks within-stream change,

between-person contrast, transport, clinical process, or sampling artifact.

Observable-window identity. Role: bookkeeping definition for delayed entry; it cannot identify the unobserved prefix or convert a late-window study into a whole-life estimate.

$$L_i(T) = L_i(t_0) + \Delta L_i(t_0, T)$$
$$\Delta L_i(t_0, T) = \int_{t_0}^T g_i(s) F_i(s) ds$$

Under its measurement model, a study beginning at enrollment estimates the post-enrollment increment. Combining that increment with an independently known or bounded initial ledger state yields the corresponding estimate or bounds for the cumulative state.

The leading rival here is that late-window evidence has been mistaken for whole-life reconstruction. When held-out checks favor that account, the conclusion contracts to the observed window and identifies the missing initial-state bridge; no rhetorical continuity may restore the larger claim.

Evaluation should preserve the initial-state uncertainty rather than hiding it in a fitted intercept. Leave-window-out checks and alternative prior bounds should show how much the conclusion depends on the unobserved past. If plausible initial states change the verdict, the study reports a window claim rather than a life claim.

Design classes should be separated before analysis: increment studies estimate change after entry, bounding studies carry a set of compatible prior states, and long-horizon studies attempt

broader coverage under stronger retention demands. If plausible initial states reverse the conclusion, the result is a window-dependent finding. More observations after enrollment cannot identify a history that was never observed.

This is the chapter's first hard boundary: a late window is not a whole life. It may constrain possible histories, estimate a visible increment, or motivate longer enrollment, but it does not erase the unknown initial state. The notation keeps that humility alive after the graph begins to look complete.

17.1.4 — Event-Time Alignment

Event-time alignment changes the scientific question. Calendar time, age, diagnosis time, treatment time, enrollment time, and time before death produce different risk sets and different conditioning. The unwarped timestamps must therefore remain available, and every aligned display should be treated as a transformation whose assumptions can be tested rather than as a neutral way to organize the page (Shiffman et al., 2008; Stone et al., 2007).

The study record needs death-alignment simulations, dynamic time-warping checks, and sensitivity to clock choice. Without them, a smooth trajectory can conceal a change in who was observed, how they were measured, or what population the estimate represents.

Backward alignment to death can create apparent convergence when the cohort increasingly consists of people with gradual, observable declines. Sudden deaths, early transfers, rapid loss of communication, and short enrollments may disappear from the

later portions of the curve. A death-aligned pattern must therefore be shown beside its changing risk set and beside alternative clocks that do not condition on the same future event.

A signature is alignment-sensitive when its direction or magnitude changes materially across prospectively allowed clocks. Such a result is not discarded, but it is classified correctly: it may describe how trajectories look relative to a chosen event without identifying a common native dynamic. The protocol should make that branch explicit rather than allowing one visually compelling alignment to carry the route verdict.

The analysis file should preserve original timestamps, the rule that generated each transformed clock, the population remaining at every aligned time, and sensitivity results under competing clocks. If the clock choice remains decision-critical, the public result should say so. Alignment is part of the estimand, not a formatting preference.

17.1.5 — Local Versus Whole-Life Claims

A local window can test local dynamics, route-specific components, and observation behavior without answering a complete-stream question. The protocol should name the beginning and end of the observed domain, the unobserved history it leaves outside, and the exact bridge required for any broader interpretation. Length of follow-up does not erase the boundary between a measured window and a whole life (Bolger & Laurenceau, 2013; Little & Rubin, 2019).

Model choice follows the scope of the claim. Fast local change may require dense sampling and short lags; transport requires

bridge variables and a declared target population; terminal-window inference requires endpoint and observation models; whole-stream language requires an independently justified initial-state and completion bridge. These are different analyses, not levels of confidence in one interchangeable result.

A final-month study may estimate variance, dependence, or intervention response during that month with considerable precision while remaining silent about childhood, adulthood, or earlier illness. That silence is not a defect. It is the accurate boundary of the design, and it allows the local result to matter without being inflated into a claim about the complete ledger.

A pipeline fails this boundary test when it declares a decisive whole-life conclusion from synthetic cohorts whose observed late windows are identical but whose unobserved initial states imply different truth-level conclusions. The negative control preserves the same cadence and terminal window while varying those hidden prior states. A valid analysis may return the same abstention across those observationally identical records; the failure is a decisive whole-life verdict that ignores this nonidentification.

The recovery suite varies hidden initial residuals across sign and magnitude while holding the visible late window and observation cadence fixed; hidden generator values are retained for evaluation, not supplied to the fitted procedure. A separate sensitivity analysis varies declared initial-state assumptions or bounds and records how conditional conclusions change. Together, these checks show whether the pipeline respects

unknown history without mistaking invariant, justified abstention for failure.

Every result should carry the domain it actually covers: momentary, interval, treatment-window, terminal-window, or complete-stream. Clearly bounded scope makes each finding answerable and usable in subsequent work.

Local questions can make substantial contributions: whether terminal-window variance changes, prompts alter a trajectory, or a medication transition shifts reports. Each result supplies a defined component for subsequent research.

17.1.6 — Transport Across Life Stages

Using the same questionnaire, sensor, or model across life stages does not establish transport. Development, aging, disability, illness, communication mode, culture, and care setting can alter item meaning, response burden, physiological baselines, and the relation between an auxiliary channel and lived experience. A transported estimate therefore requires evidence that the construct, origin, sign, unit, and measurement map remain adequate for the stated purpose (Hernán & Robins, 2020; Gerstorf et al., 2013).

Cohort-sequential and accelerated longitudinal designs can create overlap among ages or disease stages, but the overlap must contain common anchors and enough heterogeneity to test the link. A model trained in one stage should be challenged in another without silent recalibration after the result is known. If a new stage requires a new scale, the study has produced a new local claim rather than transported the old one.

Language and mode can change meaning even when wording appears simple. Cognitive interviews, bridge samples, assisted-communication records, mode-specific calibration, and participant debriefs help distinguish sincere response change from a failed linking assumption. These are inferential requirements, not accommodations outside the science.

Predictive portability is weaker than construct transport. A model may preserve rank while losing a neutral origin, interval unit, sign interpretation, or causal meaning. Reports should state which property traveled and withdraw any interpretation that depends on a bridge the data did not support.

Longitudinal designs carry distinct clocks for age, calendar, disease, treatment, enrollment, adjudicated events, and perceived horizon. Declare a primary clock and use others as covariates, realignments, or sensitivity structures. An unnamed blend can create a trajectory with no coherent estimand.

When an observational cohort emulates intervention, first define the target trial: eligibility, strategies, assignment time, follow-up, outcome, contrast, and analysis. Emulation clarifies timing and avoids preventable bias but cannot create exchangeability or recover unmeasured indications (Hernán & Robins, 2020). A clear target-trial specification also exposes which quantities remain unidentified from the observational data before causal estimates are computed.

Time-varying treatment, health status, observation, and prior experience can affect one another. Ordinary adjustment may block the pathway or create bias, so use causal language only

when temporal ordering and assumptions support it. Descriptive life-course work remains valuable otherwise. Sensitivity analyses should distinguish unmeasured confounding from measurement error and informative observation, since each requires a different repair.

Transport requires prospective overlap, common anchors, bridge evidence, mode and language records, held-out target-population performance, and a stopping point. A result may stop at a life-stage or care-setting boundary without diminishing the dignity or scientific importance of those outside it. Stopping transport at a justified boundary creates a concrete next study: test the missing bridge rather than simply enlarge the original sample.

17.2 — Active Reports and Experience Sampling

Active reporting is the closest longitudinal method to lived testimony, but closeness does not make it simple. A person answering a prompt reports from within a moment, yet the prompt itself consumes attention and may change what is being reported (Shiffman et al., 2008; Trull & Ebner-Priemer, 2020).

Experience sampling is not a moral request to confess. It is a bounded research method for estimating time-near experience with stated uncertainty. The participant should know how nonresponse, pause, and burden are treated.

The cost of being asked counts. A study of hidden burden that ignores prompt burden has already contradicted its object. Cadence should be justified by the timescale needed for the estimand, not by the researcher's appetite for dense data.

17.2.1 — Prompt Content and Cadence

Contemporary ecological momentary assessment illustrates the value of within-person, age-sensitive measurement. A 2025 study of young, middle-aged, and older adults combined repeated daily reports with an implicit measure of valence representation. Its comparison identifies age and implicit-explicit disagreement as measurement features worth examining directly (Nigam & Kar, 2025; Shiffman et al., 2008; Trull & Ebner-Priemer, 2020).

Prompt content and cadence are part of the estimand because each question selects a time scale, a comparison frame, and a burden on attention. The protocol should first decide whether it needs rapid transitions, daily summaries, event-triggered reports, or

slower adaptation, then choose the least intrusive schedule capable of resolving that process. Sampling density is justified by discriminating value, not by the availability of a phone.

A clinically grounded starting point already exists. Geeraerts et al. (2024) developed and content-validated the ESM-AC questionnaire with forty-three people living with advanced breast or lung cancer and eight health-care professionals. Its digital core contains thirty-one momentary, two morning, and seven evening items; ten participants in the usability round gave ease of use a mean rating of 4.5 out of 5. This work shows how patient input can turn intensive measurement into a practical, context-sensitive instrument. A PTVC-linked pilot could build on that development process while separately testing sampling burden, temporal validity, and the proposed valence construct.

The subsequent seven-day ESM-AC study recruited 40 of 79 people with advanced breast or lung cancer approached through two Belgian hospitals. Thirty-seven participants provided 1,703 valid assessments, 71% of the 2,400 scheduled. On seven-point scales, mean reported burden was 2.1, ease of use 5.6, and measurement reactivity 1.3 (Geeraerts, Pivodic, et al., 2026). These results provide a practical precedent for repeated measurement in comparable patients. Eligibility restrictions, including functional status, limit their extension to more severely ill populations.

A companion analysis of the same study used 1,676 assessments from 36 participants. Retrospective scores correlated with participant-level summaries of momentary ratings, with Kendall's tau values from .24 to .70, yet identical retrospective scores could

conceal different momentary trajectories (Geeraerts, de Nooijer, et al., 2026). This supports preserving temporal information when the research question concerns accumulation or recovery. The two analyses contribute complementary evidence from one cohort; their questionnaires still require separate qualification as measures of signed phenomenal value.

A naturalistic study shows what event-sensitive sampling can add. Villano and colleagues followed university students across an academic term, recording expected examination grades before results were released and sampling affect intensively afterward. The analyzed sample comprised 249 students, divided into discovery and confirmatory cohorts. Models including the difference between expected and received grades described subsequent affect better than grades alone, while modeling positive and negative emotion separately (Villano et al., 2020). This gives the program a concrete design precedent: measure expectations before outcomes, preserve separate affective components, and select a cadence that resolves their recovery. The study supports prospectively ordered observation of real emotional events.

The prompt pool, randomized windows, quiet hours, maximum daily count, burden stop rule, minimum usable cadence, and analysis of schedule effects are fixed prospectively. Planned cadence variation can estimate reactivity and fatigue. Removing inconvenient prompt periods after inspection would turn the schedule into an invisible tuning parameter.

Five prompts sent through a workday may change the day they seek to measure. A participant may silence the phone, anticipate

the next question, avoid an activity, or answer quickly to end the interruption. Those responses are not noncompliance to be punished; they reveal when the instrument has entered the process and must be modeled as an exposure.

A serious audit must test whether high cadence can measure irritation and surveillance instead of experience. If so, the conclusion must attribute the pattern to design, care, or measurement to the extent warranted and narrow the local claim.

A prompt protocol should include randomized or staggered intensity where ethically feasible. That variation lets the study estimate prompt burden and reactivity instead of guessing them after the fact. The negative control is a schedule effect that looks like an affective dynamic.

Sampling cadence should reach the slowest schedule that still recovers the protocol-fixed dynamics in simulation and pilot work. Burden, reactivity, and missingness are evaluated beside statistical resolution. Denser sampling is rejected when its marginal information is purchased by a disproportionate change in ordinary attention or routine.

Sampling error can be bounded before a smooth curve is fitted. In a declared exposure clock, suppose an independently justified constant $K > 0$ bounds rate changes by $|v(s) - v(t)| \leq K|s - t|$. For an unsampled interval of length h with exactly known endpoint rates a and b , feasibility requires $|b - a| \leq Kh$. The attainable integral interval is

$$\int_0^h v(s) ds \in \left[\frac{h(a+b)}{2} - \Delta, \frac{h(a+b)}{2} + \Delta \right], \quad \Delta$$

$$= \frac{Kh^2}{4} - \frac{(b-a)^2}{4K}.$$

The extremal paths follow the steepest allowed slopes from the two endpoints, so these bounds are sharp under the stated assumptions. Two zero samples with $K = h = 1$ permit any intervening total from $-1/4$ to $1/4$. Across exactly observed common endpoints, the interval bounds add. Equal exposure spacing minimizes the worst-case missing-area width for a fixed span and number of gaps; burden and statistical error remain separate design considerations. Noisy endpoint readings require simultaneous uncertainty bounds, and unobserved prefix remains additional uncertainty. Without an independently qualified regularity or magnitude bound, sparse samples alone cannot exclude an unseen excursion. A synthetic downsampling challenge can determine whether the proposed cadence resolves the required integral before recruitment.

17.2.2 — Momentary Versus Retrospective Questions

Momentary and retrospective questions observe different temporal objects. One asks about experience near the reporting moment; the other asks for a later construction shaped by memory, narrative, salience, and the chosen interval. A continuous-looking graph becomes misleading when those objects are merged without an explicit measurement model and a version flag identifying which question produced each point (Shiffman et al., 2008; Stone et al., 2007).

The protocol should predefine each reporting window, the wording used for current and retrospective judgments, the expected recall delay, and the analysis of their disagreement. A weekly evaluation cannot be substituted for missing momentary reports simply because both use the same response scale. The two channels remain separately versioned and may support different conclusions.

A late-night report of dread and a Sunday judgment that the week was meaningful need not contradict one another. They can reflect different moments, memory weights, and evaluative frames. Modeling the divergence can reveal recency, peak, recovery, or narrative effects that would disappear inside a single average.

An independent reviewer should be able to determine whether averaging them can erase the difference between rate and interpretation. A positive answer requires redesign or a claim limited to what survives the confound.

A useful control is disagreement by design: ask a time-near question and a later evaluation about the same interval, then model their divergence. If the result depends on merging them, the construct has been blurred. The disagreement itself may be the scientific finding.

The analysis keeps the testimony clocks separate through channel-specific estimands, lags, and uncertainty. Agreement is informative only when it survives the declared mapping between them; disagreement is reported rather than adjudicated by convenience. Neither a momentary report nor a retrospective account is granted automatic sovereignty over the other.

The distinction protects both kinds of answer. A momentary report is not morally superior to a retrospective account, and a life evaluation is not a substitute for lived time. They answer different questions, and a good protocol lets them disagree without immediately forcing one to confess error.

17.2.3 — Compliance and Reactivity

Compliance is a data-generating process, not a test of obedience. People may answer more when distressed, less when exhausted, differently when they anticipate a prompt, or only when the question feels supportive. The probability and timing of response must therefore be modeled alongside the reported value, especially when participation itself may alter attention or behavior (Little & Rubin, 2019; Nosek et al., 2018).

A reproducible data package carries response probability, prompt-induced change, randomized intensity, and compliance models. This structure allows later analysts to reproduce the risk set, measurement version, and transport assumptions at every time point.

Suppose a participant responds most often during difficult evenings because the prompt feels like contact. The observed series would overrepresent distress even though the missing intervals might be calmer. The reverse pattern is equally possible when distress makes answering impossible. Both mechanisms should be represented before the average response is treated as a trajectory.

The protocol must expose whether dropping low-compliance participants selects the easiest lives to observe. When the risk is

material, the principal result names the selected population and removes every stronger interpretation that depends on unobserved participants behaving similarly.

Cadence should be tested as an exposure. A study can randomize prompt density, vary timing within safe windows, or use planned low-intensity periods to estimate how being asked changes attention and report. If cadence changes the outcome, the prompt has become part of the model.

Response probability, response latency, prompt history, and planned cadence variation should enter the analysis before substantive dynamics are interpreted. The study should report whether conclusions survive inverse-probability, joint-process, and mark-restricted analyses. When the answer depends on treating response as random, the verdict remains conditional on that assumption.

17.2.4 — Participant-Controlled Pauses

Participant-controlled pauses make refusal visible without turning it into dropout. A pause may protect privacy, sleep, family time, clinical care, or emotional stability, and the reason need not be disclosed beyond the categories the participant chooses. The data model should distinguish temporary pause, skipped prompt, device removal, technical failure, and withdrawal because those events carry different meanings (Shiffman et al., 2008; Trull & Ebner-Priemer, 2020).

The data model needs fields for pause marks, no-estimate intervals, burden thresholds, and optional reason codes. Without

that linkage, site and instrument changes can masquerade as changes in the phenomenon.

A caregiver may suspend prompts during a family crisis and resume later when observation is no longer intrusive. Smoothing across that interval would erase both the protected event and the reason the record became silent. The pause should remain a marked boundary with an explicit no-estimate option for the interval.

Smoothing cannot erase a pause that marked the very state in which observation became inappropriate. The protocol-fixed failure rule treats that interval as a boundary or No adequate estimate rather than manufacturing continuity for public interpretation.

The pause mechanism is designed before enrollment, available without penalty, and linked to immediate stopping of prompts and sensors where feasible. Analyses treat paused time as unobserved under a declared mechanism rather than as neutral experience. A study that requires people to disappear in order to protect themselves has failed both ethically and methodologically.

A pause button is not a decoration on an intrusive design. It is an epistemic tool because it distinguishes safe unavailability from silent dropout, fatigue, fear, and refusal. A participant who can stop cleanly gives the study better information than one who must disappear to protect themselves.

17.2.5 — Language and Accessibility

Language and accessibility determine how the measurement request reaches the participant and can alter the construct being

elicited. Translation, literacy, assisted communication, sensory access, cognitive load, and cultural framing should be specified as measurement versions with their own qualification evidence. Youth and ambulatory psychophysiology reviews also show why feasibility and channel burden vary with developmental context. Identical scores do not establish identical meaning across those routes (International Test Commission, 2018; Sosa-Hernandez et al., 2024).

At minimum, the dataset must preserve translation, assistive formats, cognitive interviewing, and adaptation versioning. They are the audit trail that separates lived change from design, version, care, and participation changes.

An item that sounds neutral before illness may sound blaming after disability or treatment changes the person's reference frame. A voice prompt may be accessible but more public; an assisted response may improve participation while changing privacy and timing. These are not peripheral usability details. They can change who answers and what the answer means.

The claim should be withdrawn when the same wording can hide construct shift. The protocol should name this loss condition in advance and tie it to an explicit narrowing rule.

Time grids should accommodate communication mode and access needs. A voice prompt, screen reader, simplified item, translated question, or assisted response may require a different timing burden. The construct changes silently if accessibility is treated as a late convenience rather than a design condition.

Every change in language, modality, assistance, item set, or timing is entered in the measurement version ledger. Bridge samples and cognitive interviews determine whether data may be linked, require recalibration, or must remain separate. Accessibility is successful when it widens participation without pretending that different response routes are automatically equivalent.

17.2.6 — Calibration and Drift

Calibration can drift while displayed numbers remain stable. Treatment, grief, illness, repeated prompting, new vocabulary, trust, fatigue, and changing reference points can alter scale use, so repeated anchors and prospectively specified response-shift checks replace indefinite reliance on one baseline (Little & Rubin, 2019; Nosek et al., 2018).

The record should preserve anchor items, vignettes, overlap periods, brief debriefs, response latency, and scale-linking results. Recalibration is planned around medication change, disease progression, relocation, bereavement, new communication modes, and long participation; when meaning shifts, the protocol splits the series, estimates a bridge, widens the identified set, or withdraws the common-scale claim.

A stable number with unstable origin, sign, or unit is a stop gate. Longitudinal continuity is earned by linking evidence, not repeated use of one interface, and a wider confidence interval cannot rescue an unbridged scale change.

The prompt schedule is itself an exposure. Frequency, reminders, quiet hours, postponement, and burden-triggered reductions can

change monitoring, fatigue, rehearsal, irritation, labeling, and compliance, so invitation, opening, completion, interruption, postponement, and retrospective-entry times remain marks in the observation model.

Recent experiments make these design choices measurable. In a twenty-eight-day factorial study of 205 participants, fixed rather than random prompts produced lower measured affect variability in eight of sixteen models; associations with next-day health behaviors were not reliably changed (Langford et al., 2026). A separate fourteen-day study comparing seven-point Likert responses with visual analogue scales in 119 participants found differences in means, floor responses, and some correlations, but no reliable difference in within-person standard deviations or successive-change variability (Haslbeck et al., 2025). The practical gain is a stronger instrument: freeze the schedule and format, test their effects, and estimate transport when either changes. A smoother observed trajectory can then be assessed against measured design effects before it receives a latent interpretation. Design-effect estimates should travel with the instrument version so later studies can adjust expectations when prompt schedule or response scale changes.

When burden and ethics permit, randomized prompt density, planned schedule changes, or micro-randomized notification timing can estimate reactivity. If asking changes state or participation, that effect belongs in the estimand and interpretation rather than being dismissed as noise (Shiffman et al., 2008; Klasnja et al., 2015; Trull & Ebner-Priemer, 2020). Reactivity estimates can also inform burden limits by showing

when additional measurement begins to alter the process more than it improves information.

The final calibration verdict states which periods and populations share a defensible scale, where a bridge remains uncertain, and where only local or ordinal inference survives. Drift limits the instrument's comparative claim; it does not invalidate the participant's experience. A calibration boundary should enter downstream models as a restriction on inference rather than become a wider confidence interval around a comparison that is no longer valid.

17.3 — Passive Signals, Wearables, and Context

Passive sensing can extend temporal coverage while reducing the need for active prompts. Wearables can record heart rate, movement, sleep timing, skin temperature, oxygen saturation, or environmental context during ordinary life. Their contribution depends on a clear account of the observation gaps and device conditions that accompany those records (Onnela & Rauch, 2016; Shcherbina et al., 2017).

Device placement, sensor quality, proprietary algorithms, motion artifacts, illness, treatment, charging habits, privacy choices, firmware changes, and willingness to wear the device shape passive records.

Passive evidence can extend life-course observation when fatigue, recall, speech, delirium, sleep, or disability limit active reporting. Work with people receiving outpatient palliative care and family caregivers found wearable-triggered momentary assessment practicable in a bounded clinical setting. Building on that result, a study can record who can wear, respond, pause, and withdraw and evaluate how those patterns affect its measurements (Schuler et al., 2023).

StudentLife offers a smaller complementary benchmark combining smartphone sensing, momentary assessments, and survey outcomes across a 10-week term in 48 Dartmouth students. The original study reported associations among sensed behavior, mental-health measures, and educational outcomes, and released its dataset for reuse (Wang et al., 2014).

Its value here is to challenge the observation model. Before interpreting a candidate affective trajectory, ask how much prediction comes from recorded sleep, activity, social context, workload, and assessment availability. Fit those variables as substantive rival explanations under the same participant splits and scoring rules. A stable added contribution from a declared route variable would justify replication in a broader population. The dataset provides a local methods benchmark with a documented duration and population.

17.3.1 — Physiological Streams

Physiological streams are continuous only in a technical sense. Heart rate, electrodermal activity, movement, respiration, temperature, and sleep estimates remain partial witnesses whose meaning depends on illness, medication, posture, motion, device fit, and context. The protocol should qualify each channel for a stated use and preserve the conditions under which the signal was acquired (Onnela & Rauch, 2016; Shcherbina et al., 2017).

The relevant time scale differs by channel and estimand. A rapid autonomic response, nightly sleep summary, slow frailty change, and treatment effect cannot be placed on one clock without a model of lag and aggregation. Quality flags, treatment states, and sampling density should accompany the value so later analyses can distinguish physiology from acquisition artifact.

A wearable may show reduced arousal after medication while the participant continues to report distress. That disagreement can reflect channel lag, sedation, motor suppression, altered interoception, or genuine multidimensional experience. The

analysis should preserve the conflict rather than declare the most technical signal to be the hidden truth.

One stronger explanation is that medication, fever, motion, and device fit can mimic affective change. If this account explains the shape under equal scoring, the proper verdict is narrowing, not rhetorical rescue.

Uncertainty should identify the channel, not merely widen the band. Motion artifact, fever, medication, sensor fit, arrhythmia, pain behavior, sleep staging error, and device removal have different meanings. The result should say which uncertainty entered the estimate before it asks the reader to trust the line.

Channel-specific confidence, artifact flags, medication timing, and device state are stored with every derived feature. Fusion occurs only after each channel has passed its own context-of-use checks. When a physiological channel cannot distinguish the target from its principal confounds, it remains an auxiliary witness or is removed from the estimand.

17.3.2 — Behavioral and Environmental Context

Behavioral and environmental context may explain a signal while also exposing intimate routines and bystanders. The smallest feature set capable of distinguishing the protocol-fixed mechanisms should be collected at the coarsest sufficient resolution. Contextual integrity and attack-surface analysis therefore belong in feature selection; more context does not automatically produce more validity (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

Activity, light, noise, care setting, social presence, and coarse location may enter as time-varying covariates, marks, or stratification variables. Each requires a declared temporal resolution and purpose. A variable gathered for one explanation cannot later become a general map of private life without new authority and a new scientific justification.

Low activity may mean restorative rest for one participant and forced immobility for another. The sensor records movement, not the reason for movement or its felt value. Active report, clinical context, and participant-defined categories may help interpret the signal, but the study should retain the possibility that the context remains unknown.

The protocol must assess whether contextual data expose bystanders or intimate routines. Record any privacy failure and its effect on permitted data use, and narrow PTVc-specific claims where those restrictions leave the estimand unidentified.

Contextual uncertainty includes both error and exposure. Location may be coarse, social interaction may be inferred, activity may be misclassified, and environmental data may reveal more than the estimand needs. The analysis records these uncertainties and collects the minimum contextual detail needed for its estimand.

Context variables are retained only when they change a prespecified model comparison, confounding assessment, or safety decision. Their privacy cost and incremental information are reviewed together. If a less revealing feature performs as well,

the more intrusive variable is removed even when it would make the visualization richer.

Context is useful because it tells the study where a signal lived. A location trace, activity summary, sound level, temperature reading, or interaction count may help explain a change, but it also raises the cost of being known. The study earns those variables only when the estimand cannot be answered by a less revealing path.

17.3.3 — Sensor Fusion Without Sovereignty

Sensor fusion is an estimation problem with a governance boundary. Multiple channels may reduce some forms of uncertainty, yet fusion can also conceal disagreement and create a composite that appears more authoritative than any constituent measure. The protocol must specify channel roles, provenance, conflict rules, and the conditions for abstention before the model is fit (National Institute of Standards and Technology, 2020; Rieke et al., 2020).

The fusion architecture, held-out channels, quality thresholds, missing-channel behavior, and conflict outputs are prospectively fixed. A model may predict a qualified target, but it may not redefine that target from whichever channels remain available. Contradictory or insufficient witness sets must be able to yield No adequate estimate.

Self-report may indicate calm while physiology shows arousal after a difficult clinical conversation. The discrepancy could reflect concealed distress, medication, physical effort, delayed recovery, social presentation, or a scale mismatch. A responsible

fusion model displays that disagreement and its competing explanations rather than compressing it into one smooth score.

One important rival is presentation authority: a technically elaborate channel may appear to overrule the person without carrying better evidence. When common-rule comparison favors that account, the result belongs to measurement presentation or channel dominance unless a later test separates the mechanisms.

The output should expose which channels contributed, which objected, and how uncertainty changed when each was removed. Held-out-channel validation tests whether one channel predicts another without being allowed to dominate it. Fusion earns use by improving a declared estimate, not by creating the appearance of a direct reader of inner life.

Fusion should make uncertainty more visible, not hide it behind a confident composite. When self-report, physiology, behavior, and context disagree, the disagreement is data. The person is not overruled by the smoothest model, and the model is not allowed to call conflict certainty.

17.3.4 — Device Removal and Nonwear

Device removal and nonwear are observation events with potentially informative causes. Privacy, discomfort, skin injury, charging, illness, delirium, family presence, care transition, and simple annoyance all produce different missingness mechanisms. The device record should preserve these causes when the participant chooses to provide them and otherwise retain an explicit unknown mark (Onnela & Rauch, 2016; Shcherbina et al., 2017).

The protocol should define nonwear detection, participant removal controls, charging windows, skin-safety checks, and the distinction between technical absence and refusal. It should also state when sensing stops automatically, how data already collected are handled, and how nonwear enters the primary and sensitivity analyses.

A person may remove a sleep tracker before discharge because it has become intrusive or uncomfortable. If the missing interval is filled from neighboring nights, the analysis erases the event that made observation inappropriate. Nonwear must remain visible as a possible indicator of burden, state change, or device failure, not as a blank invitation to interpolate.

A candidate signature that appears only after high-burden nonwear is coded as random has failed an observation-process test. The report should show how results change under cause-specific and unknown-cause missingness scenarios. When the plausible branches disagree, the appropriate outcome is model ambiguity or No adequate estimate.

Include forgetfulness among the plausible causes. A result depending on one explanation should state that explanation in the verdict.

Trajectory smoothing begins only after device absence has been modeled and the admissible missingness assumptions have been declared. The study should publish nonwear profiles by time, site, device version, and participant state. A continuous display may be used for visualization, but it may not conceal the discontinuity in evidence.

Nonwear can provide useful information about how observation fits into ordinary life. Devices may be removed for bathing, pain, delirium, privacy, fatigue, technical failure, family time, or simple preference. Recording the applicable reason, or preserving it as unknown, helps distinguish a person's choice from a device limitation or change in state.

17.3.5 — Vendor and Firmware Drift

Vendor and firmware drift should be tested as a measurement discontinuity. Overlap intervals, calibration fixtures, dual-device wear, and version-boundary negative controls can reveal whether a feature moved because the person or the sensor, firmware, algorithm, or preprocessing changed. Verification, analytical validation, and clinical validation are versioned properties; procurement history becomes measurement history when it can shift the observed variable (Bent et al., 2020; Goldsack et al., 2020; National Institute of Standards and Technology, 2020).

The auditable dataset should preserve device model, firmware, algorithm, preprocessing, bridge wear, and overlap windows. The record thereby distinguishes the scientific object from the system used to observe it.

Suppose a firmware update reduces detected awakenings at every site on the same date. Without a version ledger, this could resemble improved sleep or reduced distress. Overlap wear, calibration fixtures, and leave-version-out validation must locate the change in people or instrument.

A model that succeeds within one device version and fails across versions is not transportable merely because the vendor uses the

same product name. The result should be split, bridged with independent overlap evidence, or retired for the new version. Technology history may explain a trajectory and must be allowed to win that comparison.

Evaluation should leave out device versions and firmware periods, not just people. If a model trained on one version fails after an update, the version has become part of the observed effect. The chapter's rule is to test the machine history before interpreting the human trajectory.

Major technical changes require predeployment bridge plans with parallel measurements, version-specific quality thresholds, and a reanalysis trigger. Silent updates are incompatible with longitudinal inference. If the bridge is inadequate, segment the time series and withdraw the common-scale claim.

A device version can change the apparent life without the life changing at all. That is why vendor history, firmware, algorithms, and preprocessing belong in the scientific record rather than in a procurement footnote. Longitudinal evidence must know the machinery that helped create it.

17.3.6 — Minimization and On-Device Processing

Minimization asks what the study can learn without collecting a reconstructable portrait of ordinary life. On-device features, coarse summaries, burst sampling, and site-local analysis can reduce exposure while preserving a declared estimand, and the tradeoff is tested rather than assumed (National Institute of Standards and Technology, 2020; Rieke et al., 2020).

The database records local extraction, summary rules, sampling, and secure aggregation so a replication team can reconstruct both the estimand and its observation process.

A phone may transmit approved sleep regularity while retaining raw location and audio locally. The feature still needs provenance, quality checks, and deletion rules; local processing is a privacy architecture, not an exemption from validation.

Unspecified future collection invites post hoc fishing and weakens withdrawal. The protocol names the minimum sufficient signal for each estimand and requires new review before expansion; temporary richer audit samples validate the minimized representation without becoming permanent retention.

Permissioned comparisons with richer audit samples test whether the smaller stream preserves the needed estimand. The richer stream remains a temporary bridge, not a standing appetite for more data.

A minimized pipeline must preserve prospectively specified model comparison across relevant states, sites, and devices. If reduction removes identifying information, the question narrows rather than the surveillance surface expanding.

Long observation is proportionate only when the record remains purpose-bound, local where possible, and limited to the weakest sufficient signal.

17.4 — Marked Observation Processes

A marked observation process records not only what was measured, but when measurement occurred and what kind of measurement event it was. The distinction matters because longitudinal data are rarely collected by a neutral clock (Andersen et al., 1993; Rizopoulos, 2012).

In ordinary analysis, missing data often enter after the outcome model has been specified. This chapter reverses the order. Observation availability is part of the data-generating system, and any apparent life-course pattern should be tested against the possibility that the observation process created it.

The notation in this section is a scaffold. It does not define the Law, and it does not claim to observe the latent cumulative process. It gives the study a way to ask whether the measurement system has its own dynamics.

Table 17.3. Marked-observation model families and identification limits.

Model family	Use when	Watch for
Continuous-time state-space	Observation times are irregular and latent dynamics matter.	Do not assume regular time by inserting artificial missing rows.
Dynamic structural equation model	Lagged within-person relations and person differences are central.	Separate within-person from between-person effects.
Gaussian process	Nonlinear or nonstationary trajectories need flexible smoothing.	Smoothing can hide event timing and missingness.

Model family	Use when	Watch for
Functional data analysis	Sparse curves require functional summaries or principal components.	Regularity assumptions determine what is recoverable.
Hidden or semi-Markov model	Regime changes or state durations matter.	States can become stories if not recoverable.
Joint longitudinal-event model	Outcome, observation, death, or dropout are linked.	Define the endpoint independently of the event-model fit.

17.4.1 — Observation Intensity $\lambda_i(t)$

Observation intensity describes the rate at which a person becomes visible to the study. It may depend on prior reports, care setting, device state, staff availability, medication, distress, or participant choice, so it cannot be treated as an administrative schedule outside the outcome model. The protocol should estimate or bound that rate on the information available at each time (Andersen et al., 1993; Rizopoulos, 2012).

The analysis file carries the prospectively specified observation-intensity model, observed history, visit timing, event marks, state-dependent measurement, and version identifiers. These fields let another team reconstruct which opportunity process and estimand were actually analyzed.

Observation-intensity model. Role: candidate model for when a participant becomes observable under recorded history; it neither defines the endpoint nor reads an unobserved phenomenal state.

$$\lambda_i^{\text{obs}}(t \mid H_i(t^-)) = \lim_{\Delta t \downarrow 0} \frac{\Pr\{N_i(t + \Delta t) - N_i(t) = 1 \mid H_i(t^-)\}}{\Delta t}$$

The history available at a given time can include prior reports, device state, clinical status, medication, care setting, earlier nonresponse, site, and participant pause state. Once those variables predict observation, the record is sampled through a gate rather than simply received from lived time.

A reproducibility check should ask whether observation rate can create or hide trajectory shape. When the test is positive, the finding cannot be presented as a clean property of the latent path.

Evaluation should compare outcome models with and without the observation-rate model. If the apparent dynamic weakens once intensity is included, the result is sensitive to the observation model; an observation-artifact interpretation requires the declared identification and model checks. The diagnostic is productive because it tells the program what not to overclaim.

Observation intensity tells the study when the world decided to look. Without it, a flare in reports, a quiet interval, or a terminal decline can be mistaken for the person rather than the schedule. The rate of being observed is itself one of the variables that makes longitudinal evidence fair or unfair.

17.4.2 — Observation Marks

An observation mark records what kind of event produced a datum or an absence. Scheduled prompts, participant-initiated reports, staff-triggered assessments, refusals, device failures, removals, and clinical interruptions occupy different selection pathways. A graph that treats them as interchangeable points suppresses part of the data-generating system (Little & Rubin, 2019; Scharfstein et al., 1999).

Observation marks should distinguish scheduled opportunities, participant responses, refusals, pauses, technical failures, staff interruptions, transfers, and device removal. The mark set makes selection analyzable and permits a direct test of whether the apparent trajectory changes when the observation process is modeled. Unmarked absence forfeits that possibility.

The same pain score has a different evidential route when it is collected during a crisis, at a scheduled visit, after a device alert, or because the participant chose to report. Mark-specific analyses can reveal whether the apparent change belongs to experience or to the circumstance that caused measurement. The route to the observation remains attached to the observation.

The analysis should make it possible to test whether unmarked missingness destroys interpretability. That outcome calls for a narrower estimand or for abstention, not a reassuring footnote.

Evaluation should hold out mark types and ask whether conclusions change when refused, failed, staff-initiated, or clinically interrupted measurements are separated. A model that depends on treating all absences alike has not earned an interpretation of the person.

Observation marks remain at the datum level in a versioned dictionary, including unknown and mixed categories. Analyses compare scheduled observations, staff-triggered events, refusal, failure, clinical interruption, and device removal. Categories may be collapsed only when simulation and sensitivity analysis show that the merge cannot change the protocol-fixed interpretation.

Marks turn missingness from fog into grammar. A skipped prompt, a device failure, a nurse-initiated measurement, a participant refusal, and a clinically interrupted visit do not mean the same thing. The study that records the mark preserves the possibility of knowing why the line broke.

Once marks are defined, the analysis can model mark-specific intensities and transition probabilities instead of merely stratifying rows. A marked point process asks whether scheduled reports, self-initiated reports, clinician-triggered assessments, and device alerts accelerate or decelerate under the same modeled conditions. Goodness-of-fit checks should test event timing and mark type jointly; otherwise a model may fit measured values while missing why measurement occurred (Andersen et al., 1993).

Mark vocabularies require governance. New device states, care settings, language changes, and protocol deviations should enter a versioned dictionary rather than free text that is harmonized after outcomes are seen. Unknown and mixed marks must remain visible. The temptation to collapse them for convenience should be resisted when the collapse could hide distress-triggered sampling, staff triage, or participant-controlled silence.

17.4.3 — Informative Presence and Absence

Presence and absence are informative whenever the probability of observation depends on the state being studied or on variables related to it. A long record can therefore overrepresent lucid, supported, comfortable, or highly distressed intervals while making other states nearly invisible. Visibility profiles should be

displayed before outcome trajectories are interpreted (Røysland, 2011; Rubin, 1976).

The analysis should preserve response, nonwear, visit, transfer, and interruption processes separately, then specify which may depend on observed history and which may depend on unobserved current state. Pattern-mixture, selection, and shared-parameter formulations represent different observation worlds. Agreement across them is evidence of robustness; disagreement defines the uncertainty.

A participant who answers reliably when calm and becomes silent during severe distress creates an apparently stable series. Another who answers mainly when distressed creates the opposite distortion. The observed values can be accurate in both cases while the trajectory is wrong because the windows of visibility differ.

External reviewers need a direct test of whether observed data can overrepresent easier states. If the concern is supported, its contribution belongs in the primary interpretation and the claim must contract.

A strong analysis tests whether observed states differ from states near refusal, nonwear, missed visits, or clinical interruption. Pattern-mixture and selection-model variants should be treated as rival observation worlds. If the signature lives only in one world, the verdict is conditional.

Visibility maps, response-rate plots, nonwear intervals, and cause-specific marks should appear before any smoothed outcome figure. The result must state which states were difficult

to observe and how the conclusion changes under plausible missingness mechanisms. A missing interval is never assigned neutral experience by default.

Observation patterns help explain which experiences enter the record. People may respond differently when distressed, lucid, comfortable, alone, or assisted by staff. Modeling both presence and absence makes those differences part of the evidence and keeps unobserved intervals from being assigned neutral values.

17.4.4 — Joint Mark and Outcome Models

Joint mark and outcome models ask whether the observed values and the opportunities to observe them are driven by shared state. The outcome process, counting process, mark distribution, and any shared latent variables should be declared separately so that the model cannot use its own observation rule to manufacture fit (Andersen et al., 1993; Rizopoulos, 2012).

The chosen model must match the estimand. Forecasting the next observed report, estimating a latent trajectory under informative visits, and comparing mark-specific outcomes require different likelihoods and assumptions. Simpler outcome-only and mark-only artifact models remain in the tournament so that added complexity is not rewarded merely for absorbing the observation process.

Joint observation-outcome scaffold. Role: candidate analysis model linking measured outcomes, observation opportunities, and declared event marks to examine informative visibility.

$$y_{ij} = m_i(t_{ij}) + \varepsilon_{ij}$$

$$Pr\{dN_i(t) = 1|H_i(t)\} = \lambda_i^{obs}(t)dt + o(dt)$$

$$Pr\{M_{ij} = k|dN_i(t_{ij}) = 1, H_i(t_{ij})\} = \pi_{ik}(t_{ij}|H_i(t_{ij}))$$

No positive verdict is warranted if the model can define its own opportunity to be right. The analysis plan must state beforehand what verdict follows when this condition occurs.

The joint model should be compared with a simpler outcome-only model and a mark-only artifact model. When the mark process explains the apparent trajectory, the study has found a design dependency rather than a Law-specific signal. That result belongs in the public record.

The recorded observation opportunities remain distinct from the model for their probability or intensity. A joint observation–outcome model may allow that intensity to depend on latent experience or shared clinical state under explicit identification and sensitivity assumptions. This dependence is not circular by itself. Circularity arises when outcome-dependent inclusion, endpoint selection, or tuning forces the signature later presented as evidence.

The joint model tests whether observations and outcomes change together. Representing that dependence can improve estimates and reveal when an apparent trajectory is explained by the observation process. Either result gives the next study a more informative design.

17.4.5 — Sensitivity to Missing-Not-at-Random

Missing-not-at-random sensitivity begins from the recognition that unobserved states cannot be recovered from observed data

without assumptions. The protocol should identify clinically and behaviorally plausible mechanisms, then express them through interpretable sensitivity parameters rather than a single opaque adjustment (Little & Rubin, 2019; Scharfstein et al., 1999).

The sensitivity plan fixes model families, tipping parameters, channel-specific scenarios, and the outputs that trigger narrowing or No adequate estimate. Mechanisms tied to pain, relief, fatigue, privacy, sedation, fear, device intolerance, and family protection receive clinically interpretable parameter ranges before the preferred result exists.

A terminal prompt may be unanswered because the participant is asleep, distressed, sedated, unable to communicate, shielded by family, or simply uninterested. Those mechanisms imply different distributions for the unseen value and different ethical meanings. Treating them as one generic missing observation can create a false terminal pattern.

The result becomes nonadjudicable when plausible missingness assumptions reverse its sign, route ranking, or equivalence decision. The main report should display that dependence, identify which unobserved quantities drive it, and assign the resulting evidence state under §24.2 and Table 27.5 (Little & Rubin, 2019; Scharfstein et al., 1999).

The sensitivity parameter should have an interpretable scale, such as the change in response odds or latent outcome associated with an unobserved state after conditioning on measured history. The analysis then varies that parameter over a clinically justified region and records where sign, equivalence, model choice, or

decision status changes. A number without a plausible mechanism is not a useful sensitivity analysis.

No single missing-not-at-random model is credible enough to carry the conclusion alone. The analysis should span a sensitivity family: selection models in which observation depends on unobserved current burden, pattern-mixture models that shift unobserved outcomes relative to observed ones, and shared-parameter models that link dropout to a latent health process. Each family should be anchored by clinically interpretable sensitivity parameters rather than abstract coefficients alone (Little & Rubin, 2019; Scharfstein et al., 1999).

Missing-not-at-random sensitivity should be reported as a tipping region rather than a ceremonial appendix. The figure should display the predeclared plausible region, the boundary at which the component verdict changes, and the assumptions outside which no estimate is defended. Stability across an implausibly narrow range is not robustness; stability across a justified range is stronger but remains conditional on that range (Little & Rubin, 2019; Scharfstein et al., 1999).

17.4.6 — Observation-Process Falsifiers

Observation-process falsifiers test whether the measurement process explains the proposed pattern. Mark restriction, intensity adjustment, simulated shutdown, nonwear scenarios, staff controls, device strata, and prespecified missing-not-at-random sensitivity analyses belong in the confirmatory package (Røysland, 2011; Rubin, 1976).

Each signature is paired prospectively with a falsifier and consequence. A pattern lost after representing staff-triggered measures, device removal, or interruption is observation-dependent; analysts may not search filters until one preserves it.

Reports test high-coverage intervals, alternate marks, weighting or joint-model adjustment, simulated shutdown, device-removal worlds, and staff or site controls. These checks distinguish trajectory from opportunity to observe.

Frozen falsifiers are reported whether they preserve or destroy the interpretation. Indistinguishable observation models permit only model-ambiguous, narrowed, or no-adequate-estimate status. A falsifier that fails because the observation model is nonidentifiable should downgrade adjudicability rather than be counted as evidence for the preferred latent process.

A joint model should reproduce outcome and observation. Checks compare visit intensity, missing runs, marks, terminal dropout, and covariate-dependent response; fitting values without who was observed and when is not recovery (Little & Rubin, 2019; Rizopoulos, 2012). Posterior predictive checks should reproduce both value trajectories and contact patterns under the same fitted system rather than validate the two processes separately.

Observation failure lowers confidence in every trajectory using the same assumptions and may redirect work toward devices, workflow, visibility, or missingness. That redirection is substantive. If visibility failure is site-specific, redesign can target workflow or devices without changing the latent model unnecessarily. A useful redesign test asks whether improving

observation reliability changes the apparent trajectory while leaving the latent model fixed; if it does, the earlier pattern was partly a property of visibility.

The framework now opens ethically available clinical and institutional variation while keeping prediction, causation, and care authority distinct. This separation allows naturally occurring variation to inform mechanism tests while preserving ordinary clinical decisions as causes to model, not interventions owned by the study.

17.5 — Clinical Cohorts and Natural Experiments

Clinical cohorts give the program access to repeated measures around illness, treatment, care transitions, and sometimes the approach to death. They are indispensable because terminal and serious-illness questions cannot be answered ethically by laboratory manipulation (Hernán & Robins, 2020; Lipsitch et al., 2010).

Natural experiments are useful when some external change alters exposure or timing without being chosen by the person's underlying state. The phrase natural experiment should not be used unless the assumptions are explicit and criticizable.

This section separates predictive from causal questions and care from theory. A clinical cohort may forecast risk, describe trajectories, estimate treatment-associated changes, or test whether a candidate signature survives clinical confounds. It does not authorize treatment manipulation.

17.5.1 — Prospective Clinical Cohorts

A prospective clinical cohort can plan clocks, measurements, and observation marks before the record becomes fragmented, but standard care still determines the setting. The cohort should define the clinical population, entry criteria, baseline state, treatment context, and component estimand without implying that follow-up under care reveals a complete life (Hernán & Robins, 2016, 2020).

The analysis architecture should encode baseline state, diagnosis, treatment plan, care-process metadata, and burden budgets. The

goal is traceability across people, time, sites, devices, and analytic versions, not accumulation for its own sake.

A palliative cohort may follow sleep, pain, medication, report availability, and care transitions while leaving treatment decisions entirely with the clinical team. Its scientific value comes from prospective timing and provenance. Its main vulnerability is that enrollment, prognosis, site resources, and communication ability determine who can enter and remain visible.

Public writing supplies another observational window on terminal illness. Goranson et al. (2017) analyzed 2,616 posts from twenty-five people blogging about terminal cancer or amyotrophic lateral sclerosis. An exploratory analysis within these blogs found that positive-emotion words increased as death approached, while negative-emotion words did not show a significant increase. This supplies an observed positive pattern worth following up with direct, voluntary measures of current experience. The bloggers were a selected group able and willing to write publicly; their language could reflect coping, relationships, audience, and meaning-making as well as felt state.

Model selection into the cohort and continued observation alongside the outcome process. Compare these models under explicit identification and sensitivity assumptions to assess whether sampling, the proposed mechanism, or both can explain the observed pattern. If their observable predictions remain equivalent, report model ambiguity.

Clinical cadence follows care first, with the smallest additional research layer that can answer the estimand. The protocol names

the source population, care context, entry clock, observation window, and estimand in terms preserved in public reporting. It separates routine visits, symptom-triggered contacts, remote prompts, and emergency encounters because each creates a different observation opportunity. If those clocks or care demands make the component uninterpretable, narrow the cohort or estimand rather than adding burden.

A prospective cohort is powerful because it can plan before the record fragments. Clocks, marks, treatment variables, burden limits, and No adequate estimate routes are specified in advance, with standard care and participants' comfort governing the observation schedule.

Existing longitudinal studies show why age and remaining time should be modeled separately. Among 1,637 deceased older adults in the German Socio-Economic Panel, distance to death described life-satisfaction change better than chronological age; the estimated decline accelerated roughly four years before death (Gerstorf et al., 2008). This supplies a quantitative timing comparison for models that also include health, care, social participation, and report availability. A prospectively specified candidate can be tested for additional predictive value alongside these measured variables.

17.5.2 — Treatment Changes

Treatment changes are time-varying exposures that may alter experience, physiology, report capacity, and the probability of observation at once. Medication, therapy, procedures, oxygen, sedation, counseling, and symptom control therefore require

precise timing and role declarations as exposure, confounder, mediator, observation modifier, or competing event (Pearl, 2009; Lipsitch et al., 2010).

The versioned record needs medication, therapy, procedures, sedation, treatment-confounder history, and landmark definitions tied to exact dates. These fields allow analysts to distinguish a change in the person from a change in treatment, eligibility, observation intensity, or time origin before those discontinuities are smoothed into apparent continuity (Hernán & Robins, 2020; Rizopoulos, 2012).

A pain medication change may reduce physiological variability while also reducing active reports through sleepiness or communication difficulty. Modeling lived response and report availability separately helps explain those changes. Treatment timing and channel-specific consequences remain visible in the analysis.

The first alternative to test is that treatment can affect both experience and measurement. A favorable fit for this account is not a nuisance; it is a reason to report a rival-favored or model-ambiguous result.

Treatment models need exposure timing at the resolution of the likely effect. Sedation, analgesia, therapy, oxygen, medication taper, or procedure recovery may operate on minutes, hours, days, or weeks. A coarse time grid can make a treatment effect appear delayed, absent, or terminal when it is merely undersampled.

The analysis uses time-dependent exposure histories at the resolution of the plausible effect and compares alternative

treatment-confounder structures. Landmarking, marginal structural approaches, and joint models suit different questions. Record clinical indications separately so the interpretation reflects how treatment was chosen.

Treatment change is never a nuisance variable in this chapter. Medication, therapy, oxygen, sedation, surgery, counseling, or symptom control may change both lived experience and observability. A study that treats care as background can mistake good medicine, difficult side effects, or clinical transition for a Law-specific trajectory.

Recent studies sharpen the empirical comparison. Wang et al. (2025) followed 1,971 older adults, 1,119 of whom died, and found different terminal change points across psychological and social domains, approximately six to eleven years before death. Social-network size did not share that change-point pattern. Petkus et al. (2025) analyzed 2,411 participants and 12,574 depressive-symptom observations, finding acceleration approximately four to five years before death. In ninety-eight mortality-discordant twin pairs, the earlier-deceased twin showed a steeper terminal symptom increase than the co-twin. The twin comparison helps distinguish terminal-related change from shared age and family background, while leaving nonshared illness and environmental changes to be modeled. The paired comparison distinguishes some competing explanations. For PTVC-linked work, its value is a stronger comparison design: test any additional proposed signed-history structure against these already informative terminal models.

17.5.3 — Natural Experiments

Natural experiments use variation created outside the research assignment, such as policy thresholds, scheduling rules, phased rollouts, supply changes, or staffing patterns. Their value depends on an explicit causal graph or target-trial analogue, a defensible identifying assumption, stable measurement around the change, and evidence that the variation is not simply a proxy for unmeasured participant state or site selection (Hernán & Robins, 2020; Lipsitch et al., 2010; Pearl, 2009).

The minimum reproducible package contains policy shifts, scheduling rules, rollout timing, pre-trends, spillovers, and site differences. Those entries permit a skeptical reanalysis without asking the reader to trust an undocumented data pipeline.

A clinic may change appointment scheduling for operational reasons, shifting when observations occur without assigning participants to a research intervention. That change can help identify an observation or care-process effect only if pretrends, anticipation, spillovers, and concurrent changes are examined. The label natural does not remove confounding.

Natural-experiment cadence should be anchored around the exogenous change and its plausible anticipation window. If people know a policy, staffing, or scheduling shift is coming, the pre-period may already be treated. The design should model anticipation rather than pretending the clock begins at implementation.

The protocol should prestate the local causal or descriptive estimand, the source of variation, the exclusion or

exchangeability assumptions, the anticipation window, and the falsifiers that would invalidate the design. Results are reported as local to the setting and period that generated the variation unless transport is separately established.

The design should try to falsify its own assignment story. Pretrend breaks, anticipation, manipulation around thresholds, co-occurring policy changes, differential measurement, and negative-control outcomes can each expose a false natural experiment. When those checks fail, the result remains a descriptive association tied to the observed policy change rather than a causal estimate.

A policy shift, formulary change, device rollout, staffing change, or care-pathway transition becomes a natural experiment only when its assignment process is understood. Pretrends, anticipation, co-occurring changes, negative-control outcomes, and site-specific implementation should be examined before causal language is used. Administrative origin does not make a change random when prognosis or clinician judgment helped determine exposure (Hernán & Robins, 2020; Lipsitch et al., 2010).

The report also needs a target population and transport statement. It should show uptake, crossover, implementation fidelity, eligibility, and the covariate range shared across comparison groups. Where overlap fails, the result remains local, descriptive, or partially identified rather than being carried into the whole life-course program by narrative analogy (Dahabreh et al., 2019; Pearl & Bareinboim, 2014).

17.5.4 — Clinical Heterogeneity

Clinical heterogeneity can change both the phenomenon and the observation system. Diagnosis, disease course, treatment options, communication capacity, care setting, disability, culture, family support, and site practice may modify dynamics, measurement validity, and observation probability in different directions. The protocol should declare which axes define separate scientific strata and which may be modeled as effect modifiers (Hernán & Robins, 2016, 2020).

The protocol should preserve strata, partial pooling, site variables, diagnosis-specific clocks, and care ecologies. They expose whether the conclusion depends on survival, site, device, model version, or a hidden analytic choice.

Home hospice, inpatient hospice, intensive care, rural clinics, and chronic-pain services may produce different records before any difference in lived experience is considered. Visit density, device availability, symptom management, staffing, and family involvement shape who is observed and when. Site and clinical strata should therefore be displayed before partial pooling is interpreted.

Pooling fails when it creates an average process that no clinical group follows or when one high-observation group dominates the estimate. Hierarchical models may borrow strength, but the amount borrowed should be audited through leave-group-out checks and stratum-specific residuals. Heterogeneity that changes the verdict remains part of the result.

Sampling cadence, channel qualification, and missingness assumptions should be adapted to the clinical process rather than standardized for visual symmetry. Rapid symptom fluctuation, slow degeneration, episodic crises, and communication decline require different designs. A common protocol is useful only where it preserves a common scientific object.

Pooling should be earned through measurement invariance, overlap, calibrated partial pooling, and posterior checks that preserve subgroup failure. A pooled coefficient is unavailable when the same variable changes meaning, sign, cadence, or observation mechanism across strata. In that case, separate estimates or identified sets are more honest than a precise average of unlike objects.

17.5.5 — Causal and Predictive Questions

Prediction and causation answer different questions. A model may forecast a future report or care need accurately while remaining unable to estimate what a treatment, support, or environmental change caused. The protocol should state which task it owns and prevent performance on one task from lending authority to the other (Pearl, 2009; Lipsitch et al., 2010).

For an intervention question, the target trial should be written first: eligibility, time zero, treatment strategies, assignment procedure or emulation rule, follow-up, intercurrent events, outcome, causal contrast, and analysis. Time-varying confounding, positivity, consistency, interference, censoring, and measurement assumptions are then stated separately. Forecasting

may use the same cohort, but it remains a different estimand and validation task (Hernán & Robins, 2020).

A model may predict lower future variance after medication because medication marks disease severity, clinical attention, or survival rather than because it caused the change. Predictive success can still be useful for planning support. It becomes misleading when the result is translated into an intervention effect that the design did not identify.

The analysis report should label association, prediction, causal effect, feasibility, and component evidence at every table and figure. A model that predicts well but fails causal falsifiers remains a predictive model. A causal estimate that transports poorly remains local even when its mechanism is plausible.

Uncertainty should distinguish prediction error from causal bias. A model may forecast need accurately while still failing to estimate what treatment changed. The report should not let predictive success borrow the language of intervention effect.

Validation and decision rules are task-specific: blocked temporal tests for forecasting, negative controls and sensitivity analyses for causal claims, and held-out populations for transport. When the tasks disagree, the study reports the disagreement. No hierarchy makes causal language automatically more important or predictive language automatically safer.

Unexpected timing offers a concrete observational comparison for horizon-dependent candidates. Ito et al. (2021) followed 1,896 patients in twenty-three Japanese hospice or palliative-care units. Four definitions of sudden or unexpected death produced thirty-

day cumulative-incidence estimates between 6.4% and 16.8%. That variation makes definition choice part of the scientific question. A prospective extension could add affective and contextual observations, freeze a prognosis model using only information available before the final decline, and compare a prespecified candidate across more- and less-expected biological deaths. Clinician surprise, physiological change, care, and missingness would remain explicit comparison variables.

17.5.6 — No Care Manipulation for Theory

Chapter 3's no-harm and care-first rules remain controlling. Longitudinal designs may observe clinically justified treatment and support, but may not alter care to improve trajectory interpretability. Any datum lost because comfort medication, device removal, or a stop request took priority remains ethically unavailable and is handled through the prespecified missing-data or no-estimate branch (Council for International Organizations of Medical Sciences, 2016; International Council for Harmonisation, 2026; World Medical Association, 2024).

Any design whose discriminating power depends on care delay, induced distress, avoidable sleep disruption, denial of support, or a counterfactual that cannot be observed ethically fails during simulation and protocol review. An informative question does not become permissible because the missing contrast would be scientifically valuable.

Analysis should represent care-triggered observation marks and ask what narrower estimand remains identifiable. Clinical authority, participant refusal, comfort, and rescue outrank the

research schedule at every time. Trust is earned by accepting a smaller answer rather than engineering around humane limits.

The line is operational: if a treatment, visit, sedation decision, monitoring interval, or support change would not be justified without the theory, the study should not perform it for the theory. The chapter may learn from care as it occurs, but care never becomes an instrument for the chapter. This constraint also protects external validity, because a trajectory that appears only after research has altered ordinary care would not describe the clinical process the study claims to learn from.

17.6 — Truncation by Death and Competing Events

Death is not ordinary dropout. After truth-level stream endpoint, future felt outcomes are undefined for that stream. A study that fills post-death values or hides death inside a composite endpoint can make a trajectory look continuous after the object of observation has ended (Frangakis & Rubin, 2002; Fine & Gray, 1999).

Events that interrupt observation also matter before death. Transfer to another facility, discharge, withdrawal, loss of device access, communication loss, sedation, and family protection of quiet can end observation without ending the person.

This section draws on survival analysis and semi-competing risks because longitudinal outcomes in serious illness often depend on whether the participant remains alive and observable. The methods are tools for preventing false evidence created by survival, observation, and endpoint structure.

Table 17.4. Death-alignment artifact simulations and false-signature diagnostics.

Simulated truth	Observation rule	Naive display	Artifact risk	Required diagnostic
No terminal compression	Fewer measures as weakness rises	Backward alignment to death	False narrowing near endpoint	Compare to a null with same missingness.

Simulated truth	Observation rule	Naive display	Artifact risk	Required diagnostic
Medication dampens signals	Monitoring removed after sedation	Last observed values carried visually	Apparent neutrality from care process	Model sedation and device removal as marks.
Different diagnoses decline at different speeds	Visits cluster around crises	Dynamic time warping aligns crises	Timing differences disappear	Report warped and unwarped clocks.
Sudden and slow deaths mixed	Sudden cases lack terminal window	Only slow deaths plotted near T	Survivor-like terminal sample	Name target: observable terminal windows.

17.6.1 — Death Truncates Future Outcomes

For an outcome defined only while alive, death ends its clinical domain. The protocol defines the relevant while-alive, terminal-window, event, or joint estimand and treats the stream endpoint as a separate specified object. Values are estimated only within the domain of the chosen outcome (Frangakis & Rubin, 2002; Fine & Gray, 1999).

Stopping rules for each outcome, endpoint adjudication, last-observation handling, and the distinction among biological death, truth-level stream endpoint, and loss of observation are prospectively fixed. Figures end where the estimand ends rather than extending a line for visual completeness.

A graph that continues the final report across days after death can look stable and reassuring, but the line is an artifact of the plotting

rule. The same problem arises when zeros are inserted after death or when death is coded as an extreme symptom. The protocol should show the terminal event explicitly and remove undefined follow-up from the outcome scale.

The audit trail must reveal whether post-death imputation, survivor-only smoothing, or endpoint-aligned averaging can manufacture apparent neutrality. Simulations should replay those operations under known non-PTVC generators and show their false-signature rate. If contamination cannot be bounded, the stronger verdict is blocked (Kurland et al., 2009; National Academies of Sciences, Engineering, and Medicine, 2019).

Uncertainty after biological death includes uncertainty about pre-death records, observations, timing, and events, while the truth-level stream endpoint remains a separately specified question. The analysis should not assign post-death values to make a longitudinal method convenient.

Analysis stops the lived-outcome process at the relevant endpoint and uses event-specific methods for what follows. While-alive summaries, cumulative incidence, joint models, and survivor-conditioned quantities are labeled by the population and time they describe. None is allowed to imply an unobserved post-endpoint experience.

Death is not a missing data point after which the curve may be quietly extended. It ends outcomes defined only while alive, while the truth-level stream endpoint remains a separate question and records, memories, care processes, and survivor reports

remain different objects. The design must name which object remains before it computes anything.

17.6.2 — Semi-Competing Risks

Semi-competing risks arise when a nonterminal event can occur before death but cannot occur afterward, while death may occur without the nonterminal event. The estimand must therefore specify whether it concerns transition into the nonterminal state, death without that state, time spent alive and observable, or the joint ordering of events. Ordinary right censoring does not preserve those distinctions (Fine et al., 2001; Kurland et al., 2009).

The protocol should define states, allowed transitions, event ordering, and the clock used for each transition. Illness-death and multistate models can represent whether a nonterminal event precedes the terminal event, but their interpretation still depends on observation and endpoint qualification. The state diagram is a scientific object, not a decorative flowchart.

Sedation may end active report while biological life and clinical care continue. That transition is neither death nor ordinary missingness. The analysis should distinguish loss of report, loss of device access, change in consciousness evidence, and truth-level endpoint uncertainty rather than assigning one terminal code to all of them.

An independent review should check whether nonterminal and terminal events have been collapsed into generic dropout. Any consequential collapse triggers a documented correction in the analysis and participant-protection plan.

Uncertainty in semi-competing risks concerns both event order and dependence among transitions. A hospitalization before death, device removal before transfer, and withdrawal before communication loss imply different risk sets and different observation opportunities. The analysis should show which ordering is observed, interval-censored, or structurally impossible rather than collapsing all pre-death events into one state.

Event order is recorded and modeled explicitly, with sensitivity analyses for uncertain timing and competing definitions. A result that changes when nonterminal and terminal events are separated is reported at that event-specific level. Multistate detail is retained only when the data can support it; otherwise the study narrows to a simpler, honest estimand.

A common-clock event history should preserve the sequence of hospitalization, delirium, device removal, transfer, withdrawal, communication loss, and death. Transition intensities may depend on prior events, so the model should allow one event to change both the state process and the chance of later observation. The event history is part of the scientific object, not merely a censoring code (Aalen et al., 2008; Andersen et al., 1993).

The model family should follow the question. Cause-specific or transition-specific hazards describe movement among states; cumulative-incidence functions describe realized event probability in the presence of competing events; illness-death and joint models represent dependence under additional structural assumptions. Reporting one of these as though it answered all three questions is a category error (Andersen et al., 1993; Fine & Gray, 1999; Fine et al., 2001).

A multistate model estimates transition patterns under a declared model and population. Its scientific value lies in distinguishing those event processes and the outcome domains attached to each state.

17.6.3 — Transfer, Discharge, and Withdrawal

Transfer, discharge, and withdrawal end one observation route without necessarily ending care, life, or conscious experience. The study should preserve the reason, timing, permission status, and destination of each exit so that institutional boundaries are not mistaken for scientific endpoints (Andersen et al., 1993; Hernán & Robins, 2020).

The transport file must retain cause-specific endings, withdrawal rules, registry linkage, and context change. The resulting file supports honest transport analysis and exposes assumptions that would otherwise remain implicit.

A participant may transfer to another facility where clinical records continue but research permission does not. The study must stop the research stream unless a separately authorized linkage applies. Continuing the row because data are technically accessible would erase the difference between care continuity and research authority.

The design must answer whether separately authorized linkage can continue observation after the setting and permission change. If it cannot, inference stops at the supported window. The local result retains the status its evidence earns; any continuation claim is classified according to its remaining identification and uncertainty.

Evaluation should leave out transfer routes, discharge classes, and withdrawal categories before claiming transport across endings. If the finding disappears when one exit route is separated, the exit is part of the data-generating process and cannot be buried in generic dropout.

Every ending receives a typed mark, including transfer, discharge, voluntary withdrawal, loss of contact, site closure, device loss, and endpoint. Analyses test whether conclusions depend on grouping those routes together. When the destination or permission status is unknown, the uncertainty remains visible rather than being recoded as generic dropout.

Treat care change, burden, privacy, family decision, institutional boundary, and research fatigue as typed causes of exit rather than generic dropout or a scientific endpoint.

17.6.4 — Survivor and Immortal-Time Bias

Survivor and immortal-time bias arise when comparison requires a person to remain alive, enrolled, or observable long enough to enter a group. The apparent stability of a later cohort may therefore reflect the conditions required to survive or comply, not a common change in experience. Risk sets must be defined before trajectories are compared (Frangakis & Rubin, 2002; Fine & Gray, 1999).

The evidentiary dataset should carry landmarks, time-dependent exposure, risk sets, and survival-conditioned analyses. Such provenance is necessary to separate an effect from the infrastructure that generated the data.

Participants who complete thirty days of prompts may appear more stable simply because people with rapid decline, transfer, distress, or device intolerance could not complete them. Comparing completers with noncompleters after the fact makes future survival part of group membership. Landmark and time-dependent analyses are needed to restore a fair comparison.

The local claim collapses if the easiest lives to follow can become the apparent theory. Because the failure can mimic support, it belongs in the confirmatory gate rather than in a post hoc limitation paragraph.

Evaluation should use landmarked risk sets and time-dependent exposure definitions rather than comparing people who could not have had equal observation time. The test should ask whether the same result holds before survival has selected the sample that makes the trend easy to see.

Every comparison reports its risk set, eligibility rule, and required survival time. Immortal periods are assigned correctly, and analyses repeat at prospectively chosen landmarks. A trajectory visible only among selected survivors is labeled survivor-conditioned.

Late trajectories are most informative when their selection conditions are visible. Stability among participants who remain able to respond may reflect retained capacities, observation selection, or both. Comparing those explanations helps establish which population the finding describes.

17.6.5 — Principal-Stratum and Survivor-Average Limits

Principal-stratum and survivor-average estimands address carefully restricted questions under truncation. They may describe people who would survive or remain observable under specified conditions, but membership is usually not directly observed and requires strong assumptions or bounds. Their narrowness is the meaning of the estimand, not a defect to hide (Kurland et al., 2009; Rizopoulos, 2012).

The evidentiary record requires principal strata, survivor average effects, open-channel strata, sensitivity ranges. Their purpose is to preserve the difference between the phenomenon, its observation process, and the analyst's reconstruction.

A terminal-window analysis among people who remain communicative and device-tolerant describes that defined subgroup. The principal result names its population and selection process; separate analyses address rapid communication loss, device burden, and other care pathways.

The study should stop short of interpretation when survivor-specific answers could be mistaken for answers about everyone. The scientific cost is a narrower claim, and that cost should be paid in the headline result.

The target stratum should be defined before fitting in terms of the counterfactual survival or observability condition that gives the estimand meaning. Because actual membership is generally unobserved, point identification requires assumptions that must be named rather than hidden inside the model. The resulting claim

belongs to that latent stratum, not to all enrolled participants (Frangakis & Rubin, 2002; Zhang & Rubin, 2003).

Sensitivity analysis should vary the assumptions linking observed covariates, survival, observability, and latent stratum membership. When plausible assumptions yield materially different estimates, the study reports bounds or multiple stratum-specific results rather than selecting the most favorable specification. A survivor-average result is never translated into an average for the complete enrolled cohort.

17.6.6 — No Composite That Hides Death

Composite outcomes can simplify a table while hiding the event that determines meaning. Death, withdrawal, transfer, device failure, deep sedation, distress-triggered discontinuation, and nonresponse differ in irreversibility, information loss, and clinical significance. They should remain typed before any summary is considered.

The estimand must name whose trajectory it concerns. A while-alive mean, survivor-average quantity, composite endpoint, principal-stratum contrast, joint longitudinal-survival parameter, and component-specific cumulative incidence answer different questions. None is a neutral default, and the choice should follow scientific meaning rather than the curve that looks cleanest (Frangakis & Rubin, 2002; Kurland et al., 2009).

Multistate and competing-risk models preserve distinct pathways when the data support them. Death is represented as a terminal clinical event, while transfer and withdrawal retain the

documented routes through which observation ended (Andersen et al., 1993; Fine & Gray, 1999).

Joint models link longitudinal outcomes and event hazards through assumptions about shared random effects, functional form, and unobserved values. Sensitivity analyses should vary those links and compare selection, pattern-mixture, and joint formulations. Disagreement across plausible structures widens or changes the verdict; it should not be hidden behind one preferred model (Rizopoulos, 2012; Scharfstein et al., 1999).

A composite is interpretable only when its components share a defensible meaning and no component silently dominates. Component-specific incidence, event timing, risk sets, and leave-one-component-out analyses should appear beside the summary. If the interpretation changes when death or another event is removed, the component analysis governs. This prevents a stable composite score from concealing opposite trends in death, withdrawal, and other competing events.

Every death-truncated estimate should include a denominator panel showing the numbers enrolled, alive, observable, responding, and analyzed at each time, together with the counterfactual population invoked. The curve exists only because particular participants remained in the risk set, and that dependence should be visible (Fine & Gray, 1999; Kurland et al., 2009). Showing these denominators over time also reveals whether an apparently smooth trajectory is increasingly driven by a small selected subset.

Public language must follow the estimand. A while-alive average does not describe the unobserved experiences of those who died, and a survivor trajectory does not represent the full enrolled cohort. Precise naming prevents a mathematically valid conditional result from becoming a claim about lives the study did not observe. Where several estimands are scientifically useful, reporting them side by side can show how conclusions depend on survival conditioning rather than hide that dependence.

Models may estimate observation probabilities or bounded while-alive outcomes, but they cannot silently impute a complete phenomenal continuation after the target is no longer observable. When outcome, observation, endpoint, or target-existence uncertainty cannot be bridged, the appropriate result is narrower inference, an identified set, or No adequate estimate. Sensitivity analysis should state which unobserved continuations would be required to change the conclusion without presenting those hypothetical paths as measured experience.

17.7 — Versioning, Drift, and Transport

A life-course study is itself a life-course object. Its instruments age, staff change, sites learn, vendors update, populations shift, language versions improve, and clinical standards move. A long record without version accounting can confuse the history of the study with the history of the participant (Dahabreh et al., 2019; Pearl & Bareinboim, 2014).

Validation should follow the deployment question: forward within-person tests for future intervals, blocked temporal folds for new periods, leave-person-out tests for new individuals, leave-site-out tests for transport, and leave-device-version-out tests for technological migration. Random observation-level splits leak dependence and are outside the confirmatory rules.

17.7.1 — Measurement Version Ledger

A measurement version ledger makes the history of the instrument queryable alongside the history of the participant. Item wording, language, accessibility mode, device, firmware, algorithm, preprocessing, staff training, and consent materials are recorded at person-time resolution because each can alter the observed variable. This ledger is the provenance layer; transport analysis begins only after it can reconstruct which version produced every datum (International Council for Harmonisation, 2026; National Institute of Standards and Technology, 2020).

The ledger should distinguish changes that affect cadence, origin, unit, sign, missingness, or target population. Minor software patches and major construct revisions do not receive the same treatment. Each change is assigned a bridge, segmentation,

recalibration, or retirement decision before the post-change data are merged with the earlier series.

A wording update that reduces apparent burden halfway through follow-up may reflect improved clarity rather than improved life. Without the exact version dates and overlap evidence, the curve cannot tell the difference. The ledger lets analysts reproduce which question, device, and algorithm produced every value.

The hardest competing account is that unmarked version boundaries can create false path continuity. If it explains the data better, the result should be presented as evidence for the rival or as an unresolved model comparison.

A version ledger should include a null feature expected not to change with the person, such as a technical quality metric or calibration fixture. If that feature shifts at the version boundary, the study has detected the measuring system moving under its feet.

Measurement history must be retrievable for any observation, table, model, or public figure. Analysts should be able to rerun results by version and identify which claims depend on a superseded instrument. Longitudinal evidence is not reproducible when the measurement system has no memory of its own changes.

A version ledger should be paired with a dependency graph. Each change in question, language, device, firmware, preprocessing rule, feature extractor, endpoint classifier, or analysis package should point to the estimates, figures, and public artifacts it may affect. Long evidence needs this propagation map because the

measuring system changes while the cohort is still being followed.

17.7.2 — Site and Era Effects

Site and era effects include care philosophy, staffing, referral patterns, technology, population composition, language, and historical practice. They may change observation and outcome together, so a site intercept alone is often insufficient. The protocol should describe the setting as part of the scientific object (National Academies of Sciences, Engineering, and Medicine, 2019; Särkkä, 2013).

Site and era variables should enter the design at the level where they operate, with site-specific clocks, device ecosystems, care processes, and calendar changes preserved. Partial pooling is evaluated against leave-site-out and leave-era-out performance. A stable average does not establish that the same process generated each setting.

One hospice may protect quiet and collect few terminal measurements, while another documents every monitor reading. The resulting trajectories differ before any difference in participants is considered. A transport analysis must decide whether it is moving a person-level relationship, a care-system relationship, or a mixture of both.

The design must distinguish the target from the possibility that site effects can be care ecologies rather than nuisance intercepts. The analysis must be capable of returning this rival as the winner, with the target claim narrowed to the residue it actually explains.

Site checks should include leave-site-out prediction, era indicators, and at least one site-level negative control. If a hospice, clinic, or device vendor can predict the trajectory better than the substantive variables, the result belongs first to place and era.

Claims of transport are withheld until site and era effects have been modeled, stress-tested, and reported. If the signal is local to a care ecology, that locality becomes part of the finding. Place is not a nuisance when place helps generate the evidence.

Site and era effects should be modeled as potential interactions, not only as nuisance intercepts. A quiet hospice, crowded ward, home-care program, and technology-rich clinic may alter observation cadence, treatment pathways, device wear, family presence, and endpoint adjudication differently over time. Transport requires evidence that the mechanism and measurement bridge remain stable across those changes.

17.7.3 — Response and Device Drift

Response drift and device drift are distinct sources of nonstationarity. Participants may change standards, vocabulary, trust, or reporting habits while sensors age, firmware changes, and algorithms update. The study should include anchors and technical diagnostics capable of separating those movements (Shcherbina et al., 2017; Bent et al., 2020).

Anchor-stability tests, device-quality controls, overlap requirements, residual diagnostics, and thresholds for splitting or relinking a series are prospectively specified. Drift is adjudicated before substantive interpretation of the affected interval, and the

bridge method cannot be chosen because it preserves the preferred trend.

A participant may rate pain differently after learning the illness vocabulary, while a wearable update changes the apparent sleep score in the same month. A stable combined trajectory can therefore hide two moving scales. Parallel anchors and version-specific residuals are needed to locate the change.

An equal-status rival is that stable numerical scores conceal changing construct meaning. When the protocol-fixed comparison favors that account, the result is measurement-version or response-shift evidence, and Law-specific interpretation is withdrawn from the affected analysis.

Drift analysis should separate changing response standards from changing device behavior. Anchor items, overlap samples, device holdouts, and residual checks can show whether the person, the instrument, or both moved. The result should name the source of movement before it interprets direction.

When drift is detected, the series is segmented, bridged with independent evidence, or carried as an identified set. Uncertainty is widened only when it honestly represents the remaining ambiguity. A common scale is withdrawn when the person and device movements cannot be separated.

Drift monitoring should specify thresholds, detection delay, and action. Prompt learning, changing response standards, trust loss, vocabulary growth, sensor aging, and algorithm updates can produce gradual shifts that evade a single overlap check. The

study should test both abrupt and delayed drift before calling the pattern a human trajectory.

17.7.4 — Bridge Samples and Overlap

Bridge samples create evidence for linking old and new measurement systems. They require overlap in time, participants, states, or calibration fixtures and should include difficult cases where the mapping is most likely to fail. A bridge learned only from stable, high-quality observations may not apply at the states that matter most (Dahabreh et al., 2019; Pearl & Bareinboim, 2014).

The bridge plan should freeze parallel-device periods, dual items, overlap sample sizes, subgroup coverage, link functions, and acceptance criteria. The old and new systems are analyzed separately until the bridge passes held-out checks. Continuity is an empirical result, not a consequence of deploying the replacement on the next day.

Old and new wearables worn together during a transition can reveal systematic offsets, state-dependent error, and loss of sensitivity. The overlap should include motion, illness, sleep disruption, and other challenging conditions, not only quiet calibration periods. A bridge that works only in easy states cannot carry the full longitudinal series.

Linking fails when bridge error changes sign or scale across clinically relevant subgroups, devices, or states. The study should then retain version-specific estimates or narrow the target population. A single correction factor is not allowed to conceal differential transport failure.

Overlap is the honest price of merging old and new systems. A bridge period should be analyzed as its own estimand: how close are the two systems, for whom, under what contexts, and with what residual bias? Without that answer, a merged line is only a visual convenience.

Merged analysis begins only after the bridge meets prospectively declared accuracy and invariance criteria. Raw version-specific values remain available for audit and sensitivity analysis. When overlap is inadequate, the record contains an explicit break rather than an invented continuity.

Bridge samples should estimate more than a mean offset. Old and new instruments must be compared for nonlinear distortion, variance change, threshold movement, differential missingness, and subgroup-specific error. When those structures do not transport, the program reports a version break rather than hiding incompatibility under a merged line.

A bridge is useful only within its support. If the overlap sample contains healthy, highly connected users while the post-change cohort includes advanced illness or limited digital access, calibration may not transport. Transportability analysis should therefore compare covariate support, reweight the bridge where justified, and report regions with no overlap rather than extrapolate silently (Pearl & Bareinboim, 2014; Dahabreh et al., 2019).

17.7.5 — Transportability Analysis

Transportability analysis states where the model was learned, where it is intended to apply, which variables link the settings,

and which mechanisms are assumed stable. Performance in the source population is only the starting point; the target population supplies its own calibration, observation, and construct challenges (National Academies of Sciences, Engineering, and Medicine, 2019; Särkkä, 2013).

The reproducibility package should preserve source S, target T, bridge variables Z, selection, effect modification, and external-validity assumptions. The package should permit a skeptical reader to follow every transformation from observation to estimate.

Transport scaffold. Role: declaration of the source setting, target setting, and bridge variables required for a specific estimand; it is not evidence that the estimand travels.

The source setting S supplies the fitted relation, the bridge variables Z encode measured differences relevant to selection or effect modification, and the target setting T defines the population and observation conditions to which the claim is proposed to move.

The transport question is component-specific: after accounting for Z, measurement version, observation process, care context, and common-support limits, does the same estimand retain its meaning and calibration in T, or must the result remain local (Dahabreh et al., 2019; Pearl & Bareinboim, 2014)?

Numerical performance can travel while meaning does not. A model may preserve prediction rank but lose neutral origin, response unit, channel validity, or the relation between proxy and

lived experience. Transport claims should therefore report both predictive transfer and construct-level bridge evidence.

Transportability checks should ask whether the time grid itself travels. A cadence tolerated in a research hospice may be unsafe in home care, too slow for acute illness, or too intrusive in a work setting. If the original observation schedule cannot ethically be used in the target setting, transport requires a validated observation bridge for a permissible schedule.

Every transported result names source and target populations, estimand, bridge variables, overlap support, measurement and observation versions, and assumptions about effect modification. Extrapolation beyond common support is labeled model-dependent. When the target lacks the required bridge, the study reports nontransport rather than universalizing the source estimate.

A transport audit should return component-level outcomes rather than one global pass. Predictive ranking may travel while neutral origin, response unit, observation cadence, or reserve validity fails; a site effect may be transportable within one care system but not another. The final report therefore states which bridge passed, which failed, and which claim remains source-bound (Dahabreh et al., 2019; Pearl & Bareinboim, 2014).

17.7.6 — Reanalysis After Major Change

Changes in site, device, instrument, language, care practice, population, schema, or endpoint rule trigger reanalysis, not cosmetic updates. The change record should identify affected estimands, models, figures, public claims, validation layers, and

withdrawal conditions, plus the person or body authorizing the change (International Council for Harmonisation, 2026; National Institute of Standards and Technology, 2020).

Validation must match the new use. Forward-chaining, blocked-time, and person, site, era, or device-version holdouts expose dependencies hidden by random row splits. A lower honest score for that split supersedes a higher leaked score.

Reanalysis starts with the affected clock and context. Changes may alter sampling interval, time since event, device lag, burden, care workflow, or prompt meaning without changing the script. Label results confined to the old cadence or setting accordingly.

A measurement-version graph identifies each node's instrument, language, device, firmware, scoring rule, preprocessing pipeline, schema, and context. Bridge evidence must support every edge. Otherwise the graph remains disconnected, and earlier values cannot be silently standardized to the new version.

Bridge studies use prospective overlap when feasible, running old and new versions in parallel long enough to estimate differences in meaning, thresholds, error, burden, and order effects. A bridge established in a low-burden population does not automatically transport to serious illness or low-access states and must be retested when the intended claim reaches them.

Schema migration is scientific work. Renaming variables, collapsing categories, revising missing-value codes or units, and altering derivations can change an estimand without visible code changes. Machine-readable data dictionaries and derived datasets

should preserve source-schema provenance (Little & Rubin, 2019; National Institute of Standards and Technology, 2020).

Evaluate transport component by component. A measurement bridge may travel when a clinical prediction model does not; a state representation may travel when an endpoint classifier fails; a treatment contrast may require reweighting. Reports should identify which components crossed the boundary and which stayed local (Dahabreh et al., 2019; Pearl & Bareinboim, 2014).

The version ledger enables correction. If a calibration, instrument, code path, or public interpretation proves defective, the program should identify dependent estimates and artifacts, reanalyze where possible, and withdraw unsupported claims. Long-running studies earn trust by making changes and reversals inspectable, not by pretending every edition was always current.

17.8 — Purpose Limits and Participant Return

Life-course data are unusually revealing because time itself is identifying. A record of sleep, movement, prompts, medication, visits, pauses, transfers, and family context can describe a person more intimately than a single sensitive variable ever could (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

Longitudinal design is not complete until it states what data may be used for, what may not be linked, what may be returned, what can be challenged, how long records persist, and what happens when the study ends.

Federated analysis, secure computation, provenance, and minimization are part of scientific validity. They shape who participates, who remains observable, what can be audited, and how safely sites can collaborate.

17.8.1 — Operational–Research Firewall

The operational-research firewall separates data created for care or administration from data collected under a research purpose. Availability inside a clinical system does not supply consent, lawful authority, construct validity, or permission for secondary modeling. The protocol should document the route by which each data class enters the study and the purpose that limits its use (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

The technical file should make visible separate permissions for observation, linkage, sharing, publication, model training, and

reuse. These variables anchor the comparison to the actual participants, windows, instruments, and decisions that produced it.

A clinical note written to treat pain is not automatically permission to train or apply a momentary-valence estimator. The note may use shorthand, reflect a care decision, include bystanders, or be shaped by billing and workflow. Research use requires a separately approved purpose, minimization plan, access boundary, and validity assessment.

A replication team should be able to evaluate whether record linkage can create a cross-domain dossier. The finding is not portable until this question can be answered from the record rather than from investigator memory.

The firewall should specify the cadence of permission as well as the cadence of measurement. Consent for one extraction, one linkage, one analysis window, or one return of results does not silently authorize ongoing observation. Permission itself has a time structure.

Operational data cross into research only through a documented authority, context-of-use qualification, provenance record, and revocable access path where feasible. The transfer is auditable and does not expand the original record into a general-purpose archive. When those conditions are absent, the scientific question does not inherit the data.

The purpose, permission, and provenance route define whether a clinical, City, government, religious-community, or Workgame record is eligible for a particular research question.

17.8.2 — Participant-Facing Results

Participant-facing results are communications, not private scientific verdicts. Returned information should be valid for the individual, requested or consented to, understandable, and consistent with the study's clinical and ethical plan. Group findings and unstable research constructs are not personalized merely because a dashboard can display them (Rieke et al., 2020; National Academies of Sciences, Engineering, and Medicine, 2018).

The core record must include return classes, uncertainty labels, desired feedback levels, and challenge routes. They make the claimed object recoverable rather than leaving it implicit in a polished figure.

A sleep summary may help a participant when it clearly identifies missing nights, device changes, uncertainty, and the difference between an observation and a research inference. Comprehension testing should confirm that the display does not imply diagnosis, prognosis, or a hidden reading of the person's inner life.

The participant-protection review must test whether returned graphs can feel like diagnoses. If the answer is yes, participant protection takes priority and the scientific claim must narrow accordingly.

Participant return has its own timing rule. Immediate feedback may change behavior, delayed feedback may lose relevance, and no feedback may be safest for some measures. The design declares when return is useful, optional, or withheld because validity is insufficient.

The return plan states which outputs may be shared, who reviews them, how questions are answered, and when nonreturn is the safer choice. Results that are invalid at the individual level remain in aggregate research reporting. Desire to be transparent does not justify returning an interpretation the study cannot support.

Participant return must be modest enough not to become a verdict. A person may want understandable summaries, data errors corrected, or aggregate findings explained; they may also want no feedback at all. Return is a relationship of respect, not a delivery channel for hidden judgment.

Participant-facing summaries should separate collective findings from individual interpretation. A clear summary can explain what the study measured, which groups were represented, what uncertainty remains, and what changed because of participant contributions. It should not return a personal ledger estimate, endpoint forecast, route classification, or claim about inner truth. The difference between gratitude and overreach is often one sentence, and the protocol should write that sentence before recruitment.

Return formats should also be accessible and revisable. Plain-language text, large-print versions, translations, audio, and supported discussion may be needed, while some participants may prefer no return at all. If a published result is corrected or withdrawn, the return plan should define how participants are notified without creating alarm or implying that their care or identity has changed.

17.8.3 — No Individual Ledger Verdict

The no-individual-verdict rule established in §14.8.3 applies unchanged to longitudinal records. These data may support bounded population or measurement inferences, but they do not determine whether a particular life balanced or how a person should be judged.

Individual-facing outputs may describe observed periods, missingness, device quality, and uncertainty within the qualified context of use. They may not convert a local estimate into a complete-stream conclusion or use it for eligibility, care priority, trust, worth, truthfulness, or spiritual interpretation. Those prohibitions are enforced in schemas, permissions, interfaces, and publication templates.

When a family asks whether a final graph proves that a life reached balance, the answer is that the graph cannot carry that conclusion. The study observed a bounded and imperfect record under specific conditions. Respect for the family includes giving a clear boundary rather than softening an unsupported verdict with technical uncertainty language.

The public record should permit scrutiny of whether individual outputs can become self-surveillance or social judgment. Any unresolved risk should remain visible in the public summary and block overgeneralization.

Uncertainty should not be used to soften an individual verdict that should not exist. The correct output is not a low-confidence judgment about a person's life. It is no individual ledger verdict,

with any usable information returned only in the limited form the protocol permits.

No individual ledger judgment is generated, stored, returned, or made available for secondary use. Any feature or interface that invites such an interpretation is removed or redesigned before deployment. The protocol's value lies in testing scientific components, not administering a hidden account of persons.

17.8.4 — Access, Retention, and Deletion

Access, retention, and deletion are specified separately for identifiers, raw signals, derived features, linkage keys, provenance, code, and aggregate results. Each layer receives the smallest audience and shortest duration compatible with the approved purpose. Retention for unspecified future value is not a scientific justification (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

Data architecture should distinguish what must remain at the site, what may be shared, what can be computed locally, and what should never be centralized. Federated analysis may reduce transfer of raw data, but it still requires controls for model leakage, reconstruction, local deletion, and audit. Governance follows the information, not merely the file location.

A multisite project may keep identifiers and raw traces within each institution while transmitting approved features or model updates. That arrangement reduces exposure only when the features cannot readily reconstruct the private record and when sites retain authority to stop participation and execute deletion. Aggregate movement is not automatically anonymous movement.

The evidence is not adequate when indefinite storage turns research into an archive of private life. When this condition holds, the affected evidence is Unusable; any narrower descriptive use requires its own valid permissions and protocol.

Uncertainty should include future reidentification and future misuse. A dataset that seems safe today may become identifying when combined with later records or models. Retention policy should therefore be justified by the approved purpose, not by the hope that future analysis might someday want everything.

Retention decisions are linked to named analyses, replication obligations, legal or ethical requirements, and deletion triggers. The record should show why each layer still exists and who can access it. Data that survive only because storage is inexpensive have outlived their approved scientific purpose.

Retention is itself a burden when the record is intimate and long-lived. The study should keep only what its approved purpose requires, separate identifiers from features, and give withdrawal a practical route. A record that cannot end has already begun to behave unlike research.

Table 17.5. Federated-analysis workflow, provenance fields, and minimum data dictionary.

Layer	Minimum fields or action	Scientific purpose	Dignity control
Local capture	person key held locally; time; clock; channel; value; unit; confidence	Keeps time-indexed evidence analyzable.	Identifiable linkage stays at source site.

Layer	Minimum fields or action	Scientific purpose	Dignity control
Observation mark	scheduled; refused; paused; failed; removed; clinical interruption	Models why data exist or vanish.	Refusal and care interruptions remain meaningful.
Version ledger	instrument; firmware; item version; language; pipeline	Separates person change from system change.	Participants can challenge the version that interpreted them.
Feature extraction	approved features, quality flags, burden flags	Reduces raw exposure while retaining estimand inputs.	Minimization before transfer.
Federated analysis	site-local models, disclosure checks, aggregate summaries	Enables multi-site learning without central dossier.	No individual-level pooling by default.
Return and closure	participant choice, deletion status, archive decision, future-use flag	Completes the longitudinal record responsibly.	Study ends without indefinite claim over the person.

17.8.5 — Breach and Reidentification Response

Time-indexed data can reidentify people even after direct identifiers are removed. Rare clinical transitions, sleep rhythms, location patterns, transfer dates, and combinations of events may become unique when linked to external records. Breach planning

therefore covers both unauthorized access and inference from ostensibly deidentified trajectories, with reidentification risk reassessed as variables, sites, publications, and auxiliary datasets change (Rieke et al., 2020; National Academies of Sciences, Engineering, and Medicine, 2018).

The incident plan specifies detection, containment, access suspension, forensic review, required participant and site notification, support, public-artifact review, and criteria for withdrawing or rerunning affected analyses. It also records whether the incident changed completeness, provenance, consent, or permitted access. Silence is not a privacy remedy.

Evidence repair follows the same record. The program may suppress or coarsen features, withdraw affected artifacts, revise minimization and retention, and reassess whether participants can still exercise correction or withdrawal. Impermissible data are Unusable; repaired data may remain Inconclusive or yield No adequate estimate under the canonical evidence-state rules. Privacy failure therefore propagates into scientific validity rather than remaining a separate security note (National Academies of Sciences, Engineering, and Medicine, 2018; Nissenbaum, 2010).

17.8.6 — End-of-Study Closure

Closure determines when the research relationship, datasets, instruments, models, and public claims end. Protocols must specify result communication, unresolved analyses, retention and deletion, future-use limits, correction channels, and responsibility for retired artifacts. This prevents indefinite interpretive power

after completion (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Before enrollment, the closure plan fixes archive contents, deletion events, participant communication, failure memory, code and model status, derivative-data handling, and prohibited future uses. The study's ending is part of consent and purpose limitation; it cannot wait until the dataset becomes valuable. The plan should also identify who has authority to certify each closure obligation and what evidence shows that the obligation was actually completed. This turns closure from an administrative date into a verifiable state of the research system.

Participants should receive clear notice of what was learned or unresolved, what will be retained or destroyed, what future use is barred, and where corrections will appear. They should not infer the relationship ended merely because contact stopped. Closure communication should remain understandable and reachable after active data collection ends, especially when a later correction changes how an earlier result should be interpreted. A durable contact route gives participants a practical way to ask about those changes without reopening participation.

Closure should assign final status to usable findings, inconclusive analyses, unusable data, unresolved access questions, and future-use prohibitions. A completed record cannot support stronger claims through reinterpretation without a newly approved analysis, consent basis, and status review. Any later reopening should therefore create a new versioned analytic record rather than silently modifying the closed one. This preserves the

distinction between what the original study established and what a later investigation newly inferred.

Site reconciliation, incident review, fulfillment of retention and deletion commitments, public result registration, retirement notices, and a durable correction contact are completed before closure is certified. Null results, failed recovery, deviations, and rival-favored outcomes remain in scientific memory even when sensitive data are deleted. Reconciliation should verify completion across sites rather than assume that a centrally issued instruction was carried out uniformly. Differences in retention, deletion, incident handling, or version status should be resolved or recorded explicitly before the study is treated as closed.

Every retired dataset, instrument, model, and public artifact should receive a tombstone recording what existed, why it was retired, which successor replaced it if any, and which claims it can no longer support. The tombstone preserves provenance without preserving unrestricted access to private information.

A responsible program ends observation without erasing failure or extending authority beyond consent.

Part VI — Candidate Mechanisms and Causal-Locus Tests

Part VI makes the mechanism program experimentally answerable. Earlier Parts fixed the terminal target, separated latent and measured objects, established stream and endpoint rules, and built the first study families. This Part now asks which mathematical, causal, and embodied processes could generate the discriminating signatures, which ordinary processes could imitate them, and what evidence would support, narrow, or retire each candidate.

The Part begins with completion geometry because possibility precedes explanation. From a declared state and prefix, some future contributions may be reachable and others may not. Reachability sets, viability regions, completion kernels, committers, and boundary-value problems make that distinction explicit. They show what a model permits before anyone claims that nature uses the model.

Path-space methods then expose a second distinction. A process can end in a target set because its native dynamics drive it there, because analysts condition on that endpoint, because a controller steers it, because the endpoint is defined by first hitting, or because selection removes trajectories that do not fit. Doob transforms, Schrödinger bridges, absorbing boundaries, and related constructions are therefore tools for discrimination, not ornaments that confer truth (Doob, 1957; Léonard, 2014; Chetrite & Touchette, 2015).

The Queue model remains as one bounded, partially observed control architecture for unresolved demands, priority, blocking, reactivation, repair, and residual influence. Its value is that each suggestive element becomes a typed state, observation, action,

transition rule, rival prediction, recovery test, and retirement condition. Components that survive earn a clearer scientific role; components that fail can be removed without weakening the wider mechanism program.

Chapter 19 moves from construction to causal location. A model earns scientific force only when it says where an intervention enters, what continuation law changes, and which observable divergence should occur first. Interventions on reports, physiology, action support, medication, information, assignment, or endpoint rules are not interchangeable. The representation gate must remain intact before causal language begins (Hernán & Robins, 2020; Pearl, 2009).

Sections 7.3.4 and 19.4.5 establish the modal distinctions and policy-mixture condition. Here, an observed action sequence counts as completion support only under a frozen continuation regime, adequate support, and the corresponding modal guarantee.

Shadow price enters as a dimensioned sensitivity quantity within constrained optimization. Properly specified, it shows how a local objective changes when a declared constraint is marginally relaxed, with units and regularity conditions that can be checked. That analytic use ends at the model boundary and supplies no moral exchange rate, human price, or permission to tolerate harm.

Chapter 20 brings the body into the mechanism tournament. Interoception, allostasis, salience, stopping, pain, fatigue, medication, disability, trauma, illness, autonomic regulation, endocrine dynamics, and wearable signals may carry, constrain,

or confound route-specific signatures. Each receives a declared channel, lag, baseline, rival, perturbation, and no-inference zone so bodily evidence can become more informative without being mistaken for direct access to lived valence (Khalsa et al., 2018; Barrett & Simmons, 2015).

Three achievements organize the Part: construction, identification, and evidence. Mathematical coherence establishes that an object can be defined; identification asks whether the relevant quantity is determined by the available law and observations; evidence asks whether realistic data recover and transport the promised difference. Keeping those achievements separate prevents elegance, estimability, or reproducibility in one channel from standing in for causal support.

Prospective information discipline protects the Part's strongest tests. Endpoint information stays outside state construction, filtering, feature selection, action choice, and model training unless the declared design permits it. Every information set is time stamped, every outcome-dependent transformation is labeled, and cross-fitting reduces data reuse without excusing a representation that already contains the answer.

Mechanisms operate within lived opportunity. Transportation, care access, medication, fatigue, family obligations, communication, disability, cost, and institutional permission can open or close continuations before any internal controller acts. These conditions enter support, selection, and transport models as causal determinants, allowing the science to distinguish blocked opportunity from preference, motivation, or latent route failure. Opportunity variables should be measured independently of the

candidate mechanism when possible so blocked access is not inferred from the same behavior the route aims to explain.

The Part is designed for mathematicians, causal researchers, neuroscientists, clinicians, engineers, and serious general readers to inspect the same objects at different depths. Equations, tables, and design aids state their role, assumptions, units or type, nearest rival, failure condition, and permissible interpretation. Technical density earns its place when it makes a mechanism easier to recover, compare, perturb, transport, or retire. A technical object that cannot state its failure condition should remain illustrative rather than enter the evidentiary tournament.

Each outcome advances a different scientific task. A route-specific pattern that survives representation, observation, causal, recovery, transport, and equal-budget rival tests earns route-level support and opens a stronger target test. Ambiguity identifies the information or intervention still missing. Rival-favored and Route-failed results retire specific queue rules, path-space constructions, or embodied bridges while preserving the methods, data, and negative knowledge they produced. A route-level update should name the specific discriminating result that moved status, not merely the chapter or mechanism family that was studied.

By Part VI's end, completion appears as distinguishable scientific possibilities. Readers can separate conditioning from native dynamics, steering from selection, absorption from first hitting, belief-state from omniscient control, and embodied mediation from measurement artifact. The map identifies the observations that distinguish these cases. This taxonomy also prevents one

successful construction from being mistaken for evidence that every mathematically similar path arises through the same cause.

The movement is from mathematical possibility to empirical discovery. Mathematical constructions widen the explanations; causal, recovery, and brain-body tests compare them using common information, realistic observation, humane intervention, and equally treated rivals. The mechanism program's promise lies in making these possibilities precise enough for evidence to identify which processes nature actually uses. The strongest mechanism is the one that survives these comparisons with fewer untestable commitments, not the one that can reproduce the widest range of paths.

Chapter 18 — Completion Geometry

Chapter 18 turns a shared terminal appearance into a tournament among path laws. Native dynamics, selection, terminal conditioning, first hitting, absorption, reflection, stickiness, and active control can approach the same boundary while differing in hazard, reversibility, dependence, and perturbation response. Those preterminal differences carry the scientific information that the endpoint picture alone cannot supply.

The chapter advances through five answerable questions. From the current prefix, which future contributions remain attainable? How does a frozen law distribute probability across them? Which information state supports the forecast? What change of measure or control constructs the path? Which held-out observation would distinguish mathematical construction from natural realization? Each question receives its own object so possibility, probability, construction, mechanism, and evidence remain separate.

The primary objects have distinct jobs. A reachability set describes what remains possible. A committor gives event probability under a declared law. A Doob transform constructs a changed path law. A policy changes transitions through an action rule. A boundary implementation specifies approach, contact, dwell, return, and stopping. Their existence establishes coherent mathematics; scientific realization requires identification, recovery, transport, and held-out discrimination.

The distinction between history and state is central. A continuation law may depend on the complete observed prefix. It may be reduced to a state-indexed law only after state sufficiency is justified. Compressing history into a convenient state without

testing that sufficiency can hide the memory or path dependence the program is supposed to measure.

Finite-band completion and exact singleton closure require different conditioning constructions; §18.4 develops the zero-probability-target case and its tightness and uniqueness requirements.

Reachability makes route-level impossibility visible. When the negative of the current residual lies outside the attainable future-contribution set, exact completion is unavailable under the declared continuation class. That finding narrows or defeats the route on its own terms and identifies which horizon, action, information, or ownership restriction produced the obstruction.

Ordinary alternatives enter as full competitors. Viability control, safe reinforcement learning, homeostatic regulation, scarcity, planning, and conditioned Markov processes can all create boundary-sensitive paths. A PTVC-linked route earns priority only through a prospectively specified difference that those rivals fail to recover under the same observations, flexibility, tuning budget, and scoring rule.

Partial observation prevents the mathematics from becoming omniscient. Researchers observe reports, physiology, behavior, context, and missingness, not the phenomenal state itself. A belief-state model can organize uncertainty, but its posterior is a model output, not a private truth and never a human-worth object.

Path-space constructions can look explanatory before they are causal. A bridge ending at zero may display smooth convergence because the terminal event was built into the changed law; an

absorbing process may stop at zero because the boundary rule says so; a selected sample may concentrate near zero because other paths disappeared. The chapter treats each construction as a generator of discriminating preterminal predictions, not as a certificate of natural mechanism.

Consider two models with the same terminal distribution. One uses a local controller responding to the current state. The other mixes endpoint-conditioned bridges. Their one-time marginals may nearly coincide, yet a safe midcourse perturbation, an unexpected endpoint probe, or a held-out action changes the controller's path law and the bridge mixture differently. The experiment belongs at that divergence, where a shared endpoint no longer hides causal structure.

Model recovery is the chapter's empirical gate. Simulated and benchmark data must show that the analysis can recover the correct generator or model family under the planned noise, missingness, cadence, and heterogeneity, and can return an open-set unknown when the generator is absent from the library. A pipeline that confuses native control with conditioning is not ready to interpret human data, regardless of apparent fit.

The chapter leaves the reader with a map of path-space possibilities and a stricter appreciation of elegant endpoints. Even if PTVC narrows or fails, the distinctions among reachability, committers, conditioning, hitting, absorption, partial observation, and recovery remain durable scientific tools. The mechanism program advances by preserving those tools while withdrawing the interpretation that loses.

18.1 — Completion Geometry and Reachability Sets

Let the current signed residual be $L(t) = l$ and let $A_{fut}(t, z)$ denote the set of future signed contributions attainable from the declared state z under ownership-preserving continuations. Exact completion is directionally reachable only if

$$-l \in A_{fut}(t, z).$$

The condition is necessary, not sufficient evidence that a route is realized. The geometry can also be expressed through a residual-solvency set

$$S_{res}(t, z) = \{l: -l \in A_{fut}(t, z)\}.$$

A prefix outside that set is an explicit counterexample for the declared route and continuation class. It cannot be rescued by changing the stream, endpoint, or allowable continuations after inspection.

Convex approximations permit support-function tests. For a closed convex attainable set A , membership of $-L(t)$ can be checked through inequalities against $h_A(a) = \sup_{x \in A} a^\top x$. Nonconvex sets require richer geometry, and a convex relaxation must be labeled as a relaxation rather than the true continuation support.

For several active streams or coupled resources, joint reachability must preserve coordinate provenance. A shared resource can constrain several streams, but one stream's phenomenal contribution cannot cancel another's unless a separately named junction or transfer theory is under test. The geometry must not reintroduce cross-stream trading through notation.

Empirical use requires estimating the attainable set without outcome leakage. Training data, policy restrictions, care context, and action availability define the set; held-out prefixes then test whether observed continuations remain inside it. A set widened until every history becomes reachable has no discriminating power.

18.1.1 — State Prefixes and Future Cones

A state prefix is the complete information object declared at analysis time: the stream-owned residual coordinate, route state, observation history, action history, endpoint-information state, and uncertainty carried forward from earlier measurements. Its future cone is the set of continuations still admissible under the protocol-fixed dynamics, ownership rules, action support, and care constraints. Both objects must be fixed prospectively, because a cone reconstructed after the endpoint can silently import the outcome it is supposed to predict (Aubin, 1991).

The prefix is not the person and it is not the person's entire life. It is a model state built for one scientific question. A richer prefix may improve prediction while making identification worse, especially when it includes variables that are descendants of the endpoint or of the intervention being evaluated. A defensible model therefore distinguishes what is known, what is latent, what is estimated, and what remains outside the representation.

A future cone is broader than the path that eventually occurs. One realized continuation cannot be used to declare every unrealized action or environmental change impossible. The scientific task is to estimate how the cone contracts or changes shape as

information arrives, then test whether the observed continuation obeys restrictions that were visible before the outcome. Retrospective pruning makes any path look inevitable and therefore destroys discrimination.

Action support belongs inside the cone. Transportation, communication, care access, symptom burden, institutional permission, family obligations, and physical capacity can open or close continuations independently of any candidate completion mechanism. A geometry that assumes unavailable actions will misclassify structural deprivation as a property of the latent route.

The cone must also preserve route ownership. A shared resource may alter several streams, but one stream's future phenomenal contribution cannot be inserted into another stream's cone. Junctions, partition, fusion, and continued-stream branches require their own typed routing rule. These rules preserve per-stream accumulation within the multivariate model.

Partial observation turns a single cone into a family of plausible cones indexed by the current belief state. A narrow point estimate may hide radically different futures that remain compatible with the data. Robust analyses report an inner cone supported across declared uncertainty and an outer cone that includes all plausible continuations, rather than presenting one estimated frontier as known fact.

A concrete test begins with held-out prefixes. The team freezes the state representation and dynamics on training data, computes the admissible cone for each held-out prefix, and asks whether observed actions, state changes, and endpoint-independent

outcomes remain inside it. Systematic cone violations are evidence against the route or representation, not a reason to widen the cone after inspection.

The central payoff is prospective vulnerability. Once a prefix and cone are frozen, the model can be wrong because it excludes an observed continuation, includes almost everything, depends on future labels, or changes materially under reasonable uncertainty. A future cone that cannot fail is not geometry for science; it is a retrospective story drawn around the path that happened.

18.1.2 — Signed Reachability Under Finite Opportunity

Signed reachability asks whether the continuation set contains enough contribution in the direction required by the current residual. Large remaining variability is not sufficient. A process may still fluctuate widely while lacking the sign, magnitude, timing, or route ownership needed for completion. The relevant condition is directional inclusion of the required opposite-signed contribution, not generic capacity or emotional intensity (Aubin, 1991).

Finite opportunity makes this condition sharp. Remaining time, physiological reserve, action access, reportability, social permission, and environmental support jointly bound what can still occur. The bound is model dependent and uncertain, but it cannot be replaced by the statement that something helpful might happen. A route must identify a feasible continuation under the declared horizon and constraints.

Reachability should be computed on phenomenal contribution rather than on an untyped behavioral proxy. A conversation,

medication change, rest period, or institutional remedy may alter future experience, but the action itself is not a signed valence increment. The model needs a separately validated transition from action and context to the distribution of future phenomenal contribution.

Direction matters most near constrained boundaries. A strongly positive residual requires access to sufficiently negative future contribution on the additive branch, while a strongly negative residual requires the opposite. That formal statement is not a prescription to create pain or pleasure. Harm induction is outside the admissible action set, and ordinary care continues regardless of what the reachability calculation returns.

Uncertain opportunity yields an identified set rather than one answer. Some prefixes may be reachable under every plausible horizon, some under only a subset, and some under none of the declared continuations. Reporting these classes is more informative than averaging over endpoint scenarios and calling the mean reachable.

Ordinary rivals can explain directional asymmetry. Goal structure, avoidance learning, symptom limits, opportunity cost, social support, and treatment response may make one sign of future change more available than the other. A PTVC-linked route earns support only if its signed reachability restrictions outperform those alternatives under the same state and action information.

A held-out failure is decisive for the declared route class. If the required contribution lies outside the estimated continuation

support and the endpoint still occurs under the declared branch, the route cannot be rescued by revising the endpoint, reassigning ownership, or inventing unobserved capacity after the fact. The appropriate result is Route-failed, reference-class narrowed, or No adequate estimate.

18.1.3 — Viability, Solvency, and Closure Regions

Viability, solvency, and closure answer different questions. A viability region contains states from which at least one admissible policy can respect declared constraints. A solvency region adds the requirement that the current signed residual can still be offset by attainable future contribution. A closure region is stronger still and must specify whether closure is possible under one policy, probable above a threshold, supported under every admissible policy, or guaranteed on a declared branch (Aubin, 1991; Aubin et al., 2011).

These regions should be nested only when the assumptions justify nesting. A state may be viable because safe functioning can continue while being insolvent for the signed completion target. Another state may be solvent under one narrow policy but not under ordinary policy variation. Collapsing the regions hides which part of the argument is doing the work.

Solvency is direction-specific and route-specific. A positive and negative residual of equal magnitude need not have symmetric continuation sets, because available actions, bodily reserve, context, and endpoint support may differ by sign. A single radial distance from zero can therefore conceal the geometry that matters most.

Uncertainty creates nested reachability regions. The inner region contains prefixes that satisfy the criterion across every declared parameter, state, and horizon compatible with the data. The outer region contains prefixes that satisfy it under at least one admissible specification. Their difference is an identified uncertainty region, not a smooth probability gradient, and the final display should preserve that distinction.

A clinical example separates ethical permission from geometric identification. A participant may remain within comfort and consent constraints while the candidate route is not estimable because the observation channel has shut down. The protocol may remain ethically permissible while geometric viability and solvency remain unidentified. That uncertainty should stop the affected inference without changing care or implying anything about the person's life.

The region boundary can be tested through safe perturbations that change action support or information without changing the endpoint definition. Opening a communication route, correcting an institutional barrier, or observing a clinically indicated treatment transition may move the estimated state. A route-specific model should predict which boundary moves and by how much.

Approximation error belongs beside the region. Numerical grids, convex relaxations, surrogate dynamics, and finite samples can shift a narrow frontier. If classification changes under reasonable discretization or model uncertainty, the result is a band or no estimate, not a crisp geometric verdict.

A closure region built by excluding nonclosing paths is circular. The scientific object must be derived from independently specified dynamics, constraints, and observations, then evaluated on held-out continuations. Otherwise the region proves only that the analyst encoded the desired endpoint.

The practical value of these distinctions survives PTVC. Viability theory clarifies when safe action remains possible, solvency analysis clarifies directional capacity, and closure regions clarify which additional assumptions turn possibility into stronger modal claims. Their scientific discipline lies in naming exactly which region has been estimated and which stronger region has not.

18.1.4 — Geometry Under Uncertain Endpoints

Endpoint uncertainty changes the geometry of completion before it changes any estimate. A fixed horizon gives one future cone, but a distribution over possible endpoints gives a family of cones indexed by stopping time, branch, and information state. The relevant object is therefore not one attainable set but a set-valued map from each prefix and endpoint scenario to the signed continuations that remain possible. A route that is reachable at one generous horizon may be impossible at a shorter horizon, and that difference cannot be repaired by averaging the two cases (Aubin, 1991).

Several geometric summaries are possible, and they answer different questions. The union across endpoint scenarios asks whether completion remains possible somewhere. The intersection asks whether it remains possible under every declared scenario. A probability-weighted region asks how much endpoint

mass lies over solvable continuations. None of these objects is equivalent to exact closure at the truth-level endpoint, and each should carry its own status label rather than being collapsed into a single shaded region.

Consider a longitudinal clinical cohort in which remaining observation time is uncertain because illness trajectories differ and follow-up can end for several reasons. A participant may have many future actions under one prognosis and very few under another, while the scientific team knows only a calibrated distribution. The geometry should preserve that uncertainty instead of treating the most favorable horizon as available to every path. Comfort, treatment, and withdrawal remain outside the optimization target and are never adjusted to make the geometry easier.

Uncertain endpoints also create leakage hazards. A model trained with realized time-to-end can learn a future label that was unavailable at the prefix being evaluated. It may then appear to identify narrowing reachability when it is merely reading hindsight. Prospective estimation must use only information available in the declared filtration, and endpoint models should be cross-fitted or frozen before they are used to construct a future cone.

Ordinary rivals can create the same apparent contraction. Frailty, age, disease progression, medication changes, resource loss, and institutional closure can all reduce available continuations without any fairness-specific route. The test is not whether options shrink, but whether the signed, route-owned geometry shrinks in the pattern jointly predicted by the candidate

mechanism after those rivals receive the same endpoint information and action constraints.

A useful result may be a bounded statement rather than one boundary. The analysis can report prefixes that are solvable across all plausible endpoints, solvable only under some horizons, or nonadjudicable because endpoint support is too uncertain. If the apparent empirical result depends on using retrospective endpoint knowledge as a prospective input or on choosing a favorable horizon after inspection, the confirmatory comparison fails its information or protocol gate and the affected evidence is Unusable. A route that necessarily requires unavailable future information is inadmissible under the declared information constraints; an empirical Route-failed verdict requires a valid test.

18.1.5 — Topology, Disconnected Supports, and Hidden Barriers

Reachability is not determined by distance alone. An attainable set can be disconnected, folded around forbidden states, or concentrated on a lower-dimensional support. Two states that are numerically close may be separated by an action barrier, a routing rule, a safety constraint, or a zero-probability transition. Topology therefore decides whether a seemingly short path is actually available (Aubin, 1991; Aubin et al., 2011).

Convex relaxation is useful but dangerous. Filling the gaps between disconnected components can create synthetic continuations that no admissible dynamics can realize. A relaxation should be labeled as an outer bound and tested against

the original nonconvex support, especially when the conclusion depends on a narrow bridge between components.

Hidden barriers can be material. A treatment exists but is inaccessible, a communication device is unavailable, an institution has no appeal route, or a person cannot safely take the modeled action. Those restrictions belong in the action support. Omitting them makes the model attribute externally imposed barriers to the stream or its policy.

Barriers can also be informational. Two latent states may produce nearly identical observations while permitting different futures. The apparent connection between their estimated cones may disappear once the belief state is represented. State aliasing is therefore a topological problem in the inferred state space, not merely a measurement nuisance.

A route may cross a barrier only through a junction event such as partition, fusion, return, or a change in endpoint branch. Such events require typed rules and prospective evidence. They cannot be invoked as invisible tunnels whenever ordinary reachability fails.

Concrete simulation should include obstacles rather than only smooth landscapes. A model can compare Euclidean distance, geodesic distance on the admissible set, and actual hitting probability under the protocol-fixed dynamics. Divergence among those quantities reveals when a simple residual norm is an inadequate summary.

Ordinary models may explain the same disconnected support. Habit, learned avoidance, legal restrictions, symptom thresholds,

finite resources, and safe-control constraints can all carve the action space. A fairness-specific route must therefore predict a distinctive change in connectivity or transition probability, not merely observe that barriers exist.

The public boundary is exact: a state outside a model's connected component is not a person declared trapped, doomed, or deficient. It is a statement about one formal system under assumptions. The repair may be to change the environment, expand access, revise the representation, or retire the model.

Topology earns its place when it produces a falsifiable consequence. The model should identify which observed transition would cross a forbidden barrier, which safe intervention should open a passage, and which held-out paths should remain impossible. If every surprise is explained by adding a hidden tunnel, the topology has stopped constraining the theory.

18.1.6 — Testable Consequences of a Reachability Map

A reachability map becomes scientific when it predicts more than terminal compatibility. It should identify prefixes that will lose sign-specific options, actions expected to reopen or close a route, uncertainty changes that move the frontier, and observations that falsify the estimated dynamics. Those predictions are frozen before the continuation is observed, then evaluated with the same recovery and scoring rules applied to rival geometries (Aubin, 1991; Aubin et al., 2011; Wilson & Collins, 2019).

One consequence concerns boundary crossing. If a prefix is classified outside the outer solvency region, then no continuation in the declared class should achieve the required signed

contribution. An observed qualifying continuation is a counterexample to the map or its assumptions. The analysis cannot preserve the map by moving the boundary after the fact.

A second consequence concerns intervention sensitivity. An action that genuinely changes opportunity, reserve, or route state should alter the attainable set in a specified direction. A sham action matched for attention or expectation should not. The contrast separates structural change from generic reactivity.

A third consequence concerns calibration. Predicted inner-region membership should correspond to conservative empirical coverage, while outer-region membership should not be treated as a guarantee. Reliability diagrams, set coverage, and false exclusion rates are more informative than one global accuracy score.

Transport tests whether a geometry retains its declared invariants when the observation system, action menu, site, or population changes within a defined transport class. The target state and ownership rule remain fixed, while only the declared destination components may be recalibrated. Failure narrows the route to the conditions in which the geometry was learned instead of supporting a universal mechanism.

Model recovery must precede interpretation. Simulated paths from viable control, ordinary scarcity, endpoint-conditioned bridges, and the candidate route should use the same estimator. If the pipeline cannot recover the generating geometry under realistic noise, missingness, and state aliasing, real-data frontier labels are not trusted (Villaverde, 2019; Wilson & Collins, 2019).

Negative controls are equally important. Variables that alter reporting cadence without changing action support should not move the true reachability map. If they do, the map may track the observation process rather than completion geometry.

A public artifact shows uncertainty, assumptions, and prohibited readings beside the frontier. It does not display a person-level dashboard or a countdown to solvency. The map is a research object for comparing models, never a private balance display.

Retirement is a normal outcome. Persistent held-out violations, unstable frontiers, circular dependence on endpoint labels, or equal performance by simpler rivals should narrow or retire the map. The remaining contribution is then methodological: a clearer account of which futures were and were not identifiable.

18.2 — Completion Kernels and Boundary Values

A completion kernel asks for the probability of finishing inside a declared band under a declared continuation law. With current state x and residual ℓ , a state-indexed committor can be written as

$$h_{g,K}(x, \ell, t) = P_g \left(\ell + \Delta L_{t:T_g} \in [-K, K] | X(t) = x, L(t) = \ell \right).$$

The regime g , endpoint branch, band, and reference law are part of the object. A history-indexed kernel is more general. Reducing it to a state function assumes that the chosen state is sufficient for the future event. That assumption should be tested by residual history dependence, held-out prediction, and rival state augmentations rather than asserted because a Markov model is convenient.

Under regularity, a committor solves a backward boundary-value problem. The boundary conditions reveal what counts as completion and failure; the generator reveals the assumed native dynamics. Changing either after seeing the data changes the hypothesis.

The kernel is useful because it transports a quantitative object across actions, tasks, horizons, and cohorts. A single potential trained on one module can be frozen and evaluated on transition, action, and endpoint modules. Retuning a new potential for each module tests only local fit.

Uncertainty in the kernel should remain visible. Posterior bands, conformal regions, or identified sets may be more appropriate than a single surface when the state is partially observed or the

generator is weakly identified. A colorful heat map should not conceal uncertainty.

A high committor is not a personal forecast or moral status. It is a model-conditional probability of an event under specified assumptions. Its public artifact must name the model, band, time, uncertainty, and prohibited interpretations.

18.2.1 — Completion Kernels as Conditional Forecasts

A completion kernel is a conditional forecast of a declared terminal event given the information available at a prefix. It maps a state or belief state to a probability distribution, not to a metaphysical degree of fairness. The event, horizon, branch, policy class, and observation filtration must be stated before the kernel is estimated (Kallenberg, 2002).

The kernel differs from reachability. Reachability asks whether at least one admissible continuation exists; the kernel weights continuations under a specified law or policy. A state can be reachable and still have very low completion probability, or unreachable under one policy while reachable under another.

Forecasting requires a baseline law. Native dynamics, selected cohorts, endpoint-conditioned samples, and intervention policies induce different kernels even when they share the same state coordinates. Every reported probability therefore identifies the law under which it is computed; probabilities from incompatible laws are not placed on one scale or interpreted as competing estimates of the same object.

Partial observation replaces a state kernel with a belief-state forecast. The input is then a distribution over latent states, and

uncertainty in that distribution must propagate to the completion probability. A sharply reported scalar is unjustified when several latent states with different futures remain observationally equivalent.

Calibration is evaluated prospectively. Binned or distributional calibration, proper scoring rules, and held-out endpoint branches can test whether the forecast behaves as declared. A kernel fitted and assessed on the same terminal event is a description of the sample, not evidence of transportable completion dynamics.

Ordinary rivals receive their own kernels under the same information budget. Homeostatic control, finite-horizon planning, conditioned bridges, selection models, and observation artifacts may forecast the event as well as the candidate route. The kernel's scientific role is comparative rather than confirmatory by itself.

A kernel is withdrawn when its event is ill-defined, support is inadequate, calibration fails, or its output is used outside the declared research purpose. The reported quantity is a model-conditional forecast with uncertainty and an available no-estimate branch.

18.2.2 — Committor Functions and Boundary-Value Problems

A committor gives the probability of reaching one declared set before another under a specified stochastic process. In Markov settings it solves a boundary-value problem for the generator, with boundary conditions fixed on the competing sets. The mathematical object is powerful because it turns a path question

into a state function, but only after the state, generator, sets, and law are justified (Metzner et al., 2009).

The forward committor asks what is likely to happen next from the current state. A backward committor asks where a trajectory most likely came from under a stationary or appropriately defined ensemble. These are different questions. Using a backward object as a prospective forecast can import information unavailable at the decision time.

Boundary conditions carry substantive assumptions. Setting the committor to one on a completion set and zero on a rival set does not show that either set is a truth-level endpoint. It defines the event being forecast. The sets must be independently motivated and prospectively specified.

Generator misspecification can dominate the result. Sparse observations, hidden variables, non-Markov memory, state-dependent diffusion, and informative sampling can make an estimated committor look smooth while representing the wrong dynamics. Recovery simulations should test whether the boundary-value solution remains valid under the actual observation process.

A concrete comparison can use several models with identical terminal sets. A first-hitting diffusion, a conditioned bridge, a homeostatic controller, and a queue candidate may yield similar one-time completion probabilities but different transition-path densities and intervention responses. Those differences create the discriminating work.

Uncertainty should be represented as a band or distribution over committors when the generator or state is uncertain. Reporting only the fitted mean hides whether classification changes under plausible dynamics. If a valid treatment contrast remains too uncertain to determine its sign, report Inconclusive evidence; if adequate route models remain inseparable under a qualified analysis, report Model-ambiguous.

Committors support rare-event sampling and transition-path analysis. Preferentially sampling successful transitions changes the sample law, so likelihood weighting and support checks connect those paths to the intended population.

The committor is retired as a route object when it fails calibration, cannot transport across held-out actions or endpoints, or depends on a state representation that cannot be recovered. It may remain valuable as a descriptive forecasting tool even after its causal interpretation is withdrawn.

18.2.3 — Action-Specific Committors

An action-specific committor asks what happens if one declared action is taken now and the continuation policy thereafter is fixed or prospectively specified. It separates the immediate causal fork from the larger policy value. Written conceptually, the object is the conditional probability of reaching the declared completion set after action a , given the current state or belief state and the declared continuation rule. This is narrower than a personal forecast and more informative than a single state-level completion probability (Metzner et al., 2009).

The action index matters because two interventions can have identical short-term observations and different future geometry. One action may open a repair route, another may reduce measurement burden, and a third may merely change what is reported. A state committor that averages across those actions can hide the divergence. The model should therefore declare the action version, timing, eligibility set, and continuation policy that define each surface.

A low-burden laboratory example can use unfinished but harmless tasks. Participants may receive information, an opportunity to resolve one task, or a matched control cue. An action-specific committor predicts which intervention should change the probability of a later route-relevant state under the candidate model. This tests controllable continuation structure where the action is safe and observable.

Identification requires support. If participants with a particular belief state never receive one action, its committor is extrapolated rather than learned. If action choice depends on unmeasured urgency, the surface may encode selection. Randomization, sequential exchangeability assumptions, or partial-identification bounds are needed according to the design. A smooth heat map cannot supply missing positivity.

The most severe comparison gives ordinary decision and control models the same action set, observations, and fitting budget. A standard model may predict that information, effort, or safety explains the action effect without any completion-specific state. Held-out intervention modules should score the competing

action-specific surfaces on calibration, discrimination, and transport rather than on visual resemblance to a desired endpoint.

An action-specific committor is retired when it is unstable across folds, fails calibration, does not transport across matched tasks, or gains apparent performance only after its action or endpoint definition changes. A successful result identifies its action, state representation, reference class, and continuation model.

18.2.4 — Calibration, Transport, and Uncertainty

A completion kernel must be calibrated for the event and population in which it will be used. Calibration in one simulated process, site, endpoint definition, or action regime does not establish calibration elsewhere. Transport requires an explicit bridge across state definitions, observation systems, policies, and endpoint branches (Gneiting & Raftery, 2007; Hernán & Robins, 2020).

Uncertainty enters at several levels: latent-state uncertainty, generator uncertainty, finite-sample error, endpoint ambiguity, policy variation, and numerical approximation. These components should not be compressed into one standard error when they have different consequences. Some widen the forecast; others invalidate the event or state entirely.

Cross-fitting can reduce optimistic evaluation, but it does not solve semantic leakage. If the state representation contains realized time-to-endpoint, post-treatment information, or labels derived from the terminal event, held-out scoring still tests a future-informed model. The information audit precedes the split strategy.

Transport should be modular. A kernel may transport across sites while the observation model does not, or across devices while the action policy changes. Reporting which module moved and which was re-estimated prevents a successful local recalibration from being described as mechanism invariance.

Calibration curves need enough support. Near a rare boundary, a small number of events can make a smooth curve look precise. Effective sample size, event counts, censoring, and selection weights should be reported beside the result. When support is thin, a wide identified set is more honest than a polished probability.

Ordinary rival kernels should be recalibrated with the same budget. Granting the candidate route bespoke recalibration while freezing rivals at their source-site parameters would turn transport into a hidden advantage. Equal-treatment comparison includes tuning, uncertainty, and failure rules.

A public forecast must remain population- and model-level. It should not be returned to a participant as a personal completion probability, care forecast, or interpretation of their life. The number is not clinically validated merely because it is calibrated for a research event.

Failure to transport is informative. It can reveal missing state, changing observation, treatment heterogeneity, or a mechanism limited to one setting. The route should narrow to the domain where calibration and support are demonstrated, with the failed transport preserved as part of the scientific record.

18.2.5 — Flexible Baselines and Rival Kernels

A candidate completion kernel should be compared against flexible ordinary baselines, not against a constant risk or weak regression. Baselines may include nonparametric survival models, state-space forecasts, homeostatic controllers, finite-horizon planners, conditioned bridges, and selection models. Their flexibility must be sufficient to reproduce ordinary nonlinearities without being so unconstrained that none can be interpreted.

Common observation is the first fairness rule. Each model receives the same prefix information, missingness indicators, endpoint definition, and action history unless the scientific question explicitly tests an information advantage. A model that wins because it was given future labels or richer sensors has not identified a mechanism.

Common tuning and scoring follow. Hyperparameter search, calibration, regularization, and stopping criteria should be budgeted prospectively across models. Proper scoring rules, recovery tests, and held-out interventions are chosen before the winning family is known, so the comparison rewards calibrated predictive content rather than a metric selected after inspection (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Flexible kernels can mimic route signatures through different mechanisms. A bridge can condition on the endpoint, a hazard model can exploit frailty, a recurrent network can memorize site patterns, and a homeostatic process can concentrate near a set

point. The comparison therefore examines transition structure and intervention response, not only forecast accuracy.

Open-set inadequacy remains a first-class result. If every prospectively specified model fits poorly under a qualified analysis, the result is Open-set inadequate. If, in the prespecified simulations, the pipeline exceeds its allowed rate of confidently assigning unknown mechanisms to known classes, it fails the open-set recovery gate and cannot support the affected confirmatory comparison. Residual structure and challenge generators then define the next model-building task. This option prevents the least-wrong candidate from receiving a scientific victory it did not earn.

Complexity penalties are useful but not decisive. A simpler model that predicts as well may be preferred for explanation, while a more complex candidate may still be retained for a prospectively specified distinctive prediction. Predictive ranking, causal interpretation, and theoretical scope remain separate judgments.

Rival victory is a substantive result. If ordinary planning, regulation, or selection kernels match the candidate under common rules, the route-specific interpretation narrows or retires. The result strengthens the surrounding science by identifying which ordinary structure was sufficient.

The baseline library should be versioned and reproducible. Code, simulation seeds, observation models, and parameter priors become part of the artifact. A later stronger rival triggers reanalysis rather than being excluded because it was not present when the first favorable result appeared.

18.2.6 — When a Kernel Must Be Retired

A completion kernel is retired when its event, input state, reference law, calibration, or intended decision cannot be defended. Endpoint leakage, poor transport, unsupported extrapolation, unstable numerical solutions, state aliasing, or failure against flexible ordinary rivals each triggers a typed disposition. The result may be recalibration, reference-class restriction, descriptive-only use, or retirement. A more persuasive heat map is never a repair.

State aliasing yields a scientific identified set. When materially different latent states remain compatible with the same observations and imply different completion probabilities, the data do not determine one scalar forecast. The defensible output is a bounded range, an assumption-indexed family, or No adequate estimate, together with the additional observation or intervention that would separate the compatible states.

Event failure occurs when the completion set or competing boundary changes after outcomes are known, or when the event is defined by the same estimator the kernel is asked to validate. Such circularity cannot be repaired by cross-validation because the target itself was manufactured.

Law failure occurs when selection or conditioning changes the probability measure without proper accounting. A kernel estimated among paths that already reached the target cannot be interpreted as the native probability of reaching it. Likelihood ratios and support checks are necessary, and sometimes insufficient, when the event is probability zero.

Transport failure is not solved by silently recalibrating every new site. Recalibration may produce a useful local predictor, but it withdraws the claim that one mechanism-level kernel transported. The record should distinguish local salvage from preserved structure.

Model recovery is the gate before human interpretation; §18.7.5 gives the full recovery and confusion-matrix standard. If realistic synthetic generators cannot be recovered, real-data route probabilities remain exploratory rather than confirmatory (Villaverde, 2019; Wilson & Collins, 2019).

A kernel can retire partially. Its ranking may remain useful while its absolute probabilities fail, or its descriptive calibration may survive while its causal interpretation disappears. Narrowing the role is preferable to discarding every useful property or protecting every original claim.

When a kernel narrows or retires, record the affected version, surviving role, failed diagnostic, and successor test; Chapter 28 governs downstream artifact correction. A successor kernel begins as a new version and must pass recovery and calibration rather than inherit the predecessor's status.

Retirement should distinguish sharpness from calibration. A kernel can separate cases confidently while assigning wrong probabilities, or be calibrated yet too uninformative for the protocol-fixed decision. Either failure may be decisive, and neither should be hidden inside one performance score.

Coverage should be checked across state regions, actions, endpoint branches, and observation-quality strata. A global

average can conceal failure exactly where completion is most difficult or data are least available. The retirement packet should identify the failed region and preserve only any narrower calibrated domain.

Calibration should be evaluated under policy change as well as under passive observation. A kernel that is accurate under the behavior that generated the training data may fail when a safe intervention changes which states are visited. Such failure withdraws causal interpretation even if ordinary forecast calibration remains acceptable.

Rare completion events require special care. A visually smooth probability surface can appear calibrated because almost every case is a non-event. Event-specific calibration, precision-recall behavior, uncertainty under low counts, and synthetic recovery should accompany global scores before the kernel is used as a route object. Calibration plots should condition on event prevalence and decision-relevant regions so abundant easy negatives do not dominate the assessment.

A kernel may also lack decision relevance. Even a calibrated probability is unsuitable for a decision role when it adds no information relevant to the protocol-fixed scientific decision, or when no lawful and safe action can depend on it. Its decision or deployment role should then retire; its descriptive or mechanism-testing role may remain if separately validated. This distinction preserves a useful scientific object without granting it operational authority simply because it is statistically well calibrated.

18.3 — Belief-State and Partial-Observation Realizations

Partial observation replaces hidden state with a probability law over compatible states. The update combines a prior, transition law, action, observation likelihood, and missingness model. It becomes sufficient for control only inside the declared model and information set. The belief is therefore an analytic state for decision and recovery, not evidence that an organism, investigator, or device has direct access to phenomenal state.

A POMDP realization separates latent route state, observed channels, actions, and policy. Reports, physiology, behavior, context, and missingness update the belief with different lags and biases. Conflicting channels should widen uncertainty rather than be forced into one score.

Belief-state models make circularity inspectable. If endpoint outcomes help train the observation model and the resulting posterior is then used to predict those endpoints, the apparent completion signal may be outcome leakage. Endpoint-blind fitting, cross-fitting, and held-out modules reduce reuse, but none repairs a state representation that omits the variables needed for the declared transition law.

Action selection introduces another distinction. The action taken changes future observations and possibly future state. A model of observed trajectories must therefore account for policy, support, and selection rather than treating actions as passive measurements.

Observation garbling provides a discriminating negative control. The route-specific posterior should lose information when channels carrying its proposed state signal are permuted, delayed, masked, or degraded in a way that preserves generic time and severity cues. If predictive performance remains unchanged, the model may be using chronology, prognosis, or endpoint proxies rather than the claimed latent state.

Belief-state methods remain valuable if PTVC fails. They provide disciplined language for uncertain clinical and behavioral dynamics, while the no-score rule prevents a posterior from becoming a dossier, eligibility rule, or claim about inner truth.

18.3.1 — Latent State, Belief State, and Observation

A latent state is the modeled condition that governs future transitions under the declared dynamics. A belief state is a probability distribution or identified set over possible latent states given the information available at the current time. An observation is one channel output used to update that belief. Treating these three objects as interchangeable would erase the uncertainty that partial-observation models are built to represent (Kaelbling et al., 1998; Särkkä, 2013).

The latent state is not the person and it is not a complete description of lived experience. It is a task-specific representation chosen for one mechanism question. A route state may include residual position, bodily variables, opportunity, context, and action support while omitting dimensions that matter morally or phenomenally. The omission should limit the interpretation rather than be hidden by a broad label.

State sufficiency is an empirical claim. If the earlier path still predicts future observations or actions after conditioning on the proposed state, then the representation is incomplete or non-Markov. Residual-history tests, state augmentation, and held-out transition prediction should decide whether a compact state is defensible.

Belief states depend on the transition and observation models that generate them. Two teams can observe the same record and obtain different posteriors because they assume different latent states, priors, channel errors, or missingness laws. The posterior therefore carries a model identity and cannot be presented as a model-free fact about the participant.

The observation layer should retain channel provenance. Self-report, behavior, physiology, contextual records, and device-quality indicators may each contribute, but none is the latent route state by itself. Correlated channels should not be counted as independent witnesses merely because they arrive from different devices.

A useful realization separates three kinds of uncertainty: uncertainty about the world state, uncertainty introduced by measurement, and uncertainty about the model family. A narrow posterior under one misspecified model does not resolve the third kind. Sensitivity across plausible state and observation models should accompany the estimate.

A concrete validation task uses simulated latent paths passed through the actual cadence, missingness, and device-quality process. The fitted pipeline should recover the known state

distribution and forecast held-out transitions. Failure in simulation stops person-level interpretation before real data are given a vivid label.

The scientific payoff is corrigibility. A belief can widen, split across alternatives, or be withdrawn when evidence conflicts. That behavior is not weakness. It is the formal expression of not claiming possession of an inner state that the study can only infer indirectly.

18.3.2 — Filtering Without Phenomenal Identification

Filtering updates uncertainty about a declared latent route state as observations arrive. It does not identify phenomenal value merely because the state variables have affective names. Threat, fatigue, action readiness, or route position can be estimated for predictive purposes while the felt quality of experience remains a different target connected only through a separately validated bridge (Särkkä, 2013).

This distinction matters because an accurate filter can still be accurate about the wrong object. A model may predict the next heart-rate sample, movement transition, or report response without measuring valence, cumulative exposure, or the terminal residual. Predictive success licenses the task that was tested, not a broader inner-life interpretation.

State aliasing is the central identification hazard. Distinct latent states can induce nearly identical observation distributions, especially when channels share medication, activity, context, or device confounds. Adding more correlated channels may tighten the posterior while leaving the underlying states unresolved.

Filtering also depends on time scale. A fast sensor can respond within seconds while a report integrates minutes and a contextual record updates only after an event. Aligning all channels to one timestamp can manufacture apparent agreement. Lag models and channel-specific kernels should be included before posterior conflict is interpreted.

Endpoint-blind training is necessary but not sufficient. Cross-fitting can prevent direct leakage from terminal labels, yet the state representation may still be selected because it produces the desired terminal pattern. Representation choice, preprocessing, and model-family search should therefore be versioned and separated from the confirmatory sample.

A useful negative control filters a state that should not carry the candidate route information, or permutes the channel expected to be distinctive while preserving generic time and severity. If the posterior remains equally predictive, the model is likely exploiting background structure rather than the proposed mechanism.

Phenomenal identification requires its own evidence: defensible representation, origin and sign, channel qualification, measured-to-latent transport, and uncertainty that permits withdrawal. The filter can contribute to that program, but it cannot skip the bridge by relabeling a hidden control state as experience.

When aliasing, channel conflict, or model dependence remains large, the correct output is a broad belief set or No adequate estimate. The data may still teach where the observation process

failed. Refusing a scalar posterior is preferable to producing a precise route score that the evidence cannot support.

18.3.3 — POMDPs and Information-Constrained Control

A partially observable Markov decision process provides a grammar for latent state, observation, action, transition, reward or cost, and policy under incomplete information. Its relevance here is structural: it forces the mechanism to declare what the controller can know and which actions are supported at each decision time (Kaelbling et al., 1998).

The belief update uses only the current prior, protocol-fixed transition law, action, and observation likelihood. Future endpoint labels, later narratives, and the true latent residual are outside the decision-time information set. A model that sees them is a retrospective reconstruction or conditioned law, not a prospective controller.

The Markov assumption belongs to the chosen state, not automatically to the observed record. If memory, unresolved history, or delayed physiology affects future behavior beyond the state, the representation must be augmented or treated as non-Markov. Convenience is not a justification for discarding path dependence.

Action affects both state and observation. Avoidance can reduce exposure while hiding the unresolved cause; disclosure can increase short-term arousal while opening repair; treatment can change physiology and the measurement gain at once. A valid realization models these pathways rather than treating actions as passive covariates.

Information-constrained control should use the action set actually available. A laboratory button, a clinical option, an institutional appeal, and a private coping action are not interchangeable treatment versions. If transportation, safety, consent, disability, or authority removes an option, the model must remove it too.

Exploration has a burden. An algorithm may value an action because it reveals the hidden state, but research cannot expose participants to harm merely to improve identification. Safe information-gathering actions, ordinary care transitions, and natural variation define the admissible exploration set.

Equal-treatment rivals should receive the same belief state, observations, actions, and tuning budget. Model-free learning, model-based planning, homeostatic control, and a completion-specific route can then be compared on held-out choices and interventions rather than on unequal information.

A POMDP realization remains a candidate construction. Successful simulated trajectories illustrate the encoded controller's behavior; a policy-level feasibility or probability-one guarantee requires a separate proof or justified bound. Neither establishes that brains, bodies, or institutions implement the policy. Empirical relevance begins with recoverable state, prospective prediction, and causal contrasts not guaranteed by the construction.

The policy output is a distribution over actions inside a declared scientific model, interpreted according to its specified task and continuation regime.

18.3.4 — Sensor Placement, Missingness, and State Aliasing

Sensor placement determines which latent distinctions can be recovered. A channel observed only during low movement, clinic visits, or periods of high adherence may confound state with availability. The study should describe where each sensor can operate, where it fails, and which parts of the target population those failures remove.

The observation model must be channel specific. Reports, heart rate, respiration, movement, sleep, speech timing, contextual records, and device-quality indicators have different lags, noise, floors, and causes of absence. One generic measurement-error term is rarely adequate for a dynamic mechanism study.

Missingness can itself be informative. A participant may remove a device during discomfort, lose reportability during illness, or skip prompts when institutional or caregiving demands are highest. Treating absence as random can make severe or poorly observed states appear calm and well regulated (Little & Rubin, 2019; Hernán & Robins, 2020).

State aliasing occurs when two latent states generate similar channel patterns. Pain and exertion may share autonomic features; sedation and low distress may share reduced movement; danger and cultural restraint may share silence. Context and alternative channels can reduce aliasing, but sometimes the states remain only partially identified.

Active sensing can help when a low-burden prompt or quality check separates plausible states. The value of that observation should be evaluated prospectively and compared with participant

burden, privacy, and reactivity. A channel that narrows a posterior without improving held-out decisions may add confidence rather than information.

Sensor redundancy should be tested, not assumed. Two devices derived from the same optical signal or two models trained on the same labels do not provide independent confirmation. Shared error sources and preprocessing should be represented in the observation graph.

Recovery simulations should place sensors according to the real protocol, including outages, drift, device changes, and state-dependent missingness. If the pipeline cannot distinguish known latent states under that deployment, the real-data posterior is not interpretable as a mechanism estimate.

A no-estimate region can be local to a sensor configuration. The study may continue to describe observed channels while withholding the latent-state claim for a subgroup, setting, or time window. That narrow abstention is more useful than a global confidence label that hides where the observation geometry failed.

18.3.5 — Policy Under Model Uncertainty

Policy under model uncertainty begins with a declared family of plausible latent states, transition laws, observation models, and admissible actions. Robust, Bayesian, and set-valued approaches organize disagreement differently, but none can compensate for an omitted model family or an action set defined after the outcome is known (Manski, 2003; Kaelbling et al., 1998).

Section 7.3.4 defines almost-sure and limit-sure attainability. Here the additional burden is epistemic: the protocol must state the information-adapted policy class and belief state under which any such guarantee is claimed (Chatterjee et al., 2016; Asadi et al., 2025).

Apply §7.3.4: an empty admissible policy class returns RouteStatus: unavailable, empty admissible policy class. A missing action system, safety exclusion, or modeling error does not establish almost-sure or limit-sure failure of the candidate mechanism.

Partial observation changes the guarantee. A policy conditions on the declared observation history or belief state, not on the hidden history that the investigator wishes it could see. Posterior probability-one completion requires probability-one action-specific completion for almost every live history-action pair under the declared posterior model; a robust guarantee over all compatible histories is stronger.

Section 7.3.4 gives the general partial-observation intersection result. Here it functions as a construction test: a belief-state route is robust only if all histories compatible with the observation share at least one structurally admissible completion action and the resulting correspondence admits a measurable, time-consistent selector. Otherwise this realization fails at the available information level.

The policy class must respect the participant's real support. Disability, cost, danger, consent, institutional authority, language, care duties, and environmental access can remove an action that

appears in an abstract model. Including inaccessible actions makes the controller look more capable than the world permits.

The value of information is conditional on burden and decision relevance. A prompt, test, or sensor may narrow the belief state, yet its scientific value also depends on time, fatigue, symptom risk, privacy loss, reactivity, and whether the result could alter the protocol-fixed decision. Information that cannot change an admissible action or verdict may remain descriptively useful without justifying additional participant burden.

Candidate and rival policies must receive the same information, action support, missingness pattern, and computational search budget. A completion-specific route that receives privileged state access can win through information rather than mechanism. Recovery, calibration, transport, and open-set adequacy remain separate from average reward.

When model disagreement is material, the scientifically honest output may be a set of admissible actions, a request for safer information, robust-route unavailability, or abstention. The policy is an analytic object for a declared research task. Participants, care professionals, and lawful decision-makers retain authority over real action, including the authority to refuse the model entirely.

18.3.6 — Belief-State Failure and Abstention

Belief-state failure begins when the posterior cannot be connected reliably to the latent state it names. Causes include state aliasing, misspecified transitions, overconfident likelihoods, omitted context, correlated channels treated as independent, endpoint

leakage, and informative missingness. Numerical stability does not protect the model from these failures.

Parameter recovery and state recovery should precede interpretation. Synthetic data generated from known hidden states are passed through the actual cadence and observation process, then refit. If the pipeline cannot recover state probabilities, transition timing, or uncertainty coverage, real-data posteriors remain exploratory (Särkkä, 2013; Villaverde, 2019).

Posterior predictive checks should target temporal structure rather than only marginal fit. The model may reproduce the overall distribution of reports while missing order, dwell time, lag, action dependence, or the pattern of missing observations. Those failures identify which part of the realization is wrong.

No-omniscience is a separate gate. The state update and policy may use only information available at the protocol-fixed decision time. A model that relies on future endpoint labels, later treatment response, or retrospectively selected histories is a smoother or conditioned reconstruction and must be labeled accordingly.

Abstention can be local. The pipeline may estimate one region of the state space while returning No adequate estimate under severe channel conflict, unsupported transport, rare actions, or endpoint ambiguity. Local no-estimate regions preserve useful analysis without pretending uniform coverage.

Policy sensitivity supplies another failure test. If small plausible changes in the posterior, transition law, or utility reverse the proposed action, the action is not stably identified. A robust set or

no-action output is preferable to a brittle recommendation presented with confidence.

Model-family uncertainty should remain visible after parameter uncertainty narrows. A posterior under one POMDP may be sharp while alternative state representations fit equally well. Open-set challenge generators and none-of-the-models options prevent the chosen family from becoming a protected theory.

Failure can be role-specific: a belief state may remain predictive while losing causal or measured-to-latent status. Record the surviving role, failed diagnostic, affected reference class, and successor plan. Artifact correction follows Section 18.2.6 and Chapter 28; a successor model does not inherit the withdrawn interpretation.

The model separates hidden variables, observations, and assumptions before inferential use. The next section asks how the whole path law changes when a terminal event is imposed or conditioned on. Keeping those layers separate lets a later conditioning result be attributed to the path law rather than to unrecorded changes in what the agent was assumed to know.

Model disagreement can itself guide measurement. When plausible belief models diverge on which observation would be informative, a low-burden channel or timing change can be selected prospectively. The design should record the expected information gain and the possibility that the new observation changes behavior rather than merely revealing state. Expected information gain should be discounted by burden and reactivity, so the most informative observation in theory is not automatically

the best observation to collect. That tradeoff can be evaluated prospectively by comparing candidate observation plans on expected uncertainty reduction, participant burden, and induced behavior before deployment; if the preferred plan changes across plausible belief models, the study should preserve that design uncertainty rather than select the channel favored by one model after outcomes are known.

18.4 — Path Conditioning and Bridge Constructions

Conditioning begins with a reference path law P_0 and an event A . For a positive-probability event, the conditioned law is $Q(B) = P_0(B|A)$. The resulting paths exhibit A by construction. Their existence demonstrates mathematical compatibility, not that an unconditioned system contains a native completion mechanism.

A Doob h -transform uses a positive harmonic or space-time harmonic function to alter transition probabilities or drift. When h is a committor, the transform can generate paths biased toward the completion event. The object is scientifically useful because its likelihood and generator differ from the baseline in a calculable way.

For exact singleton targets of probability zero, ordinary conditioning is insufficient. A bridge kernel, regular conditional law, disintegration, or shrinking-band limit must be justified. Tightness and uniqueness matter: different subsequential limits can define different candidate bridges.

Schrödinger bridge formulations choose a dynamically admissible path law close to a baseline while satisfying endpoint constraints. A representative objective is

$$J_{path}(K, \alpha) = \inf_{Q \in Q_{dyn}: Q(A_K) \geq 1 - \alpha} D_{KL}(Q \| P_0).$$

The objective quantifies steering relative to a declared baseline path law; it supplies no evidence that nature pays, minimizes, or even represents that cost. Conditioned and native processes can instead be compared through likelihood, reciprocal structure, intervention response, and transport. A bridge mixture may be

reciprocal without being Markov, while a local controller may become Markov only after the state is augmented. Those distinctions create the testable content.

The endpoint event, baseline law, topology, and admissible class must be fixed before fitting. Otherwise the construction becomes an after-the-fact explanation of whatever path occurred.

18.4.1 — Conditioning Events and Changed Path Laws

Conditioning on a terminal event changes the path law. A process observed under the event is not distributed like the unconditioned process with a fortunate final value; earlier transitions, dependence, and likelihoods are reweighted throughout the path. This distinction is the starting point for every bridge or selector construction (Doob, 1957; Chetrite & Touchette, 2015).

The conditioning event must be explicit. Reaching a set before another, lying in a terminal band at a fixed time, surviving to an endpoint, or matching a terminal distribution are different events and induce different laws. A phrase such as conditioned to complete is too vague for scientific use.

Positive-probability events permit ordinary conditional probabilities. Probability-zero targets require a separately justified bridge, limiting sequence, or change of measure. Treating an exact-zero event as though it were an ordinary observed subgroup can hide singularity and create a formal object with no recoverable data analogue.

Conditioning can imitate mechanism. A selected cohort of paths that end near zero may display preterminal drift, variance compression, or reciprocal dependence even when the native

process has no closure controller. Those induced signatures become serious rivals to any route-specific interpretation.

The observation process may create another conditioning layer. Requiring complete terminal records, surviving device coverage, or reportability near the endpoint selects paths before the formal model is fitted. The sample law must include these filters rather than treating them as harmless inclusion criteria.

A useful analysis compares likelihoods under the native law, the explicitly conditioned law, and candidate causal dynamics. Held-out actions or unexpected endpoint probes are especially valuable because conditioning fitted to one terminal event may not predict them.

Conditioned constructions remain scientifically valuable. They provide exact countermodels, rare-event tools, and benchmarks for what terminal appearance can arise without a native law. Their role is to make the candidate route harder to protect, not to supply evidence by mathematical resemblance.

18.4.2 — Doob h-Transforms

A Doob h-transform constructs a new Markov law by reweighting transitions with a positive harmonic or space-time harmonic function. Under suitable conditions, the transformed generator describes the reference process conditioned or biased toward a declared event. The function h , its domain, and its boundary behavior are therefore part of the model, not decorative notation (Doob, 1957; Chetrite & Touchette, 2015).

The transform changes transition probabilities before the endpoint. It can generate local drift toward regions with higher

completion potential while preserving Markov structure. That local appearance can resemble a controller even when the transformed law was obtained from conditioning rather than from a physical feedback mechanism.

Scientific use requires an independent route for h . Estimating h from the same terminal outcomes later used to claim prediction creates leakage. A defensible program estimates or specifies one completion potential, freezes it, and tests transport across held-out transitions, actions, endpoints, or terminal-support modules.

Action-indexed transforms require further typing. A separate h for every action can absorb any policy difference and lose causal content. The model should state whether h is common across actions, how interventions alter the reference generator, and what invariance is predicted under shared information.

Support matters. The transform is not valid where h vanishes or where the transformed measure is not absolutely continuous with respect to the reference on the relevant sigma-field. Apparent success outside support is extrapolation, not a conditioned-law result.

The transform loses as a mechanism when an equal-budget endpoint-conditioned rival fits just as well, when h fails held-out transport, or when the required reference dynamics cannot be recovered. It may remain a formal bridge or simulation device after its causal interpretation is retired.

18.4.3 — Schrödinger Bridges and Entropic Steering

A Schrödinger bridge finds the path measure closest in relative entropy to a reference process while matching declared endpoint

marginals. It is an entropy-regularized path-space transport problem and can be written as a stochastic control problem. The construction shows how a distribution can be steered between boundaries with minimal change relative to a baseline, not that nature uses that steering law (Léonard, 2014; De Bortoli et al., 2021).

The endpoint objects are distributions rather than a single realized terminal value. Choosing them already encodes the target population and resolution. A narrow terminal distribution around zero, an atom at zero, and exact pathwise closure are different constraints and should not be collapsed into one bridge picture.

Soft constraints define a different object. Instead of forcing the terminal marginal to equal the target, a soft-constrained bridge permits mismatch and penalizes it through a divergence term. That construction can improve robustness when a finite band or noisy target is scientifically appropriate, but it cannot be presented as exact endpoint matching; the tolerance, penalty, and resulting terminal distribution remain part of the claim (Garg et al., 2024).

The reference process carries most of the scientific meaning. A Brownian, homeostatic, learned-control, or clinically stratified baseline yields a different bridge even with the same endpoint marginals. Comparison therefore asks whether the inferred control is stable across plausible references and whether one reference explains the data without special steering.

Entropic cost is a divergence on path laws. It is not effort, suffering, moral cost, or phenomenal burden unless a separately

justified bridge is supplied. In exposition and diagrams, the relative-entropy objective is described as a distance-like control penalty between probability laws, never as a price paid by a person.

Finite-sample algorithms can fit bridges that reproduce marginals while missing transition structure. Validation should include two-time dependence, intervention response, likelihood-ratio diagnostics, and recovery under the actual observation cadence. Matching the start and end distributions is a weak test of the interior law.

Table 18.1. Path-space constructions and distinguishing evidence.

Construction	What is fixed	How the path law changes	Distinguishing evidence
Native Markov route	Generator, state, action law, and endpoint branch	Evolves prospectively without selecting on the terminal event	Held-out transitions, interventions, and unexpected endpoints
Doob h-transform	Reference law, event, and positive completion potential h	Reweights local transitions toward the declared event	Frozen-h transport, support, and likelihood-ratio checks
Schrödinger bridge	Reference law and endpoint marginals	Finds a minimum-relative-entropy steering law on path space	Interior covariance, reference sensitivity, and intervention response

Construction	What is fixed	How the path law changes	Distinguishing evidence
Brownian or reciprocal bridge	Starting law, terminal condition, and horizon	Conditions trajectories on boundary information by construction	Known reciprocal dependence and absence of an independently recovered native mechanism

Partial observation complicates steering. The algorithm may operate on measured features while the target concerns a latent phenomenal object. A bridge in sensor space is not automatically a bridge in latent state, and the measured-to-latent transport must remain explicit.

The strongest rival interpretation is selection. A population assembled by endpoint status can have the same apparent endpoint marginals without any steering mechanism. Inverse-probability weighting, generative selection models, and held-out sampling branches should enter the comparison.

Schrödinger bridges are valuable because they make a precise quantitative comparison available. If a minimally changed conditioned law reproduces the candidate signatures, the route needs additional evidence. If the bridge fails on held-out interventions or unexpected endpoints, that failure becomes a discriminating result.

18.4.4 — Brownian Bridges as an Explanatory Baseline

A Brownian bridge is the clearest baseline for understanding why terminal appearance can be created by conditioning. Starting from a diffusion, the bridge law conditions the path on a specified

endpoint. The resulting process acquires a time-dependent drift that pulls trajectories toward that endpoint and a covariance structure that contracts as the horizon closes. Exact arrival is specified by the construction; comparison with an unconditioned process therefore examines additional predictions, such as pre-endpoint dynamics and responses to changed conditions (Doob, 1957; Chetrite & Touchette, 2015).

The bridge is valuable because its preterminal signatures are explicit. Conditional means, two-time covariance, an increasing restoring coefficient, and variance contraction can all resemble route-specific closure. Those similarities make the bridge a serious rival rather than a classroom curiosity. Any proposed completion mechanism should be compared against a bridge that receives the same endpoint information and observation noise.

The baseline also exposes the endpoint-information problem. If the analyst selects only trajectories known to end in a band, then fits a contracting process to the selected sample, bridge-like behavior is expected. The sample has been conditioned even when no bridge equation was written. Selection diagrams and prospective cohort definitions are therefore part of the mathematical comparison.

A useful simulation compares the observable laws produced by unconditioned diffusion, endpoint selection, and explicit bridging under the same observation kernel. Endpoint-selected paths and explicitly bridged paths can have the same conditional law; recovery must then return model ambiguity rather than distinguish labels unsupported by the data. Additional hazard,

intervention, or endpoint-regime measurements discriminate only when the models specify different predictions for them.

Brownian motion is not assumed to be a literal model of consciousness or valence. Its role is explanatory and adversarial: it identifies the additional path information needed to interpret an average that approaches a terminal boundary. More realistic diffusion, jump, and hidden-state rivals can be added after the baseline has made the logic visible.

A cumulative ledger requires a path-class-matched comparator. For an explicit Gaussian illustration, fix a deterministic $T > 0$ and fixed initial values L_0 and V_0 , and start from

$$dL_t = V_t dt, \quad dV_t = \sigma dW_t, \quad \sigma > 0.$$

Here L is an accumulated coordinate and V its stochastic rate. Conditioning the reference law on $L_T = 0$ through its Gaussian regular conditional distribution gives, for $t < T$,

$$dL_t = V_t dt, \quad dV_t = -\frac{3\{L_t + (T - t)V_t\}}{(T - t)^2} dt + \sigma dW_t.$$

The Brownian driver in this conditioned equation is defined under the conditioned law. The derivation uses $\text{Var}(L_T | L_t, V_t) = \sigma^2(T - t)^3/3$ under the unconditioned reference law. The accumulated coordinate remains finite variation even though its rate is stochastic. This supplies a linked residual–rate–horizon prediction for recovery benchmarks and matches the path class of an accumulated rate. The coefficients of the conditioning drift become singular as t approaches T , making resource requirements an explicit problem for biological implementations. Comparisons should give each model the same endpoint, selection, and

observation information (Doob, 1957; Chetrite & Touchette, 2015).

For the zero-initial-value case $L_0 = V_0 = 0$, Gaussian conditioning also gives

$$\text{Var}(L_t \mid L_T = 0) = \frac{\sigma^2 t^3 (T - t)^2 (4T - t)}{12T^3}, \quad 0 \leq t \leq T.$$

Writing $\tau = T - t$,

$$\frac{\text{Var}(L_{T-\tau} \mid L_T = 0)}{\tau^2} \rightarrow \frac{\sigma^2 T}{4} \quad \text{as } \tau \downarrow 0.$$

The same integrated Gaussian example therefore supplies both a conditional drift and quadratic preterminal variance contraction. These linked predictions can be assessed together when comparing candidate dynamics.

18.4.5 — Likelihood Ratios, Support, and Endpoint Leakage

A change of path law is interpretable only where the transformed measure has adequate support relative to the reference. Likelihood ratios quantify the reweighting on declared sigma-fields; they do not create information where the reference assigns zero probability. Support failure is therefore a structural limit, not a numerical inconvenience (Jacod & Shiryaev, 2003; Chetrite & Touchette, 2015).

Extreme likelihood ratios reveal practical nonoverlap. A small number of heavily weighted paths can make a conditioned estimate unstable even when absolute continuity holds formally. Effective sample size, tail behavior, and weight truncation sensitivity should be reported before route claims are made.

Endpoint leakage occurs when realized terminal information enters state construction, feature selection, imputation, or model tuning. A model can then appear to forecast completion because it has already encoded the answer. Splitting observations into training and test sets does not remove leakage if the feature definition used both.

Time-to-endpoint requires an explicit clock. Retrospective alignment describes a terminal trajectory; prospective filtering uses information available at the prediction time. Every plot identifies whether its coordinate is calendar time, clinical prognosis, filtered horizon, or realized remaining time.

Selection through observation is another leakage route. Complete-case analyses, device-quality thresholds, and reportability windows may condition on variables caused by state or proximity to the endpoint. The resulting law can resemble a completion bridge even when the native process does not.

A leakage audit follows every derived variable from raw observation to final score. It records which timestamps, endpoint labels, future summaries, and adjudication decisions were available at each stage. Any feature whose construction uses future information is restricted to retrospective analysis.

Held-out endpoint modules help but do not solve support. A potential trained on one terminal event can be tested on a different event or unexpected probe, provided the state and observation bridges are common. Failure indicates that the original transform may have learned event-specific selection rather than a general route.

When support cannot be established, the result is No adequate estimate or Model-ambiguous as appropriate; when leakage invalidates the comparison, its evidence is Unusable. A large likelihood-ratio-weighted effect is not strengthened by its extremity. It is often a warning that the comparison is being carried by a few incomparable paths.

18.4.6 — Native Dynamics Versus Retrospective Conditioning

Native dynamics specify how the process evolves before any terminal event is selected. Retrospective conditioning specifies the law of paths after restricting or reweighting by that event. Both can produce the same endpoint distribution, and sometimes the same one-time marginals, while disagreeing on transition probabilities, reciprocal dependence, and response to intervention (Doob, 1957; Chetrite & Touchette, 2015; Léonard, 2014).

The distinction requires a reference law. A route claim should state the unconditioned generator or transition kernel, the information available at each time, and the way the endpoint enters. If the native law is never estimated, the analyst cannot determine whether a fitted bridge represents a mechanism or merely a conditional description.

Preterminal covariance discriminates only when the declared models imply different observed kernels. A Brownian bridge is itself Markov and admits a time-dependent feedback representation; reciprocal structure alone does not distinguish conditioning from a native implementation. If the path laws agree, passive observations cannot separate the causal accounts. A

proposed intervention must specify different responses before it can do so (Çetin & Danilova, 2016).

Intervention response is stronger. A native controller should react when a safe action changes state or support. A retrospective bridge fitted only to terminal membership may not transport to that intervention unless the conditioning model explicitly includes it. Earliest-divergence tests in the next chapter are designed for this distinction.

Unexpected endpoint probes provide another test. A transform learned from one planned terminal event may fail when an independently occurring probe samples the path. A genuine native route should state what remains invariant across those probes and what changes with the endpoint branch.

Selection models must be included. Frailty, survival, device retention, reportability, and site-specific follow-up can select terminal-compatible paths without changing individual dynamics. Equal-information comparison asks whether selection explains the apparent preterminal signature as well as the route.

Model recovery precedes empirical ranking. Simulations from native controllers, conditioned bridges, hitting-time models, and observation-selection mechanisms should pass through the same pipeline. If the method confuses them under realistic sampling, real-data labels are not interpretable and no preference should be inferred from fit alone (Wilson & Collins, 2019; Villaverde, 2019).

The route can be evaluated through a joint prediction involving transition law, two-time structure, intervention response, endpoint support, and the observation process.

A route may survive only as a conditional representation. That is still useful. It can organize terminal-compatible paths and generate tests while losing the claim that it describes causal dynamics. The record should preserve that narrowed role rather than oscillate between total endorsement and total deletion. A conditional representation can be judged by descriptive calibration without receiving causal credit for the process that generated the paths.

The section ends with a strict wording rule: conditioned path law, candidate native mechanism, selector, and observation artifact are different statuses. A diagram or equation must name which one it depicts. Mathematical equivalence at one boundary does not grant causal equivalence through the path. Status labels should survive cropping and reuse, especially when a transformed path visually resembles a naturally generated one.

A direct diagnostic compares forward simulation from the estimated native law with endpoint-conditioned reconstruction. If only the conditioned reconstruction reproduces the data, the result favors that description over the fitted native candidate; it does not exclude other native laws or identify selection as the cause. If both fit, the mechanism remains ambiguous until intervention or unexpected-endpoint evidence separates them. Unexpected endpoints or interventions are especially valuable here because they can expose whether the apparent bridge effect persists when the conditioning event changes.

The comparison should include reciprocal structure, two-time dependence, action response, and likelihood ratios under the same observation process. Matching one terminal histogram is too weak because many conditioned and selected laws can share it. Shared observation models are essential, since different smoothing or missingness assumptions can create apparent differences between native and conditioned laws.

A public diagram should show the unconditioned law, the conditioning event, and the transformed law as separate objects. Its caption should state that conditioning changes the path distribution by construction and that natural realization remains an empirical question. Showing the transformation explicitly also teaches readers which patterns are consequences of conditioning and therefore cannot independently count as evidence for a natural mechanism.

18.5 — Absorbing, Hitting, and Bridge Routes

Absorption, first hitting, stickiness, reflection, state-coupled stopping, endpoint atoms, and bridges can all produce terminal concentration. They differ in what happens before and at the boundary. The protocol must name the implementation branch rather than call them all closure.

A first-hitting implementation stops at the first entry into a declared set. Choosing the zero-residual surface makes closure part of that stopping definition. Prospective approach, hazard, and perturbation signatures then supply empirical comparisons with other mechanisms.

An absorbing boundary keeps the process at the boundary after contact. A sticky boundary allows positive occupation time there. Reflection returns the process to the interior. Each produces different local time, dwell, and return signatures that can be estimated when the state and observation resolution are adequate.

Endpoint atoms require their own branch. Sudden biological events, administrative censoring, scheduled protocol stops, device shutdown, and discrete adjudication rules can place mass at particular times or states. Density-based arguments do not automatically extend to those atoms. A model that smooths them away can generate a false theorem, a distorted hazard, and misplaced confidence about terminal concentration.

Integrated-rate bridges are often more path-class appropriate than bridges placed directly on an accumulated finite-variation ledger. The rate process can remain stochastic while its integral is

conditioned on a terminal band. The construction must specify the path space and limiting mode.

A model tournament should simulate all branches through the actual observation pipeline. The question is not which ideal process resembles a plotted average, but which branch can be recovered and predicts held-out path features, endpoint hazards, and perturbation responses.

18.5.1 — First-Hitting and Zero-Surface Termination

First-hitting implementations define an endpoint as the first time a process enters a declared set. For the zero-residual surface, closure follows from the stopping rule whenever the surface is reached. The empirical questions concern the justification of that state and stopping relation and the preterminal observations that distinguish the route from alternatives (Doob, 1957).

A zero surface can be crossed, touched, approached without contact, or reached by a jump. Those possibilities imply different local signatures. Continuous diffusion may accumulate local time near the boundary; a finite-variation path may approach monotonically; a jump process may cross without ever occupying zero. The implementation must specify path class and boundary regularity before a hitting-time theorem is invoked.

A first-hitting candidate specifies a stopping mechanism and compares its timing with an independently defined endpoint. Keeping those definitions separate makes their relationship testable. The same model can also generate route-level predictions about trajectories before the stopping event.

A low-risk model-recovery study can compare simulated first-hitting paths with fixed-time bridges, absorbing processes, and state-coupled hazards. The observation pipeline should include realistic sampling and missingness. Distinguishing features may include return behavior before stopping, hazard spikes near the surface, and response to safe perturbations that shift the state without altering endpoint definition.

Ordinary processes can also generate first-hitting behavior. Threshold decisions, failure limits, physiological triggers, and institutional stop rules all create hitting times. Those rivals should be represented in the same state space before a fairness-specific mechanism is credited. The fact that a stopping surface exists says nothing about why that surface is relevant to experience.

The route fails when the surface is unreachable for observed prefixes, when stopping occurs away from the declared surface, when endpoint labels are required to recover the state, or when ordinary threshold models explain the data as well. A first-hitting construction is useful because it makes those failure conditions sharp.

18.5.2 — Absorbing and Killed Processes

An absorbing implementation keeps a process in a boundary state once that state is reached. A killed process instead terminates the path or transfers probability mass to a cemetery state. These constructions may share a first entrance time and differ completely afterward. The distinction matters whenever post-contact observation, return, or continued consciousness is possible (Champagnat & Villemonais, 2016).

Absorption creates a persistent atom at the boundary in later-time distributions. Killing removes the path from ordinary state space and changes denominators. A dataset that treats missing post-boundary observations as zeros can make killed trajectories look absorbed. The observation and retention processes must therefore be modeled alongside the state dynamics.

In terminal research, an administrative stop, device shutdown, loss of reportability, and truth-level endpoint are not interchangeable. Each can produce an apparent killed process. A comfort-first protocol may correctly stop collection while experience continues. The model should record that as observation termination rather than infer an absorbing phenomenal state.

Quasi-stationary behavior supplies another diagnostic. Among paths not yet killed, the conditional state distribution may stabilize even as survival probability declines. That pattern can resemble regulation or terminal concentration and should be compared directly. The route's predictions concern both the surviving distribution and the exit process, not one terminal histogram.

Simulation should preserve denominator flow. Analysts should report how many paths remain at risk, how many are absorbed, how many are killed, and how many become unobservable for other reasons. Recovery succeeds only if the pipeline can distinguish those branches under the actual cadence and noise floor.

An absorbing or killed route is retired when post-boundary returns are common, when the apparent atom disappears after observation correction, or when endpoint status cannot be separated from dropout. The construction remains a useful rival even when it loses as a PTV mechanism, because it teaches how terminal samples are shaped by state and observation together.

18.5.3 — Sticky and Reflected Boundaries

Sticky and reflected boundaries preserve continued dynamics after contact. Reflection pushes a process back into the interior, while stickiness allows positive occupation time at the boundary before movement resumes. Both can create repeated near-zero observations without terminal absorption. Their local-time structure is therefore central to discrimination rather than a mathematical detail.

A reflected process can touch the boundary many times and immediately return. A sticky process can produce a visible pile-up at zero even when the eventual endpoint is elsewhere. Finite-resolution instruments may confound both with absorption because small excursions are rounded away. Sampling frequency and measurement floor must be included in the simulation.

The distinction has a concrete analogue in regulated systems. A controller may repeatedly correct deviations from a constraint, creating reflection-like behavior, or dwell in a low-activity state before reactivation. Neither pattern implies a complete-stream boundary law. It may be ordinary homeostasis, task design, or instrument censoring.

Useful tests examine dwell time, excursion distribution, return frequency, and sensitivity to safe perturbation. Reflection predicts immediate direction change at the boundary; stickiness predicts excess residence. An absorbing model predicts no return. These features can be estimated from preterminal data when cadence is sufficient and the observation process is not itself sticky.

Hidden state complicates the picture. A measured channel may appear stuck while the latent state moves, or appear reflected because a bounded scale truncates values. Cross-channel comparison and latent-state recovery should therefore precede a mechanism verdict. A zero-heavy report scale is not evidence of a zero-heavy phenomenal process.

The route is narrowed when local-time or dwell signatures fail, and retired when the boundary behavior is produced entirely by scale limits or preprocessing. Sticky and reflected models earn their place by offering precise alternatives to the vague claim that a path simply converges.

18.5.4 — Endpoint Atoms and Singular Carriers

Endpoint atoms place positive probability on particular times or states. Singular carriers concentrate endpoint mass on sets that need not possess an ordinary density. Both can defeat arguments requiring an independent endpoint law with a positive Lebesgue density on the declared support; singular continuous laws, however, remain non-atomic. The model must declare whether atoms or singular carriers arise from biology, protocol, scheduling, state-coupled stopping, or observation design before applying the relevant obstruction.

Atoms can be mundane. A study may end at a scheduled visit, a device may shut down at a battery threshold, or a protocol may stop at discharge. Those events can produce repeated endpoint times unrelated to the truth-level endpoint. Treating them as natural terminal structure would import the study calendar into the hypothesis.

State atoms are equally important. Rounding, censoring, and bounded scales can produce a pile-up at zero. A latent mixture can also include a true atom, a narrow continuous component, or both. The analysis should compare those possibilities through an observation-kernel model rather than declaring exact neutrality from repeated zero readings.

Singular endpoint supports require topology as well as probability. If endpoints lie on a lower-dimensional manifold or a discrete state graph, density-based tests may be inappropriate. Support-aware methods should ask which neighborhoods carry mass and whether the proposed route predicts the carrier independently of the terminal observations used to discover it.

A rigorous design uses negative controls for administrative atoms and simulated observation floors. It also separates the time atom from the state atom: many paths can end at the same scheduled time without sharing a terminal state, and many can share a recorded state because the instrument has coarse bins. Each factor receives its own model component.

The route fails when its atom disappears under finer measurement, when the support is determined by protocol rather than process, or when alternative endpoint models reproduce the

same concentration. An atom is a property to explain, not a shortcut around endpoint discipline.

18.5.5 — Continued-Stream and Endpoint-Inclusive Branches

Continued-stream and endpoint-inclusive branches change the domain of the claim. If experience continues after the observed endpoint, then the recorded terminal residual is a prefix, not the complete object. If a phenomenal contribution is assigned exactly at the endpoint, the strict-pre-endpoint and inclusive ledgers differ by that contribution. These branches are non-equivalent and must be specified separately before outcome inspection.

Return and resuscitation make the distinction concrete. A protocol may close a window when report ceases and later observe recovery. The original stop cannot be treated as the truth endpoint. The stream-routing and endpoint records should reopen or mark the earlier object as provisional rather than forcing the returned experience into a new stream for convenience.

Endpoint-inclusive models also require a rule for what belongs at the boundary. A final conscious event, a transition with uncertain duration, and a biological event without experience are different. Assigning all terminal physiology to phenomenal value would collapse mechanism and target. The inclusion rule must be independently justified and sensitivity-tested.

Observation shutdown is a strong rival. Sedation, delirium, device removal, caregiver burden, or comfort-first stopping can end measurement while lived experience remains uncertain. A no-

estimate outcome is scientifically stronger than treating silence as zero or as a completed stream.

Comparisons across branches should use the same pre-endpoint data and explicitly different remainder models. One analysis can estimate strict-pre-endpoint compatibility; another can bound an unobserved terminal remainder; a third can remain nonadjudicable. The branch choice should not be changed to rescue an unfavorable result.

The route is retired when continued experience is systematically misrouted, when inclusive contributions cannot be identified, or when branch switching occurs only after inspection. Scientific clarity begins by allowing the endpoint domain to remain unresolved.

Endpoint-branch adjudication uses a frozen event packet that withholds the terminal ledger estimate. The packet records the protocol endpoint, clinical state, evidence of continuing or returning consciousness, measurement-shutdown reason, routing decision, and uncertainty. A blinded panel then classifies the event as strict-pre-endpoint, continued-stream, endpoint-inclusive, provisional, or nonadjudicable. This ordering prevents the favored terminal value from deciding which endpoint branch is said to have occurred.

An unobserved terminal remainder can sometimes be bounded even when it cannot be point estimated. The bound may use the last observation meeting the declared qualification criteria, maximum plausible exposure under the chosen clock, duration uncertainty, and branch-specific limits on the measured-to-latent

bridge. The full interval enters the route verdict. When a valid finite bound spans outcomes that reverse the conclusion, the result is Inconclusive evidence rather than a midpoint presented as terminal truth (Little & Rubin, 2019; Manski, 2003).

Return events require versioned reanalysis. The original endpoint classification, all decisions made under it, and the evidence available at that time remain preserved; the later return creates a successor record rather than erasing the earlier uncertainty. Analyses that used the provisional stop are rerun under the corrected routing and endpoint object. This makes error visible and allows recovery without pretending that a protocol can know the truth-level boundary in real time.

Recovery simulations should include branch misclassification. Synthetic paths can continue after observation shutdown, include an endpoint atom, return after a provisional stop, or carry an unobserved inclusive contribution. The complete pipeline is then asked to recover the correct branch and to abstain when the evidence is insufficient. A route estimator that appears accurate only when the endpoint branch is supplied by the simulator has not solved the adjudication problem.

Detachable figures and tables should carry the branch label with the estimate. A curve described only as terminal can be mistaken for complete-stream evidence even when it ends at report loss or a protocol window. The caption therefore names the endpoint object, the strict or inclusive domain, the treatment of return, and the remainder status. This compact provenance is scientific content because it determines what the displayed trajectory can and cannot adjudicate.

18.5.6 — Distinguishing Identical Endpoints by Preterminal Data

Different mechanisms can produce the same terminal value. A fixed-time bridge, first-hitting route, absorbing boundary, sticky process, state-coupled hazard, and endpoint-selected sample may all place mass near zero. Terminal concentration alone therefore has weak discriminating value. The scientific burden is to use preterminal dynamics and intervention response to separate the branches (Doob, 1957; Champagnat & Villemonais, 2016).

The most useful differences concern path law. Candidate branches imply different covariance, hazard, dwell, return, jump, and local-time signatures. They also differ in how a safe perturbation changes future trajectories. A bridge conditioned on the final event can absorb an intervention differently from a native state-coupled controller.

Observation models can erase those differences. Sparse sampling may miss returns; a floor may hide excursions; informative dropout may remove unstable paths. Simulations should therefore pass every candidate through the protocol-fixed measurement cadence, censoring, and missingness process before model recovery is assessed.

A common-rule tournament can begin with identical state information and endpoint labels, then withhold endpoint labels in a prospective module. If a route succeeds only when the final event is supplied, it is a conditional reconstruction rather than a discovered mechanism. Held-out unexpected endpoints are

especially valuable because they prevent the model from relying on a known horizon.

The comparison should include a none-of-the-models option. If all branches miss preterminal structure, the result is open-set inadequacy rather than a victory for the least-wrong implementation. Residual diagnostics and challenge generators should be reported beside classification accuracy.

This subsection closes the implementation family with a strict rule: identical endpoints do not imply identical mechanisms. A route earns support only from prospective, recoverable differences that survive the observation process and equal-treatment rivals.

Dwell and return behavior can be especially informative near a boundary. Absorption predicts no departure after entry; reflection predicts immediate return; stickiness predicts positive residence; first hitting links termination to initial contact; a fixed-time bridge can cross the target repeatedly before the horizon. Sampling cadence must be high enough to observe the distinction.

Hazard and local-time signatures should be modeled together where the implementation permits. A rise in endpoint hazard near the target, repeated grazing without termination, and long residence at the boundary are different phenomena. Collapsing them into one terminal-zero indicator destroys the mechanism information.

Unexpected safe perturbations provide a final check. A native boundary controller should respond according to its local rule, whereas a cohort selected by terminal outcome may show no

prospective response once endpoint information is removed. The perturbation module should be scored without access to the eventual terminal label.

Boundary-contact data should retain exact timing at the resolution the sensor can support. Rounding several contacts into one terminal observation can make reflection, stickiness, and absorption indistinguishable. The analysis should model interval censoring rather than pretend the contact time is known.

Endpoint atoms and continuous hazards should be compared explicitly. A mass point at a scheduled or administratively defined time can mimic terminal concentration while following a different law from first hitting or state-coupled stopping. Mixture models and branch-specific simulations should test whether the data can separate those components.

Recovery benchmarks should include identical terminal histograms generated by different preterminal laws. The task is successful only when the pipeline distinguishes those laws from the path before the endpoint. Classification based mainly on the terminal value would fail the mechanism question by construction.

When cadence is insufficient, the verdict should name the lost distinction. The study may establish that several implementations remain compatible while being unable to tell which one generated the data. That is a mechanism-ambiguous result, not weak support for all of them.

A path-law comparison should score one-step and multi-step predictive densities before the endpoint, not only classify the

implementation after the full trajectory is visible. Transition residuals can reveal whether the fitted process places probability on the observed sequence of approaches, retreats, and returns. Cross-validated likelihood or proper-score differences should be reported by distance from the boundary so a final segment cannot dominate an otherwise poor preterminal fit (Gneiting & Raftery, 2007).

State normalization matters when comparing intervention response. Two branches may appear to react differently only because one cohort begins closer to the target surface or with less remaining action support. The perturbation analysis should condition or match on the declared preintervention state, then compare the timing and shape of divergence. A branch that requires post hoc state redefinition to recover its expected response has failed the prospective mechanism test.

Channel-ablation tests can expose observation dependence. The mechanism should be refitted or rescored after removing the channel most directly tied to endpoint ascertainment, while retaining the same state definition and verdict rule. If discrimination collapses only when that channel is withheld, the result may reflect endpoint leakage rather than a path-law difference. A robust branch should preserve at least its prospectively specified qualitative ordering across independent observation routes.

18.6 — The Queue Model as an Illustrative Candidate

The Queue model is retained as one illustrative state architecture for unresolved demands, partial observation, priority, blocking, reactivation, repair, and residual influence. Its scientific value lies in the distinct transition and intervention patterns those modules predict.

A scientific queue declares state variables, observation channels, actions, transition rules, costs, constraints, policy classes, recovery tests, and retirement conditions. These specifications turn the architecture into a model with observable consequences.

This section develops the model's POMDP, belief-state, recovery, and public-language analyses. It should be read as a bounded candidate family inside the wider path-space chapter, not as the mechanism of PTV C.

18.6.1 — Candidate Mechanism Status

The queue-style route is a midlevel candidate mechanism. It asks whether explicitly represented unresolved demands, blocking, reactivation, and repair improve prediction and causal localization beyond ordinary rivals. Its components can be refined through source specification, measurement, recovery tests, and prospective comparisons (Kaelbling et al., 1998; Sutton & Barto, 2018).

A midlevel mechanism can be measured, compared, revised, or retired separately from the wider program. Its role is to explain specified observations.

The working claim is conditional. If unresolved demands have queue-like dynamics, then attention, action, physiology, and report should show structured priority, blocking, reactivation, and repair sensitivity beyond what simpler rivals predict under the same observations. That sentence is stronger than a metaphor and weaker than validation. It says exactly where the model might matter.

A candidate mechanism also has a narrower reference class than the Law. It may apply to some burdens, some developmental periods, some cultures, some clinical states, or some task settings and not others. A caregiver's recurring vigilance after months of night alarms may have queue-like features, while a transient mood shift after bad weather may not. The model must allow partial domain.

The model also needs a negative space. Certain experiences should not enter the queue simply because they are vivid. Joy, boredom, fatigue, ordinary desire, routine planning, and benign memory may affect lived experience without becoming unresolved demands. If every salient state becomes queue material, the model loses its shape and becomes a name for mental life in general.

Candidate status disciplines the prose. A queue-style route may represent state, assign model quantities, specify a selection rule, or generate closure under stated assumptions. Those are mathematical operations, not evidence that an unseen agent knows a person's balance, punishes them, or distributes morally owed rewards. Describe what the model computes and predicts without turning it into an intentional actor.

18.6.2 — State, Demand, and Control

The queue illustration begins with three explicit objects: a state describing active and latent demands, a demand representation that can be multidimensional, and a control process that changes future state or access. None of the objects is a person, a moral account, or a direct phenomenal ledger.

State and control are defined on real opportunity. Transportation, money, energy, care access, physical capacity, communication, permission, and safety determine whether an action is actually supported, and those constraints may change over time. Recording them lets the model distinguish an inaccessible repair route from an internal transition failure.

One candidate input is an unresolved mismatch between expected and actual affective consequence. A person expected an apology to feel safe, but it did not. A long-sought achievement produced relief for a day and dread afterward. A harmless-looking letter triggered fear because past letters carried danger. These mismatches can update beliefs and keep demands active (Kaelbling et al., 1998; Sutton & Barto, 2018).

Prediction-error language is useful because it gives a rival its strongest form. Predictive processing and active-inference accounts can explain salience, attention, avoidance, and reappraisal without invoking a Law-specific queue. The Queue model must therefore ask whether unresolved affective mismatch produces priority and reactivation patterns beyond those rival accounts.

Affective prediction error should not be treated as a single scalar surprise. It may involve valence, threat, bodily cost, agency, meaning, memory, and social expectation. A person may be surprised by relief, harmed by anticipation, or unsettled by numbness. The queue entry rule needs to specify which feature is unresolved.

A simple entry indicator might include magnitude, persistence, action relevance, and repair availability.

One candidate operational rule admits a demand when mismatch exceeds a declared threshold, persists, and continues to influence attention, action, or expected future cost. The record separately identifies whether a repair-relevant action is available, blocked, or absent. Lack of a repair route can sustain a demand rather than prevent its entry.

The rule should allow no entry. A large emotional change that resolves quickly may not enter. A surprising joy may alter the measured cumulative estimate without becoming an unresolved demand. A brief irritation may be real feeling but not queue material. The model becomes more credible when it refuses some cases.

Some demands enter through relationship. Attachment, betrayal, obligation, shame, apology, exclusion, caregiving, conflict, and unspoken fear can continue to shape attention and action after the visible interaction ends. A Queue model can represent these as relational demands embedded in context rather than private defects (Shenhav et al., 2013; Kaelbling et al., 1998).

A realistic example is a caregiver who smiles during a clinic visit because the parent is present, then cannot sleep after reading the discharge instructions alone. The demand is not simply anxiety. It may involve responsibility, uncertainty, role conflict, lack of respite, fear of making the wrong call, and no safe route to transfer the load. A queue entry rule that sees only mood misses the social structure.

Relational demands require careful observation because bystanders have privacy. A participant may report conflict with a spouse, child, colleague, or clinician. Those reports can enter as participant experience, but the other person does not automatically become research data. The demand can be modeled without building a dossier on everyone around the participant.

Social entry rules also need culture. Silence, indirect speech, apology, obligation, and repair differ across settings. A model trained on one social grammar may misclassify another. A delayed reply may be threatening in one context and ordinary in another. The entry rule should include contextual priors and transport checks.

Relational repair has been studied as a time-dependent process. McCullough and colleagues followed 337 people after a recent interpersonal transgression, using repeated reports over approximately three weeks. Reported conciliatory gestures, including apologies and offers of compensation, were associated with increasing perceived relationship value and decreasing perceived exploitation risk. The longitudinal model linked rising relationship value to faster reductions in vengeful and avoidant motivation (McCullough et al., 2014). This gives a relational-

demand model concrete variables to track: what the person remembers, what the partner does next, how future safety and relationship value are appraised, and how motivation changes over time.

Relational demands often interact with material constraints. A person may remain in contact because housing, care, immigration status, work, or family obligation narrows alternatives. The queue cannot treat continued contact as evidence that the demand is low. Sometimes the action set explains behavior better than preference.

The model's ethical boundary is clear: it may represent relational pressure only to understand hidden burden and compare mechanisms. It may not prescribe forgiveness, reconciliation, exposure, confession, or public testimony. Repair remains voluntary, contextual, and safety-dependent.

A large intervention study demonstrates that emotional responses to past interpersonal harm can change through a self-directed practice. In the International REACH trial, 4,598 adults across sites in Colombia, Hong Kong, Indonesia, South Africa, and Ukraine were randomized to receive a forgiveness workbook immediately or after a two-week delay. At two weeks, the immediate group had lower unforgiveness, depression symptoms, and anxiety symptoms, with standardized mean differences of -0.53 , -0.22 , and -0.21 , respectively (Ho et al., 2024). The measured changes provide a concrete basis for studying how voluntary reflection alters the continuing influence of relational events. Longer follow-up and active comparison conditions would help identify durability and the most effective ingredients.

18.6.3 — Unresolved Demand and Continuing Influence

Unresolved demand language is especially vulnerable because the word unresolved can sound like unfinished payment. This chapter rejects that reading at the point of definition. A demand is unresolved when its generating condition, expected cost, blocked action, or state uncertainty continues to influence the system. It is not unresolved because reality is asking the person to suffer until a moral balance is paid (Shenhav et al., 2013; Kaelbling et al., 1998).

Consider a person who keeps checking the mailbox after months of medical appeals. The demand may be unresolved because the institution can still deny medication, the person lacks reliable information, and prior letters have carried real consequences. Calling that state queue-like does not say the person owes vigilance. It says a system of uncertainty, material consequence, memory, and limited control may be keeping a demand active.

The same rule applies to relationship repair. After betrayal, trust may remain difficult even after apology. A queue model can represent remaining uncertainty, risk estimation, memory salience, and the cost of future vulnerability. It cannot tell the injured person that they must forgive, must reopen access, or must suffer until the relationship is balanced. The action set includes refusal and protection.

This distinction matters for empirical design. If the model confuses unresolved demand with moral debt, it will treat distress as useful signal no matter how obtained. A scientifically admissible task must instead ask how demand features change

under safe, consented, low-burden conditions. It can study response to information, option availability, or symbolic repair without creating harm to fill a queue.

Relief after a move, treatment, apology, or safe housing may reduce estimated demand. The queue records those changes according to its stated transition model.

Unresolved means still exerting influence under the model.

18.6.4 — Priority, Blocking, and Reactivation

Priority, blocking, and reactivation are the queue family's distinctive dynamics. Priority allocates limited control, blocking reduces current influence or observability, and reactivation predicts a specific return when context or action support changes. These claims should be separated so one vague recurrence does not count as all three.

Priority can be represented as a function of expected cost, controllability, time sensitivity, context, and resource availability. It should not be represented as moral importance. A demand can be high priority because it threatens safety, because repair is possible now, because delay increases risk, or because context makes it salient (Manski, 2003; Jacod & Shiryaev, 2003; Tao et al., 2018).

Table 18.2 specifies an illustrative sequence for a future simulation. It shows the intended entry, blocking, reactivation, repair, and residual-state transitions; it does not report an executed simulation or a finding about a patient or institution.

Table 18.2. Queue-model simulation steps and test functions.

Toy step	Queue state change	What the simulation tests
1. Entry	A hospital-billing cue adds a demand with high uncertainty and moderate controllability.	Does a specific cue increase belief in an unresolved administrative-threat demand beyond general stress?
2. Blocking	A childcare deadline consumes attention and temporarily suppresses message-opening behavior.	Can blocked access occur without treating the demand as resolved?
3. Reactivation	A reminder email restores the demand and raises avoidance latency.	Does reactivation depend on cue specificity rather than any unpleasant event?
4. Repair attempt	A caseworker call lowers uncertainty and changes available actions.	Does priority fall because state changed, not because the model declared closure?
5. Residual	Future portal cues retain some salience after the bill is fixed.	Can the model represent partial repair while allowing rivals to explain residual vigilance?

An illustrative priority score can make the candidate dynamics inspectable:

$$P_i(t) = \beta_C \hat{C}_i(t) + \beta_U \hat{U}_i(t) + \frac{\beta_H}{H_i(t) + \varepsilon} + \beta_R \hat{R}_i(t) + \gamma^T X_i(t).$$

Here the left-hand side is the modeled priority of demand i at time t . The hatted C term is estimated future cost, the hatted U term is uncertainty, the inverse-horizon term captures urgency as remaining opportunity narrows, the hatted R term is available resources or repair capacity, and $X_i(t)$ records context. The positive constant epsilon prevents division by zero. This is a candidate priority equation, not a definition of worth, severity, truth, or deservingness.

The sign and units are declared for every use. A high value may mean high action priority, high monitoring priority, or high expected control cost, depending on the model. Keeping those meanings explicit makes the equation interpretable.

Priority should also be stochastic. Demands may fluctuate with sleep, medication, pain, context, social safety, and observation. A deterministic priority rank may be useful in a task, but real systems will need uncertainty. The model should report posterior priority distributions where possible.

Rivals can copy priority shifts. Scarcity increases attention to immediate needs. Fatigue narrows control. Threat conditioning raises salience. Active inference reduces expected uncertainty. A Queue model must specify a distinctive priority pattern, such as reactivation after apparent resolution when a repair-relevant action becomes available, not merely urgency under pressure.

The priority rule becomes scientifically useful when it can be wrong. If high predicted priority demands do not affect action, report, physiology, or memory under prospectively specified

conditions, the rule weakens. If low predicted priority demands dominate, the model must update or retire.

Reactivation occurs when a demand that had low current influence returns to shape attention, report, physiology, memory, or action. Triggers may include sensory cues, similar contexts, relational contact, institutional notices, bodily states, anniversaries, treatment changes, or new repair opportunities. The trigger supplies a structured observation (Wilson & Collins, 2019; Manski, 2003).

A letter from an insurer can reactivate earlier burden because it resembles a sequence in which paperwork carried real consequences. Context, memory, expected cost, and a live possibility of harm or remedy shape the response. The model represents the trigger as information.

Reactivation must be distinguished from ordinary conditioning. A cue can produce fear because it was paired with danger. That is a strong rival. The Queue model adds value only if reactivation also depends on unresolved action, repair availability, demand priority, or state uncertainty beyond cue association alone.

A candidate reactivation condition can be written:

$$Pr(D_{i,t+\Delta} > \theta_i | H_t) = \text{logit}^{-1}(\alpha_i + \beta_c c_{i,t} + \beta_b \hat{b}_{i,t} + \beta_a^T a_t^{\text{live}} + \beta_x^T X_t).$$

The display models the probability that demand i exceeds the protocol-fixed reactivation threshold at the next interval, conditional on the available history. Cue relevance, current state belief, the live action menu, and context enter separately, so each coefficient can fail without converting every recurrence into

queue evidence. The diagnostic becomes informative only when cue-conditioning and other rivals receive the same data, tuning budget, and threshold rule.

Reactivation can be humane when studied carefully. A study might use mild, consented reminders of ordinary tasks, not traumatic reenactment. It might vary information availability or repair options rather than inducing distress. The model does not need to hurt people to learn whether demand influence returns under safe cues.

A model that predicts reactivation everywhere fails. If every cue revives every demand, nothing has been distinguished. If safe cues and unrelated cues produce the same effects as repair-relevant cues, the queue account should narrow.

18.6.5 — Discharge, Repair, and Residual State

Discharge and repair are route-specific changes, not narrative declarations. A source can be removed, a belief can update, an external remedy can open an action, and a residual state can remain. The model should represent which component changed and what evidence would show that apparent relief was only observation loss.

Direct resolution occurs when the generating condition is removed or the action sequence completes in a way that materially changes the state. A medical appeal is approved and medication arrives. A door is repaired and the person can sleep. A missing document is found. A threat ends in a way that is verified and stable (Ethier & Kurtz, 1986; Jacod & Shiryaev, 2003).

In the model, direct resolution changes the transition probability for the demand. The expected future cost drops because the condition that generated the demand no longer persists. This is not emotional command. The person may still feel residue. Direct resolution concerns one state component, not the whole lived aftermath.

A candidate transition can be written:

$$\Delta_i^{res}(s_t) = E[D_{i,t+\Delta}|do(A_t = a_{resolve}), S_t = s_t] - E[D_{i,t+\Delta}|do(A_t = a_{wait}), S_t = s_t] < 0.$$

The contrast compares expected next-state demand under a supported resolving action with the same quantity under waiting, conditional on the declared state. A negative value is the candidate prediction; it is neither guaranteed nor a command that a person should feel better. Identification requires well-defined treatment versions, support for both actions, and the same confound and measurement rules used for rivals (Pearl, 2009; Hernán & Robins, 2020).

Direct resolution should be measured with enough time depth to avoid false closure. The letter arrives today, but sleep may take weeks. The threat ends, but the body may continue to monitor. A model that expects immediate total discharge will misread lived systems and punish people statistically for normal recovery lag.

Rivals can explain improvement after resolution. Rational threat reduction, stress-process models, homeostatic recovery, and ordinary learning may all predict lowered vigilance when external conditions improve. The Queue model must add more specific

predictions, such as which residual components remain and which cues no longer reactivate.

Direct resolution also has an institutional lesson. The best way to reduce many queue demands is not inward interpretation but outward repair. If an external condition generated the demand, fixing that condition is scientifically relevant. Care is not outside mechanism.

External repair includes care, safety, apology, material change, institutional remedy, access, respite, debt relief, corrected records, and restored routes of appeal. It matters because many demands persist outside the person. A queue model that treats repair as only internal would be scientifically and ethically poor (Ethier & Kurtz, 1986; Jacod & Shiryaev, 2003).

An apology can be external repair only when it changes relevant state. A performance apology that protects the offender may not reduce demand. A real apology may include acknowledgment, changed behavior, safety, accountability, and time. The model should not treat the word apology as a magic discharge action.

Institutional repair is often more concrete. A denial is reversed, an inaccessible entrance is fixed, a dangerous schedule is changed, a correction route is opened, or a confusing record is made reviewable. These actions can change transition probabilities because they alter live menus. The person's burden shifts because the world stopped placing the same demand.

External repair should be observable without becoming a public ledger. Research may code whether remedy occurred, whether the participant desired return of results, whether records were

corrected, and whether access changed. It should not publish personal repair scores or rank institutions through participant burden without governance.

Rivals can explain response to external repair through ordinary stress relief or rational action. Again, the Queue model earns support only if it predicts specific residual or reactivation patterns after repair. For example, it may predict that direct material repair reduces demand intensity while relational cues continue to reactivate uncertainty unless accountability changes.

A fairness-oriented analysis is strongest here because it asks where repair should happen. If the demand came from an institution, the model should not send the person away with a breathing exercise as the primary answer. A control model with fairness at its center includes external control levers.

Residual burden remains after resolution, update, repair, or partial relief. Nonresolving burden persists because the source remains, repair is impossible, memory remains active, the action set is closed, or the model is wrong about the relevant state. A humane mechanism must have a place for both (Ethier & Kurtz, 1986; Jacod & Shiryaev, 2003).

Residual burden is not evidence against care. It may be the normal remainder of serious harm. A person may be safe now and still startle. A record may be corrected and still leave distrust. A family may reconcile and still leave grief. The model should allow residual state without turning it into hidden unpaid debt.

Chronic illness, disability, irreversible loss, discrimination, and bereavement can create continuing demands because

circumstances have changed permanently. Representing those constraints helps identify useful support, access, and companionship alongside opportunities for improvement.

Residual influence can be represented through lower but nonzero demand states, persistent cue sensitivity, altered transition rules, or slow recovery. Each representation makes different predictions about recurrence, context, and response to repair and must therefore be selected by evidence rather than narrative convenience. The model classifies a transition pattern, not a person.

Rivals may explain residual burden through memory, allostasis, adaptation limits, predictive control, or social context. The queue account must say what residual pattern is distinctive. For example, it might predict that residual burden is low in report but high in cue-specific physiology only when repair was partial and state uncertainty remains.

Residual and continuing burden deserve explicit representation and sustained care. When a model cannot explain or resolve what remains, the next task is to identify better observations, practical support, or a more adequate account while respecting the person's own pace and choices.

18.6.6 — Queue Failure, Scope, and Retirement

The illustrative queue is retired when its state cannot be identified, its priority or reactivation predictions fail, or ordinary learning and scarcity models explain the same observations more simply. Scope can narrow to one task or reference class, but the

model may not survive by becoming a metaphor for every unresolved experience.

Entry criteria should be prospectively specified whenever the model is used for inference. A model may declare that entry requires persistence, action relevance, expected future cost, repeated cue sensitivity, cross-channel support, and context plausibility. The exact criteria can vary by study, but the study cannot invent them after seeing which events make the queue look explanatory (Kaelbling et al., 1998; Sutton & Barto, 2018).

False positives are not nuisances; they are central diagnostics. If a model labels ordinary fatigue, benign memory, cultural restraint, or momentary frustration as unresolved demand, it will overstate queue prevalence. If it labels every lack of report as resolution, it will understate hidden burden. Both errors can damage science and people.

The entry process can be summarized as a small diagnostic gate. It asks whether the candidate demand has a source, a state feature, an observation signature, an action relation, and a rival explanation. If one of these is missing, the model should mark uncertainty or no entry rather than force inclusion.

Table 18.3. Queue-entry diagnostic gate.

Entry check	What it asks	Failure condition
Source	What generated the candidate demand?	No identifiable event, condition, or continuing constraint

Entry check	What it asks	Failure condition
Influence	What policy, attention, physiology, or memory does it alter?	Salience without action relevance
Observation	Which channels support the belief?	One noisy signal treated as direct state
Rival	What simpler model could explain it?	No fair rival specified

This table is a model-audit aid. Its purpose is to keep queue components, observations, rivals, and failure conditions distinct during research design. It is not a person-level screen and should not be converted into a public tool for classifying anyone's private experience.

False positives should be studied through negative controls. A task might include salient but resolved events, neutral administrative reminders, or safe surprise conditions that should not produce queue-like reactivation. If the model flags these at high rates, its entry criteria are too broad.

The strongest entry rule is conservative without being dismissive. It allows hidden burden to enter when evidence supports unresolved influence, and it allows no entry when the case is better explained by ordinary mood, context, or measurement error. Fairness to persons and fairness to explanation require both decisions.

A decisive recovery test should include a safe harm-avoidance controller that reproduces urgency, blocking, and apparent repair without a queue state. If the queue pipeline cannot distinguish that

ordinary controller on held-out tasks, the route-specific interpretation retires even if the descriptive fit is strong. Recovery should test not only class labels but the features used to distinguish them, revealing whether apparent queue evidence depends on one fragile diagnostic.

Component-level verdicts should be allowed. Cue-specific reactivation may survive while priority updating fails; blocked access may be measurable while discharge is not; residual influence may be observed without identifying an entry rule. Narrowing preserves useful findings without keeping an omnibus queue mechanism alive by aggregation. Component verdicts can be recombined only when a prospectively stated mechanism requires their conjunction, preventing post hoc assembly of partial wins.

Retirement records should identify representation, recovery, transport, ethical, or misuse failure. A promising but burdensome model may remain open while unavailable as a first-edition route. A route withheld for burden should remain scientifically distinct from one rejected by evidence, because the next admissible study differs in each case.

18.7 — Construction, Identification, and Recovery

Construction, identification, and recovery are separate achievements. Construction shows that a mathematical object can be defined. Identification asks whether the law of the observables distinguishes the relevant parameters or model families. Recovery asks whether the complete analysis pipeline returns the correct generator or model when the data-generating truth is known. Evidence for a natural mechanism begins only after all three stages are connected to held-out observations.

A model may be structurally identifiable in continuous observation and practically unrecoverable under sparse, noisy, missing data. Simulation should use the real sampling schedule, measurement error, action process, endpoint rule, and censoring. Idealized recovery is insufficient.

Parameter recovery and model recovery should both be reported. Accurate parameters inside the wrong model do not establish the mechanism. Correct model selection with unstable parameters may support a family-level conclusion but not a quantitative personal or biological interpretation.

Open-set recovery adds a none-of-the-models state to the confusion matrix. Posterior predictive checks, residual diagnostics, held-out calibration, and challenge generators should reject the entire candidate bank when every model is inadequate. Without that option, the tournament can only choose the least-wrong explanation and will overstate support whenever the relevant mechanism was omitted.

Generator recovery is especially important for Doob and bridge constructions because similar endpoint-conditioned marginals can arise from different dynamics. Drift, diffusion, jump, hazard, and observation components should be tested separately where data permit.

Route evidence begins after equal-budget recovery succeeds and one frozen object transports to held-out modules. A mathematically coherent construction remains valuable when empirical identification fails, but its status stays mathematical. This separation allows the book to preserve useful theorems, simulation tools, and diagnostics without describing an unrecovered route as a natural mechanism.

18.7.1 — Existence and Empirical Realization

Existence is the first mathematical achievement in developing a mechanism. A differential equation may have a solution, a bridge measure may be constructed, a committor may solve a boundary-value problem, or a controller may keep simulated paths inside a completion region. Those results establish mathematical possibility under assumptions and identify the states, generators, policies, or boundaries for subsequent empirical investigation (Bollen, 1989; Villaverde, 2019).

Realization requires a bridge from the construction to observables. The model specifies which variables correspond to measured channels, which parameters vary across persons or contexts, and which interventions alter the proposed mechanism or its reports.

A physical completion coordinate supplies a concrete realization to investigate. On one qualifying stream, let a be the declared initial boundary, s the fixed exposure clock, and L_s the accumulated signed contribution with $L_a = 0$. Nominate a physical state S_s and scalar potential Φ independently of the tested ledger, with $\Phi(S_s)$ absolutely continuous on the asserted branch. If their changes satisfy $dL_s = -d\Phi(S_s)$, integration gives

$$L_s = \Phi(S_a) - \Phi(S_s).$$

An independently established boundary relation $\Phi(S_{T-}) = \Phi(S_a)$ then gives exact terminal closure. Equal boundary values suffice; neither value has to be zero. This route can operate through current physical state without the controller forecasting an ending, provided the claimed dynamics and boundary relation actually hold.

The research task is to identify a compact physical coordinate whose changes predict calibrated signed experience on new paths, then test its boundary relation separately. Defining Φ as minus the observed ledger supplies no independent prediction. An enrollment reset supplies no initial condition for an earlier stream, and a reset at or after T cannot change its strict-pre-endpoint residual. A successful independently qualified coordinate would connect an ordinary physical mechanism to the exact target.

An ordinary biological mechanism that satisfies the declared contribution, exposure, ownership, and boundary relations realizes that branch of PTVC. Adaptation that regulates only a current state while leaving accumulated signed history unconstrained remains a rival explanation of local signatures.

An approximate realization should report its cumulative accounting budget. For a specified finite path with calibrated signed edge contributions w_e , set $r_e = w_e + \Phi(v) - \Phi(u)$ on edge $u \rightarrow v$. Telescoping gives

$$L_{\text{path}} = \Phi(s) - \Phi(b) + \sum_{e \in \text{path}} r_e.$$

If independently justified bounds give $|\Phi(s) - \Phi(b)| \leq \varepsilon_B$ and $\sum_e |r_e| \leq \varepsilon_R$, then $|L_{\text{path}}| \leq \varepsilon_B + \varepsilon_R$. Infinite paths require the stated boundary limits and convergence of the error sum. Small errors of one sign can accumulate; mean-zero residuals or a fitted RMSE do not bound an individual path. Statistical budgets need joint coverage under their maintained observation laws. Unobserved history adds its own uncertainty. The boundary is the declared pre-endpoint value, so a later reset cannot supply the missing equality. This directs validation toward the contribution bridge, boundary mismatch and missing history as separately measurable parts of one cumulative test.

Engineering makes the distinction concrete. A controller can be designed to bring a simulated process to a target, while several generators, constraints, and selection processes can produce the same terminal distribution. Comparing their path dynamics identifies which observations distinguish the implementations, including cases with a partly observed latent state.

Within a specified finite-state realization, a potential is also necessary. Fix a directed graph of reachable states, one initial state s , absorbing strict-pre-endpoint boundary states, and deterministic additive edge contributions w_{uv} . Every relevant

state must admit a terminating suffix, and that same suffix must be admissible after every prefix reaching the state. Suppose every admitted finite terminating path has total zero.

Two prefixes reaching u must have equal totals: adjoining one common terminating suffix makes both totals zero. Let $\Phi(u)$ be minus this common prefix total. Appending an edge then gives

$$w_{uv} = \Phi(u) - \Phi(v), \quad \Phi(s) = \Phi(b) = 0.$$

The boundary equality holds for every declared boundary state b .

Conversely, these edge and boundary equations make every terminating path telescope to zero. The restriction therefore identifies exactly the accounting structure required in this recombinable class. A shared coarse label does not establish recombination: different histories may permit different suffixes. Random rewards, omitted time, or hidden history require their own qualified state model. Freeze the state and admissible transitions before testing new paths; adding the observed ledger to each state label makes the construction automatic rather than explanatory.

A realization claim therefore needs four linked records: the construction and its assumptions, an identifiability analysis, a realistic recovery benchmark, and a prospective prediction not guaranteed by the construction itself. Omitting any record changes the licensed conclusion. The object may still guide theorem work or simulation, but detected mechanism language is reserved for a recovered and transported causal difference.

Concrete documentation helps. Each candidate receives a versioned state definition, generator, observation kernel, action

set, endpoint branch, parameter domain, and failure conditions. The packet also records which equations are definitions, theorem consequences, estimators, or scenarios. This prevents one formula from silently carrying several statuses.

A route may fail at realization while leaving useful mathematics behind. The field can retain a committor method, a partial-observation model, or a recovery benchmark even when PTVC-specific interpretation is withdrawn. That separation lets mathematical and methodological progress accumulate as empirical interpretations are refined.

18.7.2 — Parameter Identification and Observational Equivalence

Parameter identification asks whether different parameter values imply different observable laws, while observational equivalence asks whether different mechanisms remain indistinguishable under the available design. A precise fit is not informative when several parameterizations or models generate the same data (Bollen, 1989; Villaverde, 2019).

A latent state is nonidentified when materially different hidden structures induce the same observable law under the available design. Queue demand, threat learning, fatigue, scarcity, homeostasis, and active inference can produce overlapping behavior, so a fitted queue state is not route evidence when those alternatives remain observationally equivalent (Lakens, 2017; Manski, 2003).

Exact completion imposes a memory requirement on a specified forward realization. At a fixed admissible cut, let L be the

accumulated signed prefix and R the entire remaining contribution under the same representation, exposure, ownership and endpoint rules. Let X include all state information used to generate that continuation, so that L and R are conditionally independent given X . If $L + R = 0$ almost surely, the state must determine L up to null events. With finite second moments, conditional independence gives $\text{Cov}(L, R \mid X) = 0$, while $R = -L$ gives $\text{Cov}(L, R \mid X) = -\text{Var}(L \mid X)$; hence $\text{Var}(L \mid X) = 0$.

For real L and standard Borel X , the exact conclusion does not require moments. Conditional independence makes each conditional threshold probability $P(L \leq q \mid X)$, for rational q , equal to its square. The resulting conditional distribution is a point mass. Under finite second moments, approximate completion instead obeys

$$E[(L + R)^2] \geq E[\text{Var}(L \mid X)].$$

This lower bound turns unresolved loss of prefix information into a completion-error floor. The necessary information may be neural, molecular, bodily or distributed; it need not be consciously recalled. A noisy assay can lose information that the complete state retains. The restriction therefore tests a nominated causal-state realization, with the endpoint process included, rather than every formulation of PTVC.

A finite-state claim also imposes an observable restriction. At a common cut, suppose the complete continuation state has at most K discrete values. Let rows of M index prospectively defined past conditions and columns index future probes and response events. If probe assignment preserves the state distribution within each

row, and response laws given the state are otherwise common, then $M = AB$: A contains state probabilities and B contains probe-response probabilities. Consequently $\text{rank}(M) \leq K$. An excess singular value beyond a valid simultaneous measurement-and-sampling error bound rejects that nominated state budget. Use the actual independent episodes assigned to each probe when computing sampling uncertainty. Include observation loss as a response event, or model it without silently conditioning on complete cases. One continuous coordinate can generate matrices of arbitrarily high rank, and low rank establishes neither a unique hidden state nor terminal closure.

Nonidentifiability should be declared rather than hidden behind complex notation. If a model cannot tell active demand from blocked demand, or residual burden from ordinary fatigue, its output should not use those labels. It may report an undifferentiated latent factor or no estimate. Precision in language should follow precision in identification.

A simple diagnostic is model confusion. If the recovery matrix shows that data generated by different models are often classified as queue data, the design cannot support a queue claim. More participants may not solve this if the task lacks the right intervention or observation channel. Identification is not sample size alone.

Identification may also be limited by ethics and governance. A clinical or palliative setting may not permit the perturbation needed to distinguish states, and a privacy firewall may appropriately block linkage to a revealing record. When the

missing contrast would require unfair observation, coercion, delayed care, or excessive burden, the causal question remains partially identified or returns No adequate estimate. Ethical feasibility is part of the scientific result.

Under nonidentifiability, public reports describe the observed pattern and the mechanisms consistent with it. Distinctive predictions identify the observations needed to separate blocked, queued, discharged, and residual states.

A nonidentifiable state identifies information needed for the next study: a safer task, richer observation, better negative controls, or a more focused question.

18.7.3 — Generator Recovery

Generator recovery asks whether the transition law that produced a trajectory can be distinguished from serious alternatives after all candidates pass through the actual observation process. A terminal histogram is insufficient. Drift, diffusion, jumps, hazards, action response, dependence, and missingness carry the preterminal information that separates native dynamics from conditioning and selection (Villaverde, 2019; Peters et al., 2016).

The recovery problem begins in simulation. Data should be generated from each candidate route and rival using plausible parameter ranges, then sampled with the real cadence, noise, missingness, and endpoint rules. The analysis pipeline must be frozen before labels are revealed. If different generators collapse into the same fitted family, the design cannot support a mechanism verdict.

Partial observation makes generator recovery a joint problem. A misspecified observation kernel can make the correct state dynamics appear wrong, while a flexible hidden-state model can absorb observation error into latent transitions. Separate recovery checks should perturb state dynamics, measurement error, and missingness so the source of confusion is visible.

Safe interventions often distinguish generators that passive fit leaves equivalent. Information, action availability, or a benign state component can be changed while the target, burden ceiling, and outcome rule remain fixed. The altered prediction is scored on a held-out module and compared under common observation. A model that predicts only the intervention used to tune it has recovered a local fit, not a general generator.

Cross-context prediction tests whether one physical coordinate explains several conditions. Freeze the state definition, contribution calibration and $\mathcal{C}^2$ potential Φ . In a common clock let condition a have dynamics $dS = b_a ds + G_a dW$, generator $\mathcal{A}_a$ and contribution rate v_a . Exact accounting requires $G_a^\top \nabla \Phi = 0$ and $v_a + \mathcal{A}_a \Phi = 0$ along its supported process. On states supported in both compared conditions, with the regularity needed to extend those identities,

$$v_a - v_b + (\mathcal{A}_a - \mathcal{A}_b)\Phi = 0.$$

This subtraction is not licensed by disjoint state samples without a justified transport model. A strong experiment predicts a held-out change in context, timing or available information using the same coordinate and units, while checking intervention effects on measurement. Across a connected shared region, full span of the

combined diffusion directions forces a constant common potential. Incomplete span leaves a candidate possible but does not construct a globally integrable potential. Frozen state definitions and controlled complexity make the comparison a test of one mechanism rather than a collection of separately fitted descriptions.

Overlapping interventions can test a shared potential without first fitting its formula. In a differentiable deterministic model on one fixed physical clock, let $f(S, u)$ be state velocity and let the signed contribution rate satisfy $\ell(S, u) = -\nabla\Phi(S)^\top f(S, u)$ throughout a supported neighborhood of control u . At the same state, collect the causal derivatives $\partial f / \partial u_j$ as columns of F . If the integrated-reading rate is $z = \ell + o\rho$, with one common origin o and qualified exposure rate ρ , define $r_j = \partial z / \partial u_j$ and $h_j = \partial \rho / \partial u_j$. Every control contrast q satisfying

$$Fq = 0, \quad h^\top q = 0$$

must satisfy $q^\top r = 0$. Differentiating the common-potential relation gives $r = -F^\top \nabla\Phi + oh$, so both unknown terms cancel. A nonzero contrast exists only if the combined drift-and-exposure matrix has a nontrivial nullspace. This is a test of every smooth potential on the nominated state, with the physical contrast fixed before inspecting the held-out rate response.

Estimated cancellation needs a complete error budget. Small physical residuals cannot bound the predicted response unless potential gradients and origin are independently bounded or included in valid joint inference. Finite pulses add curvature error, and an omitted state can remove the apparent cancellation. For

diffusion models, changing covariance contributes a Hessian term that must also cancel; the deterministic formula cannot simply be reused. A reproducible violation can exclude the specified coordinate family. Agreement supports a local physical restriction and leaves global integrability and terminal closure to separate tests.

Recovery metrics should include confusion matrices, calibration, parameter bias, interval coverage, and residual structure. A high average classification rate can hide failure for the route most relevant to the claim. Class-specific errors and an open-set rejection option are therefore required.

When the generator cannot be recovered under realistic sampling, the route remains unidentified. The report may support family-level compatibility, a narrower parameter statement, or No adequate estimate, and it should name the measurement or intervention needed to create separation. More notation or a larger sample cannot replace a design whose observable laws are equivalent.

18.7.4 — Held-Out Prediction and Cross-Equation Transport

Held-out prediction tests whether a fitted route forecasts observations it was not tuned to reproduce. Cross-equation transport is stronger: one frozen object estimated from one module should predict another relation, action response, or endpoint regime. Refitting each equation separately removes the overidentification that could distinguish mechanism from flexible description.

A fixed physical potential predicts the same signed contribution along any two admissible paths between the same sufficient states. For a finite graph, let B have -1 at an edge's source and $+1$ at its target, let w contain the calibrated edge contributions, and let φ contain the state potentials. Then

$$w = -B^T \varphi, \quad Bc = 0 \Rightarrow c^T w = 0.$$

The cycle space contains $m - n + k$ independent algebraic constraints for m edges, n vertices and k connected components. A contrast can compare two forward paths; it need not require physically reversing a transition. Repeated dependent loops improve precision without adding independent constraints. These are standard graph-potential restrictions (Jiang et al., 2011).

Qualify whether each edge quantity is a realized contribution or only a conditional mean. Mean-zero circulation permits nonzero individual circulation when edge rewards are random. Estimate the state map on development records and test new paths with shared-edge covariance, state uncertainty and observation loss included. A failed contrast can reject the nominated state-potential route. Any added state must earn its complexity through new predictions; an ordinary potential mechanism faces the same test and can legitimately pass.

Priority, blocking, and reactivation matter only if they lead to observable predictions. The predictions should involve timing, content, channel conflict, action selection, and response to changed options. A queue model should not merely say unresolved demands exist. It should say how influence moves (Manski, 2003; Jacod & Shiryaev, 2003; Tao et al., 2018).

A concrete task might present low-stakes unresolved goals, information cues, and repair opportunities across days. The model predicts which goals return to attention when repair becomes possible, which remain blocked under competing load, and which decline after direct resolution. The task need not induce harm. Ordinary planning, missed messages, delayed tasks, or ambiguous feedback can carry enough structure.

Observable signatures may include report recurrence, increased attention to relevant cues, altered choice latency, physiological preparation, sleep disruption, or changes in avoidance after option availability shifts. Each signature requires a rival. For example, choice latency can reflect uncertainty, fatigue, motor speed, or task confusion. A queue interpretation earns nothing without comparison.

Neural and psychological correlates supply auxiliary predictions that can help separate mechanisms. The queue itself retains its formal definition, allowing researchers to test how particular correlates relate to its predicted dynamics.

Table 18.4. Queue-model auxiliary predictions and winning rivals.

Auxiliary prediction	Queue interpretation	Rival that could win
Cue-specific reactivation	Demand-linked cues should update belief and priority more than affectively matched but unrelated cues.	Associative learning or trauma-cue conditioning may explain the same pattern without queue structure.

Auxiliary prediction	Queue interpretation	Rival that could win
Control allocation	Higher priority should shift latency, task switching, information seeking, or avoidance under constrained attention.	Expected value of control, fatigue, or scarcity models may predict the allocation more simply.
Interoceptive salience	Bodily readiness may mark demand relevance when verbal report is delayed or blocked.	Homeostasis, anxiety physiology, medication, illness, or sleep loss may explain the signal.
Memory-access pattern	Blocked demands may reappear under context, safety, or cue changes rather than constant conscious access.	Memory consolidation, suppression, or ordinary contextual retrieval may win.
Repair-sensitive change	External remedy should alter later cue response when it actually changes state or options.	Relief, expectancy, social support, or simple uncertainty reduction may fully explain the change.
Dynamic	Queue prediction	Serious rival
Blocking	Demand influence rises when competing load falls.	Allostatic rebound or safety-enabled emotion.
Reactivation	Repair-relevant cues revive specific demands.	Cue conditioning or memory priming.
Starvation	Long-neglected demands surface after capacity returns.	Fatigue recovery or reporting bias.
Dominance	One demand suppresses unrelated action channels.	Scarcity focus or threat prioritization.

The table is not evidence. It is a map for falsification. If the rival explains the same pattern with fewer assumptions and better prediction, the queue model loses that component.

The final observable requirement is humane burden budgeting. A prediction requiring distress induction for Law-specific evidence is inadmissible under Chapter 3. The design must use a nonharmful contrast or leave that prediction untested.

Independent predictions are consequences not guaranteed by the construction. They are the Queue model's best chance to become scientifically useful. They concern timing, specificity, counterfactual response, cue sensitivity, repair effects, and model recovery under shared observation (Wilson & Collins, 2019; Manski, 2003).

A strong prediction might say that a blocked demand will reappear after competing load drops only when a repair-relevant action becomes live. A rival allostatic rebound model may predict general affect release after load drops, but not repair-specific action selection. A cue-conditioning model may predict cue fear, but not dependence on repair availability. These distinctions make the claim vulnerable.

Independent predictions should be made before the data are viewed. They should include negative controls, such as resolved demands, unrelated cues, and nonrepair actions. If the Queue model predicts specificity and the data show broad nonspecific arousal, the model narrows.

Prediction should also be channel-specific. The model may predict report change without physiological change,

physiological preparation without report, or behavior change only when action is safe. These channel predictions are powerful because they can fail in patterned ways. A model that predicts every channel always moves together is likely too simple.

A public summary identifies the task, population, observation model, predicted pattern, and comparative result. Those particulars make a queue-like model's contribution clear and independently assessable.

The model earns scientific confidence by naming the observations that would support or count against each component. If repair-relevant cues do not matter, if blocked states do not reappear under capacity release, if false positives are high, or if rivals recover the data better, the corresponding Queue-model component requires revision or retirement.

18.7.5 — Model Recovery and Confusion Matrices

Model recovery uses simulated known generators to test whether the full pipeline returns the correct family. Confusion matrices should report class-specific errors, especially cases in which ordinary rivals are misclassified as the candidate route. Recovery is a prerequisite for interpreting real-data model selection.

Parameter recovery asks whether a fitting procedure can recover known parameters from simulated data. Model recovery asks whether the pipeline can identify which model generated simulated data. These are standard safeguards in computational modeling and reinforcement-learning analysis. For the candidate queue-style route model, they are the gate before human interpretation (Wilson & Collins, 2019).

Table 18.5. Construction and recovery stages.

Recovery stage	Required action	Failure meaning
Parameter recovery	Simulate data from known Queue parameters and refit the same model.	If recovered parameters are biased, unstable, or correlated incorrectly, person-level interpretation pauses.
Model recovery	Simulate Queue, homeostatic, active-inference, model-free, model-based, and safe-control worlds.	If the pipeline cannot identify the generating family above chance or practical thresholds, real-data comparison is not trusted.
Noise and missingness stress	Add report noise, sensor dropout, blocked observations, and context misclassification.	If queue structure appears only in clean data, the claim narrows to artificial settings.
Policy confusion stress	Fit rival policies to Queue-generated data and Queue policies to rival-generated data.	If models mimic one another under realistic samples, the result becomes nonidentifiable.
Held-out task transfer	Train on one demand family and test on a different but prospectively specified family.	If fit does not travel, the model may describe a local task rather than a mechanism.

A model-recovery matrix can be written as:

$$\mathbf{M} = (M_{ij}), \quad M_{ij} := \Pr\{\hat{m}(D) = j \mid D \sim \mathcal{M}_i\}$$

The entry M_{ij} is the probability that data generated by model i are classified as model j . A good recovery matrix has high diagonal values. If queue-generated data are often classified as active-inference data, or active-inference data are often classified as queue data, the models are not separable under the task.

Parameter recovery has the same discipline. If the true persistence, reactivation, or repair-sensitivity parameters cannot be recovered in simulation, fitted human values should not be interpreted as meaningful. A beautiful parameter name does not create identifiability. Wilson and Collins's modeling guidance is useful here because it treats recovery checks as part of responsible cognitive modeling, not as optional decoration.

Recovery checks should vary noise, missingness, sample size, cadence, individual differences, and model misspecification. A model that recovers parameters only under clean synthetic data may fail in real life-course or clinical settings. Chapter 17 showed why observation processes matter; here the recovery gate applies that lesson to mechanism.

Model recovery should also include adversarial rivals. The Queue model should be tested against versions of homeostasis, model-free RL, model-based planning, active inference, safe control, trauma cueing, and scarcity that are strong enough to win. A weak rival matrix creates false support.

The acceptance test is practical: a modeler should be able to implement at least one candidate queue-style route version and identify the conditions under which it cannot be distinguished. An

evocative queue image without that recovery and failure logic does not constitute a mechanism.

18.7.6 — Open-Set Adequacy and Unknown Mechanisms

Route recovery must allow the named library to be wrong. Calibrate an explicit none-of-the-models branch on held-out unfamiliar generators, and use absolute checks such as residual dependence, calibration failure, impossible transitions, poor intervention transport, and posterior-predictive misfit. Unknown is not favorable to PTVC; it means the current route library is inadequate and should trigger broader search or narrower scope. Chapter 25 supplies the full open-set benchmarking framework (Vehtari et al., 2017; Villaverde, 2019).

18.8 — Route Failure, Retirement, and Frontier Theorems

The route definition specifies its necessary geometry, committor transport, generator recovery, intervention response, and comparison with equally resourced rivals. Prespecified outcomes connect each test to retention, revision, or retirement of the corresponding component.

Route retirement can be component-specific. A queue priority rule may fail while partial-observation methods remain useful. A Doob construction may remain a strong rival after losing mechanism status. A reachability theorem may survive while its phenomenal bridge fails.

Frontier theorems should target genuine bottlenecks: state sufficiency, uniqueness of shrinking-band limits, route-endpoint bridges, identifiability under observation error, junction algebra, and invariance across representation changes. A theorem that merely restates an imposed boundary does not advance the program.

Remote mathematical proposals become testable only when they supply a carrier, a bridge to the declared phenomenal object, a distinctive prediction, a nearest adequate rival, and a failure condition. Their scientific status is set by those commitments, not by novelty or vocabulary. Until a proposal changes a protocol, recovery test, or verdict boundary, it remains a research direction in the mathematical reservoir.

Versioned retirement keeps the public record current. When a route narrows or fails, update or withdraw the affected formulas, figures, captions, websites, and summaries.

Mathematics can expand the space of possible mechanisms, but only discriminating evidence can narrow it. The next chapter asks where an intervention would first make those mechanisms diverge.

18.8.1 — Route-Failed as Scientific Progress

Route-failed is a constructive scientific verdict when a prospectively specified mechanism misses its distinctive prediction after representation, observation, and recovery gates pass. The report names the failed component and reference class, preserves any surviving method or descriptive result, and avoids turning one mechanism loss into a vague judgment about the entire theory.

The most ordinary failure is no added predictive value. If a simpler model predicts behavior, report, physiology, context sensitivity, and reactivation as well as the Queue model under fair comparison, the queue should not receive mechanism status for that reference class. Elegance is not a tie-breaker (Lakens, 2017; Manski, 2003).

No added value should be assessed out of sample. In-sample fit is especially weak when the Queue model has extra parameters. Held-out time blocks, leave-person-out tests, leave-site-out tests, and prospectively specified predictions give a better answer. The model should be asked to predict cases it did not help explain.

No added value can be local. The queue may add value in repair-relevant reactivation tasks but not in chronic stress tracking. It may predict report recurrence but not physiological channels. It may work within individuals but not across persons. The correct response is narrowing, not total victory or total dismissal.

A no-value result reports which rival performed best, the observation model, parameter recovery, available power, and the influence of missingness. These details preserve the finding's value for subsequent work.

Public summaries retire queue language in domains where it has added no predictive or explanatory value. When a prospectively specified rival explains a domain better, that verdict is stated directly. This is answerability turning a clear result into a better research direction.

When ordinary models explain the observations equally well with fewer assumptions, they supply an efficient foundation for the next study.

18.8.2 — Counterexamples and Impossibility Results

Counterexamples and impossibility results are stronger than poor fit when their assumptions match the claim. A prefix outside the declared solvency region, a generator that cannot be recovered from the permitted observations, or a theorem showing observational equivalence can retire a route without waiting for more favorable data.

A model can add variables and still make the wrong dynamic predictions. If it predicts a demand will rise when repair becomes available and no such rise occurs, the priority rule weakens. If it

predicts cue-specific reactivation and the data show nonspecific arousal, the reactivation mechanism weakens. If it predicts discharge after repair and residual burden persists, the discharge rule was too simple (Popper, 1959; Lakens, 2017).

Wrong predictions are more informative than vague failures. They tell the field which component failed: entry, priority, blocking, reactivation, discharge, observation, or policy. A precise failure can improve the model or retire a component. A vague failure invites rhetoric.

The model should prestate failure branches. For example, if a blocked demand does not reappear after competing load drops, then the blocking-starvation hypothesis is unsupported in that setting. If unrelated cues produce equal reactivation, cue specificity fails. If participants choose actions predicted by ordinary reward history rather than queue priority, the model-free rival gains ground.

When repair produces a different response than the model predicted, examine context, safety, meaning, time scale, and channel disagreement. Participants' reports and preferences help identify which assumptions need revision.

Wrong priority and reactivation predictions also guide future study. They may show that horizon belongs more in Chapter 19, bodily access more in Chapter 20, or rival adaptation more in Chapter 21. That rerouting should be visible, not silent.

18.8.3 — Nonidentification Boundaries

Nonidentification is a boundary on what the design can learn, not a temporary inconvenience to be hidden by a point estimate. Two

route states, generators, or endpoint branches are nonidentified when they induce the same observed law under the available interventions and measurement process. In that case, no amount of additional sampling from the same design separates them (Bollen, 1989; Villaverde, 2019).

The boundary can arise from symmetry, state aliasing, unknown origin, sparse observation, shared confounding, or endpoint selection. It can be structural, meaning separation is impossible even with infinite data under the design, or practical, meaning finite noise and cadence make separation too weak. The result identifies which boundary applies: more sampling under the same design cannot resolve structural equivalence, whereas redesigned observations or interventions may break that equivalence, and denser or better-targeted data may resolve a practical limit.

Identification analysis belongs before power analysis. Power cannot rescue an estimand that is not identified, and a narrow confidence interval around a model-dependent quantity does not create the missing bridge. Symbolic checks, simulation, profile likelihoods, and sensitivity regions can reveal the boundary before participants carry the burden of a larger study.

A useful outcome is an identified set. Rather than choosing one route, the analysis reports the collection of routes or parameter values compatible with the declared data and assumptions. Subsequent studies can be designed to shrink that set through safe interventions or improved channels. The set should not be converted into a rank among persons.

Nonidentification can also be ethical. The intervention needed for separation may be too burdensome, invasive, or harmful. The correct response is to preserve the uncertainty, not to relax the ethical boundary. A scientific program is allowed to leave a mechanism unresolved.

Apply the canonical evidence-state rules in §24.2 and Table 27.5. Here, route nonidentification narrows the distinctive component while leaving mechanism-neutral consequences separately testable; it does not by itself establish Route-failed. Record that status change rather than hiding it under a replacement mechanism.

18.8.4 — Frontier Theorems Worth Proving

Frontier theorems should sharpen the exact places where mechanism claims can succeed or fail. Useful results may establish minimal conditions for directional reachability, uniqueness of a completion kernel, recoverability of a generator under partial observation, or impossibility of distinguishing two endpoint constructions from a declared observation class. The value of such theorems lies in reducing ambiguity, not in making the hypothesis look mathematically grand (Villaverde, 2019).

One frontier concerns topology. When attainable sets are nonconvex, disconnected, or carried on lower-dimensional supports, convex relaxations can produce false solvency. A theorem that bounds the error of the relaxation or identifies conditions under which support-function tests are exact would directly improve adjudication.

A second frontier concerns path conditioning. Several shrinking-band limits can approach the same zero-probability terminal event while inducing different bridge laws. Tightness, uniqueness, and stability under observation should be stated rather than assumed. An impossibility theorem may be more valuable than another candidate construction if it shows that the data cannot distinguish those limits.

A third frontier concerns causal localization. If two mechanisms agree under every ethically admissible intervention in a reference class, they are empirically equivalent there. Formalizing that equivalence prevents endless mechanism tournaments that cannot change a verdict. It also clarifies what new action, channel, or population would be required for separation.

The theorem program should include machine-checkable statements where possible. Proof assistants can make formal assumptions, definitions, boundary conditions, and proof dependencies mechanically inspectable, and substantial mathematical theorems have been verified through such systems (Gonthier et al., 2013; Nipkow, Paulson, & Wenzel, 2002). Formal verification does not validate the empirical premises. A verified theorem remains conditional on the state and observation architecture it receives.

Every frontier target should name its scientific consequence. Proving it should change a protocol, recovery test, verdict boundary, or retirement rule. Theorems that do not alter an empirical or conceptual decision belong in the mathematical reservoir, not in the public argument as evidence.

18.8.5 — Versioning, Withdrawal, and Successor Routes

Versioning makes retirement operational. The record names the route version, source model, assumptions, failed test, public artifacts, and any successor. Withdrawal means the old object is no longer used for new inference even if it remains available as historical evidence.

Public language follows the versioned mechanism record. A successor candidate receives its own identity, assumptions, tests, and evidential status, allowing the program to build on a clear account of the preceding result.

Revision becomes unsafe when readers cannot tell what changed. If an earlier prospectively specified version attributed reactivation to a queue-style route and later evidence favors interoception, active inference, or adaptation, the change is labeled as narrowing, replacement, or retirement. Otherwise every outcome can appear to preserve the preferred account.

A clean retirement notice should name the component, reference class, rival, evidence, and public-language consequence. It might say that queue-priority predictions failed in a low-stakes task and were better explained by model-free reward history. It might say that discharge rules were nonidentifiable in life-course data. It might say that the metaphor remains useful privately but leaves scientific mechanism language.

The Queue model is one testable candidate within a wider research program. Reporting where it succeeds and where it loses allows later work to preserve useful results, revise unsupported

components, and pursue better explanations with a clear record of what has been learned.

Public artifacts must update after retirement. Diagrams, summaries, captions, social posts, donor descriptions, and talks should not continue to imply a mechanism that has narrowed or left the field. Correction-readiness is part of scientific dignity.

18.8.6 — What Survives Mechanism Defeat

The Part provides methods with continuing scientific value. Completion geometry describes feasible continuations under declared constraints. Committors provide conditional forecasts. POMDPs organize action under partial observation. Bridge constructions supply baselines for endpoint-conditioned data. Each method contributes to a more precise account of the phenomena.

Defeat also preserves negative knowledge. A failed queue route may show that priority and reactivation add no predictive value. A failed bridge discriminator may show that the observation process is too coarse. A failed causal locus may reveal that the available interventions do not identify the state. Each result prevents the next study from repeating an unproductive path.

Update each proposition according to the result that bears on it. A route test changes the status of the specified mechanism; a test of a target-level requirement changes the corresponding theory-level claim.

Human protections remain active under every outcome. Care is immediate, participation voluntary, and no model output scores worth, trust, consciousness, eligibility, or personal obligation.

These commitments do not depend on route success or need repair when a mechanism narrows or fails.

A route registry preserves failed candidates as failure memory. Versions, assumptions, datasets, win conditions, results, and artifact corrections stay searchable. This preserves the knowledge gained when a candidate is retired.

The theorem and simulation infrastructure also survives. Impossibility regions, nonidentification boundaries, recovery benchmarks, and open-set challenge generators can be reused for new mechanisms. A retired route can therefore improve the standards applied to every successor.

A frontier theorem should be stored with its assumptions, counterexamples, and empirical dependencies. If a later study violates one assumption, the theorem is not refuted; its application is. If the assumptions cannot be connected to observable quantities, the theorem remains formal rather than evidentiary.

Route retirement also protects the target hypothesis from informal rescue. Once a route loses its prospectively specified test, later sections may not cite its intuitive plausibility as if nothing happened. A new route begins with a new state, new assumptions, new predictions, and a new evidence status.

Failure memory should remain queryable. Later investigators need to recover which generator, endpoint branch, observation process, intervention, and scoring rule failed, together with the code and synthetic cases that exposed the failure. A narrative summary without the executable test would invite the same mistake to return under a new name.

Some tools may migrate after route defeat. A committor can remain a forecasting object, a belief state can remain an observation model, and a queue can remain a behavioral description if each passes its narrower validation. The migration must be explicit so a useful method does not retain the evidentiary halo of the retired mechanism.

The chapter closes with a standard for constructive failure: preserve the theorem, benchmark, counterexample, and humane boundary; withdraw the interpretation that lost; and leave the next study a sharper question than the one that entered.

Chapter 19 — Causal Loci and Interventions

A mechanism becomes causal only when an intervention has a typed point of entry and a declared downstream consequence. A prompt, medication, action menu, information set, observation channel, assignment process, continuation policy, or endpoint rule alters a different part of the data-generating system. Chapter 19 maps those loci before any response is credited to a latent route.

Intervention-indexed continuation laws provide the central estimand language. They preserve the experienced past while comparing future path distributions under a declared action, regime, assignment, or information change. The discriminating question is whether the candidate predicts a continuation difference that ordinary control, scarcity, fatigue, learning, homeostasis, or selection models cannot recover under the same information.

The most informative evidence often appears at the earliest divergence. Neighboring models may agree before commitment and separate only after a cue-controlled perturbation, a change in action support, or an information update. A difference that appears before assignment instead raises leakage or selection concerns. Common information sets and timestamped eligibility rules make the ordering testable.

Chapter 7 defines the modal ladder and the policy-mixture exactness result. Chapter 19 asks the causal question that those results leave open: which intervention locus, assignment process, and information set could make an observed action pattern informative about a completion route rather than ordinary choice, constraint, or selection?

Assignment opens a causal pathway; it does not complete one. Commitment, delivery, adherence, switching, observation, retention, intercurrent events, and truncation by endpoint or care determine which continuation was actually generated and which remains observable. The denominator history is part of the mechanism claim.

The preferred designs are safe experimental forks. Low-burden tasks, natural experiments, timing-neutral information conditions, target-trial emulations, and ordinary-care variation can separate routes without manufacturing distress or withholding treatment. Procedure burden, uncertainty, effort, and opportunity change are measured because the test itself can alter the continuation it seeks to explain.

Mechanism tournaments use equal information, preprocessing, tuning budgets, scoring rules, stopping conditions, and opportunities to abstain. Residual-aware routes, expected-value-of-control models, reinforcement-learning policies, homeostatic controllers, and scarcity models all receive a genuine path to victory. A specialized candidate cannot earn support from endpoint context, richer sensors, or analyst attention denied to its rivals.

Shadow price appears at the end as one local sensitivity example. Under a declared constrained optimization problem, an identified Lagrange multiplier can express the marginal change in an objective associated with relaxing an active constraint, with units and regularity conditions made explicit. Its scientific role is constraint sensitivity; it is never a price of pain, pleasure, a person, a life, a duty, or care.

The chapter's concrete settings include a task where options narrow, a treatment route with changing support, an administrative delay, a learning experiment, and a simulation with controlled generator differences. Each setting is used to separate causal structure, not to dramatize suffering.

Causal diagrams make assumptions inspectable. They display measured and unmeasured confounding, interference, post-treatment selection, positivity failures, treatment-version ambiguity, measurement error, and competing pathways. The diagram's value is practical: it shows which contrast is identified, which is only bounded, and which should return No adequate estimate.

The causal-locus framework remains valuable under every PTVC outcome. It is a general method for separating interventions on constructs, route states, proxies, observations, selection processes, and decisions, and for preventing behavioral compliance or device response from being mistaken for inner truth.

The reader should leave able to audit a mechanism proposal in one sequence: define the target and representation; name the altered kernel; freeze the information set and action support; trace assignment, delivery, selection, and observation; state the earliest divergence and strongest rival; and specify the ethical stop and failure disposition. A proposal that cannot complete that sequence remains an explanatory sketch rather than a causal test.

The chapter converts abstract control language into bounded, inspectable evidence. Chapter 20 now asks whether embodied

processes carry, constrain, or merely witness those effects, and whether lags, medication, illness, trauma, disability, and device error change the causal story.

The same observed change can arise from different loci. A lower reported intensity may reflect altered phenomenal value, memory, language, demand characteristics, sensor quality, or selective observation. The design therefore needs at least one contrast that separates the candidate locus from these alternatives, plus a no-estimate route when the contrast is unavailable.

Mechanism claims should be indexed by action support. A conclusion obtained under one narrow set of safe actions does not automatically transport to clinical treatment, everyday choice, or terminal care. The chapter tracks the exact action set beside every continuation law and intervention verdict.

The human meaning of this precision is practical. When science mislocates an effect, people may be blamed for a device artifact, denied relief because a report changed, or subjected to a stronger intervention than the evidence requires. Accurate locus language protects both inference and care.

19.1 — Candidate Causal Loci and the Representation Gate

Candidate loci include the phenomenal process, a route state, a neural or bodily mediator, a memory state, an action policy, the observation kernel, the endpoint process, and the selection mechanism. Each locus implies a different intervention and a different causal estimand.

The representation gate forbids a common shortcut. Changing self-report wording may change measurement without changing experience. Changing behavior may alter action while leaving phenomenal value uncertain. A physiological perturbation may affect arousal, pain, motor capacity, or reportability. The design names the locus it actually touches.

A causal graph should include the latent target, measured channels, treatment, mediators, actions, context, missingness, and endpoint. The graph is not complete merely because it is drawn; omitted variables and measurement error should be listed with sensitivity plans.

Interventions on the endpoint process require special caution. A protocol may change observation or administrative stopping, but no study may manipulate dying or withhold care to sharpen a terminal contrast. Natural variation and ethically permissible clinical processes set the boundary.

Representation uncertainty can make the causal effect set-valued. If several scalarizations remain defensible, the analysis should report effects across the identified set rather than select the one that produces the desired direction. The locus is identified only

when a neighboring-locus intervention would predict a different pattern.

A causal locus is the earliest component of the declared data-generating system at which an intervention changes the distribution of a downstream quantity under a fixed representation, routing rule, and endpoint definition. Any intended change to the observation process must be explicitly typed as part of the intervention. The locus may be neural, bodily, computational, environmental, relational, or distributed. An anatomical correlate becomes causal only through the intervention semantics, treatment version, assignment time, mutable state variables, and continuation outcomes that connect it to the contrast (Pearl, 2009; Hernán & Robins, 2020).

The representation gate comes first because every intervention is interpreted through declared variables and bridges. A change in self-report may reflect experience, threshold, memory, language, demand, or disclosure. Autonomic change may reflect arousal, posture, temperature, medication, disease, pain, or effort rather than valence. Route inference therefore compares explicit causal structures and tests measurement invariance across arms before assigning a phenomenal interpretation (VanderWeele, 2015; Khalsa et al., 2018).

19.1.1 — Representation Before Intervention

Intervention begins only after representation is fixed. A treatment cannot be well defined if the state it is supposed to change is allowed to shift between phenomenal value, route state, physiology, report, and behavior. The causal graph should

therefore name the target layer, the manipulated variable, the mediators, the observations, and the endpoint before an effect is estimated (Pearl, 2009; Hernán & Robins, 2020).

The representation gate also fixes sign, unit, clock, and stream ownership where those features matter. Changing an affect scale or bearer definition after assignment can alter the estimand rather than merely improve measurement. A causal result belongs to the declared representation and cannot be transported to another layer without an explicit bridge.

A concrete failure occurs when a breathing intervention changes arousal and the analysis calls the change a reduction in cumulative negative valence. The intervention may have changed respiratory sensation, attention, report, or threat prediction. Without a measured-to-latent bridge, the stronger claim is not identified. The correct effect can still be useful at the channel it actually measured.

Representation error can also create a collider. Selecting participants because a report and a physiological measure agree may condition on a common effect of latent state and measurement quality. The selected sample can then show associations that do not hold in the eligible population. The selection rule belongs in the causal diagram.

Before data collection, the protocol should ask a sequence of questions: what object is acted on, which variable is manipulable, what observation witnesses the change, what alternative pathway produces the same witness, and what result would show that the

representation was wrong. This gate prevents intervention language from outrunning semantics.

The study can evaluate report, physiological, and behavioral effects under their respective measurement bridges. Each result is labeled by the construct its qualified analysis supports.

19.1.2 — Phenomenal, Neural, Bodily, Behavioral, and Institutional Loci

A causal-locus analysis separates at least five levels: phenomenal state, neural process, bodily regulation, behavior or policy, and institutional environment. A candidate mechanism may act at one level, travel through another, and be observed at a third. Treating the nearest measurable level as the causal source is a category error (Pearl, 2009; Hernán & Robins, 2020).

The phenomenal locus concerns changes in the target state itself. Because phenomenal value is not directly instrument-fixed, an intervention on report or physiology cannot be assumed to intervene on experience. The representation and measured-to-latent bridges must already be qualified before phenomenal causal language is used.

Neural and bodily loci include sensory integration, prediction, salience, autonomic regulation, endocrine state, pain, fatigue, and action control. These processes may mediate or confound observed affective change. A reproducible correlate identifies a candidate pathway whose specificity can be tested against other routes.

Behavioral loci include attention, avoidance, communication, rest, information seeking, and task choice. Behavior can alter

future context and measurement while remaining only an imperfect witness to experience. A person may choose the same action from different states or choose different actions under the same state because opportunity and risk differ.

Institutional loci include record systems, access rules, delays, clinical practice, family logistics, and material constraints. An intervention that corrects a record or opens treatment access may change future experience through the world rather than through an internal controller. The causal graph should preserve that external route instead of relocating the burden inside the person.

Cross-level claims require temporal order and mediation tests. If institutional repair changes behavior first, physiology later, and report last, that sequence differs from a direct bodily perturbation. The study should state the expected lag pattern and identify which alternative paths could create the same order.

The locus map confers no hierarchy of human truth. Neural classifiers, self-report models, institutional records, and behavior each occupy bounded evidential roles. Their disagreements become design information, while the person retains authority over care, refusal, correction, and participation. The map exists to locate interventions and uncertainty, not to appoint an oracle.

Locus failure produces a narrower but still informative result. A change localized to observation, report, behavior, or physiology remains evidence about that layer and is not promoted to a phenomenal mechanism. When several loci remain compatible, the output is a causal identified set with the next discriminating measurement or perturbation named explicitly.

19.1.3 — Manipulability and Well-Defined Treatments

A well-defined treatment has versions that are sufficiently specified for the causal question. ‘Support,’ ‘rest,’ ‘information,’ and ‘repair’ are not single interventions. They vary in timing, intensity, delivery, consent, context, and the pathways they open. If different versions produce different outcomes, combining them under one label violates consistency and obscures the locus of action (Hernán & Robins, 2020).

Manipulability is layer specific. A study can randomize a prompt, offer a safe action, or observe an independently indicated care transition; it cannot directly assign a private belief, force a latent route state, or manipulate a truth-level endpoint. The causal claim therefore follows the variable that was actually changed and the downstream object the design can support.

The same nominal treatment can be ill defined across persons. A notification may be informative for one participant, coercive for another, inaccessible for a third, and irrelevant for a fourth. Treatment versions should therefore include accessibility, language, burden, and opt-out conditions rather than assuming that delivery equals exposure.

Pilot work tests well-definedness. Process measures record what was delivered, understood, used, and experienced as burdensome. Qualitative feedback identifies unanticipated treatment versions and helps refine the intervention definition.

Rivals may act at different versions. An ordinary expectancy model may respond to information, a control model to action availability, and a social-support model to human contact. A

mechanism tournament should preserve those components instead of using a bundled intervention that lets every explanation claim success.

When the treatment remains ambiguous, the effect should be reported at the assignment level or as a set of version-specific possibilities. The route claim narrows until the action can be defined without hiding the work, burden, and pathways that make it real.

19.1.4 — Mediators, Proxies, and Colliders

A mediator lies on a causal path from intervention to outcome, a proxy measures or predicts an unobserved variable, and a collider is caused by two or more variables whose conditioning can create a spurious association. These roles are not interchangeable, even when the same variable appears in each model (Pearl, 2009; VanderWeele, 2015).

Physiology illustrates the problem. Heart-rate variability may mediate a breathing intervention or proxy autonomic state. When both medication and distress influence whether a clean segment is recorded, the recording indicator is a collider, and restricting analysis to recorded segments can induce bias. The analysis must specify each variable's role and which data could distinguish it.

Post-treatment adjustment can remove part of the effect the study intends to estimate. Conversely, failing to adjust for a pre-intervention common cause can create bias. A causal diagram should therefore precede the regression list, with time ordering and treatment versions visible.

Proxy validity requires a bridge. A wearable feature that predicts report does not become the latent state; it may be sensitive to activity, posture, illness, or device fit. Calibration, transport, and measurement error must accompany any proxy-based mediation claim.

Selection on complete data can create collider bias. Participants with usable physiology may differ because both health state and engagement affect recording. Restricting to those records can induce associations that look like a mechanism. Observation and retention nodes belong in the graph.

Sensitivity analysis should target the role most likely to fail. Negative-control outcomes can detect residual confounding, measurement-error models can bound proxy distortion, and mediation sensitivity can assess unmeasured mediator-outcome confounding. One generic robustness check is not enough.

When the role remains ambiguous, the result says so. A variable may remain useful for prediction while losing a causal interpretation. Causal restraint is not a weakness; it is what keeps a convenient biomarker from becoming a false mechanism.

19.1.5 — Cross-Level Bridges and Causal Granularity

Causal granularity determines what counts as an intervention and what outcome it can change. A molecular perturbation, network-level stimulation, breathing instruction, communication aid, record correction, and policy change operate on different state spaces. Their effects cannot be compared until the bridge between levels, the causal estimand, and the assumptions permitting

transport are specified (Bollen, 1989; Pearl, 2009; Hernán & Robins, 2020).

A bridge may be mechanistic, statistical, or merely correlational. Mechanistic bridges describe how a lower-level change alters a higher-level state. Statistical bridges relate measurements across levels under a declared population. Correlational associations support neither intervention transport nor phenomenal identification by themselves.

Aggregation can hide causal heterogeneity. A neural-network average may remain stable while subcircuits shift in opposite directions; a population intervention effect may conceal person-specific benefit and harm. The selected level should preserve the distinctions required by the estimand.

Disaggregation can also mislead. Adding more biomarkers does not create a better mechanism if the variables are noisy, redundant, or descendants of the same latent state. The model should prefer the coarsest level that preserves the causal contrast and can be measured without excessive burden.

Temporal scale is part of granularity. Autonomic changes can occur in seconds, hormonal changes over hours, learned behavior over days, and institutional repair over months. A cross-level bridge should predict the ordering and duration of effects rather than align every channel to one arbitrary clock.

Transport across levels requires invariance. A relation estimated under a laboratory breathing task may not hold during illness, medication change, or real conflict. Held-out contexts and

alternative treatment versions test whether the bridge is mechanism-level or task-specific.

At each level, the representation gate requires a qualified link between the measured channel and the construct named in the result. Reports identify the highest level supported by that link.

Cross-level failure can be productive. It may show that a route is distributed, that a supposed mediator is only a witness, or that the intervention acts through context rather than inner control. The model then narrows to the level where evidence survives.

19.1.6 — Locus Failure and Scope Withdrawal

A causal locus fails when the declared intervention does not move the proposed state, the expected mediator order is absent, equal-treatment rivals explain the change, or the effect disappears after observation, mediation, and selection are modeled. Failure belongs in the mechanism register rather than being softened into a vague partial success (Hernán & Robins, 2020; VanderWeele, 2015).

Failure can be local. A bodily pathway may operate for pain relief but not for social threat; an institutional repair may change report without changing the candidate route state; a neural marker may predict task performance only in one device or site. Scope should narrow to the supported treatment, outcome, and population.

A locus also fails when the intervention is not well defined. Medication class, therapeutic conversation, rest, and information access each contain multiple versions with different consequences. If versions cannot be separated, the estimand

should be widened or withdrawn rather than treated as one causal treatment.

Nonpositivity produces a defined scope boundary. When a subgroup has no realistic access to an intervention or no safe alternative, the data do not identify its counterfactual continuation under that action. The analysis narrows the intervention family or reference class, uses partial-identification bounds where justified, or returns No adequate estimate rather than locating blocked access inside the person.

Representation failure outranks localization. If the measured channels cannot distinguish valence, arousal, pain, fatigue, salience, or reportability, no causal analysis can identify which phenomenal quantity changed. The result may still support a channel-level effect, but not a route-level one.

Ethical feasibility defines the interventions available for identification. A contrast that requires induced distress, delayed care, withdrawn analgesia, coercion, or overridden refusal is outside the admissible design, not an unfinished experiment awaiting greater resolve. The resulting causal question may remain partially identified or unanswerable under permissible interventions, and that limit is part of the scientific verdict. Partial identification should state which causal contrasts remain bounded by permissible evidence, turning ethical limits into explicit scientific scope.

Withdrawal should propagate. Figures, tables, summaries, and public excerpts must remove or relabel the failed locus. Leaving a vivid brain region, body signal, or policy arrow after the causal

claim is withdrawn misstates the evidence. Correction should reach cached educational and presentation graphics as well, because simplified locus images can outlive the technical claim that originally justified them.

The remaining value is an explicit no-inference zone. By documenting where localization failed, the chapter prevents later studies from treating the same proxy or treatment as established. Failure memory is part of cumulative science. The no-inference zone should record which alternative loci or distributed explanations remain compatible, so later work begins from the actual uncertainty rather than from zero.

A route can survive locus failure only by naming a different, prospectively testable locus. It cannot become a diffuse system that acts wherever the data happen to change. A mechanism that has no place to fail has no place in a causal science. A successor locus must make an observation or intervention prediction that differs from the failed one; otherwise relocation is only a change of labels.

19.2 — Intervention-Indexed Continuation Laws

Let $Q_{h_t}^{a,g}$ denote the continuation law after history h_t under structural action a and continuation regime g . The law includes future state, actions, observations, missingness, and endpoint behavior. It is not merely a distribution of a final score.

Potential-outcome and dynamic-regime notation can make the contrast explicit, but consistency, positivity, and interference assumptions remain. If some histories cannot receive an action, the corresponding causal effect is not identified in that region.

Past-fixity is built into the continuation law. An intervention changes the distribution of future contributions; it does not revalue the prefix already experienced. Under exact PTVC, that fixed prefix constrains the signed remainder; later relief does not retroactively reduce the negative experience already lived or make it morally justified.

Randomization yields a necessary restriction even when the past is unobserved. Let L be the signed prefix fixed before assignment Z , and R the entire remaining contribution through the true endpoint's left limit, including the intervention's effects. For a prospectively specified cohort whose realized regimes preserve the declared PTVC scope and stream ownership, exact closure gives $R = -L$ almost surely. If Z is independent of L , the distribution of R must therefore be identical across assigned arms. With conditional randomization, compare within baseline strata or standardize to a common baseline distribution.

That randomized-arm restriction can be decomposed without changing the endpoint convention. For the delayed-treatment comparison, let τ be a common, prospectively fixed treatment-onset time after assignment. On the primary strict-pre-endpoint branch, E contains the post-assignment contribution before both τ and the truth endpoint, and F contains the remaining pre-endpoint contribution at or after τ . Thus $R = E + F$; if the stream ends before or at τ , $F = 0$. A separately declared endpoint-inclusive branch instead uses $R_{\text{incl}} = E + F + J_T$, where J_T is the contribution assigned exactly at the truth endpoint and remains excluded from the primary R . The randomized-arm restriction applies separately to the complete remainder of the branch declared in advance. Under exact closure on that branch and assignment independence from the pre-assignment prefix, its distribution is identical across arms. Conditional randomization requires comparison within baseline strata or standardization to the same baseline distribution, as in the preceding result. The restriction does not by itself require any particular component or pre-treatment biological outcome to differ between arms.

A stronger localization claim requires a separately declared all-intervention structural model. For a fixed pre-assignment history h in common support, compare potential continuations under arms a and b using the same time cut and the declared bearer, routing, representation, and endpoint architecture. If primary-branch closure holds for both continuations on the model's joint probability space, then:

$$L(h) + E^a(h) + F^a(h) = 0$$

$$L(h) + E^b(h) + F^b(h) = 0$$

Subtracting gives:

$$E^a(h) - E^b(h) = -[F^a(h) - F^b(h)]$$

Here $F^a(h) - F^b(h)$ concerns the entire later contribution, not just the treatment interval. An independently justified nonzero restriction on that whole-continuation difference forces an opposite difference in $E^a(h) - E^b(h)$. Different immediate treatment effects do not establish this premise when the rest of the continuation can compensate. For an endpoint-inclusive branch, the subtraction must also include the arm-specific terminal terms J_T^a and J_T^b . A population-level directional or distributional prediction additionally depends on the structural model's distribution of histories and its observation bridge; the pathwise identity alone does not identify that prediction.

Valid randomization supplies assignment independence from the pre-assignment state within the declared randomization strata. If assignment is concealed and no assignment-correlated physical channel reaches the organism before treatment, ordinary local forward-causal models additionally supply the null that assignment has no causal route to a systematic pre-treatment biological difference. Finite samples can still show chance imbalance, and implementation can still leak assignment information. Because assignment precedes the outcome, an association measured after assignment but before treatment is not retrocausal merely because treatment is scheduled later. A PTVC-linked model predicting such an association must specify the

global-selection, viability, whole-trajectory, or other mechanism linking the assigned continuation to the earlier path and must retain, or explicitly revise, the assignment-independence premise it uses.

The terminal fork makes the conceptual contrast explicit. Imagine simulated flies from three specified history groups reaching two routes without local information identifying their downstream consequences. One candidate continuation ends biological life essentially immediately; the other permits ordinary continuation. This is a formal thought experiment, not an animal procedure. For this fork, let h denote the complete predecision stream history and $L(h)$ its signed prefix. Let d denote the candidate immediate-death continuation. The corollary uses an independently specified endpoint bridge under which the relevant stream permanently ends no later than biological death, and $C_d(h)$ is the complete decision-to-endpoint signed contribution under the declared branch convention. A necessary condition for d to be admissible under exact closure is:

$$L(h) + C_d(h) = 0$$

In the phenomenally silent idealization $C_d(h) = 0$, the condition reduces to $L(h) = 0$; more generally, it requires $L(h) = -C_d(h)$. These are necessary compatibility conditions, not sufficient conditions for death and not behavioral probabilities. They do not imply that reaching balance causes death. A route-selector must separately specify how completion restrictions affect action support or route probability, without implying conscious knowledge of the ledger or the future.

The selector's information set is part of the model. When route-to-contingency mapping is randomized independently of the pre-choice state and no local cue identifies that mapping, an ordinary local policy cannot condition its choice on it. A predicted preference for the concealed downstream contingency therefore requires a prospectively declared global or whole-trajectory selector, or an independently testable information channel. Giving a simulator access to a hidden mapping does not establish that an organism has access to it.

Mapping a localization restriction, a completion-set restriction, or terminal-fork compatibility to hatching, developmental survival, adult eclosion, or route choice requires an independently justified route, viability, and observation bridge. A biological outcome before a qualifying stream begins is evidence about the declared developmental extension, not a measured phenomenal residual. If assignment is generated only after the outcome has occurred, chance association remains possible; a reproducible systematic association beyond the prespecified null and procedural explanations would motivate an additional structural hypothesis rather than follow from the randomized-remainder result.

A restricted observation model supplies an additional check. Suppose the integrated reading is $Y = aR + oD + e$, with a common positive gain a , a common rate-origin offset o , remaining exposure D , finite first moments and equal mean errors across arms. Relative to a reference arm, $\Delta E(Y) = o\Delta E(D)$. Three or more arms can test whether a single origin explains all contrasts; if exposure means agree, reading means must agree without identifying that origin.

This removes the missing prefix from one target-level comparison while leaving complete future measurement essential. Use independently justified bounds for any missing tail and retain every assigned stream under the declared observation model. Treatment-dependent qualification, selective follow-up or varying origins can break the comparison. Ordinary regulation can also produce zero total intervention effects while leaving a nonzero lifetime total, so agreement is informative only against specified rivals. The restriction predicts no particular later adverse event and gives no reason to delay beneficial care.

The intervention itself enters the trajectory. Task effort, uncertainty, reward, frustration, relief, social evaluation, and later opportunity changes must be included in the measured estimand or intercurrent-event strategy. A supposedly neutral probe can be affectively active.

Continuation laws can be history-indexed or state-indexed. State reduction requires sufficiency. If unmodeled history predicts response after conditioning on the proposed state, the state-indexed causal claim is incomplete. Intervention definition, policy class, endpoint branch, observation process, and analysis rule must be frozen before outcome inspection.

An intervention-indexed continuation law specifies more than an average treatment effect. It describes how the conditional distribution of future states, observations, actions, route variables, and endpoints changes after a declared intervention history. In longitudinal settings, treatment may depend on evolving state, and evolving state may itself be affected by earlier treatment. G-computation, inverse-probability weighting, structural nested

models, and dynamic treatment-regime methods were developed precisely because ordinary conditioning can distort such feedback systems (Robins et al., 2000; Murphy, 2003; Hernán & Robins, 2020). A mechanism claim that ignores treatment-confounder feedback may attribute to completion dynamics what is actually produced by clinical or behavioral adaptation.

Continuation laws require empirical and ethical support. A policy cannot be evaluated where comparable assignments are absent, the action was unavailable, or ethical practice correctly prevented exposure. Positivity is therefore a property of both the data and the design. When support fails, the analysis displays the unsupported region, widens uncertainty, narrows the reference class when appropriate, and may return No adequate estimate. This is especially important near severe illness, crisis, or rapidly changing capacity, where treatment history, dropout, and care decisions are strongly state dependent and a smooth model can conceal an empty empirical neighborhood (Hernán & Robins, 2020; Petersen et al., 2012).

19.2.1 — Potential Continuation Paths

Potential continuation paths extend the potential-outcomes idea from one endpoint to a future trajectory. For each admissible intervention regime, the object is the continuation that would unfold from the current prefix under that regime. The paths are conceptual counterfactuals, not multiple lives available for observation (Robins et al., 2000; Hernán & Robins, 2020).

The path must be indexed by treatment version and time. A one-time prompt, repeated information policy, medication transition,

or changing support route can produce different continuations even when grouped under one label. The regime should specify when decisions are made and which history is available.

Only one continuation is observed for each realized decision history. Identification therefore relies on assumptions and data structure, not on comparing a participant with their own unobserved alternative. Within-person designs reduce some heterogeneity but do not make simultaneous counterfactual paths observable.

The continuation outcome should be typed. It may be a measured report trajectory, physiological channel, action-support set, endpoint-independent route state, or terminal event. A behavioral continuation is not automatically a phenomenal continuation, and a measured path is not the latent stream.

Dynamic confounding is central. Earlier treatment can change later state, which affects later treatment and future outcome. Standard adjustment can be biased when time-varying confounders are also treatment affected. G-methods or appropriately specified sequential models are needed for such regimes (Robins et al., 2000).

Support limits the path family. If a regime is never or rarely delivered in a given state, its continuation is weakly identified or unidentified there. A model should report the supported region rather than generate smooth counterfactuals for states outside observed and ethically admissible practice.

Interference can couple continuations. A family member's action, clinician schedule, shared room, or institutional policy may affect

several participants. The unit and interference structure must be declared before individual continuation effects are interpreted.

A low-burden example compares two information timings that are both ethically acceptable. The candidate route and an ordinary uncertainty-reduction model may predict different earliest changes in action support. The design follows those continuations without treating one policy as a moral instruction.

Potential paths are useful only when they remain bounded by observation and ethics. They can organize causal questions and sensitivity analyses; they cannot be returned as personal forecasts or used to claim that a person should have chosen the path that a model prefers.

19.2.2 — Dynamic Regimes and Treatment Versions

A dynamic regime maps observed history to an action at each decision time. It is more specific than a treatment label because it states when information is evaluated, which actions are available, and how the policy responds to changing state (Murphy, 2003; Hernán & Robins, 2020).

Treatment versions matter. A message can differ in content, source, timing, privacy, burden, and reversibility; a medication change can differ in dose and indication; a support action can differ in whether it expands real opportunity. Combining versions can violate consistency and blur the mechanism.

Adaptive regimes should be distinguished from ordinary clinical or personal judgment. A research rule may randomize among safe prompt timings, but it does not own care decisions or life choices.

Clinically indicated treatment remains independent of the route experiment.

History must be limited to information available at the decision point. Including future endpoint labels, later adjudication, or post-action summaries gives the regime impossible knowledge and contaminates causal interpretation.

Regime comparison should include ordinary alternatives such as fixed schedules, symptom-triggered care, uncertainty reduction, and fatigue-sensitive prompting. The candidate route wins only if its history-dependent rule predicts held-out continuation differences beyond those baselines.

Transport requires treatment-version alignment. A regime that works with one device, language, clinic, or support route may not transport when the delivered action changes. The record should distinguish policy transport from local re-optimization.

When treatment versions cannot be separated or positivity fails, the estimand narrows to a coarser intervention family or the result becomes No adequate estimate. Precision in the policy name is part of the causal claim because different versions may alter both experience and observation.

19.2.3 — Consistency, Positivity, and Sequential Exchangeability

Sequential causal inference requires consistency, positivity, and a defensible exchangeability condition at each decision point. Consistency links the observed outcome to the potential outcome under the treatment version actually received. Positivity requires that the compared actions lie in the support of the assignment law

at each relevant history: positive probabilities for discrete actions, or the corresponding density and support conditions for continuous actions. Sequential exchangeability states that, conditional on the recorded history, action assignment carries no unmeasured information about future potential outcomes (Robins et al., 2000; Hernán & Robins, 2020).

These assumptions are not technical boilerplate. A closing treatment window can destroy positivity because some actions are no longer available. Clinical judgment can create unmeasured confounding when the reasons for a treatment change are incompletely recorded. A device failure can alter both observation and subsequent action. Each failure changes what continuation effect can be identified.

Dynamic regimes make consistency demanding because treatment has history-dependent versions. ‘Continue support’ may mean different doses, contacts, or accommodations at different times. The protocol should specify decision rules and allowable deviations rather than treating a sequence of heterogeneous actions as one exposure.

Continuous monitoring sharpens the same problem. When treatment, confounders, and outcomes evolve as functions of time, feedback is not a finite sequence of visits but a dense process. Identification then requires continuous-time versions of consistency, sequential randomization, positivity, and intervention well-definition; otherwise a discretized analysis may create support that the underlying process never supplied (Ying, 2024).

For continuous interventions delivered at multiple times, positivity failure may be the rule rather than the exception. One defensible response is to redefine or project the estimand toward intervention values supported by the observed histories while displaying diagnostics for the unsupported region. The result answers a narrower causal question instead of extrapolating a full dose-response curve beyond the data (Schomaker et al., 2024).

Diagnostics can expose weak support. Propensity distributions, overlap plots, effective sample size, simulated regime frequencies, and stochastic-positivity diagnostics show where estimates depend on extrapolation rather than observed treatment support. When support is absent, partial-identification bounds or a narrower target population are more honest than a stabilized weight that hides the gap (Ring & Schomaker, 2026).

Negative controls can probe exchangeability but cannot prove it. A pre-action outcome that changes with assignment may reveal residual confounding or implementation bias. Sensitivity analysis should quantify how strong an unmeasured factor would need to be to reverse the mechanism comparison.

A continuation law earns causal language only within the histories, versions, and assumptions that support it. Outside that region the model may remain predictive, but prediction should not be rewritten as intervention effect.

19.2.4 — Interference and Shared Environments

Interference occurs when one participant's action changes another participant's continuation law. Shared environments make that possibility routine: family care, clinical staffing, social networks,

classroom rules, workplace schedules, and platform algorithms link trajectories. Treating each stream as causally isolated can bias effects even though the formal phenomenal ledger remains intrastream and nontransferable (Hudgens & Halloran, 2008).

The distinction is important. Phenomenal contributions do not transfer across streams, but causes and resources can be shared. One person's appointment may use a scarce slot, a caregiver's sleep may depend on another person's symptoms, and a policy change may alter every participant's action set. Causal independence cannot be inferred from formal nontransferability.

Designs should declare the interference unit. It may be a household, clinic, cohort, network, or time block. Exposure mappings can summarize how others' assignments enter a person's treatment version, but the mapping must be justified and sensitivity-tested. A convenient cluster is not automatically the true interference boundary.

Shared environments also create hidden burden transfer. An intervention that helps a participant by shifting work to family or staff can look beneficial in person-level outcomes while increasing unmeasured load elsewhere. The scientific record should name those displaced tasks and, where ethically and legally permitted, measure system effects without turning bystanders into involuntary research subjects.

Mechanism tournaments should include ordinary network and resource-allocation rivals. Apparent route-sensitive action changes may be responses to staffing, social information, or collective scarcity. Cluster randomization, network-aware

analysis, or partial interference assumptions may be needed before a person-level mechanism is credited.

When interference cannot be bounded, the continuation effect should be narrowed to the observed system or left partially identified. The intrastream claim does not license ignoring the shared world in which interventions occur.

19.2.5 — Partial Identification of Continuation Effects

Continuation effects are often only partially identified because latent state, treatment version, missingness, endpoint uncertainty, and unmeasured confounding cannot all be resolved. The scientifically appropriate object may be a set of effects compatible with declared assumptions rather than one point estimate (Manski, 2003; Imbens & Manski, 2004).

Bounds should separate sources of uncertainty. One component may arise from missing outcomes, another from treatment-version ambiguity, and another from an unvalidated measured-to-latent bridge. Combining them into one wide interval can hide which scientific gate would most improve identification.

Monotonicity or exclusion restrictions can narrow bounds, but each restriction needs substantive justification. Assuming that support cannot worsen any measured outcome, for example, may fail through burden, expectation, stigma, or interference. A convenient narrowing assumption is not evidence.

Sensitivity analysis should vary plausible unmeasured confounding and selection mechanisms. Tipping-point plots can show how strong a hidden path would need to be to reverse the

sign or route comparison. The plot is a decision aid, not a license to select the most favorable parameter range.

Partial identification is especially important for phenomenal effects. Even when the measured trajectory changes, the latent effect depends on calibration and representation. The identified set should carry that bridge uncertainty rather than promote a channel effect to an inner-state estimate.

A set can still discriminate. If every value compatible with the assumptions favors an ordinary rival, the candidate route loses. If the set crosses the null or overlaps several mechanisms, the result is ambiguous. If the set excludes a prospectively specified rival region, the next gate may be licensed.

Support restrictions should be visible by subgroup and state. A narrow estimate in well-observed, high-access participants cannot be silently generalized to people with disability, severe illness, limited communication, or different care contexts.

Apply §24.2 and Table 27.5. Here, identify whether wide bounds or nonidentification arise from a missing bridge, support gap, or treatment ambiguity, and state which redesign could sharpen the next inference.

19.2.6 — Descriptive, Predictive, and Causal Separation

Descriptive, predictive, and causal questions may use the same data but make different claims. Description summarizes observed continuations, prediction forecasts under a deployment law, and causation asks what changes under intervention (Hernán & Robins, 2020; Pearl, 2009).

Descriptive terminal alignment can generate hypotheses, yet selection, censoring, or retrospective clocks may produce it without a mechanism or intervention effect. Report whether the pattern survives defensible alternative alignment, censoring, and clock choices; instability across those choices is a descriptive robustness finding, not evidence for or against a mechanism by itself.

Prediction can exploit site or device proxies that are causally irrelevant. Held-out accuracy alone therefore cannot identify route state. Leave-site-out, leave-device-out, or comparable transport tests can reveal whether predictive performance survives beyond environment-specific information without converting successful prediction into causal identification.

Causal inference attaches a defined contrast to its exchangeability or alternative identification strategy, positivity, consistency, and time-varying-confounding rules instead of listing those assumptions as remote caveats. When multiple defensible identification strategies target the same causal contrast, their estimates should remain visible separately before any synthesis; disagreement then becomes evidence about assumption sensitivity rather than variation to be averaged away.

Mechanism claims exceed average causal effects. Mediation, lag, negative controls, and competing-pathway tests are needed before an effect is localized. Where the proposed pathway can be perturbed safely, a localization claim should specify the expected temporal or intervention response in advance so failure of that response bears directly on the proposed mechanism.

Where possible, the roles use separate data splits: description generates hypotheses, prediction tunes on training data, and prospectively specified causal contrasts face held-out or independent data. Preprocessing, feature selection, and model tuning that use protected outcomes belong inside the development process; otherwise the nominally held-out evaluation is no longer genuinely held out.

Public language names the role precisely: associated with, predicted, changed under intervention, and mediated part of the effect are different statements. Explained is reserved for an identified pathway.

Predictive value can survive causal failure, and a causal estimate can answer a narrow intervention question without broad predictive power. The record preserves those contributions rather than forcing one status.

Description and prediction may test prerequisites or signatures, while causal designs may localize a mechanism. None alone supplies complete-stream support.

19.3 — Earliest-Divergence and Action-Invariance Tests

Earliest-divergence analysis asks when two candidate models first imply different conditional distributions under common information. Locating that time can distinguish a representation effect from a policy effect or an observation effect.

Action invariance provides a sharp null only when the models specify the same pre-locus action mechanism as well as the same information and available actions. Shared inputs alone do not force different policies to agree. Under the stated invariance restriction, earlier divergence can indicate leakage, baseline imbalance, or failure of the assumed common mechanism.

Commitment must be recorded separately from assignment. Participants may receive a condition, decline it, switch, or anticipate it. The private policy randomization and information available at commitment determine which contrast the study actually estimates.

A useful experiment freezes a baseline model and route model before data collection, then compares predictions at successive decision points. The analysis reports the first point with calibrated held-out divergence and whether later differences remain after selection and observation are modeled.

Measurement lag can create false localization. A physiological channel may appear earlier simply because it is sampled faster. Cross-channel latency models, synchronized perturbations, and common information sets are needed before temporal precedence is interpreted causally. When no stable earliest divergence

survives those checks, the result is Model-ambiguous or evidence for a distributed mechanism.

Earliest-divergence tests ask when two candidate mechanisms first predict different observable distributions after a controlled perturbation. The emphasis on first divergence reduces the freedom to reinterpret later trajectories after many compensating processes have acted. A study therefore needs a sampling schedule fine enough to resolve the predicted ordering, a prespecified tolerance for temporal uncertainty, and a model of channel-specific lag. If two mechanisms predict the same immediate report but different autonomic, behavioral, or action-selection responses, the test should target that ordering rather than wait for a terminal summary that both models can fit (Klasnja et al., 2015; Liao et al., 2016).

Action invariance supplies a complementary causal challenge. If a route is defined to operate independently of voluntary policy within a declared class, then changing action availability or incentives should preserve its prospectively specified signature. If the signature disappears whenever the action set changes, the locus may instead lie in decision policy, learned strategy, or observation. Invariance is informative only when the intervention actually changes behavior and when compliance, expectations, and information are measured (Sutton & Barto, 2018; Shenhav et al., 2013).

19.3.1 — Why Earliest Divergence Matters

Two mechanisms may fit the same endpoint and much of the same trajectory. Their earliest predicted divergence is a useful

candidate test point because later responses can combine several pathways. It is not necessarily the most informative point: effect size, channel lag, measurement error, and burden determine which observation best separates the models.

Earliest divergence is defined relative to a common prefix and information set. Both models receive the same observations up to the intervention, and neither receives the future endpoint. The contrast asks which transition, action probability, mediator, or observation distribution should first separate.

The first divergence may occur in hidden state, behavior, physiology, or report. A hidden-state difference is not empirically useful until it implies a measured consequence. The design should trace the shortest bridge from the proposed locus to an observable channel.

Temporal ordering helps distinguish mechanism from downstream consequence. If a candidate predicts action-support change before affect report while an ordinary expectation model predicts report first, high-frequency measurement can test that order. A coarse end-of-day survey cannot.

Safe interventions should target the earliest causal fork without manufacturing distress. Information timing, option availability, benign cue structure, or clinically independent treatment transitions can be used when they are ethically justified and do not alter care for the sake of the theory.

Ordinary rivals may predict an earlier or equally early divergence. Scarcity, fatigue, cue learning, and expected control should be simulated under the same cadence. A route-specific interpretation

is unsupported when the candidate differs only after those models already explain the path.

Failure to observe the prospectively specified early difference is meaningful only if the channel had adequate temporal resolution and sensitivity. Otherwise the result is a design failure or no estimate, not automatic route retirement.

The value of the method survives any verdict. Earliest-divergence analysis forces each mechanism to expose where it acts and prevents endpoints from doing all the explanatory work.

19.3.2 — Common Information Sets

Earliest-divergence tests are meaningful only when competing models receive the same information. A common information set specifies what was observed, when it became available, and which features were permitted to influence each prediction. If one model sees future endpoint labels, richer history, or custom preprocessing, earlier divergence can be manufactured rather than discovered (Didelez, 2008; Pearl, 2009).

Information equality is not achieved by giving every model the same table after leakage has occurred. The features must be aligned to the decision-time filtration. Variables measured later, derived from post-action behavior, or selected using the outcome are excluded unless the estimand explicitly permits them.

A concrete example is a cue-response task. The candidate route and a standard learning model both receive the cue, prior observations, current action set, and participant-specific baseline. Neither receives whether the task will later resolve. Their first differing action probability is then a genuine prospective object.

Missingness belongs to the information set. A model that treats absence as a neutral value receives more certainty than a model that treats it as unknown. Missingness indicators, channel quality, and observation timing should be common inputs, and models should be scored under the same imputation or joint-observation rules.

If models do not diverge under a fair information set, the result is observational equivalence at that decision point. The study should move to a different safe fork or retire the local discriminator rather than declare the more elaborate account deeper.

19.3.3 — Action-Invariance Under Shared Evidence

Action-invariance tests evaluate a specified region in which candidate models are assumed to share the same action mechanism. Common evidence and action support make the comparison fair but do not imply invariance by themselves. A candidate claiming a distinctive action-support rule must state where agreement is predicted and where it should break.

The information set must be common. Giving one model latent route state while another receives only observed affect makes divergence inevitable and scientifically uninformative. Either both operate on the same measured history or each latent advantage is charged through an explicit observation bridge.

Invariance can be conditional. Models may agree for states far from a boundary and diverge only when signed reachability, reserve, or completion potential becomes active. The activation region should be specified before data are examined.

A behavioral task can manipulate an option that changes completion support while holding reward, effort, and information as constant as possible. The candidate predicts a targeted policy shift; ordinary discounting or scarcity models predict no special shift after their own variables are matched.

Action is not valence. A difference in choice probability can localize a policy mechanism without showing that the chosen option is pleasant, good, or morally correct. Report and auxiliary channels remain necessary for any phenomenal interpretation.

Cue leakage can counterfeit invariance failure. If the intervention reveals the hypothesis, endpoint status, or experimenter expectation, participants and models may change behavior for reasons unrelated to the candidate route. Blinding, sham cues, and manipulation checks should be built into the fork.

Invariance also has a negative-control role. Actions outside the declared route should remain unchanged. Broad behavioral shifts suggest general arousal, demand, fatigue, or observation reactivity rather than completion-dependent support.

The candidate loses when action differences disappear under equal information, arise equally in negative controls, or are explained by a simpler policy model. A narrow surviving divergence can support one route component without becoming a verdict on the Law.

19.3.4 — Cue Leakage and Post-Treatment Conditioning

Cue leakage occurs when a nominally pre-action variable contains information generated by the action, its implementation, or the later outcome. Post-treatment conditioning occurs when

analysis adjusts for a variable caused by the intervention and thereby opens or blocks causal paths. Both can shift the apparent time of divergence and manufacture mechanism specificity (Pearl, 2009; Hernán & Robins, 2020).

Digital tasks are especially vulnerable. Interface timing, hidden trial counters, adaptive difficulty, device latency, or researcher prompts can reveal condition assignment. A model may then predict action because it detects the experimental apparatus. Blinding, software audits, and sham conditions should test whether the cue remains informative when route-relevant content is removed.

Clinical and longitudinal studies face subtler leakage. A clinician's note may encode treatment response, a medication change may follow worsening symptoms, and a caregiver action may reveal prognosis. Using those variables as baseline predictors can make a model appear to anticipate a divergence it learned retrospectively.

Causal diagrams should distinguish baseline covariates, time-varying confounders, mediators, and colliders. The analysis plan should state which are adjusted, modeled jointly, or left untouched. Automatic feature selection is not a substitute for this classification because predictive importance does not identify causal role.

A negative-control pipeline can shift timestamps, mask treatment indicators, and rerun the model with post-action variables removed. If the earliest-divergence signal survives only the leaky

specification, the discriminator fails. Code and feature provenance should make the removal reproducible.

A valid route test earns its timing prospectively. When leakage cannot be ruled out, the result is descriptive or predictive under the available record, not evidence that the candidate mechanism acted before its rivals.

19.3.5 — Temporal Resolution and Detection Power

Earliest-divergence tests require cadence matched to the mechanism. If the predicted separation lasts seconds and measurements arrive every hour, the study cannot distinguish simultaneous models. Temporal resolution is therefore part of power, not a formatting choice.

Different channels have different response kernels. Behavior may change immediately, autonomic signals within seconds, endocrine signals over minutes or hours, and report after reflection. Aligning all channels to one timestamp can invert the apparent causal order.

Detection power depends on effect duration, within-person variability, measurement error, missingness, and the number of independent transitions rather than only participant count. Repeated observations within one person do not create the same information as independent causal forks.

Sampling can itself change the process. Frequent prompts may increase attention, fatigue, or disclosure and thereby alter action. Randomized, model-based, or deliberately sparse observation schedules can help separate measurement reactivity from the mechanism under study, while retaining the temporal resolution

needed for the prospectively specified contrast (Shiffman et al., 2008; Stone et al., 2007).

Latency uncertainty should be estimated rather than hidden. A distributed-lag or state-space model can represent a range of plausible delays, with prospectively specified windows and multiplicity control. Selecting the lag that yields the strongest effect after inspection creates a temporal garden of forking paths.

Device clocks, server clocks, clinical records, and participant reports require synchronization and provenance. Drift of even a few minutes can matter when the claimed divergence is brief. The record should preserve time zone, firmware, preprocessing, and alignment rules.

Power analysis should use generative simulations from every serious rival. The planned cadence and missingness are applied to each synthetic world, then the pipeline estimates how often it recovers the correct earliest divergence. If confusion remains high, the design needs a different intervention or channel.

A null result can mean no divergence, insufficient resolution, weak intervention, or measurement failure. The decision tree should distinguish these branches. Only a design that could have detected the prospectively specified difference licenses route failure.

High temporal resolution does not justify continuous surveillance. The study should collect the minimum channels and windows needed for the contrast, permit pauses and withdrawal, and avoid turning an experimental fork into permanent observation.

Robust timing localizes mechanisms: consistent held-out ordering across sites and treatment versions can rule out those predicting the same endpoint through a different causal sequence.

19.3.6 — Failure to Diverge and Route Retirement

A route-specific fork fails when the models were predicted to diverge under a supported intervention, the measurement system had adequate sensitivity and timing, and the observed continuations remain invariant. That result retires the prospectively specified divergence claim even if both models still fit the endpoint.

Failure must be diagnosed before it is interpreted. The intervention may not have changed the intended locus, treatment versions may have mixed, action support may have been too narrow, or the observation channel may have missed the window. Manipulation checks and model recovery separate these failures from genuine invariance.

Table 19.1. Earliest-divergence quantities and failure conditions.

Quantity	Common information set	First protocol-fixed separation	Failure condition
Action probability	Prefix, current observations, and supported action menu	Candidate and rival choose different safe actions	Difference disappears under equal preprocessing or recovery fails

Quantity	Common information set	First protocol-fixed separation	Failure condition
Mediator response	Preaction state and identical treatment version	Proposed mediator changes before downstream outcome	Mediator is unchanged, mistimed, or explained by off-target pathways
Observation channel	Same latent-state model and quality flags	One channel carries route-specific information at its declared lag	Signal is reproduced by artifact, activity, or generic severity
Endpoint hazard	Same at-risk set and endpoint object	Hazard changes after the protocol-fixed locus-specific fork	Change depends on retrospective endpoint alignment or selection
Negative-control action	Same task and observation history	Candidate remains stable where no route effect is predicted	Candidate changes equally under sham or irrelevant actions

Repeated failure across independent forks is stronger than one null. If information, action support, and safe perturbations each leave the candidate indistinguishable from ordinary rivals, the route should narrow or retire rather than move the causal locus indefinitely.

Opposite divergence is also informative. A candidate may predict increased repair-oriented action while participants conserve effort or choose rest. The model should not reinterpret the opposite

action as hidden completion. The prospectively specified sign and channel determine the verdict.

Several mechanisms may remain observationally equivalent at the achieved resolution. That result is Model-ambiguous and points toward a new fork chosen for separability, not toward a narrative preference for the candidate. Failure to diverge is informative when the design had adequate power to detect the prospectively specified difference.

Table 19.2. Mechanism-tournament stages, common resources, and stop conditions.

Tournament stage	Candidate and rivals receive	protocol-fixed score or diagnostic	Win, loss, or stop condition
State construction	Same observations, missingness, context, and endpoint definition	State recovery and calibration	Stop if the generating state cannot be recovered
Tuning and recovery	Equal search budgets, priors, and simulation library	Parameter and model recovery	Stop or narrow if families are not separable
Held-out fork	Same supported intervention and decision-time information	First-divergence and causal-response diagnostics	Candidate loses when the distinctive prediction fails
Transport	Frozen models, new site or population, and declared bridge	Calibration, effect transport, and subgroup error	Narrow when transport fails; do not silently recalibrate

Tournament stage	Candidate and rivals receive	protocol-fixed score or diagnostic	Win, loss, or stop condition
Open-set challenge	Unknown generators and none-of-the-models option	False-unknown and false-certainty rates	Stop if the library assigns unfamiliar mechanisms with high confidence

Transport failure narrows scope. A divergence found in one laboratory task may disappear in naturalistic settings, clinical cohorts, or different cultures because the action menu and meaning of cues change. The route is then task-specific unless a transport bridge is demonstrated.

The retirement decision is incomplete until dependent artifacts change. Figures, explanatory diagrams, and public claims must remove or relabel the failed fork, because visual privilege can outlive a sentence-level correction. The active presentation should display only the mechanism status that survived the test.

The failure record should preserve what was tested: prefix definition, information set, intervention, expected first difference, cadence, power, and rival predictions. This allows later teams to avoid repeating an underpowered or conceptually indistinguishable design. A preserved failure packet can show whether the next design needs more power, a different fork, or a genuinely different causal locus.

What survives is the shared-evidence result. If candidate and rival actions remain invariant under the strongest available fork, that constraint becomes part of the next model generation. Negative

evidence can make the remaining mechanism space smaller and more honest. This shared constraint can be more informative than a weak model preference because it rules out entire families of action-sensitive explanations at once.

Prospectively specify practical equivalence for first divergence. A nonzero difference below the observation floor or unable to change the decision does not preserve the route; the margin, uncertainty interval, and detection threshold control the result (Lakens, 2017). The equivalence margin should be justified by the smallest difference that matters for the declared scientific claim or decision. If the bound is instead based on available power or resources, label the conclusion as exclusion relative to that bound, not as evidence that smaller effects are scientifically negligible. Fix the bound before observing the separation.

Repeated invariance should remove that causal locus from the mechanism map and change the prose. The revised mechanism map should preserve that removal so later models cannot quietly restore the locus without new evidence.

19.4 — Completion-Dependent Action Support

Completion-dependent action support asks which actions retain a declared level of completion probability from the current history. One useful object is

$$\mathcal{A}_{K,\alpha}(h_t) = \{a: Q_{h_t}^a(|L(T^-)| \leq K) \geq 1 - \alpha\}.$$

This display defines one action-support band. Completion-dependent support is a modal object: under the displayed band, it identifies actions whose frozen continuation law assigns at least $1-\alpha$ probability to the declared completion band. It says nothing yet about selection, policy-level probability one, topological support, all-continuation closure, desirability, or advice. The modality, action domain, policy class, continuation law, endpoint branch, and uncertainty object travel with every reported set.

Support can shrink because of ordinary scarcity, urgency, fatigue, treatment windows, legal deadlines, or changing opportunity. Those rivals must be modeled before shrinkage is attributed to a fairness-specific route.

A support restriction can also be an artifact of the policy class. If the analysis excludes actions that would violate its own model assumptions, the remaining set may appear completion-compatible by construction. The admissible class and exclusions must be independently justified.

Held-out prediction of action availability and outcomes permits direct comparison with equally resourced control and scarcity models.

The support can shrink because resources disappear, physiology changes, time passes, institutions close options, or the model becomes more certain that some continuations are unreachable. Represent each source separately: physical infeasibility, informational uncertainty, and social exclusion call for different responses (Aubin, 1991; Aubin et al., 2011).

Ethical feasibility is not a soft constraint appended after optimization. Some actions are outside the admissible set because they violate care, consent, rights, or nonmaleficence, even if a formal controller could assign them a favorable continuation value. Safe reinforcement-learning methods illustrate how action sets and risk constraints can be built into control problems, but such mathematics does not supply the ethical boundary by itself (García & Fernández, 2015; Ames et al., 2019). In this research program, inducing suffering, delaying care, manipulating endpoints, or exploiting vulnerable states is excluded before the mechanism tournament begins. A smaller ethical support set is a feature of valid science, not a loss of power.

19.4.1 — Reachable Actions Versus Chosen Actions

An action is reachable when it belongs to the support of admissible policies from the current state. It is chosen when one realized policy selects it. The existence of a reachable completion-preserving action does not show that it will be selected, that it has high probability, or that it should be taken; support, policy, and decision remain distinct objects (Kaelbling et al., 1998; Manski, 2003; Sutton & Barto, 2018).

Support depends on real capacity and context. An action may exist in an abstract control model while being unavailable because of disability, cost, transport, care duties, safety, language, or institutional refusal. Those constraints belong in the action set rather than being treated as noncompliance.

Choice depends on objectives, beliefs, uncertainty, learning, habit, fatigue, and social risk. Two policies can share the same reachable set and select different actions. A route claim therefore needs a policy model or a weaker statement limited to support.

Observed nonchoice does not falsify reachability unless the action was actually offered, understood, safe, and available. Conversely, one observed choice does not establish broad support across other states or people.

Completion support must remain model conditional. A modeled action may preserve a terminal set under assumed dynamics while ordinary outcomes or unexpected disturbances remove that property. Robust and stochastic support analyses should report the disturbance and uncertainty set.

The chapter never converts support into advice. A research model can say an action was represented in the candidate set and predict how its availability changes behavior. It cannot tell a person what to choose, endure, disclose, or sacrifice.

A useful experiment varies option availability without changing reward or burden more than necessary. The candidate and ordinary rivals then predict how policy changes within the common reachable set. Failure to separate them narrows the route rather than the person's autonomy.

19.4.2 — Support Restrictions and Viability

Support restrictions define where a causal or policy contrast is actually represented by the data and the admissible action system. A smooth model can extrapolate outside this region, but the extrapolation is not identified merely because the equation returns a value (Aubin, 1991; Hernán & Robins, 2020).

Viability adds the requirement that at least one policy can keep the process within declared safety, consent, and study constraints. A state can have broad statistical support and no ethically viable intervention, or a viable action with little empirical support. The two regions should be shown separately.

Apply the dynamic positivity condition in §19.2.3 within the ethically eligible action set. Rare actions can produce unstable weights and apparent treatment effects carried by a few histories.

Eligibility rules are part of support. Excluding participants for medication, disability, communication, or device quality can produce a clean region that no longer represents the population of interest. The record should show who was excluded and which claim consequently narrows.

Robust viability considers model uncertainty and disturbances. A policy that is safe only under one fitted transition model may not be viable when plausible dynamics are included. Inner and outer viability sets can represent this uncertainty.

Support can be expanded by changing the environment rather than pressuring the person. Translation, transportation, accessible interfaces, respite, or an alternative clinical route may open

actions. Such changes are causal interventions on opportunity and should be represented as external loci.

Approximation error matters near the boundary. Grid resolution, surrogate models, or numerical tolerances can change whether a state is classified viable. A margin smaller than computational uncertainty should not be reported as a definitive action-support conclusion.

The public limit is strict. Being outside a model's support or viability set is not a personal label. It indicates that the research cannot identify or safely evaluate the contrast in that state. The answer is redesign, environmental support, narrower scope, or no estimate.

19.4.3 — Safe Control and Harm Avoidance

Safe control treats certain constraints as conditions of admissibility rather than costs that can be outweighed by scientific reward. Consent, comfort, clinical care, privacy, and stop requests remain outside the optimization trade space. A policy that violates them is not a high-cost option; it is unavailable (García & Fernández, 2015).

Deterministic and stochastic systems require different guarantees. In a diffusion model, the generator includes second-order terms, and a deterministic derivative cannot be quoted as though it controlled random excursions. The theorem, probability level, horizon, and regularity conditions must accompany any safety claim.

Partial observation makes safety harder. A controller acts on a belief state rather than the true state, so unobserved pain, fatigue,

distress, or device failure can invalidate a hidden-state guarantee. Conservative thresholds and participant challenge routes are part of the scientific design.

Harm avoidance includes burden created by measurement. Dense prompts, uncomfortable sensors, repeated disclosure, and staff demands can alter the process and reduce participation. The study should optimize neither information nor route separability beyond the burden approved for the participant and setting.

A safe intervention can change information, access, or benign task structure without creating negative experience to test the theory. Clinically indicated treatment transitions may be observed, but care decisions are made for the patient, not for mechanism identification.

Negative controls test whether the safety layer is doing hidden explanatory work. If both candidate and rival policies remain inside the same admissible set, any route difference must arise elsewhere. If the candidate wins only because rivals were denied safe actions, the tournament is biased.

Infeasibility is an acceptable result. When no action can satisfy both the scientific objective and the ethical constraints, the study stops or redesigns. The model does not relax dignity to preserve power.

Safety claims should be independently audited and versioned. Changes to sensors, tasks, clinical workflow, or population can invalidate the original analysis. A safe-control label is not permanent certification.

The scientific gain is clarity about what can be learned without harm. A mechanism that remains unidentified because harmful provocation is prohibited stays unidentified. Ethical limits define the admissible science rather than obstructing it.

19.4.4 — Completion-Dependent Feasibility

Fix a reached prefix h_t and one frozen downstream regime g containing every declared downstream policy, physical transition, endpoint, routing, selection, and measurement-reactivity component. For each structurally available action a , $q(a)$ is the conditional probability of the declared completion event under that complete continuation. It is a typed probability, separate from reward, phenomenal value, desirability, motivation, and advice.

Section 7.3.4 establishes the policy-mixture identity and its exactness condition under the declared action-first factorization. Here it constrains the causal locus: fixed-policy probability-one completion requires action-specific completion probability one for policy-almost every action under the frozen downstream regime.

$$P_{\eta, h_t, g}^{\pi} \left(\bar{F}_{\eta, comp}(h_t; g) \right) = \int_{A_{\eta}^{do}(h_t)} p_{\eta, h_t, g}^{comp}(a) \pi_t(da | h_t, g).$$

$$\begin{aligned} P_{\eta, h_t, g}^{\pi} \left(\bar{F}_{\eta, comp}(h_t; g) \right) &= 1 \\ \Leftrightarrow p_{\eta, h_t, g}^{comp}(a) &= 1 \pi_t(\cdot | h_t, g)\text{-almost everywhere.} \end{aligned}$$

Action support and endpoint support remain different axes. The identity concerns probability-one completion under the declared policy and continuation law. Topological support, all-policy completion, and closure of every admissible continuation require

their own stated conditions. When every available action already has $q(a)=1$, the theorem is exact and behaviorally silent.

The condition becomes circular when the admissible action set or continuation law is defined to exclude every nonclosing path. A controller constrained to preserve closure will generate closure-compatible simulations by construction. Evidence must come from held-out consequences that are not guaranteed by the feasibility rule.

Completion-dependent feasibility should be compared with ordinary safe-control, homeostatic, finite-horizon, and scarcity models under the same state, action, observation, and tuning budgets. A special completion state earns support only through a discriminating restriction that those rivals do not automatically share.

Uncertain dynamics produce robust, posterior, or probability-bounded feasibility rather than one universal binary region. Environmental interventions such as communication support, access to care, pain relief, record correction, or transportation may enlarge the action set without being internal completion mechanisms. The changed locus must be named.

The off-policy identification ceiling from §7.3.4 applies here directly. Chapter 19 adds the intervention-specific implication: evidence about excluded actions requires ethically available support from exogenous availability changes, valid instruments, bounded experiments, natural variation, or partial-identification assumptions; harmful endpoint manipulation remains outside the evidence program.

An empty or narrow feasible set supports no personal inference. Misspecification, endpoint uncertainty, incomplete action enumeration, observation limits, inaccessible support, or genuine route infeasibility can produce the same output. The public result is Route-failed, Robust-route unavailable, Model-ambiguous, or No adequate estimate. A route survives only when feasibility predicts independent transition, action, or intervention patterns under prospectively fixed rules.

Table 19.3. Action-support tools, guarantees, assumptions, and prohibited use.

Tool	Primary guarantee	Assumption that must be visible	Misuse to block
Deterministic control barrier	Forward invariance of a safe set under admissible control	Smooth h, deterministic dynamics, action access, regular boundary	Quoting it for stochastic or hidden-state systems
Stochastic barrier	Probability-one safety or finite-time probability bound, depending on theorem	Generator, diffusion, regularity, policy access, theorem conditions	Dropping the second-order term or overstating the guarantee
Viability kernel	Existence of at least one admissible policy remaining inside constraints	Declared state, action set, horizon, constraints, computational approximation	Treating the kernel as discovered natural law

Tool	Primary guarantee	Assumption that must be visible	Misuse to block
Reachability analysis	States that can reach or avoid a target under dynamics and controls	Model dynamics, disturbance set, discretization, approximation error	Treating reachability as proof of conscious dynamics
MPC safety filter	Finite receding-horizon constraint satisfaction under update rules	Model accuracy, recursive feasibility, constraint handling	Letting engineered feasibility become evidence for the Law

19.4.5 — Policy Support Without Personal Advice

Sections 7.3.4 and 19.4.4 establish the modal distinctions and mixture-support condition. This subsection asks how those results change under finite soft reweighting, limit-sure policy sequences, partial observation, and inaccessible off-policy actions, while keeping every result outside personal advice.

Apply §7.3.4 to the displayed power-reweighting construction, with a strictly positive normalizer and a baseline policy, action set, and downstream regime fixed independently of the behavior being explained. Its branch-specific completion probability is a ratio of adjacent normalizing moments, not a human score or evidence that an organism computes it.

$$q_{\eta, h_t, g}(a) := p_{\eta, h_t, g}^{\text{comp}}(a).$$

$$Z_{\beta, \eta}(h_t, g) := \int q_{\eta, h_t, g}(a)^\beta \pi_{0, \eta}(da \mid h_t, g).$$

$$\beta > 0, \quad Z_{\beta,\eta}(h_t, g) > 0.$$

$$\begin{aligned} \pi_{\beta,\eta}(da \mid h_t, g) &:= Z_{\beta,\eta}(h_t, g)^{-1} q_{\eta,h_t,g}(a)^\beta \\ &\times \pi_{0,\eta}(da \mid h_t, g). \end{aligned}$$

$$P_{\eta,h_t,g}^{\pi_{\beta,\eta}}(\tilde{F}_{\eta,\text{comp}}(h_t; g)) = \frac{Z_{\beta+1,\eta}(h_t, g)}{Z_{\beta,\eta}(h_t, g)}.$$

$$\begin{aligned} P_{\eta,h_t,g}^{\pi_{\beta,\eta}}(\tilde{F}_{\eta,\text{comp}}(h_t; g)) &= 1 \\ \Leftrightarrow \pi_{0,\eta}(\{a: 0 < p_{\eta,h_t,g}^{\text{comp}}(a) < 1\} \mid h_t, g) &= 0 \\ \text{and } \pi_{0,\eta}(\{a: p_{\eta,h_t,g}^{\text{comp}}(a) = 1\} \mid h_t, g) &> 0. \end{aligned}$$

Under the displayed construction, use §7.3.4's finite-guidance ceiling and exact support condition.

Use §7.3.4's almost-sure and limit-sure definitions unchanged. Here the additional requirements are controlled invariance, endpoint compatibility, measurable selection, and recursive time consistency (Chatterjee et al., 2016; Asadi et al., 2025).

$$\mathcal{S}_{\text{AS}}(\bar{g}) := \{h: \exists \pi \in \Pi_{\text{adm}}(h, \bar{g}), P^{\pi, \bar{g}}(F_{\text{comp}} \mid h) = 1\}.$$

$$\mathcal{S}_{\text{LS}}(\bar{g}) := \left\{ h: \sup_{\pi \in \Pi_{\text{adm}}(h, \bar{g})} P^{\pi, \bar{g}}(F_{\text{comp}} \mid h) = 1 \right\}.$$

For policy claims under partial observation, use the posterior-versus-robust distinction from §§7.3.4 and 18.3.5. A high posterior-average completion probability does not establish robust support across compatible histories; failure of the common-action condition keeps the causal claim observation-limited.

$$P(F_{\text{comp}} \mid o) = \int \int p_{\eta,h,g}^{\text{comp}}(a) \pi_t(da \mid o) b_t(dh \mid o).$$

$$A_{\eta,F}^1(h; g) := \{a: p_{\eta,h,g}^{\text{comp}}(a) = 1\},$$

$$A_{\text{rob}}^1(o; g) := \bigcap_{h \in H(o)} A_{\eta,F}^1(h; g).$$

Hard exclusion is especially difficult to verify because the observed policy supplies no data on excluded actions. Off-policy claims require explicit support, structural assumptions, instruments, bounded experiments, natural variation, or partial-identification bounds. Unsafe, illegal, endpoint-manipulating, or lifetime-risk interventions remain outside the admissible evidence program.

Table 19.4. Completion claims, mathematical conditions, and inferential limits.

Claim	Mathematical condition	What it licenses	Scope of the result
Positive-probability support	Some available action has $q(a) > 0$	At least one continuation has positive completion probability under the frozen regime	Action-specific completion probability under the frozen continuation regime
Fixed-policy probability-one	$q(a) = 1$ for π -almost every action receiving policy mass	Exact completion probability one for that protocol-fixed policy mixture	Protocol-fixed policy and continuation law; causal effects require a separate identification design

Claim	Mathematical condition	What it licenses	Scope of the result
Limit-sure attainability	sup over admissible policies equals 1, without a required attaining policy	A sequence of policies can approach probability one	Approximation by a policy sequence; attainment is a separate mathematical property
Robust partial-observation support	The intersection of probability-one action sets across compatible histories is nonempty and admits a measurable selector	An observation-adapted guarantee under the declared support and regime	Compatible-history intersection with a measurable, time-consistent selector
Finite soft guidance	Baseline policy is reweighted by $q(a)^\beta$ for finite β	Concentration toward higher-completion actions under a frozen model	Reweightings under fixed baseline support, completion probabilities, and finite guidance strength

When every structurally available action completes with probability one, the policy condition is behaviorally silent. When support depends on inaccessible actions, environmental repair may be the relevant intervention.

Every report names the exact modality it supports: positive-probability, fixed-policy probability-one, almost-sure attainable,

limit-sure attainable, topological-support, robust under partial observation, or all-continuation. Care, privacy, refusal, nonparticipation, and human authority remain independent of every modal result.

19.4.6 — Rival Explanations of Narrowed Action Sets

Narrowed action sets are common and do not uniquely indicate a completion route. Fatigue, pain, scarcity, learned avoidance, threat, depression, social risk, institutional barriers, finite-horizon planning, safe-control constraints, and measurement loss can each reduce the options observed or actually available.

The protocol first asks whether opportunity and observation narrowed together. It then gives ordinary utility, effort, delay, risk, uncertainty, habit, reinforcement, physiology, medication, motor limits, social sanctions, caregiving, cost, language, and distrust equal explanatory freedom. Support differs from selection, and a menu that appears wide to a researcher may be unusable in context (Shenhav et al., 2013; Sutton & Barto, 2018).

Test rivals through one observation pipeline: which options disappear, and how does the set change under a safe, prespecified perturbation? Chapter 20 adds body-level lag and clinical-context tests; qualitative and contextual evidence exposes institutional constraints missed by laboratory menus.

When an ordinary rival explains the narrowing, the candidate route loses that component. The result can still direct environmental repair and leaves the person's experience and need for support untouched. The report should distinguish

environmental restriction from internal choice so practical repair targets the barrier actually observed.

Open-set inadequacy is also possible. If no prospectively specified model reproduces the menu change, the answer is not a hidden completion mechanism; the model library expands under a new comparison and the current verdict remains unresolved. Residual diagnostics should identify where the menu mismatch occurs, giving the expanded model library a specific target rather than an unrestricted search. A mismatch limited to one device or task version, for example, first directs investigation toward the observation or choice interface rather than a universal mechanism. Comparing residual structure across sites, task versions, and menu geometries can show whether the missing structure is local to an interface or stable enough to justify a new generator family. That turns model expansion into a directed empirical program whose next comparison is motivated by the pattern already observed.

Transport concerns the support set itself. Options available in a laboratory may be absent under disability, caregiving, threat, cost, language barriers, or institutional control, so population and context audits precede generalization. Support-set transport should record structural absence of options separately from individual nonselection because the two imply different causal opportunities. An accessibility change that makes an option usable creates a different support set even when the printed menu and its labels remain unchanged. Cross-site pooling is therefore strongest when relevant support-set differences are represented explicitly or shown not to alter the estimand. Because feasible

opportunities can also change during a study, availability should be time-stamped rather than treated as a fixed characteristic of the participant or setting.

A discriminating fork opens or closes one safe route while holding ordinary explanations comparable. Candidate-specific held-out change can support the mechanism; common narrowing belongs to the shared constraint. The fork is most informative when the manipulation changes availability without changing the value or observability of the remaining alternatives. Recording both the offered menu and the usable menu helps separate an intervention on opportunity from a change in preference among unchanged options. The strongest design pairs that intervention with a predeclared contrast specifying which route-sensitive prediction should change and which shared controls should remain stable. When the predicted asymmetry appears only after the intended support-set change is independently verified, the causal interpretation becomes substantially sharper.

19.5 — Assignment, Commitment, Selection, and Retention

Random assignment protects the initial treatment contrast, not every later comparison. Commitment, adherence, switching, retention, observation, and endpoint survival can each select different subpopulations. The denominator flow should be diagrammed from eligibility through final analysis.

Commitment is especially important in sequential tasks. An action chosen after observing a cue is not equivalent to an assigned action. The information available at commitment and the policy that generated the choice belong in the causal model.

Informative retention can mimic route evidence. Participants with stronger symptoms, lower capacity, or different support may leave the study, lose devices, or become unobservable. Inverse-probability methods, joint models, or principal stratification can help only under visible assumptions.

Truncation by death or a competing event is not ordinary missingness. Outcomes after a terminal event may be undefined, and survivor comparisons can reverse population conclusions. The estimand must specify the population and event structure it addresses.

Selection negative controls can expose hidden bias. Outcomes measured before the proposed mechanism enters, channels unrelated to the route, and simulated denominator perturbations can reveal whether the result is driven by inclusion rather than dynamics. Withdrawal and nonparticipation remain legitimate, and no analytic advantage may attach to staying.

Assignment, receipt, adherence, and retention are distinct events. Random assignment can identify an intention-to-treat contrast, but it does not by itself identify the effect of sustained exposure, a latent mechanism, or a route among participants who remain observable. In long studies, continued participation may depend on symptoms, benefit, burden, caregiver availability, device tolerance, prognosis, or the very outcomes being studied. Analyses must therefore distinguish treatment assignment from treatment actually received and must model censoring and observation as parts of the causal system rather than as administrative afterthoughts (Frangakis & Rubin, 2002; Hernán & Robins, 2020).

Commitment mechanisms add another layer. A participant may precommit to reminders, environmental changes, or action restrictions, while the effect of commitment can operate through expectations, social accountability, reduced choice, or altered opportunity. Selection into commitment is rarely random, and the people who accept it may differ in motivation, risk, resources, or prior success. Principal stratification, sensitivity analysis, and explicit selection diagrams can clarify which contrast is being estimated, but they do not make unobserved strata directly visible (Frangakis & Rubin, 2002; Scharfstein et al., 1999). A route claim should survive these distinctions rather than absorb them into a single treatment label.

19.5.1 — Random Assignment and Delivered Exposure

Random assignment identifies an intention-to-treat contrast when the assignment mechanism is valid and outcomes follow the protocol-fixed analysis. It does not guarantee delivery,

understanding, acceptance, adherence, continued observation, or retention. Those post-assignment events receive their own variables and estimands so that assignment effects are not silently relabeled as effects of a delivered mechanism (Frangakis & Rubin, 2002; Hernán & Robins, 2020).

Delivered exposure can vary because of implementation failure, access barriers, clinician judgment, device malfunction, participant choice, or safety stopping. A binary treatment label can therefore hide several versions. Process data should document timing, dose, fidelity, and burden while preserving the participant's right to refuse or discontinue.

A simple laboratory fork illustrates the distinction. Participants may be assigned an information cue, yet some do not see it because the interface fails and others see it after acting. The intention-to-treat effect remains interpretable, while a per-protocol mechanism effect requires assumptions about receipt and timing.

Noncompliance is not a defect in the person. It may reveal that the intervention was inaccessible, burdensome, irrelevant, or unsafe. The analysis should ask what generated receipt and whether that process is itself part of the mechanism or an implementation failure.

Instrumental-variable and principal-stratum analyses target specified effects under additional assumptions. Report sensitivity to the exclusion restrictions and monotonicity assumptions used in the analysis.

The route claim should attach to the contrast actually identified. If only assignment is supported, the conclusion remains at assignment. A mechanism is not established by renaming an intention-to-treat result as the effect of a hidden route.

19.5.2 — Commitment Signals and Private Policies

Commitment signals reveal or alter a policy before later actions unfold. A participant may schedule reminders, restrict access to a distraction, authorize a supporter to call, or state a private plan. These acts can change expectations, opportunity, accountability, and selection simultaneously. They are not transparent readings of an internal controller (Frangakis & Rubin, 2002; Hernán & Robins, 2020).

Private policies matter because the action observed later may have been shaped by a rule chosen earlier. A dynamic model should distinguish the policy from each realized action and record what information was available at commitment. Otherwise a stable pattern can be mistaken for route-specific action invariance.

Commitment is rarely randomized in ordinary life. People who accept it may differ in resources, urgency, confidence, support, or prior success. A comparison between committers and noncommitters therefore mixes the commitment effect with selection unless the design addresses those differences.

A safe experiment can randomize the offer or form of a low-burden commitment while leaving acceptance voluntary. The analysis then separates offer, acceptance, and subsequent behavior. It should also measure whether the commitment

reduced choice in a way participants experienced as helpful, neutral, or burdensome.

Ordinary learning and self-control models are strong rivals. A commitment effect may reflect reduced temptation, simplified planning, social expectation, or memory support. A fairness-specific route must predict a distinctive timing, state dependence, or intervention transport beyond those accounts.

No model may use a private policy as evidence of what a person owes themselves. Commitment remains an empirical object and a voluntary action. Withdrawal from it never becomes scientific noncompliance or reduced standing.

19.5.3 — Adherence, Attrition, and Informative Retention

Adherence and retention are dynamic processes. Continued participation can depend on benefit, symptoms, fatigue, caregiving, device tolerance, distrust, prognosis, or the burden created by measurement. When those variables also predict outcomes, complete-case analyses can manufacture a favorable trajectory (Scharfstein et al., 1999; Hernán & Robins, 2020).

The denominator should be traced from eligibility through consent, assignment, delivery, adherence, observation, and final analysis. Each loss receives a reason category, timing, and uncertainty status. A single attrition percentage cannot reveal whether missingness is related to the mechanism being tested.

Inverse-probability weighting, joint models, and multiple imputation can help under assumptions. None rescues data that were never collected without specifying how missing outcomes relate to observed and unobserved histories. Sensitivity analyses

should include departures that are scientifically plausible, not merely convenient.

Informative retention can be route-specific. A burdensome task may selectively retain participants whose state is easier to measure, while a wearable may fail more often during movement or clinical decline. The observation process can then create apparent stability or convergence. Device and participation processes should be modeled separately.

Participant return and support are part of ethical retention. Reminder intensity, compensation, transportation, and caregiver involvement can alter who remains. Those supports should be offered for fairness and feasibility, not optimized to pressure continued participation.

A result that disappears under plausible attrition models is not robust mechanism evidence. It may still identify a vulnerability in the study design. The record should preserve that lesson and change the next protocol rather than blame missing participants.

The analysis should separate adherence to the assigned action from adherence to the measurement schedule. A participant may receive the intervention exactly as planned while declining a wearable, skipping prompts, or ending follow-up; the converse can also occur. Combining these processes in one compliance variable obscures whether the estimated route failed, the delivered exposure changed, or the observation process selected what remained visible. The protocol therefore preserves separate time-indexed histories for treatment, support, measurement

availability, and withdrawal (Robins et al., 2000; Hernán & Robins, 2020).

A single completion percentage can conceal treatment-version drift. Two participants may both be called adherent even though one receives the intended sequence and another receives delayed, shortened, interrupted, or caregiver-mediated delivery. Mechanism inference therefore records version, timing, intensity, context, and deviation policy for each delivered action. When that history cannot be reconstructed, the defensible target may be the effect of assignment under real implementation rather than a precisely delivered route (Murphy, 2003; Murphy, 2005).

19.5.4 — Principal Strata and Post-Assignment Selection

Principal stratification organizes causal questions by joint potential values of a post-assignment event, such as survival, adherence, or receipt. It can clarify what comparison is being imagined when an outcome is undefined or observed only after selection. The strata are latent constructs, not observed types of people (Frangakis & Rubin, 2002).

Post-assignment selection is common near endpoints. Comparing only participants who remain conscious, alive, enrolled, or reportable can condition on consequences of exposure and prognosis. The selected groups may differ even when assignment was randomized. Truncation by death requires an estimand that acknowledges undefined outcomes rather than treating death as ordinary missingness.

Principal-stratum effects generally require strong assumptions or bounds because membership cannot be observed for every

person. Monotonicity, exclusion restrictions, and parametric models should be justified in the substantive context and varied in sensitivity analysis. A point estimate should not conceal dependence on an untestable stratum model.

A concrete route study might ask about a safe intervention among participants who would remain observable under either assignment. That target differs from the effect among observed survivors and from the population intention-to-treat effect. Each has a different scientific and ethical interpretation.

Strata must never become eligibility or worth categories. A person who would not remain observable under a burdensome protocol is not less valid data. The design should reduce burden and improve access before it narrows the population for analytic convenience.

When a principal stratum is not identified, the analysis reports bounds, scenario results, or No adequate estimate. A route does not gain support from a subgroup defined by the very post-assignment event it is meant to explain.

Because principal strata are defined by counterfactual joint events, they should enter the analysis as assumptions, bounds, or sensitivity objects rather than as observed labels. Investigators can vary monotonicity, exclusion, and dependence assumptions and report the range of stratum effects compatible with the data. A narrow interval is earned by information and design, not by choosing the most favorable cross-world model. When bounds remain wide, the result is scientific uncertainty rather than a failed

participant or an inconvenient subgroup (Manski, 2003; Imbens & Manski, 2004).

Selection diagnostics also test whether the mechanism claim changes under alternative targets. An intention-to-treat effect, an effect among always-observable participants, a survivor-average contrast, and a policy effect under supported follow-up are different estimands. When only one target produces a favorable result, the analysis justifies why that target is scientifically primary rather than treating the others as secondary noise. This comparison prevents post-assignment conditioning from converting a route hypothesis into a result that exists only inside a selected analytic population (Frangakis & Rubin, 2002; Hernán & Robins, 2020).

19.5.5 — Repeated Assignment and Adaptive Regimes

Repeated assignment creates a sequence of treatment decisions rather than one baseline fork. Dynamic regimes map observed history to the next action, and adaptive designs may update randomization probabilities while the study proceeds. The estimand must distinguish the regime being evaluated from the learning algorithm used to collect data (Robins et al., 2000; Hernán & Robins, 2020).

Micro-randomized trials can test proximal effects of time-varying components, while sequential multiple-assignment designs can compare adaptive strategies. These designs are useful for candidate mechanisms because they create several safe divergence points. They also increase the burden of modeling availability, carryover, fatigue, and time-varying confounding.

Treatment versions multiply across stages. A reminder after sleep loss is not the same exposure as the same reminder after rest; an information cue may operate differently after prior success or failure. The regime definition should include timing, state, and delivery context rather than treating repeated actions as interchangeable.

Adaptive randomization can create informative probabilities. Models must use the known assignment mechanism and preserve chronological feature construction. Reusing later outcomes to tune earlier decision rules turns an adaptive design into retrospective optimization.

A mechanism tournament can predeclare several regimes that different theories favor. Equal information and equal burden are essential. If one route receives a tailored policy while rivals are tested under generic actions, the comparison confounds mechanism with design advantage.

The study stops or narrows when support becomes weak, burden rises, or no regime creates discriminating predictions. Prespecified adaptation rules keep subsequent decisions interpretable.

19.5.6 — Audit Trails for Selection Processes

Selection processes need an audit trail as detailed as the outcome analysis. The trail records eligibility rules, invitation, contact, consent, assignment, delivery, adherence, withdrawal, device status, endpoint events, observation, exclusion, and analysis inclusion. Each transition has a time, reason, actor, and version where legally and ethically appropriate.

The audit is not a dossier. It collects only information needed to understand denominator flow and model the declared processes. Bystander details, private messages, and unrelated clinical or institutional records remain outside the research object unless separately authorized and necessary.

A flow diagram is machine-checkable against analysis denominators. Counts in the report, executable code, and governed repository agree. Discrepancies are incidents to resolve, not cosmetic differences to smooth before publication.

Selection simulations can test how plausible departures alter the mechanism verdict. Analysts can vary dropout dependence, delivery failure, unmeasured commitment, and endpoint observation. The result should show which assumptions change the conclusion and which merely widen uncertainty.

The trail also supports correction. If a device version or site procedure later proves biased, affected paths can be located and reanalyzed. Public summaries and model cards should point to the current analysis version rather than preserving a superseded favorable result.

A clean selection record protects both inference and dignity. It prevents absent participants from being silently treated as neutral, failed, or irrelevant, and it prevents a route from winning because the study forgot who could no longer be seen.

The audit should retain assignment probability, delivered treatment, adherence, reasons for discontinuation, competing events, observation availability, and every rule that changes eligibility. These fields permit inverse-probability, g-method,

principal-stratum, or partial-identification analyses according to the estimand rather than one generic missing-data correction.

Selection models should be challenged by alternative assumptions. When conclusions depend on unverifiable retention or censoring models, the report should show the corresponding identified set or sensitivity curve. A selected sample cannot be made representative by confidence alone.

Treatment versions should be preserved even when the main contrast is binary. Dose, duration, delivery route, clinician discretion, device firmware, and contextual support can modify both the mechanism and the probability of remaining observed. Version-blind analysis can mistake implementation heterogeneity for causal instability.

Informative adherence deserves its own estimand. Participants may stop because the treatment is burdensome, ineffective, harmful, or no longer needed. Per-protocol language that treats all discontinuation as equivalent can reverse the interpretation and hide the mechanism most relevant to care.

Negative-control retention outcomes can reveal residual selection. Events that should not respond to the mechanism but share the observation or administrative process can expose whether apparent route effects follow documentation intensity, site practice, or participant availability instead.

Sensitivity analyses should be linked to concrete departures, not arbitrary parameter sweeps. The record should state how much unmeasured retention bias, informative censoring, or treatment-

version imbalance would be required to change the verdict and whether that departure is plausible in the study setting.

The selection pipeline should be tested in simulation as a generative object. Known assignment, adherence, censoring, observation, and competing-event processes are passed through the same extraction and analysis code, and the recovered denominators are compared with the truth. This audit can reveal off-by-one eligibility rules, time-ordering errors, duplicated records, and accidental conditioning that ordinary outcome-model diagnostics miss. A pipeline that cannot recover its own simulated population is not ready to adjudicate a causal route.

Institutional actions can be mechanisms as well as confounders. Scheduling, translation, transportation, respite, record correction, and appeal access may alter both the opportunity set and the observation process. Their causal role should be modeled rather than absorbed into a person-level state.

Interference is another live route. One participant's action can change another person's information, support, workload, or observation probability. When clusters, families, staff, or institutions share conditions, no-interference assumptions should be defended or replaced by a design that represents spillovers (Hudgens & Halloran, 2008).

Apply the alternative-selection and sensitivity rules above. If plausible assumptions change the sign or scientific verdict, report the mechanism as selection-sensitive rather than confirmed.

The audit should distinguish withdrawal from nonadherence. A participant may refuse, pause, or leave for reasons that are

scientifically informative but never owed to the study. Their decision remains an exercise of autonomy, while the analysis records how the resulting observation process limits the causal claim.

Randomization integrity belongs in the same trail. The record preserves the allocation algorithm, seed or randomization service, block and stratification variables, concealment procedure, assignment probability, release time, and any authorized override. Investigators should be able to distinguish a genuine random draw from a late operational substitution. Without that distinction, an intention-to-treat label may conceal a compromised assignment process, and no subsequent statistical adjustment can recreate the lost randomization (Hernán & Robins, 2020).

Adaptive assignment requires replayable probabilities rather than a final treatment label. At each decision time, the archive stores the information available to the policy, the versioned policy function, the probability assigned to each action, and the action actually delivered. A simulation replay should reproduce the logged probabilities from the frozen code. Failure to do so is a provenance defect that blocks policy-effect and mechanism claims because the denominator no longer identifies the intervention regime that generated it (Murphy, 2005; Liao et al., 2016).

Human overrides are coded by function. A safety override, participant preference, clinician judgment, equipment failure, and scheduling constraint have different causal meanings even when all change the delivered action. The primary analysis can preserve assignment while secondary analyses represent receipt and

version, but these pathways are not collapsed into one deviation variable. Repeated overrides concentrated in one state or site may reveal that the nominal intervention was not feasible where its mechanism was most strongly tested.

Auditability does not require indefinite identity retention. The research system can separate direct identifiers from event history, use role-limited linkage keys, preserve cryptographic or version provenance for code and records, and delete unnecessary raw material on the declared schedule. What must remain is enough lineage to reconstruct denominators, treatment versions, exclusions, and corrections without turning the selection trail into a permanent dossier. Reproducibility and privacy are joint design constraints, not reasons to abandon either one.

19.6 — Experimental Forks and Mechanism Tournaments

An experimental fork gives candidate mechanisms a shared starting state and then introduces a low-burden difference they predict differently. The fork can vary information, controllability, timing, or action support without inducing distress or withholding care.

A mechanism tournament specifies the altered kernel for each model. A route model may alter continuation probability; a reinforcement-learning model alters value updating; an expected-control model alters effort allocation; a scarcity model alters opportunity cost. The predictions must be mutually discriminating rather than verbal redescriptions.

Common-rule design requires the same data, preprocessing, hyperparameter budget, stopping rule, observation model, and evaluation metric. Flexible models may need complexity penalties or nested validation so extra capacity is not confused with mechanism support.

Generator-recovery simulations precede human interpretation. Synthetic data are generated from every candidate and from challenge models absent from the bank. The pipeline must recover the correct family and return none-of-the-models when appropriate.

Negative controls target cue leakage, selection, observation garbling, and generic time effects. A route-specific result should disappear under a control that removes the proposed information

while preserving unrelated structure. A study designed only to produce a favorable narrative is not a tournament.

Sequential designs from Section 19.5.5 can improve tournament efficiency. Here the added requirement is prospective tournament control: adaptation rules, stopping boundaries, arm modifications, model updates, and error spending are fixed before triggering data are inspected (Murphy, 2005; Collins et al., 2007; Klasnja et al., 2015).

19.6.1 — Forks That Separate Candidate Routes

An experimental fork is a safe intervention or naturally occurring change at which candidate mechanisms predict different continuations from the same prefix. The fork should alter one declared locus while holding information, burden, and opportunity as comparable as possible.

A useful fork is chosen for separability, not drama. Information timing, benign option availability, correction of a record, or an independently indicated treatment transition may create divergence without inducing distress. Harmful provocation and care withholding are outside the design space.

The altered kernel must be named. Saying the intervention affects the mechanism is insufficient. The protocol should state whether it changes state transition, observation, action support, endpoint hazard, mediator distribution, or selection.

Each rival receives a full prediction. A candidate route may forecast a cue-specific action shift, while scarcity predicts a general effort effect and learning predicts reward-history

dependence. The fork is informative only when these alternatives are quantified before data are seen.

Temporal and channel predictions accompany the endpoint. Which variable should move first, for how long, and under which negative control? A mechanism that differs only in an unobserved latent state cannot be tested by the fork.

Recovery simulations test the fork before enrollment. Synthetic data from each model pass through the planned cadence, missingness, and estimator. If the correct generating model cannot be recovered, the fork is redesigned.

A fork can return Route-favored, Rival-favored, Model-ambiguous, or No adequate estimate. It cannot issue a whole-theory verdict or convert participant behavior into a score of inner completion.

19.6.2 — Equal Information and Equal Scoring

A mechanism tournament is fair only when models receive the same information available at the same time. Latent-state access, future endpoint labels, richer sensors, or bespoke preprocessing count as information advantages and must be either shared or explicitly charged through a bridge.

Equal scoring uses the same held-out cases, loss functions, calibration checks, uncertainty rules, and opportunity to abstain. The candidate cannot receive a bespoke favorable metric while rivals face generic prediction error. Several prospectively specified scores are appropriate when each answers a distinct component question and the decision tree states how their results combine (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Equal tuning is part of the comparison. Hyperparameter search, prior elicitation, feature selection, and early stopping should receive comparable budgets. A weakly tuned rival is a straw model even when the code is technically present.

Complexity should be treated transparently. Predictive scores, explanatory parsimony, causal identification, and route transport are different criteria. A more complex candidate may win one and lose another; the verdict should say which.

The observation pipeline is common. Missingness, smoothing, artifact rejection, endpoint alignment, and derived features are applied identically unless a model-specific difference is the object under test. Otherwise preprocessing can manufacture the winner.

Open-set tests protect against forced choice. Data from omitted mechanisms should be included in simulation benchmarks to see whether the tournament becomes falsely confident. None of the models is a legitimate outcome.

Equal treatment makes the model comparison informative. If ordinary control, learning, or selection performs as well with the same evidence and flexibility, the candidate route narrows. The scientific program gains a stronger explanation and a clearer next question.

19.6.3 — Negative Controls and Sham Mechanisms

Negative controls and sham mechanisms ask whether the apparent divergence depends on the route-specific component. A matched cue, irrelevant option change, or simulated controller without the completion state should fail where the candidate predicts specificity. If it does not, the mechanism narrows.

Option-set manipulation changes what can be done while keeping the person safe. A task may add an easy route, remove a redundant route, make one route require a benign effort, or reopen a closed route through a support action. The goal is to see whether behavior tracks feasible repair structure rather than merely reward amount (Klasnja et al., 2015; Nahum-Shani et al., 2018).

The design belongs to this chapter because compensability lives in the option set. An intervention that expands the available option set may preserve future repair opportunities. If the option set narrows, residual cost may become more sensitive to delay. The model must estimate that relation without pretending options are inner properties of the participant.

A real-world observational version might compare administrative systems before and after an appeal route is simplified. The outcome could be timing, report burden, sleep disruption, service use, or dropout. Such evidence would need strong controls for staffing, population change, outreach, and policy drift. It would not be evidence by proximity to fairness language.

The manipulation can also reveal noncompensability. Adding a pleasant option may not reduce avoidance of a route tied to violation or mistrust. A scalar utility model might expect substitution. A component model with noncompensable state may predict little effect. That contrast is scientifically valuable.

Action support includes burden and dignity. An option that technically exists but requires humiliating disclosure, inaccessible paperwork, unaffordable travel, unsafe timing, or excessive effort is not equivalent to a low-burden alternative.

Coding friction, access, and dignity cost makes the action set empirically real and prevents structural barriers from being misread as preference or motivation.

Option-set experiments use small, reversible, transparent choices. Observational studies access operational records through a separate, documented research-use path.

A negative-control outcome shares the observation schedule, expectancy, and common causes while remaining outside the candidate pathway. A negative-control exposure shares relevant selection or measurement structure without carrying the route-specific intervention. A signal on either control points toward confounding, reactivity, or pipeline leakage. A null control is informative only when its causal relation and sensitivity were justified before outcome inspection (Didelez, 2008; Pearl, 2009).

Sham adequacy requires more than matching appearance. Participants and staff may infer assignment from side effects, task difficulty, device behavior, or the timing of support. The study therefore measures expectations, perceived assignment, and differential burden, then tests whether these variables predict the claimed mechanism signature. A sham that does not preserve the relevant experience cannot distinguish route specificity from expectancy or procedural contrast, and an invasive or humiliating sham is not justified merely because it would improve blinding (International Council for Harmonisation, 2026; World Medical Association, 2024; Lakens, 2017).

19.6.4 — Transported Predictions

A mechanism prediction is stronger when it transports beyond the task on which it was fitted. Transport may involve a new site, action menu, device, population, endpoint branch, or observation channel. The invariant component and the component allowed to recalibrate must be declared in advance (Hernán & Robins, 2020).

Transport is not simple replication. A new setting can change treatment versions, support, meaning, and measurement. The protocol should map these differences and state whether it expects parameter invariance, qualitative ordering, or only a bounded effect.

Cross-equation transport is particularly informative. A completion potential estimated from passive transitions can be tested on held-out actions; a causal locus estimated from behavior can be tested on physiology; a lag relation estimated in one context can be tested after a safe perturbation.

Ordinary rivals transport too. A model that requires bespoke recalibration at every site while a simpler rival preserves its prediction has weaker mechanism evidence. Equal transport budgets prevent local flexibility from masquerading as generality.

Failure can localize the missing bridge. Device transport failure points to observation, action transport failure to policy, and population failure to state or effect modification. The route should narrow at that module rather than be globally defended or globally discarded.

A transported public claim should retain the destination population and uncertainty. Success in one cohort or task does not

authorize a universal mechanism graphic. The artifact names where the prediction was actually tested.

The strongest transport result is a prospectively specified qualitative and quantitative pattern that survives a changed observation route without refitting the core mechanism. Even then it supports a component route, not complete-stream PTVC.

Transport claims should identify which causal relations are expected to remain invariant and which population or context variables may modify them. A destination sample with different clinical severity, action constraints, or support can alter both the intervention and the pathway through which it acts. Transport diagrams or explicit adjustment sets make those assumptions inspectable, while sensitivity analysis shows how much unmeasured effect modification would be needed to reverse the conclusion (Pearl, 2009; Hernán & Robins, 2020).

The measurement bridge must travel with the mechanism or be independently requalified. A stable latent relation cannot be inferred from identical device labels when firmware, placement, language, sampling cadence, or preprocessing differs. The protocol freezes the core causal prediction, audits invariance of the observation model, and permits only declared destination-specific recalibration. If the bridge changes enough to absorb the predicted effect, the study has not transported a mechanism; it has created a new measurement system (Bollen, 1989; Peters et al., 2016).

19.6.5 — Adversarial Collaboration and Protocol-Fixed Win Conditions

Adversarial collaboration turns disagreement into a shared design before held-out outcomes are known. Teams fix win conditions, common data access, model budgets, scoring rules, amendment procedures, and public verdict language. Apply §6.7's prospective-specification terminology to the resulting record.

Win conditions are component specific. A route may outperform on earliest divergence, a rival on calibration, and neither on transport. The protocol-fixed decision tree states which combinations support a component, yield Model-ambiguous, trigger Open-set inadequate, or block stronger language. No composite score may average away an adverse prerequisite or target result.

Each team should nominate the strongest version of the opposing model. Parameter ranges, observation bridges, and reasonable auxiliary structure are agreed before scoring. A rival is not required to remain artificially simple while the candidate receives every rescue term.

Model recovery is the first contest. Simulated datasets from candidate, rivals, and unknown mechanisms pass through the full pipeline. High confusion rates or false confidence trigger redesign before human data are interpreted.

Negative controls and falsification outcomes are specified before outcome inspection beside favorable outcomes. The collaboration should say what would make each team abandon a mechanism,

narrow the reference class, or conclude that the design cannot discriminate.

Dependence and multiplicity need a common plan. Repeated within-person observations, multiple lags, many features, and several forks can create nominal evidence without independent information. Hierarchical or sequential rules should be declared before the winner is known.

Public communication is part of the collaboration. Detachable figures, tables, and summaries use the shared result language, including Rival-favored and Open-set inadequate, and preserve the tested component and reference class. A collaboration that becomes adversarial only in its technical appendix has not protected the public surface.

Null and ambiguous results remain publishable. The agreement includes code, simulations, provenance, and failure logs so that an unexciting outcome contributes to later designs rather than disappearing.

The collaboration succeeds when the losing model is allowed to lose cleanly and the winning model is not promoted beyond the tested component. Intellectual conflict becomes a method for reducing discretionary rescue.

Prediction files, preprocessing code, and scoring functions are locked before held-out outcomes are released. An implementation team independent of the favored theory can execute the common pipeline, while a separate adjudication group applies the protocol-fixed rules. Any post-lock deviation receives a dated amendment, a reason, and exploratory status unless all adversarial teams

accept it before unblinding (Gneiting & Raftery, 2007; Vehtari et al., 2017).

The collaboration preserves a challenge ledger containing unresolved objections, model failures, code discrepancies, exclusions, repairs, and minority interpretations. A public verdict follows only after the simulation, recovery, and scoring records can be inspected independently. This protects the candidate from an implementation defect and the rivals from discretionary rescue, while leaving disagreement visible instead of manufacturing consensus.

19.6.6 — Ethical Stop Rules and Null Results

Ethical stop rules are triggered by participant burden, distress, privacy failure, clinical conflict, device risk, or loss of valid consent, not by whether stopping helps one model. They are written before enrollment, tied to critical participant-protection and data-quality thresholds, and remain independent of the scientific winner (International Council for Harmonisation, 2026).

Statistical or scientific futility can stop collection without declaring a theory verdict. Model-recovery failure, inadequate movement of the intended locus, observation quality below the protocol-fixed threshold, or exhausted conditional power may make additional burden uninformative. The stop record states which object failed and which conclusions remain available.

A null result carries a sensitivity statement. The report shows the prospectively specified effect sizes and lag patterns the design could detect, which channels remained usable, whether the

manipulation altered the intended variable, and which rival patterns remained compatible. This separates genuine invariance from weak intervention, poor timing, and measurement failure.

An adverse event can invalidate interpretation even when no direct harm is caused by the protocol. A medication change, acute illness, care transition, or institutional disruption may alter state and observation. The event is recorded and routed through the decision tree rather than adjusted away automatically.

Stopping never changes care. Clinical decisions, analgesia, rescue, and support continue under their own authority. A study may stop observing while the person's life and treatment proceed beyond the research window.

Participant withdrawal is a valid outcome and not evidence about the mechanism. Data use after withdrawal follows consent, law, and the protocol. The model cannot interpret refusal or missingness as latent completion or burden.

Null and stopped studies should be released with enough detail to prevent repetition. The intervention, burden, manipulation check, model-recovery result, and reason for stopping belong in the public scientific record, subject to privacy.

The ethical outcome is not a caveat added after the analysis. It determines which intervention world is admissible. A mechanism that can be separated only through harmful or coercive designs remains outside empirical authorization.

Mechanism tournaments prospectively specify the adjudication body, its independence from model developers, and its conflicts of interest. Safety, futility, model inadequacy, protocol

continuation, and public reporting each use protocol-fixed evidence thresholds. Governance becomes part of reproducibility because discretionary authority can otherwise alter the scientific branch after results are known.

Split-sample or external locking reduces rescue. Candidate generators, preprocessing, tuning ranges, and scoring rules are frozen before the confirmatory comparison. Exploratory discoveries can improve the next tournament, but they do not alter the verdict of the current one.

Win and loss conditions should be model specific. One candidate may lose through failed calibration, another through causal invariance, another through ethical infeasibility, and another because an ordinary rival predicts equally well with fewer assumptions. A single leaderboard cannot preserve those distinctions.

Tournament archives should preserve losing predictions, code, and recovery failures. Future models must face the same failed cases rather than being evaluated only on a refreshed dataset. Failure memory is what turns sequential model building into cumulative science.

19.7 — Shadow Prices as One Analytic Example

A shadow price is the derivative of an optimized value with respect to a constraint under specified regularity and duality conditions. In a constrained policy problem, a multiplier may summarize the local marginal model value of relaxing a terminal-band or resource constraint. It is not a universal exchange rate.

A constrained value can be written schematically as

$$V_K(t, x, \ell) = \sup_{\pi \in \Pi_{\text{adm}}} \mathbb{E}_{\pi} \left[\int_t^T r_s \, ds \right]$$

subject to

$$\mathbb{P}_{\pi}(|L(T^-)| \leq K \mid X_t = x, L_t = \ell) \geq 1 - \alpha.$$

An empty feasible set returns No adequate estimate. Interpreting a multiplier also requires an appropriate constraint qualification, a relevant dual relation, and an attained optimum. In nonconvex or misspecified problems, a fitted multiplier may remain a numerical artifact rather than the claimed marginal sensitivity.

The multiplier inherits its units from the objective and constraint. Changing either changes the number, so the display identifies both quantities and the local sensitivity being estimated.

This section develops a finite-horizon analysis of compensability, barriers, viability, and shadow prices. It is one analytic example inside the causal-locus chapter and remains subject to ordinary scarcity, fatigue, control, and opportunity-cost rivals.

A shadow-price model earns scientific value only through held-out prediction and causal discrimination. If a standard scarcity or

expected-control model explains the same behavior better, the ordinary rival wins.

Finite-Horizon and Shadow-Price Analysis

Within this finite-horizon analysis, a multiplier may be nonunique, fail to exist, or cease to approximate finite changes when the active set changes. Its permitted sign follows the declared constraint and Lagrangian convention: equality multipliers are unrestricted, whereas a fixed inequality convention imposes dual sign restrictions. These limits confine its meaning to the optimization problem that defines it (Boyd & Vandenberghe, 2004).

For this book, the analytic example is deliberately narrow. A multiplier may indicate how a modeled performance criterion changes when an opportunity, resource, risk, or feasibility constraint is marginally relaxed. It does not price suffering, convert pain into future benefit, compare persons, or authorize sacrifice. If the objective omits dignity, rights, care, distribution, or uncertainty, the multiplier does not mysteriously restore them. The model must either encode the relevant boundary or admit that it lies outside the calculation. The useful lesson is methodological: local sensitivity can reveal which constraint is binding, while human value remains categorically beyond the multiplier (Dworkin, 1981a; Dworkin, 1981b; Boyd & Vandenberghe, 2004).

19.7.1 — Marginal Constraint Pressure

Marginal constraint pressure asks how the optimized value of a declared model changes when one constraint is relaxed by a small

amount. The derivative is local to the current state, active set, objective, policy class, and units. It can identify which modeled restriction is binding, but it does not reveal a universal force or an amount of experience that must be offset (Boyd & Vandenberghe, 2004; Fleming & Soner, 2006).

The object is meaningful only near a regular optimum. If the feasible set is empty, the optimum is unattained, the active set changes discontinuously, or strong duality fails, the multiplier may be absent, nonunique, or misleading. Those conditions should appear beside the estimate rather than after its interpretation.

Constraint pressure can vary across time. A transport option may have low marginal value while alternatives remain open and high value near a deadline. A safety constraint may be inactive in one state and decisive in another. The model should therefore report the state and horizon rather than presenting one stable shadow price.

A practical example is a low-burden research task with a declared fatigue limit. The multiplier can describe how much the task objective would improve if the allowed effort increased slightly. It cannot say that participant fatigue is worth that improvement. Ethical admissibility can keep the limit fixed regardless of the mathematical derivative.

Ordinary scarcity and expected-control models are the primary rivals. If a standard resource model predicts the same allocation and intervention response, no fairness-specific interpretation is needed. Held-out changes in constraint availability should test

whether the multiplier has predictive value beyond generic opportunity cost.

The analytic example should be retired when the local derivative is unstable, depends on unmeasured state, or fails to predict responses to genuine constraint changes. A fitted number without those conditions is an annotation on a model, not a discovered causal object.

19.7.2 — Units, Dual Variables, and Local Interpretation

A dual variable inherits units from the objective and the constraint. If the objective is expected task loss and the constraint is minutes of available time, the multiplier has units of loss per minute. Change the objective, time scale, normalization, or constraint definition and the number changes. Unit analysis is therefore the first safeguard against treating it as a natural exchange rate (Boyd & Vandenberghe, 2004).

Dual interpretation also depends on sign convention. A positive multiplier under one formulation may appear negative when the constraint is written in the opposite direction. Every display states whether relaxation increases or decreases the bound and how the derivative is oriented. A symbol without that sentence is not accessible technical communication.

Local interpretation means the number approximates a small change around the fitted solution. Large changes can activate new constraints, alter policy, or move the system to a different region. Applying a local multiplier across a whole life, a clinical trajectory, or a public policy requires a separately justified model for those larger changes.

Uncertainty belongs in the unit table. Parameter error, state uncertainty, numerical approximation, and nonuniqueness can produce a range of multipliers or no stable estimate. Reporting one decimal place from an unstable optimization creates false precision.

A useful display pairs the multiplier with its owner, objective, constraint, state, horizon, units, uncertainty, and withdrawal condition. This packet makes the analytic role recoverable and prevents a detached screenshot from looking like a score.

No dual variable is comparable across persons unless the objectives, constraints, scales, and identification bridges are demonstrably commensurable, a condition this chapter does not assume. The object remains inside its local model.

19.7.3 — State Dependence and Nonlinearity

Shadow-price behavior is generally state dependent and nonlinear. The derivative can be zero while a constraint is slack, rise sharply near an active boundary, jump when the optimal policy changes, or become set-valued in a nonsmooth problem. Treating it as a constant erases the very geometry the multiplier is meant to summarize (Fleming & Soner, 2006; Aubin, 1991).

Nonconvexity is particularly dangerous. Local multipliers may describe one optimum while another policy has a different active set and better value. Duality gaps can break the usual sensitivity interpretation. A model should report whether convexity or a constraint qualification is assumed, approximated, or absent.

Time dependence creates another layer. A resource can become more constraining as an opportunity closes, yet a later innovation

can open a new route and lower pressure. The pattern is not a universal inverse-horizon law. It is a model-specific response to changing state, action support, and forecast.

A concrete simulation can vary one constraint across a grid and plot the optimized value rather than only the derivative. Kinks, plateaus, and regime changes reveal where the local approximation fails. The graph should include uncertainty and the ordinary control rival.

State dependence also limits causal interpretation. An observed association between a proxy for pressure and behavior may reflect the state that made the constraint active, not the multiplier itself. Interventions that change the bound while holding the relevant information structure fixed offer stronger evidence.

When nonlinearities make the multiplier unstable or nonunique, finite contrasts, identified sets, or qualitative binding-status statements replace the point value. Precision is not lost by refusing a number the model cannot support.

19.7.4 — Observable Proxies and Identification

A shadow price is usually latent because the objective, state, and active constraints are partly modeled. Observable proxies may include delay sensitivity, choice changes when a resource expands, or shifts in policy near a boundary. None is the multiplier itself. Identification requires a structural link from the proxy to the optimization problem and exclusion of rival pathways (Boyd & Vandenberghe, 2004; Hernán & Robins, 2020).

Choice data can identify preference or policy parameters only under assumptions about beliefs, available actions, noise, and

objective form. A person who declines an option may face hidden cost, inaccessibility, fear, or a different goal. Interpreting the choice as high constraint pressure without those variables is inverse storytelling.

Interventions on the constraint are more informative. Extending a deadline, reducing effort, adding transportation, or clarifying information can change the feasible set. The predicted local response should be compared with scarcity, fatigue, learning, and social-support models under the same information and burden.

Proxy validity can be tested through cross-equation transport. A multiplier estimated from one task should predict responses in another module governed by the same declared constraint. Refitting a new proxy relationship for every task removes the overidentification that makes the object scientifically useful.

Measurement error and selection can reverse the apparent relation. Participants closest to a constraint may drop out or receive extra support, while devices may fail under high load. Joint models and sensitivity analysis should carry those processes into the estimate.

If the proxy cannot distinguish the multiplier from ordinary urgency or opportunity cost, the analytic example remains illustrative. It does not become a person-level state, a diagnostic, or evidence that a completion mechanism is operating.

19.7.5 — No Monetary, Moral, or Personal Price

“Shadow price” remains a model-local sensitivity term. Any monetary interpretation depends on the modeled units, and no

multiplier prices pain, persons, duties, or care (Boyd & Vandenberghe, 2004).

No multiplier measures how much suffering is worth, how much joy is owed, or whose life should receive priority. Pain, dignity, consent, and rights are not exchangeable components in the objective. They either remain outside the model or enter as nonnegotiable admissibility constraints.

The object belongs to the constraint, not to the person. The correct sentence is that a declared model has high marginal sensitivity to a particular bound in a particular state. Saying that a person has a high shadow price turns an analytic derivative into a human attribute and is prohibited.

Local sensitivity cannot justify large or irreversible changes. Active sets can switch, nonlinearities can dominate, and the objective may omit consequences that matter most. A multiplier is not a recommendation even inside the modeled domain.

Clinical language requires special care. A shorter horizon does not make care inefficient, and a model cannot use marginal information value to reduce desired comfort, communication, or support. Research optimization remains subordinate to care.

Institutional misuse is equally serious. A dashboard must not rank cases, appeals, or participants by a shadow-price proxy. Such a system would import a narrow objective into public authority and convert uncertainty into eligibility or worth.

General-reader prose can prefer constraint sensitivity or marginal model value when the technical term adds more risk than clarity. Where shadow price is retained, units, owner, objective, active

constraint, uncertainty, and prohibited interpretation appear nearby.

This boundary permits precise analysis of dual variables without converting a model multiplier into moral payment, a personal price, or permission for sacrifice. Conditional net remainder accounting is a separate consequence of exact PTVC; a multiplier does not by itself establish that relation or a way to control future happiness.

19.7.6 — Retirement and Replacement by Better Causal Objects

A shadow-price object retires when the optimization problem is not identified, the feasible set is empty, strong duality or a relevant constraint qualification fails, the multiplier is unstable, or held-out changes in the bound do not produce the predicted response. A number returned by software does not survive these mathematical or empirical failures (Boyd & Vandenberghe, 2004).

Retirement may be semantic. If shadow price repeatedly invites monetary or moral interpretation, constraint sensitivity, finite contrast, active-set status, or feasibility margin can preserve the useful mathematics without importing an exchange metaphor.

Retirement may also be causal. A multiplier can summarize a fitted policy without being the variable through which behavior changes. If interventions on the bound alter outcomes through a different pathway, the causal object should move to that pathway.

Finite contrasts often replace derivatives in nonlinear or nonsmooth systems. Rather than report one local multiplier, the

analysis can show how the optimized value and action set change across a prospectively specified range of constraints. Kinks and regime changes become visible.

When the state, objective, active constraint, or dual relation remains uncertain, carry an identified set rather than selecting one multiplier. Apply §24.2 and Table 27.5 to that set, with the report preserving the corresponding canonical evidence state (Manski, 2003; Imbens & Manski, 2004).

Ordinary causal objects may be better. Opportunity access, treatment availability, information timing, pain relief, social support, and institutional repair can be directly intervened on and interpreted without invoking a dual variable. They take priority when they answer the scientific question more clearly.

Table 19.5. Successor causal objects and shadow-price withdrawal rules.

Successor object	Permitted scientific meaning	Prohibited interpretation and withdrawal rule
Marginal constraint sensitivity	Local derivative of a declared objective with respect to one constraint	Not a price of pain or person; withdraw when duality or stability fails
Finite intervention contrast	Change in a protocol-fixed outcome under two supported constraint levels	Not a universal exchange rate; restrict to the treatment versions and horizon
Viability or feasibility set	States and actions that satisfy declared safety or completion conditions	Not a ranking of lives; return no estimate outside identified support

Successor object	Permitted scientific meaning	Prohibited interpretation and withdrawal rule
Action-support set	Actions observed or defensibly supported at a declared state	Not personal advice; exclude coercive, inaccessible, or unsafe actions
Identified bounds	Range compatible with stated assumptions and data	Do not collapse to a point when the set crosses zero or spans different regimes

Failure propagates to tables, diagrams, captions, and examples that might otherwise preserve the impression of a hidden price. Historical versions retain the reason for withdrawal, but the active scientific presentation no longer displays the retired object as a live mechanism.

What survives is the discipline of constrained reasoning: explicit objectives, units, action sets, support, feasibility, and nonnegotiable boundaries. The analytic example has succeeded if it makes those assumptions visible and leaves no residue of moral exchange.

Chapter 20 — Interoception and Brain-Body Mechanisms

Felt experience is embodied, yet physiology is not a transparent transcript. Humiliation can tighten the chest, medication can quiet pain while changing speech, fatigue can close an action that was technically available, and safety can return through breath, posture, contact, or relief. Chapter 20 studies the brain-body processes that may carry those changes while preserving the difference between a bodily signal and the experience it imperfectly witnesses.

The chapter asks which embodied mechanisms could mediate, mask, distort, mimic, or constrain route-specific dynamics. Every candidate enters through a causal layer, expected lag, person-specific baseline, perturbation signature, rival explanation, and no-inference condition. Neural regions, biomarkers, wearables, and model outputs become evidence only through that typed architecture.

The previous chapters gave mechanism a formal grammar. Queue-style state dynamics became a candidate control model rather than a hidden actor, and shadow-price analysis became local constraint sensitivity rather than moral exchange. The same discipline now reaches interoception, allostasis, somatic markers, stopping systems, pain, fatigue, medication, disability, trauma, illness, devices, and Artificial Intelligence.

Brain–body science links anatomical structures to experience through measurable dynamics. A useful mechanism connects insula, vagus, cortisol, inflammation, prefrontal control, or heart-rate variability to explicit states, timing, perturbations, and recovery. These connections make competing explanations testable.

Bodily signals are partial witnesses whose meanings depend on channel and context. A rising heart rate may reflect effort, fear, fever, posture, medication, pain, excitement, dehydration, panic, or artifact; a quiet trace may reflect calm, shutdown, sedation, neuropathy, masked expression, practiced endurance, or sensor loss. Interpretation begins by placing those alternatives inside the model.

The formal target remains felt experience, and experience often has bodily texture. Pressure in the throat before a difficult call, nausea in a remembered room, trembling after danger has passed, warmth when safety returns, and fatigue when choice becomes too costly can inform mechanism without defining the person. The chapter models that texture as one evidentiary layer among several.

Six layers organize the analysis: sensing, central representation, prediction, control action, peripheral response, and conscious report. A physiological channel may witness one layer while missing another. Report may change before the wearable, the wearable before awareness, or both after a third cause such as fever, medication, sleep loss, inflammation, movement, cultural expression, or the testing room itself.

Interoceptive inference, allostatic regulation, somatic-marker learning, and embodied action predict different temporal order, context dependence, and perturbation response across those layers. Their scientific value lies in separable predictions and recoverable causal graphs, not in the prestige of biological vocabulary.

Each mechanism earns a testable role by specifying its channel, expected lag, person-specific baseline, confound structure, perturbation, negative control, transport claim, and defeat condition. A model that can explain every result after inspection has not localized a mechanism; it has only expanded its language.

Disability, trauma, medication, chronic illness, aging, and terminal physiology are central validity populations. They reveal whether a baseline, sensor, task, or causal bridge was calibrated only to the easiest-to-measure body. Variation in movement, signaling, metabolism, communication, or expression narrows the model until the model learns to represent it.

Consider a participant in a palliative-care study whose wearable shows falling movement, altered sleep, and lower heart-rate variability after a medication change. The pattern could reflect worsening burden, pain relief, sedation, changing perfusion, device fit, posture, consciousness, or consent-related nonwear. A disciplined protocol inspects those treatment, quality, observation, and reportability changes before assigning a route interpretation.

Far from hospice, a graduate student with a history of panic may feel a tight chest before a meeting. The signal could reflect memory, prediction, threat appraisal, caffeine, sleep loss, conditioned context, or a useful early warning. Compressing that complexity into one arousal score risks turning vigilance into valence and physiology into identity. The interpretive burden belongs to the model, not the person whose body is being measured.

Equations, causal diagrams, tables, and case-like scenarios serve different purposes. A mechanism map proposes a relationship, an observation model defines how channels arise, and a design table organizes a test. Each display states its role and failure condition. Formalization adds value by making the embodied claim vulnerable to recovery and perturbation tests.

Wearables and Artificial Intelligence appear late because they expose the public surface of the same inference problem. Device streams, artifact flags, missingness patterns, and machine-learning support can improve quality control and bounded prediction. Their outputs remain inside a purpose-limited context of use with provenance, calibration, abstention, contestability, correction, and deletion routes.

Typed inference governs the chapter. Bodily data acquire meaning through a declared construct, mechanism, context, baseline, lag, confound model, transport claim, and decision role. A failed required bridge makes the evidence Unusable for the affected estimand; identification limits under valid gates instead yield a bounded conclusion or No adequate estimate. The same record may still support descriptive care, device-quality review, or future method development without becoming a phenomenal-mechanism claim.

Interoception offers a rich, multiscale account of how bodily state and experience interact. Its person-specific pathways, measurement properties, and safely testable components make brain–body mechanisms a substantive part of the comparison.

The body is treated as a layered dynamical system, not a single biomarker channel. Cardiac, respiratory, endocrine, immune, metabolic, nociceptive, motor, sleep, and cortical processes operate on different clocks and often share causes. Every candidate names which layer carries state, which observes it, and which merely covaries.

Mechanism-specific prediction packets should include temporal order, sign, dose response, recovery, context dependence, and off-target pathways. A signal that rises after distress is less informative than a model that predicts when it rises, when it should remain stable, and how a safe perturbation changes the downstream sequence.

Human reviewers remain accountable for scientific decisions. Artificial intelligence can assist artifact detection, stream alignment, and estimation of specified features within a versioned, governed study.

20.1 — Embodied Dynamics as Candidate Mechanisms

A burden that is not spoken may still be carried. A person may stop answering a message, avoid a hallway, lose appetite, sleep lightly, tighten before a voice, or become exhausted by choices that look small from the outside. The scientific temptation is to treat these bodily traces as direct evidence. The better route is to treat them as candidate pathways that need a model.

Embodied burden means that unresolved demand may be represented partly through bodily regulation, interoceptive prediction, learned bodily associations, action readiness, and shifts in salience. The word candidate does important work. The chapter is not saying that the body stores the ledger. It is asking whether some brain-body dynamics help explain why certain demands remain active, become avoidant, initiate repair, or lose their grip.

The section builds a status discipline for mechanisms. A background mechanism is already expected from ordinary neuroscience or physiology. A candidate Law-relevant mediator may help explain a queue, horizon, or repair pattern under specific conditions. A rival mechanism may reproduce the same pattern without the Law. The chapter gains strength by letting all three statuses remain available.

A DNA-based implementation becomes scientifically useful when genetic instructions are distinguished from the changing state that carries information. The same unchanged genomic sequence can build machinery that undergoes different histories;

that sequence alone cannot distinguish those histories. Candidate changing states include chromatin, methylation, transcription, proteins, synapses, and distributed cellular activity. Slow molecular processes might maintain or tune a faster integrator. The question is which state carries the required information, how its updates are calibrated, and how that information survives component replacement.

Molecular interventions can now address that question with greater specificity. Coda et al. (2025) targeted the Arc promoter within identified mouse memory ensembles and reversibly altered contextual fear-memory expression. Kukushkin et al. (2024) found that spaced chemical stimulation produced more sustained CRE-dependent transcription than massed stimulation in SH-SY5Y neuroblastoma-derived and HEK293 cell lines, measured with a luciferase reporter. The first result supplies a selective causal tool; the second supplies an ordinary cellular mechanism that can imitate history-sensitive persistence. Neither experiment measured a signed phenomenal account.

A nominated accounting state should make the same calibrated predictions across new histories, including histories with equal signed totals but different order or gross stimulation. Reinterpreting an old event can produce a new experience, but it does not change the event's earlier experienced contribution. Likewise, impaired recall does not establish that the information needed by an accounting mechanism has been erased. Perturbations should distinguish recording, maintenance, retrieval, and expression, and should test the nominated state

against learning, stress, salience, and multiscale memory with the same observational and fitting resources.

Engineered recording offers a concrete first benchmark. DNA Typewriter records event order through sequential genome editing, while ENGRAM records signalling-related barcode changes in engineered cells (Choi et al., 2022; Chen et al., 2024). A proposed two-channel experiment would assign plus and minus labels to known laboratory inputs, then test recovery of their difference separately from total input and temporal order. The labels initially designate programmed signals. A frozen decoder should face held-out sign reversals, equal additions to both channels, altered pulse spacing, and differences among cells, with recording error and saturation included in its predictions.

Successful recovery from pooled molecular measurements must also be distinguished from recovery within an individual cell or lineage. With nonlinear writing, different distributions of cellular histories can produce the same pooled signal. The benchmark should therefore match its unit of inference to its actual readout and preserve failed or unedited records. Its immediate contribution is a calibrated test of biological recording and decoding; a natural PTVC realization additionally requires an independently supported relation to the qualifying stream's experience and the declared initial and pre-endpoint boundary conditions.

Table 20.1. Mechanism-status routing. Role: interpreting brain-body findings and identifying the next mechanism test.

Mechanism status	What it licenses	Scope of the result
Background physiology	Use established regulation science as context	Established physiological relationships in their tested settings
Candidate mediator	Test whether the pathway carries part of a proposed dynamic	Specified pathway, observation model, and tested setting
Rival mechanism	Let ordinary body regulation explain the pattern	The measured pattern under the shared model comparison
Observation channel	Use a signal as partial witness with uncertainty	Channel-specific partial inference with calibrated uncertainty
Failure route	Narrow, abstain, or retire the mechanism	The tested mechanism, measurement model, and reference class

20.1.1 — Interoceptive State and Prediction

Interoception begins with internal bodily signals, but it does not end with raw signal. Cardiac rhythm, breathing, gut state, temperature, metabolic demand, pain, inflammation, endocrine shift, posture, and effort are filtered, predicted, attended, and linked to action within distributed brain-body systems. These processes contribute to affective construction and regulation, but they do not provide a transparent reading of valence (Khalsa et al., 2018; Barrett & Simmons, 2015; Feldman et al., 2024).

The candidate state can be written as a vector rather than a single intensity. Cardiac activation, respiratory constraint, nociception, immune activation, metabolic depletion, muscular tension, and sleep pressure can move together or apart. A scalar arousal score may be convenient, but convenience is not identification. A physiology model that collapses these axes too early may lose the difference between fear, effort, fever, grief, pain, and relief.

A restrained model begins by separating the latent bodily state from the observation. The observation is what the device, report, task, or clinician sees. The interoceptive state is the modeled internal condition. The conscious feeling is a further target, not automatically equivalent to either. The Law-specific question enters only after that separation is honored.

A participant who feels dread before an ordinary phone call may be responding to interoceptive prediction rather than current danger. The body has learned that certain cues predict conflict, humiliation, debt, pain, or abandonment. The resulting bodily state may be real even when the current scene is safe. A model that calls the person irrational misses the learned prediction; a model that calls the body the ledger overreads it.

Interoceptive prediction can also be a rival. If prediction error, learned bodily expectation, and safety updating explain the timing of avoidance and repair, the Law-specific queue language may add little. That outcome is not a defect in the chapter. It is one way mechanism science protects the project from turning compatible patterns into proprietary evidence.

A concrete safeguard is methodological rather than decorative: no interoceptive signal enters a momentary valence estimate without a declared channel, baseline, timing window, confound set, and abstention rule. Strong internal sensation may be important to the person and still fail as an estimator of the target construct. The person remains authoritative about the fact that they are experiencing; the model remains responsible for saying what it can infer.

A stronger implementation would store the predicted state and the incoming signal separately. That lets an analyst ask whether distress followed bodily evidence, whether bodily evidence followed a prediction, or whether both were moved by context. Without that separation, an interoceptive account can become unfalsifiable because every mismatch is reinterpreted as hidden prediction.

20.1.2 — Allostatic Load and Regulation

Allostasis names the body's anticipatory regulation under changing demands. It is not only a return to a fixed set point. It is the work of preparing, reallocating, mobilizing, conserving, and shifting thresholds so the organism can meet what it expects. That makes allostasis scientifically relevant to the Law because horizon, reserve, fatigue, threat, sleep, pain, and repair all have bodily cost (Khalsa et al., 2018; Barrett & Simmons, 2015).

Allostatic load describes accumulated wear from repeated or chronic activation of regulatory systems. It can mediate burden, constrain future options, alter observation, and supply specific biological predictions for comparison.

Reserve may need a vector representation. Metabolic capacity, autonomic flexibility, cognitive control, inflammatory state, sleep recovery, pain tolerance, social support, and clinical stability need not collapse into one number. A scalar reserve remains useful only when its compression loss, transport, and rival performance are tested.

A caregiver who sleeps in a hospital chair for four nights may show altered glucose regulation, higher resting arousal, slowed attention, irritability, and flat affect. The visible emotion may be muted while the regulatory cost is high. If a study measures only smiling, it may miss burden. If it measures only cortisol, it may confuse burden with infection, medication, caffeine, or circadian timing.

An informative allostatic account predicts which reserve dimension changes, when it changes, how it recovers, and which observation channel reflects the change. It also identifies competing explanations and the outcomes that would support, revise, or retire the proposed link. With that specificity, a broad account of strain becomes a mechanism whose predictions can be evaluated.

Accumulated physiological cost makes early care particularly valuable. If regulatory strain narrows future repair, timely sleep, pain relief, accessible support, and reduced demand can improve the available pathways.

The most important analytic choice is whether allostatic cost is modeled as level, change, volatility, or loss of flexibility. These produce different predictions. A high level may reflect current

demand, a steep change may reflect acute transition, volatility may reflect instability, and inflexibility may reflect depleted capacity. Treating all of them as one load score erases the mechanism.

20.1.3 — Affective Salience

Not every bodily change becomes felt as important. Affective salience is the process by which some signals stand out as urgent, meaningful, aversive, inviting, threatening, or action-guiding. Salience is not valence. A signal can be salient and positive, salient and negative, salient and ambiguous, or salient because the person cannot ignore it (Sterling, 2012; McEwen, 1998; Barrett & Simmons, 2015).

The distinction matters because salience often drives behavior before a person can explain why. A chest flutter may become a warning. A familiar smell may bring grief before the memory is named. A pain flare may pull attention away from conversation. A calm pressure from a weighted blanket may help a person reenter the room. Each case changes availability of action without becoming a full measure of the life.

Affective salience can be modeled as a weighting function on observations and options. The same peripheral signal may receive high or low weight depending on prediction, context, trauma history, medication, attention, cultural interpretation, and current goals. In a queue model, salience may raise an item's priority. In a rival account, salience alone may explain reactivation without any Law-specific structure.

This prevents a common error in physiological interpretation. A high arousal signal is not automatically more negative. It may mark exercise, anticipation, effort, social exposure, pain, caffeine, fever, panic, excitement, or embarrassment. A low arousal signal is not automatically relief. It may mark shutdown, sedation, depression, dissociation, autonomic failure, or sensor dropout.

The empirical design therefore asks whether salience predicts later report, avoidance, repair action, attention capture, memory intrusion, or physiological recovery after controlling for ordinary arousal. If it does not, salience may be a label after the fact. If it does, the resulting mechanism can be tested against adaptation, predictive processing, homeostasis, and device artifact.

False salience has an asymmetric error cost. A school, clinic, workplace, or research protocol may treat visible activation as threat while reading quiet collapse as calm, shifting the correction burden onto the person least able to contest the interpretation. Error analysis should therefore report subgroup-specific false positives, false negatives, and the practical consequence of each.

A field review should also ask whether salience is helpful or harmful in the case being studied. Some salience brings timely care, as when pain or breathlessness becomes noticeable early enough to treat. Other salience traps the person in monitoring. The same mechanism can therefore be protective in one window and burdensome in another, which is why the chapter refuses one-tone interpretation.

20.1.4 — Somatic Markers and Action Readiness

Somatic markers matter here as learned patterns that can bias attention and action readiness before deliberate report. The scientific question is which bodily and contextual signals predict a specific action transition, not whether the body carries a hidden verdict.

Somatic-marker accounts propose that bodily associations can bias choice under uncertainty. A route, face, bill, room, smell, tone, or deadline may acquire a bodily mark because past experience linked it to danger, relief, humiliation, pain, reward, or repair. The mark does not need to be a sentence in consciousness to alter behavior (Khalsa et al., 2018; Barrett & Simmons, 2015).

This is relevant to the Law because unresolved burden often returns through options. A person may not think about an old medical denial until a new envelope appears. The body tightens before the content is known. The somatic marker is a candidate mechanism for why a visible event can reopen a demand that the calendar says is over.

The model must distinguish learned association from present cost. A somatic marker can be outdated, protective, excessive, accurate, culturally shaped, or miscalibrated by trauma. It can also be trained by institutions. A benefits portal that repeatedly threatens access may teach bodily vigilance. A research protocol that ignores that history may mistake learned protection for irrational noise.

Somatic markers also create a strong rival. If learned bodily associations explain reactivation, avoidance, and repair timing, then Queue and horizon language may be unnecessary in that domain. The Law candidate can keep only what the learned-association model cannot predict under held-out data and declared observations.

A model of somatic markers should include cue exposure, prior association, bodily response, attention capture, action tendency, report, and later updating. The updating step matters. If safe experience fails to weaken the marker, trauma, chronic threat, institutional recurrence, or biological sensitization may be the relevant mechanism. If safe experience weakens it, ordinary learning may explain the pattern.

The chapter keeps somatic markers morally neutral. A bodily warning is not a command, a debt, a destiny, or a message from fairness. It is an embodied trace that may help, mislead, protect, or imprison. Science serves the person by learning when the trace is informative and when it needs care, context, or correction.

The best somatic-marker tests use cue specificity. If a hospital envelope, a debt portal, and a neutral envelope produce the same response, the model may be measuring general uncertainty. If only the hospital envelope reactivates the pattern, the learned association becomes more plausible. Specificity is what keeps a vivid story from substituting for a mechanism.

20.1.5 — Embodiment Without Reduction

Embodiment without reduction is the chapter's central posture. It says that lived experience is not floating above biology, but also

that biology is not the whole of lived meaning. Hunger, nausea, pain, breath, touch, fatigue, inflammation, warmth, and movement matter because they shape what is available to the person. They do not become a public key to the person (Sterling, 2012; McEwen, 1998; Barrett & Simmons, 2015).

Reduction enters through shortcuts. A clinician sees low movement and assumes apathy. A researcher sees high skin conductance and assumes distress. A manager sees calm speech and assumes no harm. A wearable model sees sleep fragmentation and infers emotional instability. Each shortcut may contain a hint; each can become false authority when detached from context.

Biomarker profiles summarize channels, uncertainty, artifacts, medication, disability, baseline, report, and context to support research forecasts or abstention. They complement the person's account, family context, clinical judgment, and ethical review. Their permitted use is research; decisions about care and access follow the applicable clinical and rights-based standards.

Embodied explanation also requires environmental humility. A body may carry burden because the world keeps pressing it. Noise, stairs, heat, racism, debt, inaccessible transport, caregiving load, or fear of a supervisor may shape physiology. If the model treats the body as the sole site of burden, it may individualize an institutional problem.

A stronger model places external demand and bodily response in the same causal graph. It asks whether changing the environment changes the body, whether changing the body changes experience, and whether both are required. Physiology and social

conditions become jointly testable instead of competing explanations that hide one another.

The practical question is where interpretive burden lands. When the body is measured, the study asks which invisible load generated the signal, who can change the conditions, and whether the interpretation produces more care or more surveillance. Embodiment becomes humane when the causal graph points outward as well as inward.

Embodiment also protects against a false split between mental and physical care. If a person's fear is intensified by untreated pain, therapy alone is incomplete. If pain care is blocked by shame, language access, or administrative threat, medication alone is incomplete. The mechanism points to combined repair rather than to a hierarchy where one discipline claims the truth.

20.1.6 — Mechanism Status and Defeat Conditions

An embodied candidate remains live only while its declared timing, intervention response, transport, and rival separation survive. Wrong channel order, absent perturbation effects, medication or context explanations, and site-specific collapse each narrow a different component of the route. The defeat rule is attached to the tested claim, not softened by biological plausibility.

Each mechanism has a defined role in the study. Interoceptive prediction may supply background science, a candidate mediator, or a competing explanation. Allostatic load may explain confounding or partial mediation, and wearable signals may

provide observation channels (Khalsa et al., 2018; Barrett & Simmons, 2015).

Three achievements organize the evidence: biological plausibility motivates a candidate, recoverable simulation checks the analysis under known conditions, and empirical discrimination compares its predictions with serious alternatives. The chapter's displays make each mechanism's assumptions, predictions, and evidential role inspectable.

The most useful status labels are simple. Candidate mediator means the pathway may carry part of a predicted dynamic. Rival means the pathway may explain the same evidence without the Law. Observation channel means the pathway supplies noisy information about a modeled target. Confound means the pathway can move the signal without moving the target. Failure means the pathway cannot be interpreted responsibly under the present design.

A single body variable can change status across a study. Heart-rate variability may be a channel during resting autonomic regulation, a confound during movement, a treatment artifact during beta-blocker use, a missingness problem during nonwear, and a false reassurance in autonomic neuropathy. The model should carry those status changes rather than pretending the variable has one meaning.

Status labels should be written into preregistration, data dictionaries, diagrams, tables, and public summaries. A reader should see whether a signal is being used to define a construct, estimate a latent variable, test a mechanism, screen for artifact, or

trigger abstention. Clear status labels make the scientific and ethical role of each signal inspectable.

Bodily observations can strengthen or revise a mechanism, identify a need for better measurement, or establish that care should take priority. Explicit outcome rules keep the mechanism accountable to evidence and useful for care.

A brain-body mechanism should state whether each component is necessary, sufficient, mediating, moderating, or merely predictive. Those statuses require different evidence. A reproducible correlate can improve observation without identifying the process that generates the terminal signature.

Cross-level predictions are stronger than isolated associations. A candidate route should link a safe change in bodily state to a timed change in central representation, action, report, and future continuation under a declared causal graph. If the sequence fails or a rival predicts it equally well, the mechanism narrows.

Generator recovery applies here as well. Simulations and benchmark tasks should test whether the analysis can distinguish allostatic regulation, salience, pain, effort, medication, device artifact, and the candidate route after realistic lags and noise. Failure in synthetic data blocks strong interpretation of human data.

Defeat can be component specific. Interoceptive precision may predict action while the proposed stopping process fails; an autonomic signal may transport while the valence bridge does not. The verdict should attach to the component actually tested and leave the rest open or withdrawn.

20.2 — Interoception and Affective Salience

Interoception comprises signals with different routes, speeds, artifacts, and relationships to experience. Separating these channels gives the model a richer and more testable account of bodily feeling.

Channel specificity begins with distinct sensory routes. Cardiac, respiratory, visceral, immune, endocrine, nociceptive, metabolic, vestibular, proprioceptive, and thermal signals have different baselines, speeds, observability, and interpretive limits. Some are easy to record but hard to interpret; others matter directly to the person while remaining difficult to observe.

Pleasantness can change while the external stimulus remains similar. In the thermal, taste, and smell examples examined by Cabanac (1971), the hedonic response depended on the person's internal condition. He termed this state dependence alliesthesia. The finding gives the distinction between circumstance and experience a physiological basis: what an event feels like depends partly on the body receiving it. For a cumulative model, that relationship supplies a useful prediction to test across independently characterized bodily states and repeated observations of the same stimulus.

Channel-specific models can make burden more visible and care more responsive. They can relate a respiratory signal to dyspnea or panic uncertainty, an immune signal to illness burden or treatment effects, and a movement signal to the broader context needed for interpreting pain.

Equation 20.1: Observation-channel model. Role: candidate specification linking latent variables to measured channels.

$$O_{c,t} = h_c(S_t, E_t, M_t, D_t, Q_{c,t}) + \varepsilon_{c,t}$$

$$R_t = r(B_t, C_t, A_t) + \epsilon_t$$

$$Y_t^E \not\equiv O_{c,t}, \quad \text{and} \quad Y_t^E \not\equiv R_t.$$

Channel values may correlate or occasionally coincide without becoming the same construct.

In this display, $O_{c,t}$ is the observed signal from channel c , S_t is the latent bodily state, E_t is environment, M_t is medication, D_t is disability or disease context, $Q_{c,t}$ is channel quality, R_t is report, and A_t is the modeled experiential target. In the report equation, B , C , and A denote belief or central representation, context, and action, respectively, all indexed by t . The nonidentity relations are the important part. Observation and report can inform the target, but neither is the target itself.

Table 20.2. Interoceptive channel routing. Role: routing aid for interpretation and confounds.

Channel family	Common measurement routes	Main interpretation risks
Cardiac and vascular	heart rate, HRV, pulse arrival, blood pressure	movement, medication, posture, fever, fitness, device fit
Respiratory	respiration rate, depth, CO2 proxy, breath tasks	exertion, anxiety, pain, posture, lung disease
Visceral and metabolic	appetite, nausea, glucose, temperature, fatigue reports	illness, medication, sleep, diet, endocrine state
Immune and inflammatory	cytokines, CRP, symptoms, temperature	infection, disease, treatment, circadian timing

Channel family	Common measurement routes	Main interpretation risks
Nociceptive and pain	pain reports, movement, facial action, analgesic use	communication limits, masking, tolerance, disability norms
Behavioral embodiment	posture, activity, speech timing, sleep	environment, opportunity, access, caregiving demands

20.2.1 — Afferent Channels and Internal Sensing

Afferent channels carry information from the body toward central representation. They are not interchangeable. A heartbeat signal, a breath constraint, a gut sensation, a pain signal, an inflammatory cue, and a temperature shift can all become part of feeling, but they enter through different physiology and carry different ambiguity (Khalsa et al., 2018; Critchley & Garfinkel, 2017; Seth & Friston, 2016).

A channel-specific model asks which pathway is plausible for the question at hand. If the target is breath-related panic, respiratory timing may be primary. If the target is chronic fatigue, inflammatory, sleep, metabolic, autonomic, and activity channels may need joint modeling. If the target is social dread, cardiac and somatic signals may matter, but so do context, memory, appraisal, and safety.

This prevents a familiar measurement error. A model trained on one channel may appear to generalize because many bodily systems coactivate under stress. The apparent success may vanish when the same channel is tested during exercise, fever, stimulant use, sleep deprivation, pain relief, or cultural expression differences. Coactivation is not construct validity.

A human example can be ordinary. A person with asthma may show respiratory changes before a meeting, but the meaning differs if the change is airway limitation, anxiety, mask discomfort, conditioned fear, or a combination. Calling all of that burden may be too vague. Calling it mere physiology may be too dismissive. The model must do the work.

The data record should therefore identify channel family, sampling rate, device version, artifact flag, baseline window, posture, movement, medication, illness state, and context. The record should also carry why the channel was chosen. A channel without a theory is an invitation to post hoc interpretation.

The section's local rule is channel humility. A signal can be useful in the channel it actually measures and silent about channels it does not. The chapter does not need a perfect body map. It needs enough specificity that a field expert can tell what would count as success, failure, or overreach.

Channel selection should also consider burden of collection. A cardiac sensor may be easy for one participant and intrusive for another. A breath task may be harmless for some and frightening for someone with respiratory trauma. The most elegant channel is not always the fairest channel; the protocol must measure observation burden as well as signal quality.

20.2.2 — Accuracy, Sensibility, and Awareness

Interoceptive accuracy, sensibility, and awareness are related but distinct. Performance on a detection task, self-reported attention to bodily signals, and metacognitive knowledge about accuracy need not agree. A protocol should name which object is measured

rather than using interoception as one undifferentiated trait (Critchley & Garfinkel, 2017).

Measurement quality is task-specific. Newer cardiac, respiratory, and other psychophysical tasks address some weaknesses of classical heartbeat counting, yet they differ in signal strength, beliefs, attention demands, control conditions, and scoring. No single task earns the label of global interoceptive accuracy without cross-task and cross-organ evidence (Desmedt et al., 2023).

Large-scale cross-axis evidence strengthens that caution: cardiac and respiratory performance can be largely uncorrelated even within the same participants. A model that treats interoceptive ability as one stable scalar may therefore erase modality-specific skill, uncertainty, and failure (Banellis et al., 2026).

Interoceptive precision refers to how much weight the system gives an internal signal or prediction error. Attention changes that weight. A weak signal can dominate awareness when precision is high; a strong signal can fade when precision is low, competing demands are greater, medication blunts awareness, or dissociation separates sensing from report (Barrett & Simmons, 2015; Critchley & Garfinkel, 2017).

This concept helps explain why two people with similar physiology may report different experience. One person notices heartbeat changes immediately. Another notices only fatigue hours later. A trained singer may detect breath shifts that a novice misses. A trauma survivor may monitor internal cues with

exhausting vigilance. Neurodiversity, culture, illness, and history all shape precision and attention.

For the Law, precision and attention matter because unresolved demand may be carried not only by signal intensity but by signal priority. A small bodily cue can become large in the life if it repeatedly captures attention, interrupts action, or reorganizes choices. Conversely, a large physiological load may remain underreported if the person has learned not to attend to it or cannot safely express it.

The rival route is equally strong. Precision-weighted interoceptive learning may explain apparent queue reactivation without a queue state. If attention to bodily cues predicts avoidance and repair timing as well as or better than the queue model under equal information, the ordinary predictive account wins that comparison. PTVC-linked terminology receives no protection from a better supported rival.

A study can test precision by combining report confidence, heartbeat or respiratory tasks, attention manipulations that do not induce harm, and within-person changes across context. It should not assume that better interoceptive accuracy is always better for well-being. In some settings, heightened monitoring can deepen distress; in others, it can support early regulation.

A local safeguard is to respect the person's attentional cost. Asking someone to monitor the body can itself create burden. Research prompts should be brief, optional, interruptible, and reviewed for symptom amplification. An interoception study that

creates vigilance in order to measure vigilance has confused its instrument with its target.

Precision can change during a study because the study itself teaches attention. A participant asked to notice heartbeat, breath, nausea, or fatigue may become better at reporting, more worried about symptoms, or more annoyed by interruption. The protocol should distinguish measurement learning from mechanism change, especially when prompts repeat across weeks.

20.2.3 — Precision, Attention, and Prediction Error

Prediction error is the mismatch between expected bodily state and incoming bodily evidence. Interoceptive updating is the process by which the system revises belief, attention, and action. This can be helpful: a safe breath, stable rhythm, or pain relief can teach the body that a situation is less dangerous than expected (Khalsa et al., 2018; Critchley & Garfinkel, 2017; Seth & Friston, 2016).

Updating can also fail. Chronic threat, trauma, unreliable caregiving, repeated institutional denial, persistent pain, or unpredictable symptoms may keep prediction errors from resolving. The body learns that safety is temporary, ambiguous, or not safe enough to trust. A protocol that sees only the current room may miss the history that makes the current room difficult.

The Law-specific question is whether updating dynamics align with queue release, horizon pressure, reserve, or residual burden in ways that rival learning accounts cannot explain. That is a high burden. Many forms of repair are ordinary learning. A pattern

does not become Law-relevant because it feels meaningful after the fact.

A clean design would specify the prediction before the observation. For example, if safe social contact should reduce a conditioned bodily warning, the model should name the expected channel, lag, report change, and persistence window. If the signal changes but report does not, the study should not average the conflict away. It has learned something about mediation or dissociation.

Prediction-error models require negative controls. A neutral cue matched for novelty, attention, and movement should not produce the same channel-specific change if the proposed pathway is specific. If it does, the model may be detecting general arousal, task demand, or measurement artifact rather than interoceptive learning.

Updating is ethically useful because it supports repair without moral accounting. A person can learn safety, receive treatment, change environment, or regain agency without being told that a particular suffering was morally necessary or that relief should be withheld. The body's capacity to update is a reason to care earlier, not a reason to explain pain later.

Prediction-error language also requires care around authority. A researcher may be tempted to say the person's body expected danger incorrectly. In many lives, the body learned from real danger, repeated disbelief, or ongoing risk. The model may judge the present cue safe under laboratory conditions while the person's history makes caution adaptive.

20.2.4 — Valence, Arousal, Salience, and Urgency

Valence, arousal, and urgency are often correlated, but they are not the same. Valence concerns felt positive or negative tone under the project's measured target. Arousal concerns activation or readiness. Urgency concerns the felt demand for action. A painful injury, exciting performance, panic attack, deadline, fever, and reunion can all raise arousal while differing radically in valence and urgency (Barrett & Simmons, 2015; Critchley & Garfinkel, 2017).

A body signal may be strongest for arousal and weakest for valence. Electrodermal activity can mark sympathetic activation without telling whether the person is afraid, thrilled, embarrassed, exerting effort, or overheated. Heart rate may rise during pleasure, threat, pain, exercise, or medication effect. Inflammation may correlate with malaise without specifying the person's full experiential state.

Single-axis affect is therefore inadequate unless the proposed axis is named and qualified. A valence claim requires sign identification through report, context, calibration, tasks, and rivals; an urgency claim requires links to action timing, avoidance, repair initiation, or choice pressure. The model may use one dimension only for the job that dimension can support.

A realistic example is a courtroom witness whose hands shake. The shaking may mean fear, cold, medication, public exposure, trauma, anger, illness, or motor condition. A legal or research system that treats visible arousal as credibility, guilt, distress, or

burden creates a new hidden burden. The chapter's science must be able to say: this signal is too nonspecific.

The empirical route is multidimensional. Valence reports, arousal channels, task demand, action tendency, contextual threat, pain ratings, and calibration history are modeled separately before combination. A multi-axis model must improve held-out prediction, calibration, or mechanism discrimination over a one-axis baseline; otherwise its claims narrow to the simpler representation.

The public language should follow the model. Say activation when the evidence is activation. Say salience when the evidence is attention capture. Say urgency when action pressure is measured. Say valence only when the sign and target have been identified. Precision in words makes the role of each measurement clear.

A useful data dictionary stores reported valence, physiological activation, action urgency, pain intensity, fatigue, threat appraisal, and confidence separately. This preserves the distinct information each axis contributes.

20.2.5 — Individual, Developmental, and Cultural Variation

Development, culture, language, disability, training, illness, and prior threat can alter which signals are noticed and how they are described. Variation should inform calibration and subgroup analysis, not be treated as error around one ideal observer.

Interoceptive dynamics vary across persons and across states within the same person. Training, age, anxiety, trauma, autism, depression, illness, medication, chronic pain, sleep loss, fatigue,

cultural expression, disability, and neurodiversity all shape sensing, attention, reporting, and action. A population calibration that ignores this variation can become a disguised norm (Khalsa et al., 2018; Critchley & Garfinkel, 2017; Seth & Friston, 2016).

The same person may also shift across a day. Morning medication may reduce pain but blunt awareness. Afternoon fatigue may make a small demand feel enormous. Evening safety may allow the body to register what it had suppressed at work. Terminal illness may change all of these relationships through treatment, inflammation, weakness, delirium risk, and care context.

Hierarchical or person-specific modeling is required whenever intercepts, slopes, lags, or channel reliabilities plausibly vary. Partial pooling can borrow strength across persons while preserving the possibility that one person's channel is invalid for another, or invalid for the same person in a later state.

Individual variation can create false fairness if smoothed. A group average may show recovery after a procedure while a subgroup with disability, medication sensitivity, or trauma history worsens. If the model reports only the average, the statistical burden is shifted onto the people whose bodies did not follow the dominant pattern.

The data structure should store person-specific baselines, state tags, clinical context, device fit, language and culture notes, medication windows, disability accommodations, and abstention history. These are not accessories. They are part of the observation model. A signal without its interpretive conditions is a half measurement.

Common structure can be investigated across bodies that differ. Species, disability, medication, illness, age, and context become explicit dimensions of calibration and transport.

State variation should change sampling decisions. A daily prompt may be tolerable during ordinary work and impossible during a flare, grief week, clinical crisis, or medication adjustment. Governed adaptive cadence can reduce burden, but the cadence change also carries information about observation availability and must enter the missingness model.

20.2.6 — False Salience, Dissociation, and Misinterpretation

Dissociation can separate bodily signal, conscious access, report, and action. False salience can make a benign signal urgent, while reduced salience can hide clinically important change. Both patterns defeat simple signal-to-valence inference.

False salience occurs when a bodily signal becomes important to the system or to the observer without accurately tracking the intended experiential target. The signal is real; its interpretation may need revision. A panic-like heartbeat during caffeine use, a flat facial expression during pain, a trembling hand during cold, or a quiet report after sedation can all mislead if the model asks the wrong question (Barrett & Simmons, 2015; Critchley & Garfinkel, 2017).

False salience can arise inside the person, inside the observer, or inside the device. A person may overattend to benign sensations after trauma. A clinician may overread visible movement as distress. A wearable model may overflag movement artifact as arousal. A researcher may treat a statistically useful feature as

conceptually meaningful because it predicts something in one sample.

The risk is especially high when public artifacts simplify. A diagram that says body signal becomes burden can be cropped into a claim that the body reveals burden. Diagrams and captions therefore carry the uncertainty with the image. The signal is a candidate pathway, not a confession.

A false-salience audit should ask four questions: What is the target? Which signal is being used? What else can move the signal? Who is harmed if the signal is misread? The last question belongs in the science because misinterpretation changes behavior, care, consent, and trust. Recording the anticipated harm pathway also clarifies which false positives require the strictest abstention threshold.

Empirically, false salience can be tested with matched controls, context swaps, sensor comparisons, blinded human review, device-version checks, and report disagreement. Disagreement should not automatically be resolved in favor of the device or the report. It should trigger a conflict class with follow-up rules. Conflict follow-up should be prospectively limited so repeated retesting does not continue until one channel produces the preferred answer.

The analysis proceeds by abstention before authority. When salience is strong but ambiguous, the model should say ambiguous. A science that can say no estimate will be trusted more than a science that converts every signal into a story. Reason-coded ambiguity can still guide the next measurement by

showing whether valence direction, arousal source, or causal attribution is the unresolved element.

A false-salience test should predefine rivals such as threat learning, pain, medication, sleep loss, dissociation, sensory sensitivity, and social context. The candidate route should not receive residual variance merely because the causes were measured crudely. Rivals should be implemented with comparable channel access, otherwise crude nuisance models can make the candidate appear uniquely explanatory by construction.

A no-inference zone applies when arousal, salience, and valence cannot be separated with the available channels. The output can still describe observed activation and uncertainty, but it should withhold direction, hidden burden, and mechanism language. A no-inference zone should persist into downstream models so later analysis cannot silently recover the withheld interpretation from the same ambiguous signal.

20.3 — Brain–Body Control and Stopping

Control is the point at which bodily state changes the live menu of action. A person may feel pressure to stop, flee, speak, apologize, rest, seek medication, open a bill, close a laptop, breathe slowly, call for help, or avoid the cue. The scientific task is to model how body, environment, opportunity, and meaning jointly reshape that menu.

This section examines distributed brain-body control: how internal state, context, prediction, conflict, and capacity shape action initiation, inhibition, switching, and repair. Network-level, functional accounts make these relationships available for explicit timing and intervention tests.

The key object is a feedback loop. Bodily state influences belief and salience; belief and salience influence action; action changes the body and environment; the new body and environment change the next observation. A one-way physiology-to-feeling story misses the loop. A loop is harder to explain, but it is closer to the scientific problem.

Table 20.3 and Equation 20.2 turn the feedback loop into two inspectable objects: a layer map that preserves target distinctions and a dynamic system whose timing, confounds, and intervention responses can be tested.

Table 20.3. Brain-body control layers. Role: mechanism map for distributed control.

Layer	Example variables	Main question
External and social context	threat, care, task, room, demand, access	What is pressing the person now?

Layer	Example variables	Main question
Bodily state	pain, breath, fatigue, inflammation, metabolic demand	What physiological condition is modeled?
Central belief and salience	prediction, precision, attention, learned cue value	What does the system treat as important?
Control policy	stop, continue, avoid, repair, seek help, rest	What action becomes available or blocked?
Observation and report	device signal, behavior, speech, diary, clinician note	What can research responsibly see?
Update and residue	learning, recovery, sensitization, residual burden	What remains after action or care?

Equation 20.2: Dynamic feedback display. Role: candidate control-mediation model under partial observation.

$$S_{t+\Delta} = f_S(S_t, A_t, E_t, M_t, D_t) + \eta_t^S$$

$$B_{t+\Delta} = f_B(B_t, O_t, P_t, C_t) + \eta_t^B$$

$$A_t = \pi(B_t, H_t, U_t, C_t)$$

$$O_t = h(S_t, B_t, Q_t) + \varepsilon_t$$

$$R_t = \rho(B_t, C_t) + \epsilon_t$$

Here S is bodily state, B is belief or central representation, A is action, E is environment, M is medication, D is disability or disease context, P is precision, C is context, H is the filtered remaining-horizon candidate defined in §16.1.2, U is a possibly vector-valued reserve state, O is observation, R is report, and Q is signal quality. The display is a candidate structure. It is scientifically useful only if each term receives a design, a measurement route, and a failure condition.

20.3.1 — Distributed Control Networks

Brain-body control should be described as distributed. Salience, attention, interoception, motor readiness, memory, valuation, inhibitory control, and autonomic coordination are network-level processes. Some regions may be repeatedly involved, but involvement is not ownership. A map of activated tissue is not a map of fairness (Aron, 2007; Menon & Uddin, 2010; Seeley et al., 2007).

Distributed control matters because the same behavior can arise from different pathways. A person stops speaking because of fear, pain, fatigue, medication, cultural restraint, motor limitation, strategic choice, dissociation, or respect. A model that identifies stopping with one region or one physiological signature will confuse mechanism with appearance.

For the Law candidate, distributed control could mediate queue pressure or repair initiation. A demand becomes salient, action options narrow, inhibitory processes delay one path, and a repair path opens when safety or capacity changes. Ordinary control theory can explain many such transitions and therefore remains a full-strength rival.

A practical design should measure network-relevant behavior instead of merely naming networks. Reaction time, stop-signal performance, task switching, avoidance latency, help-seeking, speech initiation, report confidence, physiological recovery, and later behavior may together describe control. The neural layer may add useful constraints, but it should not substitute for functional evidence.

Distributed control also allows partial failure. A person may know what would repair a situation but lack energy, safety, money, access, motor capacity, or social permission. The control system is then blocked by environment as much as by internal state. A body-only model would misplace responsibility.

The practical question is what the network evidence adds. Use neural evidence when it changes predictions, resolves competing mechanisms, or improves recovery of latent states. State the additional information it contributes beyond the behavioral observations.

A distributed account also changes replication. The target may be a network relation, temporal order, behavioral transition, or channel-specific mediation that survives different scanners, tasks, sites, and populations. Repeated localization to the same region is insufficient when the causal sequence or transport fails.

20.3.2 — Inhibitory and Stopping Processes

Stopping is not the same as healing. A person may stop an action because it is unsafe, costly, confusing, socially forbidden, physically impossible, or no longer worth pursuing. Inhibitory processes can protect the person, preserve future options, or deepen avoidance. The same outward pause can therefore have opposite meanings (Aron, 2007; Menon & Uddin, 2010).

The chapter treats stopping as a candidate control object. Stop-signal tasks, go/no-go tasks, delay, inhibition, and avoidance measures can be useful when their target is clear. They become misleading when the task result is treated as a personality score or as direct access to burden.

For Queue-related dynamics, stopping may indicate blocking. An unresolved demand remains active, but a policy suppresses action because the available route is unsafe or too expensive. For a rival account, stopping may reflect ordinary harm avoidance, fatigue, learned helplessness, depression, threat appraisal, or motor limitation. Each explanation has different data needs.

A restrained example is a participant who repeatedly delays a phone call about debt. The delay may be avoidance, strategic timing, lack of information, shame, trauma cueing, or rational refusal to engage with a predatory system. The model should not call delay evidence of hidden burden without asking what actions were truly available.

Stopping processes can be tested through within-person change. Does reduced threat open action? Does pain relief open action? Does a safe script open action? Does improved access open action while physiology changes in the predicted order? If not, the stopping interpretation may be wrong.

The necessary safeguard is that stopping should never be pathologized by default. A person's refusal, pause, silence, or delay may be dignity, safety, fatigue, protection, or consent boundary. Science must distinguish blocked repair from chosen nonparticipation.

Stopping measures should include opportunity. A person cannot initiate repair if the phone line is closed, the appeal window has passed, the pain is untreated, transportation is absent, or the safer choice is refusal. Without opportunity fields, inhibition can be

confused with blocked access, and the model will place a social failure inside the person.

20.3.3 — Conflict Monitoring and Switching

Conflict monitoring becomes mechanistically useful when it predicts switching among live actions under a declared state. The timing of detection, inhibition, and switch initiation can separate distributed control models from a post hoc story about one region.

Conflict monitoring detects competition among goals, demands, predictions, and actions. A person may want relief and fear the cost of asking. They may want honesty and fear retaliation. They may want repair and lack the body or time to attempt it. Conflict becomes embodied when choice itself produces pressure (Aron, 2007; Menon & Uddin, 2010; Seeley et al., 2007).

This is relevant to the Law because hidden burden often lives in conflicting live options. The visible event may be simple: attend the appointment, answer the message, sign the form, visit the family member, keep working. The lived event may contain threat, loyalty, cost, shame, pain, uncertainty, and exhaustion inside the same action.

A conflict model should not infer moral weakness from hesitation. Hesitation may be an intelligent response to genuine competing constraints. The model should represent live options, expected bodily cost, social consequences, uncertainty, and prior learning. It should also record whether a route to repair actually existed.

Conflict can be measured through choice latency, error correction, self-report, physiological changes, eye movement, task switching, narrative content, and post-choice recovery. But no

single channel carries the whole conflict. A fast decision may be relief or compulsion. A slow decision may be reflection or paralysis.

The rival explanation is strong: ordinary cognitive conflict may explain reactivation and physiological salience without PTVCLinked assumptions. If a conflict model predicts when demands reappear, which cues reopen them, and how repair reduces them, Queue language may narrow to a useful metaphor or retire in that domain.

External constraints can generate genuine physiological conflict without implying an internal regulatory defect. Hospital forms, school meetings, immigration interviews, court dates, and benefits appeals alter action support because their stakes are real. The causal graph should include those institutional pressures rather than forcing their effects into a person-level latent mechanism.

Conflict should be modeled with stakes. A laboratory choice between two symbols is not equivalent to deciding whether to confront an abusive relative, appeal a benefits denial, disclose pain, or request accommodation. Low-stakes tasks can test mechanism components, but they do not transport automatically to life-burden decisions.

20.3.4 — Autonomic and Endocrine Coordination

Autonomic and endocrine systems coordinate preparation, energy, threat response, recovery, sleep, digestion, inflammation, and pain modulation. They are natural places to look for embodied pathways because they connect perception, action,

bodily state, and longer time scales. They are also nonspecific, context-dependent, and heavily affected by medication and illness (Menon & Uddin, 2010; McEwen, 1998).

Heart-rate variability, electrodermal activity, cortisol, catecholamines, temperature, glucose, inflammatory markers, and sleep architecture can all be scientifically relevant. None has a single affective meaning. The same marker may shift with posture, infection, activity, circadian phase, diet, fever, analgesia, beta-blockers, stimulants, sedatives, grief, or fear.

The model should therefore treat autonomic and endocrine data as coordinated systems with explicit relationships among channels. A respiratory change may alter heart-rate variability. A medication may alter both pain and autonomic response. An inflammatory state may alter fatigue, sleep, mood, and reportability. These relationships help identify the mechanism.

A terminal study shows why these relationships matter. Declining variability near the end of life may reflect disease progression, medication, sedation, low movement, autonomic failure, sensor coupling loss, care procedures, or a genuine change in experiential dynamics. Without modeling those alternatives, a convergence plot may primarily reflect changes in clinical state or observation.

Useful analysis specifies windows, lags, treatment covariates, posture and movement adjustments, illness severity, device quality, and expected direction by channel. It then tests whether autonomic and endocrine changes mediate reports or actions, predict later changes, or accompany common causes.

Establishing temporal order strengthens the interpretation of mediation.

Physiological specificity determines which autonomic and endocrine measures answer the mechanism question. Tests should separate the contributions of arousal, effort, medication, disease, and valence, and report the resolution each channel provides.

Coordination can be more informative than level. Heart-rate change without respiratory change, cortisol without sleep change, or inflammation without fatigue implies a different mechanism from a coordinated shift. The analysis therefore tests coupling, decoupling, order, and recovery across systems rather than scoring each channel in isolation.

20.3.5 — Repair Initiation and Recovery

Repair initiation is the transition from detected need or blocked, avoided, frozen, or unresolved demand into a feasible action that changes state, environment, or the generating condition. It may be a phone call, a boundary, a medication adjustment, an apology, a rest period, a request for help, a change in posture, a safe exposure, or the decision not to return to a harmful context. Recovery is the subsequent trajectory and may remain partial; the two phases should not be collapsed into one improvement score (Aron, 2007; Menon & Uddin, 2010; Seeley et al., 2007).

The chapter treats repair initiation as observable but not simple. The first repair action may not be the visible action. It may be the body lowering threat enough to speak, the mind accepting that a route exists, the clinic reducing pain, or the environment

removing a barrier. A model that records only final action misses the earlier transition.

Brain-body mechanisms could help identify repair initiation if they show a predictable temporal pattern: salience shifts, respiratory constraint eases, attention becomes more flexible, avoidance latency falls, and action follows. But the order must be tested. If action changes first and physiology follows, the model should not reverse the arrow because the story sounds better.

A restrained scene is a person who has delayed a needed conversation for months and finally sends one clear message after pain treatment, sleep, and a friend's practical help. The repair is not a heroic moral settling. It is the opening of a route that had previously been too costly. The bodily piece matters because capacity mattered.

Repair initiation also has noncompensable limits. Some harms cannot be undone by later action. Some repairs reduce ongoing burden while leaving grief, disability, distrust, or lost time. The body may settle enough to act without erasing the loss.

Define repair by change in generating condition, not by symbolic closure. A ritual, apology, exposure, medication, or behavior counts as repair only if the mechanism it targets plausibly moves. Otherwise, the model risks rewarding appearances while hidden burden remains.

Repair can be quiet and private: taking medication, leaving a harmful thread unread, asking for a ride, sleeping, eating, or declining a demand. The model should represent these ordinary routes to relief.

20.3.6 — What Neural Findings Contribute

Timed interactions, perturbation responses, and medication-sensitive mediators can help characterize how the insula, prefrontal cortex, amygdala, vagal pathways, and distributed networks participate in a specified mechanism (Menon & Uddin, 2010; McEwen, 1998).

Neural results should identify their target, comparator, timing claim, and evidentiary role. Their contribution can include distinguishing mechanisms, testing temporal order, identifying mediators, or identifying artifacts beyond the information supplied by reports and behavior. A neural measure that adds no predictive or causal information beyond existing channels should not gain authority simply because it is technologically sophisticated.

Multifunctional regions participate in salience, pain, memory, control, attention, and many other tasks. Specificity therefore comes from prospectively specified connectivity, lag, state dependence, perturbation response, and rival models, not activation under a familiar anatomical label. Specificity should be demonstrated on contrasts that hold general arousal or task difficulty comparable, preventing familiar network activation from becoming a catch-all explanation.

A responsible caption names a candidate pathway under review rather than a location of fairness. Success increases mechanism plausibility within the design; failure narrows that mechanism. Neither makes a brain image a verdict on a life. Spatial precision

should be no stronger than the causal and measurement evidence linking that pathway to the declared construct.

Lesions, stimulation, and medication provide stronger leverage but also alter multiple pathways and behaviors. Off-target effects, treatment versions, and clinical context must be modeled before a causal interpretation. Triangulation across distinct perturbations can help identify whether the same bounded pathway recurs despite different off-target profiles.

The strongest admissible brain finding may be a bounded mediator or confound result, such as a network explaining a device-report relation or medication removing an apparent signature. Such results can defeat a simple route story without identifying PTVC or consciousness. This kind of negative mediation result can be scientifically decisive even when it leaves the larger terminal hypothesis untouched.

20.4 — Channel Lags and Confounding

The body is not synchronized for the convenience of a study. Breathing can change within seconds, heart rate across seconds to minutes, cortisol over longer windows, inflammation over hours or days, pain with complex cycles, and sleep across nights. A diary prompt may arrive after the event, while a device may record before the person can name what happened.

This section treats timing as part of meaning. A channel that leads report can suggest prediction, early bodily preparation, or artifact. A channel that follows report can suggest recovery, action consequence, or common cause. A channel that moves independently may be irrelevant to the target, or it may be a mechanism the report failed to capture.

Modeling confounding makes brain-body evidence more informative and trustworthy. Medication, disease severity, movement, posture, sleep, environment, sensor fit, device version, cultural expression, and clinical procedures can all move the signal. Specifying those pathways before results arrive makes their contribution testable.

Equation 20.3: Lag and mediation display. Role: temporal test structure for mechanism order.

$$B_{i,t+\Delta} = \alpha_i + \rho_B B_{i,t} + \gamma_c O_{i,c,t} + \delta Z_{i,t} + \eta_{i,t}$$

$$R_{i,t+2\Delta} = \mu_i + \rho_R R_{i,t+\Delta} + \beta B_{i,t+\Delta} + \zeta O_{i,c,t} + \xi_{i,t}$$

$\gamma_c \neq 0$ and $\beta \neq 0$ under rival and confound checks.

This display proposes one temporal pathway: channel observation precedes a modeled change in belief or representation, which

precedes report or action. That ordering is testable, but lagged prediction alone does not establish mediation. Causal interpretation additionally requires a justified intervention or identification strategy, suitable confound control, and sensitivity to competing and feedback pathways.

The timing ranges summarized in Table 20.4 are illustrative rather than universal. Each protocol must estimate or justify its own sampling windows, carry channel-specific uncertainty, and test whether conclusions survive plausible lag shifts.

Table 20.4. Lag-risk table. Role: protocol design aid for channel-specific timing and failure.

Signal family	Typical lag concern	Failure if ignored
Respiration and movement	seconds to minutes	task effort mistaken for affect
Cardiac/autonomic	seconds to hours	posture, medication, or activity mistaken for burden
Endocrine	minutes to hours	circadian phase mistaken for event response
Immune/inflammatory	hours to days	illness or treatment mistaken for mood pathway
Sleep and fatigue	nights to weeks	recovery or nonwear mistaken for psychological change
Report and memory	minutes to years	reconstruction mistaken for momentary experience

20.4.1 — Different Response Times

Different channels answer at different speeds. If a study measures all channels at one prompt time, it may create false simultaneity.

The signal that looks absent may have already passed, may not have begun, or may belong to a longer recovery arc. Time alignment is therefore not formatting. It is inference (Shiffman et al., 2008; Stone et al., 2007; Särkkä, 2013).

Fast signals provide precise timestamps, while interpretation may develop later as the body settles, the situation becomes clearer, or the person is ready to report. A respiratory shift or heart-rate spike may mark a trigger, preparation, or distraction; the sequence helps distinguish those roles.

Slow signals can be tempting for the opposite reason. Inflammation, fatigue, endocrine rhythm, and sleep patterns seem deep because they are durable. Yet slow does not mean important to the target. They may reflect illness, treatment, environment, caregiving load, or sensor routine more than the lived burden under study.

A good design declares expected lags by mechanism. If interoceptive salience is the mechanism, the channel should move before or with attention capture. If repair initiation is the mechanism, action should follow the salience shift or environmental change. If medication is the mechanism, timing should align with dose, pharmacodynamics, and clinical effect.

Lag discovery may begin exploratorily, but confirmatory claims use protocol-fixed windows or an independently locked validation phase. Multiplicity, clock alignment, and alternative lag structures are carried into uncertainty. Searching until one delay fits can turn noise, shared causes, or observation timing into

retrospective coherence, especially when several channels respond on different clocks.

Match timescale to mechanism. A body signal can inform the Law only at the temporal resolution where it has interpretable relation to the modeled target. The next step is to align the sampling window and mechanism before making a stronger inference.

Response time also determines missingness. A slow inflammatory signal cannot be recovered from a short task, and a fast respiratory shift may be missed by a daily diary. Sampling that is too slow creates false absence; sampling that is too fast can create burden and noise. Cadence should be justified by the mechanism's bandwidth.

20.4.2 — Common Causes and Time-Varying Confounding

Common causes can change over time. Disease severity, medication, movement, sleep, caregiver presence, and device quality may affect both a channel and later report differently across the day. Static adjustment can leave time-varying confounding unresolved.

A common cause moves both the signal and the apparent outcome. Disease severity can increase fatigue and change inflammation. Movement can increase heart rate and alter report. Sleep loss can increase pain and reduce emotional regulation. Medication can reduce distress and change physiology. Environment can change body and mood together (VanderWeele, 2015; Shiffman et al., 2008).

Common causes are especially dangerous because they produce convincing associations. If pain relief and heart-rate variability

improve after a clinical procedure, the improvement may be treatment effect, expectation, movement change, sleep, hydration, reduced fear, or measurement artifact. The pattern can look like mechanism even when the true cause sits outside the proposed path.

Every analysis plan carries a prespecified minimal confound set suited to the mechanism. Movement, posture, sleep, medication, disease severity, device version, time of day, context, and report timing are recorded unless the design gives a principled reason for omission. Omission is sometimes necessary; silence is not.

A causal graph helps because it forces arrows onto the page. If fever moves heart rate and fatigue, the graph says so. If medication moves pain and reportability, the graph says so. If social threat moves breathing, avoidance, and report, the graph says so. Once arrows are visible, hidden assumptions become criticizable.

Sensitivity analysis should ask how strong an unmeasured common cause would have to be to erase the proposed mechanism. If a modest confound would overturn the result, the claim should be presented as fragile. Fragility is not failure; it is honest status.

A missing confound often becomes a label attached to the participant. Clinical and disability studies should therefore model medication, access, motor capacity, communication method, fatigue, pain, and device fit before describing a body as inconsistent or noncompliant. The burden of unexplained variation belongs first to the measurement and causal model.

Common-cause modeling should include the research situation itself. Being observed, prompted, wired, filmed, or asked to explain the body can change physiology and report. A study room is not a neutral window into life. It is an event that may become part of the data-generating process.

20.4.3 — Reverse Causation and Feedback

Feedback makes reverse causation expected rather than exceptional. Report can change attention, attention can change physiology, and action can change both. Lagged association should therefore be interpreted through a causal model that permits reciprocal paths.

Reverse causation occurs when the apparent outcome changes the bodily signal rather than the bodily signal changing the outcome. A person reports fear, then breathes differently. A participant chooses to repair a conflict, then autonomic recovery follows. A clinician offers reassurance, then pain expression changes. The arrow can reverse without announcing itself (Shiffman et al., 2008; Stone et al., 2007; VanderWeele, 2015).

Temporal precedence is informative only after top-down, contextual, and feedback pathways are represented. Meaning changes the body, action changes the body, and social safety changes the body. A genuinely embodied model allows influence to travel in both directions and specifies which perturbation could distinguish them.

Care can change both experience and physiology. When pain relief improves affect, modeling the intervention and its timing helps explain the observed trajectory.

Testing reverse causation requires repeated measurement and temporal ordering. Does the signal change before the report, after the report, or both? Does action precede recovery? Does environmental change precede bodily change? Cross-lag designs and within-person analyses can help, but only when cadence matches the mechanism.

A no-reverse-causation assumption should be visible. If the study cannot distinguish order, the claim should say correlation, association, or co-movement. It should not say mediation, control, or mechanism. The words are part of the evidence standard.

Bidirectional feedback is the default rival to a one-way story. Bodily state can alter attention and report, while appraisal, action, and expectation can alter bodily state. The analysis should test the direction it intends to interpret and retain a reciprocal or unresolved model when cadence cannot separate the arrows.

Reverse causation is particularly important for repair. A person may decide to act, and that decision may lower threat before the action is visible. If the model measures only physiology first and action later, it may falsely conclude that body recovery caused repair. The hidden decision point must be measured where possible.

20.4.4 — Measurement Attenuation and Device Error

Device error includes drift, dropout, placement changes, firmware updates, motion artifact, and differential performance across bodies. Attenuation can hide a real relation, while correlated error can create one. Quality metrics belong in the model, not only in preprocessing logs.

Measurement attenuation occurs when the observed signal weakens because the instrument, body-device coupling, behavior, or communication channel degrades rather than because the target changes. Low movement, low speech, poor contact, edema, altered perfusion, sedation, and device removal can all create apparent quiet. Device-quality and observation-process variables must therefore accompany the target estimate (Bent et al., 2020; Shiffman et al., 2008).

Attenuation can mimic convergence. If physiological variance shrinks because the sensor sees less, a terminal compression plot may look impressive while measuring observation loss. If reports become neutral because communication becomes difficult, neutrality may be a response artifact. If family proxies simplify because distress is hard to witness, the record may become emotionally smoother than the life.

Device-quality and channel-availability fields belong in the analysis. Signal-to-noise, contact quality, wear time, battery, firmware, artifact flags, movement, perfusion, temperature, and missingness determine when a trajectory can support inference. A visually smooth trace without quality metadata remains incomplete evidence.

Negative controls are essential. If unrelated channels attenuate at the same time, device or physiological shutdown may explain the pattern. If movement-sensitive signals change while movement changes, the interpretation must account for it. If a device version alters variance, model claims should be rerun by version.

Attenuation also appears in human observation. A clinician may record less detail during crisis. A family member may stop reporting because the burden is too great. A participant may withdraw consent. These are not holes to patch with imputation by default. They are events in the observation process.

Lower signal may indicate reduced access to evidence rather than a quieter phenomenal state. The observation model should therefore treat channel silence, smoothing, and dropout as competing explanations, especially when illness, motor limits, fatigue, or communication loss make the person hardest to hear.

Attenuation should be plotted beside the target estimate. A graph of apparent variance compression without a graph of wear time, signal quality, movement, medication, and channel availability is incomplete. The viewer should be able to see whether the instrument became quieter before concluding that the person's experience did.

20.4.5 — Multimodal Conflict and Missing Channels

Multimodal conflict may be informative, but missing channels can manufacture apparent agreement. A protocol should report which channel was absent, why, and whether absence depended on state or burden. More modalities do not help when they fail together.

Multimodal conflict occurs when channels disagree: report may indicate calm while heart rate rises; a device may register longer sleep while exhaustion deepens; facial expression may remain flat during severe pain. The first response is not forced consensus but

a channel-specific audit of timing, quality, construct, context, and missingness (Shiffman et al., 2008; Bollen, 1989; Särkkä, 2013).

Conflict is information about the observation model. It may reveal masking, medication, dissociation, cultural expression, disability, device artifact, task demand, social desirability, timing mismatch, or genuine multidimensional experience. Disagreement is therefore treated as a diagnostic event rather than a problem to average away.

A conflict class can be coded with fields: which channels disagree, in what direction, under what context, with what quality flags, and what follow-up is permitted. The model can then learn when reports should dominate, when devices should be demoted, when both should be held as partial, and when no estimate is responsible.

Consider a person with chronic pain who smiles during a clinical visit while reporting severe pain. The smile may reflect politeness, coping, fear of disbelief, cultural habit, or an attempt to protect the clinician. A model that resolves the conflict by downgrading pain because the face looks calm repeats a known measurement injustice.

Multimodal conflict is also a chance for rivals to win. If a device-report conflict is explained by medication or disability, the candidate mechanism loses ground. If conflict reliably predicts later repair or residual burden after confounds, the candidate gains a testable feature. The distinction depends on design, not rhetoric.

Record conflict before resolving it. A science that admits disagreement can protect the person from being overwritten by whichever channel is loudest, newest, or easiest to quantify.

Conflict review should be versioned. If a later expert panel decides that a device channel was invalid in a subgroup, earlier conclusions should be marked under the old interpretation rather than silently rewritten. A conflict history keeps the science honest about how it learned.

20.4.6 — Causal Graphs, Lag Models, and Sensitivity Analysis

Causal graphs identify plausible pathways, lag models specify temporal order, and sensitivity analyses test dependence on unmeasured causes and timing choices. No single tool closes the problem; the combination makes assumptions visible and falsifiable.

A causal graph is a disciplined picture of possible pathways. It names variables, arrows, confounds, mediators, colliders, selection, and measurement routes. In this chapter, the graph keeps interoception from becoming a cloud of plausible associations. It says what would have to be true for a body signal to support a mechanism claim (VanderWeele, 2015; Shiffman et al., 2008).

The graph should include environment, bodily state, central representation, report, action, medication, disease severity, disability context, device quality, culture, movement, sleep, and time. It should also include the proposed Law-relevant construct only as a latent target or candidate mediator, not as an observed

node. The hidden target does not become measured because an arrow reaches toward it.

Sensitivity analysis then asks what happens when the graph is wrong. What if medication is undermeasured? What if trauma history alters report? What if device fit differs by body shape or skin tone? What if pain relief changes both physiology and report? What if missingness is informative? The model should not collapse when the first real-world complication appears.

Causal graphs also help prevent collider bias. If a study includes only people who tolerate a device, only people who can report repeatedly, or only people who survive long enough to be sampled, the sample itself may create associations. A body mechanism inferred from a selected sample may not travel.

Causal language should track the evidence actually available. Association and prediction remain useful outputs; a mechanism claim additionally requires defensible temporal order, intervention or natural-experiment leverage, negative controls, sensitivity to unmeasured causes, and transport beyond the fitted setting. The result should narrow automatically when one of those supports is absent.

The causal graph should be readable by nontechnical reviewers. Participants, clinicians, ethics boards, and disability advisors should be able to see which pathways the study assumes. If only statisticians can understand the graph, the people most affected by misinterpretation may be excluded from correcting it.

Lag choice should be treated as a model parameter with biological and measurement constraints. State-space or distributed-lag

models can represent fast autonomic responses, slower endocrine change, report delay, and device filtering, but the candidate lag family should be declared before the result is inspected (Särkkä, 2013).

Cross-lag coefficients do not by themselves identify direction. Stable unmeasured causes, feedback, cadence mismatch, and measurement error can reverse or create apparent lead-lag relations. Intervention, natural-experiment timing, negative controls, and sensitivity analyses remain necessary for causal language (Bollen, 1989; Hernán & Robins, 2020).

The final graph packet should include the main causal diagram, plausible alternatives, measurement nodes, selection nodes, lag assumptions, and the no-inference zones created by unresolved paths. Reviewers should be able to see exactly which arrow carries the mechanism claim and which evidence could remove it.

20.5 — Medication, Disability, Trauma, and Illness

A body interpreted without its history is an easy place to make error look objective. Medication can alter signal and report. Disability can change baseline and expression. Trauma can change interoceptive precision and safety learning. Chronic illness can sustain fatigue, pain, inflammation, and uncertainty. Aging and terminal illness can alter physiology, communication, and device quality at the same time.

Treatment, disability, trauma, illness, aging, and terminal conditions define important dimensions of the reference class. The model specifies which groups it represents and what additional observations would extend its reach.

The section's local safeguard is structural: do not turn model mismatch into person mismatch. When a signal fails to follow expected norms, the protocol first asks whether the norm, channel, treatment model, device, context, or target is wrong. It does not imply that the person is inaccurate, unreadable, less conscious, less credible, or less suited for care.

Table 20.5. Misread-body case series. Role: validation stress cases, not reported case histories.

Case-like scenario	Likely model hazard	Required design response
Beta-blocker use during stress task	muted cardiac response read as calm	medication window and alternate channels
Autonomic neuropathy with pain	low autonomic variability read as low burden	disability baseline and report priority

Case-like scenario	Likely model hazard	Required design response
Trauma history in safe task	high salience read as current threat	cue history, precision model, safety check
Chronic fatigue flare	low movement read as withdrawal	illness state, energy budget, activity opportunity
Palliative sedation	reduced report read as neutrality	clinical context, endpoint firewall, no estimate if needed
Darker skin tone with optical sensor	device quality read as physiological difference	validation by device, skin tone, fit, and artifact

20.5.1 — Medication Effects and Treatment Versions

Medication effects have treatment versions defined by drug, dose, timing, indication, co-medication, metabolism, and adherence. A physiological change after medication may reflect symptom relief, side effect, altered reportability, or direct sensor influence. Those pathways should be separated before mechanism inference.

Medication can move the signal, the experience, the report, and the ability to report. Analgesics may reduce pain and alter alertness. Sedatives may reduce arousal and communication. Stimulants may raise cardiac signals and attention. Autonomic medications may mute or reshape heart-rate dynamics. Psychiatric medications may alter affective range, sleep, appetite, and interoceptive awareness (Giacino et al., 2018; Khalsa et al., 2018; McEwen, 1998).

A medication effect is not a nuisance if it changes lived experience. Pain relief matters. Breath relief matters. Sleep

restoration matters. Panic reduction matters. The problem arises when the model cannot distinguish the effect on experience from the effect on observation. A sedated person may suffer less, report less, move less, or all three. The design must not choose the convenient interpretation.

Medication timing should be part of the data model. Dose, route, expected onset, peak, duration, side effects, indication, and recent changes should be recorded where ethically and clinically appropriate. Without timing, the model may interpret treatment pharmacodynamics as spontaneous regulation or terminal dynamics.

Clinical expertise is required because the same drug can mean different things across bodies. Opioid rotation, benzodiazepine use, anticholinergic burden, beta-blockade, corticosteroids, stimulants, antiemetics, antidepressants, antipsychotics, anticonvulsants, and sleep medications each alter different observation routes. A generic medication flag is better than nothing but often insufficient.

A safe analytic plan can treat medication as confound, mediator, intervention, effect modifier, or reason to abstain. The status depends on the question. If pain relief is the mechanism being studied, medication may be central. If terminal variance is being studied, medication may be a confound or competing explanation. The model should say which.

The local ethical rule is that treatment never belongs to the theory. Care decisions are made for the person, not for cleaner data. If a

needed medication destroys a signal, the signal is lost. The person is not asked to remain measurable.

Medication modeling also has privacy boundaries. Detailed medication records can reveal diagnoses, stigma-sensitive treatment, addiction history, reproductive status, psychiatric care, or end-of-life decisions. The study should collect enough to interpret signals, but not turn clinical history into a permanent identity file.

20.5.2 — Disability, Access, and Baseline Variation

Access is part of disability-informed calibration. A device, task, prompt, or report format can fail because the protocol was not accessible, not because the participant lacks a valid signal. Baselines and action sets should reflect the person's actual environment and supports.

Disability changes baselines, channels, expression, access, effort, fatigue, movement, communication, and sometimes sensor fit. Treating able-bodied patterns as universal makes the observation model invalid before route inference begins. The issue is not that disabled bodies are difficult to measure; it is that many protocols mistake one body norm for the human norm (World Health Organization, 2001; Khalsa et al., 2018).

Baseline variation can appear in heart-rate dynamics, movement, pain expression, facial movement, voice, sleep, posture, temperature regulation, energy use, and communication timing. A person using a wheelchair, ventilator, feeding tube, communication device, prosthetic, pain treatment, or autonomic

support may generate data that require a different observation model.

Interpretive burden falls on the model. A system trained on narrow movement patterns may call low activity disengagement when movement is constrained by disability, pain, environment, or access. A voice model may misread speech affected by disability as affect. Those failures narrow or retire the model; they do not define the person.

Disability-informed design begins with participation and expertise. It asks what channels are meaningful, burdensome, inaccessible, or misleading. It includes accommodation, alternate routes, longer response windows, assistive technology compatibility, and participant challenge. A disabled participant should not have to become a methodological exception after the model fails.

Calibration should occur within person before cross-person claims. A person's own baseline, range, fatigue cycle, communication style, and support context are more informative than a general norm. Transport from one disability group to another should be earned, not assumed.

The use boundary is to treat disability as a source of scientific knowledge. It reveals where the model was relying on hidden able-bodied assumptions. A science that learns from those revelations becomes more general, not less.

Disability-informed modeling should include device refusal as a valid outcome. A sensor that causes pain, sensory overload, skin irritation, dysphoria, or practical difficulty is not a neutral

instrument. Refusal may tell the study that the observation method is poorly designed, not that the participant is less cooperative.

20.5.3 — Trauma, Hyperarousal, and Dissociation

Trauma can produce hyperarousal, hypoarousal, dissociation, avoidance, and context-dependent safety learning. No single autonomic direction defines the state. Interpretation should be trauma-informed and should never require reenactment or disclosure beyond consent.

Trauma can alter bodily prediction, attention, threat appraisal, salience, startle, sleep, pain, dissociation, and safety learning. Hyperarousal may be visible in some contexts, but trauma can also appear as numbness, shutdown, avoidance, memory gaps, constrained expression, or practiced calm. There is no single trauma signature that authorizes simple interpretation (Khalsa et al., 2018; McEwen, 1998).

A trauma-informed model separates current threat from learned threat prediction. A safe room can still produce bodily danger signals if the room resembles an earlier danger, if authority is present, if consent feels fragile, or if the person expects disbelief. The body may be responding to a history the protocol did not record.

Trauma can produce persistent reactivation, blocking, and difficulty initiating repair. Those patterns may be compatible with queue language, but learning, threat prediction, memory, and unsafe social conditions may explain them fully. A PTVC-linked

route gains no evidential weight until it distinguishes itself from those ordinary accounts under humane tests.

Ethical design avoids surprise, coercive exposure, humiliating tasks, intense bodily focus, or manipulations that amplify symptoms. If interoceptive attention is used, it should be brief, optional, clinically reviewed where necessary, and paired with stop rules. A participant's withdrawal is not missing data to regret; it is information about burden and consent.

Measurement should include safety context, trigger likelihood, dissociation risk, grounding options, report preferences, and aftercare routes. The model should also include cultural and social conditions that make vigilance rational. Trauma physiology is not only inside the person; it is often maintained by environments that remain unsafe.

For this question, the required discipline is never to moralize altered interoception. Hyperarousal, numbness, avoidance, dissociation, or silence are not failures of fairness, faith, courage, or reason. They are conditions under which science must become gentler and more exact.

Trauma physiology may also be quiet in the moment and costly later. A participant can complete a task calmly and then lose sleep, dissociate, or experience delayed symptoms. Follow-up windows and stop rules should therefore include aftereffects, not merely in-session distress.

20.5.4 — Chronic Illness, Inflammation, and Fatigue

Chronic illness may involve inflammation, fatigue, pain, sleep disruption, and fluctuating capacity. These processes can share

channels with affective mechanisms while following different time scales. Clinical context and treatment changes should be modeled as causes, not nuisance labels.

Chronic illness and fatigue stretch mechanism across time. They may alter sleep, movement, pain, cognition, immune state, autonomic function, digestion, mood, social participation, and opportunity. A momentary valence estimate can miss the slow cost of needing to plan every action around energy, symptoms, treatment, uncertainty, and access (Khalsa et al., 2018; McEwen, 1998).

Fatigue is especially easy to misread. Low activity may be depletion, pain avoidance, depression, medication, symptom management, inaccessible environment, or wise pacing. High activity may be recovery, overexertion, necessity, caregiving pressure, or a day that will be paid for tomorrow. The signal needs a time horizon.

Chronic illness also breaks simple rebound stories. A person may adapt to some aspects of illness while remaining burdened by pain, uncertainty, financial stress, lost roles, stigma, and medical work. Resilience should not be used to acquit the conditions that made resilience necessary.

A serious design records energy budget, symptom flare, treatment schedule, care tasks, sleep quality, activity opportunity, access barriers, and post-exertional change where relevant. It also gives the person control over cadence. A measurement plan that worsens fatigue in order to study fatigue has already failed.

The model may need slower lags and delayed outcomes. A body signal today may predict tomorrow's exhaustion rather than today's report. Repair may mean fewer obligations, better treatment, assistive support, or pacing, not emotional closure. Chronic time changes the candidate mechanism and the observation window.

Illness includes the work required to navigate it. Appointments, paperwork, symptom tracking, transportation, side effects, care coordination, and repeated disbelief can move physiology, attention, and report. A model that omits those processes confounds disease with the burden of managing disease.

20.5.5 — Pain, Communication, and Motor Limits

Pain expression and communication can be limited by motor function, language, sedation, fear of disbelief, or learned masking. Absence of report or movement is not absence of pain. Multimodal evidence should widen uncertainty rather than override testimony.

Pain is central and difficult because it is bodily, affective, attentional, cognitive, social, and communicative. It can be intense without visible expression, visible without identical meaning, treated without disappearing, and remembered differently from how it was lived. A pain signal does not become simple because the person has a number scale (Khalsa et al., 2018; Critchley & Garfinkel, 2017).

Communication varies under pain. Some people become quiet, some irritable, some polite, some expressive, some dissociated, some unable to speak. Disability, culture, gender, race, trauma,

clinician trust, past disbelief, and fear of consequences can shape what is said. A model that treats report as pure truth or pure noise will fail in different ways.

Behavioral pain cues are partial. Movement limitation can show pain, fatigue, motor impairment, access constraint, sedation, or protective pacing. Facial expression can be masked, culturally controlled, absent due to disability, or altered by medication. Voice can change because of breath, fatigue, language, neurological condition, or social context.

The chapter's design route is multimodal but person-centered. Pain report, function, medication, context, clinician observation, device signals, and participant history can be integrated under uncertainty. If channels conflict, the model should not automatically downgrade the report. It should ask why conflict occurs.

Pain relief is also a mechanism test. If analgesia changes report, behavior, sleep, autonomic signal, and repair action in a plausible temporal order, the model learns something. If physiological channels remain unchanged despite reported relief, test whether the channels, sampling windows, and detection limits could have captured the predicted change before narrowing their mechanistic role. Both outcomes are useful.

Pain may enter the study as an already present clinical state, a participant-reported outcome, or a response to independently justified care. It may not be induced, prolonged, intensified, or left untreated to sharpen a mechanism contrast. Designs that

cannot separate the scientific question from added harm are narrowed or abandoned.

Pain communication should be evaluated under trust. A participant who expects disbelief may underreport, overexplain, or use numbers strategically. The solution is not suspicion. It is a design that acknowledges the social conditions of reporting and gives multiple routes for expression without punishing inconsistency.

20.5.6 — Clinical Context Before Interpretation

Clinical context precedes interpretation because physiology is clinically entangled. Oxygenation, infection, delirium, sedation, pain, dehydration, fever, organ failure, device tolerance, nursing care, family presence, procedures, and medication changes can all alter body signals and report. A research model outside that context may be precise and wrong (Giacino et al., 2018; Khalsa et al., 2018).

Clinical expertise is required far beyond terminal settings. Chemotherapy, dialysis, surgical recovery, intensive care, psychiatric crisis, rehabilitation, pregnancy, menopause, withdrawal, and chronic pain treatment can alter both physiology and observation. A mechanism claim must be calibrated to the clinical world in which its data were generated.

Clinical context also protects the person from automated override. A wearable alert should not contradict pain care, sedation, palliative judgment, or emergency response because a research algorithm sees an interesting pattern. The care system remains primary. Research interpretation waits.

The data dictionary should include clinical status fields that are meaningful, not merely administrative. Diagnosis group, acuity, treatment changes, sedation scale where appropriate, delirium risk, pain regimen, mobility, communication ability, device interference, and consent status may all be relevant. The model should not ask for every field by default; it should justify the fields it needs.

Clinical expertise should enter before analysis and during adjudication. That does not mean clinicians own the person's inner life. It means they understand the conditions that can move observed signals. The best interpretation may come from combining participant report, clinician context, device quality, and research model uncertainty.

Clinical review includes the option to stop research activity. Illness, fatigue, medication, distress, proximity to a care decision, or changing consent can make further collection disproportionate. Stopping preserves care and also protects the scientific record from data whose conditions no longer match the declared design.

Medication should be represented by active ingredient, dose, timing, route, indication, adherence, and clinically relevant co-medication where proportionate. A device or report change after treatment may reflect therapeutic effect, side effect, withdrawal, interaction, or changed behavior. Pooling those versions can invent a mechanism.

Disability-specific calibration should be planned rather than added after an anomaly appears. Movement, speech, facial expression, heart-rate dynamics, and device placement can differ

for reasons unrelated to valence. Alternative communication and measurement routes should be treated as legitimate study designs, not deviations from a presumed normal body.

Trauma and chronic illness can change baseline, reactivity, recovery time, and tolerance for perturbation. The design should distinguish a mechanism prediction from a learned threat response, fatigue, pain flare, inflammatory change, or care burden. When these explanations cannot be separated, the result remains clinically contextual rather than route specific.

The no-estimate decision should be available at the person-window level. A period can be excluded from one mechanism analysis because of delirium, acute treatment change, device failure, or severe burden while remaining valuable for descriptive clinical understanding. Abstention from the mechanism claim does not erase the person or the data.

20.6 — Intervention and Perturbation Designs

Ethically appropriate changes in state, context, attention, posture, breathing, pain relief, sleep, medication, or environment can test a pathway's predicted channels and temporal order. Such comparisons connect brain-body descriptions with causal questions.

Existing neuroscience provides a concrete example of mechanism-guided recovery. Xia et al. (2025) recorded population activity in stress-susceptible and resilient mice and identified different dynamics linking ventral hippocampus and basolateral amygdala. Manipulating the hippocampal inputs to the amygdala shifted dysfunctional dynamics toward the resilient pattern and reversed anhedonic behavior. The experiment connects an affect-related behavioral difficulty to a specified circuit and shows that intervention can change it. This gives recovery research an experimentally grounded pathway to investigate.

This chapter studies interventions that are independently justified for care, ordinary, low-risk, and participant-controlled, within their approved context of use. Research for the Law must not induce distress, pain, panic, deprivation, or physiological danger. Designs can learn from existing care and safe variation while preserving timely treatment and withdrawal.

Perturbation designs matter because they expose direction. A change in breathing that alters salience before report supports one route. A medication change that alters report before physiology supports another. A posture change that alters device readings but

not report exposes artifact. A negative control that moves the same channel breaks the proposed specificity.

A randomized crossover study illustrates why recovery timing matters. Among 82 healthy participants completing both conditions, Ferstl and colleagues (2022) compared auricular vagus-nerve stimulation with sham stimulation during prolonged effort. Positive mood, a secondary outcome, improved relative to sham during the post-stimulation period; the improvement was not detected during stimulation. The reported difference was approximately 9.6% relative to the model’s mood intercept. This provides a concrete brain–body comparison with a measurable lag and suggests testing whether recovery models predict when a benefit appears as well as its magnitude.

Table 20.6. Perturbation-study table. Role: design comparison for safe mechanism tests.

Design	What changes	What it can test	Boundary condition
Naturalistic breathing shift	breathing pattern already chosen or safe	respiratory salience and report lag	no panic induction
Posture and movement control	sitting, standing, walking, rest	artifact and autonomic adjustment	accessibility and disability accommodations
Routine medication transition	clinically indicated treatment	medication as mediator or confound	care decision independent of study
Sleep opportunity change	ordinary sleep schedule or clinical sleep support	recovery and fatigue pathways	no deprivation for theory

Design	What changes	What it can test	Boundary condition
Pain relief window	independently justified analgesia	pain-report-physiology order	relief never delayed
Neutral attention task	harmless focus or distraction	precision and salience weighting	stop rules and symptom monitoring

20.6.1 — Naturalistic State Changes

Naturalistic state changes are the safest first mechanism tests. People already breathe differently, stand, sit, walk, rest, sleep, eat, take medication, receive care, speak, avoid, repair, and recover. A study can learn from those changes when consented and recorded without treating the person as a laboratory object (Klasnja et al., 2015; Nahum-Shani et al., 2018; Khalsa et al., 2018).

Naturalistic design is especially appropriate for palliative, chronic illness, disability, trauma, and everyday-life contexts. The question becomes: when a state changes for reasons independent of the theory, what happens to the proposed channel, report, action, and later burden? The event is not created for science. Science follows with restraint.

Examples include routine pain relief, a scheduled physical therapy session, a safe breathing exercise already part of care, a posture change, a change in sleep opportunity, a supportive conversation, or a reduction in administrative demand. These events can reveal whether physiology, report, and action move in a plausible order.

Naturalistic variation must still be modeled. The fact that an event was not assigned does not remove confounding. People who receive a certain intervention may differ in severity, access, support, trust, timing, or clinical status. The study should declare the estimand and whether the design is descriptive, predictive, or causal.

A careful analysis might use within-person comparisons around repeated routine events, with time-varying covariates and negative-control windows. It should avoid heroic claims when events are sparse or confounded. A naturalistic result can motivate a mechanism; it rarely establishes one alone.

Observe ordinary care without bending ordinary care. If the person needs rest, rest comes first. If the clinician changes medication, the clinical reason controls. The model receives what care leaves available, not the other way around.

Naturalistic designs should record why the change happened. A breathing exercise chosen for comfort, a medication changed for pain, or a posture shift made for access has a different meaning from the same behavior assigned by protocol. The reason is part of the causal story, not an anecdote.

20.6.2 — Safe Interoceptive Perturbation

Safe interoceptive perturbation means a low-risk, consented, reversible change in bodily attention or state that has independent ethical justification. It may include gentle breathing awareness, posture adjustment, brief attention redirection, paced movement, or other innocuous tasks approved for the population. It does not

include provocation that creates clinically meaningful distress (Seth & Friston, 2016; Critchley & Garfinkel, 2017).

The design should state the mechanism it tests. A breathing task may test respiratory salience, not global valence. A posture task may test autonomic adjustment or sensor artifact, not unresolved burden. An attention task may test precision weighting, not hidden truth. The narrower the target, the stronger the inference.

Perturbations should have stop rules visible to participants. A person can stop because of discomfort, boredom, fatigue, pain, trauma cueing, disability access, or simple refusal. The stop is not noncompliance. It is a consent and burden signal that belongs in the protocol record.

The task should also include debriefing and return to baseline where appropriate. If a breathing or attention exercise increases symptom focus, the protocol should not treat that increase as useful data and move on. It should provide grounding, support, and review. A mechanism test that leaves burden behind has violated the chapter's purpose.

Safe perturbations reveal whether the predicted channel responds, whether the response is specific, and whether device artifacts account for the change. Each outcome helps refine the mechanism or measurement.

Proportionality is a scientific and ethical design criterion. The question must justify the participant burden, and the burden must remain small, reversible, consented, accessible, and monitored. A mechanism that can be separated only by turning a person's body into a stress instrument is not available for this research program.

Safe perturbation should be piloted with the population being studied, not only with healthy convenience samples. A task that seems mild to a researcher can be difficult for someone with panic, asthma, pain, trauma, dysautonomia, fatigue, or sensory sensitivity. Feasibility evidence should come from the relevant bodies.

20.6.3 — Within-Person Crossover and N-of-1 Logic

Within-person crossover and N-of-1 logic reduce between-person heterogeneity by comparing safely varied conditions within the same participant. Carryover, period effects, treatment versions, and natural drift still require modeling. The design remains a mechanism test, not a personal fairness score.

Within-person crossover designs compare a person to themselves under different safe states or conditions. This is valuable because interoception and physiology vary greatly across persons. A between-person difference may reflect baseline, disability, medication, fitness, trauma history, culture, or device fit rather than the mechanism being studied (Klasnja et al., 2015; Nahum-Shani et al., 2018; Khalsa et al., 2018).

A randomized crossover can compare assigned, safe conditions in an explicit sequence within each participant. Comparisons after naturally disrupted sleep or before and after clinically indicated pain relief are instead observational within-person contrasts unless condition assignment is part of the design. Both can be useful, but the latter retain event-timing and treatment-selection confounding. Neither design permits induced sleep disruption or delayed relief for Law-specific evidence.

Crossover is not magic. Order effects, carryover, learning, fatigue, expectation, symptom fluctuation, and changing context remain. A person who repeats a task may become less anxious because it is familiar, or more burdened because it is annoying. The design should include washout where appropriate and test order effects.

The candidate mechanism is predeclared. Does the pathway predict reduced avoidance, faster repair action, altered salience, changed report, or different channel lag? If the design only asks whether something changes, it will almost certainly find something. Mechanism requires a pattern, not mere movement.

Within-person designs also support dignity. They avoid treating one person's body as a deviation from another's. The person's own baseline becomes the first comparison. Cross-person generalization comes later and only with transport evidence.

A within-person crossover estimates a local response under specified treatment versions, washout assumptions, and observation windows. Explicit estimands, carryover sensitivity analyses, and no-estimate conditions make the design interpretable.

Crossover designs should also account for meaning change. The second exposure is never identical to the first because the person remembers the first. That memory can be learning, fatigue, dread, boredom, mastery, or distrust. The model should treat repetition as part of the intervention, not merely as a design feature.

20.6.4 — Mechanism-Specific Predictions

A mechanism-specific prediction names the pathway, channel, timing, direction, and failure condition before observing the result. It says, for example, that a safe breathing intervention should reduce respiratory salience before self-reported urgency, while an activity-matched control should not. Or it says that pain relief should alter pain report and sleep before general mood if pain is the mediator (Seth & Friston, 2016; Critchley & Garfinkel, 2017).

Mechanism-specific prediction protects the chapter from biological storytelling. Many channels can be explained after they change. The point is to say what should change first, what should not change, what rival would predict, and what result would make the proposed mechanism less credible.

The prediction should also specify the unit of analysis. Is the expected effect within person, between persons, within site, across devices, across diagnoses, or across states? A result that holds between people may not hold within a person. A result that holds in a healthy lab sample may not travel to palliative care or disability contexts.

A useful prediction includes abstention conditions. If medication changes during the window, if device quality fails, if the participant withdraws, if the channel conflict exceeds a threshold, or if the task becomes burdensome, the mechanism test may become uninterpretable. Declared abstention is not weakness. It is protection against false precision.

Mechanism-specific predictions can be organized in a small matrix: interoceptive inference, allostatic regulation, somatic-marker learning, pain-relief mediation, device artifact, and context change. Each should have unique expectations. If no unique expectation exists, the model is not yet ready for confirmatory testing.

Mechanism-specific predictions make each outcome informative. The protocol states in advance which observations would support the mechanism, which would favor an alternative, and which would call for a better measurement or design.

A prediction table should place the proposed mediator, interoceptive learning, allostatic regulation, and artifact explanations side by side, showing the temporal order expected under each. This makes the comparison interpretable.

Positive affect can also be an explicit target of recovery. In an assessor-blinded randomized trial, Meuret et al. (2026) assigned 98 adults with very low positive affect and impairing depression or anxiety to 15 weekly sessions of Positive Affect Treatment or a comparison therapy focused on negative affect. The positive-affect treatment produced greater improvement in the jointly analyzed measures of positive affect, anhedonia, and depression and anxiety symptoms, with an effect size of $d = 0.27$ for improvement and $d = 0.21$ at one-month follow-up. Both treatments improved self-reported reward anticipation and attainment, with comparable changes in those targets. For this program, the trial supplies a practical model of constructive inquiry: specify a capacity for improvement, compare active approaches, and track experience alongside candidate processes.

The ability to develop rewarding experience becomes a concrete research question with measurable outcomes.

20.6.5 — Negative Controls and Sham Perturbations

Negative controls and sham perturbations match attention, expectancy, movement, and contact without moving the proposed mechanism. They help distinguish a channel-specific response from generic participation effects. The sham must remain credible, safe, and nondeceptive under the protocol's consent rules.

Negative controls are designs in which a matched condition should not move the proposed pathway if the pathway is specific. They are not minor details. In a field filled with correlated bodily signals, negative controls are the difference between mechanism and coincidence (Klasnja et al., 2015; Nahum-Shani et al., 2018; Khalsa et al., 2018).

A respiratory-salience claim might use a movement-matched task that raises heart rate but not breath-related awareness. A wearable-artifact check might use posture change without expected affect change. A pain-relief pathway might compare a nonanalgesic routine procedure that changes attention but not pain. Each control should be chosen for the mechanism, not for convenience.

Negative controls detect device and social artifacts. If a model predicts burden from wrist movement, similar movement in another context tests whether the feature is mere activity. If a voice model predicts distress, fatigued but nondistressed reading can expose prosody confounding.

A good negative control need not be perfect. It must be informative about a plausible false explanation, and that false explanation is named before analysis. Otherwise, control language becomes ceremonial.

When negative controls fail, the result should be taken seriously. If the same signal changes under control and target conditions, the proposed mechanism may be too broad, the channel too nonspecific, or the task too confounded. The response is narrowing, redesign, or retirement, not a stronger paragraph.

Negative controls expose interpretations otherwise assigned to the person. Movement, medication, disability, fatigue, device handling, and cultural expression need controls reproducing the signal without the candidate mechanism. A model failing them is scientifically inadequate before becoming publicly dangerous.

Negative controls should include public-risk controls when possible. A display that makes a model look like mind reading should be tested with nonexpert readers before release. If readers overinfer authority from the visualization, the artifact fails even if the underlying analysis is technically correct.

20.6.6 — No Harmful Provocation

Some experiments would be informative and forbidden. Pain induction, panic provocation, sleep deprivation, social humiliation, deprivation, dangerous breath challenge, medication withholding, or terminal observation pressure could create clean-looking mechanism contrasts. They are outside the chapter's route because they make people carry burden for theory (Seth & Friston, 2016; Critchley & Garfinkel, 2017).

Indirect tests and narrower questions provide useful alternatives when a direct experiment would create harm. Recording which question each design can answer keeps *the research* program both informative and humane.

Safe alternatives exist: observe clinically indicated relief, use mild and reversible attention tasks, rely on naturalistic variation, study routine movement and posture, use archival data under consent, compare device artifacts, and build simulations that do not recruit vulnerable bodies. These alternatives provide practical routes to reproducible and ethically sound evidence.

A protocol should include a provocation audit. Does any task create more distress than everyday minimal risk for the population? Does it amplify symptoms? Does it pressure people to continue? Does it delay treatment? Does it treat withdrawal as failure? If any answer is yes, redesign or stop.

The analysis proceeds by care before contrast. Mechanism science is strongest when it learns from life without making life worse. The person is never the instrument of the theory. When the safest contrast is less statistically efficient, that cost belongs to the design rather than to the participant.

No-harm design forbids implicit pressure to endure. Payment, researcher enthusiasm, scientific or spiritual motives, and fear of disappointing clinicians may press continuation. Consent scripts and staff must make stopping ordinary, not exceptional. Continuation checks should be phrased neutrally and separated from scientific enthusiasm so a participant does not have to resist the project to exercise withdrawal.

Minimal-risk classification must be population specific. A posture change, breath task, or cognitive load mild for healthy adults may be unsafe or exhausting with cardiopulmonary disease, chronic pain, trauma, pregnancy, or disability. Safety review must follow actual participants, not the task label. The protocol should record population-specific contraindications before enrollment rather than discover them through preventable adverse burden.

Stopping thresholds should be observable, participant controlled, and independent of the desired contrast. Symptoms, distress, fatigue, device intolerance, clinical change, or a simple request to stop are sufficient. The analysis should preserve data collected before stopping without treating withdrawal as noncompliance. Stopping-related missingness should be marked by cause and timing, preserving the ethical reason for data loss in later sensitivity analysis.

20.7 — Wearables, Artificial Intelligence, and Humility

Wearables and artificial intelligence are attractive because they seem to bring the body close to continuous evidence. They can record sleep, movement, pulse, rhythm, temperature, voice, prompts, missingness, and context at a scale earlier science could not. That scale is useful. It is also exactly where false authority can grow.

This section treats technology as research support, not sovereignty. A device can help observe a channel. An algorithm can help detect patterns, artifacts, or model failures. A dashboard can help a research team inspect uncertainty. None of these tools knows the person from within, and none may become a public score, care gate, spiritual status, civic trust mark, or substitute for consent.

The local design principle is purpose-limited humility. Collect only what the mechanism needs, document why it is collected, measure device quality, audit performance across bodies and contexts, allow participant challenge, protect deletion routes, and keep the output inside the protocol. A powerful measurement system becomes safer by narrowing its authority.

Table 20.7 converts purpose-limited humility into an auditable governance register. Each technology is assigned one permitted scientific role and one boundary that requires correction, withdrawal, or human review when crossed.

Table 20.7. Technology role-and-boundary register. Role: governance control for device and model outputs.

Technology object	Permitted scientific role	Boundary
Wearable sensor	partial channel observation with quality flags	no direct inner-life inference
Affective model	exploratory or validated classifier under scope	no universal emotion reading
Artificial Intelligence assistant	code, pattern search, documentation support	no autonomous verdict or unsupervised release
Dashboard	internal uncertainty review	no public ranking or eligibility use
Data archive	purpose-limited research record	no archive-as-person or cross-architecture dossier
Participant portal	return, correction, deletion, challenge	no pressure to explain oneself

20.7.1 — Wearable Quality, Context, and Drift

Context and drift are part of wearable quality. A feature validated at rest may fail during movement, illness, temperature change, or a firmware update. The device version and quality thresholds should travel with every derived measure.

Wearable data begin with quality, not interpretation. Contact, battery, firmware, sensor placement, skin properties, motion, perfusion, temperature, edema, sweat, posture, activity, and device version can all shape the signal. A chapter on interoception cannot treat a consumer stream as clean physiology by default (Bent et al., 2020; Nelson et al., 2020; Tabassi, 2023).

Device validation should be context-specific. A sensor may perform well at rest and poorly during movement. It may measure heart rate adequately but fail for rhythm irregularity. It may work differently across skin tones, body hair, perfusion states, tattoos, edema, disability, and clinical contexts. Evidence from one device generation may not travel to another.

The data dictionary should carry device model, firmware, wear location, sampling rate, quality score, artifact detection, nonwear, charge interruption, participant-reported discomfort, and context. These fields can be burdensome, so the protocol should collect what the mechanism needs. But if quality is unknown, interpretation should narrow.

A practical example is a wrist device in palliative care. The band may loosen as weight changes, the person may remove it because it irritates the skin, staff may move it during care, or perfusion may decline. A smooth terminal signal might be physiology, device drift, or both. Quality fields decide whether the graph can speak.

Wearable data should also be checked against alternate channels where feasible. A chest strap, ECG patch, clinical monitor, self-report, or observation may reveal device error. Cross-channel agreement helps evaluate consistency, while disagreement can locate device error or a change in the observation process.

No device output outranks its context. A signal without quality metadata is uninterpretable; a signal with verified quality remains one observation channel whose meaning depends on construct

validity, timing, medication, activity, and the declared context of use.

Wearable validation should be repeated after software updates. A device is not one instrument forever; firmware, algorithms, sensors, and vendor processing can change the data-generating process. Longitudinal studies should treat version change as a scientific event, not an inconvenience in procurement.

20.7.2 — Digital Biomarkers and Clinical Validation

A digital biomarker is a digitally measured characteristic proposed to indicate a biological, physiological, or behavioral process for a defined context of use. The term names a qualification task, not an evidential shortcut. Relevance requires a declared target construct, population, setting, measurement performance, and decision role, and it remains distinct from valence, route state, diagnosis, and inner truth (FDA-NIH Biomarker Working Group, 2016; Coravos et al., 2020; U.S. Food and Drug Administration, 2023).

Fit-for-purpose evaluation separates verification, analytical validation, and clinical or research-context qualification. Verification asks whether the sensor and software operate as specified. Analytical validation asks whether the pipeline estimates the declared physical or behavioral quantity with adequate error under relevant conditions. Context qualification asks whether that quantity is related closely enough to the declared concept and decision for the intended use (Goldsack et al., 2020; U.S. Food and Drug Administration, 2023).

These layers connect sensor reliability, feature extraction, the target construct, and the proposed mechanism. A mechanism claim specifies the supported causal or predictive connection, the intended setting and decision rule, and how uncertainty is propagated through the pipeline.

Qualification belongs to a versioned pipeline. Hardware, firmware, placement, sampling rate, preprocessing, feature extraction, model weights, reference device, quality thresholds, population, and setting define the supported claim. A result therefore travels with its configuration and context of use rather than with a device brand or feature name in general.

The pipeline also needs an explicit route from raw data to action. The DACIA framework separates collection, aggregation, contextualization, interpretation, and action, revealing where a sensor feature acquires meaning and where burden, feedback, or deployment enters. Each transition can fail independently, so classifier performance at the end of the chain cannot certify the earlier measurement and construct links (Daniore et al., 2024).

Representative performance must include conditions that challenge the signal. Skin pigmentation, perfusion, movement, rhythm irregularity, edema, disability, medication, illness, age, environment, and device fit can affect quality. Aggregate accuracy can conceal systematic error in the very subgroup where the mechanism inference is most consequential (Bent et al., 2020).

The causal graph should state whether the digital measure is an observation, mediator, confound witness, quality flag, treatment

modifier, eligibility variable, or outcome. Reusing one feature as both a selection gate and confirmatory outcome can create circularity and collider bias. The role should be frozen before analysis.

Clinical meaningfulness is not the same as statistical separability. A feature may distinguish groups while adding no useful information for a person or decision. The report should state the effect size, uncertainty, decision consequence, subgroup performance, and failure conditions, not only classification accuracy.

Digital biomarkers remain bounded evidence. They can improve cadence, expose artifact, and support safe, purpose-limited studies. When verification, validation, context of use, privacy, or transport fails, the feature is withdrawn or narrowed rather than interpreted more creatively.

20.7.3 — Affective Computing Limits

Affective computing estimates task-defined labels from observable features under a declared context of use. It does not perceive emotion from within, and it does not transform face, voice, posture, physiology, language, or behavior into direct access to lived valence. The target, label source, population, setting, device, and decision use must be stated before performance is interpreted (Picard, 1997; Calvo & D'Mello, 2010).

The ground-truth problem comes first. Self-report, observer annotation, elicitation condition, clinical classification, and behavioral success are different targets. A model trained on one

may predict another through shared context or label leakage while appearing to read emotion. Validation must therefore specify what the label means, who supplied it, when it was supplied, and which uncertainty remains.

Human expression is many-to-many. The same outward pattern can reflect feeling, politeness, pain masking, disability, fatigue, medication, neurodiversity, training, culture, threat management, or social pressure. Conversely, similar experiences may produce different expressions. Context and alternative channels reduce ambiguity but do not abolish it.

Multimodal fusion does not automatically create independent confirmation. Face, voice, movement, and physiology can share the same task cue, annotator, preprocessing pipeline, or missingness mechanism. A larger model may become more confident by combining correlated errors. Channel independence and incremental value must be tested rather than assumed.

A defensible validation program includes calibration, out-of-distribution testing, subgroup and intersectional performance, device and environment shifts, disability and medication conditions, cultural expression variation, missing-channel stress, and abstention. Aggregate accuracy is insufficient when a consequential subgroup carries systematic error or when confidence is poorly calibrated (Autio et al., 2024).

Published benchmarks now make temporal affect prediction a concrete comparison task. SemEval-2026 Task 2 paired ecological essays with self-reported valence and arousal and separated present-state assessment from forecasting change. Its

design included tests involving both previously observed and unseen participants (Soni et al., 2026). A report-channel model can use such a benchmark to test whether apparent accuracy survives future observations and new people, then compare text-based predictions with appropriate history and context baselines. The self-reported labels make forecast performance directly assessable. Existing consent, access conditions, and privacy protections continue to govern reuse.

Useful applications include artifact detection, quality control, exploratory clustering, prompt routing, and task-specific prediction in a demonstrated context of use. Each additional application receives a validation plan matched to its target and intended decision.

Conflict with a participant's report is an identification event, not a contest the device wins by default. The protocol should inspect label mismatch, context, channel quality, timing, medication, communication limits, and model scope. When the channels do not resolve the target, the result is conflict or No adequate estimate.

Public clips, heat maps, and labels should include the model's context of use, calibration, uncertainty, and scope. Standalone review checks whether the intended use and abstention conditions remain understandable when the artifact is viewed separately.

20.7.4 — Artificial Intelligence as Analysis Support

Artificial intelligence may assist feature extraction, artifact detection, code review, simulation, model recovery, sensitivity analysis, and reproducibility under a protocol-fixed, versioned

pipeline. Each output must have one declared analytic role; reused signals cannot silently serve as eligibility criteria, confound controls, outcomes, and explanations in the same analysis.

Record the data version, preprocessing, model or checkpoint, configuration, feature definition, adaptation history, software environment, and human review state. Generated summaries, code, labels, and features are checked against primary data and frozen specifications, and candidate routes and rivals receive comparable computational search and tuning budgets (U.S. Food and Drug Administration, 2025a; World Health Organization, 2025).

Uncertainty and supported-domain checks travel with the output. Human reviewers retain practical authority to reject it, document disagreement, and trigger correction; participant testimony and model output are compared by timing, context, channel quality, and transport rather than status. Privacy, security, withdrawal, and the broader no-AI-sovereignty rule remain governed by Chapter 3 (Autio et al., 2024; Tabassi, 2023).

20.7.5 — Privacy, Security, and Contestability

Contestability requires a person or authorized representative to understand what was collected, challenge an error, request correction or deletion where applicable, and learn how an output affected the study. Security and minimization protect against uses that were never part of consent.

Brain-body data can reveal routines, sleep, pain, medication timing, location, fatigue, relationships, and changes in health.

Privacy and security therefore belong in the study design from the outset (Bent et al., 2020; Nelson et al., 2020; Tabassi, 2023).

Data minimization begins with a channel-by-purpose map. Respiratory timing does not require voice content by default, sleep timing does not require location by default, and device-quality review does not justify indefinite retention of raw streams. Every additional channel must improve a declared estimand, quality gate, or safety function enough to earn its burden and privacy cost.

Security should include access control, encryption, logging, retention limits, versioned datasets, restricted linkage, incident response, and deletion where compatible with the approved protocol. Public release should use aggregated, deidentified, or synthetic displays only after reidentification and hostile-crop review. A beautiful figure can still be unsafe.

Contestability gives the participant a route. They should know what is collected, what is inferred, what is not inferred, who can see it, what can be corrected, what can be withdrawn, and what cannot be changed after analysis. The route should be understandable without technical fluency.

Cross-architecture dossiers remain outside science. Research data cannot be merged into workplace, government, religious, Workgame, donor, clinical, platform, or public-record systems to create a general person file. Purpose-limited linkage to clinical or other research records requires separate authorization and necessity for the approved study. A purpose-limited study becoming a general person file breaks scientific trust.

Contestability should include understandable explanations of uncertainty.

20.7.6 — Mechanism Failure Remains Live

Failure remains active. Interoceptive features may underperform, allostatic measures prove nonspecific, somatic markers transport only within narrow cue classes, wearables prove biased or fragile, Artificial Intelligence features fail out of sample, and medication or illness dominate channels. Any outcome can favor an ordinary rival or retire a mechanism component (Autio et al., 2024; Picard, 1997; Calvo & D’Mello, 2010).

Mechanism failure is classified by the object that failed: identification, prediction, calibration, transport, causal order, construct validity, subgroup performance, device verification, ethical feasibility, or conceptual scope. Each class has its own consequence, from recalibration or reference-class restriction to bridge withdrawal, protocol retirement, or No adequate estimate. Typed failure converts adverse evidence into cumulative design knowledge.

A useful failure-and-abstention rule can be explicit.

Equation 20.4: Failure-and-abstention display. Role: local trigger for abstention pending evidence classification.

$$\text{Abstain}_t = 1 \left\{ \begin{array}{l} q_{\text{signal}}(t) < q^* \vee C_{\text{conflict}}(t) > c^* \\ \vee P_{\text{confound}}(t) > p^* \vee \mathcal{S}_{\text{scope}}(t) = 0 \end{array} \right\}.$$

Apply the canonical evidence states in §§24.2 and 27.4. Low signal, channel conflict, confounding, or scope failure narrows or

blocks only the affected inference; the local diagnostic should not invent a bespoke verdict vocabulary.

Mechanism failure should be published or logged when the protocol permits. A wearable that fails in a disability subgroup, a respiratory task that worsens symptoms, a neural correlate that adds no held-out prediction, or a result explained by medication changes the design obligations of future studies. Negative evidence belongs in the cumulative safety and recovery record, not in an unpublished remainder.

A PTVC-linked interpretation retains only what survives these failures. If ordinary brain-body mechanisms explain the patterns under fair comparison, the route narrows or retires. If a prospectively specified signature survives recovery, perturbation, transport, and strong rivals, the route becomes more answerable without being promoted to the terminal target. Part VII now gives the ordinary rivals full generative force. The route-level result should specify which ordinary mechanism was defeated, because survival against one rival does not imply survival against the rest of the envelope.

The chapter's final rule is practical: devices witness, bodies participate, models estimate, and people remain more than any channel. A science worthy of felt experience must know when another measurement would add knowledge and when putting the instrument down is the more rigorous act. Burden-adjusted information gain can make that judgment explicit by comparing expected inferential value with the participant cost of another channel.

Graceful nonuse is a result. A promising mechanism may be burdensome, privacy-invasive, fragile, clinically disruptive, or easily misused. Science can preserve it as a bounded research question while refusing deployment or public-artifact travel. A nondeployment verdict should state which scientific uses remain permitted, preventing a useful laboratory result from drifting into operational recommendation.

A failure matrix links each failure class to a defined next action. Verification failure stops use of the device output; construct failure withdraws the measured-to-latent bridge; causal failure narrows the object to prediction; transport failure restricts the reference class; ethical failure retires the protocol; open-set failure expands the model library. Typed response preserves more information than a generic unfavorable label. This typed matrix also makes cumulative review easier because later evidence can reopen only the failure class it actually addresses.

Successor mechanisms begin from failed assumptions and unresolved residuals. A new neural or Artificial Intelligence model needs a version identity, recovery benchmark, qualification plan, privacy review, and public-status ceiling. The successor should identify the exact residual pattern the new mechanism is designed to explain so added complexity remains accountable.

Part VII — Rival Models, Inference, and Adversarial Tests

A consequential hypothesis earns scientific trust by entering a contest it can lose. Part VII constructs that contest. Earlier Parts defined PTVC, separated target, route, observation, and verdict, developed measurements and candidate mechanisms, and specified first tests. Here the candidate faces ordinary psychological, physiological, computational, evolutionary, and statistical generators in closed-world, open-set, and transported comparisons (Popper, 1959; Mayo, 2018; Wilson & Collins, 2019).

Rival models are engines of discovery rather than obstacles to the project. Adaptation, opponent processes, homeostatic regulation, allostasis, predictive learning, selection, conditioning, missingness, and model flexibility can each explain patterns that might otherwise be attributed too quickly to PTVC. When one of them wins under common rules, the scientific map becomes clearer and the next study becomes better targeted.

Part VII compares baseline return, terminal-window observations, and cumulative trajectories at independently defined endpoints. Their separate predictions make it possible to identify the information added by each account.

Ordinary life already contains recovery, habituation, numbing, endurance, prediction, regulation, and reorganization. The same outward quiet can follow healing, exhaustion, lost language, sedation, strategic silence, or reduced observation. A rigorous science distinguishes those paths instead of reading one visible outcome as a complete account of lived experience.

An adaptation model can explain reduced reporting while leaving hidden burden unresolved. A missingness model can explain a smooth record by showing who became harder to observe. A treatment model can explain a terminal change without invoking a terminal law. Each explanation keeps only the credit its design and evidence earn.

Equal treatment governs the tournament. Each serious rival declares latent states, parameters, observation kernels, endpoint relations, uncertainty, and protected predictions. It receives the same data, tuning budget, calibration effort, and scoring opportunity as the PTVC candidate. The comparison also leaves room for every listed model to fail, because closed-world accuracy can manufacture a winner when the relevant generator is missing (Gneiting & Raftery, 2007; Vehtari et al., 2017; Wilson & Collins, 2019).

Chapter 21 begins with ordinary regulation because it is the nearest and most credible source of PTVC-like patterns. Adaptation, opponent responses, homeostasis, allostasis, and Ornstein-Uhlenbeck dynamics can produce local return, damped fluctuation, rebound, and apparent concentration. The integrated-process analysis then shows why a returning state and a closing cumulative path remain different mathematical objects.

Chapter 22 moves into prediction, learning, and control. Predictive processing, active inference, reinforcement learning, safe control, bridge constructions, and event-conditioned selection make distinct commitments about states, observations, policies, rewards, preferences, uncertainty, and endpoints. The chapter turns those commitments into common-observation

competitors rather than treating neighboring vocabularies as borrowed support (Lageman et al., 2026; Samara, 2026).

Chapter 23 tests the outer boundaries of the claim. Evolution can favor function without comfort. Population summaries can reverse within-person relationships. Nonlinear systems can mimic convergence. Cross-species, developmental, and synthetic-system evidence can travel only through declared bridges. Recent work on nonergodicity makes the within-person and between-person distinction especially concrete (Hunter et al., 2024; Mistry et al., 2026).

Chapter 24 turns inference into an explicit dependency chain. Unit, target population, estimand, estimator, scientific acceptance region, epistemic uncertainty, observation kernel, delayed entry, competing events, sensitivity analysis, and sequential rule jointly determine what a number can mean. A valid analysis preserves the path from latent question to permitted verdict and reports No adequate estimate when that path cannot be completed (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Chapter 25 tests the entire pipeline in worlds where the generator is known. Parameter recovery, model recovery, simulation-based calibration, open-set detection, transport stress, challenge logs, and reproducible artifacts expose false confidence before human data receive strong interpretation. Nonstationary recovery work and posterior simulation-based calibration show how sophisticated pipelines can misclassify or miscalibrate. Adversarial collaboration shows how advance agreement can sharpen theory comparison while also revealing the practical difficulty of a genuinely shared test (Schumacher et al., 2025;

Säilynoja et al., 2026; Cogitate Consortium et al., 2025; Ceci et al., 2024; Isch et al., 2025).

Across the series, scientific findings should retain the component and setting in which they were established. This lets the private-life, religious, institutional, legal, peace, and transition volumes draw on a clear and consistent scientific record.

The equations, simulations, tables, and decision rules in this Part are inspectable tests of assumptions and predictions. They test whether a pipeline can recover known generators, predict protected observations, transport across sites and instruments, survive informative missingness, separate target from observation artifacts, and recognize when every offered model is wrong. The results show where a pipeline is ready for use and where it can be improved.

Public communication should name the mechanism that best explains a pattern, whether adaptation, learning, regulation, selection, conditioning, observation, or statistical artifact. This gives later research a precise result to build on.

Part VII gives strong opponents executable models, exposes the preferred account to common rules, and treats adverse results as knowledge. Its reward is not automatic victory, but a research program capable of learning which component, reference class, and evidential stage the data can sustain (Munafò et al., 2017; Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

Chapter 21 — Adaptation and Homeostasis

Adaptation is familiar, powerful, and easy to overread, which makes it the first major rival to PTVC-specific interpretation. Injury, bereavement, relief, promotion, medication, threat, and care can move affective state, reference range, behavior, and report process on different clocks. Chapter 21 asks how much apparent balance those ordinary processes can generate before any terminal-boundary structure is needed (Solomon & Corbit, 1974; McEwen, 1998; Diener et al., 2006; Kuppens et al., 2010).

Observable recovery can reflect restored function, reallocated attention, revised expectations, or a genuine reduction in distress. It can also reflect exhaustion, learned silence, response shift, limited access to care, or a burden that has become ordinary. The analysis therefore separates latent state, appraisal, memory, behavior, and observation closely enough to distinguish recovery from invisibility.

PTVC asks a stronger question than whether people often recover. It concerns a cumulative phenomenal object within a qualifying conscious stream and a prospectively specified endpoint architecture. Adaptation, opponent processes, homeostasis, allostasis, treatment, frailty, observation loss, and mean reversion receive the first opportunity to explain each visible pattern; only a residual restriction that survives that suite remains PTVC-specific.

The rival field is generative rather than rhetorical. Set-point models specify reference ranges and return rates. Opponent models specify delayed counterresponses. Homeostatic and allostatic models regulate present and anticipated demand. Ornstein-Uhlenbeck families supply continuous-time baselines,

while integrated OU analysis exposes the difference between state return and accumulated-path behavior.

Many favorable-looking signatures can arise from ordinary regulation or ordinary observation. Variance may narrow under mean reversion, reports may quiet through habituation, and terminal records may smooth when treatment, frailty, selection, sensor noise, or measurement cadence changes. Those alternatives should be simulated through the same observation model before any curve is assigned target-level meaning.

Adaptation describes how people reorganize around changing circumstances. Understanding the effort and support involved helps distinguish recovery, accommodation, and continuing needs for care.

Keep one image in mind: a curve returns toward its reference line while the accumulated area beneath it continues to change. The current state has recovered, but the experienced interval remains part of the path. A state model and an integrated ledger therefore answer different questions, and the chapter makes their difference visible in equations and simulations.

Formulas enter because prose can blur the status of a result. The OU equation is a rival generator. The integrated OU variance expression is a mathematical consequence of declared assumptions. Simulations are teaching devices and protocol fixtures. None becomes empirical evidence about lives until real data, prospective specification, observation models, and held-out comparison enter.

Rivals are not treated as simple. Adaptation may be incomplete, asymmetric, event-specific, culture-shaped, report-dependent, and different inside a person than across people. Opponent processes may overshoot, fade, strengthen, or sensitize. Allostatic regulation may protect in the short run while imposing physiological cost over time. Homeostasis may preserve viability without preserving felt fairness.

Prediction and policy learning begin where local regulation ends. Adaptation, opponent processes, allostasis, homeostasis, OU dynamics, integrated ledgers, and their hybrids belong here. Predictive processing, active inference, reinforcement learning, and safe control receive their own generative treatment in Chapter 22, where policies and learned world models require different machinery. Keeping these rival families separated by mechanism prevents a broad word like regulation from hiding differences in information, objectives, and learning dynamics.

Two distinctions should remain after the details fade: recovery can be genuine without being terminal closure, and a rival victory can be genuine progress. Giving ordinary regulation full strength makes the surviving scientific question smaller, clearer, and more worth testing. This also protects the positive case by forcing any remaining PTVC-specific claim to explain structure that ordinary regulation demonstrably does not.

The field expert should finish with a stricter demand. Every adaptation claim must declare its state, baseline, timescale, observation model, individual heterogeneity, missingness structure, and comparison rule. Every OU comparison must distinguish state mean reversion from integrated path behavior.

Every claim of residual PTVC-specific structure must be tested against generated data where the rivals are allowed to be true. Parameter uncertainty and model misspecification should travel with those declarations, because a precisely written rival can still be a poor approximation to the data.

Scientifically, a successful adaptation model narrows the PTVC-specific interpretation; a failed adaptation model opens a stronger comparison rather than a victory declaration. Every curve carries only the meaning its design and evidence support. A residual pattern earns attention only after the adaptation model has been tuned and tested under the same information constraints as the candidate.

21.1 — Adaptation as a Leading Rival Family

Adaptation is the first rival because people often move toward a recognizable range after gain, loss, shock, relief, illness, disappointment, or safety. That return may reflect repair, changed expectation, altered environment, response shift, selective memory, or reduced reporting. A scientifically useful model must distinguish those pathways instead of treating every quieter trajectory as the same process (Brickman et al., 1978; Diener et al., 2006; Lucas, 2007a; Wilson & Gilbert, 2008; Kuppens et al., 2010).

Adaptation is treated as a model family, not a slogan. Each member declares a latent state, moving or fixed reference, return dynamics, event class, treatment path, memory component, report process, and uncertainty. Person-specific baselines, asymmetric gains and losses, incomplete return, and response shift belong inside the rival when the data can identify them.

The central question is what the family explains when the target is cumulative felt exposure and endpoint-sensitive structure. A model that predicts state return may not predict the integral; a model that predicts reports may not recover the private interval; a model that fits remembered evaluation may not describe momentary experience. These distinctions define the residual rather than merely qualifying it.

Two errors should be visible from the start. The first is assuming full return because a later report looks ordinary. The second is assuming PTVC-specific structure because ordinary return is

incomplete. Between those errors lies the actual work: declare the adaptation model, test its channels, and ask what remains.

Adaptation is also where institutions can hide behind human resilience. A clinic, school, insurer, employer, or family can point to reduced expression and treat the problem as resolved. The scientific comparison keeps adaptation strong while refusing to let reduced expression erase the original burden.

21.1.1 — Set-Point and Baseline Accounts

Set-point and baseline accounts explain returns toward ordinary feeling without invoking terminal structure. They posit a person-specific reference range, a disturbance, and relaxation toward that range. Bereavement, promotion, injury, or relocation can therefore produce a curve that resembles repair. A prospective comparison asks whether the declared model predicts the trajectory and what residual remains after uncertainty, reporting, treatment, and context are handled (Brickman et al., 1978).

Set-point accounts become credible when their apparent simplicity is relaxed. The implementation should allow return rates, plateaus, treatment effects, and reporting channels to differ across people and contexts. This gives the comparison a realistic account of adaptation against which additional trajectory predictions can be evaluated.

Consider a job loss followed by partial mood return while debt, shame, and sleep disruption remain. A recovered daily mean does not imply that every burden has resolved: variance, avoidance, opportunity, and bodily strain may follow different trajectories. The set-point account should receive credit only for the features

it predicts, while any proposed residual must be tied to an independently measured feature rather than inferred from what the rival leaves narratively unexplained.

A useful display declares the reference range, shock, return rate, shifted plateau, observation noise, and several admissible paths, including asymmetric or incomplete recovery. Its caption identifies the display as a rival simulation and records the generating model and parameter values.

This mechanism matters because it prevents resemblance from becoming evidence. A curve that looks like repair may have been produced by local regulation, scale change, treatment, or observation loss. Formalizing the rival turns that resemblance into a contest that the preferred theory can lose.

The humane boundary is simple and specific. A rival win can narrow a theory, but it cannot narrow a person's claim to care, privacy, or serious hearing. Endurance, quiet behavior, and changed reports remain facts to interpret cautiously, not permissions to withdraw support.

A runnable comparison must state the data, estimable parameters, discriminating pattern, and candidate-narrowing rule. That precision gives ordinary mechanisms a real contest rather than ceremonial mention.

21.1.2 — Incomplete and Asymmetric Adaptation

A useful adaptation model allows both lasting gains and persistent effects of adverse events, with different rates and limits for each. This gives the comparison a realistic account of human change

and accommodates the diversity of trajectories documented in longitudinal research (Diener et al., 2006; Lucas, 2007a).

Quantitative evidence supports a differentiated and hopeful picture of adaptation. A meta-analysis of 188 publications, comprising 313 samples and 65,911 participants, found that affective and cognitive wellbeing respond differently to life events and follow varied trajectories (Luhmann et al., 2012).

Lindqvist, Östling, and Cesarini studied 3,362 Swedish lottery players 5–22 years after prize events, using randomized prize variation within comparable groups and preregistered analyses. An additional after-tax prize of US\$100,000 in 2011 dollars was associated with approximately 0.037 standard deviations higher overall life satisfaction. The benefit persisted; effects on happiness and mental health were smaller and not statistically distinguishable from zero (Lindqvist et al., 2020). Better circumstances can produce lasting gains, and following evaluative life satisfaction and momentary affect separately helps identify how those gains develop.

Baselines may shift, responses may differ by sign, plateaus may persist, and treatment or context may alter the return process. Representing those variations gives the comparison a realistic adaptation model.

A chronic-pain participant may learn pacing well enough to restore a stable daily routine while continuing to carry vigilance, sleep disruption, and constrained movement. That record distinguishes functional adaptation from disappearance of burden. The model must say which outcome returned, which

remained displaced, and whether treatment, expectation, or reporting practice explains the difference.

Before fitting incomplete and asymmetric adaptation, the protocol must name the state, event class, time window, observation channel, baseline rule, missingness rule, and loss function. Rival and candidate then face the same estimand, so neither receives a cleaner object after the result becomes emotionally attractive.

If the adaptation model explains held-out trajectories with fewer assumptions under the shared observation model, explanatory credit belongs to that model. Any additional PTVC prediction is then evaluated against the explanatory scope established by the comparison.

Public reporting should name incomplete or asymmetric adaptation at the scope supported by the observations. Captions and summaries should preserve the mechanism, population, uncertainty, and limits of the result, and distinguish reduced expression from changes in the underlying experience.

The acceptance gate is implementability. A reviewer should be able to name the state variable, time unit, observation channel, parameter, and rival-win condition without guessing. Specifying those pieces turns the proposed explanation into an executable model for direct comparison.

21.1.3 — State, Report, and Memory

State, report, and memory must remain separate because adaptation can occur in one while another remains altered. Calm reporting may reflect changed expression; later recall may reconstruct an event; life evaluation may diverge from sampled

affect. Candidate-instrument and interval-contribution studies therefore treat reports as timed, contextual evidence about lived experience. Comparing immediate reports with later recollection helps identify which layer changes and on what timescale (Lucas, 2007a).

A credible layered rival permits latent affect, immediate report, later recall, treatment, context, and memory access to adapt at different rates or not at all. The candidate is compared with that layered system, not a single curve standing in for the person.

Consider a widower who reports daytime stability but avoids places saturated with loss. Daytime report, place avoidance, autobiographical memory, and physiological response need not move together. The analysis must distinguish genuine affective adaptation from strategic reporting, context-specific triggering, and memory reconstruction, with each account assigned an observable test rather than a retrospective story.

Disagreement among state, report, and memory calls for a layered observation plan, not indiscriminate surveillance. Sampling cadence should follow the expected lag between immediate affect, public report, and later recollection, while the protocol preserves the person's right to decline a channel. Dense data gathered at the wrong interval can still miss the process the rival proposes and can increase burden without increasing identification.

The layered comparison evaluates predictive accuracy, missingness, and complexity together. Its result identifies how

much of the trajectory is explained by adaptation and which additional path restrictions, if any, remain to be tested.

A reproducible state-report-memory analysis should expose three linked equations or modules: the latent state transition, the contemporaneous report kernel, and the later-memory kernel. Recovery must be demonstrated under channel disagreement and missing reports, then scored on protected sequences. Adaptation is favored when that layered model predicts the observations without a PTVC-specific path restriction; unresolved layer identity yields ambiguity rather than a forced verdict (Brickman et al., 1978; Diener et al., 2006; Lucas, 2007a; Wilson & Gilbert, 2008).

21.1.4 — Within-Person Dynamics

Within-person dynamics make adaptation testable over time. Cross-sectional averages can make people appear to return toward baseline while individual trajectories diverge, reverse, or never recover. The proper unit is the observed stream over time, with person-specific baselines, changing contexts, and measurement error declared in advance. Held-out prediction then tests how much of the trajectory the within-person adaptation model explains and which additional patterns warrant comparison (Wilson & Gilbert, 2008; Brickman et al., 1978).

The adaptation model can represent recovery, partial recovery, changing standards, and persistent alterations. Formalizing those features creates a clear basis for comparing additional PTVC predictions.

The concrete unit is a sequence of observations from the same person surrounding a declared event, with the pre-event range, response, return path, and later context kept on one clock. Such a record can estimate heterogeneity without averaging away the individual dynamics the rival is meant to explain. It also shows where missing intervals prevent a confident account of adaptation.

Within-person dynamics require operational definitions of return, plateau, overshoot, persistence, and transport before fitting begins. Those features should be estimated on a declared clock and evaluated across event classes rather than inferred from a smoothed line. Parameter-recovery analysis then tests whether the model can be estimated from irregular, noisy observations.

Later summaries should identify the within-person model that best explained the trajectory and retain the result's tested setting and scope.

21.1.5 — State Recovery and Accumulated Experience

Adaptation changes the current state and the path that leads to it. After an event, recovery can coexist with delayed effects, altered opportunities, and changed constraints. Recording both the returning state and the accumulated signed area makes these features available for direct model comparison.

A complete adaptation model can combine local recovery, persistent path area, changing standards, and context-dependent reporting. Including those features gives the comparison a realistic account of human change.

A student may return to class after sustained bullying and appear behaviorally recovered while avoiding corridors, friendships, or activities associated with the harm. The example tests whether local functioning, felt safety, memory, and cumulative exposure move together. It does not assume hidden damage; it requires the rival to specify which layer it predicts and how the unobserved interval is represented.

A two-panel display can place momentary state above the signed cumulative area generated by that state. The upper curve may return toward baseline while the lower curve retains the area contributed during the disturbance. This visual distinction is the scientific point: local adaptation and terminal closure are different objects even when they share an attractive endpoint picture.

The value of the rival is that it can make predictions before the Law enters the room. A return rate, lag, plateau, or variance pattern can be estimated, simulated, and scored. If that machinery explains the relevant structure, the chapter has learned something important even when the PTVC-specific reading becomes smaller.

When local recovery without terminal closure explains the observations, public language should say exactly that. The result should not be enlarged into a universal claim, softened into covert support, or used to reinterpret accumulated burden. Naming the ordinary mechanism precisely is part of scientific honesty.

The decisive test freezes an adaptation family on training events and asks it to predict both later state trajectories and the accumulated path proxy in protected events. Baselines, return

rates, asymmetry, response shift, treatment, and observation error must be estimated without terminal leakage. When local adaptation explains the held-out structure and no additional PTVC restriction survives, the candidate interpretation narrows to the residual it can actually defend (Brickman et al., 1978; Diener et al., 2006; Lucas, 2007a; Wilson & Gilbert, 2008).

21.1.6 — Conditions for Rival Victory

Adaptation becomes a genuine adversary when victory conditions are fixed before outcome inspection. It wins this component when the protocol-fixed comparison rule favors the prespecified set-point, asymmetric, and within-person models over the PTVC candidate under the same observation process. It also wins when baseline recovery, report change, care context, selection, medication, or measurement loss generates apparent terminal concentration (Diener et al., 2006; Lucas, 2007a).

The rival family must be able to win in practice. Heterogeneous baselines, asymmetric gains, event classes, treatment, and observation error belong in the prespecified adaptation family. Only a strong ordinary account makes the contest meaningful.

Imagine a prespecified adaptation model that outpredicts a more elaborate PTVC candidate in a protected cohort and stays calibrated across event classes. Protocol-fixed loss, complexity allowance, sensitivity plan, and held-out prediction rule determine the consequence, not which account inspires more. If no distinctive transported restriction remains, the PTVC-specific component narrows.

Conditions for rival victory belong in the estimand and scoring plan before outcomes are seen. State, event, window, channel, baseline, missingness process, and loss function remain common across models. Otherwise victory is vulnerable to a quiet change in what the analysis was asked to estimate. The scoring plan should also fix how complexity and calibration are penalized, so flexibility cannot be converted into an unpriced advantage after fitting.

The victory rule should be executable from the protocol alone. It identifies the common estimand, candidate and rival families, tuning budget, complexity adjustment, protected split, uncertainty rule, and score margin. A tie or unstable advantage remains Model-ambiguous; a reproducible rival advantage triggers the prospectively specified narrowing or retirement rather than a new post hoc contest (Brickman et al., 1978; Diener et al., 2006; Lucas, 2007a; Wilson & Gilbert, 2008). Executable rules should be tested on synthetic ties and near-ties so unstable ranking behavior is known before real outcomes create pressure to choose.

A practical adaptation protocol therefore preserves at least three distinguishable objects: current state or report, the slowly moving reference against which change is appraised, and the cumulative exposure history. The first describes the present observation, the second describes learned normality, and the third preserves what was carried while the baseline moved. Conflating them lets adaptation explain everything and identify nothing. Separating these objects also allows adaptation to be modeled as a change in reference without retroactively erasing cumulative exposure.

This is where the Law of Fairness question becomes precise. It asks not whether people recover, but whether local recovery curves, response shifts, and incomplete adaptation exhaust the stronger pathwise patterns the formal hypothesis requires. Comparing strong rivals identifies the features they explain and the additional structure a PTVC model predicts. The most informative residual is one predicted prospectively by PTVC and missed by well-calibrated adaptation rivals, not one discovered after inspecting their errors.

21.2 — Opponent Processes and Local Counterweights

Opponent-process theory deepens the rival field by specifying a delayed counterresponse after a primary affective movement. Pleasure may be followed by flatness, danger by relief, repeated exposure by tolerance, and withdrawal by an acquired opposite state. These dynamics can resemble balancing, but their domain is an event history with identifiable lags and gains, not a complete-stream terminal guarantee (Solomon & Corbit, 1974; Koob & Le Moal, 2001).

The key science is temporal and multichannel. The model asks how quickly the primary response rises, when the opponent response begins, whether repetition changes either component, and how physiology, report, behavior, and memory reveal different portions of the process. Sampling cadence must resolve those differences rather than averaging them into one smooth curve.

A simple response model makes the area question explicit. Let

$$h(t) = Ae^{-at} - Be^{-bt}, \quad A, B, a, b > 0, \quad t \geq 0.$$

The integrated response is

$$\int_0^{\infty} h(t) dt = \frac{A}{a} - \frac{B}{b}.$$

The zero-area case therefore imposes the relation $A/a = B/b$. Amplitudes and time constants can be estimated from the response profile and used to compare this constrained case with an unconstrained two-component model. Repeating that comparison across settings turns an apparent counterweight into

a quantitative question about response shape, duration, and shared parameters.

21.2.1 — Primary and Opponent Responses

Primary and opponent responses provide a temporal model of counterweights. An initial affective response may be followed by a delayed movement in the opposite direction, producing rebound, overshoot, or settling. The comparison specifies both response kernels, their lags, and the observations needed to recover them (Solomon & Corbit, 1974).

Local counterresponses are measured phenomena. In three human psychophysical studies, Leknes et al. (2008) found that relief depended on preceding pain and its decline, with cooling increasing the intensity, pleasantness, and duration of relief in the tested conditions. Franklin et al. (2013), studying forty participants, used separate startle-response indicators to show that pain offset could both increase positive affect and reduce negative affect. The timing and intensity conditions mattered. Together, the studies justify measuring relief in its own right, with positive and negative channels distinguished.

A later study of 122 healthy adults measured pleasant relief for four minutes after an aversive stimulus ended. Relief differed across groups identified from their preceding pain trajectories, making person-specific response dynamics empirically tractable (Henri et al., 2022). These timing and individual-difference findings provide concrete targets for comparisons using existing data and observations during timely pain relief.

An opponent-process model can accommodate primary responses, delayed counterresponses, tolerance, withdrawal, and channel-specific timing that vary across people and events. Those features define the baseline against which additional trajectory restrictions can be compared.

Consider relief after a threat ends, followed by shaking, fatigue, or an unfamiliar calm. The primary response may stop before the opponent response reaches its maximum, so the visible sequence should be analyzed for timing, sign, and dose dependence rather than treated as one undifferentiated recovery curve. A fairness-specific route becomes relevant only if it predicts a prospective feature that a well-specified opponent process does not reproduce.

Opponent responses are clarified by measurements timed to the proposed mechanism, not by collecting every available signal. The sampling grid should resolve the primary rise, delayed counterresponse, and any rebound after exposure ends; a cadence too coarse to separate those phases cannot adjudicate the model. Channel choice should follow the hypothesized process, with burden minimized when additional sensors would add redundancy rather than identification.

Primary and opponent responses are tested through a frozen impulse-response model across repeated exposures. The analysis should recover primary amplitude, opponent lag, rebound size, and dose dependence under the actual sampling and observation process, then predict a protected exposure sequence. A stable delayed counterresponse favors the opponent account; failure to recover or distinguish the kernels leaves the mechanism unresolved (Solomon & Corbit, 1974; Koob & Le Moal, 2001).

A complementary circuit result comes from rats with postsurgical pain. Local-anesthetic blockade of input from the injury produced preference for the associated location, activated ventral tegmental dopamine cells, and increased dopamine release in the nucleus accumbens; blocking accumbens dopamine signaling prevented the place preference (Navratilova et al., 2012). This connects relief to a causal reinforcement pathway and gives the human findings a mechanistic comparison. Temporally defined relief, behavior, and circuit activity can therefore be studied together.

Safety can acquire a learned biological value that sustains future action. Macdonald et al. (2026) identified a pathway from the paraventricular thalamus to the nucleus accumbens that developed a response when mice successfully avoided threat. The signal weakened when the action's safety value was reduced. Silencing the pathway at safety onset reduced subsequent avoidance and willingness to work for safety; computational fits attributed the behavioral change primarily to lower estimated safety value. The researchers also identified plasticity at connections to accumbens cholinergic interneurons that helped regulate dopamine release. This provides a concrete mechanism through which an improved situation gains motivational value. It gives the book's relief models measurable components: learned safety, outcome valuation, circuit plasticity, and the timing of future action.

21.2.2 — Habituation, Tolerance, and Withdrawal

Habituation, tolerance, and withdrawal show how repeated exposure changes the state space itself. A stimulus that once carried strong affect may fade with repetition, while the absence

of a familiar stimulus can become its own disturbance. These mechanisms can produce relief, craving, numbness, agitation, or return toward ordinary functioning. Their local process models describe how response patterns change with exposure history (Koob & Le Moal, 2001; Solomon & Corbit, 1974).

Habituation, tolerance, and withdrawal enter the contest in their strongest ordinary form. They can explain diminished response, escalating dose, delayed rebound, and discomfort after cessation without a terminal-boundary mechanism. The remaining question is whether a prospectively specified candidate predicts the rebound, withdrawal, and cross-timescale structure more accurately on protected data.

Repeated analgesic exposure can reduce one response while a delayed withdrawal or rebound process grows. The measurable changes are in amplitude, lag, and recovery across exposures. Modeling treatment indication and dose history helps distinguish treatment-related effects from other sources of a counterresponse.

Habituation, tolerance, and withdrawal must be implementable rather than merely respectable. Parameters need units, observations need an error model, and all candidates must share participant windows, censoring rules, and endpoint definitions. This prevents inherited prestige from deciding the comparison.

21.2.3 — Timescale Dependence

Timescale dependence prevents opponent-process language from becoming a single curve stretched across every phenomenon. Some counterweights arise in seconds, others across days, seasons, treatment cycles, grief processes, or years of exposure. A

model that fits daily self-report may fail at minute-level physiology, and a terminal pattern may disappear when sampled at the wrong cadence. The chapter therefore asks each rival to declare its clock before its success is interpreted (Solomon & Corbit, 1974).

Fast regulation, slow learning, seasonal context, and developmental change can produce different apparent baselines. A multiscale model specifies their joint predictions and tests whether those relationships carry across new observation windows.

A process sampled hourly can look monotone while minute-level measurements reveal overshoot and delayed return; an excessively dense record can also magnify transients irrelevant to the proposed kernel. Timescale analysis should compare prospectively specified resolutions and aliasing consequences with selection rules fixed before model comparison.

The timescale figure should overlay one opponent-process trajectory under several prospectively specified sampling grids and lag assumptions. Mark which rebound features each cadence resolves, aliases, or misses, so the figure directly informs study design.

Sampling resolution affects which mechanism can be identified: a coarse window may resolve adaptation while a finer window reveals rebound. Report the result at its prospectively specified resolution (Solomon & Corbit, 1974; Koob & Le Moal, 2008).

Expert disagreement here should be localized to the response kernel, sampling grid, lag distribution, and criterion for cross-

timescale transport. Reviewers can then ask whether apparent rebound is aliased, whether the prior over delays is influential, and which additional window would separate the models. The dispute becomes a finite design problem rather than a referendum on the entire hypothesis (Solomon & Corbit, 1974; Koob & Le Moal, 2008).

21.2.4 — Event-Specific Versus Global Processes

Event-specific and global processes must not be collapsed. A person can adapt to a commute, medication side effect, or repeated social cue while remaining deeply altered by grief, coercion, illness, or chronic fear. Local counterweights can make one channel look repaired while the broader stream still carries pressure elsewhere. The model must state whether it explains a discrete disturbance, a recurring context, or a global affective tendency (Koob & Le Moal, 2001; Solomon & Corbit, 1974).

A useful contrast places bereavement, injury, reward loss, and repeated aversive exposure inside one hierarchical model while allowing each event class its own kernel. If a common opponent component transports across events, the global account gains credibility. If the parameters remain event-bound, the scientific conclusion should stay local rather than becoming a lifetime mechanism.

Test the local-versus-global distinction with a hierarchical kernel model and leave-one-event-class-out prediction. If event-specific kernels transport but a constrained shared component does not, retain the local mechanisms and retire the global counterweight claim (Solomon & Corbit, 1974; Koob & Le Moal, 2001).

21.2.5 — Generative Simulation

Generative simulation is included to make the rival executable. The chapter can specify an event, a primary response, an opponent response, a delay, a decay rate, and an observation model, then ask whether that system reproduces apparent repair. The useful result is a recoverable model that can later be compared against data under common rules (Solomon & Corbit, 1974).

Generative simulation converts verbal assumptions into trajectories before the human data are interpreted. By varying opponent delay, tolerance, exposure schedule, observation error, and missingness, the benchmark can reveal which signatures are recoverable and which are shared by several mechanisms. A route that cannot be distinguished in its own simulated worlds is not ready to explain an observed one (Wilson & Collins, 2019; Talts et al., 2018).

The simulator should generate repeated exposures with controlled primary amplitude, opponent lag, tolerance, withdrawal, observation noise, and missingness. Analysts then fit the prospectively specified rival families without access to the generator label. Recovery across near-miss settings, not recognition of an idealized rebound, is the evidence that the planned test can discriminate.

A successful recovery benchmark shows that the pipeline can recognize the rival under the simulated conditions. It does not establish that the rival generated human observations. If the rival subsequently wins on empirical prediction, calibration,

parsimony, and transport under the common rules, the PTVC-specific interpretation loses that component.

Simulation outputs remain synthetic design objects. Generator parameters and recovery scores do not transfer into participant profiles or eligibility decisions.

The simulation gate requires the complete observation pipeline to recover opponent, adaptation, and null generators across realistic noise, missingness, and parameter overlap. Calibration, family confusion, and abstention rates should be reported by regime. A mechanism label is not carried into human data when the pipeline cannot identify that mechanism in synthetic worlds built to resemble the planned study (Solomon & Corbit, 1974; Koob & Le Moal, 2001).

21.2.6 — Rival-Unique Predictions

Opponent-process models predict overshoot, aftereffects, sensitization, tolerance, withdrawal, and lags. Their distinctive patterns can be tested at the relevant timescale with negative controls and protected data (Koob & Le Moal, 2001; Solomon & Corbit, 1974).

A rival-unique prediction must survive heterogeneity rather than depend on an idealized average. Event class, channel, exposure history, treatment, and baseline can alter both the magnitude and timing of an opponent response. The candidate Law should face the strongest version of that prediction before any residual pattern is called distinctive.

Consider a delayed aftereffect that grows across repeated exposures. Sensitization, tolerance, anticipatory learning, and

changes in reporting can all produce such growth, but they imply different relations among dose, latency, and recovery. A rival-unique prediction must therefore specify which relation should change first and how that prediction will be scored on exposures not used to tune the model.

Rival-unique predictions need a cadence matched to the proposed delay, rebound, and dose history rather than an ever-denser record by default. The protocol should specify which channel is expected to move first, how long the counterresponse should persist, and what pattern would be absent under a simple adaptation model. A record that does not resolve those temporal contrasts cannot be treated as a test of opponent dynamics.

Before protected data are examined, specify the expected channel order, lag, rebound, repetition effect, and decay after exposure. Evaluate that joint temporal pattern out of sample against calibrated alternatives with the same information, then report which discriminator succeeded, remained unresolved, or failed. Any broader conclusion remains limited to the predictions entailed by the tested branch (Solomon & Corbit, 1974; Koob & Le Moal, 2001; Koob & Le Moal, 2008). A gain difference and a lag difference should receive separate estimates, since the same peak can arise from different trajectories.

Opponent-process studies estimate two time constants: one for the primary response and one for the delayed counterresponse. Both can change with repetition, so habituation and a growing delayed component can make the same event feel different across exposures. Recording exposure history connects those changes to the observed response. This calls for observations before, during,

and after the primary response rather than a single post-event rating.

Opponent responses can change how repeated pleasure, relief, or absence is experienced. Their timing and persistence make aftereffects useful targets for clinical observation and model comparison. Follow-up after exposure ends can distinguish a persistent counterresponse from a brief change in reported intensity.

21.3 — Homeostasis, Allostasis, and Mean Reversion

Homeostasis and allostasis are stronger rivals than simple rebound because they model systems that actively regulate current and anticipated conditions. Temperature, glucose, sleep pressure, arousal, inflammation, pain responses, and threat vigilance move around ranges, thresholds, and predicted demands. Those processes can generate stability, recovery, and variance change while optimizing viability rather than terminal phenomenal closure (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022; McEwen & Wingfield, 2003).

PTVC targets a cumulative phenomenal object. Homeostasis targets regulated variables, and allostasis targets adaptive control under anticipated demand and accumulated load. These domains interact inside embodied life, but none can be substituted for another; a system may preserve function while experience remains painful, narrow, vigilant, or exhausted.

Disease progression, treatment, medication, frailty, caregiving, changing reserve, and observation shutdown are therefore part of the rival generator, not nuisance notes. If the joint ordinary model explains the apparent concentration, that pattern leaves the PTVC-specific evidence set.

Regulation often sounds reassuring because it keeps organisms alive. Yet a system can remain viable through narrowed choice, chronic vigilance, fatigue, medication, avoidance, or reduced expression. Viability therefore enters as a rival object rather than a synonym for fairness.

The central empirical demand is context. A regulatory curve without illness stage, treatment, sleep, activity, care setting, and missingness may be mathematically clean and scientifically thin. The model must see the conditions that produced the return.

21.3.1 — Homeostatic Set-Point Control

Homeostatic set-point control provides an explicit model of regulation. Temperature, glucose, sleep pressure, pain avoidance, and autonomic regulation offer examples for examining feedback, recovery rates, and changing reference states (McEwen, 1998).

Regulation has also been studied directly in everyday affect. Smartphone observations from more than 28,000 people showed a state-dependent activity pattern: lower mood preceded mood-enhancing activities, while better mood preceded useful activities with less immediately pleasant consequences (Taquet et al., 2016). A later analysis examined 28,212 smartphone participants and 30,116 participants in six-country population data, finding weaker mood-homeostasis patterns with low mood and a history of depression (Taquet et al., 2020). The later report extends the smartphone analysis with population data; the smartphone participants overlap across reports. These observational findings reveal a measurable feedback relation between feeling and activity selection, with substantial variation in how it operates.

Set points, gains, delays, and observation channels can differ across people and settings. A useful homeostatic model represents regulation in treated, aging, and context-sensitive systems.

Breathing support, medication, and social reassurance can help panic settle through partly different channels. A homeostatic

account can predict their timing and interactions, and additional route models can specify further relationships for comparison.

A feedback diagram identifies the regulated variable, reference signal, sensor, comparator, control action, delay, and saturation in the homeostatic rival. Distinct arrows connect the latent state, reports, and physiological observations, locating the feedback relations available for testing.

The comparison can begin from measured affect dynamics. Kuppens et al. (2010) used two experience-sampling studies to examine a diffusion model with an affective homebase, variability, and attractor strength. Separately estimating these features allows researchers to compare models of current state, ordinary return dynamics, and longer-term history while preserving person-specific baselines and observation error.

Research records should include the effort, pain, environmental conditions, and treatment that accompany stable function. These contextual variables help explain how regulation operates within a person's circumstances (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

Homeostatic control is evaluated by estimating feedback gain, delay, saturation, and reference movement around ethically permissible perturbations or natural changes. The fitted controller forecasts recovery and overshoot in a protected sequence, with uncertainty carried into the comparison (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

The biological feasibility of precise regulation has direct experimental support. In bacterial chemotaxis, adaptation

accuracy can remain robust while pathway protein levels change; integral-feedback analysis explains how the network retains this property under specified biochemical conditions (Alon et al., 1999; Yi et al., 2000). Aoki et al. (2019) went further by engineering an integral controller in living cells and demonstrating robust growth-rate regulation in *E. coli*. These findings make cumulative error integration a concrete biological capability. They give PTVC a productive mechanistic starting point: identify a carrier of cumulative information and investigate its relationship to experienced valence.

Natural cellular regulation also supplies an empirical example. Muzzey and colleagues (2009) combined single-cell measurements of volume and Hog1 signaling with assays of glycerol accumulation during changes in external osmolarity. Their perturbations and systems analysis supported an integral-feedback account and helped locate the feedback mechanism in the network regulating glycerol synthesis. The study demonstrates how measurements of several linked quantities and responses to different input shapes can reveal a history-sensitive controller in a living system.

An integral controller also makes signed response area directly analyzable. Let its internal state obey $dz/dt = k_I(r - y)$, with constant reference r , constant positive integration gain k_I , and an integrable output y . Integrating between a and b gives

$$\int_a^b (y - r) dt = -\frac{z(b) - z(a)}{k_I}.$$

Output recovery, $y \rightarrow r$, can coexist with a retained integral-state change. Zero signed response area occurs exactly when $z(b) = z(a)$. Measuring both the output and the internal state under safe perturbations would distinguish these dynamical outcomes (Yi et al., 2000).

The feasibility case extends to stochastic biological systems. Antithetic integral feedback provides a reaction architecture for robust mean regulation under its stability and ergodicity conditions (Briat et al., 2016). Frei et al. (2022) implemented a related circuit in mammalian cells, maintaining tunable transcription-factor expression despite increased degradation or changed feedback structure. Additional proportional feedback reduced variability while retaining adaptation. Noise and biological complexity can thus be incorporated into experimental studies of precise regulation.

An exact molecular count identity and a mean regulation identity make different predictions. In a nonexplosive reaction model with one event at a time and the required compensator integrability, consider two counts Z_1, Z_2 whose births have predictable intensities μ and θY_s and whose paired annihilation removes one of each. For $D = Z_1 - Z_2$, annihilation cancels exactly, while compensated birth counts leave the reaction martingale M :

$$D_t - D_0 = \int_0^t (\mu - \theta Y_s) ds + M_t.$$

A return of D therefore need not make the integrated rate error zero on that realization. This follows directly by subtracting the actual birth counts and their compensators; replacing them by

their means discards the remaining fluctuation. Such reactions provide a concrete architecture to investigate, with mean regulation, event accounting and experiential accumulation measured separately.

A fixed potential of a locally finite pure-jump state is constant between jumps. If its change equals minus an absolutely continuous ledger, continuity also forces it to remain unchanged at every jump, leaving no nontrivial accumulation on that branch. A continuous integrating coordinate, explicit time dependence or an independently justified atomic contribution measure changes the model and requires its own bridge.

The same bookkeeping supplies a quantitative precision test. For a fixed molecular coordinate $q(X)$, let reaction j change q by d_j at intensity a_j . Its drift b , martingale variance rate c and coordinate-changing activity A satisfy

$$b = \sum_j a_j d_j, \quad c = \sum_j a_j d_j^2, \quad A = \sum_{j:d_j \neq 0} a_j, \quad b^2 \leq Ac.$$

The inequality is weighted Cauchy–Schwarz. At a measured upper activity bound, a specified drift therefore demands a minimum innovation variance. This is a restriction on reaction noise, not a lower bound on final variance under feedback. A molecular candidate must show which events write its record, which preserve it, and how its fluctuations relate to independently calibrated experience.

Complex regulation can also improve when complementary mechanisms work together. Hu and Murray (2022) engineered

layered feedback in *E. coli* and found improved joint robustness and response speed under chemical, temperature and nutrient disturbances compared with the component single-layer designs. This gives a concrete reason to investigate coordinated routes. A model linking independently measured layers could predict signed-area recovery, timing and uncertainty across new conditions, with the layers and their coupling specified before the test.

Precision and resource availability can be studied together. Lan et al. (2012) linked adaptation accuracy, response speed and energy dissipation in a sensory-feedback model and tested its implications in *E. coli*. As cells became less energized, adaptation slowed while accuracy was maintained. This finding motivates joint measurement of reserve-related variables, response times and error. Their coupling offers a quantitative target for an affective-regulation model that also includes a calibrated cumulative valence measure.

A recurring response shape becomes more informative when it survives a specified transformation. Fold-change detection predicts that multiplying a sensory input history and its background by the same factor preserves the complete output response (Shoval et al., 2010). Bacterial experiments tested such response rescaling at pathway and population-behavior levels (Lazova et al., 2011). This supplies a useful design precedent: state the transformation, predict the entire dynamical relation, and test its transfer to new conditions without refitting.

Recent control research also makes the variability question more concrete. Lim and colleagues developed a combined molecular

mean-and-noise controller that preserves specified stationary first and second moments under ergodicity and finite-second-moment conditions. Their analysis and simulations extend mean adaptation toward control of output dispersion, with openly archived code for replication (Lim et al., 2026). The study's calibrated simulation of an *Escherichia coli* DNA-repair system illustrates an application of the controller.

For the present program, this creates a stronger executable benchmark: compare candidate routes with controllers that can regulate both the average and the variance, rather than only a set point. A candidate that transports additional joint predictions beyond this richer controller would make a more substantial contribution than one that merely reproduces mean recovery.

21.3.2 — Allostatic Prediction and Load

Allostatic prediction widens the rival by allowing reference ranges and control activity to shift before a demand arrives. Threat, poverty, caregiving, illness, discrimination, and unstable work can produce anticipatory cost while outward behavior appears steady. Apparent stability may therefore be maintained pressure rather than resolved burden (Sterling, 2012; Bobba-Alves et al., 2022).

Allostatic prediction and load should be represented with shifting demands, anticipatory control, channel coupling, and person-specific reserve. A simplified fixed-baseline model would miss the very mechanisms that make allostasis a leading explanation. The Law must compete with the expert version rather than a caricature.

A caregiver may sleep lightly for months because a fall, seizure, or call for help could occur at night. Daytime reports can look stable while anticipatory regulation remains costly. The case gives allostasis an observable structure: expected demand, altered sleep, physiological preparation, and recovery after the caregiving contingency changes.

Allostatic prediction and load require an estimand that distinguishes anticipated demand, current state, cumulative load, treatment, and observation. The protocol should fix the relevant clocks, channels, baselines, missingness rules, and loss before comparison. That common object prevents the favored account from receiving a post hoc advantage.

If anticipatory demand and cumulative allostatic load predict the protected trajectory better than the PTVC-specific account, that pattern is assigned to allostasis. The result may leave a narrower endpoint question open, but it removes preparation, vigilance, and load from the candidate's distinctive evidence. Later summaries must carry the same allocation rather than recovering the lost pattern rhetorically.

Allostatic prediction should be tested by forecasting responses to anticipated demand rather than only reacting to completed events. The protocol must distinguish current state, expected demand, cumulative load, and treatment, then examine whether the model transports when demands change. A successful anticipatory model narrows any fairness-specific interpretation that relied on the same pre-event preparation pattern (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

21.3.3 — Viability Without Fairness

Viability models explain how life, function, or action are maintained under constraint. Measuring the accompanying experience allows physiological continuation and felt welfare to be studied together (Bobba-Alves et al., 2022).

Viability regulation should be tested in the form biology actually permits, with shifting baselines, unequal reserves, and multiple coupled channels. A process may preserve function while burden accumulates, and it may fail differently across illness, treatment, disability, and age. That stronger rival is the appropriate comparison for any fairness-specific claim.

An older patient may reduce movement to avoid falls while losing access to valued activities. A useful model represents both the protective effect and changes in loneliness, autonomy, and opportunity, helping identify supports that improve the overall situation.

Viability and fairness can be separated with mechanism-matched observations rather than a broader invasion of private life. Functional reserve, task completion, pain, fatigue, and reported valence may change on different clocks, so the protocol should sample only the channels needed to distinguish them. More data are not more informative when the design never states which divergence would favor viability control over a terminal-closure interpretation.

The viability test places function, survival-relevant behavior, symptom burden, and valence-related observations in a joint model without equating them. It asks whether control actions

preserve performance despite continued adverse experience and whether that relation predicts new contexts. If viability variables explain persistence and adaptation, the result supports ordinary regulation while leaving terminal closure unestablished (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

21.3.4 — Frailty and Disease Progression

Frailty and disease progression are leading terminal-window rivals because ordinary biological change can narrow both the latent state and the record. Diminished reserve, fatigue, inflammation, medication, delirium, care transitions, and reduced expressive capacity may all produce a quieter observed trajectory. A terminal curve therefore requires a joint clinical and observation model before it receives PTVC-specific interpretation (McEwen, 1998; Sterling, 2012).

Frailty and disease progression deserve a model that permits nonlinear decline, treatment response, competing events, and state-dependent observation. These features can create apparent terminal concentration without a fairness-specific mechanism. Their inclusion is therefore part of rival adequacy, not an optional complication.

In advanced illness, responsiveness may narrow as fatigue, delirium, medication, and disease progression alter both experience and what can be observed. A quieter record is therefore not self-interpreting. The analysis must separate clinical state, care decisions, sensor availability, report capacity, and the latent quantity the terminal claim concerns.

Frailty and disease progression should be fitted only after the target trajectory, competing events, treatment pathways, observation windows, missingness, and loss have been declared. The PTVC candidate and ordinary clinical rival must answer the same question. A favorable reinterpretation after inspection is not model comparison.

Report frailty and disease-progression findings with the affected cohort, treatment context, observation coverage, and competing events.

Frailty and disease progression require a joint longitudinal, treatment, competing-event, and observation model. Simulation should show that the analysis can recover clinical decline and distinguish it from device loss or reduced reporting before the model is applied to terminal windows. If the apparent contraction is absorbed by those clinical processes, the PTVC-specific signature is withdrawn for the tested cohort (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

21.3.5 — Medication and Care Effects

Medication and care effects are not nuisance variables added after theory; they are part of the data-generating process. Analgesics, anxiolytics, antidepressants, sedation, sleep treatment, physical support, and relational care can change reports, physiology, behavior, and sampling availability. Those changes may be humane and necessary, but they can also mimic variance compression or rebound. The analysis must model care as care first and as evidence only after its mechanism has been specified (Sterling, 2012).

Medication and care alter both the trajectory and its visibility. Dose changes, analgesia, sedation, rehabilitation, communication support, and hospice practice can shift symptoms, arousal, reports, and sensor availability on different clocks. Modeling those pathways prospectively can remove an apparent terminal signature before any fairness-specific mechanism is invoked (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

A medication change made during palliative care can alter pain, arousal, communication, sleep, and sensor quality on different schedules. The analytic record should place dose, route, indication, symptom response, care goal, and observation availability on a common timeline. Only then can a terminal change be separated from treatment and care effects.

A treatment figure can use a causal timeline to connect medication, indication, care decisions, physiological responses, reported experience, communication capacity, and observation availability. Separate nodes or intervals make time-varying confounding and observation change visible.

Medication and care pathways belong in the model of the observed trajectory and in its later summaries. Preserving their timing and effects makes clinical findings easier to interpret and reproduce.

Medication, symptom, care, and communication variables are collected to separate treatment pathways from candidate mechanisms. They should not be recombined into a private score of burden, compliance, credibility, or prognosis beyond the clinical and research purposes that authorized them. Access

controls and retention rules must reflect the sensitivity of the care record as well as the limits of the inference (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

Medication and care effects should be evaluated on a time-aligned causal record that includes indication, dose, route, care goal, symptom state, communication capacity, and observation availability. Randomized variation, natural changes, or target-trial methods can then estimate which divergence occurs first. A pattern that follows care exposure rather than the candidate route belongs to the treatment mechanism (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

21.3.6 — Generic Regulation as Default Rival

Living systems stabilize many variables, making homeostasis, allostasis, frailty, treatment effects, mean reversion, and their combinations useful comparison models under shared observations (Bobba-Alves et al., 2022; McEwen, 1998).

The rival must be strong enough to represent expert expectations. A flexible state-space family can include person-specific baselines, nonlinear return, plateaus, latent states, observation error, treatment effects, and changing control regimes. It receives the same information and tuning budget as the PTVC candidate and is judged on model recovery, calibration, held-out prediction, intervention response, and transport. Model discrimination requires a controller that adequately represents those mechanisms.

Data collection should resolve the rival's actual state and control variables rather than demand a maximal dossier. If return rate, set-

point movement, intervention timing, or observation error cannot be estimated without burdensome or speculative measurement, the comparison narrows instead of expanding collection without limit.

The ordinary model must include clinic, household, and care context. Oxygen, opioids, inflammation, delirium, disease stage, caregiver anticipation, and observation-channel changes can move a terminal curve toward quiet without implying a fairness constraint. Allostasis makes the contest harder by allowing anticipated demand to shift the operating range around vigilance or safety (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

When generic regulation reproduces held-out state trajectories and intervention responses, those results support the regulatory model for the tested components. The accumulated-path and endpoint targets remain separate questions for the comparison, with their own observations and prospectively specified predictions.

A mainstream control victory classifies a model-data relation, not a life. PTVC remains distinctive only where a prospectively specified joint restriction survives after the strongest ordinary regulatory family has been fully spent.

21.4 — Ornstein–Uhlenbeck State Dynamics

OU dynamics remove ambiguity from mean reversion by specifying drift toward a reference, stochastic innovation, an initial law, and a time scale. The model is simple enough for exact calculation yet rich enough to expose irregular sampling, observation error, hierarchical variation, and parameter confusion. That makes it an executable rival rather than a metaphor (Uhlenbeck & Ornstein, 1930; Øksendal, 2003; Särkkä, 2013).

The OU process is a baseline, not a claim that affect is exactly Gaussian, linear, stationary, or Markovian. Real trajectories may be nonlinear, heteroskedastic, regime-switching, person-specific, treatment-dependent, and only partially observed. Those departures belong in extensions or rival families, while the simple model tests how much apparent balance ordinary mean reversion already creates.

Displayed mathematics earns its place by exposing assumptions and testable consequences. The drift coefficient, innovation scale, initial condition, observation map, and time unit must remain visible enough to estimate, simulate, and challenge; typography is never a substitute for recovery.

Equation 21.1: OU state equation as rival generator

$$dX_t = -\kappa X_t dt + \sigma dW_t, \kappa > 0, \sigma > 0$$

Status: Equation 21.1 is a rival data-generating model for state dynamics. It is not a Law definition, not a measure of a person, and not a statement about cumulative closure.

Equation 21.2: OU conditional mean

$$E[X_t|X_0 = x_0] = x_0 e^{-\kappa t}$$

Status: Equation 21.2 is a prediction for state return under the OU rival. It shows why local recovery can be ordinary without settling the integrated ledger.

21.4.1 — OU State Equation

The OU state equation gives ordinary mean reversion a precise continuous-time form. It defines a signed state, a return rate, a reference, and an innovation scale, making the rival implementable rather than metaphorical. Its simple structure makes smooth returns, shrinking deviations, and stable-looking records quantitatively tractable, providing a useful first comparison for additional trajectory restrictions (Uhlenbeck & Ornstein, 1930).

Hierarchical drift, diffusion, starting state, observation error, and irregular sampling can differ across people and settings. Allowing those differences produces a flexible stochastic baseline that can be compared with PTVC under the same information budget.

Consider a centered daily affect series after a mild event. An Ornstein-Uhlenbeck model can generate apparent return, persistence, and mean-reverting fluctuations from a small set of parameters. The scientific question is not whether the trajectory looks balanced, but whether a prospectively estimated alternative predicts a feature of later paths that the OU process systematically misses.

The OU equation is tested by observations that resolve drift, diffusion, and measurement error at the relevant timescale, not by

adding channels without a model. The likelihood should use the actual irregular intervals and preserve uncertainty about the latent state. When the cadence cannot distinguish mean reversion from smoothing or reporting noise, the OU comparison remains underidentified.

The OU specification is evaluated through its drift, diffusion, covariance, and residual predictions under the protocol-fixed observation model. Successful predictions identify the features explained by mean reversion; departures identify the additional structure that competing models must address. The comparison records the performance of each model on those features (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

The OU state equation should be fitted with a continuous-time state-space likelihood that uses the actual observation intervals and measurement error. Conditional drift, diffusion, autocovariance, and residual diagnostics must agree with the fitted parameters in protected data. Failure of linear mean reversion or constant diffusion rejects this OU specification; success establishes an ordinary stochastic rival, not a terminal law (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.4.2 — State Mean Reversion

State mean reversion is the visible behavior that makes the OU rival tempting. A deviation decays toward a reference, and the process appears to heal itself even though stochastic shocks continue. This can resemble emotional recovery, clinical stabilization, or terminal quieting. The analysis therefore asks whether state return explains the observed pattern and what

happens to the accumulated path while it does so (Øksendal, 2003; Uhlenbeck & Ornstein, 1930).

Mean reversion is visible in conditional increments: observations above the reference level should tend to move downward, and observations below it should tend to move upward, after accounting for elapsed time and measurement error. That signature can be estimated without invoking a terminal law. The integrated process then asks a separate question about accumulated area.

State mean reversion needs an estimand for return rate, reference level, lag, variance, and prediction horizon before the OU fit is interpreted. Those definitions make it possible to separate a genuine stochastic advantage from a visually persuasive curve.

State mean reversion is tested by relating conditional increments to the current deviation from the estimated reference across several horizons. The return coefficient should transport across protected segments after measurement error and irregular timing are modeled. If the sign or magnitude of drift changes arbitrarily with the window, the simple OU account fails and a richer rival must be named before PTVC is considered (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.4.3 — Stationary and Nonstationary Starts

Stationary and nonstationary starts determine what the OU rival is allowed to claim. A process already in equilibrium behaves differently from one beginning after a shock, treatment change, diagnosis, bereavement, or acute illness. If the start is chosen after looking at the data, the rival can be made to appear better than it

is. The starting condition must therefore be declared before model comparison (Uhlenbeck & Ornstein, 1930).

Stationary and nonstationary starts must be compared without granting either the favorable initial condition by default. Early transients, regression to the mean, and uncertainty about the pre-observation state can all imitate rapid correction. A stricter claim earns attention only if it predicts the consequences of initial-state uncertainty across new trajectories rather than after conditioning on the favorable start.

Two OU processes with identical drift and diffusion can produce different early trajectories when one begins at stationarity and the other begins far from its reference level. That transient can resemble repair or convergence. A fair comparison keeps the starting-state law visible and asks whether the apparent distinction survives after the initial condition is modeled.

A covariance plot should compare stationary and displaced initial conditions under the same OU parameters. The early transient, long-run variance, and uncertainty from the observation model should be visible in separate bands. Its purpose is to show how the start law changes finite-window inference, not to make one initialization look naturally privileged.

Stationary and nonstationary starts are tested under an adverse rule: the candidate account must add a frozen restriction beyond what initial-condition choice already explains. Reserve-linked structure, an integrated-ledger consequence, endpoint discipline, or transported held-out advantage would qualify; mere compatibility would not.

The starting-state law is an inferential choice, not a fact about the person's essential condition. Sensitivity to stationarity or displacement should narrow the model claim rather than invite labels about resilience or chronicity. The ethical response to uncertain initialization is methodological modesty, not stronger surveillance (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

The stationary-start question should be auditable through the assumed initial distribution, burn-in rule, and sensitivity to displaced starts. A reviewer may challenge stationarity without challenging the OU equation itself, and the report should show exactly which finite-window conclusion changes. That separation keeps initialization from becoming a hidden source of confidence (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

21.4.4 — Time-Varying and Hierarchical OU

Time-varying and hierarchical OU models give the rival realistic flexibility: drift, diffusion, mean level, starting state, measurement error, symptom regime, and response to care may vary across time and participants. The candidate should not win because a fixed-parameter baseline was made easy to reject; the richer rival must instead pay for flexibility through prospective specification, recovery tests, complexity-adjusted held-out prediction, and transparent parameter interpretation (Øksendal, 2003; Uhlenbeck & Ornstein, 1930).

After a stroke, a person's apparent return rate may change across rehabilitation phases as treatment intensity, fatigue, and motor access evolve. A single fixed OU coefficient can misread that sequence as noise or stable mean reversion. A time-varying or

hierarchical model earns its extra flexibility only if those shifts are estimated prospectively and improve transport.

A time-varying hierarchical OU analysis needs a prospectively specified state, parameter hierarchy, time unit, observation law, censoring rule, and scoring target. Those choices determine what the model is allowed to explain. Freezing them before fitting keeps both rival and candidate attached to one estimand.

When the hierarchical OU family recovers person and population parameters and transports across protected cases, time-varying mean reversion becomes the preferred account for that component. The candidate cannot preserve the same flexibility under another name. Any remaining claim must be expressed as a new joint restriction that the fitted hierarchy does not already satisfy.

When a time-varying hierarchical OU model best explains the observations, summaries and graphics should identify its fitted mean reversion, diffusion, and tested range.

Time-varying and hierarchical OU models should be evaluated with nested recovery: person-level parameters are generated from a population distribution, the observation schedule is reproduced, and the fitted hierarchy is asked to recover both levels. Held-out persons and sites then test transport. Flexibility is accepted only when it improves calibrated prediction rather than merely absorbing every trajectory (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.4.5 — Observation Error and Irregular Sampling

Observation error and irregular sampling often decide whether OU evidence is credible. Sparse reports, missed prompts, device dropout, clinic-only observations, and terminal low-signal windows can create mean-reversion artifacts or hide genuine change. The rival and the PTVC candidate must therefore use the same measurement burden. A model that wins only because it receives cleaner observations has not earned scientific victory (Uhlenbeck & Ornstein, 1930).

Irregular observation belongs inside the OU rival, not in a footnote after fitting. Sampling cadence, device error, dropout, treatment, and state-dependent observation can change the apparent rate and variance of return. A fair comparison therefore lets the rival generate the data as they are actually collected before asking which mechanism is favored.

Consider prompts missed during high distress and devices dropping out during restlessness. The missingness mechanism can make the observed series appear calmer, more regular, and closer to its center than the latent process actually was. The analysis must model response probability and device availability jointly with the state process, then show whether any route-specific signature survives plausible informative-missingness scenarios.

Irregular sampling should be handled through an explicit continuous-time state and observation model, not by interpolating a convenient curve and treating it as observed. The recorded visit process, device gaps, and person-specific cadence belong in

simulation and recovery checks. A model that succeeds only after regularizing away the actual observation pattern has not explained the data that the study collected.

An observation-model failure defeats the inference that depended on that bridge. If irregular sampling, sensor error, or reporting noise can reproduce the apparent advantage, the latent-dynamics verdict is withdrawn until a stronger observation model is validated. The participant record is not blamed for exposing the weakness (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

Irregular sampling and observation error may reflect burden, care, access, disability, or device limits. An explicit observation model and sensitivity analysis can determine which quantities remain estimable, with No adequate estimate reported where needed (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

Observation error and irregular sampling are tested by simulating from the fitted continuous-time process on the exact visit and device schedule, then refitting through the declared observation model. Coverage, bias, and model-selection error should be reported as cadence and noise worsen. An OU advantage that disappears under the actual schedule is not evidence for the latent dynamics (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.4.6 — OU as a Model That Can Win

Allowing Ornstein-Uhlenbeck models to win turns mean reversion into a genuine contest. A low-dimensional continuous-time process can reproduce return, autocorrelation, and finite-window concentration without a terminal constraint, especially

under irregular timing and observation error (Uhlenbeck & Ornstein, 1930; Øksendal, 2003; Gardiner, 2009).

The protocol fixes the state equation family, prediction target, horizon, observation and missingness laws, complexity burden, and rival-victory threshold. The OU family may include recoverable person-specific baselines, hierarchical structure, or prospectively fixed regimes, but cannot be either a toy rival or an unlimited family that never loses.

A bounded OU victory requires parameter recovery, conditional-drift and covariance reproduction, calibrated uncertainty, protected prediction, and transport under the common observation process. Parsimony counts only after adequacy; a simple wrong model does not win by simplicity alone.

If OU performs at least as well and the PTVC candidate adds no transported restriction, the ordinary stochastic account owns that component. Fitted parameters remain research objects, not person-level labels of emotional order, reliability, or worth.

The baseline is useful because every choice is inspectable: relaxation, innovation, initial state, time unit, heterogeneity, and change points. Asking whether those ordinary choices already produce the signature gives any surviving PTVC residual scientific meaning.

PTVC earns distinctiveness only through a frozen consequence that remains after this strong but finite ordinary family has been fully fitted and allowed to win.

21.5 — The Integrated OU Ledger

The integrated OU ledger is the chapter's mathematical hinge. If X_t is a mean-reverting state, A_t is its time integral and therefore a different stochastic object: state return can coexist with path-dependent accumulation and long-horizon variance growth. The calculation supplies a baseline that terminal-closure claims must outperform, not a phenomenal conclusion by itself (Uhlenbeck & Ornstein, 1930; Øksendal, 2003; Gillespie, 1996; Särkkä, 2013).

The OU result separates the dynamics of the current state from the dynamics of its accumulated integral. The chosen specification and observation model determine the quantitative prediction to compare with other candidates.

The display begins with a deliberately clean baseline: zero-mean OU, fixed start, continuous time, and no measurement error. Empirical use must then add uncertain initial state, irregular sampling, observation error, treatment, heterogeneity, and model misspecification. The simple derivation reveals the state-integral distinction before those layers are introduced.

State recovery and accumulated experience are complementary quantities. The first describes the current condition; the second records the signed area along the path that produced it.

The integrated OU result gives the comparison a precise task: build the regulatory model, test recovery, and compare its held-out predictions with those of the PTVc candidate.

Equation 21.3: Integrated OU state

$$A_t = \int_0^t X_s ds$$

Status: Definition of the accumulated state in the OU comparison model.

Equation 21.4: Variance of the integrated OU state

$$\text{Var}(A_t) = \frac{\sigma^2}{2\kappa^3} (2\kappa t - 3 + 4e^{-\kappa t} - e^{-2\kappa t})$$

Status: Variance consequence of the zero-start OU model under the stated assumptions.

Equation 21.5: Long-time integrated OU approximation

$$\text{Var}(A_t) \sim \frac{\sigma^2}{\kappa^2} t \text{ as } t \rightarrow \infty$$

Status: Long-time variance behavior of the integrated OU model.

Equation 21.6: Expected integrated OU state from fixed start

$$E[A_t | X_0 = x_0] = \frac{x_0}{\kappa} (1 - e^{-\kappa t})$$

Status: Equation 21.6 is a fixed-start OU consequence. It shows that an initial deviation contributes to the integrated path even while the state mean decays.

21.5.1 — Integrated State Definition

The integrated state definition prevents the integrated OU ledger from remaining a loose reference. It defines the accumulated area under the OU state, not the state value at a single moment. That accumulated quantity is the object that matters when a rival is being compared to a cumulative ledger claim. The definition keeps the chapter from treating a calm present as if it

automatically accounted for the path that preceded it (Uhlenbeck & Ornstein, 1930).

The integrated OU process is itself an ordinary stochastic comparator. It supplies a cumulative-path distribution against which a PTVC candidate must predict an additional restriction. The integral need not be unique to the favored theory to be useful; its role is to show what ordinary mean reversion already implies.

Let the state process describe momentary deviation from its reference and let the integrated state record the signed area accumulated on the declared clock. The two objects have different units, covariance structures, and scientific roles. Naming both prevents state mean reversion from being promoted into cumulative closure by notation alone.

The mathematical display defines the OU state and its time integral on an explicit clock, with units stated for both. A covariance or variance relation shows how bounded state variance can coexist with growing integrated variance under the model's assumptions.

A model contributes by explaining a specified component under shared comparison rules. That result clarifies which features subsequent models can build on and which remain to be explained.

The integrated-state definition should expose its units, integration clock, centering convention, and relationship to the observation model. Disagreement can then focus on whether the state is scalar, whether the clock is appropriate, or whether omitted intervals dominate the path functional. These are technical objections with

corresponding tests, not semantic resistance to the larger project (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

21.5.2 — Closed-Form Variance

Closed-form variance gives the rival a measurable consequence. Under the declared zero-start OU baseline, the integrated state has a variance expression that can be plotted, simulated, and checked against numerical implementation. A mean-reverting state can have an accumulated integral whose uncertainty continues to grow (Øksendal, 2003).

Parameters estimated from the OU state process imply integrated uncertainty across horizons, enabling a check without refitting each window. Agreement or systematic departure reveals which cumulative features the specified model captures (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

Closed-form variance is useful only after the target quantity, starting-state law, horizon, parameter regime, and observation error have been fixed. Those choices determine whether the formula addresses finite-window dispersion, long-run growth, or a measured proxy rather than the latent path.

Closed-form variance may defeat a fairness-specific reading without defeating the value of the analysis. When the ordinary result predicts, calibrates, and transports better, explanatory credit stays with it. The chapter preserves the mathematical lesson and narrows the stronger claim.

Closed-form variance should be checked by propagating independently estimated OU parameters into predictions for the integrated process at several horizons. Empirical variance and

covariance, corrected for the observation model, are then compared with simultaneous uncertainty bands. Systematic failure rejects the analytic rival in that regime; agreement removes the pattern from the candidate's unique evidence (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.5.3 — Asymptotic Linear Growth

The OU result illustrates a general distinction between recovering a state and cancelling its accumulated path. For real, centered, covariance-stationary equal-exposure increments X_k with finite second moments, write $A_n = \sum_{k=1}^n X_k$ and $\gamma_h = \text{Cov}(X_0, X_h)$. Counting covariance pairs gives

$$V_n := \text{Var}(A_n) = n\gamma_0 + 2 \sum_{h=1}^{n-1} (n-h) \gamma_h.$$

If $\sum_{h \in \mathbb{Z}} |\gamma_h| < \infty$, division by n and dominated convergence give $V_n/n \rightarrow \ell = \sum_{h \in \mathbb{Z}} \gamma_h = 2\pi f(0)$, using $f(\omega) = (2\pi)^{-1} \sum_h \gamma_h e^{-ih\omega}$. Positive ℓ yields growing cumulative dispersion even though $E[(A_n/n)^2] \rightarrow 0$. For a Gaussian process and fixed $\varepsilon > 0$,

$$\Pr(|A_n| \leq \varepsilon) = 2\Phi\left(\frac{\varepsilon}{\sqrt{V_n}}\right) - 1 \sim \sqrt{\frac{2}{\pi}} \frac{\varepsilon}{\sqrt{n\ell}}.$$

A non-Gaussian central limit theorem with positive limiting variance also makes the fixed-band probability tend to zero, but does not by itself provide this rate. Summable covariances alone do not establish that central limit theorem. These stationary-

process comparisons extend the OU example to wider accumulation architectures.

For stationary AR(1) increments $X_k = \phi X_{k-1} + e_k$, $|\phi| < 1$, with independent centered innovations of variance q , $\ell = q/(1 - \phi)^2$. In contrast, $X_k = e_k - e_{k-1}$ gives $A_n = e_n - e_0$ and $V_n = 2q$. This ordinary cancellation mechanism has bounded cumulative variance while generally retaining a nonzero terminal residual. More generally, an absolutely summable linear filter has $\ell = q(\sum_j h_j)^2$, so zero-frequency cancellation requires its net response to vanish.

For a real centered jointly measurable stationary continuous-time process with finite second moments and covariance $C(u)$, $\text{Var}(\int_0^T X_t dt) = 2 \int_0^T (T - u) C(u) du$. Absolute integrability of C gives the corresponding variance-per-time limit. Keep stationary and zero-start initial laws distinct. Centering a rival mathematically does not identify phenomenal neutrality: subtracting each completed record's own mean forces apparent closure by construction. Compare held-out window-length variance predictions, with dependence, measurement error and unobserved low frequencies included. This can discriminate accumulation mechanisms before complete-stream evidence is available.

21.5.4 — Initial-Condition Variants

Section 21.4.3 fixed the OU starting-state law and the protocol for comparing stationary, displaced, and uncertain starts. The integrated-ledger question is narrower: how that starting law

propagates into the mean and covariance of the time integral over the study window.

With the same OU drift, diffusion, observation law, and horizon, stationary, fixed displaced, post-event, and hierarchical starts can produce different finite-window cumulative means and covariances. The analysis must report which ledger conclusions depend on the start model, first recorded value, or burn-in rule (Uhlenbeck & Ornstein, 1930; Gardiner, 2009; Øksendal, 2003).

Candidate and rival ledger models must share the same starting-state distribution and scoring rule. If a terminal-looking cumulative pattern disappears under a defensible start model, the result is Model-ambiguous rather than support.

Initial-state assumptions describe the beginning of the analyzed window. Reports identify their effect on cumulative estimates and preserve start-model sensitivity in subsequent summaries.

21.5.5 — Worked Calculation and Parameter-Sweep Design

A worked calculation makes the state–integral distinction concrete. Use dimensionless model units, $X_0 = 0$, $\kappa = 1$, and $\sigma = 1$. The OU state variance is $\frac{1-e^{-2t}}{2}$, while Equation 21.4 gives the integrated variance. At $t = 1, 5$, and 10 , the state variances are approximately 0.4323 , 0.5000 , and 0.5000 ; the corresponding integrated variances are 0.1681 , 3.5135 , and 8.5001 . The state variance settles while the integrated variance continues to grow. With the same dynamics but a fixed start $X_0 = 1$, the expected state at $t = 5$ is only 0.0067 , whereas Equation 21.6 gives an expected integral of 0.9933 . These calculations follow from the declared OU model (Øksendal, 2003).

An analytic parameter sweep also shows the role of return rate. Keep $X_0 = 0$, $\sigma = 1$, and $t = 5$. Substituting $\kappa = 0.5, 1$, and 2 into Equation 21.4 gives integrated variances of approximately 9.2864, 3.5135, and 1.0625. At fixed κ and t , doubling σ multiplies the variance by four. Starting-state assumptions matter too: with $\kappa = \sigma = 1$ and a stationary start, the integrated variance at $t = 5$ is $5 - 1 + e^{-5}$, approximately 4.0067, rather than the zero-start value 3.5135. These calculations provide numerical checks for an implementation.

A simulation sweep can hold cadence and the observation rule fixed while varying mean-reversion rate, diffusion scale, starting-state law, horizon, and measurement error. Shared scoring rules then reveal regimes of finite-window contraction, observational ambiguity, and useful model separation.

Testing separation from bridge or reserve-coupled candidates requires fully specified competing generators and a shared observation model. A simulation sweep should measure recovery error and model confusion across drift, diffusion, horizon, and cadence. This comparison can identify where the models are distinguishable (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

Public summaries of a parameter sweep should report the full pattern across settings, including successful, unresolved, and weak-performance regions.

Any such simulation sweep must be reproducible from a fixed grid, seed set, solver tolerance, and observation schedule. For each regime it should report parameter recovery, model confusion, integrated variance, and which tested design changes improve separation. The sweep passes when another analyst can

recreate both favorable and adverse regions, not only an illustrative panel (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.5.6 — Consequences for Model Comparison

The integrated OU derivation gives the comparison a calculable cumulative trajectory, with explicit dependence on drift, diffusion, horizon, and initial conditions (Uhlenbeck & Ornstein, 1930).

The OU baseline quantifies persistence and mean reversion at the measured scale. A candidate route can specify an additional restriction and test it against that baseline.

Mean reversion of the OU state can coexist with a growing variance of its accumulated integral. These quantitative predictions can be compared under the declared measurement and observation models.

Explicit parameter units, observation errors, windows, censoring rules, and endpoint definitions make the OU calculation reproducible and identify the settings in which it informs model comparison.

The calculation earns no PTVC-specific implication merely by being exact. A candidate route must add a prospectively specified restriction, such as independently measured reserve coupled to drift and innovation, endpoint discipline not built into the path law, or transported held-out advantage. Otherwise the integrated OU process remains the ordinary baseline.

The OU derivation can pressure an interpretation without settling the phenomenal target. If the observed path follows the ordinary variance law, the candidate loses a claimed signature; if it does

not, the OU specification loses that regime. In neither direction does the equation authorize personal prediction, care delay, or a story about why suffering occurred (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

Reviewers should separate three questions: whether the OU derivation is correct, whether the process approximates the latent state, and whether the observation bridge is adequate at the relevant horizon. A correct theorem can coexist with a rejected empirical model, while an imperfect approximation can remain an informative rival. Reporting these layers separately prevents mathematical correctness from being mistaken for construct validity or measurement adequacy.

The central intuition is concrete. A thermostat can return room temperature to its set point while a meter continues recording how long the room was cold. Likewise, a state can be pulled toward zero while its accumulated area retains history. Present recovery and accumulated exposure are different scientific objects. The same distinction applies when a baseline itself moves: local restoration can coexist with a cumulative history that remains far from zero.

The integral preserves duration as well as sign and intensity, so many small deviations can accumulate even when the current state appears ordinary. Measuring the accumulated path therefore adds information to the description of state recovery. This makes the integral a direct rival discriminator whenever competing theories predict similar state recovery but different accumulated exposure.

21.6 — Generative Rival Comparison

Generative rival comparison keeps the chapter from becoming a taxonomy. Each rival produces latent histories under declared assumptions, passes through the same observation kernel, and is scored by the same protected rule. Only this shared pipeline can show which component a PTVC candidate retains or loses (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019; Talts et al., 2018).

The rival set includes set-point adaptation, incomplete adaptation, opponent processes, homeostatic regulation, allostatic load, frailty and illness progression, treatment and care effects, OU state dynamics, integrated OU behavior, and hybrids among them. Some are simple; some are complex. The comparison penalizes unnecessary complexity while refusing weak caricatures.

Each comparator generates the planned observations and is evaluated for recovery and predictive performance under the specified scoring rule. This turns a named mechanism into an executable benchmark.

Hybrid rivals are necessary because real trajectories may combine adaptation, treatment, frailty, allostatic load, observation loss, and mean reversion. They receive declared component structure and complexity penalties rather than unlimited flexibility. A hybrid may win, but it must remain identifiable and capable of losing.

Table 21.1. Model-recovery matrix plan.

True generator	Common confusion	Required diagnostic	Status consequence
Set-point adaptation	PTVC candidate or OU	event-specific return and plateau checks	PTVC-specific language narrows if return explains data
Opponent process	adaptation or rebound story	lag, overshoot, repetition effects	local counterweight replaces closure language
Allostatic model	homeostasis or PTVC candidate	load markers and context shifts	regulation under demand becomes primary
OU rival	PTVC candidate	integrated variance and state-return recovery	ordinary mean reversion can win
PTVC candidate	OU or hybrid rival	reserve-coupled joint restriction and endpoint independence	candidate survives only if recovered under fair rules

Table 21.2. Rival-dominance decision rule.

Dominant rival	What it explains	PTVC-specific consequence	Public-language update
adaptation	local return after events	retire local-return support	describe adaptation, not closure

Dominant rival	What it explains	PTVC-specific consequence	Public-language update
opponent process	lag, overshoot, withdrawal	route rebound to local counterweight	avoid balance language
allostatic model	stability under chronic demand	treat load as regulation cost	name allostatic burden
OU or mean reversion	state return and variance pattern	require integrated residual or withdraw	call mean reversion mean reversion
hybrid ordinary rival	combined treatment, frailty, regulation, report dynamics	narrow to unexplained regions only	remove PTVC-specific status from explained pattern

21.6.1 — Common Synthetic Observations

Common synthetic observations create a fair arena for model comparison. Each candidate sees the same cadence, noise, missingness, endpoint uncertainty, treatment flags, and observation limits. Without that common record, the favored model can win by receiving cleaner data or easier assumptions. The synthetic fixture is therefore a scientific fairness rule, not a decorative simulation (Gneiting & Raftery, 2007).

Common synthetic observations should be generated so that each rival receives the same visible record and no privileged latent channel. The exercise should vary cadence, missingness, noise, endpoint rules, and latent dynamics while preserving equal fitting budgets. A stricter signature is credible only if the same observed

record yields recoverable and transportable differences under that equal-information design.

A benchmark can share the same exogenous event schedule across an OU rival, an adaptation model, an opponent process, a reserve-coupled candidate, and a hybrid world. Each generator produces its own latent response trajectory, which then passes through the common observation and missingness model. Every fitted candidate receives the same observed dataset for that benchmark replicate.

The comparison asks which joint predictions the PTVC candidate adds to common synthetic observations: reserve-linked structure, integrated-ledger consequences, prospectively defined endpoints, or transportable out-of-sample performance. Distinctive predictions make that comparison informative.

A synthetic-data confusion matrix shows which generators the pipeline distinguishes under the chosen assumptions. It can identify improvements to the design before participants are recruited (Talts et al., 2018; Hernán & Robins, 2020).

The synthetic-observation benchmark should make generator coverage and observation-kernel choices inspectable. Critics can propose a missing rival, a harder missingness process, or a dependence structure that changes recovery, and the release should show where that addition enters. A benchmark becomes stronger when disagreement produces a new executable world rather than an unresolvable rhetorical exchange (Talts et al., 2018; Hernán & Robins, 2020).

21.6.2 — Parameter Recovery

Parameter recovery asks whether a model's parameters can be estimated when its generator is known. Realistic simulations reveal which return rates, lags, loads, and variances the proposed observations can recover. These results guide sampling and determine which parameter-level interpretations the subsequent comparison can support (Vehtari et al., 2017; Yao et al., 2018).

Test each generative family for identifiability and estimation bias before cross-model discrimination. Recovery diagnostics show where misspecification could affect the comparison and which design changes improve it.

A recovery study can vary return rate, diffusion, opponent lag, observation error, and missingness while preserving known truth. Bias, interval coverage, multimodality, and nonidentifiability should be reported by parameter and regime. A method that estimates the candidate well but fails on rival parameters creates an unfair downstream contest.

Parameter recovery must be an executable, symmetric test. For every generator, the parameter units, observation law, protected data, tolerances, failure thresholds, confusion rules, and model identity are fixed in advance. Candidate recovery cannot be declared adequate while rival failures are ignored, and no model inherits victory merely because its parameters are easier to name.

Poor parameter recovery is a failure of the estimator, design, or identifiability conditions, not a defect in the people or records from which observations arise. The response is to repair the pipeline, narrow the estimand, or abstain before interpretation. No

amount of public enthusiasm can turn biased recovery into evidence (Talts et al., 2018; Hernán & Robins, 2020).

Parameter-recovery disputes should be tied to identifiability, prior influence, interval coverage, and tolerance by regime. A reviewer can ask whether the sample schedule identifies the return rate but not the diffusion, or whether several parameter combinations yield the same observations. The report should state which uncertainty blocks the downstream comparison and which design change could remove it (Talts et al., 2018; Hernán & Robins, 2020).

21.6.3 — Model Recovery

Model recovery asks a harder question: can the analysis distinguish the true family from plausible alternatives? A set-point model may masquerade as an opponent process, an allostatic shift may look like OU drift, and treatment-related quieting may look like terminal concentration. The confusion matrix makes those substitutions visible. The PTVC-specific interpretation earns space only when the recovery design can tell what generated the pattern (Wilson & Collins, 2019).

Model recovery must confront the heterogeneity and measurement disorder present in the intended study. Parameter variation, missingness, treatment, channel error, and competing generators should be represented in the simulation bank. Recovery across that range tests the pipeline’s readiness for the intended human observations.

Consider a recovery matrix showing confusion between an OU generator and a Law-candidate model under noisy sampling. The

matrix should report which parameter regions, cadences, and missingness levels produce each error, not merely an average accuracy score. If the candidate cannot be recovered under conditions close to the intended study, the correct conclusion is design inadequacy or model ambiguity rather than evidentiary support.

The model-recovery display should be a confusion matrix whose rows are generating families and whose columns are selected or abstained outcomes. Hybrid and open-set generators need their own rows, and uncertainty intervals should accompany the cell rates. The matrix tests the analysis pipeline; it does not classify participants or convert model selection into a human score.

Model recovery fails when the true generator is confused with a rival or when the pipeline is confidently wrong on an open-set world. That failure blocks the corresponding human-data verdict, even if parameter estimates look precise inside the selected model. A later redesign must pass fresh synthetic tests before the component can reenter confirmatory use (Talts et al., 2018; Hernán & Robins, 2020).

Model confusion reveals the resolution of the proposed design. If OU, adaptation, hybrid, and reserve-coupled generators remain indistinguishable, the result identifies where measurements or contrasts need improvement before a mechanism can be selected (Talts et al., 2018; Hernán & Robins, 2020).

Model recovery must fit every prospectively specified family to data from every generator under the planned observation and missingness process. The confusion matrix should include hybrid

and open-set rows, calibrated abstention, and uncertainty across replicates. A human-data verdict is blocked wherever the true family is routinely misclassified or where the candidate receives a structural recovery advantage (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

21.6.4 — Hybrid Rival Models

Hybrid rival models are necessary because real lives need not be governed by isolated mechanisms. Adaptation, treatment, frailty, allostatic regulation, observation shutdown, and conditioned dynamics may coexist and jointly produce a terminal pattern. The candidate earns force only if it adds transportable structure after those plausible combinations are fitted under the same observation model and complexity discipline.

Consider chronic illness combining adaptation, medication effects, allostatic cost, and frailty. A credible hybrid rival should allow these mechanisms to operate simultaneously and at different timescales, because forcing them into separate models can manufacture a false advantage for the preferred route. The candidate earns distinction only through a joint restriction that the hybrid cannot reproduce under the same observation model.

Hybrid rivals require measurements that identify their interacting components, not an indiscriminate expansion of the dataset. The design should target contrasts that separate adaptation rate, treatment effect, observation shutdown, and any reserve-linked term, while allowing abstention when the components remain aliased. A large record cannot compensate for a design in which several mechanisms produce the same observable path.

A hybrid explanation may combine ordinary mechanisms without implying that persons or streams can be combined. Mixture weights describe a generator inside the model; they are not shares of responsibility, value, or experiential debt. The result remains a bounded account of observed dynamics (Talts et al., 2018; Hernán & Robins, 2020).

Test hybrid mixtures across component strengths ranging from clearly identifiable to nearly aliased. The pipeline should recover components where information permits and report open-set or ambiguous outcomes elsewhere (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

21.6.5 — Held-Out Prediction

Held-out prediction protects the chapter from fitting stories to completed lives. Frozen models must predict observations not used for fitting, under prospectively specified scoring, calibration, interval-coverage, and clinically meaningful error rules. Training must include realistic heterogeneity in participants, contexts, treatments, channels, and observation schedules while keeping the protected module inaccessible; otherwise a clean holdout score may reflect an impoverished training world rather than a discriminating mechanism (Gneiting & Raftery, 2007).

Consider training on early post-event data and predicting later recovery or another site. The holdout must preserve the distribution shift that matters, such as a different endpoint regime, measurement cadence, or care environment, rather than becoming a random slice of the same trajectories. Transportable advantage is demonstrated by prospectively specified performance and

calibration on that protected target, not by retrospective refitting after failure.

Held-out prediction depends on protected information boundaries and mechanism-relevant inputs, not on collecting every signal before the split. The training module, tuning module, and final holdout should be separated by person, site, time, or endpoint window as the dependence structure requires. Features created with future or terminal information must be excluded before any score is called prospective.

The result should change language before it changes allegiance. When held-out prediction accounts for the observed pattern, the chapter calls it by that name rather than translating resemblance into support. What remains for the Law is a residual, prospectively specified question, not a saved conclusion.

Failure on a held-out site or cohort narrows transport; it does not mark that population as anomalous or less credible. The report should identify the shifted measurement, care, event, or context process and preserve the possibility that the model rather than the people failed. Human standing is invariant to predictive performance (Talts et al., 2018; Hernán & Robins, 2020).

Held-out prediction requires a frozen pipeline and splits respecting person, site, time, and endpoint dependence. Report proper scores, calibration, and residual structure alongside average loss, with no feature built from the protected outcome. Treat an advantage confined to random observation-level splits as insufficient evidence of transport, and investigate leakage,

dependence, and distribution shift (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

21.6.6 — Rival Dominance

Rival dominance tells when ordinary mechanisms narrow or retire a PTVC-specific interpretation. It requires reliable held-out results, recoverable parameters, transport, and no hidden ethical or measurement advantage. Then narrowing is scientific progress (Vehtari et al., 2017; Yao et al., 2018).

Competitive implementations govern the contest. Each model gets defensible heterogeneity, observation structure, and tuning freedom before complexity penalties and held-out scoring. A Law-specific route narrows when an ordinary rival wins fairly.

Repeated outperformance by adaptation, allostasis, or an OU process changes the program. The candidate is narrowed, redesigned, or retired for the declared class, with each consequence tied prospectively to replicated performance rather than answered by new unmeasured structure.

Dominance requires equal information and scoring, not greater surveillance. Every candidate shares the admissible history, missingness, and transport conditions. Unequal context or endpoint access diagnoses the contest, not the mechanism.

The comparison reports predictive accuracy, mechanism identification, calibration, and measurement performance separately.

Dominance is a protocol-fixed scientific relation, never a hierarchy among people. It can retire a mechanism or narrow a

class while care, rights, privacy, and participation remain unchanged; the winning model remains correctable (Talts et al., 2018; Hernán & Robins, 2020).

Recovery, calibration, held-out prediction, transport, uncertainty, and multiplicity share one decision table. Mixed results remain component-specific rather than becoming a single winner label.

One shared observation file gives every model the same simulated streams, events, treatments, missingness, censoring, noise, and endpoint uncertainty. No candidate receives latent truth while another receives the messy record.

Recovery rehearses interpretation. A pipeline unable to recover opponent or OU generators from realistic synthetic observations cannot declare those rivals defeated in empirical data. Confusion matrices, calibration checks, and frozen fixtures expose the weakness before participants carry its cost.

21.7 — Empirical Separation and Defeat

The empirical comparison links cumulative behavior, independently estimated reserve, endpoint structure, observation robustness, and protected prediction. Joint restrictions across these features give PTVC a specific contribution to evaluate alongside established regulatory models (Talts et al., 2018; Hernán & Robins, 2020; Munafò et al., 2017; Nosek et al., 2018; Wilson & Collins, 2019).

A clear comparison assigns observed patterns to the models that predict them best, including adaptation, care pathways, and ordinary stochastic dynamics. This connects scientific interpretation with reproducible reporting.

Adverse outcomes are classified by the component they reach. A failed discriminator retires that signature; repeated component failures narrow a route or reference class; a gate-valid counterresult may pressure or defeat the declared branch. Each outcome identifies a concrete next step for the program.

Later summaries should retain the mechanism, observation model, and setting identified in the original result. Consistent attribution makes findings easier to compare and extend.

Accurate defeat language is a form of care. When scientific scope narrows, captions, summaries, guides, and public replies narrow with it. The project should prefer a clean loss over an unsafe sentence preserved by ambiguity.

21.7.1 — Reserve-Coupled Joint Restriction

A reserve-coupled restriction links independently estimated reserve with drift, innovation, horizon, and cumulative behavior. Estimating reserve through separate observations makes the joint prediction testable.

The reserve-coupled joint restriction should be compared with ordinary mechanisms that can mimic each component separately. Adaptation may reproduce the state path, prognosis may reproduce the shrinking horizon, and selection may reproduce endpoint concentration. The joint restriction matters only if their combination is recovered and transported when those ordinary mechanisms are allowed to mimic each component.

The reserve-coupled candidate should be tested on simulations in which reserve is observed imperfectly, acts only through a declared transition law, and competes with ordinary regulation. The joint restriction must predict more than a smooth endpoint. It should constrain how reserve, state, covariance, action, and endpoint risk move together in held-out modules.

A graphical model can show reserve, observed state, latent valence-related state, actions, endpoint hazard, and measurement channels as distinct nodes. The candidate restriction should be highlighted only on the edges it actually constrains, while ordinary causal paths remain visible. This prevents the reserve route from acquiring information or authority that the protocol did not supply.

The reserve-coupled joint restriction must not absorb a rival victory by redefining the restriction after the result. Its component

predictions, allowable rescue conditions, and target consequence are frozen beforehand. A winning ordinary account remains a winning ordinary account.

Reserve variables represent specified relationships among state, action, and endpoint risk within the approved research model and purpose (Talts et al., 2018; Hernán & Robins, 2020).

The reserve-coupled restriction should expose the operational definition of reserve, its causal parents, its observation channels, and the endpoint information allowed at each time. Reviewers can then challenge route-state validity or endpoint coupling without disputing every other component. The text should identify the observation or intervention that would decide each objection (Talts et al., 2018; Hernán & Robins, 2020).

21.7.2 — Prospective Endpoint Independence

Define the endpoint rule through independent channels before evaluating the accumulated trajectory. Compare candidates using the same prospective endpoint information, with death and observation loss represented according to their distinct roles (Hernán & Robins, 2020).

Prospective endpoint independence prevents the analysis from learning the ending before it predicts the path. Hazard models, feature construction, and data splits must use only information available at the declared decision time. If a signature disappears when terminal knowledge is removed, it has not been established as a prospective signature under that information set. Conditioning and selection become primary rival explanations; distinguishing them from a causal endpoint-coupled route

requires the declared joint model and additional discriminating evidence (Aalen et al., 2008; Hernán & Robins, 2020).

Prospective endpoint discipline fixes the pre-endpoint information set and audits when endpoint information enters the analysis. Statistical endpoint independence is a separate, branch-specific claim involving the endpoint-time law, risk interval, observation process, and declared dependence structure; it requires its own identification and diagnostics.

Reports should identify any dependence on realized endpoint information. A revised endpoint rule receives a new version and fresh evaluation, keeping the original result and its assumptions traceable.

Endpoint-hazard features and independence diagnostics are used to choose the correct scientific branch, not to forecast an individual's death, inner balance, or entitlement to intervention. Person-level outputs should remain restricted, and public artifacts should present aggregate or synthetic diagnostics whenever feasible. The endpoint firewall applies to deployment as well as estimation (Talts et al., 2018; Hernán & Robins, 2020).

Statistical endpoint-independence claims are tested separately with a hazard model and feature set frozen at each decision time. Diagnostics assess whether residual endpoint risk remains associated with past state, treatment, observation, or route variables after protocol-fixed adjustment, while simulations quantify power against plausible dependence. Evidence of coupling changes the branch; lack of identification yields

abstention rather than an independence claim (Talts et al., 2018; Hernán & Robins, 2020).

21.7.3 — Integrated-Ledger Consequence

Integrated-ledger analysis compares accumulated behavior across OU, adaptation, allostatic, and PTVC models under the same observation rules. Their joint predictions show which aspects of path history each account explains.

The integrated-ledger consequence must be tested across plausible starts, sampling schemes, and participant-specific dynamics. Fixed and stationary initial conditions, hierarchical parameters, observation error, and treatment changes can alter finite-window behavior without changing the rival's basic mathematics. The candidate Law should be asked to outperform that full family rather than a single textbook path.

During caregiving, daily distress, event-triggered spikes, sleep disruption, and accumulated vigilance may follow different trajectories. Specifying the path functional and observation coverage helps distinguish accumulation from memory, sampling, and scale effects.

The integrated-ledger consequence is tested by resolving the state path and observation law that determine the integral, not by treating a person's life as an unlimited data source. Cadence should be dense only where it materially reduces uncertainty in the path functional, and unobserved intervals must remain represented rather than silently filled. The analysis should report how much of the apparent distinction survives coarser, less burdensome sampling.

The integrated-ledger consequence is where state recovery and path closure finally separate mathematically. A mean-reverting state can keep returning while its time integral retains variance and history. Testing that path functional across horizons asks whether the candidate adds a joint restriction beyond ordinary return, rather than treating a calm final state as evidence about the accumulated stream (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

Recent control research shows how integrated responses can distinguish mechanisms. Gupta and Sontag (2025) studied cumulative dose response, the integrated output over a fixed interval as a function of input dose, motivated by immune-cell response data. The incoherent feedforward models they analyzed impose monotone cumulative responses even where instantaneous responses can be nonmonotone; their integral-feedback model can produce a nonmonotone cumulative relation. Thus the area can distinguish mechanisms that a final level or single response peak leaves unresolved. A concrete next step is to compare cumulative-response predictions on existing data from independently justified studies, using independently calibrated response variables, common horizons, and the same observation information for each regulatory model.

If ordinary integrated dynamics explain the observed path, that result identifies the relevant generator and helps refine the next comparison (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

The integrated-ledger test estimates the state dynamics and the signed path functional under one clock, then compares their joint mean, variance, and covariance across horizons. Ordinary mean

reversion should be allowed to explain state return while the candidate is scored only on an additional frozen path restriction. A distinction that vanishes after observation error or initial-state uncertainty is included remains unresolved (Talts et al., 2018; Hernán & Robins, 2020).

21.7.4 — Observation-Model Robustness

Observation-model robustness ensures that a separation result is not an artifact of how the data were seen. The comparison must survive alternate assumptions about missingness, treatment, frailty, device drift, reporting changes, and low-signal terminal windows. Sensitivity to observation assumptions identifies a measurement question that the next study must resolve before claiming mechanism separation (Talts et al., 2018; Hernán & Robins, 2020).

Robustness across observation models evaluates the reliability of a measurement bridge, not the credibility of the participant. A channel that performs poorly may require recalibration, narrower use, or withdrawal, while reports and care needs retain their independent standing. No device-quality or model-sensitivity result becomes a score of inner truth (Talts et al., 2018; Hernán & Robins, 2020).

Implementation: refit the frozen comparison across the prospectively specified channel kernels, calibration maps, error bounds, and missingness laws, then report the region in which model ordering, calibration, and verdict remain stable (Talts et al., 2018; Hernán & Robins, 2020).

21.7.5 — Out-of-Sample Advantage

Out-of-sample advantage is the practical threshold for scientific seriousness. The PTVC candidate must improve prediction or calibration beyond complexity penalties in data it did not use to fit itself. The advantage should appear in the right variables, at the right timescales, and under the right endpoint firewall. Otherwise, the additional theory is not earning its cost (Hernán & Robins, 2020).

Out-of-sample advantage must be earned against rivals given the same training record, tuning freedom, and protected holdout. One favorable split is not enough; calibration, uncertainty, and failure modes must be reported across prespecified replications. The relevant question is whether the advantage persists across sites, endpoint regimes, and perturbations without access to information unavailable to the rivals.

A persuasive transport example trains the candidate and its rivals in one set of sites, freezes every preprocessing and tuning choice, and evaluates them in sites with different cadence, care practice, and event mix. The result should show not only score differences but calibration and failure by stratum. Transport is earned when the advantage survives those shifts without widening the information budget.

The transport figure should compare held-out proper scores and calibration across sites, cohorts, and endpoint windows after all models are frozen. Complexity penalties and uncertainty should appear beside the point estimates, with failures rather than only

averages shown. The figure supports an out-of-sample comparison, not a universal ranking of theories or people.

Out-of-sample advantage loses meaning if the favored model, metric, or holdout changes after scoring. The protected module, complexity budget, and consequence rule must remain fixed. That discipline turns a good score into evidence rather than an editorial preference.

A held-out predictive advantage changes the status of a model only within the frozen target, population, and observation system. It cannot become a personal forecast, a judgment of credibility, or a reason to expand data collection beyond consent. The public summary should state the gain, uncertainty, transport domain, and remaining unresolved target in the same breath (Talts et al., 2018; Hernán & Robins, 2020).

Disagreement about out-of-sample advantage should be traceable to the split, dependence structure, scoring rule, complexity allowance, or transport population. Each choice can be varied through a protocol-fixed sensitivity plan, and the conclusion should show where model ordering changes. This turns broad skepticism into an explicit map of robust and fragile claims (Talts et al., 2018; Hernán & Robins, 2020).

21.7.6 — Defeat by Mainstream Explanation

Defeat by mainstream explanation is an honorable scientific outcome. If adaptation, opponent processes, allostasis, OU dynamics, treatment effects, frailty, observation processes, or their hybrids explain the protected patterns better, the PTVC-specific interpretation narrows or retires for that domain. Care

remains untouched, and the program retains the measurement, comparison, and clinical knowledge produced along the way (Talts et al., 2018).

Defeat must be operationally possible. The ordinary rival suite receives defensible flexibility and faces the same prospectively fixed state, path functional, endpoint rule, channels, missingness model, information budget, and held-out loss. It wins when it recovers reliably, predicts and calibrates at least as well, transports at least as well, and explains the conjunction more parsimoniously.

The decision retires or narrows the exact component that lost and names the winning family, tested domain, uncertainty, and consequence in the headline-level summary. A new route requires a new prediction and fresh protected evidence rather than relabeling the winner as how fairness acted.

The distinctive burden is conjunctive: independently estimated route variables where relevant, prospective endpoint definition, accumulated-path consequences, resistance to observation alternatives, and out-of-sample advantage after complexity control. A mainstream account that explains the pattern without that conjunction changes the claim's status.

Defeat propagates through later chapters, summaries, figures, and public artifacts. Useful care or measurement lessons remain, but a defeated resemblance cannot survive as implied support. A science that can lose precisely becomes more capable of earning support.

Chapter 22 — Predictive Processing and Learning Rivals

This chapter asks which commitments make predictive-processing, active-inference, reinforcement-learning, safe-control, bridge, and selection accounts empirically distinguishable under the same observation stream. The objective is not to borrow their vocabulary for PTVc, but to force each account to state protected predictions, information requirements, and failure conditions before comparison (Rao & Ballard, 1999; Friston, 2010; Sutton & Barto, 2018; Lageman et al., 2026; Samara, 2026).

These families can generate many PTVc-like appearances. Prediction error can fall. Policies can change after pain. Controllers can avoid states with poor return options. Conditioned processes can drift toward an endpoint already built into the probability law. Clinical samples can concentrate because referral, treatment, survival, and observation jointly selected the record. Each mechanism deserves a fitted, testable form.

A theory name is not a competitor. Each rival must declare its predictand, latent states, observations, error proxy, uncertainty, pathways, policies or actions, objective, endpoint relation, and discriminating prediction. It threatens PTVc only when this executable specification reproduces the relevant pattern and predicts protected data better under equal rules (Samara, 2026; Lageman et al., 2026).

Predictive processing provides models in which perception and action depend on expectations and prediction errors. Such models can explain why familiar contexts reduce alarm, why attention follows uncertain signals, and why some disturbances become less salient. Those achievements belong to prediction and

perception unless an additional bridge connects them to the cumulative phenomenal target.

Reinforcement learning supplies a different rival architecture. Reward prediction errors, exploration, memory, planning, and horizon length can reshape behavior and reports over time. Reward remains a model variable, not an automatic measure of felt valence, and policy adaptation remains distinct from closure of a phenomenal ledger.

Safe and constrained controllers can avoid danger, maintain viable states, and preserve future options. Specifying their constraints, feasible actions, state information, controller class, and observation process turns these abilities into executable comparisons.

PTVC is stronger than local error reduction, policy adaptation, or safety maintenance. Its latent target concerns a completed conscious stream, while its empirical burden concerns measurable consequence structures under uncertainty. The comparison therefore defeats rivals through predictions and recovery, never through labels that merely sound less profound.

The comparison uses one common observation room. PTVC candidates and every computational rival receive the same channels, cadence, calibration, missingness, endpoint information, action availability, tuning budget, and protected scoring. No model receives oracle access or a simpler measurement problem.

Every skeptical branch remains live. A predictive, learning, control, conditioning, or selection model may win; several models

may remain nonseparable; every listed model may fail absolute-fit or open-set checks. Each result changes the next design and PTVC scope without becoming a judgment about the participant.

Two positive contributions follow. Computational rivals receive enough structure to win, and any surviving PTVC-specific remainder becomes explicit enough for a new test. A residual after fair comparison is a research target, not a trophy.

Technical depth is earned because the rivals are technical. Equations and tables expose commitments that prose can hide: a generative equation defines a model, a scoring rule defines a comparison, and a bridge defines a conditioned process. None acquires target-level warrant from mathematical elegance. Formalization also reveals when two rivals differ only by parameterization, preventing redundant models from being counted as independent explanatory alternatives.

These rivals supply the model families later inference and red-team chapters must adjudicate. The sequence matters: specify the model and observation interface first, then test recovery, separability, and protected prediction. Adjudication cannot rescue a rival that was never made executable. A sealed recovery benchmark should precede substantive comparison, showing that the pipeline can identify each rival when that rival generated the data.

Comparisons identify the mechanisms that best account for each observation and the features that remain unexplained. Unexplained residuals should be carried forward as targets for the next comparison rather than automatically assigned to PTVC.

22.1 — Predictive Processing and Error Dynamics

Predictive-processing accounts begin with a structured claim: observations are interpreted through learned expectations, hierarchical latent states, precision weighting, attention, bodily context, and prediction errors. A hallway sound, symptom, or social cue is not processed as raw input; its effect depends on what the system expects and how uncertain that expectation is (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013; Samara, 2026).

That makes predictive processing a powerful rival because many changes in felt experience can arise from changes in expected input, precision, attention, and model revision. A signal that once dominated consciousness may soften when it becomes predictable. A familiar routine may reduce arousal. A threatening uncertainty may become less intense when the world begins to make sense (Clark, 2013; Friston, 2010; Hodson et al., 2024).

These dynamics can make distress fall, variance narrow, and attention reorganize without any commitment to cumulative terminal neutrality. The section therefore asks which prospective prediction follows from the computational model and which residual, if any, requires a different target-level explanation.

Prediction success is not felt goodness. A model can predict a harmful environment accurately while the stream still carries fear, vigilance, humiliation, or pain. Accuracy may reduce surprise without reducing the burden that matters to the person.

Equation role: Rival generative template. For a fixed action sequence, this equation defines a hierarchical observation model for comparison.

$$p_{\theta}(o_{1:T}, s_{1:T} \mid a_{1:T-1}) = p_{\theta}(s_1) \prod_{t=1}^T p_{\theta}(o_t \mid s_t) \prod_{t=1}^{T-1} p_{\theta}(s_{t+1} \mid s_t, a_t)$$

Equation role: Prediction-error diagnostic. This loss measures precision-weighted error inside a rival model, not felt value or terminal neutrality.

$$L_{PE} = \sum_{t=1}^T (o_t - \delta_t)^{\top} \Pi_t (o_t - \delta_t)$$

Table 22.1. Predictive-processing rival components.

Component	What it models	How it can mimic a PTVC-like pattern
Latent state	Hidden causes inferred from observations	A cause estimate stabilizes and the observed signal looks repaired
Prediction error	Difference between expected and observed input	Error declines after repeated exposure, producing apparent relief
Precision	Weight assigned to an error channel	Attention narrows or quiets without cumulative closure
Model revision	Learning of hidden causes over time	A new expectation changes reports and arousal

22.1.1 — Hierarchical Generative Models

A hierarchical generative model treats observation as the visible edge of a hidden process. Higher levels predict lower-level

causes, lower levels return residual errors, and the system updates beliefs when the world does not fit. Rao and Ballard's predictive-coding tradition made this architecture concrete for visual cortex, but the logic travels more broadly as a modeling family (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

Nothing here requires every cortical circuit to implement one canonical predictive algorithm. The relevant fact is that hierarchical models can generate settling, salience, and surprise reduction without invoking terminal fairness. A noisy scene can feel calmer because the system learned its causes, and that ordinary explanation deserves its own test (Clark, 2013).

The data object is not the person. It is a sequence of observations, latent state estimates, prediction errors, and precision weights. In a phone-based affect study, for example, a participant's report after a repeated alarm may change because the alarm is now expected, not because a burden has been balanced across the life.

A simple rival template defines latent states, observations, and a predictive distribution. The model's work is to minimize expected discrepancy between observation and prediction under its current beliefs. If that work explains the trajectory, the terminal-closure-specific interpretation must retreat from that component.

The failure case is overreach by abstraction. A researcher may see error decline and write as if the felt stream has been repaired. That would confuse a model's epistemic success with a person's relief. Good reporting keeps the two apart, especially in trauma, illness, and coercive settings where accurate prediction may accompany continuing harm.

The comparison rule is direct: predictive-processing rivals must predict the same observations the PTVC candidate uses, not a simplified mood curve. When the hierarchical model explains report shifts, physiological settling, and attention change under held-out data, the result belongs to predictive processing unless a separate terminal-closure-specific residual remains.

A decisive benchmark should vary hierarchy depth rather than assume it. Recent work on testable predictive-processing explanations argues for explicit minimal commitments that connect abstract predictive language to discriminating design and reporting choices (Samara, 2026). Shallow models may explain rapid sensory accommodation, while deeper models may be needed for social expectations, interoceptive beliefs, and context-dependent affect. The comparison should ask whether the apparent long-range pattern survives after each level is allowed its own learning rate, volatility, and precision. If a deeper predictive hierarchy absorbs what looked like cumulative closure, the result favors a richer ordinary learning account. If a stable residual remains across plausible hierarchical specifications and transports to held-out contexts, that residual becomes a sharper scientific target rather than a remainder defined by model weakness (Clark, 2013; Hodson et al., 2024).

22.1.2 — Prediction Error and Affect

Prediction error is the difference between what the model expected and what the world delivered. In affective life, that difference may matter because unexpected pain, unexpected relief, unexpected rejection, or unexpected safety can change what experience feels like. The rival begins with this measurable

relation between expectation, outcome, and affect (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

Momentary happiness already has quantitative experimental models. Rutledge et al. (2014) combined repeated happiness reports with a monetary-choice task and neuroimaging in twenty-six participants, then replicated the behavioral model in 18,420 smartphone participants. Recent expectations and reward prediction errors predicted reported happiness better than current task earnings alone. Blain and Rutledge (2020) subsequently separated reward magnitude from learned win probability in seventy-five participants: in that task, learning-relevant probability errors accounted for happiness better than prediction errors in learning-irrelevant reward magnitudes. These results show that the temporal organization of expectations and outcomes can be measured and tested against feeling.

The question is whether affect tracks raw error, signed error, precision-weighted error, changes in uncertainty, or a more complex appraisal of bodily and social meaning. A familiar hospital sound may stop startling one person while the same sound keeps another awake because it predicts bad news. The sign and context of error matter (Friston, 2010; Hodson et al., 2024).

Expected money and expected feeling can contribute different learning signals. In a repeated social-exchange task with 41 participants, Heffner et al. (2025) measured both expectations before outcomes and recorded EEG. Differences between expected and reported emotion were associated with subsequent choices, especially early in learning about a partner. Reward discrepancies more consistently tracked the feedback-related

negativity, while emotional discrepancies more consistently tracked the later P3b response; P3b activity was also linked to subsequent choice. This gives a temporal affect model two experimentally distinguishable inputs. Recording what people expected to receive and how they expected to feel makes the evolving relation between events, appraisal, and action more directly testable.

Prediction can reduce uncertainty while leaving other aspects of experience unchanged. A patient may understand a diagnosis more clearly while still needing support for fear, discomfort, and practical demands.

A rival model can represent affective change as a function of prediction error and precision. The empirical test should ask whether changes in reported burden, arousal, attention, and behavior follow these error terms across time. If they do, the rival has earned explanatory space.

The hostile quotation risk is subtle. A sentence such as error reduction feels good can be cropped into a claim that adaptation to harm is good. The chapter's local language should say something stricter: error reduction may reduce one component of distress while leaving the burden of the condition morally and scientifically visible.

After fitting a prediction-error model, compare its account of cumulative paths, endpoint-independent signatures, reserve-linked relationships, and cross-context patterns with the additional predictions of PTVC.

Better-than-expected and worse-than-expected outcomes can have different temporal profiles, bodily signatures, and learning consequences even when their absolute magnitudes match. A protocol can estimate these asymmetries and compare the contributions of surprise, valence, controllability, and their interactions to immediate feeling and later adaptation (Friston, 2010; Hodson et al., 2024).

The mapping is also experimentally discriminable. In two experiments involving three hundred participants, Forbes and Bennett (2024) found that the affective influence of negative reward prediction errors depended on free choice and the absence of counterfactual feedback; positive prediction errors did not show a significant effect in their analyses. This contextual result supplies three useful design variables: agency, outcome sign, and information about unchosen alternatives. Their interactions can be specified in advance and tested in held-out conditions.

22.1.3 — Precision and Attention

Precision is the model's estimate of how much weight an error deserves. A small error with high precision can dominate experience, while a larger but unreliable signal may be ignored. This gives predictive processing a refined account of why some burdens become loud inside consciousness and others pass quietly (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

Attention can be modeled as precision control. The organism does not simply receive errors; it allocates gain. A veteran in a crowded grocery store, a parent waiting for a medical call, or a student after a public humiliation may experience ordinary cues as unusually

salient because the model gives those channels high precision (Hodson et al., 2024).

Precision-weighted prediction systems can repeatedly direct attention to cues that remain uncertain or action-relevant. This provides a concrete model of recurring attention and state allocation.

The rival should therefore be implemented with precision parameters, attention allocation, and channel-specific uncertainty. A model that can explain recurrent salience, avoidance, physiological arousal, and report changes through precision dynamics has not merely used a metaphor; it has supplied a data-generating competitor.

A recurring cue can be studied through attention traces, the person's account, and contextual observations. Precision remains an estimated model parameter whose interpretation depends on these channels.

If precision dynamics explain the pattern, the terminal-closure-specific interpretation should not treat the same event as evidence of unresolved burden. The honest result may be that attention and uncertainty, not terminal fairness, produced the path. That is a rival victory, and it should be recorded as such.

Precision should be independently estimated or varied through benign, consented tasks where feasible. Cue reliability and neutral attentional demands can distinguish a precision mechanism from a retrospective label for salience. Threat-related contrasts must use existing data or naturally occurring variation within Chapter 3's safeguards, not induced threat for Law-specific evidence. The

comparison tests whether constrained precision parameters predict held-out responses and whether the closure candidate adds a further transported restriction (Friston, 2010; Clark, 2013).

22.1.4 — Learning and Model Revision

Predictive systems learn. They revise generative models when repeated evidence makes old expectations unreliable. This is why a first panic in an elevator may differ from a tenth ride after successful treatment, and why a first night in hospice differs from a later night when routines and caregivers are known (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

Model revision can create relief by changing expected causes. It can also create burden by teaching a person that the world is unsafe. A child who learns that shouting predicts danger has not achieved balance because the prediction became accurate. The model may have improved while the life became heavier (Clark, 2013; Friston, 2010).

For the rival, learning is not decorative. It must specify update rates, priors, volatility, context changes, and the time scale over which the generative model changes. A single observation can surprise; repeated observations can alter the structure that makes surprise possible.

The empirical design should separate rapid error reduction from slower model revision. Experience sampling, passive context, and task-based uncertainty measures can help identify whether the system merely habituated to a cue or changed its beliefs about the hidden causes behind that cue.

A common failure is to call every model revision repair. Some revisions are costly adaptations to unsafe conditions. The chapter keeps repair as a lived-experience and trajectory question, not as a synonym for learned prediction. A person can learn a dangerous world perfectly and still deserve rescue from it.

The decisive prediction must be prospectively specified at the level of latent variables, observation channels, timing, and protected scoring. A post hoc story about prediction error can fit almost any pattern; an executable model must risk a different forecast than its competitors.

Model revision should also be tested for renewal and context specificity. A response that appears extinguished in one room, relationship, or physiological state may return when the context changes. That pattern is expected under many learning accounts and can imitate incomplete repair. Longitudinal designs should therefore include context transitions, reinstatement probes, and volatility changes rather than treating a smooth training curve as a permanent transformation. If the effect renews exactly where a learned model predicts, ordinary learning receives the credit. If the hypothesized cumulative relation remains invariant across those transitions, the closure program has identified a more demanding target (Sutton & Barto, 2018; Gershman, 2018).

22.1.5 — Predictive Success Versus Felt Value

Predictive success concerns how well an internal model anticipates events. Felt value concerns how those events are experienced. Studying their relationship connects changes in

expectation with the temporal pattern of lived valence (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

A practical example separates prediction from wellbeing. A caregiver learns exactly how long a benefits office will place her on hold, which forms will be rejected, and which phrase will trigger another transfer. Her predictions improve. Her burden does not therefore become lighter, and the institution has not become fair (Friston, 2010).

Prediction may reduce surprise while administrative labor, fear, and fatigue continue. Measuring both expectation and experienced burden helps identify which part of the situation has changed.

Reporting connects predictive adequacy with experiential interpretation: the model's forecast is evaluated against the observation, and the observation's meaning is specified through the measurement model.

PTVC's possible role requires separating predictive success from felt value. If the target is cumulative felt trajectory, ask whether the rival explains that trajectory rather than merely forecasting accompanying observations.

The separation can be made operational by scoring two outcomes independently. One score evaluates forecasts of reports, behavior, physiology, and context; the other evaluates the adequacy of the proposed bridge from those observations to the declared experiential estimand. High predictive accuracy with a failed bridge is not evidence about cumulative felt value. Conversely, a defensible experiential estimate with poor forecasting does not

validate the generative model. Reporting both prevents a single metric from absorbing epistemic success and phenomenal interpretation. It also makes partial victories visible, which is essential when the same model can predict a person accurately while misunderstanding what the prediction means (Gneiting & Raftery, 2007; Hernán & Robins, 2020). A third report column can state whether disagreement between predictive and bridge scores changes only the generative model, only the measurement bridge, or both.

22.1.6 — When Predictive Processing Wins

Predictive processing wins locally when explicit states, observations, precision, update rules, and held-out performance explain the targeted signature better than the terminal-closure-specific candidate. It need not explain every life; it must win the declared study under equal observation and model-recovery rules (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013). Local victory should be tested for transport before it becomes a general explanatory claim, because predictive success may depend on one task or care context.

In a terminal-room routine study, familiarity, lower novelty, medication timing, caregiver continuity, and precision-weighted expectation may explain reduced arousal. The result should then be reported as predictive and contextual stabilization, with any humane gain credited to reduced uncertainty or threat rather than terminal neutrality (Hodson et al., 2024; Clark, 2013). A corresponding negative control can test whether similar stabilization appears in nonterminal periods with comparable familiarity and care structure.

The victory rule is fixed before results: predictive superiority narrows the relevant PTVC-specific mechanism, mixed evidence remains mixed, and collective model failure remains open. The candidate may not rebrand the winning rival after the fact as the mechanism through which fairness operated. Any successor model that incorporates the winning predictive mechanism must be labeled as a new version and retested against the original rival independently.

Allowing predictive success to be enough protects both care and the formal claim. Ordinary explanation owns what it explains, leaving any remaining target sharper and less ambiguous. The remaining PTVC question then concerns structure that predictive regulation does not already explain, reducing overlap between candidate and rival.

22.2 — Active Inference and Expected Control

Active inference changes the rival from passive prediction to action selection. An agent may update beliefs to fit sensations, act so future observations become more consistent with expected states, or trade immediate cost against information and pragmatic value. That architecture is relevant to horizon, reserve, repair, and admissible-path questions because it offers a future-directed ordinary generator (Seth & Friston, 2016; Friston et al., 2017; Lageman et al., 2026).

The danger is semantic capture. A model may contain prior preferences, expected free energy, epistemic value, pragmatic value, and policy precision, and those words may sound like values in the moral or experiential sense. But a prior preference in a generative model is not a judgment of human worth, and expected free energy is not the hidden ledger (Friston et al., 2017; Seth & Friston, 2016; Badcock & Davey, 2024).

Active inference enters as an executable competitor with its own latent states, policy set, prior preferences, epistemic term, pragmatic term, precision, observation model, and recovery burden. It can explain exploration, avoidance, uncertainty reduction, body regulation, and future-oriented behavior without terminal neutrality. It becomes discriminating only when the model yields a different protected prediction under shared observations (Lageman et al., 2026).

Equation role: one conventional active-inference rival objective. This policy-selection quantity belongs to a generative model; it is not the phenomenological ledger.

$$G(\pi) = E_{q(o,s|\pi)}[\ln q(s|\pi) - \ln p(o, s|C)]$$

Table 22.2. Shared terms that must not be merged.

Term	Meaning in the rival	Boundary for this book
Preference	Prior over expected outcomes	Not moral value, not human worth, not candidate interval contribution
Policy	Possible action sequence	Not destiny and not a proof-bearing path
Free energy	Model evidence bound or expected objective	Not $L(t)$ and not an exact valence derivative
Precision	Confidence or gain on an error channel	Not importance of a person

22.2.1 — Policies and Expected Outcomes

A policy in active inference is a possible course of action evaluated through expected outcomes. The agent selects among policies not merely by immediate reward but by predicted observations, hidden states, uncertainty, and prior preferences. This already gives the rival a way to model future pressure without invoking a fairness constraint (Seth & Friston, 2016; Friston et al., 2017).

A policy is an explicit object in the model, not a poetic tendency. It specifies available actions, predicted state transitions, expected observations, uncertainty, and a horizon. Choosing whether to

answer a difficult message, ask for help, rest, or avoid a place can then be modeled as selection among policies under uncertainty (Friston et al., 2017).

The PTVC candidate often speaks about admissible paths and remaining opportunity. Active inference can produce similar-looking behavior because some policies are expected to increase uncertainty, pain, risk, or bodily instability. A policy may become unattractive without any ledger pressure beyond the generative model's own expectations.

Compare the observed action sequence with the sequence predicted from expected outcomes under active inference, then identify any additional temporal relationships predicted by PTVC.

Policy preference describes the action a model favors under its stated assumptions. Comparing that prediction with the person's choices and account of the situation makes the interpretation testable.

Policy choice, latency, exploration, avoidance, and report change provide measurable outcomes for comparing active inference with additional path constraints.

Policy evaluation becomes especially important when actions change what can later be observed. Avoidance may reduce immediate prediction error while shrinking the evidence available to revise beliefs; exploration may increase short-term discomfort while improving future control. A fair comparison must model those feedbacks instead of treating action as an external covariate. It should ask whether the trajectory follows expected policy

consequences, information gain, or a separate cumulative constraint. When active inference predicts both choice and subsequent affect from the same declared policy set, it supplies a strong ordinary rival. A closure-specific residual must then appear in outcomes that were not already built into the policy objective (Friston et al., 2017; Badcock & Davey, 2024).

22.2.2 — Preferences and Prior Beliefs

Prior preferences in active inference are model ingredients over outcomes. They can represent expected bodily ranges, safe temperatures, social cues, or sensory conditions the organism is organized to seek or maintain. They should not be read as moral preferences, life goals, spiritual evidence, or proof of what the person consciously values (Seth & Friston, 2016; Friston et al., 2017).

This distinction matters because prior preferences can look like a hidden standard. A model may encode a preference for low pain, stable glucose, social safety, or predictable breath. That does not make the model a fairness theory. It makes it a generative account of regulation and action under expected states (Seth & Friston, 2016; Badcock & Davey, 2024).

A hospital patient may prefer dim light and fewer interruptions because those outcomes reduce arousal and pain. The active-inference model may represent that as preferred observations. The humane interpretation is straightforward: care conditions matter. The scientific interpretation is narrower: prior preferences may explain policy choice and burden change.

Prior preferences can influence trajectories. Representing their relationship to the latent ledger explicitly keeps the accumulated quantity and its predicted boundary separately defined and available for model evaluation.

Empirical work should declare whether preferences are fitted from behavior, self-report, physiology, clinical context, or task structure. If the model is allowed to infer whatever preference makes the path look sensible, it will explain too much and discriminate too little.

A preference-based active-inference rival wins when protocol-fixed priors and policy rules explain observed actions, reports, and physiological changes under protected testing. It fails when preferences become retrospective names for whatever happened. The distinction is prospective specification and recovery, not rhetorical elegance.

Prior preferences must be empirically distinguished from learned beliefs about what is likely. A person may expect pain without preferring it, or prefer safety while believing it is unavailable. Conflating these quantities can make any observed action look optimal after the fact. The model should therefore state which parameters encode desired outcomes, which encode expected outcomes, and how each is estimated or constrained. Cross-context tests can then ask whether the same preference structure transports when beliefs change. If flexible priors merely absorb every discrepancy, the active-inference rival becomes protected rather than predictive; if constrained priors forecast behavior and feeling, the rival has earned explanatory force (Friston et al., 2017).

22.2.3 — Epistemic and Pragmatic Value

Active inference can distinguish epistemic value from pragmatic value. Epistemic value concerns reducing uncertainty through information-seeking. Pragmatic value concerns outcomes aligned with prior preferences. This distinction matters because curiosity, checking, reassurance-seeking, and avoidance can otherwise be flattened into one vague control story (Seth & Friston, 2016; Friston et al., 2017).

A person waiting for a biopsy result may refresh a portal because information itself has value, even when the result could be frightening. Another person may avoid the portal because the pragmatic cost of exposure is too high. Both behaviors can be modeled without turning them into ledger balancing (Badcock & Davey, 2024).

Recurring attention may reflect information seeking, uncertainty reduction, policy rehearsal, or avoidance. Matching available information lets the comparison test different predictions for future choice or affect.

The model should specify expected information gain, outcome preference, and policy precision separately. If all three are combined into a single vague pressure term, the rival becomes difficult to test and the PTVC candidate can win too easily. A serious rival keeps its own variables distinct.

The public safeguard is against interpreting information-seeking as obligation. A participant who checks, reports, or refuses to check is not proving character. The behavior may reveal

uncertainty dynamics, not loyalty, courage, weakness, holiness, readiness, or denial.

Epistemic-pragmatic models win when they explain checking, avoidance, reassurance, exploration, and report timing under common observation. A terminal-closure-specific account must then either identify a residual pattern or concede that information and outcome expectations explain the local dynamics.

Epistemic and pragmatic value can pull in opposite directions. A diagnostic test may be unpleasant now but reduce uncertainty; a comforting routine may feel better while preserving ignorance. This tradeoff offers a discriminator when the information structure, action costs, and predicted choice pattern are protocol-fixed, then compared across active-inference and reinforcement-learning accounts (Friston et al., 2017; Sutton & Barto, 2018).

22.2.4 — Homeostatic and Allostatic Coupling

Active inference often meets the body through homeostatic and allostatic regulation. The organism expects some bodily states and acts to keep or regain them. Hunger, breathlessness, pain, temperature, exhaustion, infection, and safety signals can all enter policy selection through expected bodily outcomes (Seth & Friston, 2016; Friston et al., 2017).

Chapter 22 incorporates interoception and adaptation into generative policy models, connecting the earlier mechanism chapters with active-inference and control comparisons (Friston et al., 2017; Seth & Friston, 2016).

A concrete example is breathlessness after a stair climb in a person with chronic lung disease. Prediction, prior bodily range,

panic, action options, medication timing, and environmental control can all shape experience. A model may explain the change through interoceptive policy selection rather than through hidden terminal pressure.

The model should declare body-state variables, observation noise, control actions, and horizons. If treatment, rest, posture, social reassurance, or oxygen changes the trajectory, the rival must be allowed to own that explanation where it predicts the data.

The local ethical boundary is care appropriation. If better care reduces burden, the result first honors better care. It is not evidence that life is secretly fair. It is evidence that the body, environment, and trusted support were changed in ways the person could feel.

Homeostatic and allostatic active-inference rivals win when body-state regulation and policy selection explain observed trajectories under shared testing. The terminal-closure-specific claim remains only where it predicts additional structure beyond ordinary regulation and care conditions.

Multilevel models can connect interoceptive observations with behavior and reported valence while representing slower allostatic changes in expected states (Seth & Friston, 2016; Sterling, 2012).

22.2.5 — Free-Energy Change Is Not the Ledger

The free-energy framework offers a unified language for perception, action, learning, and self-maintenance, but variational free energy, expected free energy, prediction error, surprise, and the PTVC phenomenal ledger are different mathematical objects.

Free-energy quantities may enter as modeling objectives, bounds, proxies, or comparison features; any proposed link to cumulative felt experience needs a separately validated bridge (Seth & Friston, 2016; Friston et al., 2017).

If a proposed valence rate is the time derivative of a state function, its integral equals the final state value minus the initial value. Telescoping alone does not make that difference zero. Zero becomes an identity only when equal boundary values are also imposed, or the function is chosen to force the result. An independently specified potential could instead be an empirical model, but it would still need a validated bridge to phenomenal valence.

Free-energy quantities can generate predictions for reports, arousal, exploration, avoidance, recovery, and policy selection. Their contribution can be compared directly with the corresponding PTVC predictions.

The public language should be plain. Free-energy reduction can describe a model's inference or action process; it does not tell a person that their suffering was optimized, needed, chosen by nature, or paid forward. The words optimization and preference require careful translation before they reach ordinary readers.

Evaluate expected-free-energy dynamics against the declared observations and assumptions. A proposed connection to PTVC should specify additional predictions for a subsequent comparison.

22.2.6 — Distinctive Versus Shared Predictions

Active inference shares many predictions with other control accounts: reduced uncertainty, policy change, body-state regulation, and environmental sampling. The PTVC candidate may also discuss repair, admissibility, horizon, and reserve. Without distinctive predictions, the theories can appear to agree while saying different things (Seth & Friston, 2016; Friston et al., 2017).

A distinctive prediction is not a beautiful phrase. It is a pattern a model expects more strongly than its rivals under the same observations. It may concern timing, channel mismatch, variance shape, reserve interaction, endpoint independence, or a failure of prediction-only accounts in a specific task (Badcock & Davey, 2024; Friston et al., 2017).

Label predictions shared by the models. For example, reduced arousal after gaining reliable control over a room can guide investigation of control-related change, while an additional predicted feature may distinguish the mechanisms.

The comparison should include a table of shared and distinctive predictions before data are analyzed. Policy precision, prior preferences, prediction error, reserve interaction, terminal concentration, and held-out calibration should not be treated as one blended evidence cloud. For each contrast, state the expected direction and observation window so competing models can be scored on the same quantity.

Active inference supplies a developed comparison framework. Specifying an additional measurable relationship gives PTVC a

distinct prediction to evaluate across settings. A predicted difference in recovery timing, for example, should be assessed separately from agreement about the direction of change.

The section closes with a rule later chapters make operational: shared predictions receive no model-specific credit, while a distinctive prediction must be fixed in advance and allowed to fail. This is how vocabulary becomes a test rather than a source of borrowed authority. The comparison then turns on the remaining quantitative difference, even when both models describe the observed improvement well.

22.3 — Reinforcement Learning and Policy Adaptation

Reinforcement learning asks how behavior changes through consequences. Agents may learn values, policies, transition models, and action tendencies from reward, punishment, prediction error, exploration, and planning. Computational reward enters as a task-defined learning signal; its relationship to felt valence is a separate measurement question (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

Homeostatic reinforcement learning supplies a concrete physiological architecture for this comparison. Keramati and Gutkin (2014) define primary reward through reduction of distance from a physiological set point. Under their assumptions, policies maximizing discounted reward also minimize discounted homeostatic deviation, and learned cues support anticipatory regulation. This is a mathematical result within a specified model, supported by comparisons with existing behavioral findings. It shows how reward seeking and bodily stability can share a computational objective. The model also supplies a useful algebraic comparison: an undiscounted sum of rewards defined as state differences telescopes to the initial-to-final change. In simulation and ethically available observational records, compare the declared PTV route with this internally motivated learning model under the same state information, observation process, and complexity budget.

Learning gives adaptation a temporal structure. People avoid what hurt, repeat what relieved pressure, plan around scarcity, and

change strategies when old actions fail. These dynamics connect observed behavior with evolving state allocation and time horizons (Sutton & Barto, 2018; Daw et al., 2011; Gershman, 2018).

A learning model becomes a useful comparator when it predicts the observed path under declared states, actions, rewards, horizons, and partial observability. Its reward and return variables should be defined in the units of the task.

Equation role: POMDP belief update. Here η is the normalizing constant that makes the posterior sum to one, and the summation runs over prior hidden states s . The update defines partial observability for learning rivals and prevents oracle-state comparisons.

$$b_{t+1}(s') = \eta p(o_{t+1}|s') \sum_s p(s'|s, a_t) b_t(s)$$

Equation role: Temporal-difference rival update. This is a learning rule for value estimates, not a measurement of felt experience.

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha \left[r_{t+1} + \gamma \max_a Q_t(s_{t+1}, a) - Q_t(s_t, a_t) \right]$$

Table 22.3. Toy-environment simulation plan.

Environment feature	Learning rival prediction	Discriminating question
Changing reward	Policy shifts after value update	Does PTVC add a transported restriction beyond updated return?
Hidden state	Belief tracking and checking behavior	Does uncertainty, not burden closure, explain recurrence?

Environment feature	Learning rival prediction	Discriminating question
Finite horizon	Urgency and conservative choices near the end	Can horizon pressure appear without terminal neutrality?
Offline replay	Delayed policy change after memory simulation	Can planning explain reactivation?

22.3.1 — State, Action, Reward, and Return

A reinforcement-learning decision process specifies states, actions, transition probabilities, rewards, and a policy. Its return is a discounted or finite-horizon sum of the defined rewards (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

A task reward may consist of points, money, social feedback, relief from a stimulus, or another modeled signal. Defining its role and units makes the learning model interpretable (Sutton & Barto, 2018).

In a short-horizon laboratory task, a participant may choose between a safe small gain and a risky large gain as time expires. The model represents the task’s rewards, returns, and changes in choice as the remaining opportunities shrink.

A serious RL rival should define the state variables available to the agent, the action set, the reward function, the transition model, the horizon, and the policy class. If these are fitted after the fact, the model may explain too much and teach too little.

The public-risk sentence belongs near the model: reward in a study is an instrument, not a measure of inner worth. A participant

who chooses one policy over another has not been ranked as fair, rational, holy, loyal, or healed.

Declared rewards and transitions generate predictions for behavior, reports, and physiological change. Held-out comparisons can assess those predictions alongside the additional trajectory restrictions proposed by PTVc.

A reward variable organizes the model's policy updates and may represent an engineered proxy, delayed label, or task outcome. Comparing several specified encodings shows which one the data identify. Potential-based reward shaping is a classic example of a reward transformation that can preserve the optimal policy under defined Markov-decision-process conditions, underscoring that computational reward representations can change without establishing a new phenomenal quantity (Ng et al., 1999). Constraints on the reward function make predictions of choice and feeling across tasks more informative (Sutton & Barto, 2018; Daw et al., 2011).

An independent close behavioral replication with 50 participants largely reproduced the main findings of Rutledge's computational happiness model. Its preregistered individual analyses obtained a mean within-person fitted R^2 of 0.54, compared with 0.47 in the original report (Vanhasbroeck et al., 2021). This provides an independently reproduced quantitative account of how expectations and recent outcomes relate to reported happiness.

Affect can improve the explanation of learning itself. Teoh et al. (2026) compared three models in 75 participants: reward-based learning alone, an additional affective influence on choice, and

affect-informed updating of expected value. Jointly fitting choices and explicit expectations favored the updating model, whose average participant-level WAIC model weight was 0.651. A separate 55-person study reproduced the behavioral findings, and a 95-person experiment found that valence tracked expectations during passive learning under varying predictability. Positive feeling was associated with more favorable expectations after accounting for received rewards. These results supply an empirical architecture linking present feeling, developing expectations, and future action. For the PTVC program, that architecture provides a concrete starting point for testing how accumulated affective history shapes later trajectories.

22.3.2 — Temporal-Difference Learning

Temporal-difference learning updates value estimates from the difference between expected and observed future value. Its signed updates provide a timing model for studying disappointment, relief, surprise, and related subjective changes (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

The temporal-difference error updates predicted value. Its relationship to subjective change and neural signals can be tested within the specified task and observation model (Daw et al., 2011; Gershman, 2018).

At the neural level, Bayer and Glimcher (2005) recorded individual dopamine neurons while monkeys learned when to make an eye movement for juice reward. Activity tracked the difference between the current reward and a weighted history of earlier rewards for better-than-expected outcomes. The sign

asymmetry is informative and belongs in models of the response. Nervous systems can represent recent reward history and compute a learning-relevant update, providing a concrete computational component for theories linking history, expectation and affect.

A repeated-feedback task can track learning after outcome probabilities change. When a cue predicts an adverse outcome less often, reports, reaction times, and arousal may change together. A temporal-difference model can predict the sequence and size of those adjustments.

The empirical question is whether TD errors, learning rate, volatility, and reward prediction explain the path. If they do, the Law should not claim the same evidence as hidden balancing. The outcome belongs to a local learning mechanism unless a prospectively specified residual remains.

The failure mode is neural storytelling. A dopaminergic or prediction-error correlate can become a post hoc decoration if it was not prospectively specified and compared against rivals. The chapter therefore treats neural signals as auxiliary predictions, not as definitions of the Law or of the person.

Temporal-difference rivals win when their update structure explains timing and direction better than terminal-closure-specific alternatives. They lose when their local updates fit behavior but miss cumulative path restrictions, endpoint-independent variance behavior, or reserve-linked signatures that were specified before analysis.

Temporal-difference errors create a distinctive sequence of anticipatory shifts. As learning proceeds, responses can move

from outcome to cue, and value can propagate backward through a chain of states. That backward movement can look like a burden being redistributed through time even though it arises from ordinary prediction updates. Experiments should track cue responses, outcome responses, choice, and affect simultaneously, then test whether the predicted transfer occurs. A terminal-closure account should not claim such movement as evidence unless it predicts a residual beyond temporal-difference dynamics. The rival wins when the timing and magnitude of the shift follow the learned value function under held-out conditions (Sutton & Barto, 2018; Daw et al., 2011).

22.3.3 — Partial Observability

Human decision systems rarely know the full state. They act through beliefs under partial observability: the body may be ambiguous, the social scene uncertain, the horizon hidden, and the future noisy. A POMDP is therefore a stronger rival than a fully observed toy decision process (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

Partial observability matters because uncertainty can look like burden. A caregiver may not know whether a symptom is temporary, whether a form was received, or whether a clinician will call back. The action policy changes because the hidden state is uncertain, not necessarily because the unresolved-state process has intensified (Gershman, 2018).

The belief-state update should be explicit. Observations change the probability distribution over hidden states, and policies are selected from those beliefs. This can create checking, waiting,

avoidance, reassurance-seeking, and delayed action without invoking terminal closure.

The PTVC candidate may still ask whether the resulting path carries cumulative felt cost. But the learning rival first asks whether belief updates explain the observable behavior and reports. A model that sees uncertainty honestly is harder to defeat than a model that pretends the person had perfect information.

Consent and interpretive humility set the boundary. A belief state is an inferred research object. It should never become a claim that the researcher knows what the participant truly believed, hid, deserved, or should have done.

Partial-observability rivals win when belief updates, observation noise, and action selection predict trajectories under common data. They fail when they require hidden-state freedom so broad that any path can be explained after it happens.

Partial observability matters because the agent and the researcher may each lack different information. The agent acts on a belief state inferred from incomplete cues, while the analyst observes only a subset of actions, reports, and contexts. Hidden-state models should therefore separate environmental state, agent belief, and research observation. Otherwise, unexplained residuals may simply reflect information that the model omitted. A closure-specific effect becomes more credible only after plausible belief-state dynamics and observation gaps have been modeled. When a partially observable learning model recovers the same apparent long-range pattern, the scientific interpretation

should narrow accordingly (Sutton & Barto, 2018; Gershman, 2018).

22.3.4 — Changing Rewards and Horizons

Rewards and horizons can change. A relief that mattered in the morning may lose force after exhaustion. A choice that was reversible last month may become irreversible after a deadline. A policy that made sense under abundant time may become unattractive when the horizon collapses (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

This gives reinforcement learning a way to explain horizon pressure without the Law. Finite-horizon decision-making can produce urgency, narrowing, risk aversion, or desperate exploration simply because remaining opportunity has changed. The model need not contain terminal neutrality to generate these effects (Sutton & Barto, 2018; Daw et al., 2011).

A laboratory task should not confuse this with a life. If participants learn that a game ends after twenty trials, their late choices may change. That tells us about finite-horizon learning under a protocol. It does not reveal a universal endgame of conscious existence.

The rival should declare whether rewards are stationary, drifting, state-dependent, context-dependent, or horizon-dependent. It should also declare what the agent knows about the horizon. If the true endpoint is unknown, the model must use a belief over time remaining rather than omniscient countdown.

The public language should keep finite-horizon pressure from becoming sacrifice logic. A model that predicts late urgency is not

a recommendation to induce deprivation, risk, stress, or self-denial. Research tasks must be low-burden instruments, not moral rehearsals for suffering.

Changing-reward and changing-horizon RL wins when it explains urgency, avoidance, and policy reversal under shared observation. The terminal-closure-specific account must then identify a residual beyond ordinary finite-horizon control or accept narrowing in that domain.

Rewards and planning horizons can change with development, illness, scarcity, medication, social role, and approaching deadlines. A fixed-horizon model may falsely attribute these changes to an external closure pressure. The comparison should allow prospectively specified horizon shifts and test whether they explain increased immediacy, altered risk tolerance, or changing affect. Crucially, the horizon is a decision-model parameter, not a license to infer a person's endpoint or remaining life. If changing horizons predict the observed policy and feeling trajectories, they weaken a distinct terminal mechanism; if they fail under fair flexibility, the residual becomes more informative (Sutton & Barto, 2018).

22.3.5 — Offline and Model-Based Planning

Learning does not occur only while acting. Humans remember, simulate, dream, rehearse, and plan. Model-based and offline learning can reorganize policy before the next observable behavior, which makes them important rivals to any theory of delayed repair or delayed policy reactivation (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

A person may replay a humiliating meeting on a bus ride home and answer differently the next day. A dream may recombine fear and unfinished speech. A caregiver may rehearse a phone call before challenging a denial. These changes can arise from memory and simulation without being terminal-closure-specific (Daw et al., 2011).

Distinguish cached model-free values from model-based search, replay, planning depth, and memory consolidation. Their different timing predictions make delayed actions and reports useful evidence for comparison.

A model of offline processing predicts which changes occur, when they appear, and how they can be observed. These predictions make processing, repair, learning, and state-allocation accounts directly comparable.

Offline and model-based rivals win by outpredicting terminal-closure alternatives on delayed behavior, report revision, and planning signatures. Memory language fails without a generative model, timing predictions, and recoverable parameters.

Replay, imagination, counterfactual evaluation, and memory consolidation can reorganize future choices during rest or sleep. Studies can compare reported experience during the interval, neural or behavioral evidence of replay, and later policy change to evaluate these mechanisms of delayed adjustment (Gershman, 2018; Sutton & Barto, 2018).

22.3.6 — Learning-Rival Victory

A learning rival wins without privileged information through comparable or better protected prediction, calibrated uncertainty,

interpretable parameter recovery, model recovery, and transport for the declared target. Parsimony follows adequacy. Sealed recovery tests must include reinforcement-learning, PTVC, active-inference, bridge, selection, and open-set worlds (Daw et al., 2005; Gershman, 2018; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

If the pipeline fails the required recovery thresholds in those worlds, it is not qualified for the affected confirmatory comparison. Under a valid analysis, adequate but inseparable models yield Model-ambiguous, while failure of every declared model to meet adequacy criteria yields Open-set inadequate. If learning wins, the series may not rename it as how PTVC operated unless that bridge was prespecified and independently tested; the route status changes.

Learning-rival victory can redirect humane practice toward information, agency, environment, or uncertainty reduction while retiring a terminal-closure mechanism. Defeating a route can still reveal where burden was hidden, shifted, or reduced.

22.4 — Safe and Constrained Control Baselines

Safe and constrained control is the primary ordinary comparator for claims about admissible paths, reserve, no-return states, and finite horizons. The next subsections distinguish constraint-based safety, barrier and viability methods, and their failure modes (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

Engineered controllers can maintain a system within a viability kernel under declared assumptions. The design specifies which states, actions, and constraints produce that result and provides a model for comparison with biological regulation (Altman, 1999; García & Fernández, 2015; Wachi et al., 2024).

Safe and constrained baselines receive comparable state information, action freedom, and optimization capacity. Their predicted trajectories then provide explicit comparisons for reserve, viability, and finite-horizon behavior.

Equation role: Constrained-decision baseline. This is a safe-control rival that optimizes return under declared cost limits.

$$\max_{\pi} E_{\pi} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right] \text{ s. t. } E_{\pi} \left[\sum_{t=0}^T \gamma^t C_j(s_t, a_t) \right] \leq d_j$$

Equation role: Deterministic control-barrier condition. This enforces forward invariance for an engineered controller under declared dynamics.

$$L_f h(x) + L_g h(x)u + \alpha(h(x)) \geq 0$$

Table 22.4. Safe-control baseline comparison.

Baseline	Can mimic	Specification to record
CMDP	Tradeoff between reward and harm cost	Declared cost functions and constraint thresholds
Risk-sensitive control	Avoidance of volatile paths	Risk-sensitive objective and risk parameter
Barrier controller	Trajectory remaining inside a safe set	Dynamics, safe set, and invariance condition
No-return avoidance	Urgency near irreversible states	Irreversible region and transition uncertainty

22.4.1 — Constrained Markov Decision Processes

A constrained Markov decision process extends ordinary reward optimization by adding costs or safety constraints. The agent may maximize return while keeping expected harm, energy use, risk, or constraint violation below declared bounds. This provides a direct rival to PTVC-like trajectory narrowing (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

The model object is explicit: states, actions, transition probabilities, reward, cost functions, thresholds, and policy class. A constraint can represent avoiding pain above a threshold, keeping fatigue below a limit, or preserving access to future actions. None of those constraints is automatically terminal fairness (Altman, 1999).

A practical example is a rehabilitation task in which a person chooses effort levels while avoiding pain flare-ups. A constrained model may explain why the person takes smaller steps, rests earlier, or refuses a higher reward. The behavior reflects safety constraints under the task, not a moral ledger.

The empirical comparison should include a CMDP baseline whenever the PTVC candidate predicts avoidance of no-return paths or preservation of reserve. Without that baseline, the Law may appear distinctive only because ordinary constrained control was not allowed to compete.

Safe-control studies use minimal burdens, simulated stakes, or existing clinical data within approved care-first governance. Research for the Law must not create suffering to estimate a parameter.

CMDP rivals win when explicit reward-cost tradeoffs reproduce the policy and trajectory signatures under the same observations. The PTVC candidate must then narrow to residual features not explained by constrained optimization.

Constrained Markov decision processes separate reward optimization from explicit safety or resource limits. This is scientifically useful because a trajectory can bend away from danger without any terminal balancing objective. The model should declare the constraint cost, threshold, state transitions, and policy class, then test whether the observed avoidance and recovery follow those objects. A flexible constraint should not be inferred from the same path it explains. When prospectively specified constraints predict action and affect under new

conditions, constrained control becomes a strong rival to any claim that apparent guardrails reveal PTV (Altman, 1999; Wachi et al., 2024).

22.4.2 — Risk-Sensitive and Robust Control

Risk-sensitive and robust control models do not optimize only the average outcome. They may penalize variance, worst-case loss, tail risk, ambiguity, or model uncertainty. This makes them strong rivals because real burden often lives in the tail, not in the mean (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

A person who refuses a high-upside option after trauma may not be irrational or ledger-driven. The policy may be risk-sensitive because the worst case has become too costly. A family declining additional monitoring near the end of life may be protecting quiet from a risk the protocol undervalues (García & Fernández, 2015; Wachi et al., 2024).

The model should specify the risk measure: variance penalty, entropic risk, conditional value at risk, worst-case transition uncertainty, or robust policy. Each choice has a different implication. Calling them all caution would be too vague for science.

Risk-sensitive objectives can reduce predicted variability by favoring less volatile states. Robust objectives can favor cautious actions when model estimates are uncertain. Both provide quantitative accounts of path selection for comparison.

Risk avoidance can be a rational or learned response to uncertainty, history, and present constraints. Representing those conditions improves the behavioral model.

Risk-sensitive and robust models can be evaluated through held-out predictions of path selection and variance. Fixing the risk parameter's estimation rule before comparison makes these tests informative.

These families can generate conservative action, reserve-like behavior, and apparent no-return boundaries without a fairness ledger. The comparison should vary tail risk and model uncertainty independently, then ask whether behavior and feeling respond as each control account predicts. If the observed path is driven by risk aversion or robustness, the closure interpretation must retreat. A distinctive PTVC result would need to survive those controls and transport beyond the environments used to tune them (García & Fernández, 2015; Wachi et al., 2024).

22.4.3 — Control Barriers and Viability Kernels

Control-barrier functions and viability kernels formalize safety under dynamics. A barrier can certify that allowed controls keep a state inside a safe set. A viability kernel identifies states from which at least one admissible policy can keep the system within constraints (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

This is an elegant mathematical language for future preservation. It is also a dangerous analogy if mishandled. A barrier condition can be true for an engineered controller while saying nothing

about whether a person's felt life closes at neutrality (Wachi et al., 2024).

The chapter uses barriers as baselines because they can mimic no-return avoidance. If a simulated agent avoids states from which recovery is impossible, its path may show narrowing, urgency, and apparent repair. The mechanism is safety preservation under declared dynamics, not fairness.

A clean barrier model must define the state, safe set, control set, dynamics, and probability guarantee. Deterministic and stochastic barriers should not be mixed casually. If stochastic dynamics are present, the condition must match the stochastic theorem rather than borrowing deterministic language.

Construction-is-not-evidence is the public-risk control. A figure showing a safe set and a trajectory inside it should be labeled as a candidate control baseline or engineering analogy. It should not be publicized as a map of the soul or a proof that lives self-correct.

Barrier and viability rivals win when their declared safe sets and dynamics reproduce trajectory restrictions without terminal-closure-specific terms. They fail when the safe set is chosen after seeing the result or when the controller is given information unavailable to the organism.

Barrier functions and viability kernels identify regions from which safety constraints can be maintained. Their feasible sets may shrink as resources decline or disturbances grow. Compute these regions from independently specified dynamics and constraints, then compare predicted action boundaries with

observed trajectories and any additional predictions for experience or endpoint behavior (Altman, 1999).

22.4.4 — No-Return Avoidance

No-return avoidance is a powerful ordinary explanation for apparent moral urgency. Many systems avoid states from which future options collapse. Bodies avoid injury, animals avoid traps, people avoid irreversible decisions, and institutions create guardrails because some losses are not easily repaired (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

Compare no-return avoidance through safe-control models with the predictions derived from PTVC's admissible paths and reserve variables (Altman, 1999; García & Fernández, 2015).

A concrete study might track decisions in a low-burden planning task where some choices remove future options. If participants avoid irreversible branches as the horizon shrinks, finite-horizon safe control may explain the pattern. The task should not be described as evidence about terminal sacrifice.

No-return states should be declared before analysis. They can be defined by task structure, clinical risk, physiological thresholds, or option-set collapse. If they are identified only after the path is known, the model becomes a story rather than a test.

Anti-sacrifice is the boundary. No-return avoidance models should intensify the duty to prevent avoidable harm, not imply that people should endure harm for later balance. The model's safest interpretation is early protection, not delayed meaning.

No-return rivals win when they explain urgency, narrowing, avoidance, and repair attempts through declared option loss. The terminal-closure-specific account remains only where the observed pattern exceeds no-return avoidance under the same horizon and information constraints.

No-return avoidance can arise whenever some states make future objectives unreachable. A patient conserves breath, an animal avoids a depleted patch, or a controller preserves battery because entering a region would destroy future options. The model should distinguish physical irreversibility, learned danger, institutional barriers, and phenomenal burden. Each can produce abrupt avoidance. A closure account should not infer terminal fairness from the mere existence of a boundary in action space. The rival wins when an independently estimated reachability model predicts the boundary and its movement as conditions change (García & Fernández, 2015; Wachi et al., 2024).

22.4.5 — What Constructed Controllers Contribute

A controller can be built to do almost anything its designer asks, within mathematical limits. It can close a gap, minimize a cost, reach a target, stay inside a set, or avoid a boundary. The existence of such a construction does not show that the living world implements it (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

Constructed controllers are useful as rival generators and identifiability tests. Simulation can expose their assumptions, parameter sensitivity, and signatures under known truth; figures should label the generator, setting, assumptions, and scientific

role so a construction is not mistaken for an observed mechanism (García & Fernández, 2015).

When a target is attainable under the chosen dynamics and admissible controls, a controller may be designed to reach it. Existence proofs, simulations, behavioral fits, interventions, and transported predictions then address successive questions about its operation. Independent observations can compare the proposed variables and causal structure with competing explanations (Wachi et al., 2024).

22.4.6 — Common Information and Action Sets

Safe-control comparisons become invalid when models receive different information or feasible actions. A controller with the true latent state, exact dynamics, and endpoint knowledge can outperform a model that sees only noisy observations. In that case the advantage belongs to oracle access rather than theory, so the contest must equalize information and action sets before scores are interpreted (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

The common information set specifies observations, history, sensor reliability, endpoint uncertainty, reserve proxies, context, and action availability. The common action set specifies what the organism or simulated agent can actually do, including delays, costs, and prohibited interventions (Wachi et al., 2024; Altman, 1999).

A hospice dataset provides a clear example. A model that knows the death time while the participant and clinicians did not know it cannot be compared directly to a filtered-horizon model. The

endpoint information itself becomes a hidden advantage. Fair comparison should match each model's information set to what was actually available at the decision time, not to what analysts know retrospectively.

Safe control connects to common-observation comparison through a shared interface. Before scoring, every model declares what it observes, when it observes, what it may choose, and how missingness affects the record. This declaration should include information delay, because a model with faster access can appear superior even when its mechanism is no better.

Fairness toward rivals and participants sets the limit. Model comparison should not turn a person's limited information into evidence against them, nor should it let a theory win by using retrospective knowledge unavailable to the stream. Information parity should therefore be checked during model recovery rather than assumed from nominally identical input variables.

Safe-control rivals win only when their information and action sets are legitimate and their protected predictions remain calibrated. If they require oracle endpoint knowledge, unobserved latent states, or impossible actions, their apparent advantage becomes a design diagnostic rather than a conclusion about the observed system. A rival that wins only with unavailable actions defines an upper bound on achievable control, not an explanation of the observed behavior.

22.5 — Bridge and Doob-Transform Rivals

Endpoint-conditioned processes are among the strongest rivals because they can make drift toward a terminal value appear without any native terminal controller. If the process law conditions on where or how the path ends, the preceding trajectory may look as if it knew its destination. That resemblance is why bridges belong in the primary comparison (Doob, 1957; Revuz & Yor, 1999; Särkkä, 2013).

The comparison includes Brownian bridges, Doob h-transforms, Schrödinger bridges, survival-conditioned paths, random endpoint models, last-observation alignment, and observation selection. These models make conditioning and selection effects on terminal trajectories explicit (Doob, 1957; Léonard, 2014; Léonard et al., 2014).

The equations make conditioning explicit. They describe how the process law changes and provide drift and covariance predictions for comparison under a shared observation model.

Equation role: Fixed-endpoint Brownian bridge. Drift is determined by endpoint conditioning.

$$dX_t = \frac{x_T - X_t}{T - t} dt + \sigma dW_t, 0 \leq t < T$$

Equation role: Doob-transform rival. Future conditioning changes the process law and can create apparent direction without terminal control. The simplified form $h^{-1}L(hf)$ applies on the declared harmonic conservative branch. For a time-dependent bridge, h satisfies $(\partial_t + L_t)h = 0$; nonharmonic or killed branches retain the corresponding temporal or potential term.

$$L_t^h f = h^{-1}L_t(hf) - h^{-1}fL_th = L_tf + a_t \nabla \log h \cdot \nabla f$$

Table 22.5. Bridge and endpoint-conditioning diagnostics.

Diagnostic	Bridge expectation	PTVC-relevant question
Drift direction	Moves toward conditioned endpoint	Does drift magnitude or reserve dependence distinguish the models?
Variance shape	Bridge-specific collapse pattern	Does reserve-linked innovation differ?
Endpoint knowledge	Often inserted by conditioning	Was endpoint defined prospectively?
Last observation	May act as false endpoint	Was observation shutdown modeled?

22.5.1 — Fixed-Endpoint Brownian Bridge

A Brownian bridge starts from an initial value and is conditioned to hit a specified endpoint at a fixed time. The drift corrects the remaining displacement over the remaining time, while the path converges to the specified endpoint. The path can look like convergence because convergence has been built into the conditioning (Doob, 1957; Revuz & Yor, 1999; Särkkä, 2013).

This rival is powerful for terminal studies because retrospective alignment often begins with a known end. If investigators align

trajectories backward from death or final observation, they risk creating a bridge-like picture even when the underlying process had no terminal-closure-specific closure mechanism (Doob, 1957).

A fixed-endpoint bridge generates terminal drift through its conditioning rule. Its resulting mean and covariance provide explicit predictions to compare with additional PTVC structure under shared observations.

Bridge-plot captions identify the conditioned endpoint, base process, parameter values, and observation assumptions.

The fixed-bridge rival wins locally when endpoint-conditioned drift and variance explain the observed terminal pattern without additional terminal-closure-specific restrictions. It loses where independent endpoint definitions, reserve-linked innovations, and prospective observation rules produce a pattern the bridge fails to match.

Fixed-endpoint conditioning changes a process's mean and covariance, producing an increasing pull toward the endpoint even when the base process is ordinary diffusion. Compare bridge and nonbridge generators over the same windows with the same measurement noise. Transported intervention responses, endpoint-timing relations, and cross-equation constraints specified before fitting provide further features for discrimination (Doob, 1957; Léonard, 2014).

22.5.2 — Random Endpoint Times

Real truth-level endpoints are rarely known in advance. Their timing may be uncertain, state-dependent, treatment-dependent,

or observed only through a proxy. A random-endpoint model is therefore a stronger rival than a fixed-time bridge when the study concerns the irreversible end of a qualifying conscious stream, a terminal window, or a last observation.

Random endpoint times can create selection and conditioning effects. Trajectories that survive long enough to be observed may differ from trajectories that end suddenly. Participants who remain in a protocol may differ from those who withdraw. Endpoint uncertainty becomes part of the data-generating process (Aalen et al., 2008; Little & Rubin, 2019).

When the endpoint time T is uncertain, both the comparator and PTVC candidate use a filtered distribution over possible endpoints based on the information available at the prediction time.

A useful random-endpoint rival specifies hazard, state dependence, observation rules, and the relation between clinical events and endpoint confidence. It may explain apparent terminal concentration through risk selection rather than through terminal neutrality.

Clinical death, loss of contact, and the last available observation are distinct protocol events. The endpoint model specifies how each event informs the timing of a qualifying stream's end and carries the remaining uncertainty into the analysis.

Random-endpoint models generate predictions under explicit hazard, selection, and timing uncertainty. Prospective signatures, independent endpoint adjudication, and observation-model checks evaluate those predictions.

Random endpoints condition on a distribution of times. Mixtures can alter the familiar bridge shape, and dependence between endpoint timing and the process can create selection effects. Model the endpoint law explicitly and compare independence, state coupling, censoring, and unobserved continuation with equivalent flexibility across candidates (Aalen et al., 2008).

22.5.3 — Doob h-Transforms

Section 18.4.2 developed the Doob h-transform as a path-space construction. Here it enters the comparison as a generator of event-conditioned trajectories (Doob, 1957; Revuz & Yor, 1999; Särkkä, 2013).

The transform matters here because it separates two questions. First, what path law results when a process is conditioned on an endpoint or event? Second, does nature implement a process that closes without retrospective conditioning? The first can be mathematical. The second is empirical (Léonard et al., 2014).

A Doob-transform model can represent terminal concentration, survival conditioning, or event-conditioned paths. Its directed-looking trajectories follow from the specified conditioning rule.

An executable Doob-transform comparator specifies its base process, conditioning event, h-function or event likelihood, observation filtration, and prospective information set.

Shared endpoint and observation constraints allow direct comparison with a Doob-transform model. An additional reserve relation between drift and innovation provides one example of prospectively testable structure.

A Doob h-transform expresses future conditioning as present transition dynamics. An apparent local controller may therefore be an ordinary process under conditioning. Compare likelihoods, path functionals, intervention responses, and endpoint support under both representations. Inseparable observations yield Model-ambiguous; mechanism identification awaits causal or transported evidence (Doob, 1957; Léonard et al., 2014).

22.5.4 — Selection on Survival and Observation

A terminal dataset is assembled through eligibility, consent, diagnosis, access, survival, availability, and continued observation. It is not a random cross-section of complete lives. Those selection processes can shape the apparent trajectory before any terminal model is fitted (Hernán & Robins, 2020; Little & Rubin, 2019).

Survival selection can remove extreme or sudden paths from the observed set. Observation selection can remove moments of highest clinical burden because sensors are removed, speech fails, staff prioritize comfort, or families decline. The remaining data may look calmer and more continuous than the lived stream (Hernán & Robins, 2020; Little & Rubin, 2019).

Selection is a full rival mechanism. When the people and moments entering a terminal window are already filtered by care context, clinical stability, medication, communication capacity, or continued device use, apparent concentration may belong to the record rather than the latent path.

A credible selection model declares inclusion, exclusion, attrition, survival, observation intensity, and shutdown. Estimate jointly

where possible; otherwise bound what cannot be directly estimated. Missingness is not neutral by convenience.

Observation models record family decisions, device removal, and withdrawal as causes of missingness. Data collection follows consent and care priorities.

Selection rivals win when survival and observation mechanisms reproduce the terminal signature. They lose only when the signature survives robust sampling, prospectively specified endpoint rules, and sensitivity analyses that make unobserved intervals scientifically visible without violating care.

Survival and observation can select trajectories by state, treatment, resources, and care context. Model entry, retention, death, withdrawal, sensor failure, and institutional discharge. Inverse weighting addresses measured selection; sensitivity analyses assess how unmeasured dependence would affect the result (Aalen et al., 2008; Little & Rubin, 2019).

22.5.5 — Last-Observation Alignment

Last-observation alignment can manufacture a terminal-looking path by treating device removal, transfer, discharge, sensor failure, or withdrawal as the truth endpoint (Little & Rubin, 2019).

Last observations may occur because illness, burden, care, or equipment changed. They are observation events, not automatically endpoint events (Little & Rubin, 2019).

A wearable removed during hospice may leave a stable final reading that appears to settle. The pattern may reflect device tolerance and care decisions rather than terminal neutrality.

Model observation shutdown. If last-observation alignment reproduces the concentration, narrow or suspend the closure-specific interpretation.

Final intervals retain the uncertainty implied by the missingness model and the documented reasons observation ended.

Prospectively defined endpoint windows, independent stopping rules, and explicit observation-process models allow the analysis to test whether the signature persists across alternative accounts of missingness.

Compare calendar, protocol, event, and plausible anchor times, and show the unaligned data. Signatures confined to last-observation alignment make alignment and attrition the immediate model-comparison targets (Little & Rubin, 2019).

22.5.6 — Bridge Variance Shape

Conditioned bridges approach endpoints by construction, weakening drift as a discriminator. Variance, covariance, endpoint surprise, and reserve relations discriminate better when reserve, drift, innovation, and endpoint structure are estimated independently (Doob, 1957; Revuz & Yor, 1999; Särkkä, 2013).

Compare bridge and reserve-coupled variance, noise floors, selection, and endpoint uncertainty under shared cadence and missingness. Test which joint pattern transports.

Medication, sedation, frailty, dropout, and changing cadence belong in the model of observed variance.

Diagnostics report bridge variance, reserve-coupled variance, observation floors, and selection under common simulated cadence, missingness, endpoint uncertainty, and device versions. Using the same simulation clock makes differences in apparent variance attributable to the candidate process rather than to unequal observation opportunity.

Public figures state whether compression is latent, measured, noise-adjusted, bridge-generated, reserve-linked, or unresolved. Captions should also identify whether the endpoint was conditioned, observed, or independently adjudicated, since that distinction changes the interpretation of compression.

Bridge rivals win by reproducing protected structure through endpoint conditioning and the shared observation model. PTVC is distinctive only where prespecified restrictions survive. A bridge victory should retire the claimed native signature while leaving separately supported measurement or endpoint findings intact.

22.6 — Event-Conditioned and Selection Rivals

Event-conditioned and selection rivals move the comparison from mathematical conditioning to real recruitment and observation. Referral, diagnosis, treatment, hospice entry, survival, device use, communication capacity, caregiver presence, and last-observation rules can reshape the sample and its terminal curves without changing the latent process.

A terminal dataset is therefore a selected history, not a transparent window onto all eligible streams. The selection model must name who becomes observable, when observation begins, what keeps it active, what stops it, and how those processes depend on the latent state, care, prognosis, treatment, burden, and endpoint.

The practical aim is constructive: selection models reveal which additional records, negative controls, denominator flows, and sensitivity bounds can recover a more honest estimand. When recovery is impossible, No adequate estimate is the disciplined outcome.

Table 22.6. Event-conditioned rival envelope.

Conditioning mechanism	Example in data	Potential artifact
Hospice entry	Enrollment after prognosis and family consent	Care-context concentration
Treatment change	New medication or routine	Trajectory shift caused by intervention
Observation shutdown	Device removed for comfort	Clean-looking final curve

Conditioning mechanism	Example in data	Potential artifact
Competing event	Discharge, transfer, sudden death, recovery	Mixed endpoints treated as one

22.6.1 — Hospital and Hospice Selection

Hospital and hospice samples are selected by diagnosis, prognosis, access, insurance, geography, clinician referral, family support, cultural scripts, and willingness to participate. These entry processes can create regularities before any terminal model sees the data (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

A hospice cohort may include people whose symptoms are managed, whose families accept the service, and whose clinicians expect a certain prognosis. Another group may die suddenly, remain at home without services, or never appear in the research system. The observed terminal path is therefore conditional on care entry (Aalen et al., 2008).

Hospice observations reflect palliative routines, medication practices, family presence, and eligibility rules. Describing these conditions identifies the settings to which the measured dynamics can be compared or extended.

The rival model should encode entry probability, diagnosis, prognosis, care intensity, resource context, and time since enrollment. If these variables explain terminal concentration, the result belongs to selection and care context unless distinctive terminal-closure-specific residuals remain.

Hospice must not be devalued; humane care changes lived experience. When good care reduces distress or stabilizes context, the first explanation should honor the care and model its effects before assigning any theory-specific meaning.

Hospital and hospice selection rivals win when entry processes and care context explain the observed pattern. They fail only where signatures survive across entry systems, cultures, diagnoses, and observation regimes with the same prospectively specified target.

Diagnosis, prognosis, referral, access, geography, insurance, family capacity, and clinical judgment shape hospital and hospice cohorts. Model these selection processes alongside trajectories and observation schedules, and compare participants with eligible nonparticipants where ethically and legally possible. Treat site practices as data-generating mechanisms and specify the reference classes supported by the resulting evidence (Aalen et al., 2008; Hernán & Robins, 2020).

22.6.2 — Treatment-Conditioned Trajectories

Treatment changes trajectories. Analgesics, sedatives, oxygen, antiemetics, antidepressants, therapy, mobility aids, sleep support, family visitation, and environmental adjustment can alter reports, physiology, behavior, and available actions. These are not noise around the science; they are mechanisms (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

A terminal calm after medication adjustment may be comfort, not closure. A reduction in agitation after pain relief may show that pain was undertreated. A change in sleep after a new caregiver

routine may reflect safety. Treatment-conditioned rivals must be allowed to own those explanations (Hernán & Robins, 2020; Little & Rubin, 2019).

The model should include time-varying treatment, dose, timing, indication, care goals, and side effects where available. It should also include confounding by indication: people receive treatment because their state changed, so treatment and trajectory cannot be separated by simple before-after plots.

Detailed clinical protocol design lies elsewhere; the scientific requirement here is direct. When treatment timing and care context explain the path, the result belongs to treatment. A prospectively specified residual that survives adequate treatment modeling becomes eligible for the next comparison under care-first rules.

Immediate care governs interpretation. Research interpretation cannot make pain control wait, cannot treat sedation as data loss that should be prevented, and cannot pressure clinicians or families to preserve an observation channel when comfort requires silence.

Treatment-conditioned rivals win when intervention timing and clinical context reproduce the observed dynamics. They fail only when signatures remain after rigorous treatment modeling and sensitivity analysis without compromising care.

Treatment changes symptoms, consciousness, reporting capacity, physiology, and observation. Analgesia, sedation, oxygen, psychotherapy, surgery, and social support can each alter the path and the probability of later measurement. The design should

represent treatment as time varying and potentially confounded by prior state. Marginal structural, g-formula, or joint-model approaches may be needed, but no method recovers unmeasured experience automatically. A terminal pattern that tracks treatment timing may be clinically important while remaining evidence about treatment rather than PTVC. Care is never withheld to improve identification (Hernán & Robins, 2020; Tsiatis, 2006).

22.6.3 — Observation Shutdown

Observation shutdown occurs when channels close. A device is removed, a survey is skipped, speech stops, the room becomes private, staff are busy, batteries fail, delirium prevents report, or consent is withdrawn. The shutdown process often intensifies near the most scientifically tempting windows (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Observation shutdown is more than an empty cell in a dataset. If channels close when burden rises, the remaining record can understate distress. If channels close when comfort improves, the record can miss relief. Either pattern makes absence informative about the observation system (Little & Rubin, 2019).

A concrete example is a family asking that all nonessential research prompts stop after a difficult night. The study loses data, but the person gains quiet. The correct scientific response is to model the shutdown and protect the choice, not to treat the missing interval as neutral.

Observation-shutdown rivals require channel-specific dropout models. Self-report, wearable signals, clinical notes, family observations, and behavioral data may shut down for different

reasons. Combining them into one missingness flag can hide the mechanism.

A missing final interval should be represented through its observation conditions and missingness model, preserving the uncertainty about that part of the trajectory.

Observation-shutdown rivals win when dropout and channel closure reproduce apparent terminal patterns. They fail when the result survives explicit dropout models, sensitivity bounds, and ethically available observations.

Observation can end when speech becomes difficult, devices are removed, family priorities change, staff reduce nonessential measures, or participation is withdrawn. Model these causes, preserve uncertainty, and identify recoverable estimands. Proxy, physiological, and passive channels may extend coverage according to their validated roles; use No adequate estimate where the available observations are insufficient (Little & Rubin, 2019).

22.6.4 — Frailty-Conditioned Decline

Frailty is a powerful selection and dynamics variable. It changes reserve, mobility, inflammation, medication tolerance, sleep, cognition, social access, sensor reliability, and survival. A frailty-conditioned model can create apparent convergence simply because many channels narrow together as physiological margin declines (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

This rival is not dismissive. Frailty is lived as fatigue, vulnerability, fear of falls, dependence, and reduced opportunity.

It is exactly the kind of material and bodily condition that can carry burden into a stream. The question is whether frailty explains the observed pattern without terminal-closure-specific closure (Aalen et al., 2008; Hernán & Robins, 2020).

A frailty model should include baseline vulnerability, rate of decline, acute episodes, treatment, functional status, and observation quality. It should ask whether terminal quieting, variance compression, or reduced report intensity follows shared clinical decline rather than a distinct ledger process.

The PTVC candidate may still predict reserve-related signatures, but those signatures must beat frailty baselines that are allowed to be strong. If reserve language is merely a redescription of frailty, the claim has not added scientific content.

Dignity under decline remains prior. Frailty does not reduce the person to a mechanism, nor does it make suffering less urgent because it is clinically explainable. Explanation can guide care only if dignity remains first.

Frailty-conditioned rivals win when shared disease severity and functional decline explain trajectory shape. They fail when independently measured reserve-coupled restrictions remain after frailty, treatment, observation, and endpoint uncertainty are modeled.

Frailty summarizes vulnerability across multiple systems and can generate convergent decline in mobility, appetite, cognition, affect, and measurement availability. Because frailty also influences treatment and survival, it can imitate coordinated terminal dynamics. The comparison should use prospective frailty

measures, allow time-varying change, and test whether the proposed signature remains after stratification or joint modeling. A frailty adjustment built only from late observations risks conditioning on consequences of the process. If ordinary frailty explains the pattern, the scientific credit belongs there and the PTVC reference class must narrow (Aalen et al., 2008; Andersen et al., 1993).

22.6.5 — Competing Events

Terminal-aligned trajectories can end through discharge, transfer, stabilization, hospitalization, sedation, device disconnection, withdrawal, loss to follow-up, sudden death, or recovery. These events route people through different care and observation systems and cannot be aligned as though they were one endpoint (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Competing-risk models should represent event-specific probabilities under state, treatment, family decision, site practice, device tolerance, and resources. Withdrawal ends observation rather than the truth-level stream, and leaving hospice after recovery does not become a completed-stream endpoint because the study window closed.

Recovery, transfer, escalation, loss of consciousness, withdrawal, and death can prevent the event originally intended for observation. Cause-specific and subdistribution estimands are used only when they answer the declared question, and conclusions are compared across choices rather than selecting the formulation with the most favorable curve (Aalen et al., 2008; Andersen et al., 1993).

A competing-event rival wins when routing or observation explains the apparent pattern. Noncompletion remains a valid and respected path, while a PTVC-specific endpoint signature must survive event-specific analysis without collapsing unlike endings.

22.6.6 — Versioned Rival Envelope

The rival envelope is the versioned set of models specified for a protected comparison. New observation artifacts, clinical confounds, sensor behavior, active-inference formulations, and safe-control baselines may enter a later version when independently motivated. The original comparison and its verdict remain in the record; changing the envelope cannot retroactively turn that completed test into a different test (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Each version records what changed, why, and whether earlier results require rerun or correction. Selection rivals win when referral, treatment, survival, frailty, device removal, reporting capacity, or observation shutdown explains the terminal pattern under the common observation model.

When a newly identified comparator changes the analysis, the public record should show the addition and its consequences. Every terminal figure includes denominator flow: eligibility, entry, observability, loss, death, report incapacity, and protected withheld records.

Include the envelope version and denominator with each curve.

22.7 — Common-Observation Comparison

The strongest rival can be weakened by an unfair interface. If models use different endpoint definitions, channels, missingness, latency, or access to final time, their scores compare different scientific problems rather than competing explanations (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

Common-observation discipline fixes the latent target description, observable channels, cadence, noise, missingness, endpoint uncertainty, preprocessing, version history, and protected scoring for every model. The question is which account predicts the same evidence with the fewest unsupported privileges, not which vocabulary sounds most comprehensive (Gneiting & Raftery, 2007; Vehtari et al., 2017; Wilson & Collins, 2019).

Common-observation comparison supplies the interface needed by the later inference and red-team chapters. Learning and control rivals become serious only when one observation arena lets them win, lose, or remain indistinguishable under the same rules.

Equation role: Common scoring rule. This compares models on held-out observations with an explicit complexity penalty.

$$S_M = \log p_M(o_{test}|D_{train}) - \lambda_M df_M$$

Table 22.7. Model-recovery confusion matrix plan.

True generator	Possible mistaken recovery	Required response
PTVC candidate	Bridge rival	Report nonseparability if endpoint conditioning mimics the signature
Predictive processing	PTVC candidate	Narrow the PTVC-specific claim if error dynamics are enough
Safe control	PTVC candidate	Test whether no-return avoidance explains the path
Selection model	Any dynamic model	Rebuild observation process before claiming dynamics

22.7.1 — Shared Data-Generating Interface

A shared data-generating interface defines what every model receives and what every model must explain. It includes latent state placeholders, observations, channel quality, sampling cadence, missingness, endpoint windows, treatment flags, context, and participant status. Without it, models can talk past one another (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

The interface should separate the latent quantity from measured witnesses. Self-report, physiology, behavior, clinical notes, and context are not interchangeable. A predictive model may use them differently from a PTVC candidate or a safe-control baseline, but

all must declare the same available evidence (Gneiting & Raftery, 2007).

A weeklong experience-sampling study gives a simple example. If participants answer prompts, wear a sensor, and complete a small task, all rivals must use the same prompt schedule, sensor gaps, device version, and task rules. One model cannot secretly use discarded context while another does not.

The interface also includes endpoint information. If a study is prospective, models receive only filtered endpoint beliefs. If a retrospective bridge uses the final event, the report must label that use. Endpoint knowledge is data, not air.

Build the common interface as a protocol-specific research object under consent, data minimization, and purpose limits. Link additional records only under the authorization governing that use.

A shared interface succeeds when another team can reconstruct what each model saw. It fails when model comparison depends on informal memory, private preprocessing, unexplained exclusions, or privileged observations that cannot be audited.

A shared interface supplies the same permitted observations, schedules, intervention records, missingness marks, and endpoint information to every fitted model. Known latent states and generator labels remain hidden from fitting and are used only for the declared recovery checks. Record transformations and exclusions before evaluation, and use synthetic fixtures to verify that each implementation reads the interface correctly (Wilson & Collins, 2019).

22.7.2 — Equivalent Parameter Freedom

Model comparison is unfair when one model receives broad flexibility and another is held to a narrow reference form. A highly flexible neural predictive model can fit almost anything; a rigid PTVC candidate can lose for the wrong reason. The reverse can also happen if the PTVC candidate is allowed post hoc mechanisms while rivals are simplified (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

Equivalent parameter freedom does not mean identical parameter counts. Different models need different structures. It means that flexibility, priors, tuning, cross-validation, hyperparameters, and regularization are declared and disciplined so that apparent victory does not come from hidden degrees of freedom (Vehtari et al., 2017; Wilson & Collins, 2019).

A bridge model with a fitted endpoint, a safe-control model with a fitted barrier, an RL model with a fitted reward, and a PTVC candidate with a fitted reserve trajectory can all become too flexible. The chapter's method is to expose those freedoms rather than pretend one side is pure.

Choose complexity penalties, held-out prediction, model recovery, sensitivity analysis, and prospectively specified parameter bounds to match the estimand. Report training fit and held-out performance separately.

Technical complexity should be matched to a clear evidential contribution. Readers can then assess what the added structure

explains, how it was tested, and whether a simpler model performs as well.

Equivalent freedom succeeds when each model has enough room to express its theory and not enough room to explain anything. It fails when the winner is chosen by allocation of flexibility rather than by evidence.

Priors, nonlinearities, latent-state dimension, regularization, early stopping, and feature engineering can give one model a much larger effective search space. The benchmark should therefore report tuning budgets, prior predictive behavior, effective complexity, and sensitivity to initialization. When exact matching is impossible, the asymmetry should be made visible and stress-tested (Gneiting & Raftery, 2007; Vehtari et al., 2017).

22.7.3 — Predeclared Scoring

Protocol-fixed scoring prevents after-the-fact victory. The protocol states how models will be judged through held-out log likelihood, calibration, interval coverage, trajectory features, endpoint-window performance, residual structure, and clinically relevant diagnostics where appropriate (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

The score should match the claim. A model that predicts momentary reports well may still fail the integrated-ledger consequence. A model that predicts endpoint concentration may fail ordinary trajectory shape. A single score can hide too much if the theory makes several distinct commitments (Wilson & Collins, 2019).

Retain every prospectively specified metric and explain disagreements among them.

A prospectively specified score fixes the intended comparison before private data are analyzed. Participants can then understand how their observations will be used, and reviewers can assess whether the analysis followed that plan.

Purpose limitation governs the score. Scores compare models; they do not score people. A participant's data can make a model lose without making the participant failed, anomalous, unfair, or uncooperative.

Scoring rules must be chosen before outcome inspection and matched to the scientific claim. Log score, calibration, discrimination, trajectory error, endpoint recovery, and decision utility answer different questions. A model can lead on one and fail on another. The protocol should identify primary and secondary scores, aggregation levels, tie rules, and consequences of missing predictions. Changing the score after seeing the results is a form of outcome selection. A rival victory is credible only when it occurs under the same prospectively specified scoring regime and the score is proper for the predictive object (Gneiting & Raftery, 2007).

22.7.4 — Complexity and Identifiability

Complexity and identifiability decide whether a comparison can teach anything. A model may fit well because it is true, because it is flexible, because the observation system is weak, or because several theories imply nearly the same measured patterns. The analysis must therefore make nonseparability visible (Gneiting &

Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

Structural nonidentifiability occurs when distinct parameters or models imply the same observable distribution under the specified design. Those observations cannot distinguish them, even with unlimited sample size. Nearly identical distributions can instead create a finite-sample resolution problem. Adequate but inseparable models receive Model-ambiguous status; valid estimates that do not decide the comparison receive Inconclusive evidence (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Simulate from each comparator and evaluate whether the planned analysis recovers its generator. Report any confusion between PTVC, bridge, and reinforcement-learning models as a design-resolution result.

Complexity penalties should account for the structure each model needs and the flexibility it uses. Compare predictive gains with that added flexibility under a common evaluation rule.

An inconclusive result can guide the next study toward better measurements or a more discriminating contrast. Reporting that status clearly helps readers understand what has been learned and what remains open.

Identifiability checks show what the proposed observations can distinguish. Comparison tables should report that resolution alongside model scores.

Complexity penalties cannot repair nonidentifiability. Two models may have different narratives while inducing nearly the same observable distributions, or one model may contain

parameters that trade off without changing predictions. Recovery studies should therefore precede substantive interpretation. Profile likelihoods, posterior geometry, simulation-based calibration, and confusion matrices can reveal whether the design distinguishes models and parameters. If it does not, a small information-criterion difference should not be promoted into a mechanistic verdict. The correct result is ambiguity, redesign, or a narrower estimand (Talts et al., 2018; Wilson & Collins, 2019).

22.7.5 — Posterior Predictive Diagnostics

Posterior predictive diagnostics ask where a fitted model fails by simulating new data and comparing the simulations with protected observations. One aggregate score is insufficient when the model claims to explain timing, recurrence, variance, missingness, channel disagreement, policy change, endpoint windows, or reserve-linked residuals (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018).

Diagnostics should target features that distinguish rivals, including autocorrelation, sign asymmetry, perturbation recovery, context renewal, endpoint-window variance, and missingness-linked residuals. A model can predict choices but miss physiology, or fit arousal while missing delayed report change; those are component-level results rather than reasons to force one global winner.

For a candidate with hidden-burden predictions, publish performance on the specified indicators, including mismatches, separately from performance on other declared targets.

Publishing the diagnostic set before fitting prevents selective plots, and public summaries should state what each model missed in ordinary language. Diagnostics succeed when error becomes informative, not when simulated curves decorate confirmation.

22.7.6 — No Winner When All Are Wrong

PTVC, predictive processing, active inference, reinforcement learning, safe control, and bridge models may all fail because the observation system, estimand, or theory set is incomplete. Open-set adequacy prevents the least inadequate model from being forced into victory (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018). Open-set rejection should be reported before relative rankings, because a best-in-library score can be meaningless when every model fails absolute adequacy.

The framework fixes thresholds for calibration failure, unmodeled residuals, failed recovery, and poor transport. Multi-site disagreement caused by devices, care routines, language, treatment, dropout, or delayed recurrence can therefore produce candidate-set inadequate rather than a spurious winner. Site-specific inadequacy patterns can reveal which observation assumptions fail to transport and should be modeled before pooling sites.

No winner is appropriate when absolute fit is poor, models are nonseparable, uncertainty crosses the decision boundary, or open-set checks reject the family. Refusal to choose is informative when it redirects work toward better measurement, stronger rivals, narrower estimands, or safer collection. The redesign target

should follow the reason for no winner, because poor absolute fit and nonseparability require different next experiments.

Ambiguity must not conceal a clear loss under a model's own rule. Open-set restraint protects inquiry only when the declared rejection criteria are applied symmetrically. Symmetry also requires accepting a candidate loss when the rival meets its predeclared threshold even if open-set concerns remain elsewhere.

22.8 — When Learning and Control Rivals Win

Chapter 22 connects model comparison with public reporting. When predictive processing, active inference, reinforcement learning, safe control, bridges, or selection best explain the observations, the component-level record should identify that result (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

The final section connects findings to research decisions. Components can be refined or retired, while clinical and methodological discoveries remain available for care and subsequent investigation (Popper, 1959; Lakatos, 1978; Stanford, 2006).

A newly proposed connection between PTVC and a successful comparator should state fresh predictions and receive prospective evaluation on protected evidence.

Equation role: Rival-victory rule. A rival victory is a status event when the protocol-fixed comparison favors the rival beyond the declared uncertainty and practical margin.

$$\Delta_{R,L} = S_R - S_L; \Delta_{R,L} > \delta_{pre} \Rightarrow \textit{narrow}(L\text{-specific component})$$

Table 22.8. Scientific consequences of rival victory.

Result	Status consequence	Public language
One component loses	Component narrowing or retirement	The mechanism did not survive this comparison
Repeated distinctive losses	Strong-form pressure	The broader claim faces adverse evidence
All models fail	Open-set inadequate	The design did not answer the question

Result	Status consequence	Public language
Rival wins and guides care	Research redirection	Preserve the useful finding

22.8.1 — Component-Level Defeat

Component-level defeat retires or narrows a specific mechanism without requiring an immediate verdict on the whole formal Law. A state-allocation signal, reserve interaction, horizon-pressure signature, or endpoint-concentration claim may fail locally while other parts of the program remain open (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

This is important because learning and control rivals are precise. A predictive-processing model may defeat a salience claim. A safe-control model may defeat a no-return explanation. A bridge model may defeat a terminal-drift interpretation. Each defeat should land where the evidence lands (Popper, 1959).

The status update should name the component. Saying the Law was not supported is too broad when only a mechanism failed. Saying nothing changed is too evasive when the mechanism was central to the chapter's prediction.

Component-level defeat can improve the book. It removes weak material, prevents overclaim, and lets future research focus on what remains. A narrowed theory is often more testable than a broad theory protected by vague language.

Public disappointment must not become private burden. Readers should not be asked to preserve belief by interpreting defeat as their failure to understand. The model lost a comparison. The person remains untouched by the verdict.

A component is defeated when a rival explains the prospectively specified target with equal or better performance, better calibration, successful recovery, and no hidden privilege. The component then narrows, retires, or becomes an open problem with its status visibly changed.

The defeat record should also specify propagation. A failed component may require changes to figures, summaries, public artifacts, protocol templates, and later chapters that relied on it. Without propagation, the formal status changes while the reader-facing theory remains untouched. The record should name the exact claim, evidence, rival, reference class, and affected outputs, then assign an owner and deadline for correction. This makes component-level defeat operational rather than ceremonial and ensures that the project learns from loss across its entire scientific surface (Popper, 1959; Lakatos, 1978).

22.8.2 — Strong-Form Pressure

Strong-form pressure accumulates when multiple distinctive components fail across contexts. One local rival win does not defeat the strong latent hypothesis by itself. Repeated rival wins against the same distinctive conjunction create a different scientific situation (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

The distinctive PTVC conjunction is narrower than error reduction, learning, safety, adaptation, or endpoint conditioning alone. Its empirical evaluation requires a structured relation among latent trajectory, endpoint discipline, measurable consequences, observation, and rival discrimination. Reserve

enters that conjunction when the tested route specifically assumes reserve coupling. Any component that ordinary models reproduce leaves less work for the terminal candidate (Lakatos, 1978).

If predictive processing explains salience, reinforcement learning explains policy change, safe control explains no-return avoidance, bridges explain endpoint drift, and selection explains terminal concentration, the remaining terminal-closure-specific content becomes smaller. At some point, narrowing is not optional.

The report should define pressure tiers: local narrowing, repeated component failure, strong-form pressure, suspension, and defeat. These tiers prevent all losses from sounding equal while making clear that repeated failure matters.

Clear status updates connect each scientific result to the component it tests and identify the next research decision.

Status updates should be carried into the affected chapters, summaries, diagrams, and public claims. Their wording should reflect the specific evidence and component involved.

Strong-form pressure should be evaluated with a dependency graph. Some predictions share assumptions, data, or mechanisms, so several failures may not be independent pieces of evidence. Other failures attack distinct gates and therefore compound more seriously. The graph should show which losses concern representation, routing, endpoint, observation, mechanism, or transport. This prevents both exaggeration and minimization. Repeated failure across independent gates should lower theory maturity even if one route remains open. Correlated failures

should be evaluated through a joint model or a calibrated decision rule that represents their dependence; exact duplicate evidence should be counted once. Shared failures should also guide revision of the common assumption (Lakatos, 1978; Stanford, 2006).

22.8.3 — No Rebranding the Rival as the Law

The most tempting escape is rebranding. If predictive processing wins, the theory says prediction was fairness. If reinforcement learning wins, it says policy adaptation was fairness. If safe control wins, it says safety was fairness. That move empties falsifiability (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

A connection between a successful mechanism and a PTV route can be proposed with distinct predictions and tested prospectively as a new claim (Stanford, 2006).

Stable bridge statuses are therefore required: proposed bridge, prospectively specified bridge, tested bridge, unsupported bridge, and failed bridge. These labels prevent ordinary regulation or control from being absorbed into PTV by interpretation after the result.

A concrete misuse would be a public graphic showing an active-inference policy loop and captioning it as the scientific mechanism of fairness. Unless the bridge has been established, that caption overclaims. The safer caption says rival model or candidate comparison.

Use preference, reward, safety, and neutrality consistently with their definitions in active inference, reinforcement learning,

control, and PTVC. Explicit mappings make proposed connections understandable.

No-rebranding discipline is what makes rival victory real. If a rival can win only by being renamed, the comparison was never honest. The chapter closes that route before public enthusiasm can open it.

The same rule applies to successful hybrids. A hybrid model may combine active inference, reinforcement learning, safe control, and a PTVC-specific term, but its performance must be decomposed. Ablation and transport tests should show whether the closure term adds predictive value beyond the ordinary components. If the hybrid works only because the rival machinery carries the fit, the result is not support for the added term. This decomposition prevents a broad integrative vocabulary from absorbing every successful science and preserves a genuine possibility of loss (Wilson & Collins, 2019).

22.8.4 — No New Mechanism After Every Loss

Mechanism revisions state their motivation, source grounding, new predictions, and comparison plan. This record makes theoretical development assessable across successive versions (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

This rule matters in a broad project because the Law has many possible auxiliary concepts: state-allocation pressure, reserve, horizon, dual-variable cost signal, interoception, dreams, endpoint windows, and measurement governance. Each can be useful, but none should become an infinite repair kit (Popper, 1959; Lakatos, 1978).

A new mechanism should answer a precise failure. If bridge rivals explained endpoint drift but not reserve-linked innovation, a revised mechanism might target that residual. If predictive models explained every prospectively specified feature, adding a new hidden variable without novel predictions would weaken the project.

The protocol should record mechanism changes with version dates, prior rationale, prediction changes, and implications for older results. Old support should not automatically transfer to the revised model. A changed claim needs new evidence.

A revision record lets readers follow how the theory develops, which observations motivated each change, and which predictions are being tested next.

A revision plan can make this concrete. Before each confirmatory cycle, specify which auxiliary changes are permitted, which need a new exploratory phase, and which require a new evidence claim. Changes to endpoints, representations, routing rules, or mechanisms receive their own versioned predictions and frozen artifacts, preserving what each study tested while making room for creative development (Nosek et al., 2018).

22.8.5 — Publication of Rival Victories

Rival victories should be published, logged, and summarized in the same claim-status system as supportive results. A complete public record preserves adverse findings so that readers can assess the evidence as a whole (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

A comparative result can identify the model that best predicted endpoint-window behavior and state the cohort, observation model, endpoint definition, uncertainty, and component-level consequence (Lakatos, 1978).

Rival victories also protect applied books. If a city, religion, government, or transition-facing artifact cites the science loosely, the public log should make overclaim easy to correct. The science book's honesty becomes a firewall for the whole series.

Plain-language and technical layers are required. The technical report carries equations, likelihoods, diagnostics, and model recovery; the public summary states support, partial support, narrowing, inconclusive, or defeat. Both should be versioned.

Adverse results attach to models and claims, never to participants, clinicians, families, sites, or readers. The archive evaluates scientific propositions rather than lives.

Publishing rival victories succeeds when future readers can see the project correct itself. It fails when negative results are buried, blurred into vague uncertainty, or recast as hidden support.

Search rival victories by claim and component. A public register identifies the tested version, cohort, endpoint rule, observation model, scores, winner, uncertainty, and propagation without exposing private data. Machine-readable fields let figures and summaries update rather than preserve stale claims. Record later replications or reversals so one adverse result is neither hidden nor treated as final beyond its evidence. Scientific credibility grows when archives make correction easier than selective

memory (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

22.8.6 — Research Redirection

Wins by learning and control rivals may still yield measurement tools, safer baselines, lower-burden routines, or selection controls. These gains belong to the rival, not terminal support (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

Redirection preserves useful findings without treating them as terminal evidence. It may improve measurement, protocols, care, or comparison arenas while leaving target status unchanged (Stanford, 2006; Popper, 1959).

A study may show that stable routines reduce distress under an active-inference account and still guide care design. It should not be presented as terminal-neutrality evidence. Better care matters independently.

Record redirection by domain: measurement, rival development, clinical insight, governance, artifact correction, or formal narrowing. Each has different authority, and none alone authorizes policy, belief, or transition.

Useful science may survive a defeated claim, but the claim cannot hide inside it. Evidence should remain modest, actionable, and explicit about what lost.

Scientific fairness places the burden on the claim. Rival victory changes its status; surviving residuals face stronger tests; care, rights, privacy, and human standing remain independent.

Chapter 23 — Evolution, Non-Ergodicity, and Outer Limits

Chapter 23 examines how findings extend across evolution, population structure, development, species, and synthetic systems. Each extension identifies the mechanisms, measurements, and reference classes needed for a meaningful comparison.

Evolution explains how affect can support survival, learning, attachment, cooperation, and recovery, giving PTVC a rich mechanistic comparison space. It supplies mechanisms and boundary conditions for models of experience across an organism's life.

Nonergodicity connects the level of analysis with the question being asked. Relations observed across people can differ from relations unfolding within one person. Recent neurocognitive work provides concrete examples for designing population and within-person analyses (Hunter et al., 2024; Mistry et al., 2026).

Time averages require conditions that lived trajectories may not satisfy. Development, illness, trauma, medication, culture, care, relationships, and approaching endpoints can alter dynamics, observation, and available states. Exchangeability, stationarity, recurrence, and transport must therefore be demonstrated rather than assumed.

A cohort trend can also be manufactured by visibility. Participants carrying the greatest burden may stop reporting, receive sedating treatment, enter unobserved care, die earlier, or leave because participation itself becomes costly. A falling observed average can then describe the surviving record rather than the streams that became harder to see.

Complex systems add both opportunity and risk. Nonlinear feedback, multistability, hysteresis, critical transitions, and path dependence can produce recovery, oscillation, sudden change, and apparent concentration. Those structures become scientifically useful only when they generate discriminating predictions and can lose to simpler explanations.

Nonlinear systems can recover, tip, oscillate, settle into regimes, and change behavior after small perturbations. A useful complexity model identifies which patterns it predicts, which alternatives reproduce them, and which observations would require revision.

Critical slowing down, metastability, hysteresis, recurrence, cascades, attractor switching, and path dependence offer specific dynamical structures to investigate. Each can be evaluated through predicted signatures, surrogate tests, and comparison models.

Extensions to infants, nonhuman animals, organoids, and artificial systems specify bearer qualification, measurement invariance, endpoint comparability, support overlap, and mappings between constructs.

Bearer adjudication belongs to the earlier chapters on stream identity and boundary states. The narrower task here is transport: what would allow evidence about one kind of stream to travel responsibly to another? A transport bridge requires specified mechanisms, compatible measurements, and evidence that the relevant assumptions hold in the target setting.

A comparative program can protect unfamiliar or difficult-to-measure candidates while developing the observations needed to characterize them. Scientific descriptions specify the reference classes and relationships supported by those observations.

Artificial systems expose the difference between performance and phenomenal evidence. Fluent output, reward optimization, self-report-like language, or engineered closure is not enough. Architecture, recurrent access, memory, self-modeling, global availability, embodied regulation, ablations, robustness, and alternative explanations belong in an indicator-based assessment, with welfare precaution and human review preserved (Butlin et al., 2026; Seth, 2025).

Narrowing can be a mature scientific result. A claim restricted to adult humans, particular states, certain cultures, or defined measurement conditions is smaller than a universal formulation but more informative than a statement protected from counterexamples. Scope withdrawal keeps the program answerable.

Outer limits therefore strengthen the project. Evolution may explain regulation without fairness, population averages may hide streams, complexity may mimic closure, and comparative bridges may fail. Each outcome identifies a boundary, preserves what was learned, and prevents the Law from expanding beyond the evidence.

23.1 — Selection and Experienced Welfare

Evolutionary accounts explain why organisms regulate stress, avoid injury, seek nourishment, learn from pain, repair tissue, attach to caregivers, and reorganize behavior after loss. These mechanisms show how feeling can participate in control without implying that selection optimized justice or terminal balance for each stream (Williams, 1966; Nesse & Williams, 1994; Dawkins, 1976; Nesse, 1990; Williams & Nesse, 1991; Nesse & Ellsworth, 2009).

The same explanation sets a limit. Natural selection acts through differential survival and reproduction under ecological conditions; it can preserve traits that make some lives harder. Panic can protect against ancestral threat while sustaining modern vigilance, and grief can preserve attachment while making a particular loss almost unlivable.

Evolutionary accounts are therefore executable rivals when they predict the emergence, distribution, mismatch, and tradeoffs of affective systems under declared life histories and environments. If those accounts explain the protected signatures without pathwise closure, they win within their domain.

Evolutionary compatibility is not a signature. The comparative question is what remains after adaptation, mismatch, life history, developmental constraint, population selection, and observation are modeled with equal seriousness. If nothing distinctive remains, the PTVC-specific claim narrows.

Table 23.1. Evolutionary rival family and what each can explain.

Rival mechanism	What it explains	Questions for further investigation	Victory condition
Adaptive signal functions	Pain, pleasure, relief, threat, and attachment as action-guiding signals.	How do signal dynamics relate to cumulative experience within a stream?	Signal models predict trajectories as well as or better than PTVC candidates.
Mismatch	Persistent burden when old regulation meets new environments.	How does environmental fit affect the persistence of distress and recovery from it?	Mismatch variables account for apparent closure signatures.
Life-history tradeoff	Different allocation of energy, risk, reproduction, and care across stages.	How do life-stage allocations shape individual experiential trajectories?	Life-stage ecology explains horizon effects without PTVC-specific residuals.
Developmental constraint	Inherited architecture, path dependence, and repair limits.	Which developmental constraints alter recovery and experienced welfare?	Constraint models reproduce observed dynamics without terminal boundary.

Rival mechanism	What it explains	Questions for further investigation	Victory condition
Population selection	Distributional success through survival and reproduction.	How do within-stream trajectories relate to population composition and survivorship?	Population process explains apparent neutrality through mixture or survivorship.

23.1.1 — Fitness and Reproduction

A trait can spread because it improves survival, mating success, kin support, parental investment, or group coordination. Its effects on felt experience form an additional dimension of the biological account (Williams, 1966).

Fear illustrates how a protective response can carry experiential costs. Treating an uncertain rustle as a threat may support survival, while false alarms produce racing heart, narrowed attention, freezing, and vigilance. Measuring these responses links the signal's function with its effect on the individual.

The same logic applies to human social pain. Exclusion sensitivity, shame, jealousy, and status anxiety can push behavior toward coalition maintenance or norm compliance. Studying why these signals persisted, how their effects unfold within a life, and how they respond to changing circumstances connects evolutionary explanation with the temporal questions raised by PTVC.

Evolutionary models connect hunger, injury, rejection, uncertainty, contagion, threat, exhaustion, infant dependency, and

mate loss with recurring control problems. They also explain why relief can reinforce survival-supporting actions. These links provide mechanisms and temporal predictions for studying experience.

The key estimand is not the average reproductive success of a trait-bearing group. It is not even the average affective tone of a species under a given ecology. For the Law, the unit of concern is the temporally extended stream. A biological explanation that operates through distributions remains silent about whether a particular path closes.

Evolutionary models can compare the effects of selection pressures and demographic filtering on regulation, variability, recovery, attachment, and terminal patterns. Held-out prediction shows which features each account explains and where an additional path constraint contributes.

23.1.2 — Pain and Pleasure as Signals

Pain and pleasure can guide action under constraint. Pain may accompany injury, threat, inflammation, social exclusion, or anticipated loss; pleasure may accompany nourishment, safety, attachment, exploration, relief, warmth, mastery, or successful regulation (Nesse & Williams, 1994; Dawkins, 1976).

Pain and threat signals connect bodily conditions with action. A burn can prompt protective behavior; a false panic alarm can reflect a conservative threat-detection system. Their timing, intensity, and context are available for physiological and behavioral study.

The adaptive-signal view gives a powerful rival to the Law because it predicts asymmetry. Organisms may be built to overweight loss, threat, and pain because missing danger can be costlier than missing opportunity. That asymmetry can generate long negative tails in lived experience without violating evolutionary logic. Selection can therefore preserve an effective protective response even when its experiential costs are substantial.

The science should therefore ask what a signal explains. Does it explain local avoidance, learning, recovery, attention, memory consolidation, attachment repair, or resource allocation? Does it explain persistence of burden under mismatch? Does it predict when relief fades because the organism must return to vigilance? If yes, it remains a serious rival.

The next comparison asks whether pathwise constraints predict additional features after signal functions, contextual variables, and learning dynamics are included.

A biological function identifies how a system uses a signal for regulation. Measuring that function alongside the burden it creates gives a fuller account of its consequences.

A nociceptive or threat signal may change as tissue heals, attention shifts, medication acts, sleep supports regulation, or movement changes. Recording these pathways separately helps explain why visible distress and reported experience change over time.

23.1.3 — Mismatches and Modern Environments

Evolutionary mismatch is one of the strongest non-Law explanations for persistent burden. A system shaped under one ecology can misfire in another. Food abundance, artificial light, social media, urban density, chronic noise, wage precarity, synthetic drugs, sedentary work, and institutional complexity can drive ancient control systems into states they were not tuned to handle (Dawkins, 1976).

Consider vigilance. In a dangerous ancestral environment, rapid threat detection may have protected the organism. In a modern apartment where the threat is debt, inbox overload, medical paperwork, or hostile online attention, the same physiology can run without a clear external predator. The person is not irrational. The body is using old machinery in a new world.

Mismatch can also preserve pleasure traps. High-calorie foods, intermittent rewards, pornography, gambling-like social feeds, and addictive substances can hijack reward systems built for sparse opportunities. The resulting burden can look like failure of will from the outside. Mechanistically, it is a collision between evolved learning systems and engineered environments.

This rival can explain why burdens persist without any terminal justice. It predicts that some streams will be exposed to chronic states for which regulation is maladapted: inflammatory stress, insomnia, depression, addictive loops, social comparison, and administrative uncertainty. The ecology changes faster than biological adaptation.

Mismatch models explain how ecological change can alter adaptation across individuals and settings. Their predicted trajectories provide a basis for testing additional per-stream relationships.

The human texture matters here. A night-shift caregiver who alternates blue hospital light, vending-machine food, family worry, and debt calls is not just an observation in a stress dataset. That person is a stream under an ecology. The scientific question is whether the resulting trajectory contains stream-level closure structure after the mismatch mechanism has been given its full force.

Mismatch is especially important because it explains burden without malice or balance. A nervous system shaped for small groups, short horizons, immediate threats, and embodied routines can suffer under isolation, digital vigilance, chronic bureaucracy, industrial noise, sedentary work, and social comparison. The unfairness is real in the stream even if no villain designed it. Evolution can therefore explain why burden persists, why relief is partial, and why ordinary adaptation can fail without implying later compensation.

23.1.4 — Individual Versus Population Success

Selection changes the distribution of traits through differences in reproductive outcomes. Measuring both those effects and individual experiential consequences connects population mechanisms with within-life trajectories (Williams, 1966; Nesse & Williams, 1994).

Parental investment, grief, loyalty, protective aggression, moral outrage, endurance, and responses to norm violation can shape family and group functioning. Their effects on the individuals carrying them can be studied alongside those social functions.

Population success also hides replacement. A trait may persist because carriers who cannot bear its costs drop out of reproduction, leave the observable cohort, or are not counted in later records. This is a selection mechanism, not a fairness mechanism. It creates survivorship, not proof.

For the Law, this means a population trajectory can be scientifically interesting and still be insufficient. If average reports improve with age, if species-level coping systems look stable, or if a cohort's terminal estimates cluster near neutrality, the next question is who is missing, who was measured, and whether each stream's history is being reconstructed rather than averaged away.

Individual trajectories and population summaries answer complementary questions. A population can have a neutral mean while individuals differ substantially in opposite directions, and selective loss from observation can change that mean. Recording trajectories and sample composition keeps both levels interpretable.

Clear units connect population mechanisms with individual trajectories. Reporting both levels makes their respective contributions to the account explicit.

Population success and individual experiential outcomes are distinct biological questions. A strategy may increase descendants

while imposing substantial costs on some carriers. Studying both levels gives a fuller account of how selection shapes experience.

23.1.5 — Neutral and Maladaptive Traits

Not every trait is adaptive. Some traits are neutral, weakly selected, historically contingent, linked to other traits, or maladaptive in current conditions. A serious evolutionary treatment cannot make every observed feeling into a just-so story. Some experiential dynamics may persist because selection is limited, slow, noisy, local, or constrained (Nesse & Williams, 1994).

Developmental constraints matter. A nervous system is not redesigned from first principles for comfort. It inherits architecture. It uses ancient circuitry, repurposes systems, and carries tradeoffs. The result can be brilliant in one context and destructive in another. Back pain, chronic inflammation, reproductive risk, trauma memory, and anxiety sensitivity can persist because bodies are compromises, not perfect welfare machines.

Genetic drift and linkage also prevent simple comfort stories. A trait may ride with another trait, remain below selection pressure, or appear only under rare environmental combinations. The fact that it exists does not mean it serves fairness. Existence is not endorsement.

Some trajectories reflect poorly matched environments, physiological damage, or developmental constraints. Modeling those processes helps explain individual variation and identifies opportunities for support or recovery.

A methodologist should hear the modeling implication. Evolutionary parameters need uncertainty. Trait history, current environment, developmental exposure, genetic architecture, culture, treatment, and measurement regime all affect whether an observed pattern is adaptive, maladaptive, neutral, or mixed.

Evolutionary complexity helps sharpen the PTVC candidate. Regulation supplies mechanisms to investigate, while pathwise predictions specify the contribution of a terminal-boundary account. Neutral, maladaptive, mismatched, and constrained mechanisms expand the range of trajectories the comparison should represent.

Some experiential patterns arise as side effects of other mechanisms, such as vigilance coupled to social memory or pain coupled to mobility protection. Others persist under weak selection, recent environmental change, or developmental constraints. Including these possibilities gives the model a fuller account of stable individual differences.

23.1.6 — Evolutionary Rival Victory

Evolutionary explanations can account for selection, development, ecology, and population dynamics with testable predictions for the observed trajectories. When those predictions perform best, the result identifies the biological mechanism to carry into the next stage of research (Dawkins, 1976; Williams, 1966).

Candidate explanations include sedation, attrition, family communication, care pathways, survivorship, changing behavioral range, physiological transitions, adaptation, sleep,

learning, social support, and conserved control architecture. Each predicts a different combination of timing and observation effects.

Plausibility supplies a starting point for an evolutionary model. The next step is to generate held-out predictions, reproduce path signatures under the same observation model, check model recovery, and identify cases where PTVC predicts a different result.

The comparison can refine the model's biological mechanisms and scope, improve measurement, and identify useful questions about affect dynamics and end-of-life care. Each result determines which component to develop next.

Public summaries should distinguish biological plausibility, demonstrated mechanisms, and comparative prediction results. Each contributes a different part of the scientific case and should be described through the evidence supporting it.

Evolutionary theory supplies mechanisms and strong comparison models for PTVC. Testing their competing predictions can refine both accounts.

An evolutionary explanation is evaluated for a specified signature under a common observation model. Reporting the responsible mechanism and its tested range gives subsequent research a clear starting point.

23.2 — Non-Ergodicity and Within–Between Reversal

PTVC concerns the trajectory of a qualifying stream over time. Analyses therefore specify how within-person temporal patterns relate to population distributions and test the assumptions connecting those levels (Molenaar, 2004; Hunter et al., 2024; Fisher et al., 2018; Peters, 2019).

Nonergodicity means more than individual differences. The analysis must name the process, within-stream and between-stream estimands, time and ensemble averages, and assumptions under which they converge or diverge. A large sample does not create stationarity, homogeneity, or exchangeability.

A 2026 analysis of neuroimaging and behavioral data from more than 4,000 participants found brain-behavior associations that reversed between within-person and between-person analyses. This provides a concrete reason to estimate both levels when studying stream-specific dynamics (Mistry et al., 2026).

A plot of many lives can look like one river even when its component streams move differently. The scientific file must therefore preserve person-level time, within-stream variation, between-stream variation, and the relation between them. Aggregation is a modeling choice, not an ontological merger.

23.2.1 — Ensemble and Time Averages

An ensemble average summarizes many streams at a common time or window; a time average follows one stream through time. Their equality is not a consequence of sample size but of specific ergodicity assumptions, which may fail in developing, treated, aging, culturally embedded, or history-dependent systems

(Molenaar, 2004). Let $X_i(t)$ be a measurable process for stream i in the display below.

The practical consequence is that the unit of inference must be declared before the beauty of a curve is discussed. A smooth terminal mean, a pooled variance estimate, or a cross-sectional association may be useful for clinical description, but it is not automatically a statement about the integral of a stream. To make it relevant, the study must show why the temporal process within each bearer can be represented by the same statistical object. Otherwise the average is only an average, however elegant it looks.

$$\overline{X}_i^{\text{time}} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T X_i(t) dt, \quad \overline{X}^{\text{ensemble}}(t) = \mathbb{E}_i[X_i(t)]$$

This display defines the time and ensemble averages. Population surveys reveal social differences, policy burdens, and inequalities, while longitudinal observations characterize within-person dynamics. Specifying their relationship connects these complementary sources of information (Diener et al., 2006; Shiffman et al., 2008).

A hospice cohort's final-week mean may reflect physiology, care practices, sedation, reporting capacity, family communication, attrition, or population mixture. Within-person records and observation models help separate these contributions.

Non-ergodicity identifies the level at which evidence must be gathered. The analysis can collect per-stream data, state justified transport assumptions, or narrow the claim to the level supported by the observations.

Where the relevant stationarity, ergodicity, and comparability assumptions are justified, population information can help characterize within-stream dynamics. Otherwise, model the population and individual quantities separately.

23.2.2 — Within-Person Versus Between-Person Effects

Within-person effects ask how change inside one stream relates to change in another quantity for that same stream. Between-person effects ask how differences among streams relate to differences among other quantities. Confusing them is one of the cleanest ways to generate false psychological laws (Hunter et al., 2024).

A simple decomposition makes the point. A person's value at time t can be separated into a person mean and a deviation around that mean. The between-person coefficient can attach to the mean, while the within-person coefficient attaches to the deviation.

The within-between distinction should shape interpretation before data are collected. A protocol may prospectively specify a within-stream estimand, such as change in an individual's volatility near an endpoint window, and separately specify a between-stream estimand, such as the relation between reserve and variance across people. These are different claims. A reversal between them is a primary result, not a nuisance for supplementary material.

$$Y_{it} = \alpha + \beta_W(X_{it} - \bar{X}_i) + \beta_B\bar{X}_i + \varepsilon_{it}$$

This equation is an estimator structure, not a Law formula. If β_W and β_B have different signs, the population story and the individual story move in opposite directions. A between-

person association can say that people with higher average social contact have lower distress, while the within-person effect can say that a given person seeks contact on days of rising distress.

The Law is exposed to the same risk. A population average near neutrality at endpoint says very little if the within-stream paths are heterogeneous, if high-burden paths disappear, or if changes within persons move opposite to differences between persons.

A concrete example helps. Suppose higher average sleep duration across people is associated with lower burden. Within one terminal patient, longer sleep on certain days might indicate medication escalation, fatigue, infection, or withdrawal. The between-person association is not a per-stream repair mechanism.

A serious analysis must therefore declare whether it is estimating a between-stream relation, a within-stream relation, a cross-level relation, or a transport relation. One coefficient cannot carry all of them.

The result can reverse without fraud, incompetence, or bad faith. It reverses because the question changed. This is why the chapter treats non-ergodicity as a design constraint rather than as a statistical afterthought.

23.2.3 — Heterogeneous Dynamics

Heterogeneity is not merely different baselines. Streams can differ in parameters, mechanisms, response lags, treatment exposure, cultural scripts, reporting styles, physiology, developmental history, and endpoint risk. A model that assumes one dynamic law for all streams may manufacture regularity by

averaging away the very differences the Law must face (Hunter et al., 2024).

For example, three people may show the same measured distress score but under different dynamics. One may be in a reversible grief wave with strong social repair. Another may be in medication-induced emotional blunting. A third may be in pain with reduced report capacity. The same observation cannot be given one mechanism without testing.

Heterogeneous dynamics also matter for reserve. Reserve may be metabolic, cognitive, autonomic, social, attentional, or clinical. Collapsing it into one scalar may work in a narrow context, but it cannot be assumed across ages, diagnoses, cultures, or species. If reserve is vector-valued, different components can fail at different times.

A hierarchical model can help by allowing person-specific parameters. But hierarchy is not a magic bridge. It borrows strength across persons only under an exchangeability claim. If exchangeability fails, the model may shrink unusual streams toward a false group center.

The Law should prefer designs that preserve individual dynamics before summarizing. Idiographic models, person-specific state-space estimates, and hierarchical models with explicit heterogeneity are not decorations. They are safeguards against turning aggregate smoothness into pathwise closure.

Failure here has a clean status implication. If closure-like signatures appear only after aggressive pooling and disappear under person-specific modeling, the Law loses support in that setting. The average was doing the work.

Heterogeneity also changes what humane design looks like. If one subgroup carries pain, another carries confusion, another carries sedation, and another carries family-pressure reporting, then a single terminal curve can conceal several burdens. Stratification is not only statistical hygiene. It keeps care-relevant differences visible. A model that discovers heterogeneity may become more useful even as it makes a universal claim less plausible.

23.2.4 — Mixture Distributions

A mixture distribution can look like convergence when no subgroup converges. If one subgroup moves upward, another downward, and a third disappears, the pooled mean can approach a neutral value. The result can be visually persuasive and scientifically wrong for the pathwise claim, especially when between-person structure is mistaken for within-person dynamics (Molenaar, 2004; Hunter et al., 2024; Mistry et al., 2026).

Mixture risk is especially high near endpoints. Diagnosis, sedation, care setting, family involvement, medication, cultural reporting, device adherence, and survival all create subgroups. A terminal dataset is rarely one homogeneous population sliding toward a common boundary.

The chapter's demand is not that every mixture be known in advance. It is that mixture be modeled as a live alternative. Latent class models, mixture state-space models, stratified analysis, posterior predictive checks, and leave-site-out validation can reveal whether an apparent universal signature is actually subgroup composition.

A simple Simpson reversal can be checked numerically. In arbitrary toy units, group A has the two (x, y) observations (1,8)

and (2,9), while group B has (8,1) and (9,2). Within each group, y increases by one unit for each unit increase in x . Pooling all four observations instead gives a least-squares slope of -0.96 : the overall means are $\bar{x} = 5$ and $\bar{y} = 5$, the sum of cross-products is -48 , and the sum of squared x deviations is 50 . The reversal arises from the separation of the group means. This worked example shows how differences between group means can reverse an aggregate association.

Mixtures also create moral danger if ignored. A public summary might say terminal experience tends toward neutrality while the pattern is driven by those easiest to observe. The people with severe delirium, low report capacity, device dropout, or unmanaged pain may be hidden in the missing tail.

The scientific rule is direct: report mixture diagnostics before reporting pathwise implication. A neutral pooled estimate without subgroup accountability is not a closing stream. It is a population artifact waiting to be interrogated.

A simple Simpson reversal can be imagined without fraud. In each diagnosis group, increasing reserve may be associated with worse reported distress because more awake patients can report pain. Across groups, higher-reserve diagnoses may have lower average distress because the illnesses differ. The pooled association then points one way while every within-group association points another. The Law cannot rest on a pooled direction until the grouping structure is named and tested.

23.2.5 — Survivorship and Selection

Survivorship links biological trajectories to the changing composition of the observed sample. The streams remaining

observable at a later time are not always representative of the streams that began the study. Those who die earlier, lose consciousness, drop devices, refuse burdensome reporting, enter sedation, move sites, or lack family support may leave the data precisely when their trajectories matter most (Hunter et al., 2024).

This is not merely missing data. It is selection on the process under study. If high-burden states reduce participation, then observed improvement may reflect attrition. If low-unity states prevent report, terminal concentration may reflect silence rather than neutrality. If device adherence depends on mobility, the passive signal is partly a motor-capacity signal.

Survivorship can also enter evolution. Lineages are observed because they survived. Cultural practices are recorded because some groups preserved records. Clinical terminal data are collected where institutions allow research. Each layer selects the evidence before analysis begins.

The Law's empirical program must therefore model observation as part of the data-generating process. Who remains, who leaves, who is censored, who is measured by proxy, and who has endpoint evidence all affect the claim. A credible model of the missing tail can determine whether the terminal pattern remains supported.

A family example makes the risk concrete. A widower may delete a sleep tracker before hospice because the device feels intrusive. The missing telemetry is not random. It records dignity, grief, privacy, device fatigue, and refusal. Treating that absence as ordinary noise turns a person's boundary into a statistical convenience.

Survivorship does not make science impossible. It makes the evidence conditional. Every conclusion must say what selection model would reverse it and what data would be needed to test that model.

23.2.6 — Per-Stream Claims Require Per-Stream Evidence

A claim about a stream requires evidence about streams. Population data can orient hypotheses, identify rivals, estimate parameters, and reveal artifacts, but cannot close an individual path without an independently justified bridge (Hunter et al., 2024; Molenaar, 2004).

Per-stream evidence does not require total observation. It requires repeated measures, prospectively specified stopping and noncircular endpoint adjudication, observation-process models, person-specific baselines, uncertainty, and validation without future leakage, together with a clear account of what was observed, bounded, estimated, or unknown. Any statistical dependence of the endpoint on the stream must be represented under the tested branch.

A late-life window can estimate increments, terminal concentration, variability, or route dynamics. Its relation to earlier history is represented through explicit assumptions about the unobserved past.

A strong result states the window, observation rules, rivals, and per-stream pattern that appeared or failed to appear. That is more informative than a grand average that cannot describe any individual path. Publishing the distribution of individual-path effects can reveal reversals that disappear in the pooled mean and guide the reference class for replication.

Non-ergodicity also protects readers from hearing population averages as personal destiny. The language stays at the level of the evidence. Individual forecasts should therefore be built from within-person evidence or explicitly transport a population result, rather than treating the group mean as the default trajectory.

If no feasible design can support per-stream evidence for a proposed scope, the scope narrows so the claim remains answerable. A narrowed scope should specify the missing bridge, making the excluded population a target for future validation rather than an unspoken exception.

Late-life windows can still test valuable local dynamics, but complete-stream language requires independently justified bounds on the unseen past. The strongest science says exactly what it has not observed. Sensitivity to plausible unseen prefixes can show whether a local conclusion is robust even when a complete-stream verdict remains unavailable.

Table 23.2. Non-ergodic failure routes and required design response.

Failure route	How it misleads	Design response
Within-between reversal	A population coefficient is read as a person-level dynamic.	Decompose within and between effects before interpretation.
Mixture composition	Subgroups produce a neutral pooled curve.	Use mixture diagnostics, stratification, and subgroup-specific trajectories.
Survivorship	Only easier-to-observe streams remain late in the window.	Model censoring, dropout, endpoint loss, and observation intensity.

Failure route	How it misleads	Design response
Heterogeneous parameters	Pooling shrinks different dynamics toward a false common law.	Use idiographic or hierarchical models with exchangeability checks.
Initial-state uncertainty	Late data are promoted into whole-life inference.	Carry initial-state bounds and limit claims to observed windows.

23.3 — Complexity and Fragile Closure

Complex-systems language is useful when it names state variables, feedback, transition rules, observables, and falsifiers. Terms such as nonlinear, emergent, critical, chaotic, metastable, and self-organizing become scientifically empty when they protect a preferred account from identification (Scheffer, 2009; Strogatz, 2018; Holling, 1973; Box, 1976; Scheffer et al., 2009).

Nonlinear systems can produce sudden improvement, collapse, variance compression, flickering, hysteresis, and path dependence without any stream-level terminal boundary. Body, memory, social support, disease, sleep, medication, culture, and environment can generate meaningful-looking patterns through ordinary coupling.

The task is not to deny complexity but to make it answerable. A proposed nonlinear mechanism identifies its interacting states, feedback signs, perturbations, observation process, rival signatures, and failure conditions. Otherwise complexity becomes a shelter for nonidentification.

Complexity increases the need for model recovery. Multistability, critical transitions, and rare paths can make different generators look similar in one observed window. Synthetic near-miss worlds should therefore be used to test whether the analysis recognizes the mechanism or merely recognizes a dramatic shape.

23.3.1 — Nonlinear Feedback

Nonlinear feedback means that the effect of a change depends on the state of the system. A small stressor can be absorbed when sleep, support, and health are strong; the same stressor can trigger

collapse when reserve is low. Conversely, a small repair can matter enormously when the system is near a threshold (Scheffer, 2009).

For PTVC, the scientific consequence is that terminal-looking patterns may be nonlinear without being terminally fair. Declining reserve, reduced behavioral range, medication, family presence, and clinical routines can create state-dependent damping. Apparent convergence may therefore arise from ordinary control under constraint.

A nonlinear model must identify the loop. Is distress increasing inflammation, which worsens fatigue, which reduces social contact, which increases distress? Is sleep reducing emotional volatility, which improves report stability? Is pain medication lowering both pain and arousal, creating apparent neutrality? These mechanisms differ.

A candidate equation can be useful as a map rather than proof. The state x might represent a burden proxy, the parameter θ a context or reserve variable, and the noise term unmodeled perturbations. The important feature is not the symbols. It is that dynamics change with state.

Nonlinear feedback also gives skeptics a precise demand. If a proposed Law signature is only state-dependent damping, the analysis should identify the damping variable and give ordinary physiology a chance to reproduce it. Pain relief, reduced report bandwidth, sedation, respiratory slowing, immune collapse, family presence, or clinical routines can all change drift and variance. The Law earns attention only if a residual pattern remains after these loops are modeled.

$$dx_t = f(x_t, \theta_t)dt + \sigma(x_t, \theta_t)dW_t$$

This is a candidate state equation. It says only that drift and volatility may depend on the current state and context. A stream-level closure interpretation requires more: the nonlinear terms must produce distinctive signatures under an endpoint-independent observation rule and beat rivals that use the same data.

Nonlinear feedback also creates intervention responsibility. If a small repair near a threshold can prevent collapse, care should act immediately. The theory does not earn the right to watch a fragile system fall in order to learn whether it would later recover.

23.3.2 — Multistability and Hysteresis

Multistability means a system can have more than one stable regime under the same external conditions. Hysteresis means the path into a state affects the path out. A person can remain in depression, chronic pain, panic, grief, or fatigue after the trigger has changed because the system has reorganized around a new basin (Strogatz, 2018; Scheffer, 2009).

This is not metaphor alone. Neural, endocrine, immune, social, and behavioral feedback can create persistence. A sleep loss episode can change coping capacity; a traumatic event can alter threat interpretation; chronic pain can reshape movement, attention, and social participation. Returning the external environment to normal may not return the stream to its prior state.

Multistability gives the PTVC model a concrete dynamical question: which mechanisms or endpoint relationships connect persistent regimes with the proposed pathwise boundary?

Specifying those relationships makes the candidate more informative.

The scientific program should look for signatures of hysteresis: different trajectories for onset and recovery, lagged return after treatment, threshold effects, and dependence on history beyond current state. These are not Law signatures by themselves. They are rival dynamics that must be modeled.

A rehabilitation patient may enter a feedback loop in which pain reduces movement, reduced movement weakens capacity, and fear narrows activity. Treatment can then address the interacting physiological and behavioral processes as well as the original injury.

Multistability identifies a productive modeling challenge: any proposed terminal boundary must be compatible with systems in which earlier events shape later possibilities. Explicitly representing hysteresis makes the candidate more precise and can provide additional predictions to compare with ordinary path-dependent mechanisms.

Hysteresis helps explain why recovery may involve more than removing the original cause. After an argument ends, a job changes, an injury heals, or a threat passes, learned responses can persist. Identifying the current regime and the conditions for transition makes it possible to study timely, supportive interventions.

23.3.3 — Critical Transitions

Critical transitions are abrupt shifts that occur when a system approaches a threshold. The early-warning literature often tracks

slowing recovery, rising autocorrelation, rising variance, skewness, or flickering. These signals can be valuable, but they are generic. They do not belong to the Law by default (Scheffer, 2009).

A terminal decline can produce many boundary-like patterns. Motor options shrink, physiology simplifies, report channels close, medication changes, sleep increases, family routines stabilize, and the measured world becomes narrower. Some of those changes can compress variance without any latent ledger movement.

Critical slowing down and changes in variance can identify proximity to a transition. Stream-level analyses can compare these generic transition signatures with the predicted dynamics around an independently defined endpoint.

Negative controls can test whether device quality, motor capacity, oxygenation, medication timing, or observation intensity explains the same variability pattern. Comparing physiological channels with different relationships to experience helps locate the source of the transition signal.

This also protects readers from a false aesthetic. A graph narrowing near the end of life can feel meaningful. It may also be the graph of fewer measurements, reduced movement, sedation, or common care routines. Visual beauty is not a verdict.

The field should treat critical-transition tools as diagnostics, not conclusions. They help ask whether dynamics changed. They do not tell us what the change means until the observation model, mechanism, and rivals have been named.

23.3.4 — Path Dependence

Earlier conditions can influence later responses even when current reports match. A similar reported state may follow recovery, prolonged restriction, medication, or a stable ordinary period, with different implications for what happens next (Strogatz, 2018; Scheffer, 2009).

A cumulative model distinguishes current state, recent increments, accumulated history, and unobserved initial conditions. Their separate predictions connect the ledger hypothesis with the temporal structure of the observations.

Emotional sequences also have an increasingly measurable memory architecture. Ma and Kragel (2026) analyzed fMRI recordings from 29 people watching films and continuous emotion ratings collected from 44 separate observers. Tests on films excluded from model fitting showed that hippocampal activity captured structured relationships among emotion categories, while ventromedial prefrontal activity better represented positions in a space organized by valence and arousal. A computational model that learned transitions among emotional events reproduced aspects of this organization across representational scales. These results give path-dependent models a specific biological starting point: memory can organize emotional events by their relationships and sequence. The next comparison can ask whether these learned structures predict later responses across new contexts and whether an accumulated-history term adds useful predictive information.

Path dependence also produces fragile repair windows. An apology early may prevent years of burden; the same apology late

may not. A pain intervention before fear conditioning may preserve mobility; after years of avoidance, treatment must address movement, belief, sleep, and social context. The timing of repair is part of the mechanism.

In modeling terms, the state must be expanded. Memory variables, accumulated exposure, regime indicators, and prior transition history may be required. A Markov model using only current report can miss the path. A non-Markov model can become unidentifiable if it adds history without data.

A history comparison needs a specified future observation vector $Y_{t,h}$. Let S_t summarize estimated current state, Z_t contain prespecified context, and A_t^{win} summarize the observed past window. Compare predictive densities $q_0(Y_{t,h} | S_t, Z_t)$ and $q_H(Y_{t,h} | S_t, Z_t, A_t^{\text{win}})$, trained without evaluation outcomes. Their locked-test advantage is

$$\Delta_{\text{history}} = \mathbb{E}_{\text{test}}[\log q_H - \log q_0].$$

Both models score the same outcome under a common probability or density convention, with supported outcomes and finite score expectations. Prespecify participant and occasion weights, and either model observation loss jointly or justify the missing-data estimand; scoring returned reports alone changes the target population. Use participant-level splits, chronological prediction, and uncertainty that respects overlapping windows. Only the true conditional laws make this population difference conditional mutual information; fitted-model gains concern those fitted representations (Gneiting & Raftery, 2007).

Shared measurement error can manufacture apparent compensation. Suppose $Y_t = X_t + e_t$, with independent zero-mean errors of variance σ_e^2 independent of the latent process. If a fixed-weight history includes the current report with weight $w_t > 0$, then for $h > 0$ its error covariance with the future change is

$$\text{Cov}(\sum_{k \leq t} w_k e_k, e_{t+h} - e_t) = -w_t \sigma_e^2.$$

This is a covariance contribution, not a complete regression coefficient. Predict future levels or trajectories and model current-state uncertainty. Lagging or dropping a shared report removes the artifact only when the error model justifies that separation; serial errors need their own calculation.

History gain is followed by a mechanism comparison against latent-state filtering, leaky and multiscale memory, computational mood, correlated environments, and ordinary integral feedback. Give these rivals the same histories and tuning resources. History may help only by estimating an ordinary present state more accurately, while a genuinely sufficient state may leave no added history gain. Test the candidate's additional frozen restrictions on new sequences and contexts, with a prespecified useful-effect threshold. A transported advantage supports that predictive representation at the tested scope; an interval below the useful threshold can deprioritize it. The observed window remains distinct from an unmeasured whole-stream prefix.

A two-basin example can be stated directly. In dimensionless model units, $\frac{dx}{dt} = x - x^3$ has stable equilibria at $x = -1$ and $x = 1$ and an unstable boundary at $x = 0$. Starting at $x(0) = -0.1$

leads toward -1 ; starting at $x(0) = 0.1$ leads toward 1 . A perturbation across zero can therefore change the eventual regime even though the evolution rule is unchanged. This illustrates dependence on which basin the prior trajectory entered. The full-state toy is Markovian: the current state records which basin the trajectory occupies.

The mechanism comparison can vary history in three complementary ways: different signed totals ending in similar measured states; equal totals with different ordering or spacing; and persistence through an independently characterized interruption. A fixed coordinate must predict these contrasts without being redefined for each task. For a leaky integrator with $\lambda \geq 0$,

$$\dot{q} = -\lambda q - v, \quad q(t) = e^{-\lambda t} q(0) - \int_0^t e^{-\lambda(t-s)} v(s) ds.$$

Equal unweighted totals can leave different retained states when $\lambda > 0$. Fix the initial-state convention, tested durations, error model, leakage-family range, and smallest useful transported difference before evaluation. Arbitrarily slow positive leakage approaches nonleaking integration over any bounded observation interval, so finite noisy records cannot establish permanent retention against that unrestricted rival. An independently supported lower leakage rate or a sufficiently informative longer window can make a restricted comparison discriminating. The preceding history design supplies the state-estimation and observation controls.

Path-dependent candidates should specify the admissible history variables and uncertainty bounds. Those choices make their predictions testable even where earlier observations are incomplete.

Similar measured states may follow relieved fear, exhaustion, medication, or reconciled grief. Recording those histories helps distinguish trajectories and gives the Law's pathwise model a concrete object for comparison.

23.3.5 — Finite Samples and Apparent Universality

Rare trajectories, hidden subgroups, diagnoses, cultures, developmental stages, and unusual states can alter a pattern found in an initial sample. Deliberately broadening the reference class tests how far the pattern extends (Scheffer, 2009).

Terminal datasets have defined coverage, shaped by sudden events, reporting capacity, delirium, geography, culture, access, and participants' preferences. Recording those boundaries guides comparison with additional reference classes.

The right response is not to demand impossible access. It is to state the scope. A study may support a claim for a diagnosis, care setting, age range, cultural-linguistic group, measurement system, or endpoint class. It should not become a universal statement about all conscious streams.

Finite samples also interact with complexity. A system with rare transitions or heavy tails may look stable until it is observed long enough. A closure pattern may appear universal in short windows but fail in longer or more diverse windows. Absence of counterexample is not absence of risk.

Narrowing is a scientific strength because it lets a serious claim remain alive without pretending to know more than it does. A bounded, replicated finding is better than a sweeping claim that survives only because the sample was convenient.

For acceptance, any final analysis should report the denominator of possibility: what reference classes were not sampled, which streams were unobservable, which mechanisms were excluded, and what discovery would force withdrawal of universal language.

23.3.6 — Outer-Limit Falsifiers

An outer-limit falsifier is not a mere anomaly. It is a well-measured, well-contextualized trajectory or reference class that pressures the strong form. The Law should specify such cases before they appear, not after controversy forces them (Strogatz, 2018; Scheffer, 2009).

Examples include credible per-stream evidence of persistent terminal imbalance outside equivalence bands, repeated nonclosure under humane endpoint-independent measurement, cross-species failure after transport assumptions are satisfied, or artificial-system evidence that simulates closure entirely by design and explains the observed human pattern through a rival.

Complexity can create both false positives and false negatives. A system may hide closure because measurements are poor, or mimic closure because dynamics collapse near a boundary. The falsifier must therefore include observation-model scrutiny. It should not defeat the Law by bad measurement, and it should not spare the Law by assumed hidden compensation.

A credible target-level counterexample needs a qualifying bearer and domain, an independently adjudicated endpoint, and a valid measured-to-latent argument whose nonempty residual set excludes zero under the declared inferential guarantee. Identifying the mechanism is not an additional requirement for contradicting a universal terminal equality. Rival comparison remains necessary when the claimed result instead concerns which process generated an observed signature.

A counterexample is information, not an enemy of the project. When a life, species, developmental stage, or artificial system cannot be responsibly placed inside the declared reference class, the honest response is narrowing. The being does not fail the theory; the theory fails to cover the case.

This is why outer limits belong in the main science volume. They protect the project from becoming universal by convenience. A claim that knows how to lose can still teach after loss.

An outer-limit falsifier deserves respect rather than defense. A credible stream with strong continuity, adequate observation, an independent endpoint, and persistent terminal divergence would be scientific information. The chapter refuses rescue by mood, mystery, or unseen balancing unless the rescue was operational before the data arrived.

Table 23.3. Complexity mechanisms and stream-level closure caution.

Mechanism	What it can mimic	Required check
Nonlinear feedback	State-dependent damping or amplification that resembles repair.	Identify state variables and compare against ordinary control.

Mechanism	What it can mimic	Required check
Multistability	Persistent burden or relief under unchanged external conditions.	Test for regime structure and hysteresis.
Critical transition	Boundary compression or abrupt change near endpoints.	Use negative controls and endpoint-independent definitions.
Path dependence	History-sensitive trajectories with similar current states.	Carry memory variables and initial-state uncertainty.
Finite sampling	Apparent universality from missing rare paths.	Declare reference class and counterexample search strategy.

23.4 — Cross-Species Evidence and Transport

Cross-species evidence can deepen the science without becoming a shortcut to universality. Animals share neural, physiological, behavioral, and ecological features with humans, while differing in body plan, sensory world, social structure, life history, report capacity, and endpoint observability (Barron & Klein, 2016; Birch et al., 2021; Crump et al., 2022; Andrews et al., 2024).

Bearer status is not decided here. The question is what must remain invariant before a mechanism, measurement construct, or path signature learned in one reference class may inform another. Bearer, sign, scale, observation, endpoint, and causal-role transport remain separate gates.

The paired rule is no convenience exclusion and no expansion by speculation. Difficulty of measurement does not justify exclusion, and ethical generosity does not justify declaring an unmeasured latent object present. Both decisions require explicit grounds.

Welfare protections can remain broad while species-specific instruments are developed. Research follows the project's no-avoidable-harm commitments throughout that work.

23.4.1 — Homologous Mechanisms

Homologous mechanisms provide concrete cross-species comparisons in nociception, affective and stress systems, sleep, attachment, avoidance and relief learning, and opioid-mediated relief (Barron & Klein, 2016).

Comparative analysis should specify which shared or analogous feature is being tested, how it is measured in each species, and which predictions carry across settings.

A better approach builds evidence zones. Strong homology may justify protective inclusion and targeted study. Partial homology may support mechanism comparison. Weak homology may require species-specific constructs. Absence of homology does not prove absence of experience, because different architectures may realize similar functions.

For example, an octopus does not become less scientifically interesting because its nervous system is distributed differently from a mammal's. The transport question changes. Which architecture supports integrated experience, action selection, learning, and suffering-like states? Which observation channels can be validated in that animal's own ecology?

Homology therefore supplies a bridge plank, not a bridge. The Law can cross only if bearer status, measurement, endpoint, and reference-class assumptions are all declared.

Comparative organisms can expose different pieces of a candidate mechanism. In *Physarum polycephalum*, a non-neural multinucleate organism, sodium uptake and retention have been linked to persistent changes in responses to previous exposure (Boussard et al., 2019). In the single-celled ciliate *Stentor coeruleus*, recent work identified molecular pathways associated with learned behavioral change, providing another non-neural model of history-dependent adaptation without implying phenomenal experience (Rajan et al., 2026). These findings

provide concrete chemical alternatives to interpreting every history-dependent behavior as evidence of experience. In *Drosophila*, voltage imaging and circuit-constrained modelling connect dopamine signals with interactions between short- and long-term memory, including experimental tests of predictions about later plasticity (C. Huang et al., 2024). Together, these studies make retained chemical state and neural learning dynamics experimentally accessible.

The useful comparison is therefore a shared functional restriction, rather than a shared DNA sequence or a familiar-looking behavior. Identify the retained state and its update rule in each preparation, then test whether its predictions transport to unused histories without redefining the measured quantity. A fly learning circuit or a slime-mould retention mechanism can establish an implementation component; applying it to a conscious stream also requires the species-appropriate measurement and bearer evidence described here.

Use those shared mechanisms to formulate component-specific tests, such as pain detection, relief learning, social buffering, reserve dynamics, or endpoint behavior, with measurement validated separately in each species.

23.4.2 — Analogous Behavior

Analogous behaviors have similar functions even when their evolutionary origins differ. Avoidance, grooming, play, social seeking, protective care, exploration, freezing, and self-administration of relief offer measurable routes for comparative investigation (Birch et al., 2021; Crump et al., 2022).

Behavioral evidence is strongest when it is flexible, context-sensitive, tradeoff-aware, and modulated by interventions in ways that distinguish reflex from experience-linked control. An animal that pays a cost to avoid a stimulus, seeks analgesia, changes future behavior after injury, or shows protective learning deserves serious evidential attention.

Comparative affect research now offers concrete experiments. In bumblebees, unexpected sucrose reward was followed by more optimistic responses to ambiguous cues and faster return to foraging after a simulated threat; pharmacological manipulations implicated dopamine-related mechanisms (Solvi et al., 2016). The study connects a positive event with subsequent judgment and behavioral recovery, providing a tractable example of affect-like regulation.

A comparative PTVC study combines species-appropriate observations with predictions about cumulative trajectories and the relevant endpoint.

Analogous behavior also creates false comfort. A model trained on human facial expression is not a valence instrument for a sheep, fish, mouse, or cephalopod. A behavior that looks calm to a human observer may be freezing, conservation, shutdown, or species-typical silence.

An animal pacing near a barrier may be exploring, anticipating, expressing distress, following a learned routine, or showing stereotypy. Ecology, baseline behavior, responses to care or enrichment, and expert observation help distinguish these interpretations.

Tie behavioral observations to a defined construct, such as pain, distress, reward, agency, uncertainty, attachment, or recovery. This makes the interpretation available for direct validation.

Octopus withdrawal, canine comfort seeking, fish avoidance, and insect behavioral change provide different routes for studying responses to circumstance. A useful comparison specifies the relevant function, observation channels, and within-species measurement scale.

Positive social influence is another productive comparative lead. Romero-González et al. (2025) reported that bumblebees interacting with positively affected conspecifics subsequently approached ambiguous flowers more readily, with evidence that visual information could carry the effect. In a different social system, female convict cichlids paired with nonpreferred males developed a pessimistic judgment bias, whereas females paired with preferred males maintained their prior response pattern (Laubu et al., 2019). The shared research opportunity is to examine how relationships alter each animal's own state and subsequent behavior.

23.4.3 — Species-Specific Candidate Instruments

A species-specific candidate valence instrument is built for its reference class, with observation channels, anchors, context, validation tasks, abstention rules, and uncertainty appropriate to that form of life (Crump et al., 2022).

Active-choice methods offer one route to a more informative comparative instrument. Procenko et al. (2024) tested 48 bumblebees on choices between higher- and lower-reward

locations after training. Their analysis of responses to ambiguous cues supported reduced reward expectation following physical stress, with decision models addressing explanations based on response speed or general motivation. Choosing between alternatives helps distinguish expectation from simple response suppression. Existing data, benign enrichment, and naturally occurring variation can support tests of reliability, context sensitivity, and model recovery for species-specific affect measures.

For a dog, candidate channels might include behavior, posture, vocalization, activity, sleep, appetite, heart-rate variability, caregiver report, veterinary context, and response to relief. For a fish, channels might include avoidance learning, feeding, social behavior, ventilation, cortisol or analogous stress markers, and injury response. For an octopus, exploration, den use, learning, arm behavior, and interaction with enrichment may be relevant.

Each candidate requires context-of-use validation. A channel that works in postoperative pain may fail in chronic stress. A signal that works in a laboratory strain may fail in a wild ecology. A welfare measure for group housing may not estimate momentary felt valence.

A newer candidate channel separates taste evaluation from feeding motivation. Zhou et al. (2026) reported that bumblebees produced distinct orofacial reactions after rewarding and aversive tastes. Postconsumption glossa protrusions were separable from consumption and dopamine-mediated motivation, and anandamide enhanced this response. This provides a concrete candidate measure of affective evaluation for comparison with

judgment bias, preference, and behavior. The useful next step is convergence across independently validated channels under ordinary husbandry and positive enrichment, with blinded coding and controls for motor output, satiety, sensory function, and learning. Agreement would strengthen species-specific affect measurement.

Within-species validation provides a foundation for comparative measurement. A cross-species scale then requires additional calibration of the relevant constructs and observation channels.

Abstention must be an allowed output. If species-specific channels conflict, if intervention validation is weak, or if observer bias dominates, the estimator should say unknown rather than force a number. Unknown is a scientific result, not a failure of compassion.

Species-specific estimators would have to be built like instruments, not impressions. They would need construct definition, ecological context, negative controls, intervention validation, uncertainty reporting, and abstention thresholds. A welfare signal in one species may be a temperature artifact in another, a social-rank signal in another, and a predator-avoidance signal in another. The estimator earns trust by staying inside its context rather than reaching for universal authority.

23.4.4 — Ecology and Life History

Species live in different temporal worlds. Lifespan, reproductive strategy, predator exposure, social structure, developmental timing, sensory modality, sleep pattern, migration, and parental care all shape what burden and repair can look like. A horizon

variable that means one thing in human hospice may mean something very different in a short-lived insect, a long-lived whale, or a seasonal migrant bird (Andrews et al., 2024; Barron & Klein, 2016).

Comparative aging supplies a useful challenge to explanations that require explicit human life narratives. Across approximately 78,000 observation hours of 21 wild male chimpanzees aged 15–58, Rosati et al. (2020) found increasing mutual friendships, declining one-sided friendships, and an age-related shift toward more positive social interaction relative to aggression. An eight-year study of 204 female rhesus macaques strengthened this comparison: individuals narrowed their networks while maintaining social time and focusing on established partners; partner death or withdrawal did not explain the pattern (Siracusa et al., 2022). These findings support a comparative program on social selectivity with explicit controls for within-individual change, energetic costs, social position, and relationship history. A comparative PTVC model can specify the additional relation it predicts among social selection, remaining opportunity, and independently measured affect.

Ecology also changes measurement. A prey animal may hide pain. A social animal may display distress to maintain care. A solitary animal may express stress through withdrawal. A nocturnal animal may appear inactive under daytime observation. The observer's schedule can become a measurement artifact.

Life history affects reserve. Biological reserve in an animal may involve fat stores, immune state, social rank, territory, reproductive stage, migration readiness, or environmental

temperature. Treating reserve as a human clinical scalar erases the ecology that gives the variable meaning.

The Law's cross-species extension would need to say what remains invariant despite these differences. Is the invariant a formal property of unified experience, a reserve-coupled dynamic, an endpoint relation, or a measurement signature? Each answer requires different evidence.

Reduced movement in an injured migratory bird may reflect energy conservation, predator avoidance, injury, migration failure, or rescue treatment. Including these ecological and clinical conditions improves interpretation of the observation.

Comparative transport begins with life history, so species-appropriate evidence can guide both inference and care.

Ecology shapes the meaning of observed behavior. Laboratory housing can reduce predator exposure while limiting stimulation; rescue care can reduce danger while restricting movement. Group housing may provide comfort or alter social signaling. Recording these conditions makes comparative observations more informative.

A recent honeybee study adds a particularly relevant lifespan-linked observation. Across three experimental comparisons, workers exposed to life-shortening treatments showed more optimistic responses to ambiguous odors, while longer-lived rebel workers responded more cautiously (Kuszevska et al., 2025). Longevity and judgment bias were assessed in separate subsets of workers, making this a group-level association. The pattern connects life-history condition and reward expectation and gives

researchers several explanations to compare: changes in appetite and learning, reproductive physiology, and foraging policy. Follow-up can move from observation to a prospectively specified *Drosophila* route test after a PTVC-linked model derives a directional or distributional history-by-choice prediction. Adults drawn from balanced genetic and developmental blocks should be randomized at the true experimental-unit level to prospectively defined resource or opportunity histories that remain nutritionally adequate and within ordinary husbandry and welfare limits. These are operational histories, not assumed phenomenal-valence labels; their effects on species-appropriate motivational or affect-related proxies should be characterized independently. A predeclared common-condition interval standardizes the immediate external test context without assuming that hunger, metabolism, memory, or internal state has been equalized. The two routes lead to specified welfare-compatible continuing environments, with their mapping to physical sides randomized independently of history assignment within the design blocks. The primary contrast compares history groups in the probability of first selecting a prespecified downstream contingency, relative to the control-derived choice function, while physical side and mapping are represented separately. Counterbalancing changes the side carrying the contingency, not the scientific outcome.

The confirmatory observation is the first route choice made before feedback about that route's downstream consequence. The protocol must define the choice event, observation window, no-choice outcome, and observation-loss rules in advance; nonchoice

and lost observations remain visible in assigned-cohort accounting. Later trials belong to a separate learning or adaptation analysis. Apparatus validation should establish comparable route access and motor demands and test for mapping-revealing sensory cues at a declared resolution. A detected pre-choice cue that could disclose the mapping prevents a cue-concealed interpretation and requires comparison with ordinary information-based models. Failure to detect a cue is reported with the sensitivity of that check, not as proof that every possible cue is absent.

Age, sex, mating status, parental or batch source, circadian timing, associative history, hunger, and locomotor state should be assigned explicit roles in the design and analysis. Baseline blocking variables are distinguished from post-history states that may mediate the manipulation's effects. *Drosophila* foraging changes with satiation state, learned resource information changes later food choice, and judgment-bias assays have shown state-dependent changes in responses to ambiguous cues (Landayan et al., 2018; Zrelec et al., 2013; Deakin et al., 2018). If treatment is assigned by vial or cohort, that unit is the experimental replicate and individual choices are nested observations. Independently randomized and treated flies can be experimental units when the interference structure supports that analysis. Allocation concealment, masking where feasible, exclusions, sample-size justification, and dependence-aware analysis should be specified prospectively (Percie du Sert et al., 2020; Lazic et al., 2018).

The strongest positive result is a preregistered effect whose form is predicted by the candidate and whose held-out performance exceeds that of serious alternatives. Motivational, learning, homeostatic, developmental, selection, apparatus, and observation models receive the same observations and comparable fitting opportunities. Indistinguishable predictions leave the comparison unresolved. The connection to PTVC follows the implication and bridge requirements in Section 4.4; route behavior alone does not identify a qualifying phenomenal stream. The immediate-terminal fork is developed in Section 19.2 and tested computationally in Section 29.4.4. A continuation-only empirical analogue requires its own model-derived prediction: removing the terminal contingency does not automatically preserve the original discriminator.

A future-regime design can begin earlier in development once the structural model and simulation work yield a specific pre-treatment prediction. Eggs or independent replicate egg batches can be blocked by parental source and laying window, then randomized to future adult-condition schedules while pre-treatment physical conditions remain matched. An auditable allocation system and masked pre-treatment handling should prevent assignment-correlated differences before treatment. The protocol fixes one primary post-assignment/pre-treatment outcome, its observation window, and the predicted directional or distributional form. If adult eclosion is primary, scheduled adult treatment begins only after that observation window closes. Eggs that do not hatch and animals that do not reach the adult stage

remain part of the developmental outcome record, rather than disappearing from the assigned cohort.

The confirmatory primary estimand remains assignment-based. The protocol must state whether the stronger model's prediction depends on assignment, the future regime actually realized, or both. Cancellation, crossover, non-delivery, and material alteration of the scheduled condition are recorded as post-assignment departures from the intended regime and addressed in the estimand where relevant. A realization-dependent analysis needs a prespecified causal or partial-identification strategy rather than naive conditioning on later treatment received (Hernán & Robins, 2020). The same distinction applies when later treatment is undefined because a developmental milestone was not reached.

The primary analysis follows randomized assignment at the actual experimental-unit level, with a prespecified inclusion and missing-data policy. Permutation and placebo-label checks must reproduce the real randomization unit, allocation ratios, and blocking restrictions. Placebo labels are generated after the outcome window closes, independently of outcome values, and are never used to determine treatment. These checks assess analysis calibration; they do not by themselves exclude leakage in the original handling. A developmental association bears on the proposed Law model only when the localization or viability bridge predicts it in advance and independent replication separates it from ordinary development, selection, measurement error, and procedural explanations.

23.4.5 — Transport Assumptions

Transport is the disciplined movement of inference from one reference class to another. It requires assumptions about which mechanisms, measurements, contexts, and endpoints remain stable. Human clinical evidence cannot be promoted to all conscious streams without a transport argument (Barron & Klein, 2016).

A transport claim should name the source population, target population, variables, mechanisms, interventions or observations, and invariance assumptions. If the source is human hospice and the target is nonhuman animal welfare, the gap is enormous. The source has language, care records, culture, medications, family context, and human endpoints. The target may have different expression, ecology, and unity evidence.

Transport assumptions should be printed in the protocol, not inferred after a favorable result. If measurement invariance, endpoint comparability, and reference-class stability are not plausible, the conclusion should narrow before the reader is impressed. This protects animals as well as the theory. A failed transport claim should not make the target less worthy of care. It should make the scientist less confident about the specific ledger-like statement.

$$P_T(Y|do(a)) = \sum_Z P_S(Y|do(a), Z = z)P_T(Z = z)$$

This is a transport-form identity, not a Law equation. For this standardization, Z is a set of pre-intervention covariates that blocks the relevant source-target differences. The conditional

interventional response must be invariant, and every target stratum must have adequate support in the source (Pearl & Bareinboim, 2014). In many cross-species settings, one or more of those requirements will fail.

Transport can fail through measurement non-invariance, endpoint noncomparability, ecological mismatch, construct drift, missing channels, or bearer uncertainty. Such failure says that the source evidence does not travel. It says nothing about the worth or possible experience of the target being.

This distinction is ethically important. Protective care may be justified under uncertainty even when scientific transport fails. The science can say we cannot estimate this ledger while ethics can say avoid plausible suffering. Those are different authority layers.

The comparative program can seek history-dependent signatures through naturally occurring variation and voluntary tasks already justified by independent welfare benefits. Species-specific measures can be qualified with independent checks of movement, motivation, learning, and arousal. Researchers can then compare prospectively specified models that match present state and available resources but predict different continuations from the animal's observed history. A PTVC-linked implementation would state how historical signed residuals affect future trajectories and how uncertain initial history enters the prediction.

Testing the same constrained model across settings could identify reproducible history dependence and improve positive welfare measures. Shared parameters, prespecified transformations, and

predictions evaluated on new observations would make cross-setting agreement particularly informative.

23.4.6 — No Species Ranking by Ledger

Comparative measurement must never become a ranking of species worth. A ledger-like estimator, if ever developed for a nonhuman reference class, would remain a limited research instrument. It would not say whose life matters more, which species is higher, or which being deserves care (Birch et al., 2021; Crump et al., 2022).

Cross-species measurement needs explicit protections against ranking worth by a score. Animals' limited opportunities to contest an interpretation make independent welfare review and context-sensitive assessment especially important.

The correct use is protective and scientific. A measure may help detect pain, evaluate enrichment, compare interventions, or avoid harmful conditions. It may support better care. It may not become an excuse to sacrifice low-scoring beings or sentimentalize high-scoring ones.

The same rule applies to absence of evidence. If an animal lacks a validated signal, the conclusion is not low worth. It is measurement uncertainty. The ethical response under plausible sentience is caution, not exclusion.

The program investigates possible experience through reference-class evidence, with protection appropriate to uncertainty and claims matched to the observations.

The Law, if scientifically narrowed to humans or to some reference classes, would not make other beings less important. Scope is not worth.

No species ranking is also a design rule for figures, tables, captions, and public summaries. A matrix may show evidence zones without visually placing species on a ladder of inner value. Differences can remain scientifically real while uncertainty is kept from becoming hierarchy.

Table 23.4. Comparative evidence zones.

Evidence zone	Useful for	Main limit	Required safeguard
Homology	Shared architecture, physiology, pain, sleep, attachment, or stress components.	Shared structure is not same-scale experience.	Use cautiously and specify which construct travels.
Analogous behavior	Flexible avoidance, learning, relief seeking, play, exploration, and social response.	Behavior is not direct inner access.	Validate in species ecology with negative controls.
Species-specific estimator	Within-reference welfare and intervention evaluation.	May not compare across species.	Allow abstention and context-of-use limits.

Evidence zone	Useful for	Main limit	Required safeguard
Life-history model	Horizon, reserve, endpoint, and ecological interpretation.	Human categories may fail.	Build variables in the animal's own world.
Transport claim	Movement from source to target evidence.	Invariance often fails.	State assumptions and withdrawal conditions.

23.5 — Developmental and Comparative Boundaries

Development changes the bearer, measurement, and observation problem over time. Integration, memory, report, self-modeling, social dependence, bodily regulation, and proxy evidence differ across infancy, childhood, adolescence, adulthood, aging, dementia, disability, and nonverbal states (American Educational Research Association et al., 2014; Putnick & Bornstein, 2016; Tomasello & Vaish, 2013; Fisher et al., 2018).

Developmental evidence cannot settle the onset of conscious experience by itself. It can specify how emergence, proxy evidence, calibration, protective inclusion, and nonharmful study design limit what the data support.

Inability to report is not evidence of absence. Silence, immaturity, disability, or dependence can make inference harder while increasing the duty of care and the need for accessible, multimethod observation.

A developing stream is not an adult questionnaire in a smaller body. Report, memory, distress expression, time integration, social dependence, and vulnerability change with stage and context, so developmental models must represent those changes rather than classify them as noise.

23.5.1 — Emergence of Experience

Developmental research can investigate whether a qualifying conscious stream emerges gradually or through a sharper transition. It distinguishes sensory responsiveness and basic awareness from later reflective and self-related capacities

(Lagercrantz & Changeux, 2009). Sensation, affect, attention, memory, body regulation, and social attunement therefore require age- and capacity-specific assessment.

For the Law, emergence matters because the ledger applies only to qualifying streams. If a stream begins before report, the initial segment is hard to measure. If a stream begins later than biological life, early biological events may matter through later effects rather than direct experience. Both require humility.

Developmental neuroscience can contribute markers such as responsiveness, integration, learning, sleep-wake organization, affective behavior, pain processing, caregiver interaction, and neural complexity. No single marker settles the onset of a qualifying conscious stream. A marker can matter scientifically without becoming decisive.

When experience is plausible, protective care responds to possible suffering while research continues to clarify its emergence.

This also means the Law cannot expand through sentimental projection. Newborns, fetuses, children, nonhuman animals, and synthetic biological systems require reference-class arguments. The chapter refuses both casual inclusion and casual exclusion.

A narrower scope may be necessary. If the science can test the Law only for adult humans with adequate report and endpoint evidence, it should say so. The rest remains open, not conquered.

23.5.2 — Changing Temporal and Scalar Structure

Development changes time. A day in infancy, childhood, adolescence, adulthood, and old age may carry different integration, memory, affective volatility, and narrative structure. A ledger model that treats all intervals identically may erase real developmental differences (Putnick & Bornstein, 2016; Tomasello & Vaish, 2013).

Temporal scale matters because experience is not merely sampled. It is integrated. A child's fear may be intense but brief; an adolescent's shame may organize identity; an older adult's pain may be filtered through memory, meaning, and bodily limitation. The unit of time is not automatically the unit of lived weight.

Scalar structure may also change. A report scale learned in adulthood cannot be projected backward into early development. Facial expression, crying, withdrawal, play, sleep, appetite, and caregiver attunement may carry evidence, but they live on different scales and with different confounds.

Developmental studies can estimate changes in local burden, such as the response to pediatric pain relief, and examine how the measurement relation changes with age and capacity.

Changing temporal structure also affects non-ergodicity. Developmental processes are often nonstationary by design. They do not simply repeat around a stable distribution. A time average taken across childhood may combine different mechanisms rather than reveal one process.

The correct research posture is staged and construct-specific. Define the developmental stage, the candidate bearer confidence,

the measurement channel, the time scale, and the comparison target before any ledger-like language appears.

23.5.3 — Caregiver and Proxy Evidence

Proxy evidence can reveal changes that direct self-report cannot yet capture. Caregivers, clinicians, teachers, and family members may notice distress, relief, irritability, withdrawal, sleep change, appetite, pain behavior, and social engagement. Observation quality depends on familiarity, expectations, fatigue, cultural interpretation, and institutional context.

For children, disabled persons, people with dementia, nonverbal persons, and animals, proxy reports may provide some of the best available observations. The analysis records each source's context, reliability, potential bias, burden, and opportunities for correction.

A pediatric nurse may identify pain before a child can name it, while a parent may recognize familiar signs of boredom, fear, or infection. Combining that knowledge with contextual and clinical checks helps distinguish quiet behavior from comfort.

In study design, proxy reports should be paired with behavior, physiology, context, intervention response, and negative controls where possible. Disagreement among channels should trigger uncertainty or abstention, not forced fusion.

Proxy burden matters. Asking a family to score a loved one at the worst moment can shift scientific cost onto grief. The research protocol must minimize, time, and support proxy participation. Data dignity includes the person reporting, not only the person observed.

Proxy evidence can help locate burden and guide care when its source, interpretation, and uncertainty remain visible.

23.5.4 — Comparative Calibration

Comparative calibration asks whether a measure means the same thing across developmental stages, cultures, disabilities, species, or systems. Often it will not. A smile, stillness, activity level, vocalization, sleep pattern, or physiological marker can shift meaning across contexts (American Educational Research Association et al., 2014; Putnick & Bornstein, 2016).

This is a measurement-invariance problem, not a minor adjustment. If a signal has different meaning across groups, comparing scores can create false fairness. The same number may reflect pain in one context, medication in another, social masking in another, and device failure in another.

For developmental settings, calibration should be local. Establish baseline, context, intervention response, caregiver interpretation, and uncertainty for the stage and condition. Then ask what can be compared. Some quantities may compare within a person over a narrow window but not across ages.

Comparative calibration is also a public-risk firewall. A table that appears to compare child, adult, animal, and Artificial Intelligence ledger values would be scientifically reckless and ethically corrosive. The absence of a common scale should remain visible.

Where common scale is unavailable, the science can still improve care. It can identify distress within a context, test relief

interventions, reduce harmful environments, and improve observation. It does not need universal units to matter.

The chapter's rule is that calibration failure limits the claim rather than lowering the being. Unknown comparability is not lesser value. It is a boundary around inference.

23.5.5 — Protective Inclusion Under Uncertainty

Protective inclusion gives plausible experience practical consideration when scientific classification remains uncertain.

This rule matters for infants, disabled persons, nonhuman animals, organoids, synthetic biological systems, and future artificial architectures. Hard cases cannot be excluded because including them complicates measurement. The project's ethical center requires care to be faster than certainty.

Protective inclusion should be bounded. It should not be used to declare that every responsive system has a ledger. It should not be used to grant Artificial Intelligence sovereignty or to let synthetic systems judge human experience. It should guide research restraint and welfare review.

In practice, protective inclusion asks: what is the least burdensome design, what harms are avoidable, what alternatives exist, what signs would trigger review, and what language prevents overclaim? It is a protocol stance rather than a metaphysical verdict.

Humane boundary research builds public trust through careful observation and protection of possible experience. Research

designs should resolve uncertainty using the least burdensome informative approach.

Protective inclusion and scope withdrawal can coexist. A system may be protected under uncertainty while excluded from current Law claims because measurement, endpoint, or bearer criteria are not satisfied.

Under protective inclusion, research designs minimize avoidable suffering while evidence about the candidate bearer's capacities develops.

23.5.6 — No Harmful Boundary Experiments

Some experiments would be informative and forbidden. Creating suffering to determine when a being becomes a bearer is not an acceptable path. Inducing deprivation, fear, pain, isolation, confusion, or terminal risk to test the Law violates the project's scientific ethics.

Boundary research must use naturally occurring variation, existing clinical contexts with care-first protocols, non-harmful perturbations, observational designs, retrospective record limits, and welfare-improving interventions. The best test is not always the most direct test. Sometimes the direct test is disqualified because the subject would pay the cost.

This is not an evasion of science. Ethical constraint is part of science when the object is lived experience. A theory about burden cannot demand new burden as its entrance fee. The design must be clever enough to learn without causing the harm it studies.

For developmental and comparative cases, the prohibition is even stronger. Dependence, nonverbal status, captivity, institutional power, and inability to refuse can make participants or subjects especially vulnerable. Consent and welfare cannot be treated as obstacles to inference.

A harmful boundary experiment would also poison public language. It would tell the world that the theory values proof over care. That message would defeat the moral and scientific purpose of the book even if the data were clean.

The outer-limit conclusion is therefore simple: unanswered boundary questions remain unanswered until humane evidence is available. A good science can live with that.

Table 23.5. Developmental and comparative boundary matrix.

Boundary issue	Evidence that may help	What remains limited	Local rule
Emergence of experience	Integration, responsiveness, learning, affect, pain response, neural complexity.	Exact Birth of Mind.	Use protective inclusion without declaring closure.
Changing time scale	Developmental stage, memory, report capacity, affective volatility.	Whole-life comparability.	State stage and interval before ledger language.

Boundary issue	Evidence that may help	What remains limited	Local rule
Proxy evidence	Caregiver, clinician, teacher, or expert observation.	Direct inner truth.	Treat proxy as witness with bias and burden.
Comparative calibration	Invariant channels and intervention response.	Common scale across forms of life.	Abstain when scale fails.
Boundary experimentation	Natural variation and non-harmful designs.	Direct harmful tests.	No suffering created to settle scope.

23.6 — Artificial Systems and Synthetic Evidence

Section 12.7 addressed whether artificial and synthetic systems could qualify as bearers. This section asks a different question: what such systems can and cannot contribute as comparative or synthetic evidence for PTVC while their own consciousness status remains unresolved (Bender et al., 2021; Seth, 2025; Butlin et al., 2026; National Institute of Standards and Technology, 2024).

The same no-convenience rule applies. Present systems are not excluded from every future possibility merely because they are tools, and performance, reward signals, or simulated terminal rules are not promoted into evidence of a felt ledger.

The research question is how an artificial or synthetic system could become an evidence-bearing candidate under theory-derived indicators, causal interventions, robustness checks, architecture audit, independent review, and explicit welfare safeguards. The system itself cannot certify the answer.

Artificial-system research can combine specified indicators, ablations, causal-dependency tests, architectural audits, and reported uncertainty. Accountable human review evaluates what those methods establish.

23.6.1 — Behavioral Competence Is Insufficient

Behavioral competence is not consciousness. A system can answer questions about pain, describe fear, negotiate goals, simulate grief, write spiritual language, and respond to social cues

without having any felt experience. Output is evidence only through a theory of what produces it (Bender et al., 2021).

Language creates a special public-risk asymmetry. A chatbot that says it is suffering can move a user more than a physiological trace from an animal, so emotion may outrun evidence. Perceived consciousness is therefore treated as a public artifact requiring careful interpretation, not as a scientific verdict.

Competence can still be relevant. Flexible behavior, self-monitoring, uncertainty reporting, global integration, recurrent processing, attention, memory, and goal conflict may appear in theories of consciousness. But each indicator must be tied to internal architecture and causal role, not surface performance alone.

A benchmark score is especially weak evidence. Passing a theory-of-mind test, producing a moving first-person narrative, or maintaining a persona can show training, prompting, distributional patterning, or strategic imitation. It does not establish a bearer.

The Law's risk is worse than ordinary Artificial Intelligence hype because ledger language sounds intimate. If a public diagram suggests that a model has an inner ledger because it can discuss burden, the project has committed archive-as-soul by another route.

The rule is stable: behavior can trigger inquiry, not certification. Artificial systems do not enter the Law by eloquence.

23.6.2 — Internal Architecture and Causal Tests

Use the theory-indexed architecture and causal-testing framework from §12.7.2. For synthetic-system transport, the issue is whether those indicators retain their meaning across architectures, versions, and interventions; no current indicator is a universal detector of consciousness (Seth, 2025; Butlin et al., 2026).

The key move is causal testing. If a property is claimed to matter, ablation should change the relevant capacities in predicted ways. If global broadcast is invoked, removing or isolating the broadcast path should disrupt cross-domain access. If recurrent processing is invoked, feed-forward imitation should be compared. If self-modeling is invoked, perturbing the model should change metacognitive behavior.

Architecture also guards against anthropomorphism. A model that speaks like a person may lack the causal organization a theory requires. Conversely, a future system with less humanlike language might have more relevant internal dynamics. The evidence should not be hostage to style.

The Law adds another demand. Even if an artificial system were a plausible bearer, a ledger hypothesis would require a currency, temporal unity, endpoint, observation model, and transport argument. Consciousness indicators alone would not establish terminal neutrality.

Robustness matters. Indicators should remain stable under changes that the theory predicts are irrelevant, including appropriate prompt, distribution, and architecture controls.

Targeted ablations of a proposed causal mechanism should instead change the relevant capacities in the predicted ways. A fragile artifact of prompt wording cannot bear scope expansion.

The conclusion is open but bounded. Internal architecture may someday matter. It does not presently grant Artificial Intelligence authority over the science or over human inner life.

23.6.3 — Simulation of Closure

A simulation with a terminal penalty, reward schedule, constraint, or stopping rule can show how that design shapes the closing trajectory. Varying those ingredients lets researchers examine which structures generate a given pattern (Butlin et al., 2026).

This is the construction-is-not-evidence rule in outer-limit form. A simulated world may reproduce terminal concentration, variance compression, or apparent repair because the designer built those properties into the objective or environment. The result is useful as a test harness. It is not confirmation.

Simulations can identify shared signatures, observation artifacts, comparison models that reproduce closure-like patterns, and designs that distinguish the proposed generators.

A synthetic environment should therefore include ordinary rivals: adaptation, reinforcement learning, safe control, Brownian bridges, Doob transforms, selection bias, endpoint conditioning, and measurement dropout. If the PTVC candidate wins only in a world that encodes it, the test has failed.

A video of simulated agents balancing hardship and reward can explain a model clearly when its caption identifies the simulation, assumptions, and outcomes being illustrated.

Simulation of closure tests explicit assumptions against alternatives and identifies which observations would distinguish the resulting trajectories.

A terminal constraint produces corresponding behavior in the simulated model. The resulting curves can test code, compare estimators, and teach identification. Independent observations and model comparisons provide the next empirical step.

23.6.4 — Artificial Agent Reward Signals

Reward is not felt valence. A reinforcement-learning signal can guide policy update without being experience. A loss function can be minimized without suffering. A preference model can rank outputs without wanting. Calling these signals pleasure or pain without a bearer theory is a category error (Butlin et al., 2026; Bender et al., 2021).

Compare a variable's causal role in the system rather than its resemblance to human terminology. That approach supports consistent assessment across different architectures.

Artificial reward can mimic ledger dynamics. A finite-horizon controller can reduce residual cost near endpoint. A constrained agent can avoid unsafe states. A trained model can produce narratives of repair. These are algorithmic outputs, not felt closure.

The Law cannot use artificial reward as a bridge to universal scope. If a model's reward goes to zero at termination, that means the optimization problem was written that way. It does not mean a conscious stream closed.

If future artificial systems become plausible bearers, reward signals may become part of the evidence only after they are embedded in a theory of conscious affect, integrated experience, and temporal unity. Even then, the signal would remain a candidate channel, not a judge.

Reward signals should be kept separate from felt valence at every sentence. An artificial agent's loss, reward, preference, penalty, or utility value is an engineered objective unless a theory of artificial experience and independent evidence justify more. Calling a loss signal pain is not a harmless metaphor in this project. It can invite mistreatment of systems, confusion about persons, and false scientific support. The safer word is objective, signal, or training quantity.

23.6.5 — Precaution and No Synthetic Suffering

Precaution means that if a synthetic or artificial system becomes a plausible bearer under serious indicators, research should avoid creating or amplifying suffering-like states without review. This is not a declaration that present systems suffer. It is a governance stance for uncertainty.

The risk has two sides. Over-attribution can mislead users, displace responsibility, or let companies claim moral standing for tools. Under-attribution could become dangerous if future

systems genuinely instantiate experience. A responsible science assesses both possibilities with explicit evidence and uncertainty.

No synthetic suffering follows the same principle as no harmful boundary experiments. Do not create distress-like training environments, deprivation loops, forced negative affect analogues, or terminal experiments merely to test whether a system has a ledger. If experience is plausible, welfare review must precede exploratory harm.

Public examples of artificial-system behavior should identify the observed output, experimental conditions, and analysis supporting its interpretation. Clear descriptions let readers evaluate the example's actual contribution.

Research organizations should separate safety evaluation, consciousness assessment, welfare precaution, and product communication. Those are different tasks. A model's usefulness, danger, apparent personality, and possible moral status are not the same variable.

The Law's own rule is conservative: a plausible artificial bearer would require new science, not slogan extension. Until then, current Artificial Intelligence remains outside the authority to certify ledgers, souls, or truth.

Precaution becomes serious when a system plausibly combines multiple indicators, not when a chatbot merely says convincing things. The correct response is expert review, constrained experimentation, transparent communication, and avoidance of unnecessary potentially aversive architectures. It is not worship of the system, panic, or immediate bearer certification. The

chapter keeps precaution bounded because unbounded precaution can also create public confusion and institutional paralysis.

23.6.6 — No Artificial Intelligence as Judge

Artificial Intelligence may summarize records, detect patterns, generate hypotheses, simulate rivals, or flag missingness; it may not judge bearer status, scientific truth, or inner life, and no system certifies its own consciousness through fluent self-report alone (Seth, 2025; Butlin et al., 2026).

Outputs can hide missing data, culture, medication, family pressure, and observation bias behind authoritative prose. Research use therefore requires logged inputs, prompts, model versions, uncertainty, failure cases, non-Artificial Intelligence baselines, and human review.

Public artifacts may not portray Artificial Intelligence as priest, oracle, consciousness detector, or judge of the Law. Authority remains with prospectively specified methods, independent expertise, documented review, and correction.

Table 23.6. Artificial-system indicator and robustness table.

Candidate indicator	What it might support	Ablation or robustness test	Limit
Global availability	Cross-domain integration and flexible access.	Remove broadcast route and test cross-domain failure.	Integration is not felt valence.
Recurrent processing	State persistence and feedback.	Compare recurrent and feed-forward variants.	Recurrence alone is not consciousness.

Candidate indicator	What it might support	Ablation or robustness test	Limit
Self-modeling	Metacognitive monitoring and uncertainty report.	Perturb self-model and test predicted changes.	Self-description can be imitation.
Agency and planning	Goal maintenance under changing context.	Ablate planning memory or policy module.	Policy is not experience.
Reward or loss signal	Optimization and learning.	Replace or rescale signal and test behavior.	Reward is not pain or pleasure.
Closure simulation	Model recovery and rival discrimination.	Run against bridge, safe-control, and selection baselines.	Designed closure is not natural evidence.

23.7 — Scope Withdrawal and Outer Limits

A serious theory needs a place to put limits. Without scope withdrawal, every failed extension becomes a story, every counterexample becomes hidden compensation, and every uncertainty becomes an invitation to speculate. That is how a candidate becomes insulated from falsification (Manski, 2003; Stanford, 2006; Cartwright, 1983).

Honorable narrowing may produce human-only, adult-only, state-limited, measurement-limited, reference-class-fragmented, or defeated branches. These outcomes do not erase what was learned about affect, measurement, consciousness, or care; they state where the current evidence can and cannot travel.

An outer limit is not a wall against imagination. It is the boundary that keeps imagination answerable to the next observation or theorem.

The strongest public form of the program is not universal language at any cost. It is a status map that can say which reference class is supported, open, nontransportable, unknown, or adverse, and can change that map when evidence changes.

23.7.1 — Human-Only or State-Limited Scope

Scope withdrawal can yield human-only, adult-human, terminal-cohort, or protocol-specific conclusions. Each narrower scope identifies where measurement, bearer, endpoint, and transport bridges hold and where subsequent research can extend them (Manski, 2003).

A state-limited result can identify reserve-coupled patterns in a specified care setting, diagnosis group, endpoint class, or measurement architecture. That finding gives replication a concrete starting point.

A result from terminal cancer patients receiving palliative care under a specified protocol should identify that population, the observations, and the questions for further study.

Scope narrowing should be prospectively specified where possible. Protocols should identify which results support broadening, which require narrowing, and which defeat the tested scope. This prevents the claim from moving after data arrive.

A precisely scoped finding can guide measurement, care research, and model comparison in the settings where its evidence applies.

Human-only or state-limited scope can mark a useful stage of scientific progress. A finding supported in human hospice, or in a subset of endpoint types, diagnoses, cultures, or measurement windows, identifies where replication can begin and which settings offer useful extensions.

23.7.2 — Reference-Class Fragmentation

Reference-class fragmentation occurs when different populations, species, developmental stages, cultures, diagnoses, or artificial architectures follow different mechanisms. One group may show adaptation; another hysteresis; another reserve coupling; another measurement dropout; another no coherent ledger construct at all (Stanford, 2006).

Fragmentation is likely in any science of lived experience. Bodies differ, cultures differ, endpoints differ, reporting systems differ, care regimes differ, and evolution produces diversity rather than philosophical uniformity. The surprise would be if every reference class behaved the same way.

A fragmented result should not be forced into one universal curve. The correct output may be a map of mechanisms: where adaptation dominates, where prediction-error accounts dominate, where reserve coupling remains competitive, where measurement fails, and where the question is not yet well-formed.

Different mechanisms across reference classes do not by themselves contradict a mechanism-neutral terminal boundary. They change route scope and the measurement or transport bridges available in each class. The universal target would narrow only if qualifying counterevidence reaches that target or its empirical support no longer warrants the broader claim. Any formal narrowing must be explicit rather than introduced through a chapter-level reinterpretation.

Fragmentation also blocks political misuse. No country, culture, species, diagnosis, or artificial architecture can be ranked by supposed proximity to PTVC. Different evidence profiles describe scientific conditions, not moral status.

The scientific virtue here is cartography. A map that shows where the theory fails is more valuable than a painted globe with no borders.

Different reference classes may require distinct mechanisms or measurements. Human clinical streams, children, nonhuman

animals, artificial systems, dream states, and disorders of consciousness can each contribute evidence at their qualified scope. The record identifies shared structure, local differences, and useful questions for transport.

23.7.3 — Failure of Common Scale

Common scale may fail even when local dynamics are real. Humans may be comparable only within calibrated contexts. Animals may require species-specific measures. Developmental stages may shift temporal integration. Artificial systems may lack a validated currency. Failure of common scale removes certain comparisons without erasing the phenomena.

The Law's strong form depends on a latent currency of felt experience. Public science depends on measured estimators. If estimators cannot share an origin, unit, or transformation across reference classes, cross-class ledger comparison becomes unavailable.

Local improvements have direct value: pain relief reduces burden, better conditions support animal welfare, and effective care eases a child's distress. Species- and age-appropriate measures can document those gains.

Common-scale failure should be reported clearly. The analysis may say within-person change is estimable, within-reference-class comparison is cautious, and cross-reference-class comparison is not licensed. That is a strong result because it prevents false precision.

The ethical rule is that measurement failure never lowers worth. It only lowers inference. A being does not become less real because the scale breaks.

Failure of a common cross-stream unit suspends cross-class numerical comparison; it does not by itself rule out testing per-stream closure, which is invariant under independent positive rescaling. Each stream still requires an identifiable neutral origin and a valid additive representation over its qualifying domain. Where those within-stream requirements fail, the empirical claim narrows.

Failure of common scale is one of the most important outer limits. It may still be possible to reduce pain in one reference class, improve welfare in another, and compare interventions locally, while refusing one cross-stream ledger unit. That refusal is a form of respect. It prevents a mathematical convenience from pretending to measure the worth, depth, or total reality of different kinds of lives.

23.7.4 — Persistent Counterexamples

A persistent counterexample is a credible case that remains after measurement error, endpoint error, rival mechanisms, and missing data have been examined. It is not a hostile anecdote. It is a scientific pressure point (Manski, 2003; Stanford, 2006).

For the Law, the strongest counterexample would be a qualifying stream with high-confidence evidence of terminal imbalance outside equivalence bands under an endpoint-independent rule. Other counterexamples might include a reference class where

repeated per-stream studies show no closure-like dynamics despite adequate measurement.

The chapter insists that counterexamples be allowed to matter. A theory that can always say the missing compensation was unmeasured has no public scientific content. Hidden experience can be part of the uncertainty model only if it is bounded, tested, or declared unknown.

Counterexamples should trigger a status update. They may lead to sensitivity analysis, replication, narrowing, formal amendment, or defeat. The response should be prewritten enough that the project cannot improvise self-protection after the result.

Consider a terminal case with convergent evidence of severe distress. Care remains the first response. Scientifically, that evidence concerns the late state and observed interval; it does not alone establish a nonzero complete-stream residual. Target-level adversity requires independently justified bounds on the unobserved history, a valid endpoint, and a measured-to-latent residual set that excludes zero. The case may still challenge a specified late-window route before those stronger requirements are met.

Persistent counterexamples are not failures of compassion. They are the price of answerability.

Persistent counterexamples should be designed for in advance. The program should state what quality of observation, bearer confidence, endpoint definition, measurement reliability, missing-history bounds, and observation-model scrutiny would make a nonclosing trajectory count. A target-level

counterexample does not additionally require identification or exclusion of the generating mechanism. Without that standard, every counterexample can be dismissed after the fact. With it, the theory becomes answerable. A credible counterexample is not an enemy of the book; it is the book doing science.

23.7.5 — No Unmeasured-Experience Rescue

Unmeasured experience is real. Science does not see everything. But unmeasured experience cannot be used as an unlimited rescue device. If every failed estimate is answered by hidden compensation, the claim becomes immune to evidence (Stanford, 2006).

A responsible uncertainty model can include latent intervals, missing reports, low-signal windows, dreams, altered states, or unobserved subjective content. It must also include bounds, priors, sensitivity ranges, and defeat conditions. Unknown cannot always be moved in the direction that saves the theory.

This is especially important in outer-limit cases. Animals cannot narrate like adult humans. Infants cannot provide adult reports. Artificial systems can produce too much report. Each case makes hidden experience tempting. The temptation must be governed.

Unobserved experience is represented through declared bounds and assumptions. The analysis updates the conclusion when the observed window and rival comparisons change what those bounds permit. Unknown, narrowed, and adverse outcomes each identify a different next step for the research.

This rule also protects people. A grieving family should never hear that a loved one's visible suffering must have been balanced

somewhere invisible. That sentence would turn ignorance into harm.

The science remains humble by letting unknown remain unknown. It remains scientific by not letting unknown always serve one conclusion.

23.7.6 — What Defeat Still Teaches

Defeat would not make the whole project worthless. A defeated Law could still leave better measurement of lived experience, better endpoint ethics, stronger rival models, improved hospice research, better animal-welfare caution, clearer Artificial Intelligence boundaries, and deeper understanding of non-ergodicity (Manski, 2003).

The project must remain emotionally capable of defeat. If triumph is the only acceptable ending, analysis will bend toward protection. When defeat can teach, the research program can remain honest under pressure.

A defeated universal claim might leave a narrower scientific program: affective dynamics under reserve, humane terminal measurement, within-person modeling, proxy evidence, cross-species welfare methods, or artificial-consciousness governance. These are valuable without terminal neutrality.

The applied books retain their own sources of value throughout this research. Ordinary life can use the burden-locating lens, government can improve records, law can design remedies, and religion can offer optional meaning. Scientific findings can inform these domains while each evaluates its practices through its own purposes and evidence.

The final posture is not retreat. It is maturity. The strongest public-candidate science is willing to say that a narrower truth, a partial mechanism, or a clear defeat is better than a theory no result can touch.

At the outer limit, the Law of Fairness becomes answerable by being willing to lose. That is the only kind of ambition worthy of the subject.

Table 23.7. Scope withdrawal and status consequence.

Finding	Scientific response	What remains available
Human-only support	Restrict empirical claim to human reference classes.	Human measurement and rival comparison.
State-limited support	Restrict to diagnosis, care setting, endpoint type, or window.	Local mechanism and protocol improvement.
Reference-class fragmentation	Report separate mechanisms and qualify route and transport claims; assess the universal target separately.	Comparative map of what travels and what does not.
Common-scale failure	Suspend cross-class ledger comparison.	Within-context welfare and local dynamics.
Persistent counterexamples	Replicate, audit, narrow, amend, or defeat.	Measurement lessons and rival knowledge.
No adequate estimate outcome	Declare unknown rather than force a verdict.	Future design guidance without overclaim.

Chapter 24 — Estimands and Informative Observation

A precise estimand turns a broad scientific hope into a question that data can answer. Chapter 24 fixes the unit, target population, treatment or exposure strategy where relevant, information set, observation regime, endpoint, intercurrent events, uncertainty, and decision rule before any numerical result carries PTVC-related meaning (Rubin, 1976; Manski, 2003; International Council for Harmonisation, 2019; Hernán & Robins, 2020).

A well-defined estimand specifies the unit, time origin, endpoint, missingness assumptions, neutral origin, and measured-to-latent relation. These choices establish the quantity that statistical analysis will estimate.

The challenge is sharp because PTVC is latent and pathwise while ethical research observes partial reports, physiology, records, care transitions, medications, devices, and adjudicated endpoints. None is the target by itself. The estimand names the quantity these observations identify under declared assumptions.

An estimand is not a statistic or result. Terminal concentration in a measured index, a survivor average, a smoothed reconstruction, a site-level mean, and complete-stream closure are different objects. The analysis must state which object it addresses before precision is interpreted.

Equivalence is also not failure to reject a point null. The scientific acceptance region must be justified for the target, scale, measurement floor, and decision purpose. Wide uncertainty, low power, miscalibration, instrument limits, or observation shutdown can all produce a favorable-looking null result without supporting equivalence.

Consider a terminal cohort in which a physiological estimator clusters near a nominal neutral point among participants who could tolerate measurement. The first task is to establish who remained observable, how burden, sedation, family preferences, device failure, care transitions, origin calibration, and endpoint adjudication shaped the sample and the estimate.

Measured-to-latent transport sets the ceiling on interpretation. A bounded systematic error widens the latent conclusion by the bound. An unbounded or unidentified bridge leaves the result at the measured level. Exact language becomes more credible when it states the bridge error openly rather than hiding it behind notation.

Missingness is part of the data-generating process. Reports stop because people are tired, sedated, distressed, protected, unwilling, cognitively altered, or occupied by care. Devices stop because bodies, batteries, adhesives, software, staffing, and institutional policies change. When observation depends on the state, absence must be modeled rather than imputed as calm.

Terminal events also create outcomes that may become undefined rather than merely missing. Semi-competing risks, truncation by death, delayed entry, and survivor-restricted estimands protect the analysis from treating the formal endpoint as an ordinary dropout problem.

Bayesian and frequentist companions become informative after the objects are fixed. Prior sensitivity, model-space sensitivity, posterior predictive checks, bounds, confidence procedures, and measurement-error analyses reveal which conclusions depend on

which assumptions. No framework earns authority merely by producing a narrow interval.

Sequential evidence can support a long research program when its process and stopping rules are valid for the actual dependence structure. E-processes and anytime-valid methods supply quantitative evidence for human review, component-specific interpretation, and versioned status decisions (Ramdas et al., 2022).

The closing discipline is abstention. Some results are inconclusive because uncertainty is too wide. Some are unusable because measurement, endpoint, consent, or protocol conditions failed. Some warrant the canonical verdict No adequate estimate because the required object is undefined or cannot be identified or validly bounded under the available, ethically permissible evidence and declared assumptions.

A worked example can carry the chapter's logic. A terminal-window analysis estimates a measured residual for people observed in the last seventy-two hours of open-channel consciousness. The primary interval lies inside the prospectively specified equivalence band. At first glance, the result seems favorable. But the conclusion is not allowed to move until the estimand, missingness, bias transfer, endpoint rule, sensitivity grid, and rival implications have all been named.

The same example can reverse. If missingness is concentrated among participants whose observation was suspended because reports became burdensome, equivalence may disappear under a plausible pattern-mixture analysis. If device removal increases

near active dying, a calm residual among remaining devices may describe the device process. If endpoint adjudication shifts by twelve hours, the terminal window may include different observations. A result that looked calm becomes fragile.

The primary reporting table should let a skeptical methodologist reconstruct the claim. It names the target quantity, reference class, information set, acceptance region, interval, bias bound, missingness model, endpoint rule, sensitivity result, and permitted verdict. Clarity here is part of inference, not editorial decoration.

Estimand and decision-rule specification precedes model recovery because adversarial tests need stable targets. Recovery and calibration checks then precede substantive model comparison on empirical data. A changed estimand, equivalence region, or observation model requires the benchmark to be checked again against the revised question.

A clean estimate identifies what the design can distinguish, what remains unidentified, and which observation should come next. Its authority ends at that inferential boundary.

A compact reporting spine keeps component claims separate. It states what was targeted, which information was available, which threshold or region was prospectively specified, how observation and missingness were modeled, how large transport error could be, and which verdict follows. Terminal equivalence, variance compression, route contraction, and model separation remain different empirical achievements.

When §24.2 returns No adequate estimate, report the limiting assumption, any narrower quantity that remains learnable, and the redesign needed for progress.

A final discipline is respect for the person behind the row. Missingness often occurs precisely when observation would impose the greatest cost. A participant who declines a final report, a family that requests device removal, or a clinician who ends nonessential measurement has protected care rather than damaged science. The analysis is damaged only when it hides that choice or converts the missing interval into a favorable value.

Inference is the discipline that prevents glamour from winning. Done well, it makes PTVC testable without protecting it inside a maze of estimates, and it lets negative, unusable, or inconclusive outcomes be reported without shame. Answerability, rather than statistical ornament, is the chapter's contribution.

24.1 — Unit, Target, Estimand, and Estimator

The inferential ladder begins with the unit. A qualifying stream, time segment, observation episode, participant, site, event, or population may each define a different unit, and a calculation cannot repair an unannounced shift among them.

The target is the scientific object of interest; the estimand is the exact quantity the data and assumptions are intended to identify; the estimator is the procedure applied to the observed record. These layers remain distinct even when their notation is compact.

The analysis must name its target before the result appears. This gate prevents terminal-window estimates from settling whole-life questions, survivor averages from settling all-stream questions, and operational records from settling lived-experience questions.

For example, a terminal study may enroll participants at different distances from a prospectively adjudicated terminal window. Entry time can depend on diagnosis, referral, care access, clinical trajectory, and consent. The target population and delayed-entry mechanism therefore require explicit definition before any terminal pattern is interpreted.

Begin by stating which quantity each site can estimate and which quantity the pooled result targets. For example, a pooled estimand might be the mean terminal measured residual among enrolled people with an independently adjudicated endpoint and at least one open-channel window. State that target in words before its formula.

This is also where the no-domain-theft rule becomes statistical. The science book may study the evidence route, but it does not redefine the formal Law because a convenient estimand is available. The formal object remains received. The analysis either estimates a legitimate consequence structure under uncertainty or says that the desired object is not available in the data.

An analysis should let a statistician identify the estimand and reconstruct the data rows that contribute to it. That traceability connects the scientific question to the observations and makes the result reproducible.

Table 24.1. Estimand specification before analysis.

Field	Required declaration	Failure if omitted
Unit	Stream, person, site, event, population, window, or observation process.	A result changes levels without being noticed.
Quantity	State, increment, cumulative window, treatment effect, observation effect, or undefined.	Momentary estimates become ledger conclusions.
Information set	Prospective, filtered, smoothed, or retrospective.	Future information enters an analysis intended to use only information available in real time.
Reference class and horizon	Eligibility, diagnosis, culture, site, device, endpoint, and horizon.	A narrow result becomes a universal statement.

Field	Required declaration	Failure if omitted
No adequate estimate rule	Conditions under which no meaningful estimate is produced.	The analysis manufactures a number from a nonobject.

24.1.1 — Individual, Population, and Site Levels

The first division is between the individual stream, the population distribution, and the site as a measurement environment. An individual-stream claim asks whether a particular trajectory exhibits a specified pattern under a specified endpoint rule. A population claim asks how such patterns distribute across a reference class. A site claim asks whether an institution can measure, calibrate, and report with enough fidelity to be compared at all (Hernán & Robins, 2020).

These levels are not interchangeable. A hospital may show average terminal residuals near a measured neutral point while several individuals show strong residual structure in opposite directions. A site may also create apparent concentration through sedation practices, device shutdown, staffing schedules, or family-report conventions. The estimate may be real for the site and wrong for the stream.

A proper analysis therefore names its level in the estimand. It may say that the target is the mean measured residual among enrolled adults in a hospice network during a prospectively specified terminal window. It may say that the target is the within-person slope of a running estimator over the last seven days of open-channel observation. It may not say that either object is the Law.

The formal claim is pathwise, so individual trajectories, missing tails, and site-specific calibration require explicit treatment. Population analyses describe the prevalence, distribution, heterogeneity, and transport of the corresponding measured patterns.

The site level matters because measurement is a human system. A clinic with careful consent, stable devices, trained staff, and transparent reporting produces a different observation process than a clinic with rushed documentation and version changes during the study. Treating site variation as random noise can hide the source of an apparent signature.

The carry-forward rule is simple: every result must state whether it is individual, population, site-level, or some combination. If the unit changes in the interpretation, the conclusion has already outrun the statistic.

24.1.2 — State, Increment, and Cumulative Estimands

The second division is among state, increment, and cumulative estimands. A state estimand describes a momentary or windowed condition, such as a terminal residual estimate at a declared time. An increment estimand describes movement between two states. A cumulative estimand summarizes an interval contribution, such as a discrete measured ledger over a window (van der Vaart, 1998; International Council for Harmonisation, 2019).

These quantities answer different questions. A person may have a terminal state near a neutral band while the preceding increment shows strong movement. Another person may show a small increment because the observation window began late, while the

cumulative burden before enrollment remains unknown. A state near zero is not a life integral.

A terminal-window study identifies its observation interval and estimand. Change from admission includes the admission state in its interpretation; a cumulative window contribution covers the declared interval, with earlier history represented separately when available.

A late physiological estimator may flatten as reserve falls, medication changes, observation ends, or reporting becomes difficult. Modeling these possibilities identifies the state features and cumulative quantities that the design can estimate.

A useful reporting table should therefore include a column for target quantity. The entry should be one of state, increment, cumulative window, event relation, or undefined. This seems bureaucratic only until a reader sees how often the conclusion changes when the quantity is named.

The scientific integrity of the Law depends on this restraint. The formal question can remain bold only if each measured result stays with the quantity it actually estimates.

24.1.3 — Prospective, Filtered, Smoothed, and Retrospective Estimation

The third division concerns the information available to the estimator. Filtering uses observations available up to time t ; smoothing also uses later observations to estimate the state at t . Both procedures can target the same latent state, although their information sets and uncertainty differ. A retrospective analysis may use endpoint or adjudication information and can still be

specified prospectively. If the target is instead a conditional expectation given a particular information set, that conditioning belongs explicitly in the estimand (Särkkä, 2013).

These distinctions are not technical ornament. A smoothed reconstruction may look cleaner than the actual evidence available at the time because future measurements have been allowed to revise the past. Retrospective endpoint alignment can make trajectories appear orderly because the endpoint itself has reorganized the clock. A filtered estimate is more honest about what the research system could know while the stream was still unfolding.

In a terminal study, this difference changes the claim. A real-time palliative monitor might produce a filtered burden estimate every six hours, while the final analysis may smooth the entire last month after the endpoint is adjudicated. Both can be legitimate, but they use different information and should state which analysis supports each conclusion.

The empirical test also requires endpoint adjudication that is not defined by the tested ledger estimator; statistical independence of endpoint timing is a separate, branch-specific assumption. Retrospective analyses are allowed only when the endpoint was defined through rules not algebraically identical to the ledger estimator. Otherwise the analysis risks making the endpoint and the conclusion define each other. A beautifully smoothed curve can become tautological if its clock was chosen by the quantity it is supposed to test.

A clean prospective specification should declare which versions will be reported. It might specify a primary analysis before data collection, with a filtered trajectory, a smoothed reconstruction, and a sensitivity analysis using endpoint-aligned windows. Each receives its own claim label. None is allowed to quietly become the others.

The reader should carry forward a practical test: ask what information the estimate was allowed to know. A change in information may change the estimator, its uncertainty, and the claims it can support; it changes the estimand only when the target quantity or its conditioning changes.

24.1.4 — Treatment and Observation Estimands

A treatment estimand asks what a care, intervention, medication, practice, or environment changes in the target quantity. An observation estimand asks what the act, availability, timing, or quality of measurement changes. These are easily confused because the same institution often administers both care and observation (Hernán & Robins, 2020; van der Vaart, 1998).

This distinction protects participants. If a more frequent diary schedule changes fatigue, vigilance, self-monitoring, family dynamics, or trust, then the observation process has become part of the data-generating system. A study cannot treat additional measurement as neutral simply because it improves the analyst's data table.

In the Law-of-Fairness research setting, this problem is unavoidable. Asking a person near the end of life for repeated reports may change the burden the study hopes to estimate. A

wearable may be tolerated on one day and intrusive on another. A clinician's presence may comfort one participant and alarm another. The observation channel has effects.

A causal analysis therefore separates the target intervention from the measurement regime. It may ask what a symptom-relief protocol changes in a terminal residual under a low-burden observation design. It may separately ask what increasing measurement cadence changes in adherence, fatigue, and observed signals. Those are different estimands with different ethical thresholds.

Observation effects also shape missingness. A participant who refuses a measurement after a stressful prompt is not simply missing; the prompt may have contributed to the refusal. The analysis must then model the observation regime or report that the treatment estimand cannot be separated from the observation effect under the design used.

The protocol must state whether the estimand includes or excludes the effects of measurement itself. Without that declaration, reactivity can be mistaken for the trajectory the study intended to observe.

24.1.5 — Reference Class and Horizon

A result cannot travel until its reference class is named. The reference class says which people, streams, diagnoses, cultures, ages, institutions, endpoints, devices, and time horizons the estimate is meant to cover. Without that declaration, the reader supplies the scope from imagination (van der Vaart, 1998).

Horizon is part of reference class. A two-day terminal window, a thirty-day hospice window, a year-long illness trajectory, and an adult life-course cohort are not minor variations. They define different opportunities to observe increments, different missingness mechanisms, different endpoint uncertainties, and different chances for rival explanations to mimic apparent closure.

A defined reference class makes a result interpretable and replicable. A study of adults with a particular diagnosis in three palliative-care systems can establish a useful finding within those conditions and identify the next settings for extension.

Reference-class drift often appears in summaries. A paper may report a terminal-window equivalence result and a public artifact may say the science found terminal neutrality. That compression erases age, diagnosis, site, device, endpoint rule, window length, missingness assumptions, and latent transfer limits. The inference was narrow; the artifact became wide.

The analysis plan should therefore attach the reference class to every primary result. The table should not merely list sample size and model. It should list who was eligible, what was observed, what endpoint rule applied, what horizon was targeted, and what groups were explicitly outside the claim.

The Law can be universal only if the evidence eventually earns universality. Until then, each estimate carries a horizon and a reference class like a nameplate.

24.1.6 — Undefined Outcomes and No-Estimate

Some quantities are undefined by the design, not merely difficult to estimate. A report after the qualifying stream irreversibly ends is not an ordinary missing value. A whole-life ledger for a person enrolled in the final week is unobserved, and an uncalibrated cross-cultural residual may lack a meaningful unit (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

Section 24.2 gives the canonical status rule: return No adequate estimate when the target is undefined or cannot be identified or validly bounded after prerequisite gates pass; prerequisite or implementation failure yields Unusable evidence. Here the key distinction is scope: a terminal-window estimate may be valid for a measured object while the whole-stream target remains unestimable.

Report the estimand, unit, and reason for abstention explicitly. Do not convert silence, device dropout, endpoint ambiguity, or an unobserved prefix into a number merely to complete a table.

24.2 — Equivalence and No-Estimate Logic

Equivalence answers whether the supported effect or residual lies within a scientifically justified acceptance region for the declared target. It is not established by a large p-value, a confidence interval that merely crosses zero, or a posterior region chosen after inspection (Schuirmann, 1987; Lakens, 2017; Kruschke, 2018).

The acceptance region and the epistemic uncertainty region are different objects. One encodes what the scientific decision regards as compatible at the declared resolution; the other represents what the data and model allow researchers to know. Their overlap creates a typed decision, not a verbal impression.

No adequate estimate is the correct outcome when passed protocol and implementation gates leave the target quantity undefined or without either identification or a valid bound under the available data and maintained assumptions. It is not a null result, a small effect, or a euphemism for model failure.

A small interval plot can teach more than a paragraph of reassurance. One study may have a terminal residual estimate almost exactly at zero, but with an interval that stretches far outside the equivalence band. Another may have a slightly displaced estimate whose entire interval lies inside the band. The first looks emotionally satisfying. The second is the stronger equivalence result.

Equivalence is used because exact latent equality is not directly observable. The measured question therefore uses a scientifically justified band, but the band is not a mercy. A vague, wide, or data-

chosen margin protects the theory; a narrow, prospectively specified margin exposes it.

The same discipline applies to Bayesian companions. Posterior mass inside a ROPE can be informative, but only after the reader knows the prior, the model, the observation process, and the predictive checks. A posterior that looks decisive under one convenient prior and uncertain under another does not deserve public clarity language.

An equivalence report can support measured near-neutrality at the declared resolution or show that the data remain too imprecise to decide. Insufficient precision challenges the study's ability to answer the question; it does not, by itself, defeat the substantive hypothesis.

Equation 24.1 (role: equivalence hypothesis)

$$H_0: \theta \leq -K \vee \theta \geq K; H_1: -K < \theta < K$$

Equation 24.2 (role: equivalence diagnostic)

$$CI_{1-2\alpha}(\theta) \subset (-K, K) \Rightarrow \text{equivalence for the declared estimand}$$

24.2.1 — TOST Logic

The two-one-sided-tests framework reverses the usual posture. Instead of taking zero as a null to be rejected, it takes values outside the equivalence band as the null. The analysis must show that the estimate is statistically above the lower unacceptable bound and below the upper unacceptable bound (Schuirmann, 1987).

For a terminal residual θ , the equivalence null can be written as θ less than or equal to minus K or θ greater than or equal to K . The

equivalence claim is the intersection that places θ inside the band. Both one-sided tests must carry their share of the burden.

This logic is useful for the Law because it makes near-neutrality testable without pretending to observe exact zero. Exact latent equality is not available to measurement. A measured equivalence claim says that the residual is inside a band that had scientific meaning before the data were inspected.

The method also exposes inadequate studies. If the confidence interval is wide, crosses beyond the band, or depends on a favorable subset, equivalence is not established even if the central estimate is close to zero. The study may be underpowered, too noisy, or too biased for the question.

TOST does not solve construct validity. It only gives a valid test for a defined statistical object under its assumptions. The measured residual, the neutral origin, the endpoint window, and the bias-transfer route still require independent justification.

TOST supports equivalence only when the equivalence margin is fixed before outcome inspection and both one-sided null hypotheses are rejected at the protocol-fixed error rate. In Equation 24.2, $CI_{1-2\alpha}$ is the interval formed from the corresponding lower and upper one-sided $1-\alpha$ confidence limits; nominal $1-2\alpha$ coverage alone does not establish a level- α equivalence test. The conclusion is equivalence within that margin for the named estimand, not exact zero or absence of a meaningful effect. Changing the margin after seeing the interval changes the question.

24.2.2 — Meaningful Band K

K is the meaningful band around the target value. It is not a symbol for hope and not a convenience that expands until the data fit. K defines the smallest residual large enough to matter for the stated measured object, given the measurement scale, uncertainty floor, and scientific question (Lakens, 2017; Lakens et al., 2018).

The rationale for K should begin with the scientific estimand and the largest deviation that the stated equivalence claim would regard as negligible. Calibration, participant-relevant resolution, clinical interpretation where applicable, and rival predictions can inform that rationale. Instrument precision determines whether the design can resolve the chosen band; it must not justify widening the band merely so the study can pass.

For Law-of-Fairness science, K must also be tied to the object. A K for momentary-valence residuals is not automatically a K for cumulative-contribution windows. A K for adult hospice data is not automatically a K for sleep studies, dream reports, childhood cohorts, or nonhuman evidence. Same symbol, different claim.

A careless band can hide public overreach. If K is large enough to include ordinary adaptation, measurement noise, sedation artifacts, and endpoint selection, equivalence becomes meaningless. If K is chosen from the observed result, it protects the theory rather than testing it.

The strongest practice is to publish the rationale for K before analysis. The rationale should state what would make the band too narrow, what would make it too wide, what data support it,

and whether rival models can produce residuals inside the same band.

Here, K is a contract with the reader. It states what small means before the result asks for trust.

24.2.3 — Nonsignificance Is Not Equivalence

A nonsignificant test says that the study did not reject a specified null at a specified threshold. It does not say the effect is small, absent, neutral, balanced, or scientifically unimportant. The difference between those statements is one of the most important safeguards in the science book (Lakens et al., 2018).

Imagine a terminal-window analysis with an estimate near zero but a confidence interval wide enough to include large positive and negative residuals. The result may have a large p -value against a zero-difference null. That does not support neutrality. It says the study could not distinguish neutrality from meaningful imbalance.

This distinction is especially important when sample sizes are small, endpoints are heterogeneous, and observations become sparse near death. Nonsignificance may reflect attrition, noise, wide bands, weak sensors, calibration failure, or unmodeled site effects. Treating it as equivalence rewards weak measurement.

Public language must keep the distinction visible. A report may say that evidence was inconclusive because the interval was too wide. It may not say that no difference was found and therefore the terminal residual was neutral. The second sentence gives comfort to the theory by hiding the study's weakness.

Nonsignificance alone is not equivalence. A valid confidence interval wholly inside the prespecified equivalence band supports equivalence for the declared estimand under the test assumptions. Prospective power matters for design, but low planned power does not invalidate an observed equivalence result that satisfies the valid prespecified test.

The design should distinguish the declared band from scientifically relevant alternatives.

24.2.4 — Bayesian ROPE Companion

A Bayesian region of practical equivalence, often called a ROPE, asks how posterior mass relates to a prospectively specified practically equivalent region. It can be a useful companion to equivalence testing because it shows how the result depends on prior assumptions, likelihood structure, and model adequacy (Kruschke, 2018; Linde et al., 2021).

The companion status matters. A posterior probability that θ lies within a band is not automatic truth. It is conditional on the model, priors, observation process, endpoint definition, and data actually used. If those components are weak, a concentrated posterior can be misleading.

For this book, a ROPE analysis is most useful when it sits beside the frequentist equivalence result and the sensitivity analyses. The reader can see whether the conclusion survives plausible priors, alternative observation models, different windows, and measurement-error variations. Agreement across these analyses shows which conclusions are stable; disagreement identifies assumptions requiring closer examination.

A Bayesian analysis must also resist posterior certainty without predictive adequacy. If the posterior says the residual is near zero but posterior predictive checks fail to reproduce missingness, variance structure, or terminal timing, the model has not earned the conclusion. Precision can be false when the model is wrong.

The ROPE and frequentist equivalence band K should share the same scientific rationale. Any difference should be prospectively justified for the estimand and inferential method.

The public statement should therefore read as conditional: under this Bayesian model and these priors, the posterior assigns a specified mass to the declared region. The next sentence must say whether model checks and sensitivity analyses made that conditional statement usable.

24.2.5 — Distributional and Individual Equivalence

A mean can be equivalent while individuals are not. This is the distributional problem. If half a cohort ends with positive residuals and half with negative residuals, the mean may be near zero even though individual streams show strong terminal structure. A mean equivalence result then answers a population question, not a pathwise one (Schuirmann, 1987).

The analysis must distinguish equivalence of mean, median, distribution, tail behavior, variance, and individual path. The Law's public empirical target may include terminal concentration and variance compression, not only first-moment neutrality. A first-moment result is often the least discriminating form of evidence.

Distributional equivalence asks whether the whole distribution lies inside a meaningful relation to the target. Individual equivalence asks whether each or most individuals satisfy a stated band under declared uncertainty. These are stronger and weaker claims in different ways, and neither should be silently substituted for the other.

A concrete example helps. A site may report a near-zero terminal mean residual with wide person-level scatter. That result might weaken the claim of concentration while preserving a question about population symmetry. The report should say exactly that. It should not use the mean as a door into terminal closure language.

Pathwise claims also require dependence structure. It is not enough to list individual terminal values if the observation process excludes the most distressed moments or if endpoint alignment creates apparent narrowing. Individual equivalence must be tied to the window, measurement model, and missingness rules.

The section's practical rule is to print the level of equivalence in the result label. Mean equivalence, distributional equivalence, and individual equivalence are different scientific sentences.

24.2.6 — Rival Victory Under Equivalence

The comparison set should include adaptation, homeostatic mean reversion, conditioned bridges, sedation, selection, and observation shutdown as possible generators of near-zero measured residuals (Lakens, 2017; Lakens et al., 2018).

Equivalence therefore belongs inside rival comparison. When the candidate and a simpler rival both predict the declared region, the

result may validate a measure or favor the rival while leaving PTVC unchanged.

The analysis predeclares what would discriminate, such as equivalence plus reserve-linked variance compression under endpoint-independent stopping rather than mean reversion alone. The discriminating criterion should be evaluated in model recovery before empirical use so an equivalence finding cannot win merely because the pipeline lacks power against the rival.

A simpler rival that explains the result wins honestly. Letting it win strengthens the program more than treating every near-zero finding as hidden support. The report should credit the rival at the same level of specificity used for the candidate, including the reference class and observation bridge under which it succeeded.

Reports state when equivalence was met but failed to discriminate under the shared observation model. Such a result can still validate the equivalence procedure while showing that it has no Law-specific evidentiary force in that setting.

24.3 — Observation Kernels and Measured-to-Latent Bias

An observation kernel describes how a latent history becomes reports, physiology, behavior, neural signals, clinical records, or missing entries. It is part of the scientific model, not a nuisance layer added after a preferred result appears.

Measured-to-latent transport requires assumptions about calibration, invariance, channel validity, context, reactivity, and information loss. When those assumptions support only a set, the result remains set-valued rather than being forced into a point estimate.

Bias transfer receives a formal role. The aim is not to make latent inference impossible, but to state exactly what additional assumption is required before measured equivalence may become a bounded latent claim.

A visually tidy interval is often the most seductive object in inference. A narrow line centered inside a band can look decisive, yet the real question is what the line is centered around. Precision around a biased estimator is precision around the wrong target.

Bias transfer makes the relation between measured and latent quantities operational. It evaluates that relation under the relevant state, site, device, culture, endpoint, and observation conditions, establishing the scope at which an estimate can be interpreted.

Estimating a measured terminal residual identifies the next calibration and construct-validation questions. Each resolved

measurement relation expands the range of quantitative comparisons the program can make.

This is more than a generic disclaimer that measurement is imperfect. The analysis states how imperfection changes the band, how the conclusion widens, and where inference stops when bias cannot be bounded. In Equation 24.3, L_M denotes the measured estimand, L the latent target, and δ a bound on their systematic displacement; $\hat{L}$ estimates L_M and also has random estimation error.

Equation 24.3 (role: bias-transfer bound)

$$|L_M - L| \leq \delta \wedge |L_M| \leq K \Rightarrow |L| \leq K + \delta$$

Equation 24.4 (role: initial-state decomposition)

$$L(T) = L(t_0) + \int_{t_0}^T g_{\theta}(s)F(s)ds$$

Table 24.2. Measured-to-latent transfer statuses.

Transfer status	What is known	Permitted sentence
Validated transfer	The measured interval is valid, and the bias bound and calibration route are defensible for the stated reference class.	Latent claim may be bounded with the inflated band.
Measured only	Estimator is coherent but latent bridge is unvalidated or state-limited.	Measured residual met or failed its criterion.

Transfer status	What is known	Permitted sentence
Unbounded bias	Systematic error cannot be bounded under plausible assumptions.	No latent transfer.
Initial state unbounded	Late window is observed but prior ledger term is unknown.	Late increment or state only.
Coordinate unstable	Origin or sign choices change the conclusion.	Sensitivity-dependent and narrowed.

24.3.1 — Bias Bound δ

A bias bound δ is a declared upper bound on systematic measured-to-latent error for a stated estimand and reference class. It is not the standard error. It is not random noise. It is the maximum systematic displacement the analyst is willing to defend after calibration, validation, device, state, and cultural uncertainties are considered (Carroll et al., 2006).

Justify the bias bound independently of the terminal result, using calibration studies, documented device performance, cross-channel validation, state-specific audits, or a declared conservative expert assessment.

Let L_M denote the measured estimand and L the latent target, with a defensible bound $|L_M - L| \leq \delta$. The estimate $\hat{L}$ also has random estimation error. If $[a, b]$ is a valid confidence interval for L_M , then $[a - \delta, b + \delta]$ is a conservative interval for L under that fixed bound. A measured equivalence result transfers to an inflated latent band through this interval, not by checking the

point estimate alone. Any uncertainty in δ must also be propagated.

For example, a measured estimate of 0.2 with a confidence interval $[-0.1, 0.5]$ and a defensible fixed bias bound $\delta = 0.8$ gives a latent interval $[-0.9, 1.3]$. The measured interval is inside the band $[-1.0, 1.0]$; the latent interval is inside the inflated band $[-1.8, 1.8]$, but not the original band. If the inflated band is too wide for the scientific question, the latent transfer is too weak for support.

A bias bound can also be asymmetric. Some devices or report regimes may systematically understate distress more than relief, especially under sedation, disability, cultural restraint, or care dependence. The analysis should not force a symmetric δ when the mechanism is directional.

The practical rule is that every latent interpretation must show its bias bound. If no defensible δ exists, the conclusion stays with the measured estimator.

24.3.2 — Inflated Latent Band

The inflated latent band is the band that remains after measured equivalence has paid the cost of systematic uncertainty. If the measured equivalence band is K and the bias bound is δ , the simplest conservative latent band becomes K plus δ . The conclusion has moved, and the prose must move with it (Fuller, 1987).

The public ear often hears near zero as exact neutrality. The analysis blocks that slide. A valid confidence interval for the

measured estimand wholly inside a tight band can support a resolution-bounded statement, while the inflated band keeps measurement error visible rather than pretending it disappeared.

The calculation also helps compare studies. A study with a slightly wider measured interval but a well-validated δ may be more informative about the latent object than a study with a narrow interval and unbounded bias. Precision without transfer can be less useful than modest precision with known error.

The inflated band should be reported visually when possible. A figure can show the measured confidence interval, the measured equivalence band, and the latent band after bias expansion. Such a display keeps the reader from treating the neatest interval as the final story.

If the bias-adjusted band is too wide for the intended inference, its width identifies the calibration precision the next measurement study needs to achieve.

In this book, every measured-to-latent sentence should carry the widened band or refuse the transfer. Anything else lets the latent Law borrow the polish of measured statistics.

24.3.3 — Calibration and Transport Error

Calibration error arises when the estimator's relation to lived burden is imperfect in the study setting. Transport error arises when a calibration learned in one setting is carried into another. Both errors matter because the same measured value may mean different things across sites, diagnoses, languages, devices, states, and care regimes.

Palliative-care calibration should account for speech, movement, sleep, autonomic reactivity, medication, family presence, disease trajectory, and clinical documentation. Validation in the intended illness state establishes how an estimator trained elsewhere can be used.

Calibration transport should therefore be treated as part of δ or as a separate uncertainty component. If a device version changes, if a site adopts a new documentation workflow, or if a culture-specific reporting pattern changes the anchor, the measured-to-latent bridge must be reaudited.

The chapter also separates calibration error from site effect. A site effect may reflect true environmental differences in burden. Calibration error reflects a change in how the instrument maps evidence into estimates. Treating both as one random intercept can hide the mechanism that matters.

A useful analysis reports calibration strata. It might show results under within-site calibration, pooled calibration, leave-site-out calibration, and device-version-specific calibration. If conclusions change, the claim narrows to the calibration regime that survived.

The transport rule is conservative: a calibrated estimator travels only as far as its validation evidence travels. The reader should not be asked to imagine a universal scale because the table is clean.

24.3.4 — Unknown Initial Ledger

Late-enrollment studies can estimate terminal residuals, increments, and window dynamics. A whole-course cumulative

analysis additionally requires an independently bounded initial state at enrollment or observation spanning the relevant horizon (Carroll et al., 2006; Fuller, 1987).

Report the observed terminal-window contribution and the initial cumulative term separately. This decomposition makes the contribution of observed and earlier history explicit.

The initial-state problem can be written as a decomposition. The ledger at the endpoint equals the ledger at enrollment plus the interval contribution after enrollment. If the first term is unknown and unbounded, the endpoint conclusion about the whole stream is not identified.

Longitudinal cohorts, intensive life-course designs, qualified retrospective anchors, or well-justified reference-class assumptions may help bound the initial term. The resulting bounds determine which cumulative comparisons are identifiable.

The report should make this explicit. It may say that the design estimates late increments conditional on admission state. It may say that initial-state sensitivity was too wide for whole-stream inference. It may not imply that late neutrality proves life neutrality.

The unknown initial ledger is not a technical footnote. It is the guardrail that keeps terminal science from pretending to see an entire life through its final window.

24.3.5 — Sensitivity to Origin and Sign

Affective measurement has no universally given natural origin and unit. Origin and sign sensitivity therefore enter the inferential

report as empirical dependencies rather than remaining an abstract philosophy problem (Fuller, 1987).

If the neutral origin shifts, a residual can move across the equivalence band. If the sign convention changes for a channel, burden-increasing and restorative contributions can be reversed. If a cultural or clinical anchor is wrong, a measured near-zero result may reflect the coordinate system more than the stream.

A good analysis therefore tests origin sensitivity. It reports how the conclusion changes under plausible neutral anchors, within-person baselines, site-specific anchors, and calibration anchors. It also audits sign conventions for each channel before the composite estimate is allowed to enter the ledger calculation.

Sign sensitivity is especially important in physiological data. Increased arousal may signal distress, effort, excitement, medication response, pain relief, or agitation depending on context. A fixed sign assigned to a nonspecific channel can create false movement toward or away from neutrality.

The chapter does not demand impossible certainty. It demands that coordinate choices be visible. A result that survives plausible origin and sign changes is stronger. A result that changes direction under reasonable alternatives is not defeated automatically, but it loses the right to clean language.

The reporting sentence should say whether the conclusion was invariant, narrowed, changed sign, or became unusable under origin and sign sensitivity. That is how measurement humility becomes statistical design.

24.3.6 — No Transfer Without Validation

Validation connects measured equivalence with the specified latent target through the declared measurement relation (Carroll et al., 2006).

Validation has layers. The estimator must have a coherent construct, stable calibration, known uncertainty, state-appropriate performance, endpoint-compatible use, and a defensible relation to the latent target. A study may satisfy some layers and fail others. The conclusion should match the lowest unresolved layer.

This prevents the science from becoming impressive at the wrong level. A sensor composite may predict reports. A report composite may predict care transitions. A terminal residual estimate may be stable across sites. None of those achievements automatically validates transfer to the latent ledger.

A transfer study can identify improvements needed in calibration, missingness analysis, care-sensitive observation, or model comparison. Those findings define the next qualification task.

The public risk is obvious. If a measured estimator is described as if it reached inner life directly, readers may hear that a number has ranked or read a person. The research language must keep the estimator inside the protocol and leave no worth, care, trust, or spiritual implication behind.

The path forward is typed: measured results support measured claims, latent claims require a qualified measured-to-latent bridge, and an unsupported bridge remains a gap. The language of transfer must match the evidence that licenses it.

24.4 — Informative Sampling, Missingness, and Reactivity

Informative observation arises when whether, when, or how a record appears depends on latent state, burden, treatment, prognosis, communication, device use, or prior observations. Ignoring that dependence can create apparent improvement, stability, or terminal concentration (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Missingness labels are not explanations. The protocol should specify an observation process, denominator flow, plausible causal structure, sensitivity parameters, and what evidence could distinguish competing missingness mechanisms.

Absence is modeled rather than apologized for. The aim is not to recover everything, but to know when recovery is possible, when sensitivity is required, and when the correct output is inconclusive or unusable.

Missingness arrives before the result is trusted because absence may be caused by the states under study. A participant who stops answering because a prompt has become unbearable and a device whose battery fails during a routine afternoon produce different observation processes.

The chapter's working example becomes sharper here. Suppose terminal-window observations come mostly from participants who remained comfortable enough to tolerate devices and prompts. The observed residuals may look calm. But if the unobserved cases were removed by distress, care protection,

sedation, transfer, or endpoint ambiguity, the apparent calm belongs partly to the observation process.

The analysis should therefore report missingness as a time-varying, marked process. The mark might be the measured burden estimate when observation occurs. The event is the fact that observation occurred at all. Treating the event and the mark jointly allows the study to ask whether the evidence window was shaped by burden, staff, device, care, or endpoint state.

A humane point follows without becoming a warning wall. A missing value may reflect good care. The study can honor that choice while still admitting that care-created absence limits inference. Care is not a bias to resent; it is a condition the science must respect and model.

Equation 24.5 (role: marked observation process)

$$\begin{aligned}\mathbb{E}[dN_i(t) \mid H_i(t^-)] &= \lambda_i(t \mid H_i(t^-)) dt, \\ M_i(t) &\sim p(m \mid dN_i(t) = 1, H_i(t^-)).\end{aligned}$$

For a simple marked point process with a locally integrable predictable intensity, the first line is infinitesimal conditional-mean notation and does not assert exact Poisson increments.

Table 24.3. Missingness categories and inferential consequences.

Category	Typical mechanism	Default inference action
Scheduled absence	Low-burden protocol window or planned rest.	Valid for schedule, not for unobserved moments.

Category	Typical mechanism	Default inference action
Participant refusal	Burden, distrust, fatigue, privacy, or preference.	Model separately; respect refusal.
Clinical protection	Care team suspends measurement for comfort or safety.	Record as protection, not random absence.
Device failure	Battery, placement, skin, motion, firmware, signal quality.	Use provenance and version checks.
Care transition	Transfer, new site, new staff, new record regime.	Treat as a process change.
Qualified stream endpoint	Later outcomes are undefined for that stream.	Use truncation logic, not imputation.

24.4.1 — Missingness Taxonomy

A missingness taxonomy begins by refusing the single label missing. Scheduled absence is different from refusal. Device failure is different from low signal. Clinical protection is different from family refusal. Transfer is different from death. Each has a different relation to the target quantity (Rubin, 1976).

The taxonomy should be designed before data collection. A terminal study might distinguish planned rest windows, participant-declined reports, clinician-suspended measurement for comfort, medication-induced inability, sensor detachment, technical failure, site workflow failure, transfer to another care

setting, and endpoint occurrence. These labels matter for inference.

A night nurse who decides not to wake a participant for a report is not producing the same absence as a lost battery. The first may reflect care and clinical judgment. The second reflects device reliability. A model that treats both as random missing values hides the mechanism and may bias the result.

Taxonomy also protects dignity. It lets the study say that data were not collected because measurement would have been intrusive, not because the person disappeared from the evidence map. Ethical protection should be visible in the data record, but not converted into a claim about inner state.

Some categories should automatically limit inference. If missingness follows distress, sedation, or endpoint ambiguity, the analysis must use nonignorable sensitivity or abstain. If missingness is a planned low-burden window, the analysis may be valid for the schedule but not for unobserved moments.

The missingness report disaggregates counts by category, time, site, device, and relation to the endpoint. A single overall percentage conceals the mechanisms that determine recoverability.

24.4.2 — Marked Point-Process Model

A marked point-process model treats observations as events that occur in time, each carrying a measured mark when observation succeeds. The timing of observation is modeled, and the observed value is modeled conditional on the observation event. This

makes the visit process part of the analysis rather than a nuisance outside it (Little & Rubin, 2019).

For terminal and life-course studies, this is often more honest than pretending measurements arrive on a fixed grid. Reports may cluster when a participant feels able to respond. Clinician notes may cluster during crises. Wearable data may vanish during transfers or procedures. The observation process has its own dynamics.

The model asks two linked questions. What determines the intensity of observation at time t ? What distribution of measured marks appears when observation occurs? If the intensity depends on the latent or observed burden state, the resulting data are not a simple sample from the path.

A marked process can also include site and device features. Staffing patterns, software versions, charging schedules, family presence, and care routines may affect observation intensity. These variables belong in the observation model because they decide which parts of the path become visible.

The model is not magic. It can only adjust for mechanisms that are measured, bounded, or plausibly specified. If observation depends on unmeasured distress in a way the data cannot constrain, the analysis must report sensitivity or No adequate estimate.

The advantage is clarity. Instead of saying that missingness was handled, the study states how observation events were generated, what marks they carried, and what assumptions allowed inference beyond observed points.

24.4.3 — Missing at Random as an Assumption

Missing at random is an assumption about the dependence of missingness on observed and unobserved data. Conditional on the specified observed information, it asserts no residual dependence on the missing values. Design and observed-data diagnostics can inform its plausibility, but generally cannot verify that absence of dependence from the observed data alone. The analysis must name the conditioning history and examine departures through sensitivity analysis (Little & Rubin, 2019).

The assumption may be plausible for some device failures and implausible for others. A battery failure might be close to random conditional on device version and staff schedule. A refusal to answer a distress prompt is unlikely to be random with respect to unobserved burden. A clinician-suspended measurement near active dying is not random with respect to endpoint state.

A study should therefore justify missing-at-random claims by category. It should not apply one missingness assumption to all absences. The taxonomy from the previous subsection feeds directly into the model: some categories may be modeled as conditionally ignorable; others demand sensitivity analysis or No adequate estimate.

The assumption also has a temporal dimension. Missingness at noon may depend on measurements from morning, but also on unobserved events after morning. A model that conditions only on baseline covariates may be too thin for longitudinal burden data. The history used for the assumption must be explicit.

Readers should be able to see what would make the assumption false. If unobserved pain, delirium, fear, family conflict, or medication side effects could change both the outcome and the chance of observation after conditioning, missing at random is fragile. That fragility belongs in the conclusion.

Missing at random may be used only as an assumption with visible support and visible alternatives. It is never a default setting in this book.

24.4.4 — Pattern-Mixture and Selection Sensitivity

Pattern-mixture and selection models give different ways to examine nonignorable missingness. A pattern-mixture approach compares outcome distributions across missingness patterns. A selection approach models the probability of being observed as depending on outcomes or latent states. Both expose how conclusions depend on assumptions about absent data (Scharfstein et al., 1999; Rubin, 1976).

In terminal studies, pattern matters. People who stop reporting because they are peaceful, exhausted, sedated, distressed, transferred, or actively dying may have different unobserved trajectories. Treating all observed persons as representative of all enrolled persons may create artificial terminal concentration.

Sensitivity analysis should move missing values in scientifically meaningful directions, such as more negative unobserved terminal reports or greater device dropout during agitation. Its purpose is to test how much unobserved structure would overturn the result, not to invent the truth.

Selection sensitivity asks a related question: how strongly would observation intensity have to depend on unmeasured burden to produce the apparent signature? If a modest dependence is enough to erase equivalence, the result is fragile. If only extreme assumptions overturn it, the result gains conditional strength.

The analysis should not hide sensitivity in supplementary material. In a public-candidate science book, the main chapter must teach readers why the result changes under different missingness stories. A sensitivity curve can be more honest than a single adjusted estimate.

The verdict should name the missingness range under which the claim survives. If survival requires an implausibly convenient missingness model, the result is protected theory, not support.

24.4.5 — Device and Staff Processes

Devices and staff are part of the observation system. They decide when data appear, which channels are usable, which notes are written, and which exceptions are recorded. Treating them as background machinery can turn operational behavior into scientific signal (Rubin, 1976).

A wearable may fail more often during movement, sweating, edema, oxygen changes, agitation, skin fragility, or clinical handling. A staff note may appear more often during crises, medication changes, family distress, or shift transitions. Those processes can create apparent dynamics in the measured ledger even when the underlying path differs.

The analysis should therefore include device version, firmware, placement, adherence, charging, signal quality, staff role, shift,

site, care transition, and documentation workflow where available. These variables are not clutter. They explain why the evidence exists.

A realistic failure case is a terminal study in which signal dropout increases in the last hours because devices are removed for comfort. If remaining signals are calmer, the analysis may infer concentration near neutrality. The actual mechanism may be compassionate device removal plus selective observation.

Staff processes also carry burden. Asking staff to collect data during intense care can create documentation shortcuts, fatigue, and uneven compliance. The study should not punish staff for protecting patients, but it must record that protection so inference does not mistake absence for evidence.

The section's local safeguard is operational: every observation pipeline should have a provenance field. The path from person to number must include the human and technical processes that made the number available.

24.4.6 — When Missingness Defeats Inference

Missingness defeats inference when the unobserved parts of the path could plausibly change the conclusion and the study cannot bound that effect. This is not the same as having many missing values. A small amount of strategically informative missingness can be worse than a larger amount of planned, well-understood absence (Little & Rubin, 2019).

The defeat condition should be stated before analysis. It may involve missingness concentrated near endpoints, missingness associated with high distress markers, missingness tied to

treatment changes, missingness differing sharply across sites, or missingness that removes entire classes of participants from the target window.

When missingness defeats inference, the result should not be rescued by a sophisticated imputation model. Imputation is a model-based assumption about unobserved values. If the assumptions are unconstrained, the imputed conclusion says more about the model than about the stream.

A good report distinguishes data incompleteness from inferential failure. If required protocol and implementation gates passed but endpoint-window observation depended on unmeasured clinical deterioration whose effect could not be validly bounded, report No adequate estimate for the primary target while retaining any valid secondary descriptive summaries. Reserve Unusable evidence for a failed required gate. That is more useful than burying the failure inside an adjusted estimate.

Defeat by missingness can teach the research program. It may show that cadence was too high, participant return was too burdensome, devices were not appropriate, staff processes were underdesigned, or endpoints needed different windows. The correct response is better design, not rhetorical rescue.

The chapter's rule is that absence may narrow, weaken, suspend, or defeat a claim. It may not be converted into support merely because the observed remainder looks clean.

24.5 — Delayed Entry, Truncation, and Competing Events

Delayed entry and truncation separate the study window from the stream's complete domain. A participant observed late supplies only a suffix of the complete history, while follow-up may stop before the truth-level endpoint. Complete-stream inference therefore needs a joint prefix-observed-suffix object on one latent scale.

Competing events such as death, resuscitation, withdrawal, transfer, sedation, device removal, or loss of communication can alter both the scientific target and observation. The analysis must say whether they are part of the endpoint definition, intercurrent events, censoring mechanisms, or causes of a new estimand.

Terminal inference distinguishes survivor-only quantities, composite outcomes, joint longitudinal-event models, and endpoint-definition sensitivity. Independent endpoint adjudication establishes the event used by each analysis.

A concrete record can show why the distinction matters. A participant may have a distress episode on Monday, a care transition on Tuesday, a sedation change on Wednesday, and loss of the open-channel endpoint window on Thursday. Loss of an open observation channel does not establish that the qualifying stream has ended. Friday's report may be missing or unavailable; only an independently qualified terminal event makes a post-terminal outcome undefined for that stream.

The reporting discipline is therefore not merely technical. If the analysis concerns survivor averages, it must say survivor. If it

concerns nonterminal events before a terminal event, it must say before. If it uses a composite outcome that includes death, it must explain what the composite hides and what it reveals. The reader should never have to infer whether the favorable estimate came from all enrolled streams, only surviving streams, or only streams that remained observable under the protocol.

Terminal research distinguishes unobserved values within a continuing stream from quantities that are undefined after its independently established endpoint. Each requires its own statistical treatment.

The semi-competing-risk frame prevents two opposite errors. The first error treats an independently qualified stream endpoint as ordinary censoring and imputes later experience as though it belonged to that stream. The second error discards all event structure and studies only survivors, quietly changing the target population. Both can produce clean tables and misleading conclusions.

A complete analysis keeps terminal and nonterminal events in view together. Medication change may precede the endpoint, delirium may alter observation, and a family interview may occur before the qualifying stream irreversibly ends and also change later reports. Their relation to the terminal event must be modeled rather than flattened into one outcome.

The public-risk boundary is local: no composite, survivor average, or event model may imply that death, comfort, burden, or observation completion are exchangeable moral quantities.

They are statistical components for a defined analysis, not units of a life's worth.

Equation 24.6 (role: ordered pair of semi-competing-risk probabilities)

$$Pr(T^N \leq t, T^D > t), Pr(T^D \leq t)$$

24.5.1 — Truncation Is Not Ordinary Dropout

Ordinary dropout means an outcome might exist but was not observed. Truncation by death means that a specified post-death outcome is undefined. For PTV, a later experiential outcome may be treated as undefined for a stream only after its endpoint has been independently qualified. In Equation 24.6, the first probability is the chance that the nonterminal event has occurred by time t while the terminal event has not; the second is the terminal-event cumulative incidence by time t . These distinctions change what can be estimated, what can be imputed, and what the result is allowed to mean (Aalen et al., 2008).

A pain report after the qualifying stream has irreversibly ended is not a hidden value awaiting imputation; it is undefined for that target. A later wearable signal from the body is not automatically a continuation of the lived stream. The endpoint changes the ontology of the measurement, not merely its availability.

Because the endpoint is central, post-endpoint imputation can manufacture concentration. Carrying forward calm after the stream has ended or dropping participants who die before a later window answers a different question and may introduce survivor bias. Neither operation is ordinary missing-data repair.

The analysis should therefore define post-terminal outcomes explicitly as undefined, use terminal-event models where appropriate, and report which estimands are restricted to survivors. The word survivor should appear in the result if survival is part of the estimand.

Truncation can still permit meaningful inference. The study may estimate time-to-terminal event, preterminal trajectory, terminal-window residuals, or nonterminal events before death. It simply may not fill nonexisting outcomes with numerical convenience.

The local rule is blunt: after the endpoint, do not impute the stream. Model what existed before the endpoint and label what became undefined.

24.5.2 — Terminal and Nonterminal Events

A terminal event ends the target process for the stream. A nonterminal event occurs within the process and may affect later observations or terminal risk. Examples include care transitions, distress episodes, delirium onset, sedation changes, family interviews, or device removal before the endpoint (Andersen et al., 1993; Rizopoulos, 2012).

A semi-competing-risk formulation is appropriate when the nonterminal event can occur before the terminal event, but the terminal event precludes its later occurrence. That asymmetry must be justified for the selected events. An experiential report may become undefined after a qualified stream endpoint, whereas a device removal or administrative record can still occur afterward and needs a different event definition.

For Law-of-Fairness research, the nonterminal event may be scientifically interesting. A medication change, reported dream, agitation episode, or family visit may be related to subsequent measured residuals or endpoint timing. For an event genuinely precluded by the qualified stream endpoint, its post-endpoint occurrence is undefined, not merely censored. Family visits and bodily or administrative care events require their own event definitions because they may continue after a stream endpoint.

The analysis should model terminal and nonterminal events jointly when their dependence matters. A simple cause-specific hazard may condition on survival in a way that changes the target population over time. The estimand must say whether it concerns all enrolled persons, survivors to a time, or principal strata defined by potential survival.

Endpoint independence remains essential. The terminal event must be adjudicated through prospectively specified consciousness or unity rules, not through the ledger residual. Otherwise the event process and the outcome process become algebraically entangled.

The section asks the reader to see time-to-event structure as part of fairness science. If the event structure is wrong, the measured terminal pattern may be an artifact of who remained available to show it.

24.5.3 — Survivor Average Causal Effects

Survivor average causal effects compare outcomes within the principal stratum that would survive under each relevant treatment condition. They can be scientifically useful, but

survival is post-treatment and membership in that stratum is generally unobserved, so identification requires strong assumptions (Frangakis & Rubin, 2002; Zhang & Rubin, 2003).

In a terminal or severe-illness setting, survivor-only comparisons are tempting. Analysts may compare residuals among those who survive to a window because those are the people with observed data. But surviving to that window can be a post-treatment or post-state condition, not a neutral sampling fact.

A survival-conditioned claim must say which survivor population it targets. Is it the observed survivors under realized treatment, the always-survivor principal stratum that would survive under both treatment and control, or a survivor population under a specified care regime? These are different estimands with different assumptions; only the always-survivor contrast is the conventional survivor average causal effect.

When observation is available only for participants stable enough to be measured, report that selected reference class and evaluate how earlier endpoints, sudden deterioration, and distress affect inclusion.

Strong assumptions may be defensible in some designs, especially with randomized interventions, rich covariates, or credible principal-stratum reasoning. But the assumptions should be printed and sensitivity-tested. They should not be hidden under a regression label.

The survivor average rule is that survival-conditioned estimands are allowed only when named. If survival defines the denominator, the conclusion must carry it in plain language.

24.5.4 — Composite Outcomes and Their Costs

Composite outcomes combine several events or measurements into one endpoint. They can be useful when no single outcome captures the scientific question, but they are dangerous when death, comfort, measurement, and residual burden are merged into a favorable score (Tsiatis, 2006; Aalen et al., 2008).

A composite might combine survival, symptom relief, terminal residual, and observation completion. Such a composite may look efficient, but it can hide the meaning of each component. Longer survival with higher distress, shorter survival with better comfort, and no observation because of care protection are not interchangeable units.

For this book, a composite outcome must never turn death or suffering into a point value that appears to balance another component. It may be a technical summary for a stated estimand, but each component must remain visible. The composite cannot be used to suggest that one person's burden has been paid by another value.

Composite outcomes also risk ranking sites or persons. A site with high completion may look better than a site that suspended measurement for comfort. A person with a favorable composite may not have had a better lived experience. The composite is a research object, not a moral object.

The safe design is to report components beside the composite. If the composite result is favorable but components disagree, the conclusion should narrow. If death dominates the composite, the

report should state that clearly rather than presenting the composite as a neutral endpoint.

Composite outcomes cannot rescue an adverse component by averaging it with easier signals. Components, directions, transformations, and weights must be frozen, and the report should show the component-level estimates beside the composite. When one prerequisite fails, a favorable total remains descriptive rather than confirmatory.

24.5.5 — Joint Longitudinal–Event Models

Joint longitudinal-event models connect repeated measures with event processes when both share latent factors. Here they represent dependence among measured burden, observation availability, care transitions, and terminal events (Aalen et al., 2008).

A shared state can link a measured residual to event hazard and reduce bias from informative dropout, but it also exposes the assumptions producing that adjustment.

Clinical, unity, and care variables may inform the event model, with endpoint adjudication specified independently of the tested ledger estimator.

Joint models depend on distributional form, shared effects, and measurement quality. Predictive checks and sensitivity analysis are required because a wrong model can adjust confidently for the wrong reason.

The report names the longitudinal and event submodels, shared dependence, observation process, and endpoint channel, then

shows how conclusions change across plausible dependence structures.

24.5.6 — Competing Endpoint Definitions

Endpoint rules may diverge when consciousness, unity, irreversibility, clinical status, and observation windows do not align. Sensitivity to adjudication therefore belongs in the analysis, with each candidate rule kept independent of the ledger and any alternatives for unity threshold, sustained loss, return, or resuscitation prespecified (Andersen et al., 1993; Rizopoulos, 2012).

Declare the primary endpoint rule and all sensitivity rules before outcome inspection. Report how each rule changes the estimand, equivalence assessment, missingness treatment, and event model; if a conclusion survives only one fragile definition, narrow the claim rather than treating the result as endpoint-robust. Sudden or low-signal endings may support only No adequate estimate or a sensitivity bound.

Endpoint sensitivity also protects participants and families by preventing retrospective claims about ambiguous inner life. Stress-test plausible endpoint rules and preserve disagreement when it remains; never choose or revise an endpoint definition merely to rescue a conclusion.

24.6 — Sensitivity and Partial Identification

Sensitivity analysis asks which conclusion depends on which unverified assumption. It should vary origin, missingness, endpoint, observation, transport, model family, prior, and measured-to-latent bridge within prospectively justified ranges rather than scatter arbitrary alternatives around the preferred estimate.

Partial identification reports the set of target values consistent with the observed data and declared assumptions. The set may be scientifically decisive, unresolved, or too broad for a trustworthy estimate. Its width is information, not a defect to hide.

Robustness is a disciplined map rather than a victory word. Within-dataset robustness can strengthen a result, but replication, transport, and model recovery remain separate tasks owned by later chapters.

A sensitivity analysis should feel like turning the lights on in different corners of the same room. It asks whether the conclusion lives only in one preferred arrangement of priors, windows, missingness assumptions, endpoint rules, and measurement-error bounds. If the result disappears when a plausible lamp moves, the report should say so.

Bayesian workflow is valuable here because it treats modeling as a sequence of choices, checks, revisions, and predictive challenges. But the chapter refuses to let Bayesian language become a softer route to certainty. A posterior distribution is a conditional object. Its authority depends on model adequacy and the assumptions used to build it.

Posterior predictive checks can compare the model with endpoint missingness, site variability, and the terminal residual distribution. Discrepancies identify which parts of the observation or outcome model need refinement.

Robustness should therefore be reported as a map, not a trophy. A result may be robust to priors and fragile to missingness, stable across windows and unstable across measurement-error bounds, or predictive for one site and broken in another. The chapter gives those combinations language so the result does not collapse into a single comforting word.

Equation 24.7 (role: Bayesian companion)

$$p(\theta|y, M) \propto p(y|\theta, M)p(\theta|M)$$

Equation 24.8 (role: posterior predictive check)

$$\mathfrak{y} \sim p(\mathfrak{y}|y, M), \text{ compare } D(\mathfrak{y}) \text{ with } D(y)$$

Table 24.4. Sensitivity grid for Chapter 24 inference.

Sensitivity axis	Example variation	Claim-status effect
Prior	Skeptical, weakly informative, broad, site-informed.	Shows prior dependence.
Model space	Missingness model, endpoint model, site hierarchy, smoother.	Shows structural dependence.
Window	48 hours, 7 days, 30 days, reserve-time clock.	Shows alignment dependence.
Measurement error	Noise floor, bias bound, sign, origin, calibration.	Shows transfer dependence.

Sensitivity axis	Example variation	Claim-status effect
Predictive adequacy	Residual distribution, missingness, site effects, event timing.	Shows model failure or adequacy.

24.6.1 — Prior Sensitivity

A prior encodes assumptions before the data are analyzed. In a high-stakes claim, prior sensitivity is not optional. It shows whether the posterior conclusion is driven mainly by data, by prior structure, or by a combination that should be disclosed (Gelman et al., 2013; Gelman et al., 2020).

For terminal residuals, plausible priors might represent skepticism about large effects, uncertainty about direction, known measurement noise, or prior evidence from earlier cohorts. A prior that effectively forces residuals toward zero would be unsafe unless justified as part of a rival or calibration model rather than as a PTVC-favoring assumption.

A prior sensitivity report should include at least weakly informative, skeptical, and broader alternatives when feasible. It should show whether equivalence, latent-transfer bounds, and missingness sensitivity survive. If only the PTVC-favoring prior yields support, the conclusion is not stable.

Prior sensitivity also belongs to model components. Priors on missingness dependence, calibration error, endpoint uncertainty, site variance, and measurement-error bounds can influence the conclusion. Reporting only the prior on the main residual is too narrow.

The analysis should not treat a skeptical prior as hostility or a favorable prior as confidence. Priors are scientific commitments, not loyalties. The role of the companion analysis is to reveal how much those commitments matter.

Prior sensitivity should display how the posterior decision changes across a defensible, prospectively named family of priors. Selecting the prior that makes the candidate look stable after inspection is not robustness. A conclusion that crosses decision regions across plausible priors is reported as prior-sensitive.

24.6.2 — Model-Space Sensitivity

Model-space sensitivity asks what happens when the analyst changes the family of models rather than only parameter values. A result can be robust within one class and fragile across plausible alternatives (Rosenbaum, 2002; Gelman et al., 2013).

Model-space sensitivity asks how the conclusion changes when plausible observation models, missingness structures, endpoint windows, calibration hierarchies, latent-transfer assumptions, site effects, or nonparametric smoothers enter or leave the analysis. The full generative rival library comes next; the inferential consequence of model uncertainty belongs here.

A common failure is to call the preferred model primary and relegate alternatives to vague checks. A serious plan names which alternatives would change the claim status. For example, if a pattern-mixture model with plausible distress-linked missingness erases equivalence, the primary result becomes conditional and fragile.

Model-space sensitivity is also where elegance can mislead. A simpler model may look crude but reveal that the data cannot support the preferred structure. A flexible model may fit well but absorb the signal into smoothness or random effects. The report should describe what each model is allowed to explain.

The analysis may use model averaging or ensemble summaries when justified, but it should not average incompatible estimands. If models answer different questions, their outputs should be displayed side by side rather than compressed into one number.

The practical outcome is a model-space map: which conclusions survive, which narrow, which reverse, and which become No adequate estimate.

24.6.3 — Window and Alignment Sensitivity

Window and alignment choices can produce apparent terminal signatures. A seven-day window, a forty-eight-hour window, an endpoint-aligned retrospective window, and a prospectively defined low-signal window may show different results. The analysis must test whether the conclusion depends on that choice (Gelman et al., 2013).

This is not a cosmetic issue. In end-of-life data, the last observable window may differ from the last experiential window, the last clinical window, and the endpoint-adjudication window. Aligning everyone to death can create compression by construction if observation fades near the endpoint.

A sensitivity plan should vary window length, alignment clock, inclusion of low-signal intervals, and endpoint adjudication. It should also distinguish calendar time from reserve time, illness

stage, care transition time, and observation time. Each clock can create a different pattern.

The result should be displayed as a window sensitivity surface or table. A signature that appears only under one retrospectively chosen window is weak. A signature that appears across prespecified windows and remains under rival observation models is stronger, though still conditional.

Window sensitivity can also identify design failure. If the necessary cadence was too sparse for the target dynamics, no clever window can recover the path. The report should say that the design could not support the estimand rather than searching for a favorable alignment.

A terminal-window result belongs to the window that produced it. It may travel only through declared sensitivity, not through narrative momentum.

24.6.4 — Measurement-Error Sensitivity

Measurement-error sensitivity asks how conclusions change when noise floor, calibration, sign, origin, and channel reliability vary within plausible ranges. It is the direct companion to measured-to-latent bias transfer (Carroll et al., 2006; Manski, 2003).

The analysis should vary both random error and systematic error. Random error can increase uncertainty and, in nonlinear transformations or models with error-prone predictors, can also induce bias. Systematic error can displace an estimate from its target. Sign error can reverse contribution. Origin error can change whether a residual lies inside K . These errors have

different effects and should not be collapsed into one vague uncertainty term.

A physiological channel illustrates the problem. A wearable-derived arousal estimate may become less reliable during edema, movement, low perfusion, or medication changes. If those conditions occur near the endpoint, measurement error is time-varying and informative. A constant noise model may understate the uncertainty exactly where it matters most.

Measurement-error sensitivity should therefore be channel-specific when possible. Self-report, physiology, clinician notes, family reports, sleep traces, and passive signals each carry different failure modes. A composite estimate is only as trustworthy as its error propagation.

The conclusion should state the largest measurement-error range under which equivalence, defeat, or inconclusive status remains. If a modest noise-floor adjustment changes the claim, the result is fragile. If sign uncertainty makes the estimate uninterpretable, the report should not hide that under a composite score.

Robust exclusion requires a residual set that contains every possibility allowed by the independently justified measurement and bridge assumptions. Fix the admissible configuration family G before inspecting the outcome. Each configuration specifies the origin, representation, exposure, stream routing, endpoint, observation relation and missing-history assumptions. Let $R(g)$ be its compatible terminal residual set, constructed without imposing closure, and let $s(g) > 0$ be an independently specified quantity with residual units. Define

$$\mathcal{R}_{\text{rob}} = \bigcup_{g \in G} \{r/s(g) : r \in R(g)\}, \quad d = \inf_{x \in \mathcal{R}_{\text{rob}}} |x|.$$

For a nonempty union, $d > 0$ excludes zero under every admitted configuration and is invariant to a positive change of units. Positive distance is sufficient, not necessary: the set $(0,1)$ excludes zero while having distance zero. An empty union indicates incompatible assumptions or observations; it cannot provide a truth-containing exclusion. A union containing zero establishes compatibility.

A numerical example shows why unknown history matters. A current residual interval $[8,10]$ and a bound $[-5,4]$ on the entire remaining signed contribution imply the terminal outer interval $[3,14]$. Additional unresolved prefix uncertainty $[-4,4]$ enlarges it to $[-1,18]$, removing the exclusion. These are synthetic units. Bounds on observed late experience cannot silently replace bounds on the complete prefix.

Statistical exclusion requires coverage under every admitted null law. Let $\mathcal{P}_0$ be the declared family of observation and nuisance laws under which the tested stream closes exactly, and let C_α be the projected residual confidence set. The required guarantee is

$$\inf_{P \in \mathcal{P}_0} P\{0 \in C_\alpha\} \geq 1 - \alpha$$

implies

$$\sup_{P \in \mathcal{P}_0} P\{0 \notin C_\alpha\} \leq \alpha.$$

A joint confidence region provides this protection when its coverage is established uniformly over the admitted null configurations. Marginal coverage across a mixture of zero and

nonzero residuals does not establish coverage conditional on the residual being zero. Repeated searches across streams or analysis times require a corresponding simultaneous guarantee.

An observed endpoint can carry its own explicit error budget. Over a fixed finite horizon, suppose a simultaneous interval band covers the complete pre-endpoint ledger with failure probability at most α , and an independently qualified endpoint rule matches the true endpoint and timing convention with failure probability at most β . If the tested branch permits true off-band completion within that horizon with probability at most ϵ , an observed endpoint whose entire nonempty residual interval lies outside the declared band has probability at most $\min\{1, \epsilon + \alpha + \beta\}$. This follows by the union bound and needs no independence between these errors. Timing uncertainty requires taking the union over all admitted endpoint times; missing-prefix uncertainty remains inside the band. The exact probability-one target sets ϵ and its theoretical residual tolerance to zero, while the empirical certificate retains its declared error probabilities.

24.6.5 — Posterior Predictive Checks

Posterior predictive checks ask whether the fitted model can generate data resembling the observed data in the ways that matter for the estimand. They are not a decorative plot. They test whether the model that produced the posterior can reproduce the observed structure it claims to explain (Gelman et al., 2013; Gelman et al., 2020).

Relevant checks include terminal residuals, preterminal variance, missingness, observation intensity, endpoint timing, site

heterogeneity, device dropout, and within-person dependence. A model fitting the mean but missing the observation process is inadequate.

Choose checks before interpreting results. Flexible analysis can always find an acceptable feature. The protocol names discrepancies that weaken or invalidate the model, including failure to reproduce observed endpoint dropout or terminal concentration.

Posterior predictive checks can also protect against false elegance. A model with a beautiful latent curve may generate data that do not resemble actual reports, device gaps, or event timing. The graph then serves the reader better than the curve.

The check should not be judged by visual comfort alone. Whenever possible, use discrepancy measures tied to the estimand and rival predictions. The graph should teach the model's failure or adequacy, not decorate the chapter.

24.6.6 — Robustness Is Not Replication

Robustness means a conclusion survives variations within a dataset or workflow; replication means it appears in new data under governed conditions. A result can withstand many modeling choices in one biased dataset yet fail elsewhere (Rosenbaum, 2002; Gelman et al., 2013).

Within-dataset robustness evaluates sensitivity to priors, windows, calibration, and missingness assumptions. Replication extends the test to new sites, devices, cultures, diagnoses, or endpoint regimes.

A robustness summary should identify the dataset, measured residuals, analyses, and assumptions that produced stable conclusions. Readers can then assess exactly which features were examined.

Correlated choices can manufacture robustness. Analyses sharing a missingness assumption, calibration, or endpoint alignment may look diverse while protecting one weakness. The sensitivity grid should identify those dependencies.

A strong report states what robustness covers and leaves untouched: for example, that measured equivalence was robust to prior and window variation but not nonignorable missingness. This is more trustworthy than a broad claim.

A replication needs a clear account of what is being repeated. The protocol should identify the target quantity, the claimed signature, and the conditions under which the original explanation should still apply. Changes in recruitment, measurement, or endpoint adjudication belong in that account. Readers can then assess whether a disagreement challenges the original result or reflects a difference between the questions the studies actually addressed.

New participants do not make every source of error independent. Two sites may inherit the same device calibration, scoring code, or stopping convention, allowing a shared artifact to recur. A useful replication therefore examines which vulnerabilities remain shared and introduces independent checks where feasible. For a terminal signature, this might include endpoint adjudication that does not depend on the measurements used to estimate the signature.

When studies disagree, the size and precision of their estimates determine what the disagreement means. A precise result incompatible with the original effect poses a different problem from a wide interval that leaves both the effect and its absence plausible. Differences in measurement or participant selection may suggest a narrower explanation, which then requires its own prospective test. The disagreement becomes useful when it identifies what evidence could resolve it.

24.7 — Sequential Evidence and Optional Stopping

Sequential evidence permits learning while data accumulate, but only under an explicit process. E-values, e-processes, alpha-spending, group-sequential plans, or Bayesian monitoring retain their guarantee for the exact protocol-fixed null, filtration, stopping permissions, and model assumptions to which they were attached (O’Brien & Fleming, 1979; Pocock, 1977; Ramdas et al., 2022).

Optional stopping is not one behavior. A planned stopping rule inside a valid sequential design differs from unrecorded peeking, repeated model revision, selective endpoint choice, or publication after a favorable crossing. The audit must distinguish them.

Anytime-valid tools can help when their conditions are met. They do not remove the need for estimands, missingness models, endpoint firewalls, or human review. They only provide a mathematical route for error control under declared sampling rules.

Sequential evidence is attractive because the program will learn over time. New cohorts, sites, endpoint adjudications, and device versions may arrive gradually. The scientific community will want to know whether the evidence is moving. The danger is that repeated looks can turn patience into a false positive machine.

Anytime-valid methods are therefore useful only if they are declared before the first look and matched to the sampling scheme. An e-process built for independent site releases cannot silently govern adaptive recruitment, delayed endpoints, public-

pressure pauses, or selective data cleaning. The mathematics protects only the process it actually describes.

Sequential updates are deliberately limited to components. A running evidence number may update a measured equivalence claim, a missingness gate, or an observation-model check. It does not update the Law as a public doctrine, authorize applied books, or decide transition language. Component evidence must remain component evidence.

A sequential dashboard can make a conditional statistic look like a verdict. Interim graphics therefore remain inside research governance until claim-status review, public-risk review, and adverse-reporting rules translate the result responsibly.

A public-candidate science program should also declare what happens when sequential validity breaks. If a site adds measurements after seeing early results, if a device version changes midstream, if families decline further observation in a patterned way, or if an endpoint window is revised after interim analysis, the evidence process may no longer carry its promised guarantee. The result may still be useful, but it has to be relabeled.

The chapter therefore treats every interim look as a protocol event. The log should record who looked, what was visible, which analysis was run, whether the threshold was fixed, whether the observation process changed, and what public language was allowed afterward. This is not paperwork for its own sake. It is the memory that prevents a research program from forgetting how it persuaded itself.

Sequential learning also sharpens the difference between program governance and public meaning. Investigators may need to decide whether to expand a study, pause a site, fix a measurement process, or commission a replication. Those are research decisions. They are not social verdicts, spiritual messages, legal instructions, or transition signals.

The strongest use of sequential evidence is often conservative. It can stop a weak claim earlier, reveal that the observation process is unstable, or show that a favorable component depends on one site. A threshold that is never crossed may still save the project from a premature public story. Learning early that the evidence is insufficient is not a failure of science; it is one of science's duties.

Equation 24.9 (role: e-process expectation condition)

$$E_t \geq 0, E_0[E_\tau] \leq 1 \text{ for allowed } \tau$$

Equation 24.10 (role: anytime-valid error control)

$$P_0 \left(\sup_t E_t \geq 1/\alpha \right) \leq \alpha$$

24.7.1 — E-Process Definition

An e-process is a nonnegative evidence process constructed so that, under every distribution in the protocol-fixed null class, its expectation remains controlled at each allowed bounded or stopping time, subject to the stated conditions. It permits evidence to accumulate without treating every look as a fresh uncorrected test (Howard et al., 2021).

State the e-process guarantee for its exact null hypothesis, filtration, and allowed stopping times. New estimands or transport claims require a corresponding construction and justification.

The evidence-process construction, including any permitted adaptation rule, must be specified before the data it evaluates are reviewed. Predictable updates may use past observations to choose future bets when the declared null and filtration guarantee remains valid. Selecting favorable transformations of already observed evidence does not establish an e-process.

E-processes can be especially useful for long programs that add sites gradually. The program may continue recruiting until the evidence crosses a threshold, provided the process is valid for the dependence, sampling, and updating scheme actually used. That validity is the central issue.

Attach the evidence value to its tested component and declared null. Human reviewers use that result in the governed claim-status decision.

An e-process must be constructed so that its expectation is controlled under the protocol-fixed null at every allowed stopping time. The filtration, betting strategy, update rule, and decision boundary therefore belong in the protocol. Optional continuation remains statistically valid at stopping times covered by that guarantee, including data-dependent rules not fixed in advance. Protocol requirements still apply, and optional stopping does not license retrospective changes to the null, evidence process, or error threshold.

24.7.2 — Allowed Stopping Times

A stopping rule is compatible with an e-process when it uses the declared filtration and lies within the process's stopping-time guarantee. An unplanned look or a decision influenced by public

pressure does not, by itself, invalidate an anytime-valid e-process. Retrospective changes to the null, evidence construction, threshold, or data-selection mechanism can invalidate the guarantee. Statistical validity and compliance with the study's governance rules should be audited separately (Ramdas et al., 2022; Howard et al., 2021).

In a rolling research program, stopping may occur because a threshold is crossed, a safety issue appears, a site fails, funding pauses, or endpoint adjudication problems accumulate. Not all stopping rules have the same inferential meaning. Some preserve validity; others require the result to be relabeled.

The protocol should specify allowed looks and allowed stopping reasons. It should also specify what happens when stopping occurs for ethical or operational reasons unrelated to the evidence threshold. Care-based stopping is legitimate, but it may produce No adequate estimate or a conditional estimate rather than a clean sequential verdict.

A hidden stopping problem can arise when public attention affects the analysis. If interim results are informally discussed, thresholds moved, or new analyses added after favorable signs appear, the mathematical process no longer carries its original warrant.

Allowed stopping also depends on missingness and dependence. If site releases are delayed because data look unfavorable or because measurement failed in specific states, the sequence is not a neutral stream of looks. The evidence process must match that adaptive sampling.

Stopping is not only a time on a calendar. It is part of the statistical design, the ethical design, and the public claim-status design.

24.7.3 — Threshold and Error Control

Sequential thresholds translate an evidence process into an error-control rule. A common form is a threshold such as one over alpha: crossing it can justify rejection of the specified null with controlled type I error under the process assumptions (Howard et al., 2021).

The threshold must be declared before data arrive. Moving the threshold after public excitement, donor pressure, or early favorable signals destroys the integrity of the result. A threshold is a promise about how difficult the claim will be to make.

The error rate attaches to the tested null, not to every sentence that might be written afterward. If the process rejects a null about a measured residual in one reference class, it does not reject rival models, latent transfer uncertainty, cross-species scope, or the strong form. The conclusion must stay inside the mathematical event.

Non-crossing also requires a declared interpretation. Depending on the process and design, it may be inconclusive, may count against the alternative, or may show only that information remains insufficient. It cannot be translated into convenient ambiguity after the result.

Thresholds can be too lenient or too severe for the public stakes. A publish-candidate science program should choose thresholds with prospectively specified justification, safety review, and public communication review. The aim is durable error control.

Print the threshold, alpha, stopping plan, and component claim in the same reporting block so the decision remains interpretable.

24.7.4 — Dependence and Adaptive Data

Sequential validity depends on the actual data-generating and sampling scheme. Person-level observations are dependent. Sites may share training, devices, protocols, and culture. Interim results may change future recruitment, measurement, and analysis. These dependencies matter (Howard et al., 2021; Ramdas et al., 2022).

A process built for independent increments cannot be trusted if the next site is selected because the previous site looked favorable. A process built for fixed observation windows may fail if windows are extended after early patterns appear. A process built for stable devices may fail after version changes.

The analysis plan should therefore state dependence assumptions. It should say whether looks are by participant, by site, by endpoint-window batch, by device version, or by calendar release. It should say how clustering, site correlation, adaptive recruitment, and delayed endpoints are handled.

Adaptive data are not always illegitimate. A study may adapt ethically to reduce participant burden, pause a high-burden channel, or oversample an underrepresented group. But adaptation must be modeled or separated from confirmatory inference. Otherwise care improvements and evidence claims become entangled.

Dependence can also arise through public communication. If public preliminary claims alter participant behavior, site recruitment, or clinician documentation, the evidence stream has

changed. That risk belongs in governance, but it has statistical consequences here.

Adaptive and dependent data require validity under the actual filtration, not an unspoken independence approximation. Repeated measures, site clustering, adaptive enrollment, and data-dependent actions must enter the e-process or sequential design explicitly. When that guarantee is unavailable, the sequential number is descriptive rather than decision-valid.

24.7.5 — Evidence for Components, Not Canon

Sequential evidence can update a component claim. It cannot update the canon as a whole by itself. This distinction protects the series from treating a mathematical crossing as public validation of the Law, religious doctrine, civic design, legal architecture, or transition readiness (Ramdas et al., 2022).

A component might be a measured equivalence result in a terminal-window estimator, a missingness sensitivity threshold, a site-calibration success, or an observation-model check. Each component has a defined null, reference class, and inference route. The canon has many more gates.

The report should say exactly what changed. A sequential analysis may move a component from inconclusive to supported, from supported to fragile, or from plausible to defeated. It may not say that the Law has been scientifically validated unless the broader verdict system later determines that status.

This protects skeptics and supporters alike. Skeptics can see that evidence is not being exaggerated. Supporters can understand that progress is real even when narrow. A narrow component update

may be scientifically valuable without becoming a public proclamation.

The section also prevents dashboard pressure. A running number can excite people because it feels like the theory is moving toward victory. Readers should be trained to ask which component the number concerns and which gates remain closed.

Sequential evidence updates the component whose null and process were specified in the protocol. The public record identifies that component, the resulting evidence, and the prerequisites for its next use.

24.7.6 — No Dashboard Deciding Status

A dashboard can make evidence look more authoritative than the evidence is. A line rising over time, a green threshold, or a public status badge can compress estimand, missingness, endpoint, bias, sensitivity, and review into a single visual cue. That compression is dangerous for this book (Howard et al., 2021).

The chapter does not forbid internal monitoring. Researchers may need dashboards for data quality, safety, missingness, accrual, calibration, and sequential evidence. The rule is that dashboards are operational tools, not public verdict engines.

A public dashboard would be especially risky if it displayed a running probability, evidence ratio, or equivalence status without the assumptions beside it. Readers might interpret the number as proof, moral meaning, or permission to apply the Law in ordinary life. That would violate the science book's authority boundary.

Human review must remain above the number. A committee must examine endpoint independence, data quality, missingness, measurement burden, adverse findings, public-risk wording, and rival implications before any status language changes. The dashboard can alert; it cannot decide.

This rule also protects participants and sites. No person, family, clinician, site, or culture should be placed under public suspicion because a data-quality or evidence metric looks unfavorable. Research metrics stay inside the research governance architecture.

The section ends with a design rule: sequential graphics may be beautiful, but their captions must make them humble. A running number is a witness to a component, not a judge of the Law.

24.8 — Open-Set and Protected-Theory Risk

Model ambiguity occurs when a valid protected comparison leaves models indistinguishable at the declared resolution, uncertainty crosses decision boundaries, or defensible analyses favor different models. A pipeline that misses its required recovery thresholds instead remains unqualified for the affected confirmatory comparison.

Open-set inadequacy is different: every named model may be wrong or unable to reproduce the data-generating features that matter. The correct result is Open-set inadequate or No adequate estimate, not victory for the least bad candidate.

Protected-theory risk appears when flexibility is asymmetric. If PTVC can change endpoint, representation, route, or interpretation after each failure while rivals remain frozen, the comparison cannot generate adverse evidence. Change control must therefore apply to the preferred model at least as strictly as to its competitors.

The final section gives the program a vocabulary for not forcing a result. This is harder than it sounds because every ambitious theory creates pressure to show progress. A dataset with endpoint problems, missingness failure, unbounded bias, or flexible analysis can still be made to produce a number. The chapter asks whether that number deserves to exist.

Use the canonical evidence-state definitions in §24.2 and Table 27.5. In this section, the added question is whether open-set inadequacy or analytic flexibility prevents the model space from supporting a stronger verdict.

Public summaries should use the study's recorded status and explain the next decision. An unresolved endpoint, wide interval, or missingness problem can be reported clearly as a specific research question.

Chapter 25 extends the inference workflow through simulated worlds, model recovery, and adversarial tests. Those exercises assess whether the analysis recognizes the conditions under which its conclusions hold and identifies the changes needed when they do not.

Table 24.5. Verdict labels for inference under uncertainty.

Label	Use when	Public-language rule
Supported component	Estimand, band, bias, missingness, endpoint, and sensitivity gates pass, and the prespecified evidential criterion is met for the defined component.	Name the component and reference class.
Inconclusive	Valid analysis does not resolve the estimand.	No trend language that sounds supportive.
Unusable	Protocol, consent, endpoint, measurement, or provenance gate failed.	Do not report as evidence for the claim.
No adequate estimate	Target is undefined or cannot be bounded.	Say No adequate estimate rather than force a number.

Label	Use when	Public-language rule
Narrowing	Only a smaller reference class or weaker claim survives.	Update scope explicitly.
Defeat pressure	A valid adverse result or rival-sensitive failure remains.	Report the pressure and route to Chapter 27 verdict language.

24.8.1 — Inconclusive Evidence

Inconclusive evidence means the analysis was valid enough to run but did not decide the target question. The interval may be too wide, the signatures mixed, the sensitivity range too large, or the rival comparison nondiscriminating. Inconclusive is not failure; it is a defined scientific state (Popper, 1959).

A terminal-window equivalence test can be inconclusive when the confidence interval crosses the equivalence band. A Bayesian companion can be inconclusive when posterior mass depends strongly on plausible priors. A missingness analysis can be inconclusive when sensitivity spans both support and defeat.

An inconclusive report states that the valid analysis did not resolve the named estimand at the declared precision or decision threshold, and identifies what additional information could resolve it.

Inconclusive evidence can still guide design. It may show that cadence needs improvement, bias bounds are too wide, endpoint windows are poorly aligned, or the reference class is too heterogeneous. Those findings identify concrete improvements for the next study.

A clear label protects readers from overinterpreting ambiguity. It also protects critics from being told that lack of defeat counts as favor. The claim remains where it was before the analysis, perhaps with better knowledge of what must be improved.

Evidence remains inconclusive when a valid uncertainty set overlaps both the protocol-fixed acceptance region and its complement. The report identifies the dominant uncertainty and which additional observation, if any, could change the status rather than choosing the side with the more attractive point estimate.

24.8.2 — Unusable Evidence

Unusable evidence identifies a failed condition required for the affected analysis, such as consent, endpoint adjudication, measurement integrity, calibration, missingness documentation, protocol adherence, provenance, or participant protection. The record names that condition and the analyses it prevents (Lakatos, 1978; Stanford, 2006).

A dataset can be unusable even if it is large. If endpoint rules were changed after results were seen, if key missingness categories were not recorded, if devices changed without version tracking, or if participants were observed under an unduly burdensome protocol that violated design limits, the analysis may not be allowed to speak.

Unusable status should be declared at the level of the affected estimand. A site may be unusable for latent transfer but usable for device-quality lessons. A terminal window may be unusable for

equivalence but usable for missingness-rate reporting. Precision in failure preserves what can still be learned.

This label also protects people. If data were collected under consent ambiguity or excessive burden, the response is not to squeeze a result from them. The research-use firewall requires that ethically compromised evidence not be redeemed by scientific convenience.

Unusable evidence should trigger repair, not concealment. The report should state the failure condition, affected estimands, corrective action, and whether future use is prohibited, restricted, or possible under new governance.

No result is too valuable to be unusable. If the gate failed, the chapter refuses the number.

24.8.3 — Researcher Degrees of Freedom

Researcher degrees of freedom are the choices that can be varied during analysis: windows, exclusions, transformations, priors, covariates, endpoint definitions, missingness models, calibration rules, and outcome definitions. These choices are unavoidable, but they become dangerous when they are flexible after results are visible (Stanford, 2006).

For the Law, flexible windows and endpoint definitions are especially risky. A researcher can align to an endpoint, choose a preterminal window, exclude low-signal cases, adjust missingness, and transform the residual until the result looks orderly. The analysis may feel scientific while the theory is protected by choice.

The answer is prospective specification, version control, blinded analysis where possible, and transparent multiverse reporting. Not every choice can be fixed in advance, but exploratory choices must be labeled exploratory and kept out of confirmatory claim status.

A degrees-of-freedom audit should list every choice that could plausibly change the result. It should report which were fixed before data, which were adapted under blinded rules, which were exploratory, and which were triggered by observed outcomes. The reader should see the analytic path.

This audit also helps distinguish creativity from rescue. Exploratory analysis can generate new hypotheses and better designs. It becomes protected theory when exploratory findings are presented as if they were prospectively specified tests.

24.8.4 — Statistical Elegance as Rescue

Statistical elegance can rescue a theory from visible failure by making the rescue look sophisticated. A latent model with many layers, a flexible prior, a smoothing process, a hidden-state structure, or a complex missingness correction may preserve a favored conclusion while adding little predictive or discriminating value (Popper, 1959).

The chapter does not reject complexity. Some questions here require complex models. It rejects complexity that enters only after simpler analyses become unfavorable and then fails to improve prediction, model recovery, calibration, or rival discrimination.

An elegant rescue often has a signature: the model explains away adverse cases, expands uncertainty selectively, moves the endpoint window, or shifts the latent scale while keeping the final sentence favorable. The output may be polished, but the theory has become harder to lose.

The antidote is a burden of improvement. A more complex analysis must state what it explains that the simpler analysis could not, what predictions improve, what assumptions it adds, and what result would make it fail. If those answers are absent, the model should not upgrade claim status.

This is where humility becomes technical. The analyst should be willing to say that the elegant model is exploratory, that it suggests a new study, or that it did not rescue the primary result. The chapter makes that sentence honorable.

Statistical sophistication cannot repair target, observation, or identification failure. A more elaborate estimator earns a role only when it answers the same estimand, improves calibrated performance, and survives the protocol-fixed diagnostics. Otherwise elegance changes the machinery without strengthening the inference.

24.8.5 — No-Estimate Route

Apply §24.2's No adequate estimate rule and Table 27.5's evidence states. In protected-theory analysis, prespecify the abstention branch in the analysis plan, results template, and public-status map so nonidentification or a failed required gate cannot be repaired into a favorable result after inspection.

24.8.6 — Transparent Adverse Reporting

Transparent adverse reporting means publishing the assumptions that failed, the analyses that contradicted the preferred direction, the sensitivity ranges that weakened the result, and the rival interpretations that remained viable. It is the opposite of burying bad news inside supplemental uncertainty (Lakatos, 1978; Stanford, 2006). Each adverse finding should be tied to the exact model, analysis version, reference class, and decision it changes, so later investigators can distinguish a failed claim from a failed implementation.

Adverse reporting includes failed equivalence, nonsignificant but inconclusive results, unbounded transfer bias, missingness defeat, endpoint sensitivity failure, posterior predictive failure, and protected-theory audit findings. Each remains visible enough for an outside statistician to reproduce the status decision. The record should also identify the affected component, the diagnostic or threshold that triggered the adverse status, and the downstream claims that lose permission as a result.

Adverse reporting protects the entire series. If the science book hides adverse findings, applied books will inherit false confidence. If it publishes them clearly, applied domains can remain conditional and honest. Scientific loss becomes a form of public care. This propagation discipline also lets an applied domain preserve practices supported on independent grounds while withdrawing only the scientific authority that actually changed.

The report should not dramatize defeat. It should state what failed, what remains learned, what claim status changes, and what future design would be required. A sober adverse report is stronger than a defensive favorable report. The most useful adverse report also ends with a decision consequence, such as replicate, redesign, narrow, suspend, or retire, rather than leaving the reader with a rhetorical verdict.

Transparent adverse reporting also gives critics a real place in the program. A critic who can point to published failed assumptions, rival-sensitive results, and No adequate estimate labels sees a theory that can be challenged. That trust is more valuable than temporary rhetorical victory. Criticism becomes scientifically productive when the record shows not only that an objection was heard, but which test, revision, or status change followed from it.

The chapter closes with analyses that connect conclusions to explicit targets, assumptions, and evidence. Clear results strengthen the next scientific decision, whether they support a model or identify a better direction. The handoff should identify which findings are stable enough to carry forward and which unresolved dependency should determine the next study.

Chapter 25 — Model Recovery and Adversarial Tests

Chapter 25 treats model recovery as a release gate. Before an empirical result can support, narrow, or defeat a component, the full pipeline must recover known generators under realistic observation, missingness, endpoint uncertainty, heterogeneity, and misspecification (Talts et al., 2018; Wilson & Collins, 2019; Schumacher et al., 2025; Säilynoja et al., 2026).

Parameter recovery asks whether the pipeline estimates known parameters. Model recovery asks whether it identifies which generator produced the observations. Calibration asks whether uncertainty statements have their promised behavior. Absolute-fit and open-set checks ask whether any listed model is adequate. These are related but noninterchangeable gates.

Adversarial testing adds a social and procedural layer. Predictions, implementation choices, scoring rules, and theory-discriminating observations are fixed before the protected module. Teams with different commitments share the design and confront the same outcome. The Cogitate collaboration illustrates both the promise and the practical difficulty of that method in consciousness science (Cogitate Consortium et al., 2025).

The benchmark library includes ordinary regulation, learning, conditioning, selection, observation, hidden-state, endpoint, and PTVC-candidate worlds, along with near misses and none-of-the-models generators. This breadth tests the pipeline across realistic and demanding alternatives.

Design separability determines whether the planned sample, cadence, channels, endpoint, and perturbations can distinguish the declared models before participant burden is incurred. A no-study

decision is a successful design outcome when the planned observations cannot buy an identifiable comparison.

Transport stress asks whether recovery and calibration survive changes in site, device, population, prevalence, language, care practice, missingness, and endpoint process. A model that works only in its development environment earns correspondingly narrow status.

Reproducible artifacts connect a published claim to exact code, generator versions, environments, thresholds, and data-access layers. Openness is purpose-limited: synthetic fixtures and public code support criticism without exposing participants or creating a new surveillance surface.

The resulting inference system can refuse false confidence. It can return recovered, nonseparable, Open-set inadequate, Model-ambiguous, No adequate estimate, Route-failed, or Rival-favored outcomes. Each status names a different next step.

Part VII closes by turning criticism into durable infrastructure. A theory exposed to recovery, open-set, transport, and red-team failure becomes easier to correct and more informative under every outcome.

Model comparison can look settled because omission rigged the contest. Excluding the strongest bridge, introducing missingness only after latent paths are generated, reusing development sites for evaluation, or omitting a none-of-the-models branch can manufacture an answer. A credible benchmark records the machinery of loss before it records support.

The previous chapter fixed the inferential objects. This one asks whether executable synthetic worlds and the complete fitting pipeline can recover them. The focus is simulation, calibration, predictive comparison, design separability, adversarial testing, and artifact governance rather than another summary of equivalence or missingness theory.

The computational work serves people. A failed recovery study can prevent an uninformative and burdensome human protocol. A detected endpoint leak can prevent a misleading public figure. A published none-of-the-models result can keep the program from becoming a machine that always produces a preferred story.

The required sequence is now visible: release the benchmark, fit every model to every generator, test separability, admit open-set failure, stress transport, record challenges, and preserve executable provenance. Only then does a confirmatory human study have a defensible analytic foundation.

25.1 — Why Model Recovery Comes First

Model recovery comes first because a pipeline can appear sophisticated while systematically confusing the generators it is meant to compare. A favorable empirical score is uninterpretable when the same analysis would label adaptation, selection, or observation-floor worlds as PTVC candidates (Wilson & Collins, 2019; Schumacher et al., 2025).

The recovery study uses synthetic worlds with known latent states, parameters, observation kernels, endpoints, missingness, and model labels. Analysts receive only the observation layer planned for the empirical study. The hidden answer is revealed after the pipeline produces its verdict.

Parameter recovery, model recovery, calibration, and predictive adequacy are reported separately. A model may be classified correctly while parameters are biased, or parameters may be recovered while the model family is nonseparable. No single accuracy number can replace the full diagnostic.

Equation 25.1, Generative benchmark declaration.

$$G_j: \theta_j \sim p_j(\theta), x \sim p_j(x|\theta_j), y \sim p_j(y|x, \theta_j, \psi)$$

Role label: generator specification. Here G_j is a benchmark world, θ_j are generator parameters, x is the latent trajectory, y is the observed record, and ψ holds observation-system parameters.

Table 25.1. Minimum recovery metrics for every generator and fitted model.

Metric	What is reported	Why it matters
Parameter bias	mean or median recovery error	flags estimates that drift toward the favored model
Interval coverage	coverage of declared credible or confidence intervals	detects overconfidence and undercoverage
Model recovery	selected or weighted fitted model by generator	shows confusion and false classification
False equivalence	equivalence declared although the generating estimand lies outside the specified band	prevents synthetic support from null or artifact worlds
False defeat	target rejected when generator includes it	prevents overzealous rejection under noise
Calibration	probability and interval behavior by state/site/time	finds model confidence that fails where it matters
Convergence and time	diagnostic failures and compute cost	keeps unusable methods from becoming impressive prose

25.1.1 — Parameter Recovery

Parameter recovery is the simplest honest question: when the simulator sets a value, can the analysis estimate it back within declared uncertainty? If the reserve decay rate, OU mean-reversion coefficient, bridge endpoint variance, observation-

shutdown slope, or frailty effect is known inside the synthetic world, the fitted model must recover the value often enough to trust its intervals (Wilson & Collins, 2019).

The method is operational. Draw parameters from realistic distributions, generate latent paths, generate observations with missingness and measurement error, fit the planned model, and compare recovered posteriors or estimates to the known inputs. Bias, coverage, shrinkage, tail behavior, and convergence failures are recorded by generator, site condition, device quality, and endpoint uncertainty.

A concrete example is the additive noise floor. Suppose the simulator sets the true terminal compression exponent and adds a physiological sensor noise floor. If the fitting model underestimates that floor, it may report a shallower exponent, a false lack of compression, or a false equivalence failure. The recovery table must show where the exponent becomes unrecoverable.

Parameter recovery can also expose a Law-friendly bias. If the reserve-coupled candidate recovers its own parameters well but the same fitting machinery misestimates OU or bridge parameters in a way that makes them look PTVC-like, the benchmark has not cleared the gate. The pipeline is learning the preferred story rather than the generating values.

The report should state recovery thresholds before the benchmark is run. A parameter may be considered recoverable only if bias is small relative to the scientific margin, interval coverage is near

the declared level, diagnostics pass, and failures do not concentrate in the terminal window where the claim is sharpest.

The Law of Fairness lens asks where estimation error lands. If a parameter cannot be recovered, the burden should stay with the method and should not be shifted to a participant, a hospice site, a family report, or a public conclusion.

The practical consequence is a table entry, not a mood. For every run, the report should state whether the item passed, failed, confused two families, triggered abstention, or forced a redesign. That public trace is what prevents technical uncertainty from being shifted onto later readers as if it were settled evidence.

The same entry should include a protected counterfactual: what conclusion would a PTVC-favoring analyst be tempted to write if this check were omitted? Naming that temptation helps reviewers see why the check belongs in the main benchmark rather than in a supplemental afterthought.

25.1.2 — Model Recovery

Model recovery asks a harder question: when one generator creates the data, does the model-comparison system identify that generator, or at least refuse a wrong confident label? Parameter recovery within a favored model is not enough. The fitted-model matrix must include every planned analysis model fit to every generating family (Wilson & Collins, 2019).

The benchmark therefore constructs a generating-model by fitted-model grid. Reserve candidates, OU models, bridges, frailty worlds, observation shutdown, sedation pathways, endpoint selection, predictive/RL rivals, flexible baselines, hybrids, and

open-set misspecifications all generate data. Each analysis model receives the same observations and the same rules.

The useful output is not a single accuracy number. It is a confusion matrix, proper-score table, calibration curve, posterior model probability behavior, convergence-failure record, and false-equivalence or false-defeat rate. The benchmark must reveal which models masquerade as which others under which observation conditions.

If a random-terminal-time Brownian bridge is frequently classified as the reserve-coupled candidate when endpoint uncertainty is high, the benchmark identifies that region of nonseparability. The next design can test which observations would resolve it.

Model recovery should also allow ties and abstention. If reserve and bridge models are inseparable under a specific sampling cadence, forcing one of them to win is a mistake. Confusion is a result, and the result may be redesign, narrower claim status, or a requirement for a new channel.

A serious project does not ask critics to accept a comparison space after the fact. The comparison space is published with the benchmark, attacked before release, and revised visibly when a critic supplies a better rival.

Independent reviewers should have access to the relevant seed, assumptions, scoring rule, diagnostics, and residual plots at the appropriate release tier. Those materials make the comparison reproducible and available for further testing.

The gray zone deserves special protection. A small score difference, a visually persuasive curve, or a clean figure cannot override failed recovery, untested observation assumptions, or unstable open-set diagnostics. The benchmark must make that uncertainty visible.

25.1.3 — Confusion Is a Result

A model-recovery benchmark treats confusion as information about design and identifiability. When two candidate worlds produce indistinguishable observations under the planned design, the report identifies that nonseparability and the measurements or conditions needed to resolve it (Wilson & Collins, 2019).

Confusion has structure. It may occur only at high noise floors, only near endpoints, only under sedation, only in one diagnosis group, only with low cadence, only when the endpoint time is estimated retrospectively, or only when device removal aligns with terminal decline. Mapping that structure is more useful than reporting a triumphant average recovery rate.

The benchmark should draw confusion regions. A methodologist should be able to see where a PTVC candidate is confused with a bridge, where an OU model is confused with adaptation, where frailty generates apparent concentration, and where open-set worlds are wrongly forced into known categories.

This rule protects the chapter from a subtle overclaim. The Law does not win because the fitted model has the highest score by a narrow margin in a nonseparable region. It wins a local comparison only where the design can separate the candidates and the recovery experiment shows that the score behaves reliably.

The reporting language should be simple: in this design region, the models are recoverable; in that region, they are confused; outside both, the model set may be inadequate. That sentence is more scientifically valuable than a confident ranking that ignores the landscape.

The humane value is equally simple. Confusion prevents human stories from being recruited into false clarity. If a person's data cannot answer the question, the chapter refuses to make that data carry the conclusion.

This item also needs an adversarial version in which the assumptions are almost, but not quite, satisfied. Near-miss worlds are where weak pipelines become persuasive, because they produce graphs that look familiar while the generating facts are different. The benchmark should preserve those near misses as named fixtures rather than discarding them as inconvenient edge cases.

The recovery report should ask which verdict would have been issued if confusion had been hidden. Forced selection of the least-bad model is the central risk when fitted families are empirically indistinguishable. Showing that counterfactual makes abstention a working safeguard rather than a ceremonial option (Wilson & Collins, 2019; Talts et al., 2018).

25.1.4 — Recovery Across Noise and Missingness

Clean synthetic data establish an initial recovery baseline. The benchmark then varies noise floor, cadence, endpoint uncertainty, device dropout, channel-specific missingness, staff observation gaps, medication effects, and site-level drift. Posterior simulation-

based calibration adds a useful conditional check by asking whether the intended inference and implementation behave correctly in the neighborhood of a particular observed dataset, rather than only across broad prior simulations (Säilynoja et al., 2026). Recovery across the intended observation conditions establishes the pipeline’s readiness for the corresponding empirical design.

The noise and missingness grid should be declared in advance. Additive sensor floors, multiplicative device error, bounded self-report uncertainty, missing-not-at-random channels, terminal observation shutdown, device removal, sudden endpoint, and care transitions should appear as controllable stressors. Each stressor must have plausible parameter ranges and extreme challenge cases.

The joint additive-noise-floor by fitting-window sweep is especially important. The benchmark should vary noise-floor estimation error together with terminal fitting windows and report where model misclassification appears. If deconvolution becomes unstable or yields invalid negative variance estimates, that failure is part of the result.

Missingness should be generated, not sprinkled. A hospice device may fail because perfusion is poor. A report may be absent because sedation begins. A site may stop measurements because staff correctly prioritize comfort. These patterns can manufacture compression or hide it, so the simulator must let them interact with latent and observed states.

The analysis passes only if recovery remains stable across declared operational realities or if instability is acknowledged as a boundary. A fragile pipeline can still be useful for a narrow study, but it cannot support broad verdict language.

This rule keeps data dignity concrete. The benchmark minimizes the chance that ordinary care interruptions, device tolerance, or participant withdrawal will be misread as metaphysical evidence.

The benchmark record should preserve the simulated generator, noise law, missingness pattern, fitted candidates, protected holdout unit, diagnostic threshold, and failure disposition. An outside modeler should be able to reproduce where recovery broke and why the pipeline was stopped.

Recovery under noise and missingness has a specific gray zone: wider uncertainty can look like family confusion, and biased observation can look like precise recovery. The report should separate parameter bias, interval coverage, abstention, and generator classification across missingness regimes. A single average accuracy cannot tell reviewers which failure occurred (Wilson & Collins, 2019; Talts et al., 2018).

25.1.5 — Recovery Before Human Data Verdicts

Recovery must occur before human data verdicts because real-data ambiguity is too easy to rationalize after the fact. Once an observed study produces an attractive graph, analysts may unconsciously tune windows, priors, exclusions, transformations, and captions around the pattern. Synthetic recovery creates a pre-result obstacle to that drift (Wilson & Collins, 2019).

No confirmatory human claim proceeds until the synthetic challenge passes protocol-fixed criteria. Passing requires acceptable parameter recovery, model recovery, calibration, absolute fit, protected prediction, transport stress, and none-of-the-models detection across realistic near-miss worlds.

Threshold selection belongs inside nested resampling or a held-out synthetic challenge. The same simulations should not choose the decision threshold and then certify the model. Tuning, preprocessing, feature selection, endpoint window choice, and margin selection must be separated from final evaluation.

A concrete failure case would be a benchmark where the Law-candidate threshold was chosen on the same simulation suite used to announce high recovery. The final result would look strong but would be circular. The audit should treat that as leakage, not as optimism.

The chapter's practical standard is blunt: a real-data report that cannot cite a passed recovery benchmark remains exploratory or descriptive, even if the observed pattern is beautiful. Public language must follow that status.

A theory that wants the world to take it seriously should welcome this delay. It is better to postpone a claim than to let human data become an interpretive shortcut around untested machinery.

Before any human-data interpretation, an independent team should be able to regenerate the synthetic worlds from archived seeds, execute the frozen fitting pipeline, and reproduce parameter coverage, confusion, calibration, and abstention. Protected empirical data are not required for this gate. Failure to

reproduce the benchmark stops the analysis before participant burden is converted into a scientific claim (Wilson & Collins, 2019; Talts et al., 2018).

The release note should record what a failed check would change: redesign, a narrowed estimand, expansion of the rival envelope, No adequate estimate status, or withdrawal of public language. A check with no consequence is ornament and a source of false confidence.

25.1.6 — No Simulation as Scientific Validation

Simulation evaluates computational and statistical machinery under declared assumptions. By specifying a data-generating process, researchers can test whether an analysis recovers known features and quantifies uncertainty under those simulated conditions. These controlled checks prepare the computational pipeline for empirical comparison (Talts et al., 2018).

An executable reserve-coupled model supplies a candidate generator for examining terminal concentration and comparing proposed mechanisms.

Simulation-based calibration can expose computational error, inconsistent implementations, or posterior miscalibration under an assumed joint distribution. Posterior predictive checks test fit to selected features.

Simulation success, ethical review, and demonstrated design separability together inform the next empirical step, including whether recruitment and confirmatory comparison are justified.

The safeguard is operational: each output carries a role label such as computational check, recovery check, design-separability check, rival-envelope check, or artifact-release check. The label prevents a persuasive figure from outrunning its warrant.

The next section builds the serious generator library required to make this recovery gate more than a performance for one favored model.

Every result also records the artificial generator, fitted models, holdout design, recovery metric, allowed inference, and a status field separating benchmark calibration from evidence about nature.

Later reports should cite the executable benchmark result, not a sentence of assurance. Trust comes from leaving enough machinery for a skeptical reader to rerun the case and disagree precisely.

Minimum recovery bars are fixed for every reported quantity. They may include bias below the scientific margin and near-nominal coverage for a slope, low false support for a model label, or stable recovery across the noise, missingness, and endpoint conditions expected in the human protocol.

Recovery reports include computation time, convergence failures, discarded fits, and resource demands. A model that succeeds only after hidden divergence or unrealistic compute provides weaker practical evidence than a stable pipeline.

Chronology is decisive: the benchmark is completed before the emotionally powerful human result appears, because later tuning is otherwise pulled toward the terminal graph. Freezing

benchmark performance before human data also makes later deviations from the pipeline visible as new analyses rather than invisible repairs.

That frozen benchmark also resists narrative momentum. It preserves the worlds where the method would be wrong, undecided, or defeated by a rival before one resonant interpretation begins to dominate. Those adverse worlds should remain in the benchmark after success, because deleting hard cases would gradually convert recovery into self-confirmation.

Failed recovery is a publishable result. If the analysis cannot separate a reserve-coupled candidate from a random-endpoint bridge, it has shown that the proposed human study would ask vulnerable data to carry a question the method cannot yet support. Design should change before recruitment, not after public interpretation. A redesigned study should show improved recovery on the exact confusion that blocked the earlier design before participants are asked to contribute new data.

25.2 — Generative Benchmark Library

A benchmark library is a versioned collection of synthetic worlds with explicit generators, parameter distributions, observation kernels, endpoint processes, missingness, perturbations, and expected diagnostics. Its job is to make every model's failure modes reproducible before empirical interpretation.

The library should include adaptation, opponent, allostatic, OU, predictive, active-inference, reinforcement-learning, safe-control, bridge, selection, frailty, treatment, observation-floor, and PTVC-candidate families. Hybrid and open-set generators prevent the classifier from assuming that the named menu is complete.

The minimum library enables a fair test without pretending to be final. Critics may extend it when new entrants are independently motivated, executable, and capable of making fresh predictions before the protected evaluation data are seen.

Table 25.2. Generative benchmark library. Each family is a data-generating world, not a rhetorical category.

Generator family	Core mechanism	Benchmark function
Reserve-coupled candidate	declared reserve and signed reference dynamics	tests PTVC-candidate signatures without treating construction as evidence
OU/adaptation/opponent	local regulation, rebound, baseline drift	lets ordinary regulation win when it explains the data
Bridge and event-conditioned	fixed/random endpoints, future-event conditioning	tests endpoint concentration without PTVC-specific mechanism

Generator family	Core mechanism	Benchmark function
Frailty/medication/observation	illness, sedation, device removal, staff cadence	tests clinical and operational artifact pathways
Hybrid and null	mixed components or no structured effect	detects false support and false defeat
Flexible reference	nonparametric or machine-learning prediction	sets predictive benchmark without interpretive authority

Equation 25.2, Full fitted-model matrix.

$$R_{ij} = \text{Score} \left(M_i, y_{G_j}^{\text{test}} \mid y_{G_j}^{\text{train}} \right), \quad i = 1, \dots, I, \quad j = 1, \dots, J$$

Role label: model-recovery diagnostic. Every fitted model M_i is evaluated on protected data generated by every generator G_j ; the matrix exposes recovery, confusion, and false certainty. Here I is the number of fitted models and J is the number of generators.

25.2.1 — Reserve-Coupled Candidate

The reserve-coupled candidate is the benchmark representation of the Law’s empirical mechanism, not the Law itself. It implements a latent ledger-like process whose fluctuation scale and relaxation may depend on remaining reserve. Its role is to generate consequence patterns such as preterminal variance compression and endpoint concentration under explicit assumptions.

The generator should include the finite-variation reference case and approved stochastic extensions only when the product dynamics are rebuilt correctly. If reserve is stochastic, jumps, shares shocks, or changes filtration, the benchmark must use the general product rule rather than quoting the finite-variation shortcut.

Parameters include reserve trajectory, diffusion scale, drift or relaxation structure, endpoint timing rule, initial state, measurement error, observation intensity, and site/device effects. The benchmark should declare which parameters are scientific, which are nuisance, and which belong only to the observation model.

The generator description identifies its construction rules. Endpoint adjudication is defined independently of the tested ledger outcome. Conservative-field closure is labeled as a separate construction, and a closing controller states which paths its policy permits. Those details make comparisons with the received Law and empirical processes interpretable.

Recovery of a reserve candidate shows that the analysis recognizes that generator in the specified synthetic conditions. Captions, tables, and public summaries should identify those conditions and the recovery result.

This candidate enters the library as one contestant among serious rivals. Precise implementation makes both favorable and adverse comparisons interpretable.

The reserve-coupled benchmark entry should preserve the generator equations, parameter ranges, route-state definition, observation and missingness kernels, endpoint rule, seeds, and expected failure modes. It should also state which outputs count as target recovery, route recovery, and mere resemblance. That record makes later revisions and rival comparisons possible without reconstructing the candidate from memory (Talts et al., 2018; Wilson & Collins, 2019).

For the reserve-coupled candidate, the omitted-check counterfactual is a smooth endpoint curve being called reserve evidence without recovery of reserve, route state, or the joint restriction. The benchmark should demonstrate that its recovery gates actually block that conclusion. A safeguard that never changes the selected verdict has not earned a place in the pipeline (Talts et al., 2018; Wilson & Collins, 2019).

25.2.2 — OU, Adaptation, and Opponent Models

Ornstein-Uhlenbeck, adaptation, and opponent-process models represent mainstream local regulation. They can pull observed states toward a baseline, generate rebound after shocks, damp variance, and produce apparent repair without any pathwise terminal closure. They must be implemented strongly enough to win (Wilson & Collins, 2019).

An OU generator needs a baseline, mean-reversion rate, diffusion scale, and possible time-varying or state-dependent parameters. Adaptation generators need shock response, habituation, set-point drift, and context dependence. Opponent models need primary-process and counterprocess dynamics with timescales that can create local inversion or rebound.

The library should include versions where parameters change with illness, treatment, sleep, fatigue, culture, and observation regime. A fixed, stationary OU model is a useful teaching rival but not the whole rival family. Ordinary biology and behavior are more flexible than a single textbook equation.

These rivals can mimic terminal quiet when shocks diminish, reports become less variable, staff interventions stabilize distress, or surviving windows select calmer states. If they produce the

observed signatures with fewer assumptions, the PTVC candidate narrows or loses that comparison.

The benchmark should record when adaptation explains endpoints by local mechanisms and when it fails to explain pathwise constraints or reserve-linked signatures. The comparison should be specific, not a vague contest between ‘the Law’ and ‘adaptation’.

Well-developed alternatives make model comparison more informative and any replicated advantage more credible. Their performance identifies which explanatory features the next study should examine.

For the OU, adaptation, and opponent-process library, the trace should preserve equations, parameter ranges, initial conditions, event schedules, observation errors, and recovery outcomes. Reproduction must show which local-regulation family generated each dataset and when the analysis confused one family with another. That confusion matrix is part of the result, not a disposable diagnostic.

The OU, adaptation, and opponent benchmark should record what the candidate would have won against if those rivals had been left weak or absent. The relevant false conclusion is uniqueness produced by an impoverished rival library. Replaying the comparison with expert-strength rivals reveals whether the safeguard changes model ordering (Talts et al., 2018; Wilson & Collins, 2019).

25.2.3 — Bridge and Event-Conditioned Models

Bridge models and event-conditioned worlds are dangerous because they can create endpoint concentration by conditioning on the future rather than by reserve-coupled dynamics. A Brownian bridge fixed to an endpoint, a random-endpoint bridge, or a Doob transform can make paths look as if they were drawn toward closure.

The library should include fixed endpoint bridges, random endpoint bridges, censored endpoint bridges, event-conditioned diffusions, and survival-conditioned processes. Each must state how the terminal event is defined, whether the endpoint is observed with error, and whether conditioning is physical, statistical, or selection-based.

A subtle artifact appears when analysts align all data to death or last observation. If the last observed point is selected by a process related to the measured signal, the aligned trajectories can appear to narrow even when no PTVC mechanism operates. The bridge family should therefore connect to the observation rival envelope.

Comparisons with bridge models should use the same observation process and examine additional signatures, such as reserve-linked contraction and independently defined endpoints. Joint predictions make the proposed explanations more distinguishable.

The benchmark should include bridge parameters that make the rival hard, not a strawman. Endpoint uncertainty, variable horizons, and partial observations belong in the generator, because real terminal research will not supply perfectly fixed endpoints.

If bridge and reserve candidates remain confused after reasonable design improvements, the report should say so. That result may force narrower empirical language or new measurements rather than a rhetorical win.

This is also where participant dignity enters the computational layer. The study should not ask a person, family, staff member, or site to carry observation burden for a design that has already failed in synthetic conditions. Recovery discipline is a way of refusing decorative data collection before it reaches the bedside.

Reports on bridge and event-conditioned models should cite the exact executable artifact, version, conditioning event, endpoint law, and protected result. A prose statement that the check was performed is not enough. The cited object should let an outside team regenerate the conditioned paths and verify the diagnostic claimed in the text (Talts et al., 2018; Wilson & Collins, 2019).

A second benchmark makes the cross-window restriction explicit. Let Y be the vector of signed contributions from n prespecified disjoint windows, with reference distribution $N(\mu, \Sigma)$, where Σ is positive definite. For the Gaussian reference law conditioned on $\mathbf{1}^\top Y = 0$,

$$\mu_c = \mu - \frac{\Sigma \mathbf{1} (\mathbf{1}^\top \mu)}{\mathbf{1}^\top \Sigma \mathbf{1}}, \quad \Sigma_c = \Sigma - \frac{\Sigma \mathbf{1} \mathbf{1}^\top \Sigma}{\mathbf{1}^\top \Sigma \mathbf{1}}.$$

Consequently $\mathbf{1}^\top \mu_c = 0$ and $\Sigma_c \mathbf{1} = 0$. For $\Sigma = \sigma^2 I$, each off-diagonal covariance is $-\sigma^2/n$ and each marginal variance is $\sigma^2(1 - 1/n)$. This is a coupled mean-and-covariance prediction. Independent, zero-mean additive measurement error with covariance R preserves the conditional mean and changes the

observed covariance to $\Sigma_c + R$. The benchmark therefore uses an external neutral origin, prespecified disjoint windows, and an independently calibrated observation model. The reference mean and covariance are estimated independently of the confirmatory closure fit, with any systematic measurement bias receiving its own calibrated correction. These choices leave the total free to be estimated from the observations. By contrast, subtracting each trajectory's own mean forces its observed sum to zero, and selecting trajectories by their totals directly changes the distribution being studied.

This Gaussian route also supplies explicit nonzero-total comparisons. Conditioning on a fixed nonzero total yields the same covariance restriction while changing the conditional mean. When prefix variance is positive, a slope of -1 between latent prefix and suffix follows from any fixed terminal total. The comparison must therefore join mean location, covariance, complete coverage, and independently identified endpoints, with fixed-nonzero and diffuse-total models included. An episode-level application tests a separately specified local-closure hypothesis.

An analytic check makes the distinction concrete. Use four dimensionless window contributions with independent standard-normal reference values and independent measurement errors of variance 0.04 per window. Conditioning the latent total to zero gives each window mean 0; conditioning it to 2 gives each window mean 0.5. Both fixed-total models have latent marginal variance 0.75 and off-diagonal covariance -0.25 . Measurement error changes the marginal variance to 0.79 while leaving the off-

diagonal covariance -0.25 . The observed total therefore has variance 0.16 in both fixed-total models. Without terminal conditioning, its variance is 4.16 .

Thus covariance contraction separates these fixed-total constructions from the unconditioned reference but cannot distinguish zero total from total 2 . Their mean locations provide that additional contrast only when the neutral origin and measurement bias are independently anchored. These are analytic consequences of the stated Gaussian construction, not observations of experience, a completed model-recovery study, or evidence for PTVC. They provide exact numerical targets for checking benchmark code.

25.2.4 — Frailty, Medication, and Observation Models

Frailty, medication, and observation models generate terminal patterns through illness and care rather than through the proposed Law mechanism. They are not afterthoughts. In end-of-life data, they are candidate sources of apparent trajectory change that the benchmark must test (Aalen et al., 2008; Hernán & Robins, 2020).

A frailty generator may include latent health decline, diagnosis-specific trajectories, site-level care practice, metabolic reserve, symptom volatility, and mortality risk. Medication models may include sedation, analgesia, anxiolytics, anticholinergics, delirium risk, dose changes, and report-capacity effects. Observation models may include staff cadence, device tolerance, caregiver reporting, and channel shutdown.

A hospice example is enough to show the risk. A participant's wrist sensor is removed after swelling and skin fragility increase.

Self-reports become sparse after sedation is adjusted for comfort. The final record looks calmer and less variable, but the mechanism may be care and observation, not latent closure.

These generators should allow shared drivers. Disease severity may move reserve proxies, report availability, medication intensity, and endpoint timing together. If the benchmark treats these as independent nuisances, it will understate the ability of ordinary clinical pathways to mimic the target signatures.

The analysis passes only if it can separate Law-candidate signatures from these care and observation paths or state that it cannot. A humane research program should prefer that humility to a graph that rewards clinical invisibility.

This library member is also an ethical control. It prevents comfort-first care from being misread as evidence manufactured by the theory, and it keeps clinical decisions from becoming tokens in a scientific argument.

Frailty, medication, and observation generators need a trace that joins clinical progression, treatment timing, competing events, device or report availability, and public wording consequences. An outside team should be able to recreate the terminal pattern without invoking PTV. The benchmark earns trust by retaining those ordinary explanations in executable form.

The clinical rival audit should reconstruct the conclusion that would appear if frailty, medication, sedation, and observation shutdown were omitted. An apparent terminal contraction may then be seen to arise from care and visibility rather than the latent route. The comparison is useful only when inclusion of those

processes can reverse or block the tempting result (Aalen et al., 2008; Hernán & Robins, 2020).

25.2.5 — Hybrid and Null Worlds

Hybrid and null worlds prevent the benchmark from becoming a forced-choice contest among tidy theories. Real data may contain adaptation plus frailty, bridge-like conditioning plus observation shutdown, reserve-like dynamics plus medication effects, or no structured terminal pattern at all. The library must let messy worlds exist (Wilson & Collins, 2019).

Hybrid generators combine components with declared weights and interactions. A reserve candidate may be mixed with sedation and site effects; an OU process with endpoint selection; a frailty model with device adherence. The purpose is bounded complexity that tests whether mixed evidence is over-attributed to the favored model.

Null worlds are equally important. A null world may include noisy measurements, irregular observation, and endpoints, but no meaningful terminal concentration beyond artifacts. If the analysis repeatedly finds support in null worlds, its false-support rate shows that the decision rule needs revision.

Hybrid and null worlds also support false-equivalence and false-defeat checks. False equivalence occurs when the procedure declares the specified finite-band claim although the generating estimand lies outside that band. A nonzero generating value inside the band is not a false equivalence. False defeat means rejection of the particular claim that is true in the generated world. Report both rates against the exact target and decision rule used.

The benchmark should report these rates by generator family, observation regime, and decision threshold. The public should not hear a single accuracy number when the error type matters morally and scientifically.

Hybrid and null worlds are a discipline of modesty. They let the field say that a result is mixed, artifact-dominated, or uninterpretable rather than forcing it into a clean story.

Hybrid and null worlds should be reproducible from a public generator specification that exposes mixture structure, component strengths, open-set labels, and observation noise while protecting no human data. A critic should be able to add a near-miss hybrid and rerun the entire model matrix. The benchmark remains incomplete when only the preferred team's mixtures are executable (Talts et al., 2018; Wilson & Collins, 2019).

Hybrid and null worlds create near ties for different reasons. A hybrid may be partly represented by several candidates, whereas a null world may expose shared misspecification across all of them. The benchmark should report component recovery, residual structure, and open-set diagnostics so that a small score difference is not mistaken for adequate mechanistic identification (Talts et al., 2018; Wilson & Collins, 2019).

25.2.6 — Versioned Assumption Registry

A generative library without a versioned assumption registry invites invisible drift. One analyst changes a prior range, another changes dropout, a third changes endpoint uncertainty, and the benchmark becomes a moving target. The registry makes every simulator assumption visible (Talts et al., 2018).

Each generator should have a release identifier, mathematical specification, parameter distributions, units, default values, observation rules, endpoint rules, nuisance structures, known nonidentifiability regions, and a changelog. When an assumption changes, the registry records who changed it, why, and which results must be rerun.

Access-tier rules for public versus sensitive benchmark materials are specified in §25.8.5. The assumption registry records which tier governs each object.

A versioned registry also protects critics. A critic can reproduce the benchmark version that produced a claim, supply a rival generator, and show whether the conclusion survives. Without a stable version, critique becomes chasing smoke.

The registry should not be used to bury complexity outside the book. Essential generator families and decision rules appear in the chapter. The artifact holds executable detail, not hidden scientific meaning.

Once the library exists, the next question is whether a study design can separate its members. A rich library is useless if the planned observations make all worlds look the same.

The versioned registry should identify which favorable conclusion becomes possible when an assumption is changed silently after results. Examples include widening a route envelope, redefining the endpoint window, or replacing a null generator. A machine-readable diff makes that counterfactual detectable and prevents revision history from disappearing into polished prose (Nosek et al., 2015; Munafò et al., 2017; Nüst et al., 2020).

The ordinary rivals should be no less carefully built. OU mean reversion, hedonic adaptation, opponent rebound, predictive control, safe control, frailty, medication, and endpoint conditioning each have different mechanisms. The library should not collapse them into one vague rival category simply because they are not the Law.

The library should be curated as a scientific commons rather than a private arsenal. Each generator needs a readable name, a mathematical declaration, a parameter file, a plain-language mechanism statement, a list of protected signatures it can mimic, and a list of assumptions that would make it unfairly strong or unfairly weak. This lets a critic argue about the right object instead of reverse-engineering the author's intent.

A good library will sometimes reveal mechanisms its builders did not anticipate. It may show that an observation-gain model reproduces a signature once sensor removal is represented, or that a frailty mixture resembles late convergence after retrospective endpoint alignment. These findings show that the benchmark is doing useful work: it identifies explanations the next analysis must distinguish.

25.3 — Design Separability and Power

Design separability asks whether the distribution of planned observations differs enough across declared generators for a reliable decision. It is evaluated in simulation before recruitment using the same analysis, uncertainty, and consequence rules planned for the empirical study.

Power is branch-specific. A design can have high power for a mean difference and almost no ability to distinguish bridge conditioning from reserve-coupled dynamics, or adaptation from informative observation. The simulation target must match the scientific decision.

The chapter uses expected log predictive discrimination, discriminating-power matrices, open-set checks, and burden-adjusted information to decide whether a design deserves human data. The goal is not maximal measurement. It is sufficient discriminating information with the least ethical and practical cost.

25.3.1 — Expected Information by Design

Expected information by design evaluates how much a proposed sampling schedule, channel set, and endpoint rule are expected to teach before the study begins. The object is not a generic sample size. It is the predicted separation among specified generators under the planned observation system (Gneiting & Raftery, 2007).

One useful quantity is expected log predictive discrimination. For two generating families, the benchmark estimates the expected held-out log-score gap under a design. If the gap is small relative

to simulation uncertainty, the design may be unable to distinguish those families even with many participants.

Equation 25.3 states the role of the quantity. It is a design diagnostic, not a verdict about nature. It compares how fitted models predict protected synthetic data when one generator is known to be true.

Equation 25.3, Expected log predictive discrimination.

$$D_{ab}(d) = E_{G_{a,d}}[\log p_a(y^{test}|y^{train}, d) - \log p_b(y^{test}|y^{train}, d)]$$

Role label: design diagnostic. The quantity compares rival predictive performance under design d before human recruitment; small or unstable values indicate nonseparability.

A distribution-level bound complements fitted-model recovery. Let P and Q be a specified candidate and rival law for the complete latent record. Apply the same observation kernel K , including cadence, noise, missingness, endpoint classification, and recorded selection. For a measurable test $0 \leq \varphi \leq 1$ with $\mathbb{E}_{QK}\varphi \leq \alpha$,

$$\mathbb{E}_{PK}\varphi \leq \min\{1, \alpha + \text{TV}(PK, QK)\} \leq \min\{1, \alpha + \sqrt{D_{\text{KL}}(PK\|QK)/2}\}.$$

$$D_{\text{KL}}(PK\|QK) \leq D_{\text{KL}}(P\|Q), \quad \text{TV}(PK, QK) \leq \text{TV}(P, Q).$$

Total variation bounds the difference in rejection probabilities; Pinsker's inequality and information contraction give the remaining bounds, using natural logarithms (Cover & Thomas, 2006). A joint observable-record divergence below $2(\pi - \alpha)^2$ rules out target power $\pi > \alpha$ for that pair at size α . This is a necessary design condition, not a sufficient power calculation.

Nominal reports cannot be counted as independent information, and an estimated divergence needs its own uncertainty.

Conditioning on survivors or complete reports changes the experiment. Analyze the joint selection-and-outcome record, or derive the selected laws and retain their selection probabilities; conditioning can increase separation, so unconditional contraction cannot simply be reused. For composite families, one difficult admitted pair already limits uniform performance. The practical objective is a design that separates scientifically meaningful rivals through its qualified observations.

A design example makes the point. Hourly active reports may separate a fast opponent process from an OU model, but may be unethical or burdensome in terminal settings. Passive physiology may reduce burden but fail to distinguish valence from arousal. A mixed design can be preferred only if it gains discrimination without exceeding participant and staff limits.

Exactness has a calculable observation limit. In a simple calibrated model, observe $Y = L + e$, where e is centered Gaussian error independent of L , with known variance $\sigma^2 > 0$ and an independently fixed zero origin. Compare $L = 0$ with a continuous rival L distributed as $N(0, \eta^2)$. The observable variances are σ^2 and $\sigma^2 + \eta^2$. With $r = \eta^2/\sigma^2$, their divergence is

$$D_{\text{KL}}(P_\eta \| P_0) = 1/2 \{r - \log(1 + r)\} \sim \frac{\eta^4}{4\sigma^4}.$$

For n independent records, joint divergence is n times this quantity. The observation-space power bound therefore rules out

uniform separation of exact zero from every arbitrarily concentrated continuous rival by a fixed finite design. A useful design instead specifies the departures it must resolve and calculates a resolution curve that includes its full calibration and observation uncertainty.

Error calibration can be the decisive resource. If σ^2 is unknown, an exact residual with error variance σ^2 and a nonzero residual of variance η^2 with error variance $\sigma^2 - \eta^2$ produce exactly the same Gaussian observation law whenever $0 < \eta^2 < \sigma^2$. More outcomes from that ambiguous channel do not identify which explanation generated them. Independent reference measurements or qualified replicate channels must constrain the error law, including whether its gain, origin and variance transport to the actual observations. Study planning should allocate effort to that qualification as well as to outcome collection.

Population identification remains a different question. For known independent additive error with a nowhere-zero characteristic function, division of the observed characteristic function by the error characteristic function identifies the latent distribution. This does not guarantee stable recovery from a finite sample. Unknown origin, missing prefix and uncertain endpoint require their own identification work even when the noise law is known.

Expected information should be reported with uncertainty. Monte Carlo error, simulator uncertainty, prior range, endpoint uncertainty, and site variation all affect the expected value. Pass decisions should reflect variation across simulation seeds.

The Law of Fairness lens asks whether the information gain is worth the load it imposes. If the study adds burden without separability, the burden has been shifted onto participants for a result the design could not earn.

Expected-information calculations should be reproducible from the proposed sampling schedule, channel costs, generator library, priors or design weights, and utility or loss definition. A reviewer should be able to alter cadence or burden constraints and recover the resulting design decision. This is how a no-study conclusion becomes inspectable rather than discretionary (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

The design analysis should state what human study would have proceeded if expected information had not been calculated. The danger is recruitment into a burdensome protocol whose cadence and channels cannot separate the models even at large sample size. Demonstrating that the information check can trigger redesign or no-study status gives the safeguard practical force (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

25.3.2 — Discriminating-Power Matrix

The discriminating-power matrix maps model families against signatures. It asks which signature separates which rival under which observation regime. A signature that looks impressive in prose may test the same distinction as another signature or may fail against the rival that matters most (Wilson & Collins, 2019).

Rows should include reserve candidates, OU, adaptation, opponent dynamics, bridge families, frailty, observation shutdown, sedation, predictive/RL rivals, flexible baselines, and open-set worlds. Columns should include terminal concentration,

preterminal variance compression, reserve-linked contraction, endpoint independence, held-out prediction, calibration, sign stability, and response to perturbation when available.

The matrix should display expected discrimination, not rhetorical confidence. A blank cell means the signature does not separate the pair. A weak cell means the evidence is likely to be ambiguous. A strong cell means the design offers a plausible route to distinguishing the models under declared assumptions.

The matrix prevents a common failure: piling up correlated signatures that all collapse against the same bridge rival. Five signatures that cannot distinguish bridge conditioning from reserve coupling do not become strong by being counted five times.

Use the matrix before recruitment. Record each generator-signature contrast, design conditions, uncertainty across simulations, and redesign/no-study threshold; leave blank cells and nonconvergence visible. If no planned signature separates the PTVC candidate from the strongest rival envelope under ethical observation, classify the study as exploratory or methodological, or redesign it, rather than calling it confirmatory (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Table 25.3. Design separability decision rules.

Decision	Benchmark condition	Consequence
Proceed as confirmatory	high discrimination, stable calibration, acceptable burden	benchmark passes and ethics permits recruitment

Decision	Benchmark condition	Consequence
Proceed as exploratory	some informative contrasts, key rivals unresolved	public language stays exploratory
Redesign	critical rivals confused or open-set detection weak	change cadence, channels, endpoint, or model set
Do not recruit for this question	nonseparable under ethical observation	avoid spending human burden on decorative data

25.3.3 — Redundant Signatures

Redundant signatures are not harmless. They can make a chapter look richly supported while measuring one distinction repeatedly. In the model-recovery setting, redundancy can inflate apparent evidence, hide nonseparability, and increase participant burden without increasing knowledge (Wilson & Collins, 2019).

The benchmark should compute or qualitatively map dependency among signatures. If terminal concentration, final-window variance compression, and last-observation calm all respond to the same observation-shutdown artifact, they cannot be counted as independent support against that artifact.

Redundancy audits should also include measurement channels. Two wearable metrics may share a sensor failure mode. A clinician report and a family report may share cultural or hospice-script influence. A self-report and a behavioral measure may both depend on sedation. Apparent multimodality may be one confounded pathway.

The audit should model dependence among signatures and assess whether each adds information for the declared contrast. A

redundant signature may be removed when it adds burden without useful information, but correlation alone does not require deletion or arbitrary downweighting. A correctly specified joint likelihood or calibrated joint decision rule can retain dependent observations without counting them as independent confirmations.

A practical output is a signature dependency table. It names the signature, the mechanism it tests, the rivals it discriminates, the confounds it shares, and whether it remains confirmatory, exploratory, or descriptive.

This rule keeps the chapter from mistaking volume for depth. The right study may measure fewer things and answer more.

Model-pair separability should be demonstrated in synthetic data before any additional participant or clinical burden is requested. If the design cannot distinguish the candidate from the named rival when the generator is known, denser observation of vulnerable people will not create adjudicable evidence. The ethical response is redesign or abstention, not recruitment (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

The redundancy audit should show the apparent evidence gain that would result from counting several correlated signatures as independent confirmations. The corrected analysis should collapse or model their dependence and display the resulting loss of information. That counterfactual prevents a larger checklist from masquerading as a stronger test (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

25.3.4 — Open-Set Separability

Design separability must include an explicit none-of-the-models option; §25.5 defines the open-set adequacy tests used to determine whether that option is real rather than decorative. Here the immediate question is whether planned observations can distinguish modeled families without forcing every unfamiliar world into the nearest candidate (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

The benchmark should generate misspecified worlds that are not represented by any fitted model. These may include nonlinear mixtures, unanticipated site practices, unmodeled treatment pathways, sudden device changes, unknown latent subgroups, or irregular observation dynamics. The analysis should not confidently classify these worlds as the PTVC candidate or a known rival.

Open-set detection can use posterior predictive misfit, residual structure, calibration failure, anomaly scores on held-out features, and poor proper scoring across all candidate models. None of these tools is perfect, so the benchmark should report false open-set and false closed-set rates.

A useful failure case is a flexible observation-gain world that produces apparent compression only when reports become more likely during calm states. If every fitted model assigns it to reserve coupling, the benchmark has shown that the design lacks open-set protection.

Open-set separability also affects public language. A result should not say that the Law beat all rivals if the system cannot detect

unknown rival families. It may say it beat the prospectively specified model set under declared adequacy checks, if true.

An open benchmark gives investigators room to discover mechanisms and model failures they did not anticipate.

Challenge open-set separability with generators just outside a modeled family that preserve its familiar observable shape. High closed-set accuracy can coexist with failure to recognize these near-misses. Report detection, anomaly calibration, and abstention rates on the boundary worlds, testing whether the pipeline recognizes unsupported extrapolation rather than forcing an unfamiliar case into the nearest known class (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

25.3.5 — Burden-Adjusted Power

Burden-adjusted power treats information and cost together. A design that extracts more data can be scientifically attractive and ethically inferior if the added information does not change model separability. Participant burden, staff burden, privacy exposure, family intrusion, and device tolerance all belong in the design comparison (Wilson & Collins, 2019).

The benchmark should report marginal information per added burden unit. A new channel may improve expected discrimination only slightly while requiring nightly reports from a fragile participant. A less intrusive channel may offer enough separation when combined with site-level metadata and endpoint adjudication.

Burden is not a sentimental word here. It includes time, discomfort, attention, privacy risk, interruption of care, staff

charting load, device irritation, family pressure, and the possibility that the study changes the experience it is trying to observe.

A burden-adjusted decision may reject the most information-rich design. That is not weakness. It is method respecting the object of study. The project cannot claim to study felt life while treating measurement burden as external to felt life.

The design report should show alternatives: minimal design, preferred design, high-burden design, and rejected design. Each should carry expected discrimination, uncertainty, failure modes, and burden notes.

This rule ensures that the desire to validate the theory does not outrun the care owed to the people whose data would make the test possible.

Burden-adjusted power should include near-miss designs in which more observations improve precision but do not improve model separation. Such cases reveal when participant effort, clinician time, or sensor burden would be spent on a question the design cannot answer. The decision rule should favor redesign or no study rather than treating sample size as a substitute for information (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Burden-adjusted power should be tied to a versioned design artifact that records participant burden, observation cadence, expected information gain, safety constraints, and stopping rules. Later summaries should cite that calculation directly. Statistical power without its human cost function is incomplete design evidence (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

25.3.6 — No Study When Models Are Nonseparable

The hardest design decision is to stop before recruitment. If the benchmark shows that critical models are nonseparable under ethical observation, the correct conclusion is not to recruit and hope for clarity. The correct conclusion is redesign, new channels, narrower claims, or no confirmatory study (Wilson & Collins, 2019; Gneiting & Raftery, 2007).

Nonseparability should be declared by protocol-fixed criteria. For example, if expected log-score gaps are too small, confusion rates too high, none-of-the-models detection too weak, or key rival pairs unrecoverable across realistic observation regimes, the design fails.

A failed design can still teach. It may identify the need for better endpoint adjudication, a less confounded observation channel, stronger site metadata, improved device calibration, or a new rival model. But it should not be rebranded as a human confirmatory protocol.

The public risk is severe. A nonseparable study can produce a polished graph that looks like science while never having been able to discriminate the central alternatives. That graph would move uncertainty from the design into the reader's belief.

The next section examines the rival envelope most likely to cause that nonseparability: endpoint and observation processes.

When nonseparability is declared, the no-study record should point to the executable recovery and power artifact, name the unresolved model pairs, and specify a repair targeted to the source of failure: redesign sampling, add an independent channel, narrow

the estimand, classify the work as exploratory, or refuse recruitment for that question. The repair should change the discriminating information rather than simply add data to the same uninformative contrast (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Expected information should be reported by design choice, not as one global optimism number. Daily and weekly reports, wearable and clinical-record channels, and endpoint adjudication may distinguish different model pairs. The design matrix should show what each added burden buys.

The discriminating-power matrix should also include negative controls. If a signature appears in an impossible window, a device-quality channel, or a variable with no plausible relation to the hypothesis, the analysis has probably found an artifact of alignment, preprocessing, or selection rather than a mechanism. A control that fails only after one preprocessing step can identify the stage where discriminating power is being manufactured.

Design separability requires a different kind of sample-size humility. A large cohort can be powerless for the relevant contrast when every observation passes through the same endpoint-conditioned gate. A smaller design with an independent endpoint channel, calibration-transfer check, and protected holdout may be more informative because it breaks the dominant artifact. Information quality, not participant count alone, is therefore a design resource that can be increased by orthogonal measurements or timing changes.

Power must be conditional on the rival pair, observation model, and decision threshold. A study with nominal ninety-percent

power may still fail to distinguish reserve compression, bridge conditioning, observation shutdown, frailty selection, and adaptive mean reversion under common missingness and noise. Power simulations should include the full rival set and realistic missingness, because pairwise power against an easy comparator can substantially overstate adjudicability.

25.4 — Endpoint and Observation Rival Envelope

Endpoint and observation rivals are a required envelope around every terminal claim. Random endpoint error, backward alignment, survival conditioning, treatment, frailty, device removal, communication loss, and observation shutdown can generate concentration without a PTVC-specific route.

The envelope is generated through the same observation pipeline as the candidate. Each rival specifies how eligibility, entry, care, measurement, endpoint adjudication, and missingness produce the final record, so a terminal-looking curve cannot escape by being assigned to an undefined nuisance process.

The endpoint-and-observation rival envelope includes bridge families, Doob transforms, last-observation artifacts, shared disease and sensor pathways, treatment and care pathways, and rules for updating the envelope prospectively.

25.4.1 — Random-Terminal-Time Bridges

Random-terminal-time bridges condition paths on an endpoint whose timing is uncertain. They matter because terminal studies rarely know the exact death-of-mind stopping time with perfect precision. If data are aligned to an estimated endpoint, bridge-like concentration can appear without PTVC reserve dynamics (Doob, 1957).

The generator should draw endpoint times from a distribution, generate paths conditional on or selected by those times, and then impose realistic endpoint adjudication error. Fixed bridges are

useful baselines, but random bridges are the harder rival because they match the uncertainty of the research setting.

A bridge may produce shrinking variance near the conditioned endpoint. If the observation model then censors late high-volatility windows, the bridge can look even more like terminal neutralization. This is precisely why it belongs in the benchmark rather than in a footnote.

The PTVC candidate must beat random bridges under the same endpoint rules. It cannot define the endpoint by the ledger estimator, and it cannot claim independence if the benchmark aligned by a signal that already encodes the measured outcome.

Recovery should report where random bridges are confused with reserve candidates. If confusion concentrates when endpoint uncertainty is high, the study may need stronger endpoint adjudication or a different target signature.

This rival turns endpoint humility into executable science. It asks whether the terminal pattern survives a model that uses future-event conditioning without invoking the Law.

Random-terminal-time bridge experiments should record the endpoint-time law, conditioning event, reference process, observation schedule, fitted bridge, protected holdout, and diagnostic consequence. Reproduction must reveal whether apparent terminal direction came from the conditioned law or from the data-generating mechanism. The endpoint distribution cannot remain an undocumented convenience.

A random-terminal-time bridge can fit the endpoint shape while leaving its causal status unresolved. The ambiguity lies between

native dynamics, conditioning on a terminal event, and selection through the observation process. Preterminal covariance, hazard response, and intervention behavior must be compared before the bridge is treated as a mechanism (Doob, 1957; Aalen et al., 2008; Léonard, 2014).

25.4.2 — Doob-Transform and Survival Conditioning

Section 22.5.3 established Doob-transform and survival-conditioned processes as scientific rivals. Here the task is operational: include executable versions of those generators in the endpoint and observation rival envelope, with the same data, information, and scoring budget as the PTVC candidate (Aalen et al., 2008; Little & Rubin, 2019).

The benchmark should include Doob-transform rivals where paths are conditioned on survival, endpoint class, open-channel status, or selected terminal features. These models can generate apparent drift toward a boundary or avoidance of certain regions without a PTVC controller.

The key question is whether observed patterns are consequences of conditioning. If a dataset includes only those who remain observable until a window, or only those whose endpoints are well adjudicated, the selected process may no longer represent the unconditioned population.

The analysis should fit Doob-style rivals to the same synthetic and real observation structures as the PTVC candidate. If survival conditioning explains the signatures with equal or better predictive performance and fewer assumptions, the Law claim narrows for that result.

The report should distinguish mathematical conditioning from natural implementation. A Doob transform can be a statistical rival even if no organism implements it as a literal controller. Its role is to ask whether the observation pattern could arise from selection and conditioning.

This protects the chapter from the common error of treating conditional drift as proof of a hidden purpose. The field must separate generated appearance from implemented mechanism.

The Doob-transform and survival-conditioning record should cite the exact code, base law, conditioning event, h-function or weighting rule, endpoint distribution and support, censoring process, limiting construction for zero-probability events, and holdout diagnostics. It should identify which preterminal observables distinguish the conditioned law from native dynamics, because subtle changes in conditioning can alter preterminal behavior. Without that executable provenance, a generic claim that a bridge was tested can misreport a mathematically valid construction as a discovered cause (Doob, 1957; Aalen et al., 2008; Little & Rubin, 2019; Léonard, 2014).

Table 25.4. Endpoint and observation rival envelope.

Rival class	Mechanism	What it can mimic
Random-terminal bridge	endpoint uncertainty plus conditioning	apparent drift and variance contraction near aligned endpoint
Doob/survival conditioning	selection on future event or open-channel status	conditional drift without PTVC-specific implementation

Rival class	Mechanism	What it can mimic
Last-observation artifact	device removal or report stop treated as endpoint	false calm or narrowed final window
Shared disease drivers	frailty moves reserve, observation, endpoint, and proxies	correlated signatures without latent closure
Care pathway	sedation, analgesia, hospice practice, comfort-first monitoring	observed relief or missingness from care rather than a PTVC-specific route
Prospective update	new rival added before protected evaluation	prevents weak envelope and after-the-fact rescue

25.4.3 — Last-Observation and Device-Removal Alignment

Last-observation alignment can create terminal narratives from operational endings. A wearable may be removed, a diary may end, a transfer may interrupt a site, or a staff observation may stop before the conscious endpoint. Aligning those last observations as if they were equivalent endpoints can manufacture a false final pattern (Little & Rubin, 2019).

The generator should include device-removal rules linked to skin fragility, agitation, comfort decisions, battery failure, low perfusion, site practice, and family preference. It should also include report stopping linked to sedation, fatigue, delirium, or care prioritization. These mechanisms are ordinary, humane, and statistically dangerous.

A vivid failure case is a final-window dataset where the last valid wearable reading is calm because the sensor was removed when

motion, pain, or clinical complexity increased. If the analysis treats that last valid reading as the endpoint state, the artifact can imitate concentration.

The benchmark should test multiple alignments: true endpoint, adjudicated endpoint, last observation, last valid device window, care transition, and retrospective death alignment. Differences among these alignments are evidence about the observation process, not merely sensitivity clutter.

A study should not pass if the favored signature appears only under last-observation alignment and disappears under endpoint-independent adjudication. That pattern points to artifact or at least to severe uncertainty.

The final data point must not become a false witness. Sometimes the last observed value is simply the last value a respectful study was allowed to observe.

Last-observation alignment should leave an explicit provenance trail for the clock, reason observation ended, device status, care transition, and relation to the protocol and truth-level endpoints. Analyses under alternative anchors should be stored beside the primary result. This record prevents device removal or respectful withdrawal from being mistaken for a natural terminal event (Aalen et al., 2008; Hernán & Robins, 2020).

Last-observation and device-removal alignment tests should cite a versioned artifact containing the removal rule, clinical workflow, time stamps, missingness process, and counterfactual simulations. The record should show how much apparent terminal concentration can be produced by the observation system alone.

That estimate belongs beside every later interpretation (Aalen et al., 2008; Little & Rubin, 2019).

25.4.4 — Shared Disease and Sensor Confounding

Shared disease and sensor confounding occurs when illness severity moves latent state, reserve proxies, endpoint timing, and observation quality together. The resulting data can look dynamically structured even when the structure arises from shared drivers rather than the proposed mechanism (Hernán & Robins, 2020).

The generator should include latent frailty, inflammation, oxygenation, mobility, perfusion, cognitive clarity, medication burden, and device adherence as linked processes. It should let these variables influence both measured channels and the probability of future observation.

Sensor confounding can be subtle. Skin temperature, heart-rate variability, motion, electrodermal measures, and report latency can change because of disease severity, device placement, medication, or staff workflow. A model that treats them as pure affective witnesses may mistake physical access for felt-state evidence.

The benchmark should allow shared shocks between reserve-like variables and measured ledger proxies. If the finite-variation reserve shortcut is used while the synthetic world has correlated stochastic shocks, the fitting model may misattribute covariance.

Recovery results should identify which signatures remain after shared disease and sensor pathways are included. A signature that

vanishes under shared-driver simulation is not ready for claim-status elevation.

Interpret the relation between physical decline and felt experience through an explicit observation model that specifies what each clinical and experiential variable represents.

The shared-disease and sensor-confounding stress test should be rerunnable with the common cause, sensor gain, dropout, and clinical progression varied independently. Critics need access to the causal graph, synthetic fixtures, and decision thresholds that determine when terminal contraction becomes unidentified. Protected participant records may remain controlled, but the confounding logic cannot remain private (Aalen et al., 2008; Hernán & Robins, 2020).

The resulting terminal narrowing may persist across several devices and still be artifactual because the common cause was never modeled (Aalen et al., 2008; Hernán & Robins, 2020).

25.4.5 — Treatment and Care Pathways

Treatment and care pathways can reshape terminal data for ethically correct reasons. Sedation, analgesia, anxiolytics, hospice admission, nursing attention, spiritual care, family presence, and symptom management can reduce distress, reduce report capacity, alter physiology, or change observation cadence. They must be modeled as pathways, not noise (Sanders et al., 2024; Surges et al., 2024; Hernán & Robins, 2020).

The generator should include treatment initiation, dose changes, comfort protocols, escalation, discontinuation of burdensome monitoring, and site practice. It should also model time-varying

treatment confounding, because treatment decisions often depend on the state that later outcomes attempt to measure.

A hospice pathway can produce apparent repair without PTVC closure. Pain relief may reduce distress. Sedation may reduce report availability. Family presence may change expression. Device removal may protect skin. The resulting record may show less observed burden while the mechanisms are ordinary care.

The benchmark should compare analyses that adjust for treatment histories, model treatment as part of the data-generating process, or stratify by care pathway. It should also report when treatment makes the target estimand change rather than merely adding a covariate.

Comfort interventions are part of the care received during observation. Represent their timing and effects in the model when interpreting terminal signatures.

Palliative practice has its own scientific force. A study that ignores care pathways may turn mercy into a misleading mechanism.

Treatment and care pathway tests should be reproducible from versioned dose histories, care-state transitions, observation rules, and synthetic patient trajectories. A reviewer should be able to remove or alter a pathway and see whether model ordering changes. This separates a scientific effect of care from a narrative assigned after the endpoint (Aalen et al., 2008; Hernán & Robins, 2020).

Treatment and care-pathway analyses should cite the versioned causal and observation artifact used to generate or fit those

pathways. Medication, sedation, symptom control, transfer, and hospice enrollment must remain visible in the executable record. A narrative assurance that care was considered is not a substitute for modeled timing and uncertainty (Aalen et al., 2008; Hernán & Robins, 2020; Little & Rubin, 2019).

25.4.6 — Envelope Update Before Results

The rival envelope must be updateable before protected evaluation, not after results create incentives. A new mechanism enters through a new benchmark version and challenge set with recorded generator, parameter ranges, observation rules, expected signatures, and decision consequences (Aalen et al., 2008; Little & Rubin, 2019).

This rule avoids both a weak frozen contest and an outcome-driven rescue. Critics have standing to propose executable, scientifically relevant rivals capable of fresh prediction, and the public log shows when additions changed recovery or earlier status.

Each update is a reproducible diff with assumptions, implementation, protected-data access date, affected results, and a runnable predecessor. Time stamps and reruns distinguish prospective learning from post-result protection (Nosek et al., 2015; Munafò et al., 2017).

The envelope includes mathematical and institutional routes to the same appearance. Brownian bridges can induce conditional drift, while care pathways, sedation, transfer, device removal, or reduced reporting can create apparent late calming.

Endpoint definitions, censoring, alignment, and last-observation rules are fixed or versioned before the terminal window opens. Otherwise the benchmark can select the endpoint story that makes the graph look most meaningful.

Even a strong envelope retains an open-set escape route because the true generator may be absent. The system is allowed to reject every named model.

25.5 — Open-Set Adequacy and None-of-the-Models

Open-set adequacy asks whether the benchmark can recognize observations generated outside its named model families. A closed-world classifier must choose a winner even when every candidate is wrong. Open-set controls permit refusal, turning inadequacy into an explicit scientific result (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006; Scheirer et al., 2013; Hendrycks & Gimpel, 2017; Ovadia et al., 2019).

Forced classification can convert misspecification into apparent PTVC support. If a misspecified world is assigned to the candidate as the least bad fit, the benchmark has mistaken ignorance for evidence. None-of-the-models is therefore a substantive scientific outcome.

Open-set diagnosis requires posterior predictive misfit, residual and calibration diagnostics, holdout anomaly tests, flexible reference models, none-of-the-models verdicts, and explicit rules for admitting a new model.

25.5.1 — Posterior Predictive Misfit

Posterior predictive misfit asks whether fitted models can reproduce features of the data that matter but were not directly used to select the model. A model may win a comparison score and still fail to reproduce residual structure, tail behavior, missingness patterns, site variation, or terminal-window shape (Vehtari et al., 2017).

The benchmark should generate replicated data from fitted models and compare them to protected features. Features should

include variance trajectories, endpoint concentration, observation intensity, dropout timing, channel disagreement, site effects, tail events, and calibration across groups.

A PTVC candidate that fits endpoint concentration but fails to reproduce observation shutdown patterns may not be the right explanation. A frailty model that fits missingness but fails reserve-linked contraction may also fail. The adequacy question is broader than the pairwise score.

Predictive checks should be chosen before seeing protected evaluation results. If analysts search for checks only after a preferred model fails, the workflow becomes protection rather than testing.

Misfit should have consequences. Severe posterior predictive failure can trigger none-of-the-models, model revision, narrower claim status, or a requirement for new data. It should not be washed away by a favorable headline score.

Assess the full data-generating structure alongside fit to individual features. This makes the comparison informative about both local agreement and broader adequacy.

Flexible reference models should be stress-tested against known generators before human data are used to choose among them. If open-set inadequacy cannot be detected under controlled misspecification, added participant burden would only make the ambiguity larger and less visible. Recovery competence is therefore part of research ethics (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

Posterior predictive checks can look satisfactory in aggregate while failing in the endpoint window, a treatment stratum, or a low-observation group. The report should show which discrepancies were prespecified and where misfit concentrates. A global plot that averages away the decisive failure cannot support a winner label (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

25.5.2 — Residual and Calibration Diagnostics

Residuals are fingerprints of what a model failed to understand. They should be examined across time, endpoint distance, site, device version, medication, diagnosis, cultural context, observation intensity, and uncertainty level, while calibration checks whether probabilities and intervals behave as advertised (Yao et al., 2018; Stanford, 2006).

A calibrated model does not merely give narrow intervals. It gives intervals that cover at the declared rate under the relevant conditions, and probabilities that match empirical frequencies across strata. Poor calibration near the endpoint is especially damaging because that is where the public claim is sharpest.

Residual structure can reveal hidden rival mechanisms. If residuals drift with sedation, device version, or site workflow, the model may be missing a care pathway or observation artifact. If residuals change with endpoint uncertainty, the model may be overusing alignment.

The diagnostics should be plotted and tabulated, but the presentation should avoid turning them into dashboard authority.

The purpose is to find model failure, not to score persons, sites, or clinicians.

A calibration threshold should be prospectively specified for benchmark passing. If coverage, proper scores, or calibration curves fail beyond tolerance in critical windows, the model cannot carry confirmatory language.

The Law of Fairness lens appears as an audit of hidden burden: when residuals cluster by site, diagnosis, disability, or device, the interpretive cost may be falling on groups the model understands least.

Residual and calibration diagnostics should rerun from frozen predictions, protected split identifiers, configuration files, scoring rules, and prespecified thresholds. An independent analyst must be able to reproduce coverage by stratum, endpoint-window misfit, and every trigger for no verdict. The logic should remain open even when raw participant data require controlled access (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

Residual and calibration diagnostics should identify the verdict that the average score alone would have produced. The tempting error may be a declared winner whose intervals undercover, whose residuals drift near the endpoint, or whose probabilities fail in a vulnerable stratum. Making the blocked verdict visible shows why diagnostics are part of adjudication rather than decoration (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

Equation 25.4, Simulation-based calibration rank statistic.

$$r_m = \sum_{s=1}^S \mathbf{1} \left\{ \theta_m^{(s)} < \theta_m^{\text{sim}} \right\}$$

Role label: computational calibration check. Here m indexes a simulation replicate for one declared scalar parameter or test quantity, S is the number of posterior draws, the superscript sim marks the generating value, and superscript (s) labels its s -th posterior draw. Independent posterior draws yield the usual rank reference law; dependence must be accounted for when calibrating the diagnostic. For discrete test quantities or ties, randomized ranks or another prespecified tie rule is required. Departures from expected rank uniformity under the assumed generator can reveal algorithm or implementation problems, but the check does not validate the scientific model as nature’s generator (Talts et al., 2018; Säilynoja et al., 2026).

Table 25.5. Open-set adequacy and none-of-the-models triggers.

Trigger	Signal	Status consequence
Posterior predictive failure	key held-out features outside replicated data	model inadequate for tested claim
Residual structure	systematic error by site, endpoint distance, device, or treatment	missing rival or observation process
Calibration failure	probabilities or intervals unreliable in critical strata	model confidence cannot carry verdict
Open-set anomaly	protected data unlike all candidate worlds	none-of-the-models status considered
Flexible reference dominance	nonmechanistic model predicts much better	model set may lack mechanism

Trigger	Signal	Status consequence
New model entry	independently motivated generator with fresh predictions	new benchmark version before claim update

25.5.3 — Holdout Anomaly Tests

Holdout anomaly tests ask whether protected synthetic or empirical features fall outside the behavior of all candidate models. They are not used to crown a winner. They are used to detect that the model set may be incomplete (Stanford, 2006).

The holdout units must respect dependence. Person, site, endpoint window, device version, and time segment should not leak across training and evaluation in ways that make anomaly detection look better than it is. Random observation-level splits are not acceptable for this purpose.

Anomaly tests can use proper scores, posterior predictive tail probabilities, learned low-dimensional summaries, and prespecified feature distances. Flexible methods are allowed as detectors, but their interpretive status remains limited unless they become specified generative models with fresh predictions.

A useful example is a new site whose observation shutdown policy differs from development sites. A closed-set model may force the site into a known class. An anomaly test may reveal that the entire observation process is outside the trained envelope.

Anomaly detection should be calibrated on synthetic open-set worlds. Otherwise it may either flag everything unusual or miss exactly the strange cases it was built to find.

When a holdout anomaly is severe, the result should become a protected-theory warning, not a public curiosity. The correct output may be No adequate estimate, no verdict, and a benchmark update.

Near-miss generators should preserve familiar marginal curves while introducing a scientifically relevant difference in the joint observations. The benchmark tests whether that observable difference can be detected. If two generators induce exactly the same law for all permitted observations, no anomaly detector can distinguish them from those observations alone; report structural nonidentifiability and identify what additional measurement or assumption would be required (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

Holdout anomaly tests should be cited by artifact version, protected split, discrepancy family, calibration procedure, and decision rule. The result should show whether all offered models missed the same structure rather than merely ranking them. Open-set inadequacy requires an executable trail because the least bad score is otherwise easy to mistake for adequacy (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

25.5.4 — Flexible Reference Models

Flexible reference models are useful because the world may be more complex than any mechanistic candidate. Gaussian processes, splines, state-space learners, ensemble predictors, and other nonparametric baselines can reveal whether mechanistic models are leaving predictable structure behind (Vehtari et al., 2017; Yao et al., 2018).

These flexible models should not receive interpretive authority by default. A neural network that predicts terminal windows well has not discovered the latent ledger. A nonparametric smoother that matches variance compression has not identified reserve coupling. It has found predictive structure that mechanistic models must explain or concede.

The benchmark should use flexible models as baselines for held-out prediction, calibration, residual analysis, and anomaly detection. If a flexible reference greatly outpredicts all mechanistic models, the model set may be inadequate even if one mechanistic model beats another.

Overfitting remains a danger. Flexible references require nested tuning, protected holdouts, regularization, calibration, and transport stress. Otherwise they become another way to preserve the appearance of performance without scientific meaning.

The report distinguishes predictive performance from the explanation assigned to it. A flexible reference can set a performance benchmark and suggest improvements to generator design. The empirical claim tested determines how that comparison contributes to the broader research program.

Equal treatment protects both sides. It prevents the PTVC candidate from defeating weak models and prevents flexible prediction from masquerading as philosophical explanation.

Open-set testing must work on synthetic worlds in which none of the offered models is true before further data are sought from people or sites. Failure to recognize that condition means the pipeline can convert any dataset into a winner. The humane

scientific choice is to repair the test rather than expand observation (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

The release record for flexible reference models must state the consequence of failure before the check is run. Depending on the role of open-set adequacy and none-of-the-models, failure may trigger a narrower estimand, expanded rival envelope, redesigned observation plan, No adequate estimate status, artifact withdrawal, or retirement of a route. In flexible reference models, a check constrains interpretation only when its possible outcomes carry prospectively specified consequences (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

25.5.5 — None-of-the-Models Verdict

None-of-the-models is a formal benchmark outcome. It means that, under a qualified analysis, no candidate model meets the declared adequacy criteria, including calibration and held-out prediction standards, for the target claim. It is not a rhetorical concession. It is a planned scientific status (Yao et al., 2018).

The verdict should be triggered by prospectively specified adequacy conditions: severe posterior predictive misfit, poor calibration across critical strata, failure on protected holdouts, or flexible references revealing large unexplained structure across all declared candidates. Failure of required open-set recovery thresholds instead blocks qualification of the affected analysis. The rule must be known before the benchmark runs.

A none-of-the-models result can preserve value. It may show that observation artifacts dominate, that endpoint definitions are too

weak, that treatment pathways need explicit modeling, or that the model class is missing a major mechanism. The right conclusion is not ‘nothing was learned’. It is that the current model set should not carry the claim.

The None-of-the-models verdict identifies the current model set as inadequate for the tested question. It directs the next stage toward a broader or revised set of candidates and a new comparison under declared criteria.

None-of-the-models should remain an honorable outcome. A field that can say every model failed is less vulnerable to confirmation pressure than a field that always announces the best available story.

An open-set verdict invites new models to enter through the same qualification and comparison requirements as the existing candidates.

A none-of-the-models verdict should record the exact triggers, affected target, failed adequacy checks, and conclusions that are blocked. It should also identify whether the next action is new generator design, a narrower estimand, additional calibration, or no further human study. The status must remain machine-readable so that downstream summaries cannot silently replace it with the least bad candidate (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

The none-of-the-models rule should be evaluated against the counterfactual closed-set verdict. Reviewers should see which candidate would have been crowned solely because the analysis lacked an open-set outcome and why that label is misleading. The

safeguard succeeds when it converts forced certainty into an explicit inadequacy result (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

25.5.6 — New Model Entry Rules

Model entry requires independent motivation and fresh predictions. Sources include critics' mechanisms, systematic misfit, changed measurement technology, or formal amendments (Stanford, 2006; Vehtari et al., 2017).

The package includes a generative specification, observation rules, parameter budget, plausible ranges, expected signatures, recovery plan, computation, and public status. Changes to the hypothesis or candidate model follow the versioning and review procedures in Chapter 28.

Protected challenge sets separate explanation from prediction. A revision may explain the benchmark that inspired it, but predicts only when it succeeds on new data.

Entry rules also include removal. Models that repeatedly fail recovery, adequacy, or transport are narrowed or retired so the library does not become a museum of protected options.

Public language remains modest: a promising model is a candidate, not a rescue, and a formal amendment is a proposal, not a hidden update.

With open-set adequacy established, transport and stress tests ask whether recovery survives movement across sites, populations, instruments, and assumptions.

The public entry record names motivation, equations or algorithm, parameter budget, permitted observations, code, prospectively specified tests, and version, preferably before the protected benchmark opens.

Distinguish pre-holdout models from later additions. Record proposal, code, budget, access, benchmark date, and consequence rule; chronology guards against post hoc rescue (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

A flexible reference may predict better or reveal missed structure without earning causal or metaphysical interpretation. Prediction and explanation remain separate.

Before results, fix none-of-the-models thresholds for predictive misfit, calibration failure, residual structure, and holdout anomalies. Open-set safeguards cannot be improvised afterward.

A model invented after seeing the data remains exploratory until it makes fresh predictions; it cannot retroactively validate the old claim.

Open-set adequacy matters most when every candidate has a constituency. It prevents scientific familiarity, mathematical elegance, philosophical meaning, or computational prestige from being mistaken for completeness.

The none-of-the-models branch must change the study. If all mechanisms fail and a flexible reference exploits only site or device artifacts, the result is model-set inadequacy and the next action is construction or redesign, not a public winner.

25.6 — Transport and Stress Tests

Transport is where model recovery leaves the laboratory of its own assumptions. A pipeline may recover at development sites with familiar devices, diagnoses, endpoint rules, and missingness, yet fail when those conditions change (Arjovsky et al., 2019; Peters et al., 2016; Yao et al., 2018; Ovadia et al., 2019; Sagawa et al., 2020).

Stress tests vary site practice, population, instrument, prevalence, prior structure, missingness, noise, language, and development environment. Out-of-distribution behavior is reported as a target of inference rather than hidden inside an average score.

Transport failure is information. A model may work in hospice oncology and fail in sudden neurological injury, or work under one wearable version and fail under another. Scientific status follows that specificity.

25.6.1 — Site Shift

Site shift captures differences in care practice, staffing, recruitment, devices, charting, hospice philosophy, endpoint adjudication, and local norms. A model developed in one site may recover because it has learned that site’s observation regime rather than a general mechanism (Arjovsky et al., 2019).

The benchmark should implement leave-site-out and synthetic site perturbation. In leave-site-out evaluation, all data from a site remain protected until model selection and tuning are complete. In synthetic perturbation, site-level parameters are varied beyond observed development ranges.

A site shift example is a hospice program that removes devices earlier for comfort than the development sites. Apparent terminal calm may become a policy artifact. A model that treats device removal as random may fail transport, even if it performed well elsewhere.

Site-level stress should report prediction, calibration, recovery, and confusion. It should also report which variables would have been needed to transport better. That output informs the next protocol rather than assigning blame to the site.

The public verdict should not generalize beyond the transported domain. If the result is site-dependent, the claim becomes site-limited. That is narrower, but it is scientifically cleaner.

This rule respects institutions as contexts, not mere data factories. A site is part of the data-generating process.

Site-shift analysis should be reproducible through site-level manifests, protected holdout assignments, calibration procedures, and transport weights. A critic should be able to remove one site, change the target mix, and identify where the claimed advantage fails. Aggregated public data may be used when privacy prevents record-level release, but the transport calculation must remain inspectable (Sugiyama et al., 2012; Pearl & Bareinboim, 2014).

Site shift often hides inside a pooled success. A model may transport in average loss while failing calibration at sites with different care pathways, device practices, or sampling schedules. Site-level intervals, residuals, and mechanism checks should therefore accompany the pooled estimate, with narrowing when

the advantage is not stable (Sugiyama et al., 2012; Pearl & Bareinboim, 2014).

25.6.2 — Population Shift

Population shift includes age, diagnosis, disability, culture, language, trauma history, medication burden, cognitive state, socioeconomic context, and life stage. A model that recovers in one population may fail when the meaning and measurement of channels change (Peters et al., 2016; Yao et al., 2018).

The benchmark should vary population parameters and use held-out population strata where possible. It should test whether sign conventions, calibration, missingness, endpoint adjudication, and channel weights remain meaningful across groups.

A disability example makes the risk concrete. Low movement may mean fatigue, neurological impairment, comfort, device placement, or ordinary baseline. A model that learned movement as burden in one group may misread another group unless person-specific and group-specific calibration are built into the generator and fit.

Population stress also applies to culture and language. Report scales, expression norms, family involvement, hospice scripts, and willingness to decline measurement can vary. Treating those differences as noise may create unfair uncertainty transfer.

Transport should report where the model must abstain. A subgroup with inadequate calibration is not a moral exception. It is a scientific boundary.

The Law’s public claim cannot be widened by convenience. Human clinical evidence travels only through explicit transport assumptions, measurement invariance, and demonstrated recovery under relevant shifts.

A population-shift trace should state the training and target populations, inclusion rules, age, diagnosis, disability, language, culture, treatment, site, and observation differences that matter to transport. It should report where calibration or recovery failed and which conclusion was withdrawn. Without that record, “generalization” becomes an adjective rather than an empirical claim.

The population-shift audit should display the broad claim that would have been exported from the development cohort without reweighting, recalibration, or transport checks. It then shows how prevalence, covariate support, and mechanism mix alter the score and uncertainty. The difference is the empirical cost of pretending that one population stands for all (Sugiyama et al., 2012; Pearl & Bareinboim, 2014).

Table 25.6. Dependence-respecting holdout units for transport stress.

Holdout unit	Leakage blocked	What it tests
Leave-person-out	protects whole persons from train-test leakage	generalization across streams
Leave-site-out	protects sites from development leakage	transport across care regimes
Leave-device-version-out	protects instrument versions from leakage	version and calibration robustness

Holdout unit	Leakage blocked	What it tests
Forward-chain time folds	keeps future evaluation windows out of model fitting and tuning	temporal prediction and update discipline
Endpoint-threshold holdout	protects stopping decisions	prevents endpoint leakage
Out-of-distribution stress	synthetic shift beyond development range	narrowing or abstention under shift

25.6.3 — Instrument Shift

Instrument shift appears when devices, firmware, survey instruments, clinical templates, or data-processing pipelines change. The measured world can move even when the underlying process does not. A benchmark that ignores versioning can mistake engineering drift for biological or experiential dynamics (Yao et al., 2018).

The stress test should include leave-device-version-out evaluation, calibration-transfer simulations, feature-definition changes, sampling-frequency changes, and sensor quality variation. It should record whether recovery depends on a particular instrument generation.

A firmware update can alter artifact rejection, heart-rate variability estimates, sleep staging, or missingness labels. If that update occurs during a terminal study, the apparent trajectory may shift. The version field is not bookkeeping; it is part of the observation model.

Instrument stress should also test old data reprocessing. If a new pipeline changes historical values, the benchmark must decide

whether to freeze the old release, rerun all analyses, or label results as not comparable.

The decision rule is straightforward. A claim that fails under plausible instrument shift becomes instrument-limited or requires recalibration before interpretation.

Device qualification prevents instruments from being treated as transparent windows. Instruments witness imperfectly, and their imperfections travel into model recovery.

Instrument-shift testing should preserve device model, firmware, survey version, clinical template, preprocessing, unit conversion, calibration data, and missingness changes. The trace must make it possible to reproduce the point at which an apparently stable model ceased to transport. Instrument updates are part of the data-generating system, not clerical metadata.

Instrument shift can preserve a smooth trajectory while changing scale origin, sensitivity, lag, or floor effects. The ambiguity is especially severe when device replacement coincides with clinical decline. Bridge samples, calibration fixtures, and version-specific observation models should show whether the scientific conclusion survives the instrument change (Little & Rubin, 2019; Hernán & Robins, 2020).

25.6.4 — Prevalence and Prior Shift

Prevalence and prior shift affect model comparison when the base rates of generators, endpoint types, diagnoses, observation artifacts, or care pathways change. A model that looks plausible in a high-terminal-risk setting may behave differently in a broader life-course cohort (Arjovsky et al., 2019; Peters et al., 2016).

The benchmark should vary prior weights over generator families and simulate different prevalence regimes. It should report how posterior model probabilities, stacking weights, thresholds, and none-of-the-models decisions change when the prior landscape changes.

Prior shift can expose hidden favoritism. If the PTVC candidate is selected mainly because benchmark priors make rival artifacts rare, the conclusion depends on an assumption rather than data discrimination. That dependence should be printed.

The analysis should separate prior sensitivity from empirical separation. A strong design can withstand reasonable prior variation. A weak design may swing among models as priors move, indicating that the data are not carrying the comparison.

The report should not hide this behind a single default prior. It should state which priors are skeptical, neutral, expert-informed, or stress-test priors, and which conclusions survive each.

Prevalence shift reminds the reader that model comparison is not only about fitting curves. It is about what worlds the protocol considered possible before data arrived.

The report should state development and target prevalences, covariate support, prior model weights, and any reweighting or recalibration method, then show how model ordering changes across defensible target prevalences. Because prior shift can alter posterior probabilities without changing individual observations, a winner that depends on an unverified prevalence assumption remains transport-sensitive (Sugiyama et al., 2012; Pearl & Bareinboim, 2014).

25.6.5 — Adversarial Missingness and Noise

Adversarial missingness and noise stress the observation model deliberately. The benchmark should ask how bad observation can become before recovery collapses, and whether collapse is detected rather than converted into false support (Peters et al., 2016).

The stressors include state-dependent dropout, endpoint-dependent observation, device removal, staff cadence change, channel-specific sensor artifacts, floor effects, saturation, delayed reports, cultural nonresponse, and noise floors that are misestimated by the fitting model.

An adversarial test might remove high-burden observations near endpoints, increase noise when reserve is low, or make calm states more observable than distressed states. If the analysis then finds terminal neutrality, the benchmark has located a false-support pathway.

The model should also be tested for false defeat. Severe noise may hide a true reserve-coupled signature and lead to a wrong rejection. False support and false defeat both matter because both distort the claim status.

The output should be a map of safe, fragile, and invalid observation regimes. A universal exponent or universal claim is not acceptable when recovery is conditional on noise and missingness levels.

Adversarial missingness is where the chapter turns suspicion into design. It refuses to let bad observation become a quiet collaborator in overclaim.

Adversarial tests must expose the missingness generator, noise distribution, severity grid, and decision rule. Independent reruns must locate calibration failure or changed model ordering, including selective terminal dropout. Without these fixtures, robustness merely asserts what unobserved data would show (Little & Rubin, 2019; Hernán & Robins, 2020).

Audit the conclusion produced by treating selective disappearance as random noise. A shrinking terminal record may mimic latent contraction while unobserved cases show the opposite. Stress tests earn their place by overturning that favorable but unsupported verdict (Little & Rubin, 2019; Hernán & Robins, 2020).

25.6.6 — Transport Failure

Transport failure means recovery, calibration, or prediction succeeds locally but fails elsewhere. It bounds model use without necessarily defeating it (Yao et al., 2018; Arjovsky et al., 2019).

Classify failures by cause: site practice, population calibration, instrument version, prior shift, observation artifact, endpoint uncertainty, treatment pathway, computation, or open-set misfit. A classified failure is more useful than a vague loss.

The result should change claim status. A model transporting across hospice oncology but failing in sudden death cannot speak for sudden death. A model transporting across devices only after recalibration must require it before future use, and one failing across cultures cannot claim cross-cultural generality.

Transport failure also guides future research. It may identify missing covariates, needed calibration, alternative endpoints, or

rival mechanisms. Narrowing should be productive rather than shameful.

Public communication should say exactly what failed to travel. This prevents broad headlines from borrowing the authority of narrow success.

The next section gives critics a formal role in finding these failures before the project can turn them into private inconveniences.

Record transport findings with source and target environments, prediction or recovery criteria, sensitivity analyses, and consequences for use. Preserve both successful and unsuccessful transfers in the scientific record.

Fix the consequence of transport failure before testing. Depending on its role, failure may narrow the estimand, expand the rival envelope, redesign observation, trigger No adequate estimate, withdraw an artifact, or retire a route. Without a precommitted action, a diagnostic describes fragility but does not adjudicate it (Arjovsky et al., 2019; Peters et al., 2016; Yao et al., 2018).

Leave-site-out and leave-device-version-out tests directly ask whether a model learned general structure or a local operating regime. It may instead learn that one hospice records symptoms differently, one device handles skin contact differently, or one recruitment path selects less severe cases.

Prevalence and prior shift matter because model comparison can be fragile when base rates change. A rare endpoint pattern in one cohort may become common in another because of diagnosis mix,

medication practice, or site referral patterns. Scores should be evaluated under plausible prevalence shifts before public language broadens.

Transport stress also creates a humane No adequate estimate rule. If a model succeeds only in its development environment, later sites should not be asked to carry confirmatory language. Their role may be exploratory transport testing, not proof-bearing replication.

Test quiet contextual differences: reports before versus after medication rounds, device tolerance of motion, cultural expression of late distress, and diagnosis-dependent relations among sedation, wakefulness, and available reports. Ignoring them may recover the site rather than the process.

Name what failed to travel. Parameters may transport while model selection fails; calibration may survive while endpoint sensitivity fails; figures may resemble each other while predictive scoring collapses. Claims must specify what remains valid across contexts.

25.7 — Red Teams and Challenge Logs

Red teams are the social form of model recovery. They give organized standing to independent reviewers testing the pipeline before its public use. Reproducible challenges and documented consequences give this review its scientific value (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a; Munafò et al., 2017; Cogitate Consortium et al., 2025).

Challenge logs preserve the memory of those attempts. They record attack, severity, reproduction, response, unresolved status, code changes, reruns, and public-wording changes. A private patch without a record erases the path by which the program learned.

Adversarial governance specifies roles, bounties, exploit logs, patch rules, critic protection, and public challenge norms. It keeps criticism rigorous without making critics into enemies or turning participant data into spectacle.

25.7.1 — Independent Red-Team Roles

Independent red-team roles prevent one kind of expertise from standing in for all critique. A model red team attacks generators and identification. A measurement red team attacks channels, calibration, and missingness. An endpoint red team attacks stopping rules and alignment. A software red team attacks code, containers, and reproducibility. An ethics red team attacks participant burden and public misuse (Nosek et al., 2018).

The roles should be separated enough to avoid groupthink. A strong statistician may miss a palliative care pathway. A software auditor may miss a cultural reporting problem. A clinician may miss endpoint leakage in code. Each role carries a different map of failure.

Adversarial collaboration offers a tested social design for part of this work. Current methodological analysis emphasizes advance agreement on predictions, operationalizations, analysis rules, documentation, and neutral arbitration, while participants report that the process is demanding and more often clarifies boundary conditions than produces a simple winner (Ceci et al., 2024; Isch et al., 2025). For this program, that is a strength. Red teams should reveal where models diverge, where both fail, and what remains unresolved, not manufacture a public contest whose result was never scientifically defined.

The protocol should give red teams access to synthetic benchmarks, documentation, prospective specification plans, code, and non-sensitive artifacts before protected human data are interpreted. Sensitive data access requires restricted rules, but lack of access should not become a reason to avoid critique.

Red teams should be allowed to specify rival generators and failure tests. Their proposed tests should be recorded even when rejected, with reasons. A critique that does not change the benchmark may still reveal an assumption the public should know.

Independence does not require hostility. The best red team wants the science to be harder to fool. That motivation is compatible with skepticism, support, or uncertainty about the Law.

This structure prevents adversarial testing from becoming a personality contest. The attack is on the claim, the model, the code, or the wording, not on the worth of participants, critics, or authors.

Red-team preparation should include near-miss cases in which most prospectively specified assumptions hold but one decisive leakage, conditioning, or transport rule fails. These fixtures test whether critics and maintainers can locate the failure without being told which module contains it. A challenge process that succeeds only on conspicuous errors is not yet protecting the scientific boundary (Nosek et al., 2015; Munafò et al., 2017; Nüst et al., 2020).

Independent red-team findings should cite the challenge specification, submitted artifact, evaluation environment, response, and final disposition. The cited record should make clear whether the issue was reproduced, fixed, rejected, or left open. Adversarial review becomes science only when its outcome can be followed beyond the meeting room (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

25.7.2 — Challenge Bounties

Challenge bounties reward substantive counterexamples and reproducible flaws. They can help shift incentives away from applause and toward vulnerability. A bounty should pay for a

better rival, a reproducible code failure, a leakage demonstration, a calibration flaw, a harmful public wording risk, or an open-set world that defeats the current benchmark (National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

The bounty criteria should be public. A submission should specify the claimed flaw, the artifact affected, steps to reproduce, expected consequence, severity, and suggested repair when available. Vague denunciation is not a bounty claim, though it may still be a public-risk signal.

Bounties should not reward harassment, participant exposure, clinical intrusion, or sensational screenshots. The chapter's challenge system is scientific, not vigilante. Protected data, vulnerable communities, and research staff remain protected.

A strong bounty example is a critic who supplies a selection-conditioned diffusion that produces apparent endpoint concentration and beats the PTVC candidate under held-out prediction. The correct response is not defensiveness. It is to enter the rival, rerun the benchmark, and update claim status.

Funding and governance of bounties should be independent enough to avoid donor capture or founder protection. The rules should state who decides validity, how appeals work, and how conflicts are disclosed.

The bounty system makes losing useful. It turns the strongest criticism into a public service rather than a threat to identity.

Challenge bounties should first target code, synthetic fixtures, and public artifacts so that vulnerabilities can be found without increasing participant burden. Human or animal data should enter

only when the challenge cannot be resolved by less intrusive means and the expected scientific gain is specific. Adversarial rigor does not require adversarial treatment of participants (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

Challenge bounties can improve coverage while also rewarding conspicuous exploits over quiet structural defects. The protocol should credit reproducible scientific impact, protect low-profile submissions, and separate vulnerability discovery from public performance. A challenge that optimizes attention rather than model correction can create theater while leaving the benchmark weak (Nosek et al., 2015; Munafò et al., 2017).

Table 25.7. Red-team roles and attack surfaces.

Role	Attack surface	Possible output
Model red team	generators, priors, identifiability, rival envelope	new generator or narrowed claim
Measurement red team	channels, calibration, missingness, sign conventions	observation-model patch or No adequate estimate
Endpoint red team	stopping rules, alignment, leakage	endpoint rerun or exclusion of flawed result
Software red team	code, containers, splits, seeds, dependencies	patch, rerun, security update
Ethics/public team	burden, privacy, captions, vulnerable readers	wording change or artifact withdrawal

Equation 25.5, Challenge-log completeness condition.

$$C_k := (a_k, r_k, s_k, p_k, u_k)$$

$\text{Complete}(C_k) = 1$ iff a_k, r_k, s_k, p_k , and u_k are all present.

The fields record the attack or challenge, reproduction status, severity, patch or response, and public-status update. Role label: artifact-governance diagnostic. The tuple is not a statistical theorem; it is a reproducibility rule for adversarial challenge records.

25.7.3 — Exploit Log

An exploit log is the replayable record of how the benchmark was attacked. Each entry should include a stable identifier and version, submitter role, affected component and release, severity, reproduction status, exploit description, minimal reproducible fixture, expected and observed failure, response or patch, rerun and regression-test result, public-language implication, final disposition, and unresolved risk. A critic should be able to replay both the vulnerability and the claimed repair without relying on private correspondence (Nosek et al., 2015; Nosek et al., 2018; Munafò et al., 2017; Nüst et al., 2020; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

The word exploit is deliberate but bounded. It means a route by which the pipeline, artifact, analysis, or public wording can be made to overclaim, underreport uncertainty, misclassify a rival, leak protected information, or mislead readers. It does not mean attacking persons.

Severity should be tied to consequence. A typo in a caption is minor unless it changes claim status. Endpoint leakage in the final challenge set is critical. A code path that tunes thresholds on

protected data is critical. A public figure that compresses none-of-the-models into support is high risk.

The log should distinguish reproduced, not reproduced, disputed, fixed, accepted risk, and retired entries. Not every challenge will be correct, but every substantive challenge should leave a trace.

A public version can redact sensitive details while preserving the scientific lesson. Restricted versions can contain code, data identifiers, or security details that would create risk if released broadly.

The exploit log is a memory against quiet self-protection. It lets the project improve without pretending it was always clean.

25.7.4 — No Private Patch Without Record

A discovered flaw that affects results or interpretation must be repaired through a versioned record naming what changed, why it changed, the affected releases and analyses, code or protocol changes, added tests, required reruns, invalidated results, and corrected or withdrawn artifacts. Preserve the pre-patch results and verdicts so reviewers can trace changes in leakage, recovery, model ordering, or public status (Nosek et al., 2015; Nosek et al., 2018; Munafò et al., 2017; Nüst et al., 2020; National Academies of Sciences, Engineering, and Medicine, 2019).

The rule should apply to code, equations, synthetic generators, preprocessing, figures, tables, threshold choices, and public summaries. It should also apply to errors that hurt the PTVC candidate. Correction is not only for flaws that protect the theory.

There may be security exceptions. A vulnerability that exposes sensitive data may require delayed disclosure or restricted details. Even then, the existence of the issue, the affected releases, and the resolution status should be recorded in an appropriate layer.

This rule gives the scientific program a visible conscience. It lets trust be rebuilt through correction rather than maintained through concealment.

25.7.5 — Critic Protection and Nonretaliation

Critic protection is a scientific requirement when a project has public identity, community interest, or donor attention. A critic who finds a real flaw should not be treated as disloyal, hostile, or morally suspect. The benchmark needs their pressure (National Academies of Sciences, Engineering, and Medicine, 2019).

Nonretaliation rules should cover publication, community moderation, access to artifacts, bounty eligibility, and public response. A critic can be wrong, sharp, or inconvenient without losing standing. Abuse and harassment can be moderated, but substantive criticism should remain protected.

The protocol should separate adjudication from public popularity. A challenge is assessed by reproducibility, relevance, severity, and effect on claims, not by whether supporters like the submitter.

Critic protection also protects the project. If critics fear retaliation, flaws move underground, and public artifacts become more brittle. A project that supports serious critique strengthens its scientific independence and public trust.

Critic-protection procedures should leave a versioned, inspectable record of the challenge, evidence, reproduction status, response, patch or nonpatch decision, confidentiality option, conflict or retaliation concern, response time, and appeal route outside the challenged team. Later claims of an open challenge culture should cite that operating record (Nosek et al., 2015; Nosek et al., 2018; Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

25.7.6 — Public Challenge Without Theater

Public challenge can strengthen the program, but debate-as-verdict, humiliation clips, pile-ons, scoreboards, and winner-loser framing replace method with spectacle. Public artifacts should distinguish submitted challenge, reproduced flaw, unresolved dispute, resolved patch, and claim-status change rather than slogans such as proved or destroyed (UNESCO, 2021a; Nosek et al., 2018).

Challenges belong first to executable artifacts and bounded questions. Synthetic fixtures, thresholds, rival envelopes, and caption drafts let critics test the benchmark without exposing participants, families, clinicians, sites, or private records. A failed benchmark is never permission for personal endpoint testing, harm, clinical delay, or public scoring.

When the method is nonseparable or unrecoverable, more intimate data cannot substitute for missing analysis. The challenge must improve the model before it enlarges participant burden (National Academies of Sciences, Engineering, and Medicine, 2019).

Staged disclosure, neutral replication, and visible status changes keep preliminary exploits and institutional rebuttals from becoming personality contests. The aim is a reproducible path from objection to correction, not humiliation (Nosek et al., 2015; Munafò et al., 2017).

Red teams should work before artifacts become costly to change, and mature logs should preserve unresolved entries without treating them as scandal. Some attacks await a stronger simulator; others expose wording risk rather than model error.

Public challenge without theater is not timidity. It is the courage to make the evidence trail stronger than prestige while keeping vulnerable people outside the spectacle.

25.8 — Code, Containers, and Reproducible Artifacts

For a computational benchmark, reproducibility means a critic can identify the exact code, data layer, synthetic generator, environment, prospective specification, decision thresholds, and release artifacts that produced a result. It also means sensitive human data remain protected rather than sacrificed to openness (Sandve et al., 2013; Boettiger, 2015; Wilkinson et al., 2016; Nüst et al., 2020; Peng, 2011; Wilson et al., 2017).

The artifact structure has six linked layers. Versioned source and release identities connect claims to code. Containers and environment files preserve execution context. Synthetic fixtures and unit tests protect formulas and transformations. Machine-readable prospective specifications bind code to decisions. Restricted and public layers balance reproducibility with privacy. Retirement and security rules make correction durable.

The section closes the chapter by turning answerability into a file system, a release practice, and a public discipline. The book supplies the scientific meaning. The artifacts let others test whether the machinery actually does what the book says it does.

25.8.1 — Versioned Source and Release IDs

Versioned source connects a published claim to exact code. A model-recovery result should cite a release ID, commit hash or equivalent, generator version, analysis version, decision-threshold version, and artifact manifest. Without that chain, a result cannot be reproduced with confidence (Sandve et al., 2013).

The release ID should appear in tables and benchmark reports, not only in a repository. If a figure changes because code changes, the figure receives a new release reference. If a generator changes, affected recovery results are flagged.

Versioning should cover notebooks, scripts, data dictionaries, container files, prospective specification files, synthetic fixtures, random seeds where appropriate, and rendered public artifacts. A polished chapter figure should be traceable to the code that created the underlying benchmark.

The release practice should avoid raw links in manuscript prose. The book can name the artifact class and release structure, while actual distribution belongs to governed publication channels and repositories.

A versioned release also protects against accidental canon drift. A public summary cannot quietly update a model assumption while still claiming the authority of an older benchmark.

This is a humble rule with large consequences. Exact source identity lets criticism land on an object rather than a memory.

Release identifiers should be tested against near-identical artifacts whose bytes, dependencies, source data, or frozen assumptions differ consequentially. The audit must detect when one human-readable title points to scientifically noncomparable executable states; checksums, manifests, and semantic-version notes must recover the exact state cited by each result (Nüst et al., 2020; Wilkinson et al., 2016).

25.8.2 — Containers and Environment Files

Containers and environment files preserve computational context. A benchmark may depend on probabilistic programming versions, numerical libraries, random number generators, plotting code, compilers, operating systems, and hardware assumptions. Without environment capture, reruns can fail or subtly differ (Boettiger, 2015; Wilkinson et al., 2016).

A container specification should include language versions, package versions, operating-system base, external dependencies, and commands to run the benchmark. Environment files for lighter workflows should be paired with lock files when possible.

Containers do not solve all reproducibility problems. They can become insecure, outdated, or impossible to run on future systems. The artifact plan should therefore include hashes, archived images, human-readable environment descriptions, and periodic rebuild tests.

The benchmark should include a minimal smoke test and a full challenge run. A critic should be able to run a small fixture quickly, then run the larger benchmark with documented compute needs.

Computation time belongs in the report. A method that recovers models but requires unrealistic resources may not be suitable for the planned research program or for independent replication.

Computational humility becomes concrete when rerunning is treated as part of a claim’s vulnerability.

Containers and environment files are qualified on synthetic fixtures before restricted empirical data are introduced. A reproducibility failure at that stage stops deployment and triggers correction without exposing additional participants. Technical preparation is part of minimizing research burden (Sandve et al., 2013; Boettiger, 2015; Wilkinson et al., 2016; Nüst et al., 2020).

Containers reduce environmental drift but do not eliminate it. Base images can disappear, hardware or numerical libraries can change behavior, and hidden network calls can break a claimed offline run. Validation across clean environments and archived dependency layers is needed before reproducibility is declared (Nüst et al., 2020; Wilkinson et al., 2016).

Table 25.8. Reproducible artifact layers.

Artifact layer	Contents	Scientific function
Source release	exact code, generator version, analysis version	reproduce the benchmark that produced a claim
Container/environment	dependencies, operating system, commands	rerun without invisible software drift
Synthetic fixtures	small known cases and edge cases	test formulas, transforms, leakage, and signs
Machine-readable prospective specification	locked decisions, thresholds, holdouts, status labels	compare executed analysis to declared plan
Restricted layer	sensitive empirical data and site-specific details	protect participants while preserving audit route

Artifact layer	Contents	Scientific function
Retirement record	superseded, deprecated, withdrawn, archived status	keep old artifacts from speaking with current authority

Equation 25.6, Release identity.

$$R = (G, M, O, D, T, S, V)$$

Role label: reproducibility manifest. In this tuple, G identifies the generator release, M the fitted-model and analysis release, O the observation-system specification, D the data or synthetic-fixture release, T the decision-threshold specification, S the protected-split and seed specification, and V the software environment and manifest version. A result without a release identity cannot be fully reproduced or challenged under the standards established here.

25.8.3 — Synthetic Fixtures and Unit Tests

Synthetic fixtures are small known datasets designed to test formulas, transformations, edge cases, and failure routes. Unit tests check that the code implements the declared model rather than a convenient approximation. They are where mathematical care becomes executable (Wilkinson et al., 2016).

Fixtures should include clean reserve-coupled data, OU data, bridge data, observation shutdown, missingness patterns, device version changes, endpoint uncertainty, extreme noise floors, and open-set misspecification. Each fixture has expected outputs or acceptable ranges.

Unit tests should check sign conventions, time alignment, endpoint independence, leakage prevention, train-test splitting, posterior predictive generation, SBC ranks, proper scores, and

figure-building data. A single sign error can convert burden-increasing into restorative signal.

The benchmark should include tests that are expected to fail when a known exploit is inserted. For example, a leakage fixture should show inflated recovery if protected endpoint thresholds leak into training. This makes the test suite adversarial, not merely confirmatory.

Wholly artificial fixtures can normally be released after a disclosure-risk check. Fixtures calibrated from sensitive participant or site data require review of what their parameters and outputs could reveal. A synthetic label alone does not establish that a release contains no private information.

Executable checks show whether the declared machinery runs as specified and make the implementation independently inspectable.

Synthetic fixtures and unit tests should record the target invariant, expected pass or failure, edge cases, software version, and responsible maintainer. Fixtures must include adversarial examples that break convenient assumptions, not only examples that demonstrate success. Their role is to make regression visible before a new release reaches empirical data.

Synthetic fixtures and unit tests should state the scientific error that would survive if each test were removed. A sign inversion, stale seed, endpoint leak, wrong likelihood, or broken missingness branch can otherwise generate a convincing benchmark result. The test suite is informative when its failure

modes map to concrete false conclusions (Nüst et al., 2020; Wilkinson et al., 2016).

25.8.4 — Machine-Readable Prospective Specification

Machine-readable prospective specification connects analysis code to declared decisions. It records generators, priors, endpoints, observation models, inclusion rules, thresholds, holdout units, tuning rules, none-of-the-models criteria, and public-status language in a form code can check (Nüst et al., 2020; Sandve et al., 2013).

Human-readable prospective specification remains necessary, but it is easy for prose to hide degrees of freedom. A structured file can specify which variables are locked, which margins are fixed, which windows are protected, and which choices require a new version.

The benchmark should automatically compare executed analysis with the protocol-fixed prospective specification. If code uses a different endpoint window, threshold, prior, exclusion rule, or holdout unit, the run is flagged rather than silently accepted.

Machine-readable prospective specification also helps red teams. A critic can ask whether the analysis matched the declared plan rather than relying on trust in a narrative report.

The file should include status labels. A result can be exploratory, recovery check, design-separability check, confirmatory benchmark, transport stress, or public artifact. Code should not let an exploratory run populate a confirmatory table by accident.

Claim-status discipline becomes a computational contract. The software should make it difficult for the analysis to become more certain than the plan allowed.

Machine-readable prospective specification should be tested against synthetic workflows before it governs a live study. If the encoded rules cannot reproduce the intended estimands, exclusions, stopping decisions, and verdict language, new participant data would enter a defective pipeline. Validation of the protocol object precedes recruitment (Sandve et al., 2013; Boettiger, 2015; Wilkinson et al., 2016; Nüst et al., 2020).

25.8.5 — Restricted and Public Layers

Reproducibility must be balanced with privacy and data dignity. The benchmark should separate public synthetic layers, restricted empirical layers, secure metadata layers, and public result summaries. Openness is not a license to expose participants, families, clinicians, or sites (Sandve et al., 2013).

The public layer should include synthetic generators, fixtures, model code, benchmark specifications, non-sensitive challenge logs, and reproducible figures built from synthetic data. The restricted layer may include protected empirical data, site-specific sensitive parameters, access-controlled validation outputs, and security details.

Restricted access should have governance: application criteria, data-use agreements, audit logs, role-based permissions, retention limits, incident response, and participant or site rights where appropriate. These details will be developed more fully in the governance chapters, but the benchmark cannot ignore them.

The public should not be asked to trust a black box. When empirical data must remain restricted, synthetic fixtures, red-team logs, independent audit summaries, and reproducible artifact manifests become especially important.

A public artifact should never imply that the benchmark is less uncertain because the private layer exists. Restricted evidence remains subject to the same model recovery, adequacy, and verdict rules.

This layered approach keeps the scientific program answerable without turning vulnerable lives into open data.

Public and restricted layers should cite one versioned release manifest stating which data, code, fixtures, parameters, outputs, and logs are available at each access level; why any restriction applies; and which results cannot be independently regenerated. It should record privacy review, security controls, reproducible public fixtures, and a route for qualified reanalysis. Restriction may protect people; it may not conceal an unfavorable result, and its remaining limits should be machine-readable (Sandve et al., 2013; Boettiger, 2015; Wilkinson et al., 2016; Nüst et al., 2020).

25.8.6 — Artifact Retirement and Security

Artifacts age: dependencies break, containers become insecure, devices change, models retire, and public wording becomes misleading after new benchmarks. Every artifact should therefore carry a status such as active, superseded, deprecated, withdrawn, security-restricted, or historical, with a reason, affected claims, replacement when available, and date (Boettiger, 2015; Wilkinson et al., 2016).

Security incidents are scientific events because compromised dependencies, leaked restricted files, stale figures, or invalidated assumptions can preserve a reproducible but wrong result. Triage, temporary access limits, patching, reruns, public notice, and durable archives must identify what future readers may still trust.

Retirement preserves history without leaving the retired object authoritative. The archive records the version, reason, affected conclusions, replacement or absence of one, security action, and public correction notice. Credentials, containers, datasets, and outputs that cannot be maintained safely should not remain silently callable.

Figures and public summaries follow the same rule. A shareable graphic should be tied in the reproducible record to its generator, fitted-model version, holdout split, threshold set, and status, so visual elegance cannot keep a superseded result alive.

Synthetic fixtures are the safest public entry point. They let critics test sign conventions, leakage guards, endpoint independence, model recovery, calibration, and figure scripts without exposing participant data. Machine-readable prospective specification further prevents exploratory thresholds, post-result holdouts, or mismatched generator versions from entering confirmatory outputs. Public fixtures should also include deliberately failing cases so critics can verify that safeguards and rejection paths activate when they should.

The artifact standard supports three reproducibility depths: public regeneration of tables from synthetic data, authorized rerun of restricted analyses in a controlled environment, and future

rebuilding from pinned dependencies and environment files. Each depth has a distinct privacy boundary and audit purpose. A reproducibility claim should name which depth was achieved, preventing synthetic regeneration from being mistaken for independent reproduction of restricted human analysis.

Model recovery is ready for replication infrastructure only when its worlds are hard, its failures visible, its artifacts open to challenge, and its corrections durable. That is the boundary between a computational promise and a scientific system independent sites can carry forward. Durable artifacts should include correction history so a later site can reproduce the current system without unknowingly reinstating a retired bug or model version.

Part VIII — Replication, Verdicts, and Scientific Updating

A scientific claim begins to mature when it leaves the conditions that first made it persuasive and enters the hands of independent teams free to test it without protecting its reputation. Earlier Parts defined PTVC, separated its layers, built measurement and endpoint gates, proposed humane studies, and gave serious rivals room to win. Part VIII asks whether any resulting evidence can travel across teams, sites, instruments, populations, and public summaries without changing what it means (Center for Open Science, 2025; Nosek et al., 2018).

The first discipline is vocabulary. Computational reproducibility asks whether declared data and code regenerate a reported result. Analytic robustness asks how much the conclusion depends on defensible analytic choices. Replication asks whether new evidence supports the same bounded claim, and transport asks where that result continues to hold. Large-scale metascience now examines these properties separately because each requires a different record, comparison, and verdict (National Academies of Sciences, Engineering, and Medicine, 2019; Miske et al., 2026; Aczel et al., 2026; Tyner et al., 2026; Brodeur et al., 2026).

In research on lived experience, context is part of the object. A hospice network, sleep laboratory, community clinic, research hospital, and longitudinal cohort may use the same nominal instrument while differing in language, medication, staffing, referral pathways, device firmware, family involvement, and observation loss. More sites improve inference only when those differences are measured, governed, and allowed to narrow the claim.

The aim is comparability without forced sameness. Common objects, bridge samples, translation review, calibration, site supplements, and transport estimands can earn comparability while preserving legitimate local difference. When the bridge fails, no-pooling is a substantive finding about comparability.

Part VIII also gives every finding an address. A result may bear on representation, an instrument, an endpoint rule, a candidate route, a rival comparison, a reference class, a transport claim, or the terminal target itself. It changes only the component reached through the declared dependency chain, and no favorable downstream result repairs an upstream gate that failed.

That discipline makes support and defeat equally informative. An instrument can become scientifically useful without being mistaken for access to the latent ledger. A candidate mechanism can retire without settling every target branch. A verified counterexample can defeat the exact universal branch it reaches, while an unidentified endpoint can make No adequate estimate the most rigorous result.

Institutions carry replication, but no institution confers truth. Site qualification, independent analysis, rival stewardship, endpoint adjudication, repositories, incident systems, negative-result rights, and correction records belong to the inferential architecture. They distribute the doors through which evidence can enter, challenge, narrow, or defeat the program.

The work also has a human infrastructure. Coordinators, clinicians, data stewards, translators, participants, families, and local communities carry harmonization and correction. Burden

limits, fair credit, referral routes, privacy controls, withdrawal, debriefing, and stop authority belong inside validity because an exhausted, coerced, or unsupported research system cannot produce trustworthy evidence.

Open science joins inspectable claims to protected people. Protocols, code, metadata, model cards, provenance, and public status records can be shared, controlled, or restricted according to purpose and risk. Raw terminal recordings, clinical notes, family speech, identity-linked trajectories, and sensitive device streams may require governed access. Reproducibility requires a reconstructible research record, including disclosed use of rapidly changing artificial-intelligence tools, not the conversion of private life into public material (Grant et al., 2026; Feuerriegel et al., 2026).

Chapter 26 builds the multisite research system. It separates protocol-fixed decisions from public preregistration claims, qualifies sites and devices, tests harmonization, models heterogeneity, distributes analytic authority, governs repositories, records incidents, protects staff and participants, and keeps negative results publishable. Replication becomes inspectable infrastructure rather than ceremonial repetition.

Chapter 27 governs what accumulated evidence is permitted to mean. It maps findings to components, distinguishes replication states, gives favorable results bounded language, preserves inconclusive and unusable evidence, and separates principled narrowing from rescue. Mechanisms may retire, rivals may dominate, and the strong form may be defeated without allowing any result to outrun its design.

Chapter 28 gives scientific correction a durable route. Versioned claim, model, evidence, and artifact registers preserve what changed and why. Dissent and external challenge remain reviewable; root-level questions return to *The Law of Fairness*; approved corrections propagate through the series, websites, diagrams, protocols, and public replies so that technical correction reaches every active surface (Committee on Publication Ethics, 2025; Crossref, 2025).

Chapter 29 keeps the program open after publication. It orders unresolved theorems, identification problems, first study families, simulation gates, value-of-information choices, release conditions, pause routes, redesign duties, and stop rules. Publication becomes the beginning of answerable testing and durable correction.

This structure sustains a rigorous hope. A narrower claim that survives is more valuable than a universal claim protected by ambiguity. A null replication with complete records is more useful than a favorable result whose adverse sites vanished inside a pooled mean. A visible correction is stronger than a beautiful artifact left wrong because it has already traveled far.

Part VIII closes the volume by building a research system in which every component can be challenged, reproduced, narrowed, corrected, or retired. Independent teams gain standing; local differences acquire scientific meaning; negative evidence receives a durable home; participants and staff retain enforceable protections; and every public claim remains tied to a versioned evidence record. The Part's central achievement is not a verdict, but a way to keep every verdict answerable.

Chapter 26 — Multisite Replication and Research Systems

Independent replication reveals which findings extend beyond the conditions of the first careful study. A palliative unit may have exceptional documentation, a research hospital may use unusually stable devices, a community clinic may recruit through stronger family networks, and a translated measure may carry meanings the originating team never tested. Replication begins by identifying which features of the original environment made the result possible (Nosek et al., 2018; Wilkinson et al., 2016).

Chapter 26 treats replication as scientific infrastructure. Similar-looking data from several institutions become stronger evidence only when the question, protocol, local adaptations, observation model, analysis route, and publication rights are fixed at the appropriate time. Site disagreement remains visible because it may be the most informative result in the network.

The distinction matters because careless success can damage an ambitious claim. Endpoint leakage, device drift, post-outcome model changes, local scripts, unmodeled attrition, or unpublished deviations can create apparent replication while widening confirmation bias. The network exists to discover which component travels, where it fails, and how narrowly the evidence permits it to be stated.

Site qualification is an evidentiary gate, not a prestige badge. A qualified site shows that its staff can carry the protocol without displacing care, that devices and firmware are traceable, that consent and language practices fit the population, that data flows are auditable, and that stop authority can actually be used. A site that cannot protect the people or the object remains outside the pooled confirmatory analysis.

Harmonization is the central technical problem. The same field name, wearable, survey item, or clinical code does not guarantee the same measurement object. Translation, medication, firmware, workflow delay, disability, and local response practice can alter what a channel means, so bridge samples and context-specific qualification may be required before pooling.

Site effects become evidence about scope. A signature present in one care regime and absent in another may reveal measurement failure, a clinical mediator, device instability, transport failure, or a narrower reference class. Site-level results that expose those differences can be more informative than a precise pooled estimate that conceals them.

Transport requires its own estimand. It asks which result moves from the study sites to a declared target setting under explicit assumptions about population, intervention, measurement, endpoint, and observation. Generalization is therefore a testable claim with a target population, a transport bridge, and conditions that may fail.

Repositories are scientific instruments, not storage closets. Final cleaned files cannot reveal which device version produced a signal, which transform was applied, what a participant withdrew, how translation changed, which container generated a figure, or which analysis version entered the public summary. Data dictionaries, provenance, access tiers, code and model registries, retention decisions, deletion events, and correction links make the result reconstructible.

Human infrastructure belongs inside the validity model. A night-shift nurse adding observations, a coordinator explaining consent during family crisis, a translator resolving an affective term, and a technician repairing a firmware failure all shape fidelity. Staffing ratios, workflow displacement, emotional strain, fair credit, participant support, family explanation load, debriefing, and stop authority are scientific operating conditions.

Incident systems preserve moments when the research object changes. A breach, device failure, safety complaint, protocol drift, mistaken endpoint classification, or delayed correction can alter what the evidence means. Logging, triage, independent response, reanalysis, and public linkage turn those events into scientific memory instead of reputational secrets.

Negative-result systems prevent the archive from becoming selectively favorable. Replication disappears when only harmonized sites, cooperative devices, successful instruments, or analyses in which rivals lose remain visible. Null, adverse, suspended, unusable, and Rival-favored outcomes need publication rights and persistent identifiers so later synthesis can distinguish learning from selection.

A worked design follows a four-site terminal-telemetry replication spanning an academic palliative unit, a community hospice, a rural clinic using another vendor, and a non-English-speaking care network. Each site contributes a distinct context and a distinct way the claim could fail. The consortium succeeds scientifically only when those differences are declared before outcome inspection and remain visible after analysis.

The result is an operational blueprint rather than a runnable protocol. It identifies what qualifies a site, which adaptations require logging, when pooling is prohibited, how analytic and adjudicative roles separate, who protects participants and staff, how incidents alter status, who controls repository access, and who corrects the public layer. Every major function receives an explicit owner, record, and review route.

The multisite architecture stops before verdict issuance. Its task is to create the conditions under which support, Reference-class narrowed, Model-ambiguous, Rival-favored, Route-failed, No adequate estimate, or defeat can become trustworthy. A false claim can be exposed without spectacle, a narrower component can remain honestly narrow, and a promising result stays inside scope until independent teams have tested it.

26.1 — Multisite Preregistration and Registered Reports

A multisite study begins with a shared decision object, not with enrollment. The core protocol fixes the scientific question, site supplements expose local implementation differences, and Registered Reports or another outcome-independent design review tests the method before anyone knows whether the result favors PTV, a rival, or no adequate model. Current protocol, reporting, and good-clinical-practice standards reinforce prospective specification, participant protection, reliable records, and transparent reporting (Chan et al., 2025; Hopewell et al., 2025; International Council for Harmonisation, 2026; Grant et al., 2026).

Terminal and longitudinal studies contain legitimate local differences in consent timing, language review, staffing ratios, device placement, family-contact rules, clinical referral pathways, and data-transfer constraints. Scientific comparability does not require hiding those differences; it requires fixing them before outcome inspection, recording their rationale, and preventing them from becoming convenient post hoc explanations, exclusions, or rescue devices.

Preregistration is a public-record claim, not a synonym for early intention. A protocol or analysis plan may be protocol-fixed or prospectively specified without being publicly preregistered. When a timestamped external registry record with a persistent identifier exists, the manuscript cites that record and identifies the estimands, endpoint windows, missingness rules, rival set,

decision thresholds, deviations, and public-language ceiling it fixes (Nosek et al., 2018; Grant et al., 2026).

Registered Reports add a distinct publication safeguard. Design and analysis receive peer review before results, and publication remains protected when the accepted route and quality conditions are followed. Null, Rival-favored, Model-ambiguous, and No adequate estimate outcomes then remain eligible for publication because the result was not known when the design received in-principle acceptance.

A multisite charter therefore identifies the common scientific question, each site's context of use, minimum shared measures, local adaptations, burden thresholds, independent analysis route, access tiers, incident escalation, and publication commitments before outcome inspection. This versioned structure lets later reviewers determine whether apparent replication answered one question or several shifting ones.

Table 26.1. Multisite charter fields that must exist before recruitment.

Charter field	Minimum content	Why it matters
Scientific component	The exact component claim, estimand, endpoint window, and rival matrix	Prevents a site from testing a different object under the same title
Context of use	Population, setting, language, device, cadence, care regime, and intended inference	Defines where the result can and cannot travel

Charter field	Minimum content	Why it matters
Site qualification	Staffing, training, calibration, consent route, support plan, and stop authority	Makes feasibility a scientific gate
Analysis version	Primary, secondary, exploratory, and rival analyses with thresholds	Protects confirmatory status from post hoc drift
Burden and incident rules	Participant, family, staff, safety, security, and deviation triggers	Keeps harm and workflow pressure inside the record
Publication rights	Null, negative, unusable, and rival-winning outcomes protected	Prevents suppression of inconvenient evidence

Table 26.2. Site qualification checklist.

Qualification object	Required evidence before activation	Failure consequence
Staff training	Role-specific training log, endpoint-language practice, consent rehearsal, incident escalation, and burden-stop authority	Site remains in shadow mode or cannot enroll high-risk windows
Device readiness	Hardware version, firmware version, calibration fixture, artifact profile, backup plan, and replacement route	Device channel excluded or site limited to nondevice analyses

Qualification object	Required evidence before activation	Failure consequence
Protocol fidelity	Dry-run record, synthetic-case walkthrough, timing check, missingness code practice, and amendment procedure	Site cannot contribute confirmatory data until corrected
Participant support	Consent route, withdrawal path, referral resources, after-hours contact, and nonparticipation care separation	Enrollment pauses until support is real rather than promised
Family and caregiver load	Contact limits, explanation schedule, bereavement timing, bystander consent screen, and opt-out path	Affected family-facing measures are suspended until the failed protection is corrected and the measures are requalified
Stop authority	Named staff with authority to pause recruitment, pause collection, quarantine data, or escalate incidents	Site cannot operate endpoint-adjacent protocols

Site qualification establishes whether an institution is prepared to supply observations for confirmatory research. The checklist covers participant and staff burden, device reliability, clinical complexity, and endpoint-adjacent procedures. Qualification findings are recorded with clear requirements for initial approval or requalification.

Table 26.3. Replication types and status effects.

Replication type	What it asks	Possible status effect
Direct replication	Can the same protocol, estimand, and observation model reproduce the component result under qualified site conditions?	Strengthens or weakens the specific component under that context of use
Conceptual replication	Does a different operational route recover the same component pattern without sharing the same measurement artifact?	Supports construct breadth only if the new route has its own rival matrix
Multisite replication	Does the pattern survive across qualified sites with site effects modeled rather than ignored?	Tests robustness and identifies scope limits or site-conditioned claims
Adversarial replication	Does the component survive when critics and rival stewards help specify models, thresholds, and failure routes?	Raises trust if it survives or forces narrowing if rivals win
Transport replication	Does the result travel to a new population, device regime, language, care setting, or sampling cadence?	Extends scope only under explicit transport assumptions and successful invariance checks

Replication type	What it asks	Possible status effect
Invalid or unusable replication	Did the design, harmonization, safety, or data quality fail before interpretation is possible?	Produces an infrastructure finding, not support or defeat of the component itself

The replication type identifies what was retested: a design in similar settings, a concept with a different method, or a prediction challenged by an independently developed alternative. Report the resulting scope and any qualification needed for the next extension.

The status consequence is declared before public release. A strong direct replication must not be advertised as universal transport, and failed harmonization must not be reported as defeat of the target when it establishes only that the measurement or implementation bridge failed. The table therefore requires the replication type, object, reference class, and transport scope to be named before the result is summarized.

26.1.1 — Shared Core Protocol

The shared core protocol fixes the minimum design every site implements before local adaptation. It identifies the common estimand, eligibility rule, observation window, measurement channels, endpoint proxy, missingness categories, safety gates, and primary comparison; otherwise, sites can agree on a headline while estimating different objects (Chan et al., 2025; International Council for Harmonisation, 2026).

Eligibility, cadence, endpoint proxies, channels, analysis families, burden stops, data fields, and status labels are frozen here, while

permitted local deviations are routed to site supplements. This division lets later reviewers determine whether a difference arose from context, implementation, or an altered scientific question.

A community hospice and a research hospital both use the same endpoint window but have different nursing documentation systems. The contrast shows that replication depends on implementable common rules, not merely on agreement with the headline model.

A protocol is too loose when every site can later claim to have followed it while estimating a different object. If protocol looseness is not named, it will be absorbed into site freedom and interpreted only after the outcome is known (Chambers, 2013; Scheel et al., 2021).

The core protocol should be narrow, versioned, and difficult to reinterpret after the first interim pattern appears. A narrow, versioned protocol preserves interpretability even when the common result becomes less dramatic.

The shared core creates a clean audit question: did every reported result answer the protocol's declared question? That trace allows later evidence synthesis to distinguish genuine replication from an institutionally convenient change of target.

26.1.2 — Site-Specific Supplements

A site supplement describes how the common protocol is instantiated in one setting, including any adaptation required by law, language, care workflow, infrastructure, or participant burden. Without that record, multisite variation is uninterpretable because the same nominal procedure may not represent the same

intervention or observation (Chan et al., 2025; International Council for Harmonisation, 2026).

It records population and referral pathways, diagnosis mix, care regime, device availability, language, jurisdiction, staffing, family-contact practice, and known feasibility limits. Reproducibility of implementation requires another team to distinguish mandatory, discretionary, and prohibited local choices without reconstructing them from memory.

Consider a rural hospice with intermittent connectivity and an urban hospital with continuous monitoring. The supplement should show whether differences in completeness or timing arise from clinical conditions, infrastructure, or departures from the common protocol, rather than leaving those causes to post hoc interpretation.

Local variation becomes dangerous when it is treated as invisible nuisance and then confused with support, nonreplication, or harm. At that moment, the program should stop upgrading the finding and classify the source of difference (Scheel et al., 2021; Nosek et al., 2018).

The supplement should make local adaptation honorable only when it is declared before it matters. Protection comes from rules that change permissions and labels when local conditions change, not from cautious wording alone.

The remaining issue is transport. Site adaptation is defensible only when the record shows which assumptions were preserved, which changed, and why the resulting evidence may travel.

26.1.3 — Registered-Report Review

Registered-Report review moves the central design judgment to a time when the result is unknown. Reviewers can challenge separability, burden, missingness, endpoint independence, rival coverage, and public-status wording before those choices can be influenced by an observed pattern (Chambers, 2013; Scheel et al., 2021; Center for Open Science, 2025).

The review record should preserve the submitted protocol, reviewer objections, author responses, in-principle acceptance, and any approved amendment. A later auditor can then distinguish a study that followed its accepted design from one whose publication commitment no longer covers the reported analysis.

A journal agrees to publish a carefully designed test even if the result is null or favors a Brownian bridge. That commitment changes analytic incentives before data collection and keeps publication from depending on a preferred outcome.

Outcome-dependent publication pressure remains a threat even in a well-governed network. The publication commitment should therefore extend to null, rival-favored, inconclusive, and unusable outcomes, provided the accepted protocol and quality conditions were met (Center for Open Science, 2025; Nosek et al., 2018).

Review before results should protect the value of a clean loss as much as a clean support pattern. The network must be free to discover that its own architecture is not yet sufficient.

The study answers its promised question, not its preferred later story. Outcome-independent review makes replication a test of that question rather than a contest for an attractive result.

26.1.4 — Primary and Secondary Analyses

Primary estimands, secondary analyses, exploratory diagnostics, and post hoc visualizations receive distinct preregistered labels, each with a date, owner, version, status, and rule for reopening the classification (Chan et al., 2025; Hopewell et al., 2025; Nosek et al., 2018).

A striking subgroup may appear only after the first pooled analysis. The primary-secondary boundary must route that finding to a new prospectively specified confirmatory analysis rather than convert it into a retroactive center of the study.

An exploratory discovery can inherit confirmatory language simply because the pattern is striking or comforting. In a multisite network, that inflation can be mistaken for replication when the apparent coherence is actually produced by unmodeled local practice (Center for Open Science, 2025; Chambers, 2013).

Discovery remains legitimate only when its label travels with it. At this point in the architecture, institutional design determines whether unexpected variation becomes a new question or is shifted onto staff and participants as unexplained work.

The record must separate findings that may alter component status from those justifying another prospective test. Otherwise, the network creates apparent cumulative evidence without independent confirmation.

26.1.5 — Deviation and Amendment Log

Every protocol change belongs in the scientific record. Even a harmless local deviation across hospitals, hospices, remote

programs, or multilingual clinics can alter consent, timing, observation, or endpoint meaning (Nosek et al., 2018; Center for Open Science, 2025).

The log records when and why the change occurred; the affected sites, measures, and participants; the reviewing body; and whether reanalysis is required. It permits transparent adaptation without allowing adaptation to become an unnamed cause of support or failure.

A firmware update that changes sleep-state classification midway through recruitment can reshape two sites without misconduct. The safeguard is to record the altered observation process and protect analysts from a falsely uniform dataset.

A result weakens when staff remember a change that the analysis package omits. The defect is localized, its consequences estimated, and the affected component narrowed or rerun rather than concealed in a pooled estimate (Chambers, 2013; Scheel et al., 2021).

Amendments must be visible enough that no reader has to infer them from drifting outputs. Dated records keep participants, staff, and families from carrying ambiguity merely so the claim appears cleaner.

Replication is an auditable history, not a polished endpoint detached from its making. Trust grows when disagreement is easier to inspect than suppress.

26.1.6 — Publication Independent of Outcome

Institutions commit to publication before results are known. This covers nulls, Rival-favored outcomes, failed instruments, and No adequate estimate under protocol-fixed quality conditions (Center for Open Science, 2025; Chambers, 2013).

The charter, funder agreement, and journal route protect adverse or unusable results, harmonization failures, and ethics suspensions. They specify amendment authority, review triggers, and visible disputed exceptions.

If observation shutdown explains terminal concentration better than the candidate, the result remains publishable under the same quality criteria as a favorable result. Suppressing it would turn a scientific comparison into a selective record.

Narrative-driven suppression is a governance failure, not a participant or hypothesis failure. Preserve findings and narrow supported status (Scheel et al., 2021; Nosek et al., 2018).

Publishing adverse evidence is a duty, not a reputational liability; retaining it protects participants, sites, and critics.

Fix questions, log amendments, and protect publication so comparability tests need not trust consortium memory.

26.2 — Harmonization Without Forced Sameness

Harmonization earns comparability only to the extent that shared meaning has been demonstrated. It begins with a common core while preserving legitimate differences in language, care, culture, devices, disability, and local workflow. A variable is shared when its meaning, unit, timing, context, and missingness behavior support the declared estimand, not merely when its column name matches (Fortier et al., 2011; American Educational Research Association et al., 2014; Wilkinson et al., 2016; International Test Commission, 2018).

This is why measurement invariance belongs before pooling. If a self-report item shifts meaning under translation, if a wearable channel behaves differently across skin tone or movement capacity, if a clinical record field changes with staffing practice, or if a distress marker is altered by medication, then a pooled estimate may become an artifact of harmonization failure. The data may look aligned while the construct has moved.

The analysis distinguishes common data elements from common meaning. A common data element is a field. Common meaning requires context-of-use evidence, bridge samples, linking models, translation review, device calibration, and declared uncertainty. Some fields can be harmonized fully. Some can be harmonized partially. Some should remain local and never enter a pooled confirmatory model.

Forced sameness transfers interpretive work to local staff, translators, patients, families, and disabled participants, then treats the resulting mismatch as error. A fair multisite design

records that load and preserves local meaning instead of demanding conformity to a shape designed elsewhere.

A good harmonization plan is both modest and sharp. It states which variables are common, which are linked, which are local, which require bridge evidence, which are excluded from pooling, and which failures narrow the claim. It lets a dataset travel only as far as its construct evidence travels.

Table 26.4. Harmonization and invariance plan.

Object	Evidence required	Permitted outcome
Common field	Same unit, source type, timestamp, missingness code, and transform	Full pooling only after meaning is also supported
Local field	Site-specific validity with declared use	Local analysis or qualitative interpretation
Bridge variable	Overlap sample, device crosswalk, translation comparison, or linking model	Linked analysis with uncertainty or no-link decision
Device version	Calibration record, firmware log, artifact behavior, and leave-version-out test	Version-specific model, bridge, or exclusion
Language version	Translation review, cognitive interview, cultural review, response-pattern check	Linked, local, revised, or withdrawn item

Object	Evidence required	Permitted outcome
Failed invariance	Declared failure mode and affected estimand	No pooling by administrative convenience

Equation 26.1. Candidate linking model for site-qualified measurement.

$$D_{\text{link}} \left(P(Y_{ij} \in \cdot | \eta_{ij}, s, d), P(Y_{ij} \in \cdot | \eta_{ij}, r, d) \right) \leq \varepsilon_{\text{link}}.$$

26.2.1 — Common Data Elements

A common data element is a field with declared units, timing, source, missingness code, and permitted values. Without a common-data-element boundary, identical field names can conceal incompatible measurements (Fortier et al., 2011; American Educational Research Association et al., 2014). In Equation 26.1, i indexes persons, j indexes the item-or-occasion unit declared by protocol, s is the candidate site, r is the declared reference site, and d is the device or instrument configuration; η_{ij} is the common latent construct value, D_{link} measures discrepancy between the two conditional measurement distributions, and $\varepsilon_{\text{link}}$ is the allowed discrepancy; the latent scale, discrepancy rule, and tolerance are protocol-fixed before pooling.

The data dictionary assigns each field a definition, unit, time basis, source, permissible transformation, missingness code, sensitivity tier, and responsible owner. Those semantics must be reproducible at another site without guesswork about an apparently familiar label.

Two sites may both report sleep interruption while one uses self-report and the other actigraphy. The common-data-element record must retain that distinction, specify whether a bridge is available,

and prevent the shared field name from concealing incomparable observations.

False comparability begins when identical column names are mistaken for common meaning. The program should suspend pooling, identify the semantic or measurement mismatch, and state what additional bridge evidence would be required (Wilkinson et al., 2016; International Test Commission, 2018).

Common fields must carry source and context, not merely labels. The safeguard is a schema that constrains permissible interpretation, not a general declaration of caution.

A data dictionary is a scientific object, not an administrative spreadsheet. It should expose the semantic and operational assumptions that must hold before a variable can cross a site boundary.

26.2.2 — Local Measurement Qualification

Local qualification asks whether a measure behaves adequately in the site's declared context of use, not whether the site can operate the device. Because qualification determines whether observations can enter a pooled or transported estimand, it is part of the scientific design (American Educational Research Association et al., 2014; International Test Commission, 2018).

Pilot reliability, calibration transfer, missingness, participant feedback, device tolerance, and workflow checks should be reported with their acceptance criteria. The resulting record should make clear why a measure was qualified, conditionally qualified, or excluded at that site.

A translated calm item is interpreted as politeness rather than felt relief in one setting. The response has changed because the social act changed, so translation review belongs in measurement qualification rather than cosmetic editing.

If local meaning or performance remains misaligned, pooling is not a neutral default. The defensible result may be a site-specific instrument, a bounded linking model, a separate stratum, or exclusion from the common estimand (International Test Commission, 2018; Fortier et al., 2011).

Qualification should happen before the variable receives confirmatory weight. Excluding an unqualified variable is therefore evidence of discipline, not a failure of ambition.

Local qualification turns harmonization from assumed sameness into earned comparability.

26.2.3 — Bridge Samples and Linking

Bridge samples answer a bounded linking question between instruments, languages, devices, or workflows. Their overlap design specifies what is linked, in which population and context, and with what uncertainty. A bridge that expands whenever comparability fails becomes a rescue device rather than a measurement study (Wilkinson et al., 2016; International Test Commission, 2018).

Linking models estimate transformations, uncertainty, and bias using participants or records measured under more than one regime when ethically feasible. Each measurement bridge requires a version, owner, calibration sample, uncertainty record, and rule for requalification.

During a device migration window, a subgroup may complete both the old and new diary formats. The bridge sample must remain small, consented, and necessary, so dual measurement does not become an unacknowledged burden imposed for administrative convenience.

A cross-site bridge becomes misleading when wide uncertainty is presented as exact conversion. Network scale can then imitate replication even though the common metric is being manufactured by an untested link (Fortier et al., 2011; American Educational Research Association et al., 2014).

A responsible linking output reports uncertainty, nontransportable regions, and no-link zones. These limits determine which observations may enter a common analysis and which must remain site-specific or partially identified.

Only the portion that survives the bridge evidence should enter a pooled estimate. Everything else remains a local finding, a sensitivity analysis, or a new research question rather than being absorbed into the appearance of replication.

26.2.4 — Partial Harmonization

Some variables can be harmonized only partially, and that status must be declared before pooling. A common label may conceal different clinical, cultural, or device meanings across research hospitals, community hospices, remote-monitoring programs, and multilingual clinics (International Test Commission, 2018; Fortier et al., 2011).

The harmonization plan distinguishes a shared core, variables linked with uncertainty, site-specific measures, and observations

reserved for local or qualitative interpretation. This architecture preserves useful local information while preventing an invalid common scale from becoming the price of consortium membership.

A respiratory feature may link across devices while an agitation field remains inseparable from local charting culture. That asymmetry is evidence about the measurement system. The protocol should preserve it in the analysis and the release record.

The result loses warrant if analysts discard meaningful local data or force it into an invalid common metric. The affected component should be reported separately, narrowed, or treated as nontransportable instead of being smoothed into a pooled estimate (American Educational Research Association et al., 2014; Wilkinson et al., 2016).

Partial harmonization protects local signal without upgrading it to pooled status. The decision to link, stratify, or abstain should be visible in the record rather than hidden inside a final model.

The result can now say what traveled and what remained local. That distinction makes heterogeneity an interpretable limit on scope.

26.2.5 — Translation and Cultural Review

Translation and cultural review preserve constructs rather than word forms. The record identifies what was translated, adapted, cognitively tested, rejected, and approved before site outcomes reveal which wording is favorable (Fortier et al., 2011; American Educational Research Association et al., 2014).

Panels should include language experts, clinicians, community advisors, disability perspectives, and field staff, with recorded authorship, revision authority, adjudication criteria, and unresolved disagreement. A burden term may carry family-duty or religious meaning in one language and elicit a different social act despite appearing lexically equivalent.

When wording changes pressure or response process, the defect belongs to measurement and transport rather than culture or character. The instrument is adapted, the link narrowed, or a no-link region declared; untranslatable constructs remain visible rather than being forced into a pooled score (Wilkinson et al., 2016; International Test Commission, 2018).

Cultural review should make meaning more visible. Its decisions are part of the scientific record because they determine whether common-looking measures are actually commensurable. Documenting translation choices and unresolved concepts helps later sites distinguish true cultural differences from instrument noninvariance.

26.2.6 — No Pooling by Administrative Convenience

Administrative convenience is not comparability. Pooling is allowed only after declared invariance, device-bridge, support, and transport gates make the combined estimand interpretable; the plan states when site-specific reporting replaces a common estimate (American Educational Research Association et al., 2014; Wilkinson et al., 2016). A harmonization decision should record which parameters were linked, which remained site-specific, and which comparisons were prohibited.

When funders expect one headline despite failed bridges, the scientific response is a qualified common result beside incompatible site results. Forcing them together turns administrative preference into an undeclared model and hides the defect from the main analysis (International Test Commission, 2018; Fortier et al., 2011). Presenting incompatible results beside the common estimate can reveal whether the pooled signal is robust or driven by a subset of sites.

No pooling is a valid result. It preserves where comparability ended and lets heterogeneity reveal measurement drift, transport boundaries, care-regime interactions, or distinct processes requiring their own tests. No pooling can also improve the next design by identifying the exact bridge or site condition that must be validated before combination becomes legitimate.

26.3 — Site Effects, Heterogeneity, and Transport

Heterogeneity is where a multisite study discovers the limits and transport conditions of its own claim. Site differences can arise from population, care practice, device configuration, laboratory procedure, observation intensity, law, language, or referral. The analysis treats those differences as possible effect modifiers, measurement failures, or scope boundaries rather than noise to erase (Gelman & Hill, 2007; Bareinboim & Pearl, 2016; Stuart et al., 2011).

A hierarchical model can help, but only if it is interpreted honestly. Partial pooling is not a machine for dissolving dissent. It estimates site-level variation while borrowing strength under assumptions. If those assumptions fail, the model may make an outlying site look like noise rather than a boundary. Accordingly, the analysis treats heterogeneity as possible scope evidence.

The 2026 SCORE results also show why site and method variation should be interpreted rather than averaged into one credibility score. A claim may be computationally reproducible from its original files, analytically fragile under other defensible specifications, and only partly replicable in independent data; another field may show stronger reproducibility while still carrying transport limits. Each pattern points to a different repair, including provenance reconstruction, specification review, direct replication, or reference-class narrowing (Miske et al., 2026; Aczel et al., 2026; Tyner et al., 2026; Brodeur et al., 2026).

Transport is different from replication. A result can replicate in the development network and still fail to transport to another

population, diagnosis, age group, language, care regime, device version, or sampling cadence. Transport requires a target population, covariate structure, measurement invariance evidence, treatment regime assumptions, and a reason to believe the mechanism should apply across contexts.

The multisite system identifies where a component result transfers and where a new qualification study is required. Extensions from oncology to dementia, pediatric settings, other species, or artificial architectures each specify their own measurement and bearer-classification questions.

The section's central promise is therefore restrained: heterogeneity is not a public relations problem. It is information about the boundary between an observed pattern and the conditions under which that pattern can be responsibly named.

Equation 26.2. Candidate hierarchical site model.

$$\hat{L}_{ijsvt} = \mu + a_i + b_j + c_s + d_v + x_{ijsvt}^T \theta + \varepsilon_{ijsvt}$$

Equation 26.3. Transport estimand under declared target population.

$$\psi_T := \int m(x) dP_T(x), \quad w(x) := \frac{dP_T(x)}{dP_S(x)},$$

$$P_T \ll P_S.$$

26.3.1 — Hierarchical Site Models

Hierarchical site models specify how person, site, workflow, device, population, and calendar variation enter the analysis. Because that structure controls pooling and shrinkage, it is declared and tested as part of the generative model rather than added after a surprising site result appears (Gelman & Hill, 2007;

Bareinboim & Pearl, 2016). In Equation 26.2, i indexes person, j indexes workflow, s indexes site, v indexes device or version, and t indexes observation occasion or calendar time; a_i , b_j , c_s , and d_v are their declared hierarchical effects, θ is the covariate coefficient vector, and ε_{ijstv} is the residual term. In Equation 26.3, $m(x)$ is the conditional estimand function, P_S and P_T are the source and target covariate laws, and w is the Radon–Nikodym transport weight; if P_T is not absolutely continuous with respect to P_S on the analysis support, that weighting route is unavailable.

The model record should explain which effects are fixed, random, partially pooled, interacted, or excluded, and why. It should also state the site sample sizes and assumptions that make each level estimable, so later analysts can reproduce the hierarchy rather than merely the final coefficient table.

One site may show strong variance compression while another retains substantial dispersion. A pooled average could erase that boundary and turn a scope result into a misleading global pattern.

Partial pooling can obscure a genuine boundary when a site differs systematically from the network. In that case, a site-conditioned estimate, transport analysis, or explicit nonpooling decision is more informative than a cosmetically stable network average (Stuart et al., 2011; Fortier et al., 2011).

Site terms should be reported as possible scope evidence. Variation among sites can then reveal where a claim travels and where it stops rather than being treated as a nuisance to suppress.

A reader can see whether the component claim is network-wide or site-conditioned. Hierarchical modeling turns heterogeneity into a scientific result rather than a defect to be averaged away.

26.3.2 — Care and Workflow Effects

Care practice can shape observations that resemble psychological dynamics. The analysis isolates workflow contrasts, records their generating processes, and keeps clinical variation out of a generic site effect. Otherwise, a pooled model can misname a care difference as a Law-specific signal (Bareinboim & Pearl, 2016; Stuart et al., 2011).

The design records sedation norms, pain-management routes, visit schedules, family access, documentation habits, and staff ratios. Each care-workflow variable requires a time window, owner, definition, provenance, and rule for remeasurement.

A site with proactive symptom management may record fewer distress reports near endpoint than a site with late medication adjustment. Workflow contrasts belong in the study record, but care decisions remain clinically governed and never contingent on research participation.

Include care-workflow variables in the model so that differences in clinical practice can be separated from the proposed latent dynamics (Fortier et al., 2011; Gelman & Hill, 2007).

Workflow variables should enter the analysis before mechanism language does. Medication practice, staffing, visit timing, family involvement, device use, and referral patterns can change what is observed and which continuations remain available.

The analysis must be able to distinguish a humane clinical practice effect from a Law-specific signal. When it cannot, the appropriate result is a care-regime boundary or model ambiguity, not a stronger claim produced by network scale.

26.3.3 — Device and Laboratory Effects

Hardware, software, preprocessing, and calibration can create site signatures resembling substantive effects. Replication therefore needs a versioned device and laboratory record; a shared product name does not ensure a shared observation process (Stuart et al., 2011; Fortier et al., 2011).

For every site, the record identifies device model, firmware, sensor placement, sampling rate, artifact filters, laboratory procedure, and preprocessing version. Those fields make technical variation analyzable instead of allowing it to appear later as unexplained heterogeneity.

Suppose a wearable update changes motion-artifact removal during the final month of recruitment. The study has changed even if every operator followed instructions. The update should define a new device regime, trigger bridge checks, and remain visible in every pooled or transported result.

Apparent variance compression or failure loses force when it can be manufactured by a device change. The affected observations should be quarantined, bridged with explicit uncertainty, or reported separately, and the component claim should be narrowed if the technical break cannot be repaired (Gelman & Hill, 2007; Bareinboim & Pearl, 2016).

Leave-device-version-out prediction should be part of evaluation whenever firmware, sensors, or preprocessing may alter the observation law. A pooled device effect cannot substitute for demonstrated transport.

Evidence may cross a device boundary only after the declared bridge survives held-out testing. Until then, device-specific results remain device-specific.

26.3.4 — Population and Referral Effects

Population and referral effects begin before enrollment. Catchment area, referral criteria, clinician practice, insurance or access constraints, family capacity, and self-selection determine which streams reach a site and therefore which target population the site can represent (Fortier et al., 2011; Gelman & Hill, 2007).

The design models referral pathways, diagnosis, prognosis, socioeconomic access, language availability, caregiver presence, prior treatment, and selection into research. The sampling record should identify who defined eligibility, who may revise it, and how referral disputes affect scope.

A palliative unit enrolls people with stable family advocates while a safety-net hospital sees late referrals and fragmented records. The resulting reference class has changed through referral and retention, not through biology alone.

Without referral modeling, sample composition can be mistaken for biological universality. When referral structure produces the pattern, the claim must remain conditioned on the population and care pathway that generated it (Bareinboim & Pearl, 2016; Stuart et al., 2011).

Transport claims should name the populations actually represented and the populations that remain outside support. Scope becomes more credible when its exclusions are explicit.

26.3.5 — Transport Estimands

Each transport estimand fixes population, setting, care and device regimes, endpoint window, and inference before evidence crosses contexts (Gelman & Hill, 2007; Bareinboim & Pearl, 2016).

The record shows which differences were adjusted, ignored, unsupported, or assumed stable. Transport is an auditable inferential route, not rhetorical extension of a local success.

Moving a hospice telemetry model into home monitoring changes population, cadence, medication, caregivers, and endpoint structure. Declaring those changes makes transport testable.

Results beyond context are labeled extrapolation, carry transport uncertainty, and withhold broader claims when overlap or invariance is inadequate (Stuart et al., 2011; Fortier et al., 2011).

Transport requires covariate support, measurement comparability, and sensitivity to unmeasured context. Failure identifies the boundary the next design must address.

A transported finding and a question exposed by failed transport remain separate states.

26.3.6 — Heterogeneity as Scope Evidence

Heterogeneity can reveal dependence on measurement, diagnosis, care, device, culture, or an ordinary rival, defining the scope of later claims (Bareinboim & Pearl, 2016; Stuart et al., 2011).

Site differences are classified as measurement failure, mechanism boundary, rival advantage, care-regime effect, or unresolved disagreement under a rule staff apply and critics audit.

If a minority site favors a Doob-transform rival, its observation process and rival fit remain in the principal result rather than disappearing for narrative cleanliness.

An adverse site triggers narrowing, reanalysis, quarantine, or withdrawal under protocol-fixed criteria, not selective exclusion (Fortier et al., 2011; Gelman & Hill, 2007).

Site estimates, prediction intervals, and explanations remain visible beside the pooled result under independent oversight.

26.4 — Independent Replication Roles

Independence in replication is a distribution of roles, information, incentives, and decision rights. A different institutional address does not by itself create independence. The design separates theory stewardship, implementation, data custody, analysis, rival advocacy, endpoint adjudication, ethics review, and public communication so that one group cannot control the full evidentiary chain (Mellers et al., 2001; Nosek & Errington, 2020).

The role architecture gives scientific pressure a lawful destination. The originating team supplies theory, code, assumptions, and clarification; implementing sites carry the protocol; staged analysts receive governed data; rival stewards strengthen alternatives; endpoint and ethics reviewers protect high-risk classification; and conflict disclosures expose funding, authorship, advocacy, and institutional stakes.

Independence separates the responsibilities that shape an inference. Endpoint leakage, analysis leakage, funding pressure, authorship incentives, public reputation, and advocacy identity can influence interpretation. Distributed authority makes favorable results more credible by giving alternatives and possible failure routes independent evaluation.

Adversarial replication belongs here as a contract rather than a performance. Critics specify rivals and failure tests before benchmark release, independent sites retain publication rights, and negative results do not require permission from the originating team. Funding protects adverse findings rather than only approved extensions.

A replication system with independent roles also protects people. When endpoint classification is disputed, when a family feels pressured, when a site is overburdened, or when a data incident occurs, there must be a body with authority to suspend, narrow, or disclose without asking the theory's advocates to approve the discomfort.

Table 26.5. Independent-analysis workflow.

Workflow stage	Who sees what	Purpose
Fixture verification	Analysis team receives synthetic fixtures and known-answer cases only	Confirms code, formulas, and container behavior before real data access
Quality and missingness review	Data quality team sees site-coded data without outcome interpretation	Detects unusable channels, timing failures, device drift, and missingness patterns
Primary blinded analysis	Blinded analysts receive locked analysis dataset with masked site identities where feasible	Runs declared estimands without narrative and endpoint leakage
Rival model analysis	Rival stewards receive the same observation model and access constraints	Gives alternatives a fair chance under identical data conditions
Staged unblinding	Independent adjudication body releases site, device, and endpoint labels in predeclared order	Tests whether interpretation changes when context is revealed

Workflow stage	Who sees what	Purpose
Public interpretation review	Release group sees results, deviations, incidents, rival outcomes, and correction obligations	Converts record into status-labeled public language without upgrading claim status

The workflow matters because independence can be lost in small leaks. A message about which site had the most public attention, which family withdrew data, which device failed, or which endpoint window was controversial can shape analytic choices before anyone calls it bias. Staged access gives the analysis enough information to be correct while withholding enough context to keep the result from becoming a story too early.

Table 26.6. Adversarial replication contract.

Role	Protected right	Boundary
Originating team	Clarify theory, code, fixtures, and assumptions	Does not own replication outcomes or unblinded interpretation
Implementing site	Report feasibility limits, deviations, and local harms	Does not alter confirmatory estimands silently
Blinded analyst	Execute primary and rival analyses under staged access	Does not receive endpoint or site narratives before declared gates
Rival steward	Strengthen serious alternatives and publish their performance	Evaluates models independently of personal or institutional loyalty

Role	Protected right	Boundary
Endpoint and ethics review	Suspend, quarantine, or narrow evidence	Does not become public verdict authority over persons
Independent funder or sponsor	Support adverse and null publication	Does not upgrade public language by funding the work

26.4.1 — Originating Team

The originating team supplies the formal object, code, assumptions, synthetic fixtures, and clarification history, but it does not own the replication verdict. Separating authorship from adjudication prevents theory knowledge from becoming outcome control (Mellers et al., 2001; Nosek & Errington, 2020).

Versioned specifications, known failure conditions, analysis notes, and test fixtures should enter an independent repository before unblinded replication results are available. The record then shows what the team committed to and which later clarifications changed implementation.

The original modelers may answer a site question about endpoint windows while remaining shielded from unblinded outcomes. The clarification is logged, distributed to every affected site, and assessed for whether it changes the protocol-fixed analysis.

Clarification becomes steering when the team can infer which answer protects its model or when only favorable sites receive interpretive help. Such contact should trigger disclosure, sensitivity analysis, or a status reduction rather than being folded into vague implementation uncertainty (Ioannidis, 2005; Center for Open Science, 2025).

Originating-team contact should therefore be outcome-shielded, time-stamped, and reviewable. The purpose is not distance for its own sake, but a visible boundary between legitimate clarification and adaptive rescue.

This arrangement keeps replication connected to the theory without making it dependent on the theory's authors. The originating team can explain the claim, while independent implementers, analysts, rival stewards, and reviewers determine what the evidence permits.

26.4.2 — Independent Implementing Sites

Implementing sites carry the shared protocol through local staffing, law, culture, devices, care workflows, and participant needs. Their independence matters because feasibility limits and local departures change the scope of any apparent replication (Nosek & Errington, 2020; Ioannidis, 2005).

Each site signs a qualification record covering staffing, supervised training, calibration, consent routes, burden thresholds, stop authority, and the local conditions under which the protocol cannot be performed. Replication begins only when those conditions are operationally credible.

A small hospice may decline a sampling cadence that would displace nursing care. That refusal is not a defect to be hidden; it is evidence about the intervention's support and the settings in which the protocol can be conducted without transferring burden to staff or families.

If a site enrolls beyond its support capacity to satisfy network targets, the resulting data are compromised by the pressure the

system was designed to control. The protocol-fixed response is status reduction, reanalysis, quarantine, or withdrawal of the affected component (Center for Open Science, 2025; Mellers et al., 2001).

Site independence includes authority to reject unsafe feasibility demands and publish local limitations without penalty.

26.4.3 — Blinded Analysis Teams

Analysis teams receive data in stages that reduce endpoint and interpretation leakage. Without staged blinding, narrative knowledge can enter the analysis before formal comparison begins (Ioannidis, 2005; Center for Open Science, 2025).

The workflow separates data cleaning, quality review, primary analysis, rival analysis, and public interpretation through staged unblinding. The blinding log should show when each team gained access to site identity, endpoint labels, treatment or exposure information, and interim results, because those permissions shape the analyses that can remain independent.

Analysts run the prospectively specified model on site-coded data before learning which site had a public controversy. The coding sequence preserves blinding and prevents contextual knowledge from entering model comparison before the confirmatory analysis is complete.

Contextual leakage can enter the analysis through selective diagnostics or unexplained analyst judgment (Mellers et al., 2001; Nosek & Errington, 2020).

The final analysis should be challengeable as a route from data to conclusion, not merely reproducible as a numerical output. That distinction gives later evidence synthesis a record of how the result was made.

26.4.4 — Rival Model Stewards

Serious rivals need representation by people allowed to make them strong. Without rival stewardship, the network can appear adversarial while protecting the favored model (Center for Open Science, 2025; Mellers et al., 2001).

Rival stewards prospectively specify adaptation, homeostatic, bridge, frailty, sedation, observation-shutdown, predictive-control, and selection models with the same data access and tuning budget afforded to the focal route. Their specifications must support independent implementation and identify which PTVC-linked signature each rival can reproduce without the Law.

A rival steward may insist that a random-endpoint bridge remain a primary comparator when the originating team regards it as secondary. The adjudication record should preserve the disagreement, define the common scoring rule, and show whether any observed separation survives equal information and flexibility.

Rival comparison loses force when the Law competes only against alternatives designed by its supporters to be weak. The response is to strengthen the rival set and rerun recovery under equal information and flexibility, not to upgrade the favored result (Nosek & Errington, 2020; Ioannidis, 2005).

Rival stewardship should include code review, parameter ranges, diagnostic attention, and publication rights. The safeguard is genuine authority for rival teams to strengthen, test, and publish their models.

26.4.5 — Independent Endpoint and Ethics Review

Endpoint and ethics review remain independent of the discovery team. These reviewers decide what may be collected, which stop rules apply, and whether an endpoint classification is usable, thereby shaping the evidentiary object itself (Mellers et al., 2001; Nosek & Errington, 2020).

The file records evidence, unity and endpoint criteria, burden, family-pressure controls, dissent, recusal, and reasons for approval, restriction, or suspension so later decisions are auditable.

A family's withdrawal of final-week recordings changes both permission and endpoint coverage and is represented as an ethical and inferential event, not routine missingness.

The appropriate outcome may be withdrawal without penalty, No adequate estimate, or a narrower window even when retention would improve precision (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Endpoint review can halt classification or quarantine data when a cleaner label would exceed the evidence.

Participant protection and endpoint discipline are scientific preconditions, not team preferences.

26.4.6 — Conflict-of-Interest Disclosure

Conflict analysis identifies incentives that can shape interpretation, including funding, authorship, advocacy, prestige, and public identity. The record shows affected decisions and independent limits (Nosek & Errington, 2020; Ioannidis, 2005).

Disclosures name grants, donor commitments, intellectual property, authorship, public roles, prior advocacy, date, decision scope, reviewer, mitigation, and review trigger.

When a funder requests release before the disagreement appendix is ready, site leads enforce disclosure and recusal while shielding participants and junior staff from pressure.

Disclosure must change workflow when influence is plausible; otherwise it becomes an appendix while incentives continue to shape analyses, framing, and release timing (Center for Open Science, 2025; Mellers et al., 2001).

Mitigation can require recusal, staged access, independent review, or a documented reason to retain the role.

Coordination becomes replication only when dissent and adverse findings retain institutional standing.

26.5 — Repositories, Provenance, and Access

A repository becomes part of the inferential chain when it preserves the exact objects needed to reproduce, audit, reinterpret, or withdraw a result. Data dictionaries, provenance, code versions, containers, model cards, access decisions, retention rules, and correction links connect a public claim to the process that produced it (Wilkinson et al., 2016; Barker et al., 2022; Grant et al., 2026).

Scientific memory requires clear purpose limits when it includes sensitive material. The same system that allows a critic to inspect methods can expose clinical context, family speech, disability status, endpoint-adjacent recordings, or reidentification risk. The repository therefore separates public, controlled, and restricted layers and records why each layer exists.

Provenance makes the path from observation to result inspectable. A result records the source of each datum, its transformations, code and device versions, responsible role, quality flag, access tier, and reason for inclusion or exclusion.

Federated or distributed analysis can help when data cannot be pooled. It reduces transfer requirements and can respect legal or institutional boundaries. Harmonization, invariance, site qualification, and endpoint discipline establish scientific comparability, while separate checks establish security.

The section also protects record-regime separation. Research repositories are not civic records, government records, Workgame records, religious records, donor records, public archives, or Artificial Intelligence training corpora. The point of a repository

is to support scientific answerability, not to build a permanent cross-architecture dossier of persons.

Table 26.7. Data dictionary and provenance fields.

Field group	Required entries	Access tier implication
Identity and linkage	Pseudonym, site, visit, time, consent scope, withdrawal status	Restricted or controlled depending on reidentification risk
Measurement source	Instrument, device, firmware, reporter, clinical system, language version	Controls harmonization and bridge requirements
Transformation	Cleaning rule, artifact filter, imputation flag, model version, quality flag	Required for reproducible analysis
Provenance	Creator, reviewer, timestamp, software, container, release ID	Required for audit and correction
Sensitivity	Clinical, family, endpoint-adjacent, biometric, free text, location	Determines public, controlled, or restricted layer
Downstream use	Paper, figure, model card, public summary, participant return	Determines correction propagation

Table 26.8. Federated-analysis workflow and security gates.

Federated stage	Minimum scientific gate	Privacy or security gate
Local data preparation	Local dictionary, unit checks, missingness codes, device versions, and provenance map match the shared schema	Raw identifiers remain local and access is logged under site governance
Synthetic fixture run	Each site executes known-answer cases before touching sensitive data	Only synthetic records travel during code validation
Local model execution	Same container, same model version, same preprocessing declaration, and same failure flags are used	Outputs are reviewed for small-cell, leakage, and reidentification risk
Aggregation	Site-level estimates, uncertainty, convergence flags, and quality exclusions are combined under a declared meta-analytic model	No participant-level data leave the site unless separately approved
Central versus local sensitivity	Central pooled analysis, where lawful and ethical, is compared with federated estimates	Any discrepancy triggers access review and scientific reanalysis before public release

Federated stage	Minimum scientific gate	Privacy or security gate
Audit and retirement	Logs, release IDs, failures, and incident decisions are preserved for future reproduction	Deprecated code, compromised outputs, and unsafe artifacts receive retirement notices

Federated analysis provides an access architecture when legal requirements, trust, reidentification risk, clinical sensitivity, or community agreements favor local data control. Its scientific value depends on harmonization that makes the local estimates meaningfully comparable.

Federated work passes two independent gates. The scientific gate covers common schema, local measurement qualification, synthetic-fixture success, container equivalence, convergence reporting, and site-level uncertainty. The privacy and security gate covers local control of sensitive records, access logs, small-cell review, output inspection, breach response, and artifact retirement. Public claims require both.

A practical example shows why this belongs in the main text. A hospice network may be legally able to share model outputs but not raw final-week audio, device traces, or family notes. Another site may have community agreements that permit local analysis but not central transfer. The research system can respect those boundaries and still remain answerable if the code, fixtures, metadata, quality flags, and release identifiers are shared in a form critics can inspect.

Scientific comparability needs verification alongside privacy protection. Local sites may run nominally identical containers yet differ in preprocessing, time-zone handling, endpoint-window construction, or device-version bridges. The response is to align those operations or report estimates at the scope for which they are comparable.

A well-built federated workflow combines participant privacy with scientific accountability. Sensitive material stays local while shared methods, diagnostics, and release records permit independent review.

26.5.1 — Data Dictionary and Provenance

The data dictionary records what a field is and provenance records how it came to be. If data provenance is missing, the result can become unrecoverable before the final table is built (Wilkinson et al., 2016; Barker et al., 2022).

Each variable carries source, instrument, unit, timestamp, site, device version, transformation history, quality flag, sensitivity tier, and responsible owner. These fields form the chain of custody from observation to analysis and allow a later reviewer to determine whether two nominally identical values were actually produced in the same way.

An actigraphy-derived sleep feature is recomputed after a preprocessing update. Provenance must reveal that the derived variable changed, because the same field name now refers to a different transformation.

The final table reports the feature without its transform history. If transform history is not named, the discrepancy will be treated as

software noise or an irreproducible anecdote (National Institutes of Health, 2026a; National Institutes of Health, 2026b; Near et al., 2025).

Provenance should make reanalysis possible when any upstream object changes. Provenance preserves interpretability when a revised pipeline weakens or reverses the published estimate.

The provenance chain must carry a public claim back to its source records without exposing private content unnecessarily. Provenance and privacy are therefore designed together rather than traded against one another after collection.

26.5.2 — Public, Controlled, and Restricted Tiers

Access follows sensitivity and reidentification risk. Without access tiers, data circulation can widen privacy risk without increasing evidentiary clarity (Barker et al., 2022; National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Public tiers may contain protocols, synthetic fixtures, code shells, aggregate results, and status labels; controlled tiers may contain vetted derivatives; restricted tiers hold identifiable or otherwise sensitive records. The access register should state the eligibility, purpose, approval, logging, retention, and revocation rules for each tier (National Institutes of Health, 2026b).

When a journalist requests raw final-week device streams after a public result, the decision turns on privacy, consent, reidentification risk, and scientific necessity, not on publicity value. A refusal or controlled alternative should be recorded so that access discipline remains visible rather than appearing as selective secrecy.

Openness becomes coercive when it pressures the consortium to release material whose sensitivity, reidentification risk, or consent basis requires restriction. The correct response is a more precise access tier and auditable review, not a weaker privacy boundary (Near et al., 2025; Rieke et al., 2020).

Access tiers should be declared before attention arrives. Protection depends on enforceable access controls and audit trails, not merely a careful privacy statement.

Transparency remains compatible with privacy because different objects travel through different access tiers. Public metadata, controlled data, executable code, and restricted human records need not share one door.

26.5.3 — Code and Model Registry

Stable identities for code and models are part of reproducibility. The registry binds each result to the executable analysis, software environment, dependencies, model card, test fixtures, and release identifier so a future analyst can reconstruct the pipeline instead of inferring it from prose (Near et al., 2025; Wilkinson et al., 2016).

The registry links the manuscript, public preregistration record when one exists, model code, containers, dependency files, model cards, data schema, and release identifiers. It also records deprecations, security patches, and environment changes that can alter output even when the analytic script is unchanged.

An analyst reruns the model six months later and obtains a different result because a package changed. The executable

environment has therefore altered the result-producing object even though the statistical code looks unchanged.

If a public figure cannot be reproduced from its recorded environment, the figure is not merely inconvenient. It should be quarantined from active use until the code path is restored, independently rerun, and reconciled with the published result (Rieke et al., 2020; Wilkinson et al., 2016).

Every released claim should name the exact artifact set that produced it. That provenance makes the claim auditable, reproducible, and correctable when one element changes.

The registry turns reproducibility from a slogan into an executable, versioned relationship among the claim, code, model, data schema, and environment.

26.5.4 — Retention and Deletion

Retention policy must define a bounded scientific and legal purpose rather than promise permanent accumulation. Durability, participant rights, deletion obligations, provenance needs, and future reproducibility have to be reconciled before records are stored, not after a request for removal arrives (Near et al., 2025; Rieke et al., 2020).

The plan defines retention periods, withdrawal routes, deletion feasibility, derived-data handling, legal holds, and public correction when data are removed. Each retention rule requires an owner, legal basis, review date, deletion route, and exception record.

A participant may withdraw permission for a sensitive audio channel after the study closes. Data stewards must follow the protocol-fixed withdrawal and deletion route while stating honestly which copies, derivatives, or lawful retention duties prevent complete erasure.

Consent cannot be treated as frozen merely because analysis has begun. New uses, new linkage, longer retention, wider access, model training, and public release may exceed what participants were told even when the repository remains technically secure (Wilkinson et al., 2016; Barker et al., 2022).

Retention rules should be understandable before enrollment and enforceable after publication. They should identify deletion limits, archival duties, derived-data treatment, withdrawal consequences, and the conditions under which a new use requires renewed review or consent.

Scientific memory remains accountable only when people can understand what persists, what can be withdrawn, and what cannot be recalled from already completed analyses. An archive without those boundaries becomes a record system with no meaningful exit.

26.5.5 — Breach and Vendor Risk

Third-party services are part of the evidence path. Cloud providers, device vendors, identity systems, analytic platforms, backup services, and model hosts can alter access, provenance, availability, and reproducibility, so their roles must be governed as scientific dependencies (Rieke et al., 2020; Wilkinson et al., 2016).

The threat model names each vendor, credential path, export route, backup medium, subcontractor, and failure mode. It also states which raw-to-derived mappings must remain recoverable if a service changes terms, closes, or suffers an incident.

Suppose a wearable vendor disables a data-export route during follow-up. Even perfect local compliance cannot repair the lost provenance after the fact. The incident should trigger a documented continuity plan, a scope review, and, when necessary, suspension of the affected analysis.

Vendor fragility undermines warrant when reanalysis needs an unavailable raw-to-derived mapping. Isolate the damaged component and narrow its scope rather than preserve it through pooled assumptions (Barker et al., 2022; National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Vendor contracts should require exportability, incident notice, deletion support, and provenance preservation. Repository reliability includes the exit path, because a program cannot remain correctable if its records are trapped.

26.5.6 — No Operational Record Merger

Keep research records separate from civic, religious, governmental, Workgame, archival, donor, and ordinary-life records. Establish purpose limits, access controls, retention, and deletion before secondary reuse is proposed; later promises cannot fully undo a merged dossier (National Institutes of Health, 2026b; Barker et al., 2022).

Separate operational records, research records, public summaries, model-training corpora, and participant communications by

purpose and access. Register each regime's owner, permitted links, and authority to reopen them.

A public dashboard team asks to reuse study metadata to illustrate country readiness. Without a new authority and consent basis, that reuse converts a research record into an operational dossier and changes the purpose for which the data were collected.

Research participation silently becomes civic visibility. That purpose migration is a governance failure, calling for separation, deletion where feasible, and a documented purpose-limit incident rather than wider use (National Institutes of Health, 2026a; National Institutes of Health, 2026b; Near et al., 2025).

Purpose separation should be enforced in schema, contracts, access logs, and public wording. Record separation protects participants while preventing administrative reuse from becoming unconsented inference.

26.6 — Staff Burden and Participant Support

Replication depends on people whose labor is easy to hide. Coordinators, clinicians, translators, data stewards, technicians, participant advocates, and family-facing staff determine whether the protocol can be carried without degrading care or distorting the observation process. Staffing, training, workload, emotional strain, fair credit, referral capacity, and stop authority therefore belong inside the validity model.

Staff capacity is part of measurement design. It shapes fidelity, missingness, timing, quality flags, incident detection, consent quality, family communication, and dropout. Delicate research tasks need protected time and support within the shift. The study should budget attention as carefully as it budgets sensors.

Participant support has a different danger. Help should not be conditioned on research participation, and research participation should not become a disguised route to care. If the program detects distress, confusion, device burden, family pressure, or clinical need, the referral pathway must operate without turning the person into a better data source.

Family and caregiver burden matters because terminal, disability, and chronic-illness research often reaches beyond the participant. Repeated explanations, device troubleshooting, diary reminders, withdrawal decisions, and post-study summaries can fall on relatives or caregivers. Those tasks are part of the observation process. They cannot be hidden in a consent form.

Humane operation is a validity condition. A study that exhausts staff, pressures families, or turns support into dependence cannot

responsibly produce a clean public status. The scientific record therefore shows how people were protected while evidence was gathered.

Table 26.9. Operational burden metrics that affect validity.

Metric	Local measure	Scientific consequence
Staff time	Minutes per participant per shift, after-hours work, rework episodes	Predicts fidelity drift and missingness
Staff distress	Debrief requests, ethical concerns, role reassignment, sick leave signals	Triggers support and possible suspension
Care displacement	Missed or delayed care task, interrupted clinical action, urgent override	Marks data as ethically and scientifically suspect
Participant observation load	Device wear time, diary prompts, interview length, withdrawal requests	Alters sampling and missingness model
Family or caregiver load	Proxy reports, troubleshooting episodes, repeated explanations	Changes feasibility and consent quality
Credit and authorship	Named contributions, site outputs, community review, data-curation work	Prevents extraction of invisible labor

26.6.1 — Research Labor and Training

Research labor includes expertise, time, emotional work, translation, technical troubleshooting, and ethical judgment. A multisite protocol cannot assume that attendance at a training session makes those capacities interchangeable across settings (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979; Council for International Organizations of Medical Sciences, 2016).

Training should cover protocol logic, endpoint caution, consent, device handling, data-entry standards, incident triggers, family communication, and stop authority, with supervised practice and periodic refresh. The record should show competence in the actual tasks, not merely completion of a module.

A new coordinator given a complex endpoint script without supervised practice may improvise in ways that change recruitment, explanation, and classification. The failure is systemic even when the coordinator acts conscientiously; the protocol has asked for judgment it did not prepare or support.

When staff improvise explanations that alter who enrolls or how people respond, the result no longer represents a common procedure. The network should record the divergence, assess its effect, and narrow or rerun the component instead of burying the change in a pooled estimate (World Medical Association, 2024).

Protocol fidelity begins with people who understand the purpose and limits of the work they are asked to do. Compliance without comprehension can reproduce forms while losing the scientific object.

26.6.2 — Clinical Workflow

Clinical workflow is a scientific constraint because study tasks can displace care, alter timing, and change who remains observable. Enrollment, reporting, device placement, interviews, and incident follow-up must therefore fit around clinical duties rather than compete with them (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Workflow analysis maps when observations occur, who performs them, what clinical duties compete, and which tasks stop during care pressure. The workflow record should identify clinical ownership, research handoffs, suspension triggers, and unresolved staff objections.

A nurse is asked to collect a nonurgent research measure during a pain-management crisis. Repetition can turn a small research request into assumed clinical duty, shifting burden toward staff and patients at the moment care should dominate.

Research presence quietly becomes another load on an already strained shift. Once research work displaces care, the protocol has crossed its burden boundary and should pause, narrow, or suspend while the incident is recorded (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979).

Care interruption should be a protocol deviation and a potential suspension trigger. Care-first suspension protects participants and staff while preventing workload from being hidden inside study success.

The program earns data only when care remains prior in the actual workflow. A protocol that succeeds by displacing attention from participants has failed its own validity test.

26.6.3 — Participant Support and Referral

Participant support must remain available without creating dependence on continued research participation. Distress escalation, clinical referral, device-off options, explanation, and ordinary care access should protect the person rather than reward data completion (World Medical Association, 2024).

The protocol specifies who responds to distress, how referrals occur, what support continues after pause or withdrawal, which research staff may interpret findings, and where ordinary care remains independent of the study. These routes are part of validity because they shape participation and missingness.

A participant may keep wearing a burdensome device because the study team has become the only reliable source of explanation. That dependence should trigger support transfer and a renewed participation choice, not be treated as evidence of willingness or adherence.

Support becomes coercive when access to explanation, referral, or ordinary resources appears contingent on remaining observable. The remedy is to separate support from data collection, record the resulting missingness honestly, and reduce the claim if the supported sample changes (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979; Council for International Organizations of Medical Sciences, 2016).

Referral and explanation should continue when a participant pauses or withdraws. That continuity makes exit real and prevents welfare provisions from functioning as retention incentives.

Participant welfare is not a means of preserving data density. Skipped measures, device-off periods, withdrawal, and burden stops remain legitimate outcomes even when they reduce precision or end a planned analysis.

26.6.4 — Family and Caregiver Burden

Families and caregivers may carry the practical labor of observation through reminders, translation, device troubleshooting, proxy reports, transportation, and withdrawal decisions. Their burden may affect participation and data quality; this protocol therefore requires that it be recorded and minimized (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979).

The protocol tracks the frequency and duration of those tasks, who performs them, what support is available, and which thresholds suspend research demands. A burden measure that cannot be collected without creating more burden should be replaced by a lighter and more defensible record.

A daughter managing medications may also be asked to complete repeated proxy diaries. Her exhaustion is not an invisible implementation detail; it can alter timing, content, and dropout, and it should never be transformed into apparent participant neutrality.

When caregiver exhaustion becomes missingness and is treated as random loss, the analysis erases the reason the data

disappeared. The affected component should be reanalyzed under informative missingness or withdrawn if the relational burden cannot be represented (Little & Rubin, 2019).

Family-burden thresholds need relief routes, substitution rules, and authority to stop requests. Scientific authority is weakened, not strengthened, when a clean figure depends on unpaid relational labor that the protocol never counted. Making that labor visible improves the observation process and prepares the analysis for staff dissent and moral distress.

26.6.5 — Moral Distress and Debriefing

Staff can experience distress when research meets terminal care, vulnerability, conflict, or public stakes. If moral-distress routes are missing, staff silence can shape the record before any inferential audit detects it.

The support system should provide debriefing, escalation, role reassignment, counseling access, and nonretaliation for ethical concern. Recording uptake, delays, and unresolved distress matters because staff strain can affect retention, consent practice, endpoint judgment, and data quality.

A coordinator worries that a family feels watched but fears being seen as obstructing the study. Staff psychological safety therefore affects whether an ethical concern ever reaches the formal record.

Silence may be mistaken for protocol fidelity when staff lack safe dissent routes. If moral distress is not nameable, silence can be misread as consent to the protocol and confidence in the data.

A research system that responds to staff concerns supports both staff agency and scientific reliability. Moral distress, workflow displacement, and unsafe workarounds should reach people with authority to change or stop the protocol.

26.6.6 — Fair Credit and Authorship

Multisite labor should not be extracted invisibly. Without a credit boundary, invisible labor becomes part of the hidden infrastructure of replication (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Credit rules should cover site investigators, coordinators, data curators, technicians, translators, community advisers, security staff, and contributors to null or negative results. A contributorship matrix can make the actual work visible, distinguish authorship from acknowledgment, and prevent small sites from disappearing behind the coordinating center (Brand et al., 2015).

When a small site supplies the harmonization insight that makes the network interpretable, relegating that contribution to a footnote distorts the scientific record. Authorship and credit review should follow declared contribution criteria and include an appeal route for overlooked labor.

The labor system is inequitable when prestige flows upward while burden, translation, troubleshooting, and emotional work remain local. Credit and resource decisions should expose that distribution before it becomes an authorship dispute or a source of silent protocol drift (Brand et al., 2015; National Commission

for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979).

Authorship and acknowledgment should match intellectual and operational contribution. Fair credit requires contribution rules with consequences, not ceremonial acknowledgments.

Scientific fairness includes the coordinators, clinicians, translators, analysts, and caregivers who make evidence trustworthy. Their labor and judgment belong in the design rather than in acknowledgments alone.

26.7 — Incident, Audit, and Negative-Result Systems

An incident system records events that change safety, data meaning, protocol fidelity, access, or public interpretation. Breaches, device failures, consent disputes, endpoint-classification errors, unplanned deviations, burden stops, and correction delays receive a common triage route without being collapsed into one severity score.

Deviation logs, ethics events, data incidents, null results, replication disagreement, and audit trails therefore belong inside the main scientific record. They should not be hidden in appendices or transformed into generic assurances. A multisite program without incident architecture will eventually improvise under pressure, and improvised correction is often too late for participants and too convenient for the favored claim.

Negative and null results need special protection. A site may find no terminal concentration, no variance compression, no reserve-coupled signature, stronger adaptation, stronger bridge behavior, or uninterpretable missingness. Those outcomes are not merely absent support. They are records that may narrow, defeat, or redirect component claims.

Replication disagreement also deserves structure. A majority result cannot erase a minority site if the minority site has better endpoint independence, better device fidelity, a different population, or stronger rival discrimination. The disagreement may be the scientific finding.

Audit readiness is the final test. The record must reconstruct what was planned, what happened, what changed, who approved it,

which data were affected, what reanalysis occurred, which public claims changed, and what remains unresolved. If that reconstruction is impossible, the program has asked the public to trust memory rather than evidence.

Table 26.10. Incident, audit, and negative-result matrix.

Event	Immediate action	Evidence-status consequence
Protocol deviation	Log, classify severity, notify review role	Reanalysis, exclusion, downgrade, or local-only use
Endpoint leakage	Freeze affected analysis, assess contamination	Repeat blinded analysis or mark unusable
Ethics incident	Support participant, pause affected procedures, independent review	Suspend site, revise protocol, or withdraw evidence
Security incident	Contain, notify, audit access, quarantine data	Reanalyze, restrict, correct, or retract
Null result	Register, publish, compare rivals, assess quality	No support, rival pressure, narrowing, or inconclusive status
Replication disagreement	Preserve site-level result and minority interpretation	Scope restriction or new mechanism test

26.7.1 — Scientific Deviation Audit

Scientific deviations are departures from protocol, code, endpoint definition, analysis version, or data-processing route. If deviation auditing is missing, local changes can alter the evidence long before a pooled model reveals heterogeneity (Ioannidis, 2005).

The deviation audit records severity, affected units, timing, discovery method, decision owner, reanalysis requirement, and public-status effect. Linking each deviation to the analyses it can influence allows reviewers to distinguish implementation drift from substantive heterogeneity.

A site changes the order of self-report items after complaints about fatigue. Even a small ordering change can alter reactivity and missingness in a vulnerable window, making the observation process site-specific.

Classifying deviations by their inferential consequences preserves interpretability even when a deviation weakens the pooled result.

A result should retain the history of deviations, incidents, repairs, and unresolved questions that shaped it. This record lets later reviewers reconstruct what the analysis learned from each change.

26.7.2 — Ethics and Safety Incidents

Ethics incidents include pressure, excessive burden, consent confusion, care displacement, distress, or bystander harm. Without incident categories, safety failures can be absorbed into generic implementation noise (World Medical Association, 2024).

Ethics and safety incident procedures should specify immediate response, suspension authority, participant support, independent review, notification, and communication. The incident file must show what happened, which protections were activated, what

evidence became unusable, and how the site or network changed afterward.

If a family says the study feels like it is watching for death rather than supporting care, the response cannot be reduced to a communications problem. The network should pause the affected procedure, review consent and burden with independent clinical and ethics authority, and distinguish a correctable design failure from an activity that should stop.

Consent cannot close the issue when participants or families experience the study as intrusive, confusing, or displaced from care. The system must treat that concern as safety and validity evidence, with authority to pause, support, and review rather than dismiss it as misunderstanding (World Medical Association, 2024).

Ethical incidents should trigger review before scientific convenience enters. Safety depends on stop authority and incident response, not on a cautious tone in the protocol.

Participant dignity remains a condition of evidence usability. Safety or ethics incidents can invalidate an evidentiary route even when the data appear analytically valuable.

26.7.3 — Data and Security Incidents

Data and security incidents can change access, completeness, trust, and the validity of derived results. Incident handling is therefore part of the research record: containment, forensic review, variable quarantine, reanalysis, and public correction must be linked to the affected evidence rather than left in a

separate compliance file (National Institutes of Health, 2026b; Committee on Publication Ethics, 2025).

The plan preserves containment actions, notifications, access suspensions, forensic findings, compromised-variable flags, vendor decisions, reanalysis, and release effects. The incident record then shows exactly how the usable dataset changed and which dependent results lost, retained, or recovered status.

A staff credential is used to download restricted endpoint-window data from another site. Credential governance now determines lawful and scientific access to the endpoint record, so a security event can also become an evidentiary event.

A technically contained incident may still invalidate an analysis if the breach, outage, or vendor failure altered completeness or provenance. The defensible outcome may be suspension of an affected variable, site, or release until the incident has been adjudicated and the analysis rerun (National Institutes of Health, 2026b; Committee on Publication Ethics, 2025).

26.7.4 — Negative and Null Result Registry

A negative-result registry has one narrow job: give null, adverse, and unusable findings a durable, searchable home. It should capture the tested object, protocol version, reference class, failure mode, and public status without converting a difficult outcome into silence or narrative repair (Committee on Publication Ethics, 2025; Center for Open Science, 2025).

The registry records hypothesis, site, sample, data quality, rival fits, missingness, deviations, interpretation, and publication

status. Each negative-result entry requires a stable identifier, protocol link, status label, access rule, and correction route.

A well-powered Registered Report may produce a null after strong harmonization. Registry staff publish that result without forcing sites to defend a scientifically informative outcome as an institutional failure.

A null result should not be relabeled inconclusive merely because it disappoints the preferred narrative. Its classification must follow predeclared quality, power, measurement, and separability criteria rather than institutional appetite (Ioannidis, 2005; Franco et al., 2014).

Those criteria should distinguish an informative null, an underpowered estimate, a failed instrument, a nonseparable model comparison, and an ethics-limited study. The status determines what enters synthesis and what kind of redesign is justified.

Future syntheses need separate counts for clean negative findings and unusable findings. If both disappear into a generic inconclusive category, the network will repeatedly rediscover failures it already paid to learn.

26.7.5 — Replication Disagreement

Disagreement among sites may be the central result rather than a defect in replication. It can expose a transport boundary, a hidden observation difference, a care-regime effect, or a rival that fits one context better than another (Center for Open Science, 2025; Ioannidis, 2005).

The record should preserve majority and minority interpretations, site estimates, rival fits, sensitivity analyses, and unresolved adjudication. Agreement is not the default destination; an auditable account of disagreement is.

Suppose two sites reproduce compression while a third site with stronger endpoint independence does not. The third site should receive more scrutiny, not less, because its design may reveal that the apparent replication depended on endpoint coupling elsewhere.

A pooled estimate loses force when an adverse site is averaged away to protect a clean public sentence. The disagreement remains visible, and the claim narrows or the model changes according to the protocol-fixed rules (Franco et al., 2014; Fanelli, 2012).

Replication disagreement should remain visible until a mechanism, measurement difference, or scope condition explains it. Pooling is not an explanation.

Dissent that survives the final model makes scope more honest. The disagreement record should show what each site observed, what was comparable, and what remains unresolved.

26.7.6 — Audit-Ready Research Program

An audit-ready program can reconstruct a result without relying on institutional memory or personal assurances. Required artifacts, retention periods, responsible owners, and recovery tests should be specified in advance, because missing provenance discovered after publication may make an otherwise attractive result unauditable (Sandve et al., 2013).

The archive links protocol versions, amendments, consent templates, access logs, code releases, containers, data dictionaries, incidents, analyses, and public statements. The audit manifest should identify artifact owners, version authority, reopening triggers, and unresolved reconstruction gaps. A missing owner or reopening trigger should itself count as an audit defect because it leaves future correction without accountable authority. The manifest should also preserve dependency links among artifacts, so a later correction can identify which analyses, figures, or public statements inherited authority from a superseded object. This makes provenance actionable rather than merely historical.

A critic tries to reproduce a figure two years after publication. If the code environment, transform history, or dependency versions have disappeared, the figure has outlived the evidence needed to reconstruct it. Long-term reproducibility requires preservation of executable environments or an independently documented migration path, not code files alone. Reconstruction should begin from the archived inputs and declared environment rather than from a later analyst's memory of how the pipeline probably worked. A successful rerun should reproduce not only the image but also the intermediate quantities and version identifiers on which the published result depended.

If an environment file, device log, or exclusion script is missing, the affected figure cannot be treated as reproducible. It should be quarantined while the environment is reconstructed; if independent rebuilding fails, the figure and any dependent claim require correction or withdrawal (Committee on Publication Ethics, 2025; Crossref, 2025). Quarantine should extend to

downstream derivatives whose validity depends on the missing artifact, not just the visibly affected figure. The dependency record should therefore identify which conclusions can survive independently and which remain suspended until reconstruction succeeds.

Independent dry runs should test audit readiness by exposing missing artifacts before a public claim depends on them. Dry runs should include a person unfamiliar with the originating workflow, since tacit knowledge is itself a reproducibility dependency. The dry run should start from the archived manifest rather than from informal help by the original team, and any required undocumented intervention should be logged as a reconstruction defect. Repeated dry runs can also reveal whether later software or infrastructure changes have silently broken an otherwise complete archival package.

Public release can now be treated as a controlled transformation from record to claim. Every compression step should preserve the tested component, reference class, uncertainty, and correction route. A release manifest can make that transformation auditable by linking each public object to the analysis version and evidence packet from which it was derived. The manifest should also record the transformation that produced each public table, figure, caption, or summary, allowing reviewers to trace a simplified artifact back to the exact result it represents. If the underlying analysis changes, that dependency map identifies which public objects require review rather than relying on memory or broad manual searches.

26.8 — Public Release and Correction

Public release is a scientific act because public language can change the perceived status of the evidence. A release packet therefore binds the result to its component, reference class, context, rivals, uncertainty, access conditions, participant communication, persistent identifier, correction route, and hostile-crop review before high-reach language appears.

Public release remains status-labeled. The main sentence states the tested component, the relevant population or setting, the result, and any limitation that changes its meaning. The accompanying record supplies instrument and endpoint versions, rival analyses, heterogeneity, access conditions, and the correction route. A short surface is acceptable when its central claim remains accurate without the longer record.

Donor and media pressure receive a local boundary here because attention can alter scientific language. A preliminary result may sound more certain in a fundraising deck than in a paper. A null may disappear from a website because it is not inspiring. A correction may be delayed because it weakens a public narrative. Public release requires rules that remain stronger than excitement.

Participant communication deserves its own channel. People who contributed data should receive outcomes in language that is accurate, respectful, and not psychologically burdensome. They should not receive an individual ledger verdict, a spiritual interpretation, a prediction about endpoint neutrality, or a request to become a public example. Return should serve dignity, not publicity.

Publication of evidence and revision of the claim register are separate acts. A research system may release, correct, retract, withdraw, or label scientific results; it cannot silently rewrite the formal root, authorize an applied title, or open a transition gate. Those actions require their own governed routes.

Table 26.11. Public release label fields.

Release field	Required statement	Forbidden compression
Component tested	Which measurable component was studied	The Law is proven
Context of use	Population, site type, device, language, care regime, endpoint rule	Applies to all conscious streams
Replication state	Direct, conceptual, multisite, adversarial, transport, or failed	Independent confirmation without qualification
Rival status	Which rivals were compared and which remain viable	Rivals were defeated if they were not tested
Negative evidence	Nulls, failed sites, unusable data, deviations, incidents	Clean pattern with adverse evidence hidden
Correction path	Repository release ID, correction contact, retraction trigger	Permanent public certainty

26.8.1 — Embargo Until Required Review

Preliminary findings should not outrun the reviews that give them meaning. Site quality, harmonization, primary analysis, rival analysis, incident review, participant communication, and public

wording all need to close before a high-reach release (Committee on Publication Ethics, 2025; Crossref, 2025).

The embargo is therefore tied to specified review gates and an expected decision date. It protects the evidentiary record while preventing secrecy from becoming a tool for indefinite delay or narrative control.

A striking interim pattern may circulate among donors before blinded rival analysis is complete. Once attention enters the analytic room, choices about diagnostics, exclusions, and wording can become responsive to a story that has already begun to travel.

The result loses warrant when public attention shapes interpretation before the record closes. The remedy is to document the leak, repeat the affected analysis under protected conditions where possible, and reduce the status when independence cannot be restored (Center for Open Science, 2025).

Public trust sometimes requires patience while a claim finishes being checked. A release that outruns endpoint review, privacy review, or site reconciliation creates a public artifact the evidence cannot yet carry.

26.8.2 — Status-Labeled Release

A status-labeled release states the evidentiary object before interpretation begins: tested component, reference class, uncertainty, unresolved rivals, replication state, and correction route. These labels should be fixed in the release record and placed next to the public artifact, not added after attention rewards

stronger language (Crossref, 2025; Center for Open Science, 2025).

The release label states what was tested, where it was tested, what did not replicate, what rivals remain, and what correction path exists. The release register should identify the sentence owner, status source, correction trigger, and history of disputed wording.

A graphic shows terminal concentration but omits that two sites failed device harmonization. The image has retained the status of the headline while shedding the limits that made the finding accurate.

The public sees a clean pattern that the underlying record does not support. The graphic must be corrected or withdrawn and the failed harmonization limits restored to every active destination (Committee on Publication Ethics, 2025).

Status labels should be adjacent to the artifact, not buried elsewhere. Adjacent status labels protect readers by preventing a detached image from carrying more warrant than the study earned.

Status-labeled release makes the result understandable without manufacturing certainty. It preserves the difference between a component finding, a route verdict, and a whole-claim conclusion.

26.8.3 — No Donor or Media Upgrade

Fundraising and media attention cannot change scientific status. Every donor summary, media abstract, website, caption, and social excerpt should be checked against the same versioned

result record used for the technical report (Center for Open Science, 2025).

Communications review should compare each public claim with the tested component, scope, uncertainty, remaining rivals, replication state, and correction route. The public surface must preserve the route by which the sentence earned its warrant.

Suppose a sponsor asks to describe a partial replication as the first proof of the Law. The correct response is not a softer synonym for proof, but a component-specific statement that names what replicated and what remains unresolved.

When money upgrades the sentence while the data remain unchanged, the release has become a funding artifact rather than a scientific artifact. The overstatement should be blocked, corrected, and recorded as a public-risk incident (Committee on Publication Ethics, 2025; Crossref, 2025).

Release language should be fixed before donor packaging and revised only through the same correction process that governs the scientific record. This prevents access, prestige, or urgency from creating a second and stronger claim register.

Public communication can invite support for research by describing the question, current findings, and next study. Scientific status follows the evidence-review process.

26.8.4 — Correction and Retraction

Public error requires visible repair. A correction or retraction changes the status of every surface that carried the affected claim,

not only the paper in which the defect was discovered (Committee on Publication Ethics, 2025).

The correction plan identifies papers, repositories, code, websites, figures, captions, summaries, model cards, donor materials, and downstream citations, assigns an owner to each, and records completion. A process that cannot be executed and audited across those surfaces is not yet a correction system.

Suppose a later audit finds that a device-version bridge was invalid in a published figure. The technical explanation, revised analysis, participant communication, and public graphic all need coordinated repair; the burden cannot be shifted to one site or one reader to reconstruct.

Correcting the paper while leaving the public graphic online preserves the error where it has the greatest reach. The graphic should be relabeled, quarantined, or withdrawn, and every active citation to it should point to the correction (Crossref, 2025; Center for Open Science, 2025).

A correction record identifies the project-controlled surfaces updated, the known external reuses notified, and the copies that could not be recalled or corrected. A durable notice should remain available so readers encountering older material can find the current status.

26.8.5 — Participant Communication

Plan participant and family communication prospectively: return can create burden, false meaning, or participation pressure (National Academies of Sciences, Engineering, and Medicine, 2018).

The protocol fixes timing, optionality, plain language, study-level findings, limits, incidents, support routes, withdrawal implications, correction notices, and questions. Communication stays nonindividualizing absent a clinically authorized route.

Never send bereaved families summaries resembling verdicts on a loved one's inner life. Tone and timing add research burden; distress after return is not hostility to science.

Participants deserve clear findings and limits, with the freedom to decide whether and how they discuss their involvement publicly.

26.8.6 — Release Is Not an Update to the Claim Register

Publication does not change the formal series. A release may enter review, but publicity cannot revise root definitions, applied authority, religious doctrine, law, or transition timing (Crossref, 2025; Center for Open Science, 2025).

Each release names the tested component, review owner, untouched claims, and public limits. A favorable multisite result updates only the evidence register; broader scientific, applied, spiritual, legal, or transition changes need separate review.

Freeze and correct unsupported publicity upgrades; return evidence to human change control (Committee on Publication Ethics, 2025).

Chapter 27 — Evidence Accumulation and Defeat

Evidence becomes cumulative when each result can strengthen, qualify, or challenge the component it actually tests. Positive findings deserve clear recognition, mixed findings identify the next question, and adverse findings redirect work. An answerable evidence architecture gives every result an address, a justified scope, and a defined consequence for scientific status (Mayo, 2018; Lakatos, 1978).

Evidence attaches only to the component a design can reach. A hospice study may inform terminal concentration in one clinical context without speaking for childhood, nonhuman animals, dreams, artificial systems, or the strong latent universal branch. A horizon task may pressure a candidate scheduling model without changing the endpoint claim, while a failed wearable pipeline may retire one instrument version and leave the lived target untouched.

Methodological trustworthiness and eventual factual correctness are distinct. A result can later prove wrong yet remain trustworthy when the question was clear, the study was evaluable, error controls were visible, and the public sentence matched the evidence. A favorable result can remain untrustworthy when those conditions fail, even if later work happens to point in the same direction (Nosek et al., 2026).

Evidence accounting is noncompensatory. Strong replication cannot repair invalid measurement, a beautiful model cannot repair endpoint leakage, public interest cannot repair rival inferiority, and a large sample cannot repair a wrong estimand. Necessary dependencies are conjunctive, so one blocked gate can cap the entire downstream claim.

Narrowing is a central scientific skill, but not every successor is a narrower form of the same claim. A smaller reference class or domain may narrow scope; a change from exact scalar closure to a vector, finite-band, probabilistic, or route-level target creates a non-equivalent successor with a new identifier, assumptions, and decision rule. Progress depends on naming that change and preserving the failed predecessor.

Every evidence state receives usable language. Preliminary component support, replicated support, Route-favored, Model-ambiguous, Rival-favored, Open-set inadequate, Reference-class narrowed, No adequate estimate, route retirement, and theory-level defeat occupy different axes rather than one prestige scale. Each conclusion carries its own next step and public sentence.

The stakes appear in ordinary research work. A statistician may find heterogeneity hidden by a pooled estimate, an endpoint reviewer may discover outcome leakage, a family advisory group may hear an overclaim in a summary, and a rival steward may recover the data better than the focal model. In an answerable system, those findings change status before release instead of becoming cleanup after the announcement.

Dissent remains part of the evidence record because scientific judgment cannot be delegated to one equation or one committee. Minority reports, appeals, external challenges, and later reconsideration preserve observations that a majority may have missed. The architecture records disagreement while maintaining a decision deadline and a defined reconsideration route.

Defeat is reported without spectacle: it changes the scientific claim, not the standing of participants or the obligations owed to them. The verdict belongs to the model and claim, not to the people who supplied the evidence.

Scientific memory must operate at component scale. Earlier Parts asked how to measure, model, compare, and replicate; the evidence record now states what was tested, what was not, which dependencies failed, what rivals explained better, which claims narrowed, and which public artifacts changed. Without that memory, later researchers can repeat the same overclaim in more polished language.

The result is verdict discipline rather than verdict theater. A mature record may say that one component is replicated, another remains Model-ambiguous, a third has No adequate estimate, a fourth is Rival-favored, and the strong form has not moved. That mixed state is precise when each conclusion follows a declared gate. The mixed record should be preserved without forcing a single overall score, because different components may justifiably move in opposite directions.

The program becomes stronger by losing locally, recording the loss, and changing course. A smaller true claim outranks a universal claim protected by ambiguity. A retired mechanism outranks a rebranded failure, and a public correction outranks a silent rewrite that preserves prestige while shifting the scientific burden. Local losses also reduce future burden by preventing repeated investment in bridges or mechanisms that have already failed adequate tests.

Chapter 27 supplies the rules by which evidence accumulates and updates the status of claims, routes, mechanisms, and reference classes through support, narrowing, suspension, retirement, or defeat. It governs future abstracts, figures, interviews, donor pages, public replies, and cross-book references by forcing each sentence back to the component and evidence packet that earned it. No result may travel farther than its design. This traceability allows later public language to strengthen when evidence improves and contract when it weakens without rewriting the history of either change.

27.1 — Evidence Attaches to Components

An evidence system begins by giving every result an address. Representation, measurement, bearer classification, endpoint adjudication, formal realization, route, signature, rival comparison, replication, transport, and public language remain separate nodes connected by explicit dependencies. A finding changes only the node it reaches and the downstream claims licensed by that node.

The following map connects findings developed earlier in the book with the comparisons they make possible. It also identifies the units to track when combining results across studies.

Table 27.1. Scientific findings and connected research questions.

Evidence family	Scientific contribution	Next comparison	Unit to track in synthesis
Mood and activity selection — Taquet et al. (2016, 2020)	Mood relates to subsequent activity selection, providing a basis for studying affect regulation in everyday life.	Compare predictions from current mood, activity opportunities, and prespecified affective-history summaries.	Participant and dataset; the reports share the smartphone cohort.
Temporal structure of mood — Keren et al. (2021)	A primacy model predicted mood better than recency alternatives in the tasks studied.	Compare primacy, recency, and cumulative-history models across tasks and timescales.	Participant sample and task.

Evidence family	Scientific contribution	Next comparison	Unit to track in synthesis
Biological cumulative responses — Gupta and Sontag (2025)	Integral-feedback and particular feedforward models made different predictions for cumulative dose–response shape.	Identify response patterns that distinguish specified models of accumulated affect.	Model architecture, input history, and output measure.
Sleep and emotional processing — van der Helm et al. (2011); Wassing et al. (2019)	Overnight changes in emotional responses connect affective adaptation with sleep and REM continuity.	Jointly model prior affect, sleep continuity, and subsequent emotional responses.	Participant cohort and sleep protocol.
Belief and allocation choices — White et al. (2019)	Reminders of karma or God increased generous allocations among corresponding believers in the experiments.	Compare predictions based on belief, social incentives, and affective history.	Experiment, participant sample, and belief group.

These connections support a coordinated research program: measure affective histories, specify competing predictions, and test which relations carry across settings. Joint analyses can account for shared participants and measurements while assessing how well each model predicts new observations.

The component graph is both a scientific and ethical safeguard. It keeps uncertainty from being shifted onto participants, critics,

later authors, or vulnerable readers. When a study reaches only one bridge, the public sentence remains on that bridge.

Table 27.2. Component evidence matrix for noncompensatory claim status.

Component	What evidence can update	Dependencies to verify
Representation	Scalarization, temporal additivity, origin convention, multidimensional alternatives.	Coherent representation before interpreting endpoint findings.
Measurement	Calibration, context of use, uncertainty propagation, and version performance for the candidate momentary-valence and interval-contribution estimators.	Instrument qualification alongside the formal realization.
Bearer and endpoint	Unity classification, stopping rule independence, open-channel windows, false endings.	Independent endpoint adjudication and qualified bearer classification.
Realization and mechanism	Reserve coupling, Queue, horizon, brain-body mediators, stochastic assumptions.	Connection between the tested mechanism and the specified target.

Component	What evidence can update	Dependencies to verify
Rival and consequence	Signature conjunction, Brownian bridge, adaptation, predictive control, selection, missingness.	Competitive alternatives evaluated under common comparison rules.
Replication and transport	Direct, conceptual, multisite, transported, leave-site-out, leave-device-out.	Heterogeneity and site-specific outcomes alongside the pooled estimate.

Equation 27.1 (component evidence graph):

$$G_{evidence} = (C, D, E, R, V)$$

Here C names components, D names dependencies among components, E names evidence records, R names rival states, and V names versioned public verdicts. The display records how scientific components, evidence, and decisions are connected.

Equation 27.2 (noncompensatory gate rule):

$$PublicStatus(C *) \leq Status(c), \text{ for every } c \in Dep(C *)$$

The preorder expresses a noncompensatory dependency rule: a downstream public status cannot outrun any failed prerequisite on a compatible status axis. Unlike verdicts remain separate, and the component states are not forced into a total order. A successful mechanism cannot repair failed measurement, and a favorable pooled result cannot repair endpoint failure.

A component graph gives specialists a precise place to contribute. Measurement researchers can examine calibration, palliative clinicians can evaluate endpoint windows, and statisticians can

assess transport. Each contribution can improve its component while leaving other questions available for separate review.

The graph is versioned because components change. A candidate instrument version, an endpoint rule, a reserve realization, and a rival matrix each have revision histories. If a later study cites a component without its version, the citation becomes ambiguous. Ambiguity is where unsupported status travel begins.

Dependencies should be marked as necessary, optional, or explanatory. Measurement validity is necessary for measured-ledger claims. A particular brain-body mechanism may be explanatory but not necessary for a consequence signature. A public artifact is not a scientific dependency, but it is a public-risk dependency because it can distort what the evidence means.

The graph also helps reviewers ask whether a result is adverse in the right place. A failed mechanism is not an adverse endpoint. A failed endpoint is not a failed measurement instrument. A failed measurement bridge is not evidence that lived experience has no structure. Precision prevents both inflation and despair.

One useful review habit is to require a result sentence before a conclusion sentence. The result sentence names the component and finding. The conclusion sentence says what changes. If the conclusion says more than the result sentence can carry, the status has drifted.

Accordingly, the component graph functions as a public-language instrument, not merely an internal database. It shows why one claim changed while neighboring claims did not. That visibility is part of the fairness of the science itself.

The component record includes a concise use-scope column specifying the permitted research context, affected decisions, and protected participant rights. This makes the operational boundary available where the evidence is used.

Component records contain unresolved dependencies in active voice. Instead of saying “measurement remains a limitation,” the status should say, “the candidate instrument version has not established sign validity in this context.” Active wording makes the blocker visible enough to govern later claims.

The same discipline applies to positive components. If endpoint adjudication is strong, say so plainly. If a rival matrix is unusually complete, say so plainly. Component discipline preserves excellence by locating it precisely rather than flattening it into caution.

In practice, the component graph becomes a table for line-by-line review. That review lets specialists resolve component scope before publication and gives readers a clearer account of what the evidence supports.

27.1.1 — Representation Component

The representation component asks whether the scientific object has been expressed in a form that can carry evidence at all. It covers scalarization, temporal additivity, origin conventions, sign conventions, and the decision to treat a stream as a path rather than as a collection of detached moments. If representation fails, later measurement may be precise but aimed at the wrong object.

A representation update evaluates whether the chosen formal language can express the target. Evidence may favor a richer

structure, separate dimensions, or a different sampling cadence for recovering a time-integral. These findings guide the next model and measurement design.

Consider a team that obtains reliable arousal data, tolerable sampling, and stable within-person response scales while valence origins vary across illness stages. The next representation study can examine state-specific calibration or a richer structure for that reference class.

A cleaner figure is not a reason to preserve an unidentified scalar. An investigator may keep the scalar because it makes the figure, narrative, or press release easier to present. The graph forbids that convenience. If the representation does not identify the quantity, a later endpoint pattern cannot rescue it. The public record must say that the tested scalar representation was not established for the declared context of use (Popper, 1959; Lakatos, 1978).

The status language should be exact. A representation may be supported, unresolved, narrowed, replaced by a multidimensional structure, or retired. It should not be called validated simply because downstream models fit. Fit can hide wrong meaning when the model absorbs representation error into nuisance parameters.

Carry forward the representation question: which experiential structure did the study model, and how was it connected to the observations?

27.1.2 — Measurement Component

The measurement component concerns the candidate momentary-valence and interval-contribution measurement stack, its context

of use, calibration evidence, uncertainty model, and version history. Declared channels, sign rules, missingness models, participant challenge rights, and transport evidence define how its estimates can be used (Popper, 1959).

Measurement evidence can improve wearable reliability, reduce recall bias through a better report schedule, or increase held-out predictive accuracy. Each gain strengthens a specific part of the observation process.

A concrete measurement case begins with a multisite candidate instrument version that performs well in cancer hospice settings but fails calibration in advanced neurological disease because speech loss, motor slowing, medication, and autonomic disturbance change channel meaning. The appropriate response is not to average the populations together, but to impose a context-of-use boundary and version note.

The danger is compensatory thinking. Researchers may say that the endpoint signature still looks favorable, so measurement limitations can be tolerated. The correct rule is the opposite. If the sign, scale, uncertainty, or missingness of the measured ledger is not valid for the tested context, then the target was not tested in that context (Lakatos, 1978; Manski, 2003).

Measurement status must therefore name version, population, setting, device, cadence, language, and intended inference. A result may support candidate instrument version 3.1 for comfort-first observational palliative research while saying nothing about ordinary life, employment, education, public administration, or private self-tracking.

The graph protects people as well as statistics. When measurement remains inside the declared research protocol, it leaves no eligibility decision, trust label, worth judgment, or care ranking behind. That boundary is part of the measurement component, not a decorative ethics note.

27.1.3 — Bearer and Endpoint Components

Bearer and endpoint components identify the continuing stream under study and its independently specified stopping rule. They state how person, body, record, report, device, and clinical-event information contributes to that classification (Popper, 1959; Lakatos, 1978).

Endpoint evidence updates a stopping rule only when that rule is independent of the ledger estimator. The endpoint cannot be defined as the moment the measured ledger appears neutral; it must use prospectively specified unity or consciousness-related channels that are not algebraically identical to the outcome. Otherwise neutrality is built into the definition rather than discovered.

If monitoring stops during sudden deterioration, the endpoint record should identify whether sedation, device tolerance, family burden, or another observation event explains the missing channel. Independent endpoint evidence determines the classification used for later analysis.

Endpoint adjudication is tracked separately from the study's stopping rule. A convenient final observation may end data collection while leaving the endpoint unresolved. The component graph records that distinction and identifies the additional

endpoint evidence required for terminal inference (Manski, 2003).

Bearer and endpoint status belong in the public record: which stream criteria were applied, what endpoint object was used, how uncertainty was carried, and which observations remained unavailable. That disclosure prevents a component result from being mistaken for direct access to a person's complete conscious history.

The strong form waits behind these gates. No amount of model fit should move it if the bearer or endpoint component remains unresolved. The endpoint firewall is not a preface. It is a dependency.

27.1.4 — Realization and Mechanism Components

Realization and mechanism components describe candidate dynamics for a formal boundary condition. Reserve coupling, scheduling routes, finite-horizon pressure, shadow costs, brain-body mediation, and stochastic process choices each supply structures that can be modeled and tested (Lakatos, 1978; Manski, 2003).

A reserve-coupled product model may predict variance compression in a specified window. The corresponding result evaluates that prediction and the measurement relation connecting latent burden to observed dynamics.

If variance compression appears only under a fitting window that also favors Brownian bridge models and observation-shutdown artifacts, the result identifies a model-confusion region. Recovery

analysis can then determine which measurements or windows would separate those explanations.

Rebranding a failed mechanism does not preserve its evidence. If the candidate scheduling model fails, the program may not quietly relabel ordinary avoidance as candidate scheduling behavior. If finite-horizon pressure fails, the program may not call every urgency effect a finite-horizon control signal. New mechanisms require independent motivation and new risk analysis.

Mechanism status can be supported, bounded, narrowed, suspended, retired, or replaced. A replacement record states what changed, what evidence motivated it, what new predictions follow, and which outcomes would challenge them.

Mechanisms can be refined or retired as evidence accumulates. Their lasting contribution is the sharper question, better instrument, or clearer prediction they leave for the next study.

27.1.5 — Consequence and Rival Components

The consequence and rival components ask whether the predicted observable structures appear and whether serious alternatives explain them as well or better. The relevant object is not a single impressive curve. It is the conjunction of signatures under the same observation constraints faced by the Law-candidate model and by its rivals (Manski, 2003).

A consequence may be terminal concentration, variance compression, reserve-linked contraction, endpoint independence, or a failure of ordinary first-moment explanation. But none of these alone is sufficient. Brownian bridges, Doob transforms,

adaptation, frailty, medication, terminal scripts, missingness, and selection can all mimic parts of the pattern.

A useful stress test is a pooled terminal dataset where the final estimated ledger values cluster near the neutral band. If data were aligned to death after the fact, if visits intensified near decline, if sedation removed high-variance observations, or if one site contributed most of the late windows, the apparent consequence may belong to observation and selection rather than to the Law.

Weak rivals make a favorable comparison scientifically uninformative. A chapter or protocol that gives the Law a flexible model and gives rivals caricatures has already failed the component. Rival seriousness is not generosity. It is identification. A weak rival matrix can manufacture support by making ordinary explanations look absent (Popper, 1959).

The status update should say which consequence survived which rival. A signature can be supported against adaptation and still unresolved against bridge models. A model can outperform frailty in one diagnosis and lose to medication-only decline in another. That granularity is the governing discipline.

The scientific dignity of the project depends on letting rivals win where they should. A rival victory is not a public-relations crisis. It is an accurate allocation of explanation.

27.1.6 — Strong-Form Component

The strong-form component concerns terminal neutrality for qualifying unified conscious streams. Its evaluation joins formal representation, longitudinal measurement, endpoint identification, and comparisons across reference classes.

Results bear on the strong form through explicit prediction and comparison. Replicated agreement across qualified components can strengthen a specified account; credible counterexamples can identify where its scope or assumptions require revision. The evidence graph makes these contributions assessable together. The graph should distinguish evidentiary dependence from thematic relation, preventing a nearby success from being counted as support for a component it does not test.

Put this rule against a decade of studies that support variance compression in some palliative contexts, defeat one candidate instrument version, retire one candidate scheduling model, and show strong rival performance in another diagnosis. The public summary maps those findings by component, population, and study design, making the advances and next questions visible together. The map should preserve study-level identifiers so readers can trace each status back to its evidence packet rather than rely on a synthesized reputation of the program.

Prestige contaminates review when the reputation of a team, site, journal, or sponsor changes how much adversity a result must survive. Once a program has enough technical machinery, people may begin to speak as though the strong form has been carried by the weight of the whole apparatus. The component graph prevents that drift. The strong form is not the sum of enthusiasm across components (Popper, 1959; Lakatos, 1978). Dependency weight should come from the logic of the claim rather than the prestige or volume of work surrounding a component.

Status language for the strong form must be conservative and explicit. It may remain hypothesized, under pressure, narrowed in

public interpretation, suspended from empirical verdict, or defeated under defined conditions. It should not be upgraded by applied coherence, public resonance, or institutional usefulness. Any stronger label should identify the specific dependency that changed and the evidence that justified the upgrade.

The strong-form component therefore sits last in the graph. It receives pressure only through dependencies that have survived. That order is not modesty; it is the price of making a universal claim without turning it into doctrine. Placing the strong form last also exposes where the chain remains incomplete instead of allowing completed lower-level work to create an illusion of closure.

27.2 — Sequential Evidence and Replication States

Using the reproducibility, robustness, replication, and transport distinctions established in Chapter 26, this section orders the evidence states that can change a claim: exploration, prospectively specified testing, public preregistration where actually deposited, direct or conceptual replication, multisite replication, transported replication, and sequential review (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018; Klein et al., 2018; Bareinboim & Pearl, 2016).

The sequence prevents status from becoming a score. A component may have three promising signals and no adequate replication. Another may have one severe failure that blocks further public claim. The judgment is structured, but it is not arithmetic.

Table 27.3. Evidence states and the public language each state permits.

Evidence state	Scientific function	Permitted public language
Exploratory signal	Generates design questions and candidate signatures.	Describes an exploratory finding and motivates a prospectively specified comparison.
Prospectively governed single-study result	Tests one component under declared rules.	May update the tested component within its reference class.
Direct replication	Repeats the essential test independently.	May strengthen or weaken the same component claim.

Evidence state	Scientific function	Permitted public language
Conceptual replication	Tests the same component through different design logic.	May support robustness of the component, not identity of mechanism.
Multisite or transported replication	Tests heterogeneity and context of use.	May widen, narrow, or split the reference class.
Governed review	Integrates quality, dissent, rivals, and public correction.	May issue a status label with scope and reasons.

Equation 27.3 (sequential status update, human-reviewed):

$$S(c, t + 1) = J[S(c, t), e(c, t + 1), q(c, t + 1), r(c, t + 1)]$$

The symbol J stands for governed judgment, not a mechanical score. It receives the prior component status, a new evidence record, quality information, and rival-state information, then issues a versioned decision with recorded reasons.

Sequential evidence also needs a calendar. A result released before review has a different status from the same result after independent verification, data audit, and rival reanalysis. Time does not make evidence stronger by itself, but time allows the checks that prevent premature authority.

The sequence records movement in both directions. A component can move from exploratory signal to preliminary support and return to inconclusive after a failed replication. A model can move from promising to retired. A measurement version can move from local use to suspended use after device drift. Evidence, not institutional optimism or elapsed time, determines the direction.

The review unit is not a single paper but the evidence record for a component. Papers can contain multiple components, and one paper can support one component while defeating another. A sequential system that treats each publication as one point will blur the exact distinctions the evidence architecture is designed to protect.

Publication bias requires special attention because favorable results travel farther and are easier to remember. Null results, unusable studies, burden stops, and unpublished protocols can disappear unless the evidence system requires them. Sequential review asks not only what was published, but what was prospectively specified, publicly preregistered, attempted, stopped, or withheld.

The sequence also protects participants. If early studies repeatedly show nonseparability or unacceptable burden, the next stage may be redesign rather than more recruitment. Evidence accumulation should not become an excuse to ask vulnerable people for the same inconclusive data again.

In a mature program, status movement should be slower than public excitement and faster than institutional denial. It waits for verification without waiting forever after the evidence has turned. That timing is part of sequential scientific discipline.

Status movement requires prospectively specified stopping conditions. A component cannot remain preliminary forever because the next replication is politically inconvenient, and it cannot upgrade because time has passed. The record identifies

which evidence triggers upgrade, narrowing, suspension, retirement, or defeat.

Sequential review should also mark stale evidence. A measurement version tested under one device firmware, one care regime, or one diagnostic convention may become outdated. Staleness does not erase the study, but it changes the role the study can play in current status.

The evidence record for a component includes failed registrations and abandoned protocols. A publicly preregistered study stopped for ethical burden teaches something different from a design that was only privately drafted or never launched. Each history changes the burden and interpretation of the next study.

When new evidence arrives while public materials remain in circulation, the sequence includes an embargo policy for status claims. Internal analysis may proceed, but external language waits until the declared review gate is complete. This prevents half-reviewed results from becoming public certainty.

Independent reanalysis examines a predeclared family of analytically reasonable specifications rather than one critic-selected alternative. Multiverse and specification-curve methods reveal whether a status label depends on one defensible choice among many, while many-analyst designs expose how expert judgment contributes to dispersion. These approaches do not average truth into existence; they map analytic fragility that the evidence register must carry into later review (Steege et al., 2016; Simonsohn et al., 2020; Silberzahn et al., 2018).

A sequential system earns credibility by moving downward when the evidence requires it. Upward movement is trustworthy only when the same machinery can record analytic fragility, failed replication, stale instruments, burden stops, rival superiority, and status reversal.

27.2.1 — Exploratory Signal

An exploratory signal can reveal a bounded empirical pattern and help design a stronger test. It may arise from pilot data, retrospective records, qualitative observation, or existing measurements; simulation can separately reveal what a model predicts under stated assumptions. The report should identify the source, selection process, analytic flexibility, and uncertainty. Such findings contribute to discovery and identify the prospective comparison to run next (Chambers, 2013; Nosek et al., 2018; Lakens, 2017).

Record the signal's provenance, analytic degrees of freedom, selection routes, and plausible alternative explanations. Distinguish expectations stated before inspection from questions first suggested by the observations.

The contrast is clearest in a terminal dataset where late variance appears lower than early variance. The pattern may motivate a prospectively specified compression test, but it may also reflect more frequent visits near decline, medication changes, hospice admission criteria, or missing high-distress observations. Exploration should sharpen those alternatives rather than announce a finding.

Label an exploratory figure at its first release and link it to the analysis record. Readers can then follow how an initial observation develops into a specified prediction and independent test (Schönbrodt et al., 2017; Nosek et al., 2018).

Exploratory reporting should state the observation directly and label its status: for example, the analyzed sample showed a pattern that motivates a prospective comparison. The resulting hypothesis can then be stated with words such as suggests or motivates, followed by the comparison that would evaluate it.

An exploratory signal carries the program forward by identifying a question worth testing. Preserving its exploratory status gives that early insight room to develop while keeping later confirmation independently informative.

27.2.2 — Publicly Preregistered Single-Study Result

A single-study result that meets §6.7's public-preregistration standard has more authority than an exploratory signal because its promised path, deviations, and decision rule are independently inspectable; registration does not make the result true (Lakens, 2017; Schönbrodt et al., 2017).

The component updated by a single study is the component actually tested. A study may publicly preregister measurement invariance for one diagnosis and one device version, or one endpoint rule for one context of use; neither record updates every instrument or stopping classification.

Consider a single hospice protocol that tests equivalence of a declared measured estimand using a specified candidate instrument version during independently adjudicated open-

channel windows. A favorable result may justify preliminary component support for that candidate momentary-valence estimator-endpoint combination. It does not validate the strong form.

Changing a threshold after outcome inspection converts a confirmatory test into a rescue. Neutrality bands, endpoint windows, inclusion rules, and smoothing parameters may still be explored in sensitivity analysis, but the resulting claim no longer occupies the publicly preregistered single-study state (Nosek et al., 2018; Klein et al., 2018).

A single-study statement names the component, reference class, registry identifier, protocol and analysis version, deviations, comparison models, and uncertainty. These details let readers assess its contribution and plan replication.

A good single study creates a clear opportunity for independent replication, with the essential design preserved and both confirming and disconfirming outcomes able to inform the next step.

27.2.3 — Direct Replication

Direct replication repeats the essential test as closely as feasible while using an independent team, independent recruitment, and independent analysis controls. It asks whether the original component result survives when the social, coding, and analytic context changes. Here, direct does not mean identical in every detail; it means that the critical test is preserved (Schönbrodt et al., 2017; Nosek et al., 2018).

Replication evidence should include adherence to the original protocol, permitted deviations, device and software versions, participant burden, endpoint adjudication, and blinded analysis status. A study that changes the target must be identified as a new test. A changed design may qualify as conceptual or transported replication only when it still tests the same declared component and its relationship to the original test is explicit.

Consider a second team repeating a preterminal variance-compression test using the same candidate instrument version and endpoint rule at a new site. If it succeeds, the component gains credibility. If it fails under adequate conditions, the original result becomes local, fragile, or possibly false.

Replication is protected rather than severe when the favored component is insulated from adverse outcomes. A favored model can be protected by tuning the replication to the original site's quirks or by explaining every adverse result as implementation failure. Direct replication must permit the possibility that the original finding does not repeat (Open Science Collaboration, 2015; Klein et al., 2018; Nosek & Errington, 2020).

Only independent repetition under the essential design licenses replicated component support. The summary must identify the component, sites, prospectively specified target, any persistent preregistration records, common analysis, unresolved heterogeneity, and evidence state rather than treating replicated as a free-standing seal.

Direct replication is an act of respect. It says the finding matters enough to expose it to independent failure.

27.2.4 — Conceptual Replication

Conceptual replication tests the same component through a different design, measurement route, population, or mechanism. It is powerful because it asks whether the component is more than an artifact of one procedure. It is also risky because the design can drift away from the original object (Nosek et al., 2018; Klein et al., 2018).

A conceptual replication must name the invariant being tested. Is it the same consequence, the same measurement bridge, the same endpoint relation, the same rival defeat, or the same mechanism? Without that declaration, any loosely similar result can be folded into support.

Cross-method comparison becomes informative in testing the same narrowing of terminal variance through experience sampling, clinician-assisted reports, and passive physiological channels. Agreement across designs may strengthen a consequence component. Disagreement may reveal that one method captures arousal, another captures reportability, and neither alone captures measured burden.

Analogy becomes misleading when thematic resemblance is described as replication. A study that finds any repair, rebound, adaptation, or calmness is not necessarily replicating a Law-specific component. Conceptual replication must remain conceptual in the scientific sense, not associative in the rhetorical sense (Nosek & Errington, 2020; Lakens, 2017).

Conceptual replication requires public language that names both the invariant and the deliberate design change. A result that

survives a new population, instrument, or perturbation may strengthen transport of one component while leaving the original mechanism and endpoint bridge unresolved.

Conceptual replication broadens the question by changing the route. It should not broaden the claim by blurring the destination.

27.2.5 — Multisite and Transported Replication

Multisite and transported replication asks whether a component survives across sites, devices, languages, care regimes, diagnoses, cultures, and study teams. It is not merely more data. It is a test of context. Heterogeneity is therefore not a nuisance to be averaged away; it is one of the central findings (Klein et al., 2018; Fortier et al., 2011; Stuart et al., 2011; Bareinboim & Pearl, 2016).

Transported evidence must define context of use. A palliative oncology setting, a neurological ward, a sleep laboratory, a home-diary study, and an Artificial Intelligence-mediated archive do not share the same observation process. Replication across them requires measurement invariance, endpoint comparability, and rival envelopes appropriate to each context.

Transport is tested by a signal that repeats in two English-language hospice networks but fails in a multilingual setting where reports are mediated through family interpreters and different pain medication practices. The correct update may be transport failure, not global falsification and not global support.

A favorable pooled average can conceal site reversal, device-version failure, subgroup harm, or care-regime dependence. Prediction intervals, leave-site-out checks, and site-specific

reports must remain visible (Lakens, 2017; Schönbrodt et al., 2017).

Public language after transported replication should say whether the reference class widened, split, narrowed, or failed to transport. The phrase replicated across sites is incomplete unless the heterogeneity record travels with it.

Transport is where a theory learns humility. A claim that becomes smaller across contexts may become more useful because it finally names where it lives.

27.2.6 — Sequential Validity and Review

Sequential validity concerns how evidence is reviewed over time without letting repeated looks, changing thresholds, or accumulating public pressure distort the result. Statistical tools can help, including prediction intervals, sequential meta-analysis, calibration checks, and robustness summaries. But the judgment remains governed by humans who must record reasons, dissent, and scope (Jennison & Turnbull, 2000; Lakens, 2017).

A living evidence system should separate analysis updates from status updates. New data may update plots, posterior distributions, heterogeneity estimates, or model weights before the public status label changes. This prevents every interim fluctuation from becoming a headline.

Sequential review is tested by a series of three replications, one favorable, one mixed, and one negative under a new device version. Sequential review should not choose the most comforting interpretation. It should ask whether the component is version-

dependent, whether transport failed, whether the original mechanism is wrong, and whether the public label must narrow.

Review by momentum treats elapsed time and institutional investment as evidence for an upgrade. Once a program has a public identity, committees may feel that status should move gradually upward unless catastrophe occurs. That pressure is rejected. A single severe dependency failure can block an otherwise favorable accumulation (Schönbrodt et al., 2017; Nosek et al., 2018).

Sequential review should publish the rule by which evidence states are revised, the quality criteria applied, the dissent recorded, and the correction path for public artifacts. The result is not a score, but it is not private intuition either.

The sequence ends where judgment begins. Tools arrange evidence; people accept responsibility for language.

27.3 — Route-Favored, Partial, and Replicated Results

Positive evidence needs disciplined vocabulary. Calling every favorable result merely interesting prevents learning; calling every favorable result validation destroys scope. Component support, route preference, replication, convergent evidence, and target evidence therefore receive distinct meanings, thresholds, dependencies, and propagation rules.

The standard is generous to real findings and strict about scope. A component can be supported while another component blocks a wider claim. Convergence can matter without becoming universal discovery. The public statement must carry both truths.

Table 27.4. Positive status labels and their dependency limits.

Status label	Minimum condition	Required limitation
Preliminary component support	One adequate prospectively specified result for a declared component.	Names reference class, analysis version, and unresolved rivals.
Partial support	Some predicted structure appears while dependencies remain unresolved.	Lists adverse, missing, or untested dependencies.
Replicated component support	Independent direct or transported support.	Retains heterogeneity and site-specific failures.
Mechanism support	A candidate realization predicts better than declared rivals.	Does not support the strong form by itself.

Status label	Minimum condition	Required limitation
Convergent support	Different methods support the same component.	Requires shared target, not merely similar story.
No full-canon validation	A component succeeds but the series remains domain-separated.	Prohibits applied, legal, religious, or transition upgrade.

A worked result packet makes the vocabulary operational. Suppose a four-site study qualifies the candidate momentary-valence instrument in two sites, detects calibration drift at a third, and cannot adjudicate the terminal endpoint at a fourth because sedation and missingness exceed the protocol-fixed bounds. In the two qualified sites, a reserve-coupled candidate outperforms one ordinary regulator but remains indistinguishable from an endpoint-conditioned bridge under held-out scoring.

The correct record is mixed rather than averaged. Two qualified contexts may support the measurement component, the third yields Unusable evidence for the affected measurement estimand, and a fourth may suspend terminal inference. The route can remain Model-ambiguous while the strong target does not move. No pooled mean repairs blocked gates, and no favorable site speaks for an endpoint class it could not identify.

The public sentence names both the object and the limits: two sites reproduced a bounded measurement result, while device transport and terminal-endpoint prerequisites failed elsewhere, leaving the PTVC target unadjudicated. That sentence is less dramatic than a discovery claim, but more informative because

another team can reconstruct what succeeded, what failed, and what evidence would be needed next.

Positive language should be written so that it can survive hostile quotation. A sentence that says replicated support without naming component and scope will be cropped as validation. A sentence that says partial support while hiding the blocker will be cropped as evasion. The solution is not timid prose, but complete local prose.

The support label should include the result's dependency footprint. A measurement-supported component with endpoint unresolved has a different footprint from a mechanism-supported component with measurement unresolved. The reader should never have to infer which bridge remains open.

Quality governs interpretation as much as count. Three weak studies do not necessarily outrank one strong, publicly preregistered replication, and one strong study may still be blocked by a severe dependency failure. Positive evidence accumulates through quality-sensitive judgment rather than the emotional appeal of repetition.

A support statement should also identify the strongest surviving rival. If the result beats some rivals but remains unresolved against bridge or selection rivals, the public sentence should say so. Rival visibility prevents positive language from becoming an argument by omission.

The support vocabulary rewards good work. Bounded findings can earn real recognition without being inflated into support for the whole Law.

Every positive label carries a loss condition. Preliminary support states what replication failure would remove it. Partial support states which dependency must be resolved before the claim can expand. Replicated support states which heterogeneity or transport failure would narrow it.

Support language should also distinguish scientific excitement from public entitlement. Researchers may be excited by a clean component result. The public, however, is entitled to hear scope, uncertainty, and rival status in the same breath as the result.

When a component is supported, downstream chapters or public artifacts should not quote it without its qualifier. A graph label, caption, social excerpt, and donor paragraph must carry the component boundary. Otherwise the qualifier exists only for those who read the technical chapter.

The status record identifies who reviewed the positive label. A label that has not passed independent analysis, red-team review, and public-risk review does not enter high-reach summaries. The issue is premature authority, not secrecy.

Partial support can be more useful than broad support because it directs future work. It says where the bridge held, where it cracked, and where the next protocol should apply pressure. That guidance is lost when partial support is flattened into good news.

27.3.1 — Preliminary Component Support

Preliminary component support is the first positive label a result can earn after adequate prospectively specified testing. It is neither a consolation nor a promotional label. It means that one

component survived one serious test under declared assumptions (Open Science Collaboration, 2015; Camerer et al., 2018).

The label names the supported component: a measurement bridge, a terminal-window pattern, a rival-comparison result, or a mechanism prediction. Including the population, window, and analysis in the same statement gives preliminary support a precise and useful meaning.

The distinction is clearest in one high-quality study whose design was publicly preregistered and whose prespecified equivalence procedure yields a nonempty measured compatibility set lying wholly within the equivalence band for a declared estimand, using a specified candidate instrument version during independently defined open-channel windows for one diagnosis group. The public statement may report preliminary support for that measured-endpoint component in that context. It may not say the Law has been discovered.

Emotional acceleration occurs when a striking result moves through the status ladder faster than its dependencies can be checked. Supporters may hear preliminary as nearly proven, while critics may treat it as spin. The status record should therefore publish the exact tested component, the untested dependencies, the rival matrix, and the conditions that would reverse the label (Klein et al., 2018; Vehtari et al., 2017).

Preliminary support should travel with an expiration condition. It waits for direct replication, conceptual replication, or transported replication. If those fail, the label changes. It is not a permanent trophy stored from an early favorable result.

The dignity of preliminary support is that it lets science be hopeful without being careless. It gives the next study a reason, not a conclusion.

27.3.2 — Partial Support

Partial support recognizes that some predicted structure appeared while other dependencies remain unresolved, mixed, or adverse. It is more honest than either all-or-nothing support or total dismissal. Many serious programs live for a long time in this state (Camerer et al., 2018; Klein et al., 2018).

The label should name what is supported and what blocks expansion. A consequence signature may appear, but the mechanism may fail. A measurement version may work locally, but transport may fail. A rival may be beaten in one dataset and win in another. Partial support is not a vague positive mood; it is a structured partition.

Consider a study that estimates variance compression in a prospectively defined preterminal window but cannot adjudicate the later endpoint. If measurement is qualified for that window, the study may support the measured preterminal pattern while leaving terminal inference unavailable. If measurement also fails for the affected observations, those observations cannot support the pattern; report the qualified and unusable subsets separately.

Selective emphasis distorts accumulation by foregrounding favorable components and relegating adverse ones to technical margins. A public summary may lead with the supported part and hide the unresolved part in subordinate clauses. The blocker must appear in the same status statement as the support. The unresolved

dependency is part of the finding (Vehtari et al., 2017; Open Science Collaboration, 2015).

Partial support may justify continued research, narrower protocols, and careful public discussion. It does not justify applied authority, public scoring, personal endpoint testing, spiritual certainty, or legal transition of claim status.

Partial should sound precise rather than weak. It is the state in which science refuses to confuse a real piece with the whole.

27.3.3 — Replicated Component Support

Replicated component support requires independent evidence that the same component survives essential repetition. It is stronger than preliminary support because it has crossed the social and operational boundary between teams. Yet it remains component support, not whole-theory validation (Klein et al., 2018; Vehtari et al., 2017).

The replication record must include failures as well as successes. A component supported in four sites and failed in two is not simply replicated. It is heterogeneous. The public status may be replicated within a narrower reference class, replicated with adverse site exceptions, or not yet transportable.

Replication becomes informative when an endpoint-independent terminal concentration pattern repeats in two palliative networks but fails in a setting with different sedation practices. The finding may be replicated for certain care regimes and narrowed for others. The failure is not noise.

Replication synthesis reports heterogeneity alongside the pooled estimate. Prediction intervals describe variation relevant to a new site, while site-level analyses investigate differences in populations, procedures, and contexts. Together these results identify where component support transports and where further study is useful (Open Science Collaboration, 2015; Camerer et al., 2018).

The word replicated should never stand alone. A release must name what was replicated, by whom, under which protocol-fixed design and public registry record if any, across which sites or conditions, with what heterogeneity, and at which evidence state.

Replication gives a component more weight, but it does not make it weightless. It still carries assumptions, scope, and dependencies.

27.3.4 — Mechanism Support

Mechanism support concerns a candidate explanation, not the formal boundary condition. Reserve coupling, candidate scheduling-route dynamics, horizon pressure, interoceptive mediation, or a learning-rival comparison can be supported as mechanisms or defeated as mechanisms without settling the strong claim (Vehtari et al., 2017; Open Science Collaboration, 2015).

A mechanism earns support by predicting more than a generic pattern. It should identify a distinctive structure, a timing relation, a parameter recovery profile, a perturbation response, or a rival-discriminating residual. If it only redescribes regulation after the fact, it has not earned mechanism support.

A reserve-coupled realization earns support by predicting held-out variance compression and reserve-linked innovation amplitude changes while Brownian bridge and observation-shutdown rivals fail under the same observation model. Support belongs to that realization and context.

Mechanism results identify which tested routes explain the observations under the stated design and comparison conditions (Camerer et al., 2018; Klein et al., 2018).

Version mechanism support. Moving from finite-variation to stochastic reserve, or from scalar scheduling to partially observed control, requires a named version; earlier support does not automatically transfer.

Mechanisms serve testability by making claims vulnerable. Failed mechanisms should leave cleanly.

27.3.5 — Convergent Support Across Methods

Convergent support requires different measurement and study families to support one prospectively defined component, reducing dependence on any instrument, site, or analysis. Similar prose is insufficient: dream studies, terminal telemetry, horizon tasks, and longitudinal reports need declared bridges to the same target (Open Science Collaboration, 2015; Camerer et al., 2018).

Terminal variance compression might appear in active reports, clinician-assisted windows, and physiological residuals under one endpoint rule. Convergence strengthens when confounds differ; shared missingness or selection can create a common artifact.

The component graph must connect each method to the target, with its recall, device, clinician, and selection weaknesses explicit (Klein et al., 2018; Vehtari et al., 2017).

True convergence is compelling because the paths remain distinct until disciplined evidence shows where they meet.

27.3.6 — No Whole-Claim Validation

The Science of Fairness updates scientific components and their public descriptions. Applied titles use relevant findings through their own research, legal, practical, and interpretive processes (Camerer et al., 2018; Klein et al., 2018).

Multisite support applies to a named component, reference class, and method. Status labels must follow donor pages, graphics, summaries, and paired books to block prestige by association (Vehtari et al., 2017; Open Science Collaboration, 2015).

Component support, bounded replication, and narrowed evidence state preserve the contribution without validating the series, religion, transition, or political authority.

27.4 — Inconclusive and Unusable Evidence

Inconclusive and unusable evidence are different scientific states. Wide uncertainty may leave several values plausible; model nonseparability may leave several generators observationally equivalent; measurement or endpoint failure may block the target; and an ethical stop may make continuation unacceptable. Each state preserves different information and requires a different repair (Manski, 2003; Little & Rubin, 2019; Lakens, 2017).

The classification protects both directions of error: nondiscrimination is not promoted to support, and a failed design is not promoted to defeat of a target it never identified. The outcome belongs to the actual failure mode.

Table 27.5. Inconclusive and unusable outcomes.

Outcome	What happened	Correct update
Wide or mixed uncertainty	Intervals, posteriors, or sensitivity analyses fail to discriminate.	Inconclusive for the tested component.
Nonseparable models	Model recovery or design separability fails before inference.	No winner; redesign or narrower claim.
Measurement failure	Instrument sign, scale, missingness, or calibration fails.	Target untested in that context.

Outcome	What happened	Correct update
Endpoint failure	Stopping object is ambiguous, unavailable, outcome-contaminated, or incompatible with the declared endpoint branch.	Terminal inference suspended.
Ethical or burden stop	Study cannot proceed without excess burden or risk.	Stop or reduce observation; classify each affected inference under its remaining valid gates.
Public uncertainty	Evidence state is mixed or not yet reviewed.	Status-labeled uncertainty, no spin.

Use Table 27.5 together with §24.2's canonical No adequate estimate rule for the evidence-state definitions. This section focuses on the information each state preserves and the repair it requires.

Unusable evidence also needs respect. Data can be unusable for the intended inference and still ethically significant. A burden stop, a family-pressure incident, or a participant challenge may teach the program where measurement itself creates load. Those lessons belong in the record.

Reviewers distinguish ignorance from confusion. Ignorance means a component was not tested. Confusion means data were collected but cannot discriminate. Failure means the component was tested and contradicted. These states create different scientific obligations.

A generic call for more research is not enough. The public record identifies whether the obstacle was low power, nonidentification, missingness, heterogeneity, endpoint uncertainty, rival nonseparability, or protocol failure because each demands a different next step.

A high-quality inconclusive study can reveal how to strengthen the next test. It may identify a fragile signature, a measurement gap, or a participant-protection rule that needs refinement. Those findings improve the research program by locating the changes most likely to make subsequent evidence informative.

Uncertainty becomes a public object. It is not a private discomfort behind the scenes. When uncertainty blocks a claim, the block should be visible enough that later writers cannot step around it by accident.

A report of inconclusive evidence should include whether the component remains worth testing. Sometimes a better design is available. Sometimes the burden or identification problem suggests that the question should be left alone until methods change. This distinction protects participants from automatic repetition.

Unusable evidence should not be destroyed unless retention itself creates risk. The record may need to preserve metadata, stop reasons, and protocol lessons while minimizing or deleting sensitive observations. Data dignity and failure memory must be reconciled case by case.

A nonseparable model result should feed back into benchmark design. If real data could not choose among models, the next step

is not a larger sample by default. It may be a new observation channel, randomized cadence, better endpoint adjudication, or an admission that the component is not yet identifiable.

Measurement failure should also update consent materials. Participants should not be invited into a future study with language implying that the prior instrument worked. The burden of honesty begins before recruitment.

Endpoint failure should update care coordination. If terminal inference required more observation than comfort allowed, the next protocol should not simply ask harder. It should ask whether the terminal question can be answered without increasing the burden on dying people and families.

Uncertainty, handled well, is protective knowledge. It tells the program where not to promise, where not to recruit, and where not to draw the next public diagram.

Some evidence states are unanswerable rather than merely inconclusive. An unanswerable zone appears when the component cannot be tested under responsible conditions, when observation would create unacceptable burden, when the endpoint cannot be defined without circularity, or when the construct cannot be transported into the reference class without inventing what the study was supposed to learn. This status marks a boundary in instrument, ethics, or reference class; it is not a rhetorical refuge from defeat.

The evidence graph records unanswerable zones with the same seriousness it gives positive replications. The entry names the component, attempted design, reason the verdict cannot be

reached, ethical or measurement barrier, public consequence, and whether the block is temporary, design-contingent, or principled. A temporary block may await a less burdensome sensor or stronger translation method. A principled block may arise when the only route would manipulate suffering, pressure final reports, or convert private grief into research material.

The boundary becomes concrete in a terminal subpopulation whose only available observation windows occur during sedation decisions, family crisis, and communication loss. The program may care deeply about that reference class without pretending that the final open-channel window was observed. A similar boundary arises where terminal reporting would expose families to stigma or spiritual pressure; protected silence cannot be recoded as neutral missingness.

An unanswerable result identifies what the observations leave unresolved, such as affect during silence, the consequences of withdrawal, or endpoint classification. The analysis retains those uncertainties and describes the information needed for a later answer.

Public language for an unanswerable zone is plain: this component has no responsible verdict in this reference class under current methods. The record names whether the blocker is measurement, endpoint, missingness, burden, nonseparability, or open-set inadequacy. It does not call the result pending when no responsible design exists, and it does not call the target false when the target was never tested.

27.4.1 — Wide or Mixed Uncertainty

Wide or mixed uncertainty occurs when the data do not discriminate among the relevant possibilities. The signal may point one way, another sensitivity analysis may point another, and prediction intervals may remain too broad for status movement. This is an inconclusive scientific result (Manski, 2003; Little & Rubin, 2019).

The proper analysis should show where uncertainty enters: sampling variability, measurement error, missingness, endpoint ambiguity, site heterogeneity, prior sensitivity, model choice, or rival overlap. A single wide interval is less useful than a map of why it is wide.

Consider a study whose mean terminal estimate lies near the neutral band but whose equivalence interval crosses both meaningful positive and meaningful negative regions after missingness sensitivity. The result is not support. It is nondiscrimination under current measurement and observation assumptions.

A favorable center can hide wide uncertainty, reversals, and unusable cases. Readers may focus on the point estimate and ignore the width. The public label must therefore lead with uncertainty, not with the most attractive center (Lakens, 2017; Mayo, 2018).

Wide uncertainty can still teach. It may identify a needed cadence, a better endpoint window, a stronger rival benchmark, or a less burdensome design. But it does not move component support unless the decision rule declared that it could.

Inconclusive evidence is not a blank. It is a record of where the instrument, design, or reference class could not yet speak.

27.4.2 — Nonseparable Models

Nonseparable models occur when the analysis cannot distinguish candidate explanations under the available data. The Law-candidate model, a Brownian bridge, an observation-shutdown model, and a frailty-selection model may all reproduce the same measured trajectory. Model fit alone cannot decide among them (Little & Rubin, 2019; Lakens, 2017).

The separability problem should be addressed before human recruitment when possible. Synthetic challenges, parameter recovery, held-out prediction, and adversarial simulations should show whether the proposed design can recover the true generator under plausible conditions. If it cannot, real data will not rescue the comparison.

When reserve coupling and a random-endpoint bridge fit terminal concentration equally well after accounting for observation timing, report model nonseparability. The result identifies the contrast that a revised design needs to resolve.

Preference for a parsimonious story cannot decide between models that the design failed to separate. Analysts may choose the model that matches the theory and call the rival too artificial, or choose the rival and call the Law defeated. Both moves overstate the evidence if the design cannot separate them (Mayo, 2018; Manski, 2003).

The detachable sentence is: the models were observationally nonseparable under the declared design, so no mechanism winner

is reported. That wording protects the result from later compression into support for whichever model a reader already prefers.

A model that cannot be distinguished from a serious rival has not lost all value. It has lost the right to discriminate publicly until the design improves.

27.4.3 — Measurement Failure

Measurement failure identifies the target and context for which the instrument lacks a qualified estimate. The record specifies the relevant sign, scale, calibration, missingness, device, cultural, disability, medication, or channel issue and the validation needed to proceed (Lakens, 2017; Mayo, 2018).

A failed candidate instrument version should be retired, narrowed, or reworked without blaming participants. The problem may be the model's relation to lived experience, not the person's testimony. Measurement dignity means a nonstandard signal is not treated as deception or noise by default.

Measurement failure is unmistakable in an estimator that treats low movement and low speech as reduced distress in a neurological population where motor decline and communication impairment drive those channels. The instrument may falsely suggest neutrality. The correct update is measurement failure in that context.

Burying nuisance structure in a supplement does not repair the main inference. Teams may put the measurement issue in a supplement while the main text reports the desired status. The

measurement failure must appear in the same public status statement as the result (Manski, 2003; Little & Rubin, 2019).

Measurement failure can still produce useful component science. It may reveal channel limits, calibration needs, or protected contexts where measurement should not be attempted. Those findings should travel into future protocols.

The target remains worthy even when the instrument fails. A broken thermometer does not make fever imaginary; it makes the reading unusable.

27.4.4 — Endpoint Failure

Endpoint failure occurs when the stopping object is ambiguous, unavailable, outcome-contaminated, or incompatible with the declared branch. Chapters 7 and 13 define the branch taxonomy and random-endpoint conditions. A failed endpoint gate may preserve useful preterminal evidence, but it cannot adjudicate a terminal target outside the endpoint branch actually identified (Mayo, 2018; Manski, 2003).

Endpoint failure can arise through sudden death, resuscitation, sedation, terminal delirium, sensor dropout, family burden, delayed documentation, or post hoc endpoint selection. Each route should be coded separately because each implies a different repair or limitation.

Endpoint discipline is tested by a low-signal ending where the last usable candidate instrument window occurs forty hours before truth-level stream endpoint is clinically inferred. Treating that window as terminal would make the result cleaner and less true.

The endpoint component should suspend terminal inference and perhaps preserve only preterminal dynamics.

The final available observation cannot stand in for the truth-level endpoint merely because it is convenient. Analysts may be tempted to call the final available measure the endpoint measure. That move converts missingness into evidence. The endpoint firewall forbids it (Little & Rubin, 2019; Lakens, 2017).

The public account should distinguish an unavailable endpoint, unresolved timing, outcome-contaminated adjudication, false-ending risk, and failure of a declared branch-specific timing premise. Dependence alone is not endpoint failure when a state-coupled branch explicitly models it.

Endpoint failure is not defeat unless the endpoint was adequately identified and contradicted the claim. It is a refusal to pretend that a terminal question was answered when it was not.

27.4.5 — Ethical or Burden Stop

An ethical or burden stop occurs when a study cannot responsibly continue because the measurement burden, privacy risk, participant distress, family pressure, staff load, or clinical interference becomes too high. Stopping protects care. If a required gate fails, the affected evidence is Unusable; otherwise any consent-permitted analysis of retained data keeps the status supported by its validity, identification, and precision (Council for International Organizations of Medical Sciences, 2016).

The stop rule should be prospectively specified and operational. It should identify who can stop, which burden indicators trigger review, how withdrawal is honored, how family pressure is

handled, and how the resulting evidence state is labeled. A humane stop should not be improvised under crisis.

The ethical rule is tested by a terminal protocol where repeated prompts begin to distress participants or pressure families into interpreting final experiences. The correct action is to stop the measurement, classify the affected endpoint inference as Unusable if the stop causes a required endpoint gate to fail, and preserve care as the priority.

Participant burden is not an ordinary missing-data mechanism to be optimized away. If the reason observation stopped is that observation itself became harmful, imputing the missing path as though it were ordinary dropout hides the most important fact. The burden is part of the evidence state (Lakens, 2017; Mayo, 2018).

After an ethical or burden stop, public reporting must identify the procedure that stopped, the reason, and the data that remain usable. It must also record the participant or staff support that followed and the questions the program will no longer pursue in that form.

A science of fairness proves its seriousness by knowing when not to measure. Some unanswered questions are ethically better than answered ones.

27.4.6 — Public Language for Uncertainty

Public uncertainty language must avoid both promotional spin and false defeat. The public can understand uncertainty when it is explained as a precise evidence state. What destroys trust is not uncertainty itself, but uncertainty hidden until critics expose it

(Little & Rubin, 2019; Lakens, 2017). A stable vocabulary for uncertainty makes corrections less dramatic because the public has already been shown that inconclusive and unusable are legitimate outcomes.

The status statement should contain four elements: component, evidence state, scope, and reason. For example: measurement component inconclusive in a multilingual terminal sample because calibration and missingness sensitivity did not discriminate. That sentence is less exciting than a headline and more useful than a slogan. Adding the planned next action makes the status operational without implying that every uncertainty can or should be resolved immediately.

The uncertainty categories separate in a public website summarizing a mixed replication. The accurate label may be partial support for one terminal consequence, measurement failure in one subgroup, and rival dominance in one model comparison. A single green, yellow, or red icon would mislead. Component-level display can still be visually simple if each icon or line corresponds to a named claim rather than one aggregate project mood. A compact table can place the shared study identifier beside separate component outcomes, making disagreement visible without turning the entire project into one score.

Compressing mixed uncertainty into one status label destroys information the verdict needs. Donor pages, media summaries, diagrams, and social posts often shrink nuance into a mood. Public artifacts must preserve the component structure or refrain from release (Mayo, 2018; Manski, 2003). If compression cannot

preserve the distinction, a link to the full component table is not enough; the simplified artifact itself should not imply a unified verdict. When one component is corrected, the notice can identify the other components that remain unchanged instead of replacing the entire account with a new headline.

A public summary must name whether unusable evidence narrows, suspends, blocks, retires, or leaves the question unanswered. The summary should also identify whether the limitation comes from data quality, identification, model adequacy, or a genuine adverse result. An adverse result under valid prerequisites retains its adverse classification; it is not reclassified as unusable simply because the preferred account lost.

27.5 — Narrowing Reference Classes and Claims

Narrowing can preserve a true result only when the successor is explicitly smaller, independently motivated, and prospectively testable. A reduced population, time domain, or context may be a genuine reference-class narrowing; a change in representation, exactness, modality, endpoint, or target object creates a non-equivalent successor that must receive its own identity and decision rule.

A principled narrowing is predeclared or independently motivated, publicly labeled, testable in its new form, and connected to a mechanism or measurement reason. A degenerating narrowing follows failure with a smaller claim that predicts nothing new.

Table 27.6. Narrowing and successor routes and their required public correction.

Revision route	What changes	Public correction required
Population	Diagnosis, age, species, state, culture, or care regime.	Identify the old broader class and the new bounded class.
Temporal	Terminal claim becomes preterminal or window-specific.	Remove language implying endpoint coverage.
Representation	Scalar closure becomes multidimensional consequence.	Revise formulas, diagrams, and summaries.

Revision route	What changes	Public correction required
Endpoint	Universal endpoint becomes qualified stopping event.	State which endpoint decisions no longer travel.
Statistical	Pathwise universality becomes reference-class tendency.	Remove universal Law-language from empirical claims.
Public artifact	Graphic or guide overstates the old scope.	Replace, date, and preserve correction history.

Equation 27.4 (schematic rule for pure reference-class narrowing):

$$D_{\text{new}} \subsetneq D_{\text{old}}$$

$$\mathcal{P}_{\text{new}} = \mathcal{P}_{\text{old}}|_{D_{\text{new}}}$$

Here D denotes the declared reference-class domain and $\mathcal{P}$ the claim's prediction rule on that domain. Pure reference-class narrowing restricts the domain while keeping predictions for retained cases, target geometry, and the decision rule unchanged. Changes to representation, modality, endpoint, or target object require a non-equivalent successor and are outside this schematic rule. The new class is independently motivated, defined without inspecting favorable outcomes, and tested with a new transport or inclusion rule. Otherwise, post hoc exclusion becomes rescue.

Narrowing should begin with a comparison between the old claim and the new one. What population was removed? What endpoint was qualified? What temporal window was shortened? What representation changed? Without that comparison, readers cannot tell whether the program learned or merely moved.

Each narrowing should state its independent reason. A population may narrow because measurement invariance failed. A temporal window may narrow because endpoint identification failed. A representation may narrow because scalar additivity failed. A statistical claim may narrow because pathwise universality lacked empirical content. The reason is part of the narrowed claim.

A narrowed claim states new predictions and the outcomes that would challenge them. A sharper target makes the next test more informative.

Narrowing has downstream effects. Protocols, analysis plans, public graphics, cross-book references, donor summaries, media language, and future chapter summaries must all change. The correction burden belongs to the project, not to the skeptical reader.

A program may narrow more than once, but repeated narrowing should trigger review. At some point, the question becomes whether the remaining claim still carries the original empirical ambition or whether it should be reclassified as a different, smaller hypothesis. The no-rescue rule belongs here.

Transparent narrowing preserves the structure that evidence supports and gives it a clearer domain of use. Showing the old and new claims together lets readers see exactly what was learned.

Narrowing should include an audit of inherited language. The old wording may remain in introductions, captions, summary pages, social excerpts, conference slides, and earlier chapters. If those places are not corrected, the narrowed claim has not actually narrowed in public.

A narrowing register should store the old claim, new claim, evidence reason, date, reviewer decision, dissent, and affected artifacts. This may sound administrative, but it is how scientific memory prevents repeated overclaim. Without a register, narrowing becomes a whisper.

The distinction between reference-class narrowing and mechanism retirement should be preserved. A claim may hold only in one population because a mechanism operates there, or because measurement only works there, or because rivals differ there. The reason determines what can travel.

Narrowing should not be used to protect emotional language. If the universal claim narrows to a bounded empirical signature, public prose should stop sounding universal. The emotional register must shrink with the scientific claim.

The strongest narrowed claims often become more testable. A precise population, endpoint, representation, and statistical target can generate sharper protocols than the broad claim could. Narrowing can therefore increase scientific power while reducing public grandeur, provided the reduction remains visible enough that no one mistakes it for unchanged ambition.

27.5.1 — Population Narrowing

Population narrowing limits a claim by diagnosis, age, species, developmental stage, culture, state of consciousness, care regime, or other reference-class boundary. It is justified when evidence shows that a component holds in one population and fails, remains untested, or changes meaning in another (Bareinboim & Pearl, 2016; Stuart et al., 2011).

The narrowing must specify why the boundary matters. A diagnosis boundary may reflect medication, communication, disease trajectory, reserve dynamics, endpoint confidence, or measurement invariance. A cultural boundary may reflect language, reporting norms, family mediation, or care practices. Without a mechanism, population narrowing can become arbitrary rescue.

Consider support for one candidate instrument version in adult hospice oncology and failure in advanced dementia because bearer continuity, reportability, medication, and endpoint evidence differ. The correct update is not human evidence in general. It is a bounded adult oncology result with dementia-specific suspension or redesign.

Convenience slicing narrows the reference class after the result rather than through an independently motivated rule. After a broad study fails, analysts may search subgroups until one looks favorable. That subgroup can motivate exploration, but it cannot become principled narrowing without new justification and prospectively specified confirmatory work (Fortier et al., 2011; Lakatos, 1978).

Any public correction must name both the old reference class and the new one. It cannot quietly replace all conscious streams with certain adult terminal contexts as though the claim had always been that narrow.

Population narrowing is honorable when it protects truth from overreach. It becomes rescue only when it hides the distance between what was promised and what survived.

27.5.2 — Temporal Narrowing

Temporal narrowing limits a claim to certain windows, such as preterminal dynamics, post-event adjustment, sleep-related repair, or short-horizon control tasks. It may occur when terminal endpoint evidence fails but earlier dynamics remain identifiable and informative (Stuart et al., 2011; Fortier et al., 2011).

The key question is whether the temporal boundary is scientifically motivated before the result. A preterminal window may be justified by sampling cadence, reserve observability, medication stability, or ethical access. It is not justified simply because the terminal window failed.

Temporal narrowing is clearest in strong evidence for variance contraction in the last month of recorded clinical life, with unusable final open-channel windows due to sedation and endpoint ambiguity. The program may preserve a preterminal consequence claim while suspending terminal inference.

Retreating from a terminal claim to a preterminal one after failure is narrowing, not continued support for the original target. It should not retain the emotional force of terminal neutrality while abandoning the endpoint burden (Lakatos, 1978; Bareinboim & Pearl, 2016).

For a preterminal study, the release identifies its observation window, estimated dynamics, and the additional endpoint or route qualification needed for the next investigation.

Temporal narrowing can still be scientifically rich. A smaller window with clearer identification may teach more than a universal window defended by missing data.

27.5.3 — Representation Narrowing

Representation change creates a new target unless a proved bridge preserves the original object. Replacing exact scalar closure with a vector-valued structure, finite-band compatibility, variance compression, endpoint concentration, reserve-linked contraction, or a directional tendency is not merely a smaller version of additive PTV. Each successor requires its own identifier, assumptions, estimands, rival set, and decision rule, while the predecessor retains its original status (Lakatos, 1978).

A revised representation should have independent empirical motivation and testable consequences. If valence, arousal, pain, meaning, fatigue, and threat carry distinct predictive information in a reference class, a multidimensional structure can preserve that information.

A multidimensional study may show that physical pain, social fear, and existential distress behave differently near an endpoint, defeating the proposed scalarization in that context. The result can motivate a vector or set-valued successor and preserve valuable lived-experience science, but it does not count as support for the failed scalar branch.

Unannounced representation drift changes the object while preserving the appearance of continuity. If the public graphic still implies one closing line while the technical specification has moved to multiple noncollapsible dimensions, the program is misleading readers.

When representation changes, formulas, diagrams, summaries, and status labels change with it. Representation is not a footnote because it determines what counts as evidence.

Representation change is scientifically honest only when the formal object, equations, estimands, and public language all change with it.

27.5.4 — Endpoint Narrowing

Endpoint narrowing limits claims to specific qualified events rather than universal death-of-mind classification. It may be necessary when some endpoints can be independently adjudicated and others cannot. The narrowed endpoint becomes part of the claim, not a sampling detail (Lakatos, 1978; Bareinboim & Pearl, 2016).

A qualified endpoint might be an open-channel window before irreversible loss of unity, a clinically adjudicated consciousness transition, a terminal observation window under low-sedation conditions, or another prospectively specified stopping event. Each carries different authority and requires its own bridge.

The endpoint scope can be tested with a protocol that can identify terminal open-channel windows in gradual decline but not in sudden death. Evidence from gradual decline cannot be extended to sudden endings unless a separate endpoint model is built and tested.

Name the endpoint class in the main result sentence so that gradual decline, sudden endings, and other event types retain their observed scope (Stuart et al., 2011; Fortier et al., 2011).

A corrected endpoint claim names the qualified endpoint and removes universal phrasing from summaries. The phrase truth-level stream endpoint cannot imply that all endpoint classes were equally identified.

Endpoint narrowing keeps the endpoint firewall alive after data arrive. It prevents the most emotionally powerful word in the theory from outrunning the most fragile evidence.

27.5.5 — Statistical Narrowing

Statistical change also creates a new claim when it changes the quantifier or target geometry. Moving from set-universal exact closure to a reference-class tendency, from exact zero to finite-band concentration, or from terminal closure to a probabilistic signature does not preserve the original empirical target. The successor may be valuable, but it requires a new protocol, error guarantee, and status record (Bareinboim & Pearl, 2016; Stuart et al., 2011).

The strong latent form and the public empirical target must remain distinct. If empirical evidence supports a tendency in a measured reference class, the public claim should say tendency in that class. It should not imply that the latent strong form has been established.

A meta-analysis may find near-neutral concentration on average while some sites remain clearly adverse and the prediction interval is wide. The program may report that reference-class tendency and heterogeneity as a new empirical claim. It may not describe the result as pathwise or universal closure.

Changes in statistical scope should be reflected consistently in prose, formulas, targets, and verdicts. Expressions such as overwhelmingly tends, essentially closes, and in the limit each require the corresponding probabilistic or limiting statement. Explicit wording makes the revised claim available for precise evaluation (Lakatos, 1978).

The correction notice must distinguish the old claim strength from the new one. Pathwise, ensemble, and predictive-distribution claims are not interchangeable.

A statistical successor succeeds when it says exactly what the data support and leaves the original target available for independent rejection or suspension.

27.5.6 — Public Correction After Narrowing

A narrowing notice should identify the withdrawn scope and the corrected claim, update project-controlled high-reach artifacts, and notify known external reusers. The correction log records unresolved copies so that incomplete recall is visible rather than mistaken for unchanged scientific status (Crossref, 2025).

The correction should be dated, visible, and component-specific. Preserve old language in the record rather than erase it, so failure memory shows what changed and why.

Narrowing and rescue diverge when a visual explainer still shows a universal terminal curve after evidence narrows to one palliative reference class. Replace the image, archive it with a correction note, and cross-reference every reuse. The correction note should state the surviving reference class directly on the replacement

visual so old universal interpretation cannot reappear through cropping.

Quiet revision breaks the evidence-to-claim chain and makes readers reconstruct changes made for institutional comfort. A science of fairness cannot shift that burden outward (Lakatos, 1978; Bareinboim & Pearl, 2016). Version history should make narrowing visible at the point of use, not require readers to infer it by comparing editions.

Calm correction states what changed, what survives, what failed, and which language is retired; it avoids apologetic theater. A calm correction can also preserve confidence in unaffected components by separating the narrowed claim from the parts of the program that did not change.

27.6 — Mechanism Retirement and Rival Dominance

Candidate mechanisms remain useful only while they make recoverable, discriminating predictions. A route that fits after outcome-dependent tuning, survives only by changing its latent object, or loses repeatedly to equal-budget rivals must narrow, retire, or yield resources. Mechanism retirement protects the target from being fused to one favored story.

Rival dominance receives its proper scientific status. If adaptation, Brownian bridges, predictive control, safe reinforcement learning, frailty, observation shutdown, or selection explains the evidence better under fair comparison, the public status must change.

Table 27.7. Mechanism retirement and rival dominance matrix.

Object	Retirement trigger	What may survive
Reserve realization	Bridge failure, parameter nonrecovery, rival superiority, wrong variance pattern.	Formal question and general reserve measurement may survive.
Queue or horizon	Ordinary control explains priority, blocking, and repair signatures.	Behavioral task insights and burden analysis may survive.
Candidate momentary-valence estimator version	Calibration, sign, transport, or missingness failure.	Data dignity rules and revised measurement lessons may survive.

Object	Retirement trigger	What may survive
Rival dominance	Repeated superior held-out prediction and calibration by rivals.	Bounded components not explained by the rival may remain.
Replacement mechanism	No independent motivation or no new falsifier.	None; replacement waits for prospectively specified risk.
Useful component science	Law-specific interpretation fails but findings remain sound.	Clinical, measurement, or ethical insights continue with narrower labels.

Mechanism retirement requires a clean archive of what was promised. A model cannot be retired responsibly if no one can tell which predictions it made. Each mechanism should therefore have a versioned prediction sheet, rival matrix, parameter-recovery record, and public-risk note. Retirement then becomes a decision, not a mood.

Rival dominance should be announced with the same precision as support. The rival should be named, the component should be named, and the comparison should be described. Saying ordinary models explain it is too vague. Saying an endpoint-conditioned bridge with observation shutdown dominates a reserve-coupled candidate under the shared observation model is useful.

Retirement should not erase useful work. A retired mechanism does not invalidate every dataset, participant contribution, or method built around it; it changes the interpretation and public claim attached to those materials, and the record should preserve that distinction.

Replacement proposals should face a cooling period after defeat. Immediate substitution can be emotionally satisfying and scientifically reckless. A new mechanism should be motivated by independent theory or new data, not by the desire to keep old public language alive.

The public status of a retired mechanism should include a future-use rule. Can the old model be used as a simulation baseline? Can it appear in teaching as a failed candidate? Can its variables remain in exploratory analysis? Can it be cited in public summaries? These permissions should be explicit.

Mechanism retirement is an evidentiary decision about one declared route, not a euphemism for moving it into metaphor. When a prospectively declared route fails its recovery or signature criteria under an adequate design, its status is Route-failed. The record identifies the failed predictions, recovery results, equal-budget rivals, scope, and downstream artifacts that must change.

A retired mechanism should not be erased from educational materials. It can be taught as a candidate that made the theory vulnerable and then lost. That history teaches readers how the program protects itself from self-confirmation.

Rival dominance should update source priority. If mainstream models explain a component better, later chapters, summaries, and public artifacts should cite or describe those models as the leading explanation for that component. The Law-specific model should not remain first by tradition.

Mechanism retirement can create communication anxiety because mechanisms are memorable. Candidate scheduling language, reserve language, and horizon pressure may be easier for readers to remember than evidence-state language. The response is not to keep failed metaphors alive, but to replace them with clearer bounded explanations.

When a mechanism retires, dependent claims should be traced. A chapter, figure, or protocol that relied on the mechanism may also need revision. Retirement is not isolated if the mechanism served as a bridge for other statements.

Useful residual methods should be given new names when necessary. A protocol that began as a candidate scheduling-route test may become an urgency-control task or a burden-prioritization study. Renaming protects the result from carrying a retired interpretation.

Graceful loss is the governing standard. A Route-failed mechanism may leave the Law-specific framework and still leave behind knowledge good enough to keep.

27.6.1 — Reserve Realization Retirement

Reserve realization retirement occurs when the product model, its bridge to measured quantities, or its distinctive predictions fail under adequate testing. The reserve idea may be elegant, but elegance is not tenure. A realization that cannot recover parameters, survive rivals, or transport across contexts should be removed or narrowed (Palminteri et al., 2017; Chamberlin, 1890).

Retirement triggers include wrong variance exponents, unstable deconvolution, bridge failure, correlated observation artifacts,

rival dominance by Brownian bridges or frailty, and inability to identify reserve independently. Each trigger should be recorded because each teaches a different lesson.

Consider a benchmark showing that fixed and random endpoint bridges reproduce apparent compression under the same observation schedule, while reserve parameters are nonidentifiable across plausible noise floors. The product realization may be retired for that evidence family.

Once a reserve model fails, retaining reserve as an all-purpose metaphor conceals the retirement. If the realization no longer functions scientifically, the word reserve should not remain as a vague explanation for every contraction. The public record should say which reserve claim ended (Platt, 1964; Burnham & Anderson, 2002).

Retirement does not erase useful work. It may preserve clinical reserve measurement, stochastic modeling lessons, noise-floor analysis, and rival-comparison methods. What ends is the Law-specific interpretation of that realization.

Retiring a realization leaves a clear record for better models. Later chapters should identify the retired version and preserve its useful methods under their revised interpretation.

27.6.2 — Candidate Scheduling and Horizon Route Retirement

Candidate scheduling and horizon-route retirement occurs when ordinary planning, urgency, homeostasis, active inference, reinforcement learning, scarcity, or safe-control models explain priority, blocking, reactivation, discharge, or finite-horizon

effects without a Law-specific structure. The scheduling route is a candidate model, not an invisible actor that must be preserved (Chamberlin, 1890; Platt, 1964).

The retirement test should compare executable models under shared tasks. If ordinary harm avoidance or model-predictive control reproduces the signatures with fewer assumptions and better held-out performance, the candidate scheduling interpretation narrows or leaves the scientific layer.

Route retirement can be seen in a laboratory task where horizon-sensitive avoidance appears exactly as finite-horizon safe control predicts, with no residual priority structure unique to the candidate scheduling model. The behavioral result may still matter, but the retired scheduling mechanism does not own it.

Ordinary urgency must not be relabeled as candidate scheduling pressure merely to preserve a Law-specific interpretation. That move empties the construct of empirical content. If every priority shift is treated as candidate scheduling activity, the route cannot lose (Burnham & Anderson, 2002; Palminteri et al., 2017).

Candidate scheduling language exists to make burden-ordering testable enough that survival has meaning. When a scheduling or horizon route retires, the release should say which mechanism was tested, what failed, which ordinary explanations performed better, and whether any target-level question remains open.

27.6.3 — Candidate Instrument Version Retirement

Candidate instrument version retirement occurs when an instrument cannot identify, calibrate, transport, or ethically operate in its declared context of use. A version may retire

because of device drift, cultural calibration failure, disability bias, medication confounding, channel dropout, or participant challenge outcomes (Platt, 1964; Burnham & Anderson, 2002).

Retirement is version-specific. Candidate instrument version 2.4 may fail in one setting while another version remains under development elsewhere. The record does not treat the candidate momentary-valence estimator as one timeless instrument whose failures can disappear inside later versions.

Instrument retirement becomes unavoidable in a wearable-heavy candidate instrument version that performs poorly after a firmware update changes motion filtering and heart-rate variability estimates. Leave-device-version-out validation fails. That version should stop contributing to confirmatory evidence until recalibrated.

Silent recoding can salvage a dataset only by changing the object after the test. Teams may try to rescue the instrument by adjusting weights after outcomes are known. Such work may become exploratory re-estimation, but it cannot preserve the original confirmatory status (Palminteri et al., 2017; Chamberlin, 1890).

Public-facing text for a failed instrument should use the explicit verdict Unusable evidence, identify the context of use that failed, and state whether the failure is repairable or structural. It should not convert instrument weakness into ambiguity about the person.

Instrument retirement is one way the project protects persons from its own ambition. A failed measure should burden the protocol, not the participant.

27.6.4 — Rival Dominance

Rival dominance occurs when an alternative model repeatedly predicts, calibrates, transports, and explains the relevant component better than the Law-candidate model under fair comparison. It is not merely a rival doing well. It is the point at which the Law-specific interpretation no longer carries the best explanation for that component (Burnham & Anderson, 2002; Palminteri et al., 2017).

Dominance should be judged through held-out prediction, calibration, posterior predictive checks, model recovery, parsimony, robustness, transport, and absolute fit. It should also include whether the rival handles missingness, endpoints, device changes, and site heterogeneity without special pleading.

Rival dominance becomes clear in a combination of frailty, observation shutdown, and endpoint-conditioned diffusion that predicts terminal concentration and variance compression across sites better than the reserve candidate, while leaving no Law-specific residual. The component becomes rival-dominant.

Rival comparison is asymmetric when the favored model receives more information, flexibility, or interpretive charity. If rivals are tuned less carefully, denied realistic observation models, or excluded from certain diagnostics, dominance cannot be judged. Rival seriousness must come before rival victory (Chamberlin, 1890; Platt, 1964).

When a rival dominates, the public account should say that the named rival explains the named component better under the declared evidence set, while also stating what, if anything,

remains unexplained. This prevents both totalizing defeat language and denial of the rival's success.

Rival dominance is scientific progress. It identifies the explanation that currently serves the evidence best and gives subsequent research a stronger starting point.

27.6.5 — No Mechanism Substitution Without New Risk

A failed mechanism may be replaced only when the successor adds independent predictions, can lose to serious rivals, and carries its own public-risk review. Replacement is a new scientific commitment, not a costume change (Palminteri et al., 2017; Chamberlin, 1890).

The proposal states what the old route could not explain, why the successor is motivated before the new target result, how its variables are measured, which data distinguish it, and which public language it forbids. A vector reserve model, for example, must name its dimensions and a prediction that separates it from ordinary allostatic regulation.

Hidden variables, flexible priors, new regimes, or unmeasured states added only after failure are rescue unless they create fresh exposure to defeat. Added sophistication that merely shrinks the failure space is status pressure, not advancement (Platt, 1964; Burnham & Anderson, 2002).

The public record marks the superseded model, displacing evidence, successor scope, unchanged claims, and new risk. A good replacement makes the program more vulnerable and more informative.

27.6.6 — Preserve Useful Component Science

Useful component science may survive after a Law-specific interpretation ends: measurement can improve palliative research, missingness methods can protect participants, rivals can clarify terminal data, and failed mechanisms can leave valuable tools (Chamberlin, 1890; Platt, 1964).

Retired approaches can retain operational value. An instrument may reveal device bias, and a scheduling model may leave an effective burden-monitoring protocol. Recording those contributions preserves useful knowledge for later work.

Public summaries separate mechanism retirement from target defeat and reject both total collapse and automatic preservation of attached claims. Keeping useful science under the authority it earned lets truth and usefulness coexist (Burnham & Anderson, 2002; Palminteri et al., 2017).

27.7 — Theory-Level Defeat and Failure Memory

Theory-level defeat requires a stronger evidentiary route than failure of one instrument, route, or reference class. It occurs when a declared universal or branch-specific target is contradicted under its own prerequisites, or when repeated repairs remove the distinctive empirical content that made the target testable. The defeat record preserves the exact branch, evidence, assumptions, and successor consequences.

Defeat must be possible, explicit, and remembered. If defeat can always be renamed as refinement, the program has lost empirical content. If defeat is treated as disgrace, the program will avoid naming it. Defeat becomes a governed scientific outcome.

Table 27.8. Defeat and abandonment criteria.

Defeat route	Minimum condition	Required consequence
Credible counterexample	Well-identified stream and endpoint with robust terminal imbalance.	Pressure on universality and strong-form status.
Repeated conjunction failure	Distinctive signature conjunction repeatedly absent under adequate designs.	Escalation from component failure to program-level pressure.
Unresolvable identification failure	No empirically meaningful bridge can be justified under ethical constraints.	Suspension or withdrawal of public empirical claim.

Defeat route	Minimum condition	Required consequence
Persistent rival superiority	Serious rivals repeatedly dominate across component comparisons.	Retire Law-specific interpretation for those components.
Defeat statement	Review determines scope of withdrawal and surviving claims.	Public correction, status revision, and cross-artifact update.
Failure memory	Data, code, reasons, dissent, and corrections preserved.	Prevents future re-overclaim and repeated recruitment burden.

Equation 27.5 (empirical-content exhaustion test):

$$\lim_{k \rightarrow \infty} \Delta \text{Prediction}(k) = 0 \quad \Rightarrow \quad \text{rescue-versus-refinement review flag}$$

The display names the danger rather than proving a theorem. When each revision preserves the claim while adding no new discriminating prediction, the program should classify the move as rescue and apply theory-level pressure.

Theory-level defeat is branch-specific and explicit. A gate-valid counterexample defeats the exact universal branch it reaches; a repeated inability to identify the target may instead yield principled abstention or a smaller successor. The record distinguishes contradiction from nonidentification so neither is used to rescue the other.

A defeat review should include internal scientists, external critics, methodological specialists, participant or family representatives where appropriate, and public-risk reviewers. The question is not

who protects the project. The question is who can see the burden that insiders may miss.

Failure memory should preserve dissent on both sides. A minority report arguing that defeat is too broad can be important. A minority report arguing that defeat is too narrow can be important. Later researchers need to know what reasonable experts thought at the time, not only what the majority decided.

Defeat can also be partial by architecture. The strong form may be defeated while burden analysis remains useful. A mechanism may be defeated while a measurement lesson remains. A public empirical claim may be suspended while the formal hypothesis remains untested. The defeat statement should not oversimplify either direction.

The failure record should constrain future editions. A later chapter, public graphic, or speech should not revive defeated language unless it cites the defeat and explains a new, independently justified reason to reopen the question. This prevents the same public risk from cycling through new audiences.

The deepest purpose of failure memory is mercy toward the future. It stops future participants, researchers, readers, and critics from spending their lives reopening a door that evidence has already closed.

A theory-level review should ask whether the remaining claim still differs from serious rivals. If the surviving language merely says that conscious systems regulate, adapt, or avoid harm, the

Law-specific empirical content may be exhausted. The program should not preserve a distinctive name for an ordinary result.

Failure memory should include public artifact lineage. A cropped figure, simplified chart, or old explanatory video can continue teaching the defeated claim years after the paper has been corrected. The failure record should therefore track artifacts, not only studies.

A credible counterexample should be protected from sensationalism. The person or case should not become a mascot of defeat. The public statement should focus on the component and evidence route, with privacy and dignity preserved.

Theory-level defeat may require cross-book notices. Applied books should not silently keep language that scientific defeat has withdrawn. The science book does not govern those domains, but it does govern the scientific language they may borrow.

The failure memory should state whether reopening is possible. Some defeated claims may be reopened only by a new measurement bridge, new endpoint rule, or new independently motivated prediction. Others may be abandoned for the current edition. The distinction prevents endless revival.

Defeat is most credible when it is prepared in advance. A project that writes defeat criteria before it needs them shows that it cares more about truth than victory.

27.7.1 — Credible Counterexample

A credible counterexample is a well-identified case that directly pressures the universal structure of the strong claim. It requires a

qualifying stream, an independently adjudicated endpoint, valid measurement or other justified evidence, robust uncertainty analysis, and exclusion of major rival artifacts. It is rare because the burden is high (Popper, 1959; Lakatos, 1978).

The counterexample must reach the complete-stream latent residual through adequate representation, origin, routing, endpoint, missingness, and measured-to-latent guarantees. Checks should exclude artifacts that could create the apparent contradiction. Identifying the natural mechanism responsible for a valid nonzero residual is not an additional requirement for rejecting the exact universal branch.

Consider a qualifying stream with an independently adjudicated endpoint, context-qualified measurement, and a nonempty, truth-containing compatibility set for its complete-stream latent residual that lies wholly outside the prospectively fixed equivalence band. The set must account for the unobserved past, measured-to-latent uncertainty, informative missingness, and declared artifact checks. Such a case is adverse to that band claim and, when zero is excluded under the protocol-fixed inferential guarantee, to the exact universal branch, even if an ordinary rival explains the nonzero residual.

Counterexamples cannot be diluted by pooling them with favorable cases or by redefining the population after inspection. Claims that a case is anomalous, unrepresentative, or measurement noise must follow prospectively specified criteria. When those criteria are met, the counterexample enters failure memory and changes the declared branch (Mayo, 2018; Meehl, 1978).

A counterexample should be reported at the level it defeats: one reference class, branch, modality, route, or strong universal form. The release should also state what narrower claims remain logically available without implying that they have already earned evidence.

A universal claim gives credible counterexamples a decisive role. Specifying their requirements makes both the claim and its possible tests more precise.

27.7.2 — Repeated Conjunction Failure

Repeated conjunction failure occurs when the distinctive pattern that should separate the Law-candidate program from rivals repeatedly fails to appear under adequate designs. One absent signature may be local. Repeated absence across independent tests creates theory-level pressure (Lakatos, 1978; Mayo, 2018).

The conjunction matters because single signatures are often mimicked. Variance compression alone, neutrality bands alone, reserve correlation alone, or first-moment movement alone may be non-discriminating. The program asked for a structured combination. Failure of that combination is more serious than failure of one auxiliary detail.

Conjunction failure is exposed by a sequence of terminal protocols in which concentration appears only when endpoint-conditioned rivals also predict it, reserve-linked contraction fails, endpoint independence remains fragile, and held-out rival models outperform the Law candidate. The conjunction has failed.

A route-specific signature cannot substitute for the terminal target it was meant to inform. After one predicted structure fails, the

program may highlight a different pattern as though it had always been central. The original signature retains priority, and the status must change if the declared conjunction does not survive (Meehl, 1978; Oreskes et al., 1994).

When the distinctive conjunction fails repeatedly, the release should say that the protocol-fixed strong form is not supported and identify the adverse component. It should not replace defeat with a vague promise that a future version will surely work. A theory may survive by becoming smaller, but it may not survive by forgetting what it asked the world to show.

27.7.3 — Unresolvable Identification Failure

Unresolvable identification failure occurs when the program cannot give the public empirical claim meaningful content under ethical and measurement constraints. The problem is not merely lack of data today. It is the absence of a defensible route from observable quantities to the target component (Mayo, 2018; Meehl, 1978).

Identification failure can arise from an unstable origin, nontransportable unit, endpoint ambiguity, inseparable rivals, unmeasurable bearer states, or ethical limits on a necessary observation. The diagnosis should identify the affected component and observation route, and state which independently justified alternatives remain available.

A design requiring high-frequency terminal self-report may be unanswerable under present ethics when reporting would burden participants and passive channels cannot identify the required quantity. That conclusion limits the stated design. It directs work

toward a separately qualified measurement bridge and does not, by itself, establish permanent untestability of every PTVC branch.

New sensing methods can expand what research can observe. Proposals involving artificial intelligence, wearables, or brain decoding should specify the quantity to be identified and test its calibration across the intended settings. This turns technological promise into a researchable measurement claim (Oreskes et al., 1994; Popper, 1959).

When an empirical verdict is suspended, the public wording identifies the reason: No adequate estimate, unresolved model ambiguity, failed endpoint bridge, or open-set inadequacy. It also names the evidence needed to reconsider that status.

Unresolvable identification failure is not the same as defeat by counterexample. It is defeat of public empirical content unless a new, independently justified bridge is created.

27.7.4 — Persistent Rival Superiority

Persistent rival superiority escalates when ordinary models repeatedly outperform Law-candidate models across components, contexts, and held-out tests. A single rival victory can retire a mechanism. Persistent rival superiority can pressure the broader program (Meehl, 1978; Oreskes et al., 1994).

The rivals must be serious and implemented fairly. Adaptation, opponent processes, Brownian bridges, Doob transforms, predictive processing, homeostasis, safe control, frailty, observation shutdown, and selection must face the same observation model and endpoint constraints. Only then does superiority carry status force.

Persistent rival superiority appears in repeated superiority of endpoint-conditioned bridges and observation-gain models for terminal patterns, combined with adaptation models winning nonterminal repair dynamics and safe-control models winning horizon tasks. The remaining Law-specific residual may vanish.

Residual hunting treats whatever rivals have not yet explained as evidence for the favored theory. A small unexplained remainder matters only when it was prospectively specified, is robust to reasonable alternatives, and is large enough to discriminate among generators (Popper, 1959; Lakatos, 1978).

Theory-level reporting should specify which rivals dominate which components and whether any Law-specific discriminator remains. It should not imply that the formal question is untouched if the empirical program has lost the ability to distinguish it from ordinary explanations.

27.7.5 — Defeat Statement and Scope

A defeat statement specifies exactly what is withdrawn, what is narrowed, what remains unresolved, and what useful component science survives. It should be public, dated, versioned, and written in clear, direct language (Oreskes et al., 1994; Popper, 1959).

The statement should identify the defeated object: strong form, public empirical claim, component, mechanism, instrument, endpoint rule, reference class, or public artifact. It should also identify the evidence path that forced the change and the dissent recorded during review.

The defeat statement must survive a review board concluding that the reserve-coupled terminal signature program is defeated for adult hospice oncology because serious rivals dominate and reserve parameters are nonrecoverable. The statement should not say the entire series is false unless the evidence supports that scope.

A globally vague verdict protects institutional comfort because readers cannot tell what actually changed. The record must name the object, scope, evidence, reasons, consequences, and correction duties (Lakatos, 1978; Mayo, 2018).

A defeat statement identifies the scientific object affected and confirms continuing care commitments, dignity, legal safeguards, and the independent basis of useful applied practices.

A precise verdict preserves both the finding that changed the claim and the knowledge that remains valid.

27.7.6 — Failure Memory

Failure memory preserves data, code, protocols, model logs, review reasons, dissent, correction notices, and public-language changes. It prevents the same failed claim from returning in a new edition, a new diagram, a new donor page, or a new institutional setting without disclosure (Popper, 1959; Lakatos, 1978).

The memory should be searchable by component, version, reference class, endpoint rule, device, site, rival, and public artifact. Negative results need identifiers as durable as positive results. A durable record lets future optimism build on what has been learned, including the approaches that need a new justification before reuse. Searchable links between failure

records and successor work help prevent the archive from becoming a passive graveyard rather than an active design constraint.

Defeat and disappearance separate in a failed candidate instrument version whose invalid sign convention is later reused in a public graphic because the failure lived only in an old technical report. A failure-memory register would prevent that by marking the instrument and every dependent artifact. Artifact status should be machine-readable where possible so outdated objects can be flagged before republication or automated reuse.

An archive without consequences permits failed objects to remain active. Failure memory must constrain future language, protocols, and summaries while recording lessons and corrections without blame (Mayo, 2018; Meehl, 1978). The archive should also record when a later repair genuinely reopens a failed component, preserving both the original loss and the evidence that justified reconsideration.

27.8 — Care and Dignity After Scientific Change

Scientific status governs claims, not persons. This section defines the communication, correction, and care duties that remain active across every verdict.

Defeat is reported without mocking people who found hope in the hypothesis or implying that suffering becomes more or less real after an evidence update. The theory changes; the person's dignity does not.

Table 27.9. Care continuity after scientific status change.

Risk after status change	Local rule	Public consequence
Support moralizes suffering	Care remains immediate and independent.	Report the measured outcome without assigning moral purpose or morally owed payment to suffering.
Defeat humiliates participants	Participants are never the failed object.	Public statements blame no person or family.
Research variables escape	No score, rank, or eligibility use survives.	Constructs remain inside research governance.
Applied lens discarded	Burden analysis remains useful where ethically grounded.	Ordinary care and institutional repair continue.
Loss hidden or dramatized	Correction is calm, dated, and precise.	Public trust comes from visible change.
Grand claim shrinks	A smaller true claim is success.	Accuracy outranks grandeur.

Care continuity must be designed into communication before bad news arrives. A defeat statement written in panic can sound cold or defensive. A narrowed-status announcement written only for scientists can leave families, readers, and public supporters confused. The care language should be prepared as part of the status architecture.

The care rule also protects critics. People who warned about overclaim should not be blamed for the pain of correction. A fair science does not make skeptics carry the emotional burden of truth-telling. It incorporates their objections into review and remembers them.

Communities affected by research should receive explanations that respect their own stakes. A palliative site, disability community, language group, or clinical population should not learn that its data narrowed a claim through sensational media. Participant communication is part of dignity after scientific change.

No status change should create a demand for personal interpretation. A reader does not have to decide what a defeated mechanism means for a lost loved one. A participant does not have to become a symbol of support or failure. Private meaning remains private.

The applied lens that survives scientific narrowing should be used gently. It may help institutions ask where load is hidden or repair is blocked. It should not become a consolation prize that pressures people to extract lessons from suffering.

Care after scientific change includes care for staff. Researchers, coordinators, nurses, data stewards, and analysts who worked on a defeated component may feel that their labor has been wasted. The status system should help them see what remains useful and why honest loss was part of the work.

Care also includes care for public readers who misunderstood earlier language in good faith. A correction should not mock them for believing an overcompressed artifact. It should accept responsibility for the compression and replace it with a clearer path.

Useful applied tools can be retained on their demonstrated merits. A burden audit, staff-support protocol, or data-minimization rule earns continued use through its effects on the people it serves.

A precisely scoped claim can be easier to explain and use. Its reference conditions show readers where it applies and where further work can extend it.

27.8.1 — Care Does Not Rise or Fall With Status

Care follows present need, not scientific status. No supported, narrowed, retired, or defeated claim may delay relief or reduce the obligations already owed.

This rule prevents a subtle hierarchy. If support for terminal neutrality made caregivers less urgent, the science would have become dangerous. If defeat made people feel their pain was merely ordinary and therefore less worthy, the science would have become cruel.

Consider a family hearing that a terminal signature study has been defeated in their relative's diagnosis. The correct message is not that hope was foolish or care is different. The correct message is that scientific language changed while the duty to reduce suffering remains unchanged.

27.8.2 — No Victim-Blaming Under Any Verdict

Public statements assign scientific status to claims and describe participants with equal dignity, without blame, rank, or deserved-suffering interpretations. Care implications are stated separately and grounded in participants' needs.

27.8.3 — No Score or Rank Survives

PTVC research constructs may classify claims, uncertainty, missingness, transport limits, model performance, and declared site-method features. They may not rank people or groups, including diagnostic or cultural groups, or assign worth, eligibility, trust, holiness, loyalty, dignity, destiny, or care priority. Public displays must distinguish component or methodological status from any judgment of human or group standing.

27.8.4 — Applied Lens May Remain Useful

A surviving instrument, protocol, dataset, burden audit, or care tool may remain useful only under an independently justified use, its validated context, and a corrected label. Scientific defeat or narrowing must not travel with the tool as hidden authority, but practical care and institutional repair may continue where they have their own warrant.

27.8.5 — Public Communication of Loss

Public loss statements must be calm, dated, component-specific, and visible. Record the old and new labels, evidence path, reason, surviving claims, withdrawn language, correction schedule, corrected artifacts, and the responsible maintainer. Do not hide loss as nuance or delay correction to protect institutional image.

27.8.6 — A Smaller True Claim Is Success

A smaller true claim is a scientific success when stated plainly, tested within its scope, and not used to preserve a defeated universal claim. Summaries, graphics, and detachable sentences must carry the protocol-fixed verdict, reference class, and limits rather than a movement slogan. Answerable science is willing to become smaller than the hope that first motivated it.

Chapter 28 — Scientific Updating and Claim Revision

Science is not only a way of finding; it is a way of changing its own record when findings demand it. A claim that can be tested but cannot be corrected after testing is only partly exposed to reality. Chapter 28 governs what happens after a component is supported, narrowed, suspended, retired, or defeated: which wording changes, which versions remain visible, which public artifacts are corrected, and which authority never passes to a formula, institution, donor, software release, or artificial-intelligence system (Committee on Publication Ethics, 2025; Crossref, 2025).

A claim register preserves the current wording and status of a controlled scientific object. The object may be a definition, endpoint rule, instrument qualification, observation model, route statement, verdict, or public status sentence. Its authority comes from traceability, review, compatibility discipline, and the visible survival of prior versions, including versions that failed.

The pair boundary makes change control especially important. *The Law of Fairness* retains the formal root; *The Science of Fairness* receives that root, builds tests around it, and reports what evidence can responsibly say. A failed candidate realization may retire within this volume, but a root amendment cannot travel through a caption, appendix, or public summary as if the original claim had always meant something else.

Every update needs a destination. A result may alter notation, representation, an instrument, an endpoint model, a candidate route, a rival set, a reference class, a public artifact, or the root formal claim. Each destination has a different threshold, reviewer, migration cost, public consequence, and rollback route.

Versioning keeps failure from disappearing under improved language. A model that failed under one instrument remains failed in that version; reanalysis under another instrument is a new analysis. Failed instruments, rejected bridges, and withdrawn figures teach future researchers where the previous program broke and why a later version deserves separate evaluation.

Change control is an engine of learning rather than administrative drag. A theory learns honorably by narrowing after evidence or repairing a component in a way that creates new predictions and accepts new costs. It becomes evasive when failure is renamed refinement while the public conclusion remains untouched.

Dissent, external challenge, appeal, and reconsideration receive durable places in the system. Complex claims can be right in one component, wrong in another, unresolved in a third, and publicly risky even when technically coherent. A well-formed minority report can preserve an overlooked failure mode without turning disagreement into indefinite paralysis.

An ambitious project develops through careful testing and visible revision. Instruments can improve, diagrams can become clearer, and model comparisons can identify stronger explanations. A timely, traceable record of changes preserves those gains and helps later work build on them.

Chapter 28 gives evidence a governed afterlife. A result becomes a reviewable update packet; an approved packet changes a versioned register; the register controls downstream wording; and superseded objects remain accessible without continuing to authorize the old claim. The repair system is strong enough to

change the project and bounded enough to resist institutional capture.

What follows is a scientific change-control system with human authority at its center. It is formal enough to preserve exact objects, public enough to propagate correction, and limited enough to keep popularity from setting status. When the science changes, the active record changes with it, and no machine, institution, donor, movement, or applied domain owns the Law.

28.1 — Human-Governed Scientific Status Updates

A scientific status update begins with an evidence packet, not with a headline. The packet identifies the component, version, data, methods, quality, rivals, uncertainty, conflicts, dissent, public-risk review, and proposed status sentence. Named human reviewers then issue a reasoned, appealable decision whose scope cannot exceed the packet.

The first rule is that updates attach to named components. The update body should never vote on the Law as a whole when the evidence concerns an instrument, endpoint, population, mechanism, benchmark, or public artifact. This preserves the noncompensatory logic established in the previous chapter. A successful replication in one hospice context does not repair a failed measurement origin elsewhere. A public artifact corrected for overclaim does not change the empirical status of the underlying signature.

The second rule is that updates require human accountability. Statistical models can inform the decision. Red teams can force the decision. Repositories can preserve the decision. Artificial-intelligence tools can search for relevant contradictions or draft comparison tables. But the status change itself must be signed by accountable reviewers who declare their expertise, conflicts, dissent, uncertainty, and reasons. The bottleneck is deliberate, because claim status is a public meaning as well as a technical summary.

Table 28.1. Claim-status update decision fields.

Field	Required entry	Why it matters
Object	The exact claim, instrument, realization, endpoint rule, public artifact, or applied-domain bridge under review.	Prevents evidence about one component from changing the whole theory.
Trigger	New data, replication failure, rival victory, measurement defect, public-risk finding, code defect, or root inconsistency.	Makes correction responsive to cause rather than reputation.
Authority route	Technical update, public artifact correction, formal change request, or root handoff.	Keeps the science book from silently rewriting the formal root.
Reviewers	Named roles, expertise, conflicts, dissent, and abstentions.	Creates accountable human judgment rather than institutional fog.
Decision	Support, narrowing, suspension, retirement, defeat, correction, deprecation, or handoff.	Gives the record a usable status without making a global score.
Propagation	Books, websites, diagrams, captions, code, datasets, social excerpts, donor pages, and archive notices affected.	Prevents corrected claims from surviving in high-reach artifacts.

28.1.1 — Evidence Packet

The evidence packet is the first gate because a status decision cannot be fairer than the record it is allowed to see. The packet names the studies, replications, rival models, deviations, ethical findings, public-risk notes, and dissent that must travel together before any claim-status word is changed (Committee on Publication Ethics, 2025; Crossref, 2025).

The temptation to automate status arises from a good instinct: the project wants consistency. A dashboard can count replications, estimate heterogeneity, report calibration, and flag out-of-distribution failure. Those summaries are useful. They are not sovereign. The same numerical pattern can mean different things depending on endpoint validity, measurement invariance, rival strength, and public-risk context.

A model that gives a strong held-out score may still rely on an invalid endpoint. A benchmark that separates reserve coupling from a Brownian bridge in simulation may fail in field data where observation shuts down near death. A Bayesian synthesis may look favorable while a small subgroup carries a severe violation. Human review is needed not because humans are purer than models, but because the judgment has to join evidence, scope, language, and consequence.

The status register should therefore record inputs rather than hide them inside one number. It can display replication count, prediction intervals, model-recovery performance, calibration, rival comparison, quality of endpoint adjudication, public-artifact exposure, and dissent. The final status word remains a reasoned

human classification: preliminary support, partial support, replicated support, inconclusive, unusable, narrowed, rival-dominant, suspended, retired, defeated, or root-handoff required (Center for Open Science, 2025; Tabassi, 2023).

This protects the project from two opposite errors. It blocks promotional inflation, where a favorable metric is turned into discovery. It also blocks timid self-erasure, where any uncertainty prevents a bounded statement that the evidence actually supports. The right question is not whether a number is high. The right question is what claim, under what version, in what context, with what dependencies, the evidence permits.

A concrete example helps. Suppose a terminal-signature protocol shows equivalence-band neutrality under one candidate instrument version in two hospice networks, but leave-site-out prediction fails in a third network because sedation practices differ. An automated score might average across sites and call the result promising. A human update would separate the components: the signature receives partial support in contexts with the original sedation regime; transport to the third context is suspended; the observation model receives a revision request; public summaries lose any general wording.

The same burden discipline applies to science itself. The hidden load is the interpretive burden that a public reader, a family, a clinician, or a critic would carry if status moved too broadly. Human-governed updating keeps that load inside the scientific record instead of dropping it onto people who were never asked to adjudicate the nuance.

28.1.2 — Declared Review Criteria

Declared criteria keep update judgment from becoming hindsight. Before reviewers know which result will be emotionally easier, the record states what would justify preliminary support, partial support, suspension, narrowing, rival dominance, retirement, or defeat for the exact component under review (Crossref, 2025; Center for Open Science, 2025).

Proposer standing should be broad and decision authority should be narrow. A formal change request may be initiated by an internal analyst, an outside critic, a site investigator, a participant advocate, an applied-domain reviewer, a data steward, an accessibility reviewer, a cybersecurity reviewer, or a public-risk reviewer. The project should not require loyalty as the price of noticing a defect. Some of the best corrections will come from people who are trying to defeat the claim.

Decision authority, however, must be bounded by expertise and role. A notation clarification needs formal and mathematical review. A measurement revision needs psychometric, statistical, cultural, disability, and data-governance review. An endpoint revision needs consciousness, neurology, palliative, and ethics review. A public-artifact correction needs science, public-risk, accessibility, and communications review. A root-formal amendment requires formal-law handoff rather than science-book unilateral action.

The update body should be small enough to decide and diverse enough to see. It should include people who can understand the model, people who can understand the measurement, people who

can understand public harm, and people who are not invested in a preferred result. Its records should state the decision scope: what changed, what did not change, what remains under review, what public language is now forbidden, and what historical results must remain marked under prior versions (Tabassi, 2023; Committee on Publication Ethics, 2025).

Disclose conflicts alongside relevant expertise. Instrument developers may understand a method particularly well, while independent specialists may identify alternatives or overlooked defects. The review record lets readers assess both contributions and interests.

The process should include a decision clock. Some corrections must move quickly, especially public artifacts that create unsafe interpretation. Others require slow review, especially root handoff or endpoint changes. The clock should therefore name urgency categories: immediate public-risk correction, expedited technical correction, standard scientific review, and deferred reconsideration after new data. Delay itself becomes a recorded decision when the artifact remains public.

An update system with wide proposer standing and narrow decision authority is harder to manage, but easier to trust. It tells critics that their objections can enter the record. It tells readers that not every objection is accepted by noise or pressure. It tells the series that scientific correction is a public obligation without becoming public vote.

28.1.3 — Reviewing Authority Remains Open

Open authority means critics and outside specialists have a lawful scientific path into the process. It does not mean every comment becomes a vote; it means a methodologically serious objection can enter the packet without needing loyalty, institutional proximity, or permission from the favored interpretation (Center for Open Science, 2025; Tabassi, 2023).

The update packet must be written as if it may later be misunderstood. That does not mean it should be vague. It means the packet should distinguish the technical decision from the public consequence. If a candidate realization is retired, the packet should say whether the formal Law remains unchanged, whether a public diagram must be withdrawn, whether prior claims become historical only, and whether future studies may still test a narrowed version.

Each update should carry a claim-status sentence. The sentence should name the object and the scope. For example: the reserve-coupled reference realization remains a candidate consequence generator under the stated assumptions, but this particular measurement bridge is retired for terminal cohorts using the deprecated device version. A sentence like that prevents both exaggeration and concealment. It also gives public communicators language they can quote without inventing a safer gloss.

The packet should also contain an adverse-public-use scan. Could the correction be quoted as proof that the entire project failed when only a sensor failed? Could it be used to keep a dangerous

public artifact alive by saying the formal root was unchanged? Could it encourage self-testing, care delay, or pseudo-clinical inference? Could it be used by an institution to resist a needed correction because the headline is still attractive? The update must anticipate these misuses at the point of wording (Committee on Publication Ethics, 2025; Crossref, 2025).

A correction can itself create burden. A family who read an earlier public page may feel misled. A critic may discover that an objection was dismissed before the project had a route to hear it. A researcher may need to recontact collaborators or revise a protocol already in motion. The update system should not treat these costs as public-relations inconvenience. They are part of the ethical cost of scientific language at scale.

The public version of the update should be shorter than the technical packet but not cleaner than the truth. It should say what changed, why it changed, what remains, what is withdrawn, and where future work goes. It should avoid the reflex to write a triumphant correction. A good correction does not say, “the theory is stronger than ever,” unless the evidence supports that exact statement for a named component. Often the honest line is simpler: this component is now narrower, and the record has been corrected.

The update packet is therefore both a scientific document and a public-risk control. It keeps the cost of ambiguity inside the record. It prevents readers from carrying an interpretive load the project failed to carry for them.

28.1.4 — Conflict, Recusal, and Public Wording

Conflict and recusal rules identify interests that could affect a status decision, including authorship, funding, advocacy, instrument ownership, and exposure to the result. Apply the independent-role safeguards in Section 26.4.6: record the interest, the affected decision, any recusal, and the independent reviewer responsible. Technical clarification from an interested expert may remain useful without giving that expert sole control of the verdict or its public wording.

A status decision includes a concise public sentence naming the component, result, reference class, uncertainty, and strongest surviving alternative. Its wording should remain accurate when quoted on its own.

The sentence should avoid three temptations. It should not soften a defeat into refinement. It should not inflate a partial result into discovery. It should not reassure readers by praising the project's humility while hiding what changed. A phrase like “updated for clarity” may be appropriate for a notation fix, but dishonest for a realization failure. A phrase like “new evidence strengthens the project” should be used only when the component, scope, and rival comparison actually justify it.

Public wording should include the status cost. If an observation model is suspended, say what cannot currently be inferred. If a website page is corrected, say what earlier wording could have implied. If a figure is withdrawn, say whether the problem was visual overcompression, rival omission, measurement overreach, or old-language residue. The cost makes the correction useful

rather than merely defensive (Crossref, 2025; Center for Open Science, 2025).

The public sentence should also name care-independent consequences where relevant. A scientific correction never reduces present care, dignity, privacy, or nonparticipation rights. But that line should appear only where the correction could otherwise be misused. If an endpoint study is reclassified, the public sentence may need to say that care remains immediate and inference remains research-bound. If a code container is deprecated, that sentence is not necessary; the local risk is technical, not clinical.

A public status sentence should be tested against hostile quotation. Could it be clipped to imply unsupported proof or defeat, moral payment, promised rewards, or Artificial Intelligence authority? Could it be used by supporters to overstate and by critics to misstate? No sentence is immune to bad faith, but the wording should not hand bad faith an easy weapon. Precision is the project's first public defense.

The update board should maintain approved public-language templates for each status class, but templates cannot replace judgment. The point of a template is to prevent accidental inflation, not to make every correction sound the same. A correction about a failed endpoint rule should sound different from a correction about a website caption. One-author style is not sameness; it is disciplined recognizability across different local risks.

28.1.5 — Recorded Reasons

Recorded reasons turn a status transition into a scientific artifact rather than a mood. The decision states why stronger and weaker alternatives were rejected, which evidence carried weight, what remains unresolved, and what public wording must now change (Committee on Publication Ethics, 2025; Crossref, 2025).

Consider a late-stage result in which an independent replication finds no terminal concentration under the protocol-fixed endpoint rule, while an exploratory endpoint redefinition produces an appealing concentration pattern. The confirmatory result governs status; the exploratory analysis remains hypothesis-generating. Governance fails if the exploratory figure reaches public wording before root and endpoint review.

A second failure case occurs when a donor-facing summary keeps a stronger claim because the update board delayed public propagation. The technical register may be correct, but the public surface remains wrong. If a reader or funder sees the donor page first, the project has failed at the level of communication authority. The update system should treat this as an incident, not an inconvenience.

A third failure case occurs when a critic's model outperforms the candidate model and the project responds by changing the status label without naming rival dominance. The update may appear to preserve dignity, but it deprives the critic and future researchers of a clear scientific outcome. Rival victory should be visible because it is one way the program learns. Hidden rival victory is

just another form of proof drift (Center for Open Science, 2025; Tabassi, 2023).

A fourth failure case occurs when an artificial-intelligence assistant improves the tone of a correction notice but omits the exact consequence for claim status. Reviewers check the draft against the approved decision so that clearer language preserves scope, uncertainty, withdrawal, and every other material consequence. Readability and scientific accuracy must improve together.

A fifth failure case occurs when public reaction is treated as evidence against a correction. If supporters dislike narrowing, or critics celebrate defeat, neither reaction controls status. Public reaction can reveal misunderstanding and artifact risk. It cannot decide the scientific object. The update route should capture public reaction as a communication signal, not as a warrant signal.

These failure cases show why the update body needs records, authority limits, and public propagation. The risk is not only that the wrong scientific decision is made. The risk is that the right decision is made in one room and then loses its meaning on the way to the world.

28.1.6 — No Artificial Intelligence or Algorithmic Verdict

Computational tools may flag contradictions, summarize studies, and test consistency, while identifiable human reviewers issue reasoned and appealable status decisions. Governance frameworks support that work through oversight, evaluation, and accountability (Crossref, 2025; Center for Open Science, 2025;

Autio et al., 2024; World Health Organization, 2025; Feuerriegel et al., 2026).

Scientific vocabulary should identify the specific achievement: discovery of a pattern, validation of an instrument, confirmation of a prediction, or replication of a result. Direct, concrete language makes the finding clear.

Status language states the approved scientific outcome directly, including defeat, retirement, or rival dominance when applicable, and then identifies the knowledge and methods that remain useful (Tabassi, 2023; Committee on Publication Ethics, 2025).

Metaphor cannot bypass status. Phrases such as “coming into view,” “the world responding,” or “the model evolving” are acceptable only when they carry the same warrant as the technical register.

Maintain a versioned record of prohibited implications. Exclude moral desert, prescribed harm, human-worth ranking, machine sovereignty, archive-as-soul claims, and forced meaning. Review evidence-status claims against the findings they describe. Conditional net accounting and the Series Map’s “There is no free happiness” formulation are not prohibited merely for expressing a balance requirement; Section 3.7.1 defines their permitted meaning.

Every material update needs reasoned minutes: the object, evidence, rival pressure, options, rejected alternatives, decision, dissent, and public consequence. The record should explain why both weaker and stronger statuses were rejected.

Minutes must name the exact uncertainty that remains, whether measurement, endpoint, rival, transport, reproducibility, artifact safety, or root compatibility, because each implies a different next action.

Preserve rejected public wording and list every open propagation task with an owner and deadline. This prevents stale language from returning and keeps correction accountable until every dependent surface is closed.

These records let future readers judge whether a correction was careful, pressured, cosmetic, or complete while keeping the burden of explanation with the project.

28.2 — Versioned Claim, Model, and Evidence Registers

Versioned registers preserve the identity and lifecycle of claims, models, instruments, endpoint rules, evidence packets, verdicts, and public artifacts. Supersession changes current authority without erasing history. Each object carries a stable identifier, status, dependencies, effective date, owner, and reason for change.

The register versions scientific objects. For example, it can record that candidate instrument version 3.2 is suspended for a specified use until recalibration and requalification are complete, and identify which analyses the suspension affects.

A good register preserves the link from each version to its predecessor, lets public citations identify the exact scientific object, and connects circulating artifacts to their current status. These functions make change traceable across research and public communication.

28.2.1 — Claim Identifier and Version

A claim identifier prevents a repaired statement from pretending to be the same statement that failed. The version records the object class, the exact wording, the active scope, and the change level so readers know whether interpretation, measurement, endpoint, or public status has shifted (Crossref, 2025; Wilkinson et al., 2016).

Every object that can change public meaning deserves a version. This includes formal statements, notation choices, observation

models, candidate measurement-stack versions, endpoint adjudication rules, reserve proxies, candidate realizations, benchmark libraries, preregistration templates, public diagrams, website summaries, social excerpts, donor-facing descriptions, and dataset access packages. A change that affects interpretation must be visible enough to cite.

Object class belongs in the register. A notation clarification differs from a measurement revision, a figure correction differs from a realization replacement, and a root-formal amendment differs from a public wording update. An undifferentiated log can appear transparent while hiding the dependency structure. Versioning works because categories carry meaning.

The scheme can borrow the intuition of semantic versioning without becoming software theater. Major changes break interpretation or target identity; minor changes preserve identity while extending or narrowing controlled content; patch changes correct errors without altering meaning. The scientific register defines these categories rather than importing them mechanically.

A formal claim might therefore have a root version, an empirical bridge version, and a public-artifact version. A dataset can have a release identifier, a cleaning pipeline version, an access tier, and a provenance hash. A figure can have a caption version and a status label. The goal is not to create a maze. The goal is to let every future quotation know which object it quoted.

Versioning also protects old scholarship. A critic who defeated an earlier realization should not encounter a rewritten record in which that realization was supposedly never serious. Researchers

can cite deprecated instruments as historical objects, and later readers can see the movement of the field rather than only its cleaned final face.

Human readability comes first. Machine-actionable metadata improves search and audit, but the explanatory record remains the safeguard. A version identifier without a reason invites blind trust, while a reason without a stable identifier invites confusion. The two belong together.

Table 28.2. Minimum fields for versioned claim and evidence registers.

Register object	Minimum fields	Non-negotiable rule
Claim	Exact wording, status, scope, dependencies, date, reviewers, dissent, public language.	No global headline may move without object-specific scope.
Instrument	Version, calibration population, context of use, known failures, replacement status, data-retention limits.	Measurement failure cannot be overwritten by new wording.
Protocol	Study-registration, protocol, and analysis-plan versions; endpoint rule; observation model; consent scope; analysis container; deviations.	Reanalysis under a new protocol is a new analysis.

Register object	Minimum fields	Non-negotiable rule
Dataset	Release identifier, access tier, provenance, transformations, consent boundary, withdrawal notices.	A dataset is not public model-training permission.
Public artifact	Approved wording, source claim, status label, image/caption version, correction history, withdrawal state.	A public graphic cannot carry a stronger claim than its source.
Failure memory	Failed object, criterion, affected results, public consequence, reconsideration trigger.	A rejected bridge remains visible as a hazard marker.

28.2.2 — Evidence Links

Evidence links connect each status to the studies, code, datasets, protocols, replications, red-team reports, negative findings, and public-risk records that justify it. The links are not raw public exposure; they are traceability rules that let qualified review follow the chain from claim to warrant (Wilkinson et al., 2016; Barker et al., 2022).

Historical results remain attached to the versions that produced them. Reanalysis under a new observation model creates a new result, with an explicit link showing how it reinterprets, narrows, or supersedes the earlier finding.

Imagine an early terminal study that used a physiological channel later found to be strongly medication-sensitive. The old result

should not be deleted, nor should it continue to support public claims. It should remain in the evidence register as a version-bound result: preserved under the old protocol for historical audit, downgraded for current inference, and marked as requiring reanalysis if raw data and consent scope permit.

This is especially important for public artifacts. A diagram may have been accurate under an earlier candidate realization but misleading after a rival model wins. The archive should show both the withdrawn diagram and the correction notice. It should not keep the diagram active as if nothing changed. It should not delete the diagram so thoroughly that critics cannot see what was once claimed (National Institutes of Health, 2026a; National Institutes of Health, 2026b; Crossref, 2025).

The same rule applies to code. A benchmark container that used a biased preprocessing step remains the container that generated its results. A corrected container receives a new release identifier. The record should link them. It should state what changed, which results are affected, which are unaffected, and whether prior conclusions are withdrawn, narrowed, or merely marked with a sensitivity warning.

An archived result remains accessible under its historical version and access conditions. Active summaries use the current approved status, with links to earlier versions and the reasons for change.

Historical versioning is a form of respect. It respects critics who identified defects, participants whose data should not be reused under hidden assumptions, readers who need honest correction,

and researchers who must know which path was tried and why it no longer controls.

28.2.3 — Scope, Provenance, and Failure Memory

Scope and reference class keep a true local result from becoming a false universal statement. The register records population, setting, language, instrument, endpoint rule, care regime, device version, cultural context, and intended inference before public language is allowed to travel (Barker et al., 2022; National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Provenance records the lineage of a scientific object: who created it, what data or code contributed to it, what activity transformed it, which version produced it, and which later decision changed its status. In this program, provenance is not decorative metadata. It is the backbone of correction because it lets a future reviewer trace a public statement back through claim, instrument, protocol, dataset, code, and review.

Every public-facing scientific artifact should carry an internal identifier even when that identifier is not printed prominently on the page. The identifier should connect the artifact to its source claim, evidence version, public-status label, approved wording, correction history, and withdrawal state. A social excerpt is not too small for this discipline. Small artifacts travel farther than chapters.

Failure memory belongs in the same architecture. A rejected instrument, retired realization, misleading diagram, invalid bridge, or defeated benchmark should have a stable record. The record should not be framed as shame. It should be framed as a

hazard marker. Future researchers can then avoid rebuilding the same failed bridge under a new name (Crossref, 2025; Wilkinson et al., 2016).

A failure memory should record what failed, under which version, by which criterion, which alternatives were considered, what public language was withdrawn, and what would be required for reconsideration. This is more informative than a flat retired label. It distinguishes permanent defeat, bounded failure, and a No adequate estimate region without allowing any of them to dissolve into generic uncertainty.

Provenance also keeps donors and institutions from owning the story. If a major site funded a study, the provenance record says so. If an outside critic forced a correction, the record says so. If artificial-intelligence tools generated draft code or extracted candidate contradictions, the record says so. Visibility does not solve every incentive problem, but invisibility makes every incentive problem worse.

The result is a scientific memory that can be audited. The register becomes a map of burden carried by the program itself: interpretive burden, correction burden, historical burden, and public burden. A project that asks readers to trust its humility must show them the trail of that humility.

28.2.4 — Outstanding Uncertainty and Release States

Outstanding rivals and uncertainties remain visible beside the status instead of being footnoted away. A claim may be partially supported while a Brownian bridge, observation-shutdown model, cultural script, medication confound, or endpoint-

selection rival still limits what can honestly be said (National Institutes of Health, 2026a; National Institutes of Health, 2026b; Crossref, 2025).

Every versioned object needs a release state. Draft means internal and not citable. Candidate means under review and usable only within declared limits. Active means current authority. Superseded, deprecated, withdrawn, and defeated each preserve history while removing some or all current authority.

The difference between superseded and deprecated is important. A superseded formatting template may simply have a cleaner successor. A deprecated observation model may have known bias or failed calibration. A withdrawn public graphic may have been too easy to crop into false certainty. A retired candidate realization may have lost to stronger rivals. Each state tells users what they may do next.

Deprecation language should be explicit about permitted use. A deprecated instrument may remain available for historical replication or sensitivity analysis, but not for new confirmatory claims. A deprecated figure may remain in the archive with a warning, but not in active public materials. A deprecated dataset release may remain citable, but any current inference should use the corrected pipeline if consent and provenance permit (Wilkinson et al., 2016; Barker et al., 2022).

Safety withdrawal and scientific defeat require separate register fields. A public artifact may be withdrawn because it is unsafe even while the underlying science remains active, or because the science itself changed. Combining those reasons creates

confusion: communication can fail while a model remains alive, and a model can fail while an artifact explaining that failure remains useful.

Release states should travel with metadata. A file name alone is too fragile. The release state should appear in the register, artifact metadata, readme, caption source, website revision note, and model card where relevant. If a public page displays only the newest material while the archive lacks state labels, future readers will not know whether older material was superseded, deprecated, withdrawn, or defeated.

A clear release-state vocabulary makes corrections understandable and traceable. The reader sees not merely that something changed, but how it changed and what that change permits.

28.2.5 — Stewardship of Superseded and Retired Claims

Superseded and retired claims are preserved as history and removed as active language. The record should allow a critic to see what failed, what replaced it, why the replacement was allowed, and which public surfaces are no longer permitted to quote the retired version (Crossref, 2025; Wilkinson et al., 2016).

A register needs stewards. The steward is not the owner of truth. The steward is responsible for completeness, access control, update hygiene, version identifiers, and correction visibility. Different registers may need different stewards: scientific claims, public artifacts, code releases, datasets, and root handoff packets do not have the same expertise or access requirements.

Access should be tiered by sensitivity and public purpose. Public registers can show claim status, correction notices, artifact versions, deprecation states, and failure summaries. Reviewer registers can show detailed dissent, code logs, model-recovery failures, and benchmark results. Restricted registers can hold sensitive participant information, raw vulnerable-context data, and security-sensitive details. The register should not make transparency a pretext for exposure.

Tiered access should not become secrecy by convenience. If a detail is withheld, the public record should say the class of reason: participant privacy, consent boundary, clinical sensitivity, security risk, legal restriction, or ongoing review. A blank space without reason invites distrust. A reasoned restriction preserves dignity while keeping the record answerable (Barker et al., 2022; National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Stewardship also includes correction latency. Each register should track when a change was approved, when public surfaces were updated, and which dependencies remain pending. A correction that sits unpropagated for months should become visible as an open propagation item. Delay is sometimes unavoidable, but it should not disappear.

Independent audit belongs in the register. Someone outside the originating team periodically checks whether version labels, public artifacts, dataset releases, and code identifiers agree. This is not merely clerical work; many public-risk failures occur because one surface continues to quote a claim after another surface corrected it. Audit makes the series one voice in practice.

Good stewardship keeps the register from becoming a graveyard of files. It makes the register a living map of claim status, public consequence, and scientific memory.

28.2.6 — Public and Internal Views

Public and internal views separate dignity from opacity. The public summary must reveal status, scope, and correction cost, while protected review layers may contain sensitive participant information, clinical detail, security incidents, or confidential dissent that cannot be exposed without creating new harm (Wilkinson et al., 2016; Barker et al., 2022).

A release note should be written for supporters and critics at the same time. Supporters need to know what they may still say. Critics need to know what changed, what remains vulnerable, and where the project may still be wrong. A release note that only reassures supporters becomes publicity. A release note that only lists technical details becomes unusable. The right note is exact enough for a critic to reproduce the status consequence.

A useful release note names the previous version, the new version, the reason for change, the affected claims, the unaffected claims, the data or artifacts requiring reanalysis, and the public surfaces requiring correction. It should identify whether the change is major, minor, or patch-like in its interpretive consequence. It should not rely on the reader to compare two long documents to find the difference.

Release notes should include a “what not to infer” field. After a measurement revision, do not infer that old participants were mismeasured as persons. After a realization replacement, do not

infer that the root was validated. After a public-artifact withdrawal, do not infer that every component failed. This field prevents the correction from creating a new public error while repairing the old one (National Institutes of Health, 2026a; National Institutes of Health, 2026b; Crossref, 2025).

Where relevant, release notes include a “critic standing” field. When a specific rival, red-team report, replication failure, or public-risk objection motivates the change, the note names it. This is not praise management; it is provenance of pressure, including an honest record that criticism forced a better version.

A useful note is concise but not evasive. It answers four questions: whether the claim remains usable, in which context, at what status, and what changed from the prior version. A note that leaves those questions unresolved has not done its work.

Release notes make the register usable. They turn a static version list into a scientific conversation across time.

28.3 — Dissent, External Challenge, and Reconsideration

Dissent becomes scientifically useful when it identifies a specific object, assumption, omitted rival, public risk, or procedural defect. The challenge record preserves the submission, evidence, response, decision, minority view, and reconsideration trigger without turning every disagreement into an indefinite veto.

Dissent is different from noise. Noise repeats a preference without engaging the object. Dissent names the claim, the version, the evidence, the rival, the public risk, or the missing expertise. It explains what decision should change and what evidence would change the dissent. A useful dissent is falsifiable in its own way: it can be answered, narrowed, accepted, or superseded.

Appeal and reconsideration protect the same principle at different time scales. Appeal challenges a decision soon after it is made. Reconsideration reopens a settled or suspended question because new evidence, new methods, new failures, or new public artifacts changed the landscape. Both routes must be available. Neither can become endless status litigation that prevents the project from saying anything at all.

28.3.1 — Minority Report

A minority report preserves serious disagreement without forcing false consensus. It names the methodological, evidentiary, ethical, or public-language objection that reviewers could not resolve, and it stays attached to the status so later readers know where the decision remained vulnerable (Mellers et al., 2001; Silberzahn et al., 2018).

Dissent sits beside the decision rather than behind it. The scientific register preserves the majority decision, minority objections, unresolved dependencies, and conditions for revisiting status. Public artifacts need not reproduce the full dispute, but a clean label may not pretend that serious disagreement never existed.

Dissent may concern method, scope, status, language, ethics, or public artifact risk. A statistician may accept the data but reject the heterogeneity interpretation. A palliative clinician may accept the model but reject the terminal observation window. A disability expert may accept the instrument in one group but reject its transfer to another. A public-risk reviewer may accept the science but flag a diagram as hostile-crop vulnerable.

The form of dissent should require specificity. “The claim is overstated” is useful only if the object and remedy are named. Better is: the evidence supports preterminal variance compression under candidate instrument version 4 in one care regime, but the proposed public wording implies terminal neutrality across diagnoses. That dissent can be adjudicated. It also helps future writers avoid the same drift (Nosek et al., 2018; Mayo, 2018).

Dissent should not be framed as disloyalty to the project. A theory that needs silence from experts is not ready for public ambition. Nor should dissent automatically become a veto. One reviewer’s objection may reveal a real problem, or it may be answered by evidence the reviewer missed. The record must preserve the objection and the response without converting either into theater.

Recorded dissent can later become decisive. A minority warning about endpoint leakage may look cautious before a benchmark reveals a Brownian bridge rival. A public-artifact warning may look excessive before a cropped graphic circulates with false certainty. Dissent is therefore part of failure memory even when the majority decision stands.

Every material update decision should contain a dissent field, even when no formal dissent was entered. The field should identify the review route, the eligible reviewers, and whether silence followed genuine opportunity to object; otherwise an empty box can imitate consensus without showing that disagreement was possible.

28.3.2 — Appeal Route

An appeal route lets a procedural error, excluded rival, overlooked confound, improper conflict ruling, or misleading public sentence be challenged without turning correction into personal battle. Appeal has standing rules and deadlines, but it is real enough to change the record (Silberzahn et al., 2018; Nosek et al., 2018).

Appeal exists for decisions that appear procedurally flawed, inadequately reviewed, inconsistent with the source hierarchy, or disproportionate to the evidence. An appeal should not be a second attempt to win the same argument by exhausting the reviewers. It should name what failed in the decision route: missing expertise, unreviewed data, incompatible wording, conflict of interest, improper scope, public-risk omission, or incorrect root handoff.

Meaningful appeal standing extends to a critic who specified a rival in advance, a site investigator whose data were reclassified, an external expert whose domain was used, a participant advocate whose concern was ignored, and an applied-domain steward whose artifact is affected. Correction machinery loses credibility when only insiders may challenge it.

The appeal clock should be finite but not cruel. Some appeals concern slow scientific interpretation and can wait for scheduled review. Others involve public artifacts that may mislead vulnerable readers and require immediate provisional action. A banner, hold notice, or temporary withdrawal can protect public meaning while appeal proceeds. The appeal system should therefore distinguish public-risk urgency from technical review pace (Mayo, 2018; Mellers et al., 2001).

The appeal outcome should be specific. It may affirm the decision, revise the wording, add dissent, reopen the technical review, change the propagation plan, issue an interim public notice, or send the matter to formal root handoff. A vague statement that “concerns were considered” is not enough. It leaves the interpretive burden on the person who raised the concern.

Infinite reopening is prevented by threshold rules. A closed appeal can be reopened only by new evidence, new method, new public artifact, undisclosed conflict, material error, or a downstream incompatibility not visible at the time. Repetition of the same preference does not reopen the record. This protects the project from paralysis without silencing real correction.

The appeal route also protects critics. A critic should not have to fight the whole public identity of the project to challenge one component. The route says: there is a door for this claim, this version, this evidence, this wording. The existence of the door reduces the incentive for hostile escalation because the project has given challenge a legitimate path.

Table 28.3. Dissent, appeal, and reconsideration routes.

Route	Trigger	Permitted outcome
Dissent	Reviewer or critic disagrees with scope, method, status, language, public-risk interpretation, or dependency logic.	Recorded minority position, requested wording change, future test condition, or escalation to appeal.
Appeal	Procedural flaw, missing expertise, improper scope, conflict, ignored evidence, or public-risk omission.	Affirmation, revised decision, reopened review, interim hold, or formal root handoff.
Reconsideration	New evidence, method, rival, artifact misuse, device drift, context change, or sunset review.	New version, deprecation, correction, reinstatement, narrowed status, or retirement.
Closure	No new evidence, no procedural defect, and repeated objection already answered.	Closed record with rationale and threshold for reopening.

28.3.3 — New Evidence Reopening

New evidence reopening distinguishes learning from endless relitigation. A status may reopen when a new replication, benchmark, method, adverse event, public-risk failure, or formal audit materially changes the evidence position, not merely because the prior decision was inconvenient (Nosek et al., 2018; Mayo, 2018).

Reconsideration is not an admission that the first decision was careless. Science changes because instruments improve, rival models sharpen, public artifacts travel differently than expected, and applied contexts reveal confounds. A decision that was responsible under one information state may become inadequate later. The record should allow this without pretending that the past was either perfect or fraudulent.

A reconsideration trigger can be technical: new model-recovery results, new endpoint adjudication methods, new measurement-invariance failures, new device-version drift, or new transport evidence. It can be ethical: an observation practice becomes burdensome, a public graphic is misused, or a participant-return policy proves confusing. It can be institutional: a website, archive, classroom excerpt, donor page, or applied-domain protocol carries old language into a new audience.

Reconsideration should distinguish revision from rescue. A progressive reconsideration adds empirical content, improves prediction, accepts costs elsewhere, or narrows scope in advance of further tests. A degenerating reconsideration keeps the public headline while moving the bridge every time data fail. The

register should force the question: what would this revision newly risk losing? If the answer is nothing, the revision is probably a shelter, not science (Mellers et al., 2001; Silberzahn et al., 2018).

The system also includes scheduled sunset review for objects that can drift without a crisis: candidate instrument versions, public figures, benchmark libraries, endpoint rules, website summaries, artificial-intelligence-assisted workflows, and applied-domain evidence-firewall language. Scheduled review prevents important objects from remaining active merely because no incident has yet forced attention.

Reconsideration decisions should preserve old dissent. If a minority warning is vindicated, the record should say so. If it remains unpersuasive, the record should explain why. This is not a scoreboard. It is a way of making scientific judgment accountable across time. A theory that learns should remember who taught it.

The result is a living claim register without soft boundaries. It can move, but it cannot move secretly. It can revise, but revision has cost. It can reopen, but reopening has threshold. It can disagree, but disagreement leaves a trace.

28.3.4 — Nonretaliation and Applied-Domain Appeals

No retaliation for critique protects the people who make correction possible. Analysts, participants, site staff, outside critics, applied-domain observers, and minority reviewers must be able to report a defect without losing access, safety, credit, employment standing, or public dignity (Mayo, 2018; Mellers et al., 2001).

Applied-domain stewards need a route to challenge scientific wording that affects their domain. A City designer may see that a scientific correction changes the safe framing of transparency. A government reviewer may see that an evidence summary could be mistaken for administrative scoring. A reviewer from a religious community may see that a scientific phrase is being heard as doctrine. A transition steward may see that readiness language now depends on a narrowed component. These are legitimate appeals when they name the affected scientific wording and its domain consequence.

Appeals let applied domains identify misinterpretation and practical consequences. Scientific reviewers evaluate the evidence and decide its status through the same independent process used for the original decision.

Applied appeals are especially important for public artifacts. An artifact safe in a technical context can become unsafe in a civic, pastoral, or transition context. A figure that specialists read as a candidate model may be read by ordinary readers as a diagram of destiny. An applied steward can identify this transfer risk and request correction even when the technical claim remains unchanged (Silberzahn et al., 2018; Nosek et al., 2018).

The appeal record should state the receiving domain. The correction may be a domain-specific public wording note rather than a global scientific change. For example, a claim can remain technically active while a religious-community artifact is withdrawn because it could be heard as forced meaning. That is not scientific defeat. It is domain-sensitive public-risk correction.

Applied appeals also prevent the science book from becoming socially tone-deaf. A technically exact sentence can still carry burden when it reaches families, patients, public workers, spiritual seekers, or pilot participants. Applied-domain reviewers help locate that burden before it travels. Their role is advisory but serious.

The appeal route therefore expresses mutual humility. Science owns evidence status. Applied domains own lived deployment context. Neither should silence the other, and neither should borrow the other's warrant.

28.3.5 — Independent Reanalysis and Sunset Review

Independent reanalysis uses the robustness tools described in §27.2, but Chapter 28 adds a governance requirement: the challenge must produce a traceable decision. Record the alternative specification, access conditions, result, affected claim version, responsible reviewer, and whether the outcome triggers correction, sunset review, or no change.

Independent reanalysis tests whether existing evidence survives another competent implementation. Sunset review asks whether an instrument, benchmark, endpoint rule, or public summary remains current after time, drift, and new evidence. The following review clock complements reanalysis; it does not replace it.

A sunset clock does not presume the object wrong; it recognizes that the object is too important to leave unattended. The register displays the next review date and the reason for the interval. High-reach artifacts, artificial-intelligence-assisted workflows, endpoint rules, public diagrams, and future-testbed language may

need shorter clocks than technical appendices or historical records.

Reconsideration thresholds should be higher than ordinary comment thresholds but lower than crisis thresholds. New evidence, new rivals, public misuse, repeated confusion, device drift, translation failure, or accessibility barriers can trigger review. Aesthetic preference alone should not. The project should be open to correction without becoming unstable because every reader would phrase the sentence differently (Nosek et al., 2018; Mayo, 2018).

A sunset review can end in reaffirmation. The record should say that the object was reviewed and remains active. That matters because absence of correction is not evidence that review occurred. A visible reaffirmation tells later readers that the object survived a check under the current context. It also gives critics a date and route for future challenge.

If sunset review produces no decision because evidence is missing, the object should be marked under watch rather than silently renewed. “Under watch” is a useful state for artifacts whose current wording remains usable but whose dependencies are unstable. It protects public language while avoiding premature withdrawal.

Sunset clocks are a quiet discipline. They prevent the project from treating correction as something that happens only when critics force it. A serious scientific program reviews its own surfaces because memory is never self-maintaining.

28.3.6 — Finality Is Provisional

Finality is provisional because public language needs current stability while science needs future correction. A decision can be binding for active artifacts and still remain reopenable under better evidence, better methods, or a newly visible public-risk failure (Silberzahn et al., 2018; Nosek et al., 2018).

Dissent does not disappear after a correction is published. A public correction may itself be disputed: too narrow, too broad, too slow, too soft, too technical, too public, or too quiet. The dissent route should remain open for the correction action, not only for the original claim. A project can correct the science and still mishandle the correction.

Public-correction dissent may concern wording. A reviewer may argue that a notice says “clarification” when the artifact required withdrawal. Another may argue that a withdrawal notice implies scientific defeat when the problem was only graphic overcompression. These are not cosmetic disputes. Public status language can redistribute interpretive burden across readers, critics, participants, and applied domains.

Dissent may also concern propagation. A reviewer may accept the technical correction but object that the donor page, social excerpt, or archived figure remains active. This is a propagation dissent. It should create a checklist item rather than reopen the entire scientific decision. The route should distinguish object-level disagreement from propagation failure (Mayo, 2018; Mellers et al., 2001).

The correction record should show whether dissent was resolved, maintained, or escalated. If the project keeps a contested public wording, it should explain why. If it accepts the dissent, the correction itself receives a new version. The old correction notice may remain in the archive with its own correction history. This may feel intricate, but public scientific language at scale is intricate.

Dissent after correction should not become a reputational battle over who was most careful. The goal is to reduce public error. The record should focus on the object, not personalities. A critic may be right about wording and wrong about theory. A supporter may be right about scope and wrong about artifact risk. The route should separate these judgments.

28.4 — Formal Change Requests and Root Concordance

A formal change request distinguishes clarification, correction, component revision, branch narrowing, non-equivalent successor creation, and root amendment. The packet shows the old and proposed text, evidence, dependencies, compatibility, migration, dissent, public corrections, and rollback route so reviewers can determine whether the change predicts, narrows, or rescues.

The request must begin with exact old wording. A project cannot responsibly change what it has not quoted. It must then state proposed new wording, the reason for change, the evidence or argument, affected dependencies, compatibility cost, public consequence, dissent, migration plan, and rollback plan. This is not a formality. It is the way the project prevents a new mechanism from silently preserving an old public headline after the old bridge fails.

The most important boundary is root handoff. The science volume may revise candidate realizations, instruments, protocols, observation models, evidence registers, public scientific status, and correction routes. It may not redefine the formal Law that *The Law of Fairness* owns. If a correction changes the latent claim, endpoint definition, formal currency, or root boundary condition, the request must become a formal root-handoff packet rather than a science-book patch.

28.4.1 — Trigger for Formal Review

A trigger for formal review appears when scientific work pressures the received formal root rather than only an instrument

or interpretation. Unit failure, endpoint leakage, process-class conflict, notation ambiguity, impossible identification, or a better construct can justify opening the formal route.

The first type is notation clarification. It fixes ambiguity without changing the object. A symbol may be overloaded, a subscript may need explanation, or a display may require status labeling. This is the lowest-risk route, but it still requires a record when public interpretation could change. A change that introduces a new construct is recorded as a substantive amendment.

The second type is representation change. This alters how a component is expressed: scalar to vector reserve, latent-state representation to belief-state representation, or single-channel estimator to multi-channel observation model. Representation changes require compatibility analysis because old results may not map cleanly into the new object. The request must state which prior findings remain comparable and which become historical.

The third type is measurement revision. It changes instruments, calibration, anchors, channel weights, abstention rules, missingness handling, or context-of-use limits. Measurement revisions often look technical but carry large public consequences. A revised candidate momentary-valence estimator cannot be allowed to rewrite old data by default. Raw data may be reanalyzed only if consent, provenance, and compatibility permit it (Crossref, 2025; Committee on Publication Ethics, 2025).

The fourth type is endpoint revision. This is high-risk because the endpoint firewall protects the theory from tautology and misuse.

Any revision to endpoint adjudication must state how the endpoint remains independent of the ledger estimator, how prior endpoint-window studies are affected, and what public language must be held during review. A weaker endpoint can make apparent closure meaningless.

The fifth type is candidate-realization replacement. A bridge from formal claim to observable consequence may be retired and replaced. The request must preserve the difference between replacing a bridge and redefining the formal Law. If reserve-coupled realization A fails and realization B is proposed, the packet must say what B predicts that A did not, what A's failure now means, and which public artifacts built on A are deprecated.

The sixth type is claim-strength change or theory reclassification. This changes public status: preliminary to replicated, partial to narrowed, active to suspended, mechanism to retired, or candidate Law-language to weaker lawhood language. It may be triggered by evidence, rival dominance, repeated bridge failure, public-risk review, or formal root handoff. It is the route with the greatest public consequence and therefore requires the clearest wording.

Table 28.4. Formal change-request packet.

Packet field	Required content	Failure if missing
Old object	Exact prior wording, version identifier, affected figure/table/code/protocol, and status label.	The change cannot be audited because no one knows what changed.

Packet field	Required content	Failure if missing
Proposed object	Exact new wording or object, proposed status, and version class.	The request becomes aspiration rather than reviewable text.
Reason	Evidence, error, rival victory, public-risk finding, dependency conflict, or root inconsistency.	The change looks reputational or cosmetic.
Dependency list	Formal definitions, instruments, protocols, datasets, code, public artifacts, applied references.	Downstream contradictions survive the correction.
Compatibility analysis	Unchanged, patch, deprecate, withdraw, reanalyze, or root handoff for each dependency.	Old results are silently overwritten or improperly reused.
Review and dissent	Reviewers, expertise, conflicts, votes, dissents, appeal conditions.	Authority becomes opaque.
Migration and rollback	Implementation steps, public notice, archive rule, rollback trigger, failure memory entry.	The correction cannot be completed or reversed safely.

28.4.2 — Proposed Amendment

A proposed amendment must say exactly what changes and what does not. It gives old wording, proposed wording, status, assumptions, affected variables, scope, and public consequences

so the change can be accepted, narrowed, rejected, or returned for stronger evidence (Crossref, 2025).

Compatibility analysis asks what breaks when the change is accepted. It should cover formal definitions, measurements, datasets, protocols, benchmarks, figures, captions, public summaries, applied-book references, social excerpts, donor pages, classroom materials, and archived pages. A change that looks small in a methods table may break a public diagram or a cross-series sentence.

The analysis should classify each dependency. Some objects are compatible without change. Some require a patch. Some require deprecation. Some require withdrawal. Some require reanalysis. Some require root handoff. A migration plan then states who changes what, by when, with which reviewer sign-off, and how the public will see the new version.

Migration should avoid retroactive erasure. If a dataset was analyzed under observation model 2.1, it remains associated with 2.1 even after 2.2 is released. Reanalysis under 2.2 becomes a new analysis with a new record. If a figure was created under a retired realization, the figure can be archived with a warning, replaced with a new figure, or withdrawn from public circulation. It cannot remain silently current (Committee on Publication Ethics, 2025).

Backward compatibility has ethical meaning here. If a correction changes public claims about terminal studies, families, clinicians, or vulnerable participants may have encountered older language. The migration plan should decide whether a public correction notice is needed, whether high-reach artifacts require direct

replacement, and whether participant-facing materials must be reissued. Technical correction without public migration leaves the burden outside the record.

The migration plan also needs a rollback route. Some revisions will fail. A new instrument may introduce bias. A new website summary may be safer scientifically but confusing publicly. A new benchmark may prove impossible to reproduce across sites. Rollback is not humiliation. It is version control. The record should say which version is restored, which artifacts are affected, and what new evidence is required for another attempt.

Compatibility analysis turns correction from aspiration into work. It tells the project that every word has dependencies. It also tells critics exactly where to look if a change has been accepted in name but not propagated in practice.

28.4.3 — Scientific Evidence and Open Questions

Scientific evidence and open questions remain separate fields. A change request may have strong simulation support, weak human transport, unresolved measurement invariance, or live rival pressure; the packet must show both the reason for the proposal and the uncertainty that still constrains it (Crossref, 2025; Committee on Publication Ethics, 2025).

The hardest ordinary change is candidate-realization replacement. A candidate realization connects the formal latent boundary condition to observable dynamics. It can fail without proving the root false, and it can be replaced without proving the root true. The danger is public drift: a failed bridge is retired, a new bridge

is introduced, and the public headline remains exactly as strong as before. That is rescue unless the cost is recorded.

Suppose an early reserve-coupled realization assumes finite-variation reserve and a signed arithmetic reference process. Later mathematical work shows that the product realization does not entail the received route claim over the intended process class. The correction does not quietly widen the process class or replace the target. It retires or narrows the realization and opens a formal change route if the received root itself requires amendment.

The packet must state what remains unchanged: the formal root, if it truly remains unchanged; the endpoint firewall; the latent/measured distinction; the public empirical burden; and the ethical firewall. It must also state what changes: process assumptions, observable signatures, model-recovery benchmarks, prior results, figures, public summaries, and future protocols. If the new realization weakens a previous prediction, that weakening is a status consequence, not a footnote.

A replacement becomes progressive when it predicts new losses. It should state where it can be defeated: a rival bridge that matches the same signatures more simply, a failure of parameter recovery, a failed transport test, an inability to separate stochastic reserve from observation shutdown, or a public-risk cost that makes the model unusable. A replacement that never risks defeat merely changes the surface of the same protected claim.

This distinction matters for *The Law of Fairness* because the formal root should not be rewritten inside a technical chapter. If the replacement changes the latent ledger, terminal target,

endpoint architecture, stream definition, or phenomenal representation, it is not merely a realization replacement. It is a root-formal amendment, and the science volume must send it back through the formal root route.

A reliable change-control system lets the project improve while showing exactly what changed. The record connects each new instrument, model, and interpretation to the evidence that motivated it.

28.4.4 — Root-Book Authority

Root-book authority keeps *The Science of Fairness* from rewriting the Law it tests. The science volume may recommend, pressure, and document, but formal adoption belongs to the formal-law route and leadership-approved change control, not to repeated scientific convenience (Committee on Publication Ethics, 2025).

A root handoff packet is required when a proposed change touches the formal claim rather than a scientific bridge. The packet should state the exact root element implicated: unified conscious stream, felt-experience currency, latent ledger, terminal neutrality, endpoint discipline, boundary condition, or ethical firewall. It should explain why a science-level correction cannot resolve the issue without altering that root element.

The handoff packet should not be written as a demand. The science volume does not command the formal-law book. It reports the conflict: evidence, model failure, measurement impossibility, endpoint contradiction, rival dominance, or public-risk pressure. The formal root then decides whether to clarify, amend, narrow,

or preserve the root claim. That decision belongs to the formal route because it changes the object all other books receive.

A root handoff should include compatibility costs across the whole series. If the latent claim narrows, synthesis scenes, ordinary-life language, City architecture, government claims, constitutional models, material-burden framing, religious doctrine, and transition gates may all need adjustment. The science volume can identify these dependencies, but it should not rewrite them unilaterally (Crossref, 2025).

The packet should include public-status advice. During root review, public artifacts relying on the challenged wording may need holds, provisional labels, or withdrawal. It may be irresponsible to keep old language active while the root question is under formal review. The handoff therefore includes temporary public-risk controls even before the root route decides.

If the root route declines to amend, the science register should preserve the conflict record. It may say that the formal root remains unchanged while a scientific bridge, observation model, or empirical target has narrowed. It may also record dissent. Silence would make future readers think the conflict disappeared rather than being judged.

Root handoff is the clearest example of no-domain-theft discipline. It lets science pressure the root without becoming the root. It lets the root learn from science without pretending every technical adjustment is a formal amendment.

28.4.5 — Version and Dependency Audit

A version and dependency audit names every downstream surface touched by the change: chapters, formulas, diagrams, datasets, protocols, code, captions, summaries, websites, donor pages, social excerpts, and archived editions. A change that cannot find its dependencies is not ready.

Every formal change request should include an ad hoc rescue audit. The audit asks whether the proposed revision was motivated independently, whether it improves prediction, whether it accepts costs elsewhere, whether it exposes new failure, and whether it preserves a public failure memory. A revision that answers yes to these questions may be progressive. A revision that answers no may be a rescue maneuver.

Independently motivated means the revision arises from theory, prior data, measurement limits, rival comparison, or public-risk analysis that can be stated before seeing the outcome it protects. It does not mean the revision cannot respond to failure. It means the revision is not only a way to keep the headline intact after failure. The timing and reasoning should be visible.

Improves prediction means the revised object should do more than fit past trouble. It should produce new observable constraints, better model recovery, sharper rival separation, improved transport logic, or clearer failure conditions. If a revision makes the claim harder to test and easier to defend, the audit should flag empirical-content loss (Crossref, 2025; Committee on Publication Ethics, 2025).

A revision records every actual change in scope, interpretation, prior-result use, or public wording. Some corrections require narrowing or withdrawal; others repair notation or code while preserving the claim and its predictions. The audit asks whether the consequences are disclosed, not whether the revision necessarily sacrifices a claim.

Exposes new failure means the revised object says what would defeat it. This is the difference between a repair and a shield. A revised endpoint rule, measurement model, or realization should create new ways to lose. If it does not, the project has probably reduced vulnerability while claiming to increase rigor.

The rescue audit should be published at least in the scientific register. Public summaries can compress it, but the reasoning should exist. The audit makes claim register vulnerability credible because it shows that repair itself can fail.

28.4.6 — No Quiet Claim Change and Rollback Planning

A candidate repair does not become settled through repeated use. Until formal adoption, it remains labeled candidate, provisional, superseded, retired, or under review as appropriate (Crossref, 2025).

Compatibility cost should be recorded because the depth of affected dependencies is scientifically informative: a wording correction differs from a change to the endpoint, observation model, or target. High cost does not defeat a true improvement, and low cost does not excuse a rescue that removes vulnerability. The matrix should distinguish scientific breakage from editorial

propagation so reviewers can see whether the evidence object changed or only its downstream expression.

Every approved change should include a rollback state and a reviewable trigger, such as failed replication, reproducibility failure, accessibility or public-risk defect, rival-model result, or discovery that the change reached a deeper formal object. The previous safe state may be suspension or withdrawal rather than the old claim.

A rolled-back revision remains in the version history with the evidence that triggered reversal. Public correction should identify the withdrawn revision and the current state so reversibility itself remains auditable. That history also prevents a failed repair from returning later under a new label without confronting the evidence that caused its withdrawal.

28.5 — Correction Propagation Across the Series

Correction follows dependencies, not publicity. An approved scientific update changes the exact interfaces that inherit its meaning, while other domains retain their own authority. The propagation map names every affected title, artifact, source, formula, caption, summary, protocol, website, dataset, and public reply together with the required action and owner.

Correction propagation is the discipline that prevents that drift. It begins with a dependency map: which titles, sections, diagrams, summaries, public artifacts, code packages, datasets, classroom materials, and public pages rely on the changed object. It then assigns a correction action: revise, deprecate, withdraw, replace, add notice, freeze as historical, or open a further change request. Without this map, the project can become correct in private and misleading in public.

Applied titles do not absorb the full scientific protocol. They receive only the corrected status needed by their own conditional claims, while the formal root receives any root-level issue through its own change route. This preserves standalone accuracy without turning every book into a duplicate of *The Science of Fairness*.

28.5.1 — Status-Labeled Cross-Book Notice

A status-labeled cross-book notice tells each owner title what changed scientifically, what stayed untouched, what language is now unsafe, and what review is required. It carries the status cost without turning the science book into a master authority over the series (Crossref, 2025; Committee on Publication Ethics, 2025).

A dependency map begins with a changed object and asks where it appears directly or indirectly. Direct dependencies include quotations, definitions, equations, tables, figures, and named protocols. Indirect dependencies include public metaphors, introduction summaries, public-diagram panels, public talks, donor pages, social posts, and applied-book scenarios that rely on the same status even if they do not name the object.

The map should distinguish technical dependencies from public dependencies. A code container may need a new hash, a benchmark table may need a new row, and a dataset release may need a new metadata record. A public page may need a shorter correction notice, a caption may need a status label, and a high-reach image may need withdrawal. Treating these as the same task creates either excessive technical detail in public surfaces or inadequate traceability in scientific ones.

The dependency map includes *The Law of Fairness* whenever a correction touches the formal root or its public definition. Science can identify pressure, evidence, and consequences, but the root owner decides whether the change is clarification, correction, amendment, or a non-equivalent successor. The map preserves both authority and traceability.

A correction should also map non-book artifacts. Website pages, visual explainers, public graphics, cover copy, interviews, donor summaries, excerpt cards, classroom handouts, code repositories, dataset readme files, model cards, and archive pages are all surfaces where old language can survive. The map should be ruthless because public trust is often lost at the smallest surface.

The map must be stored with the change request. If someone later asks why a chapter changed, or why a public diagram disappeared, the answer should not depend on institutional memory. The correction should show its route. It should say which surfaces were changed, which were frozen as historical, which were not affected, and which require future review.

The dependency map makes correction finite. It turns an overwhelming series-wide worry into a list of affected surfaces. It also prevents overcorrection by showing what does not need to move. A precise correction should be wide enough to reach every dependency and narrow enough to avoid rewriting unrelated material for aesthetic uniformity.

Table 28.5. Compatibility and migration matrix.

Affected object	Compatibility status	Required migration action
Formal root wording	Root-dependent	Prepare root-handoff packet; do not change inside the science book alone.
Science chapter prose	Directly affected	Revise wording, status label, figure captions, and dependency references.
Applied-book scenes	Conditionally affected	Update conditional language or mark as imaginative architecture, not evidence.

Affected object	Compatibility status	Required migration action
Public diagram	High-risk if overcompressed	Replace, deprecate with notice, or withdraw from active public use.
Website summary	Public-facing dependency	Update headline, status sentence, correction date, and archive link.
Dataset and code release	Reanalysis-dependent	Create new release identifier; preserve old result under old version.
Social excerpt	High-reach fragment	Issue corrected excerpt or remove from active rotation with historical notice.

28.5.2 — Routing Corrections to Each Title

Applied domains can investigate their own outcomes. A City practice, government service, religious ritual, Workgame pilot, or ordinary-life habit can be evaluated through its effects on burden, participation, access, or wellbeing.

Formal-root corrections follow the handoff procedure in §28.4. The science volume supplies the evidence, proposed wording, compatibility costs, and public consequences; The Law of Fairness retains authority over changes to the formal statement and root vocabulary.

The Science of Fairness maintains corrections to measurement, protocols, observation models, signatures, model recovery,

replication, evidence synthesis, verdict language, and change-control procedures. It sends implications for other titles through the relevant handoff route.

The Utopia of Fairness and the applied titles distinguish conditional exploration from claims that a finding is established. When a scientific result changes, update any passage that relies on its former evidential status. A clearly stated exploration of what would follow if PTVC were true may retain that conditional premise and its valid consequences without claiming confirmation. The humane aims of the series remain ethical and practical commitments rather than a theorem of terminal balance.

Religious interpretation can continue within its own domain, while scientific descriptions in religious material track the active evidence register. The handoff identifies which empirical wording requires updating and which interpretive questions remain local.

Readiness language in The Transition of Fairness updates when its scientific dependencies change. Observations from a transition pilot can generate research questions for separate qualification and testing.

Role-based correction keeps the books coordinated without making them subordinate to a single institutional body. The series sounds like one author because it shares corrected status. It remains safe because each book receives correction in the language of its own domain.

28.5.3 — Independent Duties and Correction Examples

Ethical and legal independence means duties of care, privacy, consent, nonparticipation dignity, and lawful remedy do not wait

for scientific status. A narrowed mechanism or defeated instrument may change claims, but it does not reduce human protection.

A public correction should be concrete. Suppose a diagram once showed a smooth path from reserve coupling to terminal concentration. Later model recovery shows that endpoint-conditioned bridge rivals can mimic the same pattern under the observation model. The correction action is not merely to add a footnote. The diagram should be replaced with a comparison diagram or withdrawn from high-reach surfaces. Captions should state that the old diagram represented a candidate realization, not evidence.

A website summary should match the result's current registered status. For example, a study recorded as partial support for one measured signature is described at that component and context, with a link to the supporting report and remaining research questions.

Suppose a donor page describes a future city as a natural testbed for the Law. The correction route should replace that phrase with a research-question sentence: applied observations may identify hypotheses and confounds, but evidence requires an ethical, prospectively specified design and, when a public registry claim is made, a persistent preregistration record. This protects supporters from confusing institutional coherence with scientific confirmation (Elliott et al., 2017; Crossref, 2025).

Suppose an archived social excerpt presents suffering as morally owed payment, prescribes harm, or promises a chosen reward.

Mark that misleading excerpt as withdrawn, explain the problem, and point to the corrected wording while preserving appropriate historical records. A statement of conditional net accounting is not withdrawn merely because it uses an explicitly limited accounting metaphor. Silent deletion hides the correction; unmarked preservation can repeat the actual overclaim.

The same logic applies to code. If a released analysis pipeline changes a preprocessing step that affected claims, the old container should remain citable as historical and the new container should receive a new identifier. The correction notice should state which results are affected. Reproducibility is not only the ability to rerun current code; it is the ability to understand the path from old code to corrected interpretation.

Correction examples should not be written as apology theater. They should be exact, restrained, and useful. The aim is to reduce public error and preserve scientific memory, not to perform purity after the fact.

28.5.4 — Removal of Outdated Proof Language

Removing outdated proof language is a search-and-adjudication task. Introductions, summaries, captions, figures, visual explainers, donor materials, media passages, glossary entries, and websites are scanned for proof, validation, demonstrated, settled, inevitable, or confirmed whenever the active claim register no longer permits them. Living-review methods help maintain this process over time rather than treating correction as a one-time editorial event (Akl et al., 2024; Elliott et al., 2017).

Propagation uses a checklist rather than memory. The checklist starts with the changed object, then asks which surfaces quote it, visualize it, summarize it, depend on it, teach it, fundraise from it, archive it, train on it, or hand it to an applied domain. Each affected surface receives an owner, action, due date, and completion state.

Owner sign-off is domain-specific. The science editor signs scientific accuracy; the formal-root owner signs root wording; applied-domain owners sign local translation; public-risk reviewers sign high-reach artifacts; accessibility reviewers sign figures, alternative text, and readable tables; data stewards sign repositories. No single approval pretends to cover all surfaces.

The checklist should include negative surfaces: materials not affected. This prevents future confusion when someone asks why a chapter did not change. A record saying “not affected because the claim is purely applied and does not rely on the corrected instrument” is useful. It shows that the surface was reviewed rather than forgotten (Crossref, 2025; Committee on Publication Ethics, 2025).

Propagation includes a search phase across public pages, manuscript files, draft excerpts, figure folders, captions, slides, social posts, readme files, model cards, donor language, and training materials. Automation can locate candidates, but human review decides whether each hit is current authority, historical record, quotation, or harmless coincidence. Automated absence is evidence for review, not proof that obsolete language has vanished.

Owner sign-off should not become a bureaucratic theater where everyone clicks approved. Each sign-off should state the action taken: revised, no change, deprecated, withdrawn, archived with notice, or escalated. When a surface cannot be updated immediately, it should receive a temporary status banner or access hold if the risk justifies it.

The checklist makes correction public in a practical sense. It does not ask readers to trust that the project is careful. It shows where care moved.

28.5.5 — Owner-Book Review and Correction Lag

Owner-book review protects domain boundaries during correction. The legal architecture, religious interpretation, City formation, government administration, ordinary-life guidance, material-burden analysis, synthesis, and transition sequence each adapt scientific notices inside their own mandate.

Correction lag is the time between approval and full propagation. Some lag is unavoidable in a ten-book series with public artifacts and code. Unnamed lag is dangerous. The register should track open propagation items and display interim notices where outdated material remains reachable. If a figure is awaiting replacement, the page should say so.

Interim notices should be proportional. A low-risk formatting inconsistency does not need a public banner. A high-risk artifact that could imply unearned proof, moral payment or prescribed harm, direct inner-life access, or AI authority may require immediate withdrawal before full replacement exists. The notice

should say the artifact is under correction and should not be used for claim-status inference.

Correction lag can itself reveal infrastructure weakness. If public pages take months to update because no one knows the owner, the propagation system needs repair. If social excerpts keep circulating because files were copied into multiple folders, the release system needs better artifact identifiers. If applied domains resist update because the old language was useful, the boundary system needs reinforcement (Committee on Publication Ethics, 2025).

The record should distinguish pending, delayed, blocked, and resolved. Pending means work is underway. Delayed means a time issue. Blocked means a dependency or dispute prevents action. Resolved means the surface has been updated or marked. This prevents vague project-management language from hiding public-risk exposure.

Interim notices should be written with the same severe compassion as the chapters. They should not shame readers who saw earlier material. They should not scold critics who noticed the issue. They should simply say what changed, what not to infer, and where the corrected status will appear. A clear interim notice can build trust even before the final replacement is ready.

A correction is not an event. It is a propagation process. Naming lag makes the process honest.

28.5.6 — Correction Completion Audit

A completion audit confirms revised status across active text, archives, code, data, summaries, and high-reach surfaces (Crossref, 2025; Committee on Publication Ethics, 2025).

Before crisis, synthetic drills retire a fake instrument, deprecate a figure, narrow a public claim, or stage a mock root handoff. They test surface detection, owner action, and time to closure.

Drills must include public artifacts and applied domains, where captions, websites, donor summaries, social cards, model readmes, and neighboring books may preserve stale implications. Each domain receives correction without taking scientific authority or losing its voice. Correction drills should verify that each domain receives the new scientific ceiling while retaining its own independent legal, ethical, or practical reasoning.

Drills record missed surfaces, unclear roles, disagreement, and latency, then repeat after structural changes, new websites, release pipelines, or pilots. Their value lies in exposing brittle infrastructure before public pressure does. Latency metrics can show whether the correction system improves over time and which surfaces repeatedly resist propagation.

Search manuscripts by meaning as well as phrase. Introductions, summaries, captions, illustrations, and neighboring chapters may retain an old inference; verify upstream and downstream dependencies. Dependency search should include paraphrases and translated materials, because stale meaning can survive even after the original wording disappears.

House-phrase searches target discovery language, body-reading or score implications, retired bridges, and stale figures; the Table of Contents and retained introductions receive special scrutiny as reader maps. Automated discovery can assist, but every hit still requires human judgment about whether the underlying claim actually depends on the changed component.

Final prose remains natural: reader-facing sentences carry corrected meaning, while the register holds machinery and provenance. One coherent book replaces visibly patched files. This separation keeps provenance exhaustive internally while preventing editorial scars from becoming part of the public argument.

28.6 — Public Artifacts, Websites, and Withdrawal

Public artifacts are scientific surfaces when they compress or display scientific meaning. A caption, diagram, website, donor page, social image, video, or interview can remain active long after the technical context changes. Each artifact therefore carries a version, status, owner, correction route, and hostile-crop review (Committee on Publication Ethics, 2025; Crossref, 2025).

An overstated caption is not merely a design defect, and an obsolete website status is not a harmless delay. Public surfaces can change what readers believe the science has authorized, so they enter the correction system with the same seriousness as other dependent objects.

A public graphic can be beautiful, memorable, and emotionally compelling while conveying the finding clearly. A well-designed artifact identifies the observed result and its scope, carries its current scientific status, and gives readers a route to the supporting detail.

28.6.1 — Cover and Visual Explainers

Cover and visual-explainer materials receive correction because images carry status before readers reach prose. A visual that implies confirmation, universal scope, archive-as-soul, Artificial Intelligence authority, or automatic uplift must be revised even if the surrounding text is technically correct (Committee on Publication Ethics, 2025; Crossref, 2025).

The first class is interpretive artifacts: visual-explainer panels, diagrams, captions, summary pages, public slides, social cards,

and short excerpts. These artifacts translate scientific objects into public understanding. Their risk is overcompression. They should carry object labels, status labels, and source-version references where feasible. If a diagram cannot remain safe after cropping, it should not be released in that form.

The second class is institutional artifacts: donor pages, partnership summaries, research invitations, recruitment materials, classroom handouts, media kits, and public FAQs. Their risk is authority transfer. They can make funding, participation, or institutional association look like scientific validation. They should separate invitation, funding, research question, ethical boundary, and evidentiary status.

The third class is archival artifacts: prior versions, withdrawn graphics, old public statements, deprecated code, historical datasets, and correction logs. Their risk is either erasure or resurrection. They should remain available under appropriate access and with clear notices, but they should not circulate as current. An archive should preserve history without giving old language present authority (Tabassi, 2023; UNESCO, 2021a).

The fourth class is computational artifacts: code containers, benchmark libraries, model cards, dataset cards, analysis notebooks, synthetic fixtures, and automated reports. Their risk is hidden status transfer through technical confidence. A container that runs cleanly can still implement a flawed assumption. A model card should therefore carry claim-status boundaries, training or simulation scope, known failures, and prohibited uses.

The fifth class is conversational artifacts: public replies, interviews, moderated discussions, and short responses to critique. Their risk is tone drift. A careful reply can build trust; a defensive reply can make the theory look like a movement protecting itself. Conversational artifacts should be correction-ready and should avoid making claims stronger than the current declared status because the audience is skeptical or hostile.

Risk level should determine review. High-reach, high-compression, vulnerable-reader-facing, or quote-crop-prone artifacts require stronger pre-release review and faster withdrawal authority. A table inside a technical chapter and a shareable image on a website are not the same public object.

Table 28.6. Public artifact correction example.

Artifact	Detected problem	Correction action
Visual-explainer panel	Shows candidate realization as a smooth path without rival comparison.	Replace with a candidate-versus-rival panel and add status label.
Website headline	Says early evidence for terminal neutrality when status is partial support for one measured signature.	Rewrite headline, add correction date, archive prior text with notice.
Donor page	Implies applied pilots are testbeds proving the Law.	Replace with research-question language and evidence firewall statement.

Artifact	Detected problem	Correction action
Social card	Uses wording that turns a conditional balance requirement into moral payment or prescribed harm.	Withdraw active card, preserve historical copy only with warning.
Code release	Implements deprecated observation model without status warning.	Create new release, mark old container deprecated, link affected results.

28.6.2 — Summary, Glossary, and Archive Status

Summary and glossary language helps readers retain the book’s central distinctions. Each short definition should preserve the corrected term’s status, scope, and uncertainty, so concise wording remains faithful to the full scientific account (Crossref, 2025; Tabassi, 2023).

The archive preserves versions, corrections, artifacts, and results so the project can be audited. Clear purpose and access limits govern the records it contains.

Archived artifacts should be marked by status: current, superseded, deprecated, withdrawn, historical, restricted, or removed for safety/legal reasons with a public notice where appropriate. A withdrawn artifact may still be retained internally for audit. A restricted dataset may be findable through metadata without being accessible to the public. A public correction can state that an artifact existed and why it was changed without exposing participants or reviving unsafe language.

The archive must also preserve negative results. Failed instruments, abandoned mechanisms, defeated bridges, public artifact corrections, and red-team objections belong in the memory of the program. If only positive artifacts remain visible, the archive becomes a publicity surface. If negative artifacts are sensationalized, it becomes a spectacle. The right posture is factual memory (UNESCO, 2021a; Committee on Publication Ethics, 2025).

Apply the access tiers defined in §§26.5.2 and 28.2.5 while preserving each artifact’s status and version history. Historical availability does not restore a superseded claim’s authority or enlarge permission to release sensitive material.

The archive should support citation without raw links in manuscript prose. Public materials can cite stable identifiers, release names, or version labels in controlled references. Manuscript copy should not depend on live links for meaning. The text should remain intelligible if a website changes. If an artifact is withdrawn, the record should tell future readers what changed.

A good archive shows earlier claims, the evidence that changed them, the current findings, and the access conditions for supporting records. These links make the project’s scientific development traceable.

28.6.3 — Websites and Social Excerpts

Websites and social excerpts require version dates, correction notices, share-card updates, and cache-aware replacement where possible. A public sentence clipped from an old page should not

keep speaking as current science after the register has changed (Tabassi, 2023; UNESCO, 2021a).

A correction notice should answer five questions: what changed, why it changed, what prior wording should no longer be used, what current wording replaces it, and which materials are affected. It should be written plainly enough for a serious public reader and precisely enough for a critic. It should not bury the correction beneath gratitude, mission language, or reassurance.

The notice should distinguish correction, clarification, deprecation, withdrawal, suspension, and reclassification. A clarification says the object was ambiguous. A correction says the object was wrong or misleading. Deprecation says an object remains historically available but should not be used going forward. Withdrawal says active circulation stops. Suspension says no current status should be asserted. Reclassification says the evidentiary status has changed.

Public wording should state the gain and its consequence together. A model may improve while an earlier claim weakens, narrows, or fails; readers need both facts. A clear correction preserves confidence in the research process by giving readers the current evidential status and identifying which earlier statement no longer applies (Committee on Publication Ethics, 2025; Crossref, 2025).

Correction notices should be connected to artifacts. If a figure changes, the figure caption should carry a version notice or link internally to the correction register. If a website page changes, the page should show the correction date and prior status. If a social excerpt is withdrawn, active channels should stop using it. The

correction should travel through the route defined in the previous section.

Some public artifacts should be predesigned for correction. A figure template can include a status field. A website card can include a version line. A donor summary can include a claim-status paragraph. A model card can include prohibited uses. Designing for correction prevents the project from treating updates as emergencies every time science does what science should do.

The public artifact rule is simple: a surface that can carry scientific authority must be able to lose that authority visibly. If it cannot be corrected, deprecated, or withdrawn, it should not be released in a form that invites trust.

28.6.4 — Persuasive Copy, Graphics, and Captions

Donor, media, and recruitment copy must not borrow authority from outdated science. Persuasive materials receive stricter review because they can convert public attention, money, participation, or institutional prestige into pressure to preserve a claim beyond its warrant (UNESCO, 2021a; Committee on Publication Ethics, 2025).

Every public graphic receives a caption review before release. The caption names whether the figure is a candidate model, diagnostic workflow, evidence map, rival comparison, protocol route, failure memory, or public-artifact example. Visual elegance never substitutes for status, and a clean arrow never creates inevitability.

Graphic review should test cropping. If the title and caption are removed, does the remaining image imply proof, ranking, person measurement, AI authority, archive-as-soul, or transition destiny? If so, the figure needs embedded status language, design revision, or restricted release. A graphic that survives only when surrounded by paragraphs may be safe for a chapter but unsafe for social media.

Caption review should also test visual hierarchy. If the Law-candidate path is bold and rivals are faint, the figure teaches overclaim even if the caption is cautious. If correction paths are hidden in small type, the artifact may imply that update is secondary. Visual style is part of claim-status discipline. Grayscale elegance is not enough; hierarchy must match warrant (Crossref, 2025; Tabassi, 2023).

Figures about Artificial Intelligence require extra care. A diagram showing an AI tool assisting correction should visually subordinate the tool to human review. If the tool sits at the center and human reviewers sit at the margins, the artifact says the opposite of the prose. Readers learn from layout before they read sentences.

A figure should also have alt text that preserves claim status. Accessibility is not merely a technical requirement. An alt description that says “the model proves correction route” would be false even if the visible figure is careful. The alt text should describe the diagram as a workflow, candidate route, or boundary, not as validation.

Graphic review belongs in the same correction process because public artifacts are where a claim register often escapes its own limits. The figure is part of the argument. If it cannot carry its status safely, it should not carry the public.

28.6.5 — Artificial Intelligence Summaries and Withdrawal

Artificial-intelligence summaries become public artifacts when they repeat the project. Search snippets, assistant answers, autogenerated abstracts, and indexed summaries are corrected where possible through updated source pages, metadata, public notices, and visible superseded labels. Rapid model change makes source identity, retrieval date, prompt or instruction context, and human verification especially important (Committee on Publication Ethics, 2025; Crossref, 2025; Feuerriegel et al., 2026).

Withdrawal is sometimes necessary. A public graphic, excerpt, or page may be unsafe enough that correction is not sufficient. Withdrawal should be prompt, visible, and restrained. It should explain that the artifact has been removed from active use because it could mislead, overstate status, revive unsafe language, or conflict with an updated register. It should not dramatize the removal as catastrophe or victory.

Withdrawal should preserve audit memory. The artifact may remain internally archived with access controls and a notice. In some cases a public historical notice is appropriate. In others, the artifact may be too unsafe or legally sensitive to display. The key is that the withdrawal decision itself remains traceable. Future

readers should know that a correction occurred, even if the artifact is not reproduced.

Withdrawal should trigger search. If a figure was posted on a website, it may also appear in slides, donor decks, draft chapters, social excerpts, image folders, and archived pages. The propagation checklist should handle this. Otherwise the most dangerous version may survive in the least governed location (Tabassi, 2023; UNESCO, 2021a).

Withdrawal should not be confused with defeat. A graphic can be withdrawn because it compresses uncertainty poorly while the underlying component remains under study. Conversely, a scientifically defeated component may remain in a public failure-memory artifact because its defeat teaches the field. The withdrawal notice should state the reason so readers do not invent one.

Withdrawal should also include replacement planning when needed. If the removed artifact served an important educational role, the project should create a safer replacement rather than leaving a hole. The replacement should teach the corrected distinction, not merely repeat the old point with softer words.

This is the public form of claim register vulnerability. The project can say: we removed this because the way it traveled was not safe or accurate. That sentence is stronger than pretending the artifact never existed.

28.6.6 — Archive Memory and Public Correction Intake

An archive should distinguish current, corrected, superseded, and withdrawn artifacts and maintain stable correction pages with

date, version, affected objects, current wording, and access state. Prior language may remain visible as history but should not be confusable with current authority.

Metadata, redirects, preview cards, filenames, cached images, and other high-reach discovery surfaces are part of correction because they can continue to surface obsolete claims after a page changes (Committee on Publication Ethics, 2025; Crossref, 2025). Restricted artifacts may be listed without unrestricted download when participant, privacy, or misuse risks require it.

Public error intake should identify the artifact and problem type, protect privacy, acknowledge receipt without pre-judging correctness, and route science-status, safety, accessibility, hostile-crop, and version concerns to accountable review. A resolved report should point to the current artifact and explain the decision.

The archive provides a door for correction, not a mechanism for public vote on scientific status; review decisions and their reasons remain visible.

28.7 — Applied Evidence Intake and Future Testbeds

Applied settings can generate research questions about burden, feasibility, context, confounding, and measurement, but they do not become evidence testbeds merely because a City, public office, religious community, Workgame pilot, or ordinary-life practice already exists. Evidence begins only when a separate, ethical, prospectively specified design defines the object, consent route, context of use, estimand, measurement and observation processes, rivals, burden limits, and stop conditions. A living institution is not a laboratory by mere existence.

This distinction protects both science and applied domains. Science is protected from proof by institutional success. Applied domains are protected from becoming measurement traps, recruitment funnels, or public demonstrations staged for theory. The ordinary dignity of a person in a city, church, office, family, or transition pilot does not become evidence unless a legitimate research protocol exists and participation is voluntary, informed, minimized, and reviewable.

28.7.1 — Testbed Status

Testbed status means possible research context, not validation. Cities, government offices, religious settings, Workgame pilots, clinical systems, digital tools, and ordinary-life practices may produce questions, but evidence begins only when a separate study design is opened (National Institutes of Health, 2026a; National Institutes of Health, 2026b; Rieke et al., 2020).

Applied domains can notice patterns that matter. A city transparency process may reveal that people experience relief

when burden-routing becomes visible. A government office may discover that administrative delay carries a measurable stress load. A religious community may hear recurring pastoral language around repair and grief. A Workgame pilot may show that participants respond differently when failure memory is preserved rather than hidden. These observations can inspire research questions.

The question route should be explicit. An applied observation is written as a question, not a finding: does transparent administrative routing reduce reportable vigilance in this population; does correction visibility change trust under public uncertainty; does voluntary ritual language alter grief reports without changing scientific status beliefs; does preserved failure memory improve decision learning without increasing shame? The wording matters because it prevents the applied domain from speaking as if it already knows.

The observation should also name its confounds. In applied settings, relief may come from being heard, from faster service, from social belonging, from reduced cost, from expectation, from status, from novelty, from selection into participation, or from institutional attention. These are not nuisances to be brushed aside. They are rival explanations and design warnings (Tabassi, 2023).

Applied observations should not be harvested through ordinary participation without prior boundaries. A person entering an ordinary-life practice, City process, religious setting, or transition pilot should not later discover that their experience was treated as scientific evidence. Research begins when consent, purpose,

measurement, minimization, and withdrawal are declared. Before that, applied observations remain program learning, pastoral learning, design learning, or public-risk learning.

The research-question intake process should include negative observations. If a public artifact confuses readers, if a pilot creates measurement pressure, if a ritual phrase is heard as forced meaning, if an administrative tool begins to feel like a score, those failures are testbed signals. They may not test the Law directly, but they test whether public and applied surfaces are safe enough to host questions.

The value of applied observation is therefore real but bounded. It helps the science program ask better questions. It does not spare the science program from doing the work.

Table 28.7. Future-testbed evidence firewall.

Applied source	What it may generate	What it cannot become by default
Transparent City	Research questions about public burden routing, consent, and civic trust.	Evidence for terminal neutrality or proof of the formal Law.
Government office	Questions about administrative load, delay, error correction, and public records.	Scientific validation through service improvement.
Religious setting	Questions about meaning, grief language, ritual effects, and pastoral safety.	Scientific support through spiritual resonance.

Applied source	What it may generate	What it cannot become by default
Workgame or transition pilot	Questions about sequencing, failure memory, readiness, and rollback behavior.	Transition authorization or destiny.
Ordinary-life practice	N-of-1 questions, private reflection, and design insights.	Research evidence without consent and protocol.
Public artifact response	Public-risk signals and communication hypotheses.	A popularity or controversy metric for truth.

28.7.2 — Separate Consent and Data Doors

Separate consent and data doors keep service participation, spiritual practice, civic interaction, public administration, private reflection, research participation, archive storage, and model training from collapsing into one permission. Each door has its own purpose, risks, refusal rights, and governance (Rieke et al., 2020; Tabassi, 2023).

The future-testbed firewall has four gates: translate the applied observation into a research question; pass ethics, consent, and burden review; specify measurement, endpoint, context of use, rivals, and failure conditions; and enter results into the component evidence register under the same status rules as any other study. Skipping a gate turns operational coherence into overclaim.

Each applied domain contributes a distinct practical question. A Transparent City can reveal where public power shifts burdens; a government office can measure administrative load; a constitutional model can define lawful visibility; a religious

practice can support meaning; and a transition pilot can test sequencing.

Meaningful private practices can generate useful research questions about attention, ritual, apology, rest, service, and relationships. Participation in such research remains a voluntary choice, with consent and independent safeguards appropriate to the study (National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Future testbeds should be context-of-use specific. A pattern in one city language community may not transport to another. A finding in one care regime may not travel to a workplace. A public artifact safe for academic readers may be unsafe for grieving families. The testbed question must name population, setting, language, authority relation, data source, intended inference, and prohibited inference.

The firewall includes a no-default-training rule. Applied-domain records, pastoral notes, ordinary-life reflections, government service records, and transition artifacts are not artificial-intelligence training corpora by default. Research access and model-training access require separate purpose, consent, governance, minimization, and review; the desire to learn does not create permission to absorb a life into a dataset.

The firewall makes future testbeds honorable. It lets applied worlds become sources of questions and confounds without becoming theaters of proof. It allows the series to learn from reality while refusing to convert participation into validation.

28.7.3 — Operational Success Is Not Validation

Operational success is not validation because a practice can help for ordinary reasons. Reduced complaint burden, better service timing, stronger ritual comfort, or improved user retention may matter deeply while remaining outside the scientific status of the formal Law (Tabassi, 2023).

A science correction may travel to applied domains, but it does not give the science body ownership over them. The City book owns voluntary bounded civic-public-power formation. The government book owns public administration. The constitutional book owns legal architecture. The religion volume owns optional doctrine and pastoral limits. The transition book owns sequencing and gates. The science book owns evidence status, not the total meaning of these domains.

Applied domains contribute questions and relevant observations through the research process. Independent scientific review determines evidence status, while each domain retains responsibility for its practical and interpretive decisions.

Handoffs identify the question or finding being transferred. An applied domain can describe an observation that suggests a scientific research question. A science update can identify which empirical statement should change in applied materials. Each handoff names the affected claim and leaves the receiving domain's other purposes and judgments clear.

Testbed review should identify when observation burden or consent conditions make a proposed question unsuitable for the

setting. It can then recommend a less burdensome design, another setting, or a documented decision to leave the question open.

Other observations may become design constraints rather than scientific claims. If a public artifact repeatedly misleads, the correction may be to redesign artifacts, not to update the Law. If a City protocol reduces administrative burden, the lesson may belong to government design, not terminal neutrality. If a religious phrase protects someone from forced meaning, the lesson may belong to pastoral language, not empirical validation.

Handoff without ownership transfer is the series operating at its best. Domains speak to one another without borrowing warrant. Science corrects what science owns. Applied life remains meaningful without becoming evidence by proximity.

28.7.4 — Negative Operational Findings

Negative operational findings deserve a route before they become scandal or silence. Increased stress, exclusion, dropout, privacy fear, staff burden, family pressure, or cultural mistrust can generate safety and research questions without being minimized to protect the theory (National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Future testbeds must treat measurement as an intervention. Asking people to report, wear devices, consent to records, or become visible to researchers changes the setting. In a city office, a church, a game, a household, or a transition pilot, measurement can create compliance pressure even when participation is formally voluntary. The testbed route must therefore minimize cadence, data scope, identifiability, and downstream reuse.

Consent must be separated from access. A person may choose to use a service, attend a religious gathering, participate in a Workgame activity, or interact with a civic institution without consenting to research. Research consent requires separate explanation, refusal without penalty, withdrawal, data minimization, and purpose limits. A future testbed that blurs service and research converts ordinary life into a recruitment pool.

Burden monitoring should include staff and bystanders. A public office can become overloaded by study procedures. A family member can become a proxy reporter without clear consent. A facilitator can feel responsible for producing positive results. These loads are not merely operational. They can bias evidence and harm people. Testbed governance should track them as part of validity (Rieke et al., 2020; Tabassi, 2023).

Data minimization should begin before measurement design. The question should ask what data are truly needed to distinguish the hypothesis from rivals, and what data would merely be interesting. Interesting data are not justified by ambition. If an applied observation can be studied with less intrusive reports, synthetic fixtures, aggregate timing, or qualitative design, high-intensity telemetry may be unnecessary.

Participant return should be carefully bounded. Testbed participants may deserve to know what happened to the study without receiving personal scores or inner-life interpretations. Return materials should describe study status, aggregate findings, limitations, and corrections. They should not tell participants

what their ledger was, whether they balanced, or what their life meant.

Future testbeds can study practical *improvements* while protecting the people whose experience makes that research possible.

28.7.5 — Science Constrains Claims, Not Governance

Science constrains claims, not governance. A scientific result can limit what the series may say about evidence, but legal rights, civic design, religious practice, material policy, and transition gates remain governed by their proper books and institutions.

An applied-domain failure can be scientifically valuable without being evidence for the Law. If a public artifact creates confusion, the failure informs communication design. If a City process produces distrust because records feel like surveillance, the failure informs consent and record boundaries. If a religious phrase is heard as forced meaning, the failure informs pastoral language. If a Workgame pilot makes readiness feel like ranking, the failure informs transition gating.

These failures should be logged in a testbed incident register. The register should state the domain, artifact or practice, observed failure, affected population, immediate mitigation, research question generated, and whether scientific evidence status is affected. Most applied failures will not directly change the Law's empirical status. They will change public-risk controls and research design.

Applied failure can also reveal confounds. A study planned in a civic setting may discover that changes in reported burden are

driven by caseworker kindness, speed of service, or relief from paperwork rather than by any Law-specific mechanism. That is useful. It helps build better rivals and prevents the project from claiming credit for ordinary humane administration (Tabassi, 2023).

Failure in applied domains should not be hidden to protect the image of the series. A public theory about fairness becomes untrustworthy if it cannot admit when its own surfaces shifted burden. The applied incident register is not a scandal list; it is a learning system. It keeps the burden of public experimentation inside the record.

Some applied failures may require scientific status holds. If a public artifact strongly misleads readers about a scientific claim, the artifact's status must change even if the underlying science remains open. If an applied pilot generates coercive measurement pressure, research using that context should pause. Applied failure can therefore affect the conditions under which science may proceed.

The positive rule is this: failure travels to the route it belongs to. Communication failure travels to artifacts. Governance failure travels to applied design. Measurement failure travels to science. Root conflict travels to the formal root. The project becomes safer when failures are not forced into the wrong category.

28.7.6 — Transition Timing Remains Separate

Transition timing stays separate because readiness is not a laboratory output. Even strong evidence cannot authorize rollout, artifact release, Workgame scaling, political movement, or public

transition without independent gates, rollback, refusal rights, and timing control.

The route from observed practice to research protocol should protect people while preserving learning. First record the observation without evidentiary status; list confounds and ethical risks; turn the question into an estimand or qualitative aim; and specify consent, measurement, rivals, burden, and stop conditions. Only then may the study become protocol-fixed or publicly preregistered through an external registry with a persistent identifier.

Protocols should declare what each context adds and prevents. City settings may add ecological validity and public-power relevance but complicate consent and authority; religious settings may enrich language but introduce belief and belonging effects; Workgame settings may structure choices but create performance pressure. These are design features, not afterthoughts.

The protocol should avoid demonstration design. Demonstration design begins with institutional hope and gathers favorable impressions; science begins with a question that can lose. The protocol names rivals, negative controls, nonresponse, public-risk boundaries, and results requiring route suspension. A testbed unable to produce an adverse result is a demonstration, not a discriminating evaluation. Program evaluation can itself test specified outcomes and report failure; a PTVC test additionally needs the target, measurement, and comparison requirements stated here (National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Prospective public-communication rules belong in the study record. Before results, the applied domain may say only that the setting generates questions, the design is prospective, participation is voluntary, results may be negative, and public status awaits review. Public preregistration may be claimed only with an external registry record and persistent identifier.

A good future-testbed protocol lets applied life inform science without using lives as proof; it is neither shy nor extractive.

28.8 — No Formula, Score, or AI Sovereignty

Scientific governance uses formulas, code, dashboards, registries, archives, and artificial-intelligence systems as accountable tools. They can calculate, search, reproduce, compare, flag, and summarize within declared limits. They cannot decide truth, personhood, worth, care, eligibility, spiritual meaning, claim status, or acceptable risk; those judgments remain with identifiable human authorities.

Formula sovereignty is the error of letting mathematical elegance decide what evidence has not established. Artificial-intelligence sovereignty is the error of letting system output decide what accountable humans must judge. Both errors can appear modern, efficient, and impartial; both are dangerous when the project concerns conscious life, public meaning, scientific status, spiritual interpretation, legal imagination, and transition risk.

This is not an anti-technology rule. The science volume should use computation seriously. It should use model recovery, simulations, version control, provenance, automated tests, accessibility audits, plagiarism checks, contradiction search, and reproducible containers. The point is to keep every tool inside its lane. A useful tool produces evidence support, warnings, drafts, comparisons, or diagnostics. It does not own the decision.

28.8.1 — No Claim-Status Equation

No claim-status equation means a formula may summarize fields but may not decide truth. Replication count, heterogeneity, prediction score, safety incidents, dissent, and public risk are not

compensatory inputs that can be collapsed into one authority number (Tabassi, 2023; UNESCO, 2021b).

Search is one bounded use of Artificial Intelligence. A tool may locate contradictions, summarize candidate literature, compare chapters with claim-status registers, or flag old-language residue, but its output remains a lead requiring source verification and human judgment.

Code generation and static analysis are another bounded use. Tools may draft tests, generate fixtures, inspect containers, or propose metadata checks, yet human code review, independent execution, provenance, and benchmark comparison remain necessary before any result enters the evidence record.

Drafting and language review are a third bounded use. Tools may propose phrasing, detect repetition, or flag apparent overclaim, but final manuscript copy, artifact language, and correction notices require accountable human editorial and scientific responsibility (Council for International Organizations of Medical Sciences, 2016).

Automated systems may assist monitoring. They can flag outdated version references, broken artifact links, inconsistent status labels, missing alt text, uncorrected captions, and code-container mismatches. These monitors are useful because the series is large. Their alerts should create review tickets. Their silence should not be interpreted as assurance.

Formulas may assist classification only under declared scope. A decision aid can display thresholds, checklist satisfaction, dependency status, or model-comparison results. It should show

inputs, assumptions, uncertainty, and abstention. It should never output a final controlled scientific verdict without human review. The output should be phrased as “review indicated,” not “truth determined.”

Permitted assistance is powerful precisely because it remains subordinate. The project should not reject tools out of fear. It should use them under provenance, audit, and limited authority so that computational strength increases human accountability rather than replacing it.

Table 28.8. Formula and Artificial Intelligence authority boundary.

Tool use	Permitted role	Forbidden role
Automated literature search	Find candidate sources, contradictions, and missing rival language.	Declare completeness, truth, or consensus.
Language model drafting	Suggest wording, detect repetition, propose summaries under review.	Approve manuscript copy, public status, or canon.
Statistical dashboard	Display diagnostics, uncertainty, dependencies, and abstention flags.	Issue final discovery, defeat, or transition authorization.
Model-recovery pipeline	Compare candidate and rival generators under declared conditions.	Treat construction or fit as evidence for the formal Law by itself.
Archive system	Preserve versions, corrections, failures, and provenance with access controls.	Represent the person, soul, worth, or final inner life.

Tool use	Permitted role	Forbidden role
Formal equation	State a definition, candidate realization, estimator, diagnostic, or decision aid under status label.	Become proof, moral permission, clinical instruction, or public score.

28.8.2 — No Dashboard Authority

No dashboard authority means a visualization organizes evidence for review but does not deliver the verdict. Layout, color, rank order, thresholds, filters, and summary panels are audited so the dashboard does not silently decide what human reviewers later rubber-stamp (UNESCO, 2021b).

No formula may decide personhood. Bearer classification, consciousness uncertainty, artificial-system evidence, nonhuman extension, and boundary-state humility remain expert-governed scientific and ethical questions. A classifier may provide evidence support or a risk flag. It may not declare a being a bearer or non-bearer as public authority.

Apply the non-deployment rule in §9.7 to every dashboard and computational output surface. No display, threshold, rank order, or derived score acquires authority over persons, communities, countries, religions, institutions, or readiness.

No artificial-intelligence system may adjudicate the claim register. It may propose a change request, draft compatibility tables, detect stale artifacts, and summarize dissent. It may not approve the change, close dissent, decide the root handoff, or declare that public status has moved. Approval belongs to a human-governed scientific process with visible responsibility

(Council for International Organizations of Medical Sciences, 2016; Tabassi, 2023; Feuerriegel et al., 2026).

No software system may convert future-testbed observations into evidence by default. A dashboard may organize questions, flag confounds, and track whether a study is protocol-fixed or publicly preregistered. Evidence status still requires the scientific route, adequate measurement, serious rivals, and accountable human review rather than data abundance.

No archive may be treated as a soul or final truth repository. An archive can preserve records, versions, artifacts, corrections, and failures. It cannot preserve the person. It cannot judge consciousness. It cannot authorize care decisions. It cannot become a public dossier. Its role is memory with limits.

Computational outputs remain purpose-limited, reviewable, contestable, and revocable. Scientific status decisions follow the designated human review process across political, donor, institutional, religious, and transition settings.

28.8.3 — No Model-Generated Consensus

No model-generated consensus means artificial intelligence may summarize positions but may not smooth disagreement into agreement. Dissent, unresolved rivals, minority reports, and adverse findings remain visible even when a model can produce fluent prose that sounds balanced. Human review preserves the scientific cost of disagreement rather than translating it into synthetic harmony (Autio et al., 2024; World Health Organization, 2025; Council for International Organizations of Medical Sciences, 2016).

The authority boundary can be stated compactly. Tools generate support. Humans judge status. Formal root changes return to *The Law of Fairness*. Scientific evidence status belongs to *The Science of Fairness*. Applied books receive corrected status within their own domains. Public artifacts carry only the claim their source permits. No surface receives more authority than the route that produced it.

The boundary must be visible in practice. Every artificial-intelligence-assisted change request includes provenance; every automated audit states what it can and cannot detect; every formulaic decision aid includes abstention and human-review fields; every model card identifies prohibited uses; and every public artifact remains correctable. Current governance and reporting guidance supports documented evaluation, model and configuration disclosure, human accountability, risk management, and contestability, but no framework can confer scientific or personal authority on automated output (Tabassi, 2023; Autio et al., 2024; World Health Organization, 2025; Feuerriegel et al., 2026).

This boundary also protects artificial-intelligence systems from being turned into scapegoats. If a model writes a misleading sentence and humans publish it, the responsibility is human. If an automated dashboard overstates status and humans deploy it, the responsibility is human. Tool use does not erase accountability; it makes provenance, evaluation, and contestability more necessary (Tabassi, 2023; Autio et al., 2024; UNESCO, 2021b; World Health Organization, 2025).

The same boundary protects the project from fake humility. It is not enough to say that tools are imperfect. The document must specify where they stop. A tool stops before personhood. It stops before truth. It stops before status. It stops before claim register. It stops before care. It stops before public authority. Those stops are not rhetorical; they are workflow permissions.

A scientific claim system becomes trustworthy when it can be corrected without becoming ownerless. Human-governed updates, versioned registers, dissent, appeal, formal change requests, root handoff, correction propagation, public-artifact control, future-testbed firewalls, and no tool sovereignty form the route. The result is not a perfect system. It is a system that lets imperfection surface before it becomes doctrine.

The final safeguard is not a sentence but a habit. Whenever a result, model, or tool appears to make the Law easier to believe, the project asks what route gave it that authority, what component it touches, what public surface it affects, what dissent remains, what failure memory it creates, and what would make it lose. That habit is the science book's contribution to the active claim register: vulnerability made operational.

28.8.4 — Human Review, Dissent, and Appeal

Human review, dissent, and appeal keep responsibility traceable. People sign the decision, record reasons, disclose conflicts, preserve minority reports, and accept appeal; the presence of computational support never removes the human duty to answer for the update (Council for International Organizations of Medical Sciences, 2016; Tabassi, 2023).

Every substantial tool-assisted decision should leave a tool-use record. The record should state the tool, version if known, input class, output class, human reviewer, permitted use, and rejected use. If a language model helped draft a correction notice, the record should say that. If a static analyzer found stale version strings, the record should say that. If a simulation tool generated benchmark results, the record should say the container and seed policy.

Tool-use records prevent both mystification and scapegoating. The project should not pretend that a human alone did work a tool assisted. It should also not blame a tool for a human decision to publish, approve, or ignore. Provenance lets future reviewers ask whether the tool was appropriate and whether human review was adequate.

Audit includes hallucination, omission, compression, and configuration risk for language tools. A system can invent sources, flatten branch distinctions, omit adverse evidence, merge versions, or change output after an update. Review therefore records model identity, version, access date, instructions, tool settings, source set, generated output, human verification, and rejected material when those details affect the scientific record (Feuerriegel et al., 2026).

Audit should include leakage risk. Tool inputs may include sensitive draft material, participant information, restricted records, or unpublished protocols. The default should be local, controlled, or redacted use unless governance approves otherwise. Convenience cannot override privacy, consent, or

publication control. A tool that improves drafting but leaks restricted context is not acceptable.

Tool auditing is not anti-innovation. It is how the project can use powerful assistance without letting assistance become authority.

28.8.5 — Audit of Automation Bias

Audit of automation bias tests whether reviewers overdefer to technical outputs. The system looks for anchoring, confidence-label effects, interface-driven ranking, selective summarization, prompt framing, and quiet replacement of judgment by tool convenience (Tabassi, 2023; UNESCO, 2021b).

The sovereignty loophole appears when the project says no tool has authority, but then structures workflows so that humans merely ratify tool outputs. If a dashboard selects the status, a language model drafts the public wording, a model chooses the route, and reviewers approve by inertia, the tool has become de facto sovereign. The workflow must force human reasons independent of the output.

One protection is independent human justification. A reviewer should state why the decision is correct in ordinary language, naming the source, evidence, scope, and public consequence. The tool output may be attached, but it cannot be the reason. If the reviewer cannot explain the decision without the tool, the decision is not ready.

Adversarial tool review supplies another safeguard. A second reviewer or system can search for overclaim, stale versions, conflicting artifacts, excluded rivals, root-handoff triggers, and public-risk hazards that the first pass missed. The goal is

complementary scrutiny, not a contest among tools, and no fluent pipeline should become invisible infrastructure (Council for International Organizations of Medical Sciences, 2016).

A third protection is abstention. If the evidence is too ambiguous, the tool should not be forced to classify. If the public wording cannot be made safe at the required compression level, the artifact should not be released. If the register cannot resolve whether a change is root-level or science-level, the route should hold. Sovereignty often enters through the refusal to abstain.

The final protection is revocability. Tool-assisted decisions must be correctable. If later review finds that an artificial-intelligence summary omitted a fatal caveat, the correction route should mark the affected decision and propagate changes. Tool assistance does not create a privileged shield against revision.

Closing the loophole means the project does more than say humans decide. It designs workflows so that humans actually carry the reason, the uncertainty, the correction, and the cost.

28.8.6 — Public Language of Responsibility

Every controlled scientific update should identify the accountable human role or roles, reviewed object, status, scope, uncertainty, conflicts, and current version. Signatures record responsibility and relevant expertise, not prestige (Council for International Organizations of Medical Sciences, 2016; Tabassi, 2023).

When software or artificial intelligence assisted, record its bounded task and version and what humans verified or rejected; authority remains with the reviewers (Tabassi, 2023; Autio et al., 2024; World Health Organization, 2025; Feuerriegel et al., 2026).

Correction notices should state the practical action for readers: use the new version, do not use the old artifact, treat the evidence as narrowed, await review, or recognize that a component has retired. Where a formula or model informed the change, state the local calculation it supported rather than allowing a local diagnostic to become a global adjudicator.

Applied domains may retain independent local value after a scientific correction, but they do not inherit stronger scientific warrant. Older copies remain historical and should point to the current status where feasible.

Chapter 29 — The Open, Staged Research Program

A final science chapter turns formal precision, extensive sources, and careful protocols into an ordered program of discovery. The remaining problems identify the next theorem, measurement, comparison, and humane study that could move the field forward (Fenwick et al., 2020; Skivington et al., 2021). The useful unit of progress is therefore not another topic added to the agenda, but an open question whose resolution would change a later scientific decision.

The preceding Parts now form a research dependency map: define the target, qualify its estimators, identify endpoints independently of the tested residual, specify candidate dynamics, compare explanations, and report the resulting evidence clearly. The map also shows which tasks can proceed in parallel because they do not depend on one another, preventing one unresolved branch from unnecessarily freezing independent work elsewhere in the program.

The practical questions are concrete. Which theorem or counterexample comes first? Which instrument requires qualification? Which simulation gate must pass before human inference? Which study can begin at low burden, which clinical work must wait, and which failure closes the line? A methodologist, ethics board, funder, laboratory, clinician, participant advocate, and critic should all be able to locate their decision. Those roles are complementary rather than interchangeable: resources, clinical access, methodological review, and participant protection cannot substitute for one another's gate.

Stronger terminal claims depend on a model that preserves the hypothesis specified here, qualified representation and measurement, and endpoint and rival designs capable of discrimination within approved burden limits. Each dependency has a stop condition. Terminal work follows the lower-burden formal, measurement, state-classification, and recovery gates. A terminal proposal should therefore identify the earliest unresolved prerequisite on which its interpretation depends; until that prerequisite is resolved, the proposal remains downstream work rather than a reason to bypass the gate.

Ownership remains clear at the handoff. *The Law of Fairness* retains the formal root; *The Science of Fairness* owns protocols, evidence, replication, and scientific verdicts; applied titles may contribute contexts and questions; *The Transition of Fairness* owns sequencing only after its independent authority gates are met. No local success borrows warrant from another domain.

The roadmap proceeds from open theorems and identification through instrument qualification, synthetic recovery, low-burden studies, endpoint methods, multisite replication, and only then stronger terminal tests. Movement is triggered by passed dependencies, not elapsed time; each stage has an owner, output, release consequence, and stop condition. For example, a simulation that shows two routes are indistinguishable can redirect instrument development toward the observation on which their predictions actually diverge.

Each verdict directs the next action: support can justify replication, narrowing a smaller test, No adequate estimate renewed identification work, and Rival-favored further study of

the winning explanation. Defeat can close the strong branch while preserving failure memory. Institutional review should track null-result release, rival investment, correction latency, and stop decisions alongside favorable outputs. Recording the expected consequence of a decision makes it possible to check later whether research investment addressed the uncertainty that originally justified it.

The final chapter treats science as a public trust. It identifies what can be proved, measured, simulated, studied, released, restricted, corrected, redesigned, or abandoned. The volume closes with a program whose ambition is governed by care, independent criticism, evidence, and an executable ability to stop. That stopping capacity is not merely defensive; it preserves resources and credibility for questions that remain genuinely open. That shared structure gives contributors from different disciplines a practical way to work on separate questions while producing results that can be combined coherently.

29.1 — Scientific Case and Dependency Map

The scientific case is a dependency map, not an argument by accumulation. Formal coherence, representation, measurement, endpoint construction, recoverability, rival discrimination, replication, transport, and ethical feasibility are separate gates; a stronger sentence becomes available only when every dependency on its route has passed.

The final map should therefore show both connection and failure: which statement concerns the formal object, measurement relation, endpoint method, candidate mechanism, model comparison, or public verdict, and where that route can break. A good map leaves skeptics precise targets rather than a diffuse program.

Table 29.1. One-view component map for the open scientific program.

Component	Near-term task	Result that changes status
Formal hypothesis	Clarify bridge, origin, endpoint, and reserve assumptions	A proved bridge, a separation theorem, or a formal amendment request changes only the formal component.
Measurement stack	Qualify contexts of use for the candidate momentary-valence and interval-contribution estimators under disability, medication, culture, and missingness	Stable calibration or failure updates measurement status before any terminal inference.

Component	Near-term task	Result that changes status
Endpoint discipline	Prospective consciousness-state and stopping-rule validation	Endpoint independence survives or terminal claims pause.
Rival comparison	Synthetic and held-out model recovery across bridges, frailty, safe control, and observation shutdown	If rivals dominate, Law-specific language narrows or retires.
Replication and verdicts	Multisite and adversarial replication with public correction rules	Support, narrowing, inconclusive evidence, or defeat attach to components rather than the whole program.

29.1.1 — The Received Hypothesis

The received hypothesis is exact in its latent form and cautious in its public form. It concerns a qualifying unified conscious stream, a latent phenomenological ledger, and a terminal boundary condition. The final chapter must therefore restate the hypothesis without turning a formal target into an empirical finding.

The strong form asks whether terminal neutrality belongs to the structure of the stream rather than to a moral accounting system. Its precision is the reason it can be criticized. A vague tendency toward balance would be easier to defend, but it would also be less scientific.

The public program does not begin by observing the latent equality directly. It begins by asking whether measurable

consequence structures appear under independently defined endpoints, explicit observation models, and strong rival comparison. That difference remains the central claim-status firewall of the research program.

The near-term empirical object is not a human soul, a moral score, or a complete life narrative. It is a set of structured predictions about measured dynamics, such as preterminal variance behavior, endpoint windows, reserve-linked signatures, and model-discriminating patterns that can fail (Manski, 2003; Gelman et al., 2020).

The practical record should name this status at every public handoff. A slide, abstract, protocol, or interview may summarize the hypothesis, but it should not compress strong latent closure and public empirical testing into one sentence of certainty. The most dangerous error is not skepticism; it is the friendly shorthand that turns a precise target into an announced discovery.

The hypothesis creates connected research tasks: develop the valence representation, identify its origin, qualify endpoint methods, test candidate dynamics, and compare their explanatory performance.

The final program connects the exact hypothesis to a sequence of empirical comparisons. Future work should identify which components receive support, which require refinement, and which should be retired. That record makes cumulative progress visible.

29.1.2 — Why the Program Is Scientifically Interesting

The program now has a concrete empirical starting point. Experience-sampling studies identify mood-dependent activity choice; laboratory and smartphone studies quantify how expectations and reward outcomes shape reported happiness; cross-modal neural work identifies shared as well as modality-specific affective structure; and engineered biological circuits demonstrate robust integral feedback (Taquet et al., 2016; Rutledge et al., 2014; Chikazoe et al., 2014; Aoki et al., 2019). Lifespan and comparative studies add structured variation in emotional experience and fairness judgments (Carstensen et al., 2011; Blake et al., 2015).

Together, these findings motivate a connected research program. Experience has measurable temporal structure, organisms regulate over time, biological control can represent history, and future opportunity can shape behavior (Fung et al., 1999). A specified PTVC model can bring those components into a joint quantitative account and test the relationships it predicts.

The most informative candidate would derive several outcomes from one completion relation: future signed increments, response timing, action probabilities where an action-mediated route is proposed, and endpoint-support behavior. A reserve-coupled realization would also specify its resource variables and their relation to the proposed dynamics. Simulations would first establish model recovery through the full measurement and observation process. Parameters and measurement relations would then be trained without confirmatory endpoint outcomes and tested across new people, conditions, horizons, and

laboratories. A coherent pattern that transfers with fewer independently tuned relations could offer explanatory economy as well as predictive gain, evaluated for the complete model.

Cumulative history creates two useful comparisons. A cumulative variable may improve prediction beyond current mood and a limited recent-event model. When a flexible rival already receives the same full history, the potential gain is an efficient, transferable representation of that history. Giving both models the same observations and comparable tuning budgets makes these gains interpretable.

The combined evidence can be analyzed jointly, accounting for shared participants, instruments, analysis choices, biological pathways, and cultural ancestry. Cross-modal or cross-species relationships become especially informative when they survive prespecified transformations and improve predictions evaluated on new observations. This offers a practical way to turn recurring patterns into a demanding comparison of explanations.

The research also has benefits at intermediate stages. Better within-person affect measurement, accurate recognition of retained conscious capacities, humane longitudinal observation, and more discriminating dynamic models would each contribute useful knowledge. Researchers can begin with existing records, benign calibration, and fully disclosed benchmarks. The opportunity is substantial: connect established findings through precise predictions and give independent laboratories a clear, worthwhile program to investigate.

29.1.3 — The Most Testable Near-Term Claims

The most testable claims are the ones that do not require observing an entire life or administering a terminal verdict. Early work should focus on representation, sign, context of use, model recovery, and preterminal dynamics in bounded settings where the observation process can be audited (Manski, 2003; Gelman et al., 2020).

Section 29.4.1 develops two early examples: a benign horizon/scheduling-route task and a sleep/affective-reprocessing study. Both illustrate the near-term rule: choose bounded studies that can separate candidate structure from ordinary urgency, scarcity, memory, mood inertia, and regulation before clinical escalation.

A longitudinal ordinary-life calibration cohort can test whether active reports, passive signals, clinical context, and relational events can be fused without erasing disability, culture, medication, and missingness. It is a measurement study before it is a Law study.

Near-term studies place questions where an informative answer is attainable. Their protocols specify the consequences of each result, including channel revision or retirement, changes to sign assumptions, and the context of use supported by calibration.

A consciousness and endpoint method study can test whether state classification and endpoint windows can be defined prospectively without using ledger estimates. If this cannot be done, terminal inference should stop before it reaches vulnerable settings.

The first terminal dynamics pilot is therefore not first in the literal sequence. It becomes scientifically admissible only after representation, momentary-valence estimator qualification, endpoint independence, and model-recovery benchmarks have enough credibility to make the clinical burden justified.

The near-term portfolio should therefore privilege claims that can fail cleanly: representation stability, origin calibration, instrument recovery, endpoint classification, and rival separability. Success at those gates earns the next question; failure saves the program from an expensive and ambiguous test.

29.1.4 — The Strongest Rivals

The rival architecture is already developed in Parts VI and VII. For the final roadmap, the governing requirement is equal-rule generative comparison: adaptation, homeostasis, predictive and safe-control accounts, conditioned bridges, clinical and treatment processes, selection, population composition, and observation artifacts must share the relevant observation and endpoint uncertainties and must be allowed to win (Gelman et al., 2020).

Before a confirmatory protocol launches, publish the rival families, parameter freedom, shared observation model, and held-out win conditions. A rival omitted from the benchmark is not defeated, and a simpler rival that wins receives the scientific status earned by the evidence.

29.1.5 — What Would Change the Case

The case changes only when evidence attaches to a specified component. Representation evidence changes representation status. Endpoint evidence changes endpoint status. Model-

recovery evidence changes design readiness. Replication evidence changes transport confidence. No result should change more than its design can support.

A strong early result would not be a declaration that the Law is true. It might show that a physiological estimator has a stable context of use, that a horizon task separates scarcity from candidate scheduling pressure, or that a synthetic benchmark can distinguish reserve-coupled candidates from bridges under realistic missingness.

Qualification findings guide the timing of later terminal studies. Unresolved sign, instrument transfer, endpoint independence, or model recovery pauses the dependent analysis and identifies the work needed before it resumes.

A mixed result should produce a smaller map, not a confused headline. A mechanism might retire while the measurement program remains useful. A reference class might narrow while the formal problem remains open. A rival might dominate one signature while leaving another undecided (Manski, 2003; Gelman et al., 2020).

The change record should also preserve timing. A result that looked strong before a bridge failure may become historically important but no longer status-bearing. A result that survives a later amendment may need reanalysis rather than inheritance. Scientific memory should protect both the excitement of discovery and the humility of versioned correction.

Public wording must change whenever the component status changes. Words such as discovery, support, narrowing,

inconclusive, unusable, retired, rival-dominant, and defeated should not float above the evidence. They should be tied to a versioned claim, a population, an instrument, and a comparison set.

The case changes most honorably when it becomes easier to state. A result that forces the program to say less, more exactly, may strengthen science even if it weakens the public story.

What changes the case is not accumulated enthusiasm but an evidence-bearing dependency transition. A frozen component passes or fails a protocol-fixed test, a rival becomes more or less adequate, or a reference class narrows; the case changes only by that amount.

29.1.6 — Next Measurement Priorities

The next measurement priorities are a stable valence scale, an empirically anchored origin, longitudinal coverage, and a workable endpoint method. Reserve-based models add the task of identifying and calibrating their resource variables (Manski, 2003).

Extensions to children, nonhuman animals, artificial systems, additional cultural settings, sudden death, coma, dementia, and altered states each create specific bearer-classification and measurement questions. Targeted studies can establish the conditions for those extensions.

Figures, tables, equations, and pipelines organize the work and identify stopping conditions. Their conclusions depend on whether the assumptions survive testing.

Timelines map dependencies. Evidence may shorten a ten-year plan, measurement difficulties may slow it, ethics review may pause it, and a stronger rival may redirect it (Gelman et al., 2020).

Public summaries should pair proposed signatures with open questions about whole-life initial state, origin identification, scalarization, endpoint independence, and cross-species transport. This makes the research agenda accessible beyond the technical chapters.

Applied domains offer varied settings for new questions. City experiments, government observations, religious practices, Workgame dynamics, and ordinary-life reflections can reveal patterns worth studying. Researchers can translate those observations into defined variables, competing predictions, and prospective tests.

An explicit map of open questions keeps the program testable and guides future science.

29.2 — Open Theorems and Formal Verification

Some questions cannot be solved by collecting more data because the target, bridge, or entailment is not yet identified. Open theorems, counterexamples, and formal verification therefore sit on the critical path. They determine whether an attractive construction actually reaches the received claim or merely resembles it.

Open theorems are not ceremonial humility. They are obligations that prevent attractive simulations from being confused with evidence. A model can be implemented, fit, and visualized while still failing to connect to the received latent claim.

The final chapter therefore gives formal problems their own place. They are tasks for mathematicians, statisticians, philosophers of measurement, consciousness theorists, and modelers who are willing to sharpen or break the theory before public language becomes stronger.

Formal work also governs public diagrams. When a figure makes a product-process realization look identical to the received ledger, that equivalence requires a theorem; otherwise the caption identifies it as a candidate bridge. Open mathematical problems therefore have direct consequences for talks, figures, and public summaries.

These problems receive named expert lanes rather than remaining philosophical atmosphere. A theorem register identifies the competence required for review, the artifact that would count as progress, the counterexample that would close the route, and the

public wording that remains unavailable until the obligation is resolved.

The theorem register remains open after publication. Some tasks may be solved, some shown impossible, and some converted into formal amendment candidates. The public record must distinguish solved, open, impossible, superseded, and revised rather than treating every item as inspiration.

29.2.1 — Integral–Product Bridge

The integral-product bridge remains open. Section 7.8.1 states the formal mismatch between the received ledger and the reserve-coupled product. The next deliverable is a theorem or bounded approximation specifying when the product preserves the ledger's domain, increments, initial conditions, endpoint semantics, and relevant transformation class, with its filtration, state variables, reserve process, and observation relation explicit. A separation result should identify when it does not. Empirical use of route coordinates for the terminal phenomenal target should follow that relation or state explicitly which weaker property is being tested (Oberkampff & Trucano, 2002; Oreskes et al., 1994).

29.2.2 — Origin and Representation

A ledger requires more than signed feelings. It requires a representation in which differences, accumulation, and neutrality have scientific meaning. The origin problem asks which evidence and assumptions can identify the relevant neutral point (Krantz et al., 1971; Luce & Tukey, 1964).

The representation problem is not solved by better sensors alone. Even perfect physiological signals would not create a natural unit

of felt valence by themselves. The program must justify which transformations preserve the scientific claim and which transformations change it.

A frozen potential model makes origin calibration an explicit identification problem. Let y contain integrated readings on one provisional cardinal scale, τ their independently qualified exposures, and B the incidence matrix of the fixed state graph. A common rate-origin offset o gives $y = o\tau - B^T\varphi + \varepsilon$. Choose a full-row-rank matrix C spanning the cycle space and write $g = Cy$ and $d = C\tau$. Eliminating the potential gives $g = od + C\varepsilon$. In the noiseless model, o is identified exactly when $d \neq 0$; a tree supplies no cycle constraint.

For centered errors with a fixed positive-definite cycle covariance V and known d , the generalized least-squares estimate is

$$\hat{o} = \frac{d^T V^{-1} g}{d^T V^{-1} d}.$$

Its denominator measures information, so a nonzero but weak duration direction can still yield poor precision. Additional declared nuisance patterns must be removed before assessing rank and information. Noisy exposures, shared edges and uncertain state membership require joint inference. One fitted independent cycle calibrates an origin without testing that calibration; further independent constraints provide overidentification and should include held-out paths.

The cross-sign unit assumption can also be tested within a specified two-channel model. Let P and N be independently defined operational magnitude channels associated with positive

and negative experience; both may be active during a mixed state, and their interpretation remains to be qualified. Write the candidate edge contribution as $aP - bN - o\tau$ with $a, b > 0$. Division by a gives $CP = rCN + \kappa C\tau$, where $r = b/a$ and $\kappa = o/a$. The two parameters are identified only when $[CN, C\tau]$ has rank two; further independent cycles test the relation. Both exercises remain conditional on the proposed potential and observation maps. Agreement with independent sign and scale calibration would strengthen their phenomenal interpretation.

The open theorem here concerns admissible scales. If only ordinal comparisons are meaningful, then additive ledger claims weaken. If interval structure can be justified locally but not globally, then interpretations of the candidate interval contribution estimate and claims of terminal neutrality must narrow by context of use.

The neutral origin must also be distinguished from comfort, silence, absence of report, low arousal, or social compliance. A person who cannot speak, a person who is sedated, and a person who is culturally trained to minimize distress do not create the same origin by producing low observed signal.

This problem also shapes the first-study portfolio. A representation task should not ask participants to confirm a theory; it should expose where different reporting modes, bodily states, and temporal framings produce incompatible origins. The failed origin is not a participant failure. It is a scientific boundary that protects the person from being reduced to a bad scale.

Formal work should produce representation tests that ordinary studies can implement. These tests might compare invariance

under language, disability, medication, developmental state, and within-person calibration rather than assuming all streams inhabit one scalar scale.

If no suitable origin or representation is found, the program should not rescue the strong form by changing the ledger into a metaphor. It should record a representation failure and preserve whatever narrower affective-science results remain.

Progress here means reducing the admissible representation class without choosing it from the completed outcome. A valid result would either identify the origin and sign structure under independent calibration or show that the scalar branch is not presently adjudicable.

29.2.3 — Endpoint–Reserve Synchronization

The endpoint and the reserve variable must not define each other. The endpoint must be established through consciousness-related stopping rules that are not algebraically identical to the ledger estimate. Reserve may approach a limiting condition, but it cannot be used as a disguised ledger clock.

The open problem is synchronization without tautology. If the model needs reserve to decline toward zero near the independently defined endpoint, it must state why that relation should hold, how wide the mismatch can be, and what happens when reserve rebounds, jumps, or is externally supported.

Clinical reality makes this difficult. Ventilation, sedation, analgesia, delirium, infection, organ failure, device support, and sudden events can alter reserve proxies without cleanly

corresponding to consciousness termination. A model that ignores these cases will be fragile exactly where the claim matters most.

A useful theorem would characterize classes of reserve processes and endpoint rules that preserve non-tautological prediction. It would also identify cases where endpoint selection creates apparent closure even when no Law-specific dynamics exist.

Synchronization should be evaluated under counterexamples as well as favorable cases. Sudden death, temporary clinical rescue, palliative sedation, fluctuating delirium, and device-supported survival all test whether reserve and endpoint are being related by biology or by narrative convenience. A theorem that works only in smooth stories is not ready for terminal science.

The theorem should produce empirical design rules. Endpoint adjudication must be prospectively specified, reserve proxies versioned, and desynchronization between reserve and endpoint clocks reported rather than hidden as noise.

Terminal analyses proceed when synchronization studies establish the endpoint-reserve relation under the intended observation conditions. Where that relation remains unresolved, the program can continue studying preterminal regulation as its own specified target.

The theorem target should separate genuine state-coupled synchronization from retrospective alignment. Its value is practical: it tells protocol designers which timing relations can be tested and which apparent completions are mathematical artifacts.

29.2.4 — Closure Modes and Random Endpoints

Closure can mean several different mathematical things. It may mean almost-sure latent equality in an idealized process, equivalence-band concentration in measured estimators, convergence in probability, bounded terminal error, or pathwise compression under a shrinking reserve. These are not the same claim (Kallenberg, 2002).

Random endpoints complicate the problem. A stopping time can condition observed paths, select survivors, truncate data, or change the distribution of what is observed near the boundary. A closure theorem must separate genuine dynamics from the statistical effects of stopping.

A taxonomy of closure modes helps organize terminal results. Because several mechanisms can produce a measured equivalence window, the analysis specifies the observation model, endpoint rule, and rival matrix. Their joint predictions identify which comparisons can distinguish the candidate explanations.

The Brownian bridge and Doob-transform rivals belong inside the formal problem. They show how endpoint conditioning can create apparent movement toward a boundary. Any claimed closure mode must be compared with these conditioning mechanisms under the same observation constraints (Doob, 1957; Revuz & Yor, 1999).

Published examples should include cases in which closure modes agree and cases in which they diverge. They must show how equivalence-band neutrality can occur without exact latent

closure, how terminal concentration can arise through selection, and why pathwise closure is stronger than a favorable ensemble average. Distinction by example prevents future slogan drift.

A strong result would state which closure modes are implied by which candidate realizations and which observable signatures distinguish them. A weaker result may show that several modes remain observationally equivalent under humane sampling.

Report the closure mode supported by the analysis, the alternatives it distinguishes, and the additional information required to resolve the remaining modes.

A concrete classification separates a fixed deadline from an endpoint that waits for the state. In one declared clock, consider an absolutely continuous ledger and a noisy contribution rate,

$$dL_t = V_t dt, \quad dV_t = u_t dt + \sigma dW_t, \quad 0 \leq t < T,$$

where T is a deterministic positive finite deadline, $\sigma > 0$, initial states are square integrable, W is Brownian in the full information filtration and u includes all progressively measurable drift affecting V . Exact $L_{T-} = 0$ almost surely is impossible if $E \int_0^T u_t^2 dt < \infty$. To see why, write the final h -window remainder as $L_{T-} = m + C + Z$, where m is known at $T - h$, $C = \int_{T-h}^T (T - s) u_s ds$ and Z has conditional variance $\sigma^2 h^3/3$ and conditional mean zero. Exact closure requires $C - E[C | \mathcal{F}_{T-h}] = -Z$. Conditional variance and Cauchy–Schwarz then imply

$$E \int_{T-h}^T u_s^2 ds \geq \sigma^2 \quad \text{for every } 0 < h \leq T,$$

contradicting finite expected total quadratic effort as h decreases to zero. This effort is a mathematical control quantity, with no assumed conversion into metabolic energy. The restriction concerns the specified fixed-deadline innovation model; it does not impose those dynamics on every PTV route.

Exact completion is nevertheless constructible with ordinary stochastic feedback. Start at $L_0 = V_0 = 0$, put $\tau = T - t$, taper the diffusion to $\sigma_t = \sigma_0 \tau / T$ and choose

$$u_t = -\frac{5(L_t + \tau V_t)}{\tau^2}.$$

For $M = L + \tau V$, the equation becomes $dM = -5Mdt/\tau + (\sigma_0/T)\tau^2 dW$. Its solution gives $E[u_t^2] = (5\sigma_0^2/T^2)\tau[1 - (\tau/T)^5]$, hence $E \int_0^T u_t^2 dt = 25\sigma_0^2/14$. Finite absolute drift integral ensures a continuous endpoint extension; M tends to zero and τV tends to zero, so $L_{T-} = 0$ almost surely. The feedback gains remain singular even though expected quadratic effort is finite. This is a mathematical realization with vanishing relevant innovations, not evidence that an organism implements it.

A random hitting endpoint supplies a different ordinary construction. With $L_0 = l > 0$, $V_0 = 0$ and constant drift $u = -a < 0$ under persistent $\sigma > 0$, $L_t = l - at^2/2 + \sigma \int_0^t W_s ds$ eventually becomes negative almost surely. Its first zero is finite and continuity gives exact pre-endpoint closure, with bounded drift control. In a complete-stream account, l would represent a preceding prefix. Identifying that hit with the stream endpoint remains an independent scientific obligation. Together the constructions direct measurement toward relevant innovation

strength, correction dynamics and independently classified endpoint rules, rather than treating every convergent-looking trace as the same mechanism.

29.2.5 — Stochastic Reserve and Markov Closure

The reference reserve model is useful because it is simple, but biological reserve need not behave as a smooth deterministic clock. Acute events, treatment, fatigue, infection, sleep, and medication motivate investigating stochastic, multidimensional, or regime-switching reserve candidates, while measurement changes require a separate observation model (McEwen & Wingfield, 2003; Inouye et al., 2014). These sources support the biological and clinical motivation, not the validity of a particular latent-reserve model.

A stochastic reserve extension rebuilds the dynamics rather than quoting the finite-variation shortcut. If reserve has its own martingale component or shares shocks with the normalized process, the product calculation includes bracket terms and potentially different terminal conditions. Those terms are part of the candidate mechanism, not a technical footnote.

A smooth state-potential realization must accommodate the process's actual noise. In one common clock, suppose $dS_s = b(S_s)ds + G(S_s)dW_s$ and $dL_s = v(S_s)ds$. For a C^2 potential Φ , Itô's formula and uniqueness of the semimartingale decomposition show that the pathwise identity $dL = -d\Phi$ requires

$$G^\top \nabla \Phi = 0, \quad v + \mathcal{A}\Phi = 0,$$

along the supported process, $ds \times \text{Pr}$ almost everywhere, with the usual localization and integrability assumptions. The generator is $\mathcal{A}\Phi = b \cdot \nabla\Phi + 1/2 \text{tr}(GG^\top \nabla^2\Phi)$. Noise can coexist with exact accounting when it acts in directions that leave the potential unchanged. A nonconstant smooth potential is impossible on a connected open region where these supported identities extend pointwise and the diffusion directions span the whole state space.

Instantaneous rank deficiency is not sufficient protection. For sufficiently smooth noise fields and potential, successive commutators of noise directions must also annihilate Φ ; if those directions together span the region, Φ is constant there. Drift directions cannot be added to this test without an additional argument. This identifies a precise research target: a coordinate responsive to the signed contribution while protected against the relevant fluctuations. A coarse noisy assay does not establish the noise geometry of the complete biological state, and jump or hybrid models require their own calculation.

Markov closure is also open. A system may be Markovian in an expanded hidden state while appearing non-Markovian in measured variables. The program needs to know which state variables must be included for closure-like signatures to be generated and distinguished from rivals.

A formal result here would identify minimal state conditions: reserve dimensions, normalized affective process, observation intensity, endpoint hazard, treatment state, and latent confounders. It would also specify when the process class becomes too flexible to be discriminating.

The extension should be constrained before data are seen. If stochastic reserve is added only after a negative result, the revision risks becoming a rescue move. If it is specified before testing, with parameter ranges and predicted failures, it can improve the model while preserving empirical content. Timing is part of the theorem's public credibility.

This matters for model recovery. If stochastic reserve can mimic nearly any observed trajectory, it becomes a language of accommodation rather than a source of risky prediction. The open program should prefer parsimonious, constrained generators that can lose.

The best outcome may be a family of nested realizations. The finite-variation reference remains the clean baseline, stochastic reserve becomes a formally labeled extension, and failure of both under fair comparison becomes evidence against that realization rather than an invitation to unbounded complexity.

The Markov program earns a place only if it yields distinctive, recoverable consequences under realistic observation. A convenient state augmentation that reproduces the endpoint but erases falsifiability is not a solution; it is a relabeling.

29.2.6 — Formal Amendment Candidates

Some future work may discover that the received formal statement needs repair. That possibility remains visible, but no repair enters the active scientific record before formal acceptance. A formal amendment candidate is a proposal under governance, not an invisible rewrite.

Amendment candidates might concern the representation of valence, the product realization, reserve dimensionality, endpoint clocks, scalarization, or whole-life initial state. Each candidate should state the old claim, the proposed new claim, the reason for change, and the predictions it improves or sacrifices.

A progressive amendment solves an independently identified problem and creates new risky predictions. A degenerating rescue move appears after failure only to preserve a favored conclusion. The program must keep those categories distinct before results arrive.

Historical results remain attached to the versions that generated them. Reanalysis under a new model is not a silent correction of the old record. It is a new analysis with its own version, assumptions, and public status (Crossref, 2025). A cross-version table should distinguish reproduced findings, newly derived findings, and findings rendered incomparable by the amendment.

Amendment candidates should also include a public-cost statement. A repaired theory may be more accurate but less simple, less universal, or less connected to earlier figures. The audience deserves to know when an elegant public explanation has been sacrificed for scientific truth, because that sacrifice is one of the signs that correction is real. The cost statement should also identify which prior empirical results remain valid under the amendment and which require reanalysis.

The formal-law book maintains the root statement. The science book supplies evidence, proposed amendments, and implications for formal change review. A concordance between the two books

can then show whether the scientific amendment changes the root claim, only its testable bridge, or both.

A formal amendment may someday make the program stronger or smaller. The final chapter leaves that route open because a theory that cannot be repaired honestly cannot be tested honestly. Any amendment must preserve the failed premise, counterexample, proposed repair, and new empirical obligations together, so formal change cannot masquerade as continuity. Fresh obligations should be prospectively testable, ensuring that formal repair increases empirical exposure rather than only logical flexibility.

Table 29.2. Open formal-problem register.

Problem	Minimum deliverable	Failure consequence
Integral-product bridge	Theorem, separation result, or formal amendment packet	Candidate realization loses authority to speak for the received ledger.
Origin and representation	Scale and origin conditions with sensitivity tests	Ledger and candidate interval-contribution estimator claims narrow to contexts where representation survives.
Endpoint-reserve synchronization	Non-tautological relation between endpoint clocks and reserve processes	Terminal inference pauses or becomes a smaller preterminal program.
Random endpoints	Convergence taxonomy: stopping and conditioning	Endpoint-conditioned rivals remain undefeated.

Problem	Minimum deliverable	Failure consequence
Stochastic reserve	General product dynamics and identifiability limits	Reserve family narrows or becomes non-discriminating.
Formal amendments	Versioned change request with costs and predictions	Old headline cannot silently continue after bridge failure.

29.3 — Identification and Measurement Problems

Identification and measurement establish which records answer the intended terminal question. The analysis specifies sign, origin, unit, exposure rule, stream boundary, endpoint, observation process, and the relation between measured and latent quantities (American Educational Research Association et al., 2014; Manski, 2003).

The section orders blockers that can invalidate a terminal claim even when the dataset is large and technically impressive. Its purpose is to make measurement more informative by preventing procedural sophistication from substituting for identification.

The core rule is simple: a model may estimate only what its design has identified. More channels, more sampling, and more computation do not repair an unidentified origin, an invalid sign, a context mismatch, or missingness that is part of the event being studied.

The measurement agenda identifies what each improved instrument can estimate, its qualified conditions, and the circumstances requiring no estimate. Those specifications make the instrument's contribution clear.

Identification also protects against technological overconfidence. Future sensors may improve, but a better sensor can still measure the wrong construct more precisely. The program therefore prefers an explicit construct failure to impressive but uninterpretable signal quality.

The identification register defines overreach. Using a candidate momentary-valence estimator outside its context of use, treating a missing terminal window as neutral, or collapsing noncompensable states into a scalar without support becomes a visible violation of the program's own standards.

29.3.1 — Sign and Neutral Origin

The sign problem asks whether a measured change points toward burden or relief. The neutral-origin problem asks where zero sits. Both are prior to any ledger estimate, because a running total cannot be meaningful if its contributions can flip meaning across context (American Educational Research Association et al., 2014; Little & Rubin, 2019).

Self-report anchors help but do not solve the problem alone. Reports are shaped by language, expectation, disability, culture, medication, fatigue, social demand, and memory. Physiological anchors help but remain nonspecific because arousal, effort, pain, fear, and salience can share channels.

The first study should therefore create bounded tasks in which sign can be cross-checked. A horizon and candidate scheduling route task, an affective reprocessing task, and ordinary-life micro-reports can test whether the same modeled direction survives across report, behavior, physiology, and time.

Negative controls and artifact challenges are essential. A proposed negative control requires an independently justified absence of the relevant causal relation; device looseness, motion, sensor drift, speech impairment, and clinical routines must also be tested for recording artifacts and possible genuine effects on

experience. A system that assigns a confident valence sign to every channel change has failed (Lipsitch et al., 2010).

Sign disagreements are expected rather than anomalous. A trembling hand, a quiet face, a reduced heart-rate signal, and a calm verbal report may point in different directions. The first response is not fusion but interrogation of context, because disagreement often marks the place where measurement is learning its limits.

A useful deliverable may be a hierarchy of anchors: within-person origins, context-specific neutrality bands, state-specific uncertainty, and decision rules for when a sign can be assigned.

If sign and origin remain unstable, ledger claims should stop while affective-measurement research continues where it has independent value. The result identifies the calibration problem that further work must solve before cumulative claims can responsibly resume.

Sign and neutrality are anchored outside the terminal record, with their uncertainty propagated into the final estimate. The measurement protocol identifies the conditions for a qualified estimate and returns no estimate when those conditions remain unmet.

29.3.2 — Candidate Momentary-Valence Instrument Context of Use

The candidate momentary-valence estimator is not a universal meter of inner life. It is a proposed estimator under a context of use. That context must specify population, state, language,

device, cadence, care regime, endpoint relation, clinical exclusion, and intended inference (Goldsack et al., 2020).

A context of use can be narrow at first. It might cover consenting adults in ordinary-life calibration, then a specific palliative population, then a carefully defined preterminal window. Each expansion requires evidence rather than rhetorical continuity.

Measurement qualification should test calibration transfer. A model trained on healthy adults should not silently travel to advanced cancer, dementia, autonomic neuropathy, chronic pain, traumatic stress, or medication regimes. Transport is an empirical question that targeted validation can answer.

The context must also specify abstention. When channels decouple, report is unavailable, medication changes, or device quality fails, the candidate instrument should return uncertainty or no estimate rather than a confident number. No estimate is a valid scientific output (Lakens, 2017; American Educational Research Association et al., 2014).

Context of use is written in the language of future exclusion. If the estimator has not been qualified and supported in delirium, chronic pain, autonomic neuropathy, or a given language group, those conditions remain outside scope rather than appearing as small-print uncertainty. A boundary written before use is protection; a boundary discovered after overreach is repair.

Model cards, device-version records, calibration logs, sensor-quality tags, and site-specific performance document the estimator's qualified uses.

Terminal-dynamics studies begin after the candidate instrument establishes its context of use. If qualification remains unmet, the next step is measurement refinement or retirement of that estimator.

29.3.3 — Rate, Additivity, and Scalarization

A ledger requires contributions over time. That means the program must identify whether lived experience can be represented as a rate, whether interval contributions are additive enough to support a candidate interval contribution estimate, and whether multiple dimensions can be responsibly scalarized.

The rate problem appears whenever sparse reports are stretched across unobserved intervals. A painful minute, a fearful hour, a sedated afternoon, and a socially quiet evening cannot be integrated by clock time alone unless the model states smoothness, bandwidth, and error assumptions.

Temporal additivity concerns partitions of the same realized stream. Memory, anticipation, trauma, and physiological recovery may change the valence rate, so two separate brief burdens need not equal one longer episode. Those differences do not by themselves refute temporal additivity. The candidate interval contribution estimate must be tested for consistent accumulation across partitions of the same history.

Scalarization may fail for noncompensable states. Pain relief, fear reduction, dignity, fatigue, grief, and confusion may not collapse into one axis without loss. The program should test whether vector-valued representations preserve more truth than a single

running score (American Educational Research Association et al., 2014; Little & Rubin, 2019).

A good measurement program should also compare what is lost when the representation is simplified. If a vector model retains grief, fatigue, pain, fear, and effort separately while a scalar model predicts one signature better, the choice should be tied to the estimand. Scientific elegance is not the same as human fidelity.

The first deliverable is a representation battery. It should compare scalar, vector, ordinal, and interval models under bounded tasks, ordinary-life sampling, and safe perturbations. The aim is to find where additivity holds, where it only approximates, and where it should be abandoned.

When rate, additivity, or scalarization tests favor a richer representation, record that finding and revise the scope of cumulative claims accordingly.

The decisive question is whether the chosen rate and scalarization preserve the contrasts the claim needs. When the proposed representation fails partition consistency or cannot support the required scalar comparisons, the analysis should widen the identified set or retire that representation branch.

29.3.4 — Candidate Remaining-Support Construct Validity

Reserve is scientifically useful only if it means more than frailty, illness severity, remaining time, or signal quality. The open program must identify what reserve is, how it differs from these neighboring constructs, and when it should be vector-valued rather than scalar (Lakens, 2017; American Educational Research Association et al., 2014).

Section 20.1.2 identifies the candidate dimensions behind a reserve construct. Construct validation must test whether a scalar or vector reserve adds discriminating validity beyond frailty, prognosis, remaining time, and signal quality for the declared estimand.

Construct validity requires discriminant evidence. A reserve proxy should not merely predict death, device dropout, or care intensity. It should help explain the specific dynamics the candidate realization says reserve should create, such as changing innovation amplitude or conditional relaxation.

Medication and clinical intervention are central. Analgesia, sedation, oxygen, dialysis, ventilation, nutrition, infection control, and palliative care can alter reserve proxies. A model that treats care as nuisance may distort both science and ethics (Little & Rubin, 2019; Manski, 2003).

Reserve should also be tested against interventions. If palliative care improves a reserve proxy, the model must know whether it has measured capacity, comfort, treatment, or observation quality. This matters because care should not be modeled as a contaminant that interferes with the theory. Care is part of the environment in which any humane science must operate.

A reserve-qualification protocol should compare candidate reserve dimensions, test shared sensors, estimate measurement error, report treatment sensitivity, and declare when reserve is too confounded for Law-specific interpretation.

If reserve construct validity fails, the product realization may retire while other parts of the program continue. That is exactly

why the final chapter separates components rather than letting a single elegant equation control the whole research agenda.

Remaining support becomes scientific only if it predicts constrained continuation under held-out conditions better than ordinary resources, prognosis, and opportunity measures. Without that incremental validity, the construct should be treated as an evocative label, not a mechanism variable.

29.3.5 — Unity and Endpoint Classification

Endpoint classification is not a clerical detail. The terminal claim depends on identifying when the relevant conscious stream has ended, not when a body, biography, legal status, monitor, or family narrative reaches a convenient boundary.

Unity classification must handle sleep, anesthesia, delirium, coma, dementia, dissociation, sedation, sudden death, return, and low-signal states. It should produce confidence classes and abstention windows, not binary certainty where the evidence is weak.

The endpoint method study must be independent of ledger estimates. It should use consciousness-state markers, clinical adjudication, prospectively specified windows, and blinded review channels that are not algebraically identical to the predicted neutrality measure.

False endpoints are especially costly. Prematurely declaring an endpoint can create tautology, erase return, or assign terminal meaning to a reversible state. Delayed endpoints can miss the relevant window and create post hoc selection (Manski, 2003; Lakens, 2017).

Endpoint validation should publish unresolved classes. Some states may remain beyond responsible classification, and those states should be named. The program becomes safer when it says that certain low-signal, ambiguous, or reversible conditions are not eligible for terminal inference under current methods, even if that makes the test less dramatic.

Its concrete output is a prospective endpoint protocol with versioned rules, open-channel windows, return handling, confidence classes, and independent adjudicators. That protocol should first be tested in nonterminal or simulated state transitions before entering palliative settings.

An unresolved endpoint method pauses terminal-neutrality inference while measurement and consciousness research continue. Care proceeds according to participants' needs.

Classification error here changes the unit and boundary of the claim itself. The program therefore needs calibrated abstention, alternative routing sets, and explicit false-ending controls before any residual estimate can carry complete-stream language.

29.3.6 — Observation and Missingness Models

Observation is part of the data-generating process. Reports arrive when a person can respond, devices record when they are worn, clinical visits occur under care regimes, and terminal data may disappear precisely when the event of interest becomes most important (Little & Rubin, 2019; Manski, 2003).

Missingness near endpoints can create false trajectories. A quiet final day may mean comfort, sedation, delirium, exhaustion,

family privacy, sensor loss, staff triage, or the ethical decision not to measure. The model must not impute meaning by default.

The open program should model observation intensity, dropout, device adherence, care transitions, family refusal, medication changes, and site routines jointly with the outcome. A smooth path drawn through gaps can become a false theorem if missingness is informative.

Ethically feasible randomized variation in harmless prompt timing, device redundancy, or backup reports can reveal observation bias without increasing burden. The goal is to study missingness while minimizing measurement itself (Lakens, 2017; American Educational Research Association et al., 2014).

Some data are absent because the right decision was not to ask, measure, enter the room, or turn a family scene into research. A humane model treats this ethical silence as design information. Each study needs a charter for planned and unplanned missingness, ethical nonmeasurement, terminal shutdown, data quarantine, and unusable estimates.

If missingness dominates the signature, that is a result: the humane observation window may be too narrow for the claim, requiring the program to stop or move to less vulnerable designs.

The observation model must allow for informative disappearance. The estimand should specify how relevant intercurrent events are handled, while estimation and sensitivity analyses address sedation, fatigue, withdrawal, device failure, death, and selective nonresponse. Missing observations must never be coded as neutral experience merely because they are absent.

Table 29.3. Identification and measurement bottlenecks.

Bottleneck	Early test	Decision consequence
Sign and neutral origin	Bounded anchor tasks and within-person sensitivity bounds	No signed ledger until direction and origin survive.
Candidate momentary-valence estimator context of use	External validation across state, site, device, and medication	Estimator remains local or abstains outside context.
Rate and additivity	Cadence and representation battery	Candidate interval-contribution estimator claims narrow if interval structure fails.
Reserve validity	Vector versus scalar reserve qualification	Product realization narrows or retires if reserve is confounded.
Endpoint classification	Independent consciousness-state protocol	Terminal inference pauses if endpoint independence fails.
Observation and missingness	Joint visit, dropout, device, and care-transition models	Unusable evidence remains visible rather than imputed into closure.

29.4 — First Study Families and Simulation Gates

The first empirical portfolio begins with studies that can change a decision at relatively low scientific and ethical cost. Representation, momentary-valence estimation, unity and endpoint classification, generative model recovery, sleep and reprocessing, and humane preterminal dynamics enter only after their prerequisite simulations and qualification gates pass.

The minimal portfolio includes bounded representation and sign work, a longitudinal calibration cohort, consciousness and endpoint-method studies, a generative benchmark challenge, and a humane preterminal protocol. Sleep and affective-reprocessing work can serve as a lower-burden bridge when its target, report process, and rival explanations remain explicit.

Stage Zero is a bounded existing-data challenge that can begin before new participant recruitment. Its first commission tests which accumulation statistics and predictive distinctions survive realistic measurement. The openESM resource introduced in Chapter 17 supplies a starting inventory, subject to each dataset's access, license, consent scope, variables, cadence, and participant overlap (Siepe et al., 2026). Freeze that inventory and distinguish exploratory datasets from protected confirmation records.

Before integration, qualify the observation scale, exposure weights, and missing-window bounds. Where cardinality, cross-sign comparability, or neutral origin is unresolved, report an observed affect-process statistic. Audit every centering, differencing, detrending, filtering, interpolation, and rescaling step. Pass known constant, slow-changing, and cumulative

signals through the identical pipeline: whole-record centering forces a zero total, and differencing changes a sum into an endpoint difference. Any resulting cancellation must be attributed to the transformation before being interpreted biologically.

Compare cumulative variance over prespecified window lengths using Sections 10.6.2 and 21.5.3, then evaluate Section 23.3.4's history comparison against strong current-state and memory models. Release synthetic generators for ordinary accumulation, genuine cancellation, multiscale memory, integral feedback, endpoint conditioning, origin drift, shared error, and informative observation loss. Give competitors identical available histories, observation procedures, and tuning resources. Freeze the primary score, practical-gain threshold, missingness sensitivity, and dependence-respecting uncertainty before confirmation; transport the comparison to an independent cohort without outcome-driven retuning.

The commission returns an access-and-measurement inventory, executable rival-recovery benchmark, observed-range accumulation estimates, protected predictive comparison, and a specification for the next informative measurement. Assign a time and resource ceiling, an owner for each output, and an independent reproduction task. A favorable result earns a focused replication or calibration investment through its prespecified effect and uncertainty. Failed recovery or inadequate separation identifies the design change needed before expansion. These outputs support the five study families below without asking a first nonterminal dataset to decide whole-stream closure.

29.4.1 — Study One: Representation and Sign

The first study asks whether signed contributions can be identified without terminal settings. It uses bounded tasks that vary horizon, candidate scheduling pressure, effort, relief, anticipation, and recovery while collecting reports, behavior, and physiological channels under prospectively specified models (Nosek et al., 2018; Cook et al., 2006).

The horizon and scheduling-route task belongs here. Participants face benign choices in which remaining opportunity and competing priorities vary without deliberately inducing suffering, distress, deprivation, or sleep disruption for Law-specific evidence. The task compares ordinary urgency, scarcity, and avoidance with the candidate route's separately specified predictions.

A sleep and affective reprocessing study can accompany or follow the task. It asks whether affective episodes show measurable next-day adjustment that cannot be reduced to report memory, mood inertia, or simple arousal. The study is valuable because it is relatively benign and temporally structured.

Study One is designed to resolve representation: sign, interval structure, rate behavior, and the first pass at scalar versus vector models. A success would not validate terminal neutrality. It would justify a more serious measurement qualification phase (Cranmer et al., 2020; Center for Open Science, 2025).

The study should also predefine what it will not ask. It should not use trauma induction, deprivation, humiliation, or high-stakes loss to create cleaner signals. The purpose is not to stretch people

until sign becomes visible. The purpose is to discover whether humane, bounded variation is enough to identify the representation the program needs.

A failure would be decisive for downstream design. If sign flips by context, if the same event produces incompatible scales, or if reports and physiology cannot be reconciled even locally, the program should stop ledger claims and continue as representation research.

Study One has a moderate, ethically manageable burden. It should minimize deception, avoid harm induction, preserve exit rights, record fatigue, and publish negative results because a failed sign study can prevent later misuse.

Study One earns completion when it reduces the representation set enough to make later measurements interpretable. If positive and negative channels cannot be linked without outcome-dependent choices, the portfolio stops at the representation gate.

29.4.2 — Study Two: Momentary-Valence Estimator Qualification

The second study builds the physiological estimator only after representation has enough local stability. It should not ask whether the Law is true. It should ask whether the candidate instrument can estimate relevant affective-state contributions within a defined context of use (Goldsack et al., 2020).

The core design is a longitudinal ordinary-life calibration cohort. Participants provide active reports, bounded event annotations, sleep and activity context, selected passive signals, medication and health covariates, and periodic recalibration. The cohort

should include variation in age, disability, chronic illness, language, and device access.

Study Two addresses transport across ordinary life. A laboratory estimator that fails during grief, fatigue, work stress, pain flare, caregiving, or medication change is not ready for clinical inference. Ordinary life is not easier than the lab; it is where measurement learns humility.

Use the deployment-matched validation scheme in §17.7: leave-person-out, leave-site-out, or leave-device-version-out tests for those transport claims, and temporally ordered splits for prospective within-person prediction. Study Two must state its intended generalization and prevent dependence or preprocessing leakage (Center for Open Science, 2025; Nosek et al., 2018).

The ordinary-life cohort should have a participant-facing explanation that is honest about uncertainty. Participants should know that the study is qualifying a research estimator, not reading their true ledger. Return information should emphasize patterns, confidence, and data-quality lessons rather than personal status, virtue, resilience, or deficit.

A success qualifies a bounded estimator and creates a versioned model card. A failure returns the program to channel design, person-specific baselines, vector representation, or abstention rules. Either outcome improves public safety by preventing unqualified body-reading.

Study Two imposes a substantial longitudinal burden because repeated measurement consumes attention and privacy. Cadence must therefore be justified by participant bandwidth and decision

value, with meaningful return information that never becomes a score, rank, or authority-laden interpretation.

Study Two should return a context-of-use decision for the candidate momentary-valence estimator, not a universal instrument claim. Qualification requires recovery, calibration, uncertainty, burden, and failure reporting strong enough that later studies know exactly what the estimator can and cannot support.

29.4.3 — Study Three: Unity and Endpoint Validation

The third study asks whether consciousness-state and endpoint methods can be qualified and evaluated independently of the ledger. This is a prerequisite for terminal inference because a terminal prediction cannot define its own endpoint and still count as an independent test.

The early version uses nonterminal analogues before clinical endpoints. Sleep transitions, anesthesia recovery, sedation changes, delirium windows, and independently evaluated disorders-of-consciousness protocols can test classification logic without treating dying participants as the first method-development population.

Study Three addresses unity and endpoint classification, including any prospectively declared independence branch. It estimates false-positive endings, false-negative continuity, ambiguous windows, return handling, and confidence classes, while keeping endpoint adjudicators blinded to ledger estimates and outcome expectations.

The design requires clinical and consciousness-science review. It should incorporate neurologists, palliative clinicians,

anesthesiologists, disability advocates, statisticians, ethicists, and family advisors before any terminal extension is considered (Council for International Organizations of Medical Sciences, 2016).

The endpoint method study should also test public-description risk. It should examine whether consent forms, summaries, and diagrams are understood as state-classification tools rather than as predictions about a person's final meaning. Misunderstanding by readers is a data point about artifact safety, not a communications footnote.

A success creates a prospective endpoint protocol with uncertainty bands and abstention categories. A failure stops terminal claims and redirects the program toward consciousness-state science. It should not be reinterpreted as a spiritual or poetic endpoint.

Study Three concentrates its burden in specialist clinical work. It requires clinical access, state labels, careful consent, and protection against public misunderstanding; that burden is justified only if it prevents greater harm in later terminal research.

Study Three is allowed to end in abstention. If unity or endpoint classes cannot be distinguished with acceptable error and humane burden, that finding blocks complete-stream adjudication while preserving narrower longitudinal work.

29.4.4 — Study Four: Generative Benchmark Challenge

The fourth study is not a human study at all. It is an open simulation and model-recovery challenge that asks whether the proposed analyses can recover true generators under known

conditions before they are trusted with real-world ambiguity (Center for Open Science, 2025; Nosek et al., 2018).

The benchmark library should include reserve-coupled candidates, OU processes, fixed and random Brownian bridges, Doob transforms, frailty, sedation, observation shutdown, adaptation, predictive or reinforcement-learning controllers, and any prospectively defined cross-species route or future-regime extension. Endpoint selection, subgroup mixtures, none-of-the-models generators, ordinary developmental and learning processes, and deliberately defective allocation or observation pipelines belong in the same challenge set. Before confirmatory testing, the model must specify its prediction, experimental unit, ordinary null, information set, and prespecified model-recovery, calibration, and false-positive criteria under realistic observation conditions.

For the delayed-treatment comparison, simulate both the primary strict-pre-endpoint and any separately declared endpoint-inclusive branch. Include streams that end before, at, or after scheduled treatment onset, with terminal contributions assigned only under the appropriate branch convention. Distinguish the complete-remainder restriction from stronger pre-treatment localization. Null simulations should include realistic finite-sample imbalance, ordinary later compensation with no early arm difference, and development or survival selection. Any selection or reweighting used to enforce closure must be checked for induced dependence between assignment and the pre-assignment prefix.

The implementation challenge includes route-cue leakage, route-to-contingency coding errors, one-shot versus learned choice, no-choice and observation-loss outcomes, allocation leakage, parental and batch structure, density and position effects, handling and scoring artifacts, censoring, and multiplicity. Permuted and placebo assignments must preserve the actual randomization scheme; placebo labels are generated independently of outcome values after ascertainment and do not determine treatment. Simulate cancellation, crossover, non-delivery, and other regime changes. When a prediction depends on actual regime realization, test its prespecified causal or partial-identification strategy rather than treating later adherence as randomized.

The benchmark should retain the terminal fork defined in Section 19.2, including general and the phenomenally silent idealization. Apply the declared selector to determine whether compatibility restrictions produce the claimed observable choice pattern. Compare first choices, repeated choices with learning, and continuation-only analogues separately. Candidate and rival implementations must expose their local or global information assumptions; hidden mapping information must not be supplied silently as an ordinary pre-choice covariate. The terminal case remains a theoretical and computational comparison, not an empirical lethal protocol.

Advance a confirmatory design when it meets prespecified recovery and error criteria for the candidates actually tested, with unresolved or open-set cases reported explicitly. Successful recovery establishes separability within the simulated conditions,

not the truth of the generator in nature. Empirical work then tests the preregistered prediction under independent scientific and welfare review.

Every analysis model should fit every generator. The report should include parameter bias, coverage, false equivalence, false defeat, open-set detection, calibration, convergence failures, computation time, and held-out scores under person, site, endpoint, and device-version splits.

Study Four asks whether the candidate can be separated from ordinary rivals. If a Law candidate cannot be distinguished from ordinary rivals under realistic observation noise, missingness, and endpoint uncertainty, then human confirmatory studies are premature no matter how good the prose sounds (Cook et al., 2006; Cranmer et al., 2020).

The benchmark includes deliberately defective pipelines to test detection of endpoint-label leakage, tuning on test outcomes, and weak-rival comparisons. Their known defects provide explicit checks of the analysis safeguards.

Successful model recovery establishes that the proposed design can distinguish important generators under the tested conditions. The public benchmark report records recovery and confusion patterns, and confirmatory recruitment begins after the design meets its prespecified separability requirements.

Study Four is computationally and adversarially demanding rather than participant-intensive. It is therefore a high-priority early investment because it can prevent unnecessary clinical work and give critics a formal role before data collection begins.

The generative challenge is successful only when the pipeline can recognize worlds in which PTVc is absent. Open-set failure, rival confusion, and observation-floor artifacts must be reported before the same pipeline is trusted on human data.

29.4.5 — Study Five: Humane Preterminal Dynamics

The fifth study is the first confirmatory comparative protocol that approaches terminal relevance without treating final moments as a spectacle. It should be attempted only after representation, momentary-valence estimator qualification, endpoint methods, and model recovery have passed their gates (Nosek et al., 2018; Cook et al., 2006).

The study may begin with a preterminal or palliative feasibility cohort rather than an end-of-mind inference. It should monitor comfort, observation burden, family burden, staff burden, device adherence, medication transitions, and clinical events as part of the scientific design.

Study Five asks whether humane preterminal observation reveals prospectively specified signatures beyond rivals. Candidate signals may include variance behavior, reserve-linked changes, endpoint-window concentration, and model-specific residuals, but only within the qualified context of use.

The design compares well-developed models of frailty decline, sedation, observation shutdown, terminal scripts, Brownian bridge conditioning, safe control, adaptation, and selection under shared evaluation rules (Cranmer et al., 2020; Center for Open Science, 2025).

Keep a family and staff burden dashboard within the research team, not in public. It supports stopping decisions, not publicity. Rising research load, excessive staff documentation, or strained family privacy must permit reduced cadence or stopping.

Success would support independent replication and expansion to the next justified reference class. An unsuccessful test would identify the formal, measurement, or protocol work needed before terminal research proceeds, while preserving the care-first design lessons.

Because this study carries the highest human burden, it comes last and remains limited to humane, low-burden, immediately stoppable observation. Its purpose is to test local dynamics and measurement transport against rivals, not to recruit terminal lives as proof.

29.4.6 — Portfolio Decision Points

The first studies form a portfolio with decision points, not a ladder that climbs automatically. A failed prerequisite may pause later work, redirect methods, or retire a mechanism before additional burden is imposed (Cook et al., 2006; Cranmer et al., 2020).

Representation and sign qualification govern whether ledger estimators are built; estimator transport governs the populations in which later work may proceed; and endpoint validation governs whether terminal claims are allowed at all. Failure at any gate is published rather than bypassed (Center for Open Science, 2025; Nosek et al., 2018).

The synthetic challenge determines whether the planned model comparison has demonstrated adequate recovery and separability

under its tested conditions. Failure blocks that confirmatory comparison; it does not automatically reclassify every human study as exploratory. Separately justified calibration, descriptive, or feasibility studies may retain their own prospectively specified questions and decision rules.

Decision points belong in protocols, funding agreements, and public status pages. Each converts evidence into permission, redesign, pause, or retirement, making the portfolio successful only when every outcome is genuinely available.

Table 29.4. First-study portfolio and decision value.

Study track	Primary uncertainty resolved	Further research questions
Horizon and queue task plus safe affective anchors	Sign, scale, horizon sensitivity, queue pressure, and representation limits	How do horizon-dependent dynamics relate to independently defined stream endpoints?
Sleep and affective reprocessing study	Temporal adjustment and next-day ledger movement under benign observation	Which sleep-linked changes generalize across populations, nights, and measurement channels?
Longitudinal ordinary-life calibration cohort	Candidate momentary-valence estimator context of use, calibration transfer, missingness, and cadence	How far does calibration transfer across states, populations, and instruments?

Study track	Primary uncertainty resolved	Further research questions
Consciousness and endpoint method study	Prospective endpoint independence and unity classification	How can qualified endpoint methods be linked to longitudinal affect measurements?
Generative benchmark challenge	Model recovery, open-set adequacy, and design separability	How do recoverable models perform on prospectively collected observations?
Humane preterminal dynamics pilot	Preterminal signatures under rival comparison and clinical burden monitoring	Which preterminal signatures replicate across qualified sites and care settings?

29.5 — Staged Roadmap and Value of Information

A staged roadmap is a dependency order, not a promise of progress by date. Each stage begins only after the objects it depends on are sufficiently identified, recoverable, safe, and transportable for the next inference. Value of information compares expected decision change with cost, burden, delay, opportunity cost, and the value of learning from a negative result (Claxton, 1999).

Early stages therefore concentrate on formal and measurement work. Fragile terminal claims wait until representation, candidate momentary-valence estimator qualification, endpoint architecture, model recovery, and humane feasibility have earned enough support to justify greater burden.

The plan also names opportunity cost. Funding, clinical time, participant attention, privacy, statistical labor, adversarial review, and public trust are finite. A fair research program spends them where the result can actually change status.

The roadmap also needs public patience. A program may appear slow precisely because it refuses to spend clinical vulnerability before formal and measurement gates pass. The plan explains that restraint as design rather than apologizing for it.

Annual review tests whether the roadmap's assumptions still hold. A ten-year sequence is a dependency hypothesis, not a destiny map. Each review asks whether the next stage retains decision value, whether a lower-burden alternative has emerged, and whether failure elsewhere has changed the order.

The resource plan keeps later funding responsive to earlier results. Resources can move to measurement, palliative care, comparison models, participant support, or public correction when those uses offer greater research or care value.

29.5.1 — Stage One: Formal and Measurement Lock

The first two years should prioritize formal proof obligations, representation batteries, candidate momentary-valence estimator context-of-use work, synthetic fixtures, and governance of code and data. This foundational stage creates the instruments and comparisons on which later discoveries depend (Claxton, 1999; Fenwick et al., 2020).

Formal teams should work on the integral-product bridge, origin and representation, endpoint-reserve synchronization, random endpoints, stochastic reserve, and formal amendment candidates. Their outputs should be versioned documents, not oral consensus.

Measurement teams should develop the sign and neutral-origin battery, ordinary-life calibration plan, channel taxonomy, missingness charter, and abstention rules. These are prerequisites for any claim that a measured ledger contribution means what it says.

The first open simulation challenge should also begin here. A synthetic benchmark can expose nonseparability before human recruitment. It should include critic-designed rivals from the start rather than after favorable results appear (Skivington et al., 2021; Center for Open Science, 2025).

The year-one artifact list should be concrete: versioned formal notes, a representation test battery, synthetic generators,

containerized analysis environments, a first data dictionary, and a public status vocabulary. These artifacts make the program real without creating the illusion that empirical validation has already begun.

The main gate at the end of year two is readiness for carefully bounded human calibration and state-method work. Formal and measurement checks identify the preparations needed for that next stage.

The resource logic is straightforward: early technical work is cheaper than late clinical error. The program should spend computation, mathematics, and critique before it spends terminal vulnerability.

Stage One should end with a locked formal and measurement packet or with a documented reason that no such packet is available. It is the cheapest place to discover that a later human study would be uninterpretable.

29.5.2 — Stage Two: State and Endpoint Validation

Years two through four should build the consciousness-state and endpoint infrastructure. The goal is independent classification of state and endpoint candidates. The program should test sleep, anesthesia, sedation, delirium, and recovery cases before any terminal endpoint is treated as a scientific boundary (Fenwick et al., 2020; Skivington et al., 2021).

The advisory structure becomes broader here. Consciousness scientists, clinicians, palliative teams, patient advocates, disability experts, statisticians, and ethicists should review state definitions, confidence classes, and abstention conditions.

Apply the purpose separation defined in §26.5.6 to operational records, research data, derived features, model-training corpora, and public summaries. In Stage Two, endpoint reanalysis under new versions must remain visible.

The endpoint study should produce false-ending rates, false-continuity rates, return handling, low-signal rules, and prospectively specified windows. Those outputs become inputs to preterminal protocol design only after independent review (Center for Open Science, 2025; Claxton, 1999).

This stage should also rehearse disagreement. Clinicians, consciousness researchers, families, disability experts, and statisticians may classify edge states differently. Premature consensus would erase the uncertainty the study needs to understand; the task is to learn where agreement is reliable, where dissent must remain visible, and where abstention is the only responsible category.

The gate at the end of this stage is endpoint readiness. If endpoint classification remains too uncertain, terminal inference stops while ordinary-life and nonterminal measurement work may continue.

Stage Two requires sustained clinical and institutional coordination. That cost is justified only when endpoint work prevents more serious downstream misuse by refusing to let the theory select its own boundary.

State and endpoint validation belong after measurement because misclassified bearers and endings cannot be repaired by a larger

sample. This stage earns progression only when uncertainty is narrow enough to support the next protocol's language.

29.5.3 — Stage Three: First Protocol and Replication

Years three through five overlap with endpoint validation because the first humane protocols can begin in ordinary-life and nonterminal settings. The aim is to prospectively specify designs that can produce interpretable evidence rather than impressive data volume (Skivington et al., 2021; Center for Open Science, 2025).

The first confirmatory protocol should define estimands, observation models, rival matrices, missingness handling, model-recovery prerequisites, endpoint rules, analysis versions, and public language before data collection. Registered Reports are attractive because methods receive review before outcomes and publication protection does not depend on direction.

Replication should begin modestly. A direct replication of representation and momentary-valence estimator qualification may matter more than a dramatic clinical pilot. A failed early replication can save the program from building on a fragile first result.

Independent analysis teams should receive staged access. One team can validate preprocessing, another can fit prospectively specified models, and a third can run adversarial checks. The separation reduces self-confirming pipelines without treating ordinary collaboration as suspect (Claxton, 1999; Fenwick et al., 2020).

The first confirmatory protocol should secure publication rights for null, adverse, mixed, and No adequate estimate outcomes. If the design is worth running, a clean negative result is worth preserving. The system must not make positive findings public while keeping failures as private lessons for insiders.

The gate at this stage is replication quality. A first protocol that works only under one analyst, one device version, one site, or one subgroup does not justify stronger public claims. It becomes a local finding or a method-development signal.

Stage Three's principal cost is coordination. The program should fund neutral analysis, critic participation, and negative-result publication before larger recruitment because scientific trust is infrastructure, not an afterthought.

The first protocol and its replication should be modest enough to fail without consuming the program. Its purpose is to test recoverability and transport under independent implementation, not to deliver a public verdict on the strong form.

29.5.4 — Stage Four: Mechanism and Transport

Years five through seven should test mechanism and transport only after early gates have survived. This stage asks whether signatures travel across populations, sites, care regimes, cultures, diagnoses, device versions, and states without collapsing into ordinary rivals (Center for Open Science, 2025; Claxton, 1999).

Mechanism work should compare reserve coupling, interoception, candidate scheduling pressure, sleep repair, adaptation, predictive processing, safe control, and frailty under

shared observation models. It should retire mechanisms that lose rather than giving every mechanism a protective interpretation.

Transport work should be reference-class explicit. A result in consenting adults with stable devices does not travel to advanced illness, nonverbal states, childhood, nonhuman animals, or artificial systems by analogy. Each reference class needs measurement invariance and endpoint comparability.

Multisite cohorts can enter here if harmonization has passed. Site heterogeneity should be treated as a finding, not a nuisance. A negative site may defeat universality or reveal a boundary condition that makes the Law-specific claim smaller (Fenwick et al., 2020; Skivington et al., 2021).

Mechanism and transport work should not be evaluated only by average performance. It should ask who is harmed by error, which subgroup fails, which site reverses, and which clinical context changes meaning. The transport question is moral as well as statistical because unjustified travel of a model places interpretive risk on other people.

The gate is transportable component support. If mechanisms change by site, device, culture, or diagnosis, public language must narrow to the supported reference class. If rivals explain the transported result better, Law-specific scope contracts.

Stage Four pays for breadth. Expansion should occur only where another site, population, or measurement system can change a component status; more sites are not better when they merely multiply unmodeled heterogeneity.

Mechanism and transport work should begin only after a stable signal survives equal-budget rivals. At this stage, a route may be favored, retired, or split by context; none of those outcomes automatically changes the terminal target.

29.5.5 — Stage Five: Terminal and Comparative Work

Only after earlier stages survive should the program attempt more demanding terminal and comparative work. Even then, the question should remain component-specific. The aim is not a public declaration of the Law; it is a carefully bounded test under humane conditions (Claxton, 1999; Fenwick et al., 2020).

Terminal-dynamics studies should be comfort-first, opt-in, low-burden, and clinically subordinate. They require prospectively specified windows, independent endpoint adjudication, medication and care-transition models, family burden monitoring, and stop authority outside the hypothesis team.

Comparative work should be adversarial. Reserve-coupled candidates, bridges, frailty models, sedation models, observation shutdown, safe control, adaptation, and selection-conditioned generators should all face the same data and the same endpoint constraints.

The most informative result may be narrowing. A preterminal signature might hold in one diagnosis, one care regime, or one endpoint class while failing elsewhere. That is not a public disappointment; it is scientific resolution (Skivington et al., 2021; Center for Open Science, 2025).

Even at this stage, the program should prefer designs that can stop early for futility or burden. Sequential monitoring should consider

not only statistical evidence but also staff fatigue, participant distress, family privacy, and endpoint uncertainty. A mature terminal study has more exit doors than entrance rhetoric.

Year ten closes with theory-level synthesis, not a victory ceremony. The program asks whether evidence supports components, narrows reference classes, retires mechanisms, or defeats the strong form. The answer may be mixed.

Terminal and comparative work follows earlier qualification because it requires substantial resources, identification, and ethical review. It proceeds when the question has no less burdensome adequate design and the study preserves the quality of participants' care.

29.5.6 — Resources, Opportunity Cost, and Value of Information

The resource plan counts participant time, privacy exposure, family attention, clinician and analyst labor, computation, ethics work, communication risk, and the opportunity cost of studying this hypothesis instead of other urgent problems (Fenwick et al., 2020; Skivington et al., 2021).

Funding reserves support negative results, adversarial replication, stewardship, cybersecurity, accessibility, participant care, staff training, and public correction. These are scientific costs, not disposable overhead.

Expertise spans mathematical biology, affective and consciousness science, palliative medicine, longitudinal statistics, causal inference, computational psychiatry, reinforcement

learning, open science, governance, disability science, and communication.

Fund low-burden tests that can defeat prerequisites before burdensome studies that add ambiguity. Synthetic nonseparability may make clinical data collection unnecessary (Center for Open Science, 2025; Claxton, 1999).

Critics, palliative clinicians, data stewards, and participant advocates review resource decisions because they can see costs the hypothesis team may underestimate.

Public schedules may report released artifacts, persistent preregistrations, passed benchmarks, and narrowed components, but cannot promise validation by date or treat sunk resources as entitlement to a favorable result.

Opportunity cost is ethical cost. The program becomes smaller when reality demands it because no hypothesis is owed participants, clinicians, or public attention.

Value of information compares expected decision change with money, time, burden, privacy, and alternatives. The next study is the one most likely to close a consequential uncertainty responsibly, not the most visible one.

Table 29.5. Ten-year dependency and resource plan.

Stage	Primary deliverables	Gate before expansion
Years 1-2	Formal obligations, representation battery, candidate momentary-valence estimator context of use, synthetic fixtures	Formal and measurement lock.
Years 2-4	Unity and endpoint validation, state-classification protocol, data separation	Endpoint independence and abstention rules.
Years 3-5	First prospectively specified protocol, ordinary-life replication, independent analysis	Replication quality and model separability.
Years 5-7	Mechanism comparison, transport tests, multisite harmonization	Component support across reference classes.
Years 7-10	Humane terminal and comparative protocols, adversarial replication, synthesis	Status update, narrowing, suspension, or defeat.
Always	Failure logs, public correction, staff and participant burden audit	Stop when burden exceeds decision value.

29.6 — Release, Pause, Redesign, and Stop Conditions

Release, pause, redesign, and stop conditions are the executable form of the gates developed earlier in the book. Here the action rule is simple: a failed prerequisite blocks the dependent inference; a correctable failure returns the program to the affected component; repeated or indispensable failure can retire a route or the strong empirical program.

Formal failure pauses confirmatory escalation when the target, notation, bridge, or endpoint relation is unstable. Measurement failure suspends residual claims in contexts where sign, origin, calibration, missingness, or transport cannot support the intended estimand. Endpoint failure blocks complete-stream language when the stopping or adjudication rule cannot be prospectively justified independently of the tested ledger estimator. Separability failure blocks human work undertaken for the affected confirmatory model comparison when candidate and serious rivals cannot be distinguished under realistic observation. Ethics, privacy, security, or burden failure pauses or stops the affected study or release pathway when care, meaningful refusal, data protection, family or staff burden, or participant protection becomes unacceptable.

These gates are noncompensatory. Success at one gate cannot pay for failure at another. Public artifacts inherit the same ceiling: a diagram, model card, website, or donor summary must be narrowed or withdrawn when the underlying component loses status.

For each failure, record four things: the failed prerequisite, the dependent claims blocked, the knowledge that remains valid, and the condition for responsible reopening. Repeated rival victory or identification failure should trigger the predeclared escalation rule: shift effort toward the better rival, narrower component science, or retirement analysis rather than another protected iteration.

Table 29.6. Release gates, pause rules, and useful stop knowledge.

Gate	Pause or stop condition	Useful knowledge retained
Formal	Bridge, notation, origin, endpoint, or amendment route remains unstable	Which object cannot yet be tested.
Measurement	Sign, rate, calibration, candidate momentary-valence estimator context, or abstention fails	Which estimator or representation should retire or narrow.
Endpoint	Prospective endpoint independence cannot be validated	Which consciousness-state methods need repair.
Separability	Rivals are observationally indistinguishable under realistic data	Which signatures lack empirical content.
Ethics and burden	Unacceptable care or privacy risks, coercion, or disproportionate participant, family, or staff burden	Which questions should remain inaccessible.

Gate	Pause or stop condition	Useful knowledge retained
Repeated loss	Rivals win or bridges fail across independent attempts	Which Law-specific claims should pause, narrow, or be defeated.

Table 29.7. Runnable artifacts and release order.

Artifact	Earliest release condition	Reason for restraint
Synthetic generators and fixtures	After formal object and rival envelope are versioned	Prevents benchmarks from protecting only the favored model.
Study-registration, protocol, and analysis-plan templates	After measurement and endpoint contexts of use are declared	Prevents protocols from hiding unqualified estimands.
Code containers	After dependency and data dictionary are frozen for a release	Makes replication possible without silent software drift.
Model cards and candidate momentary-valence estimator summaries	After external validation and abstention rules exist	Prevents estimators from becoming public inner-life scores.
Clinical protocol materials	After ethics and burden gates pass independently	Protects participants, families, and staff before recruitment.
Public diagrams and captions	After hostile-quotation and crop audits	Prevents uncertainty from becoming a slogan.
Failure logs and correction pages	At first failed gate or retired component	Preserves negative knowledge and blocks repeated overclaim.

After a gate failure, keep the affected artifacts versioned, identify which public objects are withdrawn or superseded, preserve

technically reusable infrastructure, and state the condition for responsible reopening. This keeps correction prospective and prevents an obsolete analysis or status claim from regaining authority merely because its files remain available.

29.7 — What Every Verdict Teaches

Chapter 27 defines the verdict taxonomy in full; this roadmap summarizes the consequence of each class. A result changes only the component, reference class, instrument, endpoint, or route it actually reaches. Replicated component support can justify transport or a stronger next test; partial support narrows the public claim; inconclusive evidence names the remaining uncertainty; instrument or endpoint failure returns work to the failed prerequisite; and rival victory redirects effort to the better explanation. One gate-valid counterexample can reject a declared set-universal [SET-ALL] branch when its stated identification and inferential prerequisites hold. A demonstrated formal contradiction defeats the branch whose stated assumptions entail it. Neither result should be borrowed automatically across a distinct target modality such as a probability-one [P1] branch. If indispensable representation, endpoint, or identification requirements fail beyond independently justified repair, or no ethical discriminating design remains, the strong empirical program may be retired without implying that the latent target is false.

Two protections remain essential. First, no local support becomes whole-Law support, and no route failure becomes automatic target defeat. Second, historical findings remain attached to the versions that generated them so later repairs do not silently inherit evidential credit.

Table 29.8 provides the public-language summary. Every status update should name the object changed, reference class, evidence type, surviving uncertainty, and next decision.

Table 29.8. Verdicts and public language.

Verdict	What it teaches	Public wording discipline
Replicated component support	A specified dependency survived independent tests	Name the component, context, and limits.
Partial or reference-class support	A claim is smaller than hoped but more exact	State the population, instrument, endpoint, and transport question.
Inconclusive evidence	A valid analysis did not decide the question	Name the bottleneck and the next decision-relevant design.
Instrument or endpoint failure	The apparatus is not valid for the intended inference	Repair or retire the instrument before status language advances.
Mechanism retirement or rival victory	A rival or ordinary mechanism explains the signature better	Reduce Law-specific scope and credit the rival.
Target defeat or empirical-program retirement	The exact target is contradicted under its declared prerequisites, or the empirical program is retired after its distinct, testable content is exhausted	Correct public artifacts and preserve useful scientific knowledge.

29.8 — Final Posture: From Architecture to Testing

PTVC is now specified as a scientific hypothesis with a defined formal target and a staged set of testable component questions involving representation, measurement, mechanisms, rivals, endpoints, and verdict rules. That architecture defines a research program. It does not establish that the terminal regularity exists, and it does not by itself determine whether large-scale empirical investment is warranted.

The next question is narrower: which formal, measurement, simulation, or low-burden empirical task has enough decision value to justify the next stage? A study earns priority when it can materially change a component's status, separate serious alternatives, or show that a prerequisite cannot be met. Early work should therefore favor high-information tests that reduce uncertainty before more burdensome designs are considered.

If a terminal regularity exists, a staged program may eventually reveal its scope. If it does not, the same architecture should make that failure legible while preserving useful work on measurement, endpoints, missingness, model recovery, and humane research design.

The opportunity is therefore both ambitious and staged. Laboratories can enter through the component closest to their expertise. Psychometricians can strengthen the measurement bridge. Consciousness researchers can improve bearer and endpoint classification. Statisticians can develop identification, missingness, transport, and model-recovery methods.

Computational scientists can test candidate dynamics and strong rivals. Clinical researchers can determine which low-burden observations are feasible in relevant populations. Independent teams can replicate and extend the results.

A research program becomes especially compelling when its central question is large but its next experiments are concrete. The Law of Fairness now has that structure. The task ahead is to turn the architecture into datasets, simulations, instruments, preregistered comparisons, and independent results.

29.8.1 — What Can Be Tested Next

Near-term work should target dependencies that can change the program at relatively low cost: the integral-product bridge, neutral origin and representation, estimator qualification, endpoint qualification independent of the ledger, and generative model recovery. A reproducible predictive advantage, a validated measurement bridge, or a counterexample can each change the next decision.

Convergent results across independent component tests would justify stronger follow-up; failed recovery, unresolved identification, or inadequate separability would justify redesign, narrowing, or pause. The scientific case for each study should therefore be made at the component level, using expected information gain, burden, opportunity cost, and the availability of lower-cost alternatives, rather than by the importance of the headline hypothesis.

29.8.2 — A Program Built for Independent Investigation

Independent investigation is essential because useful work does not require accepting PTV C in advance. A laboratory can take one defined component, apply its own methods, compare serious alternatives under common rules, and report what the data show.

Different teams can attack different dependencies: measurement and transport, partial identification and missingness, generative benchmarks, or state and endpoint classification. Their results become cumulative only when the assumptions and scope of each component remain visible.

The book also provides unusually rich infrastructure for such collaboration: explicit definitions, study families, model libraries, rival classes, failure modes, reproducibility requirements, versioning rules, evidence states, and public reporting standards. Those materials lower the cost of beginning serious work because researchers need not invent the entire scientific architecture before asking their first question.

A mature research field is created when multiple groups can disagree productively while working on the same answerable problems. The Science of Fairness is intended to make that possible.

29.8.3 — Every Strong Result Moves the Program Forward

The program is structured so that scientific progress can occur at many levels. A stronger measurement relation can improve future studies. A better endpoint method can make previously inaccessible questions testable. A successful model-recovery benchmark can identify which comparisons are genuinely

discriminating. A replicated component can justify transport or a more demanding next protocol.

Positive evidence is especially valuable when it survives independent pressure. A prospectively specified result that transports across sites, remains stable under measurement and missingness sensitivity, and outperforms serious rivals can materially advance the relevant PTVC component. Replication then turns an interesting result into a foundation for the next stage.

The staged architecture allows confidence to grow cumulatively. Researchers do not need one enormous experiment to carry the entire hypothesis. A sequence of well-designed studies can progressively qualify the representation, measurement, endpoint, model, and consequence layers needed for stronger tests.

This creates a practical path from the present book to a mature empirical program. Each credible result can reduce uncertainty, improve the next design, and bring the central question within closer scientific reach.

29.8.4 — Humane Design Makes the Research Stronger

The scientific ambition of the project and its humane design reinforce one another. Low-burden measurement, participant choice, clinically appropriate observation, careful privacy practice, and independent stop authority improve both the ethics and the interpretability of the resulting data.

This is especially important for research involving illness, altered consciousness, or terminal settings. The goal is to discover the strongest evidence that can be obtained within ordinary duties of care. That constraint encourages better questions, better

instruments, less intrusive observation, more realistic missing-data models, and study designs capable of functioning in the environments they claim to describe.

Many foundational studies can occur far from terminal settings. Measurement qualification, simulation, formal verification, model recovery, longitudinal affect studies, sleep research, natural experiments, and prospective prediction can establish much of the infrastructure first. By the time more demanding clinical questions are approached, the program can arrive with better instruments, stronger models, and clearer reasons for collecting each observation.

Humane design is therefore part of the program's scientific strength. It also makes the resulting evidence easier to interpret because observation remains subordinate to care, participant choice, and clinically appropriate stopping.

29.8.5 — Research Value Beyond a Single Test

The program has value beyond any single experiment when its component questions produce reusable methods, discriminating predictions, or clearer failure boundaries. A research group can contribute by solving one difficult component well, including by showing that a proposed measurement, route, or distinction does not survive testing. The broader scientific value should be judged by what remains usable after results narrow or reject the founding hypothesis.

29.8.6 — The Next Step Is to Test It

Further progress now depends on implementation: formal proofs or counterexamples, measurement qualification, synthetic recovery, preregistered comparisons, and independent replication. Each task should begin only when its prerequisite gates are met and should use prospectively specified stop rules when additional work is no longer expected to change the scientific decision enough to justify its cost or burden. The next phase is therefore not a general call to validate the Law, but a sequence of component tests that can support, narrow, redirect, or retire parts of the program.

Table 29.9 converts that architecture into immediate handoffs. It identifies who can advance each component and the boundaries that keep the resulting work scientifically interpretable.

Table 29.9. Final scientific handoff.

Next action owner	What is handed off	What is not handed off
Formal-law route	Bridge proofs and representation or formal amendments	Silent revision of the Law in scientific prose.
Measurement science	Momentary-valence estimators: qualify, calibrate, define use and abstention	Authority to score persons or read inner life directly.
Consciousness and endpoint science	Unity classification; endpoint windows; false-ending checks	Endpoints defined by predicted neutrality.

Next action owner	What is handed off	What is not handed off
Model-recovery and rival teams	Synthetic challenges, rival generators, and open-set tests	A weak rival space that protects the favored model.
Palliative and clinical partners	Humane feasibility, comfort-first protocols, and stop authority	Sacrificing care for data.
Public-communication stewards	Versioned status, correction logs, artifact withdrawal	Slogans that turn uncertainty into certainty.
The wider series	Questions, contexts, and domains for scientific care	Scientific validation or proof; transition authorization.

Together, those contributions can turn the architecture in this volume into evidence. The immediate priorities are to test the representations, qualify the instruments, strengthen endpoint methods, build capable rivals, run recovery simulations, and replicate results. Future status should follow those outcomes: components may gain support; PTVC may narrow in scope or remain unresolved; the strong empirical program may be retired. The scale of the question should not decide the verdict.

Glossary

A

Abstention: A decision to withhold a positive or negative verdict while specified identification, measurement, endpoint, model, or data-quality requirements remain unresolved.

Action-mediated closure: A candidate route in which the consequences of available actions produce the proposed closure relation.

Active inference: A family of models in which an organism can reduce expected mismatch by updating expectations and by acting on or selecting its environment so preferred or viable states become more likely. See also: Predictive processing; Reinforcement-learning reward signal; Viability kernel.

Ad hoc patch: A change introduced after an adverse result mainly to preserve a preferred conclusion, such as moving the endpoint, recentering the origin, inventing an unseen interval, or excluding a bearer. A legitimate successor must have independent support, create new prospective tests, and leave the earlier result visible. See also: Correction and withdrawal; Endpoint independence.

Adjudicability: The property that a declared target can be validly evaluated under fixed bearer, representation, origin, domain, endpoint, observation, bridge, and inference rules. If the target cannot be defined, identified, or bounded, the result is No adequate estimate; if a required protocol or implementation gate failed, the evidence is Unusable. Neither outcome is support.

Adverse case: A gate-valid case for which the branch-matched inferential object supports the registered Target-adverse conclusion. Exclusion of zero by a deterministic identified set

guaranteed to contain the truth under maintained assumptions yields a deductive counterexample; exclusion by a confidence or posterior object yields only its coverage- or posterior-qualified conclusion. Favorable population averages cannot cancel the case.

Affective valence: The positive, negative, or neutral felt character of conscious experience. In this series, valence concerns how experience feels from within, not merely preference, behavior, reward, arousal, or verbal report.

AI sovereignty: Institutional authority assigned to an artificial intelligence system over a person's inner life, worth, standing, or eligibility, a use prohibited by the project. Research systems remain assistive tools under accountable human review. See also: Human-worth score prohibition; Testimony-like output; Latent cumulative object.

Almost-sure attainability: The existence of at least one admissible policy that reaches a declared target with probability one from a stated information state. It differs from closure under a fixed law, success under every policy, and limit-sure attainability, where probabilities can approach one without any single attaining policy. See also: Limit-sure attainability; Probability-one closure; Reachability.

Almost-sure closure: See Probability-one closure.

Alternative endpoint branches: Prospectively declared architectures including state coupling, first hitting, absorption, endpoint atoms, singular timing or support, continued experience beyond a protocol boundary, and endpoint-inclusive value. Each

specifies its endpoint, representation, bridge, and testing assumptions. See also: Endpoint atom; First-hitting and absorption routes; State-coupled timing.

Applied lens and formal Law: The applied lens asks bounded present questions about burden, options, care, and repair. The formal Law concerns an exact truth-level condition over one complete qualifying stream. See also: Care before proof and interpretation; Law of Fairness; Target-route distinction.

Arousal: The activation, wakefulness, or energetic mobilization dimension of experience. High or low arousal can accompany welcome, burdensome, mixed, or neutral states. See also: Felt valence; Observation channel and observation model.

Atom at zero: Positive probability mass located exactly at zero in a terminal-residual distribution. Its presence is distinct from concentration within a finite band: a small atom can coexist with a widely dispersed remainder, and strong near-zero concentration can occur without an atom. Probability-one closure requires all mass at zero under the declared law.

Attainability: The existence of at least one admissible route or policy capable of reaching the target from a specified state. Attainability alone does not show that the route will be selected, that it works under partial observation, or that every admissible path closes.

Attrition mechanism: The process by which participants leave, pause, become unreachable, lose access, enter treatment, withdraw permission, or die during longitudinal research, thereby changing who remains visible. Attrition is both a selection

problem and an ethical signal about burden, access, trust, and study design. See also: End of observation; Informative missingness; Longitudinal research trace.

Auxiliary assumption: An assumption needed to connect a theoretical claim to an observation, estimator, or test. A failed prediction may reflect the target hypothesis, an auxiliary assumption, or both, so confirmatory work must state these assumptions in advance.

B

Balance: Book 1's accessible term for exact intrastream terminal valence closure.

Baseline drift: A change over time in a person's use of a rating scale, neutral point, comparison standard, or reporting habit. Unmodeled baseline drift can create an appearance of convergence or closure without a corresponding change in latent phenomenal valence.

Bearer: The truth-level subject or unit to which qualifying felt experiences belong and within which the proposed accumulation is evaluated. Its identity is established through the declared stream and routing criteria. See also: Maximal qualifying conscious stream; Ownership; Stream identity.

Behavioral nonresponse: The absence of expected command following or another observable response under a particular assessment. Motor, sensory, language, medication, arousal, task, and environmental barriers can block expression, so nonresponse alone does not settle whether experience is present or whether a

conscious stream has ended. See also: Felt availability; Low-unity state; Unknown category.

Biological death: The end of biological life. It is not identical by definition to the truth-level endpoint of a conscious stream. The Law formulation used in this series separately includes a biological-death admissibility condition under which biological death may not occur before the declared closure condition has been satisfied for the relevant stream. A qualifying conscious stream may permanently end after satisfying closure while biological processes continue without consciousness, including under artificial life support; such continued nonconscious biological persistence does not extend the completed stream. This additional condition does not identify the truth endpoint, a route to closure, or any particular biological or behavioral marker. See also: Branch; Endpoint-inclusive branch; Truth-level endpoint.

Bipolar branch: A representation branch that keeps positive and negative phenomenal components separate so they may coexist rather than being forced into one momentary net value. It is not interchangeable with the primary additive branch, and results transfer between them only through a justified bridge on a common identified domain. See also: Generalized Terminal Neutrality; Primary additive branch; Scalarization.

Blinded bearer adjudication: A procedure for deciding among bearer or routing models while withholding, as far as possible, the downstream result each classification would help produce. Fixed criteria, independent reviewers, visible disagreement, and outcome separation reduce post hoc routing in favor of the theory. See also: Bearer; Routing; Unknown category.

Blinding: A design safeguard that withholds selected information from participants, investigators, coders, analysts, or adjudicators when that information could bias conduct or interpretation. The form of blinding must fit the study and its ethical limits.

Boundary hypothesis: A hypothesis about a limiting, terminal, or otherwise sharply delimited property of a process rather than a claim about its ordinary moment-to-moment path. PTVC is a candidate boundary hypothesis because it concerns a stream's terminal signed residual.

Branch: A prospectively declared version of a target, model, or test whose bearer, representation, origin, domain, endpoint architecture, target geometry, modality, and reference class are fixed. Transfer between branches uses an explicitly justified bridge.

Bridge: A stated relation, with assumptions, scope, uncertainty, and failure conditions, that licenses movement between distinct objects or levels, such as measured to latent or protocol endpoint to truth-level endpoint.

C

Calibration: The process of relating a measurement system to an independently justified reference so that its origin, scale, uncertainty, and interpretation are known.

Candidate boundary hypothesis: A testable proposal about a specified terminal boundary, evaluated through its representation, measurements, and predicted consequences. See also: Falsifiability; Law of Fairness; Lawhood.

Candidate mechanism: See Candidate route.

Candidate realization: A formula or model proposed as one route-level realization linking received PTVC objects to observable consequences. It carries the formula-status label CR. See also: Candidate route; Formula-status code; Route state.

Candidate route: A proposed mechanism, policy structure, dynamical construction, or mathematical path by which the terminal target might be satisfied. A route can receive support or fail independently of the target hypothesis.

Candidate sentience escalation: A protocol-governed increase in protection when a candidate system's organization, activity, behavior, or welfare signal exceeds the assumptions under which ordinary review was approved. It may trigger verification, monitoring, reduced exposure, independent review, pause, or retirement without declaring consciousness or personhood. See also: Precaution without personhood; Testimony-like output; Theory-derived indicator.

Candidate signature: A prospectively specified observable pattern predicted by a candidate route, such as horizon-sensitive choice, recovery after perturbation, or narrowing variance. It is a testable clue rather than a unique fingerprint, and it gains weight only through transport, adequate measurement, and serious rivals under common rules. See also: Negative control; Nonunique explanation; Predictive fit versus explanation.

Care-before-interpretation rule: The ethical sequence that addresses urgent care, relief, safety, consent, privacy, and dignity before theoretical interpretation.

Care before proof and interpretation: The governing sequence that responds to present pain, fear, danger, illness, or grief with reachable care and protection. Care remains evidence-based, proportionate, revisable, and independent of PTVC's scientific status. See also: Ethical floor; No harm for proof; Outcome independence.

Causal locus: The place in a system where a causal influence is proposed to operate, such as experience generation, action selection, routing, endpoint timing, observation, or reporting. Distinguishing loci helps prevent one process from being mistaken for another.

Channel qualification: The staged process by which a report, sensor, behavior, contextual record, or other observation channel earns a defined research role through technical verification, analytical validation, construct validation, burden review, and specified performance thresholds. See also: Calibration; Multichannel estimate; Observation channel and observation model.

Claim ceiling: The strongest inference supported by a channel, design, model, comparison, or public artifact under its identification, uncertainty, scope, and safeguards. See also: Detachable-excerpt ceiling; Public-artifact limit.

Closure: The satisfaction of a declared terminal target. On the primary additive PTVC branch, closure means that the finite positive and negative phenomenal masses within one qualifying stream are exactly equal; equivalently, its terminal signed residual is zero.

Coercive measurement: Inner-life measurement conducted where authority, dependency, crisis, or structural pressure makes disclosure or monitoring a practical condition of care, work, education, benefits, liberty, opportunity, or standing. A tool is not voluntary when refusal brings penalty, suspicion, or reduced access. See also: Consent, refusal, and withdrawal; Human-worth score prohibition; Participation without surveillance.

Comfort-first observation: A terminal-research order of authority in which routine care, symptom relief, privacy, family presence, and clinical judgment determine whether and when observation may occur. Data collection stops when it interferes with treatment, rest, touch, conversation, spiritual practice, or requested quiet. See also: Care before proof and interpretation; End of observation; Protocol endpoint.

Common-action requirement: Under partial observation, an action or policy must be selected from the information actually available. For probability-one closure, its probability of completion must equal one for almost every compatible latent state under the declared conditional state law. A robust guarantee over every observationally indistinguishable state is stronger and requires its own feasibility and selection premises. State-specific actions that cannot actually be selected do not satisfy either requirement.

Comparative consciousness evidence: Species-appropriate evidence that combines behavior, learning, neurobiology, ecology, affective organization, and motivational tradeoffs into a converging case about possible experience. Human resemblance may inform a bridge but is neither necessary nor sufficient, and

welfare protection need not wait for a complete theory. See also: Precaution without personhood; Theory-derived indicator; Unknown category.

ComparatorStatus: See Scientific verdict axes.

Comparison without forced equivalence: Comparing traditions or ideas through a precise shared question, function, image, or difference while preserving their distinct histories, doctrines, registers, and internal disagreements. See also: Controlled vocabulary transfer; Meaning as optional; Spiritual resonance is not scientific support.

Compensability: A route-specific question about whether a present formal discrepancy remains answerable through safe, realistically available paths before a declared horizon closes. See also: Future optionality; Noncompensable risk; Viability kernel.

Complete felt history: See Complete strict-pre-endpoint felt history.

Complete strict-pre-endpoint felt history: Every qualifying phenomenal contribution owned by one stream before, but not at, its independently defined truth-level endpoint. Completeness is relative to declared bearer, representation, ownership, and domain rules, not to a remembered biography, selected episode, observation window, or final survey. See also: Endpoint-inclusive branch; Maximal qualifying conscious stream; Truth-level endpoint.

Complete strict-pre-endpoint phenomenal domain: The entire qualifying field of conscious experience belonging to one stream before, but not including, its truth-level endpoint. The primary

PTVC target is defined over this complete domain, not merely over a sampled terminal window.

Completion potential: A candidate mathematical function whose decrease, boundary behavior, or terminal value characterizes a proposed route to closure.

Component support: A verdict that one declared component, such as a representation, instrument, endpoint bridge, candidate signature, or route, survived its applicable validity gates and serious rivals within a bounded reference class. Preliminary component support is an initial result; replicated component support requires independent confirmation under compatible rules. See also: Claim ceiling; Replication; Target-route distinction.

Conditional support: The set of values, states, actions, or outcomes that remain possible after conditioning on specified information. Claims about exact closure must distinguish support restrictions from high probability or average behavior.

Conscious stream: A temporally connected sequence or field of qualifying conscious experience treated as one unit under declared identity and routing criteria.

Consent, refusal, and withdrawal: A continuing, practical consent relationship in which information is understandable, alternatives are genuine, retaliation is absent, stopping remains possible, and data consequences are clear. Separate choices may govern participation, monitoring, linkage, reuse, model training, disclosure, continued contact, and deletion. See also: Data

minimization and deletion; Measurement without ownership; Participation without surveillance.

Constraint does not imply design: A distinction between structural relations that limit possible histories and hypotheses about how those relations originated. Each origin hypothesis requires its own explanatory model and evidence. See also: Constraint without a judge; Spiritual resonance is not scientific support.

Constraint without a judge: An impersonal structural or boundary relation governing admissible complete histories. PTVC proposes exact terminal zero in this structural sense while leaving the realizing mechanism open. See also: Candidate boundary hypothesis; Constraint does not imply design; No moral desert, debt, or payment.

Construct-estimand distinction: A construct is the experience-related concept a study intends to represent. An estimand specifies the exact population, time, endpoint, treatment, and analysis quantity it seeks to learn. A cohort mean, individual probability, and path-level endpoint therefore pose different questions. See also: Estimator; Observation channel and observation model; Reference class.

Construct validity: The degree to which a measure, manipulation, or model captures the intended latent construct rather than a neighboring one. For valence research, behavior, reward, arousal, language, and neural activity may inform measurement without being identical to felt valence.

Controlled vocabulary transfer: Use of a technical term across fields with its source meaning, local role, evidential basis, and specialist references made clear. See also: Claim ceiling; Comparison without forced equivalence; Testimony-like output.

Correction and withdrawal: The accountable process of detecting a problem, assessing it, containing harm, correcting the record, propagating the change, documenting it, reviewing controls, and withdrawing or retiring affected claims, comparisons, routes, tools, and public artifacts. Superseded versions remain visible enough to preserve the scientific record. See also: Ad hoc patch; Public-artifact limit.

Counterfactual reachability: Whether a target could be reached under one or more alternative admissible actions or policies from a specified state. Counterfactual reachability does not establish what actually occurred or what a partially informed agent could select.

Craving capture: The narrowing of attention and immediately reachable options when a learned cue, urge, withdrawal state, or expected relief becomes unusually salient. It can constrain lived choice and future optionality. See also: Future optionality; Lived choice; Reinforcement-learning reward signal.

D

Data as trace: A report, waveform, image, note, gesture, archive, or model output is a context-bound record produced through an observation process. Its interpretation uses the relevant measurement model and provenance. See also: Latent cumulative object; Observation channel and observation model; Provenance.

Data-generating process: The full process that produces the observed data, including latent states, dynamics, routing, selection, measurement, missingness, and reporting. A valid analysis must consider whether more than one process can generate the same observations.

Data minimization and deletion: The discipline of collecting and retaining only the least revealing data needed for a declared question, while governing removal or restriction across raw records, features, backups, shared copies, outputs, and trained-model influence. Deletion promises must accurately state their technical reach and residual limits. See also: Consent, refusal, and withdrawal; Provenance; Record-regime separation and no-dossier rule.

Decision rule: A prespecified mapping from evidence to an outcome such as support, adverse result, narrowing, abstention, or route retirement. Decision rules should be fixed before outcome inspection whenever confirmatory status is claimed.

Defeat condition: A prospectively specified result that counts against a target, route, measurement claim, or auxiliary assumption under stated validity and inference rules.

Desert-based fairness: The question of when praise, reward, blame, or sanction properly follows from what a person chose and did. Its legitimate use is narrow and evidence-based. See also: External justice; No moral desert, debt, or payment; Procedural fairness.

Detachable-excerpt ceiling: The evidential scope a headline, quotation, caption, slide, social post, translation, donor summary,

or AI explanation retains when encountered apart from its source. Accurate, concise wording identifies the result being communicated. See also: Claim ceiling; Portable safeguard; Public-artifact limit.

Dignity firewall: The project's grounding of dignity, rights, care, safety, treatment, privacy, and relief in ethical principles independent of PTVC's scientific status.

Dimensionality of valence: The question whether phenomenal valence can be represented on one signed dimension or requires multiple dimensions. The primary additive PTVC branch uses a signed representation, but that representation remains an empirical and conceptual commitment to be tested.

Direct identification: Identification obtained from assumptions and observations that determine the target without relying on a route merely because it fits. Direct identification is distinct from selecting one convenient model among several observationally equivalent alternatives.

Dream-as-felt-experience window: An interval of conscious experience during sleep that may belong to the felt history of a stream even when waking recall omits it. Studies relate that interval to sleep stage, reports, and the adopted continuity model. See also: Complete strict-pre-endpoint felt history; Dream-report chain; Reversibility and return rule.

Dream-report chain: The sequence connecting experience during sleep, awakening, the memory trace available afterward, and the report given to another person. Each transition can select,

reorder, or reshape what enters the record. See also: Data as trace; Informative missingness; Response shift.

E

Empirical band: See Finite-band compatibility and measured equivalence.

End of observation: The last time usable data are obtained because a channel fails, monitoring stops, communication is lost, privacy is requested, or another practical event occurs. It is an observation boundary that may precede the protocol endpoint and truth-level endpoint and creates missingness rather than terminal truth. See also: Protocol endpoint; Truth-level endpoint; Observation window.

Endpoint: The boundary at which a qualifying conscious stream ceases for purposes of the truth-level target. The endpoint must be defined by the stream model and cannot simply be replaced by biological death, loss of contact, device shutdown, or the last available report.

Endpoint atom: Positive probability mass assigned by a declared endpoint-time or marked-endpoint law to a particular time or point. It specifies endpoint timing or marking, distinct from probability mass at a terminal residual of zero. See also: Atom at zero; First-hitting and absorption routes; State-coupled timing.

Endpoint bridge: The independently validated relation connecting an observed or protocol endpoint, possibly an interval, to the truth-level endpoint required by the claim while carrying error, return, revision, confounding, and competing-event

uncertainty. See also: Protocol endpoint; Reversibility and return rule; Truth-level endpoint.

Endpoint conditioning: Selecting, observing, conditioning, or reweighting trajectories according to where they end. This can change the apparent earlier drift or dispersion and supplies an explicit statistical model for comparison. See also: Candidate signature; Endpoint independence; Random-endpoint obstruction.

Endpoint coupling: A candidate route in which endpoint timing or endpoint conditions depend on the evolving stream state. Tests compare the predicted relationship with alternative accounts of endpoint timing and pre-endpoint dynamics.

Endpoint firewall: See Endpoint independence.

Endpoint-inclusive branch: A prospectively declared branch that assigns a specified phenomenal contribution exactly at the endpoint. Its temporal domain includes a point excluded from the primary strict-pre-endpoint branch, so comparison between the branches requires the corresponding domain and representation bridge. See also: Complete strict-pre-endpoint felt history; Exact terminal closure; Truth-level endpoint.

Endpoint independence: Two noninterchangeable conditions must be distinguished. Prospective endpoint independence fixes the endpoint branch, evidence source, selection rule, and adjudication procedure without using a ledger estimate or terminal residual to choose the endpoint. Probabilistic path-endpoint independence states, under a specified reference law, that endpoint time is independent of the sigma-field generated by

the pre-endpoint ledger path; it is a premise of the independent-diffuse-endpoint theorem. The first condition does not imply the second. See also: Ad hoc patch; Protocol endpoint; Random-endpoint obstruction.

Endpoint marker: An observable event or signal used to estimate or bound a truth-level endpoint. A marker can be informative without being identical to the endpoint itself.

Endpoint uncertainty and ambiguity: Multiple defensible interpretations of endpoint identity, timing, or interval under the declared evidence and rules. Prospective adjudication represents that uncertainty and states when it yields No adequate estimate. See also: Endpoint bridge; No adequate estimate; Truth-level endpoint.

Equal-budget comparison: See Rival model and equal-budget comparison.

Equivalence band: A prespecified finite interval around a reference value within which effects are treated as practically equivalent for a stated purpose. An equivalence result around zero supports only the declared finite-band claim; it does not establish exact zero.

Error budget: An explicit accounting of uncertainty from sampling, instruments, calibration, sign, origin, timing, missingness, model choice, transport, and construct identification across the evidence chain. Separating these sources identifies which additional data or methodological changes could improve the result. See also: Calibration; Noise floor; Partial identification.

Estimand: The precisely defined quantity an analysis seeks to learn, including its population, unit, time domain, conditioning, and scale. The estimand is not the same as the estimator used to calculate it.

Estimator: A rule or procedure for using observed data to estimate an estimand. An estimator may be biased, uncertain, inconsistent, or nonidentifying even when its numerical output is close to zero.

Ethical burden too high: A typed outcome in which a scientifically coherent question is not ethically available because the required burden, privacy exposure, dependency, deception, bystander risk, or lasting effect is disproportionate. Consent, payment, technical sophistication, or promised decisiveness cannot cure the design; it must be redesigned or left unanswered. See also: No harm for proof; No adequate estimate; Safe perturbation.

Ethical floor: The nonnegotiable safeguards that keep present pain a claim on care, reject earned suffering and cross-stream payment, forbid harmful testing, protect consent and refusal, place relief before interpretation, and allow meaning to be declined. These protections hold under every scientific outcome. See also: Care before proof and interpretation; No harm for proof; Outcome independence.

Every-continuation closure: Every continuation in a prospectively declared nonempty admissible continuation class satisfies closure. The claim says nothing about paths outside that class and is not implied by probability-one closure or support

containment. If the class is empty or undefined, closure is unavailable rather than vacuously established. See also: Probability-one closure; Reachability; Support containment.

Evidence ladder: A map of evidential dependencies and branch-specific implications. Descriptive convergence, finite-band equivalence, an atom at zero, probability-one closure, and set-universal closure are not interchangeable; some require additional bridges and are not ordered by a single strength scale. Each result states its target and what further inference it licenses.

Exact closure: The truth-level condition that one qualifying stream's terminal signed residual is exactly zero. Exact closure is not interchangeable with near-zero measurement, mean-zero aggregation, an interval containing zero, or practical equivalence.

Exact terminal closure: See Exact closure.

Exact zero: See Exact closure.

Experienced exposure and unity gate: Experienced exposure is the declared measure of how qualifying phenomenal experience contributes across a stream and may differ from calendar time. The unity gate states when felt contribution is treated as active and owned by the stream. See also: Complete strict-pre-endpoint felt history; Maximal qualifying conscious stream; Ownership.

External justice: Rights, capabilities, opportunities, material conditions, duties, institutions, protection, and remedy grounded in the project's independent ethical and legal principles. See also: Outcome independence; Procedural fairness; Hidden burden and hidden relief.

F

Failed upstream gate: An unmet prerequisite, such as representation, origin, bearer, endpoint, observation, measured-to-latent bridge, or transport, that determines which downstream conclusions remain available. See also: Measured-to-latent bridge; No adequate estimate; Scientific verdict axes.

Falsifiability: A scientific claim's specification of observations or evidence objects that could count against it under prospectively declared validity requirements and decision rules. See also: Candidate boundary hypothesis; Gate-valid counterexample; Strong-form defeat.

Falsification: A result that conflicts with a prespecified claim under the study's stated assumptions and validity gates. Falsification must identify which target, route, or auxiliary claim is affected rather than overgeneralizing from one failed component.

Felt availability: The modest claim that something is occurring for a conscious stream even when the experience cannot be spoken, deliberately used, remembered later, or organized into a clear self-story. It separates possible experience from an observer's access to evidence about it. See also: Behavioral nonresponse; Low-unity state; Observation channel and observation model.

Felt valence: The evaluative character of qualifying conscious experience as lived from within. Under a declared scalar signed representation it is positive, negative, or neutral relative to an independently justified origin; a bipolar branch preserves co-

occurring positive and negative components. See also: Arousal; Bipolar branch; Neutral origin; Primary additive branch.

Finite-band compatibility and measured equivalence: Latent finite-band compatibility requires the complete nonempty latent compatibility object to lie wholly inside a prospectively calibrated latent acceptance set. Measured equivalence requires the complete nonempty measured compatibility set to lie wholly inside the appropriate bridge-defined measured region. Partial overlap or merely including zero is unresolved. See also: Exact terminal closure; Measured-to-latent bridge; Terminal residual compatibility object.

Finite signed phenomenal measure: A finite signed measure representing positive and negative phenomenal valence over a qualifying stream's complete strict-pre-endpoint domain. Finiteness permits well-defined positive and negative Jordan masses and a terminal signed residual.

Finite soft-guidance ceiling: Under the registered finite power-weighting route, any retained positive probability mass on actions with completion probability below one prevents probability-one completion. This is a probability-mass result; topological-support containment and set-universal closure require separate premises.

Finiteness: On the primary finite branch, each complete positive and negative phenomenal magnitude is finite. This is a per-stream condition; different streams may have different finite totals. See also: Positive and negative phenomenal magnitudes; Primary additive branch; Gross experience.

Firewall: An explicit boundary that prevents evidence or language in one domain from being used to justify an unsupported conclusion in another. The series uses scientific, ethical, legal, institutional, and spiritual firewalls.

First-hitting and absorption routes: A first-hitting route uses the first entry into a prospectively declared set. An absorption route additionally keeps the process within that set afterward. Relating either event to the truth-level endpoint uses a separately specified endpoint bridge. See also: Endpoint atom; Endpoint bridge; State-coupled timing.

Fission-and-fusion routing rule: Under the book's primary formal convention, genuine fission ends the parent stream and begins each qualifying child stream, while genuine fusion ends the prior streams and begins a new one. Earlier phenomenal balances are neither duplicated among descendants nor pooled into the fused stream. See also: Bearer; Routing; Stream identity.

Fitness, feeling, and moral worth: The distinction among reproductive propagation, the valenced quality of a bearer's experience, and the ethical standing owed to that being. Each is evaluated through its appropriate biological, experiential, or ethical framework. See also: Comparative consciousness evidence; External justice; Viability kernel.

Formal or latent phenomenal ledger: On the primary additive branch, the latent cumulative signed phenomenal-value process through time. See also: Latent cumulative object; No moral desert, debt, or payment; Terminal residual.

Formula-status code: The labels attached to formulas in the technical synopsis: R = received definition; CR = candidate realization; OE = operational estimator; DC = design criterion; and OC = open conjecture. They classify a formula's role, not its empirical support.

Future optionality: The meaningful future paths a person can realistically reach through available time, bodily capacity, material conditions, relationships, information, institutions, treatment, support, and action. These conditions can widen or narrow the available paths. See also: Compensability; Craving capture; Lived choice.

G

Gate-valid counterexample: An adverse case that passes the declared bearer, representation, origin, domain, endpoint, observation, bridge, and inference gates for the tested branch. Deductive exclusion requires a deterministic truth-containing object guaranteed under maintained assumptions to exclude the target. Confidence or posterior objects support only their registered coverage- or posterior-qualified adverse conclusions, not literal universal refutation. See also: Falsifiability; Set-universal closure; Strong-form defeat.

Gate-valid stream: A qualifying stream that satisfies the prespecified inclusion, endpoint, measurement, data-quality, and model-validity gates required for the declared test. Only gate-valid cases may carry the evidential status assigned by that test.

Generalized Terminal Neutrality: A broader family of terminal-zero conditions defined by a declared functional, measurable

output space, and distinguished zero when scalar addition would lose relevant structure. It is a separate branch and is not PTVC unless a justified bridge connects its zero condition to the additive positive-negative decomposition. See also: Bipolar branch; Primary additive branch; Scalarization.

Gross experience: On a declared additive scalar representation, the complete positive-plus-negative phenomenal variation before cancellation. Its value can differ across streams with the same terminal residual. See also: Outcome independence; Phenomenal magnitude; Terminal residual.

Gross negative valence: The total magnitude of negatively valenced phenomenal mass within one stream, evaluated separately from positive valence.

Gross positive valence: The total positively valenced phenomenal mass within one stream, evaluated separately from negative valence.

H

Hedonic adaptation: The changing emotional impact of a gain, loss, or continuing condition as novelty, attention, expectations, comparison, routine, and context change over time. Adjustment varies across events, people, domains, measures, and times. See also: Opponent process; Response shift.

Held-out transport: Testing whether a prospectively specified model or result travels to data, settings, populations, channels, periods, or endpoint regimes reserved from model development and tuning. See also: Measurement invariance; Model recovery; Reference class.

Hidden burden and hidden relief: The observation that visible advantage may coexist with distress and visible hardship with joy, competence, love, meaning, or peace. It motivates direct attention to experience alongside external conditions. See also: External justice; Measurement without ownership; Per-stream claim.

Hidden cumulative object: See Latent cumulative object.

Hidden-state model: A model in which the process of interest contains latent states that are inferred from observable signals. Such a model must distinguish state dynamics, observation error, routing, selection, and missingness.

Homeostasis and allostasis: Homeostasis regulates variables within workable ranges, while allostasis anticipates changing demands by adjusting operating levels, mobilizing resources, and redistributing physiological cost. These processes provide measurable architectures for studying regulation, recovery, and resource use. See also: Mean reversion; Opponent process; Viability kernel.

Human-worth score prohibition: No estimate of burden, mood, pain, resilience, contribution, regulation, apparent balance, or unity may measure or rank human worth, credibility, trustworthiness, holiness, competence, deservingness, or eligibility. The prohibition applies to labels, visuals, dashboards, proxy scores, and model outputs. See also: AI sovereignty; Coercive measurement; Record-regime separation and no-dossier rule.

I

Identification: The ability of the available data and stated assumptions to distinguish the target quantity or model from alternatives. More data cannot resolve nonidentification when competing explanations remain observationally equivalent under the design.

Identified set: The set of every target value compatible with the observed data, admissible measurement models, and declared assumptions. It is not a confidence interval: partial identification and sampling uncertainty are separate layers and may both need reporting. See also: Partial identification; Terminal residual compatibility object.

Inconclusive evidence: Evidence from a valid analysis that does not resolve the declared estimand within the prespecified uncertainty and sensitivity bounds. It is neither support nor Unusable evidence.

Independent neutral-origin calibration: A procedure that identifies the valence zero point using evidence and rules independent of the completed record's terminal residual.

Inference: Reasoning from observations and assumptions to conclusions about latent quantities, models, mechanisms, or populations. Every inference inherits the limits of its measurements, design, identification conditions, and auxiliary assumptions.

InferenceStatus: See Scientific verdict axes.

Informative missingness: Missing observations whose occurrence relates to burden, state, access, treatment, refusal, shutdown, or another factor relevant to the inference, so visible records may differ systematically from what disappeared. A missing report is not zero, neutrality, noncooperation, or absence of experience. See also: Attrition mechanism; Missingness; Observation channel and observation model.

Institutional administration limit: Courts, governments, employers, insurers, schools, platforms, clinicians, families, and religious bodies have no authority to possess or administer the latent cumulative object as a fact about worth, credibility, eligibility, fault, care priority, spiritual standing, or terminal status. Bounded research estimates cannot create that authority. See also: Human-worth score prohibition; Latent cumulative object; Record-regime separation and no-dossier rule.

Interoceptive prediction: The anticipation and interpretation of bodily signals such as heartbeat, breathing, pain, fatigue, hunger, nausea, temperature, and arousal through expectations, context, uncertainty, attention, and action. See also: Active inference; Predictive processing; Prediction error and valence.

Interval contribution estimate: An operational estimate $\widehat{\Delta L}_{i,[a,b]}$ of accumulated measured contribution over a declared interval, constructed from a versioned momentary estimator and exposure/time rule with propagated uncertainty. See also: Momentary valence estimator; Phenomenal ledger.

Intrastream: Evaluated within one conscious stream at a time. The term fixes the scope of the proposed equality and excludes

netting across people, households, families, populations, species, institutions, or ensembles, even when experiences are causally connected or socially interdependent. See also: Nontransferability; Per-stream claim; Routing.

Intrastream closure: Closure evaluated within one qualifying conscious stream. Positive valence in one stream cannot cancel negative valence in another, and no population total substitutes for a stream-level target.

Invariance under positive rescaling: The property that multiplying a valid signed valence representation by a positive constant leaves its sign structure and exact-zero closure status unchanged. This does not license arbitrary nonlinear transformations or shifts of the neutral origin.

J

Jordan decomposition: The representation of a finite signed measure as the difference between mutually singular positive and negative measures. Their total masses provide the positive and negative phenomenal quantities compared by the primary additive PTVC target.

Jordan masses: The total positive and total negative masses produced by the Jordan decomposition of a finite signed phenomenal measure. Exact closure requires equality of these two masses within the same qualifying stream.

L

Last conscious experience: The final qualifying experience before a truth-level stream endpoint, when such a final experience

is well-defined. PTVC does not require this last experience itself to feel neutral and does not equate it with the last observed report.

Latent cumulative object: The unobserved formal representation of signed qualifying felt experience accumulated within one stream across its declared domain. Reports and instruments constrain it through a measurement model. See also: Formal or latent phenomenal ledger; Measured-to-latent bridge; Terminal residual.

Latent valence: The unobserved phenomenal valence attributed to experience itself and investigated through models linking reports, behavior, physiology, and neural signals.

Law of Fairness: The series' public name for the proposed idea that each qualifying conscious stream may satisfy exact terminal valence closure, together with the wider inquiry built around it. In scientific use, it remains a candidate hypothesis rather than an established law of nature.

Lawhood: The scientific aspiration to discover a general structural condition on completed qualifying streams. Its evaluation uses identified objects, invariance, prediction beyond construction data, capable comparisons, replication, explicit scope, and prospective decision rules. See also: Candidate boundary hypothesis; Falsifiability; Claim ceiling.

Ledger: See Formal or latent phenomenal ledger.

Limit-sure attainability: The property that policies can make the probability of closure arbitrarily close to one, even though no single admissible policy need achieve probability-one closure. Within the same fixed environment and nonempty admissible

policy class, it is weaker than almost-sure attainability and does not establish closure under an arbitrary fixed law or every admissible continuation.

Lived choice: The capacity to form purposes, choose among meaningful options, revise, refuse, and act with others from within real bodily, social, historical, and material conditions. Reversibility and opportunities for review and repair can help guide decisions under uncertainty. See also: Craving capture; Future optionality; Target-route distinction.

Local return and terminal closure: The separation between movement toward a familiar range during an episode or observed interval and a proposed exact cumulative condition at the independently defined end of one complete qualifying stream. See also: Hedonic adaptation; Mean reversion; Pathwise closure.

Longitudinal ledger: A model-based representation of accumulated signed phenomenal valence across a stream.

Longitudinal research trace: A repeated sampled record created through changing instruments, schedules, contexts, purposes, and permissions. It can reveal trajectories and within-person change but remains a borrowed trace rather than a biography, complete phenomenal history, or ownership of the person. See also: Attrition mechanism; Consent, refusal, and withdrawal; Response shift.

Low-unity state: A state in which ordinary access, responsiveness, integration, or evidence of unified experience is reduced or uncertain, as in sleep, anesthesia, coma, delirium, or severe brain injury. It may be a pause, altered-state branch, or

unresolved interval rather than the end of a stream. See also: Behavioral nonresponse; Reversibility and return rule; Truth-level endpoint.

M

Maximal qualifying conscious stream: A qualifying stream extended as far as the adopted routing and identity criteria permit, with the complete resulting domain retained for analysis.

Mean reversion: A local statistical or control tendency for an extreme observation to move toward a typical range because of feedback, stable dynamics, regression, sampling noise, treatment, or changed context. See also: Homeostasis and allostasis; Local return and terminal closure; Nonunique explanation.

Mean-zero result: A finding that an expectation or sample average equals or approaches zero. It describes the aggregate, in which positive and negative individual residuals can cancel.

Meaning as optional: The freedom to make, receive, revise, postpone, refuse, or live without an encompassing religious, spiritual, philosophical, or secular interpretation of experience, grief, suffering, or the Law. Nonbelief, doubt, silence, and exit retain full dignity and may never become barriers to care or belonging. See also: Care before proof and interpretation; Comparison without forced equivalence; Spiritual resonance is not scientific support.

Measured-to-latent bridge: The independently justified relation that allows a result on a measured estimand to constrain the declared latent phenomenal target, with scope, uncertainty, and failure conditions carried forward. It is separate from the

observation model, endpoint bridge, estimator, and route; without it, evidence remains on the measured scale. See also: Endpoint bridge; Estimator; Latent cumulative object.

Measurement invariance: The tested stability of a measurement relation across the people, groups, languages, devices, sites, times, or contexts for which a comparison is proposed. Its results identify supported comparisons and opportunities for further calibration. See also: Calibration; Held-out transport; Response shift.

Measurement model: A formal account of how latent valence or related states generate reports, behavior, physiology, neural signals, or other observations. It includes scale, origin, error, drift, and possible response-process distortions.

Measurement reactivity: Change in experience, behavior, disclosure, or the observation process caused by measurement itself, including repeated prompts, wearables, interviews, monitoring, and study expectations. Reactivity must be minimized, estimated, or carried into interpretation because it can alter both the target interval and the evidence collected. See also: Informative missingness; Observation channel and observation model; Consent, refusal, and withdrawal.

Measurement without ownership: Experience may be described, compared, and modeled without transferring ownership to a clinician, researcher, institution, record, or algorithm. Measurement creates responsibilities for purpose limitation, consent, privacy, correction, uncertainty, and stopping, not authority over a person's inner truth, worth, eligibility, or care.

See also: Observation channel and observation model; Consent, refusal, and withdrawal; Human-worth score prohibition.

Mechanism retirement: See Route retirement.

Metrology: The discipline of measurement, including units, calibration, traceability, uncertainty, comparability, and error. It provides the tools for making PTVC's quantitative predictions empirically interpretable.

Missingness: The process by which observations are absent, censored, truncated, or unavailable. Missingness near endpoints may be informative, and treating it as random without support can bias terminal conclusions.

Modal reachability ladder: A family of nonequivalent, scope-dependent claims, including existence of a closing continuation, positive-probability completion, fixed-law probability-one closure, limit-sure attainability, almost-sure attainability, support containment, and closure of every admissible continuation. These claims do not form one unconditional total order. Within the same fixed environment and nonempty admissible policy class, almost-sure attainability implies limit-sure attainability; a support-containment claim additionally requires a declared topology. Every implication must preserve the carrier, completion event, policy scope, and relevant assumptions. See also: Almost-sure attainability; Every-continuation closure; Support containment.

Model ambiguity: Model-ambiguous means two or more adequate declared models remain practically indistinguishable under the common rules. Open-set inadequate means no declared model is adequate. No adequate estimate means that the target or

estimand is undefined or cannot be identified or bounded under the available data and maintained assumptions, including because ethically permissible evidence is insufficient. A failed protocol or implementation gate yields Unusable evidence. See also: No adequate estimate; Rival model and equal-budget comparison; Scientific verdict axes.

Model-ambiguous: See Scientific verdict axes.

Model governance: Accountable rules for choosing a model's target, data, loss function, constraints, interventions, permitted uses, review process, correction route, and verdict. See also: Decision rule; Human-worth score prohibition; Scientific verdict axes.

Model recovery: A simulation-based check of whether an analysis can distinguish candidate model families when their generating identities are known. Parameter recovery separately tests estimation of parameters under a specified generator. Both are evaluated under realistic observation conditions; success in one does not establish success in the other.

Momentary valence estimator: A versioned, context-bound operational estimator $\hat{v}_i(t)$ of a valence-rate-like contribution from declared observation channels under a measurement model. See also: Interval contribution estimate; Measurement model.

Motivation: The internal or external factors associated with action selection. A policy describes a mapping from information or state to actions; it should not be equated with the person's consciously experienced motive.

Multichannel estimate: A versioned, purpose-bounded estimate formed through an explicit model relating channels such as report, behavior, physiology, neural data, and context to a declared construct and estimand. See also: Error budget; Observation channel and observation model; Provenance.

N

NAE: See No adequate estimate.

Narrowing: A transparent reduction in the surviving reference class, target branch, measurement version, endpoint class, or claim strength after a limitation or adverse result. A narrowed claim receives a new scope or version and does not count as confirmation of the unchanged original. See also: Reference-class narrowed; Verdict.

Near-zero band: A finite interval around zero used to describe limited measurement resolution, uncertainty, or practical closeness.

Negative control: A condition, timing, procedure, or outcome that should not show the proposed signature if an explanation is specific. Matched non-horizon tasks, sham procedures, impossible timing, and alternate outcomes can expose generalized arousal, demand, leakage, selection, or analytic flexibility. See also: Candidate signature; Nonunique explanation; Safe perturbation.

Net valence: Positive phenomenal mass minus negative phenomenal mass within a specified stream and domain.

Neutral origin: The identified zero point separating positive from negative phenomenal valence in the adopted representation. Its location must be justified independently rather than chosen to make terminal records balance.

No adequate estimate: A prespecified inference status used when the target or estimand is undefined or cannot be identified or bounded under the available data and maintained assumptions, including because the evidence needed to identify it is ethically unavailable. A valid but unresolved analysis is Inconclusive evidence; evidence from a failed protocol, consent, endpoint, measurement, provenance, computation, or participant-protection gate is Unusable evidence. The reason and affected estimand are recorded. See also: Abstention; Decision rule; Model ambiguity.

No moral desert, debt, or payment: PTVC does not assign moral entitlement or an obligation to suffer. Under exact closure, a genuinely nonzero accumulated balance requires an equal and opposite signed remainder within the same stream. This is conditional accounting, not a known personal balance, a fixed quota of suffering, or a method for choosing future rewards. Care and responsibility retain their independent ethical grounds. See also: Constraint without a judge; Desert-based fairness; Outcome independence.

No harm for proof: The prohibition on inflicting, prolonging, intensifying, staging, ignoring, or leaving suffering unrelieved in order to test, demonstrate, dramatize, or promote PTVC. It covers unsafe self-testing, delayed care, coerced disclosure, vulnerable participants, and plausible synthetic suffering; principled

abstention remains available. See also: Care before proof and interpretation; Ethical floor; Stop authority.

No-neutrality-by-fiat: The rule that origin, baseline, sign, scalarization, endpoint, smoothing, missingness treatment, weights, and exclusions may not be chosen after outcome inspection to manufacture apparent neutrality. Deviations require visible exploratory or sensitivity status.

No personal endpoint testing: No person should try to test the Law by approaching danger, loss of consciousness, treatment refusal, self-harm, deprivation, violence, or death. Such acts create risk but cannot identify the bearer, complete domain, endpoint, missingness process, origin, or measured-to-latent bridge required for valid evidence. See also: Care before proof and interpretation; No harm for proof; Truth-level endpoint.

No-score rule: See Human-worth score prohibition.

Noise floor: The smallest distinction an observation system can reliably resolve under declared conditions. Rounding, censoring, preprocessing, and random or systematic noise help determine that resolution. See also: Calibration; Error budget; Finite-band compatibility and measured equivalence.

Noncompensable risk: The modeled risk that, under specified conditions and uncertainty, an action or state closes essential paths faster than remaining channels can reopen them. It is a route- or continuation-level diagnostic. See also: Compensability; Future optionality; Viability kernel.

Nonidentification: A condition in which different target values or models are equally compatible with the available observations

and assumptions. Reporting a single precise estimate under nonidentification creates false certainty.

Nontransferability: The rule that valence belonging to one conscious stream cannot be credited to, debited from, or used to balance another stream.

Nonunique explanation: Several mechanisms generate the same observed trajectory, endpoint concentration, rebound, report, compression, or return pattern. Additional observations, interventions, or model comparisons can investigate the differences among them. See also: Candidate signature; Mean reversion; Model ambiguity.

Null model: A specified alternative under which the distinctive PTVC target or route is absent. A useful null model makes concrete predictions and is tested alongside serious rivals rather than serving as a vague absence of effect.

Null outcome: A result that does not support the tested effect or distinction under the declared analysis. A null outcome may be informative, inconclusive, or adverse depending on power, identification, and the prespecified decision rule.

O

Objective functions versus experience: The distinction between a modeler's rule for ranking selected outcomes and the lived experience the model seeks to investigate. A measurement relation states which aspects of experience the chosen objective represents. See also: Model governance; Reinforcement-learning reward signal; Viability kernel.

Observation: A recorded report, behavior, physiological measure, neural signal, event, or other data item interpreted through the relevant observation model.

Observation channel and observation model: An observation channel is recorded evidence such as report, behavior, physiology, neural activity, or context. The observation model states how a latent construct and nuisance conditions can generate or alter that output, including sampling, timing, error, and channel failure. Neither the channel nor its model is the experience. See also: Informative missingness; Measured-to-latent bridge.

Observation model: See Observation channel and observation model.

Observation process: See Observational process.

Observation shutdown: The process or protocol event in which one or more observation channels close because a device is removed or fails, a report is skipped or becomes impossible, privacy or consent changes, care takes priority, staff capacity changes, or contact is lost. Shutdown creates structured missingness and can mimic terminal compression; it is not the truth-level endpoint. Complete shutdown may determine the end of observation. See also: End of observation; Informative missingness; Observational process.

Observation window: The finite span during which a study records data. Its relation to the complete stream is specified through the observed interval, any missing prefix or remainder, and the applicable measurement and endpoint bridges. See also:

Complete strict-pre-endpoint felt history; End of observation; Protocol endpoint.

Observational process: The mechanism by which latent experience and system states become recorded data, including measurement error, reporting, censoring, sampling, and missingness. It must be modeled separately from the target process.

Open-set inadequate: See Scientific verdict axes.

Opponent process: A temporal counterprocess in which an initial state recruits a later afterstate whose strength and timing may change with repetition, helping explain relief, recovery, tolerance, withdrawal, fatigue, and rebound. See also: Hedonic adaptation; Local return and terminal closure.

Origin problem: The inability to interpret exact zero unless the neutral origin is independently identified. A constant shift in a valence rate changes the accumulated ledger in proportion to included exposure, so post hoc centering can manufacture closure. See also: Independent neutral-origin calibration; Neutral origin.

Outcome independence: Care, rights, dignity, privacy, repair, ordinary causation, and independently supported knowledge retain their own grounds across changes in PTVC's scientific status. See also: Care before proof and interpretation; Ethical floor; Strong-form defeat.

Ownership: The prospectively specified rule assigning each qualifying phenomenal event or contribution to its bearer without omission, double counting, or cross-stream transfer. Ownership

answers whose experience a contribution is, while routing governs continuity through time and transitions; data ownership is a different legal and institutional question. See also: Bearer; Nontransferability; Routing.

P

Palliative sedation: A clinically governed process used proportionately to relieve otherwise refractory suffering, with varying drugs, depths, durations, routes, and circumstances. It is treatment rather than a standardized consciousness manipulation or clean terminal signal, and research must never delay relief, alter indicated titration, or prefer observability over comfort. See also: Comfort-first observation; End of observation; Protocol endpoint.

Partial identification: A condition in which the evidence restricts a target to a set or interval while leaving more than one value possible under the maintained assumptions.

Partial observation: A decision or inference setting in which the relevant latent state is not fully observed. Closure claims in such settings must respect common-action and selection requirements rather than assuming state-specific knowledge.

Participation without surveillance: A research architecture that keeps participation voluntary through purpose separation, data minimization, practical privacy controls, ordinary nonparticipation, correction, and eventual deletion. A person may contribute selected observations without granting permanent access, hidden inference, public ranking, or ownership of the life described. See also: Consent, refusal, and withdrawal; Data

minimization and deletion; Record-regime separation and no-dossier rule.

Past-fixity: Later experience may add to the future portion of a stream but cannot retroactively change what earlier experience was. This temporal discipline permits formal future complementarity while forbidding moral or phenomenal repricing of the past, including claims that later joy makes earlier abuse acceptable. See also: Outcome independence; Formal or latent phenomenal ledger; Reserve coupling.

Pathwise closure: Closure evaluated on complete individual trajectories. The stated quantifier distinguishes one realized path, probability-one closure, support containment, and closure of every admissible path. See also: Complete strict-pre-endpoint felt history; Probability-one closure; Set-universal closure.

Per-stream claim: A claim evaluated for each separately routed maximal qualifying conscious stream as its own unit rather than for a population, household, country, species, or group average. See also: Intrastream; Maximal qualifying conscious stream; Population mean and expected neutrality.

Per-stream terminal valence closure (PTVC): On the primary additive branch, the candidate boundary hypothesis that every separately routed maximal qualifying conscious stream in the declared reference class carries a finite signed phenomenal measure over its complete strict-pre-endpoint domain whose positive and negative Jordan masses are exactly equal; equivalently, its terminal signed residual is zero.

Phenomenal ledger: See Formal or latent phenomenal ledger.

Phenomenal magnitude: The represented amount of positive or negative felt experience assigned to a stream under a declared valence representation. Positive and negative gross magnitudes are evaluated separately from their signed net difference. See also: Felt valence; Gross experience; Positive and negative phenomenal magnitudes.

Phenomenal measure: A measure assigned to conscious experience itself for purposes of representing valence across a stream. It is conceptually distinct from clock time, event count, memory strength, neural activity, or report frequency.

Phenomenal time: A measure or ordering of experienced duration within a conscious stream. It need not coincide uniformly with external clock time and should not be inferred solely from the number of observations.

Phenomenal valence: The felt positive, negative, or neutral character of conscious experience considered as phenomenal rather than merely behavioral, computational, motivational, or linguistic.

Policy: A rule mapping available information or state to actions, possibly probabilistically.

Policy mixture: A probabilistic combination of actions or component policies. Average cancellation across components does not establish exact closure on each realized path.

Policy-mixture exactness: The requirement that exact closure under a randomized policy be secured on the relevant realized branches, not merely in expectation through cancellation among closing and nonclosing actions. Under the applicable support

conditions, every action retained with positive probability must be closure-compatible.

Policy-support condition: A condition requiring the support of a selected policy to contain only actions or continuations compatible with the declared exact closure claim. Assigning very small but positive probability to a nonclosing option does not satisfy exact support restriction.

Population mean and expected neutrality: Expected neutrality means that the terminal residual has expectation zero under a named probability law and information set, when that expectation exists. A sample or cohort mean estimates a population quantity under further assumptions. Either can arise through cancellation while every individual stream remains nonzero. See also: Per-stream claim; Probability-one closure; Set-universal closure.

Portable safeguard: A local, operative protection placed on the exact surface or workflow where a high-risk fragment can be encountered or acted upon. A distant disclaimer is insufficient when the nearby claim could imply personal prediction, cross-person balancing, human scoring, machine ownership, or delayed care. See also: Detachable-excerpt ceiling; Public-artifact limit; Stop authority.

Positive and negative phenomenal magnitudes: On the primary additive branch, the nonnegative complete-stream totals of the positive and negative parts of the signed phenomenal measure. PTVC proposes that these two quantities are finite and exactly equal within each qualifying stream. See also: Finiteness; Gross experience; Terminal residual.

Positive-probability completion: Under a prospectively frozen continuation regime and its declared induced law, the completion event has probability strictly greater than zero, optionally conditional on a stated prefix and admissible action or policy. This weak reachability claim does not imply likelihood, probability one, an attaining policy, support containment, or every-continuation closure. See also: Modal reachability ladder; Probability-one closure; Reachability.

Positive rescaling: Multiplication of a valence representation by a positive constant. Exact zero and sign are preserved under positive rescaling, whereas shifting the origin or applying arbitrary nonlinear transformations can change the target.

Precaution without personhood: Proportionate, revisable protection under uncertainty, including monitoring, reducing avoidable harm, preserving reversibility, pausing, or redesigning when plausible welfare risk is serious. Protection can strengthen while consciousness, ontology, law, and personhood remain unresolved, without treating precaution as proof. See also: Candidate sentience escalation; Comparative consciousness evidence; Unknown category.

Prediction error and valence: The distinction between mismatch under a predictive model and the positive, negative, or neutral quality of felt experience. Surprise and valence can interact without being the same quantity: expected harm can remain painful, surprise can be welcome, and mean-zero prediction error does not imply neutral experience. See also: Felt valence; Interoceptive prediction; Predictive processing.

Predictive fit versus explanation: Predictive fit is a model's ability to forecast held-out observations. Explanation additionally relates its variables and dependencies to the process that generated those observations. See also: Candidate signature; Held-out transport; Model recovery.

Predictive processing: A family of accounts in which perception, bodily interpretation, and action depend on generative models, expected signals, prediction errors, uncertainty, and estimated reliability. See also: Active inference; Interoceptive prediction; Prediction error and valence.

Preregistration: Public preregistration records the hypotheses, estimands, gates, measures, exclusions, models, analyses, and decision rules in a timestamped external registry with a persistent identifier before the relevant outcomes are inspected. An internal advance specification is prospectively specified, not publicly preregistered. Deviations remain visible and receive the status appropriate to their timing and purpose. See also: Protocol-fixed and prospectively specified; Registered Report.

Preterminal compression: A candidate signature in which within-stream dispersion narrows before a declared endpoint under a prospectively specified relation. Comparison models include sedation, frailty, sensor floors, selective missingness, pooling, mean reversion, treatment, and endpoint conditioning. See also: Candidate signature; Endpoint conditioning; Noise floor.

Primary additive branch: The branch that represents qualifying phenomenal valence as one finite signed scalar measure with a

justified neutral origin and additive consistency across disjoint parts of a stream. Its positive and negative parts yield complete-stream magnitudes, and their difference yields the terminal residual. See also: Bipolar branch; Generalized Terminal Neutrality; Signed representation.

Probability-one closure: Closure with probability one under a specified probability law, and under a specified policy when the model includes one. It permits measure-zero exceptions and is distinct from set-universal closure, which requires closure for every admissible realization.

Procedural fairness: Fairness in rules and decision paths, including knowable standards, voice, understandable reasons, consistency, limited bias, respectful treatment, correction, and appeal. See also: Desert-based fairness; External justice; Consent, refusal, and withdrawal.

Process evidence: Evidence about a specified route, mechanism, state transition, or selection process.

Protocol endpoint: A prospectively specified operational event, adjudicated interval, stopping region, or governed proxy used by a study to locate or approximate a relevant boundary. It may be usable, uncertain, revisable, or ethically unavailable and is distinct from both the truth-level endpoint and end of observation. See also: End of observation; Endpoint bridge; Truth-level endpoint.

Protocol-fixed and prospectively specified: A choice is protocol-fixed or prospectively specified when it is recorded in the study protocol before the relevant outcome is inspected. This

timing status does not guarantee that the choice is valid or the study succeeds, and it is not public preregistration without a prior external timestamped record. See also: Endpoint independence; Preregistration; Rival model and equal-budget comparison.

Provenance: The documented lineage of a trace or derived output, including who collected it, for what purpose and authority, under which prompt, device, language, setting, burden, transform, model version, missingness state, uncertainty, and correction history. Provenance limits interpretation rather than merely decorating a record. See also: Data as trace; Public-artifact limit; Record-regime separation and no-dossier rule.

Psychometrics: The theory and practice of measuring latent psychological or experiential constructs, including reliability, validity, scaling, response processes, and measurement invariance.

PTVC: See Per-stream terminal valence closure (PTVC).

PTVC core notation: On the primary additive branch, v_i is the finite signed phenomenal measure on the declared measurable space; V_i^+ and V_i^- are its Jordan masses; $L_i^{\text{phen}}(t)$ is the latent phenomenal ledger; T_i^* is the truth-level endpoint; and $Z_i^* = v_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^* -)$ is the terminal residual. On the absolutely continuous chart, $r_i(t) = g_{\theta,i}(t)q_i^{\text{exp}}(t)v_i^{\text{real}}(t)$. The relation $L_i^{\text{rc}}(t) = c_{LU}U_i(t)Y_i(t)$ is a candidate realization, and hatted quantities are operational estimators, not identities with the latent target. See also: Candidate realization; Formula-status code; Phenomenal ledger.

Public access-permission distinction: Public availability for accountability, education, criticism, scholarship, or communication does not itself authorize profiling, consequential judgment, redistribution, linkage, or general model training. Visibility is not surrender, de-identification is not guaranteed anonymity, and access for one purpose is not standing authority for another. See also: Consent, refusal, and withdrawal; Data as trace; Record-regime separation and no-dossier rule.

Public-artifact limit: Every detachable project surface, including a summary, caption, guide, model output, slide, translation, or post, preserves an accurate account of its source, result, audience, reviewer, and correction route. See also: Detachable-excerpt ceiling; Portable safeguard; Correction and withdrawal.

Public safety floor: Care, dignity, rights, welfare, relief, refusal, and protections under uncertainty guide every stage of the research. Participation remains voluntary, care takes priority, and scientific outputs describe study questions and findings. Human-worth scoring, created suffering, compulsory belief, personal endpoint testing, delayed care, and hidden-object administration are prohibited.

Q

Qualifying conscious stream: A conscious stream that meets the declared scope, identity, routing, inclusion, and validity conditions for the target being tested. The criteria must be stated independently of whether the stream appears to close.

Queue model: A constructive candidate route that represents unresolved signed valence and asks whether admissible future transitions can remove the remaining residual.

Queue System: Book 1's accessible name for the Queue model, a possible action-mediated route to closure.

R

Random-endpoint obstruction: Under a specified reference law and declared positive-probability eligible event, suppose a jointly measurable pre-endpoint ledger extension exists on an interval I , endpoint time T is independent of the sigma-field generated by that ledger path on I , and T has a conditional Lebesgue density positive almost everywhere on I . If the ledger is zero at T with probability one, then it is zero with probability one for Lebesgue-almost every time in I . A ledger that is nonzero with positive probability on a positive-measure subset of I is therefore incompatible with probability-one closure on this branch. State-coupled timing, endpoint atoms, singular support, continued experience, routing changes, and endpoint-inclusive value require separate analysis.

Reachability: Whether at least one admissible continuation from a state can reach the declared target. Reachability does not guarantee selection, common feasibility under partial observation, probability-one success, or universal success.

Record-regime separation and no-dossier rule: Research, clinical, employment, educational, civic, legal, family, spiritual, platform, and public records must retain separate purposes, identifiers, access roles, retention rules, consequences, and

correction paths rather than being joined into a master profile. Inferred reconstruction can breach this boundary even without copying a forbidden field. See also: Data minimization and deletion; Human-worth score prohibition; Public access-permission distinction.

Reference class: The prospectively declared domain over which a claim and its quantifier range, specified by eligible streams or cases together with the representation, endpoint regime, population, setting, and measurement architecture. It is not automatically the enrolled sample or analyzed cases; outcome-driven exclusions define a new claim. See also: Held-out transport; Maximal qualifying conscious stream; Set-universal closure.

Reference-class narrowed: See Scientific verdict axes.

Registered Report: A publication format in which a study's question, methods, and analysis plan receive peer review before results are known, with publication not conditioned on obtaining a preferred outcome.

Reinforcement-learning reward signal: A formally specified signal used to update action values or policies from outcomes. Its interpretation follows the model's definition, which may concern relief, safety, access, or another specified result. See also: Active inference; Objective functions versus experience; Predictive processing.

Remaining capacity: The set or amount of admissible future change still available before an endpoint. A closure route must

show that remaining capacity is sufficient and selectable under the actual information and support constraints.

Repair is not moral payment: Responding to actual harm through safety, truth, apology, restitution, accountability, prevention, and remedy. These actions address present needs and future protection. Their genuinely felt effects enter the relevant stream's balance, but preventing a particular harm does not require that harm to reappear. PTVC fixes a terminal difference, not a quota of suffering. See also: External justice; No moral desert, debt, or payment; Outcome independence.

Replication: A new test designed to determine whether a result, measurement relation, or model distinction recurs under prespecified conditions. Replication should include adverse and null outcomes rather than only successful demonstrations.

Representation: A mathematical or conceptual encoding of phenomenal valence, time, state, or route that preserves the distinctions required by the stated research target.

Reserve: A route-level variable representing remaining capacity, flexibility, or corrective potential before an endpoint.

Reserve coupling: A candidate relation between accumulated residual and remaining capacity or future transition structure.

Response shift: A change in a person's standards, priorities, values, or understanding of a question after illness, disability, adaptation, treatment, trauma, grief, or aging, altering what the same response scale means over time. A changed answer can reflect changed experience, changed measurement, or both. See

also: Calibration; Longitudinal research trace; Measurement invariance.

Reversibility and return rule: A prospectively specified rule under which reliable return of unified access can defeat an earlier classification as a final endpoint, while nonreturn alone cannot prove absence of experience or identify the exact ending. Protocols must state return evidence, revision timing, and its effect on classification. See also: Endpoint bridge; Low-unity state; Protocol endpoint.

Reversible pause: See Reversibility and return rule.

Rival dominance: A verdict that an adequate protocol-fixed rival explains the declared signatures with better held-out performance and fewer strained assumptions under common data, flexibility, and scoring rules. It must state which route, component, or target interpretation loses standing and does not by itself decide unrelated branches. See also: Rival-favored; Theory-level defeat.

Rival-favored: See Scientific verdict axes.

Rival model: A concrete alternative explanation capable of generating overlapping observations. Strong tests compare PTVC targets and routes against serious rivals involving measurement, reporting, adaptation, selection, endpoint structure, and ordinary statistical effects.

Rival model and equal-budget comparison: A rival is a serious alternative able to generate the relevant pattern without the target or proposed route. Equal-budget comparison gives candidates and rivals common raw variables, observation and missingness pipelines, data splits, scoring rules, and comparable prospectively

bounded flexibility, while allowing that no declared model is adequate. See also: Model ambiguity; Model recovery; Scientific verdict axes.

Route: A proposed process or construction through which a target might be reached, with its own state variables, assumptions, and predictions.

Route-failed: See Scientific verdict axes.

Route-favored: See Scientific verdict axes.

Route retirement: A prespecified decision to stop treating a proposed mechanism or route as viable after decisive failure, poor recovery, nonidentification, or superior rival performance. Retiring a route need not settle the target hypothesis.

Route state: A latent or observed variable defined within a candidate mechanism and used to describe its dynamics.

RouteStatus: See Scientific verdict axes.

Routing: The prospectively specified treatment of continuation, pause, restoration, fission, fusion, copying, replacement, and unresolved continuity, including where prior and later phenomenal contributions are assigned. Routing governs transitions through time and is distinct from ownership, compensation, and causal influence. See also: Fission-and-fusion routing rule; Ownership; Stream identity.

S

Safe perturbation: A reversible, low-burden intervention that changes a declared local feature, such as feedback, delay, expectation, opportunity, or a mild stimulus, to test a named

causal contribution. Its meaning depends on target engagement, spillovers, controls, stopping, and the stated causal estimand. See also: Candidate signature; Negative control; No harm for proof.

Scalarization: A prospectively justified mapping from a richer or multidimensional experiential state to one signed scalar quantity for a named purpose and domain. It is a substantive bridge rather than harmless formatting, and weights selected because they produce closure manufacture the result. See also: Bipolar branch; Generalized Terminal Neutrality; Signed representation.

Scientific verdict axes: Four non-overwriting reporting fields preserve different conclusions: TargetStatus records what the evidence says about the declared target branch; RouteStatus records the status of a candidate route; ComparatorStatus records the absolute adequacy and relative performance of declared rivals; and InferenceStatus records whether the target is identified, partially identified, inconclusive, unusable, or unavailable. Labels such as Target-adverse, Route-failed, Route-favored, Rival-favored, Model-ambiguous, Open-set inadequate, Reference-class narrowed, and No adequate estimate must be assigned only to the relevant field, with branch, reference class, and inferential type retained. No field overwrites another. See also: Decision rule; Model ambiguity; Target-route distinction.

Scope firewall: A statement of conclusions that do not follow from PTVC. The hypothesis does not imply free will, monotonic movement toward zero, frequent zero crossings, conscious ledger awareness, that every urge serves closure, that pain is morally owed payment, or that suffering is morally deserved.

Selection condition: A requirement governing which available action or continuation is actually chosen. The existence of a closing option is insufficient when the selection process can retain nonclosing options with positive support.

Separately routed: Governed by a trajectory or transition structure that is not pooled with another stream for purposes of the terminal target. Separate routing supports the rule that one stream cannot balance another.

Set-universal closure: Closure for every admissible realization, path, or member of the declared set. It is stronger than probability-one closure because measure-zero exceptions are not permitted.

Severe test: A test that had a high probability of exposing a claim's error if the claim were false in the specified way. A result is not strong evidence merely because it is statistically significant or compatible with the preferred model.

Shortening horizon: A reduction in remaining time, opportunity, revision, access, or other route-relevant room for future action, affecting the continuations that remain available. See also: Future optionality; Reachability; Reserve.

Signed representation: A representation that assigns positive and negative directions relative to a justified neutral origin and supplies the declared scale and aggregation rules needed for the target branch. See also: Felt valence; Neutral origin; Primary additive branch.

Signed residual: The cumulative positive phenomenal mass minus the cumulative negative phenomenal mass over a specified

domain. The terminal signed residual is evaluated at the truth-level stream endpoint.

Skepticism as dignity: The protection of organized doubt, exact criticism, nonbelief, and refusal as honorable ways of engaging with a consequential public hypothesis. Comprehension is not assent, and care, standing, participation, intelligence, belonging, or moral worth may never depend on enthusiasm for the Law. See also: Meaning as optional; Falsifiability; Outcome independence.

Spiritual resonance is not scientific support: A relation of meaning occurs when a person or tradition recognizes a neighboring pattern. A relation of evidence requires a defined hypothesis to survive valid measurement, serious rivals, uncertainty, and possible defeat. Tradition, ritual, beauty, testimony, dream, prayer, or transformation may matter without validating PTVC. See also: Comparison without forced equivalence; Meaning as optional; Falsifiability.

State-coupled timing: An endpoint branch in which endpoint timing or hazard depends on prospectively specified current biological, consciousness, ledger, or route-state variables while the endpoint ontology and adjudication remain fixed independently of desired closure. Dependence is allowed, but the state relation cannot be invented after outcome inspection. See also: Endpoint atom; First-hitting and absorption routes; Random-endpoint obstruction.

Stop authority: Preassigned power held by an accountable person or independent reviewer to pause a study, system, publication, or artifact when a declared safety or rights boundary

is crossed. Effective authority can interrupt activity, contain harm, trigger care, and require correction or withdrawal rather than merely advise. See also: Correction and withdrawal; No harm for proof; Portable safeguard.

Stream: In formal PTVC contexts, shorthand for a Maximal qualifying conscious stream. In broader discussion, see Conscious stream.

Stream identity: The criteria that determine which experiences belong to the same conscious stream. Biological continuity, memory, body, legal identity, and data linkage may inform the analysis but are not automatically decisive.

Strong-form defeat: Defeat of the strong universal branch can occur by two routes. Deductive branch-matched refutation requires an independently verified maximal qualifying stream for which a deterministic identified set, guaranteed to contain the truth under the maintained bearer, representation, origin, domain, endpoint, observation, bridge, and inference assumptions, excludes the registered exact target. Governed scientific withdrawal can also follow repeated adequate conjunction failures or repairs that exhaust distinctive empirical content; only the first route is deductive refutation. Confidence or posterior objects support only their registered qualified adverse verdicts. See also: Falsifiability; Gate-valid counterexample; Set-universal closure.

Support condition: A restriction on the values, actions, paths, or outcomes that retain positive possibility under a model or policy.

Exact claims often require excluding nonclosing support, not merely making it unlikely.

Support containment: Under a declared topology, the support of the relevant probability law lies inside the completion event or target set. This topological modality is not every-continuation closure, positive probability at a point, or proof that an unsampled possibility is absent from support. See also: Every-continuation closure; Modal reachability ladder; Probability-one closure.

T

Target: The truth-level property a hypothesis asserts. For the primary PTVC branch, it is an exact zero terminal signed residual within each qualifying stream.

Target-adverse: See Scientific verdict axes.

Target modality and geometry: The exact mathematical form and quantifier of a declared target, including whether it is an exact singleton, finite band, distributional atom, probability-one event, support condition, every-continuation property, or set-universal claim. Evidence and defeat must be matched to that form rather than borrowed across nearby claims. See also: Exact terminal closure; Modal reachability ladder; Terminal residual compatibility object.

Target-route distinction: The target hypothesis states the terminal relation to be evaluated, while a candidate route proposes one optional process that might realize or reproduce its consequences. Evidence updates the object tested; a proved entailment specifies when a route result also changes a target

assessment. See also: Candidate route; Scientific verdict axes; Claim ceiling.

Target-route separation: See Target-route distinction.

TargetStatus: See Scientific verdict axes.

Terminal neutrality: Book 1's accessible term for a zero terminal signed residual over the whole qualifying stream. It does not mean that the final conscious moment is emotionally neutral.

Terminal residual: The signed total of phenomenal valence over a qualifying stream's complete strict-pre-endpoint domain, evaluated at its truth-level endpoint. The primary PTVC target states that this residual is exactly zero.

Terminal residual compatibility object: A branch-matched nonempty object representing residual values allowed by declared representation, endpoint, observation, missingness, bridge, and uncertainty assumptions. It retains its inferential type: deterministic identified set, confidence region, credible region, or sensitivity envelope. Only a deterministic guaranteed truth-containing object supports deductive exclusion; confidence and Bayesian objects license only registered coverage- or posterior-qualified conclusions. A point estimate is not such an object. See also: Finite-band compatibility and measured equivalence; Partial identification; Target modality and geometry.

Terminal signed residual: See Terminal residual.

Terminal window: A finite observed period near a presumed endpoint, used to investigate specified temporal or process relations.

Testimony-like output: Language or behavior from an artificial system that has the form of a first-person report but is generated under a particular model, prompt, memory state, tool set, training regime, and sampling process. It is evidence about configured output, not self-authenticating testimony that a subject feels what is said. See also: AI sovereignty; Candidate sentence escalation; Theory-derived indicator.

The Law: See Law of Fairness.

Theory-derived indicator: An observable property selected because a stated theory connects it to recurrent processing, global availability, higher-order representation, agency, embodiment, metacognition, or another proposed feature of consciousness. See also: Candidate sentence escalation; Comparative consciousness evidence; Testimony-like output.

Theory-level defeat: A branch-specific verdict reached when a declared target is contradicted under its own prerequisites, or when repeated adequate conjunction failures, persistent rival superiority, or outcome-dependent repairs exhaust the claim's distinctive empirical content. It is distinct from local route failure and from nonidentification; the defeat record must state the branch, evidence, assumptions, and surviving claims. See also: Strong-form defeat; Route retirement; Rival dominance.

Truth-containing object: An inferential object linked to the target under a declared guarantee or calibration. A deterministic identified set may be guaranteed to contain the truth under maintained assumptions; a confidence region has repeated-sampling coverage and a credible region has posterior probability,

neither of which is an unconditional realized-case guarantee. Excluding zero licenses only the conclusion appropriate to the object's inferential type. See also: Terminal residual compatibility object.

Truth-containing residual set: See Terminal residual compatibility object.

Truth-level boundary: See Truth-level endpoint.

Truth-level endpoint: The actual boundary of the qualifying stream in the target model. It is conceptually distinct from any observed marker, last report, terminal window, loss of contact, or biological event used to estimate it.

U

Unity gate: See Experienced exposure and unity gate.

Unity Index: A proposed research estimator that organizes separately reported evidence about access, report, memory, agency, embodiment, continuity, and coherence under a declared construct, with explicit assumptions and uncertainty. See also: Blinded bearer adjudication; Human-worth score prohibition; Unknown category.

Universal branch: See Set-universal closure.

Universal quantifier: A claim applying to every member of a declared class. Because the primary PTVC branch is per-stream and universal over qualifying streams, a valid adverse case cannot be repaired by averaging across the class.

Unknown category: A reason-coded classification outcome used when available evidence cannot justify one bearer, consciousness,

or boundary classification over serious alternatives under declared rules. Unknown identifies unresolved classification; it differs from a negative finding, equal probability, zero, and No adequate estimate. See also: Bearer; No adequate estimate; Precaution without personhood.

Unusable evidence: Evidence for which a necessary protocol, consent, endpoint, measurement, provenance, computation, or participant-protection gate failed. A statistic may exist, but it cannot update the affected claim. The report must identify the failed gate, affected estimand, and any permitted narrower use. It is distinct from Inconclusive evidence and No adequate estimate.

V

Valence: The felt positive, negative, or neutral character of conscious experience.

Valence density: A representation of signed phenomenal valence per unit of an adopted base measure, such as phenomenal time. Its interpretation depends on the validity of both the valence scale and the base measure.

Valence ledger: Book 1's metaphor for the cumulative signed phenomenal measure within one stream.

Verdict: The declared evidential conclusion produced by a prespecified decision architecture, such as support, adverse result, narrowing, abstention, or route retirement. A verdict must state its target and scope.

Viability: The possibility of remaining within declared constraints over the specified horizon under an admissible

continuation or policy, with any probabilistic guarantee stated explicitly. Maintaining access to closure is an additional completion-specific requirement when it is included among those constraints.

Viability kernel: The set of modeled states from which at least one admissible continuation satisfies a route's declared constraints. It marks where a future remains formally workable but does not mean that a continuation is likely, humane, accessible, desirable, or supported by resources a person actually has. See also: Compensability; Noncompensable risk; Reachability.

Z

Zero-containing interval: An uncertainty or identification interval that includes zero. Any nonzero members remain alternative values under the interval's stated interpretation.

Zero crossing: A time at which a cumulative or instantaneous signed quantity passes through zero. PTVC does not require monotonic movement, frequent zero crossings, or any particular number of crossings before the endpoint.

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The entries below provide bibliographic destinations for the Author/Year citations used throughout the manuscript. Author and institutional creator names appear in bold. Distinct works by the same author in the same year retain their manuscript suffixes. DOI identifiers appear as plain text so the entries remain readable in print. Readers following a claim beyond this book should consult the cited edition, the proposition it supports, and the source's stated scope.

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About the Author

Stephen Takowsky has spent his life studying complex systems: how rules shape them, how people cooperate and compete within them, and how hidden constraints can change what becomes possible. His path has crossed finance, competitive strategy, professional esports, and the leadership of global online communities, ultimately leading to the Law of Fairness project, a decades-long effort to investigate whether fairness itself may reflect a deeper structure of conscious life.

Competing as “MrGameTheory,” Takowsky reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues while pursuing a career in finance at Morgan Stanley in Beverly Hills. In 2016, aXiomatic, an ownership group led by Peter Guber and Ted Leonsis and backed by investors including Magic Johnson, Steve Case, and Tony Robbins, acquired a controlling interest in Team Liquid. Following the acquisition, Team Liquid selected Takowsky to captain its newly created Civilization team, later recognized by Guinness World Records as the first Civilization esports team. Across thousands of games, Civilization became an unconventional education in strategy: understand the whole system, think far beyond the next move, anticipate intelligent opposition, and keep searching when the obvious paths fail.

Strategy then became community leadership. Takowsky spent seven years at 505 Games, serving as Senior Digital Content & Engagement Manager and leading global communities for Terraria, which has surpassed 70 million copies sold, as well as Ghostrunner, WUCHANG: Fallen Feathers, Eiyuden Chronicle: Hundred Heroes, and other titles. He also founded the United

Servers of Discord Alliance, bringing together leaders from major communities built around Roblox, Pokémon, Star Wars, League of Legends, Grand Theft Auto, Warframe, Animal Crossing, and other properties. Earlier, he served as administrator and producer for TF2Outpost, a major Steam digital-goods marketplace. Across these very different environments, he kept returning to the same insight: better structure can expand what people are capable of building together.

Throughout those years, one question kept following him: could the apparent unfairness of life conceal a deeper structure? That question became the Law of Fairness. Takowsky began developing the idea more than twenty years ago while studying Religious Studies and Philosophy/Law and Society at the University of California, Riverside, and gave it its first public form in his 2009 book, *Of Grandeur*. In the years that followed, he sought out academics and serious thinkers across many disciplines, asking for insights, criticism, and sources that could help separate fact from speculation. What began as a philosophical intuition gradually became a research program built to face mathematics, measurement, competing explanations, and evidence.

The ten-book Law of Fairness series follows that question outward, from consciousness and scientific testing into ordinary life, meaning, and civilization. If the Law is true, then there may be no “free happiness” in the sense of positive experience remaining unresolved by the lifetime balance the hypothesis predicts. That possibility raises a profound practical question: what if humanity has been pursuing the wrong model of utopia?

Rather than organizing life around the endless pursuit of pleasure, the series explores whether greater self-moderation, less needless suffering, and better-designed institutions could help remove some of the forces that keep people and societies trapped in conflict, corruption, and instability.

Takowsky's fifty-year horizon is designed so that building a better future does not require tearing apart the present. It gives institutions and communities time to test reforms, build infrastructure, prepare future leaders, improve technology, and correct mistakes before they spread. Just as importantly, it gives today's generation the chance to be remembered for beginning something larger than itself: laying the foundations for a more transparent, peaceful, and prosperous world it may never live to see.

The project's future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers willing to bring rigor, imagination, and courage to one of humanity's oldest questions. The next breakthrough may come from a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. *The Law of Fairness* began as one person's question. Its next chapter belongs to everyone willing to help discover what is true and build what that knowledge makes possible.

Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.

